EDUCATIVE JEE (MATHEMATICS)

Text of Second Edition

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Chapter 1

COUNTING PROBLEMS

Our mathematical education begins with counting and so it is just as well that the first chapter be devoted to counting problems. Childhood association often gives the impression that counting problems are essentially trivial. This is far from the case. The central problem in counting is to express the cardinality (i.e., the number of elements) of a given (finite) set, say S, in terms of the cardinalities of some more familiar or standard sets. This set S is sometimes a 'material' set, such as the set of all students in a class. But more frequently, it is a more 'abstract' set such as the set of all pairs of students or the set of all the ways to arrange them in a certain way or the set of certain functions from one set to another. As a result, problems of permutations and combinations come under counting. The Main Problem given here is of this type. Counting problems can also be posed as problems in probability. This will be taken up in Comment No. 4.

The four basic elementary counting techniques, viz., decomposition, complementation, products and transformation are given in Comment No. 2. Although these techniques are elementary, they work wonders when applied ingeniously as will be illustrated in Comment No. 6. There are also more advanced counting techniques in which the desired cardinality occurs as the coefficient of a suitable power of a variable, say x, in some expression involving x. The basic idea will be explained in Comment No. 11. The remaining comments discuss other related topics, such as Venn diagrams, double counting, incidence relations and the principle of inclusion and exclusion.

In Comment No. 12 we briefly discuss mathematical puzzles. Although puzzle type questions are not asked in the conventional examinations, they make the subject lively.

In the JEE, counting problems are asked usually as short questions. However, they can form a part of a long question such as a probability problem.

Main Problem : The number of permutations of length 6 of 9 things, 5 of which are alike and the rest all different is .....

First Hint: Classify the permutations according to the number of like things in them.

Second Hint: Show that the number of permutations in which r things are alike is $\binom{4}{6-r}\frac{6!}{r!}$.

Solution: For a set X, denote by $|X|$ the number of elements in X. Let S be the set of all permutations of length 6 of 9 things out of which 5 are alike and the remaining 4 are all distinct. Each such permutation must contain at least two of the 5 like things. For $2 \leq r \leq 5$, let S_r be the set of those permutations (in S) which contain exactly r like things. Then the sets S_2, S_3, S_4 and S_5 are mutually disjoint and their union is S. So,

$$|S| = \sum_{r=2}^{5} |S_r|. \tag{1}$$

Fix r, $2 \leq r \leq 5$. A permutation in S_r has r like things and the remaining $6-r$ things come from the remaining 4 distinct things. These could be chosen in $\binom{4}{6-r}$ ways. ($\binom{4}{6-r}$ is also denoted by ${}^4C_{6-r}$, see (11) below.) Fix any one such choice. We now have r like objects and other $6-r$ distinct objects. If we distinguish the r objects (by putting marks on them, for example) then we have 6 distinct objects and these can be permuted in 6! ways. But in any one such permutation, if we reshuffle the marked objects, we get the same permutation of the original objects. This reshuffling can be done in $r!$ ways. So $\frac{6!}{r!}$ is the number of permutations in S_r for a given choice of $6-r$ of the distinct objects. Hence we get

$$|S_r| = \binom{4}{6-r}\frac{6!}{r!} \tag{2}$$

for $r = 2, 3, 4, 5$. A direct computation gives $|S_2| = \binom{4}{4}\frac{6!}{2!} = 1\times6\times5\times4\times3 = 360$ and similarly, $|S_3| = \binom{4}{3} \times 6 \times 5 \times 4 = 480$, $|S_4| = 6 \times 6 \times 5 = 180$ and $|S_5| = 4 \times 6 = 24$. Hence by (1), $|S| = 360 + 480 + 180 + 24 = 1044$. So this is the number of desired permutations.

Comment No. 1:

Although this problem is short and simple, it captures three of the four basic elementary counting techniques. In a typical counting problem, we need to find the number of elements (more formally known as the **cardinality**) of a finite set X. Superficially, the problem may ask to find the number of ways something can be done (such as putting balls into boxes or seating guests). But we can always paraphrase this in terms of finding the cardinality of a suitable set, say S (such as the set of all possible placements of the balls or the set of all

possible seating arrangements). Such a paraphrase does not necessarily simplify the problem. But it often enables you to focus on the essence of the problem. By subjecting the set S to some standard manipulations, it is often possible to find $|S|$, the cardinality of the set S. There are four such basic techniques. We briefly describe them here, one-by-one. It is to be noted, however, that usually, it is a suitable combination of techniques rather than a single technique that works best in a given problem.

(i) **Decomposition :** As the name suggests, here we decompose the set into a number of mutually disjoint subsets, say, $S_1, S_2, \ldots, S_k$ whose union is the set S. It is common sense that in such a case every element of S belongs to one and only one of the subsets and so the cardinality of S is the sum of the cardinalities of these subsets. In symbols,

$$|S| = \sum_{i=0}^{k} |S_i| = |S_1| + |S_2| + \ldots + |S_k| \tag{3}$$

More generally, if T is any subset of S, then a decomposition of S into subsets $S_1, S_2, \ldots, S_k$ induces a decomposition of T into subsets $T \cap S_1, T \cap S_2, \ldots, T \cap S_k$ and so analogous to (3) we get

$$|T| = \sum_{i=0}^{k} |T \cap S_i| = |T \cap S_1| + |T \cap S_2| + \ldots + |T \cap S_k| \tag{4}$$

It is very rare that decomposition itself gives the answer to a counting problem. It is usually only the first step. For example, we began the solution to the Main Problem by decomposing the set S into the subsets S_2, S_3, S_4 and S_5. The next task is to find the cardinalities of each of these subsets. For this, a variety of techniques may be needed. Sometimes we may have to decompose the subset S_i further. A decomposition which is obtained by further decomposing (some or possibly all) the subsets of some other decomposition is said to be a **refinement** of the latter or is said to be 'finer' than the latter. For example, let S be the population of a country. We can decompose S into the populations of the various states of that country to get a decomposition. If we further decompose some of these according to the districts, we get a finer decomposition. The same set can, of course, be decomposed in many different ways. And for a particular problem, one particular decomposition may be more advantageous than some other. A good decomposition is one where it is easy to find the cardinality of each subset and also to find their sum. A good decomposition may not always exist and even when it does, it may not always be obvious. In such cases, hitting the right decomposition is a thrilling experience. See Comment No. 6 for an excellent example of this.

A common form of a decomposition of a set is obtained by classifying its elements in terms of some attribute relevant to that problem. For example, in the solution to the Main Problem, we classified the permutations

according to the number of like things they contained. More generally, suppose $f : S \longrightarrow T$ is some function. Then we get a decomposition of S by classifying its elements according to the values of this function. That is, all elements on which the function f has the same value are put in the same class (or part) of the decomposition. Formally, suppose that the distinct elements of the set T are $t_1, t_2, \ldots, t_k$ (where $k = |T|$). For $i = 1, 2, \ldots, k$ we let S_i be the inverse image of the singleton set $\{t_i\}$. By definition, S_i consists of those elements $s \in S$ for which $f(s) = t_i$. (Unless the function f is onto, some of the S_i's may be empty. But that does not affect the truth of (3).)

(ii) **Complementary Counting :** Instead of decomposing the set S into smaller subsets, we can in fact enlarge it to a suitable superset, say T. Then we have a decomposition of T into the subsets S and $T - S$. (Here $T - S$ denotes the complement[1] of S in T, i.e., the set $\{t \in T : t \notin S\}$.) So by (3) above (or, by sheer common sense), $|T| = |S| + |T - S|$ and hence $|S| = |T| - |T - S|$. If the set T is cleverly chosen so that it is easy to find both $|T|$ and $|T - S|$, then we have a very easy method for finding $|S|$. As a simple example, suppose we want to find the number of arrangements of the letters of the word BANANA in which the two N's do not appear adjacently (JEE 2002). Let T be the set of all arrangements of the letters of the word BANANA. Then the arrangements of the desired type (i.e., where the two N's do not appear adjacently) constitute a subset, say S of T. As there are 6 letters and 3 of them are A while 2 more are N, it is easy to show (as we shall soon do) that $|T| = \frac{6!}{3!2!} = 60$. The subset $T - S$ consists of those arrangements in which the two N's appear adjacently. In that case we may as well treat NN as a single letter. We now effectively have 5 letters of which 3 are alike. So the number of their arrangements is $\frac{5!}{3!} = 20$. Put differently, $|T - S| = 20$. So, $|S| = |T| - |T - S| = 60 - 20 = 40$ is the number of desired arrangements.

As an example where we need both a suitable decomposition and complementary counting, suppose we want to find the total number of five-digit numbers divisible by 3 using the numerals 0, 1, 2, 3, 4 and 5 without repetition (JEE 1989). As usual, let S denote the set of all such numbers. Every member of S is obtained by excluding one of the numerals from 0 to 5 and permuting the remaining five. Because of the 'rule of three', in order for a number to be divisible by 3, the sum of the digits in it must be a multiple of 3. In the present case, $0+1+2+3+4+5 = 15$ is divisible by 3. So, for every member of S, the excluded numeral must be either 0 or 3. This gives a decomposition of S into two subsets, say A and B, where A consists of all five-digit numbers formed using the numerals 1 to 5 exactly once and B is the set of five-digit numbers using each of the numerals 0, 1, 2, 4 and 5 exactly once. Clearly every member of A corresponds to a

[1] The notation $T \backslash S$ is also common. Also when the ambient set T is understood, complement of S w.r.t. T is often denoted by symbols like S', $\bar{S}$, $\tilde{S}$ or $\neg S$. The notation $T - S$ is consistent with the fact that $|T - S| = |T| - |S|$.

permutation of the five symbols and so $|A| = 5! = 120$. However, to find $|B|$ we must exclude those permutations whose first entry is 0. (This is not given in the problem itself. But it is a standard convention that we do not write the leading zeros while writing a number. In counting problems, one has to know such standard conventions. Some of them have to be inferred by common sense. For example, when guests are to be seated on chairs, it is to be assumed that no two guests occupy the same chair simultaneously, even though the problem may not say so explicitly!) To find $|B|$, we use complementary counting. The number of all permutations of 5 symbols is 120. If a permutation has 0 in the first place, then the remaining four places can be filled in 4!, i.e., in 24 ways. So, $|B| = 120 - 24 = 96$. Hence finally, $|S| = |A| + |B| = 120 + 96 = 216$. This is the total number of numbers of the desired type.

Despite the simplicity of the problem we have given its solution in detail to illustrate the correct thought process. In writing the solution, the candidate is hardly expected to give so many details. Most of them can be done mentally. Such problems are generally asked only as multiple choice or 'fill into the blank' type questions. In numerical problems, the very fact that the candidate has hit the right answer is usually a sufficient indication that he has gone through the correct thought process.

(iii) **Products :** If X and Y are any two sets then their **cartesian product**, or simply the product, denoted by $X \times Y$, is defined as the set of all ordered pairs of the form (x, y) with $x \in X$ and $y \in Y$. In symbols, $X \times Y = \{(x, y) : x \in X, y \in Y\}$. Pictorially, if we represent the elements of X as points on the x-axis and those of Y by points on the y-axis, then the points of $X \times Y$ can be represented by points in the cartesian plane (whence the word 'cartesian' in the term). Suppose now that X and Y are both finite sets with m and n elements (say) respectively. Then we can decompose $X \times Y$ 'row-wise' or 'column-wise'. In the row-wise decomposition, all elements having the same y-coordinate lie in the same class, called a 'row'. Thus there are n rows with m elements each. Similarly, in the column-wise decomposition, there are m columns with n elements in each column. Either way we see that the cardinality of the product $X \times Y$ is the product of the cardinalities of X and Y. This explains the name 'product'. In symbols,

$$|X \times Y| = |X| \times |Y| \tag{5}$$

More generally, suppose S is any subset of $X \times Y$. Then we can get an analogous formula for $|S|$ by decomposing (or 'slicing') S. To be more specific, let the (distinct) elements of X be $x_1, x_2, \ldots, x_m$ and those of Y be $y_1, y_2, \ldots, y_n$. For each $i = 1, 2, \ldots m$, let $S_i = \{(x, y) \in S : x = x_i\}$. Equivalently, S_i is the intersection of S with the i-th column, $\{x_i\} \times Y$.

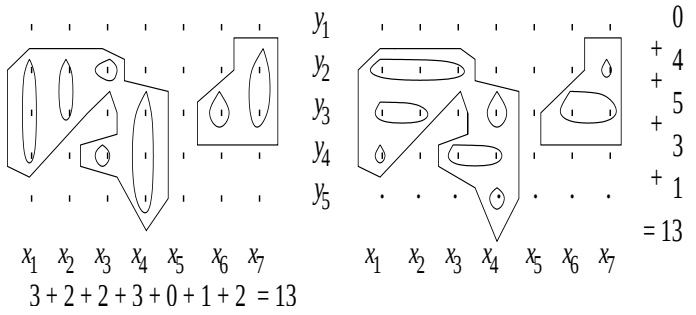
This gives a decomposition of S and so by (3) we get

$$|S| = \sum_{i=1}^{m} |S \cap (\{x_i\} \times Y)| \tag{6}$$

In an entirely analogous manner we get

$$|S| = \sum_{j=1}^{n} |S \cap (X \times \{y_j\})| \tag{7}$$

In Figures (a) and (b) below we show (6) and (7) pictorially.



(a) Column-wise slicing (b) Row-wise slicing

Although these results are hardly profound (and, in any case, are special cases of (3) as just shown), a special case of (6) is very commonly needed in counting problems, especially in its verbal form which is more popular. Although in (6), the i-th term of the summation, i.e., the cardinality of the set $S \cap (\{x_i\} \times Y)$ may change as i changes, in many situations, it is the same for all $i = 1, 2, \ldots, m$. Denote this common value by r. Then $|S| = mr$. Verbally, this can be expressed as follows. "Whenever, one thing can be done in m ways and when done in any of these m ways, some other thing can be done in r ways, then the two things can together be done in mr ways."

Instead of the cartesian product of two sets we can consider the cartesian product of any (finite) number of sets, say $X_1, X_2, \ldots, X_k$. A typical element of $X_1 \times X_2 \times \ldots \times X_k$ is an ordered k-tuple of the form $(x_1, x_2, \ldots, x_k)$, where $x_i \in X_i$ for $i = 1, 2, \ldots k$. The analogue of (5) now is

$$|X_1 \times X_2 \times \ldots \times X_k| = |X_1| \times |X_2| \times \ldots \times |X_k| \tag{8}$$

The analogues of (6) are somewhat messy to state in terms of symbols. However, the verbal version of (6) given above generalises easily if instead of two things, we have k things, to be done one after the other.

As a foremost application, we count ${}^nP_k$, the number of k-permutations, i.e., permutations[2] of length k of n objects. Each such permutation corresponds to an ordered k-tuple $(x_1, x_2, \ldots, x_k)$ of distinct elements of a given set, say S, of n elements (or 'objects' or 'symbols'). Here the first entry x_1 can be any of the n objects. So it can be chosen in n ways. Having chosen it, we have $n-1$ possibilities for x_2, viz., any of the n objects other than the one chosen for x_1. Having chosen x_2, we now have $n-2$ possible choices for x_3. Continuing like this, we get

$$ {}^nP_k = n(n-1)(n-2)\ldots(n-k+1) = \frac{n!}{(n-k)!} \tag{9}$$

where $1 \leq k \leq n$. By convention, we set ${}^nP_0 = 1$ and ${}^nP_k = 0$ if $k > n$. The first convention is consistent with (9) if we interpret a product with 0 factors (or an 'empty product') equal to 1, very much like equating the 0-th power of a number with 1. The second convention is consistent with the middle term of (9) because when $n < k$, one of the factors in it will be 0. The last term, however, is meaningless if $k > n$ since in that case $n-k$ is negative and hence $(n-k)!$ is undefined. Such degenerate cases rarely arise in practical problems. But when we are considering a summation, some of the terms may become degenerate and so to exclude them we shall have to restrict the index variable of the summation suitably. The convention we are adopting makes it unnecessary to do so since the troublesome terms are 0 anyway.

When the length of a permutation is not specified we take it to be equal to the number of objects. Thus by a permutation of 10 letters we mean a permutation of length 10. There are 10! such permutations. This number is 3628800 which is fairly large. The numbers $n!$ grow very rapidly with n. See Exercises (6.52) and (6.53) for more on this.

(iv) **Transformation :** Suppose S, T are finite sets and $f : S \longrightarrow T$ is a bijection, i.e., a one-to-one and onto function, then it is clear that $|S| = |T|$. More generally suppose that for some positive integer r, f is an r-to-one function in the sense that every element of T has exactly r preimages under f (or, put differently, for every $t \in T$, $|f^{-1}(\{t\})| = r$). Then decomposing the set according to the values of the function f we get a decomposition in which every subset has r elements. The number of subsets in this decomposition is simply $|T|$. So by (3), we have

$$ |S| = r|T| \tag{10}$$

Once again, (10) is hardly profound by itself. But it enables us to express the cardinality of the sets S and T in terms of each other. Such

[2] A permutation is also known as an **arrangement**. A formal definition of a k-permutation of n objects or symbols can be given as a one-to-one function from the set $\{1, 2, \ldots, k\}$ to the set S of the n objects. If $k = n$, then every such function corresponds to a bijection $g : S \longrightarrow S$ and we often call this bijection g as a permutation of n symbols. Colloquially, g is a reshuffling of the elements of S listed in the original order. Thus, (a, c, d, b) is a permutation of (b, c, a, d) with $g(b) = a, g(c) = c, g(a) = d$ and $g(d) = b$. Here $S = \{b, c, a, d\}$.

a transformation is especially useful when we have no easy direct way to find the cardinality of one of the two sets. By choosing the other set and the function suitably, we may be able to find the desired cardinality. (Remember the shepherd in a story who found the number of sheep in his herd by counting the number of their legs and dividing by 4?) We already used a transformation in the BANANA problem above. In order to count the number of permutations in which the two N's appear adjacently, we treated NN as a single letter. In effect, this amounts to considering a bijection between the set of those permutations of 6 letters in which the two N's appear together and the set of all permutations of 5 letters (of which NN is one).

As an example where such a bijection is far less obvious, suppose n players enter a knock-out tournament. At each round, they are paired, with one player getting a bye in case there is an odd number of players entering that round. A match is played between the two players in each pair and the winners enter the next round along with the player, if any, who got a bye. This continues till a champion is decided. Suppose we want to find the total number of matches played in the tournament. By a direct calculation, it can be seen that if the number of players entering the first round is, say 10, 25 or 87 (some random figure), then the number of matches played is, respectively, 9, 24 and 86. (For example, with $n = 25$, the number of matches played in the successive rounds is 12, 6, 3, 2 and 1, the total being 24.) This suggests that with n players entering the tournament, the number of matches played must be $n - 1$. It is not difficult to prove this guess by induction on n. But there is a clever way to find the number of matches using a suitable bijection. Note that every match has a winner and a loser. Let M be the set of all matches played and let P be the set of all players other than the champion. We define a function $f : M \longrightarrow P$ by $f(m)$ to be the player who loses the match m. Then since every player other than the champion loses at least one match, this function is onto. But it is also one-to-one, for no player can lose more than one match. (After his first defeat, the player is knocked out and plays no more matches.) Thus the function $f : M \longrightarrow P$ is a bijection. Therefore, by (10) (with $r = 1$), $|M| = |P|$. Since there are n players in all and the complement of P consists of just one player (viz., the champion), $|P| = n - 1$. So in all $n - 1$ matches are played! (It may be argued that in this solution we only get the total number of matches and not their roundwise break-up. But that's what is asked! The latter may be a means but not the goal. This is, in fact, true of many elegant solutions. They take you directly to the destination, often bypassing the steps one would normally encounter in a straightforward approach. We shall encounter more examples of such elegant solutions. See for example, Comment No. 6 below and Exercises (1.25)(b), (1.37) and (23.5).)

Formula (10) also allows us to find the number of ways to select k distinct objects from a set, say X, of n (distinct) objects or equivalently,

the number of all k-subsets of X. Such a selection is also called a k-**combination**. It differs from a k-permutation in that in the latter, the order of the objects matters. As a result, if we take a k-permutation and ignore the order of the entries in it we get a k-selection. Formally, let S and T be, respectively, the sets of all k-permutations and all k-selections of elements of the set X. Define a function $f : S \longrightarrow T$ by $f(x_1, x_2, \ldots, x_k) = \{x_1, x_2, \ldots, x_k\}$. If we permute the x_i's among themselves in any manner, the value of f remains unaffected. As this reshuffling can be done in $k!$ ways, we get that every element of T has exactly $k!$ preimages under f. So we can apply (10) with $r = k!$. From (9) we already know that $|S| = {}^nP_k = \frac{n!}{(n-k)!}$. So we get $|T| = \frac{n!}{k!(n-k)!}$. Numbers of this type appear so frequently in counting problems that it is convenient to have special symbols for them. Two standard notations are ${}^nC_r$ and $\binom{n}{k}$ (read as 'n choose k'). With these notations, we record the result as

$$ {}^nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\ldots(n-k+1)}{k!} \tag{11} $$

for $1 \leq k \leq n$. Again we set $\binom{n}{0} = 1$ which can be justified by observing that a 0-selection corresponds to selecting the empty subset. For $k < 0$ as well as for $k > n$, we set $\binom{n}{k} = 0$.

We have already encountered these numbers in the second step of the solution to the Main Problem. These numbers appear as the coefficients of the various terms in the well-known binomial theorem (to be studied in the next chapter) and hence are called the **binomial coefficients**. There are hundreds of identities involving them, collectively called the **binomial identities**, a few of which will be studied in Chapter 5. One frequently needed identity called the symmetry property says that $\binom{n}{k} = \binom{n}{n-k}$. This is immediate from (11).

Yet another application of (10) which we have already used in the solution to the Main Problem is in finding the number of permutations in which some of the objects are indistinguishable from each other, for example, balls of the same colour or coins of the same denomination. Suppose we have k types of objects and n_i objects of the i-th type for $i = 1, 2, \ldots, k$. Let n be the total number of objects and T the set of all permutations of them. We want to find $|T|$. Clearly, $n = n_1 + n_2 + \ldots + n_k$. Let us distinguish objects of the i-th type by putting some distinguishing marks (for example, the numbers from 1 to n_i) on them. Now we have n distinct objects and if we let S be the set of all their permutations then $|S| = n!$. Now define a function $f : S \longrightarrow T$ which assigns to each permutation of S, the same permutation but with the identification marks on like objects removed. Then clearly, $f(x_1, x_2, \ldots, x_n) = f(y_1, y_2, \ldots, y_n)$ if and only if the permutation $(y_1, y_2, \ldots, y_n)$ is obtained from $(x_1, x_2, \ldots, x_n)$ by reshuffling objects of the same type within themselves. For each $i = 1, 2, \ldots, k$, the objects of the i-th type can be reshuffled among themselves in $(n_i)!$ ways. So, for a given permutation $(x_1, x_2, \ldots, x_n) \in S$, we can get $n_1!n_2!\ldots n_k!$

permutations all of which will go to the same permutation in T under f. Put differently, this is the number of preimages every element of T has under f. So by (10) (with $r = n_1!n_2!\ldots n_k!$) we get $|T|$ = number of permutations of the given k types of objects (with n_i objects of the i-th type for $i = 1, 2, \ldots, k$) as $\dfrac{n!}{n_1!n_2!\ldots n_k!}$. We already used this in the third step of the solution to the Main Problem.

Note that the numbers ${}^nP_k$ and ${}^nC_k$ depend only on the integers n and k and not on the particular set whose elements are permuted or selected. All the results obtained so far, as well as their derivations are very standard. As a little less commonly needed result, suppose that S is a set with n elements. Let $P(S)$ denote the set of all subsets of S (including the empty set and the entire set S). For example, if $S = \{a, b\}$, then $P(S) = \{\emptyset, \{a\}, \{b\}, S\}$. We prove the following result.

Theorem: If $|S| = n$ then $|P(S)| = 2^n$. Verbally, the number of subsets of a set with n elements is 2^n.

Proof : The simplest proof is by induction on n. If $n = 0$, then $S = \emptyset$ and $P(S) = \{\emptyset\}$ has $2^0 = 1$ element. Assume $n > 1$ and that the result holds for all sets with $n - 1$ elements. Let S be a set with n elements. Fix any element, say x of S and let $T = S - \{x\}$. Then $|T| = n - 1$. Classify the subsets of S according to whether they contain x or not. Those that do not contain x may be looked upon as subsets of T and vice versa. So by induction hypothesis, there are 2^{n-1} such subsets. Next consider subsets of S that contain x. If A is such a subset then $A - \{x\}$ is a subset of T. Conversely, if $B \subset T$ then $B \cup \{x\}$ is a subset of S containing x. This gives a bijection between the set of subsets of S containing x and the set $P(T)$. By induction hypothesis, $|P(T)| = 2^{n-1}$. So the number of subsets of S containing x is also 2^{n-1}. Put together, $|P(S)| = 2^{n-1} + 2^{n-1} = 2^n$. This completes the inductive step and hence the proof.

There is an alternate proof which is more instructive. Let Y be the set of all binary sequences of length n. Here by a binary sequence we mean a sequence whose only terms are 0 and 1. Equivalently, a binary sequence of length is an ordered n-tuple, say $(y_1, y_2, \ldots, y_n)$ where every y_i is either 0 or 1. Clearly there are 2^n such sequences since each of the n terms can be chosen in 2 ways independently of the others. We now establish a bijection $f : P(S) \longrightarrow Y$ as follows. Call the elements of the set S as $s_1, s_2, \ldots, s_n$. Now given a subset A of S, we let $f(A)$ be the binary sequence whose i-th term is 1 if $s_i \in A$ and 0 if $s_i \notin A$. It is easy to recover the subset from its corresponding binary sequence $(y_1, y_2, \ldots, y_n)$ (say). All we have to do is to let $A = \{s_i : y_i = 1\}$. Thus f is a bijection. Therefore $|P(S)| = |Y| = 2^n$.

The set $P(S)$ is often called the **power set** of the set S - the reason being that its cardinality is always a power of 2.

Comment No. 2:

With the four basic counting techniques at our disposal we can tackle the elementary problems about counting. The techniques themselves are elementary. In fact, they are so elementary that generally they are used without explicit mention of their names (just as we use commutativity of addition without ever mentioning it). But applying them may require some ingenuity. For example, choosing the right decomposition for which to apply (3) or the right bijection for which to apply (10) may not always be so obvious.

As a very straightforward problem, suppose we want to find the number of ways 52 cards can be dealt among four players with each player getting 13 cards (JEE 1979). Call the players as P_1 to P_4. Then P_1's hand can be chosen in $\binom{52}{13}$ ways. Once it is chosen, we are left with 39 cards, from which P_2's hand can be chosen in $\binom{39}{13}$ ways. Continuing in this way, the total number of deals is $\binom{52}{13}\binom{39}{13}\binom{26}{13}$. Using (11), this equals $\frac{52!}{39!\times 13!}\frac{39!}{26!\times 13!}\frac{26!}{13!\times 13!}$ which simplifies to $\frac{52!}{(13!)^4}$. Actual evaluation of this ratio is not expected. (There is no hard and fast rule about this. But generally, if the arithmetic needed in the final numerical answer involves only three digit figures, then you should carry it out. Calculations involving factorials upto 10! generally fall in this category.)

In the preceding problem, suppose however, that we want to divide the 52 cards into four piles with 17 cards in three piles and just one card in the fourth. Reasoning as above, it is tempting to think that here the answer is $\binom{52}{17}\binom{35}{17}\binom{18}{17} = \frac{52!}{(17!)^3}$. But this is not quite so. This would be the right answer if there were four players and we were giving 17 cards to each of P_1, P_2 and P_3 and one card to P_4. But as the problem stands, there are no players. We are merely dividing the cards into 4 piles. This is equivalent to saying that the hands of the first three players can be permuted among themselves. This can be done in 3! i.e., in 6 ways. So the correct answer is $\frac{52!}{6(17!)^3}$.

A similar problem is to find the number of ways in which an even number, say $2n$ of players can be divided into n pairs. Denote this number by a_n. The first pair can be formed in $\binom{2n}{2}$ ways. After it is formed, the second one can be formed in $\binom{2n-2}{2}$ ways and so on. But once again, the order of the pairs is immaterial and so the correct answer is $a_n = \frac{\binom{2n}{2}\binom{2n-2}{2}\cdots\binom{4}{2}\binom{2}{2}}{n!}$ which simplifies to $\frac{(2n)!}{2^n n!}$. It is easy to show that this is precisely the product of all odd integers from 1 to $2n-1$. Thus, $a_1 = 1, a_2 = 3, a_3 = 15$ and so on. In such problems, the data involves an integer parameter, viz., n. As a result induction is often a convenient technique. Let us do the present problem by induction. Consider one player, say A. In each of the a_n possible pairings of $2n$ players, A will be paired with one of the remaining $2n-1$ players. Classify the pairings according to whom A is paired with. If A is paired with B (say) then the remaining $2n-2$ can be paired in a_{n-1} ways. Thus we get

$$a_n = (2n-1)a_{n-1} \tag{12}$$

An equation like this is called a **recurrence relation**. It does not give a_n explicitly. But it expresses a_n in terms of a_{n-1}, which in turn can be expressed in

terms of a_{n-2}. This process recurs again and again (hence the name). To express a_n directly in terms of n is known as **solving** the recurrence relation. There is a systematic theory for this which we shall not go into. But in the present case the solution of (12) is very easy. We merely apply it again and again with decreasing values of n to get, successively, $a_n = (2n-1)a_{n-1} = (2n-1)(2n-3)a_{n-2} = (2n-1)(2n-3)(2n-5)a_{n-3} = \ldots = (2n-1)(2n-3)\ldots 3a_1$. Since a_1 clearly equals 1, we get the same answer as before, viz. that a_n is the product of the odd integers from 1 to $2n-1$. Alternately, we could have guessed this answer by calculating a_n for some small values of n and then proved it by induction on n. In that case, (12) does the work of the inductive step.

Comment No. 3:

As a straightforward application of counting by decomposition, we do the following problem.

Five balls of different colours are to be placed in three boxes of different sizes. Each box can hold all five balls. In how many different ways can we place the balls so that no box remains empty? (JEE 1981)

We classify the placements according to the numbers of balls put into the three boxes. Number the boxes from 1 to 3 and let n_i be the number of balls in the i-th box. Then (n_1, n_2, n_3) is an ordered triple in which each $n_i \geq 1$ (since every box is to hold at least one ball) and $n_1 + n_2 + n_3 = 5$. There are 6 such triples in all, viz., $(3,1,1), (1,3,1), (1,1,3), (2,2,1), (2,1,2)$ and $(1,2,2)$. Consider placements of the type $(3,1,1)$. This means that the first box contains 3 balls and the other two one ball each. These three balls can be chosen in $\binom{5}{3} = 10$ ways. Having chosen them, one ball from the remaining two can be chosen in 2 ways. So the number of placements of the type $(3,1,1)$ is $10 \times 2 = 20$. Clearly, this is also the number of placements of the type $(1,3,1)$ and of the type $(1,1,3)$. So we have in all 60 placements in which one box has 3 balls and the other two one each. By a similar reasoning, the number of placements of the type $(2,2,1)$ is $\binom{5}{2}\binom{3}{2} = 10 \times 3 = 30$ and this is also the number of placements of the type $(2,1,2)$ and of the type $(1,2,2)$ each. So in all there are 90 ways to put 2 balls into two boxes and one ball in the third. As we have exhausted all the possibilities by now, we have $60 + 90 = 150$ ways of putting the balls into the boxes so that no box is empty.

Counting by decomposition is an instance of the general strategy 'divide and rule'. Care has to be taken to see that all the possibilities are taken into account and that no two possibilities are overlapping. In Exercise (1.29) we shall give another method of solving the problem above.

Comment No. 4:

Counting problems can often be posed as probability problems. We shall take up general probability problems in Chapter 22. Normally, in a probability problem, we have to find the probability of the occurrence of some 'event'. Suppose that some particular thing can be done in n ways in all and that the event occurs in m of these. If these n ways are equally likely, then the

desired probability is simply the ratio $\frac{m}{n}$. So in such problems the knowledge of probability needed is negligible. In such cases, the problem is equivalent to two counting problems, viz. finding n (popularly called the total number of cases) and finding m (popularly, the number of favourable cases). We illustrate this with a simple problem.

Six boys and six girls sit in a row randomly. Find the probability that

(i) the six girls sit together

(ii) the boys and girls sit alternatively. (JEE 1979)

The 6 boys and the 6 girls can be seated in 12! ways in all. All these arrangements are equally likely. (This is the meaning of the word 'randomly'.) To find the number of favourable cases for (i), the six girls sitting together form a block. This block along with the 6 boys can be permuted in 7! ways. In the block itself, the girls can be permuted in 6! ways. So the number of favourable cases for (i) is $7! \times 6!$. Hence the probability that the girls sit together is $\frac{7!\ 6!}{12!}$. As the figures involved are relatively small, we simplify this as $\frac{6!}{12\times11\times10\times9\times8} = \frac{1}{132}$. Similarly, for (ii), number the seats 1 to 12 serially from one end of the row. Then either the 6 boys occupy the 6 even numbered seats and the 6 girls the 6 odd numbered ones or vice versa. There are $6! \times 6!$ arrangements of each type. So the probability of (ii) is $\frac{2\times6!\times6!}{12!} = \frac{2\times6!}{12\times11\times10\times9\times8\times7} = \frac{2}{7}\frac{1}{132} = \frac{1}{462}$.

Comment No. 5:

As an example of a problem where the transformation that does the trick is not so obvious, suppose we want to seat m men and n women in a row so that no two women sit together (JEE 1983). If $n = 1$, this restriction is vacuous and the number of arrangements is $(m+1)!$. For $n = 2$, we can take the two women together as a single person and get the answer by complementary counting (as we did in the BANANA problem, except that here the two women are distinct from each other). But if $n > 2$, this method will fail. To make it work, we shall have to divide the unwanted arrangements (i.e., those where at least two women are together) into so many cases. But there is a better way out. First put the m men in a row in $m!$ ways. In any one such arrangement, there are $m + 1$ places where a woman can be put, viz., the $m - 1$ places between two adjacent men and the two places at the end. The restriction that no two women are to be together means that the n women must occupy n distinct places out of these $m + 1$ places. If $n > m + 1$, this is impossible to do. But if $n \leq m + 1$, then the places for women can be chosen in $\binom{m+1}{n}$ ways. Moreover, for each such choice, the n women can occupy the n places in $n!$ ways. Putting it together, the total number of ways to seat m men and n women so that no two women are adjacent is $m! \times \binom{m+1}{n} \times n! = \frac{m!\ (m+1)!\ n!}{n!\ (m-n+1)!} = \frac{m!\ (m+1)!}{(m-n+1)!}$.

In this solution we used (10) with $r = 1$ (the case where the function $f : S \longrightarrow T$ is a bijection). But we did not explicitly define the bijection f (or even the sets S and T for that matter). All this is implicit in the solution and hardly needs to be spelt out. Still, one must be in a position to do so if called

upon. In the present case, for example, S is the set whose cardinality we want, viz., the set of all arrangements of m men and n women with no two women together. The description of T is a little more intricate. Let X be the set of all arrangements of the m men in a row. Then $|X| = m!$. Now let Z be the set of $m+1$ possible places where the women can be put. (As noted above, $m-1$ of these places are between adjacent men and 2 are at the ends.) We are putting the women so that no two women go to the same place. In effect, this means that we are taking a one-to-one function from the set, say W, of the n women to the set Z of the $m+1$ places. Let Y be the set of all one-to-one functions from W to Z. The last part of our solution amounted to showing that $|Y| = \binom{m+1}{n} n!$ (which is the same as ${}^{m+1}P_n$). The central idea in the solution above was to construct a bijection between the set S and the cartesian product $X \times Y$. So the set T was the set $X \times Y$. The bijection $f : S \longrightarrow X \times Y$ was constructed by assigning to a typical element of S (i.e., to a typical arrangement of the desired type) an ordered pair whose first entry was an arrangement of the m men (obtained by ignoring the women in the arrangement we started with) and the second entry was the placement of the n women into distinct places from the set Z.

We emphasise again that it is unnecessary, and in fact rather pedantic, to give the argument so formally. Nor does it help the reader in understanding it. The real advantage of such formalism is that it guards you against mistakes of logic. Three most common pitfalls in counting arguments are that certain unwanted elements get counted, some wanted elements do not get counted while some elements get counted more than once. If you are able to paraphrase the argument rigorously by introducing appropriate sets and functions, then such mistakes can be avoided. From now onwards, we shall not bother about such paraphrases. But it will be a good exercise for you to supply them once in a while. It will sharpen your thinking process.

As yet another example of the application of a clever transformation, let us count the number of selections with repetitions. Suppose we have to choose k objects from n types of objects with an unlimited supply of objects of each type. (This may sound unrealistic. For a fixed k, it is enough to have at least k objects of each type so that we can select all k objects of just one type if we want.) Here objects of the same type are indistinguishable from each other and so what matters is only *how many* objects of each type are selected and not *which ones* are selected. Suppose we select k_i objects of the i-th type for $i = 1, 2, \ldots, n$. Then each k_i is a non-negative integer and $k_1 + k_2 + \ldots + k_n = k$. So the problem is equivalent to finding the number of distinct ordered n-tuples $(k_1, k_2, \ldots, k_n)$ of non-negative integers whose entries add up to k. Let S be the set of all such ordered n-tuples. To find $|S|$ we represent every element of S pictorially. Let $(k_1, k_2, \ldots, k_n)$ be a typical element of S. Then we draw, on a straight line, n blocks of dots with k_1 dots in the first block, k_2 dots in the next, and so on. Also in between two consecutive blocks we put a partition (or a wall). Thus in all we have $n-1$ walls. We illustrate this construction in the figure below where $n = 5$ and $(k_1, k_2, k_3, k_4, k_5) = (3, 2, 0, 1, 2)$ (so that $k = 3 + 2 + 0 + 1 + 2 = 8$). The four walls are shown by vertical bars. (Note

that between the second and the third wall there are no dots. This corresponds to the fact that $k_3 = 0$, i.e., that the third block is empty.)

Thus every element of S gives rise to an arrangement of k dots and $n-1$ walls in a row. Conversely, from every such arrangement we get an element $(k_1, k_2, \ldots, k_n)$ of S. All we have to do is to let k_1 be the number of dots that appear before the first wall, k_2 be the number of dots that appear between the first and the second wall and so on. These two processes are inverses of each other. We have thus established a bijection between the set S (whose cardinality we are after) and the set, say T, of all arrangements of k dots and $n-1$ walls in a row. Since all dots are alike and so are all walls, we know that $|T| = \frac{(k+n-1)!}{k!(n-1)!}$ which can also be written as $\binom{n+k-1}{k}$. So this is the number of ways we can select k objects from n types of objects with repetitions allowed freely.

As a numerical example, if we want to have an animal circus with elephants, lions and tigers, then (assuming that all animals of the same type are alike) we can choose 10 animals in $\binom{10+3-1}{10} = \binom{12}{10} = \binom{12}{2} = 66$ ways. If we want to include at least one animal of each type, then our freedom is restricted to choosing 7 animals of the three types and then the circus can be formed in $\binom{9}{2} = 36$ different ways.

Comment No. 6:

As a brilliant combination, due to **Andre**, of techniques of complementary counting and transformation, let us count the number, say b_n, of balanced arrangements of n pairs of parentheses. Here we have n left parentheses (also called opening parentheses) and n right parentheses (also called closing parentheses). So in all we have $\frac{(2n)!}{n!\,n!}$ arrangements of n pairs of parentheses. But not all of these are balanced. For example, an arrangement in which the starting parenthesis is ')' can never be balanced. The general rule is that in a balanced arrangement, as you go scanning from left to the right, whenever you encounter a right parenthesis, there must be some earlier left parenthesis to go with it. In simpler terms this means that at no stage should the number of right parentheses exceed the number of left parentheses. For example, for $n = 3$, ((( ))), ( ) (( )) and ( ) ( ) ( ) are all balanced arrangements but ( )) (( ) and (( ))) (are not. It can be verified by direct trial that $b_1 = 1, b_2 = 2$ and $b_3 = 5$. With a little patience, b_4 comes out to be 14. Thereafter, the problem is too unwieldy to be done by hand. A recurrence relation can be written down for b_n, but it is considerably more complex than (12).

But the problem can be solved quite easily by counting the number of unbalanced arrangements of n pairs of parentheses. Call this number c_n. Then $c_n = |S|$, where S is the set of all unbalanced arrangements of n pairs of parentheses. Let T be the set of all arrangements of $n+1$ right and $n-1$

left parentheses. (Obviously no such arrangement can be balanced.) Then $|T| = \frac{(2n)!}{(n+1)!(n-1)!}$. We construct a bijection from S to T as follows. In any unbalanced arrangement of n pairs of parentheses, there will be a first right parenthesis for which there will be no matching left parenthesis. Call this the critical parenthesis. Figure (a) below shows one such unbalanced arrangement with an arrow pointing to the critical parenthesis. After the critical parenthesis, there will be an odd number of parentheses, the number of left parentheses among them will be one more than the number of right parentheses. Let us interchange them. That is, replace every right parenthesis occurring after the critical parenthesis by a left one and vice versa. The parentheses upto and including the critical parenthesis are to be left unaffected. This way we get an element of T, i.e. an arrangement of $n+1$ right and $n-1$ left parentheses. Figure (b) below shows the new arrangement obtained from the unbalanced arrangement in (a). (In the figure, $n = 8$.)

a) (((()))()))))(()(b) (((()))())))())()

The crucial point now is to observe that that the original unbalanced arrangement of n pairs of parentheses can be recovered from the new arrangement of $n+1$ right and $n-1$ left parentheses. All we have to do is to go on scanning till we come across a critical right parenthesis (which is sure to happen as there are more right parentheses than left ones) and thereafter interchange the left and the right parentheses. This gives a bijection between the sets S and T and so c_n = number of unbalanced arrangements of n pairs of parentheses $= \frac{(2n)!}{(n+1)!(n-1)!}$. Subtracting this from $\frac{(2n)!}{n!n!}$, we get the number of balanced arrangements of n pairs of parentheses as $\frac{(2n)!}{n!(n+1)!}$ which can also be written as $\frac{1}{n+1}\binom{2n}{n}$. This number is known as the n-th **Catalan number** and often denoted by C_n. These numbers come up in many combinatorial problems, because a surprisingly large number of problems can be reduced to the problem of counting balanced arrangements of pairs of parentheses. A few of them will be indicated in the exercises (see Exercises (1.31) - (1.35)).

Comment No. 7:

In many counting problems, you will have to know a few facts from mathematics (or sometimes from real life, such as the elementary rules of some popular game) in order to decide just which things are to be included in the count. We already had an example of this where we needed the 'rule of three', i.e., the fact that a number is divisible by 3 if and only if the sum of its digits is a multiple of 3. As another example, suppose we are given 3, 4 and 5 interior points on the sides AB, BC and CA respectively of a triangle ABC and we want to find the number of triangles that can be formed with these interior points as vertices (JEE 1984). Here we have 12 points in all and so there are $\binom{12}{3}$ i.e., 220 ways of picking 3 points out of these. But they will form a triangle if and only if they are not collinear, i.e., they do not lie on the same side. So we have to exclude

such triplets where all the three points come from the same side. Thus we have to exclude $\binom{3}{3}+\binom{4}{3}+\binom{5}{3}$ i.e. $1+4+10=15$ triplets. So the number of triangles that can be formed is $220-15=205$.

Comment No. 8:

In the examples above we had to know some mathematical facts to decide which things are to be included in the count. There are problems where we have to know some mathematical facts in order to decide which things are to be identified with each other, i.e., are to be treated as not distinct. As a typical illustration, consider the problem of seating n guests around a circular table. If they were to be seated in a row then the answer is $n!$ as we know. But now we have to take the geometry of the table into account. When guests are seated around a circular table then what matters is their relative position with respect one another. If each of them is shifted clockwise (or anti-clockwise) by the same angle then the relative position remains the same. So the answer is not $n!$ but something less than that. For each seating arrangement, such a cyclic shift is possible in n different ways. The corresponding n placements are therefore to be identified with each other. So the total number of distinct relative positions is $\frac{n!}{n}=(n-1)!$. We are assuming here that the table is such that no two points on its rim can be distinguished from each other. If there is a spot marked as a place of honour (say), then every rotation of the table will give us a different arrangement and the answer will be $n!$. In fact in such a case, the problem is not substantially different from that of seating the guests in a row.

The same reasoning applies if instead of a round table, we had a table in the form of a regular n-gon and the guests are to be seated along its edges. In particular, we see that 4 guests can be seated around a square table in 3! i.e., in 6 ways. Suppose, however, that the table is not a square but a rectangle. In that case the longer sides can be distinguished from the shorter ones and so two arrangements would be equivalent only when they can be obtained from each other by a 180 degrees rotation. So the number of distinct arrangements is 12.

Let us now consider the problem of colouring the vertices of a regular n-gon by n distinct colours, each colour to be used exactly once. This may seem to be the same as the problem of seating n guests around a table. But there is a vital difference. We cannot flip a table, but we can flip the regular n-gon in space. This amounts to taking a reflection along a line passing through its centre. As a result, two colourings which are mirror images of each other are to be treated as equivalent. (Put differently, a colouring in which the colours appear in a particular order clockwise cannot be distinguished from one in which they appear in the same order anti-clockwise.) So the number of distinct colourings now is neither $n!$ nor $(n-1)!$, but $\frac{1}{2}(n-1)!$.

In all these problems we had a set, say S (such as the set of all arrangements of guests or of all coloring of vertexes), and a certain equivalence relation (see Chapter 20, Comment No. 12 for a formal definition) on this set S and the problem was to count not so much the cardinality of the set S, but rather, the number of distinct equivalence classes. When each equivalence class has the

same number of elements, say k, then the answer is, of course, $\frac{|S|}{k}$. But there are situations where this is not the case. For example, let S be the set of all colorings of the vertices of a regular pentagon in which exactly two vertices are red, exactly two are yellow and the remaining vertex is blue. Using elementary counting techniques, it follows that $|S| = \binom{5}{2}\binom{3}{2} = 30$. Here two colourings are to be treated as equivalent if they can be obtained from each other by a rotation (through multiples of 72 degrees) or by a flip. Now all equivalence classes are not of the same size. Take, for example, a colouring in which the blue vertex has both neighbours red. Such a colouring does not give rise to a new colouring upon flipping and so there are only 5 colourings equivalent to it (including itself). Similarly, a colouring in which the blue vertex has both neighbours yellow has only 5 elements in its equivalence class. The other colourings (i.e. those in which the blue vertex has one red and one yellow neighbour) fall into two equivalence classes of size 10 each. (Draw appropriate diagrams to see this.) So in all there are four equivalence classes. An *ad-hoc* reasoning like this is very prone to errors in more complicated situations (see Exercise (1.19) for two problems of a similar nature). Such problems can be handled systematically by what is called **Polya's theory of counting**. But it is beyond our scope. We only remark that such problems are very important in practical applications. In fact, Polya's theory originated in an attempt at solving a problem in chemistry. This is not surprising. For example, the problem of deciding how many distinct dihydroxybenzenes there are is equivalent to colouring the vertices of a regular hexagon (corresponding to the benzene ring) with two colours, one corresponding to the hydrogen radical and the other to the hydroxyl radical. (Mathematically, the answer comes out as 3 and chemically all these three compounds do exist. They are called catechol, resorcinol and quinol.)

Comment No. 9:

When we find the cardinality of a set by decomposing it as a union of subsets, it is vital that these subsets have no overlaps. Sometimes, a set has a natural expression as the union of certain subsets, but these subsets may not be mutually disjoint. For example, the set of students failing in an examination (with multiple subjects) is the union of the sets of those who fail a particular subject. The population of a city is the union of the sets of those residents who know a particular language out of those prevalent in the city and so on. In such cases we cannot apply formula (3). There are two ways out. One of them is to replace the given subsets by some related subsets which are mutually disjoint. But this can be clumsy as compared to the original subsets which may be more natural. Another method is to allow the subsets to overlap but to keep a systematic count of the common elements.

As the simplest case, suppose A and B are subsets of some set. If they are mutually disjoint, then we have $|A \cup B| = |A| + |B|$. Otherwise, $|A \cup B|$ will fall short of $|A| + |B|$, because the elements that are common to both A and B get counted twice in the sum $|A| + |B|$ but only once in $|A \cup B|$. If we make this

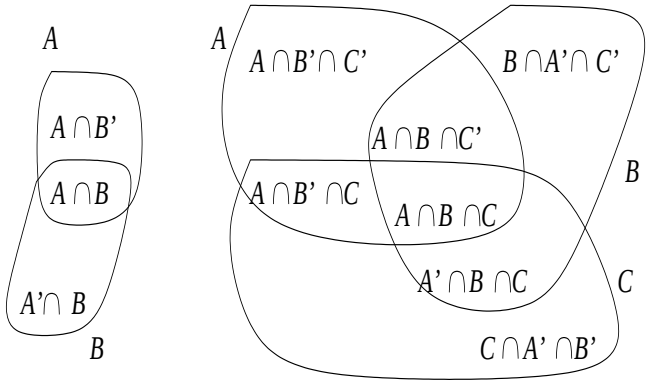
'correction' then we have

$$|A \cup B| = |A| + |B| - |A \cap B| \tag{13}$$

The significance of the $-$ sign on the R.H.S. is noteworthy. It arises because elements of the intersection $A \cap B$ have been included in each of the first two terms. To make up for this, they have to be excluded once. For this reason, (13) is a special (and the simplest) case of what is formally called the **principle of inclusion and exclusion**. The next case is to consider three subsets, say A, B and C. Then the analogue of (13) is

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \tag{14}$$

We leave it as an exercise to check that the net effect of the various inclusions and exclusions in the R.H.S. is that every element of $A \cup B \cup C$ gets counted exactly once. (For example, an element which belongs to A and B but not to C is included in the first two terms but gets excluded once because of the fourth term.) We can, more generally, write the formula for the cardinality of the union of n subsets. But the two cases given above would generally suffice for our purpose. These cases can be expressed vividly by what are called **Venn diagrams** shown below. (Such diagrams are drawn only to clarify ideas. They are analogous to the pie diagrams that you frequently see accompanying the results of surveys of various types but are generally not drawn to scale. That is, the area of the region representing a subset is not necessarily proportional to its cardinality.)



$|A \cup B|$ $|A \cup B \cup C|$

Like the other elementary counting techniques, the principle of inclusion and exclusion is so akin to common sense that it is often applied without explicitly representing the various subsets with suitable symbols. We illustrate its application in the following problem.

An unbiased die with faces numbered $1, 2, 3, 4, 5, 6$ is thrown n times and the list of n numbers showing up is noted. What is the probability that among the numbers 1, 2, 3, 4, 5, 6, only three numbers appear in this list? (JEE 2001)

As the tosses are independent of each other, the total number of possible lists is 6^n. As the die is unbiased, all these are equally likely. So the problem reduces to counting the number of those lists in which exactly 3 of the 6 numbers 1 to 6 appear. The choice of these 3 numbers can be made in $\binom{6}{3}$ i.e., in 20 ways and for each such choice, the number of lists in which only those three particular numbers appear is the same. So without loss of generality we count the number, say a_n, of those lists in which only 1, 2 and 3 appear and multiply it by 20. The number of lists that can be formed with 1, 2 and 3 is 3^n. However, from this we must exclude those in which only one or only two of the numbers appear. Clearly there are only 3 lists of the former type, viz., the three 'constant' lists. For the latter, consider those lists in which only 1 and 2 appear. The number of such lists is $2^n - 2$ (as we have to exclude the two constant lists). This is also the number of lists in which only 2 and 3 appear and also the number of those in which only 1 and 3 appear. So we have $a_n = 3^n - 3 - 3(2^n - 2) = 3^n - 3 \times 2^n + 3$. As noted before, the desired probability is $\frac{20a_n}{6^n}$, which comes out as $\frac{20}{2^n} - \frac{60}{3^n} + \frac{60}{6^n}$. For a large n the last term is insignificant as compared to the first two and so the probability is nearly $\frac{20}{2^n} - \frac{60}{3^n}$. For still larger n, even the second term is insignificant and the probability is therefore approximately $\frac{20}{2^n}$. (This last piece of commentary is, of course, not a part of the solution and would, in fact, look ridiculous if written in a conventional type of examination. But the ability to make such observations, at least to oneself, is a valuable asset. Miss no opportunity to develop it.)

The general version of the principle of inclusion and exclusion along with a few problems based on it will be given in the exercises. (Exercises (1.41) - (1.46)).

Comment No. 10:

Sometimes we can count the cardinality of a set in two different ways. As the two counts must be the same, by equating the two we get an identity. This technique is called **double counting**. As a simple example, we prove the following identity by double counting.

$$\binom{2n}{2} = 2\binom{n}{2} + n^2 \tag{15}$$

We recognise the L.H.S. as the number of ways to select 2 objects from a set with $2n$ elements. This set could be chosen arbitrarily. We take it to be a set of n women and n men. Then the selections can be classified into three types - both women, both men or a woman and a man. The terms on the R.H.S. correspond to the numbers of such selections, thereby proving the identity. A direct algebraic proof of this identity is possible and in fact quite easy. But we want to illustrate the method. In Chapter 5 we shall prove a few more identities using double counting.

A particular case of double counting deserves special mention. Suppose X and Y are finite sets with m and n elements. Let us apply a double counting argument to a subset S of the cartesian product $X \times Y$. As discussed earlier

(see Equations (6) and (7)), we can 'slice' S either row-wise or column-wise. Equating the cardinalities obtained by these two slicing can sometimes lead to interesting results. To begin with, let us take S to be the entire set $X \times Y$. Slicing S column-wise, there are n elements in each column and since there are m columns we get $|S| = mn$. But an analogous argument applied to row-wise slicing will give $|S| = nm$. So we have a proof of the commutative law for the multiplication of positive integers, viz., that $mn = nm$. Over-familiarity with this law make the result look trivial. But it is significant. Less obvious equalities are obtained by taking S to be some other subset of $X \times Y$. Suppose for example, that X is a set of m boys, say $b_1, b_2, \ldots, b_m$ and that Y is a set of n girls, say $g_1, g_2, \ldots, g_n$. Assume that some of the boys are friendly with some of the girls. (We assume that this is a symmetric relationship. No one-way friendships are allowed!) Now let S be the set of all friendly pairs, i.e. $S = \{(b_i, g_j) : b_i \text{ and } g_j \text{ are friendly with each other }\}$. For $i = 1, 2, \ldots m$, let q_i be the number of girl friends of the boy b_i. Then by (6) we get $|S| = q_1 + q_2 + \ldots + q_m$. Similarly, using (7) we get $|S| = p_1 + p_2 + \ldots + p_n$ where, for $j = 1, 2, \ldots, n$, p_j is the number of boyfriends g_j has. So by double counting we get

$$p_1 + p_2 + \ldots + p_n = q_1 + q_2 + \ldots + q_m \tag{16}$$

An especially interesting case is where all the p_j's are equal to say p and all q_i's are also equal to, say q. In that case (16) reduces to

$$np = mq \tag{17}$$

It follows that if we know any three of the numbers m, n, p, q then we can determine the fourth. Thus, for example, if there are 10 boys and each has 4 girlfriends and each girl has 5 boyfriends then the number of girls is $\frac{10 \times 4}{5} = 8$. Most people will get this answer intuitively. But the argument above gives it a rigorous formulation.

The argument above applies more generally, to what is called an **incidence relation** between two sets X and Y. This is a relation which holds for certain pairs (called incidence pairs) of elements, one coming from X and the other from Y. Friendship was an example of this. But there are others. For example, X could be a set of locks, Y a set of keys and the incidence relation may be that a particular lock can be opened by a particular key. Or X could be some set, Y could be a set of some subsets of X (or, in other words, Y is a subset of the power set of X) and we may say that (x, A) is an incidence pair if and only if $x \in A$. Yet another example which is very important in geometry is where X and Y are sets of points and lines in a plane (say) and we say that a point P is incident on a line L if and only if L passes through P.

As an application of a double counting of this type, we have the following problem.

Suppose $A_1, A_2, \ldots, A_{30}$ are thirty sets each with five elements and $B_1, B_2, \ldots, B_n$ are n sets each with three elements. Let

$$\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S \, .$$

Assume that each element of S belongs to exactly ten of the A_i's and exactly nine of the B_j's. Find n. (JEE 1981)

We first let $X = S$, $Y = \{1, 2, \ldots\}$ and consider the set, say F of those pairs (x, i) such that $x \in A_i$. Since $|A_i| = 5$ for all $i = 1, 2, \ldots, 30$ and every $x \in S$ lies in exactly 10 A_i's, by a double counting argument applied to F, we have, analogous to (17),

$$30 \times 5 = |F| = |S| \times 10 \tag{18}$$

which gives $|S| = 15$. Similarly, letting $X = S$, $Y = \{1, 2, \ldots, n\}$ and $G = \{(x, j) : x \in B_j\}$, we get

$$n \times 3 = |G| = |S| \times 9 \tag{19}$$

Since we already know $|S| = 15$, this gives $n = \frac{15 \times 9}{3} = 45$.

Comment No. 11:

We conclude the discussion on counting problems with a brief mention of how algebra can be used in counting problems. Suppose we want to find the number of ways to put seven identical balls into three distinct boxes in such a way that the first box contains 0, 1 or 3 balls, the second one contains 1, 2 or 3 balls and the third one contains 4 or 6 balls. Each such placement can be represented by an ordered triple of the form (n_1, n_2, n_3) where $n_1 = 0, 1$ or 3, $n_2 = 1, 2$ or 3, $n_3 = 4$ or 6 and $n_1 + n_2 + n_3 = 7$. Because of the relation $x^{n_1}x^{n_2}x^{n_3} = x^{n_1+n_2+n_3}$, this is equivalent to a factorisation of x^7 into factors of the form $x^{n_1}x^{n_2}x^{n_3}$, where the exponents n_1, n_2, n_3 satisfy the restrictions given. As a result, the desired number of placements is simply the coefficient of x^7 in the polynomial $(1+x+x^3)(x+x^2+x^3)(x^4+x^6) = p(x)$ (say). This observation *does not* by itself simplify the problem, because to find the coefficient of x^7 in $p(x)$ by direct expansion, the work involved will be essentially the same as that of finding the number of ways to put 7 balls into 3 boxes subject to the given restrictions. But the advantage is that we can manipulate this expression. For example, by taking out a common factor x from every term in the second factor and x^4 from every term in the third factor, we see that $p(x) = x^5 q(x)$, where $q(x) = (1 + x + x^3)(1 + x + x^2)(1 + x^2)$. Therefore, the coefficient of x^7 in $p(x)$ is the same as that of x^2 in $q(x)$. To find the latter, we may obviously omit the term x^3 from the first factor. Also in taking the product of the second and the third factors we drop all the terms from x^3 onwards. Thus the desired answer is the coefficient of x^2 in $(1 + x)(1 + x + 2x^2)$. By inspection, this comes out as 3. So there are three ways to place 7 balls into 3 distinct boxes satisfying the given restrictions.

The combinatorial equivalent of taking the factors x and x^4 out is to begin by putting one ball in the second and four balls in the third box. But a person is not likely to think of this as readily as he is to take the factors x and x^4 out, because algebraic manipulation is a more familiar process.

At the heart of this method is the law of indices $x^m x^n = x^{m+n}$. In essence what we are doing here is forming an algebraic codification of a combinatorial

process. In spirit this is the same as representing a point in a plane by cartesian coordinates. The success of such codification depends upon how well we are able to manipulate the algebraic expression. It turns out that if we confine ourselves to polynomials, then we cannot go very deep. But if we use infinite series of powers of x (called **power series**) then the method works wonders. As power series are beyond our scope, we shall not pursue this topic further. However, in Chapter 23 we shall deal with one particular infinite series, viz., the geometric series $1 + x + x^2 + x^3 + \ldots + x^n + \ldots = \sum_{n=0}^{\infty} x^n$. As an application of it, we shall show how we can get an alternate derivation of the number of selections with unlimited repetitions allowed. (Earlier we obtained this using a rather tricky bijection.) Apart from such direct applications to counting problems, the power series are also an important tool for solving recurrence relations.

The binomial theorem is also an algebraic codification of a combinatorial process. Let n be a positive integer and consider the expression $(x+y)^n$. When written out in full, this is the product $(x+y)(x+y)\ldots(x+y)$, where there are n factors, each equalling $x+y$. So the complete expansion will have 2^n terms in all. For a given r between 0 and n, a term $x^r y^{n-r}$ will arise if we take x from exactly r of these factors and y from the remaining $n-r$ factors. Since these r factors can be chosen in $\binom{n}{r}$ ways, the coefficient of $x^r y^{n-r}$ is $\binom{n}{r}$. If we put $y = 1$, then $(1+x)^n$ is a polynomial in x of degree n and $\binom{n}{r}$ is precisely the coefficient of x^r in it. In Chapter 5 we shall see many interesting applications of this simple observation.

Comment No. 12:

Puzzle type problems are rarely asked at examinations like the JEE. But the reasoning needed in some puzzles is quite mathematical and so they deserve at least a mention. The kind of mathematics they need is usually within a layman's reach and so the best place to accommodate them is in counting problems. (There are other branches of mathematics such as **graph theory** and **Boolean algebra** that are more suited to a mathematical paraphrase of certain puzzles. But they are beyond our scope.)

In the more straightforward puzzles, you are given some set S of possibilities and some pieces of information and asked to find out which possibility holds. For example, you may be given some information about 4 men and their wives and asked to find out who is married to whom. Here the set S has $24 (= 4!)$ elements. The various pieces of information correspond to various subsets of S (often called their **truth sets**) and their intersection gives the answer.

In the more challenging puzzles, you are not given the information directly. Instead you are given a mechanism to find certain information. For example, you may be allowed to ask a sequence of questions that invoke only a 'yes' or 'no' as an answer and you may also be given the freedom to frame your questions on the basis of the answers to earlier questions. We discuss here one popular puzzle in which we are given a balance and 12 coins which look identical but out of which, one coin is a fake. The fake coin differs in weight slightly from

the genuine coins. But it is not given whether it is lighter or heavier. Using the balance, we have to identify the fake coin and also whether it is lighter or heavier than a genuine coin. Naturally, we have to do this using the balance minimum number of times. (Otherwise, we could just go on weighing every pair of coins.)

Let us first paraphrase this puzzle in terms of sets. Call the coins as $C_1, C_2, \ldots, C_{12}$. For $i = 1, 2, \ldots, 12$, let L_i be the statement, 'C_i is lighter than a genuine coin' and H_i be the statement, 'C_i is heavier than a genuine coin'. Let S be the set of these 24 statements. Then exactly one member of S is true and we have to find out which one it is. Every time we use the balance, there are three possible outcomes and depending upon which of them occurs, the search will narrow down to one of three mutually disjoint subsets. For example, if we begin by weighing C_1 against C_2 then equality of the two pans will eliminate L_1, H_1, L_2 and H_2 and the true statement must be one of the remaining 20 statements. But if C_1 is heavier than C_2 then the search narrows down to the set $\{H_1, L_2\}$. Now we want to use the balance in such a way that regardless of the sequence of outcomes, we shall finally end up with a singleton subset of S, i.e., a subset with only one element.

This type of a reasoning does not immediately give the answer. But it helps in designing an efficient sequence of weighings. Every weighing reduces the search to one of three mutually disjoint subsets of the set to which the search had been reduced just before that weighing. In general, in k weighings, the original set S will get decomposed into 3^k mutually disjoint subsets (some of which may be empty). When all these subsets have at most one element in them, the solution is complete. Since $3^2 = 9 < 24$, it follows that the solution cannot be obtained in two weighings in *every* case. Since $3^3 > 24$, it *may be* possible to do the problem in three weighings if we use them efficiently. This requires that at every weighing, the three subsets should be more or less of the same size so that even if we land in the largest of these three subsets, we should not need a large number of weighings subsequently.

This makes it mandatory that at the first weighing we should have four coins in each pan of the balance. (We are assuming that the difference between the weights of a genuine and a fake coin is small enough so that there is no point in weighing unequal numbers of coins against each other.) The set S then gets divided into three subsets of cardinality 8 each. We leave you to complete the solution because we do not want to rob you of the charm of solving it. But we hope the comments here illustrate how the reasoning required is highly mathematical. There is a recently developed branch of mathematics, called the 'design and analysis of **algorithms**' which needs this type of reasoning. In a nutshell, this branch is aimed at making your computer work faster!

EXERCISES

1.1 (a) Set A has 3 elements and set B has 6 elements. What can be the minimum number of elements in the set $A \cup B$? (This was asked way back when set theory was a 'new' topic for JEE. Don't expect such

trivial questions now!) (1980)

(b) Let A and B be two sets each with a finite number of elements. Assume that there is an injective mapping from A to B and that there is an injective mapping from B to A. Prove that there is a bijective mapping from A to B. (1981)

(c) Let A be a set of n distinct elements. Find the total number of distinct functions from A to A. Also find how many of these functions are onto. (1985)

1.2 For a subset A of a set S, let A' denote its complement w.r.t. S. For any three subsets A, B, C of S prove that :

(a) $(A \cap B)' = A' \cup B'$ and $(A \cup B)' = A' \cap B'$ (**DeMorgan's laws**.)

(b) $A \times (B' \cup C')' = (A \times B) \cap (A \times C)$ (1980)

(c) $(A \circ B) \circ C = A \circ (B \circ C)$ where $A \circ B = (A \cap B') \cup (A' \cap B)$ etc. [*Hint*: Draw suitable Venn diagrams.] (1980)

1.3 In how many ways can a collection of $3n$ distinct objects be divided into n triplets, each having 3 objects?

1.4 Find the total number of ways in which six '+' and four '−' signs can be arranged in a line so that no two '−' signs can occur together. (1988)

1.5 A man has 7 relatives, 4 of them are ladies and 3 gentlemen; his wife also has 7 relatives, 3 of them are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of the man's relatives and 3 of the wife's relatives? (1985)

1.6 A committee of 12 is to be formed from 9 women and 8 men. In how many ways can this be done if at least five women have to be included in a committee? In how many of these committees : (a) the women are in majority? (b) the men are in majority? (1994)

1.7 In how many ways can three girls and nine boys be seated in two vans, each having numbered seats, 3 in the front and 4 in the back? How many seating arrangements are possible if the three girls should sit together in a back row on adjacent seats? Now, if all the seating arrangements are equally likely, what is the probability of 3 girls sitting together in a back row on adjacent seats? (1996)

1.8 Suppose a travelling salesman wants to make a round air tour of n cities, visiting each city exactly once and returning to the starting city. Assuming there is a direct flight between every two of these cities, how many such tours are there? If the starting point does not matter, i.e., if only the circular order in which the cities are visited matters, how many tours are there? What will be the answer, if further, the sense of the tour does not matter, i.e., two tours in which the cities are visited in the opposite order are not to be distinguished from each other?

1.9 Give an algebraic as well as a combinatorial proof of the identity $1 + \sum_{r=1}^{n-1} r \times r! = n!$ for every positive integer n.

1.10 Let n and k be positive integers with $k \leq n!$. Then prove that there exist unique integers $d_1, d_2, \ldots, d_{n-1}$ such that $0 \leq d_i \leq n - i$ for $i = 1, 2, \ldots, n-1$ and $k = 1 + \sum_{i=1}^{n-1} d_i(n-i)!$. [*Hint:* For $k = n!$, apply the last exercise. Otherwise, let d_1 be the largest non-negative integer such that $d_1(n-1)! < k$. Let $r = k - d_1(n-1)!$. Then $1 \leq r \leq (n-1)!$ and repeat the argument with n replaced by $n-1$ and k by r. Continue. This gives the existence of the integers $d_1, d_2, \ldots, d_n$. For uniqueness, observe that $|\{(d_1, d_2, \ldots, d_{n-1}) : 0 \leq d_i \leq n - i, \text{ for } i = 1, 2, \ldots, n-1\}| = n!$.

1.11 Given a positive integer k, let n be the smallest positive integer such that $n! \geq k$. Show that if k is expressed as in the last exercise then $d_1 > 0$. (The expression $d_1(n-1)! + d_2(n-2)! + \ldots + d_{n-1}1! + 1$ is called the **factorial representation** of k. The d_i's actually have a certain combinatorial significance. But we shall not go into it.) Express 50, 100 and 1000 in this form.

1.12 An n-digit number is a positive number with exactly n digits. Nine hundred distinct n-digit numbers are to be formed using only the three digits $2, 5$ and 7. Find the smallest value of n for which this is possible. (1998)

1.13 Ten different letters of an alphabet are given. Words with five letters are formed from these given letters. Find the number of words which have at least one letter repeated (although not necessarily consecutively). (1980)

1.14 A **palindrome** is a word which reads the same backward as forward (e.g., 'MADAM' or 'ANNA'). Find how many palindromes of length n can be formed from an alphabet of k letters.

1.15 A real-valued function f defined on a subset, say D, of real numbers is called **monotonically increasing** if $f(x) \leq f(y)$ whenever x, y are two points in D with $x \leq y$. The function is called **strictly monotonically increasing** if $f(x) < f(y)$, whenever $x < y$ in D. A sequence of length k is a function defined on the set $\{1, 2, \ldots, k\}$. Let S be the set of all monotonically increasing sequences of length k taking values in the set $\{1, 2, \ldots, n\}$ and let T be the set of all strictly monotonically increasing sequences of length k taking values in the set $\{1, 2, \ldots, n+k-1\}$. Prove that $|S| = |T| = \binom{n+k-1}{k}$. [*Hint:* Define a bijection θ from S to T as follows. Denote a sequence of length k by an ordered k-tuple, say $(a_1, a_2, \ldots, a_k)$. Let $\theta(a_1, a_2, \ldots, a_k)$ be the sequence $(a_1, a_2 + 1, a_3 + 2, \ldots, a_i + i - 1, \ldots, a_k + k - 1)$.]

1.16 (i) Using the last exercise, give an alternate proof that the number of selections of k objects from n types of objects, repetitions being allowed freely, is $\binom{n+k-1}{k}$.

(ii) Find the maximum number of 5-digit numbers (leading zeros allowed) no two of which can be obtained from each other by a permutation of the digits. [*Hint:* Among all integers which can be obtained from a given number by permutations of the digits, select the smallest one.]

1.17 Prove that the number of ways to put r identical objects into n distinct boxes is $\binom{r+n-1}{r}$. What if we further require that no box be empty?

1.18 A person has 3 sons. He owns 101 shares of a company. He wants to give these to his sons so that no son should have more shares than the combined total of the other two. In how many ways can he do so?

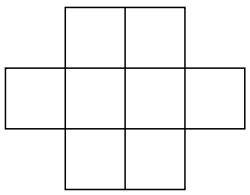
1.19 (a) Find the number of ways to seat m men and n women in a circle so that no two women are seated together.

(b) Find how many different circular bracelets can be formed using $6n$ blue and 3 red beads, where n is a positive integer.

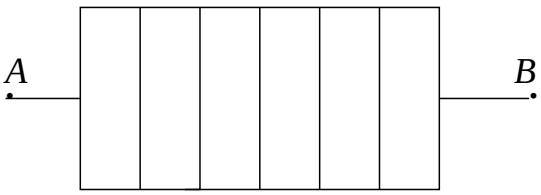
1.20 A city has m parallel roads going East−West and n parallel roads going North−South. How many rectangles can be formed with their sides along these roads? If the distance between every consecutive pair of parallel roads is the same, how many shortest possible paths are there to go from one corner of the city to its diagonally opposite corner?

1.21 In the last exercise suppose 2^k persons (where $k \leq \min\{m-1, n-1\}$) start from the southwest corner of the city. Half of these persons proceed eastward and the other half northward. They travel with the same speed and reach the next junction at the end of one unit of time. Persons arriving at each junction again bifurcate, half of them going eastward and half northward and then all keep on traveling with the same speed. If this process continues for j units of time where $0 \leq j \leq k$, find how many persons there will be at each junction, at the end of j units of time.

1.22 Six X's have to be placed in the squares of Figure (a) below in such a way that each row contains at least one X. In how many different ways can this be done? (1978)



(a)



(b)

1.23 How many paths are there from the point A to the point B in Figure (b) above if no point in a path is to be traversed more than once?

1.24 A box contains two white balls, three black balls and four red balls. In how many ways can three balls be drawn from the box if at least one black ball is to be included in the draw? (1986)

1.25 (a) Let n and k be positive integers with $n \geq \frac{1}{2}k(k+1)$. Find the number of solutions $(x_1, x_2, \ldots, x_k) : x_1 \geq 1, x_2 \geq 2, \ldots, x_k \geq k$, (all integers), satisfying $x_1 + x_2 + \ldots + x_k = n$. (1996)

(b) By a **composition** of a positive integer n we mean any finite sequence $(x_1, x_2, \ldots, x_k)$ of positive integers such that $x_1 + x_2 + \ldots + x_k = n$. (Here k can vary. Obviously, k cannot exceed n. Note that the order of the summands matters. In other words, $(2, 1, 1)$ and $(1, 2, 1)$ are two different compositions of 4.) Find the total number of compositions of n. [*Hint:* Experiment with small values of n. Come up with a guess and then prove it combinatorially by getting a bijection with a suitable set.]

(c) Given n straight line segments of lengths $1, 2, \ldots, n$, show that the number of triangles that can be formed from them equals $\frac{n(n-2)(2n-5)}{24}$ if n is even and $\frac{(n-3)(n-1)(2n-1)}{24}$ if n is odd.

1.26 What will be the answer to the problem in Comment No. 3 if the three boxes are of the same size and indistinguishable from each other (there being no other change)? Can the answer be obtained directly from that of the original problem?

1.27 Let m, n be positive integers. The number of ways to put n distinct objects into m non-distinct boxes so that no box is empty is called a **Stirling number** of the second kind and denoted by $S_{n,m}$. (To be sure, there are also Stirling numbers of the first kind. But we shall not go into them.) For all n prove that:

(i) $S_{n,m} = 0$ if $m > n$ (ii) $S_{n,n} = S_{n,1} = 1$

(iii) $S_{n,2} = 2^{n-1} - 1$ (iv) $S_{n,n-1} = \frac{1}{2}n(n-1)$.

These numbers arise naturally in many different counting problems. For example, show that the number of ways to put n distinct objects into m distinct boxes so that no box is empty is $m!S_{n,m}$. Since every such placement amounts to a surjective function from the set of objects to the set of boxes, this is also the number of all surjective functions from a set with n elements to a set with m elements. Further show that the number of ways to put n distinct objects into r non-distinct boxes (there being no restriction regarding any of the boxes being empty) is the sum $\sum_{m=1}^{r} S_{n,m}$.

1.28 For all $m \geq 2$ and $n \geq 2$, prove that $S_{n,m} = mS_{n-1,m} + S_{n-1,m-1}$. (Starting from the values of the Stirling numbers given in the last exercise, repeated applications of this recurrence relation make it possible to

determine $S_{n,m}$ for any values of m and n (with $m \leq n$). For small values of n, say $n = 8$, a table can be prepared for easy reference.)

1.29 Using the recurrence relation in the last exercise compute $S_{5,3}$. Hence solve Exercise (1.26) and then the problem in Comment No. 3.

1.30 In how many ways can 3 blue, 4 white and 2 red balls be distributed into 4 distinct boxes?

1.31 A vendor sells tickets costing 1 rupee each. 100 customers approach him one - by - one. 50 of them tender a one rupee coin each while the other 50 tender 2 rupees coin each and claim a change of one rupee each. The vendor has no money to begin with. If the customers approach him in a random order, what is the probability that he will not run out of change when needed? [*Hint:* Relate the problem to that of counting the number of balanced arrangements of parentheses.]

1.32 Suppose there are 20 players of different heights. These are to be divided into two teams, A and B, of 10 players each so that for every $i = 1, 2, \ldots, 10$, the i-th tallest player in A will be taller than the i-th tallest player in B. In how many ways can this be done?

1.33 Let n be a fixed positive integer and S be the set of all ordered $(2n + 1)$-tuples $(a_0, a_1, a_2, \ldots, a_{2n-1}, a_{2n})$ of non-negative integers such that (i) $a_0 = a_{2n} = 0$ and (ii) $|a_i - a_{i-1}| = 1$ for every $i = 1, 2, \ldots, 2n$. Prove that $|S|$ is the n-th Catalan number $\frac{1}{n+1}\binom{2n}{n}$.

1.34 Prove that for every positive integer n, the number of monotonically increasing functions from the set $\{1, 2, \ldots, n\}$ to itself with the property that $f(x) \geq x$ for all $x = 1, 2, \ldots, n$ equals the Catalan number $\frac{1}{n+1}\binom{2n}{n}$.

*1.35 The operation of addition for real numbers is **associative**. This means that for any three (not necessarily distinct) real numbers a, b and c, we have $(a+b)+c = a+(b+c)$. That is why we need not put the parentheses and can write $a+b+c$ without ambiguity. If associativity did not hold, then the expression $a+b+c$ could have two possible interpretations obtained by inserting parentheses in two different ways. Prove that in the absence of associativity, for every positive integer $n \geq 2$, the expression $a_1+a_2+\ldots+a_{n-1}+a_n$ will have $\frac{1}{n}\binom{2n-2}{n-1}$ possible interpretations. (For example, for 4 real numbers a_1, a_2, a_3 and a_4, the expression $a_1+a_2+a_3+a_4$ could have 5 different interpretations, viz., $[[[a_1+a_2]+a_3]+a_4]$, $[[a_1+[a_2+a_3]]+a_4]$, $[a_1+[a_2+[a_3+a_4]]]$, $[a_1+[[a_2+a_3]+a_4]$ and $[[a_1+a_2]+[a_3+a_4]]$ where every time we have applied $+$, the result has been enclosed within a pair of brackets. This is yet another instance where the Catalan numbers occur in the solution. However, unlike the last four problems, here the correspondence with balanced arrangements of $n-1$ pairs of parentheses is not so obvious. To get it, ignore the left brackets and all occurrences of $+$ sign. Also ignore a_1 and change each $a_2, a_3, \ldots, a_n$ to a left parenthesis

and change each right bracket to a right parenthesis. Thus, $[a_1+[a_2+a_3]]$ changes to (( )), while $[[a_1+a_2]+a_3]$ changes to ( ) ( ). Show that this gives a bijection from the set of all possible interpretations of $a_1+a_2+\ldots+a_n$ to the set of all balanced arrangements of $n-1$ pairs of parentheses.)

1.36 A set of (straight) lines in a plane is said to be in **general position** if no two of them are parallel and no three of them are concurrent. (As the name implies, the contrary is exceptional. That is, if lines are drawn at random in a plane, it is extremely unlikely that some two of them will be parallel or that some three of them will be concurrent. This is consistent with common sense because for either of these two things to occur, the lines will have to be drawn with a special effort, and not randomly. A mathematically precise interpretation can also be given to the term 'extremely unlikely' here. See Exercise (23.18). Similarly we define a set of planes in space to be 'in general position'.) Let a_n be the number of regions (some of which may be unbounded) into which a plane is decomposed by n lines in general position and b_n be the number of regions into which the space is decomposed by n planes in general position. Prove that $a_n = a_{n-1}+n$ and $b_n = a_{n-1}+b_{n-1}$ for $n>1$. Hence show that $a_n = \frac{1}{2}(n^2+n+2)$. Obtain a similar formula for b_n. [*Hint* : First experiment with small values of n. Note that the last line/plane increases the count by as much as the number of earlier regions it cuts into two subregions each.)

1.37 How many diagonals does a convex n-gon have? If no three of them are concurrent except possibly at a vertex, find the number of segments into which these diagonals are decomposed by one another. [*Hint* : Because of convexity, every four vertices determine a unique point of intersection of two diagonals. Every segment has one or two such points as its ends.]

1.38 Take an 8×8 chess board and remove two diagonally opposite corner squares. Prove that it is impossible to pair off the remaining 62 squares in such a way that every pair contains two adjacent squares, i.e., squares having one side in common. Do the problem both without and with colouring the squares. This illustrates the facility gained by the introduction of an additional device, in this case, the colouring of squares.

1.39 In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. Find the number of newspapers. (1998)

1.40 A box contains secret documents and only 5 persons are privileged to have access to them. Design a system of ordinary locks and keys for this box so that when any three but no fewer of these 5 persons come together, they can open the box.

1.41 For any three subsets A, B, C of a finite set S, prove :

$$\begin{aligned} &\text{(i)} && |A'\cap B'| &&= |S|-(|A|+|B|)-|A\cap B| \\ &\text{(ii)} && |A'\cap B'\cap C'| &&= |S|-(|A|+|B|+|C|) \\ & && && \quad +(|A\cap B|+|B\cap C|+|C\cap A|)-|A\cap B\cap C| \end{aligned}$$

where $'$ denotes complementation w.r.t. S.

1.42 Using the last exercise, find the number of ways to seat m men and 3 women in a row so that no two women sit adjacently. [*Hint:* Let the women be w_1, w_2 and w_3. Let S be the set of all $(m+3)!$ arrangements and A the subset of those in which w_1, w_2 sit together, B the subset of those in which w_2, w_3 sit together and C the set of those in which w_1, w_3 sit together. The method given in Comment No. 5 is, of course, simpler.]

1.43 Prove, by induction on n or otherwise, the following general version of the principle of inclusion and exclusion. Let $A_1, A_2, \ldots, A_n$ be subsets of a finite set S. For $r = 1, 2, \ldots, n$, let s_r be the sum of the cardinalities of the intersections of these subsets taken r at a time. Thus $s_1 = \sum_{i=1}^{n} |A_i|, s_2 = \sum_{i<j} |A_i \cap A_j|, s_3 = \sum_{i<j<k} |A_i \cap A_j \cap A_k|$ and more generally, $s_r = \sum_{1 \le i_1 < i_2 < \ldots < i_r \le n} |A_{i_1} \cap A_{i_2} \cap \ldots \cap A_{i_r}|$. Note that s_n is simply $|A_1 \cap A_2 \cap \ldots \cap A_n|$. Set $s_0 = |S|$. Then

$$\left| \bigcap_{i=1}^{n} A_i' \right| = s_0 - s_1 + s_2 - s_3 + \ldots + (-1)^n s_n = \sum_{r=0}^{n} (-1)^r s_r \qquad (20)$$

1.44 In how many ways can the four walls of a room be painted with three colours so that no two adjacent walls have the same colour?

1.45 A **derangement** is a permutation[3] in which no symbol remains in its original place. For example, (2,3,1,5,4) is a derangement of (1,2,3,4,5) but (2,3,5,4,1) is not. Let D_n be the number of derangements of n symbols. Using (20) show that $D_n = n!(\frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + (-1)^n \frac{1}{n!}) = n! \sum_{k=0}^{n} (-1)^k \frac{1}{k!}$ with the understanding that $0! = 1$. (The first two terms are actually redundant as they cancel each other. But they are included for the sake of notational uniformity.) [*Hint:* For $i = 1, 2, \ldots, n$, let A_i be the set of those permutations in which the integer i (and possibly some others) remains in its own place.]

1.46 There are four balls of different colours and four boxes of colours, same as those of the balls. Find the number of ways in which the balls, one each in a box, could be placed such that a ball does not go to a box of its own colour. (1992)

1.47 Ten cards are paired into 5 pairs of two each. They are put together, shuffled and paired back again. Find the probability that no pair in the first pairing appears in the second.

1.48 Complete the solution to the puzzle in Comment No. 12. If instead of 12 coins we had 13 coins show that it is impossible to detect the fake coin in 3 weighings in *every* case even though $3^3 > 26$.

[3] Popularly, a derangement is an arrangement in which nothing is in its 'right' place.

Chapter 2

BASIC ALGEBRA

As the title indicates, this chapter introduces you to the basic concepts in algebra. Specifically, it includes progressions and means of the three types (viz., arithmetic, geometric and harmonic), recursively defined sequences, exponentials and logarithms, polynomials, the binomial theorem, the complex numbers, matrices and finally determinants. All of these topics, except possibly the matrices, are standard. We have included matrices for two reasons. First, they provide a foundation to determinants and to systems of linear equations (to be studied in the next chapter). Secondly, they are needed so frequently in collegiate mathematics that we believe a student completing Junior College should have at least a nodding acquaintance with them even though at present they are not included in the syllabi[4] of the JEE or of some of the higher secondary boards. But we have kept the treatment optional. So, in subsequent chapters, where there are references to matrices, they can be skipped without affecting continuity.

The topics covered here are all classic and old texts contain a bounty of good problems involving them. In the JEE papers, after the introduction of calculus, progressions and logarithms are generally relegated to short questions. The topics here are so basic that questions combining them with other topics such as geometry, vectors or inequalities are frequent. So this chapter is an important prerequisite to others apart from being a rich collection of interesting problems in its own right.

[4]Matrices are included in the JEE syllabus from 2003 onwards. Many of the Higher Secondary School boards have already included them.

Main Problem : For an integer $n \geq 3$, let S_n denote the sum of the products of the integers from 1 to n taken three at a time. (For example, $S_3 = 1\times2\times3 = 6$, $S_4 = 6 + 8 + 12 + 24 = 50$ and so on.) Then $S_{10} =$ .....

First Hint: Consider the terms in the expansion of $(1 + 2 + \ldots + 10)^3$.

Second Hint: Classify these terms according to the number of times they occur.

Solution: There are in all 1000 terms in the expansion of $(1+2+\ldots+10)^3$. Terms of the form ijk with i, j, k all distinct appear 6 times each. We are interested in the sum of these terms, each such term being considered only once. Then there are terms of the form i^2j with $i \neq j$. Each such product appears three times. Finally, there are terms of the form i^3. Each such term appears only once. Hence,

$$A = 6S_{10} + 3B + C \tag{1}$$

where $A = (\sum_{i=1}^{10} i)^3$, $C = \sum_{i=1}^{10} i^3$ and $B = \sum_{i\neq j}^{10} i^2j$. Using the well-known formulas

$$1 + 2 + \ldots + n = \frac{n(n+1)}{2} \tag{2}$$

$$\text{and } 1^3 + 2^3 + \ldots + n^3 = \frac{n^2(n+1)^2}{4} \tag{3}$$

A and C come out to be respectively, $(55)^3$ and $(55)^2$. To calculate B, note that if we add to B products of the form i^2i, then we get all the terms of the product $(1^2 + 2^2 + \ldots + 10^2)(1 + 2 + \ldots + 10)$. The first factor equals 55×7 using the formula

$$1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6} \tag{4}$$

Hence $B = 55 \times 7 \times 55 - (\sum_{i=1}^{10} i^3) = 6 \times (55)^2$. Putting these values in (1),

$$\begin{aligned} S_{10} &= \frac{1}{6}(A - 3B - C) \\ &= \frac{(55)^3 - 18 \times (55)^2 - (55)^2}{6} \\ &= \frac{(55)^2(55 - 19)}{6} = (55)^2 \times 6 \\ &= 121 \times 150 = 18150 \end{aligned} \tag{5}$$

Comment No. 1:

There is nothing special about the integer 10 here. For any $n \geq 3$, exactly the same argument as above would give

$$S_n = \frac{1}{48} n^2 (n+1)^2 (n^2 - 3n + 2) \tag{6}$$

We could get (5) by putting $n = 10$ in (6). In fact in certain respects this is a better idea because in this approach the particular values come only at the end and the only numerical calculation involved is $S_{10} = \frac{100 \times 121}{48} \times 72 = 150 \times 121 = 18150$. In the solution above, the numbers enter early and the calculations are prone to numerical mistakes. Also when you see something like $(55)^3$, there is a temptation to expand it. In the present problem, this work would be a waste because all the terms have $(55)^2$ as a factor and the other factors of the terms add up to give an expression which is more manageable. When you work with a general n, the question of expanding n^3 does not arise at all.

In fact, the problem would look more natural if it asks you to prove (6). But when asked in this conventional form, it poses difficulties for the examiner. In order to save time, the candidates sometimes combine two or three intermediate steps together without giving adequate verbal justification. An examiner may get the impression that the candidate is trying to cheat. And, sometimes he is right! When the answer is given, it is not difficult to work backwards for a couple of steps and give the impression that it has really been arrived at honestly. Naturally, this can lead to controversies as it is ultimately a matter of judgement to decide whether the candidate knows the justification but has not given it or whether he simply does not know it. Partly to avoid such controversies and partly to expedite the evaluation, there is a general trend nowadays to convert the question to a form where it will have a single numerical answer. If the answer is correct, there is no need to look at the working since it is extremely unlikely that anybody could get an answer like 18150 by bluffing. The trouble, of course, is that if such questions are asked in the 'fill in the blank' form where no work is to be shown, candidates who work the problem correctly all the way but make a computational slip somewhere suffer heavily.

Comment No. 2:

Formulas (2), (3) and (4) are quite standard. They can all be proved by the method of mathematical induction. In fact, they are often given among the first illustrations of this method. Alternate proofs are available. For example, the best way to prove (2) is to write the sum on the left backwards as $n + (n-1) + \ldots + 2 + 1$. When added to the original sum term-by-term, this gives a term $(n+1)$ repeated n times. Hence $n(n+1)$ is twice the sum! (A story goes that a German schoolboy with the name **Gauss** observed this when he was 7. Later he became one of the greatest mathematicians of all times.) A similar approach is also possible for (3) if we observe that for $i = 1, 2, \ldots, n$, $i^3 + (n+1-i)^3 = (n+1)^3 - 3(n+1)^2 i + 3(n+1)i^2$ from which it follows that $2\sum_{i=1}^{n} i^3 = n(n+1)^3 - 3(n+1)^2 \sum_{i=1}^{n} i + 3(n+1)\sum_{i=1}^{n} i^2$. So, if (2) and (4)

are already proved then (3) would follow. But a similar approach for (4) is not possible. (Why?)

In Chapter 5, Comment No. 13, (see also Exercise (5.14)(a)), we shall prove these formulas using some binomial identities. A perceptive reader will be led to think that the sum of the r-th powers of the first n positive integers will be a polynomial in n. (The more perceptive will probably also guess that the coefficient of n^{r+1} in this polynomial will be $\frac{1}{r+1}$.) This guess turns out to be quite correct. There is, in fact, a succinct expression for $\sum_{i=1}^{n} i^r$ in terms of what are called **Bernoulli** numbers. But we shall not go into it.

Comment No. 3:

Formula (2) can be applied to give the sum of the first n terms of an arithmetic progression (often abbreviated to A.P.) Specifically, if the initial term is a and the common difference is d, then the sum of the first n terms is given by

$$\begin{aligned} a + (a+d) + \ldots \quad &+ \quad (a + (n-1)d) \\ &= \quad \sum_{r=1}^{n} (a + (r-1)d) = na + \frac{n(n-1)}{2} d \end{aligned} \tag{7}$$

There is a similar formula for the sum of the first n terms of a geometric progression (G.P.) with initial term a and common ratio $r \neq 1$.

$$a + ar + ar^2 + \ldots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1} \tag{8}$$

Although this formula is valid only for $r \neq 1$, the L.H.S. makes sense for $r = 1$ and equals na. The R.H.S. is meaningless for $r = 1$. But if we take its limit as $r \to 1$, then it can be shown that it tends to na. So with this understanding, (8) is valid for all r.

There is yet another kind of a progression, viz., the harmonic progression (H.P.). By definition, this is a progression whose reciprocals are in an A.P. In other words, the numbers $b_1, \ldots, b_n, \ldots$, are in an H.P. if there exists some number d such that for every $i \geq 1$, $\frac{1}{b_{i+1}} - \frac{1}{b_i} = d$. Unlike (7) and (8), there is no analogous formula for the sum of the first n terms of an H.P. In particular, there is no succinct formula for the sum $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$. This sum appears rather frequently in applications. It is called the n-th **harmonic number** and is often denoted by H_n. It can be shown that for large n, H_n is very close to $\ln n$, where ln denotes the natural logarithm, i.e., logarithm w.r.t. the base e. (See Exercise (6.53) for more on harmonic numbers.)

Possibly because of the lack of a formula for the sum of the terms of an H.P., these progressions do not appear as frequently in problems as the other two types of progressions. The best way to handle H.P.s is by taking the reciprocals of their terms. The same holds true for the corresponding three types of means, the arithmetic mean (A.M.), the geometric mean (G.M.) and the harmonic

mean[1] (H.M.). The A.M., which is another name for the concept of an average, can be defined for any collection of numbers, say $a_1, a_2, \ldots, a_n$. It is simply $\dfrac{a_1 + a_2 + \ldots + a_n}{n}$. The geometric mean requires the extraction of roots and so is generally defined only when the numbers are all positive. Specifically, it is the n-th root[2] of the product of the numbers, i.e., $(a_1 a_2 \ldots a_n)^{1/n} = \sqrt[n]{a_1 a_2 \ldots a_n}$. The harmonic mean, which appears less frequently, is defined as the reciprocal of the arithmetic mean of the reciprocals, i.e., $\dfrac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \ldots + \frac{1}{a_n}}$. There is a well-known inequality which says that among these three means, the A.M. is the largest, followed by the G.M. and then the H.M. (See Chapter 6.)

Problems involving the three types of progressions and means are generally straightforward and standard texts give plenty of them. Here we do just one problem involving these concepts.

Let $a_1, a_2, \ldots, a_n$ be positive real numbers in geometric progression. For each n, let A_n, G_n, H_n be, respectively, the arithmetic mean, the geometric mean and the harmonic mean of $a_1, a_2, \ldots, a_n$. Find an expression for the geometric mean of $G_1, G_2, \ldots, G_n$ in terms of $A_1, A_2, \ldots, A_n, H_1, H_2, \ldots, H_n$. (JEE 2001)

(As n is a fixed integer here, in the second sentence of the statement of the problem, it would save confusion, if some symbol like k or i were used instead of n to denote the index variable.) Let

$$M = (G_1 G_2 \ldots G_n)^{1/n} \tag{9}$$

Our goal is to express M in terms of $A_1, A_2, \ldots, A_n, H_1, H_2, \ldots, H_n$. For each $k = 1, 2, \ldots, n$ we can express G_k in terms of $a_1, a_2, \ldots, a_k$ by

$$G_k = (a_1 a_2 \ldots a_k)^{1/k} \tag{10}$$

So if we could express $a_1, a_2, \ldots, a_n$ in terms of $A_1, A_2, \ldots, A_n, H_1, \ldots, H_n$, we would be through. Actually, we do not need the harmonic means for this purpose, because we can express the a_k's in terms of A_j's alone. This is very easy to do. Clearly, $a_1 = A_1$. Also, $A_2 = \frac{a_1 + a_2}{2}$ gives $a_2 = 2A_2 - a_1 = 2A_2 - A_1$. Similarly, from $3A_3 = a_1 + a_2 + a_3$ we can express a_3 as $3A_3 - 2A_2$. More generally, for every $k = 2, 3, \ldots, n$ we have

$$a_k = kA_k - (k-1)A_{k-1} \tag{11}$$

[1]The origins of the peculiar names of the three types of means deserve a mention, although, of course, such information does not help you in solving problems. The arithmetic mean is the same as the average one studies in arithmetic. The geometric mean of two positive real numbers, say, a and b has a geometric significance. It is the side of a square whose area equals that of a rectangle with sides a and b. There is a geometric construction for such a square (see Exercise (11.22)). The harmonic mean has a connection with harmony, which is a concept in music. It is based on the fact that the frequency or the 'pitch' of the sound produced by a string is inversely proportional to its length (assuming that other factors such as the thickness and the tension in the string are the same.) Given two strings with the frequency of one being the double of the other, the frequency which 'harmonises' best with the two is not the A.M. but the H.M. of their frequencies.

[2]The existence of the n-th root of a real number is far from obvious. An intuitive argument is given in Comment No. 6 below.

If we put (11) (along with $a_1 = A_1$) into (10) and then (10) into (9), we get an expression for M in terms of $A_1, A_2, \ldots, A_n$. Similarly, we could have used the relations $H_1 = a_1, H_2 = \dfrac{2}{\frac{1}{a_1} + \frac{1}{a_2}}$ and, in general, $H_k = \dfrac{k}{\frac{1}{a_1} + \ldots + \frac{1}{a_k}}$ for $k = 1, 2, \ldots, n$ to express the a_k's, one by one, in terms of H_j's and hence M in terms of $H_1, H_2, \ldots, H_n$. Logically, either of these is a correct solution. In fact, these are, in a way, better solutions because they work even if the numbers $a_1, a_2, \ldots, a_n$ are not given to be in a G.P.

From this point of view the problem is somewhat ill-posed. A well-posed problem generally has a unique answer (although, there may be many different ways of arriving at it ranging from the dullest to the most elegant). Also, a well-designed problem should have no redundant hypothesis (unless the very purpose of the problem is to test the ability to recognise such redundancies). In other words, normally, every piece of the data has to be used in getting the answer. In the present problem, if we use the data that the numbers $a_1, a_2, \ldots, a_n$ are in a G.P. then we can get a much more economical expression for M. For, in this case we let r be the common ratio of the geometric progression and write $a_k = a_1 r^{k-1}$ for $k = 1, 2, \ldots, n$ (with the understanding that $r^0 = 1$). We can now express r as $\frac{a_2}{a_1}$ and hence as $\frac{2A_2 - A_1}{A_1}$. As a result,

$$a_k = A_1 \left(\frac{2A_2 - A_1}{A_1} \right)^{k-1} \tag{12}$$

Putting these into (10) and then substituting (10) into (9) we get an expression for M in terms of just A_1 and A_2. This is not surprising, because the numbers $a_1, a_2, \ldots, a_n$ are completely determined by two parameters and so anything based on them can also be expressed in terms of two parameters.

Yet another expression for M can be obtained by expressing G_k in terms of A_k and H_k for each $k = 1, 2, \ldots, n$. For this we use $a_m = a_1 r^{m-1}$ for $m = 1, 2, \ldots, k$. Multiplying,

$$\begin{aligned} a_1 a_2 \ldots a_k &= a_1^k \times (1 \times r \times r^2 \times \ldots \times r^{k-1}) \\ &= a_1^k r^{0+1+2+\ldots+(k-1)} \\ &= a_1^k \, r^{k(k-1)/2} \end{aligned}$$

Taking the k-th root, we have

$$G_k = a_1 r^{(k-1)/2} \tag{13}$$

An expressions for A_k can be obtained using the formula (8) above.

$$A_k = \frac{a_1 + a_1 r + a_1 r^2 + \ldots + a_1 r^{k-1}}{k} = \frac{a_1 (r^k - 1)}{k(r-1)} \tag{14}$$

Since $a_1, a_2, \ldots$ are in G.P. with common ratio r, their reciprocals are also in a G.P. with common ratio $\frac{1}{r}$. Therefore, analogously to (14) we have,

$$H_k = \frac{k}{\frac{1}{a_1} + \frac{1}{a_2} + \ldots + \frac{1}{a_k}}$$

$$= \frac{ka_1(\frac{1}{r}-1)}{(\frac{1}{r})^k-1}$$
$$= \frac{a_1kr^{k-1}(r-1)}{r^k-1} \quad (15)$$

Putting (13), (14) and (15) together, we have $G_k = \sqrt{A_kH_k}$ for $k = 1, 2, \ldots, n$. It now follows that

$$M = (G_1G_2 \ldots G_n)^{1/n} = (A_1H_1A_2H_2 \ldots A_nH_n)^{1/2n}.$$

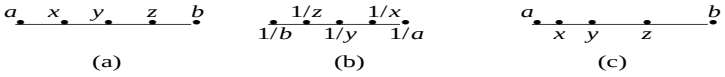
This is probably the answer the papersetters had in mind. For one thing, it crucially uses the fact that the given numbers are in a G.P. (without which assumption, the relation $G_k = \sqrt{A_kH_k}$ would be true only for $k = 1$ and 2, but not for any higher values of k). Secondly, it involves all of $A_1, A_2, \ldots, A_n$ and $H_1, H_2, \ldots, H_n$ (and each to an equal extent.)

Comment No. 4:

In problems dealing with only a few terms of a progression, it is often time-consuming and laborious to start by expressing the terms in terms of the first term and the common difference (or the common ratio as the case may be). Instead, the job can be done easily using some simple relationships among the terms of a progression. Typically one such relationship says that if x, y, z are in A.P. then $x + z = 2y$ or equivalently y is the A.M. of x and z. More generally, if $a_1, a_2, \ldots, a_{n-1}, a_n$ are in A.P. then the sums $a_1 + a_n, a_2 + a_{n-1}, a_3 + a_{n-3}, \ldots$ are all equal, each being twice the middle term of the progression in case n is odd and equal to the sum of the two middle terms in case n is even. Analogous statements hold for the terms of a G.P. if we replace sums by products. Note also that if x and y are, say, the p-th and the q-th terms of an A.P., where p and q are some positive integers then the difference $y - x$ is proportional to the difference $q - p$.

Such simple observations, if applied judiciously, can lead to short cuts. Suppose for example, that we are given that the value of $x + y + z$ is 15 if a, x, y, z, b are in A.P. while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{3}$ if a, x, y, z, b are in H.P. and from this we want to determine a and b (JEE 1978). The first piece of information gives $a + b = 2y$ and also $x + y + z = 3y = 15$. Put together, this means $a + b = 10$. In the second part of the data, it is given that if $\frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P. then $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$. So here also we have $\frac{1}{a} + \frac{1}{b} = \frac{2}{y}$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{3}{y} = \frac{5}{3}$. Together, we get $\frac{1}{a} + \frac{1}{b} = \frac{10}{9}$, i.e., $10ab = 9(a + b)$. As we already know $a + b = 10$, we get $ab = 9$. To determine a and b from these two equations systematically, we can eliminate a from the two to get a quadratic in b, viz., $b^2 - 10b + 9 = 0$. The theory of quadratic equations (and more generally, other types of algebraic equations) will be taken up in the next chapter. In the present problem, however, the two equations $a + b = 10$ and $ab = 9$ imply, by inspection, that the numbers a and b are 9 and 1. (The data is insufficient to decide which one is which.)

The statement of the problem can be slightly confusing to a beginner. Here a and b are two fixed numbers (which are to be determined). But the quantities x, y, z stand for two different things in the two parts of the data. In the first part, x, y, z can be interpreted geometrically as the intermediate points of subdivision if the interval $[a, b]$ is divided into four equal parts as shown in Figure (a) below. (Such points are also called the **nodes** or the nodal points of the subdivision.) Here we are tacitly assuming that $a < b$. In the second part of the data, too, the points x, y, z divide the interval $[a, b]$ into four parts, but not of equal lengths. Instead, here the points $1/z, 1/y, 1/x$ divide the interval $[1/b, 1/a]$ into four equal parts as shown in (b). The corresponding subdivision of the interval $[a, b]$ is shown in (c).



(a) (b) (c)

It is customary to say that in (a), x, y, z are the first, the second and the third arithmetic means between a and b, while in (c), they are, respectively, the first, the second and the third harmonic means between a and b. More generally, we can insert any number of arithmetic or harmonic means in a given interval. In a musical string instrument, these nodal points often correspond to the harmonic and not the arithmetic means. We can also insert geometric means between a and b. They are precisely the terms of a G.P. whose first and last terms are a and b. Thus, if we insert two geometric means between a and b, then they would be $a \times (\frac{b}{a})^{1/3}$ and $a \times (\frac{b}{a})^{2/3}$, or equivalently, $a^{\frac{2}{3}} b^{\frac{1}{3}}$ and $a^{\frac{1}{3}} b^{\frac{2}{3}}$.

Comment No. 5:

All the three progressions we have met are examples of what are called **recursively defined sequences**. In such sequences, the first few terms are specified by giving their actual values often called the **initial values**. Thereafter, every term is specified by expressing it, by some rule, in terms of the preceding several terms (sometimes all of them) and the formula which expresses this relationship is called a **recurrence relation**. For example, in the case of an A.P., every term after the first is obtained by adding a fixed number (the common difference, say, d) to the preceding term. Here the recurrence relation is $a_n = a_{n-1} + d$. The value of a_1 has to be specified directly. A similar comment holds for a G.P. In Chapters 4, 6, 18 and 22 we shall see more examples of such sequences.

By applying the recurrence relation again and again (or 'recursively' as it is is called), any particular term of a recursively defined sequence can be evaluated as in the following problem.

A sequence[3] of numbers $a_0, a_1, a_2, a_3, \ldots$ satisfies the relation $a_n^2 -$

[3] As in this problem, sometimes we denote the first term of a sequence by a_0 rather than by a_1. Some authors then prefer to call it the 0-th term instead of the first term which is logical to a mathematician but not to a layman. Sometimes this leads to confusion and consequent controversy. In problems asking you to find, say, the fourth term of a sequence, it is better to first make sure which convention is followed.

$a_{n-1}a_{n+1} = (-1)^n$. Find a_3 given $a_0 = 1$ and $a_1 = 3$. (JEE 1978)

Here the recurrence relation can be written as

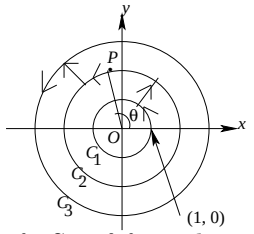
$$a_{n+1} = \frac{a_n^2 + (-1)^{n+1}}{a_{n-1}} \tag{16}$$

which is valid for all $n \geq 1$. Putting $n = 1$, $a_0 = 1$ and $a_1 = 3$ in (16), we get $a_2 = \frac{a_1^2+1}{a_0} = 10$. Next, putting $n = 2$, we have $a_3 = \frac{a_2^2-1}{a_1}$. Since the value of a_1 is given while that of a_2 has already been found, we get $a_3 = \frac{100-1}{3} = 33$.

Continuing like this we can successively determine the values of a_4, a_5, a_6 and so on. That is straightforward (perhaps too straightforward by JEE standards!). The real challenge lies in getting what is called a **closed form expression** for the general term a_n of the sequence, i.e., a formula which expresses a_n directly in terms of n without involving any of the earlier terms of the sequence (except possibly the initial terms whose values are given). For example, in the case of an A.P. with common difference d, we have $a_n = a_1 + (n-1)d$. From this we can get a_{100} directly without first having to know a_{99}. A closed form expression like this is called a solution to the recurrence relation from which it was obtained. We already encountered recurrence relations in Comment No. 2 and Exercise (1.36) of the last chapter. There we solved them by repeatedly applying them and using certain summation formulas for the latter. In general, finding closed form formulas is not always possible, and even when possible, is not always easy. For example, the n-th harmonic number H_n can be defined recursively by the simple recurrence relation $H_n = H_{n-1} + \frac{1}{n}$ with $H_1 = 1$ as the initial value. As already mentioned, there is no closed form expression for H_n. Systematic methods are available to solve certain special types of recurrence relations called the linear recurrence relations. But that is beyond our scope. Nevertheless, sometimes a recursive process is inherent in a problem. In the absence of a general method for solving recurrence relations, such problems have to be tackled by *ad-hoc* methods. Here is an example.

For each natural number k, let C_k denote the circle with radius k centimeters and centre at the origin. On the circle C_k, a particle moves k centimeters in the counter-clockwise direction. After completing its motion on C_k, the particle moves to C_{k+1} in the radial direction. The motion of the particle continues in this manner. The particle starts at $(1, 0)$. If the particle crosses the positive direction of the x-axis for the first time on the circle C_n, then find the value of n. (JEE 1997)

A diagram is helpful in such problems just to understand the problem. In fact, once you understand the problem, the solution itself is easy. Consider the ray OP joining the origin O to the position, say P, of the particle and let θ be the angle OP makes with the positive x-axis. As the particle moves k cm on the circle C_k, θ increases by one radian.



During the radial motion, there is no change in θ. So, if θ_k is the value of θ at the beginning of the motion on the circle C_k, then we have the recurrence

relation $\theta_{k+1} = \theta_k + 1$ with the initial condition $\theta_1 = 0$ (since the particle is given to start at $(1, 0)$). This recurrence relation can be solved by inspection to get $\theta_k = k-1$ for all $k \geq 1$. Now when the particle crosses the positive x-axis for the first time, θ equals 2π. Since the integral part of 2π is 6, we have $6 \leq 2\pi < 7$. So when $\theta = 2\pi$, the particle is moving on the seventh circle C_7. Thus $n = 7$. (The problem is really simple once you understand it. The solution is almost intuitive and the fancy language of recurrence relations is hardly necessary to present the solution. We have used it only to stress that there is a recursive process inherent in the problem, because the starting point of the motion on the k-th circle is determined by the terminal point of the motion on the preceding circle.)

Comment No. 6:

The three types of progressions can be converted into each other by simple transformations. Such conversions can sometimes simplify a problem. Also they save duplication of work because theorems proved for one type of progressions can sometimes be translated to theorems for another type of progression. (Of course, this does not apply if the theorem involves something which is not preserved under the transformation. For example, by taking reciprocals of the terms, we can transform an A.P. into an H.P. and vice versa. We have a handy formula, viz., (7) for the sum of consecutive terms of an A.P. But under reciprocation, sums are not preserved. That is to say, the sum of reciprocals is in general not equal to the reciprocal of the sum. In symbols, $\frac{1}{a+b} \neq \frac{1}{a} + \frac{1}{b}$. As a result, the formula (7) cannot yield us a similar formula for the sum of an H.P. In fact, as remarked earlier, no such formula exists.

Before studying the transformation of one type of progression to another, let us first see what transformations convert a progression to another progression of the same type. The simplest such operation is to multiply every term by a fixed number (or 'scalar' as it is sometimes called). In symbols, if $a_1, a_2, \ldots, a_n$ are in a progression of any of the three types, then so are $\lambda a_1, \lambda a_2, \ldots, \lambda a_n$ for any number λ. The proof is trivial. Note that this also applies to any of the means. That is, if A, G, H are, respectively, the arithmetic, the geometric and the harmonic means of $a_1, a_2, \ldots, a_n$ (which need not be in any progression), then for every λ, $\lambda A, \lambda G$ and λH are, respectively, the arithmetic, the geometric and the harmonic means of $\lambda a_1, \lambda a_2, \ldots, \lambda a_n$. (As geometric means are generally defined only for positive numbers, we take $\lambda > 0$ in the case of a geometric mean). Again the proof is straightforward.

Another elementary operation one can perform is that of **translation** or shifting. In simpler language this simply means adding a fixed number to every term. So suppose, $a_1, a_2, \ldots, a_n$ are some numbers. Then translating them by a number λ gives us the numbers $a_1 + \lambda, a_2 + \lambda, \ldots, a_n + \lambda$. It is clear that if the numbers $a_1, a_2, \ldots, a_n$ are in an A.P. then so are the translated numbers $a_1 + \lambda, a_2 + \lambda, \ldots, a_n + \lambda$. But this is no longer true for the geometric or the harmonic progression. Similarly, if A is the A.M. of the numbers $a_1, a_2, \ldots, a_n$ (not necessarily in any progression, then $A + \lambda$ is the A.M. of the numbers

$a_1 + \lambda, a_2 + \lambda, \ldots, a_n + \lambda$. A similar statement is not true for either the G.M. or the H.M. In fact, there is no easy expression for either the G.M. or the H.M. of $a_1 + \lambda, a_2 + \lambda, \ldots, a_n + \lambda$ in terms of the G.M. or the H.M. of $a_1, a_2, \ldots, a_n$. Formally, we say that the arithmetic mean is **compatible** with translation, while the other two means are not.

The compatibility of the arithmetic mean with translation is often used in finding averages. Suppose, for example, that a teacher finds that most of the students in a class have scored around 40. Then to find the class average, instead of adding the scores as they are, she first subtracts 40 from each score and adds the differences (some of which may be negative, thereby making it easier to add because of cancellations with positive differences). She then takes the average of these translated scores (which is again easier because this total is generally a small number) and adds back 40 to get the class average. This way, many teachers can find the average mentally for a class of size 20 or so.

Returning to the conversion of one type of a progression to another, we already know, by the very definition, that taking reciprocals changes an A.P. into an H.P. and vice versa. The conversion from an A.P. to G.P. and vice versa can be effected through the operations of exponentiation and logarithms. We assume familiarity with the basic properties of both these operations, specifically, the following two **laws of indices** and the corresponding properties of logarithms, viz.,

$$a^{m+n} = a^m a^n \tag{17}$$

$$(a^m)^n = a^{mn} \tag{18}$$

$$\log_a(xy) = \log_a x + \log_a y \tag{19}$$

$$\log_a(x^y) = y \log_a x \tag{20}$$

In addition, there is the following **change of base** formula which allows us to express logarithms w.r.t. an arbitrary base in terms of those w.r.t. some fixed standard base. In taking logarithms, we assume that all the numbers involved are positive and further that no base equals 1 since logarithm w.r.t. the base 1 cannot be defined.

$$\log_x y = \frac{\log_a y}{\log_a x} \tag{21}$$

(Actually, (21) follows from (20) by putting $\log_x y = m$ so that $y = x^m$ whence $\log_a y = \log_a(x^m) = m \log_a x$ giving $m = \dfrac{\log_a y}{\log_a x}$ as desired. There is a mnemonic device to remember (21). Think of $\log_x y$ as if it were a ratio $\frac{y}{x}$. Then (21) simply says $\dfrac{y}{x} = \dfrac{y/a}{x/a}$.)

A rigorous definition of exponentiation and of logarithms is far from easy. The standard approach is to start by defining the **powers** a^n when n is a positive integer. This has a 'physical' meaning, viz., the product of a with itself taken $n - 1$ times, thus $a^1 = a, a^2 = a \times a, a^3 = a \times a \times a$ and so on.

In this approach, the truth of (17) is obvious because of the associative law of multiplication (see Exercise (1.35) for definition of associativity). An essentially equivalent definition is to define a^n recursively as $a^1 = a$ and then $a^n = a^{n-1} \times a$ for $n > 1$. Then (17) can be proved for any fixed m by induction on n.

The next stage is to extend the definition a^n for all integral values of n. Here we have no choice if we want (17) to continue to hold for all integers m and n. Thus, for example, we would like a^{10+0} to equal $a^{10}a^0$. but the only way this is possible (for $a \neq 0$) is if $a^0 = 1$. Similarly, we must define a^{-2} as the reciprocal of a^2 i.e., as $\dfrac{1}{a^2}$ because that is the only way the equality $a^8 = a^{10-2} = a^{10} \times a^{-2}$ will hold true. More generally, we define a^n as $\dfrac{1}{a^{-n}}$ for every negative integer n. It can then be shown that (17) and (18) hold for all integral values of m and n.

Now comes the task of defining the power a^x when x is a rational number, say $x = \frac{p}{q}$, where p, q are integers with $q > 0$. As an illustration, if we want (18) to hold, then $\left(a^{\frac{1}{2}}\right)^2$ will have to equal a^1, i.e., a. Hence $a^{1/2}$ has to be defined as a square root of a. By a similar reasoning, $a^{p/q}$ has to be defined as the q-th root of a^p (or, equivalently, as the p-th power of the q-th root of a). But this raises the fundamental question of the existence of such roots, which is often bypassed in elementary texts. We postpone it to Chapter 16. An intuitive argument can be given as follows. We are assuming that a is positive. If $a = 1$ then certainly, 1 is a q-th root of 1 for every positive integer q. Assume $a > 1$. As the number x moves from 1 to a, the value of the power x^q goes on increasing. But $1^q = 1 < a$ while $a^q > a$ (assuming $q \geq 2$). So somewhere along the way there must be a value, say u, of x at which u^q exactly equals a. Also this value must be unique since $1 < u_1 < u_2$ would mean $u_1^q < u_2^q$ which means no two distinct numbers greater than 1 can have the same q-th power. Thus every real number $a > 1$ has a unique q-th root. Finally, if $0 < a < 1$, then we set $b = \frac{1}{a}$. Then $b > 1$ and so has a unique q-th root. Its reciprocal will then be the q-th root of a. Summing up, for every positive real number a and for every rational exponent m we can now define x^m. The laws of indices can then be extended to hold for all rational exponents.

To further extend the definition of a^m to the case where m is an irrational number, we use a similar idea. Assume first that $a > 1$. Then as m increases through rational values, so does a^m. Now suppose r is an irrational real number. Then for every rational number m, we must have either $m < r$ or $m > r$. Naturally, we would like to define a^r in such a way that in the first case, $a^m < a^r$ holds while in the second, $a^r < a^m$ holds. The crucial question now is whether there exists a real number, say b, which answers this prescription, i.e., which is bigger than all real numbers of the form a^m whenever m is a rational number less than r and at the same time is smaller than all numbers of the form a^m whenever m is a rational such that $m > r$? Moreover, is this number b unique? If the answers to both these questions are in the affirmative then this number b will be our definition of a^r. As before, the case $0 < a < 1$ is handled

by taking reciprocals. This done, the laws of indices (17) and (18) can be shown to hold for all positive a and for all real exponents.

The proof that both the questions raised above have affirmative answers requires certain basic properties of real numbers and is well beyond the JEE level. So we shall take the existence of arbitrary powers for granted. A similar comment holds for logarithms. Given a base a and a positive real number x, $\log_a x$ is, by definition, the unique real number to which a must be raised to get x. This gives us two relations: (i) $a^{\log_a x} = x$ and (ii) if $a^b = x$ then $b = \log_a x$, or in other words, $\log_a(a^b) = b$. Verbally, logarithms and exponentiation are inverse processes of each other. Saying that b is the log of x to the base a is simply another way of saying that the b-th power of a equals x just as in real life, to say 'Ram is Seeta's husband' is simply another way of saying 'Seeta is Ram's wife' (assuming monogamy, of course). Because of this mutually inverse relationship, the basic laws (19) and (20) of logarithms can be derived easily from the corresponding laws (17) and (18) respectively for exponentiation.

The base a of logarithms does not matter for most purposes. But it is generally taken to be greater than 1. We can consider logarithms with base a even when $0 < a < 1$. But then, we have to be careful because, smaller numbers have bigger logarithms. Logarithms with base 10 are called **common logarithms**. The name is appropriate because tables of these logarithms were used extensively in manual calculations till pocket calculators became cheaply available. For theoretical purposes, it is most natural to take for the base a peculiar number e, defined as the limit of the sequence $(1 + \frac{1}{n})^n$ as $n \to \infty$. (Limits of sequences will be formally defined in Chapter 6, Comment No. 20.) Logarithms with base e are called **natural logarithms**. This number e is an irrational number approximately equal to 2.718281828. It is not immediately clear why such a clumsy number would be a convenient base for logarithms. One has to know a little calculus to see why. In combinatorial problems, e.g. those dealing with binary sequences, we often encounter powers of 2. In such a context, it is more convenient to take logarithms with base 2.

When the base is understood, it may be suppressed from the notation. However this can sometimes cause confusion when two bases are equally popular. For some time $\log x$ was used both for the common logarithm (i.e., for $\log_{10} x$) and for the natural logarithm (i.e., $\log_e x$). To avoid confusion nowadays, the latter is denoted by $\ln x$ the 'ln' being the acronym of *logarithm naturelle* which in French means 'natural logarithm'. (The mathematician who introduced the base e was a Frenchman, **Napier**.) Of course, because of (21) we can convert the logarithms with any other base to the natural logarithms and vice versa.

With this preamble about powers and logarithms, let us go back to progressions. In view of (17), if $a_1, a_2, \ldots, a_n$ are in an A.P. with common difference d, then for any base b (> 0), the numbers $b^{a_1}, b^{a_2}, \ldots, b^{a_n}$ are in a G.P. with common ratio b^d. Conversely, if $a_1, a_2, \ldots, a_n$ are positive numbers in a G.P. with common ratio r then by (19), $\log_b a_1, \log_b a_2, \ldots, \log_b a_n$ are in an A.P. with common difference $\log_b r$. This way we can convert an A.P. into a G.P. and vice versa. Here is a problem where we put such a conversion to immediate use.

If x, y, z are, respectively, the p-th, the q-th and the r-th terms of an A.P. and also of a G.P., find the value of $x^{y-z}y^{z-x}z^{x-y}$. (JEE 1979)

Denote the given expression by E. Instead of finding its value directly, we find, instead, the value of its logarithm to some base b. In some problems a clever choice of b helps. But in the present problem it hardly matters. So we may as well take $b = e$. Then $\ln E = (y-z)\ln x + (z-x)\ln y + (x-y)\ln z$. Now, x, y, z are given to be the p-th, the q-th and the r-th terms of an A.P. Letting d be the common difference of this A.P., we have

$$x - y = (p-q)d, \quad y - z = (q-r)d \text{ and } z - x = (r-q)d \tag{22}$$

Further, x, y, z are also the p-th, the q-th and the r-th terms of a G.P. We let t be the common ratio of this G.P. (We avoid r since it is already used for denoting something else.) Taking logarithms, $\ln x, \ln y, \ln z$ are the p-th, the q-th and the r-th terms of an A.P. with common difference $\ln t$. So analogously to (22), we have

$$\begin{aligned} \ln x - \ln y = (p-q)\ln t, \qquad & \ln y - \ln z = (q-r)\ln t \\ \text{and} \qquad & \ln z - \ln x = (r-q)\ln t \end{aligned} \tag{23}$$

Using (22) and (23) we get,

$$\begin{aligned} \ln E &= (y-z)\ln x + (z-x)\ln y + (x-y)\ln z \\ &= d[(q-r)\ln x + (r-p)\ln y + (p-q)\ln z] \\ &= d[q(\ln x - \ln z) + r(\ln y - \ln x) + p(\ln z - \ln y)] \\ &= d\ln t[q(p-r) + r(q-p) + p(r-q)] = 0 \end{aligned} \tag{24}$$

which gives $E = 1$.

Of course, the problem could also have been done without converting the G.P. to an A.P. by writing $\frac{y}{x} = t^{q-p}$ etc. But taking logarithms simplifies things practically if not conceptually. In fact, the reason logarithms are so useful in computations is precisely that they convert products to sums. It is definitely less time-consuming to add than to multiply two numbers. The logarithms (and the exponentials) have many other uses besides conversion of progressions. The foremost is probably that certain natural growths satisfy the law that the rate of growth at any time is proportional to the quantity existing at that time. The analysis of such quantities leads naturally to logarithms (with base e) as we shall see in Chapter 19, Comment No. 5 and 6.

Comment No. 7:

To solve equations involving powers with different bases (e.g. 3^x and 5^{-x^2+1}), the best way is to bring all the terms to a common base. Here is a simple problem which illustrates the procedure.

Solve for x : $4^x - 3^{x-\frac{1}{2}} = 3^{x+\frac{1}{2}} - 2^{2x-1}$. (JEE 1978)

Writing 4^x as 2^{2x} and bringing powers of the same number on the same side we get

$$2^{2x} + 2^{2x-1} = 3^{x+\frac{1}{2}} + 3^{x-\frac{1}{2}}.$$

The first term on the L.H.S. can be written as $2^{2x-1} \times 2$ by (17) and hence a common factor of 2^{2x-1} comes out from the terms on the L.H.S. As for the R.H.S., we apply (17) to the first term to rewrite it as $3^{x-\frac{1}{2}} \times 3$ and then the factor $3^{x-\frac{1}{2}}$ comes out as common. So we get $2^{2x-1}(2+1) = 3^{x-\frac{1}{2}}(3+1)$, i.e., $3 \times 2^{2x-1} = 4 \times 3^{x-\frac{1}{2}}$. Bringing all powers of 2 to the left and all powers of 3 to the right, we get

$$2^{2x-3} = 3^{x-\frac{3}{2}}$$

By inspection, $x = \frac{3}{2}$ is a solution. But how do we *arrive at* it? Also how do we know that there is no other solution? It is tempting to try to do this by saying that a power of 2 can equal a power of 3 only when both equal 1. As we shall see in Chapter 4, Comment No. 18, such a reasoning is indeed useful in solving equations in Number Theory where we mostly deal with positive integers and their factorisations into integers. But here it is inapplicable because we do not know yet that the exponents are integers. Instead, let us express both the sides as the powers of the same number. One way to do this is to write 3 as $2^{\log_2 3}$ in the R.H.S. Then using (18) above, we get

$$2^{2x-3} = 2^{(\log_2 3)(x-\frac{3}{2})}$$

As the bases are the same, the equality of powers implies that of the exponents. So we have $2x - 3 = (\log_2 3)(x - \frac{3}{2})$. This can be solved easily to give $x = \frac{3-\frac{3}{2}\log_2 3}{2-\log_2 3}$ which simply equals $\frac{3}{2}$. Hence $x = \frac{3}{2}$ is the only solution.

Comment No. 8:

Far more common are equations in which all the terms are powers of a common variable with constant exponents. Typically, an equation of this kind looks like $c_1x^{m_1} + c_2x^{m_2} + \ldots + c_nx^{m_k} = 0$, where k is a positive integer and $c_1, c_2, \ldots, c_k, m_1, m_2, \ldots, m_k$ are some real numbers. Especially interesting is the case where the exponents $m_1, m_2, \ldots, m_k$ are all multiples of some fixed number, say m. In that case we can write m_i as n_im $(i = 1, 2, \ldots, k)$, where $n_1, n_2, \ldots, n_k$ are all integers. Therefore we can write x^{m_i} as $(x^m)^{n_i}$ or as y^{n_i}, where $y = x^m$. The original equation can now be written as an equation in y, viz., as $c_1y^{n_1} + c_2y^{n_2} + \ldots + c_ky^{n_k} = 0$, where $n_1, n_2, \ldots, n_k$ are some integers. These integers need not be all non-negative. But we may assume them to be so. For, suppose r is the smallest among them and r is negative. Then $-r$ is positive. If we multiply the equation throughout by y^{-r} that does not affect its roots. The new equation will then be $c_1y^{n_1-r} + c_2y^{n_2-r} + \ldots + c_ky^{n_k-r} = 0$, where all the exponents are non-negative integers. The expression just obtained is called a **polynomial** in the variable y. In a more standard notation, if n is a non-negative integer, then a polynomial of **degree** n in a variable x is an expression of the form $a_nx^n + a_{n-1}x^{n-1} + \ldots + a_2x^2 + a_1x + a_0$, where $a_0, a_1, a_2, \ldots, a_n$ are some numbers with $a_n \neq 0$. (a_nx^n is called the **leading term** and a_n the **leading coefficient** of the polynomial. The restriction that $a_n \neq 0$ ensures that the degree of a polynomial is uniquely defined, because if $a_n = 0$ then we can simply drop that term and consider only $a_{n-1}x^{n-1} + \ldots + a_2x^2 + a_1x + a_0$,

with further dropping if a_{n-1} is also 0). According to our definition, 0 cannot be a polynomial. However, for many purposes it is convenient to treat it as one and we do so. But it cannot be assigned any degree[4].

A polynomial whose leading coefficient is 1 is called a **monic** polynomial, the suffix *mono* meaning one. We assume familiarity with the elementary operations such as addition, subtraction and multiplication of two polynomials. Division of polynomials is in general not possible. That is, given two polynomials, say $f(x)$ and $g(x)$, the ratio $\dfrac{f(x)}{g(x)}$ need not always be a polynomial. If it is, then we say that $g(x)$ divides $f(x)$ or that it is a **factor** of $f(x)$. Even when the ratio is not a polynomial, it is still a function of x defined for all values of x for which the denominator does not vanish. A function of this type is called a **rational function** of x, the name coming from an obvious analogy with rational numbers. This can be written uniquely as $q(x) + \dfrac{r(x)}{g(x)}$, where $q(x)$, called the quotient, and $r(x)$, called the remainder, are some polynomials in x and the degree of $r(x)$ is less than that of $g(x)$. This property is called the **euclidean algorithm for polynomials**. The quotient and the remainder can be found explicitly by performing what is called the **long division**. If further, $f(x), g(x)$ have all integer coefficients and $g(x)$ is monic, then $q(x)$ and $r(x)$ also have integer coefficients. We omit the details since high school mathematics provides enough drill in it.

Comment No. 9:

We can also consider polynomials in two or more variables. For simplicity we define the relevant terms only for the case of two variables, say x and y. They can be obviously extended to the general case. A **monomial** is an expression of the form ax^ry^s, where a is some (non-zero) number and s, t are some non-negative integers called respectively, the degree of the monomial in x and y. The integer $s + t$ is called the **total degree** or simply the degree of the monomial. A polynomial in two variables is a (finite) sum of such monomials, i.e., an expression of the form $\sum_{i=1}^{k} a_i x^{s_i} y^{t_i}$. (We assume here, that 'like terms' i.e., terms in which the degrees in x as well as the degrees in y are equal, are added together.) The highest of the degrees of the monomials appearing in the sum is called the degree of the polynomial. For example, $3x^2 - 7x^3y + \pi y^2 + ex^7y^3 + 99$ is a polynomial of degree 10 (= 7+3). A polynomial in which all the terms have the same total degree, say n, is said to be **homogeneous** of degree n. Thus $x^3 - 6x^2y + y^3$ is a homogeneous polynomial of degree 3.

Polynomials of two (or more) variables are added, subtracted and

[4]We can assign it some degree by convention. But there is no way to do so in a manner which will be consistent in all contexts. For example, every constant polynomial has degree zero and so the zero polynomial also ought to have degree 0. But then the fact that the degree of the derivative of a polynomial is one less than that of the original polynomial will not hold for non-zero constant polynomials. So the degree of the zero polynomial will have to be set equal to -1.

multiplied in the expected manner. The concept of a factor can also be defined for such polynomials. Thus $x-y$ is a factor of x^2-y^2 because there is a polynomial, viz., $x+y$ such that $(x-y)(x+y)=x^2-y^2$. Expanding a product of polynomials is a drill but factoring a polynomial is an art as there is no set procedure for it. Problems involving factorisation can be challenging sometimes because one has to often add and subtract some terms and then group the terms suitably. As the simplest example, to find the factors of x^2-y^2, we add and subtract xy. Then we have $x^2-y^2=x^2+xy-xy-y^2=(x^2+xy)-(xy+y^2)=x(x+y)-y(x+y)=(x-y)(x+y)$. This example, is of course, too well-known. We omit problems of factorisation because you must have already experienced their charm. Just to refresh you. we shall give a couple of factorisation problems in the exercises. In the JEE context, problems of factorisation *per se* are rarely asked after the syllabus was widened to make room for new topics such as calculus. However, factorisation may be needed as a part of some other problem, e.g., the problem of solving an equation or a system of equations.

It should be noted that for polynomials in two or more variables, there is no equivalent of the euclidean algorithm for division. For example, we cannot divide x^3y^2 by x^4 so as to obtain polynomials $q(x,y)$ and $r(x,y)$ such that $x^3y^2=q(x,y)x^4+r(x,y)$ with the degree of $r(x,y)$ less than 4 (the degree of x^4). However, we can talk of rational functions in two or more variables. They are ratios of two polynomials.

Comment No. 10:

By far the most celebrated result about polynomials in two variables is the **binomial theorem** which says that for every positive integer n,

$$\begin{aligned}(x+y)^n &= \sum_{r=0}^{n}\binom{n}{r}x^{n-r}y^r \\ &= \binom{n}{0}x^n+\binom{n}{1}x^{n-1}y+\binom{n}{2}x^{n-2}y^2+\ldots \\ &\quad +\binom{n}{n-2}x^2y^{n-2}+\binom{n}{n-1}x^{n-1}y+\binom{n}{n}y^n \qquad (25)\end{aligned}$$

where the coefficients $\binom{n}{r}$ (also often denoted by ${}^nC_r$ or sometimes by ${}_nC_r$) are called the **binomial coefficients** and are defined combinatorially as the number of ways to choose r objects from n objects or algebraically as $\dfrac{n!}{r!(n-r)!}$ with the understanding that $0!=1$. After cancelling the common factors, the latter can also be written as $\dfrac{n(n-1)(n-2)\ldots(n-r+2)(n-r+1)}{r!}$. Thus, $\binom{n}{0}=1, \binom{n}{1}=n, \binom{n}{2}=\frac{n(n-1)}{2}$ and so on. We already used binomial coefficients in the last chapter for counting problems. Numerous identities are known about the binomial coefficients. Probably the two most important ones are:

$$\binom{n}{r} = \binom{n}{n-r} \text{ for } 0\le r\le n \qquad (26)$$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \text{ for } 1 \leq r \leq n \tag{27}$$

Both the identities can be proved either by a combinatorial or by an algebraic argument. The first one is called the **symmetry** property. Because of it, the R.H.S. of (25) is completely symmetric in x and y as you go from the left to the right. The coefficient of $x^{n-r}y^r$ is the same as that of $x^r y^{n-r}$. The second identity (27) is called the **reduction formula** or the **recursion formula** for the binomial coefficients because it expresses the binomial coefficient $\binom{n}{r}$ in terms of two binomial coefficients in which the upper index has been reduced by 1. Using (27) one can give an inductive proof of the binomial theorem.

We assume that you have already seen this (or some other) proof. The applicability of the binomial theorem can be increased considerably by substituting for x and y any expressions we want. This leads to some standard problems such as finding the constant term (or the term independent of x) in the expansion of, say $(3x^2 - \frac{2}{x})^n$ for a given n. Here a typical term will be of the form $\binom{n}{r}(3x^2)^{n-r}(-\frac{2}{x})^r$ in which the exponent of x is $2(n-r)-r = 2n-3r$. If this is to be 0, then n must be a multiple of 3, say $n = 3k$ and the value of r must then be $2k$. So the constant term in the given expression will be $\binom{3k}{2k}3^k(-2)^{2k} = \binom{3k}{2k}12^k$ which, in view of (26) can also be written as $\binom{3k}{k}12^k$. If n is not a multiple of 3, then there is no constant term. (Or we may say that the constant term is 0.)

Comment No. 11:

In Chapter 5 we shall prove a number of identities about binomial coefficients. The binomial theorem will be the principal tool of attack. The binomial theorem is also often stated in the form

$$\begin{aligned}(1+x)^n &= \sum_{r=0}^{\infty} {}^nC_r x^r \\ &= 1 + nx + \frac{n(n-1)}{2}x^2 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{r!}x^r \\ &\quad + \ldots\end{aligned} \tag{28}$$

Note that here there are infinitely many terms in the summation! When n is a positive integer, this makes little difference, because for $r > n$, ${}^nC_r = 0$ anyway. So even though the sum is ostensibly infinite, it is really finite. In fact, in this case (28) may appear to be only a special case of (25). But actually, it is no less general, because if we apply (28) (n being a positive integer), with x replaced by $\frac{x}{y}$ and then multiply both the sides by y^n we get (25). One reason this case is singled out is that since all the powers of 1 equal 1, the terms of (28) are a little less clumsy than those in (25). But this is a superficial reason. The real advantage of (28) is that under certain restrictions on x, (28) is valid even when n is not a positive integer but some other exponent such as -3 or $\frac{1}{2}$. This is done as follows. First in this case, we cannot give any combinatorial meaning to ${}^nC_r$.

However, we can *define* it as the ratio $\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}$. (Note that even if we do not require n to be a positive integer, the index variable r can take only non-negative integers as possible values.) In that case every term of the summation, viz., ${}^nC_r x^r$ certainly makes sense. But the crucial question now is how to add an infinite number of terms. This is an important question and we postpone its discussion till Chapter 23.

Another nagging question about (28) is to decide for which values of x it is valid. When n a is non-negative integer, (28) is a special case of (25) and is unconditionally valid for all values of x. But this is not so when n is not a non-negative integer. Let us illustrate what can go wrong. Suppose $n = -1$. Then for every non-negative integer r, $\binom{-1}{r} = \frac{(-1)(-2)\ldots(-r)}{r!} = (-1)^r$. So, in this case (28) reduces to

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \ldots + (-1)^r x^r + \ldots \tag{29}$$

Now let us put $x = -2$. Then the L.H.S. is -1. The terms on the R.H.S. are simply successive powers of 2, viz., $1, 2, 4, 8, \ldots, 2^r, \ldots$. They are all positive. Even though we have not yet defined how to add an infinite number of terms, no matter which definition we make, the very least we expect is that the sum of positive terms should not equal a negative number. But that's exactly what happens if we put $x = -2$ in (29). So we have to swallow that (29) is not valid for all values of x. It turns out that (29) is true for all x such that $-1 < x < 1$. In fact it turns out that for every exponent n, (28) is true for all x such that $-1 < x < 1$. But the proof is not easy. In fact, even the correct interpretation of (28) needs some care. We shall need the binomial theorem only for positive integral exponents. We remark, however, that (28) is a frequently used result. Note that when n is not a non-negative integer, the R.H.S. of (28) is not a polynomial in x since it involves the sum of an *infinite* number of powers of x. Such an expression is called a **power series** in x and is a powerful tool in the hands of a mathematician (cf. Comment No. 11 of Chapter 1).

Comment No. 12:

Returning to polynomials, in the next chapter we shall briefly study the theory of polynomial equations, i.e., equations of the form $p(x) = 0$, where $p(x)$ is a polynomial in x. We recall that these are very special cases of equations involving powers, where the base is variable and the exponents are constants as opposed to the equations we considered in Comment No. 7, where the bases were constants and the exponents varied. There are also equations of the mixed type where in some of the terms the base varies, while in some of them the exponent varies and in some both vary (e.g., a term of the form x^x). These equations are generally impossible to solve exactly except by inspection in some cases. Even proving the existence of a solution often requires methods from calculus − often advanced ones. As a relatively simple example, consider the equation $2^x + 1 - x^2 = 0$. By inspection we can identify $x = 3$ as a solution (or a 'root' as it is often called). But is this the only solution? To answer this

question one has to carefully study how the function $f(x) = 2^x + 1 - x^2$ behaves for $x < 3$ and for $x > 3$. We shall do so in Comment No. 14 of Chapter 10. It will then follow that the equation has one more root which lies between 3 and 4. The trouble is that we cannot express it in terms of powers or logarithms, say, for example as $\dfrac{\ln 8 - \sqrt[3]{2}}{4 - \ln 2}$. All we can do is find its approximate value. Various methods for doing so are available. But they are beyond our scope.

A similar comment holds for equations where trigonometric expressions are mixed with powers. In Chapter 10, we shall see how to solve purely trigonometric equations, i.e., equations where a trigonometric expression is equated to 0. But a mixed equation such as $x^2 - \sin x = 0$ can be solved only by inspection (in the present case $x = 0$ being an obvious solution) or only approximately.

Comment No. 13:

We often have a system of equations in several variables. For example, $y = mx + c$ and $x^2 + y^2 = r^2$ is a system of two polynomial equations in the two unknowns[5] x and y. Naturally, solving a system is more complicated than solving a single equation in a single unknown. In the next chapter we shall see how to solve a particularly simple type of system, called a linear system. For other types of systems there is no golden method and we often have to be content with approximate solutions.

A related problem with systems is that of **elimination**. As a general rule you can eliminate $n - 1$ quantities from a system of n equations. For example, from the system just considered, viz., $y = mx + c$ and $x^2 + y^2 = r^2$, we can easily eliminate y by directly substituting the first equation into the second to get $x^2 + (mx + c)^2 = r^2$. Eliminating x requires slightly more work and gives $(y - c)^2 + m^2y^2 = r^2m^2$. Here we solved one of the equations for the variable to be eliminated and substituted it into the other. More generally, if we are given a system of n equations in $n - 1$ variables, it is tempting to think that we can eliminate them by solving, say, the first $n - 1$ equations so as to get the values of these $n - 1$ variables and then substitute them into the last one. But this is usually not possible. In fact, just as there is no golden method for solving a system, there is no golden method for elimination. Each problem has its own challenge and needs methods tailored to that problem, taking advantage of special features such as symmetry inherent in that problem or the use of some standard identities. Here is an illustration.

Eliminate x, y, z from the system of equations:

$$x^2(y + z) = a,\ y^2(z + x) = b,\ z^2(x + y) = c,\ \text{ and }\ xyz = d$$

If we could solve any three of these four equations to get the values of x, y, z in terms of a, b, c and d, then we can substitute these values into the fourth

[5] Here m and c are not variables. They are constants whose values are not specified. Such quantities are often called **parameters**. When nothing is specified, deciding which symbols represent variables and which ones represent parameters has to be done by convention and context. As a general rule, symbols such as x, y, z always denote variables. Symbols such as a, b, c are almost exclusively reserved for constants.

equation and get the answer. But this would be cumbersome. Instead, we can take advantage of the cyclic symmetry of the first three equations. Indeed, if we multiply them together and use the well-known expansion of $(y+z)(z+x)(x+y)$ we get,

$$x^2y^2z^2(x^2y + xy^2 + y^2z + yz^2 + z^2x + zx^2 + 2xyz) = abc.$$

Regrouping the terms in the parentheses as $x^2(y+z)+y^2(z+x)+z^2(x+y)+2xyz$ and using the given equations the answer comes out as

$$d^2(a + b + c + 2d) = abc\ .$$

As with factorisation, problems of elimination are rarely asked *per se* in an examination like the JEE. However, elimination is an important tool for other types of problems, especially for problems of finding the loci of plane curves, as we shall see in Chapter 9. In a locus problem, we have to find an equation satisfied by the coordinates, say x and y of a point moving in the plane according to some condition. The data of the problem is often of a type that does not readily give such an equation. We then introduce one or more auxiliary variables and write a system of equations involving x, y and these auxiliary variables. Elimination of these auxiliary variables then gives the equation of the locus.

Comment No. 14:

The search for solutions of equations often leads to something new. In fact, many of the important developments in mathematics owe their origins to solving some equations. Take the concept of logarithms. Suppose a and b are fixed positive real numbers. Then $\log_a b$ is nothing but a solution of the equation $a^x = b$. Note that if b is negative then this equation has no real solution. In that case we try to extend the real number system so that the new number system we get will contain a solution, i.e., contain some x such that a^x will equal b. Such an extension indeed exists. However, instead of constructing it through equations of the form $a^x = b$, it is more customary and convenient to define it through the solution of the equation $x^2 + 1 = 0$. This equation, too, has no real solution because the square of a real number can never be negative. So we 'imagine' a solution and call it i. Then by definition, $i^2 = -1$. Once we include this 'imaginary'[6] number in our extended number system, then we also

[6]The adjective is historically significant but somewhat inaccurate. Its origin lies in the fact that the real numbers deal with matter. Given any (positive) real number, say a, we can at least in theory, produce a piece of wire whose length is a cm. or an object which weighs a grams. So, in this sense the real numbers have a real existence. A number like i, on the other hand, has no such existence and so it is imaginary in this sense. It should be noted however, that a number, whether real or imaginary, is an abstract concept. Just because we can *associate* a piece of wire to a real number a does not mean that the piece of wire *is* the real number a. And if this be the criterion to decide whether something is real or imaginary, then there are some physical quantities, e.g., the impedance of a coil, which can be handled most conveniently by treating them as imaginary numbers. So there is really no distinction of this type between the numbers 1 and i. All numbers, are, after all, creations of the mind. And so, the number 1 is as 'imaginary' as $\sqrt{2}$ or i.

have to include all (real) multiples of it, i.e., all numbers of the form yi, where y is a real number. Such numbers are called 'purely imaginary' and are generally denoted by iy instead of yi. A **complex number** is one which is obtained by adding a real and a purely imaginary number. Typically, such a number is denoted by z and written as $x + iy$, where x and y are real numbers, called respectively, the real and the imaginary parts of z. (Note that the imaginary part y itself is not an imaginary number! In fact, it is real. It is the number iy that is purely imaginary.) Every real number x can be expressed as a complex number by writing it as $x+i0$. This way, we can say that the real numbers form a subset of the set of complex numbers. Or in other words, that the complex number system is an extension of the real number system.

Complex numbers are added, subtracted and multiplied in the expected manner, keeping in mind that the product of i with itself is the real number -1. We assume that you are familiar with these definitions. However, just for the record's sake, suppose $z_1 = x_1+iy_1$ and $z_2 = x_2+iy_2$ are two complex numbers. Then $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$ and $z_1z_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$. If $z_2 \neq 0$ (i.e., if at least one of x_2 and y_2 is non-zero), then we can also form the quotient $\dfrac{z_1}{z_2} = \dfrac{x_1 + iy_1}{x_2 + iy_2}$. To express this ratio in the 'standard' form, viz., as $a + ib$, where a, b are real, we multiply both the numerator and the denominator by $x_2 - iy_2$. Then it is clear that $\dfrac{z_1}{z_2} = \dfrac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \dfrac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\dfrac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}$.

Two complex numbers x_1+iy_1 and x_2+iy_2 are equal if and only if $x_1 = x_2$ and $y_1 = y_2$. As a result, one complex equation is equivalent to a system of two real equations. For example, to solve the equation $\frac{(1+i)x-2i}{3+i} + \frac{(2-3i)y+i}{3-i} = i$, for x and y (JEE 1980), we first multiply both sides by $(3 + i)(3 - i) = 10$ to get $(3 - i)[x + i(x - 2)] + (3 + i)[(2y + i(1 - 3y)] = 10i$. This gives a system of two equations, viz., $4x + 9y = 3$ and $2x - 7y = 13$. Solving simultaneously, we get $x = 3$ and $y = -1$.

However, in many problems it is not a good idea to resolve a complex equation into two real equations as we shall soon see.

With these basic operations, we can do with complex numbers all the algebraic manipulations that we do with real numbers. In particular, we can consider polynomials and rational functions in one or more complex variables. An important concept associated with complex numbers is that of **complex conjugation**. If $z = x + iy$ is a complex number, then its complex conjugate, denoted by $\bar{z}$, is defined as the complex number $x - iy$. We can express x and y in terms of the complex conjugate by $x = \dfrac{z + \bar{z}}{2}$ and $y = \dfrac{z - \bar{z}}{2i}$. It is clear that z is real if and only if $z = \bar{z}$ and purely imaginary if and only if $z = -\bar{z}$. Note also that $\bar{\bar{z}} = z$, i.e., taking the conjugate of the conjugate gives back the original complex number. Put differently, the operation of complex conjugation is its own inverse. For any complex number $z = x + iy$, $z\bar{z}$ equals $x^2 + y^2$ by a direct computation. This is a non-negative real number, which is zero only

if $z = 0$. The unique non-negative square root of this number is called the **absolute value** or the **modulus** of z and is denoted by $|z|$. Thus we have, $|z|^2 = z\bar{z} = x^2 + y^2$ by definition. Clearly $|z| = |\bar{z}|$. The formula for the ratio $\frac{z_1}{z_2}$ given above can be expressed more compactly as $\frac{z_1}{z_2} = \frac{z_1\overline{z_2}}{|z_2|^2}$.

An important property of complex conjugation is that it commutes with the basic algebraic operations. In simpler terms, this means that the complex conjugate of a sum is the sum of the complex conjugates with similar equalities for differences, products and quotients (wherever defined) of two complex numbers. in symbols, if z, w are any two complex numbers, then, $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $\overline{z-w} = \bar{z} - \bar{w}$ and $\overline{(\frac{z}{w})} = \frac{\bar{z}}{\bar{w}}$, provided $w \neq 0$. All these four formulas can be checked by direct computation. As an immediate application of the second formula, we get that $|zw| = \sqrt{zw\overline{zw}} = \sqrt{zw\bar{z}\bar{w}} = \sqrt{z\bar{z}w\bar{w}} = |z||w|$. In words, the absolute value of a product is the product of the absolute values. A similar assertion is true for quotients. That is, $|\frac{z}{w}| = \frac{|z|}{|w|}$. Note in particular, that for any non-zero complex number z, the ratio $\frac{z}{\bar{z}}$ always has the absolute value 1. Note, however, that in general we do not have $|z + w| = |z| + |w|$ as seen by taking $z = 1$ and $w = i$. It can in fact be shown that $|z + w|$ is, in general, less than $|z| + |w|$. We shall take this up in Chapter 6 on Inequalities. (See Comment No. 11 there.)

These simple properties of absolute values and complex conjugates give a solution to the following problem.

For complex numbers z and w prove that

$$|z|^2 w - |w|^2 z = z - w \tag{30}$$

if and only if $z = w$ or $z\bar{w} = 1$. (JEE 1999)

Rewriting the L.H.S. of (30) as $z\bar{z}w - w\bar{w}z$, we see that (30) is equivalent to

$$z(\bar{z}w - 1) = w(z\bar{w} - 1) \tag{31}$$

Now, (30) is trivially true if $z = w$. Also if $z\bar{w}$ equals 1, then taking complex conjugates, $\bar{z}w$ also equals 1 and so (31) holds as both the sides vanish. Thus we see that (30) holds when either $z = w$ or $z\bar{w} = 1$. Conversely assume that (30) holds and hence that (31) is true. Suppose $z\bar{w} \neq 1$. Then taking complex conjugates, $\bar{z}w \neq 1$. Also $|\bar{z}w - 1| = |z\bar{w} - 1|$ is non-zero. So taking absolute values of both the sides of (31), we get

$$|z| = |w| \tag{32}$$

We want to show that $z = w$. If this is not so then $\bar{z} - \bar{w}$ is also non-zero. Moreover, $|\bar{z} - \bar{w}| = |z - w|$ which is non-zero. So rewriting (30) as $zw(\bar{z} - \bar{w}) = z - w$ and taking absolute values of both the sides, we get

$$|z||w| = 1 \tag{33}$$

(32) and (33) together imply that $|z| = |w| = 1$. Putting this into (30) gives $z = w$. Thus when (30) holds either $z\bar{w} = 1$ or $z = w$. Both the implications are now proved. (If we put $z = x + iy$ and $w = u + iv$ then (30) can be replaced by a system of two equations in the real variables x, y, u and v. But it is too complicated. So it is not always a good idea to replace one complex equation with a system of two real equations.)

Comment No. 15:

A simple but important consequence of the fact that complex conjugation commutes with sums and products is noteworthy. Suppose $p(z) = c_n z^n + c_{n-1}z^{n-1} + \cdots + c_2 z^2 + c_1 z + c_0$ is a polynomial. Here the variable z as well as the coefficients are complex. By $\bar{p}(z)$ we denote the **conjugate polynomial** $\overline{c_n}z^n + \overline{c_{n-1}}z^{n-1} + \ldots + \overline{c_1}z + \overline{c_0}$ so called because its coefficients are precisely the conjugates of the corresponding coefficients of the polynomial $p(z)$. Then for every complex number w we have $\bar{p}(\bar{w}) = \overline{c_n}\bar{w}^n + \overline{c_{n-1}}\bar{w}^{n-1} + \ldots + \overline{c_1}\bar{w} + \overline{c_0} = \overline{p(w)}$. In particular, if w is a root of $p(z)$ then we see that $\bar{w}$ is a root of the conjugate polynomial $\bar{p}(z)$ and vice versa. Thus we see that the roots of the conjugate polynomial are precisely the complex conjugates of the roots of the original polynomial. Now what happens if the coefficients $c_n, c_{n-1}, \ldots, c_2, c_1, c_0$ are all real? Then clearly $\bar{p}(z)$ is the same as $p(z)$. We thus get an important result which we record as a theorem.

Theorem 1: If $p(z)$ is a polynomial with real coefficients in a complex variable z, and w is a complex root of $p(z)$ then so is $\bar{w}$. Put differently, the complex roots of a polynomial with real coefficients occur in conjugate pairs.

We already saw that the conjugate pair i and $-i$ are roots of the polynomial $z^2 + 1$. More generally, suppose we have a quadratic polynomial, say $az^2 + bz + c$, where the coefficients a, b, c are all real. If $b^2 - 4ac < 0$, then the roots are complex conjugates of each other as we shall see in the next chapter.

From the fact that complex conjugation commutes with sums and products, it also follows that if $p(z)$ is a polynomial with real coefficients then for any complex z, $p(z) + p(\bar{z})$ is real since the two summands are complex conjugates of each other. Also in this case, $p(z) - p(\bar{z})$ is purely imaginary for every complex number z. As a very special case we see that for any positive integers n_1 and n_2, the expression $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$ is real (JEE 1996). This follows because since $i^3 = -i$, $1 + i$ and $1 + i^3$ are complex conjugates of each other which makes the sum of the first two terms real (by taking $p(z) = z^{n_1}$). Similarly, $i^5 = i$ and $i^7 = -i$ makes the sum of the last two terms real.

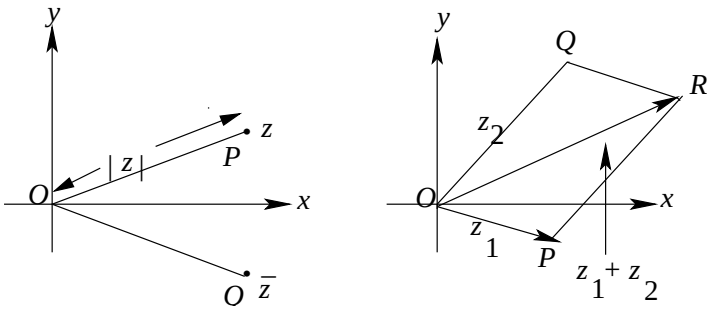
Comment No. 16:

A complex number $z = x + iy$ is completely determined by its real and imaginary parts, viz., x and y respectively. But these two real numbers also determine a unique point (x, y) in a cartesian plane.[7] So we can as well represent

[7] A **cartesian plane** is a plane in which a frame of rectangular coordinate axes has been fixed. Coordinates of various points in such a plane are with reference to this plane. The same plane may have a different frame of axes. In that case, the two cartesian planes are

a complex number by a point in a cartesian plane. Specifically, we let $z = x + iy$ represent the point (x, y). Thus the real and the imaginary parts correspond, respectively, to the **abscissa** and the **ordinate** of the corresponding point in the plane. (These are old names for the two coordinates. Nowadays, they are simply called the x- and the y- coordinates respectively.) A diagram in which complex numbers are identified with points of a plane is called an **Argand** diagram.

A correspondence like this will be of little value unless the basic concepts associated with complex numbers have some vivid, easy geometric interpretations. It turns out that this is indeed the case. In the Argand diagrams below we show the geometric interpretations of the concepts of absolute value, complex conjugation and the addition of complex numbers. Let O be the origin and $P(x, y)$ be the point representing the complex number $z = x + iy$. Let $Q(x, -y)$ represent the complex conjugate $\bar{z} = x - iy$ of z. Then from Figure (a) it is clear that Q is precisely the reflection of P in the x-axis. Similarly, $|z|$, i.e., the absolute value of z, is the length of the segment OP. More generally, if P_1, P_2 are any two points represented by the complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ respectively, then $|z_2 - z_1| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ is the length of the segment P_1P_2.



(a) complex conjugation and absolute value

(b) complex addition

To interpret complex addition geometrically, let the points P, Q and R represent the complex numbers z_1, z_2 and $z_1 + z_2$ respectively. Then OR is a diagonal of the parallelogram with OP and OQ as a pair of adjacent sides as shown in (b). This is not only analogous to, but essentially the same as the **parallelogram law** for vector addition, because we can as well represent the complex numbers $z_1, z_2, z_1 + z_2$ by the vectors $\overrightarrow{OP}$, $\overrightarrow{OQ}$ and $\overrightarrow{OR}$ respectively.

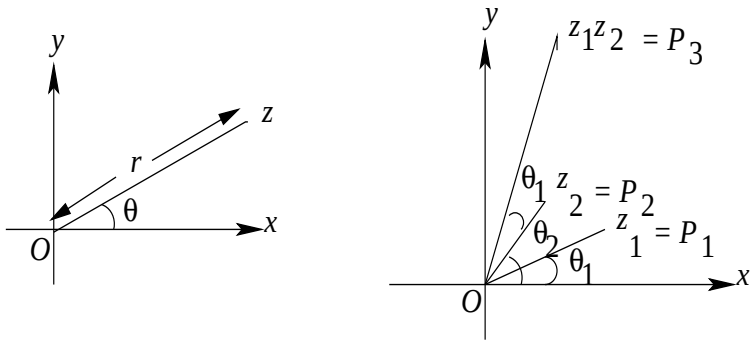
different and the same point of the plane will, in general have different coordinates in the two planes. Formulas for conversion of coordinates w.r.t. one frame of reference to those w.r.t. another are given in standard texts. We shall not need them here as we are dealing with just one fixed cartesian plane. By the way the word 'cartesian' comes from **Descartes**, the mathematician who introduced these coordinates. They revolutionised geometry by making the methods of algebra available to tackle problems in geometry. We shall see ample evidence of this in Chapters 8 and 9.

To get a geometric interpretation of complex multiplication, we need another kind of coordinates, called the **polar coordinates** to describe the position of a point in the plane. We fix any point O in the plane, to be called the pole and any ray through O. If the plane is already coordinatised, it is convenient to take the pole at the origin and the ray as the positive x-axis. Then the position of any point P in the plane is uniquely determined by the length, say r, of the segment OP and the angle, say θ which the ray OP makes with the fixed ray. (See Figure (c) below.) These numbers r and θ are called the polar coordinates of P. From elementary trigonometry, they are related to the cartesian coordinates by the equations

$$x = r\cos\theta \; ; y = r\sin\theta \tag{34}$$

Note that for $P = O$, $r = 0$ and θ has no unique value. For any other point P, too, θ is not uniquely defined because of the periodicity of the trigonometric functions. In fact, (r_1, θ_1) and (r_2, θ_2) will denote the same point if $r_1 = r_2$ and $\theta_1 - \theta_2$ is an integral multiple of 2π. Given the cartesian coordinates x and y of a point P its polar coordinates can be obtained as follows. r is simply $\sqrt{x^2 + y^2}$. θ is any angle which satisfies the equations[8]

$$\cos\theta = \frac{x}{r} \text{ and } \sin\theta = \frac{y}{r}. \tag{35}$$



(c) polar coordinates (d) complex multiplication

Now, suppose $z = x + iy$ is a complex number representing a point P (other than O) in the cartesian plane. Then it is clear that $|z|$ is simply the first polar coordinate of P. The second polar coordinate, viz., θ, of P is

[8]Sometimes an attempt is made to define θ as $\tan^{-1}(\frac{y}{x})$. But this leaves it open to define θ when $x = 0$. Also according to this definition, the points (x, y) and $(-x, -y)$ will have the same polar coordinates. To avoid this, the definition has to be modified so that sometimes θ equals $\tan^{-1}(\frac{y}{x})$ and sometimes it equals $\tan^{-1}(\frac{y}{x}) + \pi$. But we shall not do so. As we shall see in Chapter 10, any two solutions of the system (35) differ by a multiple of 2π. So the system (35) provides an accurate definition of θ.

called an **argument**[9] of z and is denoted by $\arg(z)$. Note that for a fixed (non-zero) z, $\arg(z)$ is not a single real number but a whole set of real numbers any two members of which differ by a multiple of 2π. For example, $\arg(1)$ is the set of all multiples of 2π, $\arg(-1)$ is the set of all odd multiples of π, $\arg(i)$ is the set $\{\frac{\pi}{2} + 2\pi n : n = 0, \pm 1, \pm 2, \ldots\}$ while $\arg(-i)$ is the set $\{-\frac{\pi}{2} + 2\pi n : n = 0, \pm 1 \pm 2, \ldots\}$. That is why, although we talk of *the* length of a complex number, we talk of *an* argument of it and not *the* argument of it. If r is the length of $z = x + iy$ and θ an argument of it, then in view of (34) we have $z = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta)$. This is sometimes called the **polar representation** of z (as opposed to $z = x + iy$ which is the cartesian representation).

Now suppose $z_1 = x_1 + iy_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = x_2 + iy_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ are two complex numbers and $z_3 = x_3 + iy_3 = r_3(\cos\theta_3 + i\sin\theta_3)$ is their product $z_1 z_2$. Then we have

$$r_3 = r_1 r_2 \tag{36}$$

$$\arg(z_3) = \arg(z_1) + \arg(z_2) \tag{37}$$

Verbally, these two equations say that to multiply two complex numbers, multiply their lengths and add their arguments. We have already proved (36). Before proving (37) let us see what it really means, since neither side is a definite real number but a certain set of real numbers. Suppose we take some particular values of $\arg(z_1)$, $\arg(z_2)$ and $\arg(z_3)$, say, θ_1, θ_2 and θ_3 respectively. Then θ_3 need *not* be equal to $\theta_1 + \theta_2$ as we would normally expect. The correct interpretation of (37) is that $\theta_1 + \theta_2$ is one of the possible values of the L.H.S., viz. of $\arg(z_3)$. So all we can say is that $\theta_3 - (\theta_1 + \theta_2)$ is a multiple of 2π.

(Sometimes it is desirable to have a single argument assigned to each non-zero complex number z. See Chapter 6, Comment No. 11 for this.)

Once the correct meaning of (37) is understood, the proof itself is a simple exercise in trigonometry. If we write z_1 as $r_1(\cos\theta_1 + i\sin\theta_1)$, z_2 as $r_2(\cos\theta_2 + i\sin\theta_2)$ and z_3 as $r_3(\cos\theta_3 + i\sin\theta_3)$ then since we already have $r_3 = r_1 r_2$, the equation $z_3 = z_1 z_2$ reduces to

$$\cos\theta_3 + i\sin\theta_3 = (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \tag{38}$$

Expanding the R.H.S. and equating the real and imaginary parts of the two sides we get,

$$\cos\theta_3 = \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 = \cos(\theta_1 + \theta_2) \tag{39}$$

$$\sin\theta_3 = \sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2 = \sin(\theta_1 + \theta_2) \tag{40}$$

From elementary trigonometry, these two equations can hold together if and only if $\theta_3 - (\theta_1 + \theta_2)$ is a multiple of 2π. But this is exactly the meaning of (37).

[9] This name is rather strange, especially when the same word has two other very different meanings in mathematics. One is a logical sequence of statements. Another usage arises in connection with functions. An argument is the variable on which a function acts. Thus, if y is a function of x, say $y = f(x)$, then we say that x is the argument of f. For a function of two variables, say $y = f(x_1, x_2)$, we say x_1 is the first argument of f and x_2 the second.

Based on (36) and (37) a geometric interpretation can be given to complex multiplication. Let P_1, P_2 and P_3 be the points in the Argand diagram which correspond to z_1, z_2 and z_3 respectively, where $z_3 = z_1 z_2$. Then as shown in Figure (d) above, the segment OP_3 is obtained by rotating the segment OP_1 counterclockwise around the point O, through an angle θ_2 (which is an argument of z_2) and then multiplying its length by the factor r_2 (which is the absolute value of z_2). The latter operation is called a **dilation** or a **contraction** depending on whether $r_2 > 1$ or $r_2 < 1$. (Informal terms for these will be 'stretching' and 'shrinking' respectively.) Thus, geometrically, multiplication by a complex number $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ amounts to a rotation followed by a dilation/contraction. If $r_2 = 1$, then the second operation is redundant. So the effect of multiplication by a number of the form $\cos\theta_2 + i\sin\theta_2$ is a counter-clockwise rotation through an angle θ_2. (If $\theta_2 < 0$, say $\theta_2 = -\frac{\pi}{3}$, then we take a clockwise rotation through the angle $\frac{\pi}{3}$, or what amounts to the same, a counter-clockwise rotation through the angle $2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$.)

Comment No. 17:

The fact that rotation around the origin can be expressed in terms of multiplication by a complex number can be applied to tackle problems in geometry as we shall see in Chapter 8. Note that the crux of the argument was the trigonometric identities in (39) and (40) for the expansions of $\cos(\theta_1 + \theta_2)$ and $\sin(\theta_1 + \theta_2)$. We can put these two identities into a single complex identity and write

$$\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2) = (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) \tag{41}$$

We now introduce a shorthand notation. For any real number θ, let us write

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{42}$$

Then (41) takes the compact form

$$e^{i(\theta_1+\theta_2)} = e^{i\theta_1} e^{i\theta_2} \tag{43}$$

(42) is often called the **Euler formula**. This is a wrong term at this stage. In a genuine formula, both the sides are already defined independently of each other and the formula asserts their equality. In the present case, on the other hand, the L.H.S. does not mean the result of raising the number e to the $i\theta$-th power. In fact, we have not[10] defined $e^{i\theta}$ at all so far. So (42) is the *very definition* of $e^{i\theta}$. Similarly, (43) does not follow from the law of indices given in Equation (17) above. The only proof of (43) comes from (41) which is proved using trigonometry.

We can however use (42) to define e^z for any complex number z. Let $z = x + iy$. We have already defined e^x in the manner outlined in Comment

[10]There is a way to define e^z for any complex number z using power series. The trigonometric functions can also be defined through power series. Once this is done, both the sides of (42) have their own independent meanings and then it is a genuine formula. But we shall not go into it.

No. 6 above. Also, by (42), we have $e^{iy} = \cos y + i \sin y$. So if we want the law of indices (17) to hold then we have no choice but to define e^z as

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y) \tag{44}$$

It is clear from this definition that e^z is a complex number whose absolute value is e^x (which is always positive no matter what the real number x is) and which has y as an argument. In case z is a real number, i.e., $y = 0$, then the R.H.S. of (44) is simply e^x. So (44) is an extension of the earlier definition of an exponential. As a concrete example, $e^{2+i\frac{\pi}{3}} = e^2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}) = e^2(\frac{1}{2} + i\frac{\sqrt{3}}{2})$ $= \dfrac{e^2}{2} + i\dfrac{e^2\sqrt{3}}{2}$. Note also that $e^{i\pi} = -1$. This simple equation is significant in two ways. First we can rewrite it as $e^{i\pi} + 1 = 0$. In this form the equation involves five numbers, viz., $1, 0, e, i$ and π. Historically, these five numbers entered mathematics at different times and in very different contexts. So it is remarkable that a single equation combines all five of them. Secondly, we can also interpret $e^{i\pi} = -1$ to say that $\ln(-1) = i\pi$. So using complex numbers we can take natural logarithms of negative numbers. The trouble is that complex logarithms are not unique. Indeed because of the periodicity of the trigonometric functions, for any integer n, $e^{z+2\pi ni}$ is the same as e^z. It is easy to show that any two logarithms of the same complex number differ by a multiple of $2\pi i$.

The law of indices for complex exponents, viz., $e^{z_1+z_2} = e^{z_1}e^{z_2}$ can be proved by directly calculating both the sides. A special case of this in which the exponent is purely imaginary is especially popular. Suppose θ is a real number. Then it follows easily by induction on n, that for every positive integer n, $e^{in\theta} = (e^{i\theta})^n$. Rewriting both the sides using (42) we get

$$\cos n\theta + i \sin n\theta = (\cos\theta + i\sin\theta)^n \tag{45}$$

This is known as **DeMoivre's rule**. The right-hand side can be expanded using the binomial theorem and the fact that the successive powers of the number i recur in a cycle of four, viz., $i, -1, -i, 1, i, -1, -i, 1, i, \ldots$. Then equating the real and the imaginary parts of the two sides of (45) we get expressions for $\cos n\theta$ and $\sin n\theta$ in terms of $\cos\theta$ and $\sin\theta$. For example, for $n = 3$ we get $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$ and $\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$. These can, of course, also be derived from the well-known trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$. But if one wants, these identities can be derived from (45). If we replace θ by $-\theta$ in (42) we get $e^{-i\theta} = \cos\theta - i\sin\theta$. Solving this simultaneously with (42) gives us the following relations which serve to express the two basic trigonometric functions in terms of the complex exponentials.

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \tag{46}$$

We can now express any trigonometric function in terms of the complex exponentials and sometimes this helps in proving certain trigonometric formulas. For example, the identity $\cos^2\theta + \sin^2\theta = 1$ follows by squaring the formulas in (46) and adding.

Comment No. 18:

The polar representation of complex numbers also makes it easy to extract their roots. Let z be a complex number and n be a positive integer. If $z = 0$ then trivially 0 is the only n-th root of z. Suppose $z \neq 0$. Write z as $re^{i\theta}$. Let $w = \rho e^{i\phi}$ be a complex n-th root of z, i.e., $w^n = z$. Then we must have $\rho^n = r$ and $n\phi = \theta$. Hence ρ is the unique positive n-th root of the positive real number r. Finding ϕ is more interesting. Surely $\frac{\theta}{n}$ is one possible value of ϕ. But there are others. Indeed, we can replace θ by $\theta + 2\pi k$ for any integer k. So for every integer k, if we take $\phi = \frac{\theta+2\pi k}{n}$ then we also get $\rho e^{i\phi}$ as an n-th root of z. In particular, taking $k = 0, 1, \ldots, n-1$ we get $\rho e^{i\frac{\theta}{n}}, \rho e^{i\frac{\theta+2\pi}{n}}, \rho e^{i\frac{\theta+4\pi}{n}}, \ldots, \rho e^{i\frac{\theta+2(n-1)\pi}{n}}$ as possible n-th roots of z. They are all distinct since the arguments of any two of them differ by at most $\frac{2\pi(n-1)}{n}$. However, if we give the integer k the higher values, viz., $n, n+1, n+2, \ldots$, we keep getting the same roots in succession. We record this as a theorem.

Theorem 2: Every non-zero complex number $z = re^{i\theta}$ has n distinct n-th roots for every positive integer n. They are given by $\rho e^{i\frac{\theta+2\pi k}{n}}$, where $k = 0, 1, \ldots, n-1$ and $\rho = r^{1/n}$.

Instead of the values 0 to $n-1$, we can, of course, give k any consecutive n integral values and still get all the n-th roots of z. Sometimes it is convenient to take the values from 1 to n instead of those from 0 to $n-1$. In the Argand diagram, all the n-th roots of z lie on a circle of radius $(|z|)^{1/n}$ centred at the origin and form the vertices of a regular n-gon inscribed in this circle. We shall see an interesting application of this in Chapter 10, Comment No. 3.

When a complex number is given in a cartesian form, before Theorem 2 is applied, it has to be converted into the polar form. Suppose, for example, that we want all cube roots of $z = 1 + i$. Then we first write z as $\sqrt{2}(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}})$ and hence as $\sqrt{2}e^{\frac{\pi i}{4}}$. (This is a little easier to do if you show $1+i$ in an Argand diagram.) Hence the three cube roots of $1 + i$ are $\sqrt[6]{2}e^{\frac{\pi i}{12}}$, $\sqrt[6]{2}e^{\frac{9\pi i}{12}}$ and $\sqrt[6]{2}e^{\frac{17\pi i}{12}}$.

An interesting special case of Theorem 2 occurs when $z = 1$. The n complex numbers $e^{\frac{2\pi k i}{n}}$ for $k = 0, 1, 2, \ldots, n-1$ are called the complex n-th **roots of unity**, 'unity' being an old name for the number 1. Note that they are successive powers of the complex number $e^{\frac{2\pi i}{n}}$. They have interesting applications to trigonometry as we shall see in Chapter 7.

Comment No. 19:

The basic idea behind the geometric representation was that a complex number $z = x + iy$ can be identified with the ordered pair (x, y), which, in turn, can be identified with with a point in the cartesian plane. The former correspondence can also be used to give a rigorous definition of complex numbers. Although the treatment of complex numbers given above is standard, it is open to a nagging question. At the very start, we assumed that there is a number i whose square is -1. Once this assumption was made, the rest of the work was smooth sailing. But how can we simply assume something that is not given to exist? Questions like this do not bother the layman and are not expected to be tackled at the JEE level. Still, for the sake of completeness, we show how we

can actually construct complex numbers from real numbers. (The construction of real numbers from rational numbers is itself a very involved task and we shall not go into it. The only reason we are giving the construction of complex numbers from the reals is to illustrate how such an extension can be achieved by defining suitable operations on suitable sets.)

In this rigorous approach to complex numbers, we *define* a complex number as an ordered pair of real numbers, i.e., an ordered pair of the form (x, y), where x and y are real numbers. This is an improvement over the earlier approach of defining it as a sum of the form $x+iy$ because now we are not relying upon any undefined concept. An ordered pair of real numbers is a perfectly well defined entity. Next we define the addition and the multiplication of two such ordered pairs by the following rules:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \quad (47)$$

$$(x_1, y_1)(x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1) \quad (48)$$

While the first formula appears quite natural the second one is impossible to swallow without some motivation. On the face of it it will be far more natural to define $(x_1, y_1)(x_2, y_2)$ as (x_1x_2, y_1y_2). But if we do so then we are faced with a strange situation. For example, we would have $(1, 0)(0, 1) = (0, 0)$. This would mean that the product of two non-zero complex numbers can be zero and surely we would not like this to happen. Of course, it still does not explain how we thought of (48) in the first place. The true explanation is that we are cheating! We already know the multiplication of complex numbers 'unofficially' and we are now only trying to put an official mark on it. (48) is just a 'clean' way of saying $(x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 - x_2y_1)$. The latter is forbidden because we do not know what i means yet.

Next comes the ritualistic verification of various properties such as the commutative and the associative laws. We skip all this. For our purpose the most noteworthy part is as follows. First note that if we take only those ordered pairs whose second entries are 0, then (47) and (48) reduce respectively to $(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0)$ and $(x_1, 0)(x_2, 0) = (x_1x_2, 0)$. This means we can very well identify a real number x with an ordered pair of the form $(x, 0)$. In particular the number -1 can now be identified with the pair $(-1, 0)$. But then a curious thing happens. If we apply (48), we see that the product $(0, 1)(0, 1)$ equals $(-1, 0)$. In other words, we have shown that the number -1 *has* a square root, viz., the complex number $(0, 1)$. We call this ordered pair i. It is then easy to verify that for any ordered pair (x, y) we have

$$(x, y) = (x, 0) + (0, y) = (x, 0) + (0, 1)(y, 0) \quad (49)$$

As we are identifying $(x, 0)$ and $(y, 0)$ with the real numbers x and y respectively, (49) can be rewritten as $(x, y) = x + iy$. We can now legitimately write every complex number in the form $x+iy$, where x, y are real numbers. Also using (47) and (48) we can *prove* that for any two complex numbers $x_1 + iy_1$ and $x_2 + iy_2$, the following formulas hold.

$$(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2) \quad (50)$$

$$(x_1 + iy_1)(x_2 + iy_2) \quad = \quad (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1) \qquad (51)$$

The rest of the journey from this point onwards is exactly as before. Unless you have a passion for logical perfection, you might not appreciate exactly what has been achieved here. We know that in the real number system, there is no room for the square root of -1. In the earlier approach we took it for granted that there is a larger number system which can accommodate the reals on one hand and also simultaneously a square root of -1 on the other. In the rigorous approach just described we have *constructed* such a system. Admittedly, the construction is clumsy and modelled after a prior knowledge of complex numbers. Still it is a perfectly logical construction.

Comment No. 20:

A more natural and convincing construction of complex numbers can be given through what are called matrices. Let m and n be positive integers. Then by a real matrix with m **rows** and n **columns** (or simply, an $m \times n$ **matrix** for short) we mean a rectangular array of mn real numbers. For example, $\begin{bmatrix} 1 & 0 & \sqrt{2} \\ -e & \pi & -\frac{7}{3} \end{bmatrix}$ is a 2×3 matrix with rows $[1 \;\; 0 \;\; \sqrt{2}]$ and $[-e \;\; \pi \;\; -\frac{7}{3}]$ and columns $\begin{bmatrix} 1 \\ -e \end{bmatrix}$, $\begin{bmatrix} 0 \\ \pi \end{bmatrix}$ and $\begin{bmatrix} \sqrt{2} \\ -\frac{7}{3} \end{bmatrix}$. A matrix in which there is only one row is also called a **row vector**. Similarly, a matrix with only one column is called a **column vector**. In each case the length of the row or the column vector is the number of entries in it. Thus a row vector of length k is simply a $1 \times k$ matrix and similarly a column vector of length k is a $k \times 1$ matrix. A matrix is also denoted by enclosing its entries in a pair of parentheses instead of a pair of brackets. But the brackets look neater. A matrix in which the number of rows equals the number of columns is called, quite appropriately, a **square matrix**.

It is customary to denote matrices by capital letters such as A, B, P etc. or sometimes by capital bold face letters $\mathbf{A}, \mathbf{B}, \mathbf{C}$ etc. We shall use the latter. However, row and column vectors are almost always denoted by small case bold face letters such as $\mathbf{a}, \mathbf{b}, \mathbf{u}, \mathbf{x}$. If a general matrix is denoted by a capital letter, say $\mathbf{A}$, then its entries are usually denoted by the corresponding small letters with double subscripts, the first for the row and the second for the column. Thus, if $\mathbf{A}$ is an $m \times n$ matrix then the entry in its i-th row and j-th column (where $1 \leq i \leq m$ and $1 \leq j \leq n$) is denoted by $a_{i,j}$ or often simply by a_{ij}. In a compact form we write $\mathbf{A} = (a_{i,j})$ or $\mathbf{A} = (a_{ij})$ to indicate that the (i, j)-th entry of A is $a_{i,j}$. In a full form we would write

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & \dots & a_{im} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix}$$

(A beginner who is seeing double subscripts for the first time is warned against reading terms containing them as 'a sub eleven', 'a sub twelve' etc. The correct reading of a_{11} is 'a sub one, one' or simply, 'a one one'. It would, in fact, be more logical to denote the entry in the i-th row and the j-th column of $\mathbf{A}$ by $a_{i,j}$ rather than by a_{ij}. But the notation above is quite standard. In actual numerical problems, m and n are rarely double digit numbers. If they were, then a_{111} would indeed be ambiguous as it could stand either for $a_{11,1}$ or for $a_{1,11}$.)

We now define the addition and the multiplication of matrices. Not every pair of matrices can be added together. Two matrices, say $\mathbf{A}$ and $\mathbf{B}$ can be added only if their sizes match i.e., they have the same number of rows and the same number of columns. When this is the case, the two matrices are added entrywise. That is, the (i,j)-th entry of $\mathbf{A}+\mathbf{B}$ is the sum of the (i,j)-th entries of $\mathbf{A}$ and $\mathbf{B}$. In a compact form, $\mathbf{A}+\mathbf{B}=\mathbf{C}$, where $\mathbf{C}=(a_{ij}+b_{ij})$. It is obvious that matrix addition, when defined, is both commutative and associative.

At the outset it would seem natural to multiply two matrices exactly the way they are added, viz., by multiplying the corresponding entries of the two matrices. But it turns out that with this multiplication, the matrices are not a particularly handy tool. Instead, we define a multiplication for matrices which, even though slightly clumsy, makes matrices a powerful tool. We first define the product of a row vector and a column vector of equal lengths, say n. Let the row vector be $\mathbf{a}=[a_1 \;\; a_2 \;\; \ldots a_n]$ and let the column vector be $\mathbf{b}=\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$. Then we define $\mathbf{ab}$ to be the sum of the products of the corresponding entries of the two vectors, i.e., the sum $\sum_{j=1}^{n} a_j b_j = a_1b_1 + a_2b_2 + \ldots + a_nb_n$.

Now suppose $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times p$ matrix, where m, n, p are some positive integers. Then the length of the rows of $\mathbf{A}$ is the same as the length of the columns of $\mathbf{B}$, each being equal to n. So we can multiply any row of $\mathbf{A}$ with any column of $\mathbf{B}$ using the definition just given. Specifically, for any $i = 1, 2, \ldots, m$ and any $k = 1, 2, \ldots, p$, the product of the i-th row of $\mathbf{A}$ and the k-th column of $\mathbf{B}$ is $\sum_{j=1}^{n} a_{ij}b_{jk}$. We form an $m \times p$ matrix whose (i,j)-th entry is this sum for $1 \le i \le m$ and $1 \le k \le p$ and call this matrix the **product** of $\mathbf{A}$ and $\mathbf{B}$. We denote it by $\mathbf{AB}$. It is an $m \times p$ matrix. As a concrete example, here is the product of a 2×3 matrix with a 3×4 matrix.

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & -3 & \pi \end{bmatrix} \begin{bmatrix} 0 & e & \sqrt{2} & 0 \\ 1 & 0 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 2e+1 & 2\sqrt{2} & -1 \\ 2\pi - 3 & -\pi & \pi - 3 & \pi \end{bmatrix}$$

To see how the matrix on the right-hand side is obtained consider the entry in, say, its second row and third column. The entries in the second row of the first matrix on the left are $0, -3$ and π while those of the third column

of the second matrix are $\sqrt{2}, 1$ and 1. So multiplying the corresponding entries and adding gives $0 \times \sqrt{2} + (-3) \times 1 + \pi \times 1 = \pi - 3$. So this is the entry in the second row and the third column of the product matrix. We leave it to you to check that the other entries of the product matrix are also obtained this way.

This definition of matrix multiplication may appear rather clumsy. But it turns out that matrix products defined this way arise naturally in applications. One instance of this will be given in the next chapter where we shall see how a system of linear equations can be expressed in terms of matrices. (Specifically see Exercise (3.24).)

It can be shown by a straightforward but somewhat laborious computation that the matrix multiplication, whenever defined, is associative. That is, suppose $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are matrices such that the matrix products $\mathbf{AB}$ and $\mathbf{BC}$ are both defined, which means that the number of columns in $\mathbf{A}$ equals the number of rows in $\mathbf{B}$ and the number of columns in $\mathbf{B}$ equals the number of rows in $\mathbf{C}$. Then the matrix products $(\mathbf{AB})\mathbf{C}$ and $\mathbf{A}(\mathbf{BC})$ are both defined and are equal. However, the matrix multiplication is not commutative. In fact, when $\mathbf{AB}$ is defined, $\mathbf{BA}$ need not even be defined and even if defined need not equal $\mathbf{AB}$, as we see by taking $\mathbf{A}$ and $\mathbf{B}$ to be the 2×2 matrices $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ respectively. A direct computation gives $\mathbf{AB} = \mathbf{B}$ but $\mathbf{BA}$ is the 2×2 zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

The theory of matrices is rich both in terms of the non-trivial theorems in it and in terms of real life applications. However, it is not our intention to develop it. We introduced them merely to show how complex numbers can be constructed using certain matrices. Admittedly, this is not the best reason to study matrices. But right now we are more concerned with the construction of the complex numbers from reals. In Comment No. 19 we already gave a construction in which a complex number was defined as an ordered pair of real numbers. But that construction was open to the objection that the multiplication of two ordered pairs was defined very artificially. With matrices, the situation is different. We already *have* the definitions of matrix addition and multiplication. We shall show that if these definitions are applied to a certain class of 2×2 matrices then we get a system in which the matrices of this class correspond naturally to complex numbers.

A typical 2×2 matrix is of the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where a, b, c, d are some real numbers. (For small-sized matrices, such notations are better than the double subscripted ones.) The addition and multiplication of any two such matrices are always defined and are given by

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{bmatrix} \tag{52}$$

$$\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{bmatrix} \tag{53}$$

Note that the multiplication is not, in general, commutative as we observed

above. However, we shall consider certain special 2×2 matrices for which the multiplication will be commutative. Let us first consider all 2×2 matrices of the form $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$. Evidently every such matrix is determined by the real number a. Moreover, if $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ and $\begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix}$ are two such matrices then from (52) and (53), their sum and product are the matrices $\begin{bmatrix} a+b & 0 \\ 0 & a+b \end{bmatrix}$ and $\begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix}$ respectively. It follows that we may very well identify a real number x with the 2×2 matrix $\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$. We now need a 2×2 matrix $\mathbf{A}$ which will play the role of the complex number i, i.e., a matrix $\mathbf{A}$ whose square $\mathbf{AA}$ will equal the matrix $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (which corresponds to the real number -1). Each of the matrices $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has this 'magic' property as can be seen by direct substitution into (53). We choose the first one to correspond to the complex number i. What about real multiples of i, that is, complex numbers of the form iy, where y is real? As we are identifying y with the matrix $\begin{bmatrix} y & 0 \\ 0 & y \end{bmatrix}$, the correct analogue of iy should be the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y & 0 \\ 0 & y \end{bmatrix}$ which works out to be the matrix $\begin{bmatrix} 0 & -y \\ y & 0 \end{bmatrix}$. It is now clear that a complex number $x + iy$ ought to correspond to the matrix $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$.

To put the construction in the 'polished' form, we now pretend that we know absolutely nothing about complex numbers. We then *define* a complex number to be a 2×2 matrix of the form $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$, where x and y are any real numbers. We apply (52) and (53) to check that the sum and the product of two complex numbers is again a complex number. We leave this verification as an exercise (Exercise (2.26)). We identify a matrix of the form $\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$ with the real number x. Also we denote the matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ by the symbol i. Then the complex number $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$ equals $x + iy$. (Alternately, we could denote $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ by i. In that case $x + iy$ will correspond to the matrix $\begin{bmatrix} x & y \\ -y & x \end{bmatrix}$. The choice is essentailly a matter of taste.)

The construction of complex numbers is just one application of matrices. They can also be used to construct other things. There are, for example, what are called real quaternions (Exercise (21.25)). They can be constructed as certain 4×4 real matrices or also as certain 2×2 complex matrices. But we

shall not go into it.

Comment No. 21:

Associated with every square matrix **A** is a number called its **determinant**, denoted by $det(\mathbf{A})$ or by $|\mathbf{A}|$. A more standard notation for the determinant[11], however, is simply to replace the brackets (or the parentheses) of the matrix by a pair of vertical lines. Thus a typical 3×3 determinant is $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$. To evaluate it we first have to know what a 2×2 determinant is. By definition, $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ equals $ad - bc$. We can now 'expand' the given 3×3 determinant w.r.t. any one of its rows or columns. We consider various 2×2 determinants formed by the entries in the remaining two rows or the columns and multiply them with the entries of the chosen row (or column), attaching suitable $+$ or $-$ sign to the product. For example, if we expand w.r.t. the first row we get the determinant (often denoted by Δ or D) as

$$\begin{aligned} \Delta &= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \qquad (54) \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{31} - a_{21}a_{33}) \\ &\quad + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \qquad (55) \end{aligned}$$

This is a sum of 6 terms corresponding to the six permutations of the three symbols $1, 2$and 3. Each term is product of the form $a_{1i}a_{2j}a_{3k}$, where $(i\ j\ k)$ is a permutation of the three symbols $1, 2$ and 3. Half the terms are assigned $+$ signs and half the $-$ sign. There is a systematic way to tell which terms have $+$ signs and which ones have $-$ signs. But we shall not go into it. The point to note is that the value of a 3×3 determinant could have been defined either in terms of 2×2 determinants as in (54) or directly as a sum of 3! terms each corresponding to a permutaion of three symbols as in (55). The former method is called the **inductive definition** of the determinant while the latter is the definition using permutations.

Each approach has its advantages and disadvantages. For example, the inductive definition can be easily applied to define the determinant of any

[11]The use of this word requires a little care because it is used with two different shades of meaning. When we talk of the determinant of a matrix, we only mean the number associated with it. But when we use expressions like the first row of a 3×3 determinant, we usually mean the underlying matrix. In other words, in such a usage the particular form of the determinant matters. Thus even though the two matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\begin{bmatrix} -a & c \\ b & -d \end{bmatrix}$ have the same determinants, viz., $ad - bc$, the two determinants $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ and $\begin{vmatrix} -a & c \\ b & -d \end{vmatrix}$ are not considered to be the same. This confusion would not arise if matrices are studied before determinants for then a determinant is always a certain number associated with a square matrix. But the standard practice nowadays is to study determinants even without mentioning the word 'matrix' and this leads to confusion. To avoid it, sometimes the phrase the 'value of a determinant' is used to indicate the number associated with a determinant (which is a square array enclosed by a pair of vertical lines).

$n \times n$ matrix in terms of those of certain $(n-1) \times (n-1)$ submatrices. A direct definition in terms of permutations will involve $n!$ terms each term being of the form $\pm a_{1i_1} a_{2i_2} \ldots a_{ni_n}$, where $(i_1\ i_2\ \ldots\ i_n)$ is a permutation of the indices $1, 2, \ldots, n$, the $+$ sign occurring for half the terms and the $-$ sign for the remaining terms. The inductive definition is sometimes more convenient when proving something by induction. On the other hand, the inductive definition gives a somewhat special role to the first row, whereas, the determinant is absolutely symmetric w.r.t. all rows and columns. A determinant can be expanded w.r.t. any row or column and we still get exactly the same value. In the definition using permutations this anomaly does not arise. For us it really does not matter which definition we follow since we shall, almost exclusively, confine ourselves to determinants of order not exceeding 3. We mention, however, that all the results about them can be generalised to determinants of arbitrary orders. (See Exercises (3.23), (3.26) and (8.25) and Chapter 9, Comment No. 6 for problems where a reference is made to determinants of order higher than 3.)

We assume familiarity with certain well-known results about determinants. The foremost is that a determinant is unaffected if its rows are completely interchanged[12] with its columns. As a result, any theorems that hold for rows have analogues for columns and no separate proof is needed. For example, once it is proved that a determinant changes its sign when two of its rows are interchanged, it also follows that it changes sign when any two columns are interchanged.

The three most frequently used formulas in simplifying a determinant involve what are called **elementary row operations**. The interchange of two rows just mentioned is one such operation. If the i-th row is interchanged with the j-th row, this operation is symbolically denoted by $R_i \longleftrightarrow R_j$. As just mentioned the determinant gets multiplied by -1 under this operation. Another elementary row operation is to multiply all entries in one row by a constant. Specifically, if all the entries in the i-th row are multiplied by λ then symbolically we write $R_i \longrightarrow \lambda R_i$. (The notation suggests that the i-th row is being replaced by its multiple by λ.) Under this row operation, the value of the determinant gets multiplied by λ. The third elementary row operation is symbolically of the type $R_i \longrightarrow R_i + \lambda R_j$ and means that we are adding λ times the j-th row to the i-th row (where $i \neq j$). Under this operation the value of the determinant is unchanged.

In view of what is said above, there are also corresponding three elementary column operations. When any of these operations are performed, they are to be performed one at a time. For example, suppose we perform $R_1 \longrightarrow R_1 + 2R_2$. This means we are adding twice the second row to the first. In this process all other rows remain intact. But suppose we next perform $R_1 \longleftrightarrow R_3$. Here R_1 stands for the first row of the *new* determinant (which is

[12] In the language of matrices, the matrix obtained by such an interchange is said to be the **transpose** of the original matrix. The definition makes sense for any matrix, not necessarily a square one. Thus if $\mathbf{A} = (a_{ij})$ is any $m \times n$ matrix, then its transpose, denoted variously by $\mathbf{A}'$, $\mathbf{A}^t$ or by $\mathbf{A}^T$ is the $n \times m$ matrix whose (i, j)-th entry is a_{ji}. The property asserted here is that any square matrix and its transpose always have the same determinant.

$R_1 + 2R_2$ of the original determinant).

As a simple illustration of these rules let us evaluate the determinant

$$\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$$

where a, b, c are respectively the p-th, q-th and r-th terms of a harmonic progression. (JEE 1997*)

Call the given determinant as D. As a, b, c are terms of an H.P., their reciprocals are terms of an A.P. So we must somehow involve these reciprocals in the determinant. Once this idea strikes, it is not difficult to come up with the right row operation. If we divide the first row throughout by abc then we get $D = abcD'$, where

$$D' = \begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}.$$

Now since $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are the p-th, the q-th and the r-th terms of an A.P. the differences $\frac{1}{a} - \frac{1}{c}$ and $\frac{1}{b} - \frac{1}{c}$ are proportional to $p - r$ and $q - r$ respectively. Call them as $\lambda(p - r)$ and $\lambda(q - r)$ for some λ. If we now subtract the third column of D' from each of the first two (i.e., perform the operations $C_1 \longrightarrow C_1 - C_3$ and $C_2 \longrightarrow C_2 - C_3$) and expand the resulting determinant we see that it is 0. So D is also 0.

As another example, let $D = \begin{vmatrix} b^2 + c^2 & a^2 & a^2 \\ b^2 & c^2 + a^2 & b^2 \\ c^2 & c^2 & a^2 + b^2 \end{vmatrix}$ (JAT 1980).

To evaluate D, as a first measure of simplification, we replace a^2, b^2, c^2 by x, y, z respectively. (Such simple tricks save a lot of unnecessary work. In the present problem, for example, the squares have really no role. If we keep them they will only complicate the writing and make it more prone to numerical errors.) Then $D = \begin{vmatrix} y + z & x & x \\ y & z + x & y \\ z & z & x + y \end{vmatrix}$. Subtracting the second and then the third row from the first gives $D = \begin{vmatrix} 0 & -2z & -2y \\ y & z + x & y \\ z & z & x + y \end{vmatrix}$. A direct expansion w.r.t. the first row now gives $D = 4xyz = 4a^2b^2c^2$.

The present problem could also have been done without the row operations using the well-known expansion of $(y + z)(z + x)(x + y)$ as $y^2z + yz^2 + z^2x + zx^2 + x^2y + xy^2 + 2xyz$. In fact this is typical of many 3×3 determinants. For a 2×2 determinant direct expansion is the best method. For determinants of order 4 or above, a direct expansion is ruled out except in very special cases (e.g. when a large number of entries are 0). But the case of a 3×3 determinant is critical. Surely, evaluation based on the properties of determinants is more elegant. But in many problems, the direct expansion (which is often looked down as a 'brute force' method) is not all that bad. In fact, in some

problems the time it takes to come up with the right trick to pull may as well be spent in a direct straightforward expansion. For example, to decide whether the two determinants $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$ and $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ are identically equal (JEE 1983), it is far better to see by direct expansion that each one equals a sum of three terms of the form $bc(c-b)$ than to look for 'clever' short cuts. (One such short cut is to take out the factor abc from the last column of the first determinant and to rewrite it as $abc\begin{vmatrix} 1 & a & 1/a \\ 1 & b & 1/b \\ 1 & c & 1/c \end{vmatrix}$. Similarly taking out the factors a, b, c respectively from the three rows of the second determinant, it equals $abc\begin{vmatrix} 1/a & 1 & a \\ 1/b & 1 & b \\ 1/c & 1 & c \end{vmatrix}$. Now a suitable permutation of the columns shows that the two determinants are equal.)

Of course, when the question specifically prohibits or requires a particular method, the mandate has to be followed. In some cases, the prohibition is not explicit. But the nature of the entries is such that a particular method is indicated or contraindicated. For example, sometimes we want to show that a certain determinant vanishes. In that case, instead of expanding it, see if the sum or some other suitable linear combination of the rows (or of the columns) can be easily shown to vanish. We shall see instances of this in Chapters 8, 9 and 10 where the concurrency of three lines in a plane is equivalent to the vanishing of a certain determinant. (Actually, as we shall see in Corollary 9 of the next chapter, if a determinant vanishes then some non-trivial linear combination of its rows must vanish with a similar statement holding for its columns. The trouble, of course, is that there is no golden rule to tell beforehand the coefficients of this magic linear combination. This is where some perceptivity and experience pays. See the problem in Comment No. 22 below for a good illustration of this.) Similarly in a determinant whose entries are trigonometric expressions, adding or subtracting one row from another may be particularly easy because of some trigonometric identities. In such problems, brute force expansion is clearly a bad idea.

Sometimes a judicious interpretation of a problem about determinants suggests what simplifications are needed. Suppose, for example, that for a fixed positive integer n,

$$D = \begin{vmatrix} n! & (n+1)! & (n+2)! \\ (n+1)! & (n+2)! & (n+3)! \\ (n+2)! & (n+3)! & (n+4)! \end{vmatrix}$$

and we are asked to show that $[D/(n!)^3] - 4$ is divisible by n. (JEE 1992)

The use of the brackets in the statement of the problem is a little unfortunate, since they are also commonly used to denote the integral part of a real number and this interpretation is tempting in the present problem since it involves integers. However, if the brackets are meant to denote the greatest

integer, this is generally made clear. At any rate, in the present problem it is abundantly clear that D is divisible by $(n!)^3$ and so $D/(n!)^3$ is an integer anyway. So no confusion need arise. Now, coming to the problem itself, the conclusion is equivalent to saying that $D/(n!)^3$ is of the form $an+4$ for some integer a. So we should begin by taking out a factor $n!$ from each of the three rows. When we do so, we see that $D/(n!)^3$ equals D', where D' is the determinant

$$\begin{vmatrix} 1 & (n+1) & (n+1)(n+2) \\ (n+1) & (n+1)(n+2) & (n+1)(n+2)(n+3) \\ (n+1)(n+2) & (n+1)(n+2)(n+3) & (n+1)(n+2)(n+3)(n+4) \end{vmatrix}.$$

It is now tempting to try to expand D' by doing further row operations. But that is not necessary. Our goal is to prove that D' is of the form $an+4$ for some integer n. Now even without expanding D' it is clear that its value is a polynomial with integer coefficients in the variable n. So if we could show that the constant term of this polynomial is 4, it would follow that $D'-4$ is divisible by n and hence D' has the desired form. So the problem reduces to showing that the constant term in D', treated as a polynomial in n, is 4. Now, the constant term of any polynomial is simply its value at 0. If we put $n=0$, D' becomes the determinant $\begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 6 \\ 2 & 6 & 24 \end{vmatrix}$. By a direct computation, the value of this determinant is 4 and that completes the solution.

As another illustration, we have the following.

If a, b, c are constants with $a^2+b^2+c^2=1$, prove that the equation

$$\begin{vmatrix} ax-by-c & bx+ay & cx+a \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix} = 0$$

represents a straight line. (JEE 2001)

A direct expansion of this determinant will be too laborious. But even without expanding it we see that it is a polynomial of degree 3 in the variables x and y. But the equation of a straight line is linear, i.e., of first degree. Of course, it may happen that because of cancellation of some terms, the degree of the polynomial represented by the determinant is not 3 but less. But that does not happen here. For example, the coefficient of x^3 will come out to be $a^3+ab^2+ac^2$, i.e., a (since $a^2+b^2+c^2$ is given to be equal to 1). Similarly, by inspection, the coefficient of y^3 comes out to be b. So the determinant will have degree 3 unless $a=0$ and $b=0$. How can it then be the equation of a straight line? The only possibility is that this polynomial factors as a product of two polynomials, one linear and the other quadratic in such a way that the quadratic factor never vanishes. As already noted, the coefficients of x^3 and y^3 are a and b respectively. The constant term in the polynomial can be obtained by putting $x=0=y$ in the determinant. That gives us a determinant $\begin{vmatrix} -c & 0 & a \\ 0 & -c & b \\ a & b & c \end{vmatrix}$ whose value, by direct computation, is $c(a^2+b^2+c^2)=c$. Hence

the determinant probably factors something like $(ax+by+c)(x^2+y^2+1)$.

The reasoning so far does not help us directly in simplifying the determinant. But it opens one possible line of attack. We try to get a factor of $ax+by+c$ from the determinant. Call the determinant as D. If we multiply the first row by a and add to it b times the second row and then c times the third row, then using $a^2+b^2+c^2=1$, the determinant equals

$$\frac{1}{a}\begin{vmatrix} x & y & 1 \\ bx+ay & -ax+by-c & cy+b \\ cx+a & cy+b & -ax-by+c \end{vmatrix}$$

We now do the same thing column-wise. That is, we multiply the first column by a and add b times the second and c times the third to get

$$D=\frac{1}{a^2}\begin{vmatrix} ax+by+c & y & 1 \\ y & -ax+by-c & cy+b \\ 1 & cy+b & -ax-by+c \end{vmatrix}$$

which is still somewhat formidable. But now that we have a hunch that $(ax+by+c)$ is likely to be a factor of D, we expand keeping this in mind and try to extract $ax+by+c$ as a factor.

$$\begin{aligned} a^2D &= (ax+by+c)(a^2x^2-b^2y^2-c^2-c^2y^2-b^2) \\ &\quad +y(b+axy+by^2)+(cy^2+ax+c) \\ &= (ax+by+c)(a^2x^2-b^2y^2-c^2-c^2y^2-b^2) \\ &\quad +(ax+by+c)+y^2(ax+by+c) \\ &= (ax+by+c)(1+y^2+a^2x^2-b^2y^2-c^2-c^2y^2-b^2) \\ &= (ax+by+c)(a^2+a^2x^2+a^2y^2) \quad (\text{since } a^2+b^2+c^2=1) \\ &= a^2(ax+by+c)(x^2+y^2+1) \end{aligned}$$

Thus we finally have $D=(ax+by+c)(x^2+y^2+1)$. Since the second factor can never vanish for any (real) values of x and y, the given equation represents the straight line $ax+by+c=0$. (In the solution above we tacitly assumed that $a\neq 0$. If $a=0$ but $b\neq 0$, we multiply the second row by b and add multiples of the other two rows to it. If a,b are both 0, then $c\neq 0$ since $a^2+b^2+c^2=1$ and then we begin by multiplying the third row by c. Consideration of such degeneracies is generally not expected in a time constrained examination. But one should be prepared to give it.)

Comment No. 22:

There is yet another property of determinants which is not needed as frequently as the row/column operations but which is theoretically important. It deals with the product of determinants. We already defined the product of matrices above. By a product of two determinants (of the same order) we merely mean the product of their underlying matrices. Thus the (i,j)-th entry of the product determinant is obtained by adding the products of the entries in the i-th

row of the first determinant with the corresponding entries in the j-th column of the second determinant. It turns out that the determinant of the product always equals the product of the two determinants. In the notation of matrices we have that for any two square matrices $\mathbf{A}$ and $\mathbf{B}$ of the same size,

$$|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}| \tag{56}$$

Note that in general, matrix multiplication is not commutative. In other words $\mathbf{AB}$ need not equal $\mathbf{BA}$ as matrices. Still, because of (56), they have the same determinants since $|\mathbf{A}|$ and $|\mathbf{B}|$ do commute with each other.

As a simple illustration of (56), suppose $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are two complex numbers. We saw in Comment No. 20 above that we can identify them with the matrices $\begin{bmatrix} x_1 & y_1 \\ -y_1 & x_1 \end{bmatrix}$ and $\begin{bmatrix} x_2 & y_2 \\ -y_2 & x_2 \end{bmatrix}$ respectively. Note that the determinants of these matrices are $x_1^2 + y_1^2$ and $x_2^2 + y_2^2$ respectively, which equal $|z_1|^2$ and $|z_2|^2$ respectively. Now the product of these two matrices is the matrix $\begin{bmatrix} x_1x_2 - y_1y_2 & x_1y_2 + x_2y_1 \\ -(x_1y_2 + x_2y_1) & x_1x_2 - y_1y_2 \end{bmatrix}$ which corresponds to the complex number z_1z_2. By a direct calculation, the determinant of this product matrix comes out to be $(x_1^2 + y_1^2)(x_2^2 + y_2^2)$ after simplification. But we could have also got this from (56) since we already know that $|z_1z_2| = |z_1||z_2|$. Alternately, we could compute the determinant of the product matrix directly and then use (56) to give an alternate proof that $|z_1z_2| = |z_1||z_2|$.

An application of (56) sometimes leads to a quick and somewhat unexpected simplification. Suppose for example, that A, B, C and P, Q, R are any numbers and we want to show (JEE 1994) that

$$\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} = 0. \tag{57}$$

Here a direct expansion would be horrendous. So let us try to prove (57) by elementary row/column operations. We remarked earlier that one way to do this is to show that a suitable non-trivial linear combination of the rows (or columns) vanishes identically. So we look for some real numbers x, y, z not all 0, such that $xR_1 + yR_2 + zR_3$ vanishes identically. This means we want $x\cos(A-\theta) + y\cos(B-\theta) + z\cos(C-\theta) = 0$ for $\theta = P, Q, R$. Obviously, then, x, y, z should depend only on A, B, C and not on P, Q, R. In that case $x\cos(A-\theta) + y\cos(B-\theta) + z\cos(C-\theta)$ ought to vanish for *every* θ. Using standard trigonometric identities for $\cos(A-\theta)$ etc. we want that the equation

$$(x\cos A + y\cos B + z\cos C)\cos\theta + (x\sin A + y\sin B + z\sin C)\sin\theta = 0$$

should hold for every θ. Putting $\theta = 0$ and then $\theta = \frac{\pi}{2}$, this is equivalent to requiring that the two equations

$$x\cos A + y\cos B + z\cos C = 0 \tag{58}$$

$$\text{and} \quad x\sin A + y\sin B + z\sin C = 0 \tag{59}$$

should hold simultaneously. As we shall see in the next chapter, this is a homogeneous system of two equations in three unknowns. It has infinitely many solutions. But we want just one in which at least one of x, y and z is non-zero. By symmetry, we may suppose $z \neq 0$. Then calling $\frac{x}{z}$ and $\frac{y}{z}$ as u and v respectively, the system reduces to

$$u\cos A + v\cos B = -\cos C \tag{60}$$
$$u\sin A + v\sin B = -\sin C \tag{61}$$

which can be solved easily to give

$$u = \frac{\sin B\cos C - \cos B\sin C}{\sin A\cos B - \cos A\sin B} = \frac{\sin(B-C)}{\sin(A-B)} \tag{62}$$
$$v = \frac{\sin C\cos A - \cos C\sin A}{\sin A\cos B - \cos A\sin B} = \frac{\sin(C-A)}{\sin(A-B)} \tag{63}$$

As the system is homogeneous, we can choose z arbitraily. Taking it as $\sin(A-B)$, we get $x = \sin(B-C)$, $y = \sin(C-A)$ and $z = \sin(A-B)$. With these values, (58) and (59) indeed hold true as we can see by direct substitution. So reversing the reasoning above, the linear combination $xR_1 + yR_2 + zR_3$ vanishes identically. Hence (57) holds. (Again, for the sake of perfection, we must ensure that x, y, z are not all 0. But if that happens then A, B, C differ from each other by multiples of π. But in that case, at least two of them differ from each other by an even multiple of π. But if, say, $A - B = 2\pi k$, then $(A-P)-(B-P)$, $(A-Q)-(B-Q)$ and finally, $(A-R)-(B-R)$ all equal $2\pi k$. So the first two rows of the determinant in (57) are equal, forcing it to vanish.)

This solution was elementary in that it only used the elementary row operations. But to find out the right row operations required a good deal of work. Let us now see how the problem can be done using (56). If we expand $\cos(A-P)$ it is $\cos A\cos P + \sin A\sin P$. Let us rewrite this as $\cos A \times \cos P + \sin A \times \sin P + 0 \times 0$. Doing so may seem silly, but there is a purpose to it. For in this form we recognise $\cos(A-P)$ as the product of the row vector $[\cos A \quad \sin A \quad 0]$ and the column vector $\begin{bmatrix} \cos P \\ \sin P \\ 0 \end{bmatrix}$. If we do this to every entry in the determinant in (57) and apply (56), we see that the determinant in (56) equals the product of the determinants

$$\begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \begin{vmatrix} \cos P & \cos Q & \cos R \\ \sin P & \sin Q & \sin R \\ 0 & 0 & 0 \end{vmatrix}$$

Evidently, each factor is 0 and so (57) holds!

Comment No. 23:

Verbally, (56) says that the determinant of a product is the product of the determinants or, in a more sophisticated phraseology, that the determinant is a **multiplicative** function. It is tempting to think that an analogous result holds for addition of matrices, i.e., if $\mathbf{A}$ and $\mathbf{B}$ are two square matrices of the same order then $|\mathbf{A}+\mathbf{B}| = |\mathbf{A}| + |\mathbf{B}|$. But this is not true in general. A simple counterexample is to take $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Then $|\mathbf{A}| = |\mathbf{B}| = 0$ but $\mathbf{A}+\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and so $|\mathbf{A}+\mathbf{B}| = 1$. In other words, although the determinant is a multiplicative function, it is not an **additive** function.

But the determinant *is* additive in each of its rows or columns one at a time. Instead of defining this formally in its full generality, we explain what it means in the case of a 2×2 determinant, because the extension to higher order determinants is then obvious. Take a typical 2×2 determinant, say $D = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$. Suppose we 'split' the first row $[a \quad b]$ as the sum of two row vectors, say $[a_1 \quad a_2]$ and $[b_1 \quad b_2]$. In other words, suppose $a = a_1 + a_2$ and $b = b_1 + b_2$. Let D_1 and D_2 be the determinants $\begin{vmatrix} a_1 & b_1 \\ c & d \end{vmatrix}$ and $\begin{vmatrix} a_2 & b_2 \\ c & d \end{vmatrix}$ respectively. Note that we are not splitting the second row and that it is common to both the determinants and the original determinant too. Then we have $D = D_1 + D_2$ or, in a full form,

$$\begin{vmatrix} a_1+a_2 & b_1+b_2 \\ c & d \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ c & d \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c & d \end{vmatrix} \tag{64}$$

which can be proved by direct computation. A similar formula will hold if instead of the first row we split the second row of D. If, however, we split the first column, then analogously to (64) we get

$$\begin{vmatrix} a_1+a_2 & b \\ c_1+c_2 & d \end{vmatrix} = \begin{vmatrix} a_1 & b \\ c_1 & d \end{vmatrix} + \begin{vmatrix} a_2 & b \\ c_2 & d \end{vmatrix} \tag{65}$$

(Although there is a perfect analogy between (64) and (65), the latter looks a bit more natural because we are more used to writing the summands in a sum horizontally from left to right rather than vertically from top to bottom. Note further that we can only split a row or a column. We cannot split entries occurring in some other straight line, for example, a line parallel to either one of the two diagonals of the matrix. Suppose, for example, that $a = a_1 + a_2$ and $d = d_1 + d_2$. Then in general

$$\begin{vmatrix} a_1+a_2 & b \\ c & d_1+d_2 \end{vmatrix} \neq \begin{vmatrix} a_1 & b \\ c & d_1 \end{vmatrix} + \begin{vmatrix} a_2 & b \\ c & d_2 \end{vmatrix}$$

as can be shown by simple examples, e.g. take $a_1 = b = c = d_2 = 0$ and $a_2 = d_1 = 1$.)

Formulas similar to (64) and (65) also hold for higher order determinants. We already saw that if any one row of a determinant is multiplied by a constant λ (say), then the determinant also gets multiplied by the same constant. Along with the additivity of the determinant in each of its rows (in the sense just defined) this is expressed by saying that the determinant is a **linear function** in each row and hence that it is a **multilinear function** of its rows. Because of the interchangeability of the rows and columns, the determinant is also a multilinear function of its columns.

What happens if we split two or more rows (or two or more columns) of a determinant? In that case we go on applying linearity in each row one at a time. For example, if in (64), the second row of D is also split, then this would induce a similar splitting in the second rows of D_1 and D_2. As a result, each would further split into two determinants so that the original determinant D will now be a sum of four determinants. In the full form, we have

$$\begin{aligned} D &= \begin{vmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{vmatrix} \\ &= \begin{vmatrix} a_1 & b_1 \\ c_1 & d_1 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 \\ c_2 & d_2 \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c_1 & d_1 \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c_2 & d_2 \end{vmatrix} \end{aligned} \tag{66}$$

(which is also the result of splitting the two columns of D, each into two).

More generally, if we split any k rows (or columns) of an $n \times n$ determinant then it will be the sum of 2^k determinants of the same order. For $k < n$, the summands corresponding to row-wise splitting will be quite different from those in the column-wise splitting. However, for $k = n$, the 2^n summands are the same for both (as is the case in (66) for $n = 2$).

In applications, when we split a determinant into 4 or 8 or more parts by using multilinearity, it is often unnecessary to show them all. Only a few sample terms need be shown. Generally, this technique is applied when most of these summands vanish for various reasons. The reasons for vanishing may be different for different summands. But if they all vanish then so will their sum, i.e., the original determinant, even though none of these reasons may apply to it. So this splitting technique is an excellent example of the 'divide and rule' strategy. We illustrate it by giving yet another proof of (57).

Call the determinant on the L.H.S. of (57) as D. We begin by expanding each term in the first column using a trigonometric identity for the cosine of the difference of two angles. Then the first column of D becomes

$$\begin{bmatrix} \cos(A-P) \\ \cos(B-P) \\ \cos(C-P) \end{bmatrix} = \begin{bmatrix} \cos A \cos P + \sin A \sin P \\ \cos B \cos P + \sin B \sin P \\ \cos C \cos P + \sin C \sin P \end{bmatrix}$$

which can be split to give

$$\begin{bmatrix} \cos(A-P) \\ \cos(B-P) \\ \cos(C-P) \end{bmatrix} = \begin{bmatrix} \cos A \cos P \\ \cos B \cos P \\ \cos C \cos P \end{bmatrix} + \begin{bmatrix} \sin A \sin P \\ \sin B \sin P \\ \sin C \sin P \end{bmatrix} \tag{67}$$

We split the second and the third columns of D in an analogous manner to get

$$\begin{bmatrix} \cos(A-Q) \\ \cos(B-Q) \\ \cos(C-Q) \end{bmatrix} = \begin{bmatrix} \cos A\cos Q \\ \cos B\cos Q \\ \cos C\cos Q \end{bmatrix} + \begin{bmatrix} \sin A\sin Q \\ \sin B\sin Q \\ \sin C\sin Q \end{bmatrix} \tag{68}$$

and

$$\begin{bmatrix} \cos(A-R) \\ \cos(B-R) \\ \cos(C-R) \end{bmatrix} = \begin{bmatrix} \cos A\cos R \\ \cos B\cos R \\ \cos C\cos R \end{bmatrix} + \begin{bmatrix} \sin A\sin R \\ \sin B\sin R \\ \sin C\sin R \end{bmatrix} \tag{69}$$

Because of these three splittings, D equals the sum of 8 determinants. Each such determinant will have three columns, one each from the right-hand sides of (67) to (69). Note that the entries in the first columns on the R.H.S. of (67) to (69) are proportional to $\cos A, \cos B, \cos C$ respectively while those of the second columns are proportional to $\sin A, \sin B, \sin C$ respectively. Since three choices are to be made, no matter which combination of columns appears in one of these 8 determinants, there will be at least two columns whose entries are proportional either to $\cos A, \cos B, \cos C$ respectively or to $\sin A, \sin B, \sin C$ respectively. In either case, that particular determinant will vanish. But then D vanishes as was to be proved.

Admittedly, this solution is not as short and elegant as the solution based on the product of two determinants. But then, the method of splitting is more general. (In fact, the standard proof of the formula (56) is by splitting each of the columns of the product matrix **AB**. But we shall not go into it.) Expressing a given determinant as a suitable product may not always be possible, and even when possible may not be easy to recognise. The idea of splitting one or more rows or columns, on the other hand, is relatively easy to click. Ultimately, whether it is a problem about determinants or any other problem in mathematics, each technique has its strong and weak points. The art lies in figuring out which technique is more suited for a given problem.

EXERCISES

2.1 The interior angles of a polygon are in arithmetic progression. The smallest angle is 120° and the common difference is 5°. Find the number of sides of the polygon. (1980)

2.2 Assume that $a_1, a_2, \ldots, a_n$ are positive and in an A.P. Prove that

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} = \frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}. \quad (1982)$$

2.3 (a) For any odd integer n, evaluate $n^3-(n-1)^3+\ldots+(-1)^{n-1}1^3$. (1996)

(b) Find the sum of the first n terms of the series (1988)

$$\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+\ldots.$$

(c) The sum of the first n terms of the series

$$1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \ldots$$

is $n(n+1)^2/2$ when n is even. Find the sum when n is odd. (1988)

(d) If the product of four consecutive terms of an arithmetic progression with integer entries is added to the fourth power of the common difference, prove that the sum is the square of an integer. (2000)

2.4 The harmonic mean of two numbers is 4. Their arithmetic mean A and their geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers. (1979)

2.5 Suppose p, q, r are three positive integers. Find a necessary and sufficient condition that there exists a G.P. whose p-th term is 27, q-th term is 8 and r-th term is 12. Find one set of values of p, q, r which satisfies this condition. (1982)

2.6 Find three numbers a, b, c between 2 and 18 such that (i) their sum is 25, (ii) the numbers $2, a, b$ are consecutive terms of an A.P. and (iii) the numbers $b, c, 18$ are the consecutive terms of a G.P. (1983)

2.7 If the first and the $(2n-1)$-st terms of an A.P., a G.P. and an H.P. are equal and their n-th terms are a, b, c respectively, which of the following statements is/are true? (1988)

(A) $a = b = c$ (B) $a \geq b \geq c$ (C) $a + c = b$ (D) $ac - b^2 = 0$.

2.8 Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = \dfrac{3}{2}$, find the value of a. (2002, modified)

2.9 Let $S_1, S_2, \ldots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1}. If the length of a side of S_1 is 10 cm, find the least value of n for which the area of S_n is less than 1 sq. cm. (1999, modified)

2.10 (a) Let a, b be positive real numbers. If a, A_1, A_2, b are in A.P., a, G_1, G_2, b are in G.P. and a, H_1, H_2, c are in H.P., show that (2002)

$$\frac{G_1 G_2}{H_1 H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a+b)(a+2b)}{9ab}.$$

(b) If $x > 1$, $y > 1$, $z > 1$ are in G.P., determine what type of a progression, if any, the numbers $\frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ are in. (1998)

(c) Let $a_1, a_2, \ldots, a_{10}$ be in A.P. and $h_0, h_1, \ldots, h_{10}$ be in H.P. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, find the value of $a_4 h_7$. (1999, modified)

(d) Let x be the arithmetic mean and y, z be the two geometric means between any two positive numbers. Find the numerical value of $\dfrac{y^3 + z^3}{xyz}$. (1997*)

2.11 Let T_r be the r-th term of an A.P. for $r = 1, 2, 3, \ldots$. If for some positive integers m, n, we have $T_m = \dfrac{1}{n}$ and $T_n = \dfrac{1}{m}$, then find which of the following equals T_{mn}. (1998)

(A) $\dfrac{1}{mn}$ (B) $\dfrac{1}{m} + \dfrac{1}{n}$ (C) 1 (D) 0.

2.12 Factorise the following expressions:

(i) $x^3 + y^3 + z^3 - 3xyz$

(ii) $a^5(b-c) + b^5(c-a) + c^5(a-b) + abc(b-c)(c-a)(a-b)$.

2.13 Show that the expression $(x + (x^2-1)^{\frac{1}{2}})^5 + (x - (x^2-1)^{\frac{1}{2}})^5$ is a polynomial in x and determine its degree. (1992, modified)

2.14 If $(m, n) = \dfrac{(1-x^m)(1-x^{m-1})\ldots(1-x^{m-n+1)})}{(1-x)(1-x^2)\ldots(1-x^n)}$, where m and n are positive integers $(n \leq m)$, show that (1978)

$$(m, n+1) = (m-1, n+1) + x^{m-n-1}(m-1, n).$$

2.15 (a) Eliminate θ from $x = a\cos\theta,\ y = b\sin\theta$.

(b) Eliminate t from $x = a\dfrac{e^t + e^{-t}}{2},\ y = b\dfrac{e^t - e^{-t}}{2}$.

(c) Eliminate x, y, z from the equations:

$$x - 2y + z = a,\ x^2 - 2y^2 + z^2 = b,\ x^3 - 2y^3 + z^3 = c \text{ and } \frac{1}{x} - \frac{2}{y} + \frac{1}{z} = 0.$$

2.16 Express $\dfrac{1}{1 - \cos\theta + 2i\sin\theta}$ in the form $X + iY$. (1978)

2.17 If $x + iy = \sqrt{\dfrac{a+ib}{c+id}}$, prove that $(x^2+y^2)^2 = \dfrac{a^2+b^2}{c^2+d^2}$. (1979)

2.18 If $iZ^3 + Z^2 - Z + i = 0$, then show that $|Z| = 1$. (1995)

2.19 Find the value of $\sum_{n=1}^{13}(i^n + i^{n+1})$. (1998)

2.20 If r and θ are the polar coordinates of a (non-zero) complex number z, what are the polar coordinates of $\bar{z}$ and of $\frac{1}{z}$? Give a geometric interpretation of division of complex numbers.

2.21 Using (46) prove that $\tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta}$.

2.22 For any two complex numbers z_1 and z_2 and real numbers a and b, simplify the expression $|az_1 - bz_2|^2 + |bz_1 + az_2|^2$. (1988)

2.23 In the Argand diagram, prove that the various powers of a complex number of the form $e^{i\theta}$, where θ is real always lie on the unit circle, i.e., the circle of radius 1 centred at the origin, and that any two consecutive powers are equally apart from each other. If θ is not a rational multiple of π then show that these powers are all distinct and keep revolving around the circle. What happens if θ is of the form $\frac{p\pi}{q}$, where p, q are integers and $q > 0$?

2.24 Let the complex numbers z_1, z_2, w_1, w_2 represent the distinct points P_1, P_2, Q_1, Q_2 respectively in the Argand diagram. Let L be the straight line through P_1 and P_2 and let L' be the straight line through Q_1 and Q_2. Prove that L and L' are parallel to each other if and only if the ratio $\frac{w_2-w_1}{z_2-z_1}$ is real and that they are perpendicular to each other if and only if this ratio is purely imaginary. Hence obtain necessary and sufficient conditions for four complex numbers to represent the vertices of a parallelogram, a rhombus, a rectangle and a square. (Instead of the ratio $\frac{w_2-w_1}{z_2-z_1}$, one can also take the product $(z_1-z_2)(\bar{w}_1-\bar{w}_2)$ which is a little more convenient.)

2.25 If n is a positive integer, then the $n \times n$ matrix $\mathbf{I}_n = (a_{ij})$ where $a_{ij} = 1$ if $i = j$ and 0 otherwise is called the **identity matrix** of order n. This matrix has 1 at all diagonal positions and 0 everywhere else. For every $m \times n$ matrix $\mathbf{A}$, it is easy to show that $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$. In this sense, the identity matrix plays for matrix multiplication the same role which the number 1 plays for ordinary multiplication of numbers. If $\mathbf{A} = (a_{ij})$ is any matrix and λ is any real number, then by $\lambda\mathbf{A}$ we mean the matrix obtained by multiplying every entry of $\mathbf{A}$ by λ. In other words, $\lambda\mathbf{A} = (\lambda a_{ij})$. Note that this is the same as the product $(\lambda\mathbf{I}_n)(\mathbf{A})$. For this reason, while dealing with square matrices of a fixed order n, it is often convenient to identify $\lambda\mathbf{I}_n$ simply with the number λ.

(a) Determine 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ satisfying the following matrix equations (JAT 1980)

$$4\mathbf{A} + 3\mathbf{B} = \begin{bmatrix} 10 & -5 \\ -25 & 15 \end{bmatrix}, \quad 2\mathbf{A} - \mathbf{B} = \begin{bmatrix} -5 & 30 \\ 20 & -10 \end{bmatrix}.$$

(b) Find the values of f and g satisfying the equation $\mathbf{A}^2 + f\mathbf{A} + g\mathbf{I} = \mathbf{O}$, where $\mathbf{A}$ is a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with $a \neq 0, b \neq 0, c \neq 0, d \neq 0$, $\mathbf{I}$ is the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{O}$ is the matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. (JAT 1980)

2.26 Show that the sum and the product of two 2×2 matrices of the form $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$ where x and y are real number are again matrices of this form.

2.27 Find the values of the following determinants.

(i) $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ (1988) (ii) $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}$ (1998)

(iii) $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$, where x, y, z are positive numbers. (1993)

(iv) $\begin{vmatrix} \frac{1}{a} & \frac{1}{a(a+d)} & \frac{1}{(a+d)(a+2d)} \\ \frac{1}{(a+d)} & \frac{1}{(a+d)(a+2d)} & \frac{1}{(a+2d)(a+3d)} \\ \frac{1}{(a+2d)} & \frac{1}{(a+2d)(a+3d)} & \frac{1}{(a+3d)(a+4d)} \end{vmatrix}$, where $a > 0,\ d > 0$. (1996)

(v) $\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}$ (vi) $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$.

2.28 Consider the set A of all determinants of order 3 with entries 0 and 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of A consisting of all determinants with value -1. Which of the following statements is true? (1981)

(a) C is empty. (b) B has as many elements as C.

(c) $A = B \cup C$. (d) B has twice as many elements as C.

2.29 (a) Let $p\lambda^4 + q\lambda^3 + r\lambda^2 + s\lambda + t = \begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda+3 \\ \lambda+1 & -2\lambda & \lambda-4 \\ \lambda-3 & \lambda+4 & 3\lambda \end{vmatrix}$ be an identity in λ, where p, q, r, s and t are constants. Find the value of t. (1981)

(b) Prove that the determinant $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix}$ is a polynomial of degree 1 in x. (1982)

(c) If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then find the value of $f(100)$. (1999)

2.30 (i) Under which of the following conditions does the determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix}$ vanish? (1997*)

(a) x, y, z are in A.P. (b) x, y, z are in G.P.

(c) x, y, z are in H.P. (d) xy, yz, zx are in A.P.

(ii) Under which of the following conditions does the determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ vanish? (1986)

(A) a, b, c are in A.P. (B) a, b, c are in G.P. (C) a, b, c are in H.P.
(D) α is a root of the equation $ax^2 + bx + c = 0$
(E) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c$.

2.31 Prove that:

(a) $\begin{vmatrix} B & A & B \\ B & B & A \\ A & B & B \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2$, where $A = a^2 + b^2 + c^2$ and $B = ab + bc + ca$.

(b) $\begin{vmatrix} b^2 + c^2 & c^2 & b^2 \\ c^2 & c^2 + a^2 & a^2 \\ b^2 & a^2 & a^2 + b^2 \end{vmatrix} = 4a^2b^2c^2$ both directly (cf. an analogous problem solved in Comment No. 21) and also by expressing the determinant as a product of two determinants. (In such problems, although the entries appear as squares for no apparent reason, they give a hint that the determinant may be factorisable into two determinants with only linear entries.)

(c) $\begin{vmatrix} 2 & c + \frac{1}{c} & b + \frac{1}{b} \\ c + \frac{1}{c} & 2 & a + \frac{1}{a} \\ b + \frac{1}{b} & a + \frac{1}{a} & 2 \end{vmatrix} = \dfrac{2(bc - a)(ca - b)(ab - c)(abc - 1)}{a^2b^2c^2}$.

(d) $\sum_{a=1}^{n} \Delta_a = c$, a constant, where $\Delta_a = \begin{vmatrix} a - 1 & n & 6 \\ (a - 1)^2 & 2n^2 & 4n - 2 \\ (a - 1)^3 & 3n^3 & 3n^2 - 3n \end{vmatrix}$.

(1989)

2.32 Let $'$ denote the transpose of a matrix. If $\mathbf{A}, \mathbf{B}$ are matrices of the same size, prove that $(\alpha\mathbf{A} + \beta\mathbf{B})' = \alpha\mathbf{A}' + \beta\mathbf{B}'$. (Here $\alpha\mathbf{A}$ is the matrix obtained by multiplying every entry of $\mathbf{A}$ by α. $\beta\mathbf{B}$ is defined similarly.) However, prove that even when $\mathbf{AB}$ is defined, $(\mathbf{AB})'$ equals not $\mathbf{A}'\mathbf{B}'$ but $\mathbf{B}'\mathbf{A}'$.

2.33 Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix with complex numbers as entries. Then the **Hermitian adjoint** (or simply the adjoint) of $\mathbf{A}$, often denoted by $\mathbf{A}^*$ is defined as the $n \times m$ whose (i, j)-th entry is the complex conjugate of a_{ji}. In other words, $\mathbf{A}^*$ is obtained from $\mathbf{A}$ by first taking the transpose $\mathbf{A}'$ and then taking the complex conjugates of all its entries. For this reason the Hermitian adjoint is also called the conjugate transpose. State and prove results analogous to those in the last exercise for Hermitian adjoints. Prove also that the determinant of $\mathbf{A}^*$ is the complex conjugate of the determinant of $\mathbf{A}$.

2.34 A square matrix $\mathbf{A} = (a_{ij})$ is called **symmetric** (**skew-symmetric**) if $a_{ij} = a_{ji}$ (respectively, $a_{ij} = -a_{ji}$) for all i, j. Equivalently, $\mathbf{A}$ is symmetric (skew-symmetric) if and only if $\mathbf{A} = \mathbf{A}'$ (respectively, $\mathbf{A} = -\mathbf{A}'$), where $\mathbf{A}'$ is the transpose of $\mathbf{A}$.

(i) Prove that the **diagonal entries** (i.e., entries of the form a_{ii}) of a skew-symmetric matrix are all 0.

(ii) Find all matrices which are both symmetric and skew-symmetric.

(iii) When a complex number $z = x + iy$ is identified with the matrix $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$, what does the symmetry (skew-symmetry) of this matrix mean in terms of the complex number z?

(iv) Prove that if $\mathbf{A}, \mathbf{B}$ are symmetric (skew-symmetric) matrices of the same order then so is $\alpha\mathbf{A} + \beta\mathbf{B}$ for any two numbers α and β.

(v) Prove that for every square matrix $\mathbf{A}$, the matrix $\mathbf{A} + \mathbf{A}'$ is always symmetric while the matrix $\mathbf{A} - \mathbf{A}'$ is always skew-symmetric. Hence show that every square matrix can be expressed uniquely as the sum of a symmetric and a non-symmetric matrix.

(vi) Suppose $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order. What can be said about the symmetry (skew-symmetry) of the product matrix $\mathbf{AB}$ in terms of the symmetry and skew symmetry of $\mathbf{A}$ and $\mathbf{B}$?

2.35 A square matrix $\mathbf{A}$ is called **Hermitian** (or **skew-Hermitian**) if $\mathbf{A} = \mathbf{A}^*$ (respectively, $\mathbf{A} = -\mathbf{A}^*$). State and prove the analogues of the various parts of the last exercise (except part (iii) of it) for Hermitian and skew-Hermitian matrices. For example, begin by showing that the diagonal entries of a Hermitian matrix are all real while those of a skew-Hermitian matrix are purely imaginary.

2.36 The **trace** of an $n \times n$ matrix $\mathbf{A} = (a_{ij})$, denoted by $\mathrm{tr}(\mathbf{A})$, is defined as the sum of its diagonal elements, i.e., the sum $\sum_{i=1}^{n} a_{ii}$. Prove that if $\mathbf{A}, \mathbf{B}$ are square matrices of the same order, then $\mathrm{tr}(\mathbf{A} + \mathbf{B}) = \mathrm{tr}(\mathbf{A}) + \mathrm{tr}(\mathbf{B})$ but that, in general, $\mathrm{tr}(\mathbf{AB}) \neq \mathrm{tr}(\mathbf{A})\ \mathrm{tr}(\mathbf{B})$. (In other words, the trace is additive but not multiplicative, exactly the opposite behaviour of the determinant.) Prove, however, that $\mathbf{AB}$ and $\mathbf{BA}$ have the same trace.

2.37 Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix. Then the determinant

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} - \lambda \end{vmatrix}$$

is a polynomial of degree n in the variable λ. This polynomial is called the **characteristic polynomial** of the matrix A and its roots (which could possibly be complex) are called the characteristic roots of $\mathbf{A}$. Prove that $\mathbf{A}$ has 0 as a characteristic root if and only if the determinant of $\mathbf{A}$ is 0. Prove further that the characteristic roots of $\mathbf{A}'$, the transpose of $\mathbf{A}$, are the same as those of $\mathbf{A}$ while the characteristic roots of $\mathbf{A}^*$, the adjoint of $\mathbf{A}$, are the complex conjugates of those of $\mathbf{A}$.

Chapter 3

THEORY OF EQUATIONS

Equations, like counting, are our childhood friends. When we are given that Anil had 7 mangos and ate 4 of them and are asked to find out how many mangos he still has, it is a problem of subtraction. But if we are given that Anil got some mangos from his father and that after getting 4 more mangos from his mother, he had 7 mangos in all and are asked to find out how many mangos he got from his father, this is *not* a problem of subtraction. Subtraction is a tool to solve it. But the problem itself is that of solving the equation $x + 4 = 7$. This may seem like an oversimplification. But in essence this is how algebra (and, indeed, many other things in mathematics) is applied to real life problems. You assign variables to the 'unknown' quantities. (These variables are often themselves called 'unknowns'.) Then you convert the data of the problem into a system of equations in these variables and solve this system to determine the values of these quantities. That is why the theory of equations is important practically. From a theoretical point of view, it is important because of the variety of techniques that are needed to determine, first of all, if the given system has a solution and if so, to obtain it exactly, and when that fails, approximately.

In this chapter we consider algebraic equations where the expressions involved are polynomials in the unknowns. A few equations involving logarithms and exponentials which are reducible to these will also be considered. Trigonometric equations will be taken up in Chapter 10. We first consider polynomial equations in a single variable. Except when the degree is 1 or 2, these equations are not easy to solve. But a lot can be said about their roots even without solving them. We shall also consider systems of linear equations in several variables. Matrices and determinants play a crucial role here. However, as our treatment of matrices is sketchy and optional, we shall deal in detail only with equations in two or three variables, which can be handled directly without matrices.

In the JEE, questions on quadratic equations and systems of linear equations in 3 unknowns are common, sometimes combined with other topics.

Main Problem: The sum of the cubes of roots of the equation $x^4 - 2x^3 + x^2 - 5x + 1 = 0$ is ..... .

First Hint: Call the roots as α, β, γ and δ. Find the values of $\alpha+\beta+\gamma+\delta$, $\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\gamma\delta+\beta\delta$, $\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta$ and $\alpha\beta\gamma\delta$. (It is customary to denote such sums by the compact notations $\sum\alpha, \sum\alpha\beta, \sum\alpha\beta\gamma$ and $\sum\alpha\beta\gamma\delta$. In each case only one term of the summation is shown and the summation is of all terms of a similar form. For example, the second sum $\sum\alpha\beta$ consists of all products of distinct roots, in the present case there being $\binom{4}{2} = 6$ such products, since there are four roots. The last sum $\sum\alpha\beta\gamma\delta$ has only one term and so the summation sign is really redundant. But it is retained for notational uniformity.)

Second Hint: Express $\sum\alpha^3$ (i.e., $\alpha^3+\beta^3+\gamma^3+\delta^3$) in terms of the sums above.

Solution: If $\alpha, \beta, \gamma, \delta$ are the (not necessarily distinct) roots of the polynomial $x^4 - 2x^3 + x^2 - 5x + 1$ then it factorises as $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$. Expanding the product and comparing the coefficients of the like powers gives the following equations:

$$\alpha+\beta+\gamma+\delta = \sum\alpha = 2 \quad (1)$$

$$\sum\alpha\beta = 1 \quad (2)$$

$$\sum\alpha\beta\gamma = 5 \quad (3)$$

$$\text{and } \alpha\beta\gamma\delta = 1 \quad (4)$$

We are interested in $\sum\alpha^3$, i.e., in $\alpha^3+\beta^3+\gamma^3+\delta^3$. We can get this from $(\alpha+\beta+\gamma+\delta)^3$. But the expansion will contain 60 more terms. It is not hard to classify them. Each of the 64 terms is the product of three factors, each of which is α, β, γ or δ. The four terms in which all the three factors are equal give $\sum\alpha^3$. A typical term in which all the three factors are distinct is of the form $\alpha\beta\gamma$. This term appears 6 times, corresponding to the 6 permutations of three symbols. In all there are 24 such terms and together they add up to $6\sum\alpha\beta\gamma$. We still have 36 more terms. They are of the form $\alpha^2\beta$, where two factors are equal and the third one is different. Each such term appears 3 times. Summing up we get

$$(\alpha+\beta+\gamma+\delta)^3 = \sum\alpha^3 + 6\sum\alpha\beta\gamma + 3\sum\alpha^2\beta \quad (5)$$

Here $\sum\alpha^2\beta$ is a sum of 12 terms. If we add $\sum\alpha^3$ to it we get $(\sum\alpha)(\sum\alpha^2)$. The second factor $\sum\alpha^2$ can be written as $(\sum\alpha)^2 - 2(\sum\alpha\beta)$. Hence, we get

$$\sum\alpha^2\beta = (\sum\alpha)[(\sum\alpha)^2 - 2\sum\alpha\beta] - \sum\alpha^3 \quad (6)$$

Substituting (6) into (5) and taking all the cubes on one side we get

$$2\sum\alpha^3 = 2(\sum\alpha)^3 + 6\sum\alpha\beta\gamma - 6(\sum\alpha)(\sum\alpha\beta) \quad (7)$$

Substituting the values of $\sum\alpha, \sum\alpha\beta$ and $\sum\alpha\beta\gamma$ from (1), (2) and (3), we finally get the value of $\sum\alpha^3$ as $\frac{1}{2}(16+30-12)$, i.e., as 17.

Comment No. 1:

The leading (i.e., the top degree) coefficient in the given polynomial was 1. (Such polynomials are called **monic polynomials.**) In general it could be some other non-zero number. In that case before applying the method we would first have to divide throughout by the leading coefficient. This simple operation does not change the roots[1] of the polynomial. In general, suppose $p(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \ldots + a_2x^2 + a_1x + a_0$ (where it is tacitly assumed that $a_n \neq 0$), is a polynomial of degree n. Here the variable x as well as the coefficients may be real or complex numbers. It is a well-known result (popularly called the **remainder theorem**) that if α is any real or complex number then the remainder left when $p(x)$ is divided by $x-\alpha$ is simply $p(\alpha)$, in other words, we can write $p(x)$ as $q(x)(x-\alpha)+p(\alpha)$, where $q(x)$ is a polynomial of degree $n-1$. In particular it follows that α is a root of $p(x)$ if and only if $(x-\alpha)$ is a factor of $p(x)$. Sometimes a higher power of $(x-\alpha)$ may also divide $p(x)$. Suppose r is the largest integer such that $(x-\alpha)^r$ divides $p(x)$. Then we say that α is a **multiple root** of **order** (or **multiplicity**) r. Roots of multiplicity 1,2 and 3 are also called **simple, double** and **triple** roots respectively. For example, $p(x) = x^6 - 4x^5 + 5x^4 - 2x^3$ has 0 as a triple root, 1 as a double root and 2 as a simple root as can be seen by direct substitution and division. The complete factorisation of $p(x)$ comes out to be $x^3(x-1)^2(x-2)$. In counting or listing the roots of a polynomial, it is customary to count or to list each root as many times as its multiplicity. For example, the roots of the polynomial just considered are $0, 0, 0, 1, 1, 2$.

Comment No. 2:

Not every polynomial has a root. For example, as a real polynomial. the polynomial x^2+1 has no roots. But as a complex polynomial, it has two roots, viz., i and $-i$ and factors as $(x-i)(x+i)$. In fact, this is true of every polynomial, say $p(x)$ of degree ≥ 1. Regarded as a complex polynomial, it always has at least one root, say α. This highly non-trivial result is known as the **fundamental theorem of algebra**. By the remainder theorem, we can now write $p(x) = (x-\alpha)q(x)$, where $q(x)$ is a polynomial of one degree lower than $p(x)$. We can apply the fundamental theorem now to $q(x)$ and get a further factorisation. If we keep going, ultimately we get a complete factorisation of $p(x)$ into linear (i.e., first degree) factors. The solution to the Main Problem begins with this idea.

[1] Traditionally, one says roots of a polynomial equation, say, $p(x) = 0$ rather than the roots of the polynomial itself. More generally, we can consider an equation of the form $f(x) = 0$, where $f(x)$ is any function of one variable, such as $\sin x$ or $e^x - x^2$. Then the solutions of the equation $f(x) = 0$ are called the **zeros** of $f(x)$. However, we shall not be so fussy and use the phrase 'root of a polynomial'to mean a zero of it. When a system of equations is involved, neither term is used. Instead, the values which satisfy the system are simply called **solutions**, which is a more general (and a safer) term to use.

The fundamental theorem of algebra has an interesting consequence for polynomials with real coefficients. Suppose $p(x)$ is such a polynomial. Treating it as a complex polynomial, we can factor it completely into factors of the form $(x-a-ib)$, where a and b are real and $a+ib$ is a complex root of $p(x)$. If $b=0$ then $a+ib=a$ is a real root of $p(x)$ and hence $(x-a)$ is a (real) linear factor of $p(x)$. Suppose, however, that $b \neq 0$. In that case, since the coefficients of $p(x)$ are all real, the complex conjugate of $a+ib$, i.e., $a-ib$ is also a root of $p(x)$, by Theorem 1 of the last chapter. So by the remainder theorem, $(x-a+ib)$ is also a factor of $p(x)$. Multiplying these two factors, we get $(x-a)^2+b^2$ as a real quadratic factor of $p(x)$. Moreover, this factor is an irreducible quadratic in the sense that it does not factor as the product of two real linear factors. Thus, as a consequence of the fundamental theorem of algebra, we see that a real polynomial can be completely factored into real factors some of which are linear and some are irreducible quadratic. If l and q denote the numbers of these factors respectively, then clearly, $l+2q=n$, where n is the degree of $p(x)$. If n is odd then l must be at least one. We therefore have the following theorem. (See the end of Comment No. 4 for a similar result about real polynomials of even degrees.)

Theorem 1: Every real polynomial of odd degree has at least one real root (and more precisely, an odd number of such roots).

Comment No. 3:

It is tempting to try to solve the Main Problem by actually identifying the four roots of $p(x) = x^4 - 2x^3 + x^2 - 5x + 1$. In fact, once we identify the roots α, β, γ and δ, we can directly evaluate not only the sum of their cubes as asked, but any expression involving them. Unfortunately, this approach rarely works. The fundamental theorem of algebra guarantees us the existence of n roots (counted according to multiplicities) of a polynomial of degree n. But it does not tell us how to find them. For $n=1$, there is only one root and it is, of course, trivial to find it. For $n=2$, there is the well-known quadratic[2] formula. For $n=3$ and 4, there are similar formulas to find the roots but they are far too complicated and are rarely used. For $n \geq 5$, the situation is even worse. Not only is no formula for the roots of a general polynomial of degree 5 or above known, it has been proved that *no such formula exists.* Not for lack of efforts, but because there is something inherently impossible about such a formula − something like making two parallel lines meet[3]. A particular polynomial equation can, of course, be sometimes solved if we can identify at least some of its roots by inspection or otherwise. For then, we can take the corresponding factors out.

[2]The terms **quadratic**, **cubic**, **quartic** and **quintic** are used for polynomial equations of degree 2,3,4 and 5 respectively.

[3]The point is actually subtler. In the case of parallel lines, the point of intersection does not exist. In the case of a polynomial of degree 5 or above, it is not that the roots do not exist. They certainly do. But a formula expressing them in terms of the coefficients of the polynomial does not exist. A better analogy would be the trisection of an angle. Given any angle θ, an angle which equals a third of it certainly exists. But we may not always be able to find it with a ruler and compass. We shall take this up again in Chapter 11.

The quotient will be a polynomial of a lower degree and hence will be more manageable. Suppose, for example, that $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ is a cubic. If we can somehow identify some root, say α, then we can write $p(x)$ as $(x-\alpha)q(x)$, where $q(x)$ is a quadratic and hence can be solved by the quadratic formula. Similarly, the polynomial equation $x^6 - 4x^5 + 5x^4 - 2x^3 = 0$ can be solved completely by factorising the L.H.S. as $x^3(x^3-4x^2+5x-2)$. The second factor vanishes if we put $x = 1$. So $x - 1$ is yet another factor. Taking it out, $x^6-4x^5+5x^4-2x^3 = x^3(x-1)(x^2-3x+2) = x^3(x-1)(x-1)(x-2)$, giving all the roots. Sometimes we may not be so lucky. That is, after identifying some roots, we may get stuck with a factor which has no obvious roots. Even such partial success can be useful sometimes.

Sometimes a polynomial equation can be solved by converting it to a simpler equation by a suitable substitution. Take, for example, the equation

$$a_4x^4 + a_2x^2 + a_0 = 0 \tag{8}$$

Here there are no odd degree terms and so if we put $x^2 = y$, then (8) becomes

$$a_4y^2 + a_2y + a_0 = 0 \tag{9}$$

which is a quadratic in y. (Here, y itself is a quadratic in x and so (8) is said to be a **biquadratic** in x.) From the quadratic formula, (9) gives

$$y = \frac{-a_2 \pm \sqrt{a_2^2 - 4a_4a_0}}{2} \tag{10}$$

By taking square roots of each of these two values of y, we get the four roots of (8). Note however, that if we are looking for real roots of (8), then we have to analyse further. First of all, the roots given by (10) have to be real (the condition for which is that the discriminant, i.e., the expression $a_2^2 - 4a_4a_0$ be non-negative). Even when this condition is satisfied, we further have to worry which of the two values of y given by (10) are non-negative. Since $y = x^2$, a negative value of y does not correspond to any (real) value of x and hence does not give rise to any solution of (8). This scruple has always to be followed whenever an equation (or more generally, a system of equations) is reduced to another. Every solution of the original equation (or system) will automatically be a solution of the latter. The converse will be true if the reduction is of a reversible type. For example, if y is obtained from x by a substitution like $y = 2x-3$, then x is uniquely determined by y as $x = \frac{y+3}{2}$. But if the reduction is not of the reversible type, then we have to carefully see exactly which solutions of the new equation come from solutions of the original equation. In the present problem, the reduction of (8) to (9) is reversible if both are thought of as complex equations but not so if they are only thought of as real equations because x^2 can be real without x being so.

Comment No. 4:

So, with polynomial equations of degree greater than 2, whatever we want to prove has to be usually done without solving the equation explicitly. Methods

from calculus help in some problems. It can, for example be shown that between any two real roots of a polynomial $p(x)$, there must lie at least one root of its derivative $p'(x)$, which is a polynomial of degree one less than $p(x)$. Similarly, Theorem 1 which was derived above from the fundamental theorem of algebra, can be proved using a well-known property of continuous functions. However, in this chapter we confine ourselves to methods which are purely algebraic, i.e., methods which do not use derivatives, integrals and, in general, anything based on the concept of a limit. The solution to the Main Problem was based on the fact that even if we do not know the roots of a given polynomial equation individually, we can easily tell their sum, the sum of their products taken two at a time and so on. The Main Problem dealt with a quartic. More generally, suppose, $p(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n$ is a monic polynomial[4] of degree n. Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of $p(x)$, counted according to their multiplicities. Then $p(x)$ also equals the product $(x - \alpha_1)(x - \alpha_2)\ldots(x - \alpha_n)$ (which, in analogy with the sigma notation for summation, is also denoted by the 'pi' notation as $\prod_{k=1}^{n}(x - \alpha_k)$). So exactly as in the solution to the Main Problem, equating the coefficients of like powers we get

$$\left.\begin{array}{rcl} \sum_{i=1}^{n} \alpha_i = \sum \alpha_1 & = & -a_1 \\ \sum_{1 \le i < j \le n} \alpha_i\alpha_j = \sum \alpha_1\alpha_2 & = & a_2 \\ \vdots & & \vdots \\ \sum \alpha_1\alpha_2\ldots\alpha_r & = & (-1)^r a_r \\ \vdots & & \vdots \\ \alpha_1\alpha_2\ldots\alpha_n & = & (-1)^n a_n \end{array}\right\} \tag{11}$$

It is customary to denote the sums on the left sides of these equations by $\sigma_1, \sigma_2, \ldots, \sigma_n$ respectively. Then (11) takes the more compact form

$$\sigma_r = (-1)^r a_r \tag{12}$$

for $r = 1, 2, \ldots, n$. Note that each σ_r is a **symmetric function** of $\alpha_1, \alpha_2, \ldots, \alpha_n$ in the sense that it remains unchanged if we interchange α_i and α_j for any two indices i and j (between 1 and n, of course). There are many other symmetric

[4]Note that the notation has been changed slightly. In the former notation, a typical polynomial of degree n was denoted by $a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$. This is keeping in line with the notation for an infinite power series of the form $a_0 + a_1x + a_2x^2 + \ldots + a_nx^n + a_{n+1}x^{n+1} + \ldots$, where we have no choice but to write the powers of x in an ascending order. In the case of a polynomial we can write them either in the ascending or in the descending order. The latter is more traditional. And once we subscribe to it, it is more natural that the subscripts of the coefficients should appear in the ascending order too. As remarked before, as far as the roots are concerned, we may always take the polynomial to be monic. However, if we do want to consider a polynomial which is not monic, we take it as $a_0x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n$. For small values of n, say $n = 2$ or 3, it is more customary to denote the polynomials by $ax^2 + bx + c$ or $ax^3 + bx^2 + cx + d$ instead of $a_0x^2 + a_1x + a_2$ and $a_0x^3 + a_1x^2 + a_2x + a_3$ respectively.

functions besides these. For example, the function $\alpha_1^m+\alpha_2^m+\ldots+\alpha_n^m$, where m is any positive integer is a symmetric function of the α_i's. A little less obviously, the function $\sum \alpha_1^2\alpha_2$ which consists of the sum of all products of the form $\alpha_i^2\alpha_j$, where i and j are distinct indices is also a symmetric function. The function $\alpha_1\alpha_2+\alpha_3^2+\alpha_4^2+\ldots+\alpha_n^2$, on the other hand is not a symmetric function. (We may say it is symmetric w.r.t. the first two variables, and also symmetric between any two of the last $n-2$ variables. But as a function of all the n variables it is not symmetric.) For $n=3$, the function $\alpha_1^2\alpha_2+\alpha_2^2\alpha_3+\alpha_3^2\alpha_1$ is not symmetric. However, it is **cyclically symmetric** because it remains unchanged if we replace $\alpha_1,\alpha_2,\alpha_3$ by $\alpha_2,\alpha_3,\alpha_1$ respectively. Cyclically symmetric functions of three variables arise naturally in trigonometry as is to be expected.

Although there are many symmetric functions of the roots, the functions $\sigma_1,\sigma_2,\ldots,\sigma_n$ considered above have a very interesting property. It can be shown (although the proof is far from easy), that every symmetric function of the roots $\alpha_1,\alpha_2,\ldots,\alpha_n$ can be expressed in terms of the functions $\sigma_1,\sigma_2,\ldots,\sigma_n$, and hence, because of (12), in terms of the coefficients $a_1,a_2,\ldots,a_n$ of the polynomial $p(x)$. In other words, the symmetric functions $\sigma_1,\sigma_2,\ldots,\sigma_n$ are, in a sense, building blocks for all symmetric functions. For this reason they are called the **elementary symmetric functions**. In the Main Problem, $p(x)$ was a polynomial of degree 4 and the essential part of the problem was to express the symmetric function $\alpha_1^3+\alpha_2^3+\alpha_3^3+\alpha_4^3$ in terms of the elementary symmetric functions $\sigma_1,\sigma_2,\sigma_3,\sigma_4$. Essentailly the same solution goes for any n.

As an application of the last equation in (11), suppose $p(x)=x^n+a_1x^{n-1}+\ldots+a_{n-1}x+a_n$ is a monic polynomial with real coefficients with n even and $a_n<0$. Then the product of all roots is negative. Since the complex roots must occur in conjugate pairs and the product of any such pair is positive, we see that $p(x)$ has at least two real roots. Moreover they must be of opposite signs.

Comment No. 5:

The elementary symmetric functions are important as building blocks for all symmetric functions. But in applications, the functions that appear more frequently are the power sum functions. Specifically, these are the functions $s_1,s_2,s_3,\ldots$, where for each positive integer m, s_m is the sum of the m-th powers of the roots of $p(x)=x^n+a_1x^{n-1}+\ldots+a_{n-1}x+a_n$, i.e.,

$$s_m=\sum_{i=1}^{n}\alpha_i^m=\alpha_1^m+\alpha_2^m+\ldots+\alpha_n^m \tag{13}$$

for $m=1,2,\ldots$. We may also set $s_0=n$ with the understanding that $\alpha_i^0=1$ for every $i=1,2,\ldots,n$. But this sometimes leads to problems if one of the roots is 0. In this new notation, the Main Problem amounts to expressing s_3 in terms of the elementary symmetric functions. In practice sometimes a situation arises where the quantities $s_1,s_2,\ldots$ can be found rather easily and from them we have to determine the polynomial $p(x)$. For this we need to express the σ_i's in terms of the s_j's rather than the other way round. The following equations, called **Newton's identities** relate the s_i's with the σ_j's and hence with the

a_j's. The proof of (14) is not easy. But (15) can be proved using the fact that for $k > n$, $\alpha_i^k + a_1\alpha_i^{k-1} + \ldots + a_{n-1}\alpha_i^{k-n+1} + a_n\alpha_i^{k-n} = \alpha_i^{k-n}p(\alpha_i) = 0$, for every $i = 1, 2, \ldots, n$ and adding these n equations.

$$s_k + a_1 s_{k-1} + a_2 s_{k-2} + \ldots + a_{k-1}s_1 + ka_k = 0 \text{ (if } 1 \leq k \leq n) \quad (14)$$

$$s_k + a_1 s_{k-1} + a_2 s_{k-2} + \ldots + a_n s_{k-n} = 0 \text{ (if } k > n) \quad (15)$$

(14) is really a system of n equations starting from $s_1 + a_1 = 0$, $s_2 + a_1 s_1 + 2a_2 = 0$, and so on. It can be used both ways. That is, if the polynomial $p(x)$ is known, i.e., if $a_1, a_2, \ldots, a_n$ are given then from them we can determine, in succession, $s_1, s_2, \ldots, s_n$. (In essence, Equation (7) in the solution to the Main Problem is the case $k = 3$ of (14).) Conversely, if $s_1, s_2, \ldots, s_n$ are known (or can be found from the data, as is the case in some applications), then one-by-one we get the values of $a_1, a_2, \ldots, a_n$.

Comment No. 6:

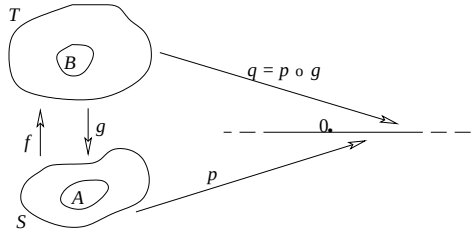
Newton's identities cannot be applied to negative exponents, that is, they do not give the values of expressions of the form $\alpha_1^{-m} + \alpha_2^{-m} + \ldots + \alpha_n^{-m}$ if m is a positive integer. If one of the roots is 0 then, of course, an expression like this is undefined. So we assume that none of the roots is 0, which is equivalent to saying that $a_n \neq 0$. In order to get the value of $\alpha_1^{-m} + \alpha_2^{-m} + \ldots + \alpha_n^{-m}$ in such a case, we first construct a polynomial whose roots are the reciprocals of the roots of the original polynomial $p(x) = x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n$. Let us put $y = \frac{1}{x}$. Then $x = \frac{1}{y}$ and so $p(x) = p(\frac{1}{y}) = \frac{1}{y^n} + a_1\frac{1}{y^{n-1}} + \ldots + a_{n-1}\frac{1}{y} + a_n$. Call this expression as $q(y)$. Note that $q(y)$ is not a polynomial in y. Still it is a well-defined function for all non-zero values of y. It is clear that $q(y) = 0$ if and only if $p(\frac{1}{y}) = 0$ which is the case if and only if $\frac{1}{y} = \alpha_i$ for some $i = 1, 2, \ldots, n$. In other words, the zeros of the function $q(y)$ are precisely the reciprocals of the roots of the polynomial $p(x)$. We can rewrite $q(y)$ as $\frac{1}{y^n}r(y)$, where $r(y) = 1 + a_1y + a_2y^2 + \ldots + a_{n-1}y^{n-1} + a_ny^n$. Note that $r(y)$ is a polynomial of degree n. Its coefficients are the same as those of $p(x)$ but in a reverse order. Since the factor $\frac{1}{y^n}$ is never 0, the zeros of the polynomial $r(y)$ are the same as the zeros of the function $q(y)$. The latter we already know. Putting it all together, we have finally found a polynomial, viz., $r(y) = 1 + a_1y + \ldots + a_ny^n$ whose roots are precisely the reciprocals of the roots of the polynomial $p(x) = x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n$.

It is now easy to find an expression for $\alpha_1^{-m} + \alpha_2^{-m} + \ldots + \alpha_n^{-m}$, where m is a positive integer. Denote this sum by t_m. Then t_m is precisely the sum of the m-th powers of the roots of the polynomial $r(y)$ and so identities analogous to (14) and (15) can be written down to determine the t_m's successively. However, in order to do so we must first convert $r(y)$ to a monic polynomial by dividing it by the leading coefficient a_n (which is assumed to be non-zero). So, instead of $r(y)$, we consider the polynomial $y^n + b_1y^{n-1} + b_2y^{n-2} + \ldots + b_{n-1}y + b_n$, where $b_i = \dfrac{a_{n-i}}{a_n}$ for $i = 1, 2, \ldots, n-1$ and $b_n = \dfrac{1}{a_n}$. With this spadework done we now merely replace a_i's with b_i's and s_j's with t_j's in Equations (14) and (15) to

get a system of equations from which we can determine $t_1, t_2, \ldots$ in succession. A couple of problems based on this will be given in Exercise (3.5).

Comment No. 7:

The method given above for constructing an equation whose roots are the reciprocals of the roots of a given equation can be easily generalised. In fact the main idea is purely set theoretic and can be explained with the help of the diagram below.



Theorem 2 : Suppose S is a set and p is a real or complex valued function defined on S. Let A be the zero set of p, i.e., $A = \{x \in S : p(x) = 0\}$. Let T be another set and assume $f : S \longrightarrow T$ is a bijection with the inverse function $g : T \longrightarrow S$. Let q be the composite function $p \circ g$, i.e., for every $y \in T$, we have $q(y) = p(g(y))$. Then q is a real (or complex) valued function on the set T. Let B be the zero set of q, i.e., $B = \{y \in T : q(y) = 0\}$. Then the function f maps the set A (i.e. the zero set of p) bijectively to the set B.

The proof is a triviality. Let $x \in A$. Then $p(x) = 0$. Let $y = f(x)$. Then $g(y) = x$, g being the inverse function of f. So, $q(y) = p(g(y)) = p(x) = 0$ which shows that $y \in B$. Conversely, suppose $y \in B$. Let $x = g(y)$. Then $p(x) = p(g(y)) = q(y) = 0$ which shows that $x \in A$. Moreover, since $x = g(y)$, we must have $y = f(x)$ as f and g are inverse functions of each other. In effect we have thus shown that f maps the subset A of S onto the subset B of T. Moreover, it does so bijectively since f is a bijection (and hence injective).

In the situation considered in the last comment, both S and T were the sets of non-zero complex numbers and the function $f : S \longrightarrow T$ was the reciprocal function, $f(x) = \frac{1}{x}$. The inverse function g was also the reciprocal function and hence equal to f. But this need not be so every time. The function p was defined by a polynomial and q was the function defined by $q(y) = p(\frac{1}{y})$.

In general we are free to choose the bijection in any manner we like. Suppose for example, that we take $y = f(x) = \frac{3x-1}{2}$. Then the inverse function $x = g(y)$ is given by $x = g(y) = \frac{2y+1}{3}$. Or we can let $S = [-\frac{\pi}{2}, \frac{\pi}{2}]$, $T = [-1, 1]$ and $f(x) = \sin x$. Then the inverse function is given by $g(y) = \sin^{-1} y$. Theorem 2 then implies that if $p(x)$ is a polynomial all of whose roots, say $\alpha_1, \alpha_2, \ldots, \alpha_n$ lie in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$, then the zeros of the function $q(y) = p(\sin^{-1} y)$ are precisely the real numbers $\sin\alpha_1, \sin\alpha_2, \ldots, \sin\alpha_n$. After a little practice one can often write down the function $q(y)$ without explicitly writing down the inverse function g. Suppose, for example, that $p(x) = a_0x^n + a_1x^{n-1} + \ldots +$

$a_{n-1}x + a_n$ is a polynomial with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ and c is some constant. Then the polynomial whose roots are $\alpha_1 + c, \alpha_2 + c, \ldots, \alpha_n + c$ is simply $q(y) = p(y - c) = a_0(y - c)^n + a_1(y - c)^{n-1} + \ldots + a_{n-1}(y - c) + a_n$. ($q(y)$ can be written in the standard form $b_0y^n + b_1y^{n-1} + \ldots + b_{n-1}y + b_n$ by expanding the various powers of $(y - c)$. But for many purposes this may not be necessary. For example, we may be interested only in the coefficients of y^{n-1} and y^n. In that case, there is no need to expand $(y-c)^k$ for $k \leq n-2$. It suffices to expand only $(y - c)^{n-1}$ and $(y - c)^n$ and that too only partially till we get the terms involving y^{n-1} and y^n. See Exercise (7.4) for an interesting application of this procedure.)

Comment No. 8:

Having discussed general polynomial equations, we now specialise to the case of quadratics, i.e., polynomial equations of degree 2. Here we, of course, have a handy formula for the roots. Many times, however, we can do without it and sometimes it is not of much help anyway. As an illustration of the first kind, we do the following problem both with and without using the quadratic formula.

If α, β are the roots of $ax^2 + bx + c = 0$ and for some constant δ, $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0$, prove that (JEE 2000)

$$\frac{b^2 - 4ac}{a^2} = \frac{B^2 - 4AC}{A^2}. \tag{16}$$

The roots of the first quadratic are $\dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and we may suppose, without loss of generality that $\alpha \geq \beta$. In that case,

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}. \tag{17}$$

We are given that $\alpha + \delta$ and $\beta + \delta$ are the roots of the second quadratic. Since $\alpha + \delta \geq \beta + \delta$, we get

$$\alpha + \delta = \frac{-B + \sqrt{B^2 - 4AC}}{2A} \quad \text{and} \quad \beta + \delta = \frac{-B - \sqrt{B^2 - 4AC}}{2A}. \tag{18}$$

Eliminating α, β and δ among these four equations will give us (16). The elimination is very easy. Subtracting the equations in (17) from each other gives $\alpha - \beta = \frac{\sqrt{b^2-4ac}}{a}$. Similarly, from (18) we get $\alpha - \beta = \frac{\sqrt{B^2-4AC}}{A}$. Equating these two expressions and squaring we get (16).

Simple as this solution is, it is open to an objection. It presupposes that the roots are real. Without that it would be meaningless to talk of which root is greater. What if the roots are complex? In that case we will not be able to compare the roots. Of course, we can still call one of them as α and the other as β so that (17) continues to hold. But we can no longer conclude that (18) will hold. All we can say is that one of the two values of $\frac{-A \pm \sqrt{B^2-4AC}}{2A}$ equals $\alpha + \delta$ and the other equals $\beta + \delta$. But we cannot be sure which one is which. Fortunately, the solution can still be salvaged. Regardless of which possibility

holds, we have $\alpha - \beta = \pm\frac{\sqrt{B^2-4AC}}{A}$ and hence $\frac{\sqrt{b^2-4ac}}{a} = \pm\frac{\sqrt{B^2-4AC}}{A}$. Squaring both the sides still gives the result.

It is clear that the solution essentially hinges upon the square of the difference of the two roots. But in order to know the difference of two things, it is not necessary to know them individually. So the use of the quadratic formula can probably be dispensed with. This indeed is the case. We can write $(\alpha - \beta)^2$ in terms of the elementary symmetric functions as $(\alpha + \beta)^2 - 4\alpha\beta$. Since $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$, we get

$$(\alpha - \beta)^2 = (-\frac{b}{a})^2 - 4\frac{c}{a} = \frac{b^2 - 4ac}{a^2} \tag{19}$$

We now apply the same reasoning to the second quadratic $Ax^2 + Bx + C = 0$ whose roots are given to be $\alpha + \delta$ and $\beta + \delta$. We then get

$$((\alpha + \delta) - (\beta + \delta))^2 = \frac{B^2 - 4AC}{A^2} \tag{20}$$

Since $(\alpha + \delta) - (\beta + \delta) = \alpha - \beta$, (19) and (20) together imply (16) as desired.

There is yet another way to do the problem. α and β are the roots of $ax^2 + bx + c = 0$. So by Theorem 2, the quadratic whose roots are $\alpha + \delta$ and $\beta + \delta$ is $a(x-\delta)^2 + b(x-\delta) + c = 0$. But these are also the roots of the quadratic $Ax^2 + Bx + C = 0$. Hence the coefficients of the corresponding powers of x in the two quadratics must be proportional, i.e.,

$$\frac{a}{A} = \frac{b - 2a\delta}{B} = \frac{c + a\delta^2 - b\delta}{C} \tag{21}$$

Now we eliminate δ. Equality of the first two ratios gives $\delta = \dfrac{bA - aB}{2aA}$. If we substitute this in the third ratio and equate it with the first ratio, viz. $\frac{a}{A}$, we get (16) after a little computation. The work involved in this method is rather laborious as compared with the earlier two methods. This is so because the quantities $b^2 - 4ac$ and $B^2 - 4AC$ have a special significance for the respective quadratics. The earlier solutions made a crucial use of it while the third method does not.

Here is another problem where it is not a good idea to apply the quadratic formula. Suppose it is given that a and b are the roots of $x^2 + px + 1 = 0$ and c and d are the roots of the quadratic $x^2 + qx + 1 = 0$ and we have to show that

$$q^2 - p^2 = (a - c)(b - c)(a + d)(b + d). \qquad \text{(JEE 1978)} \tag{22}$$

We can solve the first quadratic and write $a = \frac{-p+\sqrt{p^2-4}}{2}$ and $b = \frac{-p-\sqrt{p^2-4}}{2}$. Of course we can never know which root is which but that does not matter since the R.H.S. of (22) is unchanged if a and b are interchanged. Similarly we can get expressions for c and d. We can then substitute these values in the R.H.S. of (22) and try to simplify it to the L.H.S. But doing so would be horribly complicated since so many messy radicals will be involved. There is a much

better way out if we observe that we are not interested in a and b individually but only in the products $(a-c)(b-c)$ and $(a+d)(b+d)$. These products equal, respectively, $c^2-(a+b)c+ab$ and $d^2+(a+b)d+ab$. So to write them, all we need to know is $a+b$ and ab rather than a and b individually. By the formulas for elementary symmetric functions applied to the quadratic $x^2+px+1=0$, we know $a+b=-p$ and $ab=1$. Substituting, we have

$$(a-c)(b-c) = c^2+pc+1 \tag{23}$$

$$(a+d)(b+d) = d^2-pd+1 \tag{24}$$

(There is a slicker way to get these. Since a and b are the roots of the monic polynomial x^2+px+1, we must have $x^2+px+1=(x-a)(x-b)$ for all x. Putting $x=c$ gives (23) while putting $x=-d$ gives (24). Although the resulting saving is not very substantial here, one should always look for such short cuts. They make the solution elegant.)

Now to derive (22) we simply multiply (23) and (24). After expansion, the product of the expressions on the right hand sides is $c^2d^2+c^2+d^2+1-p^2cd+pcd^2-pc^2d+pc-pd$. Since c and d are the roots of $x^2+qx+1=0$, we have $c+d=-q$ and $cd=1$. This gives $c^2+d^2=(c+d)^2-2cd=q^2-2$. If we substitute these values, the product becomes $1+q^2-2+1-p^2+pd-pc+pc-pd$, i.e., q^2-p^2. Thus (22) holds.

Comment No. 9:

It should be clear by now that there is often a variety of methods available and picking the right one can simplify the work considerably. It should also be clear that when we want to prove something about the roots which does not require their individual values, it is generally not a good idea to apply the quadratic formula. When the problem specifically asks for solutions, then, of course, we have to use the quadratic formula. Even then the tendency to use it mechanically is to be discouraged. See if you can get the roots by inspection. This is usually possible when the coefficients are small integers. For example, to solve the quadratic $x^2+12x-85=0$, it is better to pause and see if by inspection you can factorise 85 so that the difference of the two factors is 12. Since 85 can be factorised as a product of integers only in 2 ways, this is not a very big task and we see that 17 and 5 are the desired factors. Since the coefficient of x is 12 rather than -12, the corresponding factorisation of the quadratic is $(x+17)(x-5)=0$ and the roots are -17 and 5. The quadratic formula would give the roots as $\frac{-12\pm\sqrt{144+340}}{2}$. Since $144+320=484$ is a perfect square, we get 5 and -17 as the roots. But it is criminal to do so when an easier method is available! If the problem were to solve an equation like $x^2+13x-85=0$, then, of course, the quadratic formula is the only go.

In an examination like the JEE, it is highly unlikely that a question will merely ask you to solve a quadratic! But there are plenty of equations which reduce to a quadratic with a suitable substitution. As already remarked, a biquadratic equation is an example of such equations. In Chapter 10 we shall see problems where a trigonometric equation reduces to a quadratic. In

coordinate geometry, it often happens that there are two solutions to a given problem. For example, there are two straight lines which pass through a given point P on a given line L and make a given angle with L. In such a case, one chooses a suitable parameter (e.g., the slope, say, m of the desired line), writes a quadratic in that parameter and solves it. The two roots give the two solutions to the problem. We shall see many instances of this procedure in the chapters to come. Here we take up a problem where an equation in which the unknown appears as a part of exponents can be solved by reducing it to a quadratic.

Solve for x, (JEE 1985)

$$(5+2\sqrt{6})^{x^2-3}+(5-2\sqrt{6})^{x^2-3}=10 \tag{25}$$

Since the unknown x appears only in the exponent x^2-3, the first substitution that comes to mind is to call x^2-3 as y (say). The equation then becomes

$$(5+2\sqrt{6})^{y}+(5-2\sqrt{6})^{y}=10 \tag{26}$$

In general, there is no easy way to solve equations of the type $a^y+b^y=c$, where a, b, c are some unrelated positive real numbers. However, in the present case, the numbers $5+2\sqrt{6}$ and $5-2\sqrt{6}$ are reciprocals of each other and hence so are the powers $(5+2\sqrt{6})^y$ and $(5-2\sqrt{6})^y$. If we denote the former by u, the latter is $\frac{1}{u}$. With this substitution, (26) changes to $u+\frac{1}{u}=10$, or equivalently, to

$$u^2-10u+1=0 \tag{27}$$

By the quadratic formula, the solutions of (27) are $u=\frac{10\pm\sqrt{96}}{2}$, i.e., $u=5\pm2\sqrt{6}$. Since $u=(5+2\sqrt{6})^y$, the corresponding values are the logarithms of $5\pm2\sqrt{6}$ to the base $5+2\sqrt{6}$. Note that both the values of u are positive and so both the logarithms make sense. In case (27) had a negative root, it would not give any solution of (26) and hence would have to be discarded. This is keeping in line with the remarks made at the end of Comment No. 3 to the effect that when an equation is converted to another, not every root of the new equation necessarily gives a solution to the original equation. In the present case, both the solutions of (27) are positive and give rise to one solution each of (26). Moreover these solutions are very easy to identify. Since one of the values of u is $5+2\sqrt{6}$, its logarithm to the base $5+2\sqrt{6}$ is clearly 1. The other value of u, viz., $5-2\sqrt{6}$ is the reciprocal of $5+2\sqrt{6}$ and so its logarithm to base $5+2\sqrt{6}$ is -1. So we conclude that $y=1$ and $y=-1$ are the solutions of (26).

To get the solutions of the original equation (27) we recall that $y=x^2-3$. So $y=1$ and $y=-1$ give, respectively, $x^2=4$ and $x^2=2$. Here both the values are positive and so neither has to be discarded. Thus we see that (25) has four solutions, ±2 and $\pm\sqrt{2}$.

Sometimes the nature of the given equation is such that we can predict how many solutions it can possibly have. If we can identify that many solutions by inspection, then there is no need to do anything further to solve the equation. Suppose, for example (JEE 1981), that we want to identify the solution set of

the equation

$$\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$$

Surely, we can do this by expanding the determinant and then solving the resulting quadratic either by inspection or by the quadratic formula. But it is unnecessary to take all this trouble. Even without expanding we know that the result is going to be a quadratic equation in x and so there can be at most two solutions. Now if we put $x = 2$, the first and the third rows of the determinant coincide and so the determinant vanishes. Thus $x = 2$ is a solution. Similarly, comparing the second and the third row we see that $x = -1$ is a solution. As there cannot be more than two solutions, we can identify the solution set as $\{2, -1\}$.

Comment No. 10:

It is a peculiarity of mathematics that mathematicians spend far more time than a layman in worrying about the existence and uniqueness of something rather than in actually finding it. If a car fails to climb a steep road, a layman will immediately try to fix it. A mathematician, however, will first of all ensure that the failure is really due to the car (rather than to the condition of the road or to the prevailing weather) and that the problem is such that it can be fixed by a mechanic! This sort of an insistence on the existence of something rather than on actually finding it, is indeed one of the factors why mathematics alienates some laymen. The only word of reconciliation that can be put in is that in real life too, there are situations where the existence of something matters more than its actual value. So many application forms ask you if you are married without inquiring whom to. Here what is asked is whether a spouse exists rather than his/her identity. Also, ensuring the existence and uniqueness of something is as prudent in real life as in mathematics. For example, if you invite some new male acquaintance of yours to your place, then before asking him to bring along his wife too, you better make sure that he is married and monogamous. If he is a confirmed bachelor or a divorcee, you might embarrass him. And if he has two or more wives, he might embarrass you by asking 'which one?'!

Coming to the roots of a polynomial equation, nobody expects them to be unique, except when the polynomial is of degree one. In fact, a polynomial of degree n can have as many as n distinct roots. The other question, viz., that of the existence of roots, is settled completely by the fundamental theorem of algebra if the polynomial, say $p(x)$, is treated as a complex polynomial. In applications, however, the variable x is often a real variable (such as the length of a piece of wire, the slope of a line and so on) and so are the coefficients in the polynomial. Obviously, in such cases we are interested only in the real roots of the polynomial $p(x)$. (What good is a piece of wire whose length is an imaginary number?) So the question of the existence of real roots of real polynomials acquires tremendous importance from a theoretical as well as a practical point of view.

Unfortunately, there is no simple answer to this question. The fundamental theorem of algebra is not of much use here. All it tells us, as we proved in Theorem 1, is that if $p(x)$ is a real polynomial of an odd degree then it has at least one real root (and in fact an odd number of real roots). Apart from this, there are hardly any elementary results applicable to a general real polynomial. Methods from calculus give some information but no complete answer. It can, for example, be shown that if $p(x)$ has n real roots (counted according to multiplicities) then its derivative $p'(x)$ has $n-1$ real roots. But the converse is not true. A simple counter-example is to take $p(x) = x^3 - 3x^2 + x - 3$. Then $p'(x) = 3x^2 - 6x + 1$ which has $\frac{6\pm\sqrt{24}}{6}$ as two distinct real roots. But $p(x)$ factorises as $(x-3)(x^2+1)$ which shows that it has only one real root, viz., 3 (the other two roots being $\pm i$).

So the problem of deciding the existence of real roots of a real polynomial, say $x^n+a_1x^{n-1}+\ldots+a_{n-1}x+a_n$ in terms of its coefficients $a_1, a_2, \ldots, a_n$ admits only *ad-hoc* answers. A complete answer can be given for polynomials of lower degrees. Its complexity increases considerably as n increases. For $n = 1$, i.e., for linear polynomials, the answer is trivial. Every real (monic) polynomial $x + a_1$ has a real root, viz., $-a_1$. The next case is $n = 2$ i.e., of a quadratic, which is more customarily represented by $ax^2 + bx + c = 0$. Multiplying by $4a$ and completing the square we can rewrite this as $(2ax + b)^2 = b^2 - 4ac$. Since the square of any real number is always non-negative, this gives the well-known criterion for a quadratic to have real root, viz., that $b^2 \geq 4ac$, or equivalently, $\Delta \geq 0$, where Δ is defined as $b^2 - 4ac$. This quantity is popularly called the **discriminant** of the quadratic, the name coming from the fact that it tells us when the two roots of the quadratic are unequal, i.e., can be 'discriminated' from each other. Specifically, the roots of a quadratic (whether real or complex) are distinct if and only if its discriminant Δ is non-zero. If further, the coefficients are real, then the sign of the discriminant tells us a lot more as in the following theorem called the **discriminant criterion for quadratics**.

Theorem 3: Suppose a, b, c are real and $a \neq 0$. Let $\Delta = b^2 - 4ac$. Then the roots of the quadratic $ax^2 + bx + c = 0$ are real and distinct, real and coincident, or imaginary according as $\Delta > 0$, $\Delta = 0$ or $\Delta < 0$.

The proof follows immediately from the quadratic formula (or, directly from the equality $(2ax + b)^2 = b^2 - 4ac$). This simple theorem has numerous applications especially in coordinate geometry. As a typical illustration, consider the points of intersection of a straight line L given by the equation

$$y = mx + c \tag{28}$$

with the circle C given by

$$x^2 + y^2 = r^2 \tag{29}$$

(Clearly, this is a circle of radius r centred at the origin.) To find the points of intersection we substitute (28) into (29) to get a quadratic in x, viz.,

$$(m^2 + 1)x^2 + 2mcx + c^2 - r^2 = 0 \tag{30}$$

For this quadratic, the discriminant is $4(r^2(m^2+1)-c^2)$. Theorem 3 above then tells us that (30) has two distinct real roots if $r^2(m^2+1) > c^2$, two equal real roots if $r^2(m^2+1) = c^2$ and no real roots if $r^2(m^2+1) < c^2$. Geometrically, in the first case, the line L cuts a chord off C, in the second case it touches C and in the third case it does not intersect C at all. In particular, we get the condition for tangency as $c = \pm\sqrt{r^2(m^2+1)}$. For a fixed m, this has two possible values for c. Geometrically, m represents the slope of L and hence we get that two tangents to C can be drawn that are parallel to a given direction. Also for a fixed m we see that the line $y = mx + c$ cuts a chord off the circle $x^2 + y^2 = r^2$ if and only if the inequality $c^2 < r^2(m^2+1)$ holds.[5] This, in turn, is equivalent to saying that $-\sqrt{r^2(m^2+1)} < c < \sqrt{r^2(m^2+1)}$ or equivalently that c lies in the open interval $(-\sqrt{r^2(m^2+1)}, \sqrt{r^2(m^2+1)})$.

Comment No. 11:

If we draw the graph of some real-valued function, say $y = f(x)$, of a real variable x, then it is clear that the zeros of $f(x)$ are precisely the points where the graph intersects the x-axis. (More generally, for any constant λ, the solutions of the equation $f(x) = \lambda$ correspond to the points of intersection of the line $y = \lambda$ with the graph of the function.) If we keep this simple fact in mind then the geometric interpretation of the discriminant criterion is that the graph of the function $y = f(x) = ax^2 + bx + c$ cuts the x-axis in two distinct points if $\Delta > 0$, in one point if $\Delta = 0$ and that it does not meet the x-axis at all if $\Delta < 0$. With a little more work, we can make a qualitative sketch of the graph of the function $y = f(x) = ax^2 + bx + c$ even with the rudimentary methods we have at our disposal at present. (More sophisticated techniques which reveal certain vital features of the graph of a function require the use of calculus, specifically that of the derivatives of a function. We shall discuss them briefly later.) It suffices to consider only the case where $a > 0$. For if $a < 0$, then we can first draw the graph of $y = -ax^2 - bx - c$ and then take its reflection in the x-axis.

So, assume $y = f(x) = ax^2 + bx + c$, where $a > 0$. Rewrite $f(x)$ as $a(x^2 + \frac{x}{a} + \frac{c}{a})$ and then, after completing the square, as $a(x + \frac{b}{2a})^2 - \frac{b^2-4ac}{4a}$ or as

$$y = f(x) = a(x-\gamma)^2 - \frac{\Delta}{a} \tag{31}$$

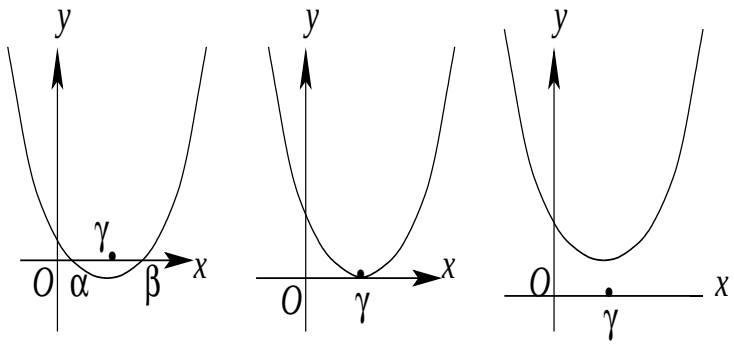
where $\gamma = -\frac{b}{2a}$ and $\Delta = b^2 - 4ac$ is the discriminant.

From (31) it is immediate that for every $h \geq 0$, $f(\gamma + h) = f(\gamma - h)$. This means that the graph of $y = f(x)$ is symmetric about the vertical line $x = \gamma$. Moreover, as h increases, so does y. The minimum value of y occurs when $h = 0$, i.e., when $x = \gamma$. Put differently, the graph of $y = f(x)$ is a vertically upward parabola with its lowest point at $(\gamma, -\frac{\Delta}{4a})$. (If $a < 0$, then the graph is a downward parabola with its topmost point at $(\gamma, -\frac{\Delta}{4a})$.)

[5] Inequalities will be studied in detail in Chapter 6. Here we need some very elementary results such as the product of two real numbers of the same sign is positive while that of two real numbers of opposite signs is negative. Problems where more elaborate methods are needed to decide when the discriminant is positive will be taken up in Chapter 6.

Let us now combine these observations with the discriminant criterion. Assume first that $\Delta > 0$. Then the quadratic has two distinct real roots, viz., $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$. Let us call $\frac{-b-\sqrt{b^2-4ac}}{2a}$ as α and $\frac{-b+\sqrt{b^2-4ac}}{2a}$ as β. Then $\alpha < \beta$. Note that γ is precisely the arithmetic mean of α and β. Geometrically, this means that the points $(\alpha, 0)$ and $(\beta, 0)$ are symmetrically located w.r.t. the the line $x = \gamma$ which is the axis of symmetry of the graph. It is clear then that the portion of the graph between these two points lies below the x-axis while the portions corresponding to $x < \alpha$ and $x > \beta$ lie above the x-axis. (This can also be seen in a purely algebraic manner. The quadratic $ax^2 + bx + c$ factors as $a(x-\alpha)(x-\beta)$. We are assuming $a > 0$. If $\alpha < x < \beta$ then the factor $(x-\alpha)$ is positive while the other factor $(x-\beta)$ is negative. So $f(x)$ is negative for such values of x. When either $x < \alpha$ or $x > \beta$, both the factors are of the same sign and hence their product is positive.)

Summing up, the graph of the quadratic in this case looks like that in (i) of the figure below. The next case is where the discriminant Δ vanishes. In this case both α and β coincide with γ. In other words, γ is a double root of the quadratic. The graph is above the x-axis at all points except at the point $(\gamma, 0)$, where it touches the x-axis. It is shown in (ii) below. The last case is where $\Delta < 0$. In this case, the graph is as in (iii) below. It lies completely above the x-axis, because even the lowest point on it, viz., $(\gamma, -\frac{\Delta}{4a})$ lies above the x-axis.



(i) $\Delta > 0$ (ii) $\Delta = 0$ (iii) $\Delta < 0$

Graph of $y = ax^2 + bx + c$ for $a > 0$

We remark once again that for $a < 0$, these graphs have to be turned upside down. These simple observations enable us to solve the following problem. (JEE 1979)

Determine for which values of k the entire graph of the equation

$$y = x^2 + kx - x + 9$$

is strictly above the x-axis.

Here the leading coefficient 1 is positive and so we can apply the analysis above directly. The discriminant Δ equals $(k-1)^2 - 36$. So the graph will

lie entirely above the x-axis if and only if we are in case (iii) above, i.e., if $(k-1)^2 - 36 < 0$. We can write Δ as a polynomial of degree 2 in k, viz., $k^2 - 2k - 35$. The roots of this polynomial are -5 and 7. So the expression $k^2 - 2k - 35$ will be negative precisely when k lies between -5 and 7, i.e., when $-5 < k < 7$. But there is a better way of arriving at the answer. Once we get $(k-1)^2 < 36$ as the necessary and sufficient condition for the graph to lie strictly above the x-axis, we can immediately paraphrase it to say that numerically $k-1$ should be less than 6, or in other words $-6 < k-1 < 6$, which gives the same answer as before, viz., $-5 < k < 7$. (The first method will be truly silly if, after writing Δ as $k^2 - 2k + 35$, you use the quadratic formula to find the roots. This is so because, the quadratic formula is based on the idea of completing a square. So, in effect, you are expanding the square $(k-1)^2$ and again completing it!)

Comment No. 12:

We thus have a complete analysis of the behaviour of a real quadratic. The next case is that of a cubic, say, $y = ax^3 + bx^2 + cx + d$ (with $a \neq 0$). Again, we assume $a > 0$ without loss of generality, as otherwise we take a reflection in the x-axis. In fact, we may suppose $a = 1$ since this affects neither the roots nor the qualitative behaviour of the graph. As a further simplification, we may suppose that $b = 0$. To see this, write $x^3 + bx^2 + cx + d$ as $(x + \frac{b}{3})^3 + (c - \frac{b^2}{3})(x + \frac{b}{3}) + (\frac{2b^3}{27} - \frac{bc}{3} + d) = u^3 + c'u + d'$, where $u = x + \frac{b}{3}$, $c' = c - \frac{b^2}{3}$ and $d' = \frac{2b^3}{27} - \frac{bc}{3} + d$. Since the substitution $u = x + \frac{b}{3}$ is reversible (with $x = u - \frac{b}{3}$ as the inverse function), finding the roots of $f(x)$ is as good as finding those of $g(u) = u^3 + c'u + d'$. Thus we may get rid of the square term by a simple algebraic trick.

Unfortunately, the subsequent work is not easy if attempted with pure algebraic tools. Being an odd degree polynomial, $f(x)$ has at least one real root. A formula for the three roots is possible but far more complicated than the quadratic formula. Even the condition for all the three roots to be real (along with the possible shapes of the graph of a cubic) is not easy to derive algebraically. With calculus, the latter can be done rather easily, although even calculus is of little help in giving a formula for the roots. (Interestingly, a cubic can sometimes be solved using trigonometry. See Chapter 10, Comment No. 1. Also see Comment No. 13 of Chapter 13 for the graphs of general cubics.)

However, as noted earlier, if we are given or can easily identify one root of a cubic then we can solve it completely. Suppose, for example, that we want to solve the equation (JEE 1989)

$$x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}. \tag{32}$$

Here the exponent of the L.H.S. is a quadratic in $\log_2 x$. If the base were some constant, say a (instead of x), then we could take the logarithms of both the sides and solve the resulting quadratic in $\log_2 x$. However, as the problem stands, taking logarithms of both the sides w.r.t. 2 will give $(\frac{3}{4}(\log_2 x)^2 + \log_2 x -$

$\frac{5}{4}) \log_2 x = \frac{1}{2}$ which is a cubic in $\log_2 x$. Writing y for $\log_2 x$, this becomes

$$\frac{3}{4}y^3 + y^2 - \frac{5}{4}y - \frac{1}{2} = 0. \tag{33}$$

By inspection we find that $y = 1$ is a solution of (35). Taking out $(y-1)$ as a factor, the L.H.S. factors as $(y-1)(\frac{3}{4}y^2+\frac{7}{4}y+\frac{1}{2})$. The roots of the second factor can be obtained by inspection (after clearing the denominators by multiplying by 4) or by the quadratic formula. They come out to be $-\frac{1}{3}$ and -2. So (33) has three real roots, viz., $1, -\frac{1}{3}$ and -2. Since $x = 2^y$, the corresponding solutions of (32) are $2^1, 2^{-\frac{1}{3}}$ and 2^{-2}, i.e., $2, \frac{1}{\sqrt[3]{2}}$ and $\frac{1}{4}$.

Note that the coefficients of the cubic (33) are not only real, but rational. This made it somewhat easier to identify one root by inspection. There is a systematic method to identify all possible rational roots of a polynomial with rational coefficients. Appropriately, it is called the **rational root test**. See Exercise (10.11) for it.

Comment No. 13:

When we cannot fully solve a cubic or even give a criterion for its roots to be all real, there is little point in trying to do the same for polynomials of higher degrees. Instead we now turn to the problem of solving a system of equations[6] in several variables. Such systems arise very frequently in applications. The reason is obvious. In real life problems, usually there are two or more variables. The data of the problem gives rise to a system about these variables and solving the system gives a solution to the problem. As a very simple illustration, suppose that we want to find a rectangle whose perimeter is p cm.s and whose area is A sq.cm.s, where p and A are some given positive real numbers. The most natural way to tackle this problem is to let x and y denote the length and the breadth of the rectangle (in centimeters). The data then gives the system of equations:

$$2x + 2y = p \quad \text{and} \quad xy = A \tag{34}$$

Solving this system is very easy. We eliminate one of the two variables, x and y. This can be done either by getting $y = \frac{A}{x}$ from the second equation and putting it into the first or by getting $y = \frac{p}{2} - x$ from the first and putting it into the second. Either way, the answer is a quadratic in x, viz.,

$$x^2 - \frac{p}{2}x + A = 0 \tag{35}$$

solving which we get $x = \frac{p \pm \sqrt{p^2-16A}}{4}$ and correspondingly, $y = \frac{p \mp \sqrt{p^2-16A}}{4}$. Thus one of the two values will be the length and the other the width of the rectangle. The data is insufficient to tell which is which. But if we adopt

[6] Formerly the equations appearing in a system used to be called, **simultaneous equations** to stress that the values of the variables in each one of them are the same. Nowadays, this is generally understood and so the adjective is rarely used except for emphasis.

the convention that the breadth cannot exceed the length, then the (unique) solution is $x = \dfrac{p + \sqrt{p^2 - 16A}}{4}$ and $y = \dfrac{p - \sqrt{p^2 - 16A}}{4}$.

This, of course, is the most simple minded way of solving the problem. Improvements are certainly possible. For example, instead of taking the sides as x and y, we could have taken them as x and $\frac{A}{x}$ or as x and $\frac{p}{2} - x$ to begin with. In effect, this means that we are using a piece of data to reduce the number of variables. This is a very healthy habit which pays off in more complicated problems. Instead of introducing a large number of variables and writing equations linking them, it is far better to incorporate these equations right at the beginning and thereby dispense with some of the variables. This gives a more manageable system to solve.

As yet another stroke of elegance, if we note that the sum of x and y is $\frac{p}{2}$ while their product is A, then both of them are the roots of a certain quadratic which may always be taken to be monic. Then we get (35) instantaneously.

Still, the solution given above illustrates various points and procedures. To begin with, it illustrates how systems of equations arise naturally in real life problems. Secondly, it shows that to solve a system of two equations in two unknowns, one method is to express one of the variables in terms of the other using one of the two equations. Substituting this into the second equation then gives an equation in a single variable. In the problem just done, either of the two equations could be used with equal ease to express one of the variables in terms of another. But this may not always be so. Consider, for example, the system consisting of Equations (28) and (29). The first equation, viz., $y = mx + c$, is already in a form that can be substituted in the second. But doing it the other way round would be complicated. Equation (29) can be treated as a quadratic in y and solved so as to express y in terms of x. But there are two roots, viz., $y = \pm\sqrt{r^2 - x^2}$ and so when we substitute this into (28) we shall have to deal with the two cases separately. In general, whenever one of the equations of a system is linear (i.e., involves only the first degree terms) it is best to substitute from it into the other equations. Systems where none of the equations is linear can demand considerable ingenuity for a solution as the following example shows.

$$\text{Solve} \ : \ \frac{x^2}{a+x} + \frac{y^2}{a+y} = a, \ \ x^2 + y^2 = 2a^2. \tag{36}$$

Expanding the first equation and substituting from the second gives

$$a^3 + xy(x+y) = a^2(x+y) + axy \tag{37}$$

The trick now is to group together the terms with $(x + y)$ as a factor to get

$$(a^2 - xy)(x+y) = a(a^2 - xy) \tag{38}$$

which implies either $xy = a^2$ or $x + y = a$. The first possibility, together with $x^2 + y^2 = 2a^2$ gives a quartic $x^4 - 2a^2x^2 + a^4 = 0$ which can be rewritten as $(x^2 - a^2)^2 = 0$, showing that $x = \pm a$ and correspondingly $y = \pm a$. The

second possibility, viz., $x + y = a$, together with $x^2 + y^2 = 2a^2$ gives $2xy = (x + y)^2 - (x^2 + y^2) = -a^2$ and hence $(x - y)^2 = x^2 + y^2 - 2xy = 3a^2$, i.e., $x - y = \pm\sqrt{3}a$. Together with $x + y = a$ this gives $x = \frac{1+\sqrt{3}}{2}a$ and $y = \frac{1-\sqrt{3}}{2}a$ or vice versa.

Reducing (36) to (37) was straightforward. But further reduction to (38), although simple, was not so obvious. Note further that if in the original system, the second equation were something else, say, $x^2 + y^2 = a^2$ then a similar factorisation would not be possible. So the method given above is of an *ad-hoc* type. In this problem, the first equation becomes a cubic (i.e., a third degree) equation when written in the standard form (that is, after clearing the denominators) while the second one is of degree two. In general, there is no easy way to solve even two second degree equations, say $a_1x^2+b_1y^2+2h_1xy+2g_1x+2f_1y+c_1 = 0$ and $a_2x^2 + b_2y^2 + 2h_2xy + 2g_2x + 2f_2y + c_2 = 0$. Geometrically, each equation represents a conic (which could be a pair of possibly coinciding straight lines, a circle, an ellipse, a parabola or a hyperbola depending upon the relationship of the coefficients as will be seen later in Chapter 10, Comment No. 19). In general two conics intersect in four points which correspond to the solutions of this system. But there is no easy way to find these points of intersection explicitly. Success is possible only in special cases.

Comment No. 14:

The situation with a system of more than two equations is still more complicated. Explicit solutions are possible only in rare cases. But some general observations can be made. Suppose we have the following system of m equations in n unknowns $x_1, x_2, \ldots, x_n$.

$$\left.\begin{array}{rcl} f_1(x_1, x_2, \ldots, x_n) & = & 0 \\ \vdots & & \vdots \\ f_m(x_1, x_2, \ldots, x_n) & = & 0 \end{array}\right\} \tag{39}$$

where $f_1, f_2, \ldots, f_m$ are polynomials in the variables $x_1, x_2, \ldots, x_n$. The general rule is that if $m = n$, i.e., if there are as many equations as the number of variables then the system can, in theory, be solved completely and has a finite number of solutions. We emphasise again that this simply means that the solutions exist and not that there is a systematic procedure for obtaining them. Such a system is called **well-determined**. If $m < n$, then the system is called **under-determined**, because there is not enough information to determine the values of the variables. Such a system has an infinite number of solutions. It is easy to see why. Since $m < n$, if we treat $x_{m+1}, x_{m+2}, \ldots, x_n$ as constants, we get a system of m equations in the m unknowns, viz., $x_1, x_2, \ldots, x_m$. So, in theory, this system can be solved to express $x_1, x_2, \ldots, x_m$ in terms of $x_{m+1}, x_{m+2}, \ldots, x_n$. But now we are free to give any values to these $n-m$ variables and still get a solution. These values can be assigned in infinitely many ways. More specifically, we say that in this situation, we have $n - m$ **degrees of freedom** because that is the number of values we can assign at our will.

The last case, viz., $n > m$, is called **over-determined** because there are

more constraints than the number of variables. In such a case a solution may not even exist. The simplest example is that of a system of three linear equations in two unknowns, say x and y. Geometrically, each equation represents a straight line. The system will have a solution if and only if the three lines are concurrent, which, in general, will not be the case. So in the case of an over-determined system, an important problem is to find necessary and sufficient conditions for the system to be **consistent**, i.e., to have at least one solution. Such problems are generally not easy and satisfactory answers are possible only in special cases.

Even this qualitative analysis is not as simple as it seems at first glance. The trouble is that the number of equations in the system may be an illusive one. What if one of the equations is a linear combination of some others? In that case such an equation is superfluous in the sense that it will automatically hold whenever the others hold. So, it can be dropped from the system. But that reduces the value of m and thereby affects our analysis. Conversely, we can add to the system (42) any number of 'dummy' equations obtained, for example, by simply taking various linear combinations of $f_1, f_2, \ldots, f_m$. So before classifying a system as well-determined, under-determined or over-determined, it is very important to ensure that each equation in the system (39) is indispensable, or that the m equations in it are 'really different'. Even a precise definition of this is not easy.

Further complications arise if we restrict the variables $x_1, x_2, \ldots, x_n$ to be real and further require that the coefficients in the polynomials $f_1(x_1, x_2, \ldots x_n)$, $f_2(x_1, x_2, \ldots, x_n), \ldots, f_m(x_1, x_2, \ldots, x_n)$ be all real. For in that case one single equation can be equivalent to several. The simplest example is that the single equation $x_1^2 + x_2^2 = 0$ is equivalent to the two equations $x_1 = 0$ and $x_2 = 0$. Stretching this idea further, the entire system (39), in case all the polynomials in it are real and the variables are to assume only real values, can be replaced by a single equation, viz.,

$$f_1^2(x_1, x_2, \ldots, x_n) + f_2^2(x_1, x_2, \ldots, x_n) + \ldots + f_m^2(x_1, x_2, \ldots, x_n) = 0.$$

This again means that the number of equations is an illusory concept, which depends more on the form than on content.

We leave aside this discussion except to remark that a certain branch of mathematics called **algebraic geometry** handles such issues among others.

Comment No. 15:

The situation is pleasant if the polynomials in (39) are all of degree one each. The system is then called **linear.** In such a case it is customary to bring the constant terms in these linear polynomials to the right-hand sides and write the system as:

$$\left.\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n & = & b_1 \\ \vdots & & \vdots \\ a_{i1}x_1 + a_{i2}x_2 + \ldots a_{in}x_n & = & b_i \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots a_{mn}x_n & = & b_m \end{array}\right\} \qquad (40)$$

where $a_{11}, a_{12}, \ldots, a_{1n}, a_{21}, \ldots, a_{mn}, b_1, b_2, \ldots, b_m$ are some real (or complex) numbers. If we use matrices then (43) can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1j} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2j} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \ldots & a_{ij} & \ldots & a_{im} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mj} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_j \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_j \\ \vdots \\ b_m \end{bmatrix} \tag{41}$$

or more compactly as

$$\mathbf{Ax} = \mathbf{b} \tag{42}$$

where $\mathbf{x}$ and $\mathbf{b}$ are $n \times 1$ and $m \times 1$ column vectors with entries $x_1, x_2, \ldots, x_n$ and $b_1, b_2, \ldots, b_m$ respectively and $\mathbf{A}$ is an $m \times n$ matrix whose (i, j)-th entry is a_{ij}.

The role of matrices in the study of systems of linear equations is much deeper than providing a convenient notation. But we shall not go into it. Such systems are important for two reasons. First, they arise in many real life problems. Suppose, for example, that we have n types of food items, say $F_1, F_2, \ldots, F_n$ and m types of vitamins, say $V_1, V_2, \ldots, V_m$. For $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$ denote by a_{ij} the amount of vitamin V_i that you can get from 1 gram of the food item F_j. If we want to have a diet consisting of these food items which provides exactly b_i grams of vitamin V_i for $i = 1, 2, \ldots, m$, then the problem is equivalent to finding a solution of the system (41). As noted above, if $m < n$ then the system will in general have infinitely many solutions. In that case, if we are also given the costs of the food items, the problem can be made more practical by requiring us to find the most economical diet which meets the vitamin requirements. There is a branch of mathematics called **linear programming** which deals with such problems. We shall mention it briefly in Chapter 13, where we shall study maxima and minima.

Another reason why linear systems are important is that unlike in the case of a general system of m polynomial equations in n unknowns, the system (40) above can be analysed completely. That is, we can tell exactly when it has a solution, a unique solution and infinitely many solutions. We summarise the answer in the following two theorems. We skip the proofs as they are rather involved. They belong to a branch of algebra called **linear algebra**. We shall need only some special cases of these theorems (to be listed later) which can be proved directly anyway. Before stating the theorems we introduce some terminology. The system above is called **homogeneous** if all the b_i's are zero. Otherwise it is called **non-homogeneous**. (The name is justified because if the equations are written in the standard form by transporting the b_i's to the left-hand sides, then the left-hand side of the i-th equation would contain all first degree terms except the term $-b_i$ which is of degree 0, as it is a constant. When b_i is 0, this term is absent.)

The following theorem completely answers the question of the existence and uniqueness of the solutions of a homogeneous linear system.

Theorem 4: Suppose the system (40) is homogeneous, i.e., $b_i = 0$ for all $i = 1, 2, \ldots, m$. Then :

(i) If $m < n$, the system has a non-trivial solution, i.e., a solution in which at least one of $x_1, x_2, \ldots, x_n$ is non-zero.

(ii) If $m = n$ then let Δ be the $n \times n$ determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & & \ldots & \vdots \\ a_{i1} & a_{i2} & \ldots & a_{im} \\ \vdots & & \ldots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{vmatrix}$$

Then (40) has a non-trivial solution if $\Delta = 0$. But if $\Delta \neq 0$, then (40) has only the trivial solution in which $x_i = 0$ for every $i = 1, 2, \ldots, n$.

(iii) If $m > n$ then among the given m equations it is possible to weed out some equations (as being linear combinations of some other equations in the system) so that we are left with only r of them, where r is some integer, not exceeding n, which depends on the coefficients a_{ij}'s, and no further weeding out is possible. (This integer r is called the **rank**[7] of the matrix **A**.) The system has a non-trivial solution if and only if $r < n$.

Moreover, in each case, a linear combination of two solutions is also a solution. In particular, if $x_1 = c_1, x_2 = c_2, \ldots, x_n = c_n$ is a solution then for every λ, $x_1 = \lambda c_1, x_2 = \lambda c_2, \ldots, x_n = \lambda c_n$ is also a solution. Hence the existence of a non-trivial solution implies that the system has infinitely many solutions.

Coming to the general case where the system (40) is not homogeneous. a solution may not always exist as the system may not be consistent. A trivial example is a system of just two equations, say

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ 5a_{11}x_1 + 5a_{12}x_2 + \ldots + 5a_{1n}x_n &= b_2 \end{aligned}$$

Here if $b_2 \neq 5b_1$, the system is inconsistent and can have no solution. However, if $b_2 = 5b_1$ then there are infinitely many solutions (assuming that $n \geq 2$ and that at least two of the coefficients are non-zero). Note further that if $x_i = c_i, i = 1, 2, \ldots n$ and $x_i = d_i, i = 1, 2, \ldots, n$ are two solutions of the system above, then their difference, i.e., $x_i = c_i - d_i, i = 1, 2, \ldots, n$ is a solution of the corresponding homogeneous system obtained by replacing b_1 and b_2 by 0.

[7]There are other ways of defining the rank of a (not necessarily square) matrix A. For example, it is the size of the largest square submatrix of A with a non-zero determinant.

The following theorem establishes the relationship between the solutions of (40) and those of the associated homogeneous system (obtained by replacing every b_i by 0). The proof is beyond our scope.

Theorem 5: Suppose the system (40) has a solution. Then every solution of it is obtained by adding to it a solution of the associated homogeneous system. In particular for $m < n$, the system (43) has either no solution or else infinitely many solutions. For $m = n$, let Δ be the determinant as in the last theorem. If $\Delta \neq 0$ then the system has a unique solution. Moreover, this solution is given by the following formula, called **Cramer's rule**.

$$x_1 = \frac{\Delta_1}{\Delta},\ x_2 = \frac{\Delta_2}{\Delta}, \ldots, x_n = \frac{\Delta_n}{\Delta} \tag{43}$$

where for $i = 1, 2, \ldots, n$, Δ_i is the determinant we get from the determinant Δ if we replace its i-th column entries by $b_1, b_2, \ldots, b_n$. If $\Delta = 0$, then the system has no solution if at least one of $\Delta_1, \Delta_2, \ldots, \Delta_n$ is non-zero. If $\Delta, \Delta_1, \ldots, \Delta_n$ all vanish then the system has either no solution or infinitely many solutions.

We remark that Cramer's rule is of little practical use. First, it is applicable only when $m = n$. Even then, evaluating an $n \times n$ determinant is a very time-consuming process for $n > 3$. Better methods are available to solve a linear system in all cases. In one such method, suitable multiples of the first equation are subtracted from the subsequent ones and the process is then repeated with the second equation, then the third one, then the fourth and so on till the system is reduced to a form in which the solutions can be identified easily. For this reason, the method is called **row reduction**. The details are beyond our scope.

Nevertheless, Cramer's rule is of theoretical importance. It not only guarantees the existence and uniqueness of the solution when $\Delta \neq 0$, it also tells us what the solution is and thereby gives information which may not be obvious otherwise. For example, suppose that all the coefficients a_{ij}'s and the constants b_j's are real. It is not immediately clear that the solutions are also real. Indeed, for non-linear systems this need not be so as we see from the fact that a quadratic with real coefficients can have complex roots. But if we look at (43), each x_i is the ratio of two determinants all of whose entries are real numbers. Since the full expansion of a determinant involves only the sum of the products of its entries, it is clear that if all the entries are real, so is the determinant. Further, since the ratio of two real numbers is real, from (43) we conclude that the solution is real (i.e., has all real entries). By an identical argument, if the coefficients and the b_j's are all rational, then the solution is rational.

Note, however, that even if all the coefficients and the b_j's are all integers, Cramer's rule does not imply that the solution will be integral, i.e., will have all integer values, because in such a case, even though both Δ_i and Δ are integers, their ratio need not be so. In fact, the problem of deciding when a system of linear equations has integer solutions comes more under number theory than linear algebra. (See Exercise (4.23) for a very special case.) The problem of

finding integral solutions of non-linear systems is an extremely challenging one. It is called **diophantine analysis** after a Greek mathematician **Diophantus** who studied it centuries ago. A simple example of a diophantine equation is the equation $x^2 + y^2 - z^2 = 0$. It is possible to find all integral solutions of this equation (e.g., $x = 3, y = 4, z = 5$ is a solution). See the end of Comment No. 3 of Chapter 8 for more on this.

Comment No. 16:

Let us now study some special cases of the theorems above when the number of variables and the number of equations are both small, specifically, at most three each. In such cases, it is customary to change the notations so that we do not have to use double subscripts. Thus, for example, a system of three linear equations in three unknowns can be denoted by

$$\left.\begin{array}{rcl} a_1x + b_1y + c_1z &=& d_1 \\ a_2x + b_2y + c_2z &=& d_2 \\ a_3x + b_3y + c_3z &=& d_3 \end{array}\right\} \tag{44}$$

The case $n = 1$ is absolutely trivial. In this case a single equation $a_1x = b_1$ has a unique solution if $a_1 \neq 0$. If $a_1 = 0$, then there is no solution if $b_1 \neq 0$. However, if b_1 and a_1 are both 0 then any value of x is a solution and so there are infinitely many solutions. Moreover, a system of two equations $a_1x = b_1$ and $a_2x = b_2$ has a solution if and only if $a_1b_2 - a_2b_1 \neq 0$ as we see by multiplying the first equation by a_2, the second by a_1 and subtracting the two. This condition can be written in the determinant form as $\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \neq 0$.

The next case, $n = 2$, although simple, is not so trivial. In this case the validity of Cramer's rule for a solution of the system

$$\left.\begin{array}{rcl} a_1x + b_1y &=& c_1 \\ a_2x + b_2y &=& c_2 \end{array}\right\} \tag{45}$$

can be proved directly by multiplying the first equation by b_2, the second by b_1 and subtracting the two. This gives the value of x as $\dfrac{c_1b_2 - c_2b_1}{\Delta}$, where $\Delta = a_1b_2 - a_2b_1$. That of y is obtained similarly. Moreover, Cramer's rule is easy to apply since a 2×2 determinant can be evaluated just by inspection. If c_1, c_2 are both 0, then the system above becomes homogeneous and always has a trivial solution, viz., $x = 0 = y$. If $\Delta \neq 0$ then the calculation above for the values of x and y shows that there cannot be any non-trivial solution. Combining this fact with properties of the roots of a quadratic, we can do the following problem.

Let α_1, α_2 and β_1, β_2 be the roots of $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ respectively. If the system of equations $\alpha_1y + \alpha_2z = 0$ and $\beta_1y + \beta_2z = 0$ has a non-trivial solution then prove that $\dfrac{b^2}{q^2} = \dfrac{ac}{pr}$. (JEE 1987)

The existence of a non-trivial solution to the system of equations $\alpha_1y + \alpha_2z = 0$ and $\beta_1y + \beta_2z = 0$ implies that $\begin{vmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{vmatrix} = 0$, i.e., $\alpha_1\beta_2 = \alpha_2\beta_1$.

Hence $\frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\beta_2}$. (We are assuming here that $\alpha_2\beta_2 \neq 0$. Otherwise we consider $\frac{\alpha_2}{\alpha_1} = \frac{\beta_2}{\beta_1}$ assuming that $\alpha_1\beta_1 \neq 0$. If both $\alpha_1\beta_1$ and $\alpha_2\beta_2$ vanish then we run into a degenerate case in which one of the two equations has 0 as a double root and the result to be proved holds in a degenerate way.)

From $\frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\beta_2}$, we get

$$\frac{(\alpha_1 + \alpha_2)^2}{(\alpha_1 - \alpha_2)^2} = \frac{(\beta_1 + \beta_2)^2}{(\beta_1 - \beta_2)^2}$$

Since we have $\alpha_1 + \alpha_2 = -\frac{b}{a}$ and $\alpha_1 - \alpha_2 = \pm\frac{\sqrt{b^2-4ac}}{a}$ and similar expressions for $\beta_1 + \beta_2$ and $\beta_1 - \beta_2$, a direct substitution in the equation above gives

$$\frac{b^2}{b^2 - 4ac} = \frac{q^2}{q^2 - 4pr}$$

from which the desired result follows by subtracting 1 from both the sides and taking reciprocals. (An alternate solution is obtained by converting the equality $\alpha_1\beta_2 = \alpha_2\beta_1$ to $\frac{\alpha_1}{\beta_1} = \frac{\alpha_2}{\beta_2}$. Call each ratio as k. Then the roots of the first equation are k times those of the second one and hence by Theorem 2, the first equation must be the same as $p(\frac{x}{k})^2 + q(\frac{x}{k}) + r = 0$, or equivalently, $px^2 + kqx + rk^2 = 0$. Comparing this with $ax^2 + bx + c = 0$ we get $\frac{p}{a} = \frac{qk}{b} = \frac{rk^2}{c}$. The desired result can now be obtained by eliminating k.)

The question of consistency of a system of three (or more) linear equations in two variables, say x and y, is interesting geometrically, because each such equation represents a straight line in the xy-plane. So the problem is equivalent to deciding if a set of three (or more) lines with given equations are concurrent. We first consider the case of three lines, say, L_1, L_2 and L_3. Let us write their equations as

$$a_1x + b_1y = c_1 \tag{46}$$

$$a_2x + b_2y = c_2 \tag{47}$$

$$a_3x + b_3y = c_3 \tag{48}$$

respectively. The first two equations are identical to those in the system (45) and so we already know that they have a unique solution if and only if $a_1b_2 - a_2b_1 \neq 0$. Geometrically this means that the lines L_1 and L_2 intersect in a unique point. (If $a_1b_2 - a_2b_1 = 0$, then the lines are either identical or parallel depending upon whether $b_1c_2 - b_2c_1$ and $a_1c_2 - a_2c_1$ also vanish or not as is easy to show). So we assume that $a_1b_2 - a_2b_1 \neq 0$. As we already know, the unique solution of (46) and (47) is then given by

$$x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1} \qquad y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1} \tag{49}$$

The condition for consistency of the equations (46) to (48) is that (48) should hold when (49) is substituted in it. A straightforward calculation shows that this is equivalent to saying that

$$a_3(c_1b_2 - c_2b_1) + b_3(a_1c_2 - a_2c_1) = c_3(a_1b_2 - a_2b_1) \tag{50}$$

If we take all the terms to the R.H.S. and regroup them then they are precisely the terms in the expansion of the 3×3 determinant consisting of the coefficients in equations (46) to (47). We have thus proved the major part of the following theorem.

Theorem 6: Assume that at least one of the three numbers $a_1b_2 - a_2b_1$, $a_2b_3 - a_3b_2$ and $a_3b_1 - a_1b_3$ is non-zero. Then the equations (46) to (48) have a common solution if and only if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0 \tag{51}$$

Proof : We have already dealt with the case when $a_1b_2 - a_2b_1 \neq 0$. The argument is similar in the other two cases. Or, one can simply change the order of the three equations and note that this may only change the sign of the determinant but will not affect its vanishing.

This theorem has numerous applications in proving the concurrency of three lines in a plane as we shall see in Chapters 8 and 9. Usually, the very nature of the lines is such that no two of them can be coincident or parallel (e.g., when they represent the three altitudes of a triangle). In that case the hypothesis of the theorem need not be verified explicitly and the concurrency is equivalent to the vanishing of the determinant of coefficients. Note that if the equations of the lines are written in the standard form (that is, as $a_1x + b_1y + c_1 = 0$ etc.) the theorem still holds since multiplying the third column of the determinant by a constant -1 does not affect its vanishing.

In case there are more than three linear equations in x and y, the theorem above has an obvious extension. Suppose we have n equations of the form $a_ix + b_iy = c_i$ for $i = 1, 2, \ldots, n$, where $n > 3$. Assume there are two distinct indices, say r and s for which $a_rb_s - a_sb_r \neq 0$. Then the system is consistent if and only if for every $i \neq r, s$, the determinant $\begin{vmatrix} a_r & b_r & c_r \\ a_s & b_s & c_s \\ a_i & b_i & c_i \end{vmatrix}$ vanishes.

Comment No. 17:

Finally, we deal with the case of systems of linear equations in three unknowns, say x, y, z. Although the results to be proved are only special cases of those mentioned in Comment No. 15 above, this special case has a special importance. For one thing, the space we can perceive with the senses is three-dimensional. Secondly, systems of equations with three unknowns arise naturally in some contexts, e.g., in connection with triangles. Also linear equations are intimately related to vectors and for three-dimensional vectors we have the concepts of the dot and the cross products. (The dot product can be defined for vectors in any dimension but the cross product is a peculiarity of the three-dimensional space. There is no analogue of it for higher dimensions.) Vectors

will be formally taken up in Chapter 21. But at present a little knowledge of the elementary properties of the dot and the cross products will help.

The case of solutions of a single equation, say $a_1x + b_1y + c_1z = d_1$ is as trivial as for two variables. But the case of two equations is not so trivial. Especially interesting is the case of two homogeneous equations which is answered in the next theorem.

Theorem 7: Every system of the form

$$a_1x + b_1y + c_1z = 0 \tag{52}$$

$$a_2x + b_2y + c_2z = 0 \tag{53}$$

has at least one non-trivial solution (and, in fact, infinitely many such solutions). If further, $a_1b_2 - a_2b_1$, $b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ are not all zero and x, y, z satisfy the system above, then x, y, z are proportional to the numbers $b_1c_2 - b_2c_1$, $c_1a_2 - c_2a_1$ and $a_1b_2 - a_2b_1$ respectively, that is, there exists some number r such that

$$x = r(b_1c_2 - b_2c_1), \;\; y = r(c_1a_2 - c_2a_1) \;\text{ and }\; z = r(a_1b_2 - a_2b_1) \tag{54}$$

Conversely any x, y, z which satisfy (54) also satisfy (52) and (53).

Proof : First assume that at least one of $a_1b_2 - a_2b_1$, $b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ is non-zero. Because of cyclic symmetry, we may assume, without loss of generality, that $a_1b_2 - a_2b_1 \neq 0$. Eliminating y between (52) and (53) gives

$$x(a_1b_2 - a_2b_1) = z(b_1c_2 - b_2c_1) \tag{55}$$

while eliminating x gives

$$y(a_1b_2 - a_2b_1) = z(c_1a_2 - c_2a_1) \tag{56}$$

Since $a_1b_2 - a_2b_1 \neq 0$, the result follows simply by setting $r = \dfrac{z}{a_1b_2 - a_2b_1}$. The converse is trivial and proved by direct substitution. (Note that the converse holds even if $b_1c_2 - b_2c_1$, $c_1a_2 - c_2a_1$ and $a_1b_2 - a_2b_1$ all vanish.)

We still have to prove the first part of the theorem in the case where $a_1b_2 - a_2b_1$, $b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ are all zero. In this case we construct a common solution of the system as follows. If all the six coefficients a_1, b_1, c_1, a_2, b_2 and c_2 vanish then we can give any values to x, y and z we please and get a solution. Otherwise, we may suppose, without loss of generality, that $a_1 \neq 0$. Let $t = \frac{a_2}{a_1}$. Then we get $a_2 = ta_1$, $b_2 = tb_1$ and $c_2 = tc_1$. Consequently, any solution of (52) will also be a solution of (53). So we shall be through if we can construct a non-zero solution of (52). And this is very easy. We can, for example, set $x = -b_1$, $y = a_1$ and $z = 0$. This solution is non-trivial since $a_1 \neq 0$. Finally, note that since the system is homogeneous, whenever $x = x_0, y = y_0, z = z_0$ is a solution, so is $x = \lambda x_0, y = \lambda y_0, z = \lambda z_0$ for any number λ. Hence the existence of one non-trivial solution implies that of infinitely many such solutions. Thus we have proved the theorem completely.

This theorem has a geometric significance. Each of the equations (52) and (53) represents a plane in the three-dimensional space, passing through the origin. Therefore the set of common solutions to (52) and (53) is the line of intersection of these two planes. This is a straight line passing through the origin and (54) gives parametric equations of that line. Here r is a parameter and every point on the line is expressible in terms of this parameter.

The theorem above can also be paraphrased in terms of vectors. Let **i,j,k** be a right-handed orthonormal system of unit vectors. Denote by $\mathbf{v}, \mathbf{u_1}$ and $\mathbf{u_2}$ respectively, the vectors $x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$, $a_1\mathbf{i}+b_1\mathbf{j}+c_1\mathbf{k}$ and $a_2\mathbf{i}+b_2\mathbf{j}+c_2\mathbf{k}$. Then (52) and (53) mean that the dot products $\mathbf{v}\cdot\mathbf{u_1}$ and $\mathbf{v}\cdot\mathbf{u_2}$ both vanish, or equivalently, that the vector $\mathbf{v}$ is perpendicular to $\mathbf{u_1}$ and to $\mathbf{u_2}$. The conclusion (54) amounts to saying that $\mathbf{v}$ is a scalar multiple of the cross product $\mathbf{u_1}\times\mathbf{u_2}$.

There is yet another way of looking at Theorem 7. Ostensibly, (52) and (53) form a system of two equations in three unknowns, viz., x, y and z. So it is not surprising that it has infinitely many solutions. However, if at least one of $b_1c_2-b_2c_1$, $c_1a_2-c_2a_1$ and $a_1b_2-a_2b_1$ is non-zero, then because of (54), all these solutions are multiples of each other. So in a sense they are not really different solutions. The explanation lies in the fact that the equations are homogeneous. Suppose, without loss of generality, that $a_1b_2-a_2b_1\neq 0$. If we divide (52) and (53) by z and call $\frac{x}{z}$ and $\frac{y}{z}$ as u and v respectively, then we get

$$\begin{aligned} a_1u+b_1v &= -c_1 \\ a_2u+b_2v &= -c_2 \end{aligned}$$

which is a non-homogeneous system of linear equations in just two variables, viz., u and v. Since $a_1b_2-a_2b_1\neq 0$, this system has a unique solution viz., $u=\dfrac{b_1c_2-b_2c_1}{a_1b_2-a_2b_1}$, $v=\dfrac{c_1a_2-c_2a_1}{a_1b_2-a_2b_1}$. From this unique solution we get the infinitely many solutions of (52) and (53), given by (54).

More generally, in any homogeneous system what matters is the relative proportions of the the values of the various variables rather than the actual values. As a result, a homogeneous equation in n variables is really speaking an equation in $n-1$ variables. This observation applies to non-linear equations as well. For example, the solutions of $x^2-3xy=y^2$ can be obtained from those of the quadratic $\alpha^2-3\alpha=1$ which results if we divide the first equation throughout by y^2 and set $\alpha=\frac{x}{y}$. The latter is an equation in just one variable.

As an immediate consequence of Theorem 7, we get a criterion for the existence of a non-trivial solution of a homogeneous system of three linear equations in three variables and an interesting corollary of it.

Theorem 8: The homogeneous linear system

$$\begin{aligned} a_1x+b_1y+c_1z &= 0 \\ a_2x+b_2y+c_2z &= 0 \\ a_3x+b_3y+c_3z &= 0 \end{aligned}$$

has a non-trivial solution if and only if $\Delta = 0$, where $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$.

Proof : First assume that the system has a non-trivial solution. Then at least one of the three variables, say the variable z has a non-zero value. Dividing by it, we see that the third column vector of Δ is a linear combination of the other two. Specifically, $(c_1, c_2, c_3)^T = -\frac{x}{z}(a_1, a_2, a_3)^T + \frac{-y}{z}(b_1, b_2, b_3)^T$. Hence by the properties of the determinants, the determinant vanishes.

Conversely suppose $\Delta = 0$. We have to show that the system has at least one non-trivial solution. We make two cases. Suppose first that at least one of the three expressions $b_1c_2 - b_2c_1$, $c_1a_2 - c_2a_1$ and $a_1b_2 - a_2b_1$ is non-zero. When the determinant is expanded w.r.t. its third row, it equals $a_3(b_1c_2 - b_2c_1) + b_3(c_1a_2 - c_2a_1) + c_3(a_1b_2 - a_2b_1)$. So if $\Delta = 0$, then $x = b_1c_2 - b_2c_1,\ y = c_1a_2 - c_2a_1,\ z = a_1b_2 - a_2b_1$ is a solution of the third equation in the system. Since this is also a solution of the first two equations as seen by direct substitution, we see that the system has a non-trivial solution. Finally assume that the three expressions $b_1c_2 - b_2c_1$, $c_1a_2 - c_2a_1$ and $a_1b_2 - a_2b_1$ all vanish. Then one of the first two equations is superfluous as being either identically 0 or a multiple of the other. Removing this redundant equation from the system, the other two have a non-trivial solution by the last theorem. As this is also a solution of the removed equation, the given system has a non-trivial solution. So the proof is complete in all cases.

Note that the existence of a non-trivial solution of the system above can be paraphrased to say that one of the columns is a linear combination of the other two, or equivalently, that some non-trivial linear combination of the columns vanishes. Indeed, the proof of the direct implication was based on this paraphrase. By taking transpose, the same thing is true of the rows of a determinant associated with a system. As a result, we get the following corollary.

Corollary 9: A 3×3 determinant vanishes if and only if some nontrivial linear combination of its rows/columns vanishes identically.

We remark once again that both Theorem 8 and Corollary 9 are true if the integer 3 is replaced by any other positive integer. But to prove them in the higher dimensions, the case-by-case arguments above become messy.

As another application of this theorem, suppose we are given that

$$x = cy + bz, \quad y = az + cx \ \text{ and } \ z = bx + ay$$

where x, y, z are not all zero. Then we can prove that (JEE 1978)

$$a^2 + b^2 + c^2 + 2abc = 1. \tag{57}$$

Rewriting the given system as $\begin{cases} 1x - cy - bz &= 0 \\ cx - 1y + az &= 0 \\ bx + ay - 1z &= 0 \end{cases}$ and applying the theorem above, we get that the 3×3 determinant $\begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix}$ vanishes. A

direct expansion of the determinant gives (57).

Sometimes the problem asks to find the conditions for the existence of non-trivial solutions which are further required to satisfy some other restriction. Here is an example. (JEE 1979)

For what value of k do the following system of equations possess a non-trivial solution over the set of rationals?

$$x + ky + 3z = 0 \tag{58}$$

$$3x + ky - 2z = 0 \tag{59}$$

$$2x + 3y - 4z = 0 \tag{60}$$

For that value of k, find all solutions of that system.

Once again, by Theorem 8 the system will have a non-trivial solution (which may or may not be rational) if and only if the determinant $\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix}$ vanishes, i.e., if and only if $6-4k+8k+27-6k = 0$ which gives $k = \frac{33}{2}$. Moreover, from the proof of Theorem 8, for this value of k, any solution of (58) and (59) will also be a solution of (60). Hence we only have to find the common solutions of (58) and (59). By Theorem 7, these are of the form (putting $k = \frac{33}{2}$)

$$x = r(\frac{33}{2} \times (-2) - \frac{33}{2} \times 3) = -\frac{165}{2}r, \;\; y = 11r, \;\; z = -33r \tag{61}$$

for some number r. Naturally, rational values of r will give solutions in which x, y, z are all rationals.

Like Theorem 7, Theorem 8 can also be looked at algebraically. Its statement can be paraphrased to say that if the variables x, y, z are eliminated from the three equations then we get $\Delta = 0$. Normally, we can eliminate only two variables among three equations. But in Theorem 8, we are eliminating three variables, viz., x, y and z in three equations. Again, the explanation lies in the fact that as the equations are homogeneous, there are really only two variables and we are eliminating them in three equations. Normally, a system of equations is said to be consistent when it has at least one solution. However, in the case of a homogeneous system, since it always has a trivial solution, consistency is taken to mean the existence of a non-trivial solution. Theorem 8 can be looked upon as giving a necessary and sufficient condition for the consistency of a homogeneous system of three linear equations in three unknowns.

We now turn to the non-homogeneous system (44). Let again $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$. If $\Delta = 0$, then by Theorem 8, the associated homogeneous system has infinitely many solutions, because suppose $x = x_0, y = y_0, z = z_0$ is a non-trivial solution. Then for every λ, $x = \lambda x_0, y = \lambda y_0, z = \lambda z_0$ is also a solution. In this case it may happen that (44) has no solution at all. However, if at all it has one solution, say, $x = x^*, y = y^*, z = z^*$, then for every λ, $x = x^* +$

$\lambda x_0,\ y = y^* + \lambda y_0,\ z = z^* + \lambda z_0$ is also a solution as we see by direct substitution. (For example, in the first equation, $a_1(x^* + \lambda x_0) + b_1(y^* + \lambda y_0) + c_1(z^* + \lambda z_0) = (a_1x^* + b_1y * + c_1z^*) + \lambda(a_1x_0 + b_1y_0 + c_1z_0) = d_1 + 0 = d_1$ and similarly for the other two equations of the system.) Thus, when $\Delta = 0$, the system (47) exhibits an extreme behaviour. Either it has no solution or else it has an infinite number of solutions. There is a criterion (cf. Exercise (3.20)) to tell which possibility holds when. Also, in case the second possibility holds, there is a systematic procedure to identify all the solutions. But it is rather complicated. In numerical problems, one solution of the non-homogeneous system can often be found just by trial. The solutions of the homogeneous system can all be obtained by applying Theorem 7 suitably because when $\Delta = 0$, by Corollary 9, one of the three homogeneous equations is a linear combination of the other two. So any solution of the other two equations (given by Theorem 7 applied to these two equations) is automatically a solution of all three.

The case $\Delta \neq 0$ leads to Cramer's rule. We omit the proof as it may be found in any standard textbook. Moreover, even for $n = 3$, Cramer's rule is not a very practicable proposition. In a given numerical problem, it is far better to find the solution by *ad-hoc* methods, taking advantage of certain peculiarities of the given problem (e.g., vanishing of some of the coefficients) as we illustrate in the following problem where solving a 3×3 linear system is clubbed together (somewhat artificially) with other gadgets such as geometric progressions and quadratic equations.

Let a, b, c, d be real numbers in G.P. If u, v, w satisfy the system

$$\left.\begin{aligned} u + 2v + 3w &= 6 \\ 4u + 5v + 6w &= 12 \\ 6u + 9v &= 4 \end{aligned}\right\} \tag{62}$$

then show that the roots of the equation

$$\left(\frac{1}{u} + \frac{1}{v} + \frac{1}{w}\right)x^2 + \left[(b-c)^2 + (c-a)^2 + (d-b)^2\right]x + u + v + w = 0 \tag{63}$$

and those of the equation

$$20x^2 + 10(a-d)^2x - 9 = 0 \tag{64}$$

are reciprocals of each other. (JEE 1999)

Before tackling the problem let us note that if α, β are the roots of a quadratic $ax^2 + bx + c = 0$, then by Theorem 2, the equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is $a(\frac{1}{x})^2 + b(\frac{1}{x}) + c = 0$, or equivalently, the quadratic $cx^2 + bx + a = 0$. Hence if we want to show that the roots of (64) are reciprocals of those of (63), then we must show that the coefficients $20, 10(a-d)^2$ and -9 are proportional, respectively, to $u + v + w$, $(b-c)^2 + (c-a)^2 + (d-b)^2$ and $\frac{1}{u} + \frac{1}{v} + \frac{1}{w}$. We are yet to find u, v, w. But the middle coefficients do not depend upon u, v, w and so their ratio can be found from the fact that a, b, c, d are in G.P. Letting r be the common ratio of the G.P., we get $b = ar, c = ar^2$ and $d = ar^3$. So,

$(b-c)^2+(c-a)^2+(d-b)^2$ equals $a^2[(r-r^2)^2+(r^2-1)^2+(r^3-r)^2]$, and hence $a^2(r-1)^2[r^2+(r+1)^2+r^2(r+1)^2]$. Similarly, $(a-d)^2=a^2(1-r^3)^2=a^2(r-1)^2[(r^2+r+1)^2]$. Hence we get

$$\begin{aligned}\frac{(b-c)^2+(c-a)^2+(d-b)^2}{10(a-d)^2} &= \frac{a^2(r-1)^2[r^2+(r+1)^2+r^2(r+1)^2]}{10a^2(r-1)^2[(r^2+r+1)^2]} \\ &= \frac{r^4+2r^3+3r^2+2r+1}{10(r^4+2r^3+3r^2+2r+1)}=\frac{1}{10} \qquad (65)\end{aligned}$$

as we see by directly expanding the polynomials. Thus the problem is now reduced to showing that

$$\frac{\frac{1}{u}+\frac{1}{v}+\frac{1}{w}}{-9}=\frac{1}{10}=\frac{u+v+w}{20} \qquad (66)$$

To do this, the best (and indeed, the only) way is to solve the system (62). Instead of applying Cramer's rule, we note that the third equation does not involve w. So elimination of w between the first two equations will give us a system of two equations in u and v. The elimination can be done simply by subtracting twice the first equation from the second to give $2u+v=0$. Along with the third equation, this gives $6v=4$, i.e., $v=\frac{2}{3}$ and then $u=-\frac{v}{2}=-\frac{1}{3}$. Putting these values in the first equation gives $3w=6+\frac{1}{3}-\frac{4}{3}=5$ and hence $w=\frac{5}{3}$. Thus the solution of (65) is $u=-\frac{1}{3}$, $v=\frac{2}{3}$, $w=\frac{5}{3}$. This gives $u+v+w=\frac{6}{3}=2$ and $\frac{1}{u}+\frac{1}{v}+\frac{1}{w}=-3+\frac{3}{2}+\frac{3}{5}=-\frac{9}{10}$. So we see that (66) holds. As noted before, this completes the solution.

The normal tendency in a problem like this is to begin by solving the system (62) and then to show the proportionality of the coefficients of the quadratics. In the solution above, this has been reversed. Although the work involved in both the approaches is exactly the same, the approach given here has a slight advantage. If you begin by solving (62) first, and make a numerical slip, it would show much later. In the approach taken here, on the other hand, the problem is first reduced to proving (66) and this serves as a kind of touchstone to check the accuracy of the calculations. In general, in any problem, instead of hastily starting from the data, it is always a good idea to reduce the goal of the problem to a level at which it can inspire how to reach it from the data. Although in the present problem there is no alternative to solving the system (62), in some problems it may happen that you may get the relevant information from the data even without fully solving the system. In that case, there is no point in wasting your precious time in doing something which is not vitally needed to reach the goal.

Note also that in (65), the L.H.S. *ought to* simplify to a constant. If, instead, it depended on r, then an equation like (66) cannot hold because the first and the last terms are some fixed numbers (as u, v, w are so) while the middle one is not. An observation like this often guards you against numerical slips in (65). In fact, this is a built-in hint for the perceptive student.

Comment No. 18:

At the end of Comment No. 9, we encountered a quadratic which could be solved without using the quadratic formula, taking advantage of the particular form in which it was given. Similarly, there are situations where instead of solving a linear system by any of the standard methods for it, some *ad-hoc* method based on some special features of the problem comes in more handy. As an example, consider the system

$$x + y + z = 1,\ x + ay + a^2z = a^3,\ \text{and}\ \ x + by + b^2z = b^3$$

where a, b are some constants. Here, a slick solution can be based on the fact that for each equation, the coefficients of x, y, z and the R.H.S. form a G.P. with common ratio $1, a, b$ respectively. As a result, we see that $1, a, b$ are precisely the roots of the cubic equation $t^3 - zt^2 - yt - x = 0$ (in the variable t). So x, y, z can be obtained using (11) as $x = ab, y = -a - b - ab$ and $z = 1 + a + b$. (It is tacitly assumed here that $1, a, b$ are all different as otherwise the system becomes degenerate.)

See Exercise (3.29) for another illustration of the (somewhat unusal) technique of designing a suitable cubic to paraphrase a problem. See also Exercises (7.28) and (9.61).

In passing, we remark once again that for non-linear systems of equations, there is no golden method. When we cannot solve such a system fully, the next best thing is to reduce it to a simpler system, e.g. a polynomial equation having some of the unknowns as its roots. Methods are available to find the roots approximately. From these values, one can sometimes determine the remaining unknowns. See Exercise (3.26)(c) for an illustration.

EXERCISES

3.1 Find the number of roots (counted according to multiplicity) of the equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$. (1984)

3.2 If $\alpha, \beta, \gamma, \delta$ be the roots of the equation $ax^4 - bx^3 + cx^2 - dx + e = 0$, prove that $a^4(\beta + \gamma + \delta - \alpha)(\alpha - \beta + \gamma + \delta)(\alpha + \beta - \gamma + \delta)(\alpha + \beta + \gamma - \delta)$ equals $-b^4 + 4ab^2c - 8a^2bd + 16a^3e$.

3.3 By completely factorising and then differentiating the polynomial $x^n + ax + b$, show that the product of the squares of the differences of its roots is $(-1)^{n(n+1)/2}\{-a^n(n-1)^{n-1} + (-1)^n b^{n-1} n^n\}$. (A proof without differentiation is also possible. The product considered here is called the **discriminant** of a (monic) polynomial, since its vanishing or otherwise tells you whether the roots are all distinct or not.)

3.4 Let s_r be the sum of the r-th powers of the roots of the equation $x^n - x^{n-1} - x^{n-2} - \ldots - x - 1 = 0$. Using Newton's identities, prove that $s_r = 2^r - 1$ for $1 \leq r \leq n$.

3.5 (a) If $a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1} + a_nx^n$ is a polynomial with $a_0 \neq 0$, prove that the sum of the reciprocals of its roots is $-\frac{a_1}{a_0}$.

(b) Let t_m denote the sum of the m-th powers of the reciprocals of the roots of the equation $1 + \dfrac{x}{1!} + \dfrac{x^2}{2!} + \ldots + \dfrac{x^n}{n!} = 0$. Show that $t_2 = t_3 = \ldots = t_n = 0$ and $t_{n+1} = \frac{1}{n!}$. (These polynomials have very few real roots. See Exercise (16.20)(b).)

3.6 Let p and q be the roots of the equation $x^2 - 2x + A = 0$ and let r and s be the roots of the equation $x^2 - 18x + B = 0$. If $p < q < r < s$ are in an arithmetic progression, determine the values of A and B. (1997)

3.7 (a) If $f(x) = x^9 - 6x^8 - 2x^7 + 12x^6 + x^4 - 7x^3 + 6x^2 + x - 3$, find $f(6)$. (1979)

(b) A polynomial in x of degree greater than three, leaves remainders 2, 1 and -1 when divided, respectively, by $(x-1), (x+2)$ and $(x+1)$. What will be the remainder when it is divided by $(x-1)(x+2)(x+1)$? (1978)

3.8 Solve the following equations for x:

(a) $\log_7 \log_5(\sqrt{x+5} + \sqrt{x}) = 0$. (1986)

(b) $\sqrt{x+1} - \sqrt{x-1} = 1$. (1978)

(c) $\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$. (1997*)

(d) $\log_{(2x+3)}(6x^2 + 23x + 21) = 4 - \log_{(3x+7)}(4x^2 + 12x + 9)$. (1987)

(e) $2\log_x a + \log_{ax} a + 3\log_{a^2x} a = 0$, where $a > 0$. (1978)

3.9 (a) Let the harmonic and the geometric mean of two positive numbers be in the ratio $4 : 5$. Determine the ratio of the two numbers. (1992)

(b) If $\log_3 2, \log_3(2^x - 5)$ and $\log_3(2^x - \frac{7}{2})$ are in arithmetic progression, determine the value of x. (1990)

(c) If $x < 0, y < 0, x + y + \frac{x}{y} = \frac{1}{2}$ and $(x+y)\frac{x}{y} = -\frac{1}{2}$, find the values of x and y. (1990)

3.10 (a) One root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the n-th power of the other. Show that $(ac^n)^{\frac{1}{n+1}} + (a^nc)^{\frac{1}{n+1}} + b = 0$. (1983)

(b) Let α and β be the roots of the equation $(x-a)(x-b) = c$, $c \neq 0$. Find the roots of $(x-\alpha)(x-\beta) + c = 0$. (1992)

(c) Let a, b, c be real numbers with $a \neq 0$ and let α, β be the roots of the equation $ax^2 + bx + c = 0$. Express the roots of $a^3x^2 + abcx + c^3 = 0$ in terms of α, β. (2001)

(d) If α, β are the roots of $x^2 + px + q = 0$ and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s. Deduce the condition that the equations have a common root. (1979)

(e) If the quadratic equations $x^2+ax+b=0$ and $x^2+bx+a=0$ $(a \neq b)$ have a common root, find the numerical value of $a+b$. (1986)

(f) Find the harmonic mean of the roots of the equation $(5+\sqrt{2})x^2 - (4+\sqrt{5})x + 8 + 2\sqrt{5}$. (1999)

3.11 (a) If l, m, n are real and $l \neq m$, determine if the roots of the equation $(l-m)x^2 - 5(l+m)x - 2(l-m) = 0$ are real and equal, real and unequal or complex. (1979)

(b) If $a < b < c < d$, determine if the roots of the equation $(x-a)(x-c) + 2(x-b)(x-d) = 0$ are real and distinct. (1984)

(c) If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, where $ac \neq 0$, must $P(x)Q(x) = 0$ have at least two real roots? (1985)

(d) Let a, b, c be positive. Which, if any, of the following statements is true about the roots of the equation $ax^2 + bx + c = 0$? (1980)

(A) they are real and negative (B) they have negative real parts.

3.12 If the product of the roots of the equation $x^2 - 3x + 2e^{2\ln k} - 1 = 0$ is 7, determine for which value(s) of k the roots are real. (1984)

3.13 If a, b, c are in G.P. determine the type, if any, of progression in which $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ must lie in order that the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root. (1985)

3.14 Prove that if α is a double root of a real quadratic polynomial $p(x)$, then the graph of $y = p(x)$ touches the x-axis at the point $(\alpha, 0)$. (A formal definition of the concept of 'touching' or tangency requires the use of a limit. For the present take it only to mean that the points on the graph of $y = p(x)$ near to the point $(\alpha, 0)$ all lie on the same side of the x-axis. More generally the result holds for a root of even multiplicity of any polynomial. But the proof is not so easy. Moreover, the graph may very well cross the x-axis at some other points, as is the case with $p(x) = x^4 - x^2$ which has 0 as a double root and 1 and -1 as simple roots.)

3.15 (a) Suppose $f(x) = ax^2 + bx + c$, where a, b, c are real and $a \neq 0$. Suppose there are real numbers $x_1 < x_2$ such that $f(x_1)$ and $f(x_2)$ are of opposite signs. Prove that there exists some $x_3 \in (x_1, x_2)$ such that $f(x_3) = 0$. More generally, suppose that m is a real number which is 'intermediate to' $f(x_1)$ and $f(x_2)$ in the sense that one of $f(x_1), f(x_2)$ is smaller and the other larger than m. (Put differently, either $f(x_1) < m < f(x_2)$ or $f(x_2) < m < f(x_1)$. Then show that there exists some $x_4 \in (x_1, x_2)$ such that $f(x_4) = m$. (This property is called the **Intermediate Value Property**. Actually, it is true of any continuous function, and not just a quadratic. But the proof requires methods from calculus. See Chapter 16, Comment No. 3. For quadratics, a purely algebraic proof is possible as shown here.)

(b) Let a, b, c be real numbers, $a \neq 0$. If α is a root of $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then show that the equation $a^2x^2 + 2bx + 2c = 0$ has at least one root γ such that $\alpha < \gamma < \beta$. (1989 modified)

3.16 Solve $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ after finding one root by inspection.

(1983 modified)

3.17 If the equations $\dfrac{cy+bz}{b-c} = \dfrac{az+cx}{c-a} = \dfrac{bx+ay}{a-b} = \dfrac{ax+by+cz}{a+b+c}$ are consistent, show that either $a+b+c=0$ or $(a-b)(b-c)(c-a) = abc$.

(JAT 1979)

3.18 (a) Prove that the system $\begin{cases} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \end{cases}$ either has no solution or has infinitely many solutions. Find a necessary and sufficient condition for the latter. [*Hint:* Once again consider the two cases: one where $a_1b_2 - a_2b_1, b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ all vanish and the other when at least one of them is non-zero.]

(b) Determine the number of solutions of $\begin{cases} x + 2y + 2z &= 1 \\ 2x + 4y + 4z &= 9 \end{cases}$. (1979)

(c) Find the number of values of k for which the system of equations $\begin{cases} (k+1)x + 8y &= 4k \\ kx + (k+3)y &= 3k-1 \end{cases}$ has infinitely many solutions. (2002)

3.19 If $x^2 + px + 1$ is a factor of $ax^3 + bx + c$, determine which, if any, of the following is true. (1980)

(A) $a^2 + c^2 = -ab$ (B) $a^2 - c^2 = -ab$ (C) $a^2 - c^2 = ab$

3.20 (a) In Theorem 8, suppose $\Delta = 0$ and at least one of $a_1b_2 - a_2b_1, b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ is non-zero. Prove that the third equation is a linear combination of the first two equations, i.e., there exist some s and t such that $a_3 = sa_1 + ta_2$, $b_3 = sb_1 + tb_2$ and $c_3 = sc_1 + tc_2$.

(b) Using (a) above and Exercise (3.18)(a), obtain a criterion for the system (44) to have infinitely many solutions. [*Hint:* First consider the case when at least one of the nine expressions of the form $a_1b_2 - a_2b_1$ is non-zero.]

3.21 (a) Show that the system of equations $\begin{cases} 3x - y + 4z &= 3 \\ x + 2y - 3z &= -2 \\ 6x + 5y + \lambda z &= -3 \end{cases}$ has at least one solution for any real number λ. Find the set of solutions if $\lambda = -5$. (1983)

(b) Find the value(s) of λ for which the system of equations

$$\left\{\begin{array}{rcl} \lambda x + y + z &=& 0 \\ -x + \lambda y + z &=& 0 \\ -x - y + \lambda z &=& 0 \end{array}\right. \quad \text{has a non-trivial solution.} \qquad (1984)$$

(c) Determine for which values of a and b the system of equations

$$\left\{\begin{array}{rcl} x + 2y + 2z &=& 3 \\ 2x + 3y + 6z &=& 5 \\ 3x + y + az &=& b \end{array}\right.$$

has (i) unique solution, (ii) infinite number of solutions, (iii) no solution. (JAT 1980)

3.22 Prove that one way implication in Theorem 6 does not require the hypothesis. That is, if the system (46) to (48) has a solution then (51) holds even if $a_1b_2 - a_2b_1$, $a_2b_3 - a_3b_2$ and $a_3b_1 - a_1b_3$ are all zero.

3.23 Prove the analogue of Theorem 6 for a system of four linear equations in three unknowns, say, $a_ix + b_iy + c_iz = d_i$, where $i = 1, 2, 3, 4$. For $i = 1, 2, 3, 4$, let Δ_i be the 3×3 determinant whose row entries are $a_j\ b_j\ c_j$ for $j \neq i$. (For example, $\Delta_2 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{vmatrix}$.) Let Δ be the 4×4 determinant $\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix}$. Assume that at least one Δ_i is non-zero. Then prove that the system has a solution if and only if $\Delta = 0$. (Again, one way implication holds even when $\Delta_1, \Delta_2, \Delta_3$ and Δ_4 all vanish.)

3.24 Consider the linear system given by (44) or (45). Suppose the variables $x_1, x_2, \ldots, x_n$ are themselves linear combinations of some other variables, say $u_1, u_2, \ldots, u_p$ given by the system

$$\left.\begin{array}{rcl} x_1 &=& b_{11}u_1 + b_{12}u_2 + \ldots b_{1p}u_p \\ \vdots && \vdots \\ x_n &=& b_{n1}u_1 + b_{n2}u_2 + \ldots b_{np}u_p \end{array}\right\} \qquad (67)$$

or, in a compact form $\mathbf{x} = \mathbf{Bu}$, where $\mathbf{B}$ is the $n \times p$ matrix whose entries are the coefficients b_{ij} of the system above and $\mathbf{x}$ and $\mathbf{u}$ are the $n \times 1$ and $p \times 1$ column vectors with entries $x_1, x_2, \ldots, x_n$ and $u_1, u_2, \ldots, u_p$ respectively. Prove that when these values are substituted into (44), the resulting system of equations in the variables $u_1, u_2, \ldots, u_p$ can be written in the compact form as $\mathbf{Cu} = \mathbf{b}$, where $\mathbf{C}$ is the product matrix $\mathbf{AB}$. (This justifies the apparently clumsy definition of matrix multiplication.)

3.25 Represent the following transformations in matrix notation. Also find their composite and express z_1, z_2 in terms of x_1, x_2. (JAT 1979)

$$\text{I:}\quad \begin{array}{rcl} x_1 &=& 2y_1 - y_2 \\ x_2 &=& y_1 + y_2 \end{array} \qquad \text{II:}\quad \begin{array}{rcl} y_1 &=& z_1 + 2z_2 \\ y_2 &=& 2z_1 + z_2 \end{array}$$

3.26 A **Vandermonde determinant** of order n is a determinant whose (i,j)-th entry is α_j^{i-1}, where $\alpha_1, \alpha_2, \ldots, \alpha_n$ are some fixed numbers. In a full form, $D(\alpha_1, \alpha_2, \ldots, \alpha_n) = \begin{vmatrix} 1 & 1 & 1 & \ldots & 1 \\ \alpha_1 & \alpha_2 & \alpha_3 & \ldots & \alpha_n \\ \vdots & \vdots & \vdots & & \vdots \\ \alpha_1^{n-1} & \alpha_2^{n-1} & \alpha_3^{n-1} & \ldots & \alpha_n^{n-1} \end{vmatrix}$. (Occasionally, the transpose of $D(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is taken as the Vandermonde determinant. They are, of course, equal.)

(a) Prove that $D(\alpha_1, \alpha_2, \ldots, \alpha_n) = \prod_{1 \le i < j \le n} (\alpha_j - \alpha_i)$. In particular, $D(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is non-zero if the numbers $\alpha_1, \alpha_2, \ldots, \alpha_n$ are all distinct. [*Hint:* Starting from the last row, from each row subtract α_1 times the preceding row. Then apply induction on n. Interestingly, in most applications, only the second assertion is needed. One such application will be given in Chapter 9, Comment No. 6. The determinant of the coefficients of the system solved in Comment No. 18 is a Vandermonde determinant. See also Exercise (21.3).]

(b) Prove that $D^2(\alpha_1, \alpha_2, \ldots, \alpha_n)$ equals the $n \times n$ determinant whose (i,j)-th entry is s_{i+j-2}, where s_k's are defined by (13). Note that $D^2(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is also the discriminant of the monic polynomial having the α_i's as the roots, as defined in Exercise (3.3) above.

(c)* As an illustration of the remark at the end of Comment No. 18, in applications to chemistry and physics, we sometimes run into unknowns $\alpha_1, \alpha_2, \ldots, \alpha_n, \beta_1, \beta_2, \ldots, \beta_n$ that satisfy a system of $2n$ equations:

$$\alpha_1^j \beta_1 + \alpha_2^j \beta_2 + \ldots + \alpha_n^j \beta_n = a_j \; ; \quad (j = 0, 1, 2, \ldots, 2n-1)$$

where $a_0, a_1, \ldots, a_{2n-1}$ are known. Show that the elementary symmetric functions $\sigma_1, \ldots, \sigma_n$ of the α's can be expressed in terms of the a's. Hence get a polynomial with known coefficients whose roots are the α's. [*Hint:* Expand the Vandermonde determinant $D(\alpha_1, \alpha_2, \ldots, \alpha_n, x)$ w.r.t. its last column.]

3.27 Let $\mathbf{A}$ be an $n \times n$ matrix and $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its characteristic roots as defined in Exercise (2.37). Prove that the determinant of $\mathbf{A}$ is precisely the product while the trace of $\mathbf{A}$ (as defined in Exercise (2.36)) is the sum of the characteristic roots of $\mathbf{A}$.

3.28 Let $\mathbf{A}$ be a 2×2 matrix with complex entries. Prove that if $\mathbf{A}$ is Hermitian then the characteristic roots of $\mathbf{A}$ are all real while if $\mathbf{A}$ is skew-Hermitian, then they are all purely imaginary. (These results are true for square matrices of any order. But the proofs are not so easy. Note that symmetric/skew-symmetric real matrices are special cases.)

3.29 If $a + b + c = 0$, prove that $\frac{a^5+b^5+c^5}{5} = \frac{a^2+b^2+c^2}{2} \cdot \frac{a^3+b^3+c^3}{3}$. [*Hint:* Let $p = ab + bc + ca$ and $q = abc$. Note that a, b, c are the roots of the cubic $x^3 + px - q = 0$. Use Newton's identities.]

Chapter 4

NUMBER THEORY

The theory of numbers has fascinated mankind in many different parts of the world for ages. There is hardly a mathematician who did not play with numbers at least for some time. Although a 'number' could be any real or complex number, in the context of number theory, it usually means an integer; in fact, a positive integer, or a 'whole number' as it is called. Especially enigmatic are the prime numbers and their unpredictable behaviour. Number theory abounds in problems which are so simple to state that anybody who has completed primary school can understand the statement of the problem but whose solutions are extremely difficult and often require very deep methods from apparently unrelated areas of mathematics. Some of them are still unsolved. A good example is the **Goldbach conjecture** which says that every even number greater than 4 is a sum of two (not necessarily distinct) odd primes (e.g., 20 = 13+7=17+3, 48= 29+19 etc.). Although this has been verified in a large number of cases, it is still not proved for *all* even numbers.

We shall mostly deal with problems involving the divisibility of one integer by another. Especially interesting is the case where the divisor is a prime power. We shall prove a famous theorem due to Kummer in this regard. That will also take us to the representation of an integer in a given base. Regarding the divisibility by a prime, there are two famous theorems, one due to Fermat and another to Wilson. Their combinatorial proofs are given in the exercises. We also do some miscellaneous problems about divisibility where induction is the principal tool of attack. This naturally leads to a brief discussion of what are called recurrence relations. Considerations of divisibility also lead to modular arithmetic and the theory of surds. We mention without proof a theorem of Lucas regarding the congruence class of a binomial coefficient modulo a prime.

In JEE, number theory problems are not asked every year. When asked, they are mostly asked as problems on induction.

Main Problem : The number of integers n in the range, $10 \leq n \leq 1000$ having the property that $\binom{n}{k}$ is odd for all integers k $(0 \leq k \leq n)$ is ..... .

First Hint: Prove that $\binom{n}{k} = \binom{n}{k-1}\frac{n-k+1}{k}$.

Second Hint: For a given n, write n as $2^r + u$, where $0 \leq u < 2^r$. Prove that if $u < 2^r - 1$ there is some k $(0 \leq k \leq n)$ such that the highest power of 2 that divides $n - k + 1$ is higher than that which divides k.

Third Hint: Prove that n has the property in question if and only if $n + 1$ is a power of 2.

Solution: For a fixed n and k (with $1 \leq k \leq n$), writing $\binom{n}{k}$ as $\frac{n(n-1)...(n-k+2)(n-k+1)}{1.2.3.....(k-1)k}$, we get

$$\binom{n}{k} = \binom{n}{k-1}\frac{n-k+1}{k} \tag{1}$$

Now let 2^r be the highest power of 2 not exceeding n. Call $n - 2^r$ as u. Then $0 \leq u \leq 2^r - 1$. If the second inequality is strict, take k to be $u + 1$. Then $1 \leq k < 2^r$. Applying (1),

$$\binom{n}{k} = \frac{x(n-k+1)}{k} = \frac{x2^r}{k} \tag{2}$$

where x is the integer $\binom{n}{k-1}$. The highest power of 2 which divides the numerator $x2^r$ is at least 2^r. But since $k = u + 1 < 2^r$, the highest power of 2 that divides the denominator k is at most 2^{r-1}. So the ratio $\frac{x2^r}{k}$ is divisible by 2, i.e., $\binom{n}{k}$ is even. Note that in this case $2^r < n + 1 < 2^{r+1}$ and hence $n + 1$ is not a power of 2.

Thus we have shown that if $n+1$ is not a power of 2, then we can find some k $(0 < k < n)$, such that $\binom{n}{k}$ is even. In other words, n does not have the property stated in the question. We now prove the converse. That is, assume n is of the form $2^m - 1$ for some positive integer m. Then we claim that for *every* k (with $0 \leq k \leq n$), $\binom{n}{k}$ is odd. This can be done by induction using (1). (Note that the induction is not on n which is fixed, but on k which varies from 0 to n.) Trivially, $\binom{n}{0} = 1$ is odd. Also, $\binom{n}{1}$ equals n which is odd (since $n + 1$ is even, being a power of 2). For the inductive step, we apply (1) to write $\binom{n}{k}$ as $\frac{x(n-k+1)}{k} = \frac{x(2^m-k)}{k}$, where $x = \binom{n}{k-1}$ is an integer which is odd by the induction hypothesis. Let 2^s and 2^t be, respectively, the highest powers of 2 which divide $(2^m - k)$ and k. Evidently both s, t are at most m and so $2^s, 2^t$ both divide 2^m. Since 2^t divides both 2^m and k, it divides their difference $2^m - k$. Hence $t \leq s$. But writing k as $2^m - (2^m - k)$, we see that 2^s divides k, whence $s \leq t$. So $s = t$. Further, as x is odd, 2^s is also the highest power of 2 which divides $x(2^m - k)$. Putting it all together, for the fraction $\frac{x(2^m-k)}{k}$, the highest power of 2 that divides the numerator $x(2^m - k)$ is the same as that which divides the denominator k. So the integer $\frac{x(2^m-k)}{k}$ is odd, i.e., $\binom{n}{k}$ is odd, thus completing the inductive step.

Summing up, we have shown that a positive integer n has the property in question (viz., that $\binom{n}{k}$ is odd for every k with $0 \leq k \leq n$) if and only if n is of the form $2^m - 1$ for some positive integer m, which is equivalent to saying that $n+1$ is a power of 2. In the given range, viz., $10 \leq n \leq 1000$, the integers with this property are $2^4 - 1\ (= 15), 2^5 - 1, \ldots, 2^9 - 1\ (= 511)$. Hence in the given range the number of integers with the given property is 6. (This can be shortened a little by considering logarithms with base 2. Since neither 10 nor 1000 has the given property, asking how many powers of 2 lie in the interval 10 to 1000 is equivalent to asking how many integers lie in the interval from $\log_2 10$ to $\log_2 1000$. The answer is precisely the integral part of the length of this interval, i.e., the integral part of $\log 1000 - \log 10$ which is the same as $\log \frac{1000}{10} = \log 100$. Since $2^6 < 100 < 2^7$, this log has 6 as its integral part. So the desired number is 6.)

Comment No. 1:

The key idea behind the solution is that whenever an integer, say z, is expressed as a ratio of two integers, say as $\frac{v}{w}$, then in order to test if z is even or odd, we must consider the highest powers, say 2^s and 2^t of 2 that divide the numerator v and the denominator w respectively. Since v is an integral multiple of w (viz., $v = zw$), we must have $2^s \geq 2^t$. If equality holds then z is odd. If strict inequality holds, then it is even. In our problem, z equals $\binom{n}{k}$. Using (1) we rewrote this as $\frac{v}{w}$, where $v = \binom{n}{k-1}(n-k+1)$ and $w = k$. Although this approach finally led to the solution, admittedly, it was a rather tricky one. Although the proof of (1) is very short and straightforward, it is not one of the more standard identities about the binomial coefficients (such as the symmetry relation $\binom{n}{k} = \binom{n}{n-k}$ or the reduction formula $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$) which are useful in many problems. How, then, does one come up with the idea that (1) will be useful in the present problem?

The answer lies in experimentation. After a little experimentation, it is not hard to come up with the right guess (viz., that a positive integer n has the property that $\binom{n}{k}$ is odd for all k, $(0 \leq k \leq n)$ if and only if it is of the form $2^m - 1$ for some m), even though proving the guess requires some work. Experimentation cannot be a substitute for a rigorous mathematical proof. But it can often inspire such a proof and the present problem is an excellent example of this. When one cannot readily think of the key idea that will lead to the solution, it often pays to do some experimentation, especially in problems involving natural numbers. To encourage this habit, it is instructive to go over the present problem again to see how one can guess the answer and, later on, the proof too.

Naturally, we begin the experimentation with lower values of n. So we fix n and then for various values of k, test if $\binom{n}{k}$ is even or odd. Naturally, we would like to give increasing values, $k = 1, 2, 3, \ldots$. (There is no point in taking $k = 0$ as $\binom{n}{0}$ is always 1.) Because of the symmetry relation $\binom{n}{k} = \binom{n}{n-k}$, we need to scan only values of k upto $\frac{n}{2}$. Also, the moment we come across some k

for which $\binom{n}{k}$ is even, we need not scan further, because we already know that n cannot have the desired property. We then replace n with some higher integer and begin scanning again with $k = 1, 2, \ldots$. It is clear that if n itself is even then $\binom{n}{1} = n$ is even and so n cannot have the desired property. So we need to conduct the experiments only for odd values of n. (This simple observation saves nearly half our work. The ability to quickly identify a few such cases where the experiment is destined to fail is a valuable asset in any experiment, whether in mathematics or in real life.)

Now, coming to the experiment itself, the first two odd values of n, viz., $n = 1$ and $n = 3$ are both favourable because with these values n does have the property in question as one checks. The next value, viz., 5 is not favourable, because $\binom{5}{2} = 10$ is even. We should therefore try the next odd value of n, viz., $n = 7$ and that's what a mechanical experimenter would do. But someone who wants to learn from the experiment, would pause a little to ponder exactly what makes $\binom{5}{2}$ even. This may seem a silly question since $\binom{5}{2}$ equals 10. But not so. If we write $\binom{5}{2}$ as $\frac{5.4}{1.2}$ we see that the second factor in both the numerator and the denominator are even. But the one in the numerator (viz., 4) is divisible by a higher power of 2 than the corresponding factor 2 in the denominator and this is the reason why $\binom{5}{2}$ is even. There is not much to conclude from this right away. But we make a mental note of it and proceed to the next odd value of n, viz., $n = 7$.

For $n = 7$, the binomial coefficients $\binom{n}{k}$ for $k = 1, 2, \ldots, 7$ are, respectively, 1, 7, 21, 35, 35, 21, 7 and 1. They are all odd. So the number 7 has the desired property like 1 and 3. A perceptive person may notice right at this stage that all these three numbers are of the form $2^m - 1$ and will be tempted to try the next such number, viz., $n = 15$. When he sees that $\binom{15}{k}$ is odd for all k he will be convinced that the guess is correct. But what about the proof? For this, let us compare what happens to the successive binomial coefficients for $n = 11$ (an unfavourable case) and $n = 15$ (a favourable case). In the first case, the binomial coefficient $\binom{11}{k}$ is a ratio of two integers. The numerator is the product $11.10.\ldots.(13-k)(12-k)$ whose factors are alternately odd and even. The denominator is the product $1.2.\ldots.(k-1)k$, where also the factors are alternately odd and even. Now, the fourth factor in the numerator is 8 which is a high power of 2 (in fact the highest power of 2 not exceeding 11). To match it, in the denominator we have only 4. So, even without calculating $\binom{11}{4}$ explicitly we can tell that it is an even integer by writing it as $\frac{x8}{4}$, where x is the integer $\frac{11.10.9}{1.2.3}$ which is the same as $\binom{11}{3}$. By a similar argument $\binom{13}{6}$ is even because now the sixth factor in the numerator is 8 which is pitted against 6 in the denominator. (Of course, in the case of 13, we don't have to wait till $\binom{11}{6}$ to conclude that 13 is not a favourable case, because already $\binom{13}{2}$ is even since the factor 12 in the numerator goes with 2 in the denominator. But the reasoning applied to the factor 8 may also apply to show that, more generally, if $n = 2^r + u$, where $0 \leq u < 2^r - 1$, then $\binom{n}{k}$ will be even when k is so chosen that $n - k + 1$ equals this highest power, viz., 2^r, i.e., when $k = n + 1 - 2^r = u + 1$.)

What happens when n is 15?. In this case, the highest power of 2 less

than or equal to 15 is 8. If in the numerator of $\binom{n}{k}$, we go down starting from 15, then to reach 8 we have to wait till the eighth factor. But the eighth factor in the denominator is also 8, which is an equally high power of 2. The same happens with the lower powers of 2, viz., 4 and 2. By the time you reach a factor in the numerator which is divisible by 4, you also get a factor in the denominator which is divisible by 4. In general, as we saw above, the highest power of 2 that divides $2^m - k$ is the same as the highest power of 2 that divides k. So in this case, the binomial coefficients will all be odd.

Once these observations are made, writing the proof is mostly a clerical matter. The identity (1) is nothing but a generalisation of the equation $\binom{11}{4} = \binom{11}{3}.\frac{8}{4}$. Of course, not everybody (especially at the Junior College level) is equally good at such clerical work. There are students who can come up with the right ideas but who fumble in formulating an acceptable proof. Such students can do very well on multiple choice or fill into the blank type questions. Questions like this one are a boon to such students. The present question, if asked in such a form, will be an excellent way to test the thinking ability of a student as isolated from his ability to put his thoughts into a mathematically precise form.

Comment No. 2:

Although the present problem is a good example of the power of experimentation, the fact remains that the key idea (regarding the highest powers of 2 which divide the numerator and the denominator of a ratio of two integers) was applied to $\binom{n}{k}$ in an unexpected manner, i.e., after first writing this ratio as $\frac{v}{w}$, where $v = \binom{n}{k-1}(n-k+1)$ and $w = k$. Since $\binom{n}{k}$ is already a ratio of two integers, viz., $n!$ and $k!(n-k)!$, it is natural to see, at least as a first attempt, if a solution can be given by considering the highest powers of 2 that divide these numbers. Such a solution is certainly possible. It crucially uses the idea of the **greatest integer function** (i.e., the function $f(x) = [x]$, where $[x]$ denotes the greatest integer not exceeding x). This function is a standard example of a discontinuous function and it appears so frequently in problems where the continuity of a given function is to be tested and almost nowhere else, that a Junior College student of mathematics can hardly be blamed if he thinks that the only use of this function is to create weird counter-examples! But it is not so. A fairly common use is that it gives a succinct definition of **rounding** a real number, say x, to the integer, say $r(x)$, which is nearest to it (with the understanding that if there are two such integers, we take the larger one). Thus, $r(3.1) = 3, r(3.6) = 4, r(3.5) = 4, r(-3.1) = -3, r(-3.6) = -4$ and $r(-3.5) = -3$. It is easy to show that $r(x)$ is simply $[x + \frac{1}{2}]$ for all real x. (See Exercise (6.28) for a few other properties of the greatest integer function.)

(The greatest integer function is also often called the **floor function** and denoted by $\lfloor x \rfloor$, rather than by $[x]$. This is done especially when we also want to consider another closely related function, called the **ceiling function**, denoted by $\lceil x \rceil$ and defined as the smallest integer greater than or equal to x. Thus, for example, $\lceil 3.1 \rceil = 4, \lceil 3 \rceil = 3$ and $\lceil -3.1 \rceil = -3$. Its properties are analogous to that of the greatest integer function. In fact we have $\lceil x \rceil = \lfloor x \rfloor + \delta$ where δ

equals 0 or 1 according as x is or is not an integer. As a situation where both the floor and the ceiling functions are involved, suppose we are putting n apples into bags with m in each. Then we shall fill $\lfloor \frac{n}{m} \rfloor$ bags completely. But we need $\lceil \frac{n}{m} \rceil$ bags to put all the n apples.)

Returning to the Main Problem we now give a solution that will use the greatest integer function in a very meaningful way.

Let 2^a, 2^b and 2^c be the highest powers of 2 that divide $k!$, $(n-k)!$ and $n!$ respectively. Then 2^{a+b} is the highest power of 2 that divides the denominator $k!(n-k)!$ and so we must have $a+b \leq c$ as otherwise the ratio will not be an integer. Further, $\binom{n}{k}$ is odd if and only if $a+b=c$.

To obtain an expression for c, note that $n!$ is the product of the integers from 1 to n. Out of these, $[\frac{n}{2}]$ are even, where $[x]$ denotes the integral part of a real number x. Each of these is divisible by 2^1 and hence contributes 1 to the exponent c. However, among these, those integers that are divisible by 4 contribute more. The number of such integers is $[\frac{n}{4}]$. They should be counted once more because they contribute 2 to the exponent c. Again, among these, $[\frac{n}{8}]$ should be counted once more. This goes on till 2^r, where 2^r is the highest power of 2 not exceeding n. Or in other words, $2^r \leq c < 2^{r+1}$.

Thus we get $c = \sum_{i=1}^{r} [\frac{n}{2^i}]$. Note that for $i > r$, $[\frac{n}{2^i}] = 0$ and so we can as well write the sum as an ostensibly infinite sum $\sum_{i=1}^{\infty} [\frac{n}{2^i}]$ which is more convenient when we are dealing with more than one integer at a time. Writing c_i for $[\frac{n}{2^i}]$ for brevity, we thus have

$$c = \sum_{i=1}^{\infty} c_i \quad \text{where } c_i = [\frac{n}{2^i}] \text{ for } i \geq 1 \tag{3}$$

$$\text{and similarly} \quad a = \sum_{i=1}^{\infty} a_i \quad \text{where } a_i = [\frac{k}{2^i}] \text{ for } i \geq 1 \tag{4}$$

$$\text{and} \quad b = \sum_{i=1}^{\infty} b_i \quad \text{where } b_i = [\frac{n-k}{2^i}] \text{ for } i \geq 1 \tag{5}$$

For every $i \geq 1$, we have $\frac{n}{2^i} = \frac{k}{2^i} + \frac{n-k}{2^i}$. The crucial step now is to observe that for any two real numbers x and y, $[x+y] \geq [x]+[y]$. This follows easily by writing $x = [x]+\{x\}$, where $\{x\}$, the **fractional part** of x, is defined as $x-[x]$. Note that $0 \leq \{x\} < 1$ with $\{x\} = 0$ if and only if x is an integer. Similarly write $y = [y]+\{y\}$. Then $x+y = [x]+[y]+(\{x\}+\{y\}) \geq [x]+[y]$ which shows that $[x+y] \geq [x]+[y]$. Inequality will hold if and only if $\{x\}+\{y\} \geq 1$. But in that case we must have $[x+y] = [x]+[y]+1$ since $\{x\}+\{y\} < 2$ is always true. Applying this fact with $x = \frac{k}{2^i}$ and $y = \frac{n-k}{2^i}$, we get that for every $i \geq 1$,

$$c_i = a_i + b_i \text{ or } a_i + b_i + 1 \tag{6}$$

with the second possibility holding if and only if the fractional parts of $\frac{k}{2^i}$ and of $\frac{n-k}{2^i}$ add up to 1 or more.

From (3) to (6) we immediately get that $a + b \leq c$, which we already know anyway. But we also get that $a+b < c$ if and only if $a_i + b_i + 1 = c_i$ for at least one i. Thus we get that for $0 \leq k \leq n$, the integer $\binom{n}{k}$ is even if and only if there is some $i \geq 1$ such that the fractional parts of $\frac{k}{2^i}$ and of $\frac{n-k}{2^i}$ add up to 1 or more. This condition is easy to characterise in terms of the remainders that are left when $k, n-k$ are divided by 2^i. Let us write k as $s2^i + u$, where s (the quotient) is an integer and u (the remainder) is also an integer such that $0 \leq u < 2^i$. (The property of integers that is used here is formally called the **division algorithm**. We could replace the divisor 2^i by any positive integer m. In that case, the remainder would be required to lie in the interval from 0 to $m-1$. This requirement makes the quotient as well as the remainder unique.) Clearly s is also the integral part of $\frac{k}{2^i}$ while its fractional part is $\frac{u}{2^i}$. Similarly, if we write $n-k = t2^i + v$ with t, v, integers and $0 \leq v < 2^i$, then the fractional part of $\frac{n-k}{2^i}$ is simply $\frac{v}{2^i}$. We thus get that $\binom{n}{k}$ is even if and only if we can find some i such that the remainders u and v of k and $n-k$ (each w.r.t. 2^i) add up to 2^i or more.

Let us now identify those n for which $\binom{n}{k}$ is even for *some* k, $(0 \leq k \leq n)$. Let 2^r be the highest power of 2 not exceeding n. Then $2^r \leq n \leq 2^{r+1} - 1$. We consider two cases depending upon whether the second inequality is strict or not.

(i) Suppose $2^r \leq n < 2^{r+1} - 1$. Call $n - 2^r$ as z. Then $0 \leq z \leq 2^r - 2$ and so it is possible to split z as a sum of two non-negative integers, say x and y each of which is at most $2^{r-1} - 1$. (For example, suppose $n = 27$. Then $r = 4, z = 27 - 16 = 11$ and we can split z as 7+4 or as 6+5 but not as 8+3 or as 9+2 or as 10+1 or as 11+0.) Now simply take $k = 2^{r-1} + x$. Then $n - k = 2^{r-1} + y$. (In the example just given, if we let $11 = 7+4$, then $k = 8 + 7 = 15$ and $n - k = 8 + 4 = 12$.) Note that k and $n-k$ are both less than 2^r but their sum n is bigger than or equal to 2^r. So taking i as r we see that k and $n-k$ are their own remainders and add up to 2^i or more. Hence in this case we can find some k such that $\binom{n}{k}$ is even.

(ii) Suppose $n = 2^{r+1} - 1$. Then for *every* k (with $0 \leq k \leq n$), $\binom{n}{k}$ is odd. To see this, first observe that for any $i > r$, the integral parts of all the three ratios $\frac{n}{2^i}, \frac{k}{2^i}$ and $\frac{n-k}{2^i}$ are 0, and hence, in the notations above, we have $c_i = a_i + b_i$. For $1 \leq i \leq r$, $n = 2^{r+1} - 1 = (2^{r+1} - 2^i) + (2^i - 1) = (2^{r-i+1} - 1)2^i + (2^i - 1)$. Since $2^{r-i+1} - 1$ is an integer, this means that the remainder of n w.r.t. 2^i is $2^i - 1$. Now given any k (with $0 \leq k \leq n$), write $k = s2^i + u$ and $n - k = t2^i + v$, where u, v are non-negative integers less that 2^i each. Adding, $n = (s + t)2^i + (u + v)$. By uniqueness of remainders, this means that $u + v$ is of the form $\lambda 2^i + (2^i - 1)$ for some integer λ. We have to show that $u + v$ cannot be 2^i or bigger. Suppose it is. Then $\lambda \geq 1$. But that would mean $u + v \geq 2^i + 2^i - 1$ contradicting that $u \leq 2^i - 1$ and $v \leq 2^i - 1$.

Summing up, we have shown that a positive integer n has the property in question (viz., that $\binom{n}{k}$ is odd for every k with $0 \leq k \leq n$) if and only if n

is of the form $2^m - 1$ for some positive integer m, which is equivalent to saying that $n + 1$ is a power of 2. The subsequent work is identical to that in the first solution.

Comment No. 3:

Thus we have two solutions to our problem. The first one was shorter but a bit tricky and the second one more natural but a bit longer. Although the problem is about binomial coefficients, there is hardly any combinatorics in it. The problem is, in fact, more about divisibility of certain integers by some fixed integer (viz., 2). Not surprisingly, both the solutions used many properties about the divisibility of integers which we simply took for granted. For example, when we talked of the highest power of an integer which divides some other integer, we are implicitly assuming that every positive integer has a unique factorisation as powers of primes. If the factorisation were not unique, it would be meaningless to talk of such highest powers. Similarly, several times we used the fact that the highest power of 2 which divides a product of two numbers is the product of the highest powers of 2 that divide the two factors. In doing so, we crucially used that 2 is a prime number. A prime number has the property that whenever it divides a product of two numbers, it divides at least one of them. If, instead of 2, we take a composite number, say 6, then this is false. For example, 6 divides 12 but it divides neither 3 nor 4. So the highest power of 6 which divides 12 is not the product of the highest powers of 6 that divide 4 and 3 (both of which are 6^0, i.e., 1 each). It turns out that the division algorithm (also often called the **euclidean algorithm**[1]) is the basic property from which all properties about divisibility such as the unique factorisation can be derived. The euclidean algorithm can be proved by mathematical induction. We shall not go into this. We only mention that unique factorisation of integers (as products of prime powers) and the basic property of primes mentioned above are frequently needed in problems of this type, where the divisibility or otherwise of an integer by a prime is to be established.

Comment No. 4:

The heart of the second solution was equation (6). However, a discerning student will hardly fail to notice that (6) was not used in its full strength. In fact we used it only to conclude that for every $i \geq 1$, $a_i \geq b_i + c_i$ with strict inequality holding if and only if the fractional parts of $\frac{k}{2^i}$ and of $\frac{n-k}{2^i}$ add up to 1 or more. *We never used the fact that in the latter case*, a_i is not only bigger than $b_i + c_i$ but is, in fact, exactly one bigger than it, i.e., $a_i = b_i + c_i + 1$. In other words, instead of using the stronger statement '$a_i = b_i + c_i + 1$' we only used the much weaker statement '$a_i \geq b_i + c_i + 1$'. Although this was good enough for a solution to the given problem, it is instructive to see if we can get something stronger by using the sharper statement. (This, by the way, is a very healthy

[1] Note the close analogy with the euclidean algorithm for polynomials in one variable mentioned in Chapter 2, Comment No. 8. There is, in fact, a common theory covering both. But we shall not go into it.

habit. Whenever you have a solution, either your own or somebody else's, see if the method used could have given some stronger or more general result. Such a search often increases the applicability of that method and enables you to solve more challenging problems. Even when the search does not lead to something concrete, the time you have spent is not wasted because the thought process you have gone through increases your maturity.)

In the present problem, it is easy to see what additional information we get if we use (6) in its full strength. With the notations introduced there, 2^a and 2^{b+c} are, respectively, the highest powers of 2 that divide the numerator and the denominator of the ratio $\frac{n!}{k!(n-k)!}$ which equals $\binom{n}{k}$. So clearly 2^{a-b-c} is the highest power of 2 that divides $\binom{n}{k}$. From (3) to (5), we have

$$a - b - c = \sum_{i=1}^{\infty} a_i - b_i - c_i \tag{7}$$

By (6), every term of this sum is either 0 or 1. So the sum equals the number of the non-zero terms in it, or equivalently, the number of those integers $i \geq 1$ for which the fractional parts of $\frac{k}{2^i}$ and of $\frac{n-k}{2^i}$ add up to 1 or more. We could record this as a nice little theorem. But unfortunately, such a theorem will not have much appeal because we are equating $a - b - c$ with something rather clumsy, viz., the number of those integers $i \geq 1$ for which the fractional parts of $\dfrac{k}{2^i}$ and of $\dfrac{n-k}{2^i}$ add up to 1 or more. If, instead, we could equate $a - b - c$ with something familiar, then the result would be quite appealing.

Fortunately, such a description is possible and fairly easy. It is based on what is called the **binary arithmetic**. Normally we write a positive integer as a sum of powers of 10, viz., 1, 10, 100, 1000, etc., which are called, respectively, units, tens, hundreds, etc. For example, the integer 5087 has five thousands, no hundreds, eight tens and seven units. This is a compact way of saying that $5087 = 5\times10^3+0\times10^2+8\times10^1+7\times10^0$. The coefficients of the various powers of 10 are called the **digits**. Note that each digit can take 10 possible values, from 0 to 9. Without this restriction, the representation would not be unique. This representation of integers is called the **decimal representation**. The integer 10 is called the **base** or the **radix** or the **scale** of this representation.

We are all so familiar with how to add, subtract and multiply numbers in the decimal representation that we think of it as the 'natural' way to write integers. But actually there is nothing very special about the base 10. (The origin of its use is generally attributed to the fact that human beings have ten fingers which are used in counting.) We could replace 10 by any integer, say q, greater than 1 and take it as the base. Then every positive integer can be written in the form $a_r q^r + a_{r-1}q^{r-1} + \ldots + a_2q^2 + a_1q + a_0$, where r is some non-negative integer, and for each $i = 0, 1, \ldots, r$, $0 \leq a_i \leq q - 1$. This representation is unique if we ignore the leading 0's. (This can be ensured by requiring that a_r, the leading digit, is non-zero. But sometimes it is convenient to allow it to be 0, for example when we want to write several integers in a common unified form.) This representation is called the q-**ary**

representation. The terms **binary** and **ternary** are used, respectively, for $q = 2$ and 3. Note that in the binary representation, each digit has only two possible values, viz., 0 and 1. As there are many physical devices which exist in two natural states (e.g., lamps, electromagnetic bits), the binary representation is especially convenient in computers and other electronic devices. The price to be paid is that the 'length' i.e., the number of digits needed to represent an integer, increases. For example 98 (in the decimal representation) is 1100010 in the binary representation (obtained by first splitting 98 into a sum of powers of 2 as 64+32+2 i.e., as $2^6 + 2^5 + 2^1$). See Exercise (24.83) for an interesting 'practical' application of the binary representation.

In any q-ary representation, the basic arithmetic can be done exactly the way it is done in the decimal representation. When the digit to be put into a particular 'place' exceeds $q - 1$, we reduce it to an integer between 0 and $q - 1$ by subtracting a suitable multiple of q, say uq from it and then 'carry' u to the next higher place. (Note that here we are using the division algorithm crucially.) For example, in the binary representation, $1100010 + 101111 = 10010001$. Here there are four carries, each equal to 1, brought to the second, the third, the fourth and the sixth place from the right, the counting of places beginning with 0 rather than with 1. (In the decimal representation, this sum is $98 + 47 = 145$ which has 2 carries.)

It is very easy to tell when a carry will occur while adding two integers, say x and y. Suppose the q-ary representations of x, y and of their sum $z = x+y$ are, respectively, $\sum_{i=0}^{r} x_i q^i$, $\sum_{i=0}^{r} y_i q^i$ and $\sum_{i=0}^{r} z_i q^i$. Let us show the addition in a tabular form, the way we first learn it in schools.

place value	q^r	q^{r-1}	$\ldots$	q^i	q^{i-1}	$\ldots$	q^2	q	1
x	x_r	x_{r-1}	$\ldots$	x_i	x_{i-1}	$\ldots$	x_2	x_1	x_0
$+\ y$	y_r	y_{r-1}	$\ldots$	y_i	y_{i-1}	$\ldots$	y_2	y_1	y_0
carry									
$=\ z$	z_r	z_{r-1}	$\ldots$	z_i	z_{i-1}	$\ldots$	z_2	z_1	z_0

It is clear that in the 0-th, i.e., in the rightmost column (or the 'units column' headed by 1) there can never be a carry. In the first column there will be a carry if and only if $x_0 + y_0 \geq q$ which is equivalent to saying that $\frac{x_0}{q} + \frac{y_0}{q} \geq 1$ and in that case the carry will be 1. But x_0 and y_0 are nothing but the remainders left when x and y are divided by q. So, $\frac{x_0}{q} + \frac{y_0}{q}$ is precisely the sum of the fractional parts of $\frac{x}{q}$ and of $\frac{y}{q}$. More generally, a carry will occur in the i-th column from the right (headed by q^i) if and only if the 'tail portions of x and y' i.e., the expressions $x_{i-1}q^{i-1} + x_{i-2}q^{i-2} + \ldots + x_2q^2 + x_1q + x_0$ and $y_{i-1}q^{i-1} + y_{i-2}q^{i-2} + \ldots + y_2q^2 + y_1q + y_0$ add up to q^i or more (which is the least integer which cannot be 'accommodated' in the columns upto the $(i-1)$-th column from the right). But these tail-ends are precisely the remainders left when x and y are divided by 2^i. Thus, once again, there is a carry in the i-th place if and only if the fractional part of $\frac{x}{2^i}$ and that of $\frac{y}{2^i}$ add up to 1 or more.

And, in that case 1 is carried to the i-th column. (Note, incidentally, that when we are adding two integers, no carry can be bigger than 1, no matter what is the base of the representation.)

We are now in a position to give a natural interpretation to the R.H.S. of (7). Earlier we saw that a non-zero term in this summation corresponds to an integer $i \geq 1$ for which the fractional parts of $\frac{k}{2^i}$ and of $\frac{n-k}{2^i}$ add up to 1 or more. But this is also precisely the condition for a carry to occur in the i-th column when the integers k and $n-k$ are added in the binary representation. Putting it all together, we get that the highest power of 2 which divides the binomial coefficient $\binom{n}{k}$ (where $0 \leq k \leq n$) is 2^j, where j is the number of carries that occur when k and $n-k$ are added in the binary representation.

We can generalise this a little. The derivation of equations (3) to (6) is valid if 2 is replaced by any prime p. The subsequent part involving the characterisation of carries in the binary representation is valid for any base q, not necessarily a prime. So for any prime p we have proved the following theorem, due to **Kummer**.

Theorem : Let n be a positive integer and p any prime. Then the highest power of p which divides $n!$ is p^a, where

$$a = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots + \left[\frac{n}{p^i}\right] + \ldots \tag{8}$$

Further, suppose $0 \leq k \leq n$. Then the highest power of p which divides the binomial coefficient $\binom{n}{k}$ is 2^j, where j is the number of carries needed when k and $n-k$ are added to each other in their p-ary representations.

Note that the number of carries needed in adding k and $n-k$ is the same as the number of carries needed in subtracting k from n. This gives an alternate formulation of the second part of the theorem which is a little more convenient to apply. We proved above that if n is of the form $2^m - 1$ for some positive integer m, then the binomial coefficients $\binom{n}{k}$ are all odd for $0 \leq k \leq n$. Kummer's theorem not only gives a very short proof of this, but in some sense it also 'unravels' why integers of this form have this property. If we write such an integer in its binary representation, we see that it has m digits and all of them are equal to 1. So when we subtract k from it, there will be no carries in any place. Hence the highest power of 2 that divides $\binom{n}{k}$ is 2^0, which is another way of saying that $\binom{n}{k}$ is odd. Actually, Kummer's theorem does something more. When n is not of the form $2^m - 1$, we already know that $\binom{n}{k}$ is even for *some* k. But now we can be far more specific. We can tell precisely for which k's this happens. Let $\sum_{i=0}^{r} z_i 2^i$ and $\sum_{i=0}^{r} x_i 2^i$ be the binary representations of n and k respectively. Then while subtracting k from n, a carry will occur if and only if $x_i > z_i$ for some i. As 0 and 1 are the only possible digits, this is equivalent to saying that $x_i = 1$ and $z_i = 0$. For example, suppose $n = 27$. In the binary representation, 27 equals 11011. So a carry will occur if k has a 1 in the second place from the right (once again keeping in mind that the count of

places begins from 0 and not from 1). The smallest such integer is 100 in the binary representation, or 4 in the decimal system. And indeed we see that $\binom{27}{4}$ which equals $\frac{27.26.25.24}{1.2.3.4}$ is even.

Comment No. 5:

Kummer's theorem is a powerful tool for problems involving the divisibility of the binomial coefficients by prime powers. Depending on the problem, some less powerful methods may also work. For example, we already saw that for the problem we started with, we could get an elementary solution using (1) and another solution using (6) but not in its full strength. In fact, the variety offered by such elementary techniques makes them more interesting, even though not always more efficient than Kummer's theorem. This is analogous to problems of triangular optimisation (see Chapter 14), where, even though the method of Lagrange's multipliers is the most efficient, it kills the charm of the problem.

As a good illustration, consider the following problem. (JEE 1998)

Suppose n is a positive integer which is divisible by a prime p and k is an integer not divisible by p. Then prove that $\binom{n}{k}$ is divisible by p.

It is not given in this problem that $0 \leq k \leq n$. But we may assume that this is the case, because for values of k outside this range, $\binom{n}{k}$ is 0 and hence is divisible by p. Now consider the p-ary expansions of n, k and $n-k$. As p divides n, the digit in the units place in the p-ary expansion of n must be 0. On the other hand, for k it is non-zero since k is not divisible by p. So when we add k and $n-k$, there will be a carry in the first place. Hence by Kummer's theorem the highest power of p that divides $\binom{n}{k}$ will be at least p, which proves the assertion. Note, however, that Kummer's theorem is not used in its full strength here. We could have as well got a solution by considering the analogues of (3) to (6) for the prime p. Here since p divides n but not k it follows that p cannot divide $n-k$ either (as otherwise it would divide the difference $n-(n-k)$, a contradiction). Consequently, in the equation $\frac{n}{p} = \frac{k}{p} + \frac{n-k}{p}$, the L.H.S. is an integer but neither of the terms on the R.H.S. is an integer. So the fractional parts of $\frac{k}{p}$ and of $\frac{n-k}{p}$ add up to 1. (In the notations introduced in (3) to (6), this means $a_1 > b_1 + c_1$.) This is good enough to show that p divides $\binom{n}{k}$.

There is also a proof by induction. But the induction is neither on n nor on k. If n is divisible by p then $n-1$ is not divisible by p and so we cannot apply the inductive hypothesis to $n-1$. The same holds for $n-2$ also. We shall have to descend all the way down to $n-p$ (which is divisible by p) to apply the inductive hypothesis. Put differently, let us write $n = mp$, where m is a positive integer. Then we shall derive the truth for the case $n = mp$ from that for the case $n = (m-1)p$. In effect, then, we are inducting on m rather than on n. The case $m = 1$ corresponds to $n = p$. As noted above, we may suppose $0 \leq k \leq p$. We may further suppose $0 < k < p$ since it is given that p does not divide k. Now, $\binom{p}{k} = \frac{p(p-1)\ldots(p-k+1)}{1.2.\ldots.k}$. Evidently, p divides the numerator but not the denominator. As p is a prime, it divides the ratio. Thus the truth of the assertion is established for $m = 1$.

For the inductive step, we apply the identity

$$\binom{(m+1)p}{k} = \sum_{r=0}^{p} \binom{p}{r}\binom{mp}{k-r} \tag{9}$$

which is proved by writing $(1+x)^{(m+1)p}$ as $(1+x)^p(1+x)^{mp}$ and equating the coefficients of x^k on both the sides. (This is a special case of (12) in the next chapter on binomial identities.) We already proved that for $0 < r < p$, $\binom{p}{r}$ is divisible by p. So every term in the summation in (9), except possibly the first and the last term, is divisible by p. Since $\binom{p}{0} = \binom{p}{p} = 1$, these terms are, respectively, $\binom{mp}{k}$ and $\binom{mp}{k-p}$. By induction hypothesis, $\binom{mp}{k}$ is divisible by p. Also, since p does not divide k it cannot divide $k-p$ either. So we can apply the inductive hypothesis replacing k by $k-p$ and get that $\binom{mp}{k-p}$ is divisible by p. Thus, every term in the R.H.S. of (9) is divisible by p. Therefore so is $\binom{(m+1)p}{k}$, thereby completing the proof of the inductive step.

There is yet another and a far cleverer solution which is based on a formula analogous to (1). As noted before, we may confine ourselves to the case where $1 \leq k \leq n$. Then we can rewrite $\frac{n(n-1)\ldots(n-k+1)}{k!}$ as $\frac{n(n-1)\ldots(n-k+1)}{k(k-1)\ldots.2.1}$ and further as $\frac{n}{k}\frac{(n-1)\ldots(n-k+1)}{(k-1)\ldots.2.1}$. If we now multiply both the sides by k we get,

$$k\binom{n}{k} = n\binom{n-1}{k-1} \tag{10}$$

Both the sides of (10) are products of two integers. By hypothesis, p divides n and hence the R.H.S. So p divides the L.H.S. Since p is a prime, it must divide at least one of the two factors of the L.H.S. But it is given that p does not divide k. So we are forced to conclude that p divides the other factor, viz., $\binom{n}{k}$ as was to be proved! (Formulas (1) and (10) are sometimes called the **extraction** formulas for binomial coefficients because they are obtained by extracting $\frac{n-k+1}{k}$ and $\frac{n}{k}$ respectively as (non-integral) factors from $\binom{n}{k}$.)

Comment No. 6:

Formula (8) in Kummer's theorem is not true if the prime p is replaced by a composite number, say q. Suppose for example, that $n = 4$ and $q = 6$. Then $[\frac{4}{6^i}] = 0$ for all $i \geq 1$. But still, 4! is divisible by 6. To get the analogue of (8) in such a case we have to consider the factorisation of q into powers of primes. The situation gets slightly complicated if a higher power of some prime divides q. However, if q is the product of distinct primes and the largest one among them is p, then it is not hard to show that the R.H.S. of (8) gives the exponent of the highest power of q that divides $n!$. This reasoning gives a solution to the following problem.

Find the number of zeros at the end of 100! (when written in the decimal representation).

Evidently, it is ridiculous to try to find the answer by actually calculating 100!. Nor is it necessary. To say that a number has r (or more) zeros at the

end (in its decimal representation) is equivalent to saying that it is divisible by 10^r. So the problem asks, in disguise, to find the highest power of 10 that divides 100!. Now, 10 is the product of the primes 2 and 5 and so for every $i \geq 0$, an integer is divisible by 10^i if and only if it is divisible by both 2^i and 5^i. Let 2^a and 5^b be the highest powers of 2 and 5 respectively which divide 100!. Then clearly the highest power of 10 which divides 100! will be 10^c, where c is the smaller of a and b. We can get the values of a and b from (8). Since $5 > 2$, $[\frac{100!}{5^i}] < [\frac{100!}{2^i}]$ for every $i \geq 1$. So b is the smaller of a and b. By (8), $b = [\frac{100}{5}] + [\frac{100}{25}] = 20 + 4 = 24$. Thus we get that 100! has exactly 24 zeros at the end.

Comment No. 7:

While we are at it, it is worthwhile mentioning, without proof, yet another powerful result, due to **Lucas**, regarding the divisibility of the binomial coefficients by primes. Let p be a prime, and let n, k be any positive integers. (We allow k to be bigger than n, in which case $\binom{n}{k}$ will be 0.) Now write n and k in p-ary representations as, say, $\sum_{i=0}^{r} n_i p^i$ and $\sum_{i=0}^{r} k_i p^i$ respectively. Here each of the digits $n_0, n_1, \ldots, n_r, k_0, k_1, \ldots, k_r$ lies in the range from 0 to $p-1$. We consider the binomial coefficients $\binom{n_i}{k_i}$ again with the understanding that $\binom{n_i}{k_i} = 0$ if $n_i < k_i$. (It may happen that even if $n > k$, $n_i < k_i$ for some i.) In case both n_i and k_i are 0, we take $\binom{n_i}{k_i} = 1$. Consider now the product $\binom{n_0}{k_0}\binom{n_1}{k_1}\ldots\binom{n_r}{k_r}$. Lucas' theorem says that the difference between this product and the binomial coefficient $\binom{n}{k}$ is always divisible by p. For example, take $n = 16, k = 4$ and $p = 3$. Then in the ternary system, $n = 121$ and $k = 011$. The product $\binom{1}{0}\binom{2}{1}\binom{1}{1}$ equals 2 while the binomial coefficient $\binom{16}{4}$ equals 1820. Their difference 1818 is indeed divisible by 3.

A special case of Lucas' theorem is noteworthy. If $n_i < k_i$ for some i, then the binomial coefficient $\binom{n_i}{k_i}$ vanishes and so does the product $\binom{n_0}{k_0}\binom{n_1}{k_1}\ldots\binom{n_r}{k_r}$. Lucas' theorem implies that in such a case, the binomial coefficient $\binom{n}{k}$ is divisible by p. In particular take $p = 2$. Then we get that $\binom{n}{k}$ is even if for some i, $n_i < k_i$. Earlier, we also got this from Kummer's theorem. But Lucas' theorem further implies that in case n is of the form $2^m - 1$ for some integer m, then $\binom{n}{k}$ is odd for all k $(0 \leq k \leq n)$. For, in this case, $n_i = 1$ for every $i = 0, 1, \ldots, m-1$ and consequently, regardless of whether k_i equals 0 or 1, each binomial coefficient $\binom{n_i}{k_i}$ is 1 for $i = 0, 1, \ldots, m-1$. Hence their product is 1. But then by Lucas' theorem, the difference $\binom{n}{k} - 1$ is even, so that $\binom{n}{k}$ is odd.

Several proofs of Lucas' theorem are known, one of them being based on the binomial theorem. But we omit it. We mentioned it just to show that the problem with which we started can be done in so many different ways, some of them short but rather tricky, while some of them based on non-trivial theorems. The motto, once again, is that there is so much to learn from a single problem.

Comment No. 8:

Let us now turn to some other problems involving divisibility of one integer,

say a, by another, say b. If we factor b as the product of powers of distinct primes, say $b = p_1^{m_1} p_2^{m_2} \dots p_k^{m_k}$, then it is clear that b divides a if and only if each $p_i^{m_i}$ divides a for $i = 1, 2, \dots, k$. This reduces the divisibility problem to that of testing divisibility by prime powers. Suppose for example that n is an odd positive integer. We claim that $n(n^2-1)$ is divisible by 24 (JEE 1983). Since the prime power decomposition of 24 is $8 \times 3 = 2^3 3$, the problem is equivalent to showing that $n(n^2-1)$ is divisible by both 8 and 3. Divisibility by 3 is easy to show. Writing $n(n^2-1)$ as $(n-1)n(n+1)$ we see that it is the product of three consecutive integers. So at least one (in fact, exactly one) of them is a multiple of 3. Thus 3 divides $n(n^2-1)$. In fact, here we do not need that n is odd. However, showing that $n(n^2-1)$ is divisible by 8, does require it, as the assertion is actually false for an even n, e.g., for $n = 2$. When n is odd, write it as $2m-1$, where m is a positive integer. Then $n(n^2-1) = (2m-1)4(m^2-m)$. Since $m^2-m = (m-1)m$ is a product of two consecutive integers, it is divisible by 2 since either $m-1$ or m is even. So the product $4(m^2-m)$ and hence also the integer $n(n^2-1)$ is divisible by 24, as was to be proved.

The problem can also be done by induction on m. The case $m = 1$ is trivially true since in that case n is also 1 and so $n(n^2-1)$ is 0 which is divisible by any integer. For the inductive step it is convenient to write $f(m)$ for $(2m-1)4(m^2-m)$. Now suppose that the assertion holds for $m = k$, i.e., that $f(k)$ is divisible by 24. We have to show that $f(k+1)$ is also divisible by 24. The standard method for this is to write $f(k+1)$ as $(f(k+1) - f(k)) + f(k)$. By induction hypothesis, $f(k)$ is divisible by 24. So we would be through if we can show that $f(k+1) - f(k)$ is divisible by 24. By a direct calculation, $f(k+1) - f(k) = (2m+1)4((m+1)^2 - (m+1)) - (2m-1)4(m^2-m) = 4m[(2m+1)(m+1) - (2m-1)(m-1)] = 24m^2$ which is indeed divisible by 24 as m is an integer. This completes the induction.

Of the two proofs, clearly the first one is superior as it really tells us *why* $n(n^2-1)$ is divisible by 24 for odd n. Although, logically, an inductive proof is a perfectly valid proof, in the inductive step in it, we are not directly proving the divisibility of $f(k+1)$. Instead, we are directly proving only the divisibility of the difference $f(k+1) - f(k)$ and then borrowing the divisibility of $f(k)$ (by induction hypothesis). But then, this is also the reason why inductive proofs are mandatory in those situations where tackling $f(k)$ directly may not be easy though it may be relatively easy to tackle the difference $f(k+1) - f(k)$. More generally, induction would also be very convenient if we can find some integer c such that we can easily prove the divisibility of $f(k+1) - cf(k)$ by the given integer. For, in that case, the divisibility of $f(k+1)$ would follow by writing it as $[f(+1) - cf(k)] + cf(k)$. In the example just given, c was 1. In some problems it may be different and choosing it appropriately may call for some perceptivity. Here are a couple of other examples of such types.

Prove that $7^{2n} + 2^{3n-3}3^{n-1}$ is divisible by 25 for every positive integer n. (JEE 1982)

Here a direct proof can be given by writing $7^{2n} + 2^{3n-3}3^{n-1}$ as $(49)^n + (24)^{n-1}$ and then further as $(50-1)^n + (25-1)^{n-1}$. If we expand this by the binomial theorem, $(-1)^n$ and $(-1)^{n-1}$ will cancel each other. All other terms

will have a factor which is either of the form $(50)^k$ or of the form $(25)^k$ for some $k \geq 1$ and hence will be divisible by 25. So their sum will be divisible by 25. Although this proof is not difficult, it requires the use of the binomial theorem. An inductive proof, however, is elementary, if somewhat less elegant. Again we write $f(n)$ for $7^{2n} + 2^{3n-3}3^{n-1}$. Then $f(1) = 49 + 1 = 50$ which is divisible by 25. For the inductive step, if we work with the difference $f(k+1) - f(k)$, i.e., with $7^{2k+2} + 2^{3k}3^k - 7^{2k} + 2^{3k-3}3^{k-1}$, we can see that it is not very convenient to prove that it is divisible by 25. But if we, instead, consider $f(k+1) - 49f(k)$ (or, in other words, take $c = 49$ in the discussion above), then the powers of 7 cancel out and we get $f(k+1) - 49f(k) = 2^{3k}3^k - 49(2^{3k-3}3^{k-1}) = 2^{3k-3}3^{k-1}(24 - 49) = -25(2^{3k-3}3^{k-1})$ which is indeed divisible by 25. So by induction, $f(n)$ is divisible by 25 for all n.

In an entirely analogous manner, we can prove that for every positive integer n, $2.7^n + 3.5^n - 5$ is divisible by 24. (JEE 1985) We remark that in this case a direct proof of divisibility will run into several cases depending on the parity of n (i.e., whether it is even or odd) and so is not as easy as in the examples above. But the method of induction works fine with $c = 7$. For, taking $f(n) = 2.7^n + 3.5^n - 5$, the expression $f(k+1) - 7f(k)$ comes out as $-6.5^k + 30$, i.e., as $-6(5^k - 5)$. Since $24 = 6 \times 4$, the proof of the inductive step is now reduced to showing that for every positive integer k, $5^k - 5$ is divisible by 4. This itself can be done by induction on k (the magic number c being 5 this time). Or we can write $5^k - 5$ as $(4+1)^k - 4 - 1$. If we expand by the binomial theorem then every term will have 4 or some higher power of 4 as a factor and hence is divisible by 4. (Better still, write $5^k - 5$ as $5(5^{k-1} - 1^{k-1})$ and factorise the second factor further.)

Comment No. 9:

Sometimes we want to prove the divisibility of some expression $f(n)$ for all integral values of n. The method of induction is applicable only for positive integers. More generally, by a shifting of variables, one can extend it to all values greater than or equal to some fixed integer, say, -20, because we can start the induction at $n = -20$ instead of the usual $n = 1$. But even then we cannot apply induction to cover all integral values of n. So in such problems, separate arguments may be needed to handle negative values of n. We illustrate the technique in the following problem. (JEE 1985)

Let r be a fixed positive integer. Prove that the product of any r consecutive integers is divisible by $r!$.

Let the consecutive integers be $n, n+1, \ldots, n+r-2, n+r-1$ and let $f(n)$ denote their product. We have to show that $r!$ divides $f(n)$ for all integers n. We first prove this by induction when n is a positive integer. Clearly $f(1) = r!$ and so the statement is true for $n = 1$. Now, for the inductive step, note that for any positive integer k, $f(k+1) - f(k) = (k+1)(k+2)\ldots(k+r-1)(k+r) - k(k+1)\ldots(k+r-1) = (k+1)(k+2)\ldots(k+r-1)(k+r-k) = g(k+1)r$, where $g(k+1)$ is the product of $r-1$ consecutive integers, $k+1, k+2, \ldots, k+r-1$. The integer r is fixed. But it can take any positive integer as a value. So if we allow it to vary, then we can apply induction on r itself. Then we can assume

that the product of any $r-1$ consecutive positive integers is divisible by $(r-1)!$. But then $g(k+1)r$ is divisible by $r!$ and this is what we need to complete the proof of the inductive step (this time the induction being on n and not on r).

The method adopted here is known as **double induction**. Here there are two parameters, say, n and r in the statement to be proved which we denote by $P(r,n)$. For a fixed value of the parameter r, $P(r,n)$ depends only on the other parameter n. In double induction, to prove $P(r,n)$ we may use not only the truth of $P(r,n-1)$, but we may also use the truth of $P(r-1,m)$ for any positive integer m (including those that are bigger than n). To establish the logical validity of this method, as just remarked, for each fixed positive integer r, $P(r,n)$ is a statement which depends on n only. We therefore denote it by, say, $P_r(n)$. Now let $Q(r)$ be the statement '$P_r(n)$ is true for all positive integers n'. Notice that $Q(r)$ is a statement involving a single parameter r. So, we can prove it by induction on r. Proving it for $r=1$ amounts to proving that $P(1,n)$ is true for all n. This we can do by induction on n, or by some other method. If we do it by induction on n, then in the inductive step, we can assume the truth of $P(1,n-1)$ to deduce the truth of $P(1,n)$ for $n>1$. Here there is no question of using the truth of $P(0,m)$ for any m, because $P(0,m)$ is not even defined. However, once we have proved $Q(1)$, that is once we have proved that $P(1,n)$ is true for *all* n, then afterwards we are allowed to assume it anytime. In particular, in proving $Q(2)$, i.e., in proving $P(2,n)$ by induction on n, we can assume the truth of $P(1,m)$ for *any* value of m. More generally, for any $r>1$, in proving $P(r,n)$ by induction on n, we can assume the truth of $Q(r-1)$, i.e., that of $P(r-1,m)$ for all m. That is what we did in the proof above. Of course, we should have begun by first showing that the product of any 1 consecutive integer is divisible by 1!. But this is ridiculously true.

There is also a combinatorial proof to show that the product of any r consecutive positive integers, say, $n,(n+1),\ldots,(n+r-1)$ is divisible by $r!$. This is equivalent to showing that the ratio of the product $n(n+1)\ldots(n+r-1)$ (which we have denoted by $f(n)$ above) to $r!$ is an integer. Now if we rewrite $f(n)$ in the reverse order, it is $(n+r-1)(n+r-2)\ldots(n+1)n$. So $\frac{f(n)}{r!}=\frac{(n+r-1)(n+r-2)\ldots(n+1)n}{r!}$ which is nothing but $\binom{n+r-1}{r}$, i.e., the number of ways to choose r objects from $n+r-1$ objects. So it has to be a whole number. Hence we get that $r!$ divides $f(n)$. This proof is unusually short and its elegance can be best appreciated when it is compared with the proof by double induction just given. Still we have given the earlier proof to acquaint you with double induction. In the next chapter we shall see more examples of elegant combinatorial proofs.

We are yet to finish the solution to the problem. We have to prove that $f(n)$, i.e., the product $n(n+1)\ldots(n+r-1)$ is divisible by $r!$ for all integers n. So far we have shown it only for positive integers. Suppose now that $n\leq 0$. If $-(r-1)\leq n\leq 0$, then one of the factors in the product $n(n+1)\ldots(n+r-1)$ is 0 and so is the product. So in this case, $f(n)$ is certainly divisible by $r!$ (in fact, divisible by anything). The only remaining case now is $n\leq -r$. In this case all the r factors in the product $n(n+1)\ldots(n+r-1)$ are negative and so

their product is the same as $(-1)^r(-n)(-n-1)\ldots(-n-r+1)$ which can be written as $(-1)^r m(m-1)(m-2)\ldots(m-r+1)$, where $m=-n$. Reversing the order of the factors, this is the same as $(-1)^r(m-r+1)(m-r)\ldots(m-1)m$ which is nothing but $(-1)^r f(m-r+1)$. Our assumption that $n<-r+1$, implies that $m-r+1$ is a positive integer. So by the earlier part of the proof, $f(m-r+1)$ and hence $(-1)^r f(m-r+1)$, which equals $f(n)$, is divisible by $r!$. We have now completed the solution.

Comment No. 10:

Sometimes we are asked to prove that a given integer a is *not* divisible by another given integer b. Again, when a is a member of some sequence, i.e., is of the form $f(n)$ for some function f, induction on n is a standard tool. We consider a couple of problems of this type.

Let $p\geq 3$ be an integer and α,β be the roots of $x^2-(p+1)x+1=0$. Using mathematical induction show that $\alpha^n+\beta^n$ is (i) an integer and (ii) not divisible by p. (JEE 1992)

Write $f(n)$ for $\alpha^n+\beta^n$. In order to prove the desired properties for $f(n)$ it is neither necessary nor of much use to identify the roots α and β explicitly. What is crucially needed is to express $f(n)$ in terms of $f(n-1)$, or, more generally, in terms of $f(m)$, where m takes one (or more) values less than n. In the present case, even without explicitly finding α and β, we know that they satisfy $\alpha^2-(p+1)\alpha+1=0$ and $\beta^2-(p+1)\beta+1=0$. Multiplying both the sides of these equations by α^{n-2} and β^{n-2} respectively, we get that for all $n\geq 2$,

$$\alpha^n-(p+1)\alpha^{n-1}+\alpha^{n-2} = 0 \quad (11)$$

$$\text{and}\quad \beta^n-(p+1)\beta^{n-1}+\beta^{n-2} = 0 \quad (12)$$

Adding, we get

$$f(n)=(p+1)f(n-1)-f(n-2) \quad (13)$$

for all $n\geq 2$. It is now very easy to show by induction that $f(n)$ is an integer. Since (13) expresses $f(n)$ in terms of not just $f(n-1)$ but in terms of $f(n-1)$ and $f(n-2)$, the induction must begin by verifying the first two and not just the first case as usual. We could verify directly that $f(1)$ and $f(2)$ are integers. But there is no harm in starting the induction with $n=0$, since $f(0)=\alpha^0+\beta^0=1+1=2$ which is certainly an integer. (Here the actual values of α and β do not matter. The only thing is that 0^0 is undefined. So we better make sure that α and β are both non-zero. But this follows since $\alpha\beta=1$.) Next, $f(1)=\alpha+\beta=p+1$ is also an integer since p is given to be an integer. The inductive step follows immediately from (13) since if $f(n-1)$ and $f(n-2)$ are integers, so is $(p+1)f(n-1)-f(n-2)$. (Again, we are using the fact that p is given to be an integer.) This completes the proof of (i).

Proving the second part of the question, viz., that $f(n)$ is not divisible by p, by induction on n, is, however, tricky. There is no difficulty in verifying the starting two cases. Indeed, as we saw, $f(0)=2$ which cannot be a multiple

of p since it is given that $p \geq 3$. As for $f(1)$, we already know it equals $p+1$ which is surely not divisible by p. However, proving the inductive step is not so easy. Suppose we know that neither $f(n-1)$ nor $f(n-2)$ is divisible by p. But still their difference can be. And in that case, by (13), $f(n)$ would be divisible by p.

To overcome this difficulty, let us rewrite (13) slightly as

$$f(n) = pf(n-1) + f(n-1) - f(n-2) \tag{14}$$

The first term on the R.H.S., viz., $pf(n-1)$ is always divisible by p and therefore does not affect the divisibility or otherwise of $f(n)$ by p. So let us leave it intact. However, the next term, viz., $f(n-1)$, can be rewritten if we apply (13) (or rather, (14)) with n replaced by $n-1$. Then we get

$$f(n-1) = pf(n-2) + f(n-2) - f(n-3) \tag{15}$$

If we replace the second term in the R.H.S. of (14) by (15) we get

$$f(n) = p[f(n-1) + f(n-2)] - f(n-3) \tag{16}$$

which is valid for all $n \geq 3$. From (16) we see that the non-divisibility of $f(n-3)$ by p implies that of $f(n)$. So we can prove the non-divisibility of $f(n)$ by p by induction on n. However, this time we shall have to verify the first *three* cases before we start the induction. We have already showed that $f(0)$ and $f(1)$ are non-divisible by p. As for $f(2)$, by (13) it equals $(p+1)f(1) - f(0) = (p+1)^2 - 2 = p^2 + 2p - 1$. Since $p^2 + 2p$ is divisible by p but -1 is not, we see that $f(2)$ is not divisible by p. Thus we have proved part (ii) of the problem too.

Comment No. 11:

In the last problem, $f(n)$ was given as a certain linear combination of the n-th powers of the roots α and β of a certain quadratic equation, viz., $x^2 - (p+1)x + 1 = 0$. From this we derived (13). The fact that the coefficients of the terms of this quadratic are all integers and the leading coefficient is 1 was crucially needed in the proof of part (i) of the problem. A relation like (13) is called a **recurrence relation** and $f(n) = \alpha^n + \beta^n$ is called a **solution** of it, satisfying the **initial conditions** $f(0) = 2$ and $f(1) = p+1$. There is a general theory for solving such recurrence relations but we shall not go into it. But we remark that part (i) of the problem can be turned around. That is, we could be given that $f(n)$ satisfies the recurrence relation (13) and the two initial conditions $f(0) = 2$ and $f(1) = p+1$ and then we could be asked to prove, by induction on n, that $f(n) = \alpha^n + \beta^n$. The proof will again depend on (11) and (12). We solve below a problem of this type. (JEE 1981)

Let $u_1 = 1, u_2 = 1$ and

$$u_{n+2} = u_{n+1} + u_n \tag{17}$$

for $n \geq 1$. Use mathematical induction to show that

$$u_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right] \tag{18}$$

for all $n \geq 1$.

Denote $\frac{1+\sqrt{5}}{2}$ and $\frac{1-\sqrt{5}}{2}$ by α and β respectively. Then $\alpha + \beta = 1$ and $\alpha\beta = -1$. So α and β are precisely the roots of the quadratic equation $x^2 - x - 1 = 0$. Therefore $\alpha^2 = \alpha + 1$ and $\beta^2 = \beta + 1$. Multiplying these equations by α^k and β^k respectively, we get

$$\alpha^{k+2} = \alpha^{k+1} + \alpha^k \tag{19}$$

$$\text{and} \quad \beta^{k+2} = \beta^{k+1} + \beta^k \tag{20}$$

for every $k \geq 1$. We want to prove (18) by induction on n. (In effect this amounts to verifying that (18) is a solution of the recurrence relation (17).) In terms of α and β, (18) can be rewritten as

$$u_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n) \tag{21}$$

The truth of (21) for $n = 1$ and $n = 2$ can be checked directly since $\alpha - \beta = \sqrt{5}$ and $\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta) = \sqrt{5}$. For the inductive step, subtracting (20) from (19) and dividing throughout by $\sqrt{5}$ gives

$$\frac{1}{\sqrt{5}}(\alpha^{k+2} - \beta^{k+2}) = \frac{1}{\sqrt{5}}(\alpha^{k+1} - \beta^{k+1}) + \frac{1}{\sqrt{5}}(\alpha^k - \beta^k) \tag{22}$$

From (22) and (17) it is clear that if (21) holds for $n = k$ and for $n = k+1$ then it also holds for $n = k + 2$. So, by induction, (21) and hence (18) is true for all $n \geq 1$.

We remark that the recurrence relation (17) is a very well-known relation, called the **Fibonacci relation**. The terms of the sequence $\{u_n\}$ are called the **Fibonacci numbers**. The first few Fibonacci numbers are 1,1,2,3,5,8,13,21,34 and so on. They arise in many different contexts. But most importantly, they arise in natural growth. Suppose some organism is capable of producing another one of its kind once every year after it attains maturity at the end of two years. If we start with a newly born organism of this type then it is easy to show that during the n-th year we shall have precisely u_n of these organisms. (We are assuming that the organisms never die. To take care of this possibility, more complicated recurrence relations are needed.) There is a huge literature on the Fibonacci numbers.

Comment No. 12:

Our second problem of non-divisibility is much more straightforward.

Using mathematical induction, prove that for every integer $n \geq 1$, $(3^{2^n} - 1)$ is divisible by 2^{n+2} but not by 2^{n+3}. (JEE 1996)

For notational convenience, denote $(3^{2^n} - 1)$ by a_n. The problem amounts to showing that a_n can be written as $2^{n+2}b_n$, where b_n is an odd integer. Clearly, $a_1 = 3^2 - 1 = 8 = 2^{1+2}.1$. So the result is true for $n = 1$ (with $b_1 = 1$). Now for the inductive step suppose $a_n = 2^{n+2}b_n$, where b_n is odd. We have to show that a_{n+1} can be expressed in a similar form. Now, $a_{n+1} = 3^{2^{n+1}} - 1 = 3^{2.2^n} - 1 =$

$(3^{2^n})^2 - 1 = (3^{2^n} - 1)(3^{2^n} + 1) = a_n(a_n + 2) = a_n^2 + 2a_n = 2^{2n+4}b_n^2 + 2^{n+3}b_n = 2^{n+3}b_{n+1}$, where $b_{n+1} = 2^{n+1}b_n^2 + b_n$. Since $n \geq 0$, 2^{n+1} and hence $2^{n+1}b_n^2$ is even. But by induction hypothesis, b_n is odd. Therefore, b_{n+1}, being a sum of an even and an odd integer, is odd. Thus we have expressed a_{n+1} in the desired form. Therefore the assertion is proved by induction.

Comment No. 13:

We remarked above at the beginning of Comment No. 8 that the problem of checking the divisibility of an integer by another integer b reduces to that of checking the divisibility by the prime power factors of b. We then did a problem where the divisibility by 24 was proved by proving divisibility by 8 and by 3. However, in general this type of a reduction is not of much help. The reason is that there are hardly any general results which guarantee divisibility by integers of the form p^m, where p is a prime and $m > 1$. The situation is much better if $m = 1$. That is, there are some results which assert divisibility by a prime. Probably the most famous among these is the following, known as **Fermat's theorem**.

Theorem : For every positive integer n, and for every prime p, the number $n^p - n$ is divisible by p. If further, n is not divisible by p then $n^{p-1} - 1$ is divisible by p.

The first part can be proved very easily by induction on n, the case $n = 1$ being trivially true. For the inductive step we expand $(k+1)^p - (k+1)$ by the binomial theorem. The first and the last in the expansion of $(k+1)^p$ are k^p and 1. Every other term has a coefficient which is divisible by p. So we can write $(k+1)^p$ as $k^p + 1 + mp$ for some integer m. But then $(k+1)^p - (k+1) = mp + (k^p - k)$ which is divisible by p because $k^p - k$ is divisible by p by induction hypothesis. This establishes the first part of the theorem. For the second, write $n^p - n$ as $n(n^{p-1} - 1)$. This is divisible by p by the first part. But p is a prime which does not divide the first factor n. So it must divide the second factor, viz., $n^{p-1} - 1$, as was to be shown.

We remark that Fermat's theorem can also be proved by a combinatorial argument (see Exercise (4.39)). But the proof given above is too simple to require any alternatives. As an immediate application of the first part of Fermat's theorem, we can show (JEE 1990) that for every positive integer n, the expression $\dfrac{n^7}{7} + \dfrac{n^5}{5} + \dfrac{2n^3}{3} - \dfrac{n}{105}$ is an integer. Bringing to a common denominator, the problem is equivalent to proving that for every $n \geq 1$, $15n^7 + 21n^5 + 70n^3 - n$ is divisible by 105. Since 105 is the product of three primes 7, 5 and 3, the problem is further reduced to showing that each of them divides the expression. We take them up one at a time. First consider 7. Surely it divides $21n^5 + 70n^3$. Also it divides $15(n^7 - n)$ by Fermat's theorem. So rewriting the expression as $15(n^7 - n) + 21n^5 + 70n^3 + 14n$ we see that it is divisible by 7. Similarly rewriting it as $15n^7 + 21(n^5 - n) + 70n^3 + 20n$ and applying Fermat's theorem to the second term shows that it is divisible by 5 while divisibility by 3 follows by rewriting it as $15n^7 + 21n^5 + 70(n^3 - n) + 69n$. Thus the expression is divisible by 105. Alternatively we could have written $15n^7 + 21n^5 + 70n^3 - n$ as

$15(n^7 - n) + 21(n^5 - n) + 70(n^3 - n) + 105n$. Then the fourth term is divisible by 105. But each of the first three terms is also divisible by 105 by Fermat's theorem (applied with $p = 7, 5$ and 3 respectively). So the expression is divisible by 105. Yet another method is to apply induction on n and show directly that the expression is divisible by 105. But this essentially amounts to duplicating the proof of the Fermat's theorem.

Comment No. 14:

As an application of the second part of Fermat's theorem, we do the following problem.

Prove that the fortysecond power of any positive integer not divisible by 7 is of the form $49m + 1$ for some integer m.

The problem is equivalent to showing that if a positive integer n is not divisible by 7, then $n^{42} - 1$ is divisible by 49. By Fermat's theorem, $n^6 - 1$ is a multiple of 7 and so we can write n^6 as $7r + 1$ for some integer r. So $n^{42} - 1 = (n^6)^7 - 1 = (7r + 1)^7 - 1$. If we expand $(7r + 1)^7$ as $\sum_{i=0}^{7} \binom{7}{i} 7^i r^i$ we see that the first term (corresponding to $i = 0$) cancels with 1. Every other term in the summation is divisible by 7^2, i.e., by 49. So $n^{42} - 1$ is divisible by 49 as was to be proved.

Note that this problem cannot be done by induction on n in the usual manner, because even if n is not divisible by 7, $n - 1$ may be and so we cannot apply the induction hypothesis to $n - 1$. We shall then have to apply it to $n - 2$ instead of $n - 1$. At any rate, proving the divisibility of $n^{42} - 1$ by 49 from that of either $(n - 1)^{42} - 1$ or $(n - 2)^{42} - 1$ is not so easy. If at all the problem is to be done by induction, we shall have to apply it in the manner of the solution to the problem at the beginning of Comment No. 5 above. If n is not divisible by 7 but is bigger than 7 then we shall apply the induction hypothesis to $n - 7$ which is also not a multiple of 7. Call $n - 7$ as k. Then $n^{42} - 1 = (k + 7)^{42} - 1$. When expanded by the binomial theorem this is of the form $k^{42} - 1 +$ a sum of multiples of 49. So by the induction hypothesis, it is divisible by 49. The trouble with this approach is that since the truth for n is derived from that for $n - 7$, to start the induction we shall have to first verify the first 7 cases. The case $n = 7$ holds true vacuously, since the hypothesis is inapplicable. But the cases $n = 1$ to 6 will have to be verified. And this is a horrendous task. The proof based on the second part of Fermat's theorem is far shorter. (It may sound paradoxical that induction which worked so smoothly in the proof of Fermat's theorem does not work so smoothly in a particular application of it. To explain the paradox, note that in the solution above, we used the second part of Fermat's theorem. The second part was a corollary of the first part which was applicable to all positive integers n and therefore could be proved easily by induction. If we try to prove the second part of Fermat's theorem directly by induction on n, we shall run into the same difficulties as we did above.)

Comment No. 15:

It is convenient to paraphrase some of the results about divisibility in terms of

what is called **modular arithmetic** or **residue arithmetic**. Let b be a positive integer. We say two integers m and n are **congruent** to each other modulo b if the difference $m-n$ is divisible by b. We use the notation $m \equiv n \ (mod\ b)$ for this. It is very easy to show that this is an equivalence relation on the set of all integers. (See Chapter 20, Comment No. 12 for a formal definition of an equivalence relation.) Also using the division algorithm mentioned above in Comment No. 2, it is easy to tell when two integers are congruent to each other. Write $m = qb + r$ and $n = sb + t$, where the remainders r and t are required to lie between 0 and $b-1$ (both included). Then it is clear that $m \equiv n \ (mod\ b)$ if and only if $r = t$. For example, $23 \equiv -57 \ (mod\ 10)$ since $23 - (-57) = 80$ is a multiple of 10. Also, $23 = 2 \times 10 + 3$ and $-57 = -6 \times 10 + 3$. (Note that -57 cannot be written as $-5 \times +(-7)$ for this purpose because of the restriction that the remainder must lie between 0 and 9.)

The equivalence classes of the relation just defined are called the **congruence classes** or **residue classes** modulo b. The residue class $(mod\ b)$ of an integer n consists of the set of all integers which are obtained by adding various multiples of b to n. For example, the mod 12 residue class of 50 consists of the integers $\{\ldots, -34, -22, -10, 2, 14, 26, 38, 50, 62, 74, \ldots\}$. Note that because of the division algorithm, exactly one member of each residue class lies in the range from 0 to $b-1$. This member, often called the **residue** represents that class. Thus 2 is the residue of 50 mod 12. We use the symbol $[2]_{12}$ (or simply [2] when the integer 12 is understood) to denote this residue class. Note that an integer n is divisible by b if and only if the residue class of n $(mod\ b)$ is [0]. This observation is hardly profound. But it suggests why residues are important in divisibility problems.

The residue arithmetic is based on the very simple observation that if b is a fixed positive integer and x, y, z, w are any integers such that $x \equiv z \ (mod\ b)$ and $y \equiv w \ (mod\ b)$ then (i) $x+y \equiv z+w \ (mod\ b)$ (ii) $x-y \equiv z-w \ (mod\ b)$ and (iii) $xy \equiv zw \ (mod\ b)$. As a result, we can add, subtract and multiply whole residue classes unambiguously in terms of their representatives. For example, take $b = 12$ and the residue classes [8] and [7] modulo 12. Then we can take 8 and 7 as their representatives. Adding them we get 15, which is congruent to 3 $(mod\ 12)$. So we write $[8] + [7] = [3] \ (mod\ 12)$. These classes could as well have been represented by, say, 20 and -29 respectively. The sum $20 + (-29)$ is -9 which is not the same as 3, but is congruent to 3 modulo 12. So it has the same residue class, viz., [3]. Similarly, $[8][5] = [4] \ (mod\ 12)$. So we can do all the arithmetic modulo 12 much the same way as we do ordinary arithmetic. In fact after some practice we may drop the brackets and write $8 + 7 = 3 \ (mod\ 12)$ etc. Note however, that the product of two non-zero residues may be 0. For example, we have $[8][6] = [0] \ (mod\ 12)$. So we have to be careful in doing division. For example, if x, y are integers and $8x = 8y$ then we instinctively 'cancel' 8 and conclude $x = y$. But in mod 12 this may be false as can be seen, for example, by taking $x = 5$ and $y = 11$.

Residue arithmetic is very convenient when we want the residue of the product of two large numbers, or, a high power of some integer. Suppose for example, that we want to multiply 827 and 353 modulo 12. We could, of

course, multiply 827 and 353 as ordinary integers to get 291931. Dividing by 12 we can write this as $24327 \times 12 + 7$. So, $827 \times 353 = 7$ $(mod\ 12)$. But we could do this much more easily if we observe that $827 \equiv 11$ $(mod\ 12)$ and $353 \equiv 5$ $(mod\ 12)$. For then, $827 \times 353 = 11 \times 5 = 55 = 7$ $(mod\ 12)$. We can make it still simpler by noting that $11 \equiv -1$ $(mod\ 12)$ and hence $11 \times 5 \equiv (-1) \times 5 \equiv -5 \equiv 7$ $(mod\ 12)$. Of course this is not as informative as the earlier laborious calculation. But if our interest is only in the *mod* 12 residue of the product 827×353 then it is good enough. Similarly suppose we want to find the last two digits of, say, 2^{83}. This is the same thing as finding the residue class of 2^{83} modulo 100. A little experimentation gives $2^{10} = 1024 = 24$ $(mod\ 100)$. Now, $(24)^2 = 576 = 76 = -24$ $(mod\ 100)$. But that means $(24)^3 = -(24)^2 = 24$ $(mod\ 100)$. If we keep on going, we get $(24)^8 = -24$ $(mod\ 100)$ and hence $2^{80} = (2^{10})^8 = (24)^8 = -24$ $(mod\ 100)$. So, $2^{82} = 2^{80}.4 = -96 = 4$ $(mod\ 100)$ and finally $2^{83} = 8$ $(mod\ 100)$. Thus the last two digits of 2^{83} are 0 and 8. (The same answer could have been obtained by noticing that $2^{12} = -4$ $(mod\ 100)$ and by many other means.)

Note, by the way, that we also get $(24)^{10} \equiv -24 \equiv 76$ $(mod\ 100)$ which implies $2^{100} \equiv 76$ $(mod\ 100)$. Thus we see that Fermat's theorem is false if the prime p is replaced by 100 which is a composite number. There is, in fact, no analogue of Fermat's theorem even when p is replaced by p^2. However, when the base is a prime p, Fermat's theorem can be paraphrased as (i) $n^p \equiv n$ $(mod\ p)$ for every positive integer n and further (ii) if $n \not\equiv 0$ $(mod\ p)$ then $n^{p-1} \equiv 1$ $(mod\ p)$. The second part is a powerful tool in computing the residues of various powers of an integer n modulo a prime p. If p divides n, then, of course, all powers of n have residue 0 modulo p. So suppose n is not a multiple of p. Then since $n^{p-1} \equiv 1$ $(mod\ p)$, modulo p, the successive powers $n^p, n^{p+1}, n^{p+2}, \ldots$ are the same as $n, n^2, n^3, \ldots$. When we get to $n^{2(p-1)}$, we again get 1 and then the cycle repeats itself. For example, suppose we want the residue of 10^{100} modulo 7. Take $n = 10$ and $p = 7$. We first reduce 10 to 3 modulo 7. The residues of the successive powers of 3 (and hence of 10) modulo 7, then are $3, 2, 6, 4, 5, 1, 3, 2, 6, 4, 5, 1, 3, 2, \ldots$. They repeat in cycles of length 6. So we subtract the largest multiple of 6, viz., 96 from 100 to get 4. Then $10^{100} \equiv 3^{100} \equiv 3^{96}3^4 \equiv (3^6)^{16}3^4 \equiv 1.(3^4) \equiv 4$ $(mod\ 7)$. With some residues, the repetition may begin even with cycles less than length 6. For example, we have $2^3 = 8 \equiv 1$ $(mod\ 7)$ and so the residues of the successive powers of 2 modulo 7 recur with a cycle of length 3. In particular, therefore, the residue of 2^{100} modulo 7 is simply 2 (since $100 = 99 + 1$ and 99 is a multiple of 3).

The theorem of Lucas, mentioned in Comment No. 7 above can also be paraphrased in terms of congruence relation. Let p be a prime and $\sum_{i=0}^{r} n_i p^i$ and $\sum_{i=0}^{r} k_i p^i$ be the p-ary expansions of two positive integers n and k respectively. Then Lucas' theorem says that $\binom{n}{k} \equiv \binom{n_0}{k_0}\binom{n_1}{k_1} \ldots \binom{n_r}{k_r}$ $(mod\ p)$.

In real life problems residue arithmetic comes in handy when things repeat in regular cycles, as for example, the days of the week. A problem based

on this will be given as an exercise. Also in the exercises we shall point out how the well-known 'rule of 3' (which gives a criterion for divisibility of a positive integer by 3) and a few other similar rules follow easily from modular arithmetic.

Comment No. 16:

We mention here another result, popularly (but incorrectly), called **Wilson's** theorem regarding the divisibility by a prime. It says that if $p > 2$ is a prime, then $(p-1)! + 1$ is divisible by p, or in terms of congruences, $(p-1)! \equiv -1 \ (mod\ p)$. For example, for $p = 5$, $(p-1)! + 1 = 25$ which is indeed divisible by 5. The standard proof is obtained by taking the product of the residues classes of the integers from 1 to $p-1$ and pairing them suitably. We shall omit it. But a combinatorial proof will be given in Exercise (4.40).

Comment No. 17:

An important concept associated with divisibility is that of the **greatest common divisor** (or simply g.c.d. for short) of two integers, say m and n. By definition, this is the largest positive integer which divides both m and n. It is commonly denoted by g.c.d.(m, n) or sometimes simply by (m, n), when this notation is not likely to be confused with the ordered pair (m, n)). The **least common multiple** or the l.c.m. of m and n (denoted by $[m, n]$) is the least positive integer which is divisible by both m and n. In middle school, you must have been thoroughly drilled in finding the g.c.d. and the l.c.m. of two given positive integers m and n by first writing down their prime factor decompositions. It is then easy to show that $(m, n)[m, n]$ always equals mn. Hence the two concepts are related and it suffices to know either one of them. Two integers m and n are called **relatively prime** or **coprime** if they have no common factors other than ± 1. In terms of their prime factorisations this means that there is no prime dividing both m and n. In terms of the g.c.d., it is easy to show that m and n are relatively prime if and only if their g.c.d. is 1. Note that if m is a prime then it is relatively prime to all integers except its own multiples.

There is another method for finding the g.c.d., say d, of two given positive integers, m and n with $n \leq m$ (say). We divide m by n using the euclidean algorithm mentioned above in Comment No. 3. (As a result, this method for finding the g.c.d. is itself called the **euclidean algorithm** sometimes.) If the remainder is 0 we stop. Otherwise we divide n by the remainder and repeat this process till we get 0 as a remainder. Then the last divisor is precisely d. The advantage of this method is that it also shows that d can be expressed as a linear combination of m and n with integer coefficients, that is, there exist (not necessarily positive) integers x and y such that $d = mx + ny$. To illustrate, suppose $m = 1092$ and $n = 195$. Then the following calculation

$$\begin{array}{rcl@{\qquad\qquad}rcl} 1092 & = & 5 \times 195 + 117 & 195 & = & 1 \times 117 + 78 \\ 117 & = & 1 \times 78 + 39 & 78 & = & 2 \times 39 + 0 \end{array}$$

gives 39 as the g.c.d. of 1092 and 195. To express it as a linear combination, we work backwards and write $39 = 117 - 1 \times 78 = 117 - (195 - 117) = 2 \times 117 - 195 =$

$2 \times (1092 - 5 \times 195) - 195 = 2 \times 1092 - 11 \times 195$. So we can take x as 2 and y as -11. Note that these are not unique. In fact, by Exercise (4.11)(b), the integers 28 ($= \frac{1092}{39}$) and 5 ($= \frac{195}{39}$) are relatively prime. It is easy to find, by inspection, many linear combinations of 28 and 5 which equal 1, e.g. $2 \times 28 - 11 \times 5 = 1$, $(-3) \times 28 + 17 \times 5 = 1$ etc. We take one such and multiply throughout by 39.

In most applications, what matters is not the actual coefficients x and y but simply their existence. Suppose, for example, that an integer m divides nk for some integer k. We saw earlier that unless m is a prime, we cannot conclude that either m divides n or that m divides k. Suppose, however, that m and n are relatively prime. Then we can write 1 as $mx + ny$ for some integers x and y. Multiplying by k we get $k = mkx + nky$. As m is given to divide nk, it divides nky. Also it divides mkx. So it divides their sum k. Another important application of the expressibility of the g.c.d. as a linear combination will be indicated in the exercises (Exercise (4.13)).

Comment No. 18:

As remarked in Comment No. 7 of Chapter 2, considerations of divisibility and prime factorisation also help in solving certain equations involving integers, where mere algebra will not suffice. Suppose for example that we want to determine a positive integer n for which $\frac{^{n+2}C_8}{^{n-2}P_4} = \frac{57}{16}$. Writing out the formulas for the expressions, this comes out as $\frac{(n+2)!}{8!(n-6)!(n-2)(n-3)(n-4)(n-5)} = \frac{57}{16}$, or equivalently, $(n+2)(n+1)n(n-1) = 57 \times 7 \times 6 \times 5 \times 4 \times 3$. This is a fourth degree equation in n and is too complicated to solve algebraically. But let us observe that the L.H.S. is a product of four consecutive integers. Since the R.H.S. has 19 as a prime factor, so does the L.H.S. So one of the four factors of the L.H.S. is a multiple of 19. If this multiple is 38 or more then the L.H.S. will be much bigger than the R.H.S. So 19 equals one of these four factors. The other factors must then lie within the range 16 to 22. But since 7 is a factor of the R.H.S. one of them must be 21, the only multiple of 7 within this range. So 20 must also be a factor. The remaining factor comes out as 18. So $n = 19$. A reasoning like this takes long to write. But it can be done very quickly if attempted mentally. Problems like this are therefore, ideal for objective type questions.

In the same vein, a single equation of the form $ax + by = c$ in the two unknowns is insufficient to determine x and y uniquely. There are, in fact, infinitely many solutions. Geometrically, they correspond to points on a straight line. But in real life problems, we may encounter such an equation where a, b, x, y are all integers. For example, x and y may be the numbers of objects of certain types purchased and a, b their costs per unit. In such a case, the number of possible solutions is drastically reduced and can be identified easily. (To obtain the solutions, we divide the equation throughout by the smaller of a and b. See p.111 of *Higher Algebra* by H. S. Hall and S. R. Knight for illustrative examples. Also see Exercise (4.24) for a sample problem of this type.)

Comment No. 19:

The concept of divisibility is also the basis of what is called the theory of **surds**. We begin by showing that certain real numbers are irrational. Take, for

example, $\sqrt{2}$. If it were rational, then after cancelling the common factors, if any, of its numerator and denominator, we can write $\sqrt{2}$ as a reduced rational, i.e., as a rational $\frac{m}{n}$, where m and n are positive integers with no common factor other than ± 1. Then we get $m^2 = 2n^2$ whence m^2 and hence m must be even, say $m = 2k$. But that gives $n^2 = 2k^2$ which means n^2 and hence n must be even. So, m, n have a common divisor, viz. 2, a contradiction. The same reasoning will apply to show that for a positive integer n, $\sqrt{n}$ is rational if and only if n is a perfect square (i.e., the square of some integer).

Let us see how the knowledge of irrationality of a number helps in solving certain equations. Suppose we have an equation of the form $a + \alpha b = c + \alpha d$, where α is some number. From this we cannot conclude that $a = c$ and $b = d$ in general. But sometimes we can. Suppose for example, that a, b, c, d are all real and α is the complex number i (or any other purely imaginary, non-zero number). Then we get the two separate equations by equating the real parts and the imaginary parts of the two sides. Similarly, suppose a, b, c, d are all rational numbers but α is irrational. Then, if $b \neq d$, we can write $\alpha = \frac{c-a}{b-d}$. But this is a ratio of two rational numbers and hence is itself a rational number, a contradiction. A similar, but slightly more complicated argument (which will be given as an exercise) shows that the single equation $a + b\sqrt{2} + c\sqrt{3} = d + e\sqrt{2} + f\sqrt{3}$, where a, b, c, d, e, f are all rationals implies three equations, viz., $a = d$, $b = e$ and $c = f$. Note, however, that an equation $a + b\sqrt{2} + c\sqrt{8} = d + e\sqrt{2} + f\sqrt{8}$ *does not* imply the three equations separately. The reason is that $\sqrt{8}$ is related to $\sqrt{2}$ by a rational. In fact, $\sqrt{8} = 2\sqrt{2}$. So all we can conclude is that $a = d$ and $b + 2c = e + 2f$. Technically, the difference is expressed by saying that the surds $\sqrt{2}$ and $\sqrt{3}$ are **linearly independent** over rationals, while $\sqrt{2}$ and $\sqrt{8}$ are **linearly dependent**. You might wonder why these terms which are used in connection with vectors are used here. There is, indeed, a very close analogy in the two situations. But we shall not go into it.

EXERCISES

4.1 If $n_1, n_2, \ldots, n_p$ are p positive integers, q of which are odd and the sum $n_1 + n_2 + \ldots + n_p$ is odd, then is q necessarily odd? (1985)

4.2 Find the sum of the integers from 1 to 100 that are divisible by 2 or 5. (1984)

4.3 Let $f(x) = Ax^2 + Bx + C$, where A, B, C are real numbers. Prove that if $f(x)$ is an integer whenever x is an integer, then the numbers $2A, A + B$ and C are all integers. Conversely, prove that if $2A, A + B$ and C are all integers then $f(x)$ is an integer whenever x is an integer. (1998)

4.4 For every positive integer m, prove that $\binom{2m-1}{m}$ is even except when m is a power of 2.

4.5 Let the three digit numbers $A28, 3B9$ and $62C$, where A, B, C are integers between 0 and 9 be divisible by a fixed integer k. Show that the

$$\text{determinant} \begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix} \text{ is divisible by } k. \qquad (1990)$$

4.6 (a) Prove the 'rule of 3' learnt in school, viz., that a positive integer written in the decimal expansion is divisible by 3 if and only if the sum of the digits is divisible by 3. State and prove similar rules for divisibility by 9 and by 11. What happens if instead of the decimal expansion, we consider expansion with some other base?

(b) Find a 4-digit number with base 7, which is halved when its last digit is brought in front.

4.7 According to a Western superstition, 13 is an evil figure and when the 13th of a month falls on a Friday, such a 'Friday the 13th' is considered as an especially bad day. Using modulo 7 arithmetic, prove that in any calendar year, there is at least one and at most three Friday the thirteenths.

4.8 (a) Prove that the number $5^{12}+2^{10}$ is not prime. (Regional Mathematics Olympiad)

(b) Prove that a number of the form $p_1^2 + p_2^2$, where p_1, p_2 are primes cannot be a prime unless one of them is 2.

(c) Prove that a number of the form $p_1^2 + p_2^2 + p_3^2$, where p_1, p_2, p_3 are odd primes cannot be a prime unless one of them is 3.

(d) Prove that there are infinitely many primes but also that there are arbitrarily long blocks of consecutive integers which contain no primes. [*Hint*: For the first part show that from every finite set of primes we can generate a new prime. This argument, due to Euclid, is to this day, one of the best examples of a mathematical argument. For the second part, note that for $k = 2, 3, \ldots, n$, $n! + k$ is divisible by k.]

(e) Prove that there are infinitely many primes of the form $4k-1$. (This is also true, but not so easy to prove, for primes of the form $4k+1$.)

(f) Prove that the number x^2+x+41 is a prime for all $x = 0, 1, 2, \ldots, 39$, but not for $x = 40$. (This remarkable property of this polynomial led to the search of what are called **prime generating functions**, i.e., functions $f(x)$ which will assume only primes as values for all positive integral values of x.)

(g) Prove that no non-constant polynomial $f(x)$ can have the property that $f(x)$ is a prime whenever x is a positive integer. (In other words, no non-constant polynomial can be a prime generating function.)

(h) If $p > 2$ is a prime then prove that $\{1.3.5.7.\ldots.(p-2)\}^2 \equiv (-1)^{(p+1)/2}$ $(mod\ p)$, that $\binom{p-2}{r} \equiv (-1)^r(r+1)$ $(mod\ p)$ for $r = 1, 2, \ldots, p-2$, and finally, that $\binom{p-1}{(p-1)/2} \equiv (-1)^{(p-1)/2} (mod\ p)$.

(i) If p is a prime of the form $4n+1$, prove that there exists an integer c such that p divides c^2+1, i.e. $c^2 \equiv -1 (mod\ p)$. (For this reason, we say that -1 is a **quadratic residue** modulo p.)

(j) If p and $p-2$ are both primes (e.g., when $p = 13$ or 31) prove that $p(p-2)$ divides $4(p-3)!+p+2$. (Two primes differing by 2 from each other are called **twin primes**. It is not known if there are infinitely many such pairs.)

4.9 (a) If $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ are distinct primes, find the number of ordered triples (x, y, z) of positive integers for which $xyz = n$. Also prove that $\ln n > k \ln 2$. (1984)

(b) Find the number of divisors of the form $4n+2$ $(n \geq 0)$ of 240. (1998)

4.10 The n-th harmonic number H_n is defined as $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$. Prove that H_n is not an integer except when $n = 1$. [*Hint:* Consider $2^{m-1} H_n$, where 2^m is the highest power of 2 not exceeding n.]

4.11 If d is the g.c.d. of two integers m and n, prove that:

(a) d is the smallest positive integer which can be expressed as a linear combination of m and n with integer coefficients.

(b) $\frac{m}{d}$ and $\frac{n}{d}$ are integers and further that they are relatively prime

(c) the least common multiple (l.c.m.) of m and n is $\frac{mn}{d}$.

4.12 Let m, n be positive integers with m relatively prime to n. Prove that:

(a) for any integers x and y, $mx \equiv my \ (mod\ n)$ implies $x \equiv y \ (mod\ n)$. (This is called the **cancellation law** for modular multiplication.)

(b) there exists k such that $mk \equiv 1 \ (mod\ n)$. (This is expressed by saying that m has a **multiplicative inverse** modulo n.)

(c) there exists a positive integer k such that $m^k \equiv 1 \ (mod\ n)$. (The least such integer is called the **order** of m modulo n.) Find the orders of 6 and 2 modulo 35.

(d) if r is the order of m modulo n, then prove that for any two positive integers s and t, $m^s \equiv m^t \ (mod\ n)$ if and only if r divides $s - t$.

(e) if n is a prime, then the order of m divides $n - 1$.

(f) if $p = 4k + 3$ is a prime then -1 is not a quadratic residue modulo p.

4.13 Let the positive integers $n_1, n_2, \dots, n_k$ be mutually coprime (i.e., every two of them are relatively prime to each other). Prove (a) to (c) below.

(a) for every $i = 1, 2, \dots, k$, there exists an integer m_i such that $m_i \equiv 1 \ (mod\ n_i)$ and $m_i \equiv 0 \ (mod\ n_j)$ for all $j = 1, 2, \dots, k, j \neq i$. [*Hint:* Note that n_i is relatively prime to the product of the remaining n_j's.]

(b) given any integers $r_1, r_2, \dots, r_k$ with $0 \leq r_i < n_i$ for $i = 1, 2, \dots, k$, there exists an integer m such that $m \equiv r_i \ (mod\ n_i)$ for every $i = 1, 2, \dots, k$. (In other words, m leaves a remainder r_1 when divided by n_1, a remainder r_2 when divided by n_2 and so on.) Prove further that m is unique modulo $n_1 n_2 \dots n_k$. (This result was known to the Chinese and is called the **Chinese Remainder Theorem**.)

(c) all even numbers which, when divided by 7 and 9, give remainders 4 and 3 respectively, are of the form $126p - 24$, where p is an integer.

(d) Find the last two digits (in base 10) of 3^{110}.

(e) If $7^{100} = 100x$ and $3^{100} = 100y$, prove that $x - y = [x] - [y]$.

(f) Let $T_{n,k} = (n+1)^k + (n+2)^k + \ldots + (n+100)^k$, where n, k are positive integers. Is $T_{n,k}$ divisible by 100 for $k = 1, 2, 3$?

4.14 A remote village is visited by a mailman every third day, by a grocer every seventh day and by a physician every tenth day. On a particular day, it was found that the mailman had visited it the day before, the grocer 3 days before and the physician 2 days before. Find the first day thenceforth when all three would visit it on the same day. (Similar considerations apply to predict when astronomical events such as eclipses will occur next.)

4.15 If the integers m, n are chosen at random between 1 and 100, find the probability that a number of the form $7^m + 7^n$ is divisible by 5. (1999)

4.16 If p is a natural number then prove that $p^{n+1} + (p+1)^{2n-1}$ is divisible by $p^2 + p + 1$ for every positive integer n. (See also Exercise (7.15).) (1984)

4.17 Use mathematical induction to prove that $(25)^{n+1} - 24n + 5735$ is divisible by $(24)^2$ for all $n = 1, 2, \ldots$. (2002)

4.18 If p, q are distinct odd primes, prove that pq divides $p^{q-1} + q^{p-1} - 1$.

4.19 Prove that both the numbers $16^{99} - 1$ and $(18)! + 1$ are divisible by 437.

4.20 If m is relatively prime to 30, prove that $m^4 - 1$ is divisible by 240.

4.21 If $n^5 - 5$ is divisible by 91 then prove that $n - 31$ is divisible by 91.

4.22 (a) If $p > 3$ is a prime and $p^2 + a^2 = b^2$, where a, b are positive integers then prove that 12 divides a and $2(a + p + 1)$ is a square.

(b) If a, b, c are positive integers every two of which are relatively prime and $a^2 + b^2 = c^2$, prove that exactly one of a and b is even. If b is even, show that there exist integers u and v such that $a = u^2 - v^2, b = 2uv$ and $c = u^2 + v^2$.

(c) A triple (a, b, c) of positive integers in which $a^2 + b^2 = c^2$ is called a **Pythagorean triple** for obvious reasons. Using (b), characterise all Pythagorean triples.

4.23 Consider an equation of the form $ax + by = c$, where the constants a, b, c and the variables x, y are all integers. Let d be the g.c.d. of a and b. Prove that this equation has a solution if and only if d divides c.

4.24 Fruits of type A, B, C are available with prices 5 paise , 1 rupee and 5 rupees per fruit respectively. A person buys 100 fruits for 100 rupees with at least one fruit of each type. Find how many fruits of each type he buys.

4.25 Prove combinatorially that for positive integers m, n, p, q, (i) $(mn + pq)!$ is divisible by $(m!)^n(p!)^q$ and (ii) $(n^2)!$ is divisible by $(n!)^{n+1}$.

4.26 For $n \geq 0$, let F_n be the n-th Fibonacci number. (Thus, $F_0 = 0, F_1 = 1, F_2 = 1, F_3 = 3, F_4 = 5$ etc.) For positive integers m, n, p, q, prove that:

(a) $F_{n+m} = F_m F_{n+1} + F_{m-1} F_n$. [*Hint:* Use induction on m.]

(b) $F_{n+1} F_{n-1} - F_n^2 = (-1)^n$.

(c) If p divides q then F_p divides F_q. [*Hint:* Use (a) and induction on the ratio q/p.]

(d) F_n and F_{n+1} are relatively prime.

(e) $F_n^2 + F_{n-1}^2$ is also a Fibonacci number.

4.27 Show that the number $\log_2 7$ is irrational. (1990)

4.28 For real numbers x and y we write $x * y$ if $x - y + \sqrt{2}$ is an irrational number. Is the relation $*$ an equivalence relation (i.e., is it reflexive, symmetric and transitive)? (1981)

4.29 Using factorisation into powers of primes, show that for a positive integer n, $\sqrt{n}$ is irrational unless n is a perfect square. State and prove a similar result for cube roots and more generally, for the k-th roots, where k is any integer greater than 1.

4.30 Suppose a, b, c, p, q, r are rational numbers. Prove that an equation of the form $a + b\sqrt{2} + c\sqrt{3} = p + q\sqrt{2} + r\sqrt{3}$ implies $a = p$, $b = q$ and $c = r$. Does the equation $a\sqrt{3} + b\sqrt{12} = p\sqrt{3} + q\sqrt{12}$ imply $a = p$ and $b = q$?

4.31 If a, b, x, y are rational numbers and $(a + b\sqrt{2})^2 = x + y\sqrt{2}$, prove that $(a - b\sqrt{2})^2 = x - y\sqrt{2}$.

4.32 Find the sum of the rational terms in $(\sqrt{2} + 3^{1/5})^{10}$. (1997)

4.33 Show that the square of $\frac{\sqrt{26-15\sqrt{3}}}{5\sqrt{2}-\sqrt{38+5\sqrt{3}}}$ is a rational number. (1978)

4.34 If $x > y > 0$, show that the expression

$$\sqrt{2}(2x + \sqrt{x^2 - y^2})(\sqrt{x - \sqrt{x^2 - y^2}})$$

can be simplified to $\sqrt{(x+y)^3} - \sqrt{(x-y)^3}$. (1980)

4.35 Determine which of the following expressions equals $\frac{12}{3+\sqrt{5+2\sqrt{2}}}$. (1980)

(a) $1 + \sqrt{5} + \sqrt{2} + \sqrt{10}$ (b) $1 + \sqrt{5} + \sqrt{2} - \sqrt{10}$
(c) $1 + \sqrt{5} - \sqrt{2} + \sqrt{10}$ (d) $1 - \sqrt{5} - \sqrt{2} + \sqrt{10}$.

4.36 Let $p > 3$ be a prime. Prove that the expression $(p-1)!(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{p-1})$ is an integer divisible by p. Deduce that $(2p)! - 2(p!)^2$ is divisible by p^4. [*Hint:* For each $m = 1, 2, \ldots, p-1$, let a_m be the unique integer (given by Exercise (4.12)(b)) such that $1 \leq a_m \leq p - 1$ and $ma_m \equiv 1 \ (mod\ p)$. As m varies so does a_m and over the same set.]

4.37 Let n be a positive integer. Denote by $\phi(n)$ the number of integers between 1 to $n-1$ which are relatively prime to n. Also denote by $d(n)$ the number of positive integers that divide n. (For example, for $n = 10$, these integers are 1, 3, 7 and 9 and so $\phi(10) = 4$.) The function ϕ so defined is called the **Euler totient function** or simply the ϕ function and occurs frequently in many problems of number theory. (We set $\phi(1) = 1$.) Similarly, define $d(n)$ to be the number of positive integers that divide n (including 1 and n itself). For example, for $n = 12$, these integers are $1, 2, 3, 4, 6, 9$ and 12 and so $d(12) = 6$. Prove the following simple properties of these functions:

(i) If p is a prime, then $\phi(p) = p - 1, d(p) = 2$ and more generally, for any positive integer r, $\phi(p^r) = p^r - p^{r-1}$ and $d(p^r) = r + 1$.

(ii) If n has a prime power factorisation as $n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$, then $\phi(n) = (p_1^{r_1} - p_1^{r_1-1})(p_2^{r_2} - p_1^{r_2-1}) \dots (p_k^{r_k} - p_k^{r_k-1})$. [*Hint:* Apply the principle of inclusion and exclusion, given in Exercise (1.43).]

(iii) If m and n are relatively prime then $\phi(mn) = \phi(m)\phi(n)$. [*Hint:* Use (ii). Functions satisfying this property are said to be **multiplicative**. This usage of the term is somewhat unfortunate because normally, a real-valued function, say $f(x)$, of a variable x is called multiplicative if $f(xy) = f(x)f(y)$ for *all* x and y in the domain of f. (For example, the absolute value is a multiplicative function on the set of all complex numbers. Or, the determinant is a multiplicative function on the set of all matrices of a given order as seen from Equation (56) in Chapter 2.) In the present case, the equality $\phi(mn) = \phi(m)\phi(n)$ does not hold if m and n are not relatively prime as can be shown by simple examples. But number theory abounds in examples of functions where the multiplicativity condition holds only for a pair of integers that are relatively prime to each other and this is enough for many applications. So for functions from $\mathbb{N}$ to $\mathbb{N}$, the term is used in the restricted sense above.]

(iv) The function d is also multiplicative. Find a formula for $d(n)$ in terms of the prime factorisation of n.

*4.38 Prove that Fermat's theorem can be generalised to say that for any integer m which is relatively prime to n, $m^{\phi(n)} \equiv 1 \ (mod\ n)$.

4.39 Let p be a prime and m a positive integer.

(a) Let S be the set of all ways to paint the vertices of a regular p-gon with m colours so that at least two colours are used. Show that $|S| = m^p - m$. (Here we are not identifying any two colourings.)

(b) Prove that if two colourings obtained by a rotation of the polygon are to be identified with each other, then every colouring in S gets identified with p colourings (including itself). Is this still true if p is not a prime?

(c) Using (a) and (b) prove that $m^p \equiv m \ (mod\ p)$. (This gives a combinatorial proof of Fermat's theorem. However a similar proof of the generalised version given in the last exercise is not so easy.)

4.40 Suppose p is an odd prime and we have p guests, say, $g_1, g_2, \ldots, g_p$. Here we shall use the subscripts modulo p. That is, for any integers i and j, g_i and g_j will denote the same guest if $i \equiv j \ (mod\ p)$. (For example, if $p = 7$ the g_3 can also be denoted by g_{10}, or by g_{17} or even by g_{-4}.) Let S be the set of all circular arrangements of these guests around a table. As proved in Comment No. 8 of Chapter 1, S consists of $(p-1)!$ distinct equivalence classes, each class consisting of p arrangements obtained by cyclic shifts of one another. Let us denote such equivalence classes by putting square brackets around ordered p-tuples denoting permutations of guests. Thus, for example, the equivalence classes $[(g_1, g_2, \ldots, g_p)]$ and $[(g_3, g_4, \ldots, g_{p+2})]$ are the same but $[(g_1, g_p, g_{p-1}, \ldots, g_3, g_2)]$ is a different element of S. Denote such equivalence classes (i.e., elements of the set S) by Greek letters, α, β etc.

We now define a binary relation R on S as follows. Let $\alpha, \beta \in S$. Then we say $\alpha R \beta$ if there exist *some* arrangements $(g_{i_1}, g_{i_2}, \ldots, g_{i_p})$ and $(g_{j_1}, g_{j_2}, \ldots, g_{j_p})$ representing α and β respectively (i.e., such that $\alpha = [(g_{i_1}, g_{i_2}, \ldots, g_{i_p})]$ and $\beta = [(g_{j_1}, g_{j_2}, \ldots, g_{j_p})]$) and an integer r such that for every $k = 1, 2, \ldots, p$, $j_k = i_k + r$. (For example, suppose $p = 7$, $\alpha = [(g_2, g_4, g_5, g_7, g_3, g_1, g_6)]$ and $\beta = [(g_2, g_4, g_7, g_5, g_3, g_6, g_1)]$. Then β also equals $[(g_6, g_1, g_2, g_4, g_7, g_5, g_3)]$ and so we see that $\alpha R \beta$ by taking $r = 4$, because $2 + 4 = 6, 4 + 4 = 1, 5 + 4 = 2$ etc. the addition being modulo 7.) Prove that:

(a) R is an equivalence relation on the set S.

(b) there are $p-1$ elements of S each of which is equivalent only to itself under R (for example, for $p = 7$, $[(g_1, g_4, g_7, g_3, g_6, g_2, g_5)]$ is one such element).

(c) every other element of S is equivalent to p elements (including itself).

(d) $(p-1)! \equiv -1 \ (mod\ p)$. (This gives a combinatorial proof Wilson's theorem.)

4.41 Suppose a function $f : \mathbb{N} \longrightarrow \mathbb{N}$ satisfies the condition that $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$. (As remarked in Exercise (4.37) (iii), this condition is stronger than multiplicativity.) Prove that :

(a) $f(1) = 1$ and f is uniquely determined by its values on primes. [*Hint:* Use unique factorisation of integers into prime powers.]

(b) if further $f(2) = 2$ and f is strictly increasing then $f(n) = n$ for all n. [*Hint:* Let p_k be the k-th prime, with $p_1 = 2$, $p_2 = 3$ etc. Use (a) and induction on k to show that $f(p_k) = p_k$. Note that for $k > 1$, the prime factors of $p_k - 1$ and $p_k + 1$ are all less than p_k and hence $f(p_k - 1) = p_k - 1$ and $f(p_k + 1) = p_k + 1$. Fit in $f(p_k)$.]

Chapter 5

BINOMIAL IDENTITIES

There are numerous identities involving the binomial coefficients. The present chapter deals with some of the more common ones among them. The focus, however, is not on the identities *per se*, but rather on the methods of proofs. We discuss four standard methods to prove binomial identities. The first is by direct computation, i.e., by directly reducing one side of the identity to be proved to the other. In complicated problems this method rarely works. One can then try mathematical induction. Yet another method is to recognise the two sides as the coefficients of the same power of a variable x in some functions of x, say $f(x)$ and $g(x)$, and then show by some manipulations (which may be purely algebraic but may, at times, also involve techniques from calculus such as differentiation and integration) that these two functions are identical. This technique becomes very powerful when these functions of x are what are called a power series in x. But as power series are beyond our scope, we shall confine ourselves to the case where they are polynomials. Not surprisingly, the binomial theorem will be the principal tool of attack.

The fourth method of proving combinatorial identities is by recognising the two sides as the cardinalities of two sets and by establishing a direct bijection between these two sets. This is known as a combinatorial argument. Such an argument is not always available. But when it is, it is, by far, the most elegant way to prove the identity. The Main Problem illustrates the various methods of proving the binomial identities.

Binomial identities, unless they are very simple, are generally suitable only as full length questions. However, problems requiring only an application and not a proof of some standard binomial identity can appear as short questions in JEE.

Main Problem : Assume q is a positive integer with $q \leq 50$. If the sum $\binom{98}{30}+2\binom{97}{30}+3\binom{96}{30}+\ldots+68\binom{31}{30}+69\binom{30}{30}$ equals $\binom{100}{q}$ then q equals ..... . (JEE 2000, modified)

First Hint: First show that for positive integers m, n with $n \geq m$,

$$\binom{n}{m}+\binom{n-1}{m}+\binom{n-2}{m}+\ldots+\binom{m+1}{m}+\binom{m}{m}=\binom{n+1}{m+1} \qquad (1)$$

Second Hint: Using (1) repeatedly or otherwise, show that

$$\binom{n}{m}+2\binom{n-1}{m}+\ldots+(n-m)\binom{m+1}{m}+(n-m+1)\binom{m}{m}=\binom{n+2}{m+2} \quad (2)$$

Solution: The terms in the L.H.S. of (1) are the coefficients of x^m in $(1+x)^n$, $(1+x)^{n-1}, (1+x)^{n-2}, \ldots, (1+x)^{m+1}$ and $(1+x)^m$ respectively. So the L.H.S. of (1) is the coefficient of x^m in the sum $(1+x)^n+(1+x)^{n-1}+(1+x)^{n-2}+\ldots+(1+x)^{m+1}+(1+x)^m$. When read backwards, this is the sum of a geometric progression with the first term $(1+x)^m$ and common ratio $1+x$. So, this sum equals $(1+x)^m\dfrac{(1+x)^{n-m+1}-1}{1+x-1}$, i.e., $\dfrac{(1+x)^{n+1}-(1+x)^m}{x}$. The coefficient of x^m in this is the same as the coefficient of x^{m+1} in the numerator, i.e., in $(1+x)^{n+1}-(1+x)^m$. Since $(1+x)^m$ is a polynomial of degree m in x, it cannot contain the term x^{m+1}. So the L.H.S. of (1) is the coefficient of x^{m+1} in $(1+x)^{n+1}$, which equals $\binom{n+1}{m+1}$, the R.H.S. of (1).

We can derive (2) from (1) by regrouping the terms on the L.H.S. of (2). Note that the r-th term appears with a coefficient r for $1 \leq r \leq n-m+1$. So we divide the terms into $n-m+1$ groups as

$$\begin{array}{rrrrrr}
\binom{n}{m}+ & \binom{n-1}{m}+ & \binom{n-2}{m}+ & \ldots & +\binom{m+1}{m} & +\binom{m}{m} \\
+ & \binom{n-1}{m}+ & \binom{n-2}{m}+ & \ldots & +\binom{m+1}{m} & +\binom{m}{m} \\
 & + & \binom{n-2}{m}+ & \ldots & +\binom{m+1}{m} & +\binom{m}{m} \\
 & & & \ldots & & \\
 & & & & +\binom{m+1}{m} & +\binom{m}{m} \\
 & & & & & +\binom{m}{m}
\end{array}$$

We now apply (1) to add the terms in each row to get that they equal, respectively, $\binom{n+1}{m+1}$, $\binom{n}{m+1}$, $\binom{n-1}{m+1}$, …, $\binom{m+2}{m+1}$ and $\binom{m+1}{m+1}$. We can add these using (1) again (with n, m being replaced by $n+1, m+1$ respectively) to get that the sum on the L.H.S. of (2) equals $\binom{n+2}{m+2}$. Putting $n=98$ and $m=30$

we see that the sum in the given problem equals $\binom{100}{32}$. Thus $q = 32$ or 68. But $q \leq 50$. So $q = 32$. (This step needs a little justification. Specifically, we must ensure that if $\binom{100}{32} = \binom{100}{q}$ and $0 \leq q \leq 50$, then $q = 32$. This follows from the fact that for $0 \leq r < s \leq 50$, $\binom{100}{r} < \binom{100}{s}$ as will be shown in Comment No. 9 below.)

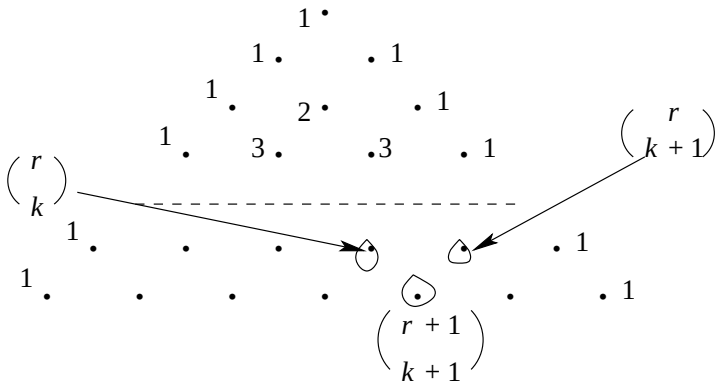
Comment No. 1:

In a more conventional form the problem would ask for a proof of (2) with (1) thrown in as an optional hint. The identity (1) as well as its proof given above are fairly well-known. However, other proofs are possible. The simplest one is probably to replace the last term in the L.H.S. by $\binom{m+1}{m+1}$ and then go on adding them from the right to the left making repeated use of the **reduction formula** (also often called the **addition formula**), viz., $\binom{r}{k} + \binom{r}{k+1} = \binom{r+1}{k+1}$. Then the sum of the last two terms is simply $\binom{m+2}{m+1}$. When we add to this the third term from the right which equals $\binom{m+2}{m}$, we get that the sum of the last three terms is $\binom{m+3}{m+1}$. We now add the fourth term from the right, viz., $\binom{m+3}{m}$ and get $\binom{m+4}{m+1}$ as the sum of the last four terms. Continuing like this, we see that the sum of all the $n - m + 1$ terms in the L.H.S. of (1) is $\binom{m+n-m+1}{m+1}$ i.e., $\binom{n+1}{m+1}$.

We can also prove (1) by induction on n for a fixed m. The statement to be proved by induction will be "For every $n \geq m$, (1) holds." The induction will start with the case $n = m$ which is true since $\binom{m}{m}$ and $\binom{m+1}{m+1}$ both equal 1. For the inductive step, assume that (1) holds for $n = k$. To prove it for $n = k + 1$, we merely add the term $\binom{k+1}{m}$ to both the sides and get that the value of the L.H.S. of (1) for $n = k + 1$ is $\binom{k+1}{m+1} + \binom{k+1}{m}$, which equals $\binom{k+2}{m+1}$, which is precisely the R.H.S. of (1) for $n = k + 1$. This completes the inductive step. There is actually hardly any difference between this proof by induction and the earlier proof obtained by adding the terms of the L.H.S. of (1) from right to left. Both the proofs began by observing that $\binom{m+1}{m+1}$ equals $\binom{m}{m}$ and the crux of both the proofs was the formula $\binom{r}{k} + \binom{r}{k+1} = \binom{r+1}{k+1}$. The difference between the two proofs is more verbal than mathematical. 'Continuing like this ...' is a layman's language. Induction merely gives it a more sophisticated form.

The crux of both these proofs lies in the repeated applications of the reduction formula $\binom{r}{k} + \binom{r}{k+1} = \binom{r+1}{k+1}$. The reason for the peculiar name is that this formula reduces the computation of the binomial coefficient $\binom{r+1}{k+1}$ to that of those binomial coefficients in which the upper index (viz., $r + 1$) is *reduced* by one. There is a nice geometric way of writing the binomial coefficients in the form of what is called the **Pascal triangle** shown in the figure below. This consists of rows of dots, the count beginning with the 0-th rather than with the first row. For every integer $n \geq 0$, there are $n + 1$ points in the n-th row. They are marked with the values of the binomial coefficients $\binom{n}{k}$ for $0 \leq k \leq n$. (Thus, there are infinitely many rows. But obviously, only the first few rows can be shown in any diagram.)

Certain properties of the binomial coefficients can be vividly interpreted in terms of the Pascal triangle. For example, the symmetry relation $\binom{n}{k} = \binom{n}{n-k}$ translates into saying that the Pascal triangle is symmetric w.r.t. the vertical line passing through its topmost vertex (or 'apex' as it is called). Similarly, the well-known identity $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (which is proved most easily by setting $x = 1$ in the binomial expansion of $(1 + x)^n$) amounts to saying that the sum of the entries in the n-th row of the Pascal triangle is 2^n.



Pascal triangle

Let us now see what the reduction formula $\binom{r}{k} + \binom{r}{k+1} = \binom{r+1}{k+1}$ means in terms of the Pascal triangle. The entry on the R.H.S. lies in the $(r + 1)$-th row of the triangle while those on the left lie in the r-th row. But because of the spacings of the dots, one of them lies to the left and the other to the right of the dot corresponding to the binomial coefficient $\binom{r+1}{k+1}$. (Three such entries are shown by circling in the diagram above.) Thus, geometrically, the reduction formula says that any entry in the Pascal triangle equals the sum of its two neighbours one to its northeast and the other to its northwest. (If the entry happens to be on the border, then one of these neighbours is absent.)

Repeated use of the reduction formula also gives the solution to the following problem. (JEE 1980)

Express the sum $\binom{47}{4} + \sum_{j=1}^{5} \binom{52-j}{3}$ as a binomial coefficient.

For the solution, we combine $\binom{47}{4}$ with the last term of the summation, viz., $\binom{47}{3}$ to get $\binom{48}{4}$. Then we combine this with the last but one term of the summation, viz., $\binom{48}{3}$ to get $\binom{49}{4}$. Continuing like this we ultimately get $\binom{52}{4}$ as the answer.

The following problem (JEE 1985) is yet another example where the use of this reduction formula does the trick.

Show that :

$$\begin{vmatrix} \binom{x}{r} & \binom{x}{r+1} & \binom{x}{r+2} \\ \binom{y}{r} & \binom{y}{r+1} & \binom{y}{r+2} \\ \binom{z}{r} & \binom{z}{r+1} & \binom{z}{r+2} \end{vmatrix} = \begin{vmatrix} \binom{x}{r} & \binom{x+1}{r+1} & \binom{x+2}{r+2} \\ \binom{y}{r} & \binom{y+1}{r+1} & \binom{y+2}{r+2} \\ \binom{z}{r} & \binom{z+1}{r+1} & \binom{z+2}{r+2} \end{vmatrix}$$

For a solution, applying the reduction formula to each term in the middle column of the determinant on the right, we split the middle column into two (leaving the other two columns unchanged). So the determinant on the right equals the sum of two determinants, viz.,

$$\begin{vmatrix} \binom{x}{r} & \binom{x}{r} & \binom{x+2}{r+2} \\ \binom{y}{r} & \binom{y}{r} & \binom{y+2}{r+2} \\ \binom{z}{r} & \binom{z}{r} & \binom{z+2}{r+2} \end{vmatrix} + \begin{vmatrix} \binom{x}{r} & \binom{x}{r+1} & \binom{x+2}{r+2} \\ \binom{y}{r} & \binom{y}{r+1} & \binom{y+2}{r+2} \\ \binom{z}{r} & \binom{z}{r+1} & \binom{z+2}{r+2} \end{vmatrix}$$

Obviously the first determinant is 0. For the second determinant, we apply the reduction formula twice to each term in the last column to get $\binom{x+2}{r+2} = \binom{x}{r} + 2\binom{x}{r+1} + \binom{x}{r+2}$ etc. Then the third column and hence the determinant splits as a sum of three determinants of which the first two vanish and the third one is precisely the determinant on the L.H.S. of the problem.

Comment No. 2:

There is, however, a proof of (1) which is qualitatively different from any of the earlier proofs. The proofs so far were purely algebraic and did not take into account the combinatorial significance of the binomial coefficients, viz., that $\binom{n}{k}$ is the number of ways to select k objects from n objects. In the language of set theory, suppose S is a set with n elements, or an n-set for short. Then $\binom{n}{k}$ is the number of all k-subsets of S, i.e., subsets having exactly k elements. A **combinatorial proof** is a proof which is obtained by counting something in two different ways and then equating the two counts. As a rather trivial example, the commutative law for multiplication of two positive integers can be proved combinatorially by considering a rectangular array of dots with m rows and n columns (say). If we count the number of dots in the array row-wise it comes to mn as there are m rows with n dots in each. But if we count the number of dots columnwise, we get nm. Since the number of dots is independent of the manner in which they are counted we must have $mn = nm$.

As a less trivial example, consider the reduction formula $\binom{r}{k} + \binom{r}{k+1} = \binom{r+1}{k+1}$ used above. A straight algebraic proof of this formula can be got by writing out each term as a ratio of certain factorials and showing that the two terms on the left, when expressed with a common denominator (viz., $(k+1)!$) do add up to the term on the right. A less direct algebraic proof would be to write $(1+x)^{r+1} = x(1+x)^r + (1+x)^r$ and to equate the coefficients of x^{k+1} of both the sides using the binomial theorem. In a combinatorial proof,

we consider a set, say S, with $r+1$ elements and the totality of its subsets having $k+1$ elements each. Then the R.H.S. $\binom{r+1}{k+1}$ gives the total number of all such subsets. We shall now divide the collection of all such subsets into two mutually disjoint classes in such a way that in the first class there are $\binom{r}{k}$ of these subsets and in the second there are $\binom{r}{k+1}$ subsets. We would then get a combinatorial proof of the formula $\binom{r}{k}+\binom{r}{k+1}=\binom{r+1}{k+1}$. The art lies in deciding how to do this classification. In the present problem this is rather easy. We fix any one element, say s_0 of the set S. (It does not matter which one we fix.) Now classify the $(k+1)$-subsets of S according to whether or not they contain this special element s_0. Those that do must get their remaining k elements from the complementary set $S-\{s_0\}$ which has r elements. So the number of such subsets is $\binom{r}{k}$. On the other hand, a $(k+1)$-subset of S which does not contain s_0 is also a $(k+1)$-subset of the k-set $S-\{s_0\}$ and vice versa. So the number of subsets of this second type is $\binom{r}{k+1}$. We now have a desired classification.

Combinatorial proofs are usually more elegant, analogous to proofs based on pure geometry. But they suffer the same drawback, viz., they are not always easy to conceive. Algebraic proofs, on the other hand, are like the proofs using co-ordinates; not very artistic but not elusive either. A more serious limitation of the combinatorial proofs is that they are applicable only to identities involving integers, and that too, mostly positive integers, because you cannot think of a set with, say, -8 elements! For example, you cannot prove the commutativity of multiplication of real numbers using a combinatorial argument. Despite these drawbacks, when you can afford to spend time, the search for a combinatorial proof can be an exciting experience. Inherent in a combinatorial proof is an association of various terms of the formula with something concrete. So, combinatorial proofs are also easy to remember and to reproduce. For example, if you know only the algebraic proof of the formula $\binom{r}{k}+\binom{r}{k+1}=\binom{r+1}{k+1}$, the algebraic manipulations in proving it are so commonplace, that you are likely to think of the formula as just one of dozens of algebraic identities and may soon forget it. But if you have seen the combinatorial proof, you will probably not forget it for a long time.

Let us now attempt a combinatorial proof of the identity (1). We immediately recognise the R.H.S. as the number of all $(m+1)$-subsets of an $(n+1)$-set, say S. We want to classify these into $n-m+1$ mutually disjoint classes so that the numbers of elements in these classes correspond to the terms of the L.H.S. of (1). There is no obvious way to do this given the set S about which we know nothing except that it has $n+1$ elements. So we resort to an important trick here, viz., we put an additional structure of our choice on the set S. As long as the set S has $n+1$ elements, the number of $(m+1)$-subsets of S does not depend on what S consists of. It could just as well be a set of elephants or a set of students or a set of some abstract objects such as the roots of a polynomial of degree $n+1$. We are, in fact, free to choose the set. We take it to be the set of all integers from 1 to $n+1$. In effect, this means that we are ordering the elements of S. Thus we are imposing an additional structure (in the present case, an order structure) on the set S. This structure is not part

of the problem. But it simplifies the solution considerably, like a catalyst in a chemical reaction. A more appropriate analogy would be the 'construction' that we sometimes employ in attacking a geometric rider. You can spend hours on the rider without getting anywhere near the solution. But just drop one perpendicular somewhere and the problem unfolds itself!

The same thing happens when we put the order structure on S, i.e., take S to be the set $\{1, 2, 3, \ldots, n, n+1\}$. Let A be an $(m+1)$-subset of S, i.e., a subset with $m+1$ elements. The trick is to consider the last (or the largest) element of A. Let this last element be k. Then the remaining m elements of A must come from the set $\{1, 2, 3, \ldots, k-1\}$. So we must have $m \leq k-1$, or equivalently, $m+1 \leq k$. We must, of course, also have $k \leq n+1$. Put together k can take values from $m+1$ to $n+1$, both included. There are $n-m+1$ such values, the same as the number of terms on the L.H.S.of (1). So, probably, classifying the $(m+1)$-subsets of S according to their last elements will do the trick. And it certainly does. Indeed, suppose $m+1 \leq r \leq n+1$ and B is an $(m+1)$-subset of S having r as its last element. Then its remaining m elements must come from the set $\{1, 2, \ldots, r-1\}$ and these can be chosen in $\binom{r-1}{m}$ ways. So this is the number of those $(m+1)$-subsets of S which have r as their last element. Since r varies from $m+1$ to $n+1$, we get that $\sum_{r=m+1}^{n+1} \binom{r-1}{m} = \binom{n+1}{m+1}$. But the terms of this summation are precisely those on the L.H.S. of (1) (read in the reverse order). Thus we have got a combinatorial proof of (1). Note that the key idea, namely the correct additional structure on the set S was not so obvious. But once we hit it, the rest of the work was almost instantaneous. No messy computations, no clumsy formulas. This is usually the charm of combinatorial proofs.

Comment No. 3:

We now have several ways of proving (1) besides the one given in the solution. Let us now see if we can effect the transition from (1) to (2) in some way other than the one given in the solution. One such method is analogous to the one given in the solution except that instead of writing the terms in a triangular form as shown there, we take out only the first row and evaluate the rest by induction on n. To be more specific, fix m and for $n \geq m$, denote the L.H.S.of (2) by S_n. The L.H.S. of (2) is the same as that of (1) except that the various binomial coefficients are appearing with increasing coefficients in (2) while in (1) they all occur with coefficient 1 each. So it is natural to take out the L.H.S. of (1) from that of (2). But if we do so, what is left is precisely S_{n-1}. That is,

$$S_n = \binom{n}{m} + \binom{n-1}{m} + \binom{n-2}{m} + \ldots + \binom{m+1}{m} + \binom{m}{m} + S_{n-1} \qquad (3)$$

Combining this with (1),

$$S_n = S_{n-1} + \binom{n+1}{m+1} \qquad (4)$$

We now have two options. If we already have a hunch (either from the question itself or by some guesswork) that S_n should equal $\binom{n+2}{m+2}$ for all $n \geq m$, then we can *verify* it by induction on n. The starting case $n = m$ is trivially true as both the sides equal 1 each. Equation (4) does the work of the inductive step. Indeed assume that (2) holds true for $n = k$, i.e., $S_k = \binom{k+2}{m+2}$. Then from (4), $S_{k+1} = \binom{k+2}{m+2} + \binom{k+2}{m+1}$ which equals $\binom{k+3}{m+2}$ using the same reduction formula which we have used earlier (and for which a combinatorial proof was given above). But that means (2) is true for $n = k + 2$, thereby completing the induction.

However, an equation like (4) has an advantage over the method of induction. In induction you have to know the answer beforehand. You can only verify it. But an equation like (4) allows you to *arrive at* the answer. Indeed, if we apply it again and again with decreasing values of n, we get the following system of equations.

$$\begin{aligned} S_n &= S_{n-1} + \tbinom{n+1}{m+1} \\ S_{n-1} &= S_{n-2} + \tbinom{n}{m+1} \\ S_{n-2} &= S_{n-3} + \tbinom{n-1}{m+1} \\ \cdots &\quad \cdots \\ S_{m+2} &= S_{m+1} + \tbinom{m+3}{m+1} \\ S_{m+1} &= S_m + \tbinom{m+2}{m+1} \end{aligned}$$

In the last equation we can replace S_m by $\binom{m+1}{m+1}$ since both equal 1. If we do so and add these equations, then, after cancelling the terms on the two sides we are left with

$$S_n = \binom{n+1}{m+1} + \binom{n}{m+1} + \binom{n-1}{m+1} + \ldots + \binom{m+2}{m+1} + \binom{m+1}{m+1}$$

The R.H.S. can be evaluated using (1) (with m, n replaced by $m+1, n+1$). Thus we get $S_n = \binom{n+2}{m+2}$ as desired.

Comment No. 4:

There is also a somewhat novel way of deriving (2) from (1). It is based on differentiation. The binomial coefficients are precisely the coefficients of suitable powers of x in the expansion of a function which is a suitable power of $(1 + x)$. All the powers $(1 + x)^r$ are differentiable functions of the real variable x. The derivative of $(1 + x)^r$ w.r.t. x is $r(1 + x)^{r-1}$ and the coefficient of x^m in it is $r\binom{r-1}{m}$. If we take the derivative of a sum of consecutive, decreasing powers of $(1 + x)$ then the coefficient of x^m in it will be a linear combination of terms of the form $\binom{r-1}{m}$ with *decreasing* coefficients. The L.H.S. of (2), on the other hand, is a linear combination of such terms with *increasing* coefficients. But

this is a superficial difficulty and can be resolved by subtracting the coefficients from some fixed number. This suggests that it may be possible to derive (2) from (1) using differentiation.

Now, coming to the details, once again, denote the L.H.S. of (2) by S_n for $n \geq m$. In the sigma notation, $S_n = \sum_{r=0}^{n-m} (r+1)\binom{n-r}{m}$. This is the coefficient of x^m in the sum $\sum_{r=0}^{n-m} (r+1)(1+x)^{n-r}$. As we shall see a little later, there is a way to evaluate this sum. But it is not so obvious. However, if we replace the coefficient $(r+1)$ by $(n-r+1)$ then the sum $\sum_{r=0}^{n-m} (n-r+1)(1+x)^{n-r}$ can be recognised as the derivative of the sum $\sum_{r=0}^{n-m} (1+x)^{n-r+1}$. This latter sum can be evaluated by applying the formula for the sum of a geometric progression. Keeping this in mind we rewrite S_n slightly by writing the coefficient $(r+1)$ as $(n+2)-(n-r+1)$. Then we can write

$$S_n = A_n - B_n \tag{5}$$

where $A_n = (n+2)\sum_{r=0}^{n-m} \binom{n-r}{m}$ and $B_n = \sum_{r=0}^{n-m} (n-r+1)\binom{n-r}{m}$. We can evaluate A_n using (1) and get

$$A_n = (n+2)\binom{n+1}{m+1} \tag{6}$$

As for B_n, we just observed that B_n is precisely the coefficient of x^m in the derivative of the sum $\sum_{r=0}^{n-m} (1+x)^{n-r+1}$. If written in the reverse order of terms, this is the sum of a geometric progression with the first term $(1+x)^m$ and the common ratio equal to $(1+x)$. (Note the advantage gained by reversing the order of the terms. If we do not reverse the order, we still get a geometric progression. But its common ratio is $\frac{1}{1+x}$ which is a little messy to handle. Of course, ultimately we shall get the same answer.) So, B_n equals the coefficient of x^m in the derivative of the function $(1+x)^{m+1}\dfrac{(1+x)^{n-m+1}-1}{1+x-1}$, i.e., of the function $\dfrac{(1+x)^{n+2}-(1+x)^{m+1}}{x}$. Call this function as $f(x)$. By a straight computation, the derivative $f'(x)$ comes out as

$$\frac{x[(n+2)(1+x)^{n+1}-(m+1)(1+x)^m]-[(1+x)^{n+2}-(1+x)^{m+1}]}{x^2}$$

The coefficient of x^m in this is the same as the coefficient of x^{m+2} in the numerator. Since $(1+x)^m$ can contain no term involving x^{m+1} and $(1+x)^{m+1}$

can contain no term involving x^{m+2}, the coefficient of x^{m+2} in the numerator is simply $(n+2)\binom{n+1}{m+1} - \binom{n+2}{m+2}$. So this is the value of B_n. From (5) and (6) it now follows that S_n is simply $\binom{n+2}{m+2}$. Thus we have proved (2) using (1).

We could have evaluated B_n without derivatives by cleverly rewriting $(n-r+1)\binom{n-r}{m}$ as $(m+1)\binom{n-r+1}{m+1}$ (cf. Equation (10) of Chapter 4). In that case, B_n is simply $(m+1)$ times the coefficient of x^{m+1} in the sum $\sum_{r=0}^{n-m}(1+x)^{n-r+1}$. Summing this geometric series, B_n is $(m+1)$ times the coefficient of x^{m+2} in the expression $(1+x)^{m+1}[(1+x)^{n-m+1}-1]$, i.e., in the expression $(1+x)^{n+2}-(1+x)^{m+1}$. This would give $B_n = (m+1)\binom{n+2}{m+2}$ which can be further rewritten as $(m+2)\binom{n+2}{m+2} - \binom{n+2}{m+2}$ and finally as $(n+2)\binom{n+1}{m+1} - \binom{n+2}{m+2}$. But this is a rather tricky way. The use of derivatives, on the other hand, is a little more natural to come to the mind because so many times we use the fact that the derivative of x^n (w.r.t. x) is nx^{n-1}.

Comment No. 5:

So far we have seen various proofs of (1) and also various ways of deriving (2) from it. Let us now see if we can prove (2) directly, i.e., without going through (1). The foremost method one can try is induction. Earlier we did prove (2) by induction on n. But that time we used (1). To get (2) directly using induction, we need to modify the statement a little. Earlier we held m fixed and the statement $P(n)$ we proved by induction on n was 'If $n \geq m$, then (2) holds'. We now let both m and n vary. The statement $Q(n)$ we shall prove by induction on n is 'For every m with $1 \leq m \leq n$, (2) holds'. There is a subtle difference between $P(n)$ and $Q(n)$. In the former, m is fixed. In the latter, it is allowed to vary. To stress this difference, let us change the notation slightly. Earlier, we denoted the L.H.S. of (2) by S_n even though it depends not only on n but on m as well. This was all right since m was fixed. Now that both n and m are variable parameters, let us denote the L.H.S. of (2) by double subscripts, say $S_{n,m}$. The statement $Q(n)$ then can be restated as 'For every m with $1 \leq m \leq n$, $S_{n,m} = \binom{n+2}{m+2}$.' (In effect, here we have two integer parameters m and n and we are applying double induction as discussed in Comment No. 9 of the last chapter.)

Now as the first step of induction, suppose $n = 1$. Then the only possible value m can take is 1 and the assertion $S_{1,1} = \binom{3}{3}$ is trivially true since both the sides equal 1. Next, for the inductive step, we shall first establish a relationship between $S_{n,m}$ and similar sums with lower values of the parameters. Specifically, we shall show that

$$S_{n,m} = S_{n-1,m} + S_{n-1,m-1} \tag{7}$$

whenever $n > 1$ and $1 < m < n$. Note that these restrictions are necessary because if $m = n$, then $m > n-1$ and so $S_{n-1,m}$ is not defined. Similarly, the second term on the R.H.S. is undefined if either m or n equals 1. The proof of (7) is based on the same reduction formula that we have used several times earlier (with slightly different notations). Applying this formula repeatedly gives

a series of equations, viz.,

$$\begin{aligned}
\binom{n}{m} &= \binom{n-1}{m} + \binom{n-1}{m-1} \\
\binom{n-1}{m} &= \binom{n-2}{m} + \binom{n-2}{m-1} \\
\binom{n-1}{m} &= \binom{n-2}{m} + \binom{n-2}{m-1} \\
\cdots &\quad \cdots \quad \cdots \\
\binom{m+1}{m} &= \binom{m}{m} + \binom{m}{m-1} \\
\binom{m}{m} &= 0 + \binom{m-1}{m-1}
\end{aligned}$$

(All except the last equation are obtained by the reduction formula. The last one holds since both the sides equal 1.)

If we multiply the first equation throughout by 1, the second equation by 2, the third by 3, ..., and the last one by $(n-m+1)$ and add, we get (7). (7) is of the same spirit as (4) which was crucial in the inductive step while proving $P(n)$ by induction. Not surprisingly, (7) does the job of the inductive step in proving $Q(n)$ which is our present goal. Assume that for some positive integer k, $Q(k)$ is true. We have to prove that $Q(k+1)$ is true. So take any integer m, with $1 \le m \le k+1$. We must show that $S_{k+1,m} = \binom{k+3}{m+2}$. This is trivially true if $m = k+1$, since in that case both $S_{k+1,k+1}$ and $\binom{k+3}{k+3}$ equal 1. Also, for $m = 1$, $S_{k+1,1}$ is the sum $\binom{k+1}{1} + 2\binom{k}{1} + 3\binom{k-1}{1} + \ldots + k\binom{2}{1} + (k+1)\binom{1}{1}$ which is simply $\sum_{r=1}^{k+1} r(k+2-r)$. Using the standard formulas $\sum_{r=1}^{k+1} r = \frac{(k+1)(k+2)}{2}$ and $\sum_{r=1}^{k+1} r^2 = \frac{(k+1)(k+2)(2k+3)}{6}$ this comes out as $\frac{(k+1)(k+2)(k+3)}{6}$ which is precisely $\binom{k+3}{3}$. Thus we have proved that $S_{k+1,m} = \binom{k+3}{m+2}$ for $m = 1$ also. It remains to prove it for $1 < m < k+1$. Since $k+1 > 1$, for any such value of m, we can apply (7) with $n = k+1$, to get $S_{k+1,m} = S_{k,m} + S_{k,m-1}$. Note that $m \le k$ and so we can apply the induction hypothesis to get $S_{k,m} = \binom{k+2}{m+2}$. As for $S_{k,m-1}$, we note first that $1 \le m-1 \le k$. Now we are assuming the truth of $Q(k)$. This means $S_{k,r} = \binom{k+2}{r+2}$ for *all* r with $1 \le r \le k$. In particular this holds for $r = m-1$. Thus $S_{k,m-1} = \binom{k+2}{m+1}$. Thus we have shown that for $1 < m < k+1$, $S_{k+1,m} = \binom{k+2}{m+2} + \binom{k+2}{m+1}$. Using the reduction formula once more, this equals $\binom{k+3}{m+2}$. Thus we have now completely established the truth of $Q(k+1)$. This completes the induction.

Comment No. 6:

There is a much shorter proof of (2) analogous to the proof of (1) given in the very first solution. We already saw in Comment No. 4 that the L.H.S. of (2)

which we denoted by S_n, is the coefficient of x^m in the sum $\sum_{r=0}^{n-m}(r+1)(1+x)^{n-r}$. This sum is an example of what is called an **arithmetico-geometric series**, because it is a combination of the terms of a geometric progression with coefficients being in an arithmetic progression. There is a simple, albeit rather tricky, way to evaluate this sum. Call this sum as S. Multiplying every term by $(1+x)$, we get $(1+x)S = \sum_{r=0}^{n-m}(r+1)(1+x)^{n-r+1}$. If we subtract S from this and divide by x we get

$$S = \frac{\sum_{r=0}^{n-m}(r+1)(1+x)^{n-r+1} - \sum_{r=0}^{n-m}(r+1)(1+x)^{n-r}}{x} \tag{8}$$

Each of the two summations in the numerator is a sum of decreasing powers of $(1+x)$. The coefficient of a typical power, say $(1+x)^k$ in the first sum is $n-k+2$. (To see this, put $n-r+1=k$.) But in the second sum, it is $n-k+1$ (obtained by putting $n-r=k$). These two terms, when added together, give only $(1+x)^k$. This is valid for $m+1<k<n+1$. Besides, the first sum has $(1+x)^{n+1}$ as a term for which there is no matching term in the second sum and the second sum has $(n-m+1)(1+x)^m$ which has no matching term in the first sum. Keeping this in mind, we see that the numerator is $(1+x)^{n+1}+(1+x)^n+(1+x)^{n-1}+\ldots+(1+x)^{m+2}+(1+x)^{m+1}-(n-m+1)(1+x)^m$. (It is a little easier to see this, if, instead of expressing S and $(1+x)S$ in the sigma notation, we write them out in full.) We now apply the formula for the sum of a geometric progression, except for the last term, of course. (As in the solution given in Comment No. 4, it is a good idea to do so after reversing the order of the terms.) We then get, after a little simplification,

$$S = \frac{(1+x)^{n+2}-(1+x)^{m+1}}{x^2} - (n-m+1)\frac{(1+x)^m}{x} \tag{9}$$

Now the desired sum S_n is the coefficient of x^m in this. This equals $A-B$, where A is the coefficient of x^{m+2} in $(1+x)^{n+2}-(1+x)^{m+1}$ and B is the coefficient of x^{m+1} in $(n-m+1)(1+x)^m$. Evidently, $A=\binom{n+2}{m+2}$ while $B=0$. Hence $S_n=\binom{n+2}{m+2}$ as desired.

Comment No. 7:

So far we have had three proofs of (2) using (1) (which itself could be proved in a number of ways) and also two direct proofs of (2). All these proofs relied, directly or indirectly, on the basic reduction formula for binomial coefficients, viz., $\binom{r}{k}=\binom{r-1}{k-1}+\binom{r-1}{k}$. We now give a combinatorial proof of (2) which is independent of this reduction formula. It is analogous to the combinatorial proof of (1) given in Comment No. 2 above. There, we took a set with $n+1$ elements, ordered its elements and then classified its $(m+1)$-subsets according to their largest elements. Now that we want to apply a combinatorial argument

to prove (2) whose R.H.S. is $\binom{n+2}{m+2}$, it is obvious that we should start with a set S with $n+2$ elements and consider the totality of its $(m+2)$-subsets. Once again, the key lies in classifying these subsets suitably. In analogy with what we did earlier to prove (2), we put an additional structure on S by ordering its elements. Effectively this amounts to taking S as the set $\{1, 2, 3, \ldots, n+1, n+2\}$. Even with this spadework, it is not obvious how to classify the $(m+2)$-subsets of S. And you are well advised to try this out on your own first. Like other problems based on hitting on the right idea, if you succeed, it will be a thrilling experience for you. Even if you don't succeed, the efforts that you have put in will enable you to appreciate the correct solution even more.

A clue can be obtained from the way we proved (1) by classifying the $(m+1)$-subsets of the set $\{1, 2, \ldots, n, n+1\}$. A typical term of the L.H.S. was of the form $\binom{r}{m}$, where $m \leq r \leq n$. This is the number of ways to choose m elements from a set with r elements and a most natural choice for such a set is the set $\{1, 2, \ldots, r\}$. So, the classification of the $(m+1)$-subsets of the set $\{1, 2, 3, \ldots, n, n+1\}$ should be done in such a way that the subsets in one of the classes correspond to all m-subsets of the set $\{1, 2, \ldots, r\}$. If the largest element of an $(m+1)$-subset is $r+1$, then the remaining m elements will have to come from $\{1, 2, \ldots, r\}$. Conversely, given any m-subset of $\{1, 2, \ldots, r\}$, we can throw in one more element, viz., $r+1$ and get an $(m+1)$-subset of $\{1, 2, \ldots, n, n+1\}$ whose largest element is $r+1$. This idea leads to the correct classification.

We try something similar now. We want to classify the totality of all $(m+2)$-subsets of the set $S = \{1, 2, 3, \ldots, n+1, n+2\}$ in such a way that the numbers of subsets in these classes correspond to the terms of the L.H.S. of (2). A typical such term is of the form $(n-r+1)\binom{r}{m}$, where $m \leq r \leq n$. The second factor here, viz., $\binom{r}{m}$ is the number of ways to choose m elements from a set with r elements and the best candidate for such a set is $\{1, 2, \ldots, r\}$. The first factor, viz., $n-r+1$ is the number of ways to choose one object from a set with $n-r+1$ elements. One such set is the set of the last $n-r+1$ elements of the set S, i.e., the set $\{r+2, r+3, \ldots, n+1, n+2\}$. So $(n-r+1)\binom{r}{m}$ is the number of ways to select an m-subset, say A, of the set $\{1, 2, \ldots, r\}$ and simultaneously an element, say x, of the set $\{r+2, r+3, \ldots, n+1, n+2\}$. Note that x can never belong to the subset A. Let us call the set $A \cup \{x\}$ as B. Then B is a subset of S and has exactly $m+1$ elements. Further, the element $r+1$ can never belong to B. Let us include this element and call the set $B \cup \{r+1\}$ as C. Then C is an $(m+2)$-subset of S. The crucial point to observe is that $r+1$ is the second largest element of the subset C. Conversely, if D is any $(m+2)$-subset of S and has $r+1$ as its second largest element, then it contains exactly one element from the set $\{r+2, r+3, \ldots, n+1, n+2\}$ and exactly m elements from the set $\{1, 2, \ldots, r\}$.

So, we have hit upon the key idea. Classify the totality of all the $(m+2)$-subsets of S according to their *second largest* elements. This second largest element can be anywhere from $m+1$ to $n+1$ (both included). And the number of $(m+2)$-subsets of S which have $r+1$ as their second largest element (where $m \leq r \leq n$), is $(n-r+1)\binom{r}{m}$, which is precisely one of the terms of the L.H.S.

of (2). Thus we have obtained a combinatorial proof of (2).

This combinatorial proof is the strongest evidence we have so far that generally such proofs involve very little computations, once you get the right idea. The time it takes to hit the key idea can, of course, vary greatly from person to person. Even for an experienced solver, it is not an uncommon experience that a combinatorial argument keeps eluding him for hours and perhaps for days and then a flash comes suddenly. While this feature makes such arguments a little risky to try in a limited-time examination, it is such arguments that increase the charm of mathematics. Moreover, by focusing on the essential ideas, the combinatorial arguments often make it easier to think of possible generalisations. In other approaches, such generalisations may get shrouded in lengthy or messy computations. For example, we took an $(n+1)$-set, classified its $(m+1)$-subsets according to their largest elements and got (1). Then we took an $(n+2)$-set, classified its $(m+2)$-subsets according to their second largest elements and got (2). It is now clear what the next identity in line is. If we take an $(n+3)$-set and classify its $(m+3)$-subsets according to their third largest elements, then we would get

$$\binom{2}{2}\binom{n}{m}+\binom{3}{2}\binom{n-1}{m}+\ldots+\binom{n-m+2}{2}\binom{m}{m}=\binom{n+3}{m+3} \tag{10}$$

More generally, for every non-negative integer k we would get,

$$\begin{aligned}\binom{n}{m}+(k+1)\binom{n-1}{m}+\ldots \quad &+ \quad \binom{n-r+k}{k}\binom{r}{m}+\ldots \\ +\binom{n-m+k}{k}\binom{m}{m} \quad &= \quad \binom{n+k+1}{m+k+1}\end{aligned} \tag{11}$$

It is clear that (1) and (2) are special cases of (11) obtained by putting $k=0$ and $k=1$ respectively. Thus we are able to put the problem into a wider perspective. It is doubtful if we could have thought of (11) with any of the other solutions to the problem (such as the one based on induction on n) and even if we did, proving (11) would have been very messy. With a combinatorial argument for (2), we could easily conceive (11). What is more, proving (11) is as easy as proving (2).

Comment No. 8:

Summing up, we have three standard methods of proving identities involving binomial coefficients. The first method is by direct algebraic manipulations using some reduction formulas or induction to simplify the work to some extent. In this method, $\binom{n}{r}$ is simply an algebraic expression, viz., the ratio $\frac{n!}{r!(n-r)!}$, or the ratio $\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}$. The second method is based on the binomial theorem, because it recognises a binomial coefficient $\binom{n}{r}$ as the coefficient of x^r in the expansion of $(1+x)^n$. The third method is the combinatorial method, in which we look at $\binom{n}{r}$ as the number of ways to choose r objects from n objects. Each method has its advantages. The results proved by the algebraic method sometimes continue to hold even when the variable n is given non-integral values,

because the expression $\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}$ makes sense for any real (or even complex) values of n even though it may not have any combinatorial significance for such values of n. (However, even when n is allowed to take any values, r still has to be a positive integer.) The method of binomial theorem has the advantage that since $(1+x)^n$ is a very nice function of the real variable x (for example, it has derivatives of all orders), the methods of calculus become applicable in addition to the algebraic manipulations. We already saw how we could derive (2) from (1) using derivatives. The third method, viz., the combinatorial one, is comparatively weak. There are, in fact, many identities for which combinatorial proofs are still not known and are sought. But usually, when it works, it is the most elegant method.

Which method is best for a particular problem will of course depend on the problem itself. Sometimes the very nature of the problem suggests which method is best suited for it. Suppose for example, that m, n and k are non-negative integers and we want to prove the well-known identity

$$\sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r} = \binom{m+n}{k} \tag{12}$$

(called **Vandermonde convolution**) with the understanding that whenever we encounter a binomial coefficient of the form $\binom{p}{q}$ with $p < q$, we take it to be 0. We recognise the R.H.S. as the number of ways to choose k objects from $m+n$ (distinct) objects. Divide these $m+n$ objects into two classes, one with m and the other with n objects. For example, we may think of choosing k persons from m men and n women. Classify these selections according to the number of men, say r, in it. Then $0 \le r \le k$ and for a fixed r, the number of women in the selection is $k-r$. The number of selections with r men and $k-r$ women is $\binom{m}{r}\binom{n}{k-r}$. (Because of the understanding we have, we do not have to worry about what happens if $r > m$ or $k-r > n$.) But then this is precisely the r-th term of the sum on the L.H.S. of (12). A proof using the binomial theorem is also easy. The R.H.S. is the coefficient of x^k in $(1+x)^{m+n}$. If we factor this as the product of $(1+x)^m$ and $(1+x)^n$, then every occurrence of the term x^k in $(1+x)^{m+n}$ arises as a product of a term x^r in $(1+x)^m$ and a term x^{k-r} in $(1+x)^n$, for some $0 \le r \le k$. Since the coefficient of x^r in $(1+x)^m$ is $\binom{m}{r}$ while that of x^{k-r} in $(1+x)^n$ is $\binom{n}{k-r}$, the L.H.S. of (12) is the coefficient of x^k in $(1+x)^m(1+x)^n$. (Actually, this proof is not substantially different from the combinatorial proof, because, as remarked at the end of Chapter 1, one of the proofs of the binomial theorem is by writing $(x+y)^p$ as the product of p factors each equal to $(x+y)$. A term $x^r y^{p-r}$ corresponds to a choice of the term x in r of these factors and of the term y in the remaining. There are $\binom{p}{r}$ such choices for a given r between 0 and p. This is a combinatorial argument.)

If one wants, (12) can also be proved by induction (JEE 1989). Note that here there are three integer parameters, viz., m, n and k. (The integer r appearing in the L.H.S. is only a dummy index of summation. Its scope extends only to the sum. So it is not considered as another integer parameter inherent in the problem.) So what we need here is a triple induction, which is the next

step further to the method of double induction discussed in Comment No. 9 of the last chapter. However, if we hold one of the parameters m and k fixed, then there are only two integer parameters and we can apply double induction on them. As the problem is symmetric between m and n, it does not matter which of the them is held fixed. We hold m fixed and apply double induction on n and k. So the statement $P(n)$ we want to prove about the integer $n \geq 0$ is that for every $k \geq 0$, (12) holds. Then $P(0)$ is trivially true since in that case all except the last term of the sum in the L.H.S. of (12) vanish and so both the sides of (12) reduce to $\binom{m}{k}$. To derive the truth of $P(n+1)$ from that of $P(n)$ we proceed as follows. Suppose $k = 0$. Then (12) holds (with n replaced by $n+1$), both sides being equal to 1. Suppose then that $k > 1$. Use the reduction formula to write $\binom{m+n+1}{k}$ as $\binom{m+n}{k} + \binom{m+n}{k-1}$. By induction hypothesis (i.e., because of the truth of $P(n)$), these terms equal, respectively, the sums $\sum_{r=0}^{k} \binom{m}{r}\binom{n}{k-r}$ and $\sum_{r=0}^{k-1} \binom{m}{r}\binom{n}{k-1-r}$ respectively. In the second summation, the index variable r can be extended to k as well since in that case $k-1-r < 0$ and so $\binom{n}{k-1-r} = 0$ anyway. Doing so, the two sums add up to the sum $\sum_{r=0}^{k} \binom{m}{r}\left(\binom{n}{k-r} + \binom{n}{k-1-r}\right)$. Using the reduction formula again, this equals $\sum_{r=0}^{k} \binom{m}{r}\binom{n+1}{k-r}$ which is precisely the sum in the L.H.S. of (12) with n replaced by $n+1$. Thus we have proved the truth of $P(n+1)$ from the truth of $P(n)$, thereby completing the induction.

Comment No. 9:

The identity (12), coupled with some other results has interesting applications. Consider for example, the relation $\binom{n}{k} = \binom{n}{k-1}\frac{n-k+1}{k}$ proved at the beginning of the last chapter. If $k \leq \frac{n}{2}$, then the factor $\frac{n-k+1}{k}$ exceeds 1 and so we get that $\binom{n}{k} > \binom{n}{k-1}$. It follows that for a fixed n, as k goes on increasing from 0 onwards so does the binomial coefficient $\binom{n}{k}$ as long as $k \leq \frac{n}{2}$. If n is even, say $n = 2m$, this means that $\binom{n}{0} < \binom{n}{1} < \ldots \binom{n}{m-1} < \binom{n}{m}$. Thereafter, because of the symmetry relation $\binom{n}{k} = \binom{n}{n-k}$, the binomial coefficients start decreasing strictly. So $\binom{n}{k}$ is maximum when $n = m$. Another consequence of this increasing/decreasing behaviour of the binomial coefficients is that if we have $\binom{n}{r} = \binom{n}{s}$, where the integers r, s lie between 0 and n then we must have either $r = s$ or $r = n - s$. This fact was tacitly used in the last step of the solution to the Main Problem.

If n is odd, say $n = 2m+1$, then the argument is similar. In that case, $\binom{n}{k}$ increases strictly with k till k becomes m. Then $\binom{n}{m} = \binom{n}{m+1}$ and after that $\binom{n}{k}$ decreases strictly as k increases upto n. Here $\binom{n}{k}$ is maximum for two values of k, viz., for $k = m$ and $k = m+1$.

Let us now combine this observation with (12). Suppose that we want to find the value of the k for which the expression $\sum_{i=0}^{k} \binom{10}{i}\binom{20}{k-i}$ is maximum (JEE

2002). By (12) we recognise this sum as nothing but the binomial coefficient $\binom{30}{k}$. So it will be maximum when $k = \frac{30}{2} = 15$.

Comment No. 10:

As a less immediate application of (12), we can do the following problem. (JEE 1982)

If

$$\sum_{r=0}^{2n} a_r(x-2)^r = \sum_{r=0}^{2n} b_r(x-3)^r \tag{13}$$

and $a_k = 1$ for all $k \geq n$, prove that $b_n = \binom{2n+1}{n+1}$.

The statement of the problem is a little clumsy. It is not given what x is. So we have to assume that it is any number and hence that (13) holds for all real values of x. Also there is no need to have $x-2$ on the L.H.S. and $x-3$ on the R.H.S. We call $x-3$ as y. Then (13) is equivalent to saying that

$$\sum_{r=0}^{2n} a_r(y+1)^r = \sum_{r=0}^{2n} b_r y^r \tag{14}$$

for all real values of y. (Such initial simplifications of a problem often pay rich dividends. In the present case, for example, (14) is evidently more manageable than (13), since we need to apply the binomial theorem only to the L.H.S. and not to both the sides as is the case with (13). We could have called $x-2$ as z, but then on the R.H.S. we would have powers of $z-1$. Our interest is in b_n, which we can easily identify as the coefficient of y^n in the R.H.S. of (14). With $x-2=z$, the R.H.S. of (13) would change to $\sum_{r=0}^{2n} b_r(z-1)^r$. In this sum we can no longer identify b_n as the coefficient of any single power of z. This shows that even among two simplifications which are mathematically equivalent, one may be preferable over the other as far as the given problem is concerned.)

Now, both the sides of (14) are polynomials of degree $2n$ in the variable y. Since their equality is given to hold for all real values of y, the coefficients of like powers of y in the two sides must be equal to each other. (Otherwise, the difference of the two sides would be a non-zero polynomial of degree not exceeding $2n$ and having every real number as a root. But no non-zero polynomial can have more roots than its degree. Justifications of this type are not expected from the candidates, unless, of course, that is the very question.)

In particular, the coefficients of y^n on the two sides of (14) are equal. On the R.H.S. it is b_n as we already noted. To find the coefficient of y^n on the L.H.S. note that the terms upto $r = n-1$ do not have y^n in them. So the coefficient of y^n in the L.H.S. of (14) is the same as the coefficient of y^n in the sum $\sum_{r=n}^{2n} a_r(y+1)^r$ which is the same as the sum $\sum_{r=n}^{2n}(y+1)^r$ since we are given that $a_r = 1$ for all $r \geq n$. The latter sum is the sum of a geometric progression with the first term $(y+1)^n$ and common ratio $y+1$. So the sum

equals $(y+1)^n \frac{(y+1)^{n+1}-1}{y}$, which is the same as $(y+1)^n \sum_{r=0}^{n} \binom{n+1}{r+1} y^r$. The coefficient of y^n in this is the sum $\sum_{r=0}^{n} \binom{n}{n-r}\binom{n+1}{r+1}$, or, with a change of index, the sum $\sum_{i=0}^{n} \binom{n}{i}\binom{n+1}{n+1-i}$. If we apply (12) (with m, n, k there replaced by $n, n+1$ and $n+1$ respectively), then this sum is simply $\binom{2n+1}{n+1}$. Hence $b_n = \binom{2n+1}{n+1}$ as was to be proved.

Comment No. 11:

As a general rule, when at least one side of a binomial identity to be proved has a combinatorial significance (i.e., it represents the number of ways to do something) one is motivated to try a combinatorial argument. Similarly, when one of the sides can be recognised as the coefficient of some power of x in some function, say $f(x)$, (usually a polynomial in x or a related function) one is inspired to get the result by subjecting this function to various manipulations such as factoring or differentiation. Sometimes considerable recasting of the terms is needed to prove an identity. Here is an excellent example of such a problem (JEE 1999).

Let n be a positive integer. Prove that for every non-negative integer $m \leq n$,

$$\sum_{k=0}^{m} \frac{\binom{2n-k}{k}}{\binom{2n-k}{n}} \frac{2n-4k+1}{2n-2k+1} 2^{n-2k} = \frac{\binom{n}{m}}{\binom{2n-2m}{n-m}} 2^{n-2m} \tag{15}$$

Here both the sides are too clumsy to permit any natural interpretation, whether combinatorial or algebraic. But a recasting and a clever splitting of each term enables us to regnise the series on the L.H.S. as a telescopic series. Specifically, for $k \geq 0$, rewrite $2n-4k+1$ in the numerator as $(2n-2k+1)-2k$. Then note that the expression $\frac{\binom{2n-k}{k}2^{n-2k}}{\binom{2n-k}{n}}$ is simply $\frac{\binom{n}{k}2^{n-2k}}{\binom{2n-2k}{n-k}}$, while the expression $\frac{\binom{2n-k}{k}2k2^{n-2k}}{\binom{2n-k}{n}(2n-2k+1)}$, upon simplification, equals $\frac{\binom{n}{k-1}}{\binom{2n-2k+2}{n-k+1}}2^{n-2k+2}$. Hence the k-th term of the L.H.S. can be expressed as $A_k - A_{k-1}$ where $A_k = \frac{\binom{n}{k}}{\binom{2n-2k}{n-k}}2^{n-2k}$. Since $A_{-1} = 0$, the sum equals A_m which is precisely the R.H.S.

Admitedly, this is a very tricky solution which is not easy to conceive in a limited time. In such cases, induction is often a safer, albeit often uninspiring, option. In an inductive proof, you are essentially passing the buck. That is, you are deriving the truth of something from that of something similar without really understanding why it is true. Although mathematically a proof based on induction is as valid as any other mathematical proof, it leaves something to be desired. Take for example, the well-known identity that $1+2+\ldots+(n-1)+n = \frac{1}{2}n(n+1)$. An inductive proof of this is very easy (and usually given as one of

the very first illustrations when the method of induction is taught). But that proof does not explain how the factors $\frac{1}{2}$, n and $n+1$ got into the answer. Let us, however, write the same sum, say S_n, backwards, i.e., as $n + \ldots + 2 + 1$. If we add the two sums term by term, we see that $2S_n$ is the sum of n terms each of which is $n+1$. Hence $S_n = \frac{1}{2}n(n+1)$. Now we really see the role of each of the three factors in the answer. But in the absence of a revealing proof like this, induction would be the only go, however uninspiring it may be.

In the present problem, the integer n is fixed and we apply induction on m. (The integer m takes only values upto n. But this need not deter us from applying induction on m. For $m > n$, the R.H.S. of (15) is 0 and so are the terms after $k = n$ in the sum in the L.H.S. Hence both the sides make sense for all $m \geq 0$ and so we can prove the statement for all non-negative integers m even though we are asked to prove it only for $m \leq n$. Alternately, the statement $P(m)$ that we want to establish by induction can be taken to be 'either $m > n$ or (15) holds'. Then for $m > n$, $P(m)$ is true by itself, without having to derive its truth from that of $P(m-1)$.)

Let us now get down to the solution. Denote the sum on the L.H.S. by S_m. Trivially, S_0 has only one term, viz., $\frac{\binom{2n}{0}}{\binom{2n}{n}}\frac{2n+1}{2n+1}2^n$ which indeed equals the R.H.S. of (15) for $m = 0$. So $P(0)$ is true. Now assume that (15) holds for some m and we want to prove it for $m+1$. In other words, assume that

$$S_m = \frac{\binom{n}{m}}{\binom{2n-2m}{n-m}}2^{n-2m} \tag{16}$$

and we want to prove that

$$S_{m+1} = \frac{\binom{n}{m+1}}{\binom{2n-2m-2}{n-m-1}}2^{n-2m-2} \tag{17}$$

Note that although the number of terms in the sum in the L.H.S. of (15) depends on m, the terms themselves don't. Each term depends only on n (which is fixed) and k (which is the index variable of summation). This is a very important observation because it implies that S_{m+1} is obtained by adding just one more term to S_m. Specifically,

$$S_{m+1} = S_m + \frac{\binom{2n-m-1}{m+1}}{\binom{2n-m-1}{n}}\,\frac{2n-4m-3}{2n-2m-1}\,2^{n-2m-2} \tag{18}$$

(If the individual terms of the summation in the L.H.S. of (15) depended on m as well, then we would not be able to make a similar statement. This is also the reason why we have opted to induct on m rather than on n. In the latter case, for a fixed m, if we denote the L.H.S. of (15) by T_n (say), then it would not be easy to express T_{n+1} in terms of T_n.)

Because of (18), deriving (17) from (16) is equivalent to proving that

$$\frac{\binom{n}{m}2^{n-2m}}{\binom{2n-2m}{n-m}} \quad + \quad \frac{\binom{2n-m-1}{m+1}}{\binom{2n-m-1}{n}}\,\frac{2n-4m-3}{2n-2m-1}\,2^{n-2m-2} = \frac{\binom{n}{m+1}2^{n-2m-2}}{\binom{2n-2m-2}{n-m-1}} \tag{19}$$

In the first term on the L.H.S. we can rewrite the denominator using (10) of the last chapter, as $\frac{(2n-2m)}{n-m}\binom{2n-2m-1}{n-m-1} = 2\binom{2n-2m-1}{n-m-1}$. Also we can cancel the power 2^{n-2m-2} from all the terms. (This is the price we pay in an inductive proof. The various powers of 2 occurring in (15), are essential to make it true. But from the inductive proof, we cannot tell just what significance they have. This is analogous to the observation made earlier that in the inductive proof of the identity $1 + 2 + \ldots + n = \frac{1}{2}n(n+1)$, we never realise exactly what significance the factor $\frac{1}{2}$ has. It is just a factor which gets carried from line to line in the inductive step. So we can term it a nuisance and cancel it. We are doing precisely the same thing here by cancelling the power 2^{n-2m-2}.)

With these simplifications, proving (19) reduces to proving

$$\frac{\binom{n}{m+1}}{\binom{2n-2m-2}{n-m-1}} - \frac{2\binom{n}{m}}{\binom{2n-2m-1}{n-m-1}} = \frac{\binom{2n-m-1}{m+1}}{\binom{2n-m-1}{n}} \frac{2n-4m-3}{2n-2m-1} \tag{20}$$

This can be verified by expanding all binomial coefficients as ratios of factorials. The first term on the L.H.S. equals

$$\frac{n!}{(m+1)!(n-m-1)!} \frac{(n-m-1)!(n-m-1)!}{(2n-2m-2)!}$$

which, after cancellation of some factors, simplifies to $\frac{n!(n-m-1)!}{(m+1)!(2n-2m-2)!}$. Applying a similar reduction to the second term, the L.H.S. of (20) equals

$$\begin{aligned} & \frac{n!(n-m-1)!}{(m+1)!(2n-2m-2)!} - \frac{2n!(n-m-1)!}{m!(2n-2m-1)!} \\ = & \frac{n!(n-m-1)![2n-2m-1-2m-2]}{(m+1)!(2n-2m-1)!} \\ = & \frac{n!(n-m-1)!(2n-4m-3)}{(m+1)!(2n-2m-1)!} \\ = & \frac{n!(n-m-1)!}{(m+1)!(2n-2m-2)!} \frac{2n-4m-3}{2n-2m-1} \end{aligned}$$

It is easy to see that the R.H.S. of (20) also equals this. Thus we have verified (20), and hence (19). As already noted, this completes the proof of the inductive step and thus the solution. Admittedly, the proof is messy and the calculations long and prone to errors. In an examination with severely limited time, it is advisable to spend a minute or two to see if such questions can be tackled by some elegant argument, involving only a reasonable computation. If not, postpone such problems till you have done others where you feel more confident.

Comment No. 13:

Here is an interesting application of the binomial identity (1). If we put $m = 1$, then the L.H.S. is simply $n + (n-1) + \ldots + 2 + 1$ while the R.H.S. is $\binom{n+1}{2} = \frac{(n+1)n}{2!} = \frac{1}{2}n(n+1)$. This is, of course, a familiar result. But

we get it so effortlessly from (1). Moreover, the method suggests that similar results will follow by giving m higher values. Indeed, by putting $m = 2$, the L.H.S. of (1) is $\frac{1}{2}\sum_{k=2}^{n} k(k-1)$. There is no harm in starting the sum with $k = 1$ since the extra term is 0 anyway. Doing so, the L.H.S. of (1) becomes $\frac{1}{2}\left(\sum_{k=1}^{n} k^2 - \sum_{k=1}^{n} k\right)$. As the R.H.S. is $\binom{n+1}{3} = \frac{1}{6}(n+1)n(n-1)$ and we already know that $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$, we now get a formula for the sum $\sum_{k=1}^{n} k^2$ as $\frac{1}{3}(n+1)n(n-1) + \frac{1}{2}(n+1)n$, which, after simplification, comes out to be $\frac{1}{6}n(n+1)(2n+1)$. All these results can, of course, be verified directly by induction on n. But the identity (1) gives a method of *arriving at* them. Yet another method for evaluating the sum $1+2+\ldots+n$ will be given in Exercise (5.14)(c).

Comment No. 14:

A few more identities will be given in the exercises. By now we have illustrated the standard elementary techniques of proving the binomial identities along with the merits and demerits of each. It should not, however, be supposed that there is a binomial identity for every possible sum. For example, if m, n are integers with $0 \le m \le n$, there is no nice expression for the sum $\binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{m}$. (This is in sharp contrast with Exercise (5.3) (i) which gives a closed form expression for the *alternating* sum, $\sum_{k=0}^{m} (-1)^k \binom{n}{k}$.) Interestingly, there is an expression for the entire sum $\binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n-1} + \binom{n}{n}$. The easiest way to obtain it is to start with the binomial expansion,

$$\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \ldots + \binom{n}{k}x^k + \ldots + \binom{n}{n}x^n = (1+x)^n \qquad (21)$$

and put $x = 1$ to get

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n-1} + \binom{n}{n} = 2^n \qquad (22)$$

This can also be proved by induction on n. For the inductive step, we write $\binom{n+1}{k}$ as $\binom{n}{k} + \binom{n}{k-1}$ for $1 \le k \le n+1$. Then the sum $\sum_{k=0}^{n+1} \binom{n+1}{k}$ breaks up as $\sum_{k=0}^{n} \binom{n}{k} + \sum_{k=1}^{n+1} \binom{n}{k-1}$. The first sum equals 2^n by the induction hypothesis. With a change of index, the second sum equals $\sum_{j=0}^{n} \binom{n}{j}$ which again equals 2^n. Since $2^n + 2^n = 2^{n+1}$, we get the result. A combinatorial proof of (22) is also possible and instructive. Let S be a set with n elements. Then from Chapter 1, we know that 2^n is the total number of subsets of S, or equivalently, 2^n is the cardinality of the power set of the set S. But every such subset must have r elements of S for a unique r, $0 \le r \le n$. Since $\binom{n}{r}$ is the number of subsets of S

having exactly r elements, the L.H.S. of (22) gives the total number of subsets of S. Equating the two counts gives (22). Of course, the proof based on (21) is the best.

If we put $x = -1$ in (21) we get,

$$\binom{n}{0} - \binom{n}{1} + \binom{n}{2} + \ldots + (-1)^k \binom{n}{k} + \ldots + (-1)^n \binom{n}{n} = 0 \tag{23}$$

Adding (22) and (23) and dividing by 2 we get

$$\sum_{k=0}^{\infty} \binom{n}{2k} = \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \ldots + \binom{n}{2k} + \ldots = 2^{n-1} \tag{24}$$

Note that in this summation we have ostensibly allowed the index variable k to take infinitely many values. But since $\binom{n}{2k} = 0$ for all $k > \frac{n}{2}$, only finitely many terms of the sum are non-zero. So the sums are really finite. The advantage of writing them in this form is that it saves us the task of computing and writing the value of the upper limit of the index of summation. If n is even, say $n = 2m$, then the sum on the L.H.S. is from $k = 0$ to $k = m = \frac{n}{2}$, while, if n is odd then the upper limit of summation is $\frac{n-1}{2}$. So if we want to write the L.H.S. of (24) as a finite sum, then we shall have to replace it by two separate identities, one for even n and the other for odd n. There is a way to combine the two by writing $[\frac{n}{2}]$, where, as usual, $[x]$ denotes the integral part of x. But then the notation becomes too clumsy to write, especially in the small type that is generally used for upper and lower indices.

Similarly, subtracting (23) from (22) and dividing by 2 gives

$$\sum_{k=0}^{\infty} \binom{n}{2k+1} = \binom{n}{1} + \binom{n}{3} + \ldots + \binom{n}{2k+1} + \ldots = 2^{n-1} \tag{25}$$

Combinatorially, the last two identities show that for a set with n elements, the number of subsets of even cardinality is the same as the number of subsets of odd cardinality, each being 2^{n-1}. A combinatorial argument using induction is also possible. But the proofs given above are the simplest. They illustrate the power of the binomial theorem, specifically, of its validity for *all* values of the variable x. By choosing these values suitably, one can get a host of results which would be difficult to obtain by other means.

There is, in fact, nothing to stop us from giving x complex values because (21) is valid even if x is complex. In particular we can put $x = i$ in it. Note that the successive powers of i (starting from the 0-th power) are $1, i, -1, -i, 1, i, -1, -i, \ldots$. They are alternately real and purely imaginary and recur in a cycle of length 4. (Put differently, this means that $i^r = i^s$ whenever $r \equiv s \ (mod\ 4)$.) This gives,

$$\binom{n}{0} + i\binom{n}{1} - \binom{n}{2} - i\binom{n}{3} + \binom{n}{4} + i\binom{n}{5} + \ldots = (1+i)^n \tag{26}$$

We write the L.H.S. in the form $a + ib$ and take the squares of the absolute values of both the sides keeping in mind that $|(1+i)^n| = (|1+i|)^n = (\sqrt{2})^n$. Then

$$\left(\sum_{k=0}^{\infty}(-1)^k\binom{n}{2k}\right)^2 + \left(\sum_{k=0}^{\infty}(-1)^k\binom{n}{2k+1}\right)^2 = 2^n \tag{27}$$

There is no easy combinatorial significance to the L.H.S. and so a combinatorial proof seems difficult. An inductive proof based on the reduction formula is fairly straightforward but not as elegant.

Comment No. 15:

The interesting thing to note about the identities (22) to (27) is that unlike (1) and (2), they have only one integer parameter, viz., n. They give the values of certain sums in which each term and also the number of terms summed depends only on n. As already remarked, there are no formulas for the 'partial' sums of these summations, where we take only the sum of the first m terms (say). These sums can therefore be called '**full binomial sums**'. There are numerous such sums and the binomial theorem is the basic tool for evaluating them.

Occasionally, we can evaluate the sum of half (or nearly half) of the terms of such full binomial sums. Suppose, for example, that n is odd, say $n = 2m+1$. Then because of the symmetry relation $\binom{n}{k} = \binom{n}{n-k}$, the $2m+2$ terms in the L.H.S. of (22) get paired, each pair having two equal terms, one occurring during the lower half of the summation, (i.e., for $0 \le k \le m$) and the other during the upper half (i.e., for $m+1 \le k \le 2m$). It follows that if we add the first $m+1$ terms, we get exactly half the sum. Thus we get that for $n = 2m+1$,

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{m} = 2^{n-1} \tag{28}$$

As a numerical verification, let $n = 7$. Then $m = 3$ and the sum $\binom{7}{0} + \binom{7}{1} + \binom{7}{2} + \binom{7}{3}$ comes out as $1+7+21+35 = 64 = 2^{7-1}$. For even n, say for $n = 2m$, there is no analogous result for half the sum. The best we can do is to say that $\sum_{k=0}^{m}\binom{n}{k} = 2^{n-1} + \frac{1}{2}\binom{n}{m}$. For $n = 6$, this becomes $\binom{6}{0} + \binom{6}{1} + \binom{6}{2} + \binom{6}{3} = 1+6+15+20 = 42 = 2^5 + \frac{1}{2}\binom{6}{3}$.

Comment No. 16:

Combining the binomial theorem with various algebraic manipulations, we can evaluate a number of full binomial sums. As a typical example, suppose we want to show (JEE 1978)

$$({}^{2n}C_0)^2 - ({}^{2n}C_1)^2 + ({}^{2n}C_2)^2 - \ldots + ({}^{2n}C_{2n})^2 = (-1)^n\ {}^{2n}C_n \tag{29}$$

We recognise the R.H.S. as the coefficient of x^n in $(1-x)^{2n}$. But this is not of much immediate help in showing that the L.H.S. reduces to it. Instead, let us

start working with the L.H.S. A typical term of it is of the form $(-1)^r({}^{2n}C_r)^2$. Using the symmetry relation for binomial coefficients, we can rewrite this as $(-1)^r\ {}^{2n}C_r\ {}^{2n}C_{2n-r}$. Now, $(-1)^r\ {}^{2n}C_r$ is the coefficient of x^r in $(1-x)^{2n}$ while, ${}^{2n}C_{2n-r}$ is the coefficient of x^{2n-r} in $(1+x)^{2n}$. Since $r+(2n-r)=2n$ we see (by a reasoning analogous to that in the proof of (12)) that the L.H.S. of (29), i.e., the sum $\sum_{r=0}^{2n}(-1)^r\ {}^{2n}C_r\ {}^{2n}C_{2n-r}$, is the coefficient of x^{2n} in the product $(1-x)^{2n}(1+x)^{2n}$ i.e., in $(1-x^2)^{2n}$. Writing y for x^2, the given sum thus equals the coefficient of y^n in $(1-y)^{2n}$, which is precisely the R.H.S. of (29). (After you gain a little practice, it is unnecessary to introduce the variable y explicitly. It is quite all right to say directly that since $x^{2n}=(x^2)^n$, the coefficient of x^{2n} in $(1-x^2)^{2n}$ is $(-1)^n\ {}^{2n}C_n$. But you better not forget to explicitly note that $x^{2n}=(x^2)^n$, even though this is very obvious. For otherwise, in a 'prove that' type question, an examiner is likely to think that you are trying to cheat.)

Note that in the solution above even though we recognised the R.H.S. of (29) as the coefficient of x^n in $(1-x)^{2n}$, this was not of much use initially. But at the end, it is this very fact which clinched the solution, except that we applied it with x replaced by x^2. Equivalently, we replaced the n and x in (21) by $2n$ and $-x^2$ respectively. The first replacement is obvious but the second one is not so. That, indeed, is the charm of the problem. Note also that the symmetry relation of the binomial coefficients played a crucial role in reducing the L.H.S. Here is another problem where similar techniques are needed. (JEE 1994)

Let n be a positive integer and

$$(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n} \tag{30}$$

Prove that

$$a_0^2 - a_1^2 + a_2^2 - \ldots + a_{2n}^2 = a_n \tag{31}$$

Once again, recognising the R.H.S. of (31) as the coefficient of x^n in the expansion of $(1+x+x^2)^n$ is not of any immediate use. Instead, let us look at the L.H.S. of (31). It consists of an alternating sum of squares. If we had a symmetry relation (analogous to that for the binomial coefficients) then we could replace these squares in a manner which will permit us to recognise the L.H.S. as the coefficient of a suitable power of x in a suitable function. To do this, we first need to check whether $a_r = a_{2n-r}$. This is not immediately obvious. But if we rewrite $1+x+x^2$ as $x(\frac{1}{x}+1+x) = x(x^{-1}+1+x)$, then $\sum_{r=0}^{2n} a_r x^r = (1+x+x^2)^n = x^n(x^{-1}+1+x)^n$. The expression $x^{-1}+1+x$ is symmetric in x and x^{-1}. So in $(x^{-1}+1+x)^n$, the coefficients of x^k and of x^{-k} will be equal for every $k \geq 0$. Hence in $x^n(x^{-1}+1+x)^n$, the coefficient of x^{n+k} will equal that of x^{n-k} for every $k \geq 0$. This means $a_{n+k} = a_{n-k}$ which is another way of saying that $a_r = a_{2n-r}$ for $0 \leq r \leq n$.

Having obtained the symmetry relation, the rest of the solution runs almost parallel to that of the last problem. We write $(-1)^r a_r^2$ as $(-1)^r a_r \times a_{2n-r}$.

We are given that a_r is the coefficient of x^r in $(1+x+x^2)^n$. Replacing x by $-x$, $(-1)^r a_r$ will be the coefficient of x^r in $(1-x+x^2)^n$. Hence the sum $\sum_{r=0}^{2n}(-1)^r a_r a_{2n-r}$ is the coefficient of x^{2n} in the product $(1-x+x^2)(1+x+x^2)$, i.e., in $(1+x^2+x^4)^n$. Since $x^{2n}=(x^2)^n$, this is also the coefficient of x^n in $(1+x+x^2)^n$. But that is precisely a_n. Thus (31) is proved.

In this solution we had to make two replacements. The work done for proving the symmetry relation $a_r = a_{2n-r}$ essentially amounts to replacing x by its reciprocal. Then we also had to replace x by $-x$ to take care of the coefficient $(-1)^r$. It is possible to combine these two replacements together, i.e., to replace x by $-\frac{1}{x}$. In that case, from $\sum_{r=0}^{2n} a_r x^r$ we would get $(1-\frac{1}{x}+\frac{1}{x^2})^n = \sum_{r=0}^{2n}(-1)^r a_r \frac{1}{x^r}$. Then the L.H.S. of (31) is nothing but the constant term in the expansion of the product $(1+x+x^2)(1-\frac{1}{x}+\frac{1}{x^2})$, or equivalently, the coefficient of x^{2n} in $(1+x+x^2)^n(1-x+x^2)^n$. The work from this point onward is exactly as above. This solution is somewhat quicker because both the replacements are done simultaneously. But it is not easy for everybody to conceive of this short cut right at the start. If you can think of it, that's great. Otherwise, proceed as most mortals do, by doing one replacement at a time.

Comment No. 17:

Sometimes a full binomial sum is hidden as a part of some other type of a problem. For example,

A is a set containing n elements. A subset P of A is chosen at random. The set A is reconstructed by replacing the elements of P. A subset Q of A is again chosen at random. Find the probability that P and Q have no common elements. (JEE 1990)

Probability problems will be taken up in Chapter 22. But in the present problem, the knowledge of probability needed is minimal. In essence, we are given an ordered pair (P,Q) of two subsets of the set A and we are asked to find the probability that these two subsets are mutually disjoint. This can be done by a counting argument. The number of possible choices of P is the total number of subsets of A (including the empty set). This number is 2^n. (See the combinatorial proof of (22) above.) Similarly, Q can be chosen in 2^n possible ways. Moreover, since P and Q are independent of each other, the ordered pair (P,Q) can be chosen in $2^n \times 2^n = 4^n$ ways. We now have to find the number, say b_n, of those ordered pairs (P,Q) for which $P \cap Q = \emptyset$, the empty set. The desired probability will then be the ratio $\frac{b_n}{4^n}$. This is all the probability theory that is needed in the present problem. The real task now is to find b_n.

To do so, first fix some subset P of A. Suppose P has r elements. Here $0 \le r \le n$. Then the complement of P in A, i.e., the set $A-P$ has $n-r$ elements. Now, a subset Q will be disjoint from P if and only if $Q \subset A-P$. There are 2^{n-r} subsets of $A-P$. Hence for a given P with r elements, there are 2^{n-r} possible choices for Q. For a fixed r, P itself can be chosen in $\binom{n}{r}$ ways. And, r varies from 0 to r. Putting it all together, b_n, the total number of

pairs of mutually disjoint subsets of A is simply the sum $\sum_{r=0}^{n} \binom{n}{r} 2^{n-r}$. A direct application of the binomial theorem gives $b_n = (1+2)^n = 3^n$. Thus the desired probability is $\frac{b_n}{4^n} = \frac{3^n}{4^n} = (\frac{3}{4})^n$. (See Exercise (24.57) for an alternate solution.)

Comment No. 18:

In Comment No. 4 above, we saw how derivatives can be used to deduce (2) from (1). This technique can also be used for evaluating full binomial sums. By definition, differentiation involves something deeper than mere algebraic manipulations, viz., the limiting process. So it is to be expected that differentiation may yield results which are not obtainable, or at least not easily obtainable by mere algebraic manipulations. For example, if we differentiate both the sides of (21) w.r.t. x we get

$$\binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \ldots + \ldots + n\binom{n}{n}x^{n-1} = n(1+x)^{n-1} \tag{32}$$

Giving x suitable values, we get several identities. For example, putting $x = 1$ yields

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \ldots + k\binom{n}{k} + \ldots + n\binom{n}{n} = n2^{n-1} \tag{33}$$

Although an algebraic proof of (33) is possible, it is rather tricky. (It begins by observing that for $k = 1, 2, \ldots n$, $k\binom{n}{k}$ equals $n\binom{n-1}{k-1}$. Applying this to each term and taking n as a common factor we get the L. H. S. of (33) to equal $n\sum_{k=1}^{n}\binom{n-1}{k-1}$ which is the same as $n\sum_{j=0}^{n-1}\binom{n-1}{j}$. Applying (22) to this sum (with n replaced by $n-1$) completes the proof.) An inductive proof of (33) can also be given using the identity $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$. Still, the derivation based on (32) is the best. (This is because a lot of hard work has already been done in going from (21) to (32).) We can go on differentiating further. For example, if we differentiate (32) w.r.t. x and then set $x = 1$, we get

$$2\binom{n}{2} + 6\binom{n}{3} + \ldots + n(n-1)\binom{n}{n} = n(n-1)2^{n-2} \tag{34}$$

Adding (33) and (34) we get

$$\binom{n}{1} + 4\binom{n}{2} + 9\binom{n}{3} + \ldots + k^2\binom{n}{k} + \ldots + n^2\binom{n}{n} = n(n+1)2^{n-2} \tag{35}$$

Comment No. 19:

Like differentiation, integration can also be used to evaluate certain full binomial sums. Let us start with (21). If we integrate both the sides w.r.t. x between 0 and 1 we get

$$\binom{n}{0}\frac{1}{1} + \binom{n}{1}\frac{1}{2} + \binom{n}{2}\frac{1}{3} + \ldots + \binom{n}{n}\frac{1}{n+1} = \frac{1}{n+1}(2^{n+1} - 1) \tag{36}$$

However, if we first replace x by $-x$ in (21) and then integrate then we get an alternating sum

$$\binom{n}{0}\frac{1}{1} - \binom{n}{1}\frac{1}{2} + \binom{n}{2}\frac{1}{3} - \ldots + (-1)^n\binom{n}{n}\frac{1}{n+1} = \frac{1}{n+1} \tag{37}$$

Repeated applications of this process give a solution to the following problem. (JEE 1997*)

Prove that

$$\frac{3!}{2(n+3)} = \sum_{r=0}^{n}(-1)^r\left(\frac{{}^nC_r}{{}^{r+3}C_r}\right) \tag{38}$$

Let us first simplify the R.H.S. a little. The binomial coefficient ${}^{r+3}C_r$ is the same as ${}^{r+3}C_3$ which equals $\frac{(r+1)(r+2)(r+3)}{3!}$. So, proving (38) is equivalent to proving

$$\frac{1}{2(n+3)} = \sum_{r=0}^{n}(-1)^r\frac{{}^nC_r}{(r+1)(r+2)(r+3)} \tag{39}$$

which is a little more manageable. In fact, in (37) we already evaluated the sum $\sum_{r=0}^{n}(-1)^r\ {}^nC_r$ as $\frac{1}{n+1}$ using integration. The denominator of (39) suggests that two more applications of integration will be needed. However, there is not much point in applying integration to (37) because (37) is an equality of numbers and not of functions. It was obtained by integrating both the sides of an equality of functions, namely,

$$\binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \ldots + (-1)^n\binom{n}{n}x^n = (1-x)^n \tag{40}$$

(which is what we get by replacing x with $-x$ in (21)). Here we took the definite integral of both the sides. It is tempting to think that if, instead, we take the indefinite integrals (also called antiderivatives) of both the sides of (40) we would get an equality of functions viz.,

$$\binom{n}{0}x - \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} - \ldots + (-1)^n\binom{n}{n}\frac{x^{n+1}}{n+1} = -\frac{(1-x)^{n+1}}{n+1} \tag{41}$$

which can then be integrated further. The trouble is that (41) is not true as it stands! At $x = 0$, the L.H.S. equals 0 but the R.H.S. equals $\frac{-1}{n+1}$. This is not surprising since two antiderivatives of the same function need not be equal. All we can say is that they differ by a constant. In the present case, this constant happens to be $\frac{1}{n+1}$. If we make this 'correction' then we indeed get an equality of functions viz.,

$$\begin{aligned}\binom{n}{0}x - \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} - \ldots \quad &+ \quad (-1)^n\binom{n}{n}\frac{x^{n+1}}{n+1} \\ &= \quad \frac{1}{n+1} - \frac{(1-x)^{n+1}}{n+1}\end{aligned} \tag{42}$$

(Another way of obtaining this is to replace the vaiable x in (40) by some other variable, say t, and then to take the definite integrals (w.r.t. t) of both the sides from 0 to x. Replacement of x by t in (40) for this purpose is only of a cosmetic type. We can as well integrate (40) as it is between 0 and x. But it looks a little awkward and confusing to integrate a function of x when one of the limits of integration also involves x.)

The path is now clear. Integrating both the sides of (42) from 0 to x we get, $\sum_{r=0}^{n}(-1)^r \frac{{}^nC_r}{(r+1)(r+2)}x^{r+2} = \frac{x}{n+1} + \frac{(1-x)^{n+2}}{(n+1)(n+2)} - \frac{1}{(n+1)(n+2)}$. Doing it once more, we finally get

$$\sum_{r=0}^{n}\frac{(-1)^r\ {}^nC_r x^{r+3}}{(r+1)(r+2)(r+3)} \quad = \quad \frac{x^2}{2(n+1)} + \frac{1-(1-x)^{n+2}}{(n+1)(n+2)(n+3)} \quad - \quad \frac{x}{(n+1)(n+2)} \tag{43}$$

If we put $x = 1$, the L.H.S. is simply the R.H.S. of (39), while the R.H.S. becomes $\frac{1}{2(n+1)} + \frac{1}{(n+1)(n+2)(n+3)} - \frac{1}{(n+1)(n+2)}$. By a straightforward computation this reduces to $\frac{1}{2(n+3)}$ which is precisely the L.H.S. of (39). So we have proved (39) and hence (38).

The application of integration in this problem was fairly straightforward (albeit requiring a little care to guard against wrong statements like (41)) because the evaluation of the integrals was quite simple, requiring nothing beyond the integrals of powers of x. In Comment No. 4 of Chapter 18, we shall present an application where more advanced methods of integration are used.

Comment No. 20:

In Comment No. 11 we gave an example of a binomial identity where induction was the only method available. Analogously, there are some full binomial sums where induction is the best, if not the only, method. This happens when the nature of the sum is such that it is easy to relate it to a similar sum with a lower value of the integer parameter n. Suppose for example, that we want to evaluate the sum $\sum_{k=0}^{n}\binom{n+k}{n}\frac{1}{2^k}$. We can recognise this sum as the coefficient of x^n in the sum $\sum_{k=0}^{n}(x+\frac{1}{2})^{n+k}$. Being the sum of a geometric progression, we can express this sum as $(x+\frac{1}{2})^n\frac{(x+\frac{1}{2})^{n+1}-1}{x-\frac{1}{2}}$. The trouble is that this is not a polynomial in x nor is it of the form $\frac{f(x)}{x^q}$, where $f(x)$ is a polynomial in x and q is some positive integer. If it were of this form then the coefficient of x^n in it will be simply the coefficient of x^{n+q} in $f(x)$. So more advanced methods (based on infinite series) are needed to find the coefficient of x^n in the sum $(x+\frac{1}{2})^n\frac{(x+\frac{1}{2})^{n+1}-1}{x-\frac{1}{2}}$. Just in case you are curious, the idea is to write the denominator as $-\frac{1}{2}(1-2x)$ and then to expand $\frac{1}{1-2x}$ as an infinite geometric series $\sum_{r=0}^{\infty}(2x)^r$. In Chapter 23, we shall briefly study such infinite series.

(Specifically, see Exercise (23.10).) Like derivatives and integrals, the infinite series also belong properly to calculus since they cannot be handled without a limiting process. But, they are also a powerful tool to sum even finite series.

But there is a more elementary way out. Call the sum as S_n. Noting that $\binom{n+k}{n} = \binom{n+k}{k}$, we can relate S_n to S_{n-1} (for $n \geq 1$) as follows.

$$\begin{aligned} S_n &= \sum_{k=0}^{n} \binom{n+k}{k} \frac{1}{2^k} \\ &= \sum_{k=0}^{n} \binom{n+k-1}{k} \frac{1}{2^k} + \sum_{k=0}^{n} \binom{n+k-1}{k-1} \frac{1}{2^k} \\ &= \sum_{k=0}^{n-1} \binom{n-1+k}{k} \frac{1}{2^k} + \binom{2n-1}{n} \frac{1}{2^n} + \frac{1}{2} \sum_{k=1}^{n} \binom{n+k-1}{k-1} \frac{1}{2^{k-1}} \end{aligned}$$

Obviously, the first sum is S_{n-1}. Write the second term as $\frac{1}{2} \times \binom{2n-1}{n} \frac{1}{2^{n-1}}$. In the last sum, change the index by putting $j = k - 1$. Then as k varies from 1 to n, j varies from 0 to $n - 1$ and so we get,

$$S_n = S_{n-1} + \frac{1}{2} \left[\binom{2n-1}{n} \frac{1}{2^{n-1}} + \sum_{j=0}^{n-1} \binom{n+j}{j} \frac{1}{2^j} \right] \tag{44}$$

Rewriting $\binom{2n-1}{n}$ as $\frac{1}{2}\binom{2n}{n}$, the expression in the brackets becomes $\sum_{j=0}^{n} \binom{n+j}{j} \frac{1}{2^j}$ which is simply S_n. So from (44) we get

$$S_n = 2S_{n-1} \tag{45}$$

In the terminology introduced in Comment No. 12 of Chapter 4, this is a recurrence relation. Although we have not developed the general theory of solving recurrence relations, this particular one is easiest to solve. In fact, by inspection, we see that $S_n = 2^n S_0$. But S_0 consists of just one term $\binom{0}{0} \frac{1}{2^0}$ which equals 1. So $S_n = 2^n$ for all $n \geq 0$. (If it appears awkward to consider $\binom{0}{0}$, you can write $S_n = 2^{n-1} S_1$ and compute S_1 directly to get the same result. Also if you do not want to solve the recurrence relation at all, you can prove by induction on n that $S_n = 2^n$, using (45) to establish the inductive step.)

EXERCISES

5.1 Prove the following identities about binomial coefficients. Give combinatorial proofs wherever you can.

(i) $\binom{n}{k}\binom{k}{r} = \binom{n}{r}\binom{n-r}{k-r}$ (For lack of a standard name, we call this the **double selection identity** because in the combinatorial interpretation of the L.H.S., we first select k objects and then again select r out of these k objects. The special cases $r = k - 1$ and $r = 1$ give the extraction formulas in Equations (1) and (10) of Chapter 4.)

(ii) $\sum_{i=0}^{r} \binom{n+i}{k} = \binom{n+r+1}{k+1} - \binom{n}{k+1}$ (This is essentially same as (1).)

(iii) $\sum_{j=0}^{\infty} \binom{p}{j}\binom{q}{n+j} = \binom{p+q}{p+n}$, where p, q are some fixed positive integers. Note the special case $n = 0$ and also the case where $p = q$.

(iv) $\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m} = \binom{n}{m}2^{n-m}$.

5.2 For $0 \le r \le k \le n$, evaluate

$$\binom{n}{k} - \binom{n}{k-1} + \binom{n}{k-2} - \binom{n}{k-3} + \ldots + (-1)^r \binom{n}{k-r}.$$

What happens if $n = r = k$?

5.3 (i) Prove that $\sum_{k=0}^{m} (-1)^k \binom{n}{k}\binom{n}{m-k}$ equals $(-1)^{m/2}\binom{n}{m/2}$ if m is even and 0 otherwise. Also evaluate the sum $\sum_{k=0}^{m} (-1)^k \binom{n}{k}$.

(ii) Let $S_n = \sum_{k=0}^{n} (-1)^k \binom{2n-k}{k}$. Prove that for all $n \ge 1$, $S_n = S_{n+1} + 2S_n + S_{n-1}$. Hence show that $S_n = 1, 0$ or -1 according as $n \equiv 0,\ 1$ or -1 (*mod* 3). (Tomescu)

5.4 (i) If $0 \le m \le n$, prove that $\sum_{k=0}^{m} \binom{m}{k}/\binom{n}{k} = \frac{n+1}{n+1-m}$. [*Hint:* Begin by using Exercise (5.1) (i).]

(ii) For positive integers m, n, show that $\sum_{k=0}^{\infty} k\binom{n}{k}\binom{m}{k} = m\binom{n+m-1}{m}$. Note the special case $m = n$.

(iii) If C_r stands for ${}^nC_r$, where n is an even integer, find which, if any, of the expressions given below equals the sum of the following series. (1986)

$$\frac{2(\frac{n}{2})!(\frac{n}{2})!}{n!}\left[C_0^2 - 2C_1^2 + 3C_2^2 - \ldots + (-1)^n(n+1)C_n^2\right].$$

(A) 0 (B) $(-1)^{n/2}(n+1)$ (C) $(-1)^n(n+2)$ (D) $(-1)^n n$.

(iv) If $C_r = {}^nC_r$, prove that $C_0 - 2^2\, C_1 + 3^2\, C_2 - \ldots + (-1)^n(n+1)^2\, C_n = 0$. (1989)

5.5 Prove (35) by induction on n. (1986)

5.6 Given that

$$C_1 + 2C_2x + 3C_3x^2 + \ldots + 2nC_{2n}x^{2n-1} = 2n(1+x)^{2n-1}$$

where $C_r = \frac{(2n)!}{r!(2n-r)!}$, prove that

$$C_1^2 - 2C_2^2 + 3C_3^2 \ldots - 2nC_{2n}^2 = (-1)^n nC_n. \quad (1979)$$

5.7 Find the coefficient of x^4 in $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$. (1983)

5.8 If $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$, then express $\sum_{r=0}^{n} \frac{r}{{}^nC_r}$ in terms of a_n. (1998)

5.9 If $(1+x)^n = c_0 + c_1x + c_2x^2 + \ldots + c_nx^n$, then show that the sum of the products of the c_i's taken two at a time, represented by $\sum_{0 \le i < j \le n} c_ic_j$, is equal to $2^{2n-1} - \dfrac{(2n)!}{2(n!)^2}$. (1983)

5.10 (a) If $(1+ax)^n = 1 + 8x + 24x^2 + \ldots$, determine a and n. (1983)

(b) If ${}^nC_{r-1} = 36, {}^nC_r = 84$ and ${}^nC_{r+1} = 126$, determine r. (1979)

(c) Given positive integers $r > 1, n > 2$, and that the coefficients of the $(3r)$-th and $(r+2)$-th terms in the binomial expansion of $(1+x)^{2n}$ are equal, which, if any, of the following statements is true? (1980)
(A) $n = 2r$ (B) $n = 3r$ (C) $n = 2r + 1$.

(d) If the coefficients of the second, the third and the fourth terms in the expansion of $(1+x)^n$ are in A.P., determine n. (1994)

(e) If in the expansion of $(1+x)^m(1-x)^n$, the coefficients of x and x^2 are 3 and -6 respectively, determine m. (1999)

(f) A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which he can select at least one book is 63, find the value of n. (1987)

5.11 Prove that $\sum_{r=1}^{k} (-3)^{r-1} \, {}^{3n}C_{2r-1} = 0$, where $k = \frac{3n}{2}$ and n is an even positive integer. [*Hint:* Write -3 as $(i\sqrt{3})^2$ and use properties of ω, the complex cube root of unity. See Comment No. 12 of Chapter 7.] (1993)

5.12 Prove (32) by a purely algebraic argument (i.e., without using differentiation or anything based on calculus).

5.13 Give a combinatorial proof of (33). [*Hint:* How about choosing a team of variable size from n players and a captain from it?]

5.14 In Comment No. 13, we used (1) to obtain an expression for the sums $1 + 2 + \ldots + n$ and then also to evaluate the sum of the squares of the integers from 1 to n.

(a) Similarly obtain an expression for the sum of the cubes of the integers from 1 to n.

(b) Can we evaluate $1+2+\ldots+n$ by differentiating the two sides of the identity

$$1 + x + x^2 + \ldots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

and then setting $x = 1$?

(c) Show that in (b), we can get the desired sum by taking the limits of both the sides of

$$1 + 2x + 3x^2 + \ldots + nx^{n-1} = \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$$

as $x \to 1$, where we find the limit of the R.H.S. by some such method as the l'Hôpital's rule, see Exercise (15.15)(a).

(d) If m, n are positive integers and $(1 + x + x^2 + \ldots + x^{2m})^n = a_0 + a_1x + a_2x^2 + \ldots$, prove that

$$a_2 + 3a_3 + 6a_4 + 10a_5 + \ldots = \frac{mn(2m+1)^n}{3!}(3mn + m - 2).$$

5.15 For every positive integer n, prove that:

(a) $\sum_{r=1}^{n} (-1)^{r-1}\ {}^nC_r 2^{n-r} r^3 = n(n^2 - 6n + 6)$.

(b) ${}^nC_1 - {}^nC_2\ {}^3C_2 + {}^nC_3\ {}^4C_2 - \ldots + (-1)^{n-1}\ {}^nC_n\ {}^{n+1}C_2 = 0$.

(c) $S_1 = S_2 = \ldots = S_{n-1} = 0$, $S_n = (-1)^n n!$ and $S_{n+1} = (-1)^n \frac{n}{2}(n+1)!$, where for every positive integer k, S_k is the sum $\sum_{r=0}^{n} (-1)^r \binom{n}{r} r^k$.

(d) $\sum_{r=0}^{n} (-1)^r\ {}^nC_r (2n - r)^{n+1} = \frac{3n}{2}(n+1)!$.

(e) the total number of ways to select n objects from $3n$ objects, of which n are of one kind, n are of some other kind and the rest are all unlike is $2^{n-1}(n+2)$.

(f) $2\ {}^nC_0 + \dfrac{2^2\ {}^nC_1}{2} + \dfrac{2^3\ {}^nC_2}{3} + \ldots + \dfrac{2^{n+1}\ {}^nC_n}{n+1} = \dfrac{3^{n+1} - 1}{n+1}$.

(g) ${}^{n+1}C_1 + {}^{n+1}C_2 s_1 + {}^{n+1}C_3 s_2 + \ldots + {}^{n+1}C_{n+1} s_n = 2^n S_n$, where, $s_n = 1 + q + q^2 + \ldots + q^n$ and

$$S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + \ldots + \left(\frac{q+1}{2}\right)^n$$

where $q \neq 1$. (1984)

5.16 For every positive integer n prove that the sum $\sum_{k=0}^{n} \binom{n-k}{k}$ is a Fibonacci number.

5.17 A stack of one rupee and two rupees coins is to be formed. Assume that the coins of each denomination are identical. Using the last exercise or directly by induction, show that the number of different stacks of total value n rupees is a Fibonacci number for every positive integer n. (Like Catalan numbers, the Fibonacci numbers also appear in many diverse combinatorial problems. The present exercise is a sample.)

5.18 Give algebraic as well as combinatorial proofs of the following identities about the Stirling numbers defined in Exercise (1.27).

(i) $S_{n+1,m+1} = \sum_{k=m}^{n} \binom{n}{k} S_{k,m}$ (ii) $S_{n+1,m+1} = \sum_{k=m}^{n} (m+1)^{n-k} S_{k,m}$.

5.19 If $(1+x)^{2n} = \sum_{k=0}^{2n} a_k x^k$ then prove that the sum of the first $n+2$ terms of the series $a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + \ldots$ is $(-1)^{n+1}\binom{2n-1}{n+1}$.

5.20 Let $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \ldots + a_k x^k + \ldots + \ldots + a_{2n} x^{2n}$. Prove that:

(a) $a_n = 1 + \dfrac{n(n-1)}{(1!)^2} + \dfrac{n(n-1)(n-2)(n-3)}{(2!)^2} + \ldots = \displaystyle\sum_{k=0}^{\infty} \frac{(2k)!}{(k!)^2}\binom{n}{2k}$

(b) $a_0 a_1 - a_1 a_2 + a_2 a_3 - \ldots - a_{2n-1} a_{2n} = 0$

(c) $a_0^2 - a_1^2 + a_2^2 - \ldots + (-1)^{n-1} a_{n-1}^2 = \frac{1}{2}[a_n + (-1)^{n-1} a_n^2]$

(d) $a_0 + a_2 + a_4 + \ldots = \frac{1}{2}(3^n + 1)$ and $a_1 + a_3 + a_5 + \ldots = \frac{1}{2}(3^n - 1)$

(e) of the four sums $a_0 + a_4 + a_8 + \ldots$, $a_1 + a_5 + a_9 + \ldots$, $a_2 + a_6 + a_{10} + \ldots$ and $a_3 + a_7 + a_{11} + \ldots$, three are equal and the fourth differs from their value by 1. (Tomescue)

5.21 For every positive integer n, prove that the number of solutions to the equation

$$x_1 x_2 \ldots x_n = y_1 y_2 \ldots y_n$$

where the variables are allowed to take only 0 and ± 1 as possible values is $3(3^{2n-1} - 3^{n-1}2^{n+1} + 2^{2n-1})$.

5.22 If $a_0, a_1, a_2, \ldots, a_n$ are in A.P. with common difference d and $C_r = \binom{n}{r}$ for $r = 0, 1, 2, \ldots, n$, prove that

$$C_0 a_0^2 + C_1 a_1^2 + C_2 a_2^2 + \ldots + C_n a_n^2 = 2^{n-1}[(a_0 + a_n)^2 + nd^2].$$

5.23 Find the sum of the series

$$\sum_{r=0}^{n} (-1)^r \, {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \ldots \text{upto } m \text{ terms}\right] \quad (1985)$$

5.24 If m, n, p, q are non-negative integers with $n \geq q$, prove that:

(i) $\sum_{k=0}^{q} k\binom{n-k-1}{n-q-1} = \binom{n}{q-1}$ (ii) $\sum_{k=0}^{p} \binom{p-k}{m}\binom{q+k}{n} = \binom{p+q+1}{m+n+1}$

(iii) $\sum_{k=0}^{p} (-1)^k \binom{p-k}{m}\binom{q}{k-n} = (-1)^{p+m}\binom{q-m-1}{p-m-n}$.

5.25 Prove that the sum of the cubes of the coefficients of $(1+x)^n$ is equal to the coefficient of $x^n y^n$ in the expansion of $\{1+x+y+2xy+x^2y+xy^2+x^2y^2\}^n$.

Chapter 6

INEQUALITIES

The word 'inequality' is not the mere negation of the word 'equality' or 'equation'. When two complex numbers (or expressions), say, a and b are unequal, there is not much more to be said about their relationship. But when they are real and unequal, exactly one of them is smaller than the other, i.e., either $a < b$ or $b < a$. Saying $a < b$ is a lot more informative than merely saying $a \neq b$. This binary relation (called the **order relation**) is a peculiarity of real numbers. There is no analogue of it for complex numbers. (See Comment No. 11.)

In schools, inequalities are taught much later than equations. The popular inequalities come through geometry, the most well-known being the triangle inequality, which says that the sum of the two sides of any triangle exceeds the third. This inequality appears in many more general forms, along with a closely related inequality, called the Cauchy-Schwarz inequality. The most well-known algebraic inequality is probably the inequality of the means, which says that the geometric mean can never exceed the arithmetic mean. There are also trigonometric inequalities which result from geometric facts.

In this chapter we confine ourselves to inequalities which are 'elementary' in that they do not require methods from calculus. The latter will be studied in Chapter 13. Closely related to inequalities is the problem of finding the maximum and the minimum value, and more generally, the entire range of a real valued function. Again, here we only consider examples amenable to elementary methods. From a theoretical point of view, the most significant application of inequalities is that they provide a rigorous definition of the concept of a limit. We illustrate this for the limit of a sequence.

Problems involving inequalities can be tricky. In the JEE, full length questions based on inequalities are not asked every year. But the A.M.−G.M. inequality is rather popular and combines conveniently with other topics to form short questions. Inequalities are also inherent in questions of determining when a quadratic has real roots, when a given conic is a hyperbola and so on.

Main Problem : The minimum value of the expression $\dfrac{(33+x)(57+x)}{8+x}$ for $x > -8$ is ..... .

First Hint: Write the expression in terms of u, where $u = 8 + x$.

Second Hint: Use the A.M.−G.M. inequality to minimise $\dfrac{1225}{u} + u$.

Solution: Following the first hint, the expression equals $\dfrac{(25+u)(49+u)}{u} = \dfrac{1225}{u} + 74 + u$ which is to be minimised for $u > 0$. As 74 is a constant, this amounts to minimising $\dfrac{1225}{u} + u$. As both the terms are positive, by the A.M.−G.M. inequality we have $\dfrac{1225}{u} + u \geq 2\sqrt{\dfrac{1225}{u} \times u} = \sqrt{1225} = 70$ with equality holding if and only if $\dfrac{1225}{u} = u$ i.e., if and only if $u = 35$. So the minimum value of the given expression is $70+74 = 144$. (It occurs when $u = 35$ i.e., when $x = 27$ although this is not asked.)

Comment No. 1:

More generally the same reasoning shows that for any three real numbers a, b, c with $a \geq c$ and $b \geq c$, the minimum value of the expression $\dfrac{(a+x)(b+x)}{c+x}$ for $x > -c$ is $(\sqrt{a-c} + \sqrt{b-c})^2$. (JEE 1979)

Comment No. 2:

There is, of course, the standard method, based on calculus, to find the maximum and the minimum of a function on a given interval. In the present case, the function is $f(x) = \dfrac{(33+x)(57+x)}{8+x}$ and the interval is the semi-infinite open interval $(-8, \infty)$. Then $f(x)$ is differentiable everywhere in the given interval and $f'(x) = \dfrac{(8+x)(2x+90) - (33+x)(57+x)}{(8+x)^2}$. Setting $f'(x) = 0$ gives $x^2 + 16x - 1161 = 0$. This is a quadratic with roots 27 and -43. Hence 27 is the only critical point of $f(x)$ lying in the interval $(-8, \infty)$. It is easy to see that $f'(x) < 0$ for $-8 < x < 27$ and $f'(x) > 0$ for $x > 27$. Hence in the interval $(-8, \infty)$, the function $f(x)$ is strictly decreasing for $x < 27$ and strictly increasing for $x > 27$. So it attains its minimum at $x = 27$. By a direct computation, $f(27) = 144$. So this is the desired answer.

Comment No. 3:

There is yet another way to look at the problem. We can rewrite the equation $\dfrac{(33+x)(57+x)}{8+x} = y$ as $(33+x)(57+x) = y(8+x)$ and hence further as a

quadratic equation in x, viz., $x^2 + (90 - y)x + (1881 - 8y) = 0$. The problem is equivalent to finding the smallest value of y for which this quadratic has at least one root bigger than -8. Once again, the substitution $u = 8 + x$ simplifies the matters a little bit, for now the quadratic equation becomes $(25+u)(49+u) = yu$, i.e., $u^2 + (74 - y)u + 1225 = 0$. We want at least one of the two roots of this quadratic to be positive. For the roots to be real, it is first of all necessary that $(74 - y)^2 \geq 4900$. Since $4900 = (70)^2$ this means either $y \leq 4$ or $y \geq 144$. As the product of the two roots is 1225 which is positive, the two roots are either both positive or both negative. If $y \leq 4$ then $74 - y > 0$ and so the sum of the roots is negative. Hence both of them are negative. However, if $y \geq 144$ then their sum is positive and so both the roots are positive. It follows therefore that 144 is the smallest value of y for which the quadratic $u^2 + (74 - y)u + 1225 = 0$ has at least one positive root and hence also the smallest value of the expression $\dfrac{(33 + x)(57 + x)}{8 + x}$ for $x > -8$.

Comment No. 4:

Of the three methods discussed so far, the second one is the most general in that it can be applied to find the maxima and the minima of many other types of functions as we shall see in Chapter 13. The other two methods are purely algebraic. They lack the strength of the limiting process (which is the very heart of calculus) and hence it is to be expected that they are not as powerful as the calculus method. But then, it is precisely for this reason that they offer a greater challenge to one's ingenuity. Note, for example, the effective use of the simple substitution $u = x + 8$ in both of them. The calculus-based method is mechanical and it makes little difference whether you work in terms of x or in terms of u. The first method is based on the A.M.$-$G.M. inequality, which is an easy consequence of the fact that the product of two real numbers of the same sign is positive and hence, in particular, that the square of any real number is always positive (or zero). The last method also needed this fact in the form of the discriminant criterion for the roots to be real. In addition, it required elementary facts from the theory of equations. So, the last method is not as elementary as the first. But it also brings a greater reward. We not only know what is the least value of the expression $\dfrac{(33 + x)(57 + x)}{8 + x}$ for $x > -8$, but we know the entire range of the function $f(x) = \dfrac{(33 + x)(57 + x)}{8 + x}$ defined on the interval $(-8, \infty)$. Actually, if we analyse the solution, we know a little more. We not only know that every real number in the interval $[144, \infty)$ has at least one preimage (in $(-8, \infty)$) under $f(x)$, we know that every real $y > 144$ has two preimages while 144 has only one preimage, viz., $x = 27$.

Comment No. 5:

The **A.M.$-$G.M. inequality** is one of the simplest and still one of the most frequently needed inequalities in elementary mathematics. Formally, it asserts the superiority of the arithmetic mean over the geometric mean. The latter is defined in terms of roots and to ensure its existence we generally confine

ourselves only to positive (or at least, non-negative) numbers. In symbols, the inequality says that for all positive real numbers $a_1, a_2, \ldots, a_n$,

$$\frac{a_1 + a_2 + \ldots + a_n}{n} \geq (a_1 a_2 \ldots a_n)^{1/n} \tag{1}$$

with equality holding if and only if $a_1 = a_2 = \ldots = a_n$.

There are many ways to prove (1). Let us first attempt a direct proof. The case $n = 1$ holds trivially. Suppose $n = 2$. (This is the most frequently needed case.) For notational simplicity denote the numbers a_1, a_2 by a, b respectively. Since both the sides of (1) are positive, proving it amounts to showing that $(a+b)^2 \geq 4ab$, or equivalently, $(a-b)^2 \geq 0$ which is true with equality holding if and only if $a = b$. So the case $n = 2$ of (1) is a direct consequence of the fact that the square of any non-zero real number is positive. In the case of three positive real numbers, say a, b, c, the inequality reduces to showing that $(a+b+c)^3 \geq 27abc$. To do this by expanding the cube is rather complicated. But there is a neat alternative. We let x, y, z be the cube roots[1] of a, b, c respectively. Then (1) reduces to proving that $x^3 + y^3 + z^3 \geq 3xyz$ and this can be done by factoring $x^3 + y^3 + z^3 - 3xyz$ as $(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$. The first factor is positive by assumption. If we write the second factor as $\frac{1}{2}[(x-y)^2 + (y-z)^2 + (z-x)^2]$, we see that it is non-negative (being a sum of squares divided by 2) and equals 0 if and only if all the three summands are 0, which can happen only when $x = y = z$ i.e., when $a = b = c$.

For $n > 3$ an algebraic proof of (1) becomes rather complicated. A proof by induction on n is possible but rather tricky. (See Exercise (6.6).) But there is a better way out. Instead of proving (1) directly, we first prove a related result and then (1) will follow as a corollary. In the solution to the main problem we showed, as a consequence of the A.M.−G.M. inequality for two numbers, that the minimum value of the sum $\frac{1225}{u} + u$ is 70 and occurs when $\frac{1225}{u} = u$. We can turn the tables around. Note that $\frac{1225}{u}$ and u are two positive real numbers with a constant product, viz., 1225. More generally, we claim that among all pairs of positive real numbers a and b whose product is a constant, say p, the sum is minimum when the two numbers are equal, i.e., when $a = b = \sqrt{p}$. To see this, merely write $a + b$ as $\sqrt{a^2 + b^2 + 2ab}$ and hence as $\sqrt{(a-b)^2 + 4p}$. Clearly this is minimum when $a = b$ and the minimum value is $2\sqrt{p}$. It follows that whenever $ab = p$, we must have $a + b \geq 2\sqrt{p}$, i.e., $a + b \geq 2\sqrt{ab}$. But this is precisely the A.M.−G.M. inequality for two real numbers.

More generally we prove the following result in which we minimise the sum of n positive real numbers whose product is a given constant.

Theorem 1: Among all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ of positive real numbers whose product is a constant, say p, the sum is minimum when all the a_i's are equal (which is equivalent to saying that each a_i equals $p^{1/n}$).

[1]The existence of cube roots, and more generally the n-th roots of real numbers for $n \geq 2$ is often taken for granted. But a rigorous proof of it requires deep properties of real numbers. See the remarks about exponentiation in Comment No. 6 of Chapter 2.

Proof: For $n = 1$, there is nothing to prove, while the case $n = 2$ was proved above. Hence assume $n > 2$. Let S be the set $\{(x_1, x_2, \ldots, x_n) : x_i > 0$ for $i = 1, 2, \ldots, n$ and $x_1 x_2 \ldots x_n = p\}$ and let $f : S \longrightarrow \mathbb{R}$ be the function defined by $f(x_1, x_2, \ldots, x_n) = x_1 + x_2 + \ldots + x_n$. We have to show that the function f attains its minimum on the set S at the point $(p^{1/n}, p^{1/n}, \ldots, p^{1/n})$ in S, or in other words that $f(x_1, x_2, \ldots, x_n) \geq f(p^{1/n}, p^{1/n}, \ldots, p^{1/n})(= np^{1/n})$ for all $(x_1, x_2, \ldots, x_n) \in S$. Proving this inequality directly is not easy. (If we could do it then we would be able prove (1) directly, but that is not easy as we saw). So we proceed by contradiction. Let $(b_1, b_2, \ldots, b_n)$ be the point in S at which the function f attains its minimum[2]. Since $b_1 b_2 \ldots b_n = p$, showing that $(b_1, b_2, \ldots, b_n) = (p^{1/n}, p^{1/n}, \ldots, p^{1/n})$ is equivalent to proving that $b_1 = b_2 = \ldots = b_n$. Suppose this is not the case. Then at least two of the b_i's are unequal and without loss of generality, we may assume that $b_1 \neq b_2$. Let $c_1 = c_2 = \sqrt{b_1 b_2}$ while for $i = 3, 4, \ldots, n$, let $c_i = b_i$. In other words, $(c_1, c_2, \ldots, c_n)$ is obtained from $(b_1, b_2, \ldots, b_n)$ by changing only the first two entries. Moreover, since $c_1 c_2 = b_1 b_2$, we have $c_1 c_2 c_3 \ldots c_n = b_1 b_2 b_3 \ldots b_n = p$. In particular it follows that $(c_1, c_2, \ldots, c_n)$ also lies in the set S. We claim that $f(c_1, c_2, \ldots, c_n) < f(b_1, b_2, \ldots, b_n)$. Clearly, $f(b_1, b_2, \ldots, b_n) - f(c_1, c_2, \ldots, c_n) = (b_1 + b_2) - (c_1 + c_2) = b_1 + b_2 - 2\sqrt{b_1 b_2}$. Since $b_1 \neq b_2$, this is positive by the A.M.−G.M. inequality for two real numbers. Thus we have found a point $(c_1, c_2, \ldots, c_n) \in S$ at which f has a smaller value that at the point $(b_1, b_2, \ldots, b_n)$. This contradicts our assumption that f attains its minimum on S at $(b_1, b_2, \ldots, b_n)$. So all the b_i's must be equal and hence $(b_1, b_2, \ldots, b_n) = (p^{1/n}, p^{1/n}, \ldots, p^{1/n})$, completing the proof.

In this theorem, the numbers $a_1, a_2, \ldots, a_n$ are not free to vary independently of each other. Instead, they are constrained by the requirement that their product be a given constant p. Such a restriction is, therefore, called a **constraint**. The theorem above minimises the sum $a_1 + a_2 + \ldots + a_n$ subject to this constraint (in addition to the constraints $a_i > 0$, of course). The nature of the constraint is such that there is little point in maximising the sum $a_1 + a_2 + \ldots + a_n$, for we can make all except one a_i,s very small (but positive) which would force the remaining one and hence the sum to be very large. There

[2]The existence of such a point is, again, one of those facts which are taken for granted but whose rigorous proof needs certain basic facts about the real number system. As we are taking the existence of the minimum for granted, to that extent our proof is incomplete. What we are showing is that *in case* the function f has a minimum on S, it must occur at the point $(p^{1/n}, p^{1/n}, \ldots, p^{1/n})$. This is *not* quite the same thing as saying that the minimum occurs at the point $(p^{1/n}, p^{1/n}, \ldots, p^{1/n})$. The latter statement is stronger because included in it is also a proof of the existence of a minimum. The difference in the two statements is rather subtle and is often missed by a beginner. To appreciate it, here is a (false) proof that 1 is the largest positive number. For, let x be the largest positive number. If $x \neq 1$ then $x > 1$. But then x^2 is a real number larger than x, a contradiction. The result is faulty because we are presupposing that a largest positive real number exists, which is simply not true. As a real-life analogy, suppose a person A dies of a gun shot and it is known that at the time the shot was fired, B was the only person in A's company. This does not quite prove that B murdered A for it is quite possible that the shot was fired accidentally or by A himself. All it proves is that in case A was murdered, then the murderer must be B. However, at the JEE level such fine differences can be safely left aside.

is a related theorem in which the constraint is of the form $a_1 + a_2 + \ldots + a_n = s$, i.e., that the numbers are required to add up to a given constant s, and subject to this constraint we want to maximise the product $a_1 a_2 \ldots a_n$. (This time there is no point in minimising the product since it can be made arbitrarily small.) The answer is provided below.

Theorem 2: Among all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ of positive real numbers whose sum is a constant, say s, the product is maximum when all the a_i's are equal (which is equivalent to saying that each a_i equals $\frac{s}{n}$).

The proof is analogous to that of Theorem 1 and hence is left as an exercise. If one wants, one can derive either of the two theorems from the other. Let us show, how Theorem 1 implies Theorem 2. Suppose s is a constant and $a_1, a_2, \ldots, a_n$ are positive real numbers such that $a_1 + a_2 + \ldots + a_n = s$. Let $q = a_1 a_2 \ldots a_n$. We have to show that $q \leq (\frac{s}{n})^n$ or equivalently, $\lambda \leq 1$, where λ is defined to be the ratio $\dfrac{q}{(s/n)^n}$. Suppose this is not so, i.e., $\lambda > 1$. Let μ be the positive n-th root of λ. Then $\mu > 1$ and $a_1 a_2 \ldots a_n = q = \lambda(\frac{s}{n})^n = \mu^n(\frac{s}{n})^n$. We 'scale down' the numbers $a_1, a_2, \ldots, a_n$ by the reciprocal of μ. Specifically, for $i = 1, 2, \ldots, n$, let $b_i = \frac{a_i}{\mu}$. Then the product $b_1 b_2 \ldots b_n$ equals the constant $(\frac{s}{n})^n$ and so by Theorem 1 (applied with $p = (\frac{s}{n})^n$), we have $b_1 + b_2 + \ldots + b_n \geq n((\dfrac{s}{n})^n)^{1/n} = s$. But this means $a_1 + a_2 + \ldots + a_n = \mu s > s$ since $\mu > 1$ contradicting the assumption that $a_1 + a_2 + \ldots + a_n = s$. Thus we see that Theorem 1 implies Theorem 2. In an entirely analogous manner, Theorem 2 implies Theorem 1.

The peculiar relationship between these two theorems is noteworthy. In Theorem 1, we are minimising the sum and the constraint is that the product is a constant while in Theorem 2, the sum is constant and we are maximising the product. This relationship is known as duality and we shall see more instances of it in Chapter 14. The two theorems are like the two approaches taken by shoppers. Sometimes we have in mind a certain quality of the product we want to buy and we look for a shop which will charge the least for that quality. On the other hand, sometimes we have a fixed budget and we try to get the maximum quality while staying within that budget.

There is a general method due to Lagrange to find the maxima and minima of functions of several variables subject to certain constraints. Both the theorems above follow as very special cases. But that theory is far too complicated to be presented here. (In Chapter 14 we shall indicate how the method works for functions of three variables.) The remarkable thing about the theorems above is the simplicity of their proofs. Both the theorems also have practical applications. It follows, for example, from Theorem 1 that among all rectangles having a given area, the square has the smallest perimeter. Theorem 2, on the other hand, implies that among all rectangles with a given perimeter, the square has the the maximum area. Note that there is no restrictions on the numbers involved other than that they be positive. So replacing the a_i's by suitable expressions which are positive we can get new results. As a typical

illustration, suppose we are given that $a_1, a_2, \ldots, a_n$ are positive real numbers whose product is a fixed number c and we want to find the minimum value of the sum $a_1 + a_2 + \ldots + a_{n-1} + 2a_n$. (JEE 2002) Here Theorem 1 cannot be applied directly. However, let $b_i = a_i$ for $i = 1, 2, \ldots, n-1$ while let $b_n = 2a_n$. Then the product $b_1 b_2 \ldots b_{n-1} b_n$ equals the constant $2c$ and so by Theorem 1, the minimum value of the sum $b_1 + b_2 + \ldots + b_{n-1} + b_n$ is $n(2c)^{1/n}$. But since $b_1 + b_2 + \ldots + b_{n-1} + b_n = a_1 + a_2 + \ldots + a_{n-1} + 2a_n$, this is also the minimum value of the sum $a_1 + a_2 + \ldots + a_{n-1} + 2a_n$.

The A.M.−G.M. inequality, viz., (1) can now be derived from either of the two theorems above. All we have to do is to call $a_1 a_2 \ldots a_n$ as p and apply Theorem 1 or call $a_1 + a_2 + \ldots + a_n$ as s and apply Theorem 2. There is another equivalent (although less frequently needed) formulation of this inequality which is called the **G.M.−H.M. inequality** and which asserts that the harmonic mean of any positive real numbers can never exceed their geometric mean and that the two are equal only when all the numbers are equal. Specifically, for any positive real numbers $a_1, a_2, \ldots, a_n$, we have

$$(a_1 a_2 \ldots a_n)^{1/n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \ldots + \frac{1}{a_n}} \tag{2}$$

with equality holding if and only if $a_1 = a_2 = \ldots = a_n$. The easiest way to prove this is to let $b_i = \frac{1}{a_i}$ for $i = 1, 2, \ldots, n$, apply (1) to $b_1, b_2, \ldots, b_n$ and then take reciprocals again (keeping in mind that taking reciprocals reverses an inequality).

Comment No. 6:

Returning to the applications of the A.M.−G.M. inequality, as direct applications we get that $\dfrac{x^n y^m}{(1+x^{2n})(1+y^{2m})} \geq \dfrac{1}{4}$ for any positive real numbers x, y and any positive integers m, n. (JEE 1989) All we need to do is to observe that each of the two factors $\dfrac{x^m}{1+x^{2m}}$ and $\dfrac{y^n}{1+y^{2n}}$ is at most $\frac{1}{2}$ because of the A.M.−G.M. inequality (applied to the numbers 1 and x^{2m} in the first case and to the numbers 1 and y^{2n} in the second).

A simple but interesting consequence of the A.M.−G.M. inequality is that the sum of a positive real number and its reciprocal is always ≥ 2 and equals 2 only when that number is 1. For example, the minimum value of $2\log_{10} x - \log_x(0.01)$ is 4 (JEE 1980) because the expression can be rewritten as $2(\log_{10} x + \log_x 10)$ and hence as $2(y + \frac{1}{y})$, where $y = \log_{10} x$. As we shall see later (see Comment No. 10 in Chapter 10) this fact is also useful in solving some other equations. The A.M.−G.M. inequality for three real numbers is especially useful in problems dealing with triangles as we shall see in Chapter 14 where we shall also study a more general inequality called the **Jensen's inequality**.

Sometimes it pays to split one (or more) terms suitably and *then* apply the A.M.−G.M. inequality. For example, for $x > 0$, proving $\frac{1}{2}x^2 + \frac{1}{x} \geq \frac{3}{2}$ is a little complicated. But by rewriting the L.H.S. as $\frac{1}{2}x^2 + \frac{1}{2x} + \frac{1}{2x}$ we get it effortlessly from the A.M.−G.M. inequality. (Also see Exercises (6.13)(b) and (c).)

Comment No. 7:

There is yet another well-known inequality, called the **Cauchy-Schwarz inequality** which is based on the fact that the square of a real number is always non-negative. It says that for every positive integer n and for any (not necessarily positive) real numbers $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ we have

$$a_1b_1 + a_2b_2 + \ldots + a_nb_n \leq \sqrt{a_1^2 + a_2^2 + \ldots + a_n^2}\ \sqrt{b_1^2 + b_2^2 + \ldots + b_n^2} \qquad (3)$$

with equality holding if and only if there exists a non-negative real number λ such that either $a_i = \lambda b_i$ for every $i = 1, 2, \ldots, n$ or $b_i = \lambda a_i$ for every $i = 1, 2, \ldots, n$.

For a proof, denote the expressions $a_1^2 + a_2^2 + \ldots + a_n^2$, $b_1^2 + b_2^2 + \ldots + b_n^2$ and $a_1b_1 + a_2b_2 + \ldots + a_nb_n$ by A, B and C respectively. Then A, B are non-negative and (3) reads as " $C \leq \sqrt{A}\sqrt{B}$ ", or equivalently, as $C^2 \leq AB$. If $A = 0$, that would force each a_i to vanish and then equality holds in (3) trivially and moreover $a_i = \lambda b_i$ for every i, where $\lambda = 0$. So assume $A > 0$, which is equivalent to assuming that at least one a_i is non-zero. Consider the function $f(t) = \sum_{i=1}^{n} (a_it + b_i)^2$. Here t is a variable taking all possible real values. When expanded, $f(t)$ equals $At^2 + 2Ct + B$ which is a quadratic in t. Note that $f(t)$, being a sum of squares is never negative and hence the quadratic equation $At^2 + 2Ct + B = 0$ cannot have two distinct real roots (or else, for values of t in between these two roots, the expression $f(t)$ would be negative). So the discriminant of this quadratic equation cannot be positive. In other words, $4C^2 - 4AB \leq 0$ which is exactly what we wanted to prove. Moreover if equality holds then the quadratic has only one real root, i.e., there is a unique t such that $f(t) = 0$ which can happen if and only if $a_it + b_i = 0$ for every $i = 1, 2, \ldots, n$. Moreover this t cannot be positive as that would force a_i and b_i to be of opposite signs for every $i = 1, 2, \ldots, n$ and since at least one a_i is non-zero, that would mean $C < 0$. But in that case equality cannot hold in (3) as the R.H.S. can never be negative. So we have $t \leq 0$ and hence putting $\lambda = -t$, we get $b_i = \lambda a_i$ for every $i = 1, 2, \ldots, n$ as desired. (A shorter proof, which bypasses the consideration of the roots of the quadratic is to note that since $f(t) \geq 0$ for all t, in particular $f(-\frac{C}{A}) \geq 0$. A direct computation of $f(-\frac{C}{A})$ gives the result. This proof is not as tricky as it appears at first sight. Since $f(t) = At^2 + 2Ct + B$ and $A > 0$, $-\frac{C}{A}$ is the point at which $f(t)$ is minimum, as we can see either by using calculus or simply by completing a square. To say that f is non-negative everywhere is equivalent to saying that its minimum is non-negative. A direct proof, based on squaring both the sides of (3) and regrouping the terms as sums of squares of the form $(a_ib_j - a_jb_i)^2$ is also possible as far as the inequality part is concerned. However, settling the equality part becomes a little messy.)

As an immediate application of the Cauchy-Schwarz inequality, we see that for any real numbers a, b and θ, $a\cos\theta + b\sin\theta \leq \sqrt{a^2 + b^2}$. We could, of course, get this by finding an angle α such that $\sin\alpha = \dfrac{a}{\sqrt{a^2 + b^2}}$ and $\cos\alpha =$

$\frac{b}{\sqrt{a^2+b^2}}$ and then rewriting the given expression as $\sqrt{a^2+b^2}\sin(\alpha+\theta)$. But the Cauchy-Schwarz inequality gives the result effortlessly. This is, of course, a relatively minor application of it. The most frequently needed application of the Cauchy-Schwarz inequality is probably the corollary that for any real numbers $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$,

$$\sqrt{(a_1+b_1)^2+\ldots+(a_n+b_n)^2} \leq \sqrt{a_1^2+\ldots+a_n^2}+\sqrt{b_1^2+\ldots+b_n^2} \qquad (4)$$

This inequality, which is proved directly by squaring both the sides and applying (3), is popularly called the **triangle inequality**. To justify the name suppose $n = 3$ and $P = (x_1, x_2, x_3), Q = (y_1, y_2, y_3)$ and $R = (z_1, z_2, z_3)$ are three points in space. Let $a_i = y_i - x_i$ and $b_i = z_i - y_i$ for $i = 1, 2, 3$. Then (4) says the same thing as that the length of the side PR of the triangle PQR is at most the sum of the lengths of the sides PQ and QR (equality holding for a degenerate triangle, where P, Q, R are collinear). The argument holds equally well in higher dimensions, i.e., for $n > 3$. In fact, in higher dimensions we need (3) to define the angle between two lines as we shall see in Chapter 21.

Comment No. 8:

Certain functions satisfy certain inequalities by their very definitions. From these basic inequalities more complicated inequalities can be proved. The most frequently occurring example of such a function is the absolute value function, which associates to each real number x, its **absolute value** (also called the **numerical value** or the **modulus**, or simply 'mod' for short) $|x|$ which is defined as the maximum of the two numbers x and $-x$. Equivalently, $|x| = \pm x$ depending upon whether x is positive or negative. (If $x = 0$ it does not matter which sign you choose!) Yet another formulation is that $|x| = \sqrt{x^2}$ because, by convention, the square root of a non-negative real number denotes the non-negative square root unless explicitly stated otherwise. If a, b are any two real numbers then $|a - b|$ equals $a - b$ or $b - a$ depending upon whether $a \geq b$ or $a \leq b$. In either case, it represents the distance between the points on the real line corresponding to the real numbers a and b and is often called the **difference** between a and b (or more precisely, the **absolute difference**). For any positive h, the inequality $|x - c| < h$ is equivalent to the double inequality $c-h < x < c+h$, which geometrically says that x lies in the interval $(c-h, c+h)$, which is centred at c and has length $2h$. This fact is used innumerably many times in calculus proofs (where the number h is often some δ or some ϵ etc.)

We skip the elementary properties (such as $|xy| = |x||y|$) and graphs of the absolute value function as they can be found in any standard textbook. However, one of the basic properties deserves to be mentioned. It says that for any two real numbers a and b, $|a + b| \leq |a| + |b|$ (with equality holding if and only if a, b are of the same sign). This is a special case of (4) and hence is called the **triangle inequality** for the absolute value. Translated in terms of differences, it says that for any three real numbers x, y and z, $|x - z| \leq |x - y| + |y - z|$. It is used frequently (and often without an explicit mention)

whenever we want to 'estimate' a difference of the form $|x-z|$, i.e., want to show that it is less than a certain value, say ϵ. Doing so directly may not be easy. So we 'break' the difference $|x-z|$ by inserting some point y of our convenience. (It is not necessary that y should lie between x and z.) If we can separately show that $|x-y| < \frac{\epsilon}{2}$ and $|y-z| < \frac{\epsilon}{2}$ (say) then the triangle inequality gives $|x-z| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. The manner in which the difference $|x-z|$ is broken depends, of course, on a particular situation. Sometimes the most obvious choice works. But sometimes it calls for a little ingenuity. For example, suppose that $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$. In proving that $\lim_{x\to c} f(x)+g(x) = L+M$, the natural breaking of the difference $|f(x)+g(x)-(L+M)|$ is as $|(f(x)-L)+(g(x)-M)|$. However, in proving that $\lim_{x\to c} f(x)g(x) = LM$, we need to break $|f(x)g(x)-LM|$ either as $|f(x)(g(x)-M)+(f(x)-L)M|$ or as $|(f(x)-L)g(x)+L(g(x)-M)|$ and then apply the triangle inequality along with the boundedness of $f(x)$ (or of $g(x)$ in the second approach). Later in this chapter we shall give the definition of the limit of a sequence in terms of absolute values.

In dealing with problems involving absolute values it is safer to resolve the expression into separate cases depending upon the sign of the expression whose absolute value is taken. Suppose, for example, that we want to solve the the equation $|x^2+4x+3|+2x+5=0$. (JEE 1988) Here the quadratic $x^2+4x+3=0$ has -3 and -1 as its roots and is negative between them. So $|x^2+4x+3| = x^2+4x+3$ for $x \le -3$ and for $x \ge -1$ and $|x^2+4x+3| = -(x^2+4x+3)$ for $-3 \le x \le -1$. We consider the two cases separately. First suppose $x \le -3$ or $x \ge -1$. Then the given equation reduces to $x^2+6x+8=0$ whose roots are -4 and -2. Of these, the root -2 does not satisfy either of the two restrictions (viz., $-2 \le -3$ or $-2 \ge -1$). Hence $x=-4$ is the only possibility. Next, suppose $-3 \le x \le -1$. Then the given equation becomes $x^2+2x-2=0$. Now the roots are $-1 \pm \sqrt{3}$. Of these only $-1-\sqrt{3}$ lies in the interval $[-3,-1]$. Putting all together, 4 and $-1-\sqrt{3}$ are the solutions of the given equation, viz., $|x^2+4x+3|+2x+5=0$.

Where more than one absolute value is involved, the breaking may have to be done at several points. Suppose, for example, that we want to identify the set of all real numbers x which satisfy the condition $|x-1|+|x-2|+|x-3| \ge 6$. (JEE 1983) Call this set as S and the expression $|x-1|+|x-2|+|x-3|$ as $f(x)$. For $x<1$, all the expressions $x-1, x-2$ and $x-3$ are negative and so $f(x) = 1-x+2-x+3-x = 6-3x$. For this to be ≥ 6, we must have $-3x \ge 0$ and hence $x \le 0$. So $S \cap (-\infty, 1) = (-\infty, 0]$. Similarly, for $x>3$, all the three terms $x-1, x-2$ and $x-3$ are positive and $f(x) = 3x-6$ which gives $S \cap (3,\infty) = [4,\infty)$. For $x \in [1,3]$, $x-1 \ge 0$ while $x-3 \le 0$ and so $f(x) = x-1+|x-2|+3-x = |x-2|+2$ and so $f(x) \ge 6$ is equivalent to $|x-2| \ge 4$ which is impossible for any $x \in [1,3]$. Hence $S \cap [1,3] = \emptyset$. Now that we have considered all the possibilities, we get $S = (-\infty, 0] \cup [4,\infty)$. (The problem could have been done more easily if we take the interpretation of the absolute value as the distance. The set S consists of those x's the sum of whose distances from the points 1,2 and 3 is 6 or more. For points in $[1,3]$, the distances from the end-points 1 and 3 add to a constant 2 (the length of the

interval) while the distance from the centre 2 of the interval can be at most 1 (half the length of the interval $[1,3]$). So the sum of the three distances is at most 3. Hence S contains no points from $[1,3]$. For points outside this interval, the distances from the end-points add up to twice the distance from the centre 2. So the condition $f(x) \geq 6$ is equivalent to $3|x-2| \geq 6$, i.e., to $|x-2| \geq 2$. As remarked above, the complementary condition, viz., $|x-2| < 2$ is equivalent to $0 < x < 4$. So S is the complement of the interval $(0,4)$. Although this solution is not significantly shorter, it is definitely more elegant and less prone to numerical mistakes.)

Using absolute values we can state the Cauchy-Schwarz inequality in a slightly stronger form than (3). Note the R.H.S. of (3) is unchanged if we replace each a_i with $-a_i$ (without changing the b_i's). The L.H.S. becomes $-\sum_{i=1}^{n} a_i b_i$. But (3) still holds. Since $|\sum_{i=1}^{n} a_i b_i|$ equals either $\sum_{i=1}^{n} a_i b_i$ or $-\sum_{i=1}^{n} a_i b_i$, we get

$$|a_1 b_1 + a_2 b_2 + \ldots + a_n b_n| \leq \sqrt{a_1^2 + a_2^2 + \ldots + a_n^2}\ \sqrt{b_1^2 + b_2^2 + \ldots + b_n^2} \qquad (5)$$

which is a more common form of the Cauchy-Schwarz inequality than (3). However, equality in (5) is not as strong as that in (3). So all we can say is that equality holds in (5) if and only if there is some λ such that either $a_i = \lambda b_i$ for all $i = 1, 2, \ldots, n$ or $b_i = \lambda a_i$ for all $i = 1, 2, \ldots, n$. (In case of (3) we could assert that λ was non-negative.)

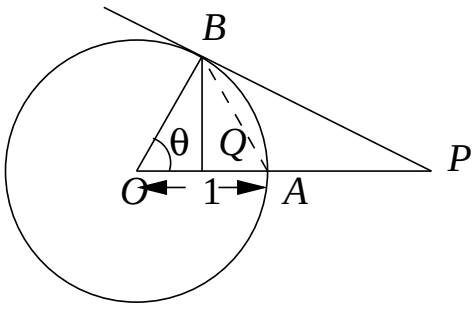
Comment No. 9:

Trigonometric functions also satisfy certain inequalities right from their definitions. The most well-known among them are that for any angle θ, $|\sin\theta| \leq 1$ and $|\cos\theta| \leq 1$ (and their consequences such as $|\sec\theta| \geq 1$). Not so immediate is the following inequality which compares the trigonometric functions of θ with θ itself. It says that for all $\theta \in [0, \frac{\pi}{2})$,

$$\sin\theta \ \leq\ \theta \ \leq\ \tan\theta \qquad (6)$$

with equality holding only for $\theta = 0$. (Actually, the first inequality holds for all $\theta \geq 0$ because $\frac{\pi}{2} > 1$ and so whenever $\theta \geq \frac{\pi}{2}$, we have $\sin\theta \leq 1 < \theta$.)

This inequality is basic in proving that $\lim_{\theta\to 0} \dfrac{\sin\theta}{\theta} = 1$, which, in turn, is pivotal in finding the derivative of the sine function and therefrom those of the other trigonometric functions. The standard proof of this inequality is geometric in nature. We draw a circle of radius 1 centred at O and take points A, B on it so that $\angle AOB = \theta$. Let the tangent at B meet the line OA at P. Let Q be the foot of the perpendicular from B to the line OA.



Now, $\sin\theta$ and θ are, by definition, the lengths of the segment BQ and of the arc BA respectively. The inequality $\sin\theta < \theta$ follows from the fact that the length of the segment BA is bigger than the length of BQ on one hand and less than the arc length BA on the other. The inequality $\theta < \tan\theta$ follows by comparing the areas of the sector BOA and of the triangle BOP. (The use of area in the proof of (6) is somewhat objectionable because the formula for the area of a sector requires results which are based on (6) such as the derivatives of the trigonometric functions. A proof which does not use areas and is based purely on showing that the arc length BA is less than the length of the segment BP is possible and may be found in *Plane Trigonometry − Part I* by S. L. Loney, page 263.)

It is also possible to identify the intervals on which the trigonometric functions are monotonically increasing/decreasing. In particular, on the interval $[0, \frac{\pi}{2}]$, the sine function is strictly increasing (i.e., for any two points x, y in $[0, \frac{\pi}{2}]$, we have $\sin x < \sin y$ whenever $x < y$) while the cosine function is strictly decreasing. The standard method for doing so is by looking at the sign of their derivatives. But in the present case, both the facts can be proved by simple geometric arguments. These facts along with the first part of (6) provide a simple solution to the following problem which is quite awkward to solve by other conventional methods.

For all θ in $[0, \frac{\pi}{2}]$, prove that $\cos(\sin\theta) \geq \sin(\cos\theta)$. (JEE 1981)

For any $\theta \in [0, \frac{\pi}{2})$, we have $0 \leq \sin\theta \leq \theta \leq \frac{\pi}{2}$. Now the cosine function is decreasing on the interval $[0, \frac{\pi}{2}]$. So we get

$$\cos(\sin\theta) \geq \cos\theta \tag{7}$$

But since $\cos\theta$ also lies in $[0, \frac{\pi}{2}]$, the same inequality gives,

$$\cos\theta \geq \sin(\cos\theta) \tag{8}$$

The result follows by combining (7) and (8). A direct comparison between $\cos(\sin\theta)$ and $\sin(\cos\theta)$ is not easy. But (7) and (8) compare them separately with $\cos\theta$ and the result falls out. The formal name for this property of inequalities is **transitivity**. More generally, when we want to prove that $x < y$ and a direct comparison seems difficult, we can insert a (finite) number of intermediate quantities, say, $a_1, a_2, \ldots, a_r$ such that $x < a_1, a_1 < a_2, a_2 < a_3, \ldots, a_{r-1} < a_r$ and $a_r < y$. Repeated application of transitivity then implies that $x < y$. The trick, of course, lies in choosing these intermediate quantities suitably. As an example, suppose m and n are positive integers with $m < n$ and we want to prove that $(1 + \frac{1}{m})^m < (1 + \frac{1}{n})^n$. Doing this directly is messy. But it is easy to show that for every positive integer k, $(1 + \frac{1}{k})^k < (1 + \frac{1}{k+1})^{k+1}$. (The proof consists of expanding both the sides by the binomial theorem and comparing them term-by-term.) Applying this fact with $k = m, m+1, \ldots$ and finally with $k = n - 1$ gives the desired result.

In essence what we have done here is to show that the function $f(n) = (1+\frac{1}{n})^n$ is a monotonically increasing function of the variable n by showing that $f(k) < f(k+1)$, i.e., by comparing the value of the function at a point with

that at the next point, instead of directly comparing $f(m)$ and $f(n)$. Here the variable n takes only positive integers as possible values. Suppose, instead, that we have a function $f(x)$ of a real variable x. In that case, there is no such thing as the 'next value' of x. However, given a and b with $a < b$ let us partition the interval $[a, b]$ by inserting some intermediate points, say, $x_1, x_2, \ldots, x_r$ such that $a < x_1, x_2 < \ldots < x_r < b$. If these points can be so chosen that $f(a) \leq f(x_1) \leq f(x_2) \leq \ldots \leq f(x_r) \leq f(b)$ then it would follow that $f(a) \leq f(b)$. Note that for any two successive intermediate points x_k and x_{k+1}, the sign of the difference $f(x_{k+1}) - f(x_k)$ is the same as the sign of the ratio $\frac{f(x_{k+1})-f(x_k)}{x_{k+1}-x_k}$, because the denominator is positive. If the distance between x_k and x_{k+1} is small, then the ratio $\frac{f(x_{k+1})-f(x_k)}{x_{k+1}-x_k}$ is approximately equal to $f'(x_k)$, i.e., to the derivative of f at x_k. This suggests that if the derivative of f is positive throughout the interval $[a, b]$, then f would be monotonically increasing on $[a, b]$. This indeed turns out to be the case, although a rigorous proof requires certain basic properties of real numbers. This gives a powerful tool to prove certain inequalities which cannot be handled by elementary arguments. For example, the inequality (6) above, which can be proved by elementary arguments, says that for $x \in (0, \frac{\pi}{2})$, x lies between $\sin x$ and $\tan x$, i.e., lies in the interval $[\sin x, \tan x]$. But it does not answer on which side of the mid-point it lies, i.e., whether x is closer to $\sin x$ or to $\tan x$. Put differently, (6) does not answer which of the two numbers $\sin x + \tan x$ and $2x$ is greater. There is, in fact, no elementary way to answer it. But using derivatives we can get the answer as we shall show in Chapters 13 and 16. In the present chapter we stick to only those inequalities which can be proved by elementary arguments.

Comment No. 10:

Yet another (although not so frequently occurring) function which satisfies certain inequalities right from its definition is the integral part function (or the floor function) defined in Chapter 4. The difference $x - [x]$ is called the **fractional part** of x and is often denoted by $\{x\}$. (However, this notation is not as standard as the one for the integral part. Moreover, $\{x\}$ is also a standard notation for the singleton set whose only element is x.) Then clearly $0 \leq \{x\} < 1$ or, in other words, $\{x\}$ always lies in the semi-open interval $[0, 1)$. Note that even if x is negative, its fractional part cannot be negative. (The fractional part of -2.17 is neither 0.17 nor -0.17. To find it we must write -2.17 as $-3 + 0.83$, giving $[-2.17] = -3$ and $\{-2.17\} = 0.83$. Conceptually, $\{x\}$, is the 'residue' left after the largest possible multiple of 1 is subtracted from x. This is analogous to the division algorithm for integers. Indeed, if m, n are integers with $n > 0$ and we write m as $qn + r$ with q, r as integers and $r \in [0, n-1)$, then it is easy to show that $\frac{r}{n} = \{\frac{m}{n}\}$.) If a real number x is written as $m + y$, where m is an integer and $y \in [0, 1)$, then we must have $m = [x]$ and $y = \{x\}$ (since $0 \leq y < 1$ implies $m \leq x < m + 1$). It is also clear that two real numbers x and y have equal fractional parts if and only if $x - y$ is an integer and that they have equal integral parts if and only if there is no integer in the semi-open interval $(x, y]$ (in case $x \leq y$) or in the semi-open interval $(y, x]$ (in case $y \leq x$). These elementary

observations often lead to solutions to problems involving the integral and the fractional parts.

Suppose, for example, that we want to find all real numbers x satisfying the equation $4\{x\} = x + [x]$. (JEE 1994) Writing x as $[x] + \{x\}$, this reduces to $3\{x\} = 2[x]$. Now, the R.H.S. is an even integer while the L.H.S. is a real number lying in the semi-open interval $[0, 3)$. Since 0 and 2 are the only even integers in this interval, we must have $[x] = 0$ or 1 and correspondingly, $\{x\} = 0$ or $\frac{2}{3}$. Thus we get $x = 0$ and $x = 1 + \frac{2}{3} = \frac{5}{3}$ as the two solutions of the given equation.

As an example of a problem where we have to show that two given numbers have equal integral parts we have the following. (JEE 2000)

For every positive integer n prove that

$$\sqrt{(4n+1)} \ < \ \sqrt{n} + \sqrt{n+1} \ < \ \sqrt{(4n+2)} \tag{9}$$

Using this or otherwise, prove that

$$[\sqrt{n} + \sqrt{(n+1)}] = [\sqrt{(4n+1)}] \tag{10}$$

Since all the three terms in (9) are positive, proving it is equivalent to proving that the squares of the terms are in the same order, i.e., $4n + 1 < 2n + 1 + 2\sqrt{n}\sqrt{(n+1)} < 4n + 2$, which, in turn, is equivalent to proving that $n < \sqrt{n}\sqrt{(n+1)} < n + 1$ which follows by squaring once again. Now, proving (10) is equivalent to showing that there is no integer in the semi-open interval $(\sqrt{(4n+1)}, \sqrt{n} + \sqrt{(n+1)}]$. By (9), this semi-open interval is contained in the semi-open interval $(\sqrt{(4n+1)}, \sqrt{(4n+2)}]$. So we shall be through if we can show that there is no integer in the latter. Suppose, on the contrary, that $(\sqrt{(4n+1)}, \sqrt{(4n+2)}]$ contains an integer, say m. Then upon squaring we get that $m^2 \in (4n+1, 4n+2]$. But the only integer in $(4n+1, 4n+2]$ is $4n+2$. So we must have $m^2 = 4n+2$, whence m^2 and hence m must be even, say $m = 2k$. But that would give $4k^2 = 4n + 2$, i.e., $2k^2 = 2n + 1$, which is a contradiction since the L.H.S. is even and the R.H.S. is odd. Hence (10) holds.

Comment No. 11:

Inequalities (or more precisely the order relation) are an important feature of the real numbers and have many uses, both theoretical and practical. Among the practical uses, inequalities provide us with estimates of quantities which we cannot measure directly. Suppose, for example, we want the numerical value of $\sin 34^\circ$. We could, of course, draw on a piece of paper a right-angled triangle with one angle equal to 34 degrees. But to measure its sides we would need highly sophisticated instruments to get even a modest degree of accuracy, say upto three places of decimals. But with inequalities the job can be done easily. Let θ equal 34° in radians. Then $\theta = \frac{34\pi}{180} \approx 0.5934$. In (6) we proved $\sin\theta < \theta$. With a little more work (using derivatives) it can be shown that $\sin\theta > \theta - \frac{\theta^3}{6}$. So $\sin 34^\circ$ lies in the interval $(0.5934) - \frac{(0.5934)^3}{6}, 0.5934) \approx (0.5586, 0.5934)$. With still further work (which is not at all difficult) we can show that $\sin\theta < \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120}$

and this is good enough to estimate $\sin 34^\circ$ upto four places of decimals. In fact we can go further and get as high a degree of accuracy as we want.

Among the theoretical uses of inequalities, a peculiar use deserves a mention as it sometimes confuses students. In pre-calculus mathematics, whenever we want to prove that two real numbers (or algebraic expressions representing real numbers), say A and B are equal, the standard method is to convert one of them into the other using a series of algebraic manipulations along with any of the equalities proved earlier. This is how we prove dozens of formulas in algebra and trigonometry. In calculus, too, this method is not outlawed. But many times a different approach is followed. That is, we show separately that $A \leq B$ and also $B \leq A$. Then we must have $A = B$. Or, sometimes, we show, by contradiction, that neither of the possibilities $A < B$ and $B < A$ can hold and thereby conclude that $A = B$. (This is how the uniqueness of limits is generally proved.) What gives sanctity to this method is what is called the **law of trichotomy**[3] which says that given two real numbers A and B, exactly one of the three possibilities holds : either $A < B$ or $B < A$ or $A = B$.

For complex numbers there is no such law because given two complex numbers, say z_1 and z_2, it is meaningless to say that one of them is less than the other. If at all you see an expression like $z_1 < z_2$ it presupposes that the complex numbers z_1, z_2 are in fact real, i.e., that their imaginary parts are 0. We can artificially define $z_1 \leq z_2$ to mean, for example, that $|z_1| \leq |z_2|$. But such a relation will not satisfy the same properties which hold in the case of real numbers. For example, we have both $1 \leq i$ and $i \leq 1$ and still $1 \neq i$. Other attempts to define an order relation for complex numbers also encounter some failure or the other. This is bound to be so because one of the least requirements of a good order relation is that the square of any non-zero element be positive and hence a sum of two squares should also be positive. But in the complex number system we have $1^2 + i^2 = 0$ and surely 0 cannot be positive! So, for complex numbers there are no inequalities!

Of course, there are many real numbers associated with complex numbers, such as their real parts, imaginary parts, absolute values, arguments and so on and there are plenty of inequalities about these associated real numbers. Take, for example the absolute value $|z|$ of a complex number z. Evidently, $|z| \geq 0$ for all z and equality holds if and only if $z = 0$. We also have the triangle inequality, viz., $|z_1 + z_2| \leq |z_1| + |z_2|$ for all complex numbers z_1, z_2 with equality holding if and only if one of them is a non-negative real multiple of the other. This can be derived either from (4) or from the complex analogue of the Cauchy-Schwarz inequality given in Exercise (6.8). Regarding the real and imaginary parts of a complex number we have the following inequality which follow by a simple geometric argument, viz., that in a right-angled triangle the hypotenuse is bigger than each of the other two sides individually but less than their sum.

$$\text{Re } z \leq |z|, \ \text{Im } z \leq |z| \text{ and } |z| \leq |\text{Re } z| + |\text{Im } z| \tag{11}$$

[3]The word literally means three branches, corresponding to the three possibilities listed. Sometimes the third possibility is merged with the other two. Then we have only two possibilities, viz., either $A \leq B$ or $B \leq A$. In that case the law is called the law of dichotomy.

As a consequence it follows that for two complex numbers z_1 and z_2, their difference $|z_1 - z_2|$ (which is a real number) is small if and only if the difference in their real parts and the difference in their imaginary parts are both small. This is very convenient in theoretical applications, for example, in proving that the continuity of a complex-valued function is equivalent to the continuity of its real part and of its imaginary part. But we shall not go into it.

As regards the argument of a (non-zero) complex number, no inequality holds, because the same complex number has infinitely many arguments, any two of which differ by an integral multiple of 2π. In practice we choose one of these arguments, call it the **principal argument** and denote it by Arg (as opposed to 'arg' which denotes any of the infinitely many arguments). There is no unanimous choice. But two conventions are common. The more common convention is to let Arg z lie in the semi-open interval $(-\pi, \pi]$. A less common convention is to let it lie in the semi-open interval $[0, 2\pi)$. We shall follow the first convention. Note that for multivalued arguments we have arg $(z_1 z_2)$ = arg (z_1) + arg (z_2) as an equality of sets in the sense that every number represented by either side is also represented by the other side. However, with principal arguments we do not always have Arg $(z_1 z_2)$ = Arg (z_1) + Arg (z_2) as an equality of numbers. For example, take $z_1 = z_2 = -\frac{1}{2} + \frac{i\sqrt{3}}{2}$. Then Arg (z_1) = Arg$(z_2) = \frac{2\pi}{3}$. But $z_1 z_2$ equals $-\frac{1}{2} - \frac{i\sqrt{3}}{2}$ and has $-\frac{2\pi}{3}$ as its principal argument. Nevertheless, since Arg $(z_1 z_2)$ differs from Arg (z_1) + Arg (z_2) by a whole multiple of 2π and all the three numbers lie in the interval $(-\pi, \pi]$, this difference has to be either 0 or $\pm 2\pi$.

If we have two complex numbers written in terms of their real and imaginary parts as, say, $z = x + iy$ and $w = u + iv$, then it follows from (11) that $|z - w|^2 \leq (x - u)^2 + (y - v)^2$. If they are represented in terms of their absolute values and Arguments as, say, $z = r_1 e^{i\theta}$ and $w = r_2 e^{i\phi}$, then analogously we should expect that $|z - w|^2 \leq (r_1 - r_2)^2 + (\theta - \phi)^2$. But obviously this is not true in general. This is to be expected because $|\theta - \phi|$ is at most 2π. By taking r_1 and r_2 equal and large and $\theta - \phi = \pi$ we get two complex numbers which are negatives of each other and for which $|z_1 - z_2|^2 = 4r_1^2$ can be made as large as we want. The following problem shows that the inequality is true if we restrict the size of r_1 and r_2.

If $|z| \leq 1$ and $|w| \leq 1$, show that (JEE 1995)

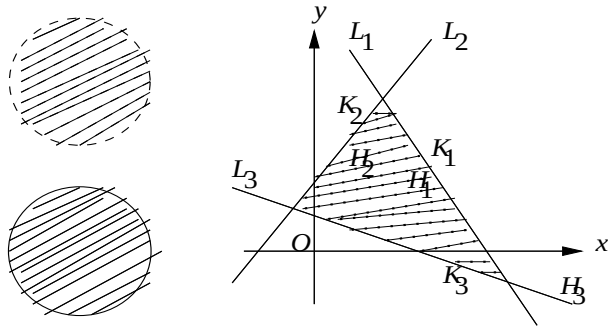
$$|z - w|^2 \leq (|z| - |w|)^2 + (\text{Arg } z - \text{Arg } w)^2 \tag{12}$$

Let the principal arguments of z and w be, respectively, θ and ϕ. As (12) is symmetric in z and w, we may assume, without loss of generality, that $\theta \geq \phi$. Let $\alpha = \theta - \phi$. Then $\alpha \geq 0$. Let r_1, r_2 denote the absolute values of z and w respectively. Then $z = r_1 e^{i\theta}$ and $w = r_2 e^{i\phi}$. So $|z - w|^2 = |r_1 e^{i\theta} - r_2 e^{i\phi}|^2 = |r_1 e^{i(\alpha+\phi)} - r_2 e^{i\phi}|^2 = |r_1 e^{i\alpha} e^{i\phi} - r_2 e^{i\phi}|^2 = |r_1 e^{i\alpha} - r_2|^2 |e^{i\phi}|^2 = |r_1 e^{i\alpha} - r_2|^2$ since $|e^{i\phi}| = 1$. So (12) is reduced to proving that $|r_1 e^{i\alpha} - r_2|^2 \leq (r_1 - r_2)^2 + \alpha^2$, it being given that $r_1 \leq 1$ and $r_2 \leq 1$. This can be done by writing $r_1 e^{i\alpha}$ as $r_1 \cos\alpha + i r_1 \sin\alpha$. Then a direct calculation gives $|r_1 e^{i\alpha} - r_2|^2 = (r_1 \cos\alpha - r_2)^2 + (r_1 \sin\alpha)^2 = r_1^2 + r_2^2 - 2r_1 r_2 \cos\alpha = (r_1 - r_2)^2 + 2r_1 r_2(1 - \cos\alpha) =$

$(r_1 - r_2)^2 + 4r_1r_2\sin^2\frac{\alpha}{2}$. As we are given that $r_1 \leq 1$ and $r_2 \leq 1$ we have $r_1r_2 \leq 1$ and the problem is now reduced to showing that $4\sin^2\frac{\alpha}{2} \leq \alpha^2$. But this follows from the first part of (6) (applied with $\theta = \frac{\alpha}{2}$). Thus (12) holds. (As remarked after (6), the first inequality in it is true for all $\theta \geq 0$. Actually, we do not need that $\alpha \geq 0$. If $\alpha < 0$, we can apply (6) with $\theta = -\frac{\alpha}{2}$ and get the result after squaring.)

Comment No. 12:

Complex numbers correspond naturally to the points in a plane just as the real numbers correspond to points on a line. As there is no order relation for complex numbers, the same holds true for points in a plane. However, their coordinates are real numbers and therefore they and any real-valued expressions obtained from them can be compared with each other. Certain subsets of the plane can be described very succinctly in terms of inequalities satisfied by such expressions. For example, the set of points (x, y) for which $x^2 + y^2 < 1$ is precisely the interior of the unit circle while the set $\{(x, y) : x^2 + y^2 > 1\}$ is its exterior i.e., the set of points outside the unit circle. Thus we see that the expression $x^2 + y^2 - 1$ divides the plane into three parts : the interior of the circle where it is negative, the exterior where it is positive and the circle itself where it is 0. (The interior is also called the **open unit disc**. Sometimes we combine the circle with it to get the set $\{(x, y) : x^2 + y^2 \leq 1\}$, called the **closed unit disc**, shown in Figure (a) below. More generally, we could consider any circle and the corresponding open and closed discs.)



(a) Open and closed discs (b) Triangle

Similarly, a linear expression of the form $ax + by + c$, where a, b, c are constants and not both a and b are 0, divides the plane into three parts. The set $\{(x, y) : ax + by + c = 0\}$ represents a straight line, say L. For all points on one side of L, the expression $ax + by + c$ is positive while for all points on the other side it is negative. The sets $\{(x, y) : ax + by + c > 0\}$ and $\{(x, y) : ax + by + c < 0\}$ are called the **open half-planes** determined by L. The sets $\{(x, y) : ax + by + c \geq 0\}$ and $\{(x, y) : ax + by + c \leq 0\}$ are called the **closed half-planes** determined by L. Note that unlike in the case of a circle,

both the half-planes are unbounded and so we cannot tell just by inspection which is which. The easiest way to distinguish between them is to fix some point $P = (h, k)$ not on L. (If L does not pass through the origin, i.e., if $c \neq 0$, then $(0, 0)$ is the easiest choice.) Then all points (x, y) for which $ax + by + c$ has the same sign as $ah + bk + c$ lie on the same side of L as the point P does. The others lie on the opposite side.

A disc is bounded by a single curve, viz., a circle. A triangle, on the other hand, is bounded by three straight lines, say L_1, L_2 and L_3. Let H_1 and K_1 be the closed half-planes determined by the line L_1. Then the entire triangle will lie either in H_1 or in K_1. Let us suppose, without loss of generality, that it lies in H_1. Similarly, let H_2, H_3 be the half-planes corresponding to L_2, L_3 respectively which contain the triangle. Then the intersection of these three closed half-planes is precisely the triangle with sides lying along the lines L_1, L_2 and L_3. (See Figure (b) above. If instead of the closed half-planes we take the open half-planes, then we get the interior of the triangle, i.e., the triangle without the boundary points. Generally when we consider a triangle we mean a closed triangle, i.e., we include the boundary points. If we don't want them we use the term 'open triangle'.)

In terms of inequalities, each H_i equals either the set $\{(x, y) : a_i x + b_i y + c_i \leq 0\}$ or the set $\{a_i x + b_i y + c_i \geq 0\}$. Thus every point (x, y) of the triangle satisfies three linear inequalities (i.e., inequalities where the functions involved are linear functions of x and y). And conversely any point of the plane which satisfies these three inequalities simultaneously lies in the triangle. Thus we see that a triangle in a plane can be specified by three linear inequalities. Similarly a convex quadrilateral is specified by four inequalities and more generally a convex n-gon by a system of n linear inequalities. This observation can be generalised to higher dimensions. There a single linear inequality determines what is called a **half space** (a generalisation of the concept of a half plane) and the intersection of a finite number of such half-spaces is called a **convex polytope**. This is the starting point of an extremely useful branch of mathematics called **linear programming**. We shall not go into it, except for a brief mention in Chapter 14. (See also Exercise (24.92).) However, here is a JEE problem (JEE 1992) where we have to express a triangle in terms of three linear inequalities.

Determine all values of α for which the point (α, α^2) lies inside the triangle formed by the lines

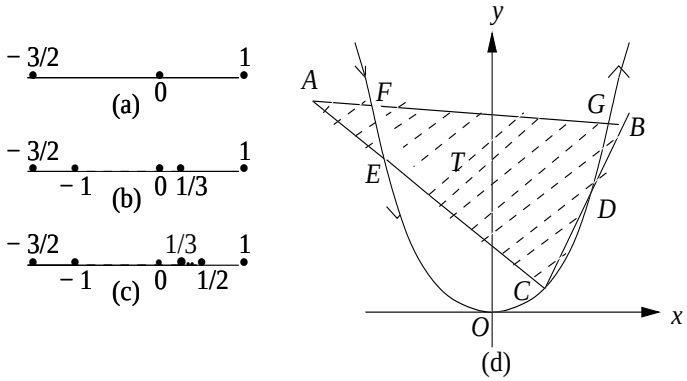
$$\begin{aligned} 2x + 3y - 1 &= 0 \\ x + 2y - 3 &= 0 \\ 5x - 6y - 1 &= 0 \end{aligned}$$

Call the lines as L_1, L_2 and L_3 respectively and let T be the triangle bounded by them. Let A, B, C be the points of intersection of L_1 and L_2, of L_2 and L_3 and of L_3 and L_1 respectively. By direct calculation, $A = (-7, 5), B = (\frac{5}{4}, \frac{7}{8})$ and $C = (\frac{1}{3}, \frac{1}{9})$. These are the vertices of T. Now T lies on the same side of L_3 as A does. Since $5 \times (-7) - 6 \times 5 - 1 = -66 < 0$ we see that for all $(x, y) \in T$, $5x - 6y \leq 1$. By a similar reasoning, for all $(x, y) \in T$, $2x + 3y \geq 1$

and $x + 2y \leq 3$. Thus $T = \{(x, y) : 5x - 6y \leq 1, 2x + 3y \geq 1 \text{ and } x + 2y \leq 3\}$. In order for (α, α^2) to lie in T it is necessary and sufficient that α satisfies the following three conditions:

$$5\alpha - 6\alpha^2 \leq 1,\ 2\alpha + 3\alpha^2 \geq 1 \ \text{ and } \ \alpha + 2\alpha^2 \leq 3 \tag{13}$$

We rewrite each of these conditions in the form of a quadratic expression in α being ≥ 0. Thus the first condition means $6\alpha^2 - 5\alpha + 1 \geq 0$. The quadratic $6\alpha^2 - 5\alpha + 1$ has leading coefficients positive and $\frac{1}{3}$ and $\frac{1}{2}$ as roots. So it is non-negative for $\alpha \leq \frac{1}{3}$ and for $\alpha \geq \frac{1}{2}$. Similarly rewriting the second condition as $3\alpha^2 + 2\alpha - 1 \geq 0$ we see that it is satisfied for all $\alpha \leq -1$ and for all $\alpha \geq \frac{1}{3}$. Finally writing the third condition as $2\alpha^2 + \alpha - 3 \leq 0$, we see that it is satisfied when α lies between the two roots $-\frac{3}{2}$ and 1, i.e., for all $\alpha \in [-\frac{3}{2}, 1]$. To finish the solution we have to take the intersection of these three sets. The safest way to do so is to draw a series of diagrams. In the first diagram we show one of these subsets by shading out its complement. Then we go on imposing the other conditions one by one and as we do so, we shrink the intersection by shading out more and more parts of it. For example, in the present problem, the third condition requires α to lie in the closed interval $[-\frac{3}{2}, 1]$ shown in the Figure (a) below. Now the second condition requires that either $\alpha \leq -1$ or $\alpha \geq \frac{1}{3}$. The first possibility implies that α must lie in the interval $[-\frac{3}{2}, -1]$. Similarly the second possibility forces α to lie in the interval $[\frac{1}{3}, 1]$ and so we shade out the portion between -1 and $\frac{1}{3}$. This gives us the diagram in Figure (b). Finally, the first condition requires either $\alpha \leq \frac{1}{3}$ or $\alpha \geq \frac{1}{2}$. So we shade out the portion from $\frac{1}{3}$ to $\frac{1}{2}$. Then the desired intersection is the set shown in Figure (c). It consists of the intervals $[-\frac{3}{2}, -1]$ and $[\frac{1}{2}, 1]$ and the isolated point $\frac{1}{3}$. So these are the values of α for which the point (α, α^2) lies in the triangle T. (After gaining a little practice, it is hardly necessary to draw a series of diagrams. In the first diagram itself you can show the parts to be removed. After still more practice you can even do without *any* diagram. But in doing so, there is a danger that you may miss out something.)

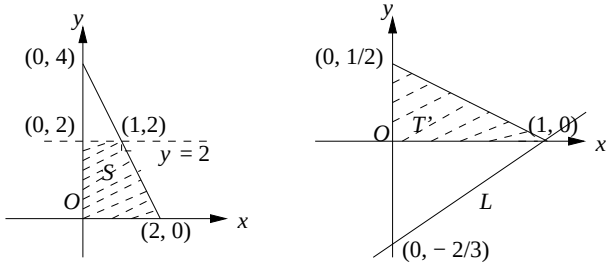


The problem can also be done geometrically by observing that points

of the form (α, α^2) constitute the parabola $y = x^2$. In the present problem the vertex $C = (\frac{1}{3}, \frac{1}{9})$ of the triangle T lies on this parabola. The other two vertices $A = (-7, 5)$ and $B = (\frac{5}{4}, \frac{7}{8})$ lie below it (since $5 < (-7)^2$ and $\frac{7}{8} < (\frac{5}{4})^2$). The parabola cuts the line CB ($= L_3$ with equation $5x - 6y = 1$) at the point $(\frac{1}{2}, \frac{1}{4})$. Call this as D. Similarly the parabola meets the line CA (with equation $2x + 3y = 1$) again at the point $(-1, 1)$ which we call as E. Also, the parabola meets the third side $x + 2y = 3$ at the points $F = (-\frac{3}{2}, \frac{9}{4})$ and $G = (1, 1)$ as shown in Figure (d) above which is not drawn to scale as that would push the point A too far to the left. As we travel along the parabola $y = x^2$ in the direction of increasing x we enter the triangle T at the point F, leave it at E, enter it momentarily at C again, then enter it again at D and finally leave it at the point G. So the portions of the parabola that lie in T are those from F to E and from D to G and the isolated point C. These correspond, respectively, to $-\frac{3}{2} \leq \alpha \leq -1$, $\frac{1}{2} \leq \alpha \leq 1$ and $\alpha = \frac{1}{3}$ which is the same answer as before. The geometric solution is definitely more vivid. But it works because the curve is a very familiar one, viz., a parabola. Also to see it clearly we need a reasonably accurate diagram. The earlier method, on the other hand, works for any curve and does not require diagrams except at the end (to take the intersection of various subsets of the real line).

Comment No. 13:

In the problem above the triangle T was the solution set of a system of three linear inequalities in two variables. We could identify T by finding its vertices for which we had to solve two equations at a time obtained by converting the corresponding inequalities to equalities. In general if we have a system of k linear inequalities in two variables x and y, then the solution set, say S, will be a convex polygon. But not every pair of inequalities will necessarily give a vertex of S. In fact the point of intersection of the corresponding straight lines need not even be a point of S as it need not satisfy all the other inequalities. Suppose, for example, that S is the solution set, shown in Figure (a) below, of the four inequality constraints $x \geq 0, y \geq 0, y \leq 2$ and $2x + y \leq 4$.



(a) (b)

Here S is a quadrilateral (in fact a trapezium) with vertices at $(0, 0), (2, 0), (1, 2)$ and $(0, 2)$. The equalities $y = 0$ and $y = 2$ have no common solution and hence do not determine a vertex of S. The equalities $x = 0$ and $2x + y = 4$ determine

the point $(0, 4)$ which is not in S. The equalities obtained from the other four pairs of inequalities do give the four vertices of S. The point to keep in mind is that even in a plane where drawing accurate diagrams is possible, it is not so easy to identify the solution set of a system of linear inequalities. Naturally, the situation is far more complex in higher dimensional spaces.

Sometimes linear equalities are clubbed together with linear inequalities. Suppose for example that we have a linear equality $a_1x_1+a_2x_2+\ldots+a_nx_n = c$, where $x_1, x_2, \ldots, x_n$ are the coordinates of a point in $\mathbb{R}^n$, the n-dimensional space and $a_1, a_2, \ldots, a_n$ and c are some constants with at least one $a_i \neq 0$. One way to handle such an equality is to replace it by two inequalities, viz., $a_1x_1 + a_2x_2 + \ldots + a_nx_n \leq c$ and $a_1x_1 + a_2x_2 + \ldots + a_nx_n \geq c$. But there is a better way out. Since we know that one of the a_i's is non-zero we can write the corresponding x_i in terms of the other variables and substitute this value in the remaining equalities or inequalities. We illustrate this technique in the following problem. (JEE 1980)

Find the solution set of the system

$$x + 2y + z = 1 \tag{14}$$

$$2x - 3y - w = 2 \tag{15}$$

$$x \geq 0, y \geq 0, z \geq 0, w \geq 0 \tag{16}$$

Ostensibly there are four variables here and so the solution set, say S, is a subset of $\mathbb{R}^4$. So identifying it by visualisation is impossible. However we can rewrite (14) as $z = 1-x-2y$ and so the inequality requirement $z \geq 0$ reduces to $1-x-2y \geq 0$, or equivalently to, $x+2y \leq 1$. Similarly because of (15), condition $w \geq 0$ becomes $2x - 3y \geq 2$. So the problem is equivalent to identifying the solution set, say T, of the four linear inequalities, viz., $x \geq 0, y \geq 0, x + 2y \leq 1$ and $2x-3y \geq 2$. To identify it, let us first drop the last inequality and consider the solution set, say T' of only the first three linear inequalities. It is a triangle with vertices at $(0, 0), (1, 0)$ and $(0, \frac{1}{2})$ as shown in Figure (b) above. Now T will be the set of those points of T' which also satisfy the fourth inequality, viz., $2x - 3y \geq 2$. The corresponding equality $2x - 3y = 2$ represents the line, say L, passing through the points $(0, -\frac{2}{3})$ and $(1, 0)$, the latter being a vertex of T'. The points of the plane which satisfy the inequality $2x-3y \geq 2$ are 'below' L. So no point of T' other than the vertex set $(1, 0)$ satisfies the inequality $2x-3y \geq 2$. In other words, T, the solution set of the system of all four inequalities is the singleton set $\{(1, 0)\}$. It is a degenerate polygon. For $x = 1, y = 0$ the values of z and w can be obtained from (14) and (15) respectively. They are both 0. So the solution set S of the given system (14) to (16) consists of just one point, viz., the point $(1, 0, 0, 0)$ in $\mathbb{R}^4$.

There is a much slicker way to do the problem. Multiplying (14) by 2 and then subtracting (15) from it gives $7y + 2z + w = 0$. By (16), y, z, w are all non-negative. So all the terms in the sum $7y + 2z + w$ are non-negative. So its vanishing forces each one of y, z and w to vanish. The value of x can now be found from either (14) or (15). It comes out to be 1. Hence $x = 1, y = z = w = 0$ is the only solution of the given system! Obviously, this slicker solution worked

only because the coefficients of the terms in (14) and (15) were such that by a simple manipulation we could get an equality in which a sum of non-negative terms was equal to 0. Such a manipulation is not possible every time and so the first method above is the standard way of doing the problem.

Comment No. 14:

We often want to prove inequalities of the form $a_n \leq b_n$, where n is a positive integer. (One of the sequences $\{a_n\}$ and $\{b_n\}$ may be a constant sometimes). Sometimes this can be done by a direct argument using some well-known inequality. For example, using the A.M.−G.M. inequality, we see that for every integer $n > 1$, $(n!)^{1/n} < \frac{1+2+\ldots+n}{n} = \frac{n+1}{2}$ and hence that $n! < \frac{(n+1)^n}{2^n}$. (Here we are using the fact that if a, b are positive real numbers with $a < b$ then $a^n < b^n$. More generally we can use any monotonically increasing or decreasing function to reduce an inequality to another. The fact that the logarithm function is monotonically increasing often helps in reducing inequalities where powers are involved to simpler inequalities where only multiples are involved.)

When a direct argument does not work, mathematical induction is the natural method to try. Normally when a statement about a sequence $\{a_n\}$ is proved by induction, in the inductive step we usually work with the difference $a_{n+1} - a_n$ (or something similar to it such as $a_{n+1} - 5a_n$). The inductive hypothesis applied to a_n coupled with the work we do on the difference $a_{n+1} - a_n$ often gives the truth of the result for a_{n+1}. This is how we prove, for example, many identities such as $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$ or the divisibility of a_n by some fixed integer (as we saw in Chapter 4). So, if we can show that $a_{n+1} - a_n \leq b_{n+1} - b_n$ for every positive integer n, then $a_n \leq b_n$ trivially implies $a_{n+1} = (a_{n+1} - a_n) + a_n \leq (b_{n+1} - b_n) + b_n = b_{n+1}$. The desired inequality then follows by induction on n (provided, of course, that we prove $a_1 \leq b_1$ separately). This application of induction is said to be of the additive type.

The trouble with this approach is that many times it is not easy to prove that $a_{n+1} - a_n \leq b_{n+1} - b_n$ and sometimes it may even be false even though the inequality $a_n \leq b_n$ itself may be true for all n as we see by taking $a_n = -\frac{1}{n}$ and $b_n = \frac{1}{n}$. As a less trivial example, suppose $a_n = (1 + \frac{1}{n})^n$ and $b_n = 3$. Expanding a_n by the binomial theorem, it can be shown (see Exercises) that $a_n \leq 3$ for all n. But as noted earlier, $a_n < a_{n+1}$ and so $b_n - a_n > b_{n+1} - a_{n+1}$. Hence an inductive proof is not likely to work.

When both a_n and b_n are positive and an inductive proof fails in an additive form sometimes it may work in a multiplicative form. That is, instead of showing that $b_n - a_n \leq b_{n+1} - a_{n+1}$, we show that $\frac{b_n}{a_n} \leq \frac{b_{n+1}}{a_{n+1}}$ or equivalently, $\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$. In that case $a_n \leq b_n$ would imply that $b_{n+1} = \frac{b_{n+1}}{b_n} b_n \geq \frac{a_{n+1}}{a_n} b_n \geq \frac{a_{n+1}}{a_n} a_n = a_{n+1}$. In effect, we are applying additive induction to prove $\log a_n \leq \log b_n$. So the technique is not radically new. But it is a little easier to think of than to think of taking the logarithms. As an illustration, we

prove (JEE 1987) that for every positive integer n,

$$\frac{(2n)!}{2^{2n}(n!)^2} \leq \frac{1}{(3n+1)^{1/2}} \tag{17}$$

Call the L.H.S. and the R.H.S. as a_n and b_n respectively. Both the differences $a_{n+1} - a_n$ and $b_{n+1} - b_n$ are clumsy to handle and so additive induction does not look promising. However, because of various cancellations, the ratio $\frac{a_{n+1}}{a_n}$ simplifies to $\frac{(2n+1)(2n+2)}{4(n+1)^2}$ and further to $\frac{2n+1}{2(n+1)}$ and so multiplicative induction seems like a workable proposition. (17) is trivially true for $n = 1$ as both the sides equal $\frac{1}{2}$. For the inductive step we need to prove that $\frac{b_n}{a_n} \leq \frac{b_{n+1}}{a_{n+1}}$ or equivalently, $\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$. As we have already shown that $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2}$, this amounts to proving that $\frac{2n+1}{2n+2} \leq \frac{(3n+1)^{1/2}}{(3n+4)^{1/2}}$, or, squaring both sides, to proving that $\frac{(2n+1)^2}{(2n+2)^2} \leq \frac{3n+1}{3n+4}$. As the denominators of both the sides are positive, this further reduces to proving that $(2n+1)^2(3n+4) \leq (2n+2)^2(3n+1)$. But this is true by direct expansion as the two sides are, $12n^3 + 28n^2 + 19n + 4$ and $12n^3 + 28n^2 + 20n + 4$ respectively. Thus, by multiplicative induction, (17) holds for all n.

Note carefully, the relationship between the inequality $\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$ and the inequality $19n \leq 20n$. The latter is stronger than the former in the sense that whenever the latter holds, so does the former. That is what we showed through a series of intermediate inequalities in which at every step we argued that in order to prove a certain inequality it would suffice to prove the next inequality. In other words, at every step, the next inequality is *sufficient* to prove the earlier one. In the problem just solved, all the steps were reversible and so the next inequality at every stage was both necessary and sufficient to prove the earlier one. But this may not always be possible. Suppose, for example, that we want to prove that $\sin\theta + \cos\theta \leq \frac{3}{2}$. Surely, this would follow if we can separately show that $\sin\theta \leq \frac{3}{4}$ and $\cos\theta \leq \frac{3}{4}$. But neither of these inequalities is true. In fact, there is no way to write $\frac{3}{2}$ as a sum $a + b$ in such a way that $\sin\theta \leq a$ and $\cos\theta \leq b$ for all θ. So this approach fails. Still the original inequality is true and can be proved by observing that $\sqrt{2} < \frac{3}{2}$, because we have already proved that as a result of the Cauchy-Schwarz inequality, $\sin\theta + \cos\theta \leq \sqrt{2}$.

Comment No. 15:

Another interesting point comes up sometimes when we try to prove certain inequalities for terms of a sequence $\{a_n\}$, especially in the case of a sequence which is recursively defined, i.e., whose general term a_n is expressed not directly as a function of n but by a recurrence relation in terms of the previous term (or sometimes several such terms) of the sequence. We already came across such

sequences in Chapter 4, Comment No. 12. Naturally, this recurrence relation has to be used either directly or indirectly in any inductive proof involving terms of that sequence. Sometimes in doing so it pays to prove more than what is asked. We illustrate this strange situation in the following problem. (JEE 2001)

Let a, b, c be positive real numbers such that $b^2 - 4ac > 0$ and let $\alpha_1 = c$. Prove by induction that

$$\alpha_{n+1} = \frac{a\alpha_n^2}{b^2 - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_n)} \tag{18}$$

is well-defined and $\alpha_{n+1} < \frac{\alpha_n}{2}$ for all $n = 1, 2, \ldots$. (Here 'well-defined' means that the denominator in the expression for α_{n+1} is non-zero.)

For $n = 1$, the denominator of α_2 is $b^2 - 2a\alpha_1 = b^2 - 2ac$ which is non-zero and in fact positive since by hypothesis, $b^2 - 2ac > 2ac > 0$. So α_2 is not only well-defined but is also positive (since a and α_1, which equals c are positive anyway). Moreover, $\alpha_2 = \frac{ac^2}{b^2 - 2ac} < \frac{ac^2}{2ac} = \frac{c}{2} = \frac{\alpha_1}{2}$. So both the assertions to be proved hold for $n = 1$. However, in order to prove the second assertion (viz., that $\alpha_2 < \frac{\alpha_1}{2}$), we need not only that the denominator of α_2 is non-zero but also that it is positive (and, in fact, greater than $2ac$).

Let us now come to the inductive step. Let us assume that for some integer $n \geq 1$, α_n is well-defined (which also implies that all the earlier α's are well-defined since we cannot define α_{k+1} without first having defined α_k). In order to show that α_{n+1} is well-defined we must show that $b_n - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_n)$ is non-zero. Let us write this expression a little differently. Using (18) with n replaced by $n - 1$ we get $b^2 - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_{n-1}) = \dfrac{a\alpha_{n-1}^2}{\alpha_n}$ and hence $b^2 - 2a(\alpha_1 + \ldots + \alpha_{n-1} + \alpha_n) = \dfrac{a\alpha_{n-1}^2}{\alpha_n} - 2a\alpha_n = \dfrac{a(\alpha_{n-1}^2 - 2\alpha_n^2)}{\alpha_n}$. (It is implicit here that $\alpha_n \neq 0$. But this follows because if $\alpha_n = 0$ then by (18) α_{n-1} would also vanish and then applying (18) backwards, $\alpha_{n-2}, \alpha_{n-3}, \ldots, \alpha_3, \alpha_2, \alpha_1$ would all vanish, contradicting the hypothesis that $\alpha_1 = c > 0$.) Hence showing that α_{n+1} is well-defined is equivalent to showing that the numerator $\alpha_{n-1}^2 - 2\alpha_n^2$ is non-zero. By inductive hypothesis we have $\alpha_{n-1} > 2\alpha_n$. From this we can conclude, by squaring, that $\alpha_{n-1}^2 - 4\alpha_n^2$, *provided* α_{n-1} and α_n *are positive* and in that case it will follow that $\alpha_{n-1}^2 - 2\alpha_n^2 > 2\alpha_n^2$ and hence that $\alpha_{n-1}^2 - 2\alpha_n^2$ is not only non-zero but positive.

Summing up, we see that even though the problem only asks us to prove that α_{n+1} is well-defined (which is equivalent to proving that $b^2 - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_n)$ is non-zero), it is convenient to prove something stronger, viz., that α_{n+1} is positive for every n (which, in view of (18), is equivalent to proving that $b^2 - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_n)$ is positive). The other part of the assertion, viz., that $\alpha_{n+1} < \frac{1}{2}\alpha_n$ also then undergoes a corresponding strengthening, viz., $0 < \alpha_{n+1} < \frac{1}{2}\alpha_n$. In fact this part is not independent of the other. It is vitally needed in proving the other.

So we *simultaneously* prove by induction that for every n, $b^2 - 2a(\alpha_1 + \alpha_2 + \ldots + \alpha_n) > 0$ and $0 < \alpha_{n+1} < \frac{1}{2}\alpha_n$. (For those who know a little bit of logic, here

we are proving a statement $P(n)$ which is the **conjunction** of two statements, say $Q(n)$ and $R(n)$. This is different from proving $Q(n)$ and $R(n)$ separately by induction.) The proof itself is not difficult. In fact we already showed how we can prove $b^2-2a(\alpha_1+\alpha_2+\ldots+\alpha_n)>0$ assuming that $b^2-2a(\alpha_1+\alpha_2+\alpha_{n-1})>0$ and $0<\alpha_n<\dfrac{1}{2}\alpha_{n-1}$. It only remains to prove that $0<\alpha_{n+1}<\frac{1}{2}\alpha_n$. The positivity of α_{n+1} follows at once from (18) since both the numerator and the denominator of the R.H.S. are positive. Finally, to prove that $\alpha_{n+1}<\frac{1}{2}\alpha_n$, note that we already proved that $b^2-2a(\alpha_1+\alpha_2+\alpha_n)=\dfrac{a(\alpha_{n-1}^2-\alpha_n^2)}{\alpha_n}$. Putting this into (18) gives $\alpha_{n+1}=\dfrac{a\alpha_n^3}{a(\alpha_{n-1}^2-2\alpha_n^2)}=\dfrac{\alpha_n^3}{\alpha_{n-1}^2-2\alpha_n^2}$. By induction hypothesis we have $0<\alpha_n<\dfrac{1}{2}\alpha_{n-1}$ which implies, upon squaring, that $\alpha_{n-1}^2-2\alpha_n^2>2\alpha_n^2$. Hence we get $\alpha_{n+1}<\dfrac{\alpha_n^3}{2\alpha_n^2}=\frac{1}{2}\alpha_n$. This completes the proof of the inductive step of the revised assertion. In particular we have now proved the original assertion, viz., that α_{n+1} is well-defined and also that $\alpha_{n+1}<\dfrac{\alpha_n}{2}$.

There is an alternate solution in which we prove the positivity of $b^2-2a(\alpha_1+\alpha_2+\ldots+\alpha_n)$ (and hence of α_{n+1}) using the formula for the sum of a geometric progression. Again simultaneously we assume that $0<\alpha_k<\frac{1}{2}\alpha_{k-1}$ and hence that $\alpha_k<\dfrac{\alpha_1}{2^{k-1}}=\dfrac{c}{2^{k-1}}$ for $k=n,n-1,\ldots,3,2$. We then have $0<\alpha_1+\alpha_2+\ldots+\alpha_n<\alpha_1(1+\dfrac{1}{2}+\dfrac{1}{4}+\ldots+\dfrac{1}{2^{n-1}})=c\dfrac{1-(\frac{1}{2})^k}{1-\frac{1}{2}}<2c$ and hence $b^2-2a(\alpha_1+\alpha_2+\ldots+\alpha_n)>b^2-2ac>2ac>0$. For the second part, viz., showing that $\alpha_{n+1}<\frac{1}{2}\alpha_n$, the inequality just proved implies by (18) that $\alpha_{n+1}<\dfrac{a\alpha_n^2}{2ac}=\dfrac{\alpha_n^2}{2c}$. So we would be through if we can show that $\dfrac{\alpha_n^2}{2c}\leq\frac{\alpha_n}{2}$ which is equivalent to showing that $\alpha_n\leq c$. But this is true because we have already proved that $\alpha_n\leq\dfrac{1}{2^{n-1}}$.

Note, incidentally, that in this problem the number a hardly plays any role. We could have as well dropped it (which, in effect, means setting it equal to 1, since it appears only multiplicatively, i.e., as a coefficient, and not additively, i.e., as a term in a sum). The condition $b^2-4ac>0$ does play a vital role. Evidently, this condition is equivalent to saying that the quadratic $ax^2+bx+c=0$ has two distinct real roots. As a result one is prone to think that there must be a way to do this problem using the theory of quadratic equations. But there probably isn't, at least not an obvious one. The sum $\alpha_1+\alpha_2+\ldots+\alpha_n$ appearing in (18), coupled with the requirement that $\alpha_k<\frac{1}{2}\alpha_{k-1}$ suggests that the formula for the sum of a geometric progression may be involved in the solution and this, indeed turns out to be the case (although not in a mandatory manner). Whenever you see a problem it is always a good idea to look for clues to solutions. The ability to perceive them is, in fact, a great asset. At the same time, one must be prepared for the possibility that some of these clues may not

lead anywhere.

As remarked earlier, in this problem we have proved by induction a statement, say $P(n)$, which is the logical conjunction of two statements $Q(n)$ and $R(n)$. In symbols, we write $P(n) = Q(n) \wedge R(n)$. It is interesting to note that the truth of $Q(n+1)$ required not only the truth $Q(n)$ but also that of $R(n)$. Similarly, $R(n+1)$ followed not solely from $R(n)$ but from the conjunction of $Q(n)$ and $R(n)$. That is why it was possible to prove that $Q(n) \wedge R(n)$ implies $Q(n+1) \wedge R(n+1)$. In other words induction fails for the statement $Q(n)$ alone and also for the statement $R(n)$ alone but works for the conjunction $Q(n) \wedge R(n)$. In case this sounds paradoxical to you, remember that a man cannot produce another man by himself, nor can a woman produce another woman all by herself. But the two together can certainly produce another man and woman!

What if we were interested only in one of the two statements $Q(n)$ and $R(n)$? In that case even if we are not interested *per se* in the other statement, we would have to introduce it for our own convenience because without it we will not be able to apply induction. The role of this auxiliary statement is comparable to that of a catalyst in a chemical reaction. Finding a suitable auxiliary statement when one is not given often calls for a certain ingenuity.

Comment No. 16:

Ingenuity is also needed in the application of certain inequalities which are so simple that what they say is little beyond common sense. Take, for example, a well-known inequality called the **pigeon hole principle** which says that when you put letters into pigeon holes, if there are more letters than pigeon holes, then at least one pigeon hole must contain two or more letters! Stated more formally, the pigeon hole principle says that a function $f : X \longrightarrow Y$, where X, Y are finite sets with $|X| > |Y|$, cannot be one-to-one. More generally, if k is a positive integer and $|X| > k|Y|$ then f must map at least $k+1$ distinct points of X to the same point of Y. This is, of course, sheer common sense. But when applied cleverly, the pigeon hole principle can work wonders when the function f is not obvious.

As illustrations, suppose first that there are 13 students in a class. Then it follows trivially that at least two of them must have their birthdays falling in the same month of the year. Here X is the set of the 13 students, Y is the set of the 12 months of an year and the function $f : X \longrightarrow Y$ assigns to each student the month in which his birthday falls. Here the function f is obvious and the conclusion is accordingly trivial. Suppose, however, that we consider the roll numbers of these students which are some positive integers (not necessarily from 1 to 13) and we want to prove that at least two students have roll numbers differing by a multiple of 12. We again let X be the set of the 13 students. But this time we take Y to be the set of residue classes modulo 12 (see Chapter 4, Comment No. 16) and define $f : X \longrightarrow Y$ by assigning to each student the residue class (modulo 12) of his roll number. As there are only 12 distinct residue classes modulo 12, the pigeon hole principle shows that there are two distinct students whose roll numbers fall in the same residue class modulo 12, which is equivalent to saying that they differ by a multiple of 12. The same

reasoning shows that if $1 \leq k < n$ and $\alpha_1, \alpha_2, \ldots, \alpha_n$ are any integers, then the Vandermonde determinant $D(\alpha_1, \alpha_2, \ldots, \alpha_n)$ (see Exercise (3.26)) is divisible by k, because there exist $i \neq j$ such that $\alpha_i - \alpha_j$ is divisible by k which makes the difference between the i-th and the j-th rows also divisible by k.

As a truly more convincing (and colourful) example, suppose we have a graph with 6 vertices, every two of which are joined by an edge. (We have not formally defined a graph. But you can take the vertices to be the nodes of an electrical network and the edges as pieces of not necessarily straight wires.) Thus in all there are $\binom{6}{2} = 15$ edges. Suppose we colour them arbitrarily with two colours, red or blue. The problem is to show that there is a monochromatic triangle, i.e., three vertices every two of which are joined by edges of the same colour. To do this, fix any vertex, say A. There are 5 edges meeting at A. As each one of them is either red or blue, by the pigeon hole principle at least three of them are of the same colour, which we may suppose to be red (as otherwise we interchange the two colours for all 15 edges). So there are three vertices, say B, C and D such that the edge joining each one of them to A is red. Now consider the three edges joining these three vertices among themselves. If at least one of them, say the one from B to C is red then we have a red triangle, viz., ABC. The only possibility left is that all three of them are blue. But in that case, BCD is a monochromatic (blue) triangle!

This beautiful problem can be generalised. Note that the result is not always true for graphs with 5 (or less) vertices. Similarly, if we have 3 colours instead of 2, then even for a graph with 6 vertices we cannot guarantee a monochromatic triangle. However, in that case it can be shown that for a graph with 17 (or more) vertices every two of which are joined by an edge, there must be a monochromatic triangle. Stronger generalisations are possible and lead to a theory called the **Ramsey theory**. The theory has surprising applications to diverse branches of mathematics, because colouring need not be interpreted literally. It could be any type of a binary relationship. For example, the problem above can be paraphrased as saying that if 6 persons are picked at random in a party then either there will be three among them, every two of whom know each other or there will be three among them every two of whom are strangers to each other. The essence of Ramsey theory is the pigeon hole principle.

Comment No. 17:

There is a dual version of the pigeon hole principle which says that if there are more pigeon holes than letters then at least one of them will be empty. Put differently, if each one of the pigeon holes is non-empty, then their number cannot exceed the number of letters. In the language of functions, if $f : X \longrightarrow Y$ is a surjective (i.e., an onto) function then $|Y| \leq |X|$ and more generally if every point of Y has at least k preimages under f, where k is some positive integer, then $|Y| \leq \frac{|X|}{k}$. The proof follows easily by letting $y_1, y_2, \ldots, y_m$ be the distinct elements of Y, where $m = |Y|$ and setting $S_i = f^{-1}(\{y_i\})$ for $i = 1, 2, \ldots, m$. Then X is the disjoint union of the subsets $S_1, S_2, \ldots, S_m$ and so $|X| = \sum_{i=1}^{m} |S_i| \geq \sum_{i=1}^{m} k = mk$.

Put differently, the dual version of the pigeon hole principle says that if $|X| < k|Y|$ and $f : X \longrightarrow Y$ is a function then at least one element of Y has less than k preimages as otherwise, the set X has no room to accommodate the mutually disjoint subsets $S_1, S_2, \ldots, S_m$ each having k or more elements. Of course the subsets need not have come from a function. They could be any subsets of X. In essence all we are saying is that a set of cardinality less than km cannot contain m mutually disjoint subsets of cardinalities k or more. Or in other words, if $|X| < mk$ and $S_1, S_2, \ldots, S_m$ are subsets of X each having cardinality k or more, then at least two of these subsets must overlap.

Here 'room' was measured in terms of the cardinality of a set. But for certain sets there are other ways of measuring how much 'room' they have, such as length, area and volume. For example, suppose m is a positive integer, h a positive real number and $[a, b]$ is an interval. If $b - a < mh$ then $[a, b]$ cannot contain m mutually disjoint intervals of length h each (or more). Similarly, a disc of radius r cannot contain more than 4 mutually disjoint discs of radius $\frac{r}{2}$. (If you draw a disc of radius r and try to fit in discs of radii $\frac{r}{2}$ in it, you will see that you cannot fit in more than two without their interiors overlapping. In other words the upper bound 4 which we got simply by considering areas is really too crude and can be improved upon. But this requires the knowledge of the geometry of the discs. It is to be expected that an inequality proved by a general, elementary argument cannot be as sharp as one proved by an argument which is more advanced and specific.)

However, if applied ingeniously, even a crude inequality obtained by an elementary argument such as above can yield results which are far from obvious. Suppose, for example, that we want to show that given any 7 real numbers there exist two numbers, say, x and y among them such that $x > y$ and $\frac{x-y}{1+xy} < \frac{1}{\sqrt{3}}$. The problem is rather puzzling but the expression $\frac{x-y}{1+xy}$ gives a clue that trigonometry, and more specifically, the formula for the tangent of the difference of two angles must be involved in the solution. Once this idea strikes us, the solution is not difficult. Call the seven given real numbers as $x_1, \ldots, x_7$ in a strictly ascending order and let $\theta_i = \tan^{-1}(x_i)$ for $i = 1, 2, \ldots 7$. Then $-\frac{\pi}{2} < \theta_1 < \theta_2 < \ldots < \theta_6 < \theta_7 < \frac{\pi}{2}$. So the interval $[\theta_1, \theta_7]$ has length less than π and contains 6 mutually disjoint open sub-intervals, viz., (θ_i, θ_{i+1}) for $i = 1, 2, \ldots, 6$. Hence at least one of them, say (θ_j, θ_{j+1}), has length less than $\frac{\pi}{6}$. Therefore $0 < (\theta_{j+1} - \theta_j) < \frac{\pi}{6}$ and hence $0 < \tan(\theta_{j+1} - \theta_j) < \tan(\frac{\pi}{6}) = \frac{1}{\sqrt{3}}$. The solution is completed by setting $x = \tan(\theta_{j+1}) = x_{j+1}$ and $y = \tan(\theta_j) = x_j$.

Comment No. 18:

The examples above are illustrations of the applications of the pigeon hole principle, which, by itself, is a most trivial inequality, saying little more than the ancient saying that a part cannot be equal to the whole. There are, of course, many other applications of inequalities. In Chapter 3 we already saw how the existence of real roots for a real quadratic reduces to a certain inequality. Specifically, the roots of a quadratic equation $ax^2 + bx + c = 0$ are real and distinct if and only if its discriminant, viz., $b^2 - 4ac > 0$. There we handled only elementary problems. Now that we have gone deeper into inequalities, we

can handle somewhat more complicated problems. Here is an illustration. (JEE 1987) Suppose we are given distinct real numbers a, b, c, d and p such that

$$(a^2 + b^2 + c^2)p^2 - 2(ab + bc + cd)p + (b^2 + c^2 + d^2) \leq 0$$

and asked to find which of the following statements about a, b, c, d is (are) true.

(A) they are in A.P. (B) they are in G.P.
(C) they are in H.P. (D) $ab = cd$.

Treat the L.H.S. of the given inequality as a quadratic in p. Then its leading coefficient is positive. So the fact that it is ≤ 0 for some value of p means that its discriminant Δ cannot be negative. In other words, we must have

$$(ab + bc + cd)^2 \geq (a^2 + b^2 + c^2)(b^2 + c^2 + d^2)$$

On expansion and rearrangement this comes to

$$0 \geq (b^2 - ac)^2 + (c^2 - bd)^2 + (ad - bc)^2$$

Since a sum of squares can never be negative, equality must hold, which, in turn, means that each of $b^2 - ac$, $c^2 - bd$ and $ad - bc$ must vanish. This is the case if and only if a, b, c, d are in G.P. Hence (B) holds.

Comment No. 19:

The entire theory of maxima and minima is based on the concept of inequalities. So problems of finding maxima and minima are essentially problems about inequalities. We shall take them up in Chapters 13 and 14. Not surprisingly, many of the results of this chapter will be needed there.

Comment No. 20:

We now come to an application of inequalities which is, by far, the most important application from a theoretical point of view. Inequalities provide us the means necessary to give a rigorous definition of the most fundamental concept of calculus, that of a limit. Derivatives and integrals are special types of limits. Here we confine ourselves to limits of sequences. Later (in Chapter 15) we shall deal with limits of functions of a real variable.

A sequence is, by definition, a function whose domain is $\mathbb{N}$, the set of all positive integers. (Occasionally we also include 0 in its domain.) The value of such a function at n is called the n-th term of the sequence and is generally denoted by symbols like a_n, x_n etc. instead of the usual notations like $f(n), g(n)$ etc. In fact we often do not even mention the function and simply say $\{x_n\}$ is a sequence. More precisely we say that $\{x_n\}_{n=1}^{\infty}$ is a sequence. Note that the symbol ∞ (read 'infinity') occurring as a superscript here is not a real number, much less an integer. So an expression like $n = \infty$ makes no sense if $n \in \mathbb{N}$. A sequence can be specified by expressing its n-th term either in a **closed form**, i.e., directly as an expression in n (such as $a_n = \frac{n^2+1}{2}$ or $b_n = 2^n - 1$) or **recursively**, i.e., in terms of some of its previous terms (e.g., the Fibonacci

sequence $F_n = F_{n-1} + F_{n-2}$ or the sequence $a_n = a_1 a_2 \dots a_{n-1} + 1$, where the first few terms of the sequence have to be specified explicitly). We have already seen many examples of both types of sequences.

Let us now see how we can give a precise meaning to the concept of a limit of a sequence, i.e., to a statement of the form 'a_n tends to L as n tends to ∞' written in symbols as '$a_n \to L$ as $n \to \infty$' or as $\lim_{n\to\infty} a_n = L$, where $\{a_n\}$ is a given sequence of real numbers and L is a real number. As a concrete example, suppose $a_n = \dfrac{n + (-1)^n}{2n}$. The first few terms of this sequence are $0, \frac{3}{4}, \frac{2}{6}, \frac{5}{8}, \frac{4}{10}, \frac{7}{12}, \frac{6}{14}, \dots$. If we calculate a few more terms we see that they are getting close to the number $\frac{1}{2}$ even though none of the terms actually equals $\frac{1}{2}$. So intuitively, $a_n \to \frac{1}{2}$ as $n \to \infty$. Similarly, if $b_n = \dfrac{n + (-1)^n}{n + 2}$ then we see that $b_n \to 1$ as $n \to \infty$.

But how do we make these statements precise? It is tempting to say that the differences $|a_n - \frac{1}{2}|$ and $|b_n - 1|$ are small if n is large. But the trouble is that there is no commonly accepted standard of smallness or of largeness. Replacing 'small' by 'very small' or by 'very very small' is no solution either, since that is equally ambiguous. Some people use phrases like 'smaller than any positive real number'. But the only (non-negative) number smaller than every positive real number is 0, and neither of the differences $|a_n - \frac{1}{2}|$ and $|b_n - 1|$ is 0 for any value of n. An attempt was made to describe this paradoxical situation by inventing the term '**infinitesimally small**' which was popular with mathematicians for more than a century. Thus the definition of $x_n \to L$ was that the difference $|x_n - L|$ was infinitesimally small if n was 'infinitely large'. No rigorous definition of the term 'infinitesimally small' was given. Still people understood it correctly and used it freely. A rigorous treatment of infinitesimals came only recently, in 1966 to be precise. But it is far too involved.

Some mathematicians were unhappy at the fact that the most basic definition in calculus should depend on such an elusive concept as an infinitesimal. So attempts were made to give a definition of limit which completely bypasses the notion of an infinitesimal. The first such successful attempt was due to Weierstrass and Heine and their definition is followed even today. The essential idea in it is to say that $x_n \to L$ as $n \to \infty$ if the difference $|x_n - L|$ can be made as small as desired by taking n sufficiently large. This, too, is imprecise. But it suggests the idea that the standard of largeness of n can be made to depend on the standard of smallness of $|x_n - L|$. The latter can be specified by a positive real number, say ϵ and the former by a positive integer, say m. So a person who says '$x_n \to L$ as $n \to \infty$' is, in essence, saying "Specify how small you would like $|x_n - L|$ to be. In other words give me a number $\epsilon > 0$ of your choice. Then I shall give you a positive number m (which I shall find looking at the number ϵ you have given me) which will guarantee that that the difference $|x_n - L|$ will be less that ϵ whenever the integer n is bigger than or equal to m." In symbols, this translates into the following definition.

Definition 1: We say that a sequence $\{x_n\}$ **tends** to (or **converges** to) L (or that L is a **limit** of the sequence $\{x_n\}$) and write '$x_n \to L$ as $n \to \infty$' (or

$\lim_{n\to\infty} x_n = L$) if for every $\epsilon > 0$ there exists some positive integer m such that whenever $n \geq m$, $|x_n - L| < \epsilon$.

In this definition, after the phrase 'for every $\epsilon > 0$', words such as 'however small' (or 'howsoever small') are added for emphasis. But it is unnecessary to do so from a mathematical point of view. Similarly, instead of saying 'some positive integer m' one sometimes says 'some sufficiently large positive integer m' for greater clarity. The statement $|x_n - L| < \epsilon$ is easily seen to be equivalent to $L - \epsilon < x_n < L + \epsilon$, which geometrically means that x_n lies in the interval of length 2ϵ centred at L. Also the statement $n \geq m$ is equivalent to saying that 'n is a positive integer other than $1, 2, \ldots, m-1$'. So the definition of $\lim_{n\to\infty} x_n = L$ can also be expressed by saying that every interval centred at L contains all terms of the sequence 'after some stage', or equivalently, that it contains all terms of the sequence except possibly the first few terms. This is linguistically less clumsy than the definition above. It should not, of course, be supposed that the initial terms of the sequence viz., $x_1, x_2, \ldots, x_{m-1}$ *cannot* lie in the interval $(L - \epsilon, L + \epsilon)$. Some of them, and sometimes even all of them, may lie in the interval $(L - \epsilon, L + \epsilon)$. All we are saying is that we are not assured of it. But we are assured that all the terms *after* the $(m-1)$-th term will lie in that interval. Naturally, the smaller this interval, the larger is the number of terms of the sequence that may lie outside it. Or, in the language of ϵ and m, the smaller the ϵ, the bigger the integer m that will be needed. The same m will never work for all $\epsilon > 0$, except in the trivial case of a sequence whose terms become equal after some stage. (Such a sequence is said to be **eventually constant**.)

Trivially, a sequence whose terms are all equal to a number, say c, converges to that number. As a less trivial example, let us show that the sequence $x_n = \frac{n}{n+1}$ converges to 1. By a direct calculation, $|x_n - 1| = \frac{1}{n+1}$. Now, given an $\epsilon > 0$, we look for an m such that $\frac{1}{n+1} < \epsilon$ for all $n \geq m$. Clearly, it will suffice to take m to be any integer[4] bigger than or equal to the number $\frac{1}{\epsilon} - 1$. For example, if $\epsilon = 3.07 \times 10^{-5}$, then we can take $m = 32573$. A larger m will also do, of course, as we are not required to find the smallest m that will work for a given ϵ. So, we can always replace 3.07×10^{-5} by any smaller positive number, say 10^{-5} whose reciprocal is easy to find. So we may take $m = 99999$ or even $1,00,000$.

The convergence of complicated sequences is rarely proved by directly showing that the condition in the definition above is satisfied. Instead, one first builds such a sequence from some very simple, standard sequences whose convergence is known beforehand. Then one applies some general results which serve to express the limit of a complicated sequence in terms of those of its

[4]The existence of such an integer is often taken for granted. But it is not always so obvious. If ϵ were rational, say $\epsilon = \frac{p}{q}$, where p and q are positive integers, then we can set $m = q$. But when ϵ is irrational, to get an integer bigger than $\frac{1}{\epsilon}$ requires a basic property of real numbers called the **Archimedian property**, which says that given any real numbers α and β, where $\alpha > 0$, there exists some integer k such that $k\alpha > \beta$. Simply stated, this is equivalent to saying that the sequence $\{\frac{1}{n}\}$ tends to 0 as n tends to infinity. We shall prove this in Chapter 16 using a very basic property of the real number system, called its completeness. Assuming the convergence of this particular sequence, we can decide the fate of many other sequences.

'building blocks'. The most frequently needed basic results about limits of sequences of real numbers are listed in the following theorem.

Theorem 3: Let $\{x_n\}, \{y_n\}, \{z_n\}$ be sequences of real numbers and L, M be real numbers. Assume that $\lim_{n\to\infty} x_n = L$ and $\lim_{n\to\infty} y_n = M$. Then,

(i) if $x_n = y_n$ for all n, then $L = M$. (This means that the limits of convergent sequences are unique.)

(ii) if $z_n = x_n + y_n$ for all $n \geq 1$, then $\lim_{n\to\infty} z_n = L + M$. (Verbally, the sum of two convergent sequences is convergent and converges to the sum of their limits, or, in a compact form, the limit of a sum equals the sum of the limits.)

(iii) if $z_n = x_n y_n$ for all $n \geq 1$, then $\lim_{n\to\infty} z_n = LM$. (Limit of a product is the product of limits.)

(iv) if $M \neq 0$, then there exists a positive integer m such that for all $n \geq m, y_n \neq 0$ and the sequence $\{\frac{x_n}{y_n}\}_{n=m}^{\infty}$ converges to $\frac{L}{M}$. (The limit of a quotient is the quotient of the limits, whenever defined.)

(v) if $x_n \leq y_n$ for all positive integers n, then $L \leq M$. (In other words, inequalities are preserved under limits. However, strictness of an inequality may be lost. That is even if $x_n < y_n$ for all n, we may not have $L < M$. All we can say is that $L \leq M$. A simple counter-example is to take $x_n = 0$ and $y_n = \frac{1}{n}$ for all $n \geq 1$.)

(vi) if $L = M$ and $x_n \leq z_n \leq y_n$ for all n, then $\lim_{n\to\infty} z_n = L$. (This is popularly called the **Sandwich theorem**, because it says that any sequence which is 'sandwiched' between two sequences converging to a common limit also converges to the same limit.)

We shall not prove these statements. The proofs are based on the properties of the absolute values, especially the triangle inequality, in the manner indicated in Comment No. 8 above. All these proofs may be found in any standard text on the subject and, at any rate, are not expected at the JEE level. Note that we have not included a statement to the effect that the limit of the difference equals the difference of the limits, or, in a precise form $\lim_{n\to\infty} x_n - y_n = L - M$ because this statement can be derived from (iii) and (ii). Applying (iii) to the constant sequence $\{-1\}$ and the sequence $\{y_n\}$ we get $\lim_{n\to\infty} -y_n = -1$. This, combined with (ii) gives $\lim_{n\to\infty} x_n - y_n = \lim_{n\to\infty} x_n + (-y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} (-y_n)$ $= L + (-M) = L - M$ as desired. More generally, the same reasoning shows that the limit of a linear combination is the corresponding linear combination of the limits, i.e., for any two real numbers α and β, $\lim_{n\to\infty} \alpha x_n + \beta y_n = \alpha L + \beta M$. (In the same vein, we could have also dropped (i) since it follows from two applications of (v). But we have listed it because of its special appeal. If limits

were not unique, there will be chaos because so many things in life are defined as certain limits.)

In evaluating limits (or in proving more complicated theorems about limits of sequences), the statements above are applied routinely and often without an explicit mention. Their applicability is further enhanced by the fact that the hypotheses in them do not have to be satisfied for *all* integers $n \geq 1$. It is all right if they are satisfied eventually, i.e., after some stage. In (i), for example, if the equality $x_n = y_n$ holds for all $n \geq 100$ (say), we still have $L = M$. (We emphasise once again that when we say that $x_n = y_n$ holds for all $n \geq 100$, we are *not* saying that it cannot hold for any of the first 99 values of n. We are completely silent about the first 99 terms and the conclusion remains unaffected no matter how they behave.) This is so, because, the limit of a sequence is unaffected by changing a finite number of terms of it.

To illustrate a simple application of the theorem above, we now prove, quite rigorously, that the sequence $a_n = \dfrac{n + (-1)^n}{2n}$ defined above converges to $\frac{1}{2}$. Actually, we shall do a little better in that we shall actually *find* the limit of $\{a_n\}$ instead of merely verifying that it equals $\frac{1}{2}$. We first divide both the numerator and the denominator to write $a_n = \dfrac{x_n}{y_n}$, where $x_n = 1 + \dfrac{(-1)^n}{n}$ and $y_n = 2$. The latter is a constant sequence which converges to 2. So, in view of Part (iv) of the theorem above, we shall be through if we can show that the numerator x_n tends to 1. Because of (ii), this is further reduced to showing that the sequence $\dfrac{(-1)^n}{n}$ converges to 0. One way to do this is to apply the Sandwich theorem. For every positive integer n, $(-1)^n$ equals either 1 or -1 depending upon whether n is even or odd. But in either case we have $-1 \leq (-1)^n \leq 1$ and hence $-\dfrac{1}{n} \leq \dfrac{(-1)^n}{n} \leq \dfrac{1}{n}$ for all n. We already know that $\frac{1}{n}$ converges to 0. By (iii) $(-1) \times \dfrac{1}{n}$ converges to $(-1) \times 0 = 0$. So by the Sandwich theorem, $\dfrac{(-1)^n}{n}$ converges to 0. As noted earlier, this shows that $a_n \to \frac{1}{2}$ as $n \to \infty$.

We leave it as an exercise to prove that the other sequence we defined above, viz., $b_n = \dfrac{n + (-1)^n}{n + 2}$ converges to 1. A few more examples will be given as exercises. Note that it is not necessary that every sequence should have a limit. Consider, for example, the sequence $\{c_n\}$, where $c_n = (-1)^n$. Its terms are alternately -1 and 1. But the sequence converges to neither -1 nor to 1. In fact, this sequence cannot converge to *any* real number L. For, if we take $\epsilon = \frac{1}{2}$, say, then the interval $L - \epsilon, L + \epsilon)$ has length 1 and hence cannot contain both -1 and 1. This means infinitely many terms of the sequence will lie outside the interval $(L - \epsilon, L + \epsilon)$. (A beginner is sometimes confused by a statement like this, because in the present case, every term has only two possible values, viz., -1 and 1. So how can we even talk of 'infinitely many terms of the sequence'? The answer is that when we say that infinitely many terms of the sequence $\{c_n\}$ lie outside the interval $(L - \epsilon, L + \epsilon)$, we only mean that c_n lies outside this interval for infinitely many distinct values of the subscript n. We do not

necessarily mean that the corresponding values of c_n are distinct. Thus the sequence $(-1)^n$ is not convergent. A sequence which is not convergent is said to be **divergent**.

In the discussion so far, we considered convergence to a finite limit L. Sometimes we encounter statements like '$x_n \to \infty$ as $n \to \infty$' or '$x_n \to -\infty$ as $n \to \infty$ '. It is tempting to try to define these statements by mechanically replacing L in Definition 1 by ∞ or $-\infty$. But this cannot be done because neither ∞ nor $-\infty$ is a real number. There is, in fact, no precise definition of either ∞ or $-\infty$. Nor is it needed. For, neither of these symbols appears by itself. Expressions containing them are defined in their entirety. Thus to say that $x_n \to \infty$ does not convey the idea that x_n is getting close to anything denoted by ∞. It is simply another way of saying that x_n is getting bigger. It is rather like this. If you ask a person a question and he says 'My boss knows', then he is referring to a particular individual, viz., his boss, who, according to that person, knows the answer to the question asked. But if the person says 'God knows', it does not mean that there is some such person as God Who knows the answer. In fact, the person who makes this statement may be a total atheist! His answer is merely another way of saying that according to him, nobody knows the answer to the question asked.

If these things are kept in mind, the following definition should not appear unnatural.

Definition 2: A sequence $\{x_n\}$ is said to tend to ∞ if for every real number $R > 0$, there exists some $m \in \mathbb{N}$ such that for all $n \geq m$, $x_n > R$.

Again, sometimes the words 'no matter how large' are added for emphasis after '$R > 0$', but this is optional. The definition of a sequence tending to $-\infty$ is similar. Or, one can adopt a common trick. That is, replace x_n by $-x_n$. Then x_n tends to $-\infty$ if and only if $-x_n$ tends to ∞. The foremost example of a sequence tending to ∞ is the sequence $\{n\}$ itself. With this as the prototype, other sequences tending to ∞ can be constructed using the analogues of the various parts of Theorem 3 above. All of them except (iv) hold true if L and M are both replaced by ∞. But the proofs require some modifications. As for (iv), nothing can be said about the convergence of $\frac{x_n}{y_n}$ even if it is given that both x_n and y_n tend to ∞. It all depends on which of them is tending to ∞ more rapidly. The situation will be taken up in an exercise. However, if L is a (finite) real number and $M = \infty$, then it is not hard to show that $y_n > 0$ for all sufficiently large values of n and further that the sequence $\frac{x_n}{y_n}$ converges to 0. Statement (ii) also continues to hold if one of L and M is a finite real number and the other is ∞. In this case $x_n + y_n$ tends to ∞ as n tends to ∞. Symbolically, this is expressed by saying that $L + \infty = \infty$ (or $\infty + M = \infty$ as the case may be). In (iii), suppose L is finite and $M = \infty$. Then the product $x_n y_n$ tends to ∞ or to $-\infty$ according as $L > 0$ or $L < 0$. However, if $L = 0$ nothing can be said as examples of all four different kinds can be given (where $x_n y_n$ tends to ∞, to $-\infty$, to a finite real number and none of these). The symbols 0∞ and $\infty 0$ have no meanings. Similarly, $\infty - \infty$ is undefined and so nothing can be said about the behaviour of $x_n - y_n$ when both x_n and y_n tend to ∞ as n tends to ∞.

A sequence whose terms are in an A.P. is of the form $x_n = a + (n-1)d$. It is clear that this sequence is convergent (with limit a) if $d = 0$. Otherwise it tends to $\pm\infty$ according as d is positive or negative. More interesting is the case of a sequence whose terms are in a G.P. because this sequence and its variations arise naturally in many real-life situations. The following theorem completely answers the question of its convergence. In future, this Theorem will be crucially needed in studying infinite geometric series. (See (5) in Chapter 23.)

Theorem 4: Let r be a fixed real number and let $x_n = r^n$ for $n \geq 1$. Then,

(i) if $|r| < 1$ (i.e., if $-1 < r < 1$) then $x_n \to 0$ as $n \to \infty$

(ii) if $r = 1$, then $x_n \to 1$ as $n \to \infty$

(iii) in all other cases the sequence x_n is divergent.

Proof: Since logarithms convert a G.P. into an A.P., it is tempting to try to prove the theorem above for $r > 0$ by taking logarithms (and then from it deduce the remaining cases). That is, we let $y_n = \log x_n = n \log r$, where the logarithms are taken w.r.t. a fixed base, say 10. Then by the observations made above for sequences whose terms are in A.P., $y_n \to 0, \infty$ or $-\infty$ according as $\log r$ is zero, positive or negative, i.e., according as $r = 0$, $r > 1$ or $r < 1$. We can now take anti-logarithms to study the behaviour of x_n as $n \to \infty$. This approach is not wrong. But to apply it rigorously we need to know that logarithms preserve limits, i.e., the logarithm of a limit is the limit of the logarithms. This is indeed the case. But the proof requires the continuity of the logarithms. (cf. Theorem 1 in Comment No. 3 of Chapter 16.)

We, therefore, prefer to give a direct proof. This proof is also standard. But we present it as an interesting application of the binomial theorem. Statement (ii) is trivial since in this case $\{x_n\}$ is a constant sequence. Before proving (i) it is convenient to dispose off the case $r > 1$ of (iii). In this case we let $h = r - 1$. Then $h > 0$ and $x_n = r^n = (1 + h)^n$ which can be expanded by the binomial theorem. When so done, it is a sum of positive terms, of which one term is nh. So $x_n \geq nh$ for all $n \geq 1$. But since $h > 0$, $nh \to \infty$ as $n \to \infty$. Hence by the analogue of (v) of Theorem 3 for infinite limits, x_n also tends to ∞ as n tends to ∞.

Next we settle the case $0 < r < 1$. In this case put $y = \frac{1}{r}$. Then, $y > 1$ and so by what we just proved, $y^n \to \infty$ as $n \to \infty$. From this it follows (using the analogue of (iv) of Theorem 3, mentioned above) that $x_n = \frac{1}{y^n} \to 0$ as $n \to \infty$. The case $r = 0$ of (i) is trivial. Finally suppose $-1 < r < 0$. Then $|r| < 1$ and so by what we just proved, $|r|^n \to 0$. But $-|r|^n \leq r^n \leq |r|^n$ for all n. So by the Sandwich theorem, $r^n \to 0$. Hence (i) holds in this case too.

We still have to settle the cases $r = -1$ and $r < -1$ of (iii). The former was already disposed of above while giving an example of a sequence with no limit. The situation $r < -1$ is left as an exercise.

A slight advantage of the second proof is that with a little modification, it also shows (see Exercise (6.51)) that whenever $|r| < 1$, $nr^n \to \infty$ as $n \to \infty$.

This is an interesting limit because the general term of the sequence is the product of two factors, the first of which (viz., n) tends to ∞ as n tends to ∞, while the second one (viz., r^n) tends to 0. So here the situation is like a race between two factors which are trying to pull the sequence in opposing directions. It turns out that the second factor r^n is more dominating and consequently the product nr^n sways in its favour.

The study of convergence of sequences is very rich both theoretically and because of their applications. A glimpse will be given in the exercises. However, it is hardly possible to go deeper into it here. We mentioned it in this chapter mostly to stress that inequalities play a crucial role in the definition as well as in the calculation of limits.

EXERCISES

6.1 (i) Let a, b, c be positive and not all equal. Show that the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is negative.

(ii) If x, y, z are real and different and $u = x^2+4y^2+9z^2-6yz-3xz-2xy$, which, if any, of the following statements is true? (1979)

(A) u is always non-negative

(B) u is always 0

(C) u is always non-positive.

(iii) Which, if any, of the following statements is true about the roots of the equation $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$ where a, b, c are real? (1980)

(A) they are positive (B) they are negative (C) they are real

(iv) If real numbers α and β are roots of $x^2+px+q=0$ and α^4 and β^4 are the roots of $x^2-rx+s=0$, what can be said about the signs of the roots of $x^2-4qx+2q^2-r=0$? (1989)

6.2 For complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, we write $z_1 \cap z_2$ if $x_1 \le x_2$ and $y_1 \le y_2$. Then for all complex numbers z with $1 \cap z$, is it necessarily true that $\dfrac{1-z}{1+z} \cap 0$? (1981)

6.3 (a) Determine whether the function $f(x) = \dfrac{x^2+4x+30}{x^2-8x+18}$ is one-to-one. (1983)

(b) Define $f : \mathbb{R} \longrightarrow \mathbb{R}$ by $f(x) = \dfrac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$. Find the interval of values of α for which f is onto. Is f one-to-one for $\alpha = 3$? (1996)

6.4 Find a necessary and sufficient condition (in terms of a, b and c) for the function $f(x) = \dfrac{(x-a)(x-b)}{x-c}$ to assume all real values. (1984)

6.5 (a) For any two real numbers x and y show that $\frac{1}{2}(x+y+|x-y|)$ equals the larger of them. Obtain a similar expression for the smaller of the two numbers.

(b) For any real number x, define its **positive part**, denoted by x^+, to be the greater of x and 0 and its **negative part**, denoted by x^-, to be the greater of $-x$ and 0. (Note that the negative part is never negative!). Prove that for every real number x, $|x| = x^+ + x^-$ and $x = x^+ - x^-$.

(c) Consider the graph of a real-valued function, say $y = f(x)$ of a real variable x. How are the graphs of the functions $|f(x)|$, $f^+(x)$ and $f^-(x)$ obtained from this graph? Illustrate your answer with the graphs of some familiar function such as $f(x) = \sin x$. (Here $f^+(x)$ means $(f(x))^+$. Similarly, for $f^-(x)$ and $|f(x)|$.)

6.6 (a) Suppose a, b are positive real numbers and $a \leq c \leq b$. Prove that $c(a+b-c) \geq ab$. [Hint : Let $m = \frac{a+b}{2}$. Write $c(a+b-c)$ and ab in the form $m^2 - x^2$ and $m^2 - y^2$ respectively, where $0 \leq x \leq y$.]

(b) Let A be the arithmetic mean of n positive real numbers $a_1, a_2, \ldots, a_n$, where $a_1 \leq a_2 \leq \ldots \leq a_n$. Prove that A is also the arithmetic mean of the $n-1$ positive real numbers $a_2, a_3, \ldots, a_{n-1}$ and $a_1 + a_n - A$.

(c) Using (a) and (b) give an inductive proof of the A.M.−G.M. inequality. [*Hint*: Using (a) prove that $A(a_1 + a_n - A) \geq a_1 a_n$.]

(d) For yet another proof of the A.M.−G.M. inequality, first assume that $n = 2^m$, where $m \geq 1$ and prove it in this special case by induction on m. For the general case, given $a_1, a_2, \ldots, a_n$, first fix any m such that $2^m \geq n$. Let A be the A.M. of $a_1, a_2, \ldots, a_n$. Set $a_r = A$ for $n < r \leq 2^m$ and apply the special case to get $a_1 a_2 \ldots a_n A^{(2^m - n)} = A^{2^m}$. Divide both the sides by $a^{(2^m - n)}$ to get $a_1 a_2 \ldots a_n \leq A^n$.

6.7 Prove Theorem 2 independently and derive Theorem 1 from it.

6.8 Prove that $\left|\sum_{k=1}^{n} z_k \overline{w}_k\right| \leq \sqrt{|\sum_{k=1}^{n} |z_k|^2} \sqrt{\sum_{k=1}^{n} |w_k|^2}$, where $z_1, z_2, \ldots, z_n$ and $w_1, w_2, \ldots, w_n$ are any complex numbers. [*Hint* : This is the Cauchy-Schwarz inequality for complex numbers but is proved slightly differently. Let $A = \sum |z_k|^2, B = \sum z_k \overline{w}_k$ and $C = \sum |w_k|^2$, where k runs from 1 to n in each summation. Then $\overline{A} = A$ and $\overline{C} = C$. Use this to show that $\sum |Cz_k - Bw_k|^2$ equals $A|C|^2 - |B|^2 C$. Note further that $C \geq 0$. As the case $C = 0$ holds trivially, assume $C > 0$.]

6.9 If $a^2 + b^2 + c^2 = 1$ determine which of the following intervals contains $ab + bc + ca$. (1984)

(A) $[\frac{1}{2}, 2]$ (B) $[-1, 2]$ (C) $[-\frac{1}{2}, 1]$ (D) $[-1, \frac{1}{2}]$.

6.10 (a) Let x and y be two real variables such that $x > 0$ and $xy = 1$. Find the minimum value of $x + y$. (1981)

(b) What is the geometric significance of the result in (a)?

(c) Among all boxes with a given volume V show that the cube has the least surface area.

(d) What will be the answer to (c) if the boxes do not have tops?

(e) State and prove the dual results of (a) and (c).

6.11 The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in A.P.. Find the intervals in which β and γ lie. (1996)

6.12 Find all real x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 3x - 4 \leq 0$. (1983)

6.13 (a) If a, b, c are positive then prove that $(a + b + c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \geq 9$. Generalise the result for $n > 3$. (1984)

(b) If a, b are positive then for any positive integers m, n, prove that $(ma + nb)^{m+n} \geq (m + n)^{m+n} a^m b^n$. [*Hint:* Write $ma + nb$ as a sum of $m + n$ terms. See the end of Comment No. 6 for an application. Also see Exercise (14.8) for a generalisation.]

(c) If x, y are positive and $x^2 y^3 = 6$, prove that $3x + 4y \geq 10$. [*Hint*: Apply (b) with $a = \frac{3x}{2}, b = \frac{4y}{3}, m = 2$ and $n = 3$.]

6.14 Find the minumum value of $\log_a x + \log_x a$ for $0 < x < a$. (1984)

6.15 For every integer $n > 1$ prove that :

(a) $(n!)^{1/n} < \frac{n+1}{2}$ (1981) (b) $(1 + \frac{1}{n})^n > 2^n \left(\frac{n!}{n^n}\right)^{3/2}$

(c) $\left(\frac{n(n+1)^3}{8}\right)^n > (n!)^4$ (d) $\prod_{k=1}^{n} \frac{2k-1}{2k} < \frac{1}{\sqrt{2n}}$

(e) $\frac{1+b+b^2+\ldots+b^n}{n+1} < \frac{1+b^n}{2}$, where $b > 0$ and $b \neq 1$.

(f) $\frac{1+a^2+a^4+\ldots+a^{2n}}{a+a^3+a^5+\ldots+a^{2n-1}} > \frac{n+1}{n}$, where $a > 0$ and $a \neq 1$.

6.16 The sum of the squares of three distinct real numbers, which are in G.P. is S^2. If their sum is αS, show that $\alpha^2 \in (\frac{1}{3}, 1) \cup (1, 3)$. (1986)

6.17 Determine the interval in which a must lie if the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3. (1999)

6.18 Determine for which values of x, $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$. (1985)

6.19 Suppose x, y are non-negative real numbers such that $x + y = 1$. Prove that $\frac{1}{2} \leq x^2 + y^2 \leq 1$.

6.20 Let S be a square of unit area. Consider any quadrilateral which has one vertex on each side of S. If a, b, c and d denote the lengths of the four sides of the quadrilateral, prove that $2 \leq a^2 + b^2 + c^2 + d^2 \leq 4$. (1997)

6.21 If a, b, c are real and the quadratic $ax^2 + bx + c = 0$ has two real roots α and β with $\alpha < -1$ and $\beta > 1$, show that $1 + \frac{c}{a} + |\frac{b}{a}| < 0$. (1995)

6.22 For a positive integer n, let $\alpha(n) = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{2^n - 1}$. Then which of the following statements is (are) true? (1999)

(A) $\alpha(100) \leq 100$ (B) $\alpha(100) > 100$
(C) $\alpha(200) \leq 100$ (D) $\alpha(200) > 100$

6.23 For $a \leq 0$, determine all real roots of : $x^2 - 2a|x - a| - 3a^2 = 0$. (1986)

6.24 (a) Find the domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$. (1983)

(b) Find the domain and the range of the function $f(x) = \dfrac{x^2}{1+x^2}$. Is the function one-to-one? (1978)

6.25 (a) Let $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$. Find all real values of x for which y is real. (1980)

(b) Identify the set of all real x for which $x^2 - |x+2| + x > 0$. (2002)

(c) Determine the set of all real x such that $\dfrac{2x-1}{2x^3+3x^2+x} > 0$. (1986)

(d) Find the sum of all real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$. (1997)

(e) Find the set of all solutions of the equation $2^{|y|} - |2^{y-1} - 1| = 2^{y-1} + 1$. (1997*)

(f) Draw the graph of $y = |x|^{\frac{1}{2}}$ for $-1 \leq x \leq 1$. (1978)

6.26 Determine the set of all real numbers a such that $a^2 + 2a, 2a + 3$ and $a^2 + 3a + 8$ are the sides of a triangle. (1985)

6.27 (a) Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$. Prove that $Rf = 4^{2n+1}$. [Recall that $R - [R]$ is the fractional part of R.] (1988)

(b) Prove that the integral part of $(\sqrt{5} + 2)^n$ is of the form $4r$ or $4r + 1$ according as n is odd or even.

(c) Find the digit in the 100-th place after the decimal in the decimal expansion of $(\sqrt{2} + 1)^{300}$.

6.28 (a) Sketch the graphs of (i) $y = [x] + [2x] + [4x],\ 0 \leq x < 3$
(ii) $y = 1 - x - [x],\ 0 \leq x < 4$. (JAT 1979)

(b) Show that there exists a real number x such that $[x] + [2x] + [4x] + [8x] + [16x] = 12345$ but that there does not exist any real number x such that $[x] + [2x] + [4x] + [8x] + [16x] + [32x] = 12345$.

(c) For positive integers m and n, prove that

$$n = [\frac{n}{m}] + [\frac{n+1}{m}] + [\frac{n+2}{m}] + \ldots + [\frac{n+m-1}{m}]$$

State and prove an analogous result for the ceiling function, defined in Comment No. 2 of Chapter 4.

6.29 Let a, b, c be constants and suppose T is a triangle in the plane with vertices $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3). If $ax_i + by_i \leq c$ for $i = 1, 2, 3$, prove that every point (x, y) of the triangle satisfies the inequality $ax + by \leq c$. [*Hint*: First show that whenever two points satisfy the inequality, so does every point on the line segment joining them.] (1986 modified)

6.30 Sketch the solution set of the following system of inequalities: (1978)
$$x^2 + y^2 - 2x \geq 0;\ 3x - y - 12 \leq 0;\ y - x \leq 0\ ;\ \text{ and }\ y \geq 0\ .$$

6.31 Find all integers x for which $(5x - 1) < (x + 1)^2 < (7x - 3)$. (1978)

6.32 Find the set of all real values of x for which $\dfrac{2x}{2x^2 + 5x + 2} > \dfrac{1}{x + 1}$. (1987)

6.33 Given $n^4 < 10^n$ for a fixed positive integer $n \geq 2$, prove that $(n + 1)^4 < 10^{n+1}$. (1980)

6.34 Find for what values of m the following system of equations has a solution satisfying the condition $x > 0, y > 0$. (1980)
$$3x + my = m,\ \ 2x - 5y = 20.$$

6.35 Suppose that the figures from 1 to 12 on a clock dial are reshuffled among themselves. Prove that there exists a pair of adjacent figures which add up to at least 14. Must there exist an adjacent pair which adds to at least 15?

6.36 Given 5 points in a triangle whose longest side has length a prove that there exist at least two among them which are at a distance $\frac{a}{2}$ or less from each other.

6.37 Given 101 integers from 1 to 200, prove that there are x, y among them such that $x \neq y$ and x divides y. More generally, prove that this holds if we take $n + 1$ (or more) integers from the set $\{1, 2, \ldots, 2n\}$, where n is any positive integer.

6.38 Integers from 1 to 16 are divided into three classes. Prove that there exist two (not necessarily distinct) integers x and y such that x, y and $x + y$ are in the same class. [*Hint:* Consider to which classes the differences of the integers in any one class belong. Repeat.] (International Mathematics Olympiad, 1978, modified)

6.39 Suppose 20 students appear for a multiple choice test in which there are 10 questions each of which has to be answered as either 'True' of 'False'. Prove that there will always be two students whose answers to at least six questions will match with each other. [*Hint:* Each student's answers give a binary sequence of length 10. Let S be the set of all 1,024 binary sequences of length 10. Number the students from 1 to 20. For $i = 1, 2, \ldots, 20$, let S_i be the set of those binary sequences which differ in at most 2 places from the one corresponding to the answers given by the i-th student. If the assertion is false show that these 20 subsets are mutually disjoint.]

6.40 (a) Show that statement (i) of Theorem 3 follows from two applications of statement (v). Can statement (vi) also be derived from (v)?

(b) If $\lim_{n\to\infty} x_n = L$, prove that $\lim_{n\to\infty} |x_n| = |L|$. Show by an example that the converse is false.

(c) Show that the result of part (ii) of Theorem 3 (viz., that the limit of a sum of two sequences is the sum of their limits) holds true for three and more generally, for any finite number of sequences. Extend Part (iv) similarly.

(d) For a positive integer n, let $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{2n-1} + \frac{1}{2n}$. Then by Part (ii) of Theorem 3 (or rather, by its extension in (c) above),

$$\begin{aligned} \lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{1}{n+1} + \lim_{n\to\infty} \frac{1}{n+2} + \ldots + \lim_{n\to\infty} \frac{1}{2n} \\ &= 0 + 0 + \ldots + 0 \quad = 0. \end{aligned} \tag{19}$$

But since $\frac{1}{n+k} \geq \frac{1}{2n}$ for all $k = 1, 2, \ldots, n$, we have $a_n \geq n(\frac{1}{2n}) = \frac{1}{2}$ for all $n \geq 2$. So by Part (v) of Theorem 3, $\lim_{n\to\infty} a_n \geq \dfrac{1}{2}$, contradicting (19). How do you explain the paradox? (The correct limit of $\{a_n\}$ as $n \to \infty$ is $\ln 2$ as can be shown using definite integrals, see Comment No. 7 of Chapter 17.)

(e) Evaluate $\lim_{n\to\infty} \dfrac{1}{1-n^2} + \dfrac{2}{1-n^2} + \ldots + \dfrac{n}{1-n^2}$. (1984)

6.41 Let $a_n = \dfrac{b_n}{c_n}$, where $b_n = p_0 + p_1 n + p_2 n^2 + \ldots + p_k n^k$ and $c_n = q_0 + q_1 n + q_2 n^2 + \ldots + q_r n^r$ for some integers k, r and real numbers $p_0, \ldots, p_k, q_0, \ldots q_r$. Assume that $p_k \neq 0$ and $q_r \neq 0$. Prove that :

(i) if $k < r$ then $a_n \to 0$ as $n \to \infty$ (i.e., c_n dominates b_n)

(ii) if $k = r$ then $a_n \to \dfrac{p_k}{q_r}$ as $n \to \infty$

(iii) if $k > r$ then as $n \to \infty$, $a_n \to \infty$ if $\dfrac{p_k}{q_r} > 0$ and $a_n \to -\infty$ if $\dfrac{p_k}{q_r} < 0$.

[*Hint*: Divide both the numerator and denominator by n^r.]

6.42 Suppose two glasses contain 100 cc. each of milk and water respectively. Suppose 5 cc. of the liquid in the first glass is poured to that in the second, the mixture is stirred and 5 cc. of it is transferred back to the first glass. Let α and β be respectively the percentages (by volume) of water in the first glass and of milk in the second glass. Calculate α and β and determine which is bigger.

6.43 In the last exercise, suppose the exchange of liquids (5 cc. at a time) continues back and forth. Let a_n be the percentage of water in the first glass after the exchange of liquids is carried out n times. Guess $\lim_{n\to\infty} a_n$ intuitively.

6.44 Continuing the last exercise, show that a_n satisfies the recurrence relation $a_n = \frac{100}{21} + \frac{19}{21}a_{n-1}$, with $a_1 = \frac{100}{21}$. Using this recurrence relation repeatedly, or by induction, show that for every $n \geq 1$, $a_n = 50(1 - (\frac{19}{21})^n)$. Using this and Theorem 4, give a rigorous proof for your guess in the last exercise. Also determine after how many exchanges of liquids, there will be more than 20 per cent water in the first glass.

6.45 Using only the recurrence relation and not the closed form expression for Fibonacci numbers, viz., $F_n = F_{n-1} + F_{n-2}$, prove that if $\lim_{n\to\infty} \frac{F_n}{F_{n-1}}$ exists then it must be $\frac{1+\sqrt{5}}{2}$. Then prove that this is indeed the case using the closed form expression for the Fibonacci numbers given in Equation (18) of Chapter 4. (This limit is popularly called the **golden ratio**. It can be shown that if a rectangle has the property that when a square on its shorter side is cut off from it, the remaining rectangle is similar to the original one, then the sides of the original rectangle must be in the golden ratio. A rectangle with this property is, therefore, called a **golden rectangle**. It is said that among all rectangles other than squares, it has the most aesthetically appealing shape. Television screens are often golden rectangles. The golden ratio and the Fibonacci numbers crop up unexpectedly in many problems, e.g., see the solution to Exercise (24.83).)

6.46 Using the Archimedian property, show that every open interval (a, b) contains at least one (in fact, infinitely many) rational number. (Actually, this property is equivalent to the Archimedian property. It is expressed by saying that the set of all rational numbers is **dense** in $\mathbb{R}$. If we symbolise an open interval with a vacant space, then intuitively, a dense set is so thickly spread over $\mathbb{R}$ that there is no vacant space in its complement. The set of all irrational numbers is also dense in $\mathbb{R}$, as can be seen from Exercise (6.49) below. On the other hand, neither the set of all integers nor the set of all positive real numbers is dense in $\mathbb{R}$.)

6.47 Let α be a real number. Using the last exercise, prove that there exists a sequence of rational numbers which converges to α.

6.48 Let $\{a_n\}$ be the sequence of rational numbers, defined recursively by $a_n = \frac{p_n}{q_n}$, where $p_1 = 1 = q_1$ and for $n > 1, p_n = p_{n-1} + 2q_{n-1}$ and $q_n = p_{n-1} + q_{n-1}$. By induction on n show that $p_n^2 - 2q_n^2 = (-1)^n$ for all n. Hence show that $a_n \to \sqrt{2}$ an $n \to \infty$. (Alternately one can show that $p_n + \sqrt{2}q_n = (\sqrt{2}+1)^n$ and hence that $p_n - \sqrt{2}q_n = (1-\sqrt{2})^n$. The last exercise shows the existence of a sequence of rational numbers converging to $\sqrt{2}$. But it is only of theoretical value. The present exercise, on the other hand, gives an explicit construction for a sequence of rationals converging to the irrational number $\sqrt{2}$. As a result, we can obtain the decimal expansion of $\sqrt{2}$ to any degree of accuracy.)

6.49 Given any positive real number ϵ, show that there exist integers m and n such that $0 < m + n\sqrt{2} < \epsilon$. Hence show that every interval (a, b) contains

a number of the form $p+q\sqrt{2}$, where p and q are integers. [*Hint:* For the first part, take a sufficiently high power of $\sqrt{2}-1$. The second assertion amounts to saying that the set of all surds of the form $p+q\sqrt{2}$, where p, q are integers is dense in $\mathbb{R}$.]

6.50 Let $a_n = n, b_n = \frac{1}{2}n, c_n = 2n, d_n = n-1$ and $e_n = n+(-1)^n$. Prove that all the five sequences tend to ∞ as $n \to \infty$ but the differences $a_n - b_n, a_n - c_n, a_n - d_n$ and $a_n - e_n$ behave totally differently as $n \to \infty$. (In other words, no consistent meaning can be given to an expression like $\infty - \infty$. Similar examples can be constructed to show that expressions like $\frac{0}{0}, 0 \times \infty$ and $\frac{\infty}{\infty}$ are meaningless. Such expressions are collectively called **indeterminate forms**.)

6.51 Let $r > 1$ and $p(n) = a_0 + a_1 n + a_2 n^2 + \ldots + a_k n^k$ be a polynomial in n of degree k (so that $a_k \neq 0$). Prove that $\dfrac{|a_0 + a_1 n + a_2 n^2 + \ldots + a_k n^k|}{r^n} \to 0$ as $n \to \infty$. [*Hint:* Apply the binomial theorem as in the proof of Theorem 4. Here the numerator is said to **grow algebraically** with order k, while the denominator is said to **grow exponentially**. The result says that even though both tend to ∞, the denominator does so more rapidly than the numerator, no matter how large k is. This is expressed verbally by saying that the exponential growth dominates algebraic growth. The enormity of the former has been known for a long time and appears even in folklore. Thus, according to an ancient story, a king who was unduly proud of his riches asked a sage what offering he would like. The sage replied, "Take an ordinary 8×8 chess board. Put one grain of rice in a corner square and thereafter go on putting twice as many in each square as in its predecessor. (Don't worry if they will all fit in or not. Just look at their numbers.) Then give me all these grains you have put in." The king ridiculed the pittance asked by the sage but was soon humbled because even before two thirds of the the board were reached, all his granaries were empty.)

6.52 In contrast to the last exercise show that no matter how large r is, $\dfrac{r^n}{n!}$ tends to 0 as n tends to ∞. (In other words, the exponential growth, although stronger than the algebraic growth, is weaker than the factorial growth. The enormity of the latter can best be illustrated by the Travelling Salesman Problem, mentioned in Exercise (1.8). As proved there, there are $\frac{1}{2}(n-1)!$ such tours. Suppose that the salesman wants to pick the shortest tour, i.e., one that will minimise the total travel time. If he were to decide which is the shortest tour by calculating the travel time of each possible tour (which is a simple addition problem), then even for a moderately large n, say $n = 30$ or so, the time it will take to make this decision will far exceed the time of even the longest tour!)

6.53 Prove that as $n \to \infty, \dfrac{n^n}{n!} \to \infty$. (In other words, although $n!$ dominates

the power r^n, for a fixed base r, it is dwarfed by n^n, where both the base and the exponent are tending to ∞. This raises an interesting question about estimating how big $n!$ is for large values of n. The question acquires practical importance because the factorials arise almost ubiquitously in combinatorial problems. If these problems are to be solved by running some algorithm for a large number of cases, then we would like to know beforehand how long the algorithm may take to finish all cases. Problems of this kind come under a branch of mathematics called **asymptotic estimation**. The name comes from the concept of an asymptote, which is a straight line which approaches[5] a curve, such as a hyperbola. In the case of a hyperbola with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, each of the straight lines $y = \pm\frac{b}{a}x$ is an asymptote. When x (and y) are numerically large, the portion of the hyperbola near to a point (x, y) on it, is very close to a portion of the corresponding asymptote. The asymptote, being a straight line, is easier to analyse than the original curve. Similarly, when we are given some sequence, say $\{x_n\}_{n=1}^{\infty}$ and we want to get an idea as to how big its terms are for large n, we try to approximate it by a more familiar and easier sequence, say $\{y_n\}$. For example, by a slight variation of Part (iii) of Exercise (6.41) above, if $a_n = \dfrac{p_0 + p_1 n + p_2 n^2 + \ldots + p_k n^k}{q_0 + q_1 n + q_2 n^2 + \ldots + q_k n^r}$, where $k \geq r$, then the sequence $\dfrac{p_k}{q_r} n^{k-r}$ is an asymptotic approximation to the more complicated sequence $\{a_n\}$.

A well-known asymptotic approximation is that of the n-th harmonic number, H_n, defined as the sum of the reciprocals of the first n positive integers, that is, $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$. (This was already defined in Chapter 2. Note that the numbers $\alpha(n)$ and a_n in Exercises (6.22) and (6.40)(d) respectively, can be expressed in terms of the harmonic numbers as $\alpha_n = H_{2^n - 1}$ and $a_n = H_{2n} - H_n$. Exercise (4.10) shows that the harmonic numbers are never integers except for $n = 1$.) By regrouping the terms suitably, in blocks of size 1,1,2,4,8, ... it follows that for every positive integer n, $H_{2^n} > \frac{n+2}{2}$. This implies that by taking n sufficiently large, we can make H_n as large as we like, or, in the language of sequences, $H_n \to \infty$ as $n \to \infty$. Actually, is not hard to show that H_n nearly equals the integral $\int_1^n \frac{1}{x} dx$ which equals $\ln n$. The difference $H_n - \ln n$ can be shown to approach a constant γ (see Exercise (24.48)), popularly called the **Euler's constant** which is approximately 0.57721.

The best known asymptotic approximation to the factorials is the **Stirling's approximation**. In a precise form it says that $\lim_{n\to\infty} \dfrac{n! e^n}{n^n \sqrt{2\pi n}} = 1$. Verbally, for large n, $n!$ is very close to $(\frac{n}{e})^n \sqrt{2\pi n}$. With this we not only know that $n!$ falls short of n^n but exactly to what extent.)

[5] A popular description of an asymptote is that it is a 'tangent at infinity'

Chapter 7

TRIGONOMETRIC IDENTITIES

No other branch of elementary mathematics is so rich in the variety of identities as trigonometry. School trigonometry is an outgrowth of geometry. The very first identity, viz., $\sin^2\theta + \cos^2\theta = 1$ is a direct consequence of the theorem of Pythagoras. The identities for the sine and the cosine of the sum of two angles are also proved by geometric arguments. Thereafter, the derivation of most identities in trigonometry (excluding those pertaining to the geometric properties of triangles) requires very little geometry and is based, almost exclusively on algebraic manipulations. Techniques such as factorisation, compodendo and dividendo of ratios can be used cleverly to prove such identities. The complex numbers also come in handy because of Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ and De Moivre's rule $\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n$. Especially interesting are the cases where θ is a submultiple of 2π, i.e., of the form $\frac{2\pi}{k}$ for some positive integer k. As noted in Chapter 2, in such a case, the complex number $\cos\theta + i\sin\theta$ and its powers are k-th roots of unity, i.e., roots of the equation $z^k - 1 = 0$ which factors as $(z-1)(z^{k-1} + z^{k-2} + \ldots + z + 1) = 0$. As a result some of the theory of polynomial equations becomes applicable. The Main Problem illustrates one such application.

We also study some other trigonometric conversions of a general type where only angles are involved. In trigonometric identities pertaining to triangles, certain lengths such as the sides or the circumradius of the triangle are also involved. We postpone such identities to Chapter 11. However, sometimes such identities can be reduced to identities involving only the angles of the triangle. We give one such example.

Trigonometric identities are asked frequently (although not necessarily every year) in JEE. But they can be tricky sometimes. If the key idea hits you, they are simple. Otherwise, they can be quite laborious.

Main Problem : The value of the expression

$$\sin 10^\circ \sin 20^\circ \sin 30^\circ \sin 40^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ \sin 80^\circ$$

when expressed as a reduced rational, equals ..... .

First Hint: Consider the complex number $z_1 = \cos 20^\circ + i \sin 20^\circ$.
Second Hint: Consider an equation whose roots are various powers of z_1.

Solution: Let $z_1 = \cos 20^\circ + i \sin 20^\circ = e^{\frac{\pi i}{9}} = e^{\frac{2\pi i}{18}}$. Then all powers of z_1 satisfy the equation $z^{18} = 1$. This equation has 18 distinct roots, viz., $1, z_1, z_1^2, z_1^3, \ldots, z_1^{17}$ which can also be written in the form $e^{\frac{2\pi k i}{18}}$ or as $\cos \frac{2\pi k}{18} + i \sin \frac{2\pi k}{18}$, where $k = 0, 1, 2, \ldots, 17$. $z^{18} - 1$ factors as $(z-1)(z^{17} + z^{16} + \ldots + z^2 + z + 1)$. For $k = 1, 2, \ldots, 17$, $\cos \frac{2\pi k}{18} + i \sin \frac{2\pi k}{18}$ must be a root of the second factor. Denoting $\cos \frac{2\pi k}{18} + i \sin \frac{2\pi k}{18}$ by z_k, we see that $z_1, z_2, \ldots, z_{17}$ are the distinct roots of the equation $z^{17} + z^{16} + \ldots + z^2 + z + 1 = 0$. As the leading coefficient is 1, we get

$$(z - z_1)(z - z_2) \ldots (z - z_{17}) \quad = \quad z^{17} + z^{16} + \ldots + z^2 + z + 1 \qquad (1)$$

Putting $z = 1$ in (1) and taking absolute values of both the sides, we get

$$|1 - z_1|\,|1 - z_2|\,\ldots\,|1 - z_{17}| = \prod_{k=1}^{17} |1 - z_k| \quad = \quad 18 \qquad (2)$$

Now, for $k = 1, 2, \ldots, 17$,

$$\begin{aligned} |1 - z_k| &= |1 - \cos \tfrac{2\pi k}{18} + i \sin \tfrac{2\pi k}{18}| \\ &= \sqrt{(1 - \cos \tfrac{2\pi k}{18})^2 + \sin^2 \tfrac{2\pi k}{18}} \\ &= \sqrt{2 - 2\cos \tfrac{2\pi k}{18}} \\ &= 2 \sin \tfrac{\pi k}{18} \end{aligned}$$

Putting these in (2) we get

$$\prod_{k=1}^{17} \sin \frac{\pi k}{18} = \frac{18}{2^{17}} = \frac{9}{2^{16}} \qquad (3)$$

The product of the first eight factors of the L.H.S. is precisely the given expression, say P, whose value is to be found. Since $\sin \frac{\pi k}{18} = \sin \frac{\pi(18-k)}{18}$ for $k = 1, 2, \ldots, 8$, we see that P is also the product of the last eight factors of the L.H.S. of (3), taken in the reverse order. The middle factor $\sin \frac{9\pi}{18}$ equals 1. Hence we get $P^2 = \frac{9}{2^{16}}$. As the factors $\sin 10^\circ, \sin 20^\circ, \ldots, \sin 80^\circ$ are all positive, so is their product P. So, $P = \frac{3}{2^8} = \frac{3}{256}$.

Comment No. 1:

The given product P can also be evaluated using simple trigonometric identities. The values of the factors $\sin 30^\circ$ and $\sin 60^\circ$ are already known to be, respectively, $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$. The factors $\sin 80^\circ, \sin 70^\circ$ and $\sin 50^\circ$ equal, respectively $\cos 10^\circ, \cos 20^\circ$ and $\cos 40^\circ$. These facts, along with the identities $2\sin\theta\cos\theta = \sin 2\theta$, $2\sin\alpha\sin\beta = \cos(\alpha-\beta) - \cos(\alpha+\beta)$ and $2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)$ give

$$\begin{aligned}
P &= \sin 10^\circ \sin 20^\circ \sin 30^\circ \sin 40^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ \sin 80^\circ \\
&= \frac{\sqrt{3}}{4} \sin 10^\circ \sin 20^\circ \sin 40^\circ \cos 40^\circ \cos 20^\circ \cos 10^\circ \\
&= \frac{\sqrt{3}}{32} \sin 20^\circ \sin 40^\circ \sin 80^\circ \qquad (4) \\
&= \frac{\sqrt{3}}{64} \sin 20^\circ (\cos 40^\circ - \cos 120^\circ) \\
&= \frac{\sqrt{3}}{64} \sin 20^\circ (\cos 40^\circ + \frac{1}{2}) \\
&= \frac{\sqrt{3}}{128} (\sin 60^\circ - \sin 20^\circ + \sin 20^\circ) \\
&= \frac{3}{256} \qquad (5)
\end{aligned}$$

This solution is more direct and simpler than the earlier one. But it depends crucially on the knowledge of the sines and cosines of some particular angles. Also, if the number of factors in the original expression is large, it may not be easy to figure out just in what manner they should be grouped. The earlier solution, on the other hand, applies equally well if the integer 18 is replaced by any integer $n > 1$. In that case, exactly the same argument would apply and instead of (3) we shall get

$$\prod_{k=1}^{n-1} \sin \frac{\pi k}{n} = \frac{n}{2^{n-1}} \qquad (6)$$

Thereafter the derivation varies slightly depending on the parity of n (i.e., depending upon whether n is even or odd). If n is even, say $n = 2m$ then the first argument given above for finding P goes through and gives

$$\prod_{k=1}^{m-1} \sin \frac{\pi k}{n} = \frac{\sqrt{m}}{2^{m-1}} \qquad (7)$$

On the other hand, if n is odd with $n = 2m+1$, then the L.H.S. of (6) is a product of $2m$ factors which pair off symmetrically from the centre, there being no middle factor. So, instead of (7) we get

$$\prod_{k=1}^{m} \sin \frac{\pi k}{n} = \frac{\sqrt{2m+1}}{2^m} \qquad (8)$$

Comment No. 2:

This problem is typical where we want to evaluate a product whose factors are expressions involving trigonometric functions of angles in an arithmetic progression. Usually, the angles are multiples of an angle of the form $\frac{\pi}{n}$ for some positive integer n. (In the present problem $n = 18$.) Sometimes the angles could be in a geometric progression as well. Take for example, the product $\sin 20^\circ \sin 40^\circ \sin 80^\circ$ appearing in (4). We could have evaluated it either by going forward to (5) using trigonometry as we did or by going backward and finding P using complex numbers as we did. Either way, the answer comes out as $\frac{\sqrt{3}}{8}$.

The crucial point to note is that in such problems, the individual factors can rarely be evaluated. In the given problem, for example, we can evaluate only two of the eight factors, viz., $\sin 30^\circ$ and $\sin 60^\circ$. The remaining factors are roots of certain cubics. For example, $\sin 10^\circ$ is a root of the cubic equation $3x - 4x^3 = \frac{1}{2}$. This cubic is not easy to solve. (If, instead of 18, we take an integer n which is a prime then all the factors are roots of a certain polynomial of degree $n - 1$.) As mentioned in Comment No. 3 of Chapter 3, there is no formula to solve a general polynomial equation of degree 5 or above. The point to keep in mind is that in evaluating a product of the type given in the Main Problem, it is usually futile to try to do it by evaluating each factor separately. *Nor is it needed!* Our concern is only with the value of the *entire* product and not with that of any of the *individual* factors. And it is sometimes possible to find the product without finding the individual factors. For example, if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the roots (counted according to multiplicities) of a polynomial $a_0x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n$, then in Chapter 3, Comment No. 4, we saw that even without identifying $\alpha_1, \alpha_2, \ldots, \alpha_n$, we know that their product $\alpha_1\alpha_2 \ldots \alpha_n$ equals $(-1)^n \dfrac{a_n}{a_0}$.

This simple idea is the key to the success of the first method. To apply it directly, we would have to first construct a polynomial whose roots are $\sin \frac{\pi k}{n}$, for $k = 1, 2, \ldots, n-1$. This is a little clumsy to do. Instead, we have observed that for any $0 < \theta < \pi$, $2\sin\theta$ is the absolute value of the complex number $1 - (\cos 2\theta + i \sin 2\theta)$ (i.e., $1 - e^{2i\theta}$) and thereby introduced complex numbers of the form $e^{i\phi}$, where ϕ is of the form $\frac{2\pi k}{n}$ for some $k = 1, 2, \ldots, n-1$. These complex numbers, along with the number 1 are called the n-th **roots of unity** because the n-th power of each of them is simply 1. ('Unity' is an old name for the number 1. The number 0 also has a similar name, naught.) An n-th root of unity is obviously also an m-th root of unity if m is a multiple of n. Because of DeMoivre's rule (which says that for every integer k and every angle θ, $\cos k\theta + i \sin k\theta = (\cos\theta + i \sin\theta)^k$), the roots of unity also come in handy in problems involving trigonometric functions of angles which are multiples of an angle of the form $\frac{\pi}{n}$. The Main Problem is a good illustration of this.

Using the same reasoning as used for deriving (1), we get that for every

$n > 1$,

$$(z - z_1)\ldots(z - z_{n-1}) = \prod_{k=1}^{n-1}(z - z_k) \quad = \quad z^{n-1} + \ldots + z + 1 \qquad (9)$$

where $z_1, z_2, \ldots, z_{n-1}$ are the n-th roots of unity (excluding 1).

In (9) we are free to replace z by any complex number of our choice. The resulting equation can sometimes lead to some interesting results. For example, putting $z = 0$ yields $z_1 z_2 \ldots z_{n-1} = (-1)^{n-1}$. Of course, we could also derive this directly by writing z_k as $e^{\frac{2\pi ik}{n}}$. The product $z_1 z_2 \ldots z_{n-1}$ then equals $e^{(\frac{2\pi i}{n})(1+2+\ldots+(n-1))} = e^{\frac{(n-1)n\pi i}{n}} = e^{(n-1)\pi i} = (e^{i\pi})^{n-1} = (-1)^{n-1}$. But the derivation using (9) is instantaneous. Similarly by putting $z = 1$ in (9) we get (JEE 1984) $(1 - z_1)(1 - z_2)\ldots(1 - z_n) = n$. If we take the absolute values of both the sides we get (6) and subsequently (7) and (8) as we already saw.

Note incidentally, that except for the number 1, no other n-th root of unity can be an $(n-1)$-th root of unity. In fact, they are hardly related. So a problem about the n-th roots of unity cannot generally be reduced to a similar problem about the $(n-1)$-th roots of unity. This means that mathematical induction, at least in its standard form (where the truth for n is to be derived from the truth for $n-1$), is not a very useful tool in tackling problems about roots of unity. One can show that if m and n are positive integers and their greatest common divisor (g.c.d.) is d, then there are exactly d complex numbers which are both m-th roots of unity and n-th roots of unity. Specifically they are all the d-th roots of unity, viz., $1, e^{\frac{2\pi i}{d}}, e^{\frac{2\pi i}{d}}, \ldots, e^{\frac{2(d-1)\pi i}{d}}$.

Comment No. 3:

We can derive several other results of a similar nature using (9). Suppose, for example, that we want to evaluate the product of the cosines (rather than the sines) of angles which are multiples of $\frac{\pi}{n}$, where $n > 1$. Specifically, suppose we want to evaluate the product, say Q_n, which equals $\cos\frac{\pi}{n}\cos\frac{2\pi}{n}\ldots\cos\frac{(m-1)\pi}{n}$ if n is even (with $n = 2m$) and which equals $\cos\frac{\pi}{n}\cos\frac{2\pi}{n}\ldots\cos\frac{m\pi}{n}$ if n is odd (with $n = 2m+1$). This can be done using the fact that $|1+(\cos 2\theta + i\sin 2\theta)| = \sqrt{(1+\cos 2\theta)^2 + \sin^2 2\theta} = \sqrt{2+2\cos 2\theta} = \pm 2\cos\theta$. If we put $z = -1$ in (9), then the R.H.S. is 1 if n is odd and 0 if n is even. The latter is consistent with the fact that in this case one of the factors of the L.H.S., viz., $(-1 - z_m)$ is 0, because $z_m = \cos\pi + i\sin\pi = -1$. So if n is even then the equality that results from (9) by putting $z = -1$ is of the form $0 = 0$ which is of little use. However, in this case, because of the identity $\cos\alpha = \sin(\frac{\pi}{2} - \alpha)$, the factors of Q_n, read in reverse order, equal, respectively, $\sin\frac{\pi}{n}, \sin\frac{2\pi}{n}, \ldots$ and $\sin\frac{(m-1)\pi}{n}$. So by (7), their product Q_n equals $\dfrac{\sqrt{m}}{2^{m-1}}$.

The case of odd n (with $n = 2m + 1$) is analogous to (8). We take absolute values of both sides of the equation obtained by putting $z = -1$ in (9) and get

$$|1 + z_1|\,|1 + z_2|\ldots|1 + z_{n-1}| = 1 \qquad (10)$$

Here $z_k = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}$, for $k = 1, 2, \ldots, n-1$ $(= 2m)$. We already showed that $|1 + z_k| = \pm 2\cos\frac{\pi k}{n}$. For $1 \le k \le m$, $\cos\frac{\pi k}{n} > 0$ and so $|1 + z_k| = 2\cos\frac{\pi k}{n}$. But for $m+1 \le k \le 2m$, $\cos\frac{\pi k}{n} < 0$ and so $|1+z_k| = -2\cos\frac{\pi k}{n}$ which also equals $2\cos\frac{\pi(2m+1-k)}{n}$. So the $2m$ factors of the L.H.S. of (10) pair off symmetrically w.r.t. the centre and we get $Q_n = \dfrac{1}{2^m}$.

Putting these together, we get that for $n = 2m$,

$$\cos\frac{\pi}{n}\cos\frac{2\pi}{n}\ldots\cos\frac{(m-1)\pi}{n} = \frac{\sqrt{m}}{2^{m-1}} \tag{11}$$

while for $n = 2m + 1$,

$$\cos\frac{\pi}{n}\cos\frac{2\pi}{n}\ldots\cos\frac{m\pi}{n} = \frac{1}{2^{m-1}} \tag{12}$$

Combining together (7) with (11) and (8) with (12) we get that for $n = 2m$

$$\tan\frac{\pi}{n}\tan\frac{2\pi}{n}\ldots\tan\frac{(m-1)\pi}{n} = 1 \tag{13}$$

while for $n = 2m + 1$,

$$\tan\frac{\pi}{n}\tan\frac{2\pi}{n}\ldots\tan\frac{m\pi}{n} = \sqrt{2m+1} \tag{14}$$

Note that because of the fact that $\tan\theta$ and $\tan(\frac{\pi}{2} - \theta)$ are reciprocals of each other, there is a very easy direct derivation of (13). However, (14) is not so easy to derive and amply illustrates the use of the complex roots of unity.

Comment No. 4:

Thus we see that by assigning suitable values to z in Equation (9), we get very interesting results. A question arises, of course, viz., how does one know beforehand which magic value to assign to z that will do the trick in a given problem. There is really no answer to a question like this. It is almost like asking a detective exactly what led him to suspect someone. Although experience plays an important role, ultimately it is an individual's ingenuity that does the job. (If every mathematical problem could be done without applying ingenuity, then mathematics will be an extremely dull subject!) In the problems we did so far, the crucial step was to cast $\sin\theta$ as $\frac{1}{2}|1 - (\cos 2\theta + i\sin 2\theta)|$ (and to cast $\cos\theta$ as $\pm\frac{1}{2}|1 + (\cos 2\theta + i\sin 2\theta)|$). Both these identities are relatively well-known.

In general, if the problem deals with a product of the form $f(\frac{\pi}{n})f(\frac{2\pi}{n})\ldots f(\frac{(n-1)\pi}{n})$, where $f(\theta)$ is a trigonometric function of θ, then we need to find a suitable complex number c so that for $k = 1, 2, \ldots, n-1$, $|c - z_k|$ can be related to $f(\frac{k\pi}{n})$. Admittedly, if the problem is to *find* the value of such a product, then the search for such a suitable complex number c may not be easy. However, if the problem asks you to *prove* that a product of this type equals some given function of n, then the answer is given and sometimes the very form of the answer may suggest what c should be. This, in fact, is an

important characteristic of a good problem. The answer itself provides a clue to the solution. This hint should not, of course, be too faint or too loud. The former makes the problem dicey, the latter dull.

Here is an excellent example of a problem where the implicit hint is just perfect.

If $n = 2m + 1$, where m is a positive integer, prove that

$$(4 + \cot^2 \frac{\pi}{n})(4 + \cot^2 \frac{2\pi}{n}) \ldots (4 + \cot^2 \frac{m\pi}{n}) = \frac{3^n - 1}{2n} \tag{15}$$

The very form of the L.H.S. suggests that this problem is of the same spirit as equation (8) or (12) which were obtained, respectively, by putting $z = 1$ and $z = -1$ in (9). So here also we try to solve the problem by assigning some suitable value c to z in the equation (9). Let us first try to do this without looking at the right hand side. A typical factor on the L.H.S. is $4+\cot^2 \frac{k\pi}{n}$. We are looking for a complex number c so that this factor can be expressed in terms of the absolute value of the complex number $c - z_k$, i.e., in terms of $|c - (\cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n})|$. If we let $c = a + ib$ then we can write this absolute value in terms of the real numbers a and b, viz., $\sqrt{a^2 + b^2 + 1 - 2a \cos \frac{2\pi k}{n} - 2b \sin \frac{2\pi k}{n}}$. Put differently, we are looking for two real numbers a and b so that for any angle θ, the expression $\sqrt{a^2 + b^2 + 1 - 2a \cos 2\theta - 2b \sin 2\theta}$ should be related (and preferably equal) to the expression $4 + \cot^2 \theta$. As the second expression is an even function of θ (i.e., its value is unchanged if we replace θ by $-\theta$), while the first expression is not so (because of the term $-2b \sin 2\theta$ in it), we get an inkling that b must be 0. This simplifies our search a little because now we have to look for only one real number, viz., a such that the expressions $\sqrt{a^2 + 1 - 2a \cos 2\theta}$ and $4 + \cot^2 \theta$ are related to each other. As the first expression is bounded while the second one is not, such an a probably does not exist.

We can make some progress if we rewrite $4 + \cot^2 \frac{k\pi}{n}$ as $\frac{3 \sin^2 \frac{k\pi}{n} + 1}{\sin^2 \frac{k\pi}{n}}$. If we do this for each of the m factors of the L.H.S. of (15) and multiply them out it becomes

$$\frac{(3 \sin^2 \frac{\pi}{n} + 1)(3 \sin^2 \frac{2\pi}{n} + 1) \ldots (3 \sin^2 \frac{m\pi}{n} + 1)}{\sin^2 \frac{\pi}{n} \sin^2 \frac{2\pi}{n} \ldots \sin^2 \frac{m\pi}{n}} \tag{16}$$

Here, because of (8) we already know how to take care of the denominator. So we need worry only about the numerator. Applying the same reasoning as before our search is now for a suitable real number a such that for every angle θ, the expressions $\sqrt{a^2 + 1 - 2a \cos 2\theta}$ and $3 \sin^2 \theta + 1$ are related to each other. The first expression can be rewritten as $\sqrt{a^2 + 1 - 2a + 4a \sin^2 \theta}$ or as $\sqrt{(a-1)^2 + 4a \sin^2 \theta}$. Now one can try various values for a. Normally one would first try the integral values. In doing so, if we put $a = 3$ we see that the first expression is $\sqrt{4 + 12 \sin^2 \theta}$, which is, indeed, intimately related to $3 \sin^2 \theta + 1$.

So far we have not looked at the R.H.S. of (15), the equation we are trying to derive from (9) by giving z some particular value c. When we do so, we notice that the R.H.S. of (9) becomes the sum of a geometric progression, viz., $c^{n-1}+c^{n-2}+\ldots+c+1$. By a well-known formula, this sum is $\frac{c^n-1}{c-1}$. And the R.H.S. of (15) also involves 3^n-1. So anybody with an iota of perceptivity will see that we should take c as 3, something which we found the hard way without looking at the answer.

Now that the magic number c is found to be 3, the rest of the work is simple. Putting $z=3$ in (9) and taking absolute values gives

$$|3-z_1|\,|3-z_2|\ldots|3-z_{n-1}| = \frac{3^n-1}{2} \tag{17}$$

For $k=1,2,\ldots n-1$, $|3-z_k| = |3-\cos\frac{2\pi k}{n}-i\sin\frac{2\pi k}{n}| = \sqrt{10-6\cos\frac{2\pi k}{n}}$ comes out to be precisely $2\sqrt{1+3\sin^2\frac{k\pi}{n}}$. Again, because of the identity $\sin(\pi-\theta)=\sin\theta$, the factors that are located symmetrically w.r.t. the centre are equal to each other and so the product of the first m factors equals that of the remaining m factors. Hence from (17) we get

$$(3\sin^2\frac{\pi}{n}+1)(3\sin^2\frac{2\pi}{n}+1)\ldots(3\sin^2\frac{m\pi}{n}+1) = \frac{3^n-1}{2^n} \tag{18}$$

We already showed that the L.H.S. of (15) can be reduced to (16). If we replace the numerator of (16) using (18) and its denominator using (8) (after squaring both the sides), we get (15) as was to be shown.

Comment No. 5:

By now you must be convinced about the power of (9) in evaluating products of the values of trigonometric expressions involving angles which are consecutive multiples of some angle. However, there are two strong limitations on the applicability of the method we have used above. First of all, it can be applied only when the angles are successive multiples of an angle of the form $\frac{\pi}{n}$ for some positive integer n. A more serious limitation is that we have to take the product of all the $n-1$ factors in order to be able to apply (9). The symmetry relation $\sin\theta=\sin(\pi-\theta)$ allows us to tackle products which are halfway (or nearly halfway) long. But if we want the product of only the first few factors, then the method is of little help. For example in the Main Problem, we found the value of the product $P=\sin10^\circ\sin20^\circ\sin30^\circ\sin40^\circ\sin50^\circ\sin60^\circ\sin70^\circ\sin80^\circ$ using (1) (which is a special case of (9)). But suppose we wanted only the product of the first five factors, i.e., the product $\sin10^\circ\sin20^\circ\sin30^\circ\sin40^\circ\sin50^\circ$. Then the method does not work. We can, of course, consider the product of the first five factors of the L.H.S. of (1), viz., $(z-z_1)(z-z_2)(z-z_3)(z-z_4)(z-z_5)$. But the trouble is that this fifth degree polynomial in z cannot be written in a form in which the coefficients of the various powers of z can be identified easily. It is only when we take the entire product $(z-z_1)(z-z_2)\ldots(z-z_{17})$ that we can write it in the simple form $z^{17}+z^{16}+\ldots+z^2+z+1$.

The situation is analogous to that discussed in Chapter 5, where we saw that by putting $x = 1$ in the binomial expansion of $(1 + x)^n$, we could prove that the sum of all the binomial coefficients in it, viz., $\sum_{k=0}^{n} \binom{n}{k}$ is 2^n and further that in case n was odd with $n = 2m+1$, then the 'half sum' $\sum_{k=0}^{n} \binom{n}{k}$ came out to be 2^{n-1}. However, if we take some other integer, say r, then there is no formula for evaluating the sum $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{r}$.

Comment No. 6:

All the applications of (9) so far were obtained by assigning some particular value to the complex variable z in it. A qualitatively different type of application would be to consider the sum of the complex numbers $z_1, z_2, \ldots, z_{n-1}$. If we equate the coefficients of z^{n-2} on the two sides of (9), we get

$$z_1 + z_2 + \ldots + z_{n-1} = -1 \tag{19}$$

(Of course, we could also have gotten this from

$$(z-1)(z-z_1)(z-z_2)\ldots(z-z_{n-1}) = z^n - 1 \tag{20}$$

by equating the coefficients of z^{n-1} on both the sides. The two approaches are hardly different since (9) is obtained from (20) by dividing both the sides by $(z-1)$.)

Let us see what this means in real terms. Recalling that $z_k = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}$ and equating the real parts of the two sides of (19), we get

$$\cos\frac{2\pi}{n} + \cos\frac{4\pi}{n} + \ldots + \cos\frac{2(n-1)\pi}{n} = \sum_{k=1}^{n-1}\cos\frac{2\pi k}{n} \quad = \quad -1 \tag{21}$$

Similarly, equating the imaginary parts of the two sides of (24) we get (JEE 1987)

$$\sin\frac{2\pi}{n} + \sin\frac{4\pi}{n} + \ldots + \sin\frac{2(n-1)\pi}{n} = \sum_{k=1}^{n-1}\sin\frac{2\pi k}{n} \quad = \quad 0 \tag{22}$$

We remark that better methods are available for deriving (21) and (22) using the following two well-known formulas for the sums of the sines and of the cosines of angles in an arithmetic progression. (In case the denominator $\sin\frac{\beta}{2}$ is 0, the formulas still remain valid if we replace the ratio $\frac{\sin\frac{k\beta}{2}}{\sin\frac{\beta}{2}}$ by k.)

$$\sin\alpha + \sin(\alpha+\beta) + \ldots + \sin(\alpha+(k-1)\beta) = \frac{\sin(\alpha + \frac{k-1}{2}\beta)\sin\frac{k\beta}{2}}{\sin\frac{\beta}{2}} \tag{23}$$

$$\cos\alpha + \cos(\alpha+\beta) + \ldots + \cos(\alpha+(k-1)\beta) = \frac{\cos(\alpha + \frac{k-1}{2}\beta)\sin\frac{k\beta}{2}}{\sin\frac{\beta}{2}} \tag{24}$$

Both these formulas are proved easily by induction on the integer k. Or one can multiply the L.H.S. of (23) by the complex number i and add it to the L.H.S. of (24). By Euler's formula (viz., $\cos\theta + i\sin\theta = e^{i\theta}$), the sum comes out to be $\sum_{r=0}^{k-1} e^{i(\alpha+r\beta)} = \sum_{r=0}^{k-1} e^{i\alpha}e^{ir\beta}$. But this is the sum of a geometric progression with the first term $e^{i\alpha}$ and common ratio $e^{i\beta}$ and therefore equals $e^{i\alpha}\dfrac{e^{ik\beta}-1}{e^{i\beta}-1}$. If we resolve this into its real and imaginary parts and simplify, we get (23) and (24). But the direct, inductive proofs are simpler.

Evidently, (21) and (22) are special cases of (24) and (23) respectively obtained by putting $k = n-1$ and $\alpha = \beta = \frac{2\pi}{n}$. Note that this method will also give us the sum of the first k terms of the L.H.S. of (21) and (22) for any k $(1 \le k \le n-1)$, whereas the earlier method, based on the complex n-th roots of unity worked only for the entire sums in the L.H.S. of (21) and (22). This only shows that a tool such as (9) which works fine for certain problems may not be the best one for some other types of problems. That is why it pays to have at one's disposal a variety of tools and, given a particular problem, to pick up the one that is best suited for it.

Comment No. 7:

We derived (19) from (9) (or from (20)) using the formula for the sum of the roots of a polynomial and then applied it to get (21) and (22). There is an alternate way to get (21) and (22) directly if we observe that the numbers $z_1, z_2, \ldots, z_{n-1}$ are simply successive powers of z_1 and so the L.H.S. of (19) is simply $z_1 + z_1^2 + \ldots + z_1^{n-1}$. But this is the sum of a geometric progression with the first term z_1 and common ratio also equal to z_1. So it equals $z_1\dfrac{z_1^{n-1}-1}{z_1-1}$ i.e., $\dfrac{z_1^n - z_1}{z_1-1}$. But $z_1^n = 1$. We thus get that $z_1 + z_1^2 + \ldots + z_1^{n-1} = -1$. Considering that z_1^k is the same as z_k, this is no different than (19), but the method of obtaining it is slightly different. In effect, we have put $z = z_1$ in the equation

$$1 + z + z^2 + \ldots + z^{n-1} = \frac{z^n - 1}{z - 1} \tag{25}$$

which is valid for all complex numbers $z \neq 1$. (21) and (22) now follow by taking the real and imaginary parts.

This alternate method may not appear to hold any advantage. And actually, it doesn't if our sole goal is to prove (21) and (22) (for which better methods are available through (23) and (24) as noted above). But there are certain things we can get by doing certain operations on (25) which are not easy to get otherwise. One such operation is differentiation. By definition, differentiation involves something deeper than mere algebraic manipulations, viz., the limiting process. So it is to be expected that differentiation may yield results which are not obtainable, or at least not easily obtainable by mere algebraic manipulations. In Chapter 5 we proved a number of identities by differentiating

the binomial expansion of $(1+x)^n$ w.r.t. x. Let us now see what we can get by differentiating (25). Here z is a complex, and not a real variable. But the definition and the basic properties of derivatives of a complex-valued function of a complex variable are the same as those of the derivatives of real-valued functions of real variables. In particular the quotient rule holds good. So differentiating both the sides of (25) w.r.t. z and multiplying them by z gives

$$z + 2z^2 + 3z^3 + \ldots + (n-1)z^{n-1} = \frac{nz^n(z-1) - z(z^n-1)}{(z-1)^2} \tag{26}$$

which is valid for all complex $z \neq 1$. In particular, we can put $z = z_1 = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ in (26). Noting that $z_1^n = 1$, we get

$$z_1 + 2z_1^2 + 3z_1^3 + \ldots + (n-1)z_1^{n-1} = \frac{n}{z_1 - 1} \tag{27}$$

Let us now take the real parts of both the sides. For the L.H.S., we already know that $z_1^k = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}$. As for the R.H.S. note that

$$\begin{aligned} \frac{1}{z_1 - 1} = \frac{1}{(\cos\frac{2\pi}{n} - 1) + i\sin\frac{2\pi}{n}} &= \frac{(\cos\frac{2\pi}{n} - 1) - i\sin\frac{2\pi}{n}}{(\cos\frac{2\pi}{n} - 1)^2 + \sin^2\frac{2\pi}{n}} \\ &= \frac{(\cos\frac{2\pi}{n} - 1) - i\sin\frac{2\pi}{n}}{2 - 2\cos\frac{2\pi}{n}} \end{aligned}$$

So equating the real parts of the two sides of (27), we get, (for $n > 1$),

$$\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n} + \ldots + (n-1)\cos\frac{2(n-1)\pi}{n} = \sum_{k=1}^{n-1} k\cos\frac{2k\pi}{n} = -\frac{n}{2} \tag{28}$$

Combining (21) and (28) we thus get a solution to the following problem. (JEE 1997)

Prove that $\displaystyle\sum_{k=1}^{n-1}(n-k)\cos\frac{2k\pi}{n} = -\frac{n}{2}$, where $n \geq 3$ is an integer.

Comment No. 8:

There is yet another way we can apply formula (9) when the integer n is odd, say, $n = 2m+1$. If we take a factor z^m out from each term of the R.H.S. of (9) and regroup the terms, it becomes $z^m[(z^m + z^{-m}) + (z^{m-1} + z^{-(m-1)}) + \ldots + (z^2 + z^{-2}) + (z + z^{-1}) + 1]$. The expression in the brackets can be expressed as a polynomial in $z + z^{-1}$. For example, we write $z^2 + z^{-2}$ as $(z + z^{-1})^2 - 2$, $z^3 + z^{-3}$ as $(z + z^{-1})^3 - 3(z + z^{-1})$ and so on. Let us write w for $z + z^{-1}$ and $f(w)$ for the expression in the brackets. Then $f(w)$ is a polynomial of degree m in w and the R.H.S. of (9) is simply $z^m f(w)$. Since $z_1, z_2, \ldots, z_{2m}$ are the roots of the R.H.S. of (9) we see that for $k = 1, 2, \ldots, m$, $z_k + (z_k)^{-1}$ is a root of $f(w)$. These are, in fact the distinct roots of $f(w)$. (For $k = m+1, m+2, \ldots 2m$ too, $z_k + (z_k)^{-1}$ is a root of $f(w)$. But it is the same as $z_r + (z_r)^{-1}$, where

$r = 2m + 1 - k$.) But $z_k + (z_k)^{-1} = e^{\frac{2\pi ki}{n}} + e^{-\frac{2\pi ki}{n}} = 2\cos\frac{2\pi k}{n}$. Note further that the coefficients of $f(w)$ are all real. It is, in fact, not hard to show that they are integers. But we shall not need it.

Summing up, we have found a real polynomial $f(w)$ of degree m whose roots are the real numbers $2\cos\frac{2\pi k}{n}$, for $k = 1, 2, \ldots m$. Although computing the coefficients of the various powers of w in $f(w)$ is rather complicated, it is easy to show that the coefficients of w^m and of w^{m-1} are 1 each. As a result, the sum of the roots is -1 and this gives an alternate derivation of (21) for the case when n is odd. (If n is even, say, $n = 2m$, then (21) is trivial anyway, since the middle term $\cos\frac{2m\pi}{n}$ is -1 and terms which are symmetrically located w.r.t. the middle term cancel each other.) Similarly, we get an alternate proof of (22).

But the real advantage of this method is something else. Suppose we have some trigonometric function, say $u = h(w)$ of the real variable w. If the function h is invertible, then as we saw in Chapter 3, Theorem 2, $f(h^{-1}(u))$ is a function of u whose zeros are precisely $h(2\cos\frac{2\pi k}{n})$, for $k = 1, 2, \ldots, m$. If the function $h(w)$ is a reasonably simple function, then the function $f(h^{-1}(u))$ $(= g(u)$, say) can be expressed as a polynomial in u, or at least can be related to a polynomial in the variable u. Knowing the coefficients of this polynomial, we can find the sum of its roots etc. We illustrate this technique in the following problem.

Prove that the roots of the equation

$$x^3 - 24x^2 + 80x - 64 = 0$$

are $\sec^2\frac{\pi}{7}, \sec^2\frac{2\pi}{7}$ and $\sec^2\frac{3\pi}{7}$. Hence evaluate $\sec^4\frac{\pi}{7} + \sec^4\frac{2\pi}{7} + \sec^4\frac{3\pi}{7}$.

Taking, $n = 7$ and using the notations above, the polynomial $f(w)$ comes out as $w^3 - 3w + w^2 - 2 + w + 1$, i.e., $w^3 + w^2 - 2w - 1$. We already know that the roots of this polynomial are $2\cos\frac{2\pi}{7}, 2\cos\frac{4\pi}{7}$ and $2\cos\frac{6\pi}{7}$. From this we want to construct a polynomial whose roots are $\sec^2\frac{\pi}{7}, \sec^2\frac{2\pi}{7}$ and $\sec^2\frac{3\pi}{7}$. For this we first find a function $u = h(w)$ such that $\sec^2\frac{\pi k}{7} = h(2\cos\frac{2\pi k}{7})$ for $k = 1, 2, 3$. This can be done if we can express $\sec^2\theta$ in terms $2\cos 2\theta$, which is easy. We simply write $\sec^2\theta = \dfrac{2}{2\cos^2\theta} = \dfrac{2}{1+\cos 2\theta} = \dfrac{4}{2+2\cos 2\theta}$. Thus, we take $u = h(w) = \frac{4}{2+w}$. Then the inverse function $h^{-1}(u)$ is given by $w = \frac{4-2u}{u}$. Therefore, the composite function, say $g(u)$ is $f(h^{-1}(u)) = \left(\frac{4-2u}{u}\right)^3 + \left(\frac{4-2u}{u}\right)^2 - 2\left(\frac{4-2u}{u}\right) - 1 = \frac{p(u)}{u^3}$, where $p(u) = (4-2u)^3 + (4-2u)^2u - 2(4-2u)u^2 - u^3$. Note that $g(u)$ itself is not a polynomial in u. But $p(u)$ is and its zeros are precisely those of $g(u)$. Thus $p(u)$ is a polynomial whose roots are precisely $\sec^2\frac{\pi}{7}, \sec^2\frac{2\pi}{7}$ and $\sec^2\frac{3\pi}{7}$. If we replace u (which is a dummy variable anyway) by x and expand, $p(x)$ indeed comes out to be $-x^3 + 24x^2 - 80x + 64$. This is the negative of the polynomial given in the question and hence has the same roots.

To complete the solution we merely note that if α, β and γ are the roots of $x^3 - 24x^2 + 80x - 64 = 0$, then from the formulas (11) in Chapter 3, $\alpha + \beta + \gamma = 24$ and $\alpha\beta + \beta\gamma + \gamma\alpha = 80$. This gives $\alpha^2 + \beta^2 + \gamma^2 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha) = 576 - 160 = 416$. The result follows by taking

$\alpha = \sec^2 \frac{\pi}{7},\ \beta = \sec^2 \frac{2\pi}{7}$ and $\gamma = \sec^2 \frac{3\pi}{7}$.

In the solution above we first found the function $u = h(w)$ by expressing $\sec^2\theta$ in terms of $2\cos 2\theta$ and then found the inverse function $w = h^{-1}(u)$. We have chosen to do so because we wanted to apply the method of Theorem 2 in Chapter 3, to obtain an equation whose roots are certain given functions of the roots of a given polynomial. But obviously, there is a certain waste here which could be avoided if we directly express $2\cos 2\theta$ in terms of $\sec^2\theta$. This can be done by the well-known formula $\cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$ which gives $2\cos 2\theta = \dfrac{2-2\tan^2\theta}{\sec^2\theta} = \dfrac{4-2\sec^2\theta}{\sec^2\theta}$. Writing w for $2\cos 2\theta$ and u for $\sec^2\theta$, this gives $w = \frac{4-2u}{u}$, exactly as before.

Let us now see how we can modify this method to cover the case when n is even, say, $n = 2m$. In that case, we see that -1 is also an n-th root of unity and so $z+1$ is a factor of the R.H.S. of (9), i.e., of the polynomial $z^{2m-1}+z^{2m-2}+\ldots+z^2+z+1$. The other factor is $z^{2m-2}+z^{2m-4}+\ldots+z^4+z^2+1$. Note that in this factor there are no odd degree terms. If m is odd, say $m = 2k+1$, we write this as z^{2k} multiplied by a polynomial in z^2+z^{-2} (and hence a polynomial in $z+z^{-1}$ if we want). However, if m itself is even, then we can take out a factor z^2+1 and apply the method to the other factor. We can continue like this until we reach the highest power of 2 which divides n. At that stage the method above will be applicable to the last factor. If n itself is a power of 2, say $n = 2^r$, then this way we get a factorisation of the R.H.S. of (9) as the product $(1+z)(1+z^2)(1+z^4)\ldots(1+z^{2^{r-1}})$. (See Exercise (7.9) for a complete factorisation of z^n-1 into polynomials with integer coefficients.)

Comment No. 9:

So far we studied how complex roots of unity can be used to tackle problems which are analogous to the problem with which we started. However, we remarked in Comment No. 1 that a purely trigonometric solution is possible and, in fact, more direct. Trigonometric methods have the advantage that they can be applied even when the angles are not in an arithmetic progression. For example, in the solution to the Main Problem, while going from (4) to (5), we evaluated the expression $\sin 20^\circ \sin 40^\circ \sin 80^\circ$ as $\frac{\sqrt{3}}{8}$. If instead of this product, we had one more factor in front, i.e., we had the product $\sin 10^\circ \sin 20^\circ \sin 40^\circ \sin 80^\circ$ to evaluate then the best we can do is to write it as $\frac{\sqrt{3}}{8}\sin 10^\circ$. No further simplification is possible since the numerical value of $\sin 10^\circ$ cannot be expressed in an easy form. This does not detract from the trigonometric method, because in this case the method of roots of unity does not work either. In fact, no method will work since the problem is inherently impossible, i.e., does not admit a succinct numerical answer.

The key to success in the trigonometrical method is to figure out how the factors should be grouped together. This is, indeed, the charm of the method and so no hard and fast rule can be laid down to find the right grouping (if it exists). The general guiding rule is to convert some of the sines to cosines

and vice versa and to convert a product of two factors (by means of any of the four standard identities of the form $2\sin A\cos B = \sin(A+B)+\sin(A-B)$) to a sum of two terms of which at least one term is the sine or cosine of some standard angle such as 30°, 45° or 120°. We already illustrated this in Comment No. 1. In more complicated problems, the other four identities expressing the sums of sines/cosines as products (i.e., identities of the type $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$) are also needed. (Knowledge of the values of the trigonometric functions of angles which are multiples of 30 and 45 degrees suffices in most problems where a trigonometric simplification requires such values. In rare problems, the values of $\sin 18°$ etc. and more generally other multiples of 18° are needed. It is a good idea to remember them. Combining these values with others suitably, we can evaluate the sines and cosines (and hence all other trigonometric functions) of angles which are multiples of 3° by writing 15 as $45-30$, 12 as $72-60$ and finally, 3 as $18-15$. But these are needed so infrequently, that it is not a good idea to tax your memory by remembering them. If at all needed, they can be found on the spot.)

From a practical point of view, if the problem involves the product of n factors where the value of the integer n is either not specified or is very large, then the trigonometric method is unlikely to work and recourse must be had to the roots of unity. However, if the value of n is relatively small, then we have a choice. Sometimes the problem can be done by both the methods (as was the case with our Main Problem). It is always a good idea to give a moment to see if by a suitable regrouping, the trigonometric method may work. Here is a good illustration of such a problem. (JEE 1984)

Find the numerical value of the expression

$$(1+\cos\frac{\pi}{8})(1+\cos\frac{3\pi}{8})(1+\cos\frac{5\pi}{8})(1+\cos\frac{7\pi}{8})$$

Here alternate multiples of $\frac{\pi}{8}$ are involved and so the problem is not quite of the type we have discussed so far. It is tempting to 'simplify' each factor individually by noting that $1+\cos 2\theta = 2\cos^2\theta$. But this conversion will force us to work with multiples of $\frac{\pi}{16}$ and thereby will complicate rather than simplify the problem. If instead, we use the identity $\cos(\pi-\theta) = -\cos\theta$ then we can multiply the first and the last factor together to get simply $\sin^2\frac{\pi}{8}$. Similarly the product of the middle two factors is $\sin^2\frac{3\pi}{8}$. So the given product equals $\frac{1}{4}(2\sin\frac{\pi}{8}\sin\frac{3\pi}{8})^2$, i.e., $\frac{1}{4}(\cos\frac{\pi}{4}-\cos\frac{\pi}{2})^2$ which is simply $\frac{1}{8}$.

If one insists, the problem *can* be done using roots of unity. Call the desired product as A. Using the simplification mentioned above, A equals $16B^2$, where B is the product $\cos\frac{\pi}{16}\cos\frac{3\pi}{16}\cos\frac{5\pi}{16}\cos\frac{7\pi}{16}$. If we multiply B by the three 'missing' factors, viz., $\cos\frac{2\pi}{16}$, $\cos\frac{4\pi}{16}$ and $\cos\frac{6\pi}{16}$ then we can apply (11) with $n=16$. Thus we get $B=\frac{\sqrt{2}}{64C}$, where $C=\cos\frac{\pi}{8}\cos\frac{2\pi}{8}\cos\frac{3\pi}{8}$. Applying (11) again, but with $n=8$ this time, we get $C=\frac{1}{4}$. So, $B=\frac{\sqrt{2}}{16}$ and thus finally, $A=16B^2=\frac{1}{8}$.

We have presented this second solution, not because it is a better solution, but because it will amply illustrate (by contrast) how economical the

trigonometric solution is. The point is not to get addicted to any particular formula or method. The possibility of short cuts, even though sometimes they may be of an *ad hoc* type, should always be kept in mind.

Comment No. 10:

When it comes to problems involving angles that are not in an arithmetic progression, the method based on roots of unity can rarely be used, except sometimes by accident. (For example, tracing the derivation of (4) in Comment No.1 above backwards, we can convert $\sin 20^\circ \sin 40^\circ \sin 80^\circ$ to the long expression $\frac{32}{\sqrt{3}} \sin 10^\circ \sin 20^\circ \sin 30^\circ \sin 40^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ \sin 80^\circ$ and evaluate it, as we did, using the 18-th roots of unity. It is doubtful if anybody will think of this as his first choice.) So, for problems of this type, the trigonometric method is the only way. If the angles are in a geometric progression with common ratio 2, then repeated application of trigonometric identities involving the double of an angle can be useful. We illustrate this with two examples.

Show that

$$16 \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} = 1. \qquad \text{(JEE 1983)}$$

Call this product A. If we multiply A by $\sin \frac{2\pi}{15}$ then we can apply the identity $2 \sin\theta \cos\theta = \sin 2\theta$ again and again (with $\theta = \frac{2\pi}{15}, \frac{4\pi}{15}, \frac{8\pi}{15}$ and $\frac{16\pi}{15}$ successively) to get that $A \sin \frac{2\pi}{15} = \sin \frac{32\pi}{15}$. But $\sin \frac{32\pi}{15} = \sin(2\pi + \frac{2\pi}{15}) = \sin \frac{2\pi}{15}$. So $A = 1$ as was to be proved.

In some problems the given product does not involve angles in a geometric progression. But it can be changed to one which does. For example,

Find the numerical value of $K = \sin(\frac{\pi}{18}) \sin(\frac{5\pi}{18}) \sin(\frac{7\pi}{18})$. (JEE 1993)

Here $K = \sin 10^\circ \sin 50^\circ \sin 70^\circ$ is not in the desired form. But because $\sin\theta = \cos(\frac{\pi}{2} - \theta)$, K equals $\cos 20^\circ \cos 40^\circ \cos 80^\circ$ which is in the desired form. Multiplying by $\sin 20^\circ$ and following essentially the same method as above, we get $8K \sin 20^\circ = \sin 160^\circ$. But $\sin 160^\circ = \sin 20^\circ$. So $K = \frac{1}{8}$. (An alternate solution is possible. If we supply the 'missing' factor $\sin 30^\circ = \frac{1}{2}$, then we get $\frac{K}{2} = \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$ which can be further written as $\frac{P}{Q}$, where the numerator P equals the product $\prod_{k=1}^{8} \sin \frac{\pi k}{18}$ and the denominator Q equals the product $\prod_{k=1}^{4} \sin \frac{\pi k}{9}$. Using (7) (with $n = 18$) and (8) (with $n = 9$) respectively, $P = \frac{3}{256}$ and $Q = \frac{3}{16}$. So $K = \frac{2P}{Q} = \frac{1}{8}$ as before. Obviously, this solution is more complicated.)

Comment No. 11:

When we have a sum (rather than a product) of trigonometric functions of angles that are in a geometric progression, there is no general formula for expressing the sum in a succinct form. But taking into account the particular features of the given problem, certain *ad-hoc* methods can be devised for the

solution. As an example where a linear combination of the tangents of angles in a geometric progression is involved, consider the following problem.

Prove that (JEE 1988)

$$\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha \tag{29}$$

It is unthinkable to do this by writing every term in terms of $\tan\alpha$. We can do so easily for $\tan 2\alpha$ and, with a little perseverance, for $\tan 4\alpha$. But for $\cot 8\alpha$ it will be horrendous. So there must be an easier way. And, indeed there is. If we bring the term $\cot\alpha$ on the R.H.S. to the L.H.S. and observe that $\tan\alpha - \cot\alpha = \dfrac{\tan^2\alpha - 1}{\tan\alpha}$ is the same as $-2\cot 2\alpha$, then we get a sort of a chain reaction. That is, by adding the next term, viz., $2\tan 2\alpha$ to what we already got (viz., $-2\cot 2\alpha$), we would get $-4\cot 4\alpha$. Add one more term and we get $-8\cot 8\alpha$ which cancels with $8\cot 8\alpha$. Thus we see that using the identity $\tan\theta - \cot\theta = -2\cot\theta$ again and again with $\theta = \alpha, 2\alpha, 4\alpha, \ldots$, we can prove, more generally, that for every positive integer n

$$\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + \ldots + 2^{n-1}\tan(2^{n-1}\alpha) + 2^n\cot(2^n\alpha) + = \cot\alpha$$

In a problem like this, how does one think of the right identity and especially the right form in which it is to be used? In the present problem, for example, the identity used is $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$. This is a very standard identity and it does not take much to figure out that in the present problem it will be needed somewhere. But the trouble is that if we apply it directly to the second term of the L.H.S. of (29), it does not help. But when we cast this identity into an equivalent form, viz., $\tan\theta - \cot\theta = -2\cot 2\theta$, it worked wonders. But this is not the standard form in which the identity is normally stated. So how does one think of putting it into that form? This is, again one of those questions asking for the trade secret! But once again, the general answer applies, viz., the form of the question itself sometimes contains a hint to its solution. In the present problem, for example, it is clear from the very form of the problem that the expression in the L.H.S. of (29) need not have stopped at $8\cot 8\alpha$. If we replace it by $8\tan 8\alpha + 16\cot 16\alpha$ then also the result should hold. But since none of the other terms on either side of (29) is affected by this change, the only way this can happen is if $8\cot 8\alpha$ equals $8\tan 8\alpha + 16\cot 16\alpha$, i.e., if $\cot 8\alpha$ equals $\tan 8\alpha + 2\cot 16\alpha$. We could, of course, replace 8α by θ. Then the success of the problem would hinge upon whether $\cot\theta$ equals $\tan\theta + 2\cot 2\theta$. But this is indeed true. In fact it is just another way of putting the very standard identity, $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$. Once this idea strikes you, there is nothing left in the problem. Note, however, the role of the particular coefficients $1, 2, 4, \ldots, 2^{n-1}$ which made the trick work. If instead, we had merely $\tan\alpha + \tan 2\alpha + \tan 4\alpha + \ldots + \tan 2^{n-1}\alpha$, there is no simple answer. Similarly, when we have a sum of the sines or cosines of angles in a geometric progression, there is no general method. But sometimes, the expression can be converted to

a product which we can evaluate. Suppose for example that we want to evaluate

$$\sin\frac{2\pi}{7}+\sin\frac{4\pi}{7}+\sin\frac{8\pi}{7}.$$

Call the expression as S. Writing the middle term as $\sin\frac{3\pi}{7}$ and the last term as $-\sin\frac{\pi}{7}$ and using the identity for $\sin C-\sin D$, we get

$$\begin{aligned} S &= \sin\tfrac{2\pi}{7}+2\cos\tfrac{2\pi}{7}\sin\tfrac{\pi}{7} \\ &= 2\sin\tfrac{\pi}{7}(\cos\tfrac{\pi}{7}+\cos\tfrac{2\pi}{7}) \\ &= 4\sin\tfrac{\pi}{7}\cos\tfrac{3\pi}{14}\cos\tfrac{\pi}{14} \end{aligned}$$

Writing $\sin\frac{\pi}{7}$ as $\cos(\frac{\pi}{2}-\frac{\pi}{7})=\cos\frac{5\pi}{14}$ we get S as $4\cos\frac{\pi}{14}\cos\frac{3\pi}{14}\cos\frac{5\pi}{14}$. This can now be written as $\frac{4P}{Q}$, where $P=\prod_{k=1}^{6}\cos\frac{k\pi}{14}$ and $Q=\prod_{k=1}^{3}\cos\frac{k\pi}{7}$. Using (11) (with $n=14$) and (12) (with $n=7$), P and Q come out, respectively, as $\frac{\sqrt{7}}{2^6}$ and $\frac{1}{2^3}$. Hence the given sum S comes out to be $\frac{\sqrt{7}}{2}$.

Comment No. 12:

Among the complex roots of unity, the complex cube roots of unity deserve a special mention. The number $e^{\frac{2\pi i}{3}}$ which also equals $-\frac{1}{2}+i\frac{\sqrt{3}}{2}$ is denoted universally by ω and is called the complex cube root of unity, which is somewhat a misnomer, because there is one more complex cube root of unity, viz. $\omega^2=e^{\frac{4\pi i}{3}}=-\frac{1}{2}-i\frac{\sqrt{3}}{2}$. Note that ω and ω^2 are complex conjugates of each other. Probably the most frequently needed algebraic property of ω is that $1+\omega+\omega^2=0$. This can be proved directly by adding the three complex numbers or can be derived as a special case of (9) (with $n=2$ and $z=\omega$). This fact combined with the periodic recurrence of the powers of ω enables us to calculate easily some expressions whose direct calculation will be quite messy. Suppose, for example, that we want the value of $(1+\omega-\omega^2)^7$. (JEE 1998) A direct calculation will involve 3^7 terms. But if we replace $1+\omega$ by $-\omega^2$, the expression becomes $(-2\omega^2)^7$, i.e., $-128\omega^{14}$. Since $\omega^{12}=1$, this equals $-128\omega^2$. Similarly to evaluate (JEE 2002) the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$, we replace $-1-\omega^2$ and ω^4 by ω each. Subtracting the first row from each of the other two, the value of the determinant comes out as $(\omega-1)^2-(\omega^2-1)^2=(\omega-1+\omega^2-1)(\omega-\omega^2)=-3(\omega-\omega^2)$.

As a noteworthy application of the fact that $1+\omega+\omega^2=0$, we can completely factor a sum of two cubes as a product of linear factors. Specifically, $a^3+b^3=(a+b)(a+b\omega)(a+b\omega^2)$. Equivalently, we have factored a^2-ab+b^2 as $(a+b\omega)(a+b\omega^2)$. Similarly we have $a^2+ab+b^2=(a-\omega b)(a-\omega^2 b)$. Here is an interesting application of this factorisation.

It is given that n is an odd integer greater than 3, but n is not a multiple of 3. Prove that x^3+x^2+x is a factor of $(x+1)^n-x^n-1$. (JEE 1980)

The hypothesis about n implies that it is of the form $6m\pm1$ for some positive integer m. (The ability to quickly see this is an asset. The reasoning

goes as follows. Since n is not a multiple of 3, it is of the form $3r \pm 1$ for some r. But if r is odd then $3r \pm 1$ would be even, contradicting that n is odd. So r itself is of the form $2m$ for some positive integer m. In the present problem, since the real crux is something else, elaborate justifications of such minor steps are not insisted upon.) Now the polynomial $x^3 + x^2 + x$ factors as $x(x^2+x+1) = x(x-\omega)(x-\omega^2)$. So showing divisibility by x^3+x^2+x is equivalent to showing divisibility by each of $x, x-\omega$ and $x-\omega^2$, which, in turn, is equivalent to verifying that $0, \omega$ and ω^2 are roots of the polynomial $(x+1)^n - x^n - 1 = f(x)$ (say). (Again, the ability to reduce the desired conclusion like this is an asset. The present problem is, therefore, a good example of testing these abilities on the part of the candidates.)

Now, coming to the solution itself, it is clear that $f(0) = 1-1 = 0$ and so 0 is a root of $f(x)$. Here we do not need the hypothesis about n. For the other two verifications, however, we do need it. First consider $f(\omega) = (\omega+1)^n - \omega^n - 1$. Recalling that $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, we have $\omega+1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ which is also equal to $e^{\frac{i\pi}{3}}$. It follows that $(\omega+1)^6 = 1$. (An easier way to see this is to note that $(\omega+1)^2 = \omega^2+2\omega+1 = (\omega^2+\omega+1)+\omega = \omega$ and hence $(\omega+1)^6 = \omega^3 = 1$.) Thus we have $(\omega+1)^n = (\omega+1)^{6m\pm1} = (\omega+1)^{\pm1}$. Also since $\omega^3 = 1$, $\omega^n = \omega^{6m\pm1} = \omega^{\pm1}$. So showing $f(\omega) = 1$ is reduced to verifying that (i)$(\omega+1) - \omega - 1 = 0$ and (ii) $(1+\omega)^{-1} - (\omega)^{-1} - 1 = 0$. (i) is, of course, trivial. (ii) can be rewritten as $\frac{1}{\omega+1} - \frac{1}{\omega} - 1 = 0$ which follows by a direct calculation, using $\omega^2 + \omega + 1 = 0$.

To finish the solution, we still have to verify that $f(\omega^2) = 0$. We could do this in an analogous manner to showing that $f(\omega) = 0$. But that is hardly necessary. As observed earlier, ω^2 is the complex conjugate of ω. Since $f(x) = (x+1)^n - x^n - 1$ is a polynomial with real coefficients, its roots occur in conjugate pairs. So $f(\omega) = 0$ implies $f(\omega^2) = 0$, thereby completing the solution.

Geometrically, in the Argand diagram, the complex numbers $1, \omega$ and ω^2 are vertices of an equilateral triangle centred at 0. In the next chapter we shall see how complex numbers can be applied to do certain problems in geometry. Not surprisingly, then, ω also figures in problems about equilateral triangles. More generally, for every integer $n > 2$, the n n-th roots of unity are at the vertices of a regular n-gon inscribed in the unit circle (i.e., the circle of radius 1 centerd at the origin). An interesting application of this will be given in Chapter 10, Comment No. 3. See also Exercises (10.36) and (10.37).

Comment No. 13:

So far we have dealt mostly with problems where the angles were submultiples of π or their multiples. Roots of unity formed a powerful tool to tackle such problems. We now turn to other problems of trigonometric reductions where the angles given are any general angles. The simplification in such cases has to be carried out using various trigonometric identities, expressing products as sums or vice versa. However, other considerations such as cyclical symmetry play a crucial role. Here is a good example.

Prove that for all x, y, z,

$$\begin{aligned}\sin x \sin y \sin(x-y) &+ \sin y \sin z \sin(y-z) + \sin z \sin x \sin(z-x) \\ &+ \sin(x-y)\sin(y-z)\sin(z-x)\end{aligned}$$

vanishes identically. (JEE 1978)

Call the given expression as E. It would be horrendous to do the problem by brute force, because the expansion of the last term alone will be a sum of eight terms each having six factors!. However, expanding the first term is not so unwieldy. It becomes

$$\begin{aligned}\sin x \sin y \sin(x-y) &= \sin x \sin y\ (\sin x \cos y - \cos x \sin y) \\ &= \tfrac{1}{2}\sin^2 x \sin 2y - \tfrac{1}{2}\sin 2x \sin^2 y \\ &= \tfrac{1}{4}(1-\cos 2x)\sin 2y - \tfrac{1}{4}\sin 2x(1-\cos 2y) \\ &= \tfrac{1}{4}(\sin 2y - \sin 2x) + \tfrac{1}{4}\sin(2x-2y)\end{aligned}$$

By cyclical symmetry we can immediately write down the expansions of the second and the third term of E. If we add the three, the terms involving $\sin 2x$ and $\sin 2y$ etc. cancel each other and we are left with $E = \frac{1}{4}(\sin(2x-2y) + \sin(2y-2z) + \sin(2z-2x)) + \sin(x-y)\sin(y-z)\sin(z-x)$. The last formidable term is still the same and it is foolish to expand it. Instead, we introduce new variables A, B, C with $A = y-z, B = z-x$ and $C = x-y$. (We could, of course, begin by taking A as $x-y$ etc. Mathematically it makes little difference. But if we want to be consistent with cyclical symmetry and let the natural order (A, B, C) correspond to the natural order (x, y, z), then we must take A as $y-z$ etc. If you see old books on mathematics, you will find that this consistency is maintained throughout.) Then $A + B + C = 0$ and the given expression E is now simply $\frac{1}{4}(\sin 2A + \sin 2B + \sin 2C) + \sin A \sin B \sin C$. If we write C as $-A - B$ we get

$$\begin{aligned}E &= \tfrac{1}{4}(\sin 2A + \sin 2B + \sin 2C) - \sin A \sin B\ (\sin A \cos B + \cos A \sin B) \\ &= \tfrac{1}{4}(\sin 2A + \sin 2B + \sin 2C) - \tfrac{1}{2}(\sin^2 A \sin 2B + \sin^2 B \sin 2A) \\ &= \tfrac{1}{4}[\sin 2A\ (1 - 2\sin^2 B) + \sin 2B\ (1 - 2\sin^2 A) + \sin 2C] \\ &= \tfrac{1}{4}[\sin 2A \cos 2B + \cos 2A \sin 2B + \sin 2C] \\ &= \tfrac{1}{4}[\sin(2A+2B) + \sin 2C] \\ &= \tfrac{1}{4}[\sin(-2C) + \sin 2C] \ \ (\text{since}\ \ 2A + 2B = -2C) \\ &= 0\end{aligned}$$

as desired.

Comment No. 14:

Finally, there are numerous problems which require us to prove some relationship between the trigonometric functions of the angles of an arbitrary triangle. Many of these relationships hold for any three angles that add up to π. However, some of them require the further restriction that all the three angles lie between 0 and π. Identities where the sides a, b, c of a triangle ABC are involved can be converted to identities involving only the angles A, B, C

and the circumradius R, because of the relations $2a = R\sin A$ etc. If the identity depends only on the shape and not on the size of the triangle, then R will cancel out from it and it will be reduced to an identity involving only A, B, C. Depending on the nature of it, one can apply various methods. For example, sometimes one can reduce the identity to be proved to some well-known one (such as $\tan A + \tan B + \tan C = \tan A \tan B \tan C$). As a general rule, however, it is best to eliminate one of the variables, say C, by writing it as $\pi - (A + B)$. We illustrate this technique with a problem.

Prove that in a triangle ABC with circumradius R,

$$2R^2(1 + \cos A \cos B \cos C) = \frac{1}{4}(a^2 + b^2 + c^2) \tag{30}$$

(It is a standard practice to denote the sides opposite to the angles A, B, C by a, b, c respectively. This need not be explicitly stated. Also, generally the JEE question papers stick to the old practice of equating a side with its length and an angle with its measure. In modern school textbooks, we say instead, that the measure of angle A is 53° and write $m(\angle A) = 53^\circ$ etc. and that the length of the side BC is a and write $\ell(BC) = a$ etc. In JEE, it is perfectly all right to write $\angle A = 53^\circ$ or even, $A = 53^\circ$.)

Now coming to the solution, we first replace a, b, c by $2R\sin A, 2R\sin B$ and $2R\sin C$ respectively. Then R drops out. If we further use $C = \pi - (A + B)$, then we can replace $\cos C$ by $\sin A \sin B - \cos A \cos B$ and $\sin C$ by $\sin A \cos B + \cos A \sin B$. Then (30) is reduced to proving

$$\begin{aligned} 2(1 \; &+ \; \cos A \cos B \sin A \sin B - \cos^2 A \cos^2 B) \\ &= \quad \sin^2 A + \sin^2 B + (\sin A \cos B + \cos A \sin B)^2 \end{aligned} \tag{31}$$

If we expand the square on the R.H.S. then the term $2\sin A \sin B \cos A \cos B$ cancels out and thus the problem further reduces to proving that

$$2 - 2\cos^2 A \cos^2 B = \sin^2 A + \sin^2 B + \sin^2 A \cos^2 B + \cos^2 A \sin^2 B \tag{32}$$

The easiest way to prove this is to replace $\cos^2 A$ and $\cos^2 B$ by $1 - \sin^2 A$ and $1 - \sin^2 B$ respectively. If we do so then both the L.H.S. and the R.H.S. of (32) reduce to $2\sin^2 A + 2\sin^2 B - 2\sin^2 A \sin^2 B$ each. Hence (32) is true and as a result, (31) and hence finally (30) is also true.

Note that we have not proved (30) in the standard fashion of starting with the L.H.S. and reducing on and on by various manipulations till we get the R.H.S.. Instead, we have reduced both the sides so that proving (30) is equivalent to proving (31). Even after this, we did not follow the L.H.S. to R.H.S. approach. Instead we reduced it to proving (32). This is the approach which is often needed in complicated problems where two expressions are to be proved equal to each other and there is no obvious way to reduce one of them directly to the other. If one wants, the solution can always be written later on in the standard L.H.S. = R.H.S. format. But this may often look artificial. In the problem just solved, for example, we can start

by replacing $\cos C$ with $\sin A \sin B - \cos A \cos B$ in the L.H.S. of (30) to reduce it to $R^2(2 + 2\cos A \cos B \sin A \sin B - 2\cos^2 A \cos^2 B)$. Now we rewrite $2\cos A \cos B \sin A \sin B$ as $2 \sin A \cos B \cos A \sin B$ and recognise it as the middle term in the expansion of $(\sin A \cos B + \cos A \sin B)^2$ which equals $\sin^2(A+B)$ and hence $\sin^2 C$. So if we add and subtract $\sin^2 A \cos^2 B + \cos^2 A \sin^2 B$, the L.H.S. of (30) now reduces to $R^2(\sin^2 C + 2 - 2\cos^2 A \cos^2 B - \sin^2 A \cos^2 B - \cos^2 A \sin^2 B)$ which simplifies to $R^2(\sin^2 C + 2 - \cos^2 A - \cos^2 B)$ and hence to $R^2(\sin^2 C + \sin^2 A + \sin^2 B)$. Now we are home because all we need to do is to replace $R \sin A, R \sin B$ and $R \sin C$ by $\frac{1}{2}a, \frac{1}{2}b$ and $\frac{1}{2}c$ respectively to get the R.H.S. of (30). Q.E.D.!

This is how the 'polished' solutions are often written. When so written they certainly look neat. But unfortunately, they often fail to tell you how they were thought of. For example, in the last paragraph, after the L.H.S. was reduced to $R^2(2 + 2\cos A \cos B \sin A \sin B - 2\cos^2 A \cos^2 B)$, it is difficult to see what motivates one to add and subtract $\sin^2 A \cos^2 B + \cos^2 A \sin^2 B$ so as to get a square. When one sees an expression like $x^2 + y^2$ it is quite natural to think of adding and subtracting $2xy$ so as to complete a square. Similarly, when we see something like $x^2 + xy$ we often add and subtract $\frac{y^2}{4}$ instinctively to complete the square. But when you only see $2xy$ you are not likely to add and subtract $x^2 + y^2$ to rewrite it as $(x + y)^2 - x^2 - y^2$, unless you already know that this is the way to success. The motivation comes from the fact that when we reduce (30) to (31), the term $2 \sin A \sin B \cos A \cos B$ cancels from both the sides.

An even more uncanny way of directly reducing the expression $2R^2(1 + \cos A \cos B \sin A \sin B - \cos^2 A \cos^2 B)$ to the R.H.S. of (30) is to rewrite the number 1 as the product $(\sin^2 A + \cos^2 A)(\sin^2 B + \cos^2 B)$ and hence as the sum $\sin^2 A \sin^2 B + \sin^2 A \cos^2 B + \cos^2 A \sin^2 B + \cos^2 A \cos^2 B$. When this is done, the L.H.S. of (30) reduces to $R^2(2 \sin^2 A \sin^2 B + 2 \sin^2 A \cos^2 B + 2 \cos^2 A \sin^2 B + 2 \cos A \cos B \sin A \sin B)$. We then complete the square to reduce this further to $R^2(2 \sin^2 A \sin^2 B + \sin^2 A \cos^2 B + \cos^2 A \sin^2 B + \sin^2(A + B))$. The first three terms in the bracket add up to $\sin^2 A + \sin^2 B$ while the third term equals $\sin^2 C$ and hence the L.H.S. of (30) equals $R^2(\sin^2 A + \sin^2 B + \sin^2 C)$. The subsequent reduction is the same as before.

While as a mathematical argument this proof is unassailable, it is probably the most unnatural way of proving (30). Writing 1 as $(\sin^2 A + \cos^2 A)(\sin^2 B + \cos^2 B)$ is like pulling a trick out of the blue. When you attempt to prove an identity like (30), you are hardly likely to think of such tricks. Nor is it expected. It is perfectly all right to first show that (30) is equivalent to (31), then to (32) and then finally prove (32). The subsequent work done in casting the proof in the L.H.S. to the R.H.S. format is largely of a cosmetic type and is not insisted upon at an examination like JEE which is aimed at testing the power of reasoning rather than the clerical skill of a student.

We omit more examples of identities like (30) because the standard textbooks contain plenty of them. However, in the exercises, a few more such identities will be given. We shall revisit this topic in Chapter 11 where we study applications of trigonometry to proving properties of triangles.

EXERCISES

7.1 Find the numerical value of $\sqrt{3}\text{cosec}\,20^\circ - \sec 20^\circ$. (1988)

7.2 Prove that:

(a) $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

(b) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

(c) $\sin^2 12^\circ + \sin^2 21^\circ + \sin^2 39^\circ + \sin^2 48^\circ = 1 + \sin^2 9^\circ + \sin^2 18^\circ$

(d) $\text{cosec}\,10^\circ + \text{cosec}\,50^\circ - \text{cosec}\,70^\circ = 6$

(e) if $\alpha = \frac{\pi}{14}$, then $\cos\alpha + \cos 3\alpha + \cos 9\alpha = \frac{1}{2}\sqrt{7}$ and $\cos 2\alpha + \cos 6\alpha + \cos 18\alpha = \frac{1}{2}$

(f) for every integer $n > 1$, $n! = 2^{\frac{1}{2}n(n-1)} \prod_{m=2}^{n} \prod_{k=1}^{m-1} \sin \frac{k\pi}{m}$

(g) if n is odd, then $\sum_{k=1}^{m} \cos^2 \frac{(2k-1)\pi}{n} = \frac{n-2}{4}$, where $m = \frac{n-1}{2}$.

7.3 Using $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$ show that:

(i) $\cos 12^\circ + \cos 60^\circ + \cos 84^\circ = \cos 24^\circ + \cos 48^\circ$

(ii) $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = 1$

(iii) $\cos \frac{2\pi}{15} + \cos \frac{4\pi}{15} + \cos \frac{8\pi}{15} + \cos \frac{14\pi}{15} = \frac{1}{2}$

(iv) $\tan 9^\circ$ is a root of the equation $x^4 - 4x^3 - 14x^2 - 4x + 1 = 0$.

7.4 Give an alternate derivation of (10) by constructing a polynomial whose roots are $1 + z_1, 1 + z_2, \ldots, 1 + z_{n-1}$.

7.5 Find the least positive integer n for which $(\frac{1+i}{1-i})^n = 1$. (1980)

7.6 Let n be a positive integer. A complex number α is called a **primitive** n-th root of unity, if n is the smallest positive integer such that $\alpha^n = 1$. Prove that for any integer k, $e^{\frac{2\pi k i}{n}}$ is a primitive n-th root of unity if and only if k is relatively prime to n. Hence show that the number of primitive n-th roots of unity is $\phi(n)$, where ϕ is the Euler totient function defined in Exercise (4.37).

7.7 More generally, suppose α is a (not necessarily primitive) n-th root of unity. The smallest positive integer m for which $\alpha^m = 1$ is called the **order** of α. If m is the order of α prove that m divides n and further that α is a primitive m-th root of unity. In particular, if n is a prime, then every n-th root of unity, other than 1, is a primitive n-th root of unity. For every divisor $d > 1$ of n prove that among the n-th roots of unity there are exactly d ones that are also the d-th roots of unity.

7.8 Let a complex number α, $\alpha \neq 1$, be a root of the equation $z^{p+q} - z^p - z^q + 1 = 0$, where p and q are distinct primes. Show that either $1 + \alpha + \alpha^2 + \ldots + \alpha^{p-1} = 0$ or $1 + \alpha + \alpha^2 + \alpha^{q-1} = 0$ but not both. (2002)

7.9 Let n be a positive integer, $m = \phi(n)$ and $\alpha_1, \alpha_2, \ldots, \alpha_m$ be the primitive n-th roots of unity. Let $\Phi_n(z) = (z - \alpha_1)(z - \alpha_2)\ldots(z - \alpha_m)$. It can be shown (by a non-trivial argument) that even though the α's are complex numbers, the polynomial $\Phi_n(z)$ is a polynomial of degree $\phi(n)$ with integer coefficients. It is called the n-th **cyclotomic polynomial**. These polynomials are very useful in algebra. Prove the following simple properties of the cyclotomic polynomials:

(i) If n is a prime then $\Phi_n(z) = z^{n-1} + z^{n-2} + \ldots + z^2 + z + 1$.

(ii) $\Phi_1(z) = z - 1$, $\Phi_4(z) = z^2 + 1$, $\Phi_6(z) = z^2 - z + 1$ and $\Phi_8(z) = z^4 + 1$. Can you guess $\Phi_n(z)$ if n is a power of 2?

(iii) $z^n - 1 = \prod_{d|n} \Phi_d(z)$, where the product is taken over all positive divisors of n, including 1 and n itself. (This also gives a complete factorisation of the R.H.S. of (9), augmenting the factorisation given in Comment No. 8.)

(iv) $\Phi_n(z) = \dfrac{z^n - 1}{\prod_{1 \leq d < n} \Phi_d(z)}$ the product in the denominator being taken over all positive divisors of n other than n itself. (This is the same as (iii). But in this form, it gives an inductive procedure for computing the cyclotomic polynomials, instead of obtaining them directly from their roots.)

(v) Using (iv), find $\Phi_{15}(z)$. [*Hint:* Perform long division for polynomials.]

(vi) Using $\Phi_{15}(z)$ show that the equation whose roots are $\tan \frac{r\pi}{15}$ as r ranges over integers from 1 to 14 which are relatively prime to 15 is

$$x^8 - 92x^6 + 134x^4 - 28x^2 + 1 = 0.$$

7.10 Starting from suitable cyclotomic polynomials, prove that:

(a) $4\cos^2 \frac{\pi}{7}$ is a root of the cubic equation $x^3 - 5x^2 + 6x - 1 = 0$

(b) $\sec^2 \frac{\pi}{14} + \sec^2 \frac{3\pi}{14} + \sec^2 \frac{5\pi}{14} = 8$

(c) $\cot^2 \frac{\pi}{11} + \cot^2 \frac{2\pi}{11} + \cot^2 \frac{3\pi}{11} + \cot^2 \frac{4\pi}{11} + \cot^2 \frac{5\pi}{11} = 15$

(d) $\tan^4 \frac{\pi}{16} + \cot^4 \frac{\pi}{16} + \tan^4 \frac{2\pi}{16} + \cot^4 \frac{2\pi}{16} + \tan^4 \frac{3\pi}{16} + \cot^4 \frac{3\pi}{16} = 678.$

7.11 Find the values of the following expressions:

(i) $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ (1999)

(ii) $1.(2 - \omega)(2 - \omega^2) + 2.(3 - \omega)(3 - \omega^2) + \ldots + (n-1)(n - \omega)(n - \omega^2)$, where ω is an imaginary cube root of unity. (1996)

7.12 (a) If $x = a + b, y = a\gamma + b\beta$ and $z = a\beta + b\gamma$, where γ and β are the complex cube roots of unity, show that $xyz = a^3 + b^3$. (1978)

(b) Factorise $A^3 + B^3 + C^3 - 3ABC$ into three linear factors using ω.

(c) Let n be a fixed positive integer and let

$$\begin{aligned} A &= 1 + \tbinom{n}{3}x^3 + \tbinom{n}{6}x^6 + \ldots \\ B &= \tbinom{n}{1}x + \tbinom{n}{4}x^4 + \tbinom{n}{7}x^7 + \ldots \\ C &= \tbinom{n}{2}x^2 + \tbinom{n}{5}x^5 + \tbinom{n}{8}x^8 + \ldots \end{aligned}$$

Prove that $A^3 + B^3 + C^3 - 3ABC = (1 + x^3)^n$.

7.13 Continuing Exercise (5.20), suppose $(1+x+x^2)^n = a_0+a_1x+a_2x^2+\ldots+a_{2n}x^{2n}$. Prove that each of the sums $a_0 + a_3 + a_4 + \ldots$, $a_1 + a_4 + a_7 + \ldots$ and $a_2 + a_5 + a_8 + \ldots$ equals 3^{n-1}.

7.14 (a) If $p < 0$ and α, β, γ are the cube roots of p, then for any x, y, z find the value of the ratio $\frac{x\alpha+y\beta+z\gamma}{x\beta+y\gamma+z\alpha}$. (1989)

(b) Express the roots of $(x-1)^3 + 8 = 0$ in terms of ω. (1979)

7.15 (a) Prove that $x^2 + x + 1$ divides $x^{n+1} + (x+1)^{2n-1}$.

(b) Can you use (a) to give an alternate solution to Exercise (4.16), where you have to show that for any natural number p, $p^{n+1} + (p+1)^{2n-1}$ is divisible by $p^2 + p + 1$?

7.16 For a positive integer n, let

$$f_n(\theta) = \tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta)\ldots(1+\sec 2^n\theta).$$

Which of the following statements is (are) correct? (1999)

(A) $f_2(\frac{\pi}{16}) = 1$ (B) $f_3(\frac{\pi}{32}) = 1$ (C) $f_4(\frac{\pi}{64}) = 1$ (D) $f_5(\frac{\pi}{128}) = 1$.

7.17 If $\alpha + \beta + \gamma = 2\pi$, which of the following statements is (are) true? (1979)

(a) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(b) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} + \tan\frac{\beta}{2}\tan\frac{\gamma}{2} + \tan\frac{\gamma}{2}\tan\frac{\alpha}{2} = 1$

(c) $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2} + \tan\frac{\gamma}{2} = -\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

7.18 (a) If $f(x) = \cos(\ln x)$, find the value of $f(x)f(y) - \frac{1}{2}[f(\frac{x}{y}) + f(xy)]$. (1983)

(b) Show that the graph of the function $\cos x\cos(x+2) - \cos^2(x+1)$ is a straight line. Find its slope. (1997)

7.19 If $\tan A = \frac{1-\cos B}{\sin B}$, must $\tan 2A$ and $\tan B$ be equal to each other? (1983)

7.20 If $\alpha + \beta - \gamma = \pi$, prove that $\sin^2\alpha + \sin^2\beta - \sin^2\gamma = 2\sin\alpha\sin\beta\cos\gamma$. (1980)

7.21 (a) Suppose that $\sin^3 x \sin 3x = \sum_{m=0}^{n} c_m \cos mx$ is an identity in x, where $c_0, c_1, \ldots, c_n$ are constants and $c_n \neq 0$. Find the value of n. (1981)

(b) Find the number of triples (a_1, a_2, a_3) satisfying $a_1 + a_2 \cos 2x + a_3 \sin^2 x = 0$ for all x. (1987)

(c) Let n be an odd integer. If $\sin n\theta = \sum_{r=0}^{n} b_r \sin^r \theta$, find b_0, b_1. (1998)

(d) Prove that for every integer $n \geq 0$, there are polynomials $T_n(x)$ and $U_n(x)$ of degree n each such that $T_n(\cos\theta) = \cos n\theta$ and $U_n(\cos\theta)\sin\theta = \sin(n+1)\theta$. [*Hint:* Expand $\cos n\theta + i \sin n\theta = (\cos\theta + i \sin\theta)^n$ by the binomial theorem. These polynomials are called **Chebyshev polynomials** of the first and second kind and have many interesting applications. Compute these polynomials explicitly for $n \leq 3$.]

(e) Prove that the sequence of Chebyshev polynomials of the first kind satisfies the recurrence relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$$

for $n \geq 1$. (This gives a method for computing $T_n(x)$ starting from $T_0(x)$ and $T_1(x)$. Obtain a similar recurrence relation for the sequence $\{U_n(x)\}_{n=0}^{\infty}$.)

7.22 Find which of the parameters a, p, d and x does not affect the value of the following determinant. (1997)

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

7.23 Simplify the following expression. (1986)

$$3[\sin^4(\frac{3\pi}{2} - \alpha) + \sin^4(3\pi - \alpha)] - 2[\sin^6(\frac{\pi}{2} + \alpha) + \sin^6(5\pi - \alpha)].$$

7.24 If $\cos(x-y), \cos x$ and $\cos(x+y)$ are in H.P., then find the value of $\cos x \sec \frac{y}{2}$. (1997*)

7.25 If x is not an integral multiple of 2π, use mathematical induction to prove that $\cos x + \cos 2x + \ldots + \cos nx = \cos\dfrac{(n+1)x}{2} \sin\dfrac{nx}{2} \operatorname{cosec}\dfrac{x}{2}$. (1994)

7.26 If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are any angles, prove that the sum of all 2^n terms of the form $\cos(\pm\alpha_1 \pm \alpha_2 \ldots \pm \alpha_n)$ obtained by taking all possible combinations of the signs equals $2^n \cos\alpha_1 \cos\alpha_2 \ldots \cos\alpha_n$.

7.27 Prove that for any three angles α, β, γ, the expression

$$\cos 3\alpha \sin(\beta - \gamma) + \cos 3\beta \sin(\gamma - \alpha) + \cos 3\gamma \sin(\alpha - \beta)$$

equals the expression $4 \sin(\beta - \gamma) \sin(\gamma - \alpha) \sin(\alpha - \beta) \cos(\alpha + \beta + \gamma)$.

7.28 Prove that $\tan^2\theta + \tan^2(\theta + \frac{\pi}{3}) + \tan^2(\theta + \frac{2\pi}{3}) = 9\tan^2 3\theta + 6$. [*Hint:* First construct a cubic polynomial which has $\tan\theta, \tan(\theta + \frac{\pi}{3})$ and $\tan(\theta + \frac{2\pi}{3})$ as roots.]

7.29 If A, B, C, α, β and γ are angles such that $A + B + C = \pi$ and $\cos A = \cos\beta\sin\gamma$, $\cos B = \cos\gamma\sin\alpha$ and $\cos C = \cos\alpha\sin\beta$ then prove that $\tan\alpha\tan\beta\tan\gamma = 1$.

Chapter 8

GEOMETRY

Geometry invokes mixed reactions in schools. Typically, a pupil who is a wiz at arithmetic and algebra dreads geometry initially because geometry demands different types of abilities. But once he comes to terms with it, he usually falls in love with it, precisely for the same reason. In Mahatma Gandhi's words[1] "When, however, with much effort I reached the thirteenth proposition of Euclid, the utter simplicity of the subject was revealed to me. A subject which only required a pure and simple use of one's reasoning powers could not be difficult. Ever since that time geometry has been both easy and interesting to me."

The geometry that is meant here is the classical euclidean geometry unaided by anything like coordinates, vectors or trigonometry. For this reason it is also called '**pure geometry**'. Till 1978 it was an important part of the JEE syllabus and every year a few beautiful problems from pure geometry were asked. Later, to make room for calculus, pure geometry took a back seat. Today even if a problem from pure geometry is asked, the use of coordinates, vectors etc. is permitted. And many times these techniques do simplify the problem. Regrettably, sometimes in this process, the very charm of the problem is gone. It is to be noted, moreover, that even to justify these techniques, some pure geometry is needed. For example, in coordinate geometry, we calculate the slope of a line from any two points on it. To ensure that you always get the same value requires the properties of similar triangles.

We begin by illustrating how a clever choice of coordinates and vectors can sometimes drastically simplify a pure geometry problem. In Comment No. 12, we show how complex numbers can be used in geometry through the Argand diagrams. In Comment No. 17, we turn the tables around by showing how a problem about complex numbers can be solved by reducing it to a problem in geometry. Problems of this type are occasionally asked in JEE. Comment No. 18 deals with a problem of solid geometry.

[1] From p.13 of Gandhi's famous autobiography, '*The Story of My Experiments with Truth*'. Unlike Lokamanya Tilak, Gandhi was not known for his mathematical ability. But in a way this gives a greater relevance to his comments.

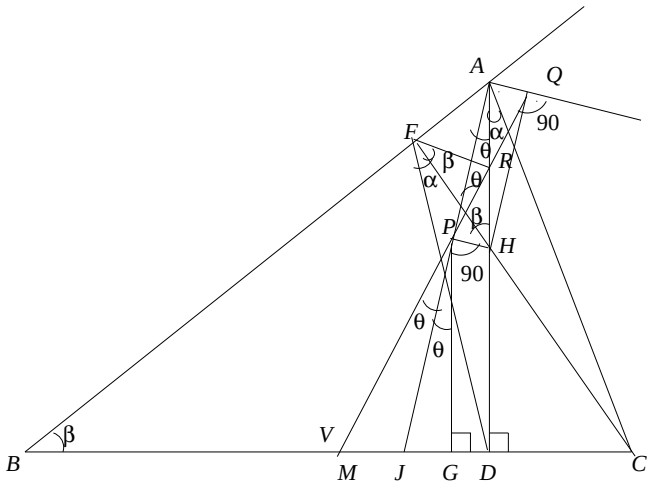
Main Problem : Perpendiculars HP and HQ are drawn from the orthocentre H of a triangle ABC to the angle bisectors at the vertex A. The line PQ meets the line BC at the point M. Then the ratio $BM : MC$ equals ..... .

First Hint: Take the angle bisectors at A as the coordinate axes.

Second Hint: Take the points B and C in the form $B = (b, mb)$ and $C = (c, -mc)$.

Third Hint: Determine the coordinates of H and thence those of P and Q.

Solution: AB, AC are equally inclined to the axis AP. So if m is the slope of AB, that of AC is $-m$. Taking B, C as suggested, the equations of the altitudes through B and C are, respectively, $y - mb = \frac{1}{m}(x - b)$ and $y + mc = -\frac{1}{m}(x - c)$. Solving simultaneously gives $H = (\frac{(1-m^2)(b+c)}{2}, \frac{(m^2-1)(b-c)}{2m})$.



As P and Q are the feet of the perpendiculars drawn from H to the coordinate axes, we get $P = (\frac{(1-m^2)(b+c)}{2}, 0)$ and $Q = (0, \frac{(m^2-1)(b-c)}{2m})$. The equation of the line PQ comes (using the two-intercepts form) as $2(b-c)x - 2m(b+c)y = (1-m^2)(b^2-c^2)$, while that of the line BC is $(b-c)(y-mb) = m(b+c)(x-b)$. Solving simultaneously we get M as $(\frac{b+c}{2}, \frac{m(b-c)}{2})$, which is precisely the mid-point of BC. So $BM : MC = 1$.

Comment No. 1:

In a more conventional form, you could be asked to *prove* that the line PQ intersects the side BC at its mid-point M. In that case, a solution by pure geometry is also possible using the nine-point circle of the triangle ABC. Let PQ and and the internal angle bisector AP meet BC at V and J respectively. From P, drop a perpendicular PG to BC. Let the diagonals AH and PQ of

the rectangle $APHQ$ bisect each other at R. Let AD and CF be the altitudes through A, C respectively. Without loss of generality, assume $\angle C \geq \angle B$ as in the figure. Let α, β, θ equal $\angle DAC, \angle B$ and $\angle JAD$ respectively. Then $\theta = \frac{1}{2}(\angle C - \angle B) = \frac{1}{2}(90^\circ - \alpha - \beta)$. (It is tacitly assumed here that $\angle C \leq 90^\circ$. Minor modifications can be made when $\angle C$ is obtuse.) Using the fact that FR and PR are medians of the right angled triangles AFH and APH respectively and cyclicity of the quadrilaterals $FDCA$ and $BDHF$ it is easy to show that $\angle RVD$ and $\angle RFD$ each equals $\alpha + \beta$. Hence, the points F, R, D, V lie on a circle. But the only circle passing through F, R, D is the nine point circle of the triangle ABC. As this circle cuts the side BC only at D and M, and $V \neq D$ (except in the degenerate case where $AB = AC$), V must coincide with M. (In the degenerate case, V, D, M all coincide.)

Comment No. 2:

The solution using only pure geometry is artistic but rather tricky and hence not very reliable, especially in a time-constrained examination because the time it takes to hit upon the key idea can vary considerably from individual to individual and even for the same individual, from problem to problem. A solution based on coordinates is comparatively surer. The calculations can be simplified considerably with a suitable choice of coordinate axes. In the present problem, for example, if we start with an arbitrary system of coordinates and take the vertices of the triangle as three points in general, say, $A = (x_1, y_1), B = (x_2, y_2)$ and $C = (x_3, y_3)$, then the coordinates of the orthocentre H come out to be quite complicated. Taking A as the origin will simplify them to some extent. Still, the subsequent work involved in finding the coordinates of the points P and Q is quite laborious as there is no easy formula for the coordinates of the foot of the perpendicular drawn from a given point to a given line. But if the line happens to be one of the coordinate axes, then this calculation is instantaneous. And this is exactly the advantage gained by taking as coordinate axes the angle bisectors at A (which are perpendicular to each other anyway). So the present problem is an excellent example of how an ingenious choice of a coordinate system simplifies a geometric problem.

Comment No. 3:

Of course, what will be a suitable system of coordinates for a given problem in geometry depends on the features of that problem. The general rule of thumb is to take the origin at a point which plays a pivotal role in the problem. For example, in the problem above, of the three vertices of the triangle ABC, the vertex A has a dominating role to play. So it pays to take A as the origin. Further, there is already a pair of mutually perpendicular lines at A, viz., the two angle bisectors at A and the problem involves dropping perpendiculars on them. So it will simplify matters considerably if these lines are taken as the coordinate axes. Indeed this is the key idea behind the solution.

On the other hand, in a problem where all the three vertices of a triangle are on par with each other, it may not be a good idea to take one of its vertices as the origin. A good illustration is the problem of showing that the three altitudes

of a triangle, say, ABC are concurrent. (JEE 1998) This is a well-known result and several pure geometric proofs of it are available in high school texts. Here all the three vertices are on par and there is not much simplification in taking one of them as the origin. In fact, in such a problem, it is sometimes advantageous *not* to take any particular point as the origin and instead let A, B, C be *any* three points, say, $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) respectively. Then the slope of the line BC is $\dfrac{y_2 - y_3}{x_2 - x_3}$ and hence that of the altitude through A is $-\dfrac{x_2 - x_3}{y_2 - y_3}$. So the equation of the altitude through A is, after minor simplification,

$$(x - x_1)(x_2 - x_3) + (y - y_1)(y_2 - y_3) = 0$$

This may seem a little complicated. But the advantage is that in finding the equations of the altitudes through B and C, this work need not be duplicated. Instead, we can use cyclical symmetry. That is, if we replace the subscripts 1,2,3 throughout by the subscripts 2,3,1 respectively in the equation above, we get the equation of the altitude through B. Similarly replacing the subscripts 1,2,3 throughout by 3,1,2 respectively, we get the equation of the third altitude, viz., that through C. These two equations are, respectively,

$$\begin{aligned} & (x - x_2)(x_3 - x_1) + (y - y_2)(y_3 - y_1) &= 0 \\ \text{and} \quad & (x - x_3)(x_1 - x_2) + (y - y_3)(y_1 - y_2) &= 0 \end{aligned}$$

The concurrency of the three altitudes is equivalent to saying that these three equations have a common solution. The most straightforward way of doing this is to solve any two of them simultaneously and show that the values of x and y so obtained also satisfy the third equation. If we follow this approach, then the common solution of the first two equations comes out, after some simplification, as

$$x = \frac{\sum (x_1x_2 + y_1y_2)(y_2 - y_1)}{\sum x_1y_2 - x_2y_1}, \quad y = \frac{\sum (x_1x_2 + y_1y_2)(x_1 - x_2)}{\sum x_1y_2 - x_2y_1}$$

(Here the notation $\sum$ is used to denote a sum of three terms, of which one is given and the other two are obtained from it by a cyclic permutation of the indices. For example, $\sum x_1y_2 - x_2y_1$ means $x_1y_2 - x_2y_1 + x_2y_3 - x_3y_2 + x_3y_1 - x_1y_3$ in the full form.)

Instead of verifying by direct substitution that these values also satisfy the third equation, we merely note that these values are cyclically symmetric in the three indices (i.e., that they remain unchanged if 1,2,3 are replaced throughout by 2,3,1 respectively). But the system of the three equations is also cyclically symmetric. So any cyclically symmetric solution of any one of the three equations is automatically a solution of the other two. Even with this short cut, this method is admittedly laborious. Its gain is that it not only proves the concurrency of the three altitudes but also identifies the coordinates of the orthocentre. But the given problem does not ask for it. It merely asks to show that the three altitudes are concurrent without necessarily identifying the point of their concurrency. So it is reasonable to expect that there must be

a simpler way out. And indeed there is, as we show now. (This point should always be kept in mind while trying a particular approach. See if you are not unnecessarily aiming too high. Although there are exceptions, normally a more modest goal demands less work.)

To show that the system above has a common solution without actually solving it, note that it is a system of three linear equations in the two unknowns x and y. So by Theorem 6 in Chapter 3, it has a common solution (or, in other words, it is consistent) if and only if the determinant of its coefficients, viz.,

$$\begin{vmatrix} x_2 - x_3 & y_2 - y_3 & -x_1(x_2 - x_3) - y_1(y_2 - y_3) \\ x_3 - x_1 & y_3 - y_1 & -x_2(x_3 - x_1) - y_2(y_3 - y_1) \\ x_1 - x_2 & y_1 - y_2 & -x_3(x_1 - x_2) - y_3(y_1 - y_2) \end{vmatrix}$$

vanishes. We could, of course, do this by brute force i.e., by expanding the determinant. But that would be too laborious. In fact, in that case, the work involved is not substantially less than solving any two of the three equations simultaneously! The elegance of the present method lies in that there is a very quick way of showing that the determinant vanishes. Simply add the second and the third rows to the first and you get a row which is identically zero!

The use of general coordinates also sometimes enables us to say something about the nature of the coordinates of some point, without actually finding them. As a good example, suppose that the vertices of a triangle ABC are all rational points. Here by a **rational point**, we mean a point of both whose coordinates (w.r.t. a fixed frame of reference) are rational numbers. We claim that the orthocentre, say H, of ABC is also a rational point. This would, of course, follow from the expressions we have given above for the coordinates of H in terms of those of the points A, B and C. Since the sum, the difference, the product and the ratio of two rational numbers (when defined) are also rational numbers, it is clear that if $x_1, y_1, x_2, y_2, x_3, x_3$ are all rational then so are x and y as given above. The point is that we could also see this without obtaining those gigantic expressions. The basic idea is that the point of intersection of two lines with equations of the form $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is a rational point if the coefficients a_1, b_1, c_1, a_2, b_2 and c_2 are all rational. This follows easily by writing the point of intersection as $(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1})$. We apply this fact to the two altitudes of the triangle ABC whose equations were written above. The coefficients of x and y as well as the constant terms in both are rational since $x_1, y_1, x_2, y_2, x_3, y_3$ are all rational. So, the orthocentre H, being the point of intersection of these two altitudes is a rational point.

But we can do even better. We need not even explicitly write down the equations of the altitudes in order to conclude that the coefficients and the constant terms in them are rational. We obtained these equations using the point−slope formula. Suppose L is a line passing through a rational point, say (h, k) and has slope m which is also rational. Then its equation is $y - k = m(x - h)$, i.e., $mx - y + k - mh = 0$. Thus we see that L is a 'rational line', i.e., a line whose equation can be written in the form $a_1x + b_1y + c_1 = 0$, where a_1, b_1 and c_1 are all rational numbers. (Note that we are not saying that *every* point

on a rational line is a rational point. This is, in fact, never true. The x-axis, for example, is a rational line. But it has many points, e.g., $(\sqrt{2}, 0)$ which are not rational. It is obvious that a line passing through two rational points is a rational line, since its slope is a rational number (or else it is a vertical line with an equation of the form $x = c$ for a rational c).

Let us now apply these simple observations to the altitudes of a triangle ABC with rational vertices. Then the line BC has a rational slope, say m. But then the slope of the the altitude through the vertex A is $-\frac{1}{m}$ which is also a rational number. Since A is a rational point, we conclude that the altitude through A is a rational line. The same reasoning shows that the other two altitudes of the triangle ABC are also rational lines. Hence, by what we proved above, their point of intersection, viz., the orthocentre H is a rational point.

By a similar reasoning, if A, B, C are rational points, then the three medians are rational lines and hence the centroid is a rational point. For the circumcentre, we observe that the perpendicular bisector of a line segment whose end-points are rational points is a rational line. Thus we have shown that if the vertices of a triangle are rational points then so are its centroid, orthocentre and circumcentre. Note, however, that the incentre need not be a rational point. (JEE 1998) This is so because an angle bisector of two rational lines need not be a rational line. (A simple counter-example is to take the lines as, say, $y = 0$ and $y = x$.)

The fact that the circumcentre of a triangle with rational vertices must be a rational point gives a quick solution to the following problem. (JEE 1997)

Prove that there can be at most two rational points on a circle with centre at $(0, \sqrt{2})$.

This follows immediately, because if there are three rational points, say A, B and C on such a circle then $(0, \sqrt{2})$, being the circumcentre of the triangle ABC would be a rational point. But this would mean that $\sqrt{2}$ is rational, a contradiction. (See Chapter 4, Comment No. 17 for a well-known proof of the irrationality of $\sqrt{2}$. However, in a problem like this, a candidate is not expected to give a proof of the irrationality of $\sqrt{2}$. The focus of the problem is geometry and not number theory. Actually, $\sqrt{2}$ has no special role in the problem. Any irrational number in its place would do.)

The study of rational points on various curves is an important (and highly challenging) area of mathematical research. It is hardly possible to even mention here any of its significant results. Suffice it to say that very recently, a problem posed by **Fermat** which remained unsolved for three centuries despite intense efforts, was finally solved through this type of research. It is very easy to find three positive integers a, b, c such that $a^2 + b^2 = c^2$. Because of the Pythagoras theorem, such a triple (a, b, c) is called a **Pythagorean triple** (see Exercise (4.22)). The simplest example of a Pythagorean triple is $(3, 4, 5)$. $(5, 12, 13)$ is another Pythagorean triple. Indeed, if u, v are any two positive integers with $u > v$, then $(u^2 - v^2, 2uv, u^2 + v^2)$ is easily seen to be a Pythagorean triple. As there is no other restriction on u and v, this in particular means that there are infinitely many Pythagorean triples. Now what happens if we replace the exponent 2 by some higher value? Can we, for example, find positive integers

a, b, c such that $a^3 + b^3 = c^3$? Fermat claimed, without proof, that this is never possible, i.e., for no integer $n > 2$ can we find three positive integers a, b, c such that $a^n + b^n = c^n$. We can rewrite this equation as $p^n + q^n = 1$, where $p = \frac{a}{c}$ and $q = \frac{b}{c}$. Then p and q are rational numbers and so Fermat's statement is equivalent to saying that there are no non-trivial rational points on the curve whose equation is $x^n + y^n = 1$.

The fact that there are infinitely many Pythagorean triples gives a solution to the following problem.

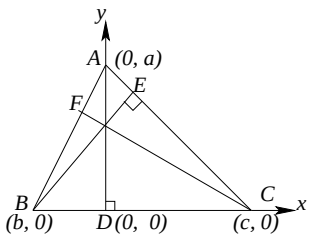
Prove that if a circle of rational radius has three rational points on it, then there are infinitely many rational points on it.

For a solution, suppose (h, k) is the centre and r the radius of the circle. The presence of three rational points on the circle implies that (h, k) is a rational point as noted above. Also r is given to be rational. Now let (a, b, c) be any Pythagorean triple. Then $(h + \frac{ra}{c}, k + \frac{rb}{c})$ is a rational point on the given circle. Clearly distinct Pythagorean triples give rise to distinct points. Since there are infinitely many Pythagorean triples, there are infinitely many rational points on the given circle.

Comment No. 4:

If we insist upon proving the concurrency of the three altitudes, say, AD, BE and CF of a triangle ABC by choosing a particular frame of coordinate axes, then, as observed above, there is little to be gained by taking any one of the vertices as the origin. Note, however, that at each of the three points D, E and F, there is already a pair of mutually perpendicular lines, viz., a side of the triangle and the altitude on it. This fact can be used advantageously. Let us, for example, take D as the origin, the line BC as the x-axis and the line AD as the y-axis. Then the coordinates of A are of the form $(0, a)$ for some real number a, while those of B and C are of the form $(b, 0)$ and $(c, 0)$ respectively. The slope of AC is $-\frac{a}{c}$ and hence that of BE is $\frac{c}{a}$. Hence the equation of BE is $ay = c(x - b)$. Interchanging the roles of B and C gives the equation of CF as $ay = b(x - c)$. The third altitude, viz., AD is the y-axis with equation $x = 0$.

To prove the concurrency of these three altitudes we can once again consider the determinant of coefficients. But in the present case, the common point of BE and CF can be calculated very easily by direct calculation. It comes out as $(0, -\frac{bc}{a})$ which obviously lies on the y-axis, i.e., on the line AD. (Actually, the y-coordinate of this point is not needed here, but it gives a handy formula for the distance of the orthocentre from D.)



Comment No. 5:

There is also an elegant way using vectors[1] to prove the concurrency of

[1]Vectors will be studied in greater detail in Chapter 21 where we shall see more examples where vector methods are applied to tackle problems of geometry. For the moment all that is

the three altitudes of a triangle ABC. Let O be the point of intersection of the altitudes through B and C. It suffices to show that the line OA is perpendicular to the side BC. Now let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ be, respectively, the position vectors of the points A, B, C w.r.t. O. (Or, in other words, $\mathbf{a}$ is the vector $\overrightarrow{OA}$ etc.) Then, $\mathbf{b}-\mathbf{a}$ is a vector parallel to the line AB. But AB is perpendicular to OC. Hence the dot product $(\mathbf{b}-\mathbf{a})\cdot\mathbf{c}$ vanishes, i.e., $\mathbf{b}\cdot\mathbf{c}=\mathbf{a}\cdot\mathbf{c}$. By a similar reasoning, perpendicularity of CA with OB gives $\mathbf{c}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{b}$. These two equations together imply $\mathbf{b}\cdot\mathbf{a}=\mathbf{c}\cdot\mathbf{a}$. But this is equivalent to saying that OA is perpendicular to BC as was to be shown.

Comment No. 6:

The reason the solution based on vectors is so short is basically that the feature of the dot product provides a succinct characterisation of perpendicularity in terms of vectors. In coordinate geometry, perpendicularity can be characterised in terms of products of slopes. But the expressions for the slopes become rather messy if expressed in terms of the coordinates of points (which are real numbers). But if we represent points of the plane by complex numbers then the idea of perpendicularity can also be expressed succinctly in terms of complex numbers and their real parts. Let z and w denote the complex numbers representing points P and Q in an Argand diagram with the origin O represented by the complex number 0. Then the lines OP and OQ are perpendicular to each other if and only if the complex number $z\bar{w}$ is purely imaginary, i.e., its real part $\mathrm{Re}(z\bar{w})=0$. (A bar over a complex number denotes its complex conjugate.) If we write z and w in terms of their real and imaginary parts, say, $z=x+iy$ and $w=u+iv$ then this means $xu+yv=0$. This is no different than saying that the slopes of the lines OP and OQ are negative reciprocals of each other. But it looks much more compact and is convenient to handle. More generally, if the complex numbers z_1, z_2, w_1, w_2 represent points P_1, P_2, Q_1, Q_2 in a plane, then the lines P_1P_2 and Q_1Q_2 are perpendicular to each other if and only if $(z_1-z_2)(\bar{w_1}-\bar{w_2})$ is purely imaginary. (See Exercise (2.24).)

With this background, it is not surprising that the concurrency of the three altitudes of a triangle ABC can also be proved using complex numbers. As in the case of the solution based on vectors, take the origin O to be the point of intersection of the altitudes through B and C. We have to prove that OA is perpendicular to BC. Represent the vertices A, B, C by the complex numbers a, b, c respectively. Perpendicularity of OB with CA gives that $b(\bar{a}-\bar{c})$ is purely imaginary while that of OC with AB gives that $c(\bar{a}-\bar{b})$ is purely imaginary. Subtracting, $(b-c)\bar{a}-(b\bar{c}-\bar{b}c)$ is purely imaginary. If we denote $b\bar{c}$ by z, then $\bar{b}c$ is simply $\bar{z}$. But $z-\bar{z}$ is purely imaginary for any complex number z. So $(b\bar{c}-\bar{b}c)$ is purely imaginary. Thus we get that $(b-c)\bar{a}$ is purely imaginary and hence BC is perpendicular to OA.

Comment No. 7:

We gave two proofs of the concurrency of the altitudes using coordinates. In

needed is the concept of the position vector of a point and some elementary properties of the dot product.

one of them the origin was unrestricted while in the other it was taken to be the foot of one of the altitudes. In the solutions based on vectors and on complex numbers, on the other hand, it was taken to be the point of intersection of two altitudes. This suggests that we could have given yet another solution using coordinates in which the origin is taken at the point of intersection of two altitudes. This is indeed possible. All we have to do is to take A, B, C as $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) respectively, consider the slopes of the lines OA, OB, OC and of the sides BC, CA, AB and apply the condition for perpendicularity. This approach, is however, not very obvious. We could think of it only after seeing the solutions based on vectors and complex numbers.

Comment No. 8:

The problem of proving the concurrency of the three altitudes of a triangle is an excellent example of the abundance of methods to solve problems in geometry. Whenever a problem can be done by several methods, it is a good idea to compare the relative merits and demerits of those methods and to try to analyse the reason why a particular method works better for that problem. Such an analysis can often help you choose the right method in tackling a new problem when there are several possible methods that give some hope. Obviously, complex numbers can be used only for problems in a plane. The method of coordinates is, by and large, the most general and straightforward, sometimes to the point of being dull. But with a clever choice of the coordinate system, there is plenty of room for applying brains in it. The method of vectors comes handy when the problem involves parallel or perpendicular lines. Trigonometry is yet another useful tool to tackle problems in geometry. But we postpone its discussion. (See Chapters 10 and 11.)

Comment No. 9:

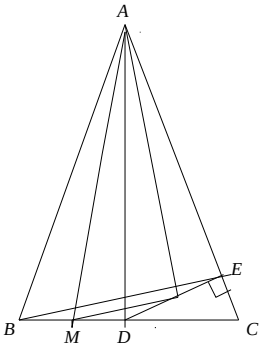
The Main Problem is a good illustration of a problem which can be solved rather easily with a suitable choice of coordinates, but whose pure geometry solution is rather tricky. Here is another problem of a similar spirit. (JEE 1989)

Suppose ABC is a triangle in which $AB = AC$. Suppose D is the mid-point of BC, E is the foot of the perpendicular drawn from D to AC and F is the midpoint of DE. Prove that AF is perpendicular to BE.

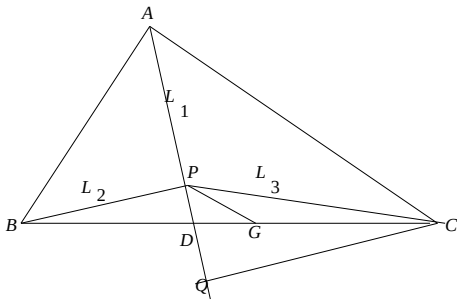
For a purely geometric solution, note that ΔAED and ΔADB are similar with F and M being mid-points of corresponding sides. (See the figure on the next page.) So $\angle MAD = \angle FAE$ and hence $\angle FAM = \angle EAD$. Also ΔAEF and ΔADM are similar, giving $\dfrac{AF}{AM} = \dfrac{AE}{AD}$. So ΔFAM and ΔEAD are similar. But then $\angle AFM = \angle AED = 90°$. The solution is completed by noting that FM is parallel to BE.

For a solution using coordinates, choose axes so that $D = (0, 0), C = (a, 0), B = (-a, 0), A = (0, b)$. Let $F = (h, k)$. Then $E = (2h, 2k)$. Perpendicularity of DE and AC gives $ah = bk$. Also since E lies on AC, we get $a(2k - b) + 2bh = 0$. A little calculation now gives $\frac{k-b}{h}\frac{2k}{2h+a} = -1$ which means $AF \perp BE$.

A solution using vectors is also possible. We must prove $\overrightarrow{BE} \cdot \overrightarrow{AF} = 0$. Since $\overrightarrow{BE} = \overrightarrow{BD} + \overrightarrow{DE}$ and $\overrightarrow{AF} = \overrightarrow{AD} + \overrightarrow{DF}$, $\overrightarrow{BD} = \overrightarrow{DC}$ and $\overrightarrow{AD} \cdot \overrightarrow{DC} = 0$, $\overrightarrow{BE} \cdot \overrightarrow{AF}$ equals $\overrightarrow{BD} \cdot \overrightarrow{DF} + \overrightarrow{DE} \cdot \overrightarrow{AD} + \overrightarrow{DE} \cdot \overrightarrow{DF}$. Further writing $\overrightarrow{AD}$ as $\overrightarrow{AE} + \overrightarrow{ED}$ and $\overrightarrow{DC}$ as $\overrightarrow{DE} + \overrightarrow{EC}$ and noting that $\overrightarrow{DE} \cdot \overrightarrow{AE} = 0 = \overrightarrow{DE} \cdot \overrightarrow{EC}$, this further reduces to $\overrightarrow{DE} \cdot \overrightarrow{DF} + \overrightarrow{DE} \cdot \overrightarrow{ED} + \overrightarrow{DE} \cdot \overrightarrow{DF}$ which vanishes since $\overrightarrow{ED} = -\overrightarrow{DE} = -2\overrightarrow{DF}$.



Comment No. 9



Comment No. 10

Comment No. 10:

Here is a more challenging problem. (JEE 2000)

ABC and $A'B'C'$ are triangles in the same plane. If the perpendiculars from A, B and C to $B'C', C'A'$ and $A'B'$ respectively are concurrent, prove that so are the perpendiculars from A', B' and C' to BC, CA and AB respectively.

For a solution based on pure geometry, let the perpendiculars from A, B and C to $B'C', C'A'$ and $A'B'$ lie along the lines L_1, L_2 and L_3 respectively, meeting at P. Let the perpendiculars from $A^{'}, B^{'}, C^{'}$ to BC, CA, AB lie along $L_1^{'}, L_2^{'}, L_3^{'}$ respectively. Let the line through C parallel to L_2 meet L_1 at Q and let the line through Q parallel to AB meet BC at G. Let L_1 meet BC at D.

Similarity of the triangles PDB and QDC gives $PD.DC = DQ.BD$. But similarity of the triangles BDA and GDQ gives $BD.DQ = DA.DG$. Hence $PD.DC = DA.DG$ or $\dfrac{DP}{DG} = \dfrac{DA}{DC}$, whence $PG//AC$. To finish the proof, show that $\Delta A^{'}B^{'}C^{'}$ is similar to ΔCPQ and further that its vertical angles are divided by $L_1^{'}, L_2^{'}, L_3^{'}$ in exactly the same way as the vertical angles of ΔCPQ are divided by CG, PG, QG. Since CG, PG, QG are concurrent so are $L_1^{'}, L_2^{'}, L_3^{'}$.

For a proof using co-ordinates, there is no obvious particular choice of coordinates that will simplify the solution. In fact, in view of the symmetry of the problem it is not a good idea to take any one of the points involved as the origin. It is better to take general coordinates. So let the coordinates of the points $A, B, C, A^{'}, B^{'}, C^{'}$ be, respectively, $(x_1, y_1), (x_2, y_2), (x_3, y_3)$, $(x_1^{'}, y_1^{'}), (x_2^{'}, y_2^{'}), (x_3^{'}, y_3^{'})$. Then the equation of L_1 is $(x - x_1)(x_2^{'} - x_3^{'}) + (y - y_1)(y_2^{'} - y_3^{'}) = 0$. Similarly write the equations of $L_2, L_3, L_1^{'}, L_2^{'}, L_3^{'}$. Concurrency

of L_1, L_2, L_3 implies that the determinant

$$\begin{vmatrix} x_2^{'} - x_3^{'} & y_2^{'} - y_3^{'} & x_1(x_2^{'} - x_3^{'}) + y_1(y_2^{'} - y_3^{'}) \\ x_3^{'} - x_1^{'} & y_3^{'} - y_1^{'} & x_2(x_3^{'} - x_1^{'}) + y_2(y_3^{'} - y_1^{'}) \\ x_1^{'} - x_2^{'} & y_1^{'} - y_2^{'} & x_3(x_1^{'} - x_2^{'}) + y_3(y_1^{'} - y_2^{'}) \end{vmatrix}$$

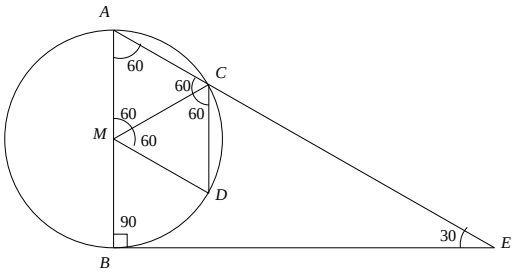
vanishes. Adding the rows, this is equivalent to the vanishing of the sum $\sum x_1(x_2^{'} - x_3^{'}) + y_1(y_2^{'} - y_3^{'})$. After re-arranging, this sum is easily seen to equal the sum $-(\sum x_1^{'}(x_2 - x_3) + y_1^{'}(y_2 - y_3))$, whose vanishing implies the vanishing of a similar determinant, proving concurrency of $L_1^{'}, L_2^{'}, L_3^{'}$. (As explained earlier, in problems like this, $\sum$ is used to denote a sum of three terms, of which one is given and the other two are obtained from it by cyclic permutation of the indices. For example, $\sum x_1 y_2^{'}$ means $x_1 y_2^{'} + x_2 y_3^{'} + x_3 y_1^{'}$.)

A proof using vectors is the shortest. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{a}^{'}, \mathbf{b}^{'}, \mathbf{c}^{'}$ be the position vectors of $A, B, C, A^{'}, B^{'}, C^{'}$ from P. Since L_1 is perpendicular to $B^{'}C^{'}$, we get $\mathbf{a}.(\mathbf{b}^{'} - \mathbf{c}^{'}) = 0$, or $\mathbf{a}.\mathbf{b}^{'} = \mathbf{a}.\mathbf{c}^{'}$. Similarly $\mathbf{b}.\mathbf{a}^{'} = \mathbf{b}.\mathbf{c}^{'}$ and $\mathbf{c}.\mathbf{a}^{'} = \mathbf{c}.\mathbf{b}^{'}$. Now assume that $L_1^{'}, L_2^{'}$ meet at R. Let $\mathbf{r}$ be the position vector of R (from P). Since $A^{'}, R$ lie on $L_1^{'}$ which is $\perp$ to BC we get $(\mathbf{a}^{'} - \mathbf{r}).(\mathbf{b} - \mathbf{c}) = 0$, or $\mathbf{r}.(\mathbf{b} - \mathbf{c}) = \mathbf{a}^{'}.\mathbf{b} - \mathbf{a}^{'}.\mathbf{c}$. Similarly $\mathbf{r}.(\mathbf{c} - \mathbf{a}) = \mathbf{b}^{'}.\mathbf{c} - \mathbf{b}^{'}.\mathbf{a}$. Adding and using the earlier equations, $\mathbf{r}.(\mathbf{a} - \mathbf{b}) = \mathbf{c}^{'}.(\mathbf{a} - \mathbf{b})$, whence $RC^{'} \perp AB$.

Comment No. 11:

In all the examples we considered so far, a solution based on pure geometry was subject to hitting a certain idea and hence rather tricky. Sometimes, however, there is something in the problem itself which makes it easier to hit upon the key idea. In that case, a pure geometry solution is not only elegant but also practicable. In the context of the Joint Entrance Examination, after the introduction of calculus into the syllabus, pure geometry is somewhat de-emphasised. As a result, there are hardly any problems which are *required* to be solved by pure geometry. Still, one should keep an open mind and see if the problem itself provides a clue to applying methods of pure geometry. Here is a good illustration of such a problem. (JEE 1980)

AB is a diameter of a circle. CD is a chord parallel to AB and $2CD = AB$. The tangent at B meets the line AC produced at E. Prove that $AE = 2AB$.



Here the trick is to observe that the triangle ABC is a right angled triangle with hypotenuse AE. So the assertion that $AE = 2AB$ would follow if

we can show that $\angle AEB = 30°$, or equivalently, that $\angle BAC = 60°$. The latter is obviously easier since we can look at $\angle BAC$ as $\angle MAC$, where M is the centre of the circle. Join MC, MD. Then CD, MC and MD all equal the radius of the circle. So the triangle MCD is equilateral and hence $\angle CMD = 60°$.

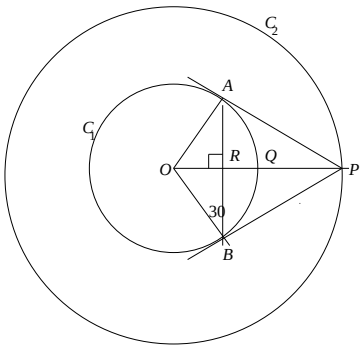
Further, since CD is given to be parallel to AB, $\angle AMC = \angle DCM$. Hence $\angle AMC = 60°$. But $MA = MC$. So MAC is an isosceles triangle with one angle equal to $60°$. Hence it is equilateral and so $\angle MAC = 60°$. As noted before, this implies $AE = 2AB$.

For a proof using coordinates one can take the radius of the circle as 1 without loss of generality, M as the origin, MN as the x-axis and MA as the y-axis. Since CD is of unit length and is bisected by the x-axis, the point C has y-coordinate $\frac{1}{2}$. This gives $C = (\frac{\sqrt{3}}{2}, \frac{1}{2})$. Also $A = (0, 1)$ and the tangent at B is $y = -1$. The solution can be completed by finding the equation of the line AC and then solving it simultaneously with $y = -1$ so as to get the coordinates of E as $(2\sqrt{3}, -1)$. By a direct calculation, AE has length 4, which is twice that of AB.

Here the reason the pure geometry solution worked so nicely was the property of a $30° - 60° - 90°$ triangle, viz., that in a such a triangle the hypotenuse is twice the shortest side. Usually, in problems involving such a triangle, a pure geometry solution is the best. Here is another example of a problem based on this property. (JEE 1998)

C_1 and C_2 are two concentric circles, the radius of C_2 being twice that of C_1. From a point P on C_2, tangents PA and PB are drawn to C_1. Prove that the centroid of the triangle PAB lies on C_1.

Let O be the common centre of the two circles. By symmetry, $OP \perp AB$. Let OP cut C_1 and AB at Q and R respectively. Then in the right-angled triangle OAP, $OA = \frac{1}{2}OP$. So $\angle OPA = 30°$. Hence in the right-angled triangle ORA, $\angle OAR = 30°$. So $OR = \frac{1}{2}OA$. Hence $RP = \frac{3}{2}OA = \frac{3}{2}QP$, showing that Q is the centroid of ΔABP.



For a solution using co-ordinates, we again take C_1 as $x^2 + y^2 = 1$. Let $P = (x_2, y_2)$ and let $R = (x_1, y_1)$ be the mid-point of AB. Then the equation of AB as the chord of contact of P, is $xx_2 + yy_2 = 1$. But in terms of the mid-point (x_1, y_1), the equation of AB is $xx_1 + yy_1 = x_1^2 + y_1^2$. Comparing the two equations, $\dfrac{x_1}{x_2} = \dfrac{y_1}{y_2} = \dfrac{x_1^2 + y_1^2}{1} = k$ (say). Thus O, R, P are collinear and $x_2^2 + y_2^2 = 4$ determines $k = \dfrac{1}{4}$. So $R = \left(\dfrac{x_2}{4}, \dfrac{y_2}{4}\right)$. Since Q, the point of

intersection of OP with C_1 is $\left(\frac{x_2}{2}, \frac{y_2}{2}\right)$, it follows that $QP = 2QR$. So Q is the centroid of ΔABP.

Note that the solution using coordinates would also work with only minor changes if C_1, C_2 are concentric ellipses with equations $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ respectively. It would thus appear that the method of coordinates is superior to that of pure geometry. This is to be expected because the latter is based on certain special simplifying features such as the properties of a $30° - 60° - 90°$ triangle. With coordinates, on the other hand, it hardly matters which particular triangle you are dealing with. You mechanically write down various equations, solve them and get the answer.

In the present case, however, there is an ingenious way to show how the solution for the problem about concentric circles can also be made applicable to the similar problem about concentric ellipses. It is based on the idea of what is called a **linear transformation** of one plane to another. Suppose we are given two ellipses, say E_1 and E_2 in the (x, y)-plane with equations $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ respectively. Also let C_1 and C_2 be the circles in the (u, v)-plane with equations $u^2 + v^2 = 1$ and $u^2 + v^2 = 4$ respectively. Let T be the transformation defined by $u = \frac{x}{a}, v = \frac{y}{b}$. Then T maps points of the (x, y)-plane onto those of the (u, v)-plane. Moreover it takes the ellipses E_1, E_2 to the circles C_1, C_2 respectively. A transformation like T takes straight lines to straight lines as is very easy to show. It need not preserve angles. For example, the angle between the x-axis and the line $y = x$ is $45°$. Their images under T are respectively, the u-axis and the line $bv = au$. And the angle between them is $\tan^{-1} \frac{a}{b}$ which is not equal to $45°$ (unless $a = b$). Still, it can be shown that the transformation preserves tangency and also the centroids of triangles. Consequently, if we solved the problem above for the pair of concentric circles in the (u, v)-plane by the method of pure geometry as we did above, then because of the transformation T, we get a solution to the corresponding problem about the ellipses E_1 and E_2 in the (x, y)-plane.

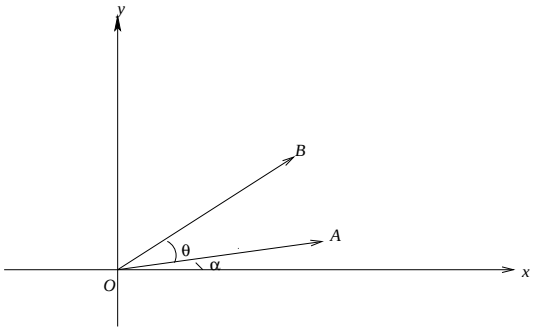
Such sophisticated arguments are hardly expected at the level of the Joint Entrance Examination. But surely, you may encounter them in your subsequent studies and so there is no harm in knowing them. What is more important is the idea of transforming a given problem into a similar but simpler problem. If your mind is trained to look for such transformations, you stand to benefit in many other problems which are of a totally different nature. For example, a complicated integral can be reduced to a manageable one by a clever substitution. (See Chapter 24, Comment No. 5(iv) for more on the use of transform technique.)

Comment No. 12:

We now describe a special feature of the complex numbers which makes them a more handy tool than coordinates or vectors in certain problems. To

understand this feature, suppose AOB is an isosceles triangle with $OA = OB$ and let $\theta = \angle BOA$. We can express this in terms of vectors, or in terms of coordinates or in terms of complex numbers also. Let us see the merits and demerits of each. In terms of vectors we shall write a system of two equations, viz., $| \overrightarrow{OA} | = | \overrightarrow{OB} |$ and $\overrightarrow{OA} \cdot \overrightarrow{OB} = | \overrightarrow{OA} | \, | \overrightarrow{OB} | \cos\theta$. Unfortunately, this system cannot be easily solved so as to express the vector $\overrightarrow{OB}$ in terms of the vector $\overrightarrow{OA}$ and the angle θ. We can say that $\overrightarrow{OB}$ is obtained from $\overrightarrow{OA}$ by a rotation around O through the angle θ. But there is no easy expression for a vector obtained in this way. In other words, even though we can easily express θ in terms of the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$, there is no handy way to express the vector $\overrightarrow{OB}$ in terms of the vector $\overrightarrow{OA}$ and the angle θ.

With coordinates the situation is better. Taking O as the origin for simplicity, suppose $A = (x_1, y_1)$ and (x_2, y_2). Then we are given two equations, viz., (i) $x_1^2 + y_1^2 = x_2^2 + y_2^2$ and (ii) $\cos\theta = \dfrac{x_1x_2 + y_1y_2}{\sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2}}$, which are obtained, respectively, from the length formula and the angle formula. We can solve this system of two equations so as to express x_2 and y_2 in terms of x_1, y_1 and θ. The easiest way to do this is by using a little trigonometry. Let r denote the common value of $\sqrt{x_1^2 + y_1^2}$ and $\sqrt{x_2^2 + y_2^2}$. Let α be the angle the line OA makes with the (positive) x-axis as shown in the figure below.



Clearly, $x_1 = r\cos\alpha$ and $y_1 = r\sin\alpha$. (In the language of polar coordinates this is the same thing as saying that the polar coordinates of A are r and α.) Then $\theta + \alpha$ is the angle between the line OB and the positive x-axis and so we get $x_2 = r\cos(\theta + \alpha)$ and $y_2 = r\sin(\theta + \alpha)$ which readily gives

$$x_2 = \cos\theta\, x_1 - \sin\theta\, y_1 \tag{1}$$

$$y_2 = \sin\theta\, x_1 + \cos\theta\, y_1 \tag{2}$$

(Here it is assumed that the rotation around O which takes A to B is counter-clockwise. If it is clockwise, we replace θ by $-\theta$. In terms of matrices introduced in Chapter 2, Comment No. 20, these two equations are together equivalent to a single matrix equation, viz.,

$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \tag{3}$$

But that is not needed here.) In case the point O is not the origin but some point (x_0, y_0) then instead of (1) and (2) we have,

$$x_2 = x_0 + \cos\theta\,(x_1 - x_0) - \sin\theta\,(y_1 - y_0) \tag{4}$$

$$y_2 = y_0 + \sin\theta\,(x_1 - x_0) + \cos\theta\,(y_1 - y_0) \tag{5}$$

Thus we are able to express the coordinates of the point B in terms of θ and the coordinates of A (along with those of O in case it is not the origin). This is certainly an advantage over vectors. But the expressions (1) and (2) are rather complicated and (4) and (5) are still more so.

With complex numbers, the situation is the best. Suppose first that the origin is at O and let the complex numbers z_1, z_2 represent the points A, B respectively. Then the statement that B is obtained from A by a (counter-clockwise) rotation through an angle θ can be expressed very succinctly by writing $z_2 = e^{i\theta} z_1$. This is obtained by writing z_1 as $re^{i\alpha}$, where r and α are respectively, the absolute value and the argument of z_1. In fact this single equation is equivalent to (1) and (2) together as one can see by writing $e^{i\theta}$ as $\cos\theta + i\sin\theta$ and then resolving into the real and imaginary parts. Nor is the proof substantially different. (The difference is more verbal than mathematical. We are merely renaming the polar coordinates r and α as the absolute value and the argument.) But see the advantage gained because of complex multiplication. In case the point O is represented not by 0, but by some complex number z_0, then we have

$$z_2 = z_0 + e^{i\theta}(z_1 - z_0) \tag{6}$$

which is equivalent to (and can be derived from) (4) and (5) together. (Once again, if the rotation around O which takes A to B is clockwise then in (6) we replace $e^{i\theta}$ by $e^{-i\theta}$. Note that regardless of what θ is, these two numbers are always the reciprocals of each other.)

The utter simplicity of (6) makes complex numbers the perfect tool in problems involving rotations. Even when a problem does not ostensibly involve any rotation, certain concepts can be expressed in terms of rotations. For example, as we just saw, instead of saying that AOB is an isosceles triangle with $OA = OB$, we can say that B is obtained from A by a rotation around the point O. If, further, the triangle is equilateral then we can put $\theta = \frac{\pi}{3}$ in (6). Taking cubes of both the sides and keeping in mind that $(e^{\frac{i\pi}{3}})^3 = e^{i\pi} = -1$, we get that if the complex numbers z_0, z_1, z_2 are the vertices of an equilateral triangle, then

$$(z_2 - z_0)^3 + (z_1 - z_0)^3 = 0 \tag{7}$$

If we factor the L.H.S. as a sum of two cubes, one of the factors is $z_1 + z_2 - 2z_0$. If this is 0, it would mean that z_0 is the mid-point of the segment joining z_1 and z_2. In particular, z_0, z_1, z_2 would be collinear, a contradiction. Hence the other factor of the L.H.S. must be 0., i.e., $(z_2 - z_0)^2 - (z_2 - z_0)(z_1 - z_0) + (z_1 - z_0)^2 = 0$. If we expand and simplify this we get (JEE 1983, with $z_0 = 0$),

$$z_0^2 + z_1^2 + z_2^2 = z_0 z_1 + z_1 z_2 + z_2 z_0 \tag{8}$$

Conversely, from (8) we easily get (7) which can also be written as $\left(\frac{z_2 - z_0}{z_1 - z_0}\right)^3 = -1$. The number -1 has only three complex cube roots, viz., $-1, e^{\frac{i\pi}{3}}$ and $e^{-\frac{i\pi}{3}}$. So, $\frac{z_2-z_0}{z_1-z_0}$ must equal one of these three complex numbers. In the first case, $z_0 = \frac{z_1+z_2}{2}$, which, along with (8) is easily seen to imply that $(z_1 - z_2)^2 = 0$, leading to the degenerate case where $z_1 = z_2 = z_3$. In either of the two remaining cases, (6) holds with θ equal to either $\frac{\pi}{3}$ or $-\frac{\pi}{3}$. In either case the triangle is equilateral. Thus (8) is a characterisation of equilateral triangles. (Another way to prove the converse is to solve (8) as a quadratic in z_0, giving $z_0 - (\frac{z_1+z_2}{2}) = \pm\frac{i\sqrt{3}}{2}(z_1 - z_2)$. Geometrically, this means that the point z_0 lies on the perpendicular bisector of the line segment joining z_1 to z_2 and is at a distance $\frac{\sqrt{3}}{2}|z_1 - z_2|$ from it. But this means that the triangle with vertices at z_0, z_1 and z_2 is equilateral. Yet another method will be given in the exercises.)

There are many other ways to derive (8) for an equilateral triangle with vertices at z_0, z_1, z_2. The simplest is probably to write $\overline{z_0} - \overline{z_1}$ as the ratio $\frac{|z_0-z_1|^2}{z_0-z_1}$. We then write $\overline{z_1} - \overline{z_2}$ and $\overline{z_2} - \overline{z_0}$ as similar ratios and simply add the three ratios, keeping in mind that their numerators are equal. This gives

$$\frac{1}{z_0 - z_1} + \frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_0} = 0$$

which, after simplification, gives (8).

But the derivation of (8) we have given using (6) is instructive because (6) can be applied to other situations as well. Here are a couple of illustrations.

Suppose the complex numbers z_1, z_2, z_3 are the vertices of an equilateral triangle with circumcentre z_0. Prove that $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$. (JEE 1981)

Without loss of generality suppose that z_1, z_2, z_3 are encountered as the circumcircle is traversed counter-clockwise. Then we can apply (6) twice, once with $\theta = \frac{2\pi}{3}$ and then with $\theta = -\frac{2\pi}{3}$ to get

$$z_2 - z_0 = e^{\frac{2\pi i}{3}}(z_1 - z_0) \text{ and } z_3 - z_0 = e^{-\frac{2\pi i}{3}}(z_1 - z_0)$$

Squaring and adding these two equations we get

$$z_2^2 + z_3^2 - 2z_0(z_2 + z_3) + 2z_0^2 = (e^{\frac{4\pi i}{3}} + e^{-\frac{4\pi i}{3}})(z_1^2 + z_0^2 - 2z_0 z_1) \tag{9}$$

Since $e^{i\pi} = e^{-i\pi} = -1$, we have $e^{\frac{4\pi i}{3}} + e^{-\frac{4\pi i}{3}} = -e^{\frac{\pi i}{3}} - e^{-\frac{\pi i}{3}} = -2\cos\frac{\pi}{3} = -1$. Substituting this in (9) we get

$$z_2^2 + z_3^2 - 2z_0(z_2 + z_3) + 2z_0^2 = -(z_1^2 + z_0^2 - 2z_0 z_1) \tag{10}$$

and hence

$$z_1^2 + z_2^2 + z_3^2 = -3z_0^2 + 2z_0(z_1 + z_2 + z_3) \tag{11}$$

Since the triangle is equilateral, z_0 is also its centroid and so $z_0 = \frac{1}{3}(z_1+z_2+z_3)$. Putting this in (11) gives the result. The presentation above can be made to

appear a little neater with the use of ω, the complex cube root of unity. For, we could have written $z_2 - z_0 = \omega(z_1 - z_0)$ and $z_3 - z_0 = \omega^2(z_1 - z_0)$. Squaring, $(z_2 - z_0)^2 = \omega^2(z_1 - z_0)^2$ and $(z_3 - z_0)^2 = \omega(z_1 - z_0)^2$. The relation $1 + \omega + \omega^2$ gives $(z_1 - z_0)^2 + (z_2 - z_0)^2 + (z_3 - z_0)^2 = 0$ which is the same as (11).

The next two illustrations of the use of (6) are even simpler. (JEE 1986)

Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles, right-angled triangle with right angle at C. Show that

$$(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2). \tag{12}$$

Here we apply (6) with $\theta = \frac{\pi}{2}$ (and calling z_0 as z_3) to get $z_2 - z_3 = i(z_1 - z_3)$. Squaring, $z_2^2 + z_3^2 - 2z_2z_3 = -z_1^2 - z_3^2 + 2z_1z_3$. Adding $-2z_1z_2$ to both the sides and rearranging the terms gives (12). (For a slightly simpler solution, see the beginning of Comment No. 15.) Equally simple is the following problem.

Show that the area of the triangle in the Argand diagram formed by the complex numbers z, iz and $z + iz$ is $\frac{1}{2}|z|^2$.

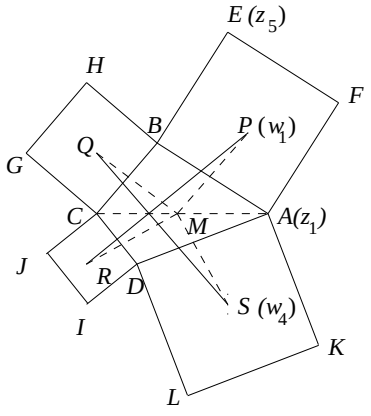
For a solution, observe that $z, z + iz, iz$ and 0 are the vertices of a parallelogram and so the desired area is the same as that of the triangle with vertices $0, z$ and iz. Since $i = e^{i\pi/4}$, from (6) we see that this triangle is an isosceles, right-angled one. Each of the equal pair of sides has length $|z|$. Hence its area is $\frac{1}{2}|z|^2$.

Comment No. 13:

This feature of the complex numbers is also convenient in proving some results in geometry where the method of coordinates turns out to be inconvenient. Here is a good illustration. We state the problem in a compact, verbal style with no symbols. This style was very much in vogue till a century ago. Those not used to it have to struggle a little just to find out what the problem means!

Prove that the centres of the squares described outwardly on the sides of any plane quadrilateral are the vertices of a quadrilateral whose diagonals are equal and at right angles to each other.

In symbols, suppose $ABCD$ is a plane quadrilateral. Now let $ABEF$ be the square described outwardly. This means that the square and the quadrilateral $ABCD$ lie on the opposite sides of the line AB. Similarly, let $BCGH$, $CDIJ$ and $DAKL$ be the squares described outwardly on the other three sides of the quadrilateral. Let P, Q, R, S be the centres of these four squares. Then we have to show that the diagonals PR and QS of the quadrilateral $PQRS$ are equal (in length) and at right angles to each other, irrespective of the shape of the original quadrilateral $ABCD$.



The problem can be done by pure geometry in a tricky manner as will be shown later. A proof using vectors or coordinates would be complicated because there is no easy way to express the coordinates of P, Q, R and S in terms of those of the vertices A, B, C and D with which we started.

However, with complex numbers, the work can be done quite easily. We show it in detail for the point P which is the centre of the square $ABEF$ as shown in (a). Denote the points A, B, C, D by the complex numbers z_1, z_2, z_3, z_4 respectively and the points P, Q, R, S by the complex numbers w_1, w_2, w_3, w_4 respectively. Since P is the midpoint of the diagonal AE of the square $ABEF$, the moment we can get the complex number, say z_5, representing E, we shall get w_1 as $\frac{z_1+z_5}{2}$. Now, BE is obtained by rotating BA around the point B through an angle of 90° degrees. So applying (6) with a slightly different notation, $z_5 = z_2 \pm i(z_1 - z_2)$, the sign depending upon whether the rotation is counter-clockwise or clockwise. In the diagram above it is shown as counter-clockwise. But we really cannot tell it from the problem itself as it would depend on which side of the line AB lies the quadrilateral as you move from A to B. If it is on the left, the rotation taking A to E is counter-clockwise. If the quadrilateral $ABCD$ lies on the right as you move from A to B then this rotation would be clockwise. We can assume, without loss of generality that the first possibility holds as otherwise we can reverse the sense in which the boundary is traversed by considering the quadrilateral $ADCB$ instead of $ABCD$ (which effectively amounts to switching the labels of B and D).

Thus we take z_5 as $z_2 + i(z_1 - z_2)$. As noted before, $w_1 = \frac{z_5+z_1}{2}$ and hence

$$w_1 = \frac{z_1 + z_2 + i(z_1 - z_2)}{2}$$

To get w_2, instead of duplicating the reasoning, we can use cyclical symmetry. Thus, w_2 can be obtained from the formula above by replacing z_1 by z_2 and z_2 by z_3. That is,

$$w_2 = \frac{z_2 + z_3 + i(z_2 - z_3)}{2}$$

$$\text{Similarly,} \quad w_3 = \frac{z_3 + z_4 + i(z_3 - z_4)}{2} \quad \text{and} \quad w_4 = \frac{z_4 + z_1 + i(z_4 - z_1)}{2}$$

It is immediate that

$$w_1 - w_3 = \frac{z_1 + z_2 - z_3 - z_4 + i(z_1 + z_4 - z_2 - z_3)}{2} \tag{13}$$

$$\text{and} \quad w_2 - w_4 = \frac{z_2 + z_3 - z_4 - z_1 + i(z_1 + z_2 - z_3 - z_4)}{2} \tag{14}$$

Clearly this implies that

$$w_2 - w_4 = i(w_1 - w_3) \tag{15}$$

By taking absolute values of the two sides of (15) we see that the diagonals QS and PR are equal in length. Also from the characterisation of perpendicularity

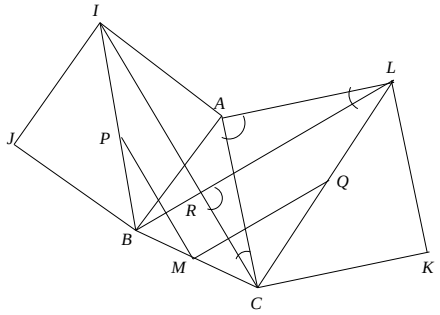
in terms of complex numbers given in Comment No. 6, we see that the lines QS and PR are at right angles to each other, because $(w_2 - w_4)(\overline{w_1} - \overline{w_3}) = i(w_1 - w_3)(\overline{w_1} - \overline{w_3}) = i|w_1 - w_3|^2$ is purely imaginary. Thus the diagonals of $PQRS$ are equal in length and at right angles to each other.

Had we worked with the coordinates of A, B, C, D, we would have got the real equivalents of (13) and (14) and hence also that of (15). But they would have been a lot more complicated. Moreover, concluding from it that QS and PR are equal and at right angles to each other would not have been easy. We would have needed separate arguments for equality of length and for perpendicularity. With complex numbers we could derive both from the single equation (15). Figuratively, the use of complex numbers allowed us to kill two birds with a single stone!

We now mention a pure geometry solution of the problem. It is based on the following two subsidiary results (or 'lemmas' as they are called.)

Lemma 1: Let M be the midpoint of the side BC of a triangle ABC and P, Q be the centres of the squares described outwardly on the sides AB, AC respectively. Then $MP = MQ$ and $MP \perp MQ$.

Proof: Let the squares on the sides be $AIJB$ and $ACKL$. Then P and Q are the midpoints of the diagonals BI and CL respectively. So it suffices to show that the segments CI and BL are equal in length and perpendicular to each other. The first assertion is immediate from the congruency of ΔCAI with ΔLAB. From this we also have $\angle ICA = \angle BLA$. Let BL meet IC at R. Since $\angle RCA = \angle RLA$, the points R, C, L, A are concyclic. So $\angle CRL = \angle CAL = 90^\circ$ which shows that $CI \perp BL$.



Lemma 2: Suppose each of the triangles AOB and $A'OB'$ is isosceles and right-angled at O. Then $AA' = BB'$ and $AA' \perp BB'$.

Proof: This is similar and left to you. (It is vital that the vertices A, O, B and A', O, B' are in either both clockwise or both anticlockwise order. Again, complex numbers would make a mincemeat of the Lemma since if we let $0, z, z'$ represent O, A, A' respectively, then $\pm iz, \pm iz'$ represent B, B' respectively.)

Now to complete the solution of the original problem, let us go back to the earlier diagram and augment it by letting M be the midpoint of the diagonal AC and joining it to P, Q, R, S (shown by dotted lines.) By Lemma 1, ΔPMQ is isosceles and right-angled at M. The same also holds for ΔRMS. Hence by Lemma 2, $PR = QS$ and $PR \perp QS$.

Comment No. 14:

A limiting case of the problem just solved deserves a mention. Suppose in the quadrilateral $ABCD$, the vertex D starts moving towards the vertex A along the side DA. Then the length of the side DA will tend to 0 while the

segment CD will tend to the segment CA. Therefore, in the limiting case, the quadrilateral $ABCD$ will reduce to a triangle ABC. The outwardly described square on the side AD will degenerate to a point A. Thus its centre S will tend to the point A and hence the limiting position of the segment QS will be QA. Therefore the assertion of the last problem would reduce to saying that the segments PR and QA are equal in length and perpendicular to each other. Using only words and no symbols, this result can be stated as follows.

The line segment joining a vertex of a triangle to the centre of the outwardly described square on the opposite side is equal and at right angles to the line segment joining the centres of the outwardly drawn squares on the other two sides.

It is, of course, also possible to prove this directly in a manner analogous to the solution of the last problem. This is left as an exercise. This exercise, along with Exercise (8.10), will give you a good drill in the use of complex numbers (specifically, use of the formula (6)) in solving geometric problems.

Comment No. 15:

There is a generalisation of (6) which is applicable to any triangle (not just one which is isosceles at z_0). Let λ be the complex number $\frac{z_2-z_0}{z_1-z_0}$ and let r and θ be, respectively, the length and the argument of λ. Then $r = \frac{|z_2-z_0|}{|z_1-z_0|}$ is the ratio of the lengths of the two sides of the triangle with included angle θ. This gives an alternate and a slightly easier derivation of (12). For, we can write $z_1 - z_2$ as $\sqrt{2}e^{i\pi/4}(z_3 - z_2)$ on one hand, and as $\sqrt{2}e^{-i\pi/4}(z_1 - z_3)$ on the other (or vice versa) and multiply these two equations together.

The equation $z_2-z_0 = \lambda(z_1-z_0)$ can also be rewritten as $z_2 = (1-\lambda)z_0+\lambda z_1$. We have thus shown that if the complex numbers z_0 and z_1 represent two (distinct) points A and B in the complex plane and if $\lambda = re^{i\theta}$ is a complex number then the complex number z_2 given by

$$z_2 = (1 - \lambda)z_0 + \lambda z_1 \tag{16}$$

represents a point C which is the vertex of a triangle in which the ratio $\frac{AC}{AB}$ equals r and the line AC is obtained from the line AB by a counter-clockwise rotation through angle θ.

This simple equation gives a quick solution to the following problem. (JEE 1985, modified)

If a, b, c and u, v, w are complex numbers representing the vertices of two triangles such that $c = (1 - \lambda)a + \lambda b$ and $w = (1 - \lambda)u + \lambda v$, where λ is a complex number, prove that the two triangles are similar.

For a solution, let A, B, C, U, V, W be the points represented by the complex numbers a, b, c, u, v, w. Let $\lambda = re^{i\theta}$. Then by (16), the ratios $\frac{AC}{AB}$ and $\frac{UW}{UV}$ both equal r. Also the included angles are θ for both the triangles. So the two triangles are similar.

Actually, because we distinguish between clockwise and anti-clockwise rotations, we have shown slightly more than similarity, viz., that the two triangles also have the same orientation, i.e., if the point C lies on the left (or right) as

you go from A to B, then the point W also lies on the left (respectively, on the right) as you go from U to V. Such a similarity is sometimes called **oriented similarity**. It is stronger than ordinary similarity. If, for example, D is the reflection of the point C in the line AB, then the two triangles ABC and ABD are similar (in fact, congruent). But they are not orientedly similar. (This fact is sometimes expressed by saying that a reflection in a straight line reverses orientation.) The problem just solved could have been slightly paraphrased. As before, let a, b, c, u, v, w represent the points A, B, C, U, V, W respectively. Assume that $c = (1-\lambda)a + \lambda b$ and $w = (1-\mu)u + \mu v$. Then the triangles ABC and UVW are orientedly similar to each other if and only if $\lambda = \mu$. It is clear that when $\lambda = \mu$, we must have

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ u & v & w \end{vmatrix} = 0 \tag{17}$$

as one can see by splitting the last column and hence the determinant as the sum of two vanishing determinants, viz.,

$$\begin{vmatrix} 1 & 1 & 1-\lambda \\ a & b & (1-\lambda)a \\ u & v & (1-\lambda)u \end{vmatrix} + \begin{vmatrix} 1 & 1 & \lambda \\ a & b & \lambda b \\ u & v & \lambda v \end{vmatrix}$$

Conversely, if (17) holds, we let $w' = (1-\lambda)u + \lambda v$. Then writing u as $u + 0$, v as $v + 0$ and w as $w' + (w - w')$ we split the last row of the determinant and get that it equals

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ u & v & w' \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 0 & 0 & w-w' \end{vmatrix}.$$

The first determinant vanishes by what we already showed. The second determinant, when expanded, equals $(b-a)(w-w')$. Thus if (17) holds we must have either $a = b$ or $w = w'$. The first possibility is ruled out as A, B are distinct points. Thus we have proved that (17) is a necessary and sufficient condition for the triangles ABC and UVW to be orientedly similar to each other.

Comment No. 16:

It is interesting to note that (16) remains valid even when $\theta = 0$ or π. In that case, λ is a positive real number and the point C lies on the ray AB for $\theta = 0$ and on the opposite ray for $\theta = \pi$. If we allow λ to take negative values as well, then we can get points on the opposite ray even with $\theta = 0$. In that case, (16) gives the parametric equation (in complex form) of the entire line AB. Various (real) values of λ correspond to different points on it. The points A and B correspond to $\lambda = 0$ and $\lambda = 1$ respectively, while $\lambda = \frac{1}{2}$ gives the midpoint of the segment AB. In general, (16) gives a point on the line AB which divides the segment AB in the ratio $\lambda : (1-\lambda)$, with the understanding that a negative ratio indicates an external division.

Equation (16) is also very convenient in representing the reflection of a point in a line. With the notations introduced while obtaining (16), suppose that D is the reflection of the point C in the line AB. Then the ratio $\frac{AD}{AB}$ is the same as the ratio $\frac{AC}{AB}$ which equals r. But the line AD is obtained from the line AB by a clockwise rotation through an angle θ. So to get the complex number corresponding to the point D, we must replace θ by $-\theta$, or equivalently, replace λ (which equals $re^{i\theta}$) by its complex conjugate $\overline{\lambda}$ (which equals $re^{-i\theta}$). Thus we get that the reflection of the point $(1-\lambda)z_0 + \lambda z_1$ in the line passing through z_0 and z_1 is simply $(1-\overline{\lambda})z_0 + \overline{\lambda}z_1$.

This simple idea gives a simple solution to the following interesting problem. Before stating the problem it is worthwhile noting the non-parametric complex form of the equation of a straight line. (A parametric form was already given in (16).) With real coordinates the equation of a straight line can be written in the form $px + qy = c$, where p, q, c are real constants. As usual we let $z = x + iy$. Let b be the complex number $p + iq$. Then $px + qy$ is simply the real part of $\overline{b}z$, which equals $\frac{1}{2}(\overline{b}z + b\overline{z})$. So the equation of the line can be written as $\overline{b}z + b\overline{z} = 2c$ or, calling $2c$ as our new c, as $\overline{b}z + b\overline{z} = c$. Here b is a non-zero complex number and c is real. Note that the vectors represented by the complex numbers b and ib are, respectively, perpendicular and parallel to the line. When necessary, we may suppose $|b| = 1$ without loss of generality.

The problem can now be stated. (JEE 1997*)

Let z_2 and z_3 be the reflections of each other into a line L whose equation is $\overline{b}z + b\overline{z} = c$. Show that $c = \overline{z_2}b + z_3\overline{b}$.

Here, in order to apply (16), we let z_0 and z_1 be any two distinct points on the line L. Then by what we said above, there is a complex number λ such that $z_2 = (1-\lambda)z_0 + \lambda z_1$ and $z_3 = (1-\overline{\lambda})z_0 + \overline{\lambda}z_1$. Therefore the expression $\overline{z_2}b + z_3\overline{b}$ equals $(1-\overline{\lambda})(b\overline{z_0} + \overline{b}z_0) + \overline{\lambda}(b\overline{z_1} + \overline{b}z_1)$. But since the points z_0, z_1 lie on the line L, each of the expressions $b\overline{z_0} + \overline{b}z_0$ and $b\overline{z_1} + \overline{b}z_1$ equals c. Therefore $\overline{z_2}b + z_3\overline{b} = (1-\overline{\lambda})c + \overline{\lambda}c = c$ as was to be shown. Note that we did not have to specify the chosen points z_0 and z_1 on L. For a fixed z_2, a different choice of z_0 and z_1 may change the value of λ. But the argument remains the same.

For a direct solution (bypassing (16)) observe that z_3 is characterised by the conditions : (i) the midpoint $\frac{1}{2}(z_1 + z_2)$ of the segment joining z_2 to z_3 lies on L and (ii) this segment is perpendicular to the line L. The first condition gives

$$\overline{b}(z_2 + z_3) + b(\overline{z_2} + \overline{z_3}) = 2c \tag{18}$$

Condition (ii) is equivalent to saying that $z_2 - z_3$ is a real multiple of b, or in other words, that the complex number $(z_2 - z_3)/b$ equals its own complex conjugate. This gives $(z_2 - z_3)/b = (\overline{z_2} - \overline{z_3})/\overline{b}$, i.e. $\overline{b}(z_2 - z_3) - b(\overline{z_2} - \overline{z_3}) = 0$. Subtracting this from (18) gives $\overline{b}z_3 + b\overline{z_2} = c$ as desired.

Thus we see that the reflection of a point z in a line L with equation $\overline{b}z + b\overline{z} = c$ is $\frac{c - \overline{z}b}{\overline{b}}$ and hence also that the foot of the perpendicular from z to L is $\frac{1}{2}(z + \frac{c - \overline{z}b}{\overline{b}}) = \frac{c + \overline{b}z - b\overline{z}}{2\overline{b}}$. See Exercise (24.98) for an interesting application of the former. The corresponding formulas in cartesian coordinates are fairly

complicated. (See Exercise (9.34)(a).) So this is yet another instance where complex numbers are more convenient for plane geometry.

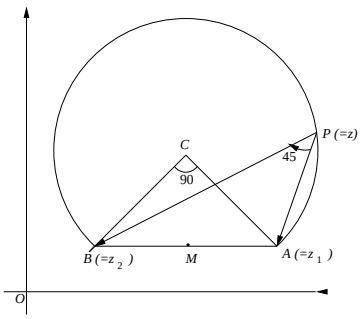
Comment No. 17:

So far we studied the applications of (6) (which is a special case of (16)) to problems in geometry. Sometimes the tables can be turned around. That is, a problem about complex numbers can be solved by converting it into a geometric problem and solving the latter by geometric means. Here is a good illustration of such a problem. (JEE 1990)

Let $z_1 = 10+6i$ and $z_2 = 4+6i$. If z is a complex number such that the argument of $(z-z_1)/(z-z_2)$ is $\pi/4$, then prove that $|z-7-9i| = 3\sqrt{2}$.

The given condition can be rewritten as $z - z_1 = re^{i\pi/4}(z-z_2) = \frac{r+ir}{\sqrt{2}}(z-z_2)$, where r is some positive real number. One can then substitute the given values of z_1 and z_2 and resolve into real and imaginary parts so as to get a system of two equations in x, y and r. Eliminating r from these two equations will give the desired result in the the real form, viz., $(x-7)^2 + (y-9)^2 = 18$. But the calculations involved in this approach will be laborious.

However, if we interpret the given condition geometrically, the problem can be done much more easily. Rewriting the given condition as $\dfrac{z-z_2}{z-z_1} = \dfrac{1}{r}e^{-i\pi/4}$ we see that the ray joining z to z_2 is obtained from the ray joining z to z_1 by a clockwise rotation through 45°. In other words, if z, z_1, z_2 represent the points P, A, B respectively, then $\angle APB = 45°$.



By a well-known result in geometry, P lies on an arc of a circle passing through A and B such that the chord AB subtends an angle of 45° at every point of the arc as shown in the figure above. (There are, in fact, two such arcs which are reflections of each other in the line AB. But because PB is obtained from PA by a clockwise rotation through 45° we must choose the upper arc.) Now let C be the centre of this circle. Then again by school geometry, $\angle ACB$ is a right angle. Also C lies on the perpendicular bisector of the segment AB. Let M be the midpoint of AB. Then CMA is a right angled, isosceles triangle. $A = z_1 = 10+6i$ and $B = z_2 = 4+6i$ gives M as $7+6i$ and then C as $(7+6i)+3i = 7+9i$. So, $|z-7-9i| = PC = CA = \sqrt{9+9} = 3\sqrt{2}$.

As another example, we have (JEE 2002, modified)

If z, w are complex numbers such that $|z-1-i| = 10$ and $|w-4-5i| = 2$, find the maximum and minimum possible values of $|z-w|$.

Again, a purely algebraic solution would be complicated. However, we observe that z lies on a circle C_1 of radius $r_1 = 10$ and centre at $z_1 = 1+i$ and w lies on a circle C_2 of radius $r_2 = 2$ and centre at $4+5i$. The distance, say d, between the centres is $|(4+5i)-(1+i)| = |3+4i| = 5$. Since $r_1 > d + r_2$, C_2

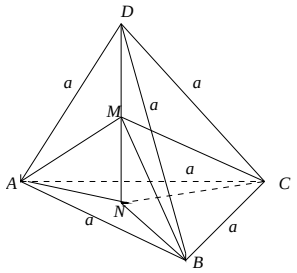
lies completely inside C_1. The maximum distance between a point P on C_1 and a point Q on C_2 will therefore occur when they both lie on the line joining the centres but on opposite sides of the centre of C_1. Hence the maximum value of $|z-w|$ is 10+5+2, i.e., 17. By a similar reasoning, the minimum possible value is $10-5-2$, i.e., 3.

Comment No. 18:

For problems of three-dimensional geometry (popularly called 'solid geometry'), the method of complex numbers naturally fails. Coordinates and vectors are, of course, available. However, sometimes pure geometry is the easiest way to tackle problems of solid geometry. Here is an example. (JEE 1980)

$ABCD$ is a regular tetrahedron. M is the midpoint of the altitude DN. Prove that MA, MB, MC are mutually perpendicular.

As the tetrahedron is regular, each face is an equilateral triangle and the foot of each altitude is the centre of the corresponding face. Let a denote the length of each edge of the tetrahedron. Then N being the centre of the equilateral triangle ABC with side a, we have



$$AN = BN = CN = \frac{2}{3}\frac{\sqrt{3}}{2}a = \frac{a}{\sqrt{3}} \tag{19}$$

Also, AND is a right-angled triangle (see the figure above) and so by (19),

$$DN^2 = AD^2 - AN^2 = a^2 - \frac{a^2}{3} = \frac{2}{3}a^2 \tag{20}$$

Since M is the midpoint of DN, $MN = \frac{1}{2}DN$. So, from the right-angled triangle ANM and (19), (20), we get

$$AM^2 = AN^2 + MN^2 = \frac{a^2}{3} + \frac{1}{4}DN^2 = \frac{a^2}{3} + \frac{a^2}{6} = \frac{a^2}{2} \tag{21}$$

By regularity of the tetrahedron, MB and MC equal MA each. So $MA^2 + MB^2 = \frac{a^2}{2} + \frac{a^2}{2} = a^2 = AB^2$. So MA and MB are perpendicular to each other. Similarly, MB, MC and MC, MA are perpendicular to each other.

EXERCISES

8.1 ABC is a triangle with $AB = AC$. D is any point on the side BC. E and F are points on the sides AB and AC respectively such that DE is parallel to AC and DF is parallel to AB. Prove that $DF+FA+AE+ED = AB+AC$ (1980)

8.2 If A, B, C, D are four points in a plane with position vectors **a, b, c, d** respectively such that $(\mathbf{a}-\mathbf{d})\cdot(\mathbf{b}-\mathbf{c}) = (\mathbf{b}-\mathbf{d})\cdot(\mathbf{c}-\mathbf{a}) = 0$. Determine how the point D is related to the triangle ABC. (1984)

8.3 Which, if any, of the following conditions, ensure the concurrency of the three lines $px+qy+r=0$, $qx+ry+p=0$ and $rx+py+q=0$?(1985)

(A) $p+q+r=0$ (B) $p^2+q^2+r^2=pq+qr+rp$ (C) $p^3+q^3+r^3=0$

8.4 Does the straight line $5x+4y=0$ pass through the point of intersection of the straight lines $x+2y-10=0$ and $2x+y+5=0$? (1983)

8.5 From a point O inside a triangle ABC, perpendiculars OD, OE, OF are drawn to the sides BC, CA, AB respectively. Prove that the perpendiculars from A, B, C to the sides EF, FD, DE are concurrent. (1978)

8.6 Two vertices of a triangle are $(5,-1)$ and $(-2,3)$. If the orthocentre of the triangle is at the origin, find the coordinates of the third vertex. (1979)

8.7 Find the orthocentre of the triangle whose vertices are located at the points $(at_1t_2, a(t_1+t_2)), (at_2t_3, a(t_2+t_3))$ and $(at_3t_1, a(t_3+t_1))$. (1983)

8.8 (a) If three complex numbers are in A.P., must they lie on a circle in the complex plane? (1985)

(b) Find a necessary and sufficient condition that the points z_1, z_2, z_3, z_4 in the complex plane are the vertices of a parallelogram taken in order. (1983)

(c) If z_1 and z_2 are two non-zero complex numbers such that $|z_1+z_2| = |z_1|+|z_2|$, find the possible values of Arg z_1− Arg z_2.

8.9 $ABCD$ is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy $BD = 2AC$. If the points D and M represent the complex numbers $1+i$ and $2-i$ respectively, then find the two possible complex numbers represented by A. (1993)

8.10 Prove that the centres of the equilateral triangles described outwardly on the sides of any triangle are vertices of an equilateral triangle. (This result is called **Napoleon's theorem**.)

8.11 Let z_1 and z_2 be the roots of the equation $z^2+pz+q=0$, where the coefficients p and q may be complex numbers. Let A and B represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha \neq 0$ and $OA = OB$, where O is the origin, prove that $p^2 = 4q\cos^2(\frac{\alpha}{2})$. (1997)

8.12 (a) Identify the loci given by the following equations for a complex z.

(i) $|\frac{z-5i}{z+5i}| = 1$. (1981) (ii) $|\frac{1-iz}{z-i}| = 1$. (1983)

(iii) $|z-i|+|z+i| = 3$ (iv) $|z-i|-|z+i| = 3$

(b) Let z_1, z_2 be fixed complex numbers and $\lambda \neq 1$ be a positive real number. Show that the locus of a complex number z satisfying $|z-z_1| = \lambda|z-z_2|$ is a circle. Can you identify a diameter of it?

8.13 Rewriting Equation (8) as $(z_0 - z_1)^2 = (z_2 - z_0)(z_1 - z_2)$ and taking the absolute values of both the sides, give an alternate proof that (8) implies that the triangle with vertices at z_0, z_1 and z_2 is equilateral,

8.14 Equation (8) can also be written as $(z_0 - z_1)^2 + (z_1 - z_2)^2 + (z_2 - z_0)^2 = 0$. What is wrong if we conclude from this that $z_0 = z_1 = z_2$ since a sum of squares can be zero only when each square vanishes?

8.15 Determine real numbers a, b such that $a+i, 1+ib$ and 0 form an equilateral triangle. (1989)

8.16 Suppose Z_1, Z_2, Z_3 are the vertices of an equilateral triangle inscribed in the circle $|Z| = 2$. If $Z_1 = 1 + i\sqrt{3}$, find Z_2 and Z_3.

8.17 If $z_1 = a+ib$ and $z_2 = c+id$ are complex numbers such that $|z_1| = |z_2| = 1$ and Re $(z_1\overline{z_2}) = 0$, and $w_1 = a+ic$ and $w_2 = b+id$, which of the following statement(s) is (are) true? (1985)

(A) $|w_1| = 1$ (B) $|w_2| = 1$ (C) Re $(w_1\overline{w_2}) = 0$

8.18 Let z_1 and z_2 be complex numbers such that $z_1 \neq z_2$ and $|z_1| = |z_2|$. If z_1 has a positive real part and z_2 has a negative imaginary part, which of the following possibilities *may* hold about $\dfrac{z_1 + z_2}{z_1 - z_2}$? (1986)

(A) it is zero (B) it is real and positive
(C) it is real and negative (D) it is purely imaginary.

8.19 If $\sin\alpha + \sin\beta + \sin\gamma = \cos\alpha + \cos\beta + \cos\gamma = 0$ then prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$.

8.20 Let $A_1, A_2, \ldots, A_m$ and $B_1, B_2, \ldots, B_n$ be the vertices of a regular m-gon and a regular n-gon inscribed in concentric circles of radii a and b respectively. Prove that the sum of the squares of the lengths of all the line segments joining a vertex of the m-gon to a vertex of the n-gon is $mn(a^2 + b^2)$.

8.21 The concept of a reflection can also be defined for a circle, say C. For simplicity, let C be the circle $\{z : |z| = r\}$, centred at O. For a point P (other than O) represented by a complex number z, let P^* be the point represented by z^*, where z^* is such that $\bar{z}z^* = r^2$. Then P^* is called the **reflection** of P into (or w.r.t.) the circle C. Prove that:

(a) P and P^* lie on the same ray from the origin and further $OP.OP^* = r^2$. (This gives a purely geometric description of P^*.)

(b) $P^{*^*} = P$ (i.e., the reflection of the reflection gives the original point back) and $P^* = P$ if and only if P lies on C.

(c) if P is outside C then P^* is the midpoint of the chord of contact of P w.r.t. the circle C while if P is inside C then P^* is the point of intersection of the tangents at the ends of the chord through P which is perpendicular to the line OP.

(d) Is P^* the reflection of P in the optical sense, that is, do the light rays originating at a point P outside C appear to have come from P^* after they are reflected by the circle C?

(e) if the centre of C is at a point z_0 (instead of at 0), then the reflection of z into C is the point $z_0 + \dfrac{r^2}{\bar{z} - \bar{z_0}}$.

(f) if C_r is the circle of radius r (> 1) centred at the point ir then the reflection of i into C_r is $\dfrac{-ri}{r-1}$ which tends to $-i$ as $r \to \infty$. What is the geometric significance of this?

8.22 Suppose z_1 and z_2 are two complex numbers regarded as vectors in the plane. Prove that their dot and cross products are given, respectively, by (i) $z_1 \cdot z_2 = \text{Re}\ (z_1\overline{z_2})$ and (ii) $z_1 \times z_2 = \text{Im}\ (\overline{z_1}z_2)\mathbf{k}$, where $\mathbf{k}$ is a unit vector perpendicular to the complex plane. (More specifically, $\mathbf{k}$ is the unit vector in the direction of the thumb of the right hand if its index finger points along the (positive) real axis and the middle finger along the (positive) imaginary axis.)

8.23 Using the last result and the formula for the area of a triangle as a cross product of vectors lying along two of its sides, prove that the area (disregarding the sign) of a triangle with vertices at z_1, z_2 and z_3 is $\frac{1}{2}|z_3 - z_2|^2 \text{Im}\left(\dfrac{z_3 - z_1}{z_3 - z_2}\right)$. Deduce that z_1, z_2, z_3 are collinear if and only if the ratio $\dfrac{z_3 - z_1}{z_3 - z_2}$ is real.

8.24 Prove that three complex numbers z_1, z_2, z_3 are collinear if and only if the determinant $\begin{vmatrix} z_1 & \overline{z_1} & 1 \\ z_2 & \overline{z_2} & 1 \\ z_3 & \overline{z_3} & 1 \end{vmatrix}$ vanishes. [*Hint:* Begin by writing z_3 as $(1-\lambda)z_1 + \lambda z_2$ for a complex λ or use the last exercise.]

8.25 Prove that four complex numbers z_1, z_2, z_3, z_4 are either collinear or concyclic if and only if the following 4×4 determinant vanishes.

$$\begin{vmatrix} z_1\overline{z_1} & z_1 & \overline{z_1} & 1 \\ z_2\overline{z_2} & z_2 & \overline{z_2} & 1 \\ z_3\overline{z_3} & z_3 & \overline{z_3} & 1 \\ z_4\overline{z_4} & z_4 & \overline{z_4} & 1 \end{vmatrix}$$

8.26. Three line segments OA, OB, OC in space of lengths a, b, c respectively are at right angles to each other. Let $\Delta_1, \Delta_2, \Delta_3$ and Δ be the areas of the triangles OBC, OCA, OAB and ABC. Prove that $\Delta^2 = \Delta_1^2 + \Delta_2^2 + \Delta_3^2$. [*Hint:* Let N be the foot of the perpendicular from O to the face ABC of the tetrahedron $OABC$. Determine Δ from V, the volume of the tetrahedron, and the length of ON. Taking axes along OA, OB, OC, note that N lies on the plane whose equation is $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$.]

Chapter 9

COORDINATE GEOMETRY

Historically, coordinates were introduced to help geometry. And so well did they do this job, that the very identity of geometry was changed. The word 'geometry' today generally means coordinate geometry. If one means the classical geometry, one has to use some qualifiers such as 'pure' geometry.

A similar transition has taken place in the JEE. Till 1978, pure geometry was an integral part of the syllabus. Nowadays, a pure geometry problem is hardly asked and even if asked, an option is given to use coordinates. Problems on coordinate geometry, on the other hand, are asked invariably. In fact, they are among the 'sure shot' questions for most candidates because they are relatively straightforward and often demand only a few standard formulas for their solutions. They are a boon to those who are very adept at algebraic manipulations but lack the ability to hit upon the key idea by visual perception. For example, to prove the concurrency of three lines, one need not cleverly split some angle hidden somewhere in the diagram and equate one part of it with something using the parallelism of some two lines and the other part with something using properties of a cyclic quadrilateral! All one needs to do is to write down the equations of the three lines and show that a certain determinant vanishes. The latter can, of course, demand some algebraic skills. But these are relatively standard and far less elusive than those demanded by pure geometry.

Problems in coordinate geometry generally fall into two types, those where you have to find something (such as the area of some triangle, the coordinates of some point or the equation of some curve) and those where you have to prove a geometric property of a given curve. The Main Problem and the problems of finding the loci belong to the first type. The majority of the JEE questions is also of this type. An example of the second type of problems is given in Comment No. 19.

Main Problem: Show that $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents a pair of straight lines. Straight lines are drawn through $(0, 1)$ which are parallel to these lines. Then the area of the parallelogram formed by these four lines is ..... sq. units. (JEE 1980)

First Hint: Factorise the expression as the product of two linear factors.

Second Hint: Identify any three vertices of the parallelogram.

Third Hint: Take twice the area of the triangle formed by these three vertices.

Solution: That the given equation represents a pair of straight lines can be shown by the well-known determinant criterion, viz., by showing that the determinant

$$\begin{vmatrix} 2 & \frac{5}{2} & 3 \\ \frac{5}{2} & 3 & \frac{7}{2} \\ 3 & \frac{7}{2} & 4 \end{vmatrix}$$

vanishes, which can be done by directly expanding the determinant. But the subsequent work requires the identification of these two lines. So we may as well begin by factorising $2x^2 + 5xy + 3y^2 + 6x + 7y + 4$ into two linear factors. This can be done largely by inspection as $(x + y + 1)(2x + 3y + 4)$ in the present problem. (In case you cannot do it by inspection, there is a systematic method for arriving at this factorisation. See Comment No. 15 below for details.)

Hence the given equation represents two straight lines, say L_1 with equation $x+y+1 = 0$ and L_2 with equation $2x+3y+4 = 0$. Solving simultaneously, their point of intersection is $(1, -2)$. Let L_1' and L_2' be the lines through $(0, 1)$ which are parallel to L_1 and L_2 respectively. Then their equations are $x+y-1 = 0$ and $2x + 3y - 3 = 0$ respectively. We have to find the area of the parallelogram with sides along L_1, L_2, L_1' and L_2'. One way to do this would be to find the lengths of the sides and the angle between two adjacent sides. But there is an easier way. We already know two (diagonally opposite) vertices of the parallelogram, viz., $(1, -2)$ and $(0, 1)$. If we could identify one more vertex, then the triangle formed by these three vertices will have half as much area as the parallelogram. To find a third vertex of the parallelogram, we solve the equations of L_1' and L_2 simultaneously. This gives $(7, -6)$ as the point of intersection. Hence the area of the parallelogram is the absolute value of the determinant $\begin{vmatrix} 0 & 1 & 1 \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix}$. By direct computation this comes out as 14 square units. (We could have as well taken the common point of L_2' and L_1. That would give $(-6, 5)$ as the fourth vertex of the parallelogram. The final answer would, of course, be the same. Although this work is redundant, in numerical problems it is a good idea to get the answer in more than one way whenever easily possible and tally them.)

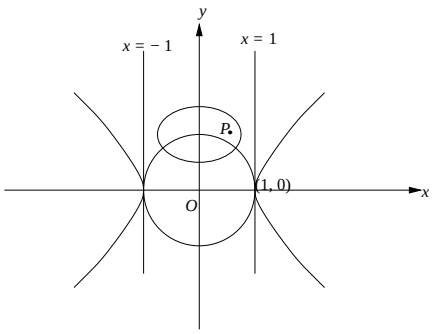
Comment No. 1:

Note that no diagram has been drawn. As a general rule, in geometric problems a good diagram not only inspires ideas for a solution but also often makes the solution easier to understand. But one of the advantages of coordinate geometry is that by converting everything into algebraic expressions, the need for a diagram can often be obviated. For example, in the present problem, the equations of the four lines L_1, L_2, L_1' and L_2' had to be determined algebraically. And once these equations are known, we can get the vertices of the parallelogram by solving them two at a time which is again a purely algebraic process. So there is no need to draw a diagram here.

This is certainly not to detract from the general importance of diagrams. Even in coordinate geometry problems, sometimes an answer becomes evident from a well-drawn diagram. The same answer may require a lot of work if attempted algebraically. Here is a good illustration. Suppose we are given an ellipse with eccentricity $\frac{1}{2}$ and one focus at $P(\frac{1}{2}, 1)$. It is given further that one directrix of the ellipse is the common tangent, nearer to the point P, to the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. We want to find the equation of the ellipse in the standard form. (JEE 1996) Obviously, in this problem the directrix is specified in a very clumsy way. To find it algebraically, we shall have to first find the common tangents to the given circle and the hyperbola.

One way to do this is to take a typical line of the form $y = mx + c$ and apply the so-called 'condition for tangency' for the circle as well as for the hyperbola. This will give us two equations in the two unknowns m and c. Solving them simultaneously we shall get common tangents to the two curves. However, this method applies only to tangents that are not vertical (i.e., for which m is a finite real number). The case of a vertical common tangent will have to be considered separately. A slightly different method is to start with a typical point on the circle $x^2 + y^2 = 1$, say the point $(\cos\theta, \sin\theta)$. The tangent at this point has the equation $x\cos\theta + y\sin\theta = 1$. We have to find the values of θ for which this line also touches the hyperbola $x^2 - y^2 = 1$. This can be done using the condition for tangency for a hyperbola. Or, if you prefer, you can get it directly by substituting $x = \frac{1 - y\sin\theta}{\cos\theta} = \sec\theta - y\tan\theta$ into $x^2 - y^2 = 1$ to get a quadratic in y, viz., $(\tan^2\theta - 1)y^2 - 2\tan\theta\sec\theta\ y + \tan^2\theta = 0$. (We are supposing here that $\tan\theta$ is defined, i.e., $\cos\theta \neq 0$. If $\cos\theta = 0$, our substitution $x = \frac{1 - y\sin\theta}{\cos\theta}$ in invalid and we have to try $y = \frac{1 - x\cos\theta}{\sin\theta}$. Or we can consider this degenerate case separately by noting that in this case the point $(\cos\theta, \sin\theta)$ is $(0, \pm 1)$, where the tangent to the circle is horizontal. But a hyperbola of the form $x^2 - y^2 = 1$ can have no horizontal tangent.) The tangency requirement is equivalent to saying that the roots of this quadratic coincide. This gives $\tan^2\theta\sec^2\theta = (\tan^2\theta - 1)\tan^2\theta$. This can hold only when $\tan\theta = 0$ since $\tan^2\theta - 1$ can never equal $\sec^2\theta$. This gives $\theta = 0$ and $\theta = \pi$ as two possibilities. The corresponding points on the circle are $(1, 0)$ and $(-1, 0)$. The corresponding tangents are the vertical lines $x = 1$ and $x = -1$. Out of these two, clearly the one closer to the point $P = (\frac{1}{2}, 1)$ is the line $x = 1$. So this is the directrix of the ellipse. Once the directrix is known, the equation of the ellipse can be written down from its eccentricity (which is given to be $\frac{1}{2}$) and

focus (which is given to be at $P = (\frac{1}{2}, 1)$. The distance of a point (x, y) from the line $x = 1$ is $|x - 1|$. So the desired equation is $(x - \frac{1}{2})^2 + (y - 1)^2 = \frac{1}{2}(x - 1)^2$, or, $\frac{3}{4}x^2 + y^2 - \frac{1}{2}x - 2y + 1 = 0$.



Thus we see that no matter which approach we take, it requires considerable work to identify the directrix, even though the subsequent work is quite simple. In such a case, a diagram such as the one above is definitely advantageous. From it we see instantaneously that the common tangents are $x = \pm 1$ and that the one nearer to P is $x = 1$.

It can, of course, be argued that a diagram is no substitute for a rigorous proof and hence that the argument above cannot be bypassed. If, instead of the circle $x^2 + y^2 = 1$ we had a circle of the form $x^2 + y^2 = r^2$, where $0 < r < 1$, then it would have four common tangents with the hyperbola. Identifying them from a diagram is hardly possible. But the algebraic method above will go through without much change. So the diagram cannot be relied upon as a general method even though in the present problem it works marvellously. While these objections are valid, one should at least give a diagram a chance. As for its validity as a proof, in a fill-in-the-blank type question (as the present problem was originally asked) or in a multiple choice question, no reasoning is to be shown. So as long as your answer is correct, it does not matter how you arrived at it.

Comment No. 2:

In fact, in some problems, you can get the answer with no algebraic computation and with no diagram either if you correctly understand the focal point of the problem. Here is a very simple but instructive example. Suppose we want to determine the number of values of c such that the straight line $y = 4x + c$ touches the curve $\frac{x^2}{4} + y^2 = 1$. (JEE 1998) We can, of course, apply the condition for tangency for an ellipse to get an equation in c whose roots will give us all tangents of the form $y = 4x + c$. But the problem does *not* ask us to that. It only asks us *how many* values of c satisfy a certain condition and not *which ones* they are. And this can be done with no computations if we interpret the problem a little differently. Lines of the form $y = 4x + c$, where c is a parameter are all parallel to each other and hence have the same direction. The equation $\frac{x^2}{4} + y^2 = 1$ represents an ellipse and the problem, in essence, amounts to asking

how many tangents to an ellipse are parallel to a given line or direction. The answer, 2, is a well-known result. (A rigorous proof of the result itself, will, no doubt, require some algebraic work. But the problem merely asks you to apply the result and not to prove it. Its truth can also be visualised geometrically. A straight line will generally cut a conic in two points. Since an ellipse is a bounded curve, it is clear that for very large or very small values of c, the line $y = 4x + c$ will not meet the ellipse at all, i.e., the points of intersection will be imaginary. As c changes, the line shifts parallelly. For values of c close to 0, it cuts the ellipse in two distinct points. So for some positive value of c the line must cut the ellipse in two coincident points, i.e., will touch it. Similarly there is some negative value of c for which $y = 4x + c$ touches the ellipse.)

A related problem (JEE 1999) is to find the points on the ellipse $4x^2 + 9y^2 = 1$ at which the tangents are parallel to the line $8x = 9y$. This time we are asked to actually find the points of contact and not just how many of them there are and so the problem does demand some work. Any line parallel to the line $8x = 9y$ has slope $\frac{8}{9}$ and hence an equation of the form $y = \frac{8}{9}x + c$, where c is a parameter. So once again the problem can be done by applying the condition for tangency. Note, however, that we are not interested in the equations of the tangents *per se* as in the points of contact. So, let us consider the equation of the tangent at a typical point, say (x_1, y_1) on the ellipse. It is $4xx_1 + 9yy_1 = 1$. Its slope is $-\frac{4x_1}{9y_1}$. Equating this with $\frac{8}{9}$ gives $x_1 = -2y_1$. Solving simultaneously with $4x_1^2 + 9y_1^2 = 1$ we get $y_1^2 = \frac{1}{25}$ and hence $y_1 = \pm\frac{1}{5}$. Correspondingly, $x_1 = \pm\frac{2}{5}$. So the desired two points are $(\frac{2}{5}, -\frac{1}{5})$ and $(-\frac{2}{5}, \frac{1}{5})$. (Actually, it is not necessary to remember even the formula for the equation of a tangent at a point of the ellipse. All we want is its slope and that can be obtained simply by differentiating $4x^2 + 9y^2 = 1$ w.r.t. x to get $\frac{dy}{dx} = -\frac{4x}{9y}$. Hence the slope of the tangent at (x_1, y_1) is $-\frac{4x_1}{9y_1}$. The rest of the work remains the same.)

Comment No. 3:

It should be clear by now that in short, numerical problems of coordinate geometry, it is not a particularly good idea to mechanically apply complicated formulas such as the condition for tangency or the equation of the chord of contact and so on. These things are certainly useful in some problems. In fact, in Comment No. 11 of the last chapter, we solved one problem using the equation of the chord of contact and also the equation of a chord in terms of its mid-point. However, for shorter questions intended to test basic understanding rather than manipulative skills, the key to success often lies in correctly focusing on what is asked and invoking the appropriate result, and often using only that part of it which is relevant to the problem.

Sometimes the insufficiency of the data precludes alternate approaches and forces us to try a short cut. Suppose, for example, that we are given that the diagonals of a parallelogram $PQRS$ are along the lines $x + 3y = 4$ and $6x - 2y = 7$ and we are asked to identify what type of a quadrilateral $PQRS$ is. (JEE 1998) Solving the two equations simultaneously will give us the centre of

the parallelogram. But from that we cannot determine the vertices. However, even without solving the two equations together, we can note that the lines represented by them have slopes $-\frac{1}{3}$ and 3 respectively. As these numbers are negative reciprocals of each other, the diagonals of $PQRS$ (which is already known to be a parallelogram) are at right angles to each other. Therefore, by a well-known result in geometry, $PQRS$ must be a rhombus. (Beware! A nasty paper-setter can try to fool you by giving the coordinates of P and Q and thereby tempting you to find the coordinates of the remaining vertices, even though none of this information is needed in the present problem!)

As another example (JEE 1999), let L_1 be a straight line passing through the origin and L_2 be the line $x+y=1$. We are given that the intercepts made by the circle $x^2+y^2-x+3y=0$ on L_1 and L_2 are equal and we are asked to find which of the following equations can represent L_1.

(A) $x+y=0$ (B) $x-y=0$ (C) $x+7y=0$ (D) $x-7y=0$.

We can calculate the length of the intercepts made by the circle on various lines by finding the points where the circle intersects the lines. However, if we recall, from elementary geometry, that two chords of a circle are equal if and only if they are equidistant from its centre, the work is simplified considerably. By inspection, the centre of the circle is at $(\frac{1}{2},-\frac{3}{2})$. Its distance from L_2 is $\frac{|\frac{1}{2}-\frac{3}{2}-1|}{\sqrt{2}}=\sqrt{2}$. So the problem reduces to testing which of the four lines (A) to (D) are at a distance $\sqrt{2}$ from the point $(\frac{1}{2},-\frac{3}{2})$. By a direct calculation, the distances of these four lines from the point $(\frac{1}{2},-\frac{3}{2})$ are $\frac{1}{\sqrt{2}}, \sqrt{2}, \frac{10}{\sqrt{50}}=\sqrt{2}$ and $\frac{11}{\sqrt{50}}$ respectively. So (A) and (C) are possible equations (among the given ones) which can represent L_1. (Note that the data is insufficient to identify L_1 uniquely. So in this problem, we cannot possibly arrive at a unique answer and then see which one among the given answers matches it. We have to examine each of the four possibilities separately and see which ones among them satisfy the given condition.)

Comment No. 4:

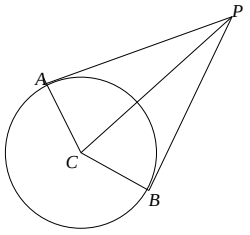
In the two problems just solved, the key idea that led to the solution came from pure geometry and was relatively easy to strike at as the problems dealt with a parallelogram and a circle respectively and both these objects are studied rather thoroughly in school geometry. Classically, the other important curves such as ellipses, parabolas and hyperbolas were also studied in pure geometry and most of their important properties (for example, the focusing property of a parabola and an ellipse proved in the exercises) had been established long before cartesian coordinates were invented. However, nowadays, these curves are studied exclusively in coordinate geometry. That sometimes gives a wrong impression that every problem about them must be solved using coordinates. The fact is that unless a question specifically stipulates that a particular method is allowed or prohibited, there is an inherent freedom of choice. Even when coordinates are used, certain short cuts using pure geometry can simplify the work sometimes as in the following problem. (JEE 1997*)

An ellipse has OB as a semi-minor axis and F, F' as its foci. The angle FBF' is a right angle. Find the eccentricity of the ellipse.

In the standard form the equation of the ellipse can be written as $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then the point B can be taken as $(0, \pm b)$. By symmetry, we may take it as $(0, b)$. The foci F, F' can be taken as $(-ae, 0)$ and $(ae, 0)$ respectively where e is the eccentricity given by $b^2 = a^2(1 - e^2)$. The perpendicularity of the lines FB and $F'B$ can be used to equate the product of their slopes with -1 as is often done in coordinate geometry. But pure geometry can spare us this calculation. The two foci are reflections of each other into the minor axis. So we always have $OF = OF'$, where O is the centre of the ellipse. Hence BO is a median of the triangle FBF' which is given to be right-angled at B. By a well-known result in geometry, $OB = OF = OF'$. This gives $b = ae$. Putting this into $b^2 = a^2(1 - e^2)$ gives $2e^2 = 1$ and hence $e = \frac{1}{\sqrt{2}}$. (Note that we cannot determine the values of a and b from the data. Nor are they needed. The eccentricity of an ellipse depends on its shape and not on its size. So we might as well take $\frac{b}{a}$ as a single variable. Then we have a system of two equations in two unknowns, viz., e and $\frac{b}{a}$.)

Comment No. 5:

Pure geometry is, of course, not a golden tool to tackle all problems of coordinate geometry. If it were, there would be no need to introduce coordinates in the first place! Like any other tool, it has to be used with discretion. Sometimes two problems may be essentially similar in terms of what they ask. But the data may be different and accordingly, pure geometry methods may suit one of them but not the other. As an illustration, suppose we are given the equation of a circle with centre C. The tangents from a point P outside the circle touch it at A and B and we are asked to find the area of the quadrilateral $PACB$. As the triangles PAC and PBC are mutually congruent, this is equivalent to finding the area of either one of them and doubling it.



From the equation of the circle, we can determine (by completing the squares) the coordinates of its centre C and also its radius, say r. For example, if the equation is $x^2+y^2-4x-2y-11 = 0$, then rewriting it as $(x-2)^2+(y-1)^2 = 16$ we get $C = (2, 1)$ and $r = 4$. Now suppose P is given as the point $(4, 5)$. Then the distance CP is $\sqrt{(4-2)^2 + (5-1)^2} = \sqrt{20}$. As PAC is a right-angled triangle, we can now determine PA as $\sqrt{PC^2 - CA^2} = \sqrt{20 - r^2} = 2$. Hence the easiest way to find the area of the triangle PAC is to take advantage of the fact that it is right-angled. The answer is $\frac{1}{2} \times 4 \times 2 = 4$. So the area of the quadrilateral $PACB$ is twice this, i.e., 8 square units. (JEE 1985)

Suppose, however, that we are not given the coordinates of P. Instead, we are given the points A and B on the circle. In this case we cannot determine

CP very easily. One way to find it is to determine P first by writing down the equations of the tangents at A and B and solving them simultaneously. For example, suppose the equation of the circle is $x^2+y^2-2x-4y-20=0$ and A, B are given as $(1,7)$ and $(4,-2)$ respectively. (JEE 1981) Then C comes out as $(1,2)$. (This can be done by inspection. There is no need to complete the squares since we are not interested in the radius right now.) Now the equations of the tangents to the circle at A and B come out, after simplification, as $y=7$ and $3x-4y=20$ respectively. (These can be obtained either from the readymade formula for the equation of a tangent to a circle at a given point on it, or by observing that the tangent is perpendicular to the corresponding radius.) Solving, we get $P=(16,7)$. Now that we know the coordinates of P we can find the area of the triangle APC by the previous method, i.e., by finding CP and then AP. But that is silly. Since we already have the coordinates of C, P and A, the best way to find the area of the triangle APC is by evaluating the determinant $\begin{vmatrix} 16 & 7 & 1 \\ 1 & 7 & 1 \\ 1 & 2 & 1 \end{vmatrix}$, which comes out to be 75. So this is twice the area of the triangle APC and hence the area of the quadrilateral $PACB$. (Again, to be safe it is good to verify that the area of the triangle BPC also comes out to be $\frac{75}{2}$.)

Thus we see that even though the two problems are basically the same, in the first one, the pure geometry solution is quicker while in the second one, the coordinates (and, in particular, the equation of the tangent) come in handy. (If one is bent upon using pure geometry in the second problem one can do so. Let $M=(\frac{5}{2},\frac{5}{2})$ be the mid-point of AB. Then the right-angled triangles AMC and PAC are similar. Since CA and MA can be easily found from the data, one can get CP from this. But this approach is certainly not so obvious. Similarly, if one wants, the first problem can be done rather laboriously by coordinates. The equation of the chord of contact of the point $P=(4,5)$ w.r.t. the given circle $x^2+y^2-4x-2y-11=0$ is $4x+5y-2(x+4)-(y+5)-11=0$, i.e., $x+2y=12$. Putting $x=12-2y$ into the equation of the circle gives a quadratic in y, viz., $5y^2-42y+85=0$ which has two roots, viz., $y=5$ and $y=\frac{17}{5}$. The corresponding values of x are 2 and $\frac{26}{5}$. So we can take A as $(2,5)$ and B as $(\frac{26}{5},\frac{17}{5})$. The area can now be evaluated using determinants.)

Comment No. 6:

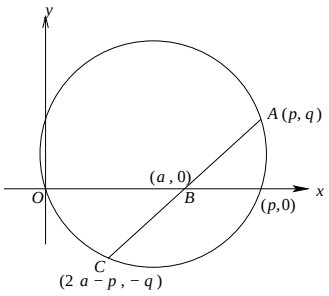
So far we saw problems where recalling the right fact from pure geometry did the trick. Sometimes it is the results from algebra that are needed crucially in the solution. This is to be expected because the very crux of coordinate geometry is to tackle geometry through algebra. In the last chapter we already saw how the concurrency of three lines reduces to vanishing of the determinant of the coefficients in their equations. Similarly, in Chapter 3 (Comment No. 10) the condition for tangency was derived from the criterion for a quadratic to have two coincident roots. The following problem is based on the criterion for a quadratic to have two distinct real roots.

Find a necessary and sufficient condition (in terms of p and q) so that

two distinct chords, drawn from the point (p, q) on the circle $x^2 + y^2 = px + qy$ are bisected by the x-axis. (JEE 1999, modified)

A natural start is to take a typical chord through (p, q) as $y - q = m(x - p)$ and to express its point of intersection with the x-axis in terms of m. A more direct (and hence preferable) start is to let $(a, 0)$ be the point where a typical chord through (p, q) meets the x-axis and work in terms of a rather than m. To say that the chord is bisected by the x-axis is equivalent to saying that the point $(2a-p, -q)$ is the other end of the chord, i.e., lies on the given circle. This means $(2a-p)^2+q^2 = p(2a-p)-q^2$ i.e., $2a^2-3pa+p^2+q^2 = 0$. (An alternate way to get this is to consider the equation of the chord in terms of the mid-point $(a, 0)$ as $T = S_1$, which in the present case becomes $ax - \frac{p}{2}(x+a) - \frac{q}{2}(y+0) = a^2 - pa$ and to note that (p, q) lies on this.) This can be solved as a quadratic in a with roots $\frac{3p \pm \sqrt{p^2-8q^2}}{4}$. The x-axis cuts the circle at $(0, 0)$ and $(p, 0)$. Assuming $p > 0$, the condition in the problem is equivalent to saying that the roots are distinct and that both lie between 0 and p. A necessary and sufficient condition for this is that $p^2 > 8q^2$. (The argument is similar if $p \leq 0$.)

In the original problem, the answer had to be picked from four choices, viz., (A) $p^2 = q^2$, (B) $p^2 = 8q^2$, (C) $p^2 < 8q^2$ and (D) $p^2 > 8q^2$. In that case there is an easier way to do the problem by drawing a diagram. Let A be the point (p, q). Then O lies on the circle and its centre is at $(\frac{p}{2}, \frac{q}{2})$. Since this is also the mid-point of OA, it follows that OA is a diameter of the circle. (In the accompanying diagram p, q are assumed to be positive.)



From the diagram it is clear that in order that a chord drawn from A be bisected at the x-axis, p must be considerably bigger than q. Exactly how big it should be cannot be determined from the diagram alone. For this we need the calculation given above. But if the problem is to pick one of the given alternatives, then it has to be (C) or (D) because the condition has to be in the form of an inequality (as it corresponds to something having two distinct solutions). And by what we have observed from the diagram, it cannot be (C). So by elimination, (D) must be the right answer!

In some problems, the appropriate fact from algebra has to be chosen in a more clever manner. Suppose for example (JEE 1989), that we are given that $(m_i, \frac{1}{m_i}), i = 1, 2, 3, 4$ are four distinct points on a circle and we are asked to prove that $m_1m_2m_3m_4 = 1$. The method that comes to mind immediately is to write the equation of the circle passing through any three of these points and then to use the fact that the fourth point lies on this circle to get an equation involving m_1, m_2, m_3 and m_4. A less complicated method to do the same is to take the equation of the circle in the general form as $x^2+y^2+2gx+2fy+c = 0$. For each $i = 1, 2, 3, 4$, the point $(m_i, \frac{1}{m_i})$ satisfies this equation. So we have

$$m_i^2 + \frac{1}{m_i^2} + 2gm_i + 2f\frac{1}{m_i} + c = 0,$$

or equivalently,

$$m_i^4 + 1 + 2gm_i^3 + 2fm_i + cm_i^2 = 0 \tag{1}$$

for $i = 1, 2, 3, 4$. This is a system of four equations from which we can eliminate the three unknowns g, f and c. Since the system is linear (as far as g, f and c are concerned), the result can be written down in terms of a 4×4 determinant, (see Exercise (3.23)), viz.,

$$\begin{vmatrix} m_1^4+1 & 2m_1^3 & 2m_1 & m_1^2 \\ m_2^4+1 & 2m_2^3 & 2m_2 & m_2^2 \\ m_3^4+1 & 2m_3^3 & 2m_3 & m_3^2 \\ m_4^4+1 & 2m_4^3 & 2m_4 & m_4^2 \end{vmatrix} = 0$$

From this we want to show that $m_1m_2m_3m_4 = 1$. This can be done as follows. Divide the second and the third column of this determinant by 2 each and call the resulting determinant as D. Then we know $D = 0$. By splitting the first column into two parts, D equals $D_1 + D_2$, where

$$D_1 = \begin{vmatrix} m_1^4 & m_1^3 & m_1 & m_1^2 \\ m_2^4 & m_2^3 & m_2 & m_2^2 \\ m_3^4 & m_3^3 & m_3 & m_3^2 \\ m_4^4 & m_4^3 & m_4 & m_4^2 \end{vmatrix} \text{ and } D_2 = \begin{vmatrix} 1 & m_1^3 & m_1 & m_1^2 \\ 1 & m_2^3 & m_2 & m_2^2 \\ 1 & m_3^3 & m_3 & m_3^2 \\ 1 & m_4^3 & m_4 & m_4^2 \end{vmatrix}.$$

If we take out the factors m_1, m_2, m_3 and m_4 from the four rows of D_1 and then reshuffle the columns we see that D_1 equals $-m_1m_2m_3m_4D_2$. So $D = D_1 + D_2 = (1 - m_1m_2m_3m_4)D_2 = 0$. Now if we permute the columns of D_2 we see that it is nothing but a Vandermonde determinant (see Chapter 3, Exercise (3.26)) with parameters m_1, m_2, m_3, m_4. But these numbers are all distinct as the points $(m_i, \frac{1}{m_i}), i = 1, 2, 3, 4$ are given to be distinct. So $D_2 \neq 0$. Thus the vanishing of D forces the other factor, viz., $(1 - m_1m_2m_3m_4)$ to vanish. Hence $m_1m_2m_3m_4 = 1$ as was to be shown.

There is, however, a better way of deriving the result from (1). Replacing m_i by a variable m and rearranging the terms, we can write it as a fourth degree polynomial equation, viz.,

$$m^4 + 2gm^3 + cm^2 + 2fm + 1 = 0 \tag{2}$$

Then (1) is equivalent to saying that m_1, m_2, m_3 and m_4 are the roots of (2) and so $m_1m_2m_3m_4$ is nothing but the product of these four roots. From the general theory of polynomial equations this is the ratio of the constant term to the coefficient of m^4. In the present case, it equals 1.

This problem is an excellent example of how certain relatively advanced topics from algebra such as the theory of equations and of determinants are sometimes needed in problems of coordinate geometry. In fact, the present problem can very well be thought of as more a problem in algebra than a problem in coordinate geometry. Its only relationship with coordinate geometry is that it uses the fact that the general equation of a circle is of the form $x^2 + y^2 + 2gx + 2fy + c = 0$.

Comment No. 7:

Let us now study some problems which genuinely belong to coordinate geometry, i.e., which are not disguised versions of either problems from pure geometry or those from algebra. Broadly, these problems can be classified into two types : those where we are asked to find something (such as the coordinates of some point, or the equation of some curve or the locus of some point) and those where we have to prove something (usually an equality, but sometimes an inequality as well, as we saw in the first problem of the last comment). We have already encountered examples of both the types. The problem just done belongs to the second category while the Main Problem belongs to the first. But now we approach them more systematically rather than in an *ad hoc* manner.

The simplest among problems of the first type are those where you are asked to find the coordinates of some point which is specified by some conditions. Let the (unknown) point be (a, b). Then to find a and b we need a system of two equations in a and b. These two equations are to be found from the data. As a simple illustration (JEE 1978), suppose it is given that the area of a triangle is 5, two of its vertices are at $(2, 1)$ and $(3, -2)$ and that the third vertex lies on the line $y = x + 3$. We want to find the third vertex. Let the third vertex be (a, b). Here we are already given one equation satisfied by a and b, viz., $b = a + 3$. The other equation comes from the fact that the area of the triangle is 5, which in terms of determinants means that the absolute value of the determinant $\begin{vmatrix} a & b & 1 \\ 2 & 1 & 1 \\ 3 & -2 & 1 \end{vmatrix}$ is 10, i.e., $|3a + b - 7| = 10$. This runs into two possibilities: $3a + b = 17$ or $3a + b = -3$. Solving the first one simultaneously with $b = a + 3$ gives $a = \frac{7}{2}, b = \frac{13}{2}$. The second possibility gives $a = -\frac{3}{2}, b = \frac{3}{2}$. So the third vertex of the triangle is either $(\frac{7}{2}, \frac{13}{2})$ or $(-\frac{3}{2}, \frac{3}{2})$.

The use of the symbols a and b was primarily to avoid confusion. After some practice, we might as well take the third vertex as (x, y) and write down the two equations in terms of x and y directly. However, in some problems some other equations in x and y are given which are not satisfied by the point to be found. In that case, it is a good idea to take the point as (a, b) instead of (x, y).

Comment No. 8:

Sometimes the conditions given are not sufficient to determine a point uniquely. All that they imply is that the point must lie on some curve. The equation of this curve is then called the **locus** of that point. Traditionally, the term means the path traced out by a moving point. Suppose t denotes the time. The position of a particle moving in a plane at time t can be specified by specifying its x and y coordinates separately as functions of t, say $x = f(t)$ and $y = g(t)$. Then by eliminating t in these two equations, we get an equation in x and y which is satisfied by all points traced out by the moving particle. Suppose, for example (JEE 1999), that

$$x = t^2 + t + 1 \tag{3}$$

$$y = t^2 - t + 1 \tag{4}$$

Subtracting (4) from (3) gives $t = \frac{1}{2}(x - y)$. Putting this into (3) gives $4x = (x - y)^2 + 2(x - y) + 4$, or, in a standard form,

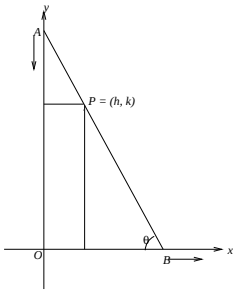
$$x^2 - 2xy + y^2 - 2x - 2y + 4 = 0 \tag{5}$$

which can be recognised as a parabola (since $h^2 = ab$, where $a, b, 2h$ are the coefficients of x^2, y^2 and xy respectively). (3) and (4) are called **parametric equations** of the locus while (5) is a non-parametric equation. Conversion of a non-parametric equation to a pair of two parametric equations is theoretically possible, but not always easy, and, at any rate, not unique. For example, (5) can also be parametrised by $x = t^6 + t^3 + 1, y = t^6 - t^3 + 1$ and in many other ways.

The parametric equations make it easier to trace the curve by directly plotting the points corresponding to a few values of the parameter. The non-parametric form, on the other hand, has certain advantages because it is unique. For example, it is easier to identify the type of the curve if it is specified in the non-parametric form as we saw in the example above. And, in case the locus happens to be of some standard form, then we immediately know some of its geometric properties, which may not be so obvious from the parametric form, especially if the parametrisation is an unusual one.

It is customary to call every equation in x and y as a locus of a point moving in a plane, even though there may not be any motion involved and the equation might not have been obtained by eliminating some parameter from two equations. An important class of problems in coordinate geometry consists of those which ask for the locus of a point which 'moves' in the plane so as to satisfy certain given conditions. The standard method to tackle such problems is to denote the moving point, by (h, k) (or by some symbols such as (α, β) etc.). These variables h and k are often called the **current coordinates** of the moving point. (Care should be taken to see that the symbols used for the current coordinates do not already appear in the given problem to denote something else!) The conditions imposed on the moving point can then be translated into a system of equations (often involving one or more auxiliary variables). This system can then be reduced to a single equation involving h and k. Replacing h by x and k by y gives the locus.

As a straightforward illustration, suppose a ladder of length l rests against a vertical wall. Its lower end starts sliding along the horizontal floor and the upper end along the wall. Find the locus of the point on the ladder which divides it in the ratio 1 : 2 (from top to bottom). (JEE 1978, modified)



Denote the top and the bottom of the ladder by A and B respectively. Taking the axes along the floor and along the wall as shown, let (h, k) be the

current coordinates of the point, say P, which divides AB in the ratio $1:2$. Using similar triangles, we get $B = (3h, 0)$ and $A = (0, \frac{3}{2}k)$. Since the length of the ladder is given to be l we have $9h^2 + \frac{9}{4}k^2 = l^2$. Replacing h and k by x and y respectively, we get the equation of the locus of P as

$$9x^2 + \frac{9}{4}y^2 = l^2 \tag{6}$$

Sometimes the problem asks you to identify the type of the curve the locus represents. In the present problem, its equation can be rewritten as $\frac{x^2}{(l/3)^2} + \frac{y^2}{(2l/3)^2} = 1$. This can be recognised as an ellipse whose minor axis is horizontal and major axis vertical.

In this problem the ladder and hence the point P on it were actually moving. But the problem could have been phrased in a manner which has little to do with motion of any kind. Let $\theta = \angle OBA$ be the angle the ladder makes with the floor. As the ladder moves, θ varies. In fact the value of θ determines the position of the ladder uniquely. If we denote the line AB by L, then L also changes accordingly. Here θ is a parameter which can assume all possible values in the interval $[0, \frac{\pi}{2}]$. The set of all possible lines that arise from various values of θ is therefore called a **one parameter family** of lines. The line L is like a variable which varies over this one parameter family of lines. As θ varies so do L, hence P and hence, finally, h and k. So, if we want we can get the locus of P in a parametric form with θ as a parameter. Since $OB = l\cos\theta$ and $OA = l\sin\theta$, by similar triangles we get $h = \frac{1}{3}l\cos\theta$ and $k = \frac{2}{3}l\sin\theta$. These are parametric equations of the locus. If we eliminate θ among them we get (6). Instead of θ we could have taken some other parameter, for example, m, the slope of the variable line L. Or we could have taken one of its intercepts as a parameter. These parameters are, obviously, inter-related. (In this problem, expressing h and k individually in terms of the parameter was easy. In many problems it is not so. Nor is it necessary. All we need is a pair of equations involving h, k and the parameter. See, for example, Exercise (9.24)(xii).)

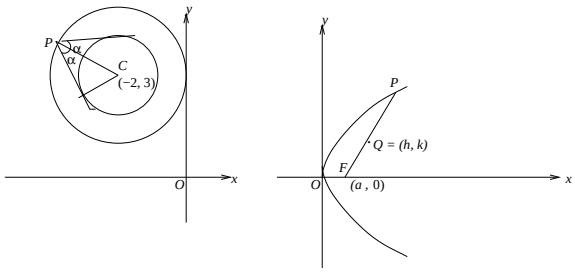
It is very important to have only one parameter, however. Sometimes we may introduce two (or more) parameters for our own convenience. But they have to be related to each other by some equation. In the present problem, we could for example, take the equation of L in the intercepts form as $\frac{x}{p} + \frac{y}{q} = 1$, where p and q are the intercepts on the x and the y axis respectively. Here ostensibly there are two parameters, viz., p and q. But they are not independent of each other. The fact that the length of the ladder remains constant puts a restriction on them, viz., $p^2 + q^2 = l^2$. So effectively, there is only one parameter. The problem can be done equally easily with these two parameters. For, we have $B = (p, 0)$ and $A = (0, q)$. So $p = 3h$ and $q = \frac{3}{2}k$ (again by similar triangles). The equation $p^2 + q^2 = l^2$ gives us the locus (6). Even for a simple locus like that of a point (h, k) which moves so that its distance from a fixed point (a, b) is r (say), we can take h and k themselves as two parameters, related to each other by the given constraint.

Whenever we have a one parameter family of curves, a point which is associated with a typical member of that family in a common manner will move as

the member changes and its locus can be found either by expressing its current coordinates in terms of one or more parameters or sometimes directly.

Comment No. 9:

When the locus is a familiar curve such as a straight line or a circle, it can often be found by pure geometry. For example, the locus of the centre of a (variable) circle of a fixed radius r_1 which externally touches a given circle with centre C and radius r is a circle with centre C and radius $r+r_1$. As a less trivial example, the locus of a point P such that the tangents from it to a given circle are at a given angle is also a circle, because this angle depends only on the distance between P and the centre of the given circle. As a specific numerical example (JEE 1996), suppose the given circle is $x^2+y^2+4x-6y+9\sin^2\alpha+13\cos^2\alpha=0$ and the tangents to it from P are to be inclined at an angle 2α to each other. Completing the squares, the centre of the circle, say C is at $(-2,3)$ while its radius is $2\sin\alpha$. From Figure (a) below, it is clear that the tangents from P are inclined to each other at an angle 2α if and only if the distance CP equals 2. So, the locus is a circle of radius 2 with centre $(-2,3)$. Its equation is $(x+2)^2+(y-3)^2=4$, i.e., $x^2+y^2+4x-6y+9=0$.



(a) (b)

Even when the locus is not a circle but some other familiar curve such as a conic, the knowledge of the geometric properties of that curve can sometimes lead to a quicker determination of analogous geometric properties of the locus than is possible by the brute force method of coordinates. As an example, suppose P is a point moving on the parabola $y^2=4ax$ with focus at F and we want to study the locus of the mid-point of the segment FP. Call the mid-point as Q. To do the problem algebraically, let $Q=(h,k)$. Since $F=(a,0)$, we have $P=(2h-a,2k)$. As P is given to lie on the parabola $y^2=4ax$, we immediately get $4k^2=4a(2h-a)$. This gives the locus of Q as $y^2=a(2x-a)$. This is a parabola. By rewriting its equation as $y^2=4\times\frac{a}{2}(x-\frac{a}{2})$ we see that its vertex is at the point $(\frac{a}{2},0)$ and its axis is the x-axis. Also since its latus rectum is $4\times\frac{a}{2}$, we see that its focus must be at $(\frac{a}{2}+\frac{a}{2},0)$, i.e., at $(a,0)$ (which is also the focus of the original parabola) and that its directrix is the line $x=\frac{a}{2}-\frac{a}{2}$ i.e., the line $x=0$. (See Figure (b) above.)

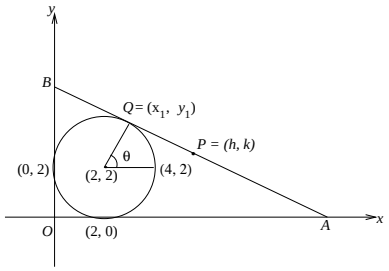
Suppose, however, that we are *given* (JEE 2002) that the locus of Q is a parabola and are merely asked to find its directrix. Then it is hardly necessary

to go through the work above. A quicker method is to observe that every point on a parabola is equidistant from its focus and its directrix. Since FQ is half of FP, if we can find a vertical line whose distance from Q is half its distance from P, we would get the directrix of the new parabola. If at all such a line exists, then by taking P to be O (which is a point on the original parabola), we see that it must be the line $x = 0$, i.e., the y-axis. It can now be *verified* that this line is indeed the directrix of the new parabola. But that is not necessary as far as the given question is concerned, where the problem was to identify the directrix, *assuming* that the locus is a parabola. This argument is not as strong as the earlier one, where we actually found the locus without being told what type of a curve it is. But an argument like this is quite all right (and, in fact, preferable because it saves time) for short questions, where you do not have to answer how you arrived at the answer.

Comment No. 10:

In some problems, the curve represented by the equation of the locus may also include points which are not on the original path. For example, Equation (6) above represents an ellipse and is the locus of a certain point P on a sliding ladder. But obviously, for all positions of the ladder, P lies in the first quadrant and so the actual path traced out by it is not the entire ellipse but only a quarter of it. In problems of finding a locus, after you obtain its equation, you are not normally expected to further analyse exactly which part of the curve represented by it is actually covered by the moving point. But sometimes such hairsplitting is a part and parcel of the problem. Here is a problem where we illustrate this point. We are given that the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two of its sides along the coordinate axes. We are asked to show that the locus of the circumcentre of the triangle is of the form $x + y - xy + \lambda(x^2 + y^2)^{1/2} = 0$ for some constant λ and further to find the value of λ. (JEE 1987, modified)

Here we have three tangents to the given circle, of which two are the coordinate axes (whose tangency follows since the given circle has centre $(2, 2)$ and radius 2). The third tangent is the variable line here. Let it cut the x- and the y-axes at A and B respectively. Since OAB is a right angled triangle, its circumcentre, say P, is simply the mid-point of AB. (Again, we are pulling a fact from pure geometry. But even if you have forgotten it, it would not matter because the equations of the perpendicular bisectors of the segments OA and OB can be written down very easily in the present problem.)



Let $P = (h, k)$ be the mid-point of AB. Then clearly, $A = (2h, 0)$ and $B = (0, 2k)$. The line AB is given to touch the circle $x^2 + y^2 - 4x - 4y + 4 = 0$. To get an equation in h and k from this, we can proceed in a variety of ways.

(i) Let the line AB touch the circle at the point $Q = (x_1, y_1)$. Then the equation of the tangent at (x_1, y_1) is $xx_1+yy_1-2(x+x_1)-2(y+y_1)+4 = 0$ or $(x_1-2)x+(y_1-2)y-2x_1-2y_1+4 = 0$. This is the same line as AB whose equation (in the two intercepts form) is $\frac{x}{2h} + \frac{y}{2k} = 1$ or, $kx + hy = 2hk$. So comparing the coefficients we get

$$\frac{x_1 - 2}{k} = \frac{y_1 - 2}{h} = \frac{x_1 + y_1 - 2}{hk} \tag{7}$$

Since (x_1, y_1) lies on the given circle, we also have

$$(x_1 - 2)^2 + (y_1 - 2)^2 = 4 \tag{8}$$

The desired locus is obtained by eliminating x_1 and y_1 from (7) and (8). One way to do this is to set each of the ratios in (7) equal to r to get $x_1-2 = kr, y_1-2 = hr$ and $x_1+y_1-2 = hkr$ giving $hr+kr+2 = hkr$, or $r = \frac{2}{hk-h-k}$. On the other hand (8) gives $k^2r^2 + h^2r^2 = 4$. Substituting for r, we get $(hk - h - k)^2 = h^2 + k^2$. Replacing h by x and k by y,

$$(xy - x - y)^2 = x^2 + y^2 \tag{9}$$

which would normally pass as the locus of the circumcentre, i.e., of the point P. But the problem asks us to recast this in a certain form. Taking the square roots of both the sides of (9) we get

$$xy - x - y = \pm(x^2 + y^2)^{1/2} \tag{10}$$

The problem requires us to determine which sign holds in the R.H.S. Now, we are given that the circle is *inscribed* in the triangle OAB. So the line AB has negative slope and the point of contact Q must lie on the quarter of the circle between the points $(4, 2)$ and $(2, 4)$. But in that case both $2h$ and $2k$ must be at least 4 each. So, $h = 2+a$ and $k = 2+b$, where a, b are positive. An easy calculation shows that $hk - h - k = a + b + ab$ is also positive. Hence the L.H.S. of (10) is positive for all positions of P and so the $+$ sign must hold on the R.H.S. In other words the constant λ must equal 1. (A slicker way is to identify any one point, e.g. $(2+\sqrt{2}, 2+\sqrt{2})$ on the locus and substitute $x = 2 + \sqrt{2},\ y = 2 + \sqrt{2}$ into (10). But this argument is valid only if it is known beforehand that the *same* sign holds in (10) for all points of the locus. This may look intuitively obvious because the locus is a continuous curve. But a rigorous justification is far from trivial. It needs a fact from theoretical calculus, called the Intermediate Value Property for continuous functions of two variables. But such subtilities are beyond the JEE level.)

(ii) Instead of taking the point of contact Q in the form (x_1, y_1) we can use the standard parametrisation of a circle with centre $(2, 2)$ and radius 2 to take Q in a parametric form as $(2 + 2\cos\theta, 2 + 2\sin\theta)$. Then instead of (7) we would get

$$\frac{\cos\theta}{k} = \frac{\sin\theta}{h} = \frac{\sin\theta + \cos\theta + 1}{hk} \tag{11}$$

from which we get the locus of P in a parametric form, viz.,

$$x = \frac{\sin\theta + \cos\theta + 1}{\cos\theta} \text{ and } y = \frac{\sin\theta + \cos\theta + 1}{\sin\theta} \tag{12}$$

To eliminate θ, we solve (12) (or equivalently, (11)) for $\cos\theta$ and $\sin\theta$ to get $\cos\theta = \frac{y}{xy-x-y}$ and $\sin\theta = \frac{x}{xy-x-y}$ and use $\sin^2\theta + \cos^2\theta = 1$. This gives (9) and hence (10). The sign determination can be done as before. But we can also do it directly in terms of θ, for, the restriction on Q, the point of contact of AB, is equivalent to saying that $0 < \theta < \frac{\pi}{2}$. A direct computation gives $xy - x - y = \frac{(\sin\theta+\cos\theta+1)}{\sin\theta\cos\theta}$ which is positive for the specified range of θ. Therefore the positive sign must hold in (10). As before, this implies that $\lambda = 1$.

(iii) There is yet another way to do the problem in which we bypass the point of contact Q completely. Instead we write the equation of the line AB in the form $y = mx + c$, where $m = -\frac{k}{h}$ and $c = 2k$. We then simply apply the condition of tangency of this line with the given circle, viz., $x^2 + y^2 - 4x - 4y + 4 = 0$. After a little simplification we get $c^2 - 4c = 8m - 4mc$, or, in terms of h and k, $4k^2 - 8k = -\frac{8k}{h} + \frac{8k^2}{h}$. Simplifying and replacing h, k by x, y we get

$$xy - 2x - 2y + 2 = 0 \tag{13}$$

as the locus of P. It is not immediately obvious that this is the same as either (9) or (10). However, (13) is quite simple to analyse as it is. It represents a hyperbola (since the coefficients of x^2 and y^2 are both 0 while that of xy is not). A hyperbola has two branches. As the point P moves continuously in the plane, it can move only on one of these two branches. Once this idea strikes it is not difficult to derive (10) from (13). We simply multiply it by xy (which is non-zero since $x > 2$ and $y > 2$) and add $x^2 + y^2$ to get

$$x^2y^2 - 2x^2y - 2xy^2 + 2xy + x^2 + y^2 = x^2 + y^2$$

As the L.H.S. is simply $(xy - x - y)^2$ we get (9), or equivalently, (10). The sign determination is done as before.

This problem is a good example of how the same locus can be found in several ways. In the first approach x_1 and y_1 were two auxiliary variables and we needed three equations to eliminate them. In the second approach, there was only one parameter, viz., θ. In the third approach, there was no auxiliary variable and we got the answer directly by applying the condition for tangency. Each approach has its merits and demerits. In a particular problem, one approach may be better than another.

Comment No. 11:

In finding the equation of the locus of a moving point, the nature of the curve it represents is not generally known beforehand. We now come to the

third type of 'finding something' type problems of coordinate geometry, where we are asked to find the equation of some known type of a curve, say C, (such as a straight line or a circle or a pair of straight lines) from some facts given about that curve (such as that it passes through a given point or touches some other given curve). One of the methods to solve such problems is to start with a general equation of the given type of curve. For example, if C is given to be a circle then we take its equation in the general form as $x^2+y^2+2gx+2fy+c=0$. We then determine the constants appearing in it (in this example, the constants g, f and c) from the given conditions. To do this, we need to first write a system of equations in which there are as many equations as there are constants. This system is then solved to get the desired curve C (sometimes several solutions are possible).

However, this procedure is often cumbersome. With a little ingeneous use of the given facts about the curve C, we can often obviate the need for some of the constants and thereby get a much simpler system. We illustrate this in the following problem.

A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point $(-10, 2)$ in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle. (JEE 1990)

Let C be the desired circle. Call the lines $y = x$ and $x + y = 0$ as L_1 and L_2 respectively. We are given that C touches L_1 at P. However, the condition $OP = 4\sqrt{2}$ does not determine P uniquely. By inspection P has to be either $(4, 4)$ or $(-4, -4)$. (If you want to get this rigorously, solve the equations $x^2+y^2 = (4\sqrt{2})^2 = 32$ and $y = x$ simultaneously.) Now let the equation of C be $x^2 + y^2 + 2gx + 2fy + c = 0$. Then its centre, say Q is $(-g, -f)$ while its radius is $\sqrt{g^2 + f^2 - c}$. Depending upon which of the points $(4, 4)$ and $(-4, -4)$ lies on C, we get $32 + 8g + 8f + c = 0$ or $32 - 8g - 8f + c = 0$. Hence

$$f + g = \pm\frac{c + 32}{8} \tag{14}$$

As there are three constants we need two more equations. The condition that the point $(-10, 2)$ lies in the interior of C will only give us an inequality (specifically, $104 - 20g + 4f + c < 0$) and not an equation. However, the fact that C touches L_1 gives that the perpendicular distance of Q from L_1 equals the radius of C, i.e., $\frac{|f-g|}{\sqrt{2}} = \sqrt{g^2 + f^2 - c}$. After squaring and a little simplification, this means,

$$(f + g)^2 = 2c \tag{15}$$

Similarly, the fact that the chord cut off by C on the line $x + y = 0$ equals $6\sqrt{2}$ means that the distance of Q from this line is $\sqrt{(g^2 + f^2 - c) - (3\sqrt{2})^2}$, which gives, after simplification,

$$(f - g)^2 = 36 + 2c \tag{16}$$

We have to solve (14), (15) and (16) simultaneously. Squaring (14) and then combining it with (15) gives $(c+32)^2 = 128c$ which has only one solution, viz., $c = 32$. Putting this back in (14) gives $f + g = \pm 8$. But subtracting (16) from (15) gives $fg = -9$. By inspection (or by conversion to a quadratic in f), $f = \pm 9$ and correspondingly $g = \mp 1$ or vice versa. This gives four possibilities for the centre Q, viz., $(9,-1), (-1,9), (1,-9)$ and $(-9,1)$. The radius in all four cases is the same, viz., $\sqrt{g^2+f^2-c} = \sqrt{50} = 5\sqrt{2}$. As the point $(-10,2)$ is given to lie in the interior of C, its distance from the centre must be less than the radius. Out of the four possibilities, the centre must then lie at $(-9,1)$. Hence we get the equation of C as $(x+9)^2+(y-1)^2 = 50$ or $x^2+y^2+18x-2y+32 = 0$.

We now show how the same problem can be done more efficiently. The essential idea is basically the same, viz., to start with the equation of the circle C in a form which involves some constants and then to determine these constants using the given conditions about C. However, instead of starting with the general equation of a circle which involves three constants, we apply the given information about C to reduce the number of constants right at the beginning. Since C is given to touch L_1, its centre must lie on a line which passes through the point of contact and is perpendicular to L_1. As above, the point of contact of C with the line L_1 is either $(4,4)$ or $(-4,-4)$. Therefore the centre Q of C must lie either on the line $x+y = 8$ or on the line $x+y = -8$. Let r be the radius of C. The line $x+y = 8$ makes an angle of 135° with the x-axis. Therefore a point on this line which is at a distance r from $(4,4)$ must be of the form $(4 \pm r\cos 135^\circ, 4 \pm r \sin 135^\circ)$, i.e., $(4 \mp \frac{r}{\sqrt{2}}, 4 \pm \frac{r}{\sqrt{2}})$. So these are two possible locations of Q. Similarly, if Q lies on the line $x+y = -8$ then it must be either $(-4+\frac{4}{\sqrt{2}}, -4-\frac{r}{\sqrt{2}})$ or $(-4-\frac{4}{\sqrt{2}}, -4+\frac{r}{\sqrt{2}})$. Note that we have got the radius as well as the coordinates of the centre of C expressed in terms of a single variable, viz., r. To determine the value of r we need just one equation involving r. This is provided by the condition that the circle C cuts a chord of length $6\sqrt{2}$ on the line L_2 (with equation $x+y = 0$). Reasoning as before, this means that the perpendicular distance of Q from L_2 is $\sqrt{r^2-(3\sqrt{2})^2} = \sqrt{r^2-18}$. We already know that Q is either $(4 \pm \frac{r}{\sqrt{2}}, 4 \mp \frac{r}{\sqrt{2}})$ or $(-4 \pm \frac{r}{\sqrt{2}}, -4 \mp \frac{r}{\sqrt{2}})$. Regardless of which of these possibilities holds, the perpendicular distance of Q from the line L_2 comes out to be $\frac{8}{\sqrt{2}}$. Equating this with $\sqrt{r^2-18}$ gives $r^2 - 18 = 32$ and hence $r = 5\sqrt{2}$ (as r is positive). So there are four possible circles each with radius $5\sqrt{2}$ and centres at $(4 \pm 5, 4 \mp 5)$ and $(-4 \pm 5, -4 \mp 5)$ i.e., at $(9,-1), (-1,9), (1,-9)$ or at $(-9,1)$. As before, the condition that the point $(-10,2)$ lies in the interior of C implies that only the last possibility can hold. This way we get the same answer as before but with much less laborious work.

It is instructive to compare the two solutions. Consider the collection of all possible circles in the plane. Each member of this collection has an equation of the form $x^2+y^2+2gx+2fy+c = 0$. Here there are three constants, viz., g, f, c. These constants are also called 'parameters'. But this usage of the term is not to be confused with the parametric equations of a curve. Technically we say

that the totality of all circles in the plane is a 3-parameter family. This is also expressed by saying that a circle in the plane has three **degrees of freedom**. There are various subfamilies of this collection. For example, all circles in the plane which pass through a given point, say (a, b), form a subfamily. Any such circle has an equation of the form $(x-h)^2 + (y-k)^2 = (a-h)^2 + (b-k)^2$, i.e., $x^2 + y^2 - 2hx - 2ky + 2ah + 2bk - a^2 - b^2 = 0$. Note that now there are only two parameters, viz., h and k. So the family of all circles passing through a given point is a 2-parameter family. Similarly, the family of all circles which touch a given line L is a 2-parameter family. A still smaller family is the family of all circles which touch a given line at a given point. This is a 1-parameter family because every member of it is determined by just one parameter, viz., its radius. (Actually, there are two circles of a given radius which touch a given line at a given point. They lie on opposite sides of the line. But still we consider the family of all circles touching a given line at a given point as a 1-parameter family. This poses no serious difficulties. Or we can take the algebraic distance of the centre from the line (which is positive on one side of the line and negative on the other) as a parameter instead of the radius.)

We already encountered 1-parameter families of curves in problems of finding loci of moving points. Continuing this line of thinking, a 0-parameter family of circles is a family which contains just one or possibly several but always a finite number of circles. In the problem above, the family of circles which touch the line $y = x$ at a point which is $4\sqrt{2}$ units away from the origin is a 0-parameter family. It has four members. The problem essentially asks you to identify this 0-parameter family and then finally to choose one of the four members of it, using the piece of data that the point $(-10, 2)$ lies in the interior of the circle. Let us compare the two solutions above in the light of this terminology. The first solution consists of identifying the desired circle C from a 3-parameter family of circles. In other words, the journey was from a 3-parameter family to a 0-parameter family. In the second solution, on the other hand, we first used the given conditions to show that C must belong to a 1-parameter family of circles, viz., the family of all circles which touch the line $y = x$ at either $(4, 4)$ or at $(-4, -4)$.

The advantage which a 0-parameter family of curves has over a k-parameter family (for $k > 0$) is that the former is finite while the latter is infinite. When a search has been narrowed down to a finite number of cases, it can be completed by examining each of those cases one by one (as we did above to find out which of the four circles had the point $(-10, 2)$ in its interior). This is not possible for a k-parameter family if $k > 0$. Therefore our goal is to start from a suitable k-parameter family and first reduce the search to a 0-parameter family.

The topic of parametrised families of plane curves will come up again in Comment No. 17 of Chapter 19, where we shall see that such families represent certain differential equations.

Comment No. 12:

In view of what is said above, it might appear that in problems where we

have to find a curve of a given type which answers a given description, we should start with a k-parameter family, where k is as small as possible, the best choice being a 1-parameter family. But this is not a hard and fast rule. Sometimes the calculations involved with a single parameter become messy if attempted in a routine manner and the job can be done better with two (or more) parameters chosen suitably.

As an illustration, suppose that the straight lines $3x + 4y = 5$ and $4x - 3y = 15$ intersect at the point A. Points B and C are chosen on these two lines such that $AB = AC$. We are asked to determine the possible equations of the line BC passing through the point $(1, 2)$. (JEE 1990)

Call the lines as L_1 and L_2 respectively. Solving their equations simultaneously, A comes out as $(3, -1)$. One way to solve the problem is to take parametric equations of the lines L_1 and L_2. As $(3, -1)$ is a point on L_1 with equation $3x+4y = 5$, every point on L_1 is of the form $x = 3-4s, y = -1+3s$ for some value of the parameter s. In particular, this is the case for B. So we take B as $(3 - 4s, -1 + 3s)$, where the value of s is to be found. Similarly, by taking parametric equations of L_2 we may suppose $C = (3 + 3t, -1 + 4t)$, where again, t is an unknown to be determined. The equation of the line BC can now be expressed in terms of these two parameters, viz., s and t. However, to determine the values of these two parameters, it is hardly necessary to write the equation of BC explicitly. All we need is two equations in these two unknowns. One of these is provided by the condition that $AB = AC$ which gives $25s^2 = 25t^2$, i.e., $s = \pm t$. One more equation is given by the condition that the line BC passes through the point $(1, 2)$. Equating two expressions for the slope of the line BC we get $\frac{3s-3}{2-4s} = \frac{4t-3}{2+3t}$, which, upon simplification, becomes

$$25st - 6s - 17t = 0 \tag{17}$$

Putting $t = s$ in this gives $25s^2 = 23s$, whence $s = 0$ or $s = \frac{23}{25}$. The solution $s = 0$ has to be discarded since that would make B and C coincide with A and hence the line BC will not be defined. The case $s = \frac{23}{25}$ gives the slope of BC (which we already know to be $\frac{3s-3}{2-4s}$) as $\frac{1}{7}$. Hence the equation of BC in this case is $y - 2 = \frac{1}{7}(x - 1)$. Note that we did not have to find either B or C, since our interest was only in the slope of the line BC. We still have to dispose of the other possibility, viz., $t = -s$. Putting this into (17) gives $25s^2 = 11s$. Again we discard the solution $s = 0$. The other solution, viz., $s = \frac{11}{25}$ determines the slope of the line BC as -7 and hence its equation as $y - 2 = -7(x - 1)$. So there are two possible equations of the line BC.

In this solution we had two parameters, viz., s and t. Let us now see what happens if we try to solve the problem by expressing the line BC as a member of a 1-parameter family. Identifying such a family is fairly easy. As the line BC is given to pass through the point $(1, 2)$, it is uniquely determined by its slope, say m. Hence the equation of BC is of the form $y - 2 = m(x - 1)$. Here m is a single parameter whose possible value(s) is (are) to be found from the data. The most straightforward way to do so would be to express the coordinates of B and C in terms of m and then to use the condition that

$AB = AC$ so as to get an equation in m. This approach is conceptually simple but computationally clumsy. To find B we have to solve simultaneously the equations $y - 2 = m(x - 1)$ and $3x + 4y = 5$. Doing so, B comes out as $(\frac{4m-3}{4m+3}, \frac{2m+6}{4m+3})$. Similarly, solving $y - 2 = m(x - 1)$ and $4x - 3y = 15$ together we get C as $(\frac{3m-21}{3m-4}, \frac{-11m-8}{3m-4})$. We are given that $AB = AC$. As before, the point A is determined by solving $3x + 4y = 5$ and $4x - 3y = 15$. It comes out to be $(3, -1)$. A direct application of the distance formula then gives, after some simplification, that $(AB)^2 = 25(\frac{2m+3}{4m+3})^2$ and $(AC)^2 = 25(\frac{2m+3}{3m-4})^2$. Equating the two gives $4m + 3 = \pm(3m - 4)$. So $m = -7$ or $m = \frac{1}{7}$. This is the same answer as before but the computations are laborious and highly prone to slips of hand if done hastily.

However, the same 1-parameter family can be used a little more cleverly to get rid of these difficulties. As $AB = AC$ it follows that the line BC is perpendicular to the angle bisector of the angle between the lines L_1 and L_2. Put differently, if m is the slope of BC then the line, say L, through A having slope $-\frac{1}{m}$ must bisect one of the angles between the lines L_1 and L_2. As there are two angle bisectors, there are two possible equations for BC. The condition for bisection of an angle can be written down easily in terms of slopes. The slopes of L_1 and L_2 are $-\frac{3}{4}$ and $\frac{4}{3}$ respectively. The angle between L and L_1 must equal the angle between L_2 and L. (Note that here we are considering the signed angles and using the fact that L lies between L_1 and L_2. Hence the rotation about the point A which takes L_1 to L will be opposite to that which takes L_2 to L, i.e., one of them will be clockwise and the other anti-clockwise. As a result, the angle between L_2 and L will be numerically equal to the angle between L_1 and L, but they are of opposite signs. So we equate the angle between L and L_1 to that between L_2 and L, and not with that between L and L_2.) Equating the tangents of these angles, we see that a line through A with slope $-\frac{1}{m}$ will be an angle bisector if and only if

$$\frac{\frac{4}{3} + \frac{1}{m}}{1 - \frac{4}{3}\frac{1}{m}} = \frac{-\frac{1}{m} + \frac{3}{4}}{1 + \frac{1}{m}\frac{3}{4}} \tag{18}$$

which simplifies to $(4m + 3)^2 = (3m - 4)^2$ and hence to $4m + 3 = \pm(3m - 4)$. Or we can retain it as a quadratic as $7m^2 + 48m - 7 = 0$. In either case, we get $\frac{1}{7}$ and -7 as the two possible values of m, which is exactly the same answer as before, but arrived at with much less effort. The work could have been shortened even further by using readymade formulas for the angle bisectors for the angles between a pair of straight lines (see Equation (24) below). But it is better to rely more on the principles and a few basic formulas (such as the formula for the tangent of the difference of two angles which was used in writing (18) above) than tax your memory with a huge collection of formulas many of which are needed only in highly specialised contexts.

Comment No. 13:

There is yet another (although not substantially different) way to do the problem above. As observed above, the line BC is perpendicular to one of the

angle bisectors of the lines L_1 and L_2. So the problem is as good as solved if we can identify the two angle bisectors of the lines L_1 and L_2. So instead of taking the line BC as a member of a 1-parameter family of curves, we might as well consider the 1-parameter family of all lines passing through $A = (3, -1)$, which is the point of intersection of L_1 and L_2. We take the slope, say m of such a line as a parameter. (This should not be confused with the usage of m in the earlier solution, where it represented the slope of the line BC.) We can now find the values of m for which the line $y + 1 = m(x - 3)$ is an angle bisector of L_1 and L_2. The calculations are very similar to those which led to (18) above and give

$$\frac{\frac{4}{3} - m}{1 + \frac{4}{3}m} = \frac{m + \frac{3}{4}}{1 - \frac{3}{4}m} \tag{19}$$

which further yields $m = -7$ and $m = \frac{1}{7}$ as possible values. Correspondingly, the slope of BC is either $\frac{1}{7}$ or -7.

In this approach, we considered the family of all straight lines passing through the point of intersection of the lines L_1 and L_2 (whose equations are given) and expressed it as a 1-parameter family by taking the slope of such a line as the parameter. This, indeed, is the best choice because the condition for angle bisection can be stated most easily in terms of the slopes of the various lines involved. However, we mention here yet another way to parametrise the family of all straight lines passing through the point of intersection of two lines L_1 and L_2. First we rewrite the equations of L_1 and L_2 in the form $3x + 4y - 5 = 0$ and $4x - 3y - 15 = 0$. Now we let E_1 and E_2 denote the linear expressions $3x + 4y - 5$ and $4x - 3y - 15$ respectively. The line L_1 is precisely the set of those points of the plane where the expression E_1 vanishes. Similarly, the line L_2 is precisely the set of all points where E_2 vanishes. Note that both E_1 and E_2 are linear expressions in the variables x and y. So for any real number λ, the expression $E_1 + \lambda E_2$ is also linear and hence the equation $E_1 + \lambda E_2 = 0$ represents a straight line, say L, in the plane. Note that L must pass through A, the point of intersection of L_1 and L_2, since $E_1 + \lambda E_2$ vanishes whenever E_1 and E_2 both vanish, which is what happens at the point of intersection. Conversely, it is not hard to show that any line L, other than L_2, which passes through A has an equation of the form $E_1 + \lambda E_2 = 0$ for some real number λ. All we have to do is to take the equation of L in a form $ax + by + c = 0$ and find a suitable λ such that $3 + 4\lambda, 4 - 3\lambda$ and $-5 - 15\lambda$ are proportional to a, b and c respectively. This amounts to solving the system of three linear equations, viz., $3 + 4\lambda = ar, 4 - 3\lambda = br$ and $-5 - 15\lambda = cr$ for the two unknowns λ and r. The determinant criterion for the existence of a solution for this system reduces to saying that $3a - b + c = 0$, which is exactly the condition that the line L passes through $A = (3, -1)$, (As noted before, for no value of the parameter λ, will the equation $E_1 + \lambda E_2 = 0$ represent the line L_2. If we want L_2 to be also included in our 1-parameter family, then we can consider $(1 - \lambda)E_1 + \lambda E_2$ instead of $E_1 + \lambda E_2$. Then we get L_2 for $\lambda = 1$. But when we know beforehand, as in the present problem, that the line we are looking for cannot possibly equal L_2, it is perfectly all right to take E as $E_1 + \lambda E_2$, which is a little easier computationally

than $(1-\lambda)E_1+\lambda E_2$.)

Let us now revisit the problem above in the light of this new parametrisation of the family of all straight lines passing through A, the point of intersection of L_1 and L_2. Each of the two angle bisectors of the angle between L_1 and L_2 will have an equation of the form $E_1+\lambda E_2=0$ for some value of the parameter λ. Writing this expression in full, it is $(3+4\lambda)x+(4-3\lambda)y-5-15\lambda=0$. This is a straight line with slope $\frac{4\lambda+3}{3\lambda-4}$. Equating the angles which this line makes with L_1 and L_2 we get, exactly the way we got (19), that

$$\frac{\frac{4}{3}-\frac{4\lambda+3}{3\lambda-4}}{1+\frac{4}{3}\,\frac{4\lambda+3}{3\lambda-4}}=\frac{\frac{4\lambda+3}{3\lambda-4}+\frac{3}{4}}{1-\frac{4\lambda+3}{3\lambda-4}\,\frac{3}{4}} \tag{20}$$

which simplifies to $-\frac{1}{\lambda}=-\lambda$, giving $\lambda^2=1$, i.e., $\lambda=\pm1$. The corresponding values of the slopes of the angle bisectors are -7 and $\frac{1}{7}$. The corresponding slopes of the line BC are $\frac{1}{7}$ and -7 as we obtained in the earlier solutions.

It can be argued that (20) is not only more complex than (19), but that the derivation of (20) was essentially through (19) since we first calculated the slope of the line $E_1+\lambda E_2=0$ anyway. So we really gained nothing by taking λ as the parameter instead of m. This objection is quite valid, because, as remarked earlier, as far as the present problem is concerned, the slope is the best parameter for the family of all lines passing through the point of intersection of the lines L_1 and L_2. Still, we have mentioned the second approach because it can be generalised when we want to consider a 1-parameter family of certain types of curves which pass through the common point(s) of two given curves, say C_1 and C_2 of a similar type. Suppose, for example, that C_1 and C_2 are two circles with equations, say, $E_1=0$ and $E_2=0$, where $E_1=x^2+y^2+2g_1x+2f_1y+c_1$ and $E_2=x^2+y^2+2g_2x+2f_2y+c_2$. We let $E=E_1+\lambda E_2$, where λ is any real number. Then the equation $E=0$ represents a circle, say C, which passes through the common points, if any, of C_1 and C_2. (For $\lambda=-1$, E becomes a linear expression in x and y and then C represents a straight line often called the **radical axis** of the two circles C_1 and C_2. This is all right since for many purposes a straight line can be considered as a special case of a circle, specifically, a circle passing through the so-called 'point at infinity'. But we shall not go into it.) Conversely, every such circle, with the exception of C_2 is of this form. (Once again, if we want to include C_2 as well we consider $(1-\lambda)E_1+\lambda E_2$ instead of $E_1+\lambda E_2$, which is anyway a better choice since the coefficients of x^2 and y^2 then come out to be 1 as we would normally like to have in the equation of a circle.) Note that this representation of the family of all circles passing through the common points, say A and B, of C_1 and C_2 has been obtained *without* identifying these common points.

A slightly different (and simpler) parametrisation of the family of circles considered above is also possible. Let $E_3=E_1-E_2=2(g_1-g_2)x+2(f_1-f_2)y+c_1-c_2$. Then $E_3=0$ is the equation of the line, say L, joining A and B. (In case C_1 and C_2 touch each other, then A and B coincide and L is a common tangent at this point.) Now for any real number μ, the equation $E_1+\mu E_3=0$ i.e., the equation $x^2+y^2+2\mu(g_1-g_2)x+2\mu(f_1-f_2)y+\mu(c_1-c_2)=0$ represents

a circle. Moreover this circle passes through the points of intersection of the circle C_1 and the line L. But these points are precisely A and B. So we could as well parametrise the family of all circles passing through the common points of C_1 and C_2 by taking μ as a parameter and writing a typical member as $E_1 + \mu E_3 = 0$.

As a slight variation of this, suppose, we are given two points, say $A = (x_1, y_1)$ and $B = (x_2, y_2)$, in the plane and we want to consider the 1-parameter family of all circles passing through A and B. One way to do so would be to observe that the centre of any such circle must lie on the perpendicular bisector of the segment AB. Parametrising this perpendicular bisector, the centre is of the form $(x_0 + t(y_1 - y_2), y_0 + t(x_2 - x_1)$, where (x_0, y_0) is the midpoint of AB and t is a parameter. Taking this as the centre and setting the radius equal to the distance of the centre from A (or from B) we get the equation of a typical circle passing through A and B.

However, for certain applications, this approach is not very convenient. Instead, a slight variation of the approach given above works better. Instead of taking two circles, say C_1 and C_2, passing through A and B, we take only one such circle C_1 with equation $E_1 = 0$. Next, we write the equation of the line L joining A and B in the form $E_3 = 0$. Then any circle passing through A and B has its equation of the form $E_1 + \mu E_3 = 0$. It does not matter which circle C_1 we choose as long as it passes through A and B. (If we change C_1 to, say C_1', E_1 will change to E_1' (say). But E_3 will not change and by changing the value of μ to some μ' we can ensure that $E_1 + \mu E_3 = E_1' + \mu' E_3$, because $E_1 - E_1'$ is a multiple of E_3.) One canonical choice is to take C_1 as the circle having AB as a diameter. For any point P on this circle, the lines PA and PB are at right angles to each other. So the equation of C_1 is $E_1 = 0$ where $E_1 = (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$. Thus we see that the 1-parameter family of all circles passing through two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$ is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \mu E_3 = 0 \tag{21}$$

where $E_3 = 0$ is the equation of the line L passing through A and B. (A limiting case of this in which A and B coincide is interesting. In that case L becomes the tangent. Thus, given a line L with equation $E_3 = 0$ and a point $A = (x_1, y_1)$ on it, the 1-parameter family of all circles which touch L at A is given by

$$(x - x_1)^2 + (y - y_1)^2 + \mu E_3 = 0 \tag{22}$$

Of course, this can also be derived directly. See Exercise (9.49)(a).)

As a direct application of (21), we have the following problem. (JEE 1993)

Consider a family of circles passing through two fixed points $A(3, 7)$ and $B(6, 5)$. Show that the chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.

Let L be the line passing through A and B. The equation of L comes out as $2x+3y-27 = 0$. Applying formula (21) above, a typical circle passing through A and B has an equation of the form $x^2+y^2-9x-12y+53+\mu(2x+3y-27) = 0$

for some value of the parameter μ. The common chord of this circle with the given circle has equation $-9x-12y+53+\mu(2x+3y-27) = -4x-6y-3$, i.e., $(2\mu-5)x+(3\mu-6)y-27\mu+53 = 0$. We have to show that this line passes through a fixed point, say (h,k), regardless of the value of μ. This amounts to showing that there exist some real numbers h and k such that the equation $(2\mu-5)h+(3\mu-6)k-27\mu+53 = 0$ is satisfied for all real values of μ. We rewrite this equation as $(2h+3k-27)\mu-5h-6k+56 = 0$. As h and k are fixed, the only way this equation can hold for infinitely many values of μ is that the coefficient of μ and the constant term both vanish. This means $2h+3k-27 = 0$ and $-5h-6k+56 = 0$. Solving, $h = 2$ and $k = \frac{23}{3}$. So all the common chords pass through the point $(2, \frac{23}{3})$ and hence are concurrent.

Comment No. 14:

It should be abundantly clear by now how a clever choice of a parametrised family of a given type of curves can work wonders. Still, we emphasise again that in many problems it is *not* a good idea to express the desired curve as a member of a parametrised family of curves. Instead, some methods, which are tailored for the particular type of curve, coupled with a few elementary facts from pure geometry work much better as we already saw in several problems. It is strongly suggested that initially you try each problem by as many methods as you can, even though some of these methods are very laborious. After gaining some experience, you will be able to predict in advance which method will work better for a given problem. For example, if we want to find the equation of a line L which is given to pass through two points, say $A = (x_1, y_1)$ and $B = (x_2, y_2)$, then instead of starting with a 2-parameter family of lines of the form $y = mx+c$ and determining the parameters m and c from the equations $y_1 = mx_1 + c$ and $y_2 = mx_2+c$, it is far better to apply the two points formula and get the equation of L directly as $\dfrac{y-y_1}{x-x_1} = \dfrac{y_2-y_1}{x_2-x_1}$. Depending on the data, other formulas for the equation of a straight line such as the two intercepts form (which was used in the problem involving the ladder in Comment No. 8 above) or the pedal form (which gives the equation of a line in terms of the length and the inclination of the perpendicular on it from the origin) may be more suitable. In the next chapter we shall see instances where it is convenient to take the equation of a line in a parametric form as $x = x_0 + r\cos\theta,\ y = y_0 + r\sin\theta$, where (x_0, y_0) is a point on the line and θ is the angle it makes with the positive x-axis.

In the case of the equation of a circle, the most commonly used form is the centre−radius formula, although a better method is available for the equation of a circle for which two ends of a diameter are given as we saw in the derivation of (21) above. Sometimes the centre is specified in a manner which entails some work, as in the following problem. (JEE 1979)

Find the equation of a circle which passes through the point $(2, 0)$ and whose centre is the limit of the point of intersection of the lines

$$3x + 5y = 1 \tag{23}$$

$$(2+c)x + 5c^2y = 1 \tag{24}$$

as c tends to 1.

This problem is a combination of coordinate geometry and a little calculus. If we put $c = 1$ in (24), the two lines become identical and the centre is not uniquely determined. So we first have to solve the system above for $c \neq 1$ and *then* take the limit as $c \to 1$. If we solve (23) and (24) together for $c \neq 1$, we get $x = \dfrac{c^2-1}{3c^2-c-2} = \dfrac{c+1}{3c+2}$ which tend to $\frac{2}{5}$ as $c \to 1$. The value of y can now be found from (23) as $-\frac{1}{25}$. So the centre of the circle is at $(\frac{2}{5}, -\frac{1}{25})$. As it is given to pass through $(2, 0)$, its radius equals $\sqrt{(2-\frac{2}{5})^2 + (0+\frac{1}{25})^2} = \sqrt{\frac{64}{25} + \frac{1}{625}} = \sqrt{\frac{1601}{625}}$. Hence the equation of the circle is $(x-\frac{2}{5})^2 + (y+\frac{1}{25})^2 = \frac{1601}{625}$.

Comment No. 15:

For the other types of conics, such as ellipses, hyperbolas and parabolas, it takes as many as five conditions to determine an ellipse or a hyperbola uniquely. In the case of a parabola we need four. This is obviously a very complicated task if attempted in its full generality. Fortunately, the problems are generally based on the assumption that these curves are in their standard forms which reduces the number of parameters to two in the case of an ellipse and a hyperbola and just one in the case of a parabola. Occasionally, there is a shift of the centre or of the vertex as we saw in the case of the problem involving a parabola in Comment No. 9 above. But there is never a rotation of the axes. As a result, identifying these curves is relatively easy.

As for problems involving pairs of straight lines, when we are given a pair of straight lines in a numerical problem, as in the Main Problem, the best thing is to factorise the equation into two linear factors. Usually this can be done by inspection. If not, here is a systematic method. The basic idea is to shift the origin to the point of intersection of the two lines, say (p, q). In that case the new coordinates, say X and Y are related to the old ones by $x = X + p$ and $y = Y + q$. With these substitutions, the equation in the Main Problem becomes $2(X+p)^2+5(X+p)(Y+q)+3(Y+q)^2+6(X+p)+7(Y+q)+4 = 0$ i.e., $2X^2+5XY+3Y^2+(4p+5q+6)X+(5p+6q+7)Y+(2p^2+5pq+3q^2+6p+7q+4) = 0$. Since the new origin is the point of intersection of the two lines, the linear terms must vanish, i.e., the coefficients of X and Y must be 0. This gives a system of equations : $4p + 5q + 6 = 0$ and $5p + 6q + 7 = 0$ solving which we get $p = 1$ and $q = -2$. With these values, the expression $2p^2+5pq+3q^2+6p+7q+4$ vanishes as it ought to (or else the original equation will not represent a pair of straight lines.) Hence the (new) equation of the pair of straight lines now simplifies to $2X^2 + 5XY + 3Y^2 = 0$. This could have been written down even without finding p and q, since the coefficients of X^2, Y^2 and XY must match those of x^2, y^2 and xy in the original equation, regardless of what p and q are and the constant term must drop out if the new origin is at the point of intersection of the two lines. It is very easy to factorise $2X^2 + 5XY + 3Y^2$ by inspection as $(X + Y)(2X + 3Y)$. (In case, it is still not so, write it as $3(\frac{2}{3}X^2 + \frac{5}{3}XY + Y^2) = 3(Y - m_1X)(Y - m_2X)$ and determine m_1 and m_2 as roots of a suitable quadratic, viz., $m^2 + \frac{5}{3}m + \frac{2}{3} = 0$, which, in essence, is the

same as $3m^2 + 5m + 2 = 0$. That would also give the slopes of the two lines.) So the two lines are $X + Y = 0$ and $2X + 3Y = 0$, which, in the old coordinates become $x + y + 1 = 0$ and $2x + 3y + 4 = 0$. (Or, without determining p and q first, we factorise $2x^2 + 5xy + 3y^2 + 6x + 7y + 4$ as $(x + y + \alpha)(2x + 3y + \beta)$ and determine α and β by solving the system $2\alpha + \beta = 6$ and $3\alpha + \beta = 7$. They come out as $\alpha = 1$ and $\beta = 4$. The point (p, q) can now be determined by solving $x + y + 1 = 0$ and $2x + 3y + 4 = 0$ simultaneously. This is, in fact, the best method to factorise the given expression when we cannot do it by inspection. In some problems, e.g., in finding the angle between the two lines, we are interested only in the slopes of the two lines. In that case we need not factorise the given expression, for we can get the slopes by simply factorising $2x^2 + 5xy + 3y^2$ which is much easier.)

When the equation of a pair of straight lines, say L_1 and L_2, through the origin is given in a general form as $ax^2 + 2hxy + by^2 = 0$, it is often advisable *not* to factorise it. Many times the information we need can be obtained without an explicit factorisation. Suppose, for example, that we want to find the angle, say α, between the lines L_1 and L_2. If we denote their slopes by m_1 and m_2 respectively, then the equation of the pair can be written as $(y - m_1x)(y - m_2x) = 0$, i.e., as $y^2 - (m_1 + m_2)xy + m_1m_2x^2 = 0$. Comparing this with the given equation $ax^2 + 2hxy + by^2 = 0$, we see that $m_1 + m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$. From this, we can determine m_1 and m_2 individually as the root of the quadratic $bm^2 + 2hm + a = 0$. But that will involve messy radicals. Nor is it needed. The formula for $\tan\alpha = \frac{m_1 - m_2}{1 + m_1m_2}$ can be rewritten as $\pm\frac{\sqrt{(m_1+m_2)^2 - 4m_1m_2}}{1+m_1m_2}$ and this comes out as $\pm\frac{2\sqrt{h^2-ab}}{a+b}$. (The two signs correspond to the two angles between L_1 and L_2, the positive sign to the acute and the negative to the obtuse angle. The lines are orthogonal, i.e., at right angles to each other if and only if $a + b = 0$. The quadratic for m also shows that the two lines will be real and distinct, real and coincident or imaginary, according as $h^2 - ab$ is positive, zero or negative.)

When it comes to finding the equation of a pair of straight lines, once again, in numerical problems it is better to find the equations of the two lines separately and then to multiply the linear expressions which vanish on them to get the desired equation. However, in non-numerical problems, it is not a good idea to do so. Suppose for example, that we want to find the equation of the pair of angle bisectors of the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$. Call the lines as L_1 and L_2 and let m_1 and m_2 represent their slopes respectively. As we saw just now, $m_1 + m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$. Let $y = mx$ be an angle bisectors of the two lines. Then equating the tangents of half the angles between L_1 and L_2, we get

$$\frac{m_1 - m}{1 + m_1m} = \frac{m - m_2}{1 + mm_2} \tag{25}$$

which upon simplification becomes a quadratic in m, viz., $m^2(m_1 + m_2) + 2m(1 - m_1m_2) - (m_1 + m_2) = 0$, or, putting the values of $m_1 + m_2$ and of m_1m_2, $hm^2 + (a - b)m - h = 0$. The slopes of the two angle bisectors are the roots, say m_3 and m_4, of this quadratic in m. We are not interested in either of

them *per se*. We are interested in the equation of the pair of the angle bisectors, which is simply $(y - m_3x)(y - m_4x) = 0$, i.e., $y^2 - (m_3 + m_4)xy + m_3m_4x^2 = 0$. Since $m_3 + m_4 = \frac{b-a}{h}$ and $m_3m_4 = -1$ (which is consistent with the fact that the two angle bisectors are always at right angles to each other), the equation of the angle bisectors reduces, after a little simplification, to

$$h(x^2 - y^2) + (b - a)xy = 0 \tag{26}$$

If the point of intersection, say (p, q), of the two lines is other than the origin, this formula requires a slight modification which is obtained by shifting the origin to (p, q), obtaining the equation w.r.t. the new coordinates and transferring it back to the old coordinates.

Sometimes, even in numerical problems, if the pair of lines happens to be mutually at right angles then its equation can be determined using (26) above, i.e., by recognising it as the pair of angle bisectors between two suitable lines. Suppose, for example, that PQR is a right-angled, isosceles triangle, right-angled at P $(2, 1)$. The equation of the line QR is given as $2x + 3y = 1$ and we are asked to find the equation representing the pair of lines PQ and PR. (JEE 1999)

One way to do this problem would be to find the equations of the two lines separately and multiply. But in the present problem we can take advantage of the fact that that PQ and PR are given to be at right angles to each other. So, if we can identify two lines, say L_1 and L_2 through P so that PQ and PR are precisely the angle bisectors of the angles between L_1 and L_2, then we can use (26). In the present problem, one such choice of L_1 and L_2 suggests itself. Since PQR is also given to be isosceles, we can take L_1 to be the line through P which is parallel to the line QR and L_2 to be line through P which is parallel to QR. (In effect, L_1 and L_2 are the angle bisectors of PQ and PR and further PQ and PR are the angle bisectors of these angle bisectors!) The slope of L_1 is that of QR which is -2 while that of L_2 is $\frac{1}{2}$. So if we shift the origin to $P = (2, 1)$ then the equation of the pair L_1 and L_2 in the new coordinates is $(Y + 2X)(Y - \frac{1}{2}X) = 0$, i.e., $2X^2 - 3XY - 2Y^2 = 0$. Hence, using (26), the equation of PQ and PR, in the new coordinates is $3(X^2 - Y^2) + 8XY = 0$. To get it in the old coordinates, we merely put $X = x - 2$ and $Y = y - 1$. Then the equation becomes $3(x - 2)^2 - 3(y - 1)^2 + 8(x - 2)(y - 1) = 0$, i.e., $3x^2 + 8xy - 3y^2 - 20x - 10y + 25 = 0$.

Comment No. 16:

There is a somewhat novel technique, called **homogenisation** which can be used to find the equation of a pair of straight lines which join the origin O to the two points of intersection, say P and Q, of a straight line L (not passing through O) and a conic C. It is based on the fact that the equation of a pair of straight lines passing through O is always of the form $Ax^2 + 2Hxy + By^2 = 0$. The terms appearing in the L.H.S. are all of the same degree (in this case 2). Such an equation is therefore said to be **homogeneous** (of degree 2). Now let the equation of the given conic C be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \tag{27}$$

This equation is, of course, not homogeneous (unless g, f and c are all 0). Since the line L does not pass through the origin, its equation can be written in the form

$$lx + my = 1 \tag{28}$$

Note that this is also not homogeneous. But the L.H.S. is homogeneous (of degree 1). We can also rewrite (28) as $(lx+my)^2 = 1$ or as $(lx+my)^3 = 1$ etc. In doing so it may appear that we are unnecessarily complicating the simple looking equation (28). But the advantage is that the expressions $(lx + my)^2$ and $(lx + my)^3$ are homogeneous (of degrees 2 and 3 respectively). The former can be used cleverly as we are about to see.

The coordinates of the points P and Q satisfy both (27) and (28) and hence they also satisfy any equation that can be obtained from these two. In particular if we multiply the linear terms (viz., $2gx$ and $2fy$) in (27) by $lx+my$ and the constant term c by $(lx + my)^2$ then we get that the coordinates of P and Q satisfy the equation

$$ax^2 + 2hxy + by^2 + 2gx(lx + my) + 2fy(lx + my) + c(lx + my)^2 = 0 \tag{29}$$

Note that we are *not* saying that (27) and (29) represent the same curve. Indeed, this is quite false. All we are saying is that the points P and Q lie on the curve, say C_1, whose equation is (29). But the L.H.S. of (29) is a homogeneous expression of degree 2. So this curve C_1 must be a pair of straight lines passing through the origin O. But the points P and Q lie on C_1. So C_1 is nothing but the pair of straight lines OP and OQ. Therefore (29) is the equation we were looking for.

As with (26), by shifting the origin suitably the technique can be applied to find the equation of a pair of straight lines joining a given point, say A, to the points, say P and Q, in which a given line L intersects a given conic C. The beauty of this method is that we do not have to determine the coordinates of the points P and Q. To illustrate how this saves messy computations, suppose $x = 9$ is a chord of contact of the hyperbola $x^2 - y^2 = 9$ and we have to find the equation of the corresponding pair of tangents. (JEE 1999) The most direct way to do so would be to identify the points, say P and Q, in which the line $x = 9$ cuts the hyperbola. They come out to be $(9, 6\sqrt{2})$ and $(9, -6\sqrt{2})$. We can then find the tangents at these points as the lines $9x - 6\sqrt{2}y - 9 = 0$ and $9x + 6\sqrt{2}y - 9 = 0$. By taking the product, we get the equation of the pair of tangents as $(9x - 9)^2 - 72y^2 = 0$. Although this calculation is not prohibitively complicated let us see how the method of homogenisation works here. Let the two tangents meet at A. As the hyperbola as well as the line L (with equation $x = 9$) are symmetric about the x-axis, the point A must lie on the x-axis. Even without finding the points P and Q explicitly, we know that their x-coordinates are 9. So the equation of the tangent at either one of them will be of the form $9x + ... = 9$ and so they will meet the x-axis at the point $(1, 0)$. Let us call this point A. Shifting the origin to A, the equation of the hyperbola becomes $(X+1)^2 - Y^2 = 9$, i.e., $X^2 - Y^2 + 2X - 8 = 0$ while that of the chord of contact

becomes $X = 8$, i.e., $\frac{1}{8}X = 1$. We now homogenise the former with the help of the latter to get $X^2 - Y^2 - 2X \times \frac{1}{8}X - 8 \times (\frac{1}{8}X)^2 = 0$ or, $9X^2 - 8Y^2 = 0$ as the equation of the pair of tangents in the new coordinates. Putting $X = x - 1$ and $Y = y$, this changes to $9(x-1)^2 - 8y^2 = 0$ which is the same answer as before, but obtained without computing P and Q.

It may be argued that in the second method, we used symmetry to find A and that with some other chord of contact the determination of A would not have been so easy. But this is not so, because there are other ways to determine the point A. For example, let $A = (x_1, y_1)$. The equation of the chord of contact of A then is $xx_1 - yy_1 = 9$. But the equation of the chord of contact is given as $x = 9$. Comparing the two equations, we get $x_1 = 1$ and $y_1 = 0$. This is, in fact, a better way of finding A because it would work no matter what chord of contact we were given. But it is always a good idea to look for simplifying features such as symmetry even though they may be useful only in special contexts.

As yet another application of the method of homogenisation, let us find the locus of the foot, say P, of the perpendicular drawn from the origin to a chord of a circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which subtends a right angle at the origin. (JEE 1988) Denote the two ends of the (variable) chord by A and B and let (h, k) be the current coordinates of P. Then the slope of AB is $-\frac{h}{k}$ and hence its equation is $y - k = -\frac{h}{k}(x - k)$ which we rewrite as $\dfrac{hx + ky}{h^2 + k^2} = 1$ for the purpose of homogenisation. Then the equation of the pair of lines OA and OB is obtained by homogenising the equation of the given circle. It comes out as

$$x^2 + y^2 + (2gx + 2fy)\left(\frac{hx + ky}{h^2 + k^2}\right) + c\left(\frac{hx + ky}{h^2 + k^2}\right)^2 = 0$$

Since the chord AB subtends a right angle at the origin, the lines represented by this equation are perpendicular to each other. As noted above, the condition for this is that the sum of the coefficients of x^2 and y^2 should vanish. This means $2 + \dfrac{2gh + 2fk}{h^2 + k^2} + \dfrac{c(h^2 + k^2)}{(h^2 + k^2)^2} = 0$, which upon simplification becomes, $h^2 + k^2 + gh + fk + \frac{c}{2} = 0$. As usual, the locus of P is obtained by replacing h, k with x, y respectively. Note that it is also a circle, whose centre is halfway between the origin and the centre of the original circle. Once this is known, then it can also be proved by pure geometry (which we omit). But it is doubtful, if one can predict beforehand that the locus will be a circle, without using coordinates. In the present problem, P was given as the foot of the perpendicular from the origin to the chord AB. If it were some other point on the chord, the argument can be modified accordingly. For example, if P were to be the mid-point of AB, then using the formula for the equation of a chord of a conic in terms of its mid-point, we get the equation of AB as $\dfrac{(h + g)x + (k + f)y}{h^2 + k^2 + gh + fk} = 1$.

Such problems can, of course, also be done without homogenisation. After finding the equation of the line AB, we solve it simultaneously with the equation of the circle so as to determine A and B. We then calculate the slopes of OA and OB. Setting the products of these slopes equal to -1 we get the

equation of the locus. This method is more direct and conceptually simpler but unnecessarily laborious as most of the work involved in it is superfluous. Note that we are not interested in the coordinates of A and B *per se*. We want only the slopes of OA and OB. That is also not quite correct. We want only the product of these two slopes rather than their individual values. And by getting the equation of the pair of lines OA and OB, we get this information effortlessly, without having to find either of the two lines individually. The advantage of the technique of homogenisation should be quite apparent now.

Comment No. 17:

The technique of homogenisation can also be applied to find the equation of a pair of straight lines OP and OQ when P and Q are points of intersection of a conic C and a circle centred at the origin. The equation of such a circle is of the form $x^2 + y^2 = r^2$ which can be rewritten as

$$\frac{x^2 + y^2}{r^2} = 1 \tag{30}$$

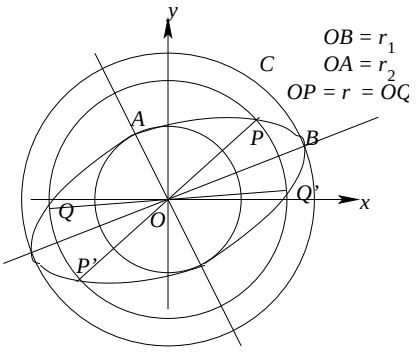
This is very analogous to (28) in that the L.H.S. is homogeneous (the difference being that the degree here is 2 while the degree of the expression in the L.H.S. of (28) was 1). Not surprisingly, (30) can be used in a manner which is analogous to (28) as we illustrate in the following problem. (JEE 1979)

Find the condition in terms of A, H and B that the equation

$$Ax^2 + 2Hxy + By^2 = 1 \tag{31}$$

represents an ellipse. Also when so, find the area of the ellipse.

Let C denote the curve represented by (31). This curve is symmetric w.r.t. the origin in that if a point $P = (x, y)$ lies on C, so does its reflection w.r.t. the origin, i.e., the point $P' = (-x, -y)$. (For this reason, a conic such as C is also called a **central conic**, in the present case the centre being the origin.) Now a circle will normally intersect C in four points and so there will be four lines joining these points of intersection to the origin. But if the circle happens to be centred at O (as is the case with the circle represented by (30)), then these 4 points will pair off in two pairs, each consisting of two points which are reflections of each other w.r.t. the origin. So in such a case, there will be only two lines joining O with the points of intersection (see the lines PP' and QQ' in the figure).



These two lines can be obtained by homogenising (31) using (30). Thus, we multiply the constant term 1 on the R.H.S. by the L.H.S. of (30) to get

$$(A - \frac{1}{r^2})x^2 + 2Hxy + (B - \frac{1}{r^2})y^2 = 0 \tag{32}$$

as the equation of the pair of lines joining the common points of (30) and (31). The nature of these lines will change as the parameter r changes. Suppose the curve C is an ellipse with r_1 and r_2 as the lengths of the semi-major and semi-minor axes respectively. Then it is clear that for $0 < r < r_2$ and for $r > r_1$, the circle (31) will not meet C and hence the lines represented by (32) will be imaginary. For $r_2 < r < r_1$, there will be four distinct points of intersection and so the two lines will be real and distinct. For $r = r_1$ the circle touches the ellipse externally and the two lines coincide as now there are only two diametrically opposite points common to the ellipse and the circle. Similarly, if $r = r_2$ then the circle touches the ellipse internally and again the two lines coincide. From (32), the condition for the two lines to coincide is $H^2 = (A - \frac{1}{r^2})(B - \frac{1}{r^2})$ which can be rewritten as

$$(AB - H^2)r^4 - (A + B)r^2 + 1 = 0 \tag{33}$$

This equation can be considered as a quadratic in r^2. Write t for r^2. Since r^2 can only be positive, the condition that (29) represents an ellipse is that the quadratic $(AB - H^2)t^2 - (A + B)t + 1 = 0$ have two distinct positive real roots. The discriminant $(A+B)^2 - 4(AB - H^2)$ equals $(A-B)^2 + 4H^2$ which is always positive (except when $A = B$ and $H = 0$, in which case (29) will represent a circle if $A > 0$ and the empty set if $A \leq 0$). So the two roots, say t_1 and t_2 (corresponding to r_1^2 and r_2^2) are certainly real and distinct. To say that they are both positive is equivalent to saying that their sum $t_1 + t_2$ and their product $t_1 t_2$ are both positive. But $t_1 t_2 = \frac{1}{AB - H^2}$ which will be positive if and only if $AB - H^2 > 0$. This in particular means that A and B are of the same sign. Hence, in the presence of this condition, the positivity of $t_1 + t_2$ (which equals $\frac{A+B}{AB-H^2}$) amounts to saying that both A and B are positive. So the necessary and sufficient condition that (29) represents an ellipse is that $AB > H^2$ and $A > 0$. Now to find the area of the ellipse we use a well-known formula (to be proved in Chapter 17, Comment No. 5) that the area of an ellipse equals π times the product of its semi-major and semi-minor axes. We are denoting these by r_1 and r_2 respectively, and they in turn are the square roots of t_1 and t_2. So the area of the ellipse represented by (29) equals $\pi r_1 r_2 = \pi\sqrt{t_1 t_2} = \frac{\pi}{\sqrt{AB - H^2}}$.

Another solution based upon bringing C to a standard form by a change of coordinate axes will be given in Comment No. 18 of the next chapter.

Comment No. 18:

Sometimes the problem asks a sketch of a set defined by some equation in terms of the coordinates of a point (x, y). If this set happens to be of a standard type such as a straight line or a circle, then usually it is easy to sketch it once we identify some vital features of it (such as slope, or radius). But in some problems the given set may be of an unusual type. In that case, to sketch it accurately, we have to go strictly by its definition and it may become necessary to decompose the set into various subsets. Here is a good illustration of this. (JEE 2000)

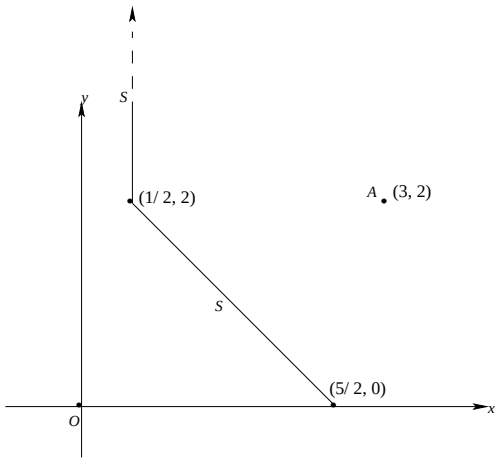
A new distance $d(P, Q)$ is defined between any two points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ of a coordinate plane by $d(P, Q) = |x_1 - x_2| + |y_1 - y_2|$. Let

$O = (0, 0)$ and $A = (3, 2)$. Prove that the set of points in the first quadrant which are equidistant from the points O and A (according to the new definition of distance) is the union of a finite line segment and an infinite ray. Sketch this set in a labelled diagram.

Denote the set in question by S. Then a point $P = (x, y)$ is in S if and only if x, y are both non-negative and further $d(P, O) = d(P, A)$. With the given definition of the new distance, the last condition translates into

$$|x| + |y| = |x - 3| + |y - 2| \tag{34}$$

Since the absolute value of an expression has two possible definitions depending on the sign of that expression and (34) involves 4 absolute values, logically it will split into 16 ($= 2^4$) cases. In the present case, as x and y are given to be non-negative, many of these cases do not arise. In fact, because of this hypothesis, the L.H.S. of (34) is always $x + y$. The R.H.S. splits into 4 cases depending upon the signs of $x - 3$ and of $y - 2$. We tackle them one by one:



(i) Assume $0 \le x \le 3$ and $0 \le y \le 2$. Then (34) becomes $x+y = 3-x+2-y$, i.e., $x + y = \frac{5}{2}$. This is the equation of a straight line whose portion in the first quadrant is the segment from the point $(\frac{5}{2}, 0)$ to the point $(0, \frac{5}{2})$. But this entire segment is not in S. The condition $x \le 3$ is satisfied at all points of it. But because of the requirement $y \le 2$, only the portion of this segment from the point $(\frac{5}{2}, 0)$ to the point $(\frac{1}{2}, 2)$ will be in S.

(ii) Assume $x \ge 3$ and $0 \le y \le 2$. Here (34) reduces to $x + y = x - 3 + 2 - y$ i.e., to $y = -\frac{1}{2}$. As this is inconsistent with $y \ge 0$, this case contributes nothing to the set S.

(iii) Assume $0 \le x \le 3$ and $y \ge 2$. Then (34) becomes $x+y = 3-x+y-2$, i.e., $x = \frac{1}{2}$ which represents a vertical straight line. Because of the restriction $y \ge 2$, only the portion of it above the point $(\frac{1}{2}, 2)$ will lie in the set S. This portion is an infinite ray originating at the point $(\frac{1}{2}, 2)$.

(iv) Finally, assume $x \geq 3$ and $y \geq 2$. Then (34) becomes $x + y = x + y - 5$ which has no solutions anywhere in the plane. So this case also contributes nothing to S.

Putting together all the four cases we see that the set S consists of the union of the line segment from the point $(\frac{5}{2}, 0)$ to the point $(\frac{1}{2}, 2)$ and the vertical ray from this point. It is sketched in the figure above.

The distance function defined in this problem is an example of what is technically called a **metric**. The usual distance function defined by the familiar formula $d(P, Q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ is also a metric, called the **euclidean metric**. Had we taken the euclidean metric, the set of points equidistant from O and A would have been the perpendicular bisector of the segment OA. But w.r.t. the given metric, it turns out to be quite different. (Even more bizarre things can happen. See Exercise (9.52) (b).) This shows that many geometric concepts depend on the metric used. There are many other ways to define a metric for points of the coordinate plane. One of them will be given in the exercises. (Exercise (9.53).)

Comment No. 19:

After dealing at length with problems where we have to find something, we conclude with problems asking to prove some property. Usually, these properties are geometric in nature. Where only straight lines and circles are involved, pure geometry proofs are often possible, although coordinates is a safer option (albeit a dull one many times) as we already saw. For parabolas, ellipses and hyperbolas too, pure geometry proofs are possible, but not very fashionable these days as we remarked earlier. Sometimes, the very statement of a property is in terms of a frame of coordinates and so the use of coordinates is a natural choice. Here is an illustration. (JAT 1979)

Show that the ordinate of the common point of any two tangents to a parabola is the arithmetic mean of the ordinates of their points of contact.

We take the equation of the parabola in the standard form $y^2 = 4ax$. Call the two points of contact as $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ and let $P = (h, k)$ be the point of intersection of the tangents at P_1 and P_2. We have to prove that $k = \frac{y_1 + y_2}{2}$. We can do so by writing the equations of the tangents in terms of x_1, y_1, x_2 and y_2 and then using the relations $y_1^2 = 4x_1$ and $y_2^2 = 4x_2$ to get rid of x_1 and x_2. Instead, it is more convenient to use the parametric equations of a parabola, i.e., take P_1 as $(at_1^2, 2at_1)$ and P_2 as $(at_2^2, 2at_2)$ for some t_1, t_2. Then the equations of the tangents at P_1 and P_2 are, in a simplified form,

$$t_1 y = x + at_1^2 \tag{35}$$

$$\text{and } t_2 y = x + at_2^2 \tag{36}$$

respectively. By a direct calculation, the point of intersection $P = (h, k)$ of these two tangents comes out as $(at_1 t_2, a(t_1 + t_2))$. (For the given problem we need only the y-coordinate of P which is obtained simply by subtracting (36) from (35).) So we have $k = a(t_1 + t_2)$. As we already have $y_1 = 2at_1$ and $y_2 = 2at_2$, we get $k = \dfrac{y_1 + y_2}{2}$ as desired.

For proving geometric properties of ellipses and hyperbolas too, parametric representations are convenient. The standard parametrisation of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $x = a\cos\theta, y = b\cos\theta$ while that of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x = a\sec\theta, y = b\tan\theta$. Not surprisingly, the properties to be proved often reduce to some trigonometric identities. So we postpone such problems to the next chapter (Comment No. 16), where we shall discuss applications of trigonometry to geometry. (The hyperbola can also be parametrised by $a = \cosh t, y = \sinh t$, where $\cosh t$ and $\sinh t$ are appropriately named as the **hyperbolic** cosine and sine respectively and are defined by $\cosh t = \dfrac{e^t + e^{-t}}{2}$ and $\sinh t = \dfrac{e^t - e^{-t}}{2}$. The parametrisation is based on the identity that for all t, $\cosh^2 t = 1 + \sinh^2 t$, which is proved by a direct calculation. The hyperbolic functions have many interesting properties and applications. They arise naturally in the parametrisation of some well-known curves such as a catenary, i.e., a heavy chain suspended at its two ends hanging freely under gravity. But as far as the parametrisation of a hyperbola is concerned, the earlier parametrisation, viz., $x = a\sec\theta, y = b\tan\theta$ is more convenient.)

EXERCISES

9.1 (a) If $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix}$, must the two triangles with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and $(a_1, b_1), (a_2, b_2), (a_3, b_3)$ be congruent? (1985)

(b) If $A = (6, 3), B = (-3, 5), C = (4, -2)$ and $P = (x, y)$, prove that the ratio of the area of the triangle PBC to that of ABC is $\left|\dfrac{x + y - 2}{7}\right|$. (1983)

9.2 If $P = (x, y)$, $F_1 = (3, 0), F_2 = (-3, 0)$ and $16x^2 + 25y^2 = 400$, find the value of $PF_1 + PF_2$. (1998)

9.3 Find the number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$. (1998)

9.4 If a, b, c are in A.P., show that the straight line $ax + by + c = 0$ will always pass through a fixed point. Find this point. (1984)

9.5 If x_1, x_2, x_3 as well as y_1, y_2, y_3 are in G.P. with the same common ratio, what type of a curve do the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) lie on? (1999)

9.6 (a) Are the points $(-a, -b), (0, 0), (a, b)$ and (a^2, ab) collinear? (1979)

(b) Do the lines $2x + 3y + 19 = 0$ and $9x + 6y - 17 = 0$ cut the coordinate axes in concyclic points? (1988)

(c) Is the line $x+3y=0$ a diameter of the circle $x^2+y^2-6x+2y=0$? (1989)

(d) Can a tangent be drawn from the point $(\frac{5}{2},1)$ to the circumcircle of the triangle with vertices $(1,\sqrt{3}),(1,-\sqrt{3})$ and $(3,-\sqrt{3})$? (1985)

9.7 Find the area of the triangle formed by the positive x-axis, and the normal and the tangent to the circle $x^2+y^2=4$ at the point $(1,\sqrt{3})$. (1989)

9.8 If $P(1,2), Q(4,6)$ and $R(5,7)$ and $PQRS$ is a parallelogram, find S. (1998)

9.9 One side of a rectangle lies along the line $4x+7y+5=0$. Two of its vertices are $(-3,1)$ and $(1,1)$. Find the equations to the other three sides. (1978)

9.10 Two vertices of an equilateral triangle are $(-1,0)$ and $(1,0)$ and the third vertex lies above the x-axis. Find the equation of its circumcircle. (1997*)

9.11 P is a variable point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, with foci F_1 and F_2. If A is the area of the triangle PF_1F_2, find the maximum value of A. (1994)

9.12 (a) Find the condition on a and b so that two chords, each bisected by the x-axis can be drawn from $(a,\frac{b}{2})$ to the circle $2x(x-a)+y(2y-b)=0$. (1992)

(b) Find the intervals of values of a for which the line $y+x=0$ bisects two chords drawn from a point $(\frac{1+\sqrt{2}a}{2},\frac{1-\sqrt{2}a}{2})$ to the circle $2x^2+2y^2-(1+\sqrt{2}a)x-(1-\sqrt{2}a)y=0$. (1996)

9.13 (a) The abscissas of the two points A and B are the roots of the equation $x^2+2ax-b^2=0$ and their ordinates are the roots of the equation $x^2+2px-q^2=0$. Find the equation and the radius of the circle with AB as a diameter. (1984)

(b) If the circle $x^2+y^2=a^2$ intersects the hyperbola $xy=c^2$ in four points $(x_1,y_1),(x_2,y_2),(x_3,y_3),(x_4,y_4)$, which of the following statements is(are) true? (1998)

(A) $x_1+x_2+x_3+x_4=0$ (B) $y_1+y_2+y_3+y_4=0$
(C) $x_1x_2x_3x_4=c^4$ (D) $y_1y_2y_3y_4=c^4$

9.14 Find the point of intersection of the tangents to the parabola $y^2=4x$ at the ends of its latus rectum. (1994)

9.15 Identify the types of curves with following descriptions:

(a) represented by the equation $\frac{x^2}{1-r}-\frac{y^2}{1+r}=1$, where $r>1$. (1981)

(b) the locus of a point S which satisfies relation $SQ^2+SR^2=2SP^2$, where $P=(1,0), Q=(-1,0)$ and $R=(2,0)$. (1988)

(c) the locus of the centre of a circle C touching a circle C_1 internally and a circle C_2 externally, it being given that C_2 lies inside C_1. (2001)

9.16 If the tangent at the point P on the circle $x^2+y^2+6x+6y=2$ meets the straight line $5x-2y+6=0$ at a point Q on the y-axis, find the length of PQ. (2002)

9.17 (a) A straight line through the origin O meets the parallel lines $4x+2y=9$ and $2x+y+6=0$ at points P and Q respectively. Find the ratio in which the point O divides the segment PQ. (2002)

(b) The lines $3x-4y+4=0$ and $6x-8y-7=0$ are tangents to the same circle. Find the radius of the circle. (1984)

9.18 In each of the following, determine the type of triangle formed

(a) with vertices at $(0,\frac{8}{3}),(1,3)$ and $(82,30)$. (1986)

(b) with sides along $x+y=0, 3x+y-4=0$ and $x+3y-4=0$. (1983)

9.19 Find the area of the triangle formed by the tangents from $(4,3)$ to the circle $x^2+y^2=9$ and the line joining their points of contacts. (1987)

9.20 Determine which quadrant of the plane contains the orthocentre of the triangle formed by the lines $x+y=1, 2x+3y=6$ and $4x-y+4=0$. (1985)

9.21 Points $(1,3)$ and $(5,1)$ are two opposite vertices of a rectangle and the other two vertices lie on the line $y=2x+c$. Find c and the remaining vertices. (1981)

9.22 Find the equations of the tangents drawn from the origin to the circle $x^2+y^2-2rx-2hy+h^2=0$. (1988)

9.23 If $a>2b>0$, find the positive value of m for which $y=mx-b\sqrt{1+m^2}$ is a common tangent to $x^2+y^2=b^2$ and $(x-a)^2+y^2=b^2$. (2002)

9.24 Determine the loci of the following :

(i) a point which divides a chord with slope 4 of $xy=1$ in the ratio $1:2$ (1997*)

(ii) the foot of the perpendicular drawn from P to AB, where $OAPB$ is a rectangle in which A, B lie on the x and y-axes respectively and AB has a constant length c (1983)

(iii) the mid-point of a chord drawn from the origin to the circle $(x-1)^2+y^2=1$ (1985)

(iv) the mid-point of the chord of the circle $4x^2+4y^2-12x+4y+1=0$ that subtends an angle of $\frac{2\pi}{3}$ at the centre. (1993)

(v) the mid-point of a chord of the circle $x^2+y^2=4$ which subtends a right angle at the origin. (1984)

(vi) the mid-point of a chord of the circle $x^2+y^2=r^2$ by a line that passes through a fixed point (h,k). (1983)

(vii) a point P such that $|AP-BP| = 6$, where $A = (0, 4)$ and $B = (0, -4)$. (1983)

(viii) the point of intersection of the lines AQ and BP, where $A = (7, 0)$, $B = (0, -5)$, P lies on the x-axis, Q lies on the y-axis and PQ is perpendicular to AB. (1990)

(ix) the vertex R of a rectangle $PQRS$ whose side PQ lies parallel to the line $y = mx$ and whose vertices P, Q and S lie on the lines $y = a$, $x = b$ and $x = -b$ respectively. (1996)

(x) the centre of a circle which externally touches the circle $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis. (1993)

(xi) the centre of a circle which passes through a given point (a, b) and cuts the circle $x^2 + y^2 = k^2$ orthogonally. (1988)

(xii) the foot, say R, of the perpendicular from O to the normal to a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

9.25 Show that the locus of a point that divides a chord of slope 2 of the parabola $y^2 = 4x$ internally in the ratio $1 : 2$ is a parabola. Find the vertex of this parabola. (1995)

9.26 Let a given line L_1 intersect the x and y axes at P and Q respectively. Let another line L_2, perpendicular to L_1, cut the x- and y-axes at R and S respectively. Show that the locus of the points of intersection of the lines PS and QR is a circle passing through the origin. (1987)

9.27 The angle between the pair of tangents drawn from a point P to the parabola $y^2 = 4ax$ is $45°$. Show that the locus of the point P is a hyperbola. (1998)

9.28 Consider the family of circles $x^2 + y^2 = r^2$, $2 < r < 5$. If in the first quadrant, the common tangent to a circle of this family and the ellipse $4x^2 + 25y^2 = 100$ meets the coordinate axes at A and B, then find the equation of the locus of the mid-point of AB. (1999)

9.29 A straight line L through the origin meets the lines $x+y = 1$ and $x+y = 3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn, parallel to $2x - y = 5$ and $3x + y = 5$ respectively. Lines L_1 and L_2 intersect at R. Show that the locus of R as L varies is a straight line. (2002)

9.30 Let t_1, t_2 and t_3 be the lengths of the tangents drawn from a point (x_1, y_1) to the circles $x^2 + y^2 = a^2$, $x^2 + y^2 = 2ax$ and $x^2 + y^2 = 2ay$ respectively. The lengths satisfy the equation $t_1^4 = t_2^2 t_3^2 + a^4$. Show that the locus of (x_1, y_1) consists of $x + y = 0$ and $x^2 + y^2 = a(x + y)$. (JAT 1980)

9.31 (a) Two sides of a rhombus $ABCD$ are parallel to the lines $y = x + 2$ and $y = 7x + 3$ and its diagonals intersect at the point $(1, 2)$. If the vertex A is on the y-axis, find the possible coordinates of A. (1985)

(b) A square is inscribed in the circle $x^2+y^2-2x+4y+3=0$. Its sides are parallel to the coordinate axes. Find its vertices. (1980)

(c) Given the four lines with the equations $x+2y-3=0$, $2x+3y-4=0$, $3x+4y-7=0$ and $4x+5y-6=0$, determine if (i) they are all concurrent, (ii) they are the sides of a quadrilateral. (1980)

(d) The vertex A of a square $ABCD$ is at $(38,1)$ while its diagonal BD lies along the line $L: 8x=15y$. Determine the other vertices. [*Hint:* Parametrise L in terms of the algebraic distance of a point on it from the origin. Find the centre, say M, of the square and then the distance AM. Note that $BM=DM=AM$.]

9.32 One of the diameters of the circle circumscribing the rectangle $ABCD$ is $4y=x+7$. If A and B are the points $(-3,4)$ and $(5,4)$ respectively, then find the area of the rectangle. (1985)

9.33 Find the equation of a circle of area 154 sq. units given that each of the two lines $2x-3y=5$ and $3x-4y=7$ is a diameter of it. (1989)

9.34 (a) Given a line L with an equation $ax+by+c=0$ and a point $P=(x_1,y_1)$, find formulas for (i) the foot of the perpendicular from P to L and (ii) the reflection of P into L. (Both the formulas are clumsy. When needed, it is best to find these points directly. The complex versions are more handy, see Comment No. 16 of Chapter 8.)

(b) The equations of the perpendicular bisectors of the sides AB and AC of a triangle ABC are $x-y+5=0$ and $x+2y=0$ respectively. If the point A is $(1,-2)$, find the equation of the line BC. (1986)

9.35 From a point A common tangents are drawn to the circle $x^2+y^2=\frac{a^2}{2}$ and the parabola $y^2=4ax$. Find the area of the quadrilateral formed by the common tangents, the chord of contact of the circle and the chord of contact of the parabola. (1996)

9.36 Identify all possible values of r for which the circles $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+8=0$ intersect in two distinct points. (1989)

9.37 Find the coordinates of the point at which the circles $x^2+y^2-4x-2y+4=0$ and $x^2+y^2-12x-8y+36=0$ touch each other. Also find equations of common tangents touching the circles in distinct points. (1993)

9.38 (a) Let $a_ix+b_iy+c_i=0$ $i=1,2$ be equations of two lines meeting at a point P. Prove that the equations $\dfrac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=\pm\dfrac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$ represent the angle bisectors of these lines with the $+$ sign holding for the obtuse and the $-$ sign holding for the acute angle bisector, in case $a_1a_2+b_1b_2>0$ and the opposite happening in case $a_1a_2+b_1b_2<0$. [*Hint:* Consider the angle between the rays from P parallel to the vectors $(-b_1,a_1)$ and $(-b_2,a_2)$ and the cosine of this angle.]

(b) Find the equation of the line which bisects the obtuse angle between the lines $x - 2y + 4 = 0$ and $4x - 3y + 2 = 0$. (1979)

9.39 Lines $L_1 = ax+by+c = 0$ and $L_2 = lx+my+n = 0$ intersect at the point P. Find the equation of a line L different from L_2 which passes through P and makes the same angle with L_1 as L_2 does. (1988)

9.40 (a) The vertices of a triangle are $A(-1,-7), B(5,1)$ and $C(1,4)$. Find the equation of the bisector of the angle ABC. (1993)

(b) Let $P = (-1,0), Q = (0,0)$ and $R = (3, 3\sqrt{3})$ be three points. Find the equation of the bisector of the angle PQR. (2002)

9.41 Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point $(1,-10)$. Determine the equation of the third side. (1984)

9.42 Obtain an alternate derivation of (26) based on the fact that every point on an angle bisector of two lines is equidistant from them.

9.43 Show that for every real number λ, the pair of straight lines given by $a^2x^2 + 2h(a+b)xy + b^2y^2 + \lambda(x^2+y^2) = 0$ are equally inclined to the pair given by $ax^2 + 2hxy + by^2 = 0$. (JAT 1979)

9.44 Find the centre of a circle passing through the points $(0,0)$ and $(1,0)$ and touching the circle $x^2 + y^2 = 9$. (1992)

9.45 Show that the chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ are concurrent and find the point through which all them pass. (1997)

9.46 Lines $5x+12y-10$ and $5x-12y-40$ touch a circle C_1 of diameter 6 with its centre in the first quadrant. Find the equation of the circle C_2 which is concentric with C_1 and cuts intercepts of length 8 on these lines. (1986)

9.47 Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from the origin O to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact, find the length of OA. (2001)

9.48 Find the equation of the circle passing through $(1,1)$ and the points of intersection of : (i) $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$ (1980) and (ii) $x^2 + y^2 + 13x - 3y = 0$ and $2x^2 + 2y^2 + 4x - 17y - 25 = 0$. (1983)

9.49 (a) Give a direct derivation of (22).

(b) Find the equation of the circle whose radius is 5 and which touches the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at the point $(5,5)$. (1978)

9.50 The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. Find the equation of the circle with AB as a diameter.

9.51 A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L and the coordinate axes is 5. Find the equation of the line L. (1980)

9.52 (a) In the problem done in Comment No. 18, identify the set, say D, of all points in the plane whose distance from the origin is at most 5. What would the set S of points equidistant from O and A have looked like had we considered *all* such points in the plane? Sketch both these sets.

(b) Show that if $A = (2, 2)$, then the set S contains two unbounded regions. (So, a change of the metric can thwart our intuition that the set of equidistant points must be a 'thin' curve with no area!)

9.53 Repeat (a) of the last exercise for yet another distance function defined by $d(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$, where $P = (x_1, y_1)$ and $Q = (x_2, y_2)$.

9.54 Each of the four inequalities given below defines a region in the xy-plane. Determine which of these four regions fails to have the property that for any two points (x_1, y_1) and (x_2, y_2) in the region, the point $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ is also in the region. (1981)

(a) $x^2 + 2y^2 \leq 1$ (b) $\max\{|x|, |y|\} \leq 1$ (c) $x^2 - y^2 \leq 1$ (d) $y^2 - x \leq 0$

9.55 If the sum of the distances of a point from two perpendicular lines is a constant, identify the type of curve represented by its locus. (1992)

9.56 (a) Find the area enclosed within the curve $|x| + |y| = 1$. (1981)

(b) Find the area bounded by $y = |x| - 1$ and $y = -|x| + 1$. (2002)

9.57 Find the slope of a chord of the parabola $y^2 = 4ax$ which subtends a right angle at the vertex of the parabola and is normal to the curve at one of its end-points. (1982)

9.58 Through the vertex O of a parabola $y^2 = 4x$, chords OP and OQ are drawn at right angles to each other. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the mid-point of PQ. (1994)

9.59 Prove the focusing property of parabola, i.e., show that every light ray parallel to the axis of a parabola will pass through its focus after being reflected by the curve. (Use the law of reflection. This property explains why parabolic mirrors are used in focusing and also in search lights.)

9.60 Points A, B, C lie on the parabola $y^2 = 4ax$. The tangents to the parabola, taken in pairs, intersect at points P, Q, R. Determine the ratio of the area of the triangles ABC and PQR. (1996)

*9.61 Let $A = (at_1^2, 2at_1)$, $B = (at_2^2, 2at_2)$ and $C = (at_3^2, 2at_3)$ be three points on the parabola $y^2 = 4ax$. If the orthocentre of the triangle ABC is at the focus, prove that $t_1 + t_2 + t_3 + t_1t_2t_3 = 0$ and $t_1t_2 + t_2t_3 + t_3t_1 = -5$. Show further that in this case the circumcircle of ABC touches the y-axis.

Chapter 10

TRIGONOMETRIC EQUATIONS

The equations studied in Chapter 3 were algebraic in that they involved only polynomials in the unknowns. An equation in which a trigonometric function of an unknown is involved is called a trigonometric equation. For example, $\sin\theta + 2\cos\theta = -\frac{3}{2}$ is a trigonometric equation. However the equation $\sin 37^\circ x = 3$ is not a trigonometric equation because here $\sin 37^\circ$ is a constant and no trigonometric function of the variable x is involved.

Because of the periodicity of the trigonometric functions, a trigonometric equation has, in general, an infinite number of solutions. We describe the set of all solutions of it by specifying what is called a general solution, which contains a parameter which can be given any of a certain set of values. Any particular value of the parameter determines a particular solution. The standard procedure to solve a trigonometric equation is to first find its general solution and then determine a particular solution from the additional requirements, if any, stipulated or implicit in the problem (e.g., that the solution lie in a given interval). Periodicity also has a bearing in defining the inverse trigonometric functions. These are studied in Comment No. 15.

A problem related to trigonometric equations is that of finding the range of a trigonometric function. This is studied in Comment No. 11. and 12. Sometimes a problem may not specify a trigonometric equation directly. But such an equation may emerge from it after doing some work. The Main Problem is of this type. It illustrates how trigonometric equations can be applied to problems in geometry. Trigonometry can also be applied to geometry through coordinates. This is illustrated in Comment No. 16 and 17.

In the JEE, trigonometric equations can appear both as short or long questions. They are often combined with other topics such as quadratic equations, complex numbers, exponentials, systems of linear equations etc.

Main Problem : Suppose $A_1, A_2, \ldots, A_n$ are the vertices of a regular n-gon. If $\dfrac{1}{A_1A_2} = \dfrac{1}{A_1A_3} + \dfrac{1}{A_1A_4}$, then the value of n is ..... . (JEE 1994) (Here A_1A_2 denotes the length of the straight line segment with end points A_1 and A_2. Similarly for A_1A_3 and A_1A_4.)

First Hint: Let $\theta = \frac{\pi}{n}$. Obtain a trigonometric equation for θ.
Second Hint: Solve the equation using trigonometric identities.

Solution: Letting r be the radius of the circle circumscribing the polygon, we have $A_1A_2 = 2r\sin\theta$, $A_1A_3 = 2r\sin 2\theta$ and $A_1A_4 = 2r\sin 3\theta$. So the given condition reduces to $\dfrac{1}{\sin\theta} = \dfrac{1}{\sin 2\theta} + \dfrac{1}{\sin 3\theta}$, i.e., $\sin 2\theta \sin 3\theta = \sin\theta(\sin 2\theta + \sin 3\theta)$. Using the formula $\sin 2\theta = 2\sin\theta\cos\theta$, this becomes $2\sin\theta\cos\theta\sin 3\theta = \sin\theta(\sin 2\theta + \sin 3\theta)$. Obviously, $\theta = \frac{\pi}{n}$ is an acute angle since $n > 2$. So $\sin\theta \neq 0$ and so $2\cos\theta\sin 3\theta = \sin 2\theta + \sin 3\theta$. The L.H.S. equals $\sin 2\theta + \sin 4\theta$. Thus we get $\sin 4\theta = \sin 3\theta$. As θ is an acute angle, this is possible only if $4\theta = \pi - 3\theta$, i.e., if $\theta = \frac{\pi}{7}$, giving $n = 7$. (See Comment No. 3 below for a figure and Comment No. 4 for some further justification.)

Comment No. 1:

It is tempting to use the formulas for $\sin 2\theta$ and $\sin 3\theta$ and reduce the first equation to an algebraic equation for $\cos\theta$, viz., $8\cos^3\theta - 4\cos^2\theta - 4\cos\theta + 1 = 0$. But this is a cubic equation (in $\cos\theta$) with no rational roots (see Exercise (10.11)) and is therefore not easy to solve. It is, in fact, the other way round, which is easier, i.e., a cubic equation can sometimes be solved by converting it to a trigonometric equation. See S. L. Loney's *Plane Trigonometry - Part II*, pp. 178 - 179. As an example, consider the following problem.

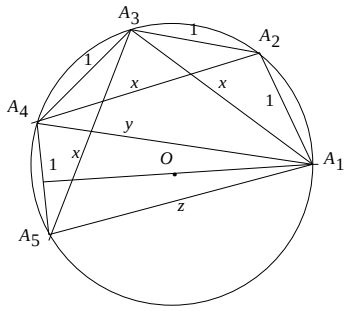
Let $-1 \leq p \leq 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval $[1/2, 1]$ and identify it. (JEE 2001)

The formula $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ suggests that a change of variable may simplify the problem. Indeed, if we put $x = \cos\theta$, then the values of x in the interval $[1/2, 1]$ correspond uniquely to those of θ in the interval $[0, \frac{\pi}{3}]$. Also the equation reduces to $p = \cos 3\theta$. As 3θ varies over the interval $[0, \pi]$ and the cosine function is one-to-one on this interval with range $[-1, 1]$, for every $p \in [-1, 1]$, the equation has a unique solution. It is given by $\theta = \frac{1}{3}\cos^{-1} p$. In terms of x, the solution is identified as $\cos(\frac{1}{3}\cos^{-1} p)$. Note that even without the trigonometric substitution, we can prove the existence and the uniqueness of the solution using methods of calculus. Consider the function $f(x) = 4x^3 - 3x - p$ which is continuous everywhere. Since $f(\frac{1}{2}) = -1 - p \leq 0$, while $f(1) = 1 - p \geq 0$, $f(x)$ must vanish at least once in $[1/2, 1]$. Further, since $f'(x) = 12x^2 - 3 > 0$ for all $\frac{1}{2} < x < 1$, $f(x)$ is strictly increasing on $[1/2, 1]$ and so cannot have more than one zeros. Put together, the given equation has a unique root in the interval $[1/2, 1]$. But when it comes to identifying this root, there is no easy way to do the problem without trigonometry.

Thus we see that in general it is *not* a good idea to try to solve a trigonometric equation by converting it to an algebraic equation, unless this reduced equation is something simple to solve, such as a linear or a quadratic one or a cubic with one obvious root. In a time-constrained examination it is very important to quickly recognise that a particular approach, which appears promising initially, is *not* going to work. The present problem is an excellent example of this.

Comment No. 2:

There is a purely geometric solution based on what is called **Ptolemy's theorem** which asserts that if $ABCD$ is a cyclic quadrilateral then $AC.BD$ equals $AB.CD + AD.BC$. (Note that the Pythagoras theorem is a special case of this, when $ABCD$ is a rectangle.) Without loss of generality suppose that every side of the given regular n-gon is of unit length. Let x, y, z denote, respectively, the lengths A_1A_3, A_1A_4 and A_1A_5. Then we are given that $1 = \frac{1}{x} + \frac{1}{y}$, i.e., $xy = x + y$. Also by regularity, $A_3A_5 = x$ and so, Ptolemy's theorem, applied to the quadrilateral $A_1A_3A_4A_5$ gives $xy = x + z$. Together, we get $y = z$.

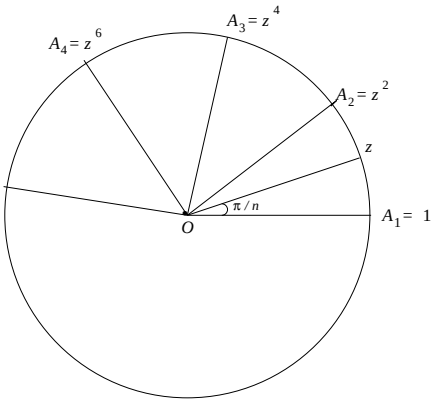


But this means that the vertices A_4 and A_5 are at the same distance from the vertex A_1. The only way this can happen is that the polygon is symmetric about the line passing through A_1 and the mid-point of A_4A_5. That implies that there are as many vertices between A_1 and A_4 as there are between A_1 and A_5 (counted in the reverse sense). So the polygon has 7 vertices, i.e., $n = 7$.

Comment No. 3:

A solution using complex numbers is also possible, although somewhat unusual. It was remarked in Chapter 8, Comment No. 6 that complex numbers come in handy where mutually perpendicular lines are involved. This is not the case in the present problem. But there is another feature of complex numbers that makes them very convenient in the present problem. If z is a complex number of unit absolute value, then z as well as all its powers lie on the unit circle. Moreover, two consecutive powers are equally spaced. This makes it very convenient to represent the vertices of a regular polygon by powers of a complex number of absolute value 1. Thus, in our problem, without loss of generality, we may suppose that the vertices of the polygon lie on the unit circle in the complex plane and that A_1 represents the complex number 1. Then, if we represent the vertex A_2 by the complex number z the successive vertices will be powers of z, i.e., $A_3 = z^2, A_4 = z^3$ and so on. Note that the n-th power z^n must be the vertex A_1 again and then the subsequent powers again represent $A_2, A_3, \ldots$. So we must have $z^n = 1$, i.e., $z = e^{2\pi i/n}$. (In the language of Chap-

ter 7, z and all its powers are the n-th roots of unity.) We can in fact do the problem with this representation of the vertices. But then during the course of the solution we shall encounter the square root $\sqrt{z}$. To avoid it, we let z be the complex number $e^{i\pi/n} = \cos\frac{\pi}{n} + i\sin\frac{\pi}{n}$. Then the vertices of the n-gon are $A_1 = 1, A_2 = z^2, A_3 = z^4, \ldots$ etc.



The given condition translates as $\dfrac{1}{|z^2-1|} = \dfrac{1}{|z^4-1|} + \dfrac{1}{|z^6-1|}$ or as, $1 = \dfrac{1}{|z^2+1|} + \dfrac{1}{|z^4+z^2+1|}$. Since $|z| = 1$, this can be further reduced as $1 = \dfrac{1}{|z+\frac{1}{z}|} + \dfrac{1}{|z^2+1+\frac{1}{z^2}|}$. Now, $\dfrac{1}{z}$ is simply $\bar{z}$, the complex conjugate of z and so $z + \dfrac{1}{z} = z + \bar{z} = 2$ Re z which is positive since evidently $n > 4$, which makes $\frac{\pi}{n}$ an acute angle and thus makes z lie in the first quadrant. Hence $|z + \dfrac{1}{z}|$ is simply $z + \dfrac{1}{z}$. By a similar reasoning, $|z^2 + 1 + \dfrac{1}{z^2}|$ equals $z^2 + 1 + \dfrac{1}{z^2}$. Hence the given condition translates as $1 = \dfrac{1}{z+\frac{1}{z}} + \dfrac{1}{z^2+1+\frac{1}{z^2}}$. Upon simplification this comes to $z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$. Multiplying by $z + 1$ we get $z^7 = -1$, i.e., $\cos(\frac{7\pi}{n}) = -1$ and $\sin(\frac{7\pi}{n}) = 0$ which implies $n = 7$.

Comment No. 4:

The Main Problem is an application of solutions of trigonometric equations to problems in geometry. Specifically we reduced the problem to that of solving the equation $\sin 3\theta = \sin 4\theta$. The pure geometry solution was elegant but tricky. The basic idea behind the solution using complex numbers was the same as that in the trigonometric solution. Instead of working with the sines and cosines of various multiples of θ, we worked with the powers of the complex number $z = \cos\theta + i\sin\theta$. We also gave an example where a cubic equation was solved by reducing it to a trigonometric equation.

The origin of the trigonometric equations lies in the periodicity of the trigonometric functions. As a result, it is possible to tell exactly when two

angles have the same sine, the same cosine etc. So in essence, there are three basic trigonometric equations. They and their solutions are listed below.

$$\begin{aligned} \sin\beta &= \sin\alpha \text{ if and only if } \beta = n\pi + (-1)^n\alpha && (1)\\ \cos\beta &= \cos\alpha \text{ if and only if } \beta = 2n\pi \pm \alpha && (2)\\ \text{and } \tan\beta &= \tan\alpha \text{ if and only if } \beta = n\pi + \alpha && (3)\\ &\text{for some integer } n \end{aligned}$$

Here α and β can both be variables or one of them can be a constant. The solutions given on the R.H.S. of these equations are often called their **general solutions** because the integer n is unspecified. As there are infinitely many integers, all the three equations have infinitely many solutions. Usually, we are interested only in those solutions (often called **particular solutions**) which satisfy some additional restriction such as that they lie in a given interval. We then eliminate the other solutions by some reasoning which is often of an *ad hoc* nature. In the solution to the Main Problem, for example, we finally had to solve the equation $\sin 4\theta = \sin 3\theta$. Here the angle θ was already known to be of the form $\frac{\pi}{n}$ for some positive integer n. By (1) above, $\sin 4\theta = \sin 3\theta$ implies $4\theta = k\pi + (-1)^k 3\theta$ for some integer k. This is the general solution. (Note that we are using k instead of n here because n is already used for the number of sides of the polygon.) For even k, this means $\theta = k\pi$. But $\theta = \frac{\pi}{n}$. This would mean that $nk = 1$ contradicting that k is even. So k must be odd. That would give, by a similar reasoning, that $nk = 7$. But the only (positive) factors of 7 are 1 and 7. As n is the number of sides of a polygon, evidently $n > 1$. So we must have $n = 7$. Strictly speaking, this justification ought to be included in the solution. Without it, we can only say that $n = 7$ is one possible solution. But we cannot say conclusively that it is the only solution. (Doing so is implicit in the word 'determine' in the statement of the question.) But when the major focus of the problem is something else, such justifications are sometimes skipped or are given only briefly. And, in the case of a problem where you do not have to show your work, it is enough if you guessed $n = 7$ as a solution. Still, one must be prepared to give such justifications. If you have the time, give them by all means. If you lack the time, at least mention that there are several possible solutions but that the other possibilities can be eliminated.

Comment No. 5:

Discarding the unwanted solutions of trigonometric equations (1) to (3) after solving them is a relatively routine part of the game. The real crux lies *before* solving them. Surely, nobody is going to give you a trigonometric equation in any of the three forms above and ask you to solve it! Almost invariably some work has to be put in to reduce the given equation to one (or more) of the equations above. And, sometimes even this original equation is not given directly but has to be derived from the data of the problem, as was the case in the Main Problem, where with a suitable choice of θ, we reduced the data to the equation $\dfrac{1}{\sin\theta} = \dfrac{1}{\sin 2\theta} + \dfrac{1}{\sin 3\theta}$, i.e., to the equation $\sin 2\theta \sin 3\theta = \sin\theta(\sin 2\theta + \sin 3\theta)$.

The real trigonometric work comes after this. We first reduced it to the equation $2\sin\theta\cos\theta\sin 3\theta = \sin\theta(\sin 2\theta + \sin 3\theta)$. This actually reduces to two possibilities, viz., (i) $\sin\theta = 0$ and (ii) $2\cos\theta\sin 3\theta = \sin 2\theta + \sin 3\theta$ which further reduced to $\sin 4\theta = \sin 3\theta$. Both these possibilities fall under (1) above. We discarded the first possibility tersely (because θ was already known to be of the form $\frac{\pi}{n}$ for some integer $n > 1$). But if the problem was merely to solve the trigonometric equation $\sin 2\theta\sin 3\theta = \sin\theta(\sin 2\theta + \sin 3\theta)$, we would have to include the solutions of $\sin\theta = 0$ (which are all integral multiples of π) in the answer.

Similarly, in the problem of identifying a root of the cubic equation $4x^3 - 3x - p = 0$, we used the substitution $x = \cos\theta$ to convert it to a trigonometric equation and then solved the latter by reducing it to a simpler equation, viz., $\cos 3\theta = p$.

Summing up, in a problem based on applications of trigonometric equations, there are four steps :

(i) Write a trigonometric equation (or, sometimes, a system of such equations),

(ii) Reduce this equation to one or more of the three basic equations (1) to (3) above,

(iii) Take the general solutions of these basic equations, and

(iv) Discard the unwanted solutions using the restrictions given or implied by the problem.

Step (iii) is trivial and (iv) is routine as observed earlier. Step (ii) requires the knowledge of the appropriate trigonometric identities. More importantly, it requires some ingenuity to apply the right identities. In some problems, Step (i) may not be present as the trigonometric equation to be solved is given to you directly. Even when present, the work required for Step (i) is non-trigonometric in nature. In some problems, it may require some knowledge of facts from other branches of mathematics, such as solving quadratic equations, properties of determinants etc. So the real trigonometrical work lies in going from Step (ii) to Step (iii).

Comment No. 6:

We first consider problems where the first step is absent, i.e., the trigonometric equation is already given in the problem. The task then is to reduce it to a number of the basic equations (1) to (3) above. The simplest such equations are those where one of the two sides of (1) to (3) is given as a constant, say p. For example, suppose that the equation is of the form $\sin\theta = p$. Then it will have a solution if and only if $-1 \le p \le 1$. To actually identify it, fix any angle β such that $\sin\beta = p$. It is customary to take $\beta = \sin^{-1} p$, in which case $-\frac{\pi}{2} \le \beta \le \frac{\pi}{2}$. Further, if p is such that $\sin^{-1} p$ is a familiar angle, then it is expected that its value is specified in the answer. For example, the general solution of the equation $\sin\theta = -\frac{1}{2}$ is $n\pi + (-1)^{n+1}\frac{\pi}{6}$, where n is an integer. Similarly, one can

obtain the general solution of the equation $\cos\theta = p$ (provided $|p| \leq 1$) and of the equation $\tan\theta = p$ (for any real p).

Next in the level of simplicity are equations of the form

$$a\sin\theta + b\cos\theta = p \tag{4}$$

where a, b, p are some constants and a and b are not both zero. The standard trick in such problems is to set $c = \sqrt{a^2+b^2}$ and let α be the unique angle in the interval $(-\pi, \pi]$ such that[1] $\cos\alpha = \frac{a}{c}$ and $\sin\alpha = \frac{b}{c}$. Then the equation (4) reduces to

$$\sin(\theta+\alpha) = \frac{p}{c} \tag{5}$$

whose general solution we already know, viz., $\theta = n\pi + (-1)^n \sin^{-1}(\frac{p}{c}) - \alpha$, where n is an integer.

As a straightforward illustration (JEE 1981) the general solution of the equation
$\sin x + \cos x = 1$ is $x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$. Here, $a = b = 1$ and so α comes out to be $\cos^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$. Note that (5) and hence (4) can have a solution if and only if $|\frac{p}{c}| \leq 1$, i.e., if and only if $-\sqrt{a^2+b^2} \leq p \leq \sqrt{a^2+b^2}$. This simple observation is very helpful in finding the maxima/minima of trigonometric functions without using calculus. It also gives a solution to the following problem. (JEE 2002)

Determine the number of integral values of k for which the equation $7\cos x + 5\sin x = 2k+1$ has a solution.

Since $7^2 + 5^2 = 74$, the question amounts to finding the number of odd integers in the interval $[-\sqrt{74}, \sqrt{74}]$. Since $\sqrt{74}$ lies between 8 and 9, the answer is the same as the number of odd integers in the interval $[-8, 8]$. It comes out as 8.

Yet another simple modification of the basic equations (1) to (3) is noteworthy. In each of these three equations, the same trigonometric function appears on both the sides. If, instead, in the given equation, the sine of some angle is given to equal the cosine of some other angle, such an equation can be converted into (1) or (2) by the simple rule $\sin(\frac{\pi}{2} - \theta) = \cos\theta$ or its analogues. This is the key idea in the solution to the following problem. (JEE 1985)

Find the smallest positive real number p for which the equation

$$\cos(p\sin x) = \sin(p\cos x) \tag{6}$$

has a solution $x \in [0, 2\pi]$.

[1] These two conditions determine the angle α uniquely. But even the first condition alone determines it except possibly for sign. That is, we can take $\alpha = \pm\cos^{-1}(\frac{a}{c})$, where the $+$ sign is taken if $b \geq 0$ and the $-$ sign is taken if $b < 0$. Another popular method is to set $\alpha = \tan^{-1}(\frac{b}{a})$ if both a, b are positive and modify it suitably otherwise. See Comment No. 16 below for more discussion. Also the choice of the interval $(-\pi, \pi]$ in which α is to lie is rather arbitrary. Any semi-open interval of length 2π would do. $[0, 2\pi)$ is an equally popular choice. Note also that α is the same as an argument of the complex number $a + ib$ and $c = |a + ib|$. So, what we are doing here amounts to writing the complex number $a + ib$ in the form $ce^{i\alpha}$.

Rewriting $\cos(p\sin x)$ as $\sin(\frac{\pi}{2}-p\sin x)$, and using (1), the general solution to (6) is $\frac{\pi}{2} - p\sin x = n\pi + (-1)^n p\cos x$, where n is any integer. Depending upon whether n is even or odd, this reduces to

$$p(\sin x + \cos x) = \frac{\pi}{2} - n\pi \quad \text{if } n \text{ is even} \tag{7}$$

$$p(\sin x - \cos x) = \frac{\pi}{2} - n\pi \quad \text{if } n \text{ is odd} \tag{8}$$

The maximum and the minimum values of both $\sin x+\cos x$ and $\sin x-\cos x$ are $\sqrt{2}$ and $-\sqrt{2}$. The minimum positive value of p for which (7) can hold will occur if $n = 0$ and $\sin x + \cos x = \sqrt{2}$. That gives $p = \frac{\pi}{2\sqrt{2}}$. This is also the minimum positive value of p for (8) (occurring when $n = 1$ and $\sin x - \cos x = -\sqrt{2}$). So $\frac{\pi}{2\sqrt{2}}$ is the minimum positive value of p for which (6) can hold.

Comment No. 7:

Let us now consider a few problems where the given trigonometric equation has to be converted into one or more of the three basic equations with some trigonometric manipulations. Typically, in such problems the given equation involves multiples of some angle. So the formulas for the trigonometric functions of 2θ and 3θ in terms of those of θ have to be frequently used. In some problems, little else is needed as was the case with the problem of solving the cubic in Comment No. 1 above. The moment you put $x = \cos\theta$, the identity for $\cos 3\theta$ hands you the solution. As another example (JEE 2002), suppose we want to find the length of a longest interval on which the function $3\sin x - 4\sin^3 x$ is increasing. This is equivalent to asking to find the length of a longest interval on which $\sin 3x$ is increasing. Put $y = 3x$. Then the longest intervals on which $\sin y$ is increasing as a function of y are of length π. (More specifically, such intervals are of the form $[2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}]$, where n is any integer. But in the present problem what matters is only their length.) Since $x = \frac{1}{3}y$, when y varies over an interval of length π, x varies over an interval of length $\frac{\pi}{3}$. So this is the answer.

As an example of a problem where some subsequent argument is needed after invoking the formula for $\sin 3\theta$, suppose in a triangle ABC it is given that $\angle A > \angle B$ and further that the measures of the angles A and B satisfy the equation $3\sin x-4\sin^3 x+k = 0,\ 0 < k < 1$ and we are asked to find the measure of the angle C. (JEE 1990) Because of the identity $3\sin x-4\sin^3 x = \sin 3x$, the hypothesis is merely a twisted way of saying that $\sin 3A$ and $\sin 3B$ are equal to each other and lie in the interval $(-1, 0)$. So by (1), $3A = n\pi + (-1)^n 3B$ for some integer n. So either $3A+3B$ is an odd multiple of π or $3A-3B$ is an even multiple of π. As A and B are angles of a triangle, $0 < A+B < \pi$ and so the first case we must have $3A+3B = \pi$. This gives $A+B = \frac{\pi}{3}$ and hence $\angle C$ comes out as $\frac{2\pi}{3}$. But there is another possibility, viz., that $3A-3B$ is an even multiple of 2π. Since A is given to be bigger than B, this means $3A - 3B = 2m\pi$ for some positive integer m. Again, because of the restriction that A and B lie in the interval $(0, \pi)$, the values of m bigger than 1 are ruled out. Still, $3A - 3B = 2\pi$ remains a possibility. To eliminate it, note that both $\sin 3A$ and $\sin 3B$ are

given to lie in the interval $(-1, 0)$. This means that $(2r+1)\pi < 3B < (2r+2)\pi$ and hence $(2r+3)\pi < 3A < (2r+4)\pi$ for some integer r. As in the triangle ABC, $\angle A$ is given to be bigger than $\angle B$, B is an acute angle. So we must have $r = 0$. But in that case $3A$ exceeds 3π which is a contradiction. Thus we have eliminated the second possibility, viz., $3A - 3B = 2m\pi$ completely. So only the first possibility holds and gives $C = \frac{2\pi}{3}$ as shown before.

Of course, not all problems are so direct. As in the case of the solution to the Main Problem, the identities for factoring $\sin A \pm \sin B$ and $\cos A \pm \cos B$, and the identities for writing products like $\sin A \cos B$ are needed all the time. In addition, the work can be further simplified by the use of some other, simple tricks.

As a good illustration (JEE 1996), suppose we want the general solution of the equation $\tan^2\theta + \sec 2\theta = 1$. We can express both the terms on the L.H.S. in terms of $\sin\theta$ or in terms of $\cos\theta$. But that would give us a polynomial equation of degree 4. Instead, we observe that $\cos 2\theta$ and hence $\sec 2\theta$ can be expressed in terms of $\cos^2\theta$. If we add 1 to $\tan^2\theta$, it also becomes easily expressible in terms of $\cos^2\theta$. So adding 1 to both the sides of the equation and putting $\lambda = \cos^2\theta$, we get $\frac{1}{\lambda} + \frac{1}{2\lambda - 1} = 2$. When expanded, this is a quadratic in λ, viz., $4\lambda^2 - 5\lambda + 1 = 0$. (As an equation in $\cos\theta$ this is a fourth degree equation. But because there are no odd degree terms, it can be expressed as a quadratic in $\cos^2\theta$. This is precisely the advantage gained in expressing every term in terms of $\cos^2\theta$.) The roots are $\lambda = 1$ and $\frac{1}{4}$. Hence $\cos\theta$ equals 1, -1, $\frac{1}{2}$ or $-\frac{1}{2}$. Applying (2) to each of these four possibilities, we get the general solution as (i) $\theta = 2n\pi$ or (ii) $\theta = 2n\pi \pm \pi$ or (iii) $\theta = 2n\pi \pm \frac{\pi}{3}$ or (iv) $\theta = 2n\pi \pm \frac{2\pi}{3}$, where n is an integer. Note that (i) and (ii) can be combined together into $\theta = m\pi$, where m is any integer. Similarly (iii) and (iv) together are equivalent to saying that $\theta = m\pi \pm \frac{\pi}{3}$, where m is an integer.

In this problem all the four roots of the equation in $\cos\theta$ gave solutions since they lay in the interval $[-1, 1]$. In some problems, this does not happen and we have to discard some of the roots. For example, suppose we want the number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$. (JEE 1998) Solving this quadratic we get $\sin x = 2$ or $\sin x = \frac{1}{3}$. The first possibility has to be discarded. The second one gives exactly two solutions in each of the three intervals $[0, \pi]$, $[2\pi, 3\pi]$ and $[4\pi, 5\pi]$. As there are no solutions either in $[\pi, 2\pi]$ or in $[4\pi, 5\pi]$, the original equation has 6 solutions in $[0, 5\pi]$.

Comment No. 8:

In the solution to the Main Problem we used the identity $2\sin A\cos B = \sin(A + B) + \sin(A - B)$. This and three similar identities are very useful when we want to convert products into sums. The solution to the following problem (JEE 1993) is based almost entirely on the repeated applications of these identities.

Determine the smallest positive value of x (in degrees) for which

$$\tan(x + 100^\circ) = \tan(x + 50^\circ)\tan x \tan(x - 50^\circ) \tag{9}$$

For brevity denote 50° by θ. Then bringing $\tan x$ on the L.H.S., (9)

becomes

$$\frac{\sin(x+2\theta)\cos x}{\cos(x+2\theta)\sin x} = \frac{\sin(x+\theta)\sin(x-\theta)}{\cos(x+\theta)\cos(x-\theta)} \tag{10}$$

Using the identities mentioned above, this reduces to

$$\frac{\sin(2x+2\theta)+\sin 2\theta}{\sin(2x+2\theta)-\sin 2\theta} = \frac{\cos 2\theta - \cos 2x}{\cos 2\theta + \cos 2x}$$

and further (by cross-multiplying and cancelling common terms, or, what comes to the same, by taking componendo and dividendo and then cross-multiplying), to

$$\sin 2\theta \cos 2\theta + \sin(2x+2\theta)\cos 2x = 0$$

Another application of the identities for converting products into sums gives

$$\sin 4\theta + \sin(4x+2\theta) + \sin 2\theta = 0 \tag{11}$$

We now convert the sum $\sin 4\theta + \sin 2\theta$ to $2\sin 3\theta\cos\theta$. Putting back $\theta = 50^\circ$, this becomes $\cos 50^\circ$ (since $\sin 150^\circ = \sin 30^\circ = \frac{1}{2}$). So (11) becomes

$$\sin(4x+100^\circ) = -\cos 50^\circ = -\sin 40^\circ = \sin(-40^\circ) \tag{12}$$

From (1) (converted in terms of degrees), the general solution to (12) is

$$4x + 100 = 360n + (-1)^{n+1}40 \tag{13}$$

where n is an integer. The least positive value of x will be obtained if we put $n = 1$ in (13), which gives $x = 75^\circ$.

When the given trigonometric equation involves a large number of sines and cosines, the key to success is in choosing the right grouping to which the trigonometric identities should be applied. As an illustration, (JEE 1989) suppose we want the general solution of the equation $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$. If we combine the sine of x with that of $2x$ we get fractional angles (i.e., angles of the form $\frac{x}{2}$) and that would only make things worse. So we combine the first and the third term on each side to get $2\sin 2x\cos x - 3\sin 2x = 2\cos 2x\cos x - 3\cos 2x$. Both the sides have $2\cos x - 3$ as a factor which can never be 0. Cancelling it, we get $\sin 2x = \cos 2x$ or, $\sin 2x = \sin(\frac{\pi}{2} - 2x)$. By (1), the general solution is $2x = n\pi + (-1)^n(\frac{\pi}{2} - 2x)$. For an even n, say $n = 2m$, this gives $x = \frac{m\pi}{2} + \frac{\pi}{8}$, where m is an integer. For an odd n, the solution becomes $2x = n\pi - \frac{\pi}{2} + 2x$ which is impossible. So the general solution is $\frac{m\pi}{2} + \frac{\pi}{8}$ with m an integer.

Comment No. 9:

As an example where the conversion to the basic equations is not so obvious, we have (JEE 1978)

Solve for x, $(-\pi \leq x \leq \pi)$, the equation

$$2(\cos x + \cos 2x) + \sin 2x(1 + 2\cos x) = 2\sin x \tag{14}$$

There is no obvious regrouping of the terms which will lead to a simplification. The only implicit hint is that if we write $\sin 2x$ as $2\sin x \cos x$ then the factor 2 can be cancelled from each term. Doing so, (14) becomes

$$\cos x + \cos 2x + \sin x \cos x + 2\cos^2 x \sin x - \sin x = 0$$

The identity $2\cos^2 x - 1 = \cos 2x$ reduces this to

$$\cos x + \cos 2x + \sin x \cos x + \cos 2x \sin x = 0 \tag{15}$$

The L.H.S. can be factored as $(\cos x + \cos 2x)(1 + \sin x)$. Hence the solutions of (15) are all solutions of

$$\cos 2x = -\cos x = \cos x \tag{16}$$

as well as all solutions of

$$\sin x = -1 \tag{17}$$

By (2), the general solution of (16) is $2x = 2\pi n \pm x$, where n is an integer. But we want x to lie in the interval $[-\pi, \pi]$. In that case, $2x = 2\pi n + x$ is possible only for $n = 0$, giving $x = 0$. However, $2x = 2\pi n - x$ is possible for $n = -1, 0$ and 1, giving, respectively, $x = -\frac{2\pi}{3}, 0$ and $\frac{2\pi}{3}$. The only solution of (17) in $[-\pi, \pi]$ is $x = -\frac{\pi}{2}$. Putting it all together, the solutions of (14) are $x = -\frac{2\pi}{3}, -\frac{\pi}{2}, 0$ and $\frac{2\pi}{3}$.

Comment No. 10:

The reduction of the given equation to the basic equations is generally done by trigonometric identities combined with algebraic manipulations such as factoring. Occasionally an argument based on inequalities is needed. The basic idea is this. Suppose $x + y = 0$. In general we cannot determine x and y from this single equation. But suppose it is known that both x and y are non-negative (for example, when they are squares of some real numbers). Then $x + y$ can vanish only when both x and y vanish. As a result, a single equation implies two separate equations.

As an example (JEE 1980), suppose we want to solve (for $0 < x \leq \frac{\pi}{2}$) the equation

$$2\cos^2(\frac{x}{2})\sin^2 x = x^2 + x^{-2} \tag{18}$$

If we write the R.H.S. as $(x - \frac{1}{x})^2 + 2$ we see that it is always bigger than or equal to 2. The L.H.S. on the other hand, is at most 2. So the only way (18) can hold is if both the sides are equal to 2. For the R.H.S. this means $x - \frac{1}{x} = 0$, i.e., $x = 1$ (since $x > 0$). But for $x = 1$, the L.H.S. is not equal to 2. So (18) has no solution.

As another example (JEE 1998), suppose we want to find those values of x, where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ is maximum. Clearly, the maximum is at most 2. But since $f(0) = 2$, it is exactly 2. Now, $1 - \cos x$ and $1 - \cos(\sqrt{2}x)$ are both non-negative numbers. So the equation $f(x) = 2$ is equivalent to the system of two simultaneous equations, viz., (i) $\cos x = 1$ and (ii) $\cos(\sqrt{2}x) = 1$. By (2) the general solution of (i) is $x = 2n\pi$, where n is an integer while the general solution of (ii) is $x = \frac{2n\pi}{\sqrt{2}}$. Certainly $x = 0$ is a common solution to both. Suppose there were another common solution, say y. Then there would be integers m and n such that $y = 2m\pi$ and $y = \frac{2n\pi}{\sqrt{2}}$. But as $y \neq 0$ this would give $\sqrt{2} = \frac{n}{m}$. But this contradicts the well-known fact that $\sqrt{2}$ is not rational. (A proof of irrationality of $\sqrt{2}$ was given in Comment No. 17 in Chapter 4.) So $x = 0$ is the only point where the function $f(x)$ attains its maximum value. (If instead of $\sqrt{2}$ we have a rational number, say k then the function $f(x) = \cos x + \cos kx$ also has maximum 2. To find out where it is attained, write k in its reduced form as $\frac{p}{q}$, where p, q are integers with no common factor other than ± 1. Then $f(x) = 2$ if and only if there exist integers m, n such that $x = 2m\pi$ and $x = \frac{2\pi nq}{p}$. This gives $pm = nq$. As p and q have no common factor, this is possible if and only if m is a multiple of q. Thus the points x, where $f(x) = 2$ are all multiples of $2\pi q$.)

Comment No. 11:

When we solve an equation of the form $f(x) = p$, where x lies in some set, say S (usually an interval), we are looking for all elements $x \in S$ for which $f(x) = p$. Sometimes we are not so concerned about the actual solutions as about the mere *existence* of the solution. That is, we only want to know whether the equation $f(x) = p$ *has* a solution in S, and not how many solutions it has and which ones they are. Naturally, the answer to this question will depend on three things, viz., the function f, the set S and the number p. When the function f is fixed, it depends only on S and p. For a given f and S, the set of all values of p for which the equation $f(x) = p$ has at least one solution in S is denoted by $f(S)$ and is called the **image** of the set S under f or the **range** of the function f on the set S. The latter term is used especially when S is the entire domain of the given function f (in which case we also say simply, the range of f without specifying the set S). For a given p, the values of x for which $f(x) = p$ are sometimes called the **pre-images** of p under f. For example, let $f(x) = x^2 + 1$ and S the interval $[-5, 3]$. Then the image of S under the function f is the interval $[1, 26]$. (Note, incidentally, that although, by definition, for every p in the image, there is at least one pre-image, the number of pre-images of p under f may vary as p varies. In the present case, for example, if $1 < p \leq 10$ then the equation $x^2 + 1 = p$ has two solutions in S, that is, p has two preimages. But if $p = 1$ or $10 < p \leq 26$, then it has only one.)

Finding the image of a set S under a given function f is not always easy. If the function f is monotonically increasing and the set is an interval, say $[a, b]$, then $f(S)$ is, of course, the interval $[f(a), f(b)]$. If f is monotonically decreasing on $[a, b]$, then $f(S) = [f(b), f(a)]$. But in other cases it is not so easy.

If f is continuous at every point of $[a, b]$, then there is a theorem of calculus, called the Intermediate Value Property (see the end of Comment No. 3, Chapter 16), which asserts that the image $f(S)$ will also be an interval, say $[c, d]$. Here, c and d are, respectively, the minimum and the maximum values of $f(x)$ for $a \leq x \leq b$. To find these values generally requires methods from calculus, unless the function f is of a simple type such as the function $f(x) = x^2 + 1$ in the example above.

For trigonometric functions, the problem of finding the image can often be handled without calculus. This is so because the graphs of the three basic functions, viz., $\sin x$, $\cos x$ and $\tan x$ are quite well-known. For example, we already saw that because of (4) and (5), the range of the function $f(x) = a\cos x + b\sin x$ is the interval $[-c, c]$, where $c = \sqrt{a^2 + b^2}$. Here x varies over the entire real line. However, because of the periodicity of the sine function, as long as the interval S is of length at least 2π, the image will still be $[-c, c]$. If the interval S is smaller, then to find its image, one has to study the behaviour of f on S carefully.

As a somewhat similar problem (JEE 1979), let us find the range of the function $5\cos\theta + 3\cos(\theta + \frac{\pi}{3}) + 3$. The last term, 3, is a constant and plays no role except for shifting the range by 3 units to the right. So, let us first omit it and take $f(\theta)$ as $5\cos\theta + 3\cos(\theta + \frac{\pi}{3})$, which, after expansion, equals $a\cos\theta + b\sin\theta$, where $a = 5 + 3\cos\frac{\pi}{3} = \frac{13}{2}$ and $b = 3\sin\frac{\pi}{3} = \frac{3\sqrt{3}}{2}$. Then $a^2 + b^2 = \frac{169}{4} + \frac{27}{4} = 49$. So the range of $f(\theta)$ is $[-7, 7]$. Adding 3, the range of the given function is $[-4, 10]$.

As a less trivial example (JEE 1992, JEE 1997) let us find the range of the function $f(x) = \dfrac{\tan x}{\tan 3x}$. (In the context of the JEE, this problem is unusual in that it has appeared twice in almost identical forms. Normally, in JEE, old questions are not repeated, although, of course, similar questions may be asked again.) Here the numerator and the denominator have periods π and $\frac{\pi}{3}$ respectively. So $f(x)$ is periodic with period π. Therefore, to find the range we may suppose x varies over an interval of length π. We take it to be the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$. As x varies over this interval, $\tan x$ varies over the entire real line. So if we call $\tan x$ as y, then the ratio $\dfrac{\tan x}{\tan 3x}$ can be expressed in terms of y as $\dfrac{1 - 3y^2}{3 - y^2}$ using the identity $\tan 3x = \dfrac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$. So the problem can be looked upon as a problem of finding the range of the function $h(y) = \dfrac{1 - 3y^2}{3 - y^2}$ which can be written as $3 - \dfrac{8}{3 - y^2}$. This function is defined for all real $y \neq \pm\sqrt{3}$. Also $h(0) = \frac{1}{3}$. Further h is an even function (i.e., $h(-y) = h(y)$ for all y) and so we need to consider only $y > 0$. As y increases from 0 to $\sqrt{3}$, y^2 increases from 0 to 3. Hence $3 - y^2$ decreases from 3 to 0 and so its reciprocal increases from $\frac{1}{3}$ to ∞. Therefore $h(y)$ decreases from $\frac{1}{3}$ to $-\infty$. Thus, under h the image of the semi-open interval $[0, \sqrt{3})$ is the semi-open interval $(-\infty, \frac{1}{3}]$. As y crosses

the point $\sqrt{3}$, $h(y)$ becomes positive and numerically very large. Thereafter, as y increases, $3 - y^2$ decreases. So its reciprocal increases. Hence $h(y)$ decreases and tends to 3 as $y \to \infty$. So the image of the interval $(\sqrt{3}, \infty)$ under h is the interval $(3, \infty)$. Putting together, the range of the function $h(y)$ is the union of these two intervals, i.e., the set $(-\infty, \frac{1}{3}] \cup (3, \infty)$. Note, however, that this is not the range of the original function $f(x) = \dfrac{\tan x}{\tan 3x}$. The number $\frac{1}{3}$ is in the range of h (as it equals $h(0)$). But the original function $f(x) = \dfrac{\tan x}{\tan 3x}$ is not defined at $x = 0$. (If this sounds paradoxical, note that h was obtained from f by cancelling the factor $\tan x$, which is 0 at $x = 0$. It is not hard to show that $\lim\limits_{x \to 0} \dfrac{\tan x}{\tan 3x} = \dfrac{1}{3}$. But that is not the same thing as saying that $f(0) = \frac{1}{3}$.) So the range of $f(x)$ is the set $(-\infty, \frac{1}{3}) \cup (3, \infty)$. (The fact that the range is not a single interval may seem to contradict the Intermediate Value Theorem mentioned above. But it is not so, because here the function f is not continuous on the entire real line. For example, it is discontinuous at 0 and also at $x = \pm\frac{\pi}{6}$.)

Since the function $h(y) = \dfrac{1 - 3y^2}{3 - y^2}$ is a ratio of two second degree polynomials, we can also find its range using elementary properties of quadratic equations. (Similar problems were considered in Chapter 6.) By definition, a real number p is in the range of $h(y)$ if and only if the equation $h(y) = p$ has a solution. With $h(y) = \dfrac{1 - 3y^2}{3 - y^2}$, this reduces to saying that the quadratic in y viz., $(3 - p)y^2 + 3p - 1 = 0$ should have a real root. The condition for this is that its discriminant be non-negative. In the present case, this simply means $(p - 3)(3p - 1) \geq 0$. So both the factors should be of the same sign. This means either $p \leq \frac{1}{3}$ or $p \geq 3$. Equality in the first case would mean $y = 0$ is the only solution. But that corresponds to $\tan x = 0$ and in that case $\tan 3x$ is also 0. So in this case the original function $f(x) = \dfrac{\tan x}{\tan 3x}$ is undefined. So $\frac{1}{3}$ is not in the range of f (although it is in the range of the function h). As for the second inequality, viz., $p \geq 3$, if equality holds, then the quadratic equation $(3-p)y^2 + (3p-1) = 0$ degenerates into $3p - 1 = 0$ which is impossible since $p = 3$. We now have the range of $f(x)$ exactly as before, viz., the set $(-\infty, \frac{1}{3}) \cup (3, \infty)$.

This method of finding the range of $h(y)$ is better than the earlier one because it is shorter and also applicable, more generally, for any $h(y)$ which is of the form $\dfrac{ay^2 + by + c}{ky^2 + ly + m}$, where a, b, c, k, l, m are any constants. Still, we have given the earlier solution because it is based on the increasing/decreasing behaviour of a trigonometric function. This knowledge is important in other problems.

The problem can also be done by writing $f(x)$ as $\dfrac{\sin x \cos 3x}{\cos x \sin 3x}$ which reduces to $\dfrac{\sin 4x - \sin 2x}{\sin 4x + \sin 2x}$. Writing $\sin 4x$ as $2 \sin 2x \cos 2x$ and cancelling the

factor $\sin 2x$, we further get $f(x) = \dfrac{2\cos 2x - 1}{2\cos 2x + 1} = 1 - \dfrac{2}{2\cos 2x + 1}$. This is a periodic function with period π. So it suffices to find the image of any interval of length π, say, the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Further, since $f(x)$ is an even function, we need to take only the image of the right half, viz., $[0, \frac{\pi}{2}]$. The function $2\cos 2x$ decreases from 2 to 0 as x increases from 0 to $\frac{\pi}{3}$. Correspondingly, $f(x)$ decreases from $\frac{1}{3}$ to $-\infty$. Between $\frac{\pi}{3}$ to $\frac{\pi}{2}$, $2\cos x + 1$ decreases from 0 to -1 and correspondingly, $f(x)$ decreases from ∞ to 3. Even though $1 - \dfrac{2}{2\cos 2x + 1} = \frac{1}{3}$ for $x = 0$, we have to exclude this value from the range of f since the original function is undefined at 0. (This time we cancelled the factor $\sin 2x$ which also vanishes at 0.) Hence the range of the function $\dfrac{\tan x}{\tan 3x}$ is the set $(-\infty, \frac{1}{3}) \cup (3, \infty)$, the same answer as before.

Comment No. 12:

Finding the range of a function of two variables is still more difficult and, in fact, beyond the scope of JEE. However, if the two variables are related to each other in such a manner that either one determines the other, then effectively, we have a function of a single variable and its range can be found. Here is a problem of this kind. (JEE 1997*)

Let A, B, C be three angles such that$A = \frac{\pi}{4}$ and $\tan B \tan C = p$. Find all possible values of p such that A, B, C are the angles of a triangle.

In simpler language, the problem amounts to finding the range of the function $\tan B \tan C$, where B and C are positive angles whose sum is $\frac{3\pi}{4}$. Since $C = \frac{3\pi}{4} - B$, we get $\tan B \tan C = \tan B \tan(\frac{3\pi}{4} - B) = f(B)$ (say). Here the variable B takes values in the interval $(0, \frac{3\pi}{4})$ and we have to find the image of this interval under the function f. Upon simplification, $f(B)$ comes out to be $\tan B \dfrac{1 + \tan B}{\tan B - 1}$ or $\dfrac{x^2 + x}{x - 1}$ if we write x for $\tan B$. Call this as $f(x)$. Here B varies over the interval $(0, \frac{3\pi}{4})$ and correspondingly, x varies over the set S which is a union of two intervals viz., $(0, \infty) \cup (-\infty, -1)$. We have to find $f(S)$, i.e., the image of the set S under the function f. This time it is not easy to identify the intervals in which f is increasing/decreasing without using calculus. So we follow the second method, viz., that of identifying those real numbers p for which the equation $\dfrac{x^2 + x}{x - 1} = p$ has at least one solution in S. This is equivalent to asking for which values of p the quadratic $x^2 + (1 - p)x + p = 0$ has at least one real root in the set S. The condition for the existence of real roots is that $(1 - p)^2 - 4p \geq 0$, which simplifies to $(p - 3)^2 - 8 \geq 0$. For this to hold there are two possibilities : (i) $p \geq 3 + 2\sqrt{2}$ or (ii) $p \leq 3 - 2\sqrt{2}$. In the first case at least one root of the quadratic equation $\dfrac{x^2 + x}{x - 1} = p$ must lie in the set S. For, otherwise both the roots will lie in the interval $[-1, 0]$ which would then mean that their product would lie in the interval $[0, 1]$. But the product of the roots is p which is at least $3 + 2\sqrt{2}$, a contradiction. So if $p \geq 3 + 2\sqrt{2}$, then p is definitely in $f(S)$. However, in case (ii), we only have $p \leq 3 - 2\sqrt{2}$. So this case

has to be further divided into three subcases. If $p < 0$, then the product of the two roots is negative and so not both can be in the interval $[-1, 0]$. Hence p is in the set $f(S)$. Next, suppose $p = 0$. Then the roots of $f(x) = p$ are -1 and 0, neither of which is in the set S. So $0 \notin f(S)$.

Finally, suppose that $0 < p \leq 3 - 2\sqrt{2}$. Then the two roots, say x_1 and x_2 of the quadratic $x^2 + (1-p)x + p = 0$ are of the same signs as their product p is positive. But x_1 and x_2 cannot be positive, as otherwise their sum which equals $p - 1$ will also be positive which is not the case for $p \leq 3 - 2\sqrt{2}$. So the only possibility left is that both x_1 and x_2 are negative. In that case, for one of them to lie in the set $S = (-\infty, -1) \cup (0, \infty)$, it is necessary that the smaller root, viz., $\dfrac{p - 1 - \sqrt{p^2 - 6p + 1}}{2}$ should be less than -1. But this would mean that $p - 1 - \sqrt{p^2 - 6p + 1} < -2$ and hence that $p + 1 < \sqrt{p^2 - 6p + 1}$.As both the sides are positive, this inequality implies $(p+1)^2 < p^2 - 6p + 1$, i.e., $8p < 0$ contradicting that $p > 0$.

Summing up, we have shown that for the equation $f(x) = p$ to have a solution (in $\mathbb{R}$), either $p \geq 3 + 2\sqrt{2}$ or $p \leq 3 - 2\sqrt{2}$. However, for $f(x) = p$ to have at least one solution in the set S, we must have either $p > 3 + 2\sqrt{2}$ or $p < 0$. So the image of the set S under f is the union of two intervals, viz., $(-\infty, 0) \cup (3 + 2\sqrt{2}, \infty)$. Going back to the original problem, this is also the set of possible values of $\tan B \tan C$ in a triangle ABC in which the angle A equals $\frac{\pi}{4}$. (See Exercise (13.26) for a simpler, calculus based solution.)

Comment No. 13:

Let us now see a few problems where the trigonometric equation to be solved is not given to you directly but has to be written down from some other data. In the Main Problem, for example, a geometric condition gave rise to a trigonometric equation whose solution finally gave the solution to the geometric problem. In many JEE problems, the trigonometric equation does not originate naturally from any particular application. Still it is given indirectly so as to test the knowledge of some other facts. As a very simple example (JEE 1988), suppose we want to find those values of x for which the complex numbers $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are complex conjugates of each other. This condition is equivalent to a system of two equations, viz., (i) $\sin x = \cos x$ and (ii) $\sin 2x = \cos 2x$. We can rewrite these as $\tan x = 1$ and $\tan 2x = 1$ respectively. There is no common solution because whenever $\tan x = 1$, $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ is undefined. Hence the original condition is not satisfied by any value of x.

As a less trivial example (JEE 1984), suppose that we want to find the values of x in $(-\pi, \pi)$ which satisfy the equation

$$8^{1 + |\cos x| + \cos^2 x + |\cos^3 x| + \ldots \text{to infinity}} = 4^3 \tag{19}$$

Here the exponent of the L.H.S. is a geometric series with first term 1 and common ratio $|\cos x|$. By a well-known formula, (see Equation (5) in Chapter 23 for a proof) the exponent equals $\frac{1}{1 - |\cos x|}$. (Here we have to assume that $|\cos x| < 1$, i.e., $-1 < \cos x < 1$. In other words, the value $x = 0$ must be

excluded from the interval $(-\pi, \pi)$.) The L.H.S. of (19) is a power of 8 while the R.H.S. is a power of 4. To solve the equation, we must express both the sides as powers of the same number. In the present case, it is very convenient to take this common base as 2. Then (19) reduces to

$$\frac{3}{1 - |\cos x|} = 6 \tag{20}$$

which is equivalent to $1 - |\cos x| = \frac{1}{2}$. The solution is $|\cos x| = \frac{1}{2}$. To solve this we have to solve (i) $\cos x = \frac{1}{2}$ and (ii) $\cos x = -\frac{1}{2}$. In the interval $(-\pi, \pi)$, (i) has solutions $\pm\frac{\pi}{3}$, while the solutions of (ii) are $\pm\frac{2\pi}{3}$. So these are the four solutions of (19).

As a somewhat analogous problem (JEE 1982), suppose we want to solve $e^{2\sin x} - 4e^{\sin x} - 1 = 0$. Treating this as a quadratic in $e^{\sin x}$, we get $e^{\sin x} = \ln(2 \pm \sqrt{5})$. Since $2 - \sqrt{5}$ is a negative number, its logarithm is undefined. So we must have only $\sin x = \ln(2 + \sqrt{5})$. But the number $2 + \sqrt{5}$ exceeds 3 and hence is definitely bigger than e (which lies between 2 and 3). So $\ln(2 + \sqrt{5}) > 1$. Hence $\sin x = \ln(2 + \sqrt{5})$ has no solutions either. So the original equation has no solutions. Note that the argument requires the knowledge of the approximate value of the number e.

In the next problem, the trigonometric equation is to be obtained from a determinant. (JEE 1988) Suppose we want those values of θ lying between $\theta = 0$ and $\theta = \frac{\pi}{2}$, which satisfy the equation

$$\begin{vmatrix} 1 + \sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1 + \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1 + 4\sin 4\theta \end{vmatrix} = 0$$

Subtracting the third row from the first two reduces the determinant to

$$\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ \sin^2\theta & \cos^2\theta & 1 + 4\sin 4\theta \end{vmatrix}$$

which can be expanded as $1 + 4\sin 4\theta + \cos^2\theta + \sin^2\theta$, i.e., as $2 + 4\sin 4\theta$. So the given equation reduces to $\sin 4\theta = -\frac{1}{2}$ whose general solution is $\theta = \frac{n\pi}{4} + (-1)^{n+1}\frac{\pi}{24}$, where n is an integer. The values between 0 and $\frac{\pi}{2}$ will be given only by $n = 1$ and $n = 2$. They are $\theta = \frac{7\pi}{4}$ and $\theta = \frac{11\pi}{24}$.

Determinants arise in many connections, e.g., in solving a system of linear equations. Not surprisingly then, a trigonometric equation can emerge from a system of linear equations as in the following problem. (JEE 1993)

Let λ and α be real. Find the set of all values of λ for which the following system of linear equations has a non-trivial solution.

$$\begin{aligned} \lambda x + \quad & (\sin\alpha)y + \quad (\cos\alpha)z \quad = 0 \\ x + \quad & (\cos\alpha)y + \quad (\sin\alpha)z \quad = 0 \\ -x + \quad & (\sin\alpha)y - \quad (\cos\alpha)z \quad = 0 \end{aligned}$$

For $\lambda = 1$, find all values of α.

From Chapter 3, we know that the necessary and sufficient condition for the existence of a non-trivial solution is

$$\begin{vmatrix} \lambda & \sin\alpha & \cos\alpha \\ 1 & \cos\alpha & \sin\alpha \\ -1 & \sin\alpha & -\cos\alpha \end{vmatrix} = 0$$

By a direct expansion this comes out to be $-\lambda+\cos^2\alpha-\sin^2\alpha+2\sin\alpha\cos\alpha = 0$, or, $\cos 2\alpha + \sin 2\alpha = \lambda$. From (4) and (5), this will have a solution if and only if λ is in the interval $[-\sqrt{2}, \sqrt{2}]$. Finally for $\lambda = 1$, the equation becomes $\cos 2\alpha + \sin 2\alpha = 1$ which can be rewritten as $\sin(2\alpha + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$ whose general solution is $\alpha = \frac{n\pi}{2} + (-1)^n\frac{\pi}{8} - \frac{\pi}{8}$, where n is an integer. (Note that the problem *does not* ask for the solutions of the given system of equations in x, y, z.)

Another natural source of equations is in finding the points of intersections of given curves. As a very simple example (JEE 1982), suppose we want to identify the points of intersection of the two curves $y = \sin 3x$ and $y = \cos x$ for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$. Evidently this amounts to finding all possible solutions of the equation $\sin 3x = \cos x$ lying in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Writing $\cos x$ as $\sin(\frac{\pi}{2} - x)$, we apply (1) to get $3x = n\pi + (-1)^n(\frac{\pi}{2} - x)$. For $n = 2m$ this gives $x = \frac{m\pi}{2} + \frac{\pi}{8}$. For the given range of x, $m = 0$ and $m = -1$ are the only possibilities, yielding $x = \frac{\pi}{8}$ and $x = -\frac{3\pi}{8}$ respectively. For $n = 2m$, we have $x = m\pi + \frac{\pi}{4}$ which is possible only for $m = 0$ giving $x = \frac{\pi}{4}$. Thus there are three possible values of x. But since the problem asks for the points where the graphs intersect, we have to find the corresponding values of $\cos x$ (or, of $\sin 3x$, which is the same). Since $\cos\frac{\pi}{8}$ and $\cos\frac{3\pi}{8}$ are not among the standard values, we leave them as they are. However, we know $\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$. So the answer is $(\frac{\pi}{4}, \frac{1}{\sqrt{2}}), (\frac{\pi}{8}, \cos\frac{\pi}{8})$ and $(-\frac{3\pi}{8}, \cos\frac{3\pi}{8})$.

Comment No. 14:

Thus we see that a trigonometric equation can arise from as diverse topics as infinite series, quadratic equations, complex numbers and systems of linear equations. There is, in fact, no end to it. Instead of giving a trigonometric equation in a straightforward manner, it is first twisted, often artificially, so as to make it come out of some other problem. In doing so, the paper-setters are likely to make a mistake once in a while, because the twisted version may not only yield the original trigonometric equation the paper-setters had in mind, but also some unwanted by-product which is not easy to resolve. This does not happen very frequently in reputed examinations such as the JEE where the papers are set with great care. But nobody is infallible, after all. So here are a couple of examples of such wrongly set questions.

Consider, for example, the problem (JEE 1996) of finding all values of θ in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ satisfying the equation $(1-\tan\theta)(1+\tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$. Writing x for $\tan^2\theta$, this reduces to finding all non-negative zeros of the function $f(x) = 2^x + 1 - x^2$. By inspection, 3 is a zero of $f(x)$ for which the corresponding values of θ come out to be $\frac{\pi}{3}$ and $-\frac{\pi}{3}$. If this were the only zero

of $f(x)$ then the problem would be a reasonable one for two points (in a 100 points paper for three hours). But the fact is that f has at least one more zero, say x_0, as can be seen with a few facts from calculus (which will be studied in Chapter 16). We have $f'(x) = (\ln 2)2^x - 2x$. Using $e > 2.6$ and hence $e^3 > 16$ it follows that $f'(3) = 2(\ln 16 - 3) < 0$. Hence for sufficiently small positive δ, $f(3+\delta) < f(3) = 0$. But by a direct calculation, $f(4) = 1 > 0$. So, by the Intermediate Value Property, f has another zero somewhere in the interval $(3, 4)$. Or, one can observe that since both $f(2)$ and $f(4)$ are positive there is an even number of zeros of f in between them. Now, 3 is already a zero but not a double zero of f, as can be seen from $f'(3) \neq 0$. As otherwise $\ln 2$ will equal $3/4$. But $\ln 2$ is irrational.[2] So, f must have at least one more zero in $(2, 4)$. An intelligent student is likely to realise the existence of x_0. But how is he going to *find* it? There is, in fact, no known way to identify the other zero exactly. Methods are available to find it approximately. But they are rather advanced and need calculational aids, preferably a programmable calculator to get reasonably close to the exact answer.

The dilemma faced by a sincere student in such a case is worse than that in the case of an outrightly wrong problem. In the latter case, if a student is sure of himself, he can confidently say that the problem is incorrect. But in a problem such as this one, especially when it appears in an examination which is reputed to contain some challenging problems, a bright student is quite likely to think that there is some ingenious way of identifying the other zero x_0. He will then spend his precious time to attempt something that is destined to fail. But there is no provision to give him any bonus marks. A mediocre student, on the other hand, who, identifies 3 as a zero of $f(x)$ but simply misses the existence of x_0 stands a better chance as he does not waste his time!. This is an unfortunate, albeit unintended, mockery of the intelligence and the hard work of the bright student.

Yet another improperly set problem (JEE 1987) is to find all possible values of x for which the expression $\dfrac{\sin(\frac{x}{2}) + \cos(\frac{x}{2}) - i\tan x}{1 + 2i\sin(\frac{x}{2})}$ is real. Multiplying both the numerator and the denominator by $1 - 2i\sin(\frac{x}{2})$, the problem reduces to identifying those x for which the product $(\sin\frac{x}{2} + \cos\frac{x}{2} - i\tan x)(1 - 2i\sin\frac{x}{2})$ has its imaginary part equal to 0. This reduces to solving the equation

$$(\sin\frac{x}{2} + \cos\frac{x}{2})2\sin\frac{x}{2} + \tan x = 0\ . \tag{21}$$

Writing $\tan x$ as $\dfrac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2} - 1}$, (21)reduces to

$$2\sin\frac{x}{2}\left[2\sin\frac{x}{2}\cos^2\frac{x}{2} - \sin\frac{x}{2} + 2\cos^3\frac{x}{2}\right] = 0\ . \tag{22}$$

It is very easy to see where the first factor, viz., $2\sin\frac{x}{2}$ vanishes. But doing so for the second factor is far from easy. Only approximate solutions can be

[2] A rigorous proof of this fact is, however, far from easy.

obtained, and that too, by fairly advanced methods. So there is no way to solve (21) in three minutes (which is the proportionate time allotted to the question in view of the credit for a fully correct answer)!

The only practical advice that can be given to a candidate in such cases is not to spend too much time, especially when the problem carries only one or two points.

Comment No. 15:

The fact that an angle is not uniquely determined by its sine or by its cosine etc. also has a bearing in defining the inverse trigonometric functions (also called **inverse circular functions**). The terminology and the notations differ slightly from author to author. For example, suppose x is a real number with $|x| \leq 1$. Then $\sin^{-1} x$ (also denoted by $\arcsin x$ sometimes) is defined as the unique angle lying in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ whose sine is x. Some authors, however, use $\sin^{-1} x$ to denote *any* angle, say θ, whose sine is x. The unique value of θ lying in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is then called the **principal value**[3] of $\sin^{-1} x$ and is denoted by $\mathrm{Sin}^{-1} x$. Nowadays, the first convention is more or less standard and is generally followed in the JEE question papers. But considering that the paper-setters are not the same every year, once in a while there can be an exception. For example, suppose we are asked (JEE 1986) to find the principal value of $\sin^{-1}(\sin \frac{2\pi}{3})$. In the more standard terminology, the words 'principal value' are redundant. Their presence does not cause any confusion in this problem. The answer is very simple. Let $\theta = \sin^{-1}(\sin(\frac{2\pi}{3}))$. Then $\sin\theta = \sin(\frac{2\pi}{3})$. From the basic equation (1), $\theta = n\pi + (-1)^n \frac{2\pi}{3}$ for some integer n. But θ has to lie in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. This is possible only for $n = 1$, giving $\theta = \frac{\pi}{3}$. Hence (the principal value of) $\sin^{-1} \sin \frac{2\pi}{3}$ equals $\frac{\pi}{3}$. Note that we did not have to use the actual value of $\sin \frac{2\pi}{3}$ (which is $\frac{\sqrt{3}}{2}$). The method is equally valid for any α in the second quadrant (i.e., for $\frac{\pi}{2} \leq \alpha \leq \pi$) and gives $\sin^{-1}(\sin\alpha) = \pi - \alpha$.

Note also that some authors denote the inverse trigonometric functions by arcsin, arccos etc. This is a somewhat better notation, because, logically, if $\sin^2 x$ means $(\sin x)^2$, then $\sin^{-1} x$ ought to mean $(\sin x)^{-1}$, i.e., $\frac{1}{\sin x}$. But we are using it for something totally different. Despite such anomalies, these notations are too commonly accepted to change and we shall follow them. So our convention is that $\sin^{-1}, \tan^{-1}, \mathrm{cosec}^{-1}$ and $\cot^{-1}$ (whenever defined) all lie in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ while $\cos^{-1}$ and $\sec^{-1}$ lie in the interval $[0, \pi]$.

The crucial point to observe is that if $\alpha = \sin^{-1} x$ then we certainly have $\sin\alpha = x$. But the converse is not true. A similar remark holds for the other inverse trigonometric functions. And this sometimes necessitates caution in routinely translating identities about trigonometric functions into the corresponding identities about the inverse trigonometric functions. This difficulty

[3] This terminology is keeping in line with a similar terminology about complex numbers. Every non-zero complex number z can be written in the form $re^{i\theta} = r\cos\theta + ir\sin\theta$, where r and θ are real. Here r is the absolute value of z and is uniquely determined by z. But θ, called an argument of z can take infinitely many values, any two of which differ by a multiple of 2π. The unique value of θ lying in the interval $(-\pi, \pi]$ is called the **principal value** of the argument of z and often denoted by Arg z.)

does not arise in the case of powers and logarithms since the functions involved there are one-to-one. For example, if a is a fixed positive base, then the identity $a^{m+n} = a^m a^n$ is completely equivalent to the identity $\log_a(xy) = \log_a x + \log_a y$. In fact either can be derived from the other. This is not quite so for trigonometric identities. Take for example, the well-known identity $\tan(\alpha + \beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$. Suppose we have $\alpha = \tan^{-1} x$ and $\beta = \tan^{-1} y$. Then we certainly have $\tan(\tan^{-1} x + \tan^{-1} y) = \tan(\alpha+\beta) = \frac{x+y}{1-xy}$. However, from this we cannot hastily conclude that $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy}$. (A counter-example is provided by $x = y = \sqrt{3}$. For, in that case, $\tan^{-1} x = \tan^{-1} y = \frac{\pi}{3}$. But $\tan^{-1}\frac{x+y}{1-xy} = \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}$ which is negative and cannot equal $\frac{2\pi}{3}$.) All we can say is that $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy} + n\pi$ for some integer n. Considering that $\tan^{-1}$ take values between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, this integer n has to be either $-1, 0$ or 1. Depending upon the signs of x, y and $1 - xy$, a distinction can be made as to which value of n applies when. We omit it. But the point to note is that in tackling problems involving inverse trigonometric functions, care should be taken to make several cases where necessary.

Take for example, the problem (JEE 2002) of showing that the expression $\cos\,\tan^{-1}\sin\,\cot^{-1} x$ equals $\sqrt{\dfrac{x^2+1}{x^2+2}}$ for real x. Calling $\cot^{-1} x$ as θ, we have $\cot\theta = x$ and so $\sin\theta = \dfrac{1}{\operatorname{cosec}\theta} = \pm\dfrac{1}{\sqrt{x^2+1}}$. Now let $\phi = \tan^{-1}\dfrac{1}{\sqrt{x^2+1}}$. Then the given expression equals $\cos(\pm\phi)$ which is always $\cos\phi$, regardless of which sign is chosen. Note further that $-\frac{\pi}{2} \le \phi \le \frac{\pi}{2}$ and hence $\cos\phi \ge 0$. Therefore, $\cos\phi = \dfrac{1}{\sec\phi} = \dfrac{1}{\sqrt{\tan^2\phi + 1}}$. Substituting $\tan^2\phi = \dfrac{1}{x^2+1}$ completes the solution. Although the calculation is straightforward, the argument would be incomplete if we take only the + sign in finding $\sin\theta$.

However, in a problem (JEE 1981) where we are asked to find $\tan\theta$, where θ equals $\tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$ we can simply apply the formula for $\tan(\alpha+\beta+\gamma)$ and get 0 as the answer. For, here we are *given* that $\tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} = \alpha$ etc. and nowhere do we have to *find* the value of any inverse trigonometric function.

Comment No. 16:

In the Main Problem we reduced a geometric problem to one involving a trigonometric equation. However, trigonometric equations are not the only channel through which trigonometry can be applied to problems in geometry. On the contrary, because of the vast multitude of trigonometric identities about triangles, for geometric problems about triangles, trigonometry is a very powerful tool. In fact we shall devote the next chapter entirely to such applications. But here we consider an indirect manner in which trigonometry can be applied to solve problems of geometry. We already know how coordinates can be used to solve problems of pure geometry. Sometimes a little trigonometry can con-

siderably simplify the calculations involving coordinates. This is so because the parametric equations of the three frequently occurring curves, viz. a circle, an ellipse and a hyperbola can be written using trigonometric functions. Suppose, for example, that the problem deals with a typical point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. We can represent P as (x_1, y_1). Here x_1 and y_1 are two real numbers which are related to each other by the equation $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$. In the course of the solution we shall have to work with these two numbers and then somewhere use the relation between them to get the desired result. This complicates matters. But if we write P in its parametric form as $(a\cos\theta_1, b\sin\theta_1)$ then we have already used the relation. Now we have only one real number, viz., θ_1 to deal with. This economy of notation can simplify the matters considerably in some problems. And the reason it is possible is because of a little trigonometry, specifically, the identity $\cos^2\theta + \sin^2\theta = 1$. (The parametric equations of a hyperbola are based on the identity $\sec^2\theta - \tan^2\theta = 1$.)

As a good illustration (JEE 2000), suppose an equilateral triangle ABC is inscribed in the circle $x^2 + y^2 = a^2$. Assume that the perpendiculars drawn from A, B and C to the major axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $a > b$), meet the ellipse at points P, Q and R respectively in such a way that P, Q, R lie on the same side of the major axis as A, B, C respectively. We want to prove that the normals to the ellipse at the points P, Q and R are concurrent.

We first give a solution where we take A, B, C as $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) respectively. The major axis of the ellipse is simply the x-axis (since $a > b$) and so the perpendicular from A to it is the line $x = x_1$. This meets the ellipse at two points, viz., $(x_1, \pm\frac{by_1}{a})$. But since P is given to lie on the same side of the x-axis as A, we must take the + sign. Thus $P = (x_1, \frac{by_1}{a})$. Similarly Q and R come out to be $(x_2, \frac{by_2}{a})$ and $(x_3, \frac{by_3}{a})$ respectively. The equations of the normals to the ellipse at P, Q, R are, respectively,

$$a^2y_1x - abx_1y = x_1y_1(a^2 - b^2) \quad (23)$$

$$a^2y_2x - abx_2y = x_2y_2(a^2 - b^2) \quad (24)$$

$$a^2y_3x - abx_3y = x_3y_3(a^2 - b^2) \quad (25)$$

Proving concurrency of these three normals is equivalent to proving $\Delta = 0$, where $\Delta = \begin{vmatrix} y_1 & x_1 & x_1y_1 \\ y_2 & x_2 & x_2y_2 \\ y_3 & x_3 & x_3y_3 \end{vmatrix}$. Expanding w.r.t. the first column,

$$\Delta = x_2x_3y_1(y_3 - y_2) + x_3x_1y_2(y_1 - y_3) + x_1x_2y_3(y_2 - y_1) \quad (26)$$

Since ABC is equilateral and inscribed in the circle $x^2 + y^2 = a^2$, we have

$$(x_1 - x_2)^2 + (y_1 - y_2)^2 = (x_2 - x_3)^2 + (y_2 - y_3)^2 = (x_3 - x_1)^2 + (y_3 - y_1)^2 \quad (27)$$

$$\text{and} \quad x_1^2 + y_1^2 = x_2^2 + y_2^2 = x_3^2 + y_3^2 \ (= a^2) \quad (28)$$

Expanding the terms in (27) and cancelling using (28) we get

$$y_2(y_3 - y_1) = -x_2(x_3 - x_1), \qquad y_3(y_1 - y_2) = -x_3(x_1 - x_2)$$

$$\text{and} \qquad y_1(y_2 - y_3) = -x_1(x_2 - x_3) \quad (29)$$

This could also have been obtained by noticing that the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are perpendicular to the vectors $\overrightarrow{BC}$, $\overrightarrow{CA}$ and $\overrightarrow{AB}$ respectively, and taking the dot products. If we substitute (29) into (26) we see directly that $\Delta = 0$, thereby proving that the normals to the ellipse at P, Q and R are concurrent.

Although this calculation is not prohibitively complicated, let us now see what simplification would occur if we use a little trigonometry. Let θ be the angle which OA makes with the positive x-axis. Then we can take A as $(a\cos\theta, a\sin\theta)$. Since ABC is given to be equilateral and has O as its centre, the angle between OA and OB and that between OA and OC are both $120^\circ = \frac{2\pi}{3}$. But they are in opposite senses. So we can take, without loss of generality, B as $(a\cos(\theta + \frac{2\pi}{3}), a\sin(\theta + \frac{2\pi}{3}))$ and C as $(a\cos(\theta - \frac{2\pi}{3}), a\sin(\theta - \frac{2\pi}{3}))$. This way we have completely bypassed (27) to (29). The points P, Q, R on the ellipse corresponding to the points A, B, C come out to be, respectively, $(a\cos\theta, b\sin\theta)$, $(a\cos(\theta + \frac{2\pi}{3}), b\sin(\theta + \frac{2\pi}{3}))$ and $(a\cos(\theta + \frac{2\pi}{3}), b\sin(\theta + \frac{2\pi}{3}))$. Instead of (23) to (25), the equations of the normals to the ellipse at P, Q, R are

$$a\sin\theta x - b\cos\theta y = \frac{1}{2}(a^2 - b^2)\sin 2\theta \tag{30}$$

$$a\sin(\theta + \frac{2\pi}{3})x - b\cos(\theta + \frac{2\pi}{3})y = \frac{1}{2}(a^2 - b^2)\sin(2\theta + \frac{4\pi}{3}) \tag{31}$$

$$a\sin(\theta - \frac{2\pi}{3})x - b\cos(\theta - \frac{2\pi}{3})y = \frac{1}{2}(a^2 - b^2)\sin(2\theta - \frac{4\pi}{3}) \tag{32}$$

respectively, so that now the problem reduces to showing that the determinant

$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin(\theta + \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) & \sin(2\theta + \frac{4\pi}{3}) \\ \sin(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{2\pi}{3}) & \sin(2\theta - \frac{4\pi}{3}) \end{vmatrix}$$

vanishes. This can be done very easily by adding the third row to the second. Since $\cos\frac{2\pi}{3} = -\frac{1}{2} = \cos\frac{4\pi}{3}$, the entries of the new second row are $-\sin\theta, -\cos\theta$ and $-\sin 2\theta$, which are precisely the negatives of the corresponding entries of the first row. (Alternately, one can add all the three rows. Then the three entries in the sum are nothing but the imaginary part of $e^{i\theta}(1 + \omega + \omega^2)$, the real part of $e^{i\theta}(1 + \omega + \omega^2)$, and the imaginary part of $e^{2i\theta}(1 + \omega^2 + \omega)$, where ω is the complex cube root of unity. Since $1 + \omega + \omega^2 = 0$, all the three entries are 0 and hence the determinant vanishes.)

The basic ideas behind the two solutions are, of course, the same. But use of trigonometry allowed us to represent the points A, B, C in terms of a single variable θ. As a result, vanishing of the determinant could be done by applying elementary row operations. In the earlier approach, it had to be done by brute force expansion of the determinant.

We have posed this problem as a problem in coordinate geometry by giving the equations of the ellipse and that of the circle. But we could have dispensed with the coordinates and stated the problem as a purely geometric

problem. All we have to do is to start with any ellipse and consider the circle with the major axis of the ellipse as a diameter. This circle is called the **auxiliary circle** of the ellipse. It helps in understanding the properties of the ellipse. For example, for a typical point $P = (a\cos\theta, b\sin\theta)$ on the ellipse, the angle θ has no natural significance. But consider the corresponding point, say $A = (a\cos\theta, a\sin\theta)$ on the auxiliary circle. Then θ is precisely the angle between the radius OA and the major axis.

Comment No. 17:

The use of trigonometry in problems of coordinate geometry involving circles, ellipses and hyperbolas is not surprising since all these three curves can be parametrised using trigonometric functions. But there are other applications too. Consider the family of all straight lines passing through a given point, say, $P = (x_0, y_0)$. As we saw in the last chapter (Comment No. 13), this is a 1-parameter family and a typical member, say L, of it has equation of the form $y - y_0 = m(x - x_0)$, where m is the slope of L. Indeed, in problems involving the slope, m is the best parameter. For certain problems, it is convenient to express m as $\tan\theta$, where θ is the angle which the line L makes with the (positive) x-axis. When so done, a typical point Q on the line can be expressed as $x = x_0 + r\cos\theta,\ y = y_0 + r\sin\theta$. Here the parameter r has a geometric significance. It is the algebraic distance between P and Q. (Here r is positive for points on L lying on one side of P and negative for points on the other side of P. Each of these two portions of the line L is often called a **ray** with P as the initial point. The geometric distance is $|r|$ and can never be negative no matter which of the two rays Q lies in.) This parametrisation is especially convenient in certain cases, as in the following problem. (The second solution to the problem in Comment No. 11 of the last chapter also used this idea.)

Consider a curve $ax^2 + 2hxy + by^2 = 1$ and a point P not on the curve. A line drawn from the point$P$ intersects the curve at points Q and R. If the product $PQ.PR$ is independent of the slope of the line, then show that the curve is a circle. (JEE 1997*)

Take P as (x_0, y_0) and let L be a typical line through P. Suppose the slope of L is m. Then the equation of L is $y - y_0 = m(x - x_0)$ and solving it simultaneously with $ax^2 + 2hxy + by^2 = 1$ will give us the coordinates of the points Q and R. We can then calculate the distances PQ and PR in terms of m. But it would simplify things if we parametrise the line L so that a typical point on it is expressed directly in terms of its (algebraic) distance from P. For this we write $m = \tan\theta$ as above. Then Q and R are of the form $(x_0 + r_1\cos\theta, y_0 + r_1\sin\theta)$ and $(x_0 + r_2\cos\theta, y_0 + r_2\sin\theta)$ respectively, where r_1 and r_2 are the (algebraic) distances of Q and R from P. We are given that the product of the geometric distances PQ and PR, i.e., the expression $|r_1r_2|$ is independent of the angle θ. However, the words used in the question are a line drawn 'from' P rather than a line 'through' P. This makes it implicit that the points Q and R lie on the same side of the point P. (Or, in other words, they lie on the same ray.) Hence, r_1 and r_2 are either both positive or both negative. In either case, r_1r_2 is positive. So the expression $PQ.PR$ equals r_1r_2

and this is given to be independent of the slope of the line L, i.e., independent of the value of θ. Now r_1 and r_2 are nothing but the roots of the quadratic that results by putting $x = x_0 + r\cos\theta$ and $y = y_0 + r\sin\theta$ in the equation of the curve, i.e., the quadratic $a(x_0 + r\cos\theta)^2 + 2h(x_0 + r\cos\theta)(y_0 + r\cos\theta) + b(y_0 + r\cos\theta)^2 = 1$. Expanding this quadratic, the product of its roots comes out to be $\dfrac{ax_0^2 + 2hx_0y_0 + by_0^2}{a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta}$. Here the numerator is independent of θ. As the ratio is given to be independent of θ, we get that the denominator viz.,

$$a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta \tag{33}$$

is also independent of θ. From this we have to conclude that the curve is a circle, i.e., that $a = b$ and $h = 0$. We can do this by giving various values to θ in (33). For example, for $\theta = 0$, the expression (33) equals a while for $\theta = \frac{\pi}{2}$ it equals b. So we get $a = b$. This reduces the expression (33) to $a + 2h\sin\theta\cos\theta$ i.e., to $a + h\sin 2\theta$. It is now clear that the only way this can be independent of θ is if $h = 0$. (To do it rigorously, equate the values that you get by putting $\theta = 0$ and $\theta = \frac{\pi}{4}$, for example.) Thus the curve is $a(x^2 + y^2) = 1$ which is a circle centred at the origin (assuming $a > 0$ as otherwise the curve becomes the empty set).

The converse is a well-known result in pure geometry. If the curve is a circle C and a line L through a fixed point P cuts it in points Q and R then the product of the algebraic distances $PQ.PR$ is constant, i.e., depends only on the circle C and the point P and not on the line L. This constant which is often called the **power** of the point P w.r.t. the circle is easily shown to be equal to $d^2 - r^2$, where d is the distance of P from the centre of the circle and r is the radius of the circle. Consequently, the power of the point P w.r.t. the circle C is negative, zero or positive according as the point P is inside, on or outside the circle C. In the case of a circle C whose equation is $x^2 + y^2 + 2gx + 2fy + c = 0$, the power of a point (x_0, y_0) w.r.t. C comes out to be the expression $x_0^2 + y_0^2 + 2gx_0 + 2fy_0 + c$. In Comment No. 13 of the last chapter we mentioned the radical axis of two circles C_1 and C_2 with equations $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ as the straight line with equation $(g_1 - g_2)x + (f_1 - f_2)y + c_1 - c_2 = 0$. It is clear that this is precisely the set of all points whose powers w.r.t. C_1 and C_2 are equal to each other. In case the two circles intersect, their radical axis is simply the line containing their common chord. But even when the circles C_1 and C_2 have no point in common, their radical axis is still defined. It is a straight line perpendicular to the line joining the centres of the two circles.

Comment No. 18:

One of the most standard applications of trigonometry to coordinate geometry is to rotate the coordinate axes suitably (keeping the origin fixed) so as to simplify the equation of a given curve. The formulas for the transformation are quite well-known and so we omit their proof (which may be found in any standard textbook on coordinate geometry). Specifically, suppose that the axes are rotated through an angle α. Let (x, y) and (X, Y) denote the coordinates of

a point w.r.t. the old and the new frames of reference. Then x and y are given in terms of X and Y by the equations:

$$x = \cos\alpha\, X - \sin\alpha\, Y \quad \text{and} \quad y = \sin\alpha\, X + \cos\alpha\, Y \tag{34}$$

while X and Y are given in terms of x and y by

$$X = \cos\alpha\, x + \sin\alpha\, y \quad \text{and} \quad Y = -\sin\alpha\, x + \cos\alpha\, y \tag{35}$$

(The easiest way to remember (34) and (35) is to let z and Z be the complex numbers $x+iy$ and $Z = X+iY$ respectively. Then (34) and (35) are expanded versions of the relations $z = e^{i\alpha}Z$ and $Z = e^{-i\alpha}z$ respectively. Using matrices, (34) and (35) can be written as $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$ and $\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ respectively. Each can be derived from the other using the fact that these two matrices are inverses of each other.)

If the equation of a curve C is given as $f(x,y) = 0$, where $f(x,y)$ is an expression involving x and y, then by replacing all occurrences of x and y throughout by $\cos\alpha\ X - \sin\alpha\ Y$ and $\sin\alpha\ X + \cos\alpha\ Y$ respectively, we get $f(\cos\alpha\, X - \sin\alpha\, Y, \sin\alpha\, X + \cos\alpha\, Y) = 0$ as the equation of the same curve in the new coordinates. If the angle α is chosen suitably, then this new equation may be easier to analyse than the old one. As a good illustration, we give an alternate solution to the problem done in Comment No. 17 of the last chapter. (JEE 1979) We are given a curve C with an equation of the form

$$Ax^2 + 2Hxy + By^2 = 1 \tag{36}$$

We want to find the condition on A, B and H that C is an ellipse and to determine its area when this condition is satisfied. Earlier we did the problem by homogenising (36). Let us now do it by changing the axes in such a manner that the new equation of C will not involve any XY term.

A direct substitution of (34) and (35) into (36) transforms it into

$$A'X^2 + 2H'XY + B'y^2 = 1 \tag{37}$$

where

$$A' = A\cos^2\alpha + 2H\cos\alpha\sin\alpha + B\sin^2\alpha \tag{38}$$

$$H' = H(\cos^2\alpha - \sin^2\alpha) + (B - A)\cos\alpha\sin\alpha \tag{39}$$

$$\text{and} \quad B' = A\sin^2\alpha - 2H\cos\alpha\sin\alpha + B\cos^2\alpha \tag{40}$$

From (39) it is clear that if α is so chosen that $\tan 2\alpha = \frac{2H}{A-B}$, then H' will vanish. For this choice of α, we take $\sin 2\alpha = \frac{2H}{\sqrt{4H^2+(A-B)^2}}$ and $\cos 2\alpha = \frac{A-B}{\sqrt{4H^2+(A-B)^2}}$. Putting these values in (38) and simplifying, we get,

$$\begin{aligned} A' &= B + \frac{A-B}{2}(1+\cos 2\alpha) + H\sin 2\alpha \\ &= \frac{1}{2}[A + B + \sqrt{4H^2 + (A-B)^2}] \end{aligned} \tag{41}$$

We could similarly obtain an expression for B'. But there is a slicker way. From (38) and (40) it is clear that $A' + B' = A + B$. From this and (41) we get

$$B' = \frac{1}{2}[A + B - \sqrt{4H^2 + (A - B)^2}] \tag{42}$$

So, if $\alpha = \frac{1}{2}\tan^{-1}\frac{2H}{A-B}$, then the equation of the curve C becomes $A'X^2 + B'Y^2 = 1$, where A' and B' are given by (41) and (42) respectively. This will be an ellipse if and only if A' and B' are both positive, which, in turn, will be the case if and only if $A' + B'$ and $A'B'$ are both positive. From (41) and (42), $A'B' = AB - H^2$. Its positivity implies, in particular, that A and B are of the same sign. But $A + B = A' + B'$ as we saw above. So, in the presence of positivity of $A'B'$, that of $A' + B'$ is equivalent to A and B both being positive. Summing up, (36) represents an ellipse if and only if A, B and $AB - H^2$ are all positive. As for the area bounded by the ellipse, it is obviously independent of the coordinate system. The equation of C in the new system is $A'X^2 + B'Y^2 = 1$, which can be written as $\frac{X^2}{(1/\sqrt{A'})^2} + \frac{Y^2}{(1/\sqrt{B'})^2} = 1$. This is an equation of an ellipse in its standard form. Hence the area equals $\pi\frac{1}{\sqrt{A'}}\frac{1}{\sqrt{B'}}$ (by a formula which can be proved using integrals). As we already know that $A'B' = AB - H^2$, we see that the area of the ellipse is $\frac{\pi}{\sqrt{AB-H^2}}$ which is exactly the same answer we got earlier using the technique of homogenisation.

During the course of the solution, we remarked that regardless of what α is $A + B = A' + B'$. Verbally, no matter how we rotate the axes, the sum of the coefficients of x^2 and of y^2 remains the same. Technically this is expressed by saying that $A + B$ is an **invariant** under rotation. A less obvious invariant is $AB - H^2$. This amounts to saying that no matter what the angle of rotation α is, we have $A'B' - H'^2 = AB - H^2$. This can be done by a direct computation using (38) to (40). However, in case the curve represented by (36) is an ellipse, there is another way to see this, because, as we just saw, $AB - H^2$ is intimately related to the area. In fact, the two determine each other. The area is an intrinsic geometric attribute of the curve. It can be measured by a formula. But its definition does not require that formula. So any two formulas expressing it must be equal. Hence $AB - H^2 = A'B' - H'^2$.

Although in the problem above we were interested only in the area of the ellipse, along the way we also got information about the directions and the lengths of the major and minor axes of the ellipse. (Specifically, their lengths will be $\frac{1}{\sqrt{A'}}$ and $\frac{1}{\sqrt{B'}}$ which can be expressed in terms of A, B and H using (41) and (42). As for their directions, one of them will lie along the line $y = \tan\alpha\, x$ in the old coordinates and the other will be perpendicular to it.) The eccentricity and the foci of the ellipse can also be determined easily. We already know them in the new coordinate system. All we have to do is to apply (34) and (35) to get them in the old coordinates. In other words, by choosing the angle of rotation suitably (so as to get rid of the xy term) we got a complete analysis of the ellipse represented by (36). A similar analysis is possible in case (36) represents a hyperbola (which is what happens when A' and B' are of opposite signs, which reduces to $H^2 > AB$).

Comment No. 19:

Note that the L.H.S. of (36) contains no linear terms. If it does, we first shift the origin (i.e. replace x, y by suitable $x' + u, y' + v$ respectively) to make the coefficients of x' and y' vanish. After that we rotate the axes suitably around the (new) origin to get rid of the XY term. We then get a complete analysis of the type of curve represented by a general second degree equation in x and y. We record the result, without proof, as a summary in the following theorem.

Theorem: Consider the general equation of second degree, viz.,

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \tag{43}$$

where at least one of the coefficients a, b, f, g, h is non-zero. Let Δ be the value of the determinant $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ (i.e., $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$). Then:

(i) If $\Delta = 0$, (43) represents a pair of straight lines (which are real and distinct, real and coincident or imaginary according as $h^2 - ab$ is positive, zero or negative).

(ii) If $\Delta \neq 0$, (43) represents an ellipse (which may reduce to a circle and may also degenerate to a point), a parabola or a hyperbola according as $ab - h^2$ is positive, zero or negative. (Note that in the case of a parabola, $h^2 = ab$ implies that the second degree terms form a perfect square, say $(px + qy)^2$, as we may assume $a > 0$ without loss of generality.)

All these curves are called **conic sections** or simply conics because they can be obtained as the intersections of the surface of a (double) right circular cone with a plane. Again we omit the details but remark that the answer depends upon the relationship of the angle, say β, which the plane makes with the axis of the cone and the angle of the cone, say α. (This is the acute angle between the axis and a **generator** of the cone, i.e., a line that revolves around the axis to generate the cone.) Then depending upon whether (i) $\beta < \alpha$, or (ii) $\beta = \alpha$, or (iii) $\alpha < \beta < \frac{\pi}{2}$ or (iv) $\alpha < \beta = \frac{\pi}{2}$, the curve is a hyperbola, a parabola, an ellipse or a circle respectively. If the plane passes through the vertex of the cone then the conic degenerates to a pair of straight lines in (i), to a straight line (a generator of the cone) in (ii) and to a point in (iii) and (iv).

Comment No. 20:

In solving a problem about a particular conic which is not in a standard form, it is generally *not* a good idea to start from (43) and mechanically apply the reduction in Comment No. 19. It is often far better to make use of the data to first simplify the initial equation as far as possible. Here is an illustration (taken from p. C - 86 of *Problems Plus in IIT Mathematics* - A. Das Gupta, where it is solved by a very different method).

A parabola is drawn touching the x-axis at the origin and having its vertex on the line $y = k$. Prove that the axis of the parabola is a tangent to the parabola $x^2 + 8k(y - 2k) = 0$.

Here the curve is already given to be a parabola and so the second degree terms in its equation form a perfect square. So its equation can be taken to be in the form $(ax+by)^2+2gx+2fy+c=0$. The fact that the origin lies on it gives $c=0$. The data further implies that $y'=0$ when $x=0, y=0$. This gives $g=0$. Hence the equation of the first parabola can be taken as $(ax+by)^2+2fy=0$. Dividing throughout by $\sqrt{a^2+b^2}$, we normalise it further to

$$(\cos\theta\ x+\sin\theta\ y)^2+2\lambda y=0 \tag{44}$$

where λ and θ are some constants. This suggests that we should rotate the axes through θ. Before doing so, we shift the origin to the vertex, say (u,k), of the first parabola. Then, if X, Y are the new coordinates of the point whose old coordinates are (x,y), we have $x=(\cos\theta)X-(\sin\theta)Y+u$ and $y=(\sin\theta)X+(\cos\theta)Y+k$. This reduces (44) to $X^2+2AX+2\lambda\cos\theta\ Y+B=0$ where $A=u\cos\theta+k\sin\theta+\lambda\sin\theta$ and $B=(u\cos\theta+k\sin\theta)^2+2\lambda k$. For this to be in a standard form, A and B must both vanish. Vanishing of A gives $u\cos\theta+k\sin\theta=-\lambda\sin\theta$. As B also vanishes, we get $\lambda=-2k\ \mathrm{cosec}^2\theta$. The axis of the first parabola in the new coordinates is $X=0$. In the old coordinates this becomes $(\cos\theta)(x-u)+(\sin\theta)(y-k)=0$, i.e. $(\cos\theta)x+(\sin\theta)y=-\lambda\sin\theta=2k\ \mathrm{cosec}\ \theta$. The data of the problem is insufficient to determine θ uniquely, because the data gives a 1-parameter family of parabolas. So θ is a parameter which cannot be determined. But regardless of its value, it follows (by directly substituting for y) that the line $(\cos\theta)x+(\sin\theta)y=2k\ \mathrm{cosec}\ \theta$ meets the second parabola $x^2+8k(y-2k)=0$ in two coinciding points and is therefore a tangent as was to be shown. (We could have expressed u in terms of θ. But we needed only the axis and not the vertex of the first parabola.)

Similarly, in a problem involving one or both the asymptotes of a hyperbola, it is better to take its equation as $(a_1x+b_1y+c_1)(a_2x+b_2y+c_2)=k$. The advantage is that the lines $a_ix+b_iy+c_i=0$, $i=1,2$ are the asymptotes. (See Exercise (10.40).)

EXERCISES

10.1 Find all solutions of : (a) $\cos^7 x+\sin^4 x=1,\ -\pi<x<\pi$. (1997)
(b) $4\cos^2 x\sin x-2\sin^2 x=3\sin x$. (1983)

10.2 If $\tan\alpha=\frac{m}{m+1}$ and $\tan\beta=\frac{1}{2m+1}$, find the possible values of $\alpha+\beta$. (1978)

10.3 A polygon of nine sides each of length 2 is inscribed in a circle. Find the radius of the circle. (1987)

10.4 (a) Find the number of solutions of the equation $\tan x+\sec x=2\cos x$ in the interval $[0,2\pi]$. (1993)

(b) Find the number of real solutions of the equation $\sin(e^x)=5^x+5^{-x}$. (1990)

(c) Find the number of real solutions of the equation: $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$. (1999)

10.5 (a) Identify the set $\{x \in [0, \pi] : 2\sin^2 x - 3\sin x + 1 \geq 0\}$. (1987)

(b) On which of the following intervals is the function $f(x) = \sin^4 x + \cos^4 x$ increasing? (1999)

(A) $(0, \frac{\pi}{8})$ (B) $(\frac{\pi}{4}, \frac{3\pi}{8})$ (C) $(\frac{3\pi}{8}, \frac{5\pi}{8})$ (D) $(\frac{5\pi}{8}, \frac{3\pi}{4})$.

(c) Which of the following intervals, if any, contain the smallest positive root of the equation $\tan x - x = 0$? (1987)

(A) $(0, \frac{\pi}{2})$ (B) $(\frac{\pi}{2}, \pi)$ (C) $(\pi, \frac{3\pi}{2})$ (D) $(\frac{3\pi}{2}, 2\pi)$.

10.6 Find which of the following intervals has the property that for every value of p in it, the equation $(\cos p - 1)x^2 + \cos p\, x + \sin p = 0$, in the variable x has real roots. (1990)

(A) $(0, 2\pi)$ (B) $(-\pi, 0)$ (C) $(-\frac{\pi}{2}, \frac{\pi}{2})$ (D) $(0, \pi)$.

10.7 Find the solution set of the system of equations $x + y = \frac{2\pi}{3}$, $\cos x + \cos y = \frac{3}{2}$, where x and y are real. (1986)

10.8 Which of the following is a necessary and sufficient condition for $\sec^2\theta = \frac{4xy}{(x+y)^2}$ to be true? (1996)

(A) $x + y \neq 0$ (B) $x = y, x \neq 0$ (C) $x = y$ (D) $x \neq 0, y \neq 0$.

10.9 (a) Does there exist a value of θ between 0 and 2π which satisfies the equation $\sin^4\theta - 2\sin^2\theta - 1 = 0$? (1984)

(b) Find the values of θ for which the following system of linear equations in x, y, z has non-trivial solutions. (1986)

$$(\sin 3\theta)x - y + z = 0,\ (\cos 2\theta)x + 4y + 3z = 0,\ 2x + 7y + 7z = 0.$$

10.10 Find all non-zero complex numbers Z satisfying $\overline{Z} = iZ^2$. (1996)

10.11 Suppose $a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$ is a polynomial with integer coefficients and has a rational root $\frac{p}{q}$, where the integers p and q are relatively prime. Prove that p divides a_0 and q divides a_n. (This result, known as the **rational root test** is useful in finding all possible rational roots of polynomials with integer coefficients. It is also applicable if the coefficients are all rational, because we can then multiply throughout by a suitable integer.)

10.12 Using the last exercise, show that the cubic equation $8x^3 - 4x^2 - 4x + 1 = 0$ has no rational solution. [*Hint:* The work is simplified if you put $y = 2x$.]

10.13 Prove that the only solution of the equation $2\cos\theta - \sqrt{3}\sin\theta = \sin 2\theta$ in the interval $[0, \pi]$ is $\theta = \frac{\pi}{6}$.

10.14 (a) Find the range of $\sin^2\theta + \cos^4\theta$ for all real θ. (1980)

(b) Determine the interval in which the values of $f(x) = 3\sin(\sqrt{\frac{\pi^2}{16} - x^2})$ lie. (1983)

(c) Find the domain of the function $f(x) = \sin^{-1}\left(\log_2 \frac{x^2}{2}\right)$. (1984)

(d) Find the domain and the range of the function $f(x) = \sin \ln(\frac{\sqrt{4-x^2}}{1-x})$. (1985)

(e) Given $A = \{x : \frac{\pi}{6} \leq x \leq \frac{\pi}{3}\}$ and $f(x) = \cos x - x(x+1)$, find $f(A)$. (1980)

10.15 If α, β are values of θ satisfying the equation $a\tan\theta + b\sec\theta = c$ and $\alpha - \beta$ is not a multiple of π then prove that (i) $\tan(\alpha+\beta) = \dfrac{2ac}{a^2-c^2}$ and (ii) $\tan(\alpha-\beta) = \dfrac{\pm 2b(a^2+c^2-b^2)^{\frac{1}{2}}}{2b^2-a^2-c^2}$.

10.16 Find the numerical values of

(a) $\sin\theta$, it being given that $\tan\theta = \frac{-4}{3}$. (1979)

(b) $\tan[\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}]$. (1983)

(c) $\cos(2\cos^{-1}\frac{1}{5} + \sin^{-1}\frac{1}{5})$. (1981)

(d) $\tan 2\alpha$, where $\cos(\alpha+\beta) = \frac{4}{5}, \sin(\alpha-\beta) = \frac{5}{13}$ and α, β lie between 0 and $\frac{\pi}{4}$. (1979)

(e) $\tan(2\tan^{-1}\frac{1}{5} - \frac{\pi}{4})$. (1984)

10.17 Let $A = 2\tan^{-1}(2\sqrt{2}-1)$ and $B = 3\sin^{-1}\frac{1}{3} + \sin^{-1}\frac{3}{5}$. Determine which of A and B is greater. (1989)

10.18 Find the set of possible values of $\tan A \tan B$ given that A, B are positive and $A + B = \frac{\pi}{4}$.

10.19 By mathematical induction, prove that:
$$\tan^{-1}\tfrac{1}{3} + \tan^{-1}\tfrac{1}{7} + \ldots + \tan^{-1}\tfrac{1}{n^2+n+1} = \tan^{-1}\tfrac{n}{n+2}. \quad (1999)$$

10.20 Sum the series (i) $\sum_{k=1}^{n} \cot^{-1}(2k^2)$ and (ii) $\sum_{k=1}^{n} \cot^{-1}(\frac{k^2}{2})$ $(n > 2)$.

10.21 OA, OB are radii of a circle C inclined at an angle of $60°$ with each other. C' is a circle with OA as a diameter. Upon OB a point P is taken so that a circle can be drawn with centre P to touch C internally and C' externally. Prove that $OP = \frac{4}{5}OB$.

10.22 $ABCD$ is a convex quadrilateral in which $AB = AD, \angle BAC = 2\angle ACD$ and $\angle DAC = 2\angle ACB$. Prove that $BC = CD$. (Regional Mathematics Olympiad)

10.23 If $P = (\cos\theta, \sin\theta)$ and $Q = (\cos(\alpha-\theta), \sin(\alpha-\theta))$, where $0 < \alpha < \frac{\pi}{2}$ is a fixed angle, which of the following operations takes P to Q ? (2002)

(A) clockwise rotation around the origin through an angle α
(B) anti-clockwise rotation around the origin through an angle α
(C) reflection in the line through the origin with slope $\tan\alpha$
(D) reflection in the line through the origin with slope $\tan\frac{\alpha}{2}$

10.24 A line L has intercepts a and b on the coordinate axes. When the axes are rotated through a given angle, keeping the origin fixed, the same line has intercepts p and q. Which of the following statements is true? (1990)

(A) $a^2+b^2=p^2+q^2$ (B) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{p^2}+\frac{1}{q^2}$
(C) $a^2+p^2=b^2+q^2$ (D) $\frac{1}{a^2}+\frac{1}{p^2}=\frac{1}{b^2}+\frac{1}{q^2}$

10.25 The point $(4,1)$ undergoes the following three transformations successively:

I. Reflection through the line $y=x$
II. Translation through a distance of 2 units along the positive x-axis
III. Counter-clockwise rotation through an angle $\frac{\pi}{4}$ about the origin.

Find the final position of the point. (1980)

10.26 P is a point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, where $0<b<a$. A line passing through P and parallel to y-axis meets the circle $x^2+y^2=a^2$ at Q, where P,Q lie on the same side of the x-axis. Find the locus of the point which divides PQ in a fixed ratio. (2001)

10.27 Prove that, in an ellipse, the perpendicular from the focus upon any tangent and the line joining the centre of the ellipse to the point of contact meet on the corresponding directrix. (2002)

10.28 $P=(a\sec\theta, b\tan\theta)$ and $Q=(a\sec\phi, b\tan\phi)$ are points on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, where $\theta+\phi=\frac{\pi}{2}$. Normals at P and Q meet at (h,k). Find k. (1999)

10.29 Find the points of intersection of the line $4x-3y-10=0$ and the circle $x^2+y^2-2x+4y-20=0$. (1983)

10.30 A tangent to the ellipse $x^2+4y^2=4$ meets the ellipse $x^2+2y^2=6$ at P and Q. Prove that the tangents at P and Q of the ellipse $x^2+2y^2=6$ are at right angles. (1997)

10.31 Prove the focusing property of an ellipse. That is, show that a light ray originating at one of its foci will pass through the other focus after being reflected by the ellipse. (This is also true of sound and is used in the construction of whispering galleries.)

10.32 Let d be the distance from the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ to the tangent drawn at a point P on the ellipse. If F_1, F_2 are the two foci of the ellipse, then show that $(PF_1-PF_2)^2=4a^2(1-\dfrac{b^2}{d^2})$. (1995)

10.33 A line through $A(-5,-4)$ meets the lines $x+3y+2=0, 2x+y+4=0$ and $x-y-5=0$ at the points B, C and D respectively. If $(\frac{15}{AB})^2+(\frac{10}{AC})^2=(\frac{6}{AD})^2$, find the equation of the line. (1993)

10.34 If the circle $C_1: x^2+y^2=16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to $\frac{3}{4}$, find the coordinates of the centre of C_2. (1988)

10.35 Let $S=0$ be the equation of a conic, where S is an expression of the form $ax^2+2hxy+by^2+2gx+2fy+c=0$. Let $P(x_1,y_1)$ be a fixed point and let AB be the chord of the conic through P which has P as its mid-point. Using the parametrisation $x=x_1+r\cos\theta, y=y_1+r\sin\theta$, show that the equation of AB is $axx_1+byy_1+h(xy_1+yx_1)+g(x+x_1)+f(y+y_1)+c=S_1$, where $S_1=ax_1^2+2hx_1y_1+by_1^2+2gx_1+2fyy_1+c$. What happens if you try to derive this by taking the equation of AB in the form $y=mx+c_1$?

10.36 From any point O on the circumference of a circle of radius a, lines are drawn making angles $\frac{\pi}{2n}, \frac{2\pi}{2n}, \ldots, \frac{(n-1)\pi}{2n}$ with the diameter through O. Prove that the product of the lengths intercepted on them by the circumference is $a^{n-1}\sqrt{n}$.

10.37 Show that the equation of the circumcircle of the triangle formed by the lines $y=\pm kx$ and $x\cos\alpha+y\sin\alpha-p=0$ is

$$(\cos^2\alpha-k^2\sin^2\alpha)(x^2+y^2)-p(1+k^2)(x\cos\alpha-y\sin\alpha)=0.$$

*10.38 A triangle is inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and has its centroid at the centre of the ellipse. Prove that the locus of the circumcentre of the triangle is $a^2x^2+b^2y^2=\{\frac{1}{4}(b^2-a^2)\}^2$. [*Hint:* Use Exercise (8.19).]

10.39 Assume that the orbit of the earth around the sun and that of the moon around the earth are both circular and in the same plane and that both the bodies move with uniform angular speeds in the same sense. If the angular speed of the moon around the earth is twelve times that of the earth around the sun, find how many solar eclipses occur during one solar year, i.e., during one revolution of the earth around the sun.

10.40 (a) Prove that the equation of a hyperbola whose asymptotes lie along the lines $a_ix+b_iy+c_i=0$, $i=1,2$, is of the form $(a_1x+b_1y+c_1)(a_2x+b_2y+c_2)=k$ for some $k\neq 0$. (As a corollary, hyperbolas having a given pair of lines as asymptotes form a one parameter family.)

(b) Find the equation of a hyperbola which passes through the points $(-1,0)$ and $(-2,1)$, has the line $y=2x+5$ as the tangent at the point $(-2,1)$ and has the line $3x+2y+1=0$ as an asymptote.

(c) Let C be the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $E=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}$ where $P=(x_1,y_1)$ is a fixed point in the plane of C. Prove that the two tangents from P to C touch the same branch of C if $0<E<1$ and different branches if $E<0$. What happens if $E=0$?

Chapter 11

SOLUTION OF TRIANGLES

The very meaning of the word 'trigonometry' is 'measurement of a triangle'. Associated with every triangle ABC are several angles (including not only the vertical angles A, B, C but many others e.g., the angles subtended by the sides at the orthocentre, the centroid etc.) and certain lengths (including, the lengths a, b, c of the three sides and also other associated lengths such as the altitudes, the circumradius, the inradius, the three ex-radii and so on). Traditionally, the phrase 'solution of a triangle' was used for the determination of the triangle, being given the values of some of these variables. For this to be possible, it is necessary to know various relationships among these angles and these lengths. There are numerous formulas giving such relationships. So the problems in this chapter are easy to understand but test the ability to pick just the right formulas from a vast collection.

Occasionally, the data of the problem is insufficient to determine the triangle uniquely, but is sufficient to find the values of some quantities associated with the triangle. The identities about triangles can also be used for such problems. Also sometimes the problem is cast in a different form and has to be first reduced to a problem about some triangle. The Main Problem is of this type. Sometimes the problem asks not only the solution of a triangle but also a geometric construction for it. Problems of geometric constructions (not necessarily confined to triangles) were common in the JEE prior to 1980. Although they are not so common nowadays, one should not miss the charm of coming up with an elegant geometric construction. One can appreciate it even more when one sees that certain constructions are *impossible*.

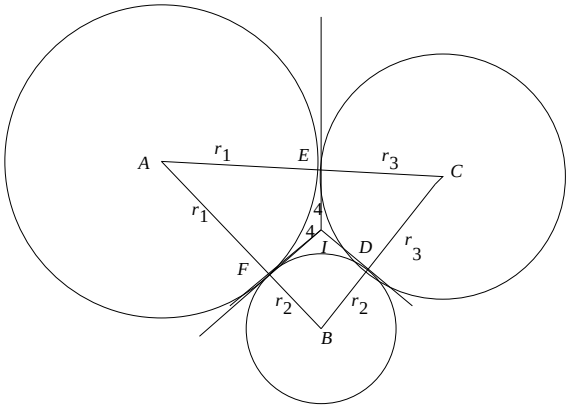
Main Problem : Three circles touch one another externally. The tangents at their points of contact meet at a point whose distance from a point of contact is 4 units. Then the ratio of the product of the radii to the sum of the radii of the circles is ..... (JEE 1992)

First Hint: Consider the triangle with vertices at the centres of the circles.

Second Hint: Show that the inradius of this triangle is 4 units.

Third Hint: Express the product as well as the sum of the radii of the circles in terms of the area and the inradius of the triangle.

Solution: Assume that the point of concurrence of the common tangents at the points of contact is I. Call the centres of the three circles as A, B and C and let their radii be r_1, r_2, r_3 respectively. Then the lines AI, BI, CI are the internal angle bisectors of the triangle ABC. Hence I is the incentre and r, the inradius of ABC, is given to be 4 units in length. We have to find the numerical value of the ratio $\dfrac{r_1 r_2 r_3}{r_1 + r_2 + r_3}$.



Call the points of contacts of the three circles as D, E, F as shown in the figure. Clearly, $r_1 = AE = AF$, $r_2 = BD = BF$ and $r_3 = CD = CE$. Hence $r_1 + r_2 + r_3$ is the semi-perimeter of the triangle ABC (often denoted by s). So if we let Δ be its area, then

$$r_1 + r_2 + r_3 = s = \frac{\Delta}{r} \tag{1}$$

It is also clear that $r_1 = r \cot \frac{A}{2}$, $r_2 = r \cot \frac{B}{2}$ and $r_3 = r \cot \frac{C}{2}$. There is an alternate formula for the area, viz.,

$$\Delta = \frac{abc}{4R} \tag{2}$$

where a, b, c are the sides and R the circumradius of the triangle. The circumradius, in turn, is related to the sides by the formulas

$$R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} \tag{3}$$

Finally, R and r, the inradius, are related by the formula

$$r = 4R \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2} \tag{4}$$

Combining these four formulas we get

$$\begin{aligned} \frac{r_1 r_2 r_3}{r_1 + r_2 + r_3} &= \frac{r^4 \cot\frac{A}{2} \cot\frac{B}{2} \cot\frac{C}{2}}{\Delta} \\ &= \frac{r^4 \cot\frac{A}{2} \cot\frac{B}{2} \cot\frac{C}{2} 4R}{abc} \\ &= \frac{r^4 \cot\frac{A}{2} \cot\frac{B}{2} \cot\frac{C}{2}}{2R^2 \sin A \sin B \sin C} \\ &= \frac{r^4}{16R^2 \sin^2\frac{A}{2} \sin^2\frac{B}{2} \sin^2\frac{C}{2}} = \frac{r^4}{r^2} = r^2 \end{aligned} \tag{5}$$

(For a simpler derivation, equate $\dfrac{r_1 r_2 r_3}{r_1 + r_2 + r_3}$ with $\dfrac{r^2 \cot\frac{A}{2} \cot\frac{B}{2} \cot\frac{C}{2}}{\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2}}$ and use (9) below. An even simpler method is to use the Heron's formula, viz. (18) in Chapter 14, which in the present case becomes simply $\Delta^2 = s r_1 r_2 r_3$, with (1) above.) As r is given to be 4 units, the answer is 16 sq. units.

Comment No. 1:

In the last chapter, we solved a purely geometric problem by converting it to a trigonometric equation. Trigonometric equations are not the only way through which trigonometry can be applied to solve problems of geometry. The basic definitions in trigonometry, viz., $\sin\theta$, $\cos\theta$ and $\tan\theta$ are geometric in nature, namely, as ratios of certain sides of a right-angled triangle[1]. So basically, it is not surprising that trigonometry is a useful tool in doing problems of pure geometry. It is not as commonly applicable as coordinates. Still, a few illustrations can be given. For example, the concurrency of the three altitudes of a triangle can be proved using the well-known Ceva's theorem in geometry and a little trigonometry. Let AD, BE, CF be the altitudes of a triangle. Then the ratio $BD : DC$ equals $\dfrac{c\cos B}{b\cos C}$. The ratios $CE : EA$ and $AF : FB$ can be expressed similarly. Clearly the product of these three ratios is 1. So, by Ceva's theorem, the lines AD, BE, CF are concurrent. As another (but more challenging) example of a similar nature, the following problem, (JEE 2000)

[1] There are other ways to define $\sin\theta$ and $\cos\theta$. For example, one definition is that they are respectively, the sums of the infinite series $\sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n+1}}{(2n+1)!}$ and $\sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n}}{(2n)!}$. If we differentiate these series term-by-term, we immediately get that $\frac{d}{d\theta}\sin\theta = \cos\theta$ and $\frac{d}{d\theta}\cos\theta = -\sin\theta$. The ease with which these and many other results can be proved gives these definitions a certain popularity. But in a rigorous approach, considerable spadework has to go in to show the validity of term-by-term differentiation. Such definitions are called **analytical** because they are given strictly in terms of real numbers, making no appeal to any diagrams. We shall not follow them.

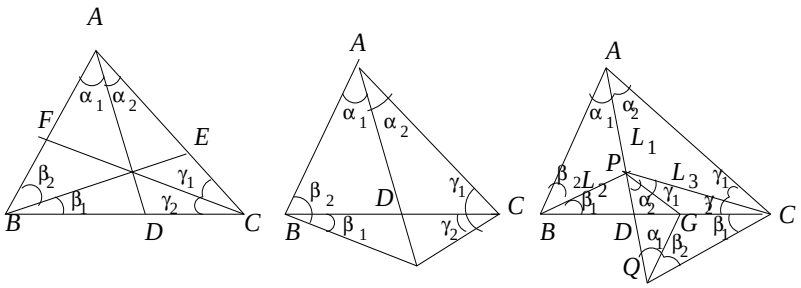
mentioned in Comment No. 10 of Chapter 8, can also be solved using Ceva's theorem and a little trigonometry.

ABC and $A'B'C'$ are triangles in the same plane. If the perpendiculars from A, B and C to $B'C', C'A'$ and $A'B'$ respectively are concurrent, prove that so are the perpendiculars from A', B' and C' to BC, CA and AB respectively.

For a solution, we first prove a subsidiary result. Suppose that lines L_1, L_2, L_3 through the vertices of a triangle ABC divide the vertical angles into two parts each as shown in Figure (a) below, so that $\alpha_1 + \alpha_2 = A$, $\beta_1 + \beta_2 = B$ and $\gamma_1 + \gamma_2 = C$. Then we claim that the lines L_1, L_2, L_3 are concurrent if and only if

$$\sin\alpha_1 \sin\beta_1 \sin\gamma_1 = \sin\alpha_2 \sin\beta_2 \sin\gamma_2 \tag{6}$$

(In Figure (a), all the lines divide the respective angles internally. But the result holds true even if some of them divide the vertical angles externally, provided we take some of the angles above as negative as illustrated in Figure (b), where we take β_1 and γ_2 as negative (so that β_2 and γ_1 are bigger than the angles B and C respectively). The condition (6) still holds and the proof requires only minor modifications.)



(a) (b) (c)

To prove this subsidiary result, let L_1, L_2, L_3 meet the opposite sides at D, E, F respectively. Applying the sine rule to the triangles ABD and ADC we get, respectively, that $BD : AD = \dfrac{\sin\alpha_1}{\sin B}$ and $DC : AD = \dfrac{\sin\alpha_2}{\sin C}$. Dividing the two we get the ratio $BD : DC$ as $\dfrac{\sin\alpha_1 \sin C}{\sin\alpha_2 \sin B}$. Similarly the ratios $CE : EA$ and $AF : FB$ come out, respectively as $\dfrac{\sin\beta_1 \sin A}{\sin\beta_2 \sin C}$ and $\dfrac{\sin\gamma_1 \sin B}{\sin\gamma_2 \sin A}$. Taking the product of these three ratios and applying Ceva's theorem gives (6) as a necessary and sufficient condition for the concurrency of the lines L_1, L_2 and L_3.

Now to prove the main result we go back to the notations and the figure in the pure geometry solution to the same problem given in Comment No. 10 in Chapter No. 8. Assume that the lines L_1, L_2, L_3 divide the angles A, B, C

into parts α_1 and α_2 etc. as shown in (a) above. Then we know that (6) holds. Now consider how the lines L_1', L_2', L_3' divide the vertical angles of the triangle $A'B'C'$. The thing to keep in mind is that the angle between two lines is equal to the angle between their perpendiculars. Here $L_1' \perp BC$ and $A'B' \perp L_3$. So the angle between L_1' and $A'B'$ equals the angle between BC and L_3 which we are calling as γ_2. By a similar argument, the angle between L_1' and $A'C'$ is the same as the angle between BC and L_2, which we are calling as β_1. So the line L_1' divides the vertical angel A' of the triangle $A'B'C'$ into parts γ_2 and β_1. Similarly the angles B' and C' are divided by L_2' and L_3' as shown in (c) above. But then again using the subsidiary result proved above (applied to the triangle $A'B'C'$ this time) we get that L_1', L_2' and L_3' are concurrent.

Comment No. 2:

As already mentioned in Comment No. 14 of Chapter 7, trigonometry abounds in identities about triangles. So it is hardly surprising that for problems about triangles, trigonometry is a powerful tool. The Main Problem is an illustration of this, where, in the solution, we used as many as four of the standard formulas involving the inradius, the circumradius and the area of the triangle. Note that the data of the problem is not sufficient to determine the values of r_1, r_2, r_3 individually. For, if it were, then we could also determine the three sides of the triangle (since we have $a = \dfrac{r_2 + r_3}{2}$ etc.). But that would mean that the inradius alone determines a triangle completely which is not the case. In fact, we cannot even find $r_1r_2r_3$ and $r_1 + r_2 + r_3$ separately from the data. It is only their ratio, viz., $\dfrac{r_1r_2r_3}{r_1 + r_2 + r_3}$ that can be found from the data. This expression is cyclically symmetric in r_1, r_2 and r_3. So one can predict that the solution will depend on those identities which are cyclically symmetric in nature. An awareness of this fact helps in choosing the right identities which may be useful in a given problem. Otherwise, one can get lost in a maze of numerous identities and keep working in circles. For example, there are three well-known formulas for the inradius of a triangle. Two of these appeared in (1) and (4) above. They are symmetric in nature. The third formula

$$r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2} \tag{7}$$

does not give the value of r as a symmetric function. So it is not likely to be of much use here. If at all it is to be used, it will have to be cast in a symmetric form, viz., $r^3 = (s-a)(s-b)(s-c)\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}$. (This is, basically, what we did with formula (3). It should be noted, however, that in doing so the formula may lose some of its strength. For example, saying that $R^3 = \dfrac{abc}{8\sin A\sin B\sin C}$ is not as strong as (3). Luckily, for our purpose it was enough. The symmetric consequence of (7) just mentioned, viz., $r^3 = (s-a)(s-b)(s-c)\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2}$ is far too complicated to be useful.)

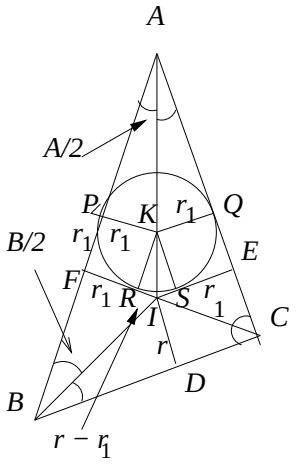
Comment No. 3:

Here is another problem (JEE 2000) where a cyclically symmetric result about a triangle is proved using trigonometry.

Suppose ABC is a triangle with incentre I and inradius r. The feet of the perpendiculars drawn from I to the sides BC, CA and AB are D, E and F respectively. If r_1, r_2, r_3 are, respectively, the radii of the circles inscribed in the quadrilaterals $AFIE, BDIF$ and $CEID$, prove that

$$\frac{r_1}{r-r_1} + \frac{r_2}{r-r_2} + \frac{r_3}{r-r_3} = \frac{r_1 r_2 r_3}{(r-r_1)(r-r_2)(r-r_3)} \tag{8}$$

Let K, L, M be, respectively, the centres of the circles inscribed in the quadrilaterals $AFIE, BDIF$ and $CEID$. (The figure shows only the first one out of these three circles.) Clearly, they lie on the angle bisectors AI, BI, CI respectively. From K draw perpendiculars KP, KQ KR and KS to AB, AC, IF and IE respectively. All these perpendiculars have length r_1. So the rectangle $KPFR$ is a square of side r_1. Since $IF = r$, we get $IR = IF - RF = r - r_1$. So the ratio $\dfrac{r_1}{r-r_1}$ equals $\dfrac{KR}{IR} = \cot \angle IKR$. But since KR is parallel to AF, $\angle IKR = \angle IAF = \frac{A}{2}$.



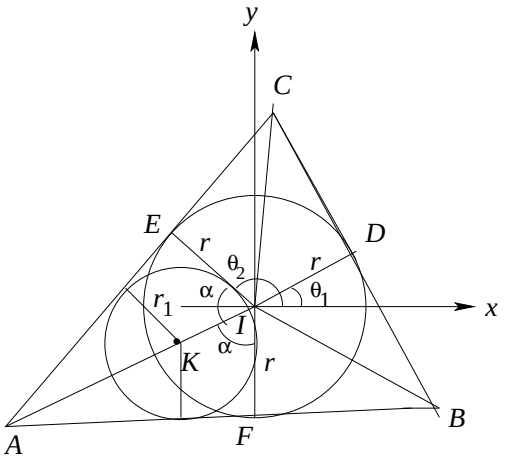
Thus we have shown that $\dfrac{r_1}{r-r_1} = \cot \frac{A}{2}$. By a similar reasoning, we get $\dfrac{r_2}{r-r_2} = \cot \frac{B}{2}$ and $\dfrac{r_3}{r-r_3} = \cot \frac{C}{2}$. Therefore, proving (8) is equivalent to proving

$$\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2} \tag{9}$$

But this is a standard identity about triangles. It can be proved by writing $\cot \frac{C}{2}$ as $\tan(\frac{A}{2} + \frac{B}{2}) = \dfrac{\tan \frac{A}{2} + \tan \frac{B}{2}}{1 - \tan \frac{A}{2} \tan \frac{B}{2}} = \dfrac{\cot \frac{A}{2} + \cot \frac{B}{2}}{\cot \frac{A}{2} \cot \frac{B}{2} - 1}$. (An alternate way to prove (9) is to use the formula $\cot \frac{A}{2} = \sqrt{\dfrac{s(s-a)}{(s-b)(s-c)}}$ and similar formulas for $\cot \frac{B}{2}$ and $\cot \frac{C}{2}$. But to do so in an examination puts a little more pressure

on your memory. Good memory is definitely an asset and can save you some precious time. But if you go solely by remembering a large number of formulas, with little idea about their derivation, then you get stuck if the given problem does not fall under any of the formulas you have remembered. Also, as happens in the proof of (9) above, you are likely to miss shortcuts. The other extreme is to remember very few basic formulas and to derive the rest as and when needed. Academically, this is a healthier approach. But it can cost you a lot in terms of time. Depending upon one's ability and inclination, one can choose a suitable balance.)

The solution given above was a direct application of trigonometry to a problem in geometry. As remarked in Comment No. 16 of the last chapter, we can also apply trigonometry through coordinates. We present another solution of this problem which is of this spirit. We take the incentre I as the origin. Then the points D, E, F lie on the circle $x^2+y^2=r^2$ and hence can be taken to be $(r\cos\theta_1, r\sin\theta_1)$, $(r\cos\theta_2, r\sin\theta_2)$ and $(r\cos\theta_2, r\sin\theta_2)$ respectively, where θ_1, θ_2 and θ_3 are the angles between the positive x-axis and the radii ID, IE and IF respectively. As before, let K be the centre of the circle inscribed in the quadrilateral $AFIE$. Then K lies on the angle bisector of the angle EIF. So the coordinates of K are of the form $(\lambda\cos\theta, \lambda\sin\theta)$, where $\theta = \dfrac{\theta_2+\theta_3}{2}$ and λ is some positive real number. Now, r_1 is the perpendicular distance of K from IE. The equation of the line IE is simply, $y\cos\theta_2 - x\sin\theta_2 = 0$. So we get,



$$r_1 = \lambda\sin\theta\cos\theta_2 - \lambda\cos\theta\sin\theta_2 = \lambda\sin(\theta-\theta_2) \tag{10}$$

But r_1 is also the perpendicular distance of K from the line EA which is a tangent to the circle $x^2+y^2=r^2$ at the point $(r\cos\theta_2, r\sin\theta_2)$. Hence the equation of the line AE is $x\cos\theta_2 + y\sin\theta_2 = r$. This gives r_1 as $r - (\lambda\cos\theta\cos\theta_2 + \lambda\sin\theta\sin\theta_2) = r - \lambda\cos(\theta-\theta_2)$. So we have,

$$r - r_1 = \lambda\cos(\theta-\theta_2) \tag{11}$$

From (10) and (11), we get $\dfrac{r_1}{r-r_1} = \tan(\theta-\theta_2) = \tan\dfrac{\theta_3-\theta_2}{2}$ (since $\theta = \dfrac{\theta_2+\theta_3}{2}$). Let us call $\dfrac{\theta_3-\theta_2}{2}$ as α. Then α is half the angle between the radii IE and IF (taken counter-clockwise as in the figure).

By a similar reasoning, $\dfrac{r_2}{r-r_2} = \tan\beta$ and $\dfrac{r_3}{r-r_3} = \tan\gamma$, where β is half the angle between IF and ID and γ is half the angle between ID and IE. Clearly $\alpha+\beta+\gamma = \pi$. (Here we are assuming that $0 \le \theta_1 < \theta_2 < \theta_3 < 2\pi$. In forming γ we take $\dfrac{2\pi+\theta_1-\theta_3}{2}$ and not $\dfrac{\theta_1-\theta_3}{2}$ which is negative.) (8) now amounts to showing that

$$\tan\alpha + \tan\beta + \tan\gamma = \tan\alpha\tan\beta\tan\gamma$$

But this is a standard result holding for any three angles which add up to π, in particular, for angles of a triangle.

Comment No. 4:

Even in a problem where the result to be proved is not cyclically symmetric, the methods of trigonometry can be applied. Here is an example. (JEE 1978)

Suppose p_1, p_2, p_3 are the altitudes through the vertices A, B, C of a triangle ABC with area Δ. Prove that

$$\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p_3} = \frac{2ab}{(a+b+c)\Delta}\cos^2\frac{C}{2} \tag{12}$$

The R.H.S. looks rather formidable. But if we bring the factor Δ in its denominator to the L.H.S. then the problem unfolds itself. Since $\Delta = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$, the problem amounts to showing

$$a+b-c = \frac{4ab}{a+b+c}\cos^2\frac{C}{2} \tag{13}$$

If we write $2\cos^2\frac{C}{2}$ as $1+\cos C$, and multiply both the sides of (11) by $a+b+c$, then the problem is reduced to proving

$$(a+b)^2 - c^2 = 2ab + 2ab\cos C \tag{14}$$

which is true by the cosine formula (viz., that $\cos C = \dfrac{a^2+b^2-c^2}{2ab}$). Thus we have proved (12). (In a formal proof, one should reverse the steps, i.e., starting from the cosine formula, one should first prove (14), then divide both the sides by $a+b+c$ to get (13) and then divide by Δ to get (12). But it is all right to give the argument as above along with a remark that the operations performed at each stage are reversible.)

A similar and a much simpler problem (JEE 1979) is to prove that with the notations above, $2Rp_1 = bc$, where R is the circumradius of the triangle ABC. This follows by considering the area Δ which equals $2ap_1$. Formula (2) above immediately gives the answer. (Incidentally, this problem was asked (in a verbal form) for 5 points in a 3 hour, 100 marks examination. The Main Problem, which is also in a verbal form but requires considerably more work, carried only 4 points in a similar examination. This is an illustration of the changing trend in the Joint Entrance Examination. Speed has acquired more and more importance.)

Here is yet another problem (JEE 1980) where neither the given information nor the conclusion asked is cyclically symmetric.

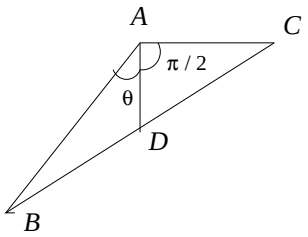
If the median AD of a triangle ABC is perpendicular to its side AC, prove that

$$\cos A \cos C = \frac{2(c^2 - a^2)}{3ac} \tag{15}$$

For the solution, we first find $\cos A$ and $\cos C$ using the given condition. As the triangle CAD is right-angled, we immediately get

$$\cos C = \frac{b}{a/2} = \frac{2b}{a} \tag{16}$$

To find $\cos A$, we let $\theta = \angle BAD$. Then $A = \frac{\pi}{2} + \theta$ and hence $\cos A = -\sin\theta$. The latter can be found by applying the sine rule to the triangle ABD. We have, $\dfrac{\sin\theta}{a/2} = \dfrac{\sin(\frac{\pi}{2} + C)}{c} = \dfrac{\cos C}{c}$. So $\cos A = -\sin\theta = -\dfrac{a\cos C}{2c}$. Using (16) we thus get



$$\cos A = -\frac{b}{c} \tag{17}$$

Multiplying (16) and (17), we get

$$\cos A \cos C = -\frac{2b^2}{ac} \tag{18}$$

So, proving (15) is equivalent to proving $a^2 = 3b^2 + c^2$. For this we use (16) again and the cosine formula to get $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab} = \dfrac{2b}{a}$. A direct calculation now gives $a^2 = 3b^2 + c^2$, thereby proving the result.

Although the solution is quite simple, there is a little catch to it. We already used (16) to derive (18). The general tendency is to think that we need something else to prove that $a^2 = 3b^2 + c^2$. It is then tempting to invoke something like the well-known Apollonius theorem which gives that $b^2 + c^2 = \frac{a^2}{2} + 2(AD)^2$. But then we would have to express AD in terms of a, b and c. And this leads to a lengthy calculation. The thing to keep in mind is that (16) is not only a consequence of the given condition (viz., that AD is perpendicular to AC), but it is, in fact, equivalent to it. Moreover, this is the only piece of information given about the triangle. So any conclusion about the triangle (other than the general properties of a triangle) ought to be derivable from (16). This indeed turns out to be the case as we saw.

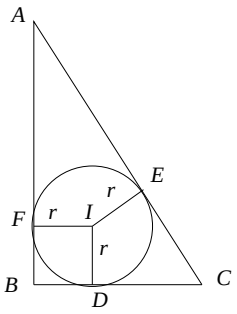
Comment No. 5:

As with coordinates, the use of trigonometric formulas simplifies the solution to a geometric problem because through it we can freely use algebraic manipulations. In a pure geometry proof, only very elementary algebraic manipulations are used. Instead, some ingeneous constructions are employed and that's why a pure geometry solution is often more elegant. When the problem is of a simple nature, it is a good idea to try it by pure geometry first. Here is a good example. (JEE 1979)

If a circle is inscribed in a right-angled triangle ABC with the right angle at B, show that the diameter of the circle is equal to $AB + BC - AC$.

With the usual notations, here we are asked to prove that $2r = a + c - b$. There are several ways to do this using trigonometry since we have various formulas for the inradius. In fact, formula (7) gives an instantaneous solution since in the present case, $\tan\frac{B}{2} = \tan 45^\circ = 1$. So a formula which was not of much use in the solution to the Main Problem makes mincemeat of the present problem. The simpler and more popular formula (1) will also give an answer if we observe that being a right-angled triangle, the area Δ is simply $\frac{1}{2}ac$ in the present case. So, $r = \dfrac{2\Delta}{a+b+c} = \dfrac{ac}{a+b+c}$. The solution is completed by multiplying both the numerator and the denominator by $a+c-b$ and observing that $(a+b+c)(a+c-b)$ is simply $2ac$ as $a^2+c^2 = b^2$. A solution based on formula (4) is also possible but requires considerably more trigonometric manipulations. So once again we see that where more than one formula is available, picking the right one can save a lot of labour.

The point, however, is that the present problem can be done very easily without *any* of these formulas, i.e., using only pure geometry. Let I be the incentre of the triangle ABC. Drop perpendiculars ID, IE and IF on BC, CA and AB respectively. Observe that since $\angle B$ is a right angle, the quadrilateral $BFIE$ is a square of side r. Also, $AE = AF$ and $CD = CE$. Hence $AB + BC - AC$ is simply $BD + BF = 2r$ as was to be proved.

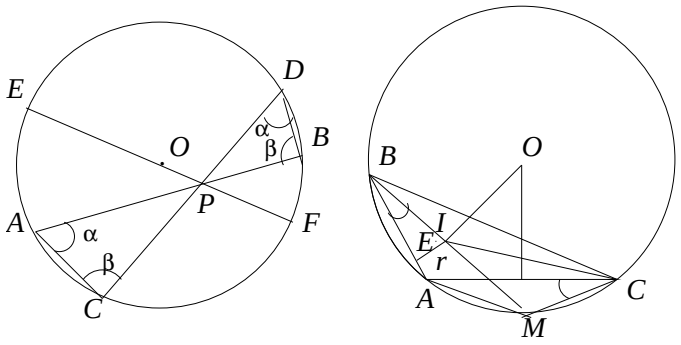


It may happen that a part of a problem can be done more elegantly with pure geometry but the rest of it is amenable more easily to trigonometry. Here is an example of such a problem. (JEE 1999)

Let ABC be a triangle with O and I as its circumcentre and incentre, respectively. If R and r are the circumradius and the inradius, respectively, then prove that $(IO)^2 = R^2 - 2Rr$. Further show that the triangle BIO is a right-angled triangle if and only if b is the arithmetic mean of a and c.

The first part of the problem is a standard result. A trigonometric proof of it is rather long and we omit it. (See, for example, p. 244 of *Plane Trigonometry - Part I* by S. L. Loney.) However, if we rewrite the result to

be proved as $R^2 - (OI)^2 = 2Rr$, then a pure geometry solution suggests itself because the L.H.S. is a familiar expression, called the **power** of the point I w.r.t. the circumcircle, a concept which we already mentioned at the end of Comment No. 17 of the last chapter. More generally, if P is any point inside a circle with centre O and radius R then it is easy to show that for any chord AB of the circle passing through the point P, the product $PA.PB$ is independent of the chord, i.e., the same for all chords passing through P. Further, the value of this product is $R^2 - (OP)^2$. (If P lies outside the circle then too is the product $PA.PB$ a constant. But its value is $(OP)^2 - R^2$.)



(a) (b)

The constancy of the product $PA.PB$ follows from the fact that for any two chords, say AB and CD, passing through P, the triangles APC and DPB are similar to each other (see Figure (a) above) whence $PA : PC = PD : PB$. To show that this constant equals $R^2 - (OP)^2$, consider the diameter, say EF, passing through P and observe that $PE = R + OP$ while $PF = R - OP$.

Now to derive the first part of the given problem from this result, draw the circumcircle of the triangle ABC and let the line BI meet it again at the point M as in Figure (b) above. Then because of the result just mentioned, we have

$$BI.IM = R^2 - (OI)^2 \tag{19}$$

Let us calculate BI and IM separately. For the former, drop a perpendicular IE from I to the side AB. Then $IE = r$ and so $BI = \dfrac{r}{\sin\frac{B}{2}}$. As for IM, note first that MIC is an isosceles triangle since the angles $\angle MIC$ and $\angle MCI$ both equal $\frac{B}{2} + \frac{C}{2}$. So $IM = MC$. But R is also the circumradius of triangle BMC, one of whose angles is $\frac{B}{2}$. So by (3), applied to the triangle BMC, we have $MI = MC = 2R\sin\frac{B}{2}$. As we already know that $BI = \dfrac{r}{\sin\frac{B}{2}}$, (19) implies that $R^2 - (IO)^2 = 2Rr$, as was to be proved. (This proof uses only elementary trigonometry. Calculation of BI is the very definition of $\sin\frac{B}{2}$. Calculation of MI used the sine rule for a triangle. This too could be done exclusively in

terms of pure geometry if one wants. But that is not very vital. What matters is the simplicity of an argument and not whether it puritanically avoids some technique.)

For the second part of the problem, the triangle BIO is right-angled (at I) if and only if $(BI)^2 = R^2 - (OI)^2$. As we already know $BI = \dfrac{r}{\sin\frac{B}{2}}$ and $R^2 - (OI)^2 = 2Rr$, this is equivalent to saying that $r = 2R\sin^2\frac{B}{2}$. (We could have also got this by equating BI with IM.) We want to show that this is equivalent to $2b = a + c$. We could do so theoretically by expressing everything (viz., r, R and $\sin\frac{B}{2}$) in terms of the sides a, b and c of the triangle ABC. But that would be too complicated. Instead, we adopt a simple trick. The condition $2b = a + c$ is unaffected if we replace all the three terms by proportional terms. By (3), a, b, c are proportional to $\sin A, \sin B, \sin C$. Hence proving $2b = a + c$ is equivalent to proving $2\sin B = \sin A + \sin C$. The latter is more manageable because we have so many identities to convert the trigonometric functions of one angle to another. In fact, we have the following chain of reversible steps. (The symbol $\Longleftrightarrow$ indicates a two-sided implication and is read as 'if and only if'.)

$$
\begin{aligned}
\angle BIO = \frac{\pi}{2} \quad &\Longleftrightarrow \quad r = 2R\sin^2\frac{B}{2} \quad \text{(as shown already)} \\
&\Longleftrightarrow \quad 2\sin\frac{A}{2}\sin\frac{C}{2} = \sin\frac{B}{2} \quad \text{(by (4))} \\
&\Longleftrightarrow \quad \cos\frac{A-C}{2} - \cos\frac{A+C}{2} = \sin\frac{B}{2} \\
&\Longleftrightarrow \quad \cos\frac{A-C}{2} = 2\sin\frac{B}{2} \\
&\qquad\qquad \left(\text{since } \frac{A+C}{2} = \frac{\pi}{2} - \frac{B}{2}\right) \qquad (20) \\
&\Longleftrightarrow \quad 2\cos\frac{A-C}{2}\cos\frac{B}{2} = 2\sin B \\
&\Longleftrightarrow \quad 2\sin\frac{A+C}{2}\cos\frac{A-C}{2} = 2\sin B \\
&\qquad\qquad \left(\text{since } \frac{B}{2} = \frac{\pi}{2} - \frac{A+C}{2}\right) \\
&\Longleftrightarrow \quad \sin A + \sin C = 2\sin B \qquad (21) \\
&\Longleftrightarrow \quad a + c = 2b \quad \text{(by (3) as noted above)} \qquad (22)
\end{aligned}
$$

Thus we have shown that the triangle BIO is right-angled (at I) if and only if b is the arithmetic mean of a and c, completing the solution. Unlike the first part, the proof of the second part required a good amount of trigonometry.

Every step in the derivation above appears quite straightforward when you read the solution. But, when you try to get the solution yourself, after reaching (20), the next manipulation, viz., to multiply both the sides by $2\cos\frac{B}{2}$, may not strike you immediately, unless you are already motivated by (21). In other words, it is easy to go forward upto (20) and also easy to go backward from

(21) to (20). But unless you have realised that (22) is equivalent to (21), you are not likely to go forward from (20) to (21). It is one thing to supply the justifications for the intermediate steps once somebody else tells you what the intermediate steps are. It is quite another to think of the right intermediate steps yourself. Even after you have observed that (21) and (22) are equivalent to each other, it might not be so obvious how to go from (20) to (21). The right thing to do in such cases is to start working forward from what is given and, simultaneously, work backward from the goal (in this case (22) or (21)) till you reach a common meeting point (such as (20)). In the final or the 'polished' form of the solution, it is customary to pretend as if you are always going forward. But there is nothing wrong if you are honest in your presentation. Thus, in the present problem, instead of giving the solution as above, it is quite all right to show separately that each of the two conditions, viz., (i) $\angle BIO = \frac{\pi}{2}$ and $2b = a + c$, reduces to (20). As the steps are all reversible, this is a perfectly valid (and, in fact, more honest) proof. (See also the comments on the proof of Equation (30) in Chapter 7.)

There are many other characterisations of the condition that b is the arithmetic mean of a and c. By itself, the statement is just another way of saying that a, b, c are in an A.P. Because of the sine formula, this is equivalent to saying that $\sin A, \sin B, \sin C$ are in A.P. Also, if p_1, p_2, p_3 are the altitudes through A, B, C respectively, then $ap_1 = bp_2 = cp_3 = 2\Delta$. Hence a, b, c are in A. P. if and only if p_1, p_2, p_3 are in H.P. (JEE 1998) Yet another characterisation comes through the ex-radii r_1, r_2, r_3 of the triangle. Using the formulas $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$ and $r_3 = \frac{\Delta}{s-c}$, it follows that r_1, r_2, r_3 are in an H.P. $\Longleftrightarrow$ $s-a, s-b, s-c$ are in an A.P. (JEE 1983), which in turn, is equivalent to saying that $-a, -b, -c$ are in A.P. and finally that a, b, c are in A.P. The second part of the problem given above could have been paraphrased in terms of any of these characterisations. For example, we could have been asked to show that the triangle BIO is right-angled (at I) if and only if r_2 is the harmonic mean of r_1 and r_3. Such a 'twist' is sometimes given to a problem in an attempt to create a 'new' problem.

Comment No. 6:

The common theme in most of the problems above was that some piece of information was given about a triangle ABC and we were asked to derive some consequence of it. In the Main Problem, for example, we were given the length of the inradius and asked to find the value of a certain ratio. In some cases, we were given that some angle associated with the triangle was a right angle and asked to prove some relationship about its sides. We remarked that a single piece of information like this is not sufficient to determine the triangle. But in every case, it was sufficient to derive the desired result. To determine a triangle uniquely[2] we need three pieces of information about it, such as all three sides of

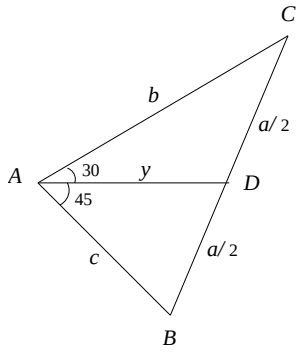
[2]Here uniqueness, of course, means uniqueness upto congruence. That is, any two triangles satisfying the given conditions are congruent to each other. In other words we do not distinguish between mutually congruent triangles.

it, or two sides and the included angle. If we are given two sides say a and b, of a triangle ABC and not the included angle C but one of the other two angles, say A, then sometimes the triangle is not uniquely determined as there can be two possible triangles which fit the prescription. Even in such cases we say that the triangle is determined by the given conditions, because this is not as bad as being given the three angles of a triangle, in which case there are infinitely many triangles (all similar to each other) having prescribed angles. This is to be expected because any two angles determine the third and so even if superficially we are given three pieces of information, in reality there are only two.

So, to determine a triangle, we need three **independent** pieces of information. This is a technical term. In plain terms, it says that each piece of information should imply something that cannot be implied by the other two. These three pieces of data may be given in hundreds of ways and it takes some knowledge of trigonometry to decide if they are independent or not. For example (JEE 1982), suppose we are given the values of $a, \sin A$ and $\sin B$. Then there are two possible values of angle A (one acute and the other obtuse) and similarly for the angle B. A and B cannot both be obtuse. In each of the remaining three cases, the triangle ABC is uniquely determined because now we know all its angles and one of the sides. Similarly, if we are given the values of $a, \sin B$ and R (the circumradius) then from a and R we can first determine $\sin A$ (using (3)). From $a, \sin A$ and $\sin B$ we can then determine the triangle as above. But if we are given $a, \sin A$ and R, then the last piece of information is superfluous since R is already determined by a and $\sin A$. So there are infinitely many triangles satisfying the data. (They are obtained by drawing a circle of radius R and any chord of length a in it. Any triangle with this chord as a base and the third vertex in one of the arcs of the circle will have a constant value of A.)

As an example of a problem where three pieces of information are given, suppose that the median AD of a triangle ABC has length $\dfrac{1}{\sqrt{11-6\sqrt{3}}}$ and that it divides the angle A into angles of $30°$ and $45°$. We want to determine the length of the side BC. (JEE 1985)

For a solution, denote the lengths of the sides BC, CA, AB by a, b, c respectively, as usual. Even though the length of the median AD is given numerically, it is a good idea to denote it by a symbol, say y and put $y = \dfrac{1}{\sqrt{11-6\sqrt{3}}}$ only at the end after getting the answer in terms of y. This saves the trouble of having to write the clumsy figure $\dfrac{1}{\sqrt{11-6\sqrt{3}}}$ again and again.



To determine the triangle, we need a system of three equations in

the three unknowns a, b and c. One of these will come from applying the sine rule to the triangles ABD and ADC. Doing so gives us, respectively, $\frac{a/2}{y} = \frac{\sin 45^\circ}{\sin B} = \frac{1}{\sqrt{2}\sin B}$ and $\frac{a/2}{y} = \frac{\sin 30^\circ}{\sin C} = \frac{1}{2\sin C}$. Putting these together, we get $\sin B = \sqrt{2}\sin C$. But since $\sin B, \sin C$ are proportional to b, c respectively, we get

$$b = \sqrt{2}c \tag{23}$$

We get two more equations by applying the cosine formula for the angles BAD and DAC. They are,

$$y^2 + c^2 - \frac{a^2}{4} = 2yc\cos 45^\circ = \sqrt{2}yc \tag{24}$$

$$y^2 + b^2 - \frac{a^2}{4} = 2yb\cos 30^\circ = \sqrt{3}yb \tag{25}$$

Subtracting (24) from (25) we get $b^2 - c^2 = \sqrt{3}yb - \sqrt{2}yc$ which can be solved simultaneously with (23) to get

$$c = \sqrt{2}(\sqrt{3}-1)y \quad \text{and} \quad b = 2(\sqrt{3}-1)y \tag{26}$$

To find a (which is our real interest) we can substitute either of these values into (24) (or (25)) and solve for a. Or one can apply Apollonius theorem to get $b^2 + c^2 = 2y^2 + \frac{a^2}{2}$. (This equation does not convey any new information. It just makes it a little easier to obtain a.) Either way, a^2 comes out to be $3b^2 - 4y^2 = (44 - 24\sqrt{3})y^2$. Now it is time to put the given value of y, viz., $y = \frac{1}{\sqrt{11 - 6\sqrt{3}}}$. Doing so gives $a^2 = 4$ and hence a, the length of the side BC is 2.

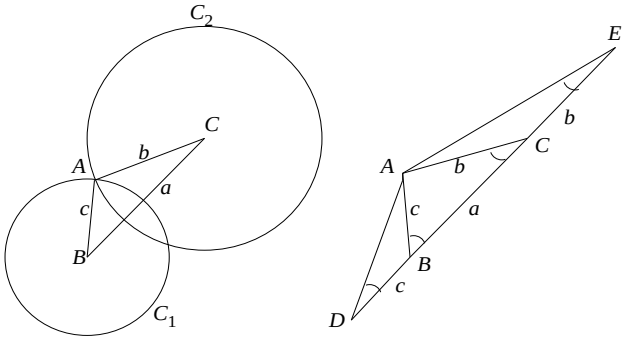
Comment No. 7:

As pointed out earlier, determination of a triangle requires three independent conditions. Sometimes we are asked to determine a triangle from just two conditions. For example (JEE 1984), one can show that if $\cos A + \cos B + \cos C = \frac{3}{2}$ and $a = 4$ then $b = c = 4$. This may seem impossible to do. But as we remarked in the last chapter, sometimes a single equation is equivalent to several. For example, the equation $x^2 + y^2 = 0$ implies $x = 0$ and $y = 0$, although the equation $x^2 + y^2 = 1$ is insufficient to determine x, y uniquely. The reason is that here 0 is the least value of a certain function. Similarly, $\cos x + \cos y = 2$ can hold if and only if $\cos x = 1$ and $\cos y = 1$. We shall take up this problem in Chapter No. 14 (specifically, see Comment No. 9 and 10 there).

Comment No. 8:

Sometimes a problem may give three (independent) pieces of information about a triangle and ask you to give a geometric construction for drawing it. There is a mechanical way of doing this. First you determine the three sides of

the triangle (from the data) using various identities and doing various algebraic manipulations. Having known the three sides of a triangle, say ABC, there is a very standard construction for drawing the triangle as shown in Figure (a) below. First draw a line segment BC of length a (which is given, or has been found from the data). Now draw circles C_1 and C_2 centred at B and C and with radii c and b respectively (whose values are either given or have been found). These circles will intersect in two points. If we call either of these two points as A we get a desired triangle.



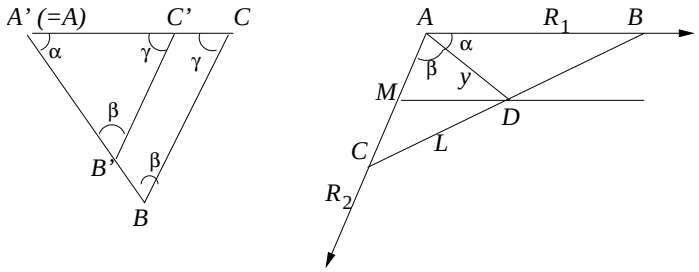
(a) (b)

But this method is not very exciting. The work in it is purely algebraic, similar to the solution of a geometric problem using coordinates. Just as pure geometry solutions are often more elegant (and challenging) than solutions using coordinates, a direct geometric construction for a triangle is often far more elegant and challenging than the method described above. Suppose, for example, that we are given the angles and the perimeter, say p of a triangle ABC. To determine the triangle trigonometrically, we can use (3) to get $p = a + b + c = 2R(\sin A + \sin B + \sin C)$. Since $p, \sin A, \sin B$ and $\sin C$ are all known, this equation gives us R. And then, using (3) again, we can find a, b, c individually. Then by the construction in Figure (a) above we can draw the desired triangle.

But there is a far more elegant geometric construction for drawing ABC. (JEE 1979) To understand it, extend the line BC on both the sides and mark points D and E so that $BD = BA$ and $CE = CA$ as shown in Figure (b) above. Then the triangles ABD and ACE are isosceles. So $\angle ADB = \angle DAB$. But the sum of these two angles is $\angle B$ which is given. So $\angle ADB = \frac{B}{2}$, which is known. Similarly, $\angle AEC = \frac{C}{2}$ which is also known. The segment DE has length $c + a + b = p$ which is also given. So we can draw the triangle ABC as follows. First draw a line segment DE of the given length. From D and E, draw lines L_1, L_2 respectively, making angles $\frac{B}{2}$ and $\frac{C}{2}$ with the line DE. (Standard constructions are available for drawing such lines.) Take A as the point of intersection of these lines. Next, draw the perpendicular bisectors of the segments AD and AE. (Again, there is a standard construction for drawing the perpendicular bisector of a given segment.) Take B, C to be the points of

intersection of these perpendicular bisectors with the line DE. Then ABC is a desired triangle.

There is an alternate and a simpler (but less ingenious) construction for the triangle ABC. Since all the angles are known, we first draw a triangle $A'B'C'$ which is similar to the desired triangle. Let the perimeter of this triangle be p'. Call A' as A. On the line $A'B'$ mark off a point B so that the ratio $A'B' : A'B$ equals the ratio $p' : p$ (see Figure (c) below). Standard constructions for marking such a point are available. They are based on elementary properties of similar triangles. Next draw a line through B parallel to the line $B'C'$ and let it intersect the line $A'C'$ at C. Then ABC is a desired triangle.



(c) (d)

(Constructions of this type are useful when the data is such that it is easy to construct a triangle which is similar to the desired triangle. For example, suppose we are given the three altitudes of a triangle. Taking their reciprocals, we know the proportions of its sides and so we can draw a triangle similar to the desired triangle.)

Let us now go back to the problem above, where we determined the sides of a triangle ABC, being given the length of its median AD and the angles $\angle BAD, \angle DAC$. Call these as y, α and β respectively. In the problem done above, y was given as $\dfrac{1}{\sqrt{11-6\sqrt{3}}}$ while α and β were $45°$ and $30°$ respectively. The solution we gave did not use the particular value of y except at the very end. However, we did use the particular values of α and β right from the beginning to simplify the calculations. In contrast to this, we shall give a geometric construction for the triangle ABC which will work equally easily for any values of y, α and β. (We, of course, have to assume that α, β are positive and that their sum is less than π.)

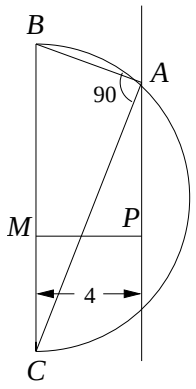
Now for the construction of the triangle ABC first draw a line segment AD of the given length. Then from the point A draw half-rays R_1, R_2 on opposite sides of AD making angles α and β with AD as shown in Figure (d) above. The problem of constructing the triangle ABC now reduces to that of drawing a line, say L, through the point D whose intercept between the rays R_1 and R_2 is bisected at D. The construction we shall give for this easily generalises to find a line whose intercept between R_1 and R_2 is divided at D in a given ratio, say, $\lambda : \mu$. (In our problem, this ratio is $1 : 1$.) To draw this line, first draw through

D a line parallel to the ray R_1 and let it meet the ray R_2 at the point M. Mark a point C on R_2 such that $AM = MC$. Join CD and let it meet the ray R_1 at B. Since M is the mid-point of AC and MD is parallel to AB, it follows that D is the mid-point of the segment BC. So ABC is the desired triangle.

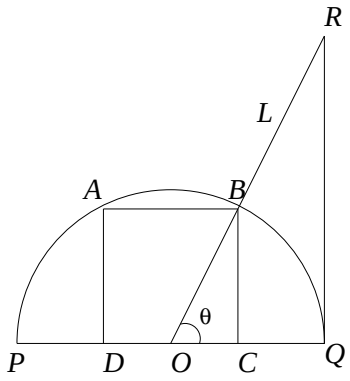
Here is one more example of a problem asking for a construction of a triangle about which three pieces of information are given.

Construct a right-angled triangle given that the median and the altitude to the hypotenuse are 6 cm and 4 cm respectively. Justify your construction. (JEE 1978)

For an algebraic solution, suppose ABC is right-angled at A and has AD and AM as the altitude and median respectively, to the hypotenuse. Then the hypotenuse BC is twice the median AM and hence 12 cm in length. That is, $a = 12$. To determine the other two sides b and c, we need two equations. One of them comes from the Pythagoras theorem, i.e., $b^2 + c^2 = a^2 = 144$. To get another equation in b and c we consider the area of the triangle ABC, which is $\frac{1}{2}bc$ on the one hand, and $\frac{1}{2}AD.BC$ i.e., 24 cm^2 on the other. Equating the two, we have $bc = 48$. The two equations together imply $b^4 - 144b^2 + (48)^2 = 0$. This can be solved as a quadratic in b^2 to give $b^2 = 24(3 + \sqrt{5})$ and $c^2 = 24(3 - \sqrt{5})$ (or vice versa). Taking positive square roots of these quantities gives the numerical values of b and c. Knowing a, b, c we can now draw the desired triangle as described above. (Of course, drawing a circle whose radius is exactly $\sqrt{24(3+\sqrt{5})}$ or $\sqrt{24(3-\sqrt{5})}$ is itself a messy job. But it can be done.)



(e)



(f)

On this clumsy background, it is easy to appreciate the elegance of a purely geometric construction. Note that as $\angle A$ is given to be a right angle, the vertex A must lie on a circle with diameter BC. Also since the altitude through A is 4 cm A must lie at a perpendicular distance 4 cm from the line BC. So we first draw the segment BC and the circle with BC as a diameter. We now need to draw a line, say L which is parallel to BC and at a perpendicular distance 4 cm from it. For this, first draw a perpendicular to BC at any point on it, for example, at M (see Figure (e) above). For this a standard construction is

available. On this perpendicular mark off a point P at a distance 4 cm from M. Then L will be the perpendicular line to MP at the point P. The line L will intersect the circle (with BC as a diameter) in two points. Either of these two points can be taken as the vertex A. Then ABC is a desired triangle. (In Figure (e) above we show only a semi-circle with BC as a diameter. The other semi-circle will also give two desired triangles, of which one will be the mirror image of the triangle ABC we have shown.)

Constructing a triangle directly from the given data (instead of first finding its sides using trigonometry) offers an infinite variety of challenges. Moreover, the method is not limited only to the construction of triangles. The techniques can be applied to other constructions as well. Here is an example.

The diameter PQ of a semi-circle is 6 cm Construct a square $ABCD$ with points A, B on the circumference and the side CD on the diameter PQ. Describe briefly the method of construction. (JEE 1980)

Let O be the centre of the semi-circle. By symmetry, O has to be the mid-point of CD. (For a rigorous justification, note that AD and BC are halves of equal chords of the circle and hence are equidistant from the centre O.) Let θ be the angle BOC. Then $\tan\theta = \frac{BC}{OC} = \frac{BC}{\frac{1}{2}CD} = 2$. So B lies on the line, say L, passing through O and having slope 2. It is easy to construct L (without knowing where B is). Simply draw a perpendicular to PQ at the point Q and mark a point R on it so that $QR = 2OQ = 6$ cm and R lies on the same side of Q as the semi-circle (see Figure (f) above). Then L is the line OR. Let B be the point where it meets the given semi-circle. Determine C as the foot of the perpendicular from B to the diameter PQ. The points A and D can now be determined as the mirror images of B and C respectively, in the perpendicular bisector of the segment PQ.

It is noteworthy that in all these constructions the only curves drawn are straight lines and circles. So a ruler and a compass are the only instruments needed. The ruler is also not the usual marked ruler, but merely a straight edge whose sole purpose is to draw the line passing through two points which are either given or have been previously constructed. It suffices because we only draw those segments whose lengths are given or have been determined earlier in the construction. For example, in the problem just solved, 6 cm was a given length. It means we are given a segment of that length. The entire subsequent construction goes through if we start from this segment. Nowhere do we need a ruler. For example, to find the mid-point O of the segment PQ, if we have a ruler, we can mark it as a point 3 cm away from either end. But we can as well find it without a ruler, using only a compass and a straight edge, by a well-known simple construction. (Specifically, draw circles of radii PQ with centres at P and Q and let K, L be their points of intersection. Then O is the point of intersection of the lines PQ and KL.)

Because of this, these constructions are called **straight edge and compass constructions** (or sometimes, less accurately, **ruler and compass** constructions). The search for such constructions has fascinated mathematicians for centuries. What makes them intriguing is that problems which look analogous

may behave very differently when it comes to giving straight edge and compass constructions for them. There is, for example, an easy construction for dividing a given segment into any number of (equal) parts. There is also a construction for dividing an angle into two equal parts, i.e., for bisecting it. Surprisingly, no construction was found, despite intense attempts, to *trisect* a given angle, i.e., to divide it into three equal parts. Similarly, in Figure (d) above, we gave a construction for drawing a line through D whose intercept between the rays R_1 and R_2 is bisected at D and remarked that a similar construction would apply even if we wanted this intercept to be divided in some other given ratio at D. But no construction is possible to draw a line whose intercept between R_1 and R_2 is of a given length. Try it and you will be convinced that it is impossible.

Of course, just because you or nobody could give a construction for something does not mean that such a construction does not exist. Maybe more efforts will lead to it. It was finally proved a couple of centuries ago, that for angle trisection, no construction exists. That is, there is something inherently impossible in trying to trisect an angle with a straight edge and compass only, no matter by whom and how cleverly it is attempted − something like trying to find the point of intersection of two parallel lines. The proof consists of an algebraic analysis of what is amenable to such constructions and then showing that trisecting an angle would involve something that does not fall under this. The proof is well beyond our scope. But we mention it because such proofs are a landmark in the history of human thought. Two similar problems which were finally proved to be inherently impossible are (i) squaring a circle, in which one is to draw a square whose area is that of a given circle and (ii) doubling a cube, i.e., to draw a line segment the cube on which will have double the volume of a cube on a segment of a given length. It is to be noted that the crux here is to solve these problems exactly. Straight edge and compass constructions are available for all these problems that would give solutions with a given degree of accuracy.

Possibly because such discoveries lack any practical applications, they are yet to reach the common man, even an educated common man. That is why, every now and then, the newspapers carry sensational stories announcing that some wisecrack has solved an age-old problem by giving a construction for trisecting an angle!

Comment No. 9:

Trigonometric formulas about triangles are useful in expressing the various attributes of a triangle (such as its altitudes, circumradius, inradius, three exradii etc.) in terms of three given (independent) attributes of the triangle. Such conversions can often throw some light on the nature of the derived attributes, provided we have some information about the original attributes. In Comment No. 3 of Chapter 8, we saw examples where, knowing that certain points are rational points (i.e., points with both coordinates rational), we can deduce that certain other points associated with the given points are also rational points. In trigonometry also we can sometimes draw conclusions of a similar nature as illustrated in the following problem. (JEE 1994)

Consider the following statements concerning a triangle ABC.

(i) The sides a, b, c and the area Δ are rational.

(ii) $a, \tan\frac{B}{2}, \tan\frac{C}{2}$ are rational.

(iii) $a, \sin A, \sin B, \sin C$ are rational.

Prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

It is important to realise that the problem *does not* ask you to prove any of these statements. The statements given are not of the type, say, that $a = 2R\sin A$, which holds for every triangle. Indeed, none of these statements is necessarily true for every triangle. What the problem asks is the inter-relationship between these statements. The first part (viz., (i) $\Rightarrow$ (ii)), read as (i) implies (ii), asks you to prove that in case (i) holds for a triangle, then so does (ii), or, in other words, that the truth of (ii) can be derived from that of (i) (using any of the identities that hold for an arbitrary triangle). Similarly, (ii) $\Rightarrow$ (iii) means that we can prove (iii) assuming that (ii) holds. The last part, (iii) $\Rightarrow$ (i) is to be interpreted similarly, that is, (i) can be deduced from (iii). In proving each of these implications, we are not allowed to assume anything else. For example, in proving that (ii) $\rightarrow$ (iii) we cannot assume that (i) holds. (It may happen to be true. But we cannot *assume* it. We can only assume (ii).)

Once this is clearly understood, the solution itself is not difficult. We prove the three implication statements, one-by-one. In order to stress exactly what is assumed and what is to be proved at a given point in the solution, it is customary to prove each implication statement in a different paragraph, headed by the implication statement itself. We follow this practice. The order in which the implication statements are proved makes little difference logically. So we follow the order given in the statement of the problem.

(i) $\Rightarrow$ (ii): Here we are given that a, b, c and Δ are all rational and from this we have to show that $a, \tan\frac{B}{2}$ and $\tan\frac{C}{2}$ are rational. The rationality of a is, of course, already given. For the remaining two, we need to express $\tan\frac{B}{2}$ and $\tan\frac{C}{2}$ in terms of a, b, c. By a well-known formula, $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$, where, as usual, s is the semi-perimeter, i.e., $\frac{1}{2}(a+b+c)$. Since a, b, c are rational, so is s and hence the ratio $\dfrac{(s-a)(s-c)}{s(s-b)}$ is also rational. All we are using here is that the sum, the difference, the product and the quotient of two rational numbers (whenever defined) is also a rational number. However, the square root of a rational number need not be rational. So from the rationality of a, b, c alone we cannot conclude that $\tan\frac{B}{2}$ is rational. But we are also given that the area Δ is rational, something which we have not used so far. Using the formula $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$, we can express $\tan\frac{B}{2}$ as $\dfrac{\Delta}{s(s-b)}$. As s, b and Δ are all rational, it is now clear that $\tan\frac{B}{2}$ is rational. By a similar argument, $\tan\frac{C}{2} = \dfrac{\Delta}{s(s-c)}$ is rational. Thus we have proved (i) $\Rightarrow$ (ii).

(ii) ⇒ (iii): Here we are assuming that $a, \tan\frac{B}{2}$ and $\tan\frac{C}{2}$ are rational and have to prove that $a, \sin A, \sin B$ and $\sin C$ are all rational. Once again, the rationality of a is already given. That of $\sin B$ and $\sin C$ follows by writing them in terms of $\tan\frac{B}{2}$ and $\tan\frac{C}{2}$ as $\sin B = \dfrac{2\tan\frac{B}{2}}{1+\tan^2\frac{B}{2}}$ and $\sin C = \dfrac{2\tan\frac{C}{2}}{1+\tan^2\frac{C}{2}}$ respectively. The rationality of $\sin A$ cannot, however, be derived this way because it is not given that $\tan\frac{A}{2}$ is rational. However, we can write $\sin A = \sin(\pi - B - C) = \sin(B + C) = \sin B\cos C + \cos B\sin C$. We have already proved that $\sin B$, $\sin C$ are rational. Since $\cos B = \dfrac{1-\tan^2\frac{B}{2}}{1+\tan^2\frac{B}{2}}$ and $\tan\frac{B}{2}$ is rational, we see that $\cos B$ is rational. Similarly, from the rationality of $\tan\frac{C}{2}$ we get that of $\cos C$. So now we are in a position to say that $\sin A$ which equals $\sin B\cos C + \cos B\sin C$ is also rational, thereby completing the proof of the implication statement (ii) ⇒ (iii). (It is tempting to try to prove the rationality of $\sin A$ by writing $\sin A = \dfrac{a\sin B}{b}$. Here, a is given to be a rational, while we have already proved that $\sin B$ is rational. But the rationality of b is not given, as we are assuming (ii) and not (i). Nor have we proved it from (ii). So this attempt is unsuccessful.)

(iii) ⇒ (i): Finally, assume (iii) holds, i.e., $a, \sin A, \sin B, \sin C$ are rational. We can *now* apply the sine rule to get that $b = \dfrac{a\sin B}{\sin A}$ and $c = \dfrac{a\sin C}{\sin A}$ are rational. As for the area Δ, we can use $\Delta = \frac{1}{2}bc\sin A$ to conclude that it is rational since $\sin A$ is given to be rational while, b and c have been proved to be rational.

The solution is now complete. Because of what we have proved it now follows that all the three statements (i) to (iii) are equivalent to each other. That is, the truth of any one of them will imply that of any other. As there are three statements, there are 6 possible implication statements. (More generally, when there are n statements, there are $n(n-1)$ possible implications. Actually, there are n^2 of them. But we are ignoring those which are of the type (i) ⇒ (i).) We have proved three of these. No proof is needed for the remaining three implications, since they follow from what we have proved. For example, we have not proved (i) ⇒ (iii) directly. But we have proved (i) ⇒ (ii) and (ii) ⇒ (iii). Putting these together, we get (i) ⇒ (iii). (There may, of course, be a direct way of proving (i) ⇒ (iii). And sometimes it may be a better one. Academically, it is always instructive to try to prove each possible implication directly. But logically we do not need it.)

In the present problem we have proved the equivalence of the three statements in a particular cyclical manner, as demanded by the question. If the problem merely asks to prove that the given statements are equivalent to each other, then we are free to change the order of the given statements and then prove the implications in a cyclic order. This freedom pays since it is not equally easy to prove all possible implications directly. For example, in the present problem, we proved that (ii) ⇒ (iii). But proving the converse, i.e., proving (iii) ⇒ (ii) directly is not so easy. This is so because the formula

$\sin B = \dfrac{2\tan\frac{B}{2}}{1+\tan^2\frac{B}{2}}$ allows you to deduce the rationality of $\sin B$ from that of $\tan\frac{B}{2}$, but not vice versa. To get $\tan\frac{B}{2}$ from $\sin B$ we would have to solve a quadratic equation in $\tan\frac{B}{2}$, viz., $\sin B\tan^2\frac{B}{2} - 2\tan\frac{B}{2} + \sin B = 0$. Here, even if $\sin B$ is rational, the roots of this quadratic need not be so, since they involve a radical sign. As an actual example, suppose $B = 30°$. Then $\sin B = \frac{1}{2}$ is rational. But $\tan 15°$ which comes out as $2-\sqrt{3}$ is not rational. In other words, we cannot derive the rationality of $\tan\frac{B}{2}$ from that of $\sin B$ and so a direct proof of the implication (iii) $\Rightarrow$ (ii) is not going to be easy. An indirect proof is, of course, possible. Indeed, we already have it by combining (iii) $\Rightarrow$ (i) and (i) $\Rightarrow$ (ii) both of which we proved above.

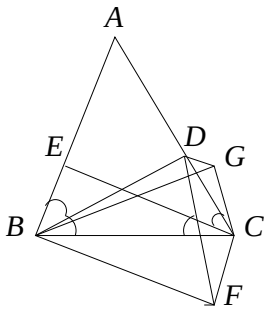
On the face of it, it may sound paradoxical that even though the rationality of $\sin B$ does not imply the rationality of $\tan\frac{B}{2}$ and similarly the rationality of $\sin C$ does not imply the rationality of $\tan\frac{C}{2}$, we can still deduce (ii) from (iii), albeit indirectly. The explanation is that, when we assume (iii) we are *simultaneously* assuming that a, $\sin B$ and $\sin C$ are all rational. This is much stronger than assuming the rationality of these three numbers individually. A real-life analogy perhaps, is that suppose we have two pairs of tennis players, say $\{A, B\}$ and $\{C, D\}$. It may happen that in the single matches neither A nor B can beat either of C and D. But in the doubles, A and B playing together can trounce the pair $\{C, D\}$.

Sometimes it so happens that one of the given statements is 'friendly' with all others in the sense that it is relatively easy to establish its equivalence with each of the others separately. In such a case, instead of proving the equivalence in a cyclic manner, we can prove separately that each of the remaining ones is equivalent to it. For n statements, this means $(n-1)^2$ implications. So it is not as economical as the cyclic manner which requires only n implications. But in a particular situation, it may be more convenient. The present problem is not of this type, because the implication (ii) $\Rightarrow$ (i) is not easy to prove directly either. So, out of the three given statements, none is friendly with all others.

Comment No. 10:

The trigonometric identities assert the equality of a trigonometric expression with another. So far we saw how these equalities can be applied to prove geometric results, especially those about triangles. Interestingly, occasionally some elementary trigonometric inequalities can do the job better than equalities. Here is an example. It is well-known and fairly easy to show (using congruency of certain triangles) that the angle bisectors of the equal angles of an isosceles triangle are equal. The converse is also true but not so easy to prove. Pure geometry proofs of it are, in fact, quite tricky, especially so when compared with the straightforward proofs of similar assertions about altitudes and medians (i.e., showing that equality of two altitudes or of two medians of a triangle implies that it is isosceles). We include two such proofs here because of their unusual nature and also because this result, called the **Steiner-Lehmus theorem**, is rarely covered in school geometry books nowadays.

So assume that the angle bisectors BD and CE of a triangle ABC are equal. We have to show that $\angle B = \angle C$. If not, assume, without loss of generality that $\angle B > \angle C$. Through B and C draw lines to get a parallelogram $EBFC$. Join D and F. Now compare the triangles DBC and ECB. The sides DB and BC of the former are equal to the sides EC and CB of the latter. But the included angle (viz., $\angle\frac{B}{2}$) of the former is greater than the corresponding included angle (viz., $\angle\frac{C}{2}$) of the latter. As both these angles are acute, the sides opposite to them are also related by $CD > EB$.



(This is one of the very few results of pure geometry which deal with inequalities. A trigonometric proof is immediate if one applies the cosine rule to the triangles DBC and ECB. For a pure geometry proof, take a point G, on the same side of the line BC as the point D lies, so that $\angle GBC = \angle\frac{C}{2}$ and $\angle GCB = \angle B$. Since $\angle GCB > \angle ACB$, G must lie outside the triangle ABC. Moreover, the triangles EBC and GCB are congruent by the $A-S-A$ property, whence $BG = EC = BD$. So the triangle BGD is isosceles. As G lies outside the triangle ABC we have $\angle GDC < \angle GDB = \angle DGB < \angle DGC$. Since in any triangle, the side opposite to a larger angle is larger, we get $CD > GC$. But by congruency of the triangles EBC and GCB, we have $GC = EB$. Hence we get $CD > EB$. In the diagram above, the triangle BDG hardly looks isosceles. But some such distortion is inevitable, because the diagram is supposed to represent something which cannot exist, viz., a non-isosceles triangle in which two angle bisectors are equal.)

Now, since $EBFC$ is a parallelogram, we have $EB = CF$. So we get

$$CD > CF \tag{27}$$

On the other hand, let us compare $\angle BDC$ and $\angle BFC$. The former equals $\angle A + \angle\frac{B}{2}$. The latter equals $\angle BEC$ (as $EBFC$ is a parallelogram) and hence equals $\angle A + \angle\frac{C}{2}$. Since we are assuming that $\angle B > \angle C$, we get

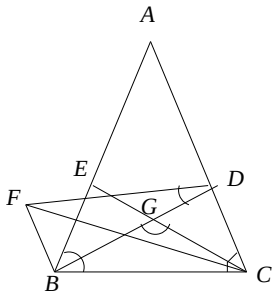
$$\angle BDC > \angle BFC \tag{28}$$

Now comes the tricky step. As $EBFC$ is a parallelogram, $BF = EC$. Since we are assuming $EC = BD$ we get that the triangle BDF is isosceles with $\angle BDF = \angle BFD$. If we subtract these equal angles from $\angle BDC$ and $\angle BFC$ respectively, then in view of (28) we get

$$\angle CDF > \angle CFD \tag{29}$$

Once again we use the fact that in any triangle, the side opposite to a larger angle is larger. So from (29) we get $CF > CD$ which directly contradicts (27), thereby proving the equality of $\angle B$ with $\angle C$.

Our second geometric proof is shorter, but somewhat weird. Assume that the equal angle bisectors BD and CE meet at G. Take a point F so that $BF = BE$ and $FD = BC$. (F will in fact lie on the line ED. But we don't know that yet.) Since $BD = CE$, by $S - S - S$ we get $\Delta BFD \equiv \Delta EBC$ whence $\angle FBD = \angle BEC$ and $\angle FDB = \frac{1}{2}\angle C$. Hence $\angle FBC = \angle FBD + \frac{1}{2}\angle B = \angle BEC + \frac{1}{2}\angle B = \angle BGC = \angle GDC + \frac{1}{2}\angle C = \angle GDC + \angle FDB = \angle FDC$. Note further that $\angle BGC = 90^\circ + \frac{1}{2}\angle A$ is obtuse. So $\angle FBC$ and $\angle FDC$ are equal and obtuse. Therefore $S - S - A$ is applicable to conclude that $\Delta FCB = \Delta CFD$. Hence $CD = FB = BE$. The rest is easy.



Let us now tackle the problem trigonometrically. An immediate advantage is that unlike pure geometry, trigonometry offers handy formulas for the lengths of the angle bisectors. Using these formulas we get $BD = \frac{2ac}{a+c}\cos\frac{B}{2}$ and $CE = \frac{2ab}{a+b}\cos\frac{C}{2}$. (cf. p. 246 of *Plane Trigonometry - Part I* by S. L. Loney.) From $BD = EC$ we get, after a little simplification,

$$(ac + bc)\cos\frac{B}{2} = (ab + bc)\cos\frac{C}{2} \tag{30}$$

Using the standard formulas $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$ and $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$ (where $s = \frac{1}{2}(a + b + c)$) and squaring both the sides gives $c(a + b)^2(a + c - b) = b(a + c)^2(a + b - c)$. Bringing all terms on one side and factoring we get

$$(b - c)(a^3 + a^2b + a^2c + 3abc + b^2c + bc^2) = 0 \tag{31}$$

which forces $b = c$ since the second factor is positive. For an easier solution, which is based on inequalities rather than on equalities, assume that $b \neq c$. Then, without loss of generality, we may suppose $b > c$ and hence also that $\angle B > \angle C$. As $c < b$ and $a > 0$, we have $ac + bc < ab + bc$, i.e., the first factor of the L.H.S. of (30) is smaller than that of the R.H.S. Further, since $0 < \frac{C}{2} < \frac{B}{2} < \frac{\pi}{2}$, the second factor of the L.H.S. of (30) is also smaller than the second factor of the R.H.S. As all the factors are positive, we get a contradiction.

Comment No. 11:

Some algebraic results can also be proved by converting them into trigonometric problems with suitable substitutions. In Cmoment No. 1 of Chapter 10, we already saw how certain cubics can be solved trigonometrically. See Exercise (11.33) for another application of trigonometry to algebra. For an example involving an inequality, the A.M.-G.M. inequality $a + b \geq 2\sqrt{ab}$ where a, b are positive can be proved by letting $c = a + b$ and writing a as $c\sin^2 A$ and b as $c\cos^2 A$ for some acute angle A. Then the inequality reduces to showing that $1 \geq \sin(2A)$ which is true. Here, a direct algebraic proof is, of course, better. But there are situations where this is not so. See Exercises (11.39) and (14.12).

EXERCISES

11.1 With usual notation, if in a triangle ABC, $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$; then prove that $\frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$. (1984)

11.2 (a) In a triangle, the lengths of the two larger sides are 10 and 9 respectively. If the angles are in A.P., find the possible values of the length of the third side. (1987)

(b) ABC is a triangle such that $\sin(2A + B) = \sin(C - A) = -\sin(B + 2C) = \frac{1}{2}$. Find A, B, C given that they are in arithmetic progression. (1990)

11.3 Let the angles A, B, C of a triangle ABC be in A. P. and let $b : c = \sqrt{3} : \sqrt{2}$. Find the angle A. (1981)

11.4 In a triangle ABC, $a : b : c = 4 : 5 : 6$. Find the ratio of the radius of the circumcircle to that of the incircle. (1996)

11.5 A circle passes through three points A, B and C with the line segment AC as its diameter. A line passing through A intersects the chord BC at a point D inside the circle. If $\alpha = \angle DAB$, $\beta = \angle CAB$ d is the distance between A and the mid-point of the segment DC, prove that the area of the circle is $\frac{\pi d^2 \cos^2 \alpha}{\cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta \cos (\beta - \alpha)}$. (1996)

11.6 If in a triangle ABC, $\cot A, \cot B, \cot C$ are in A.P., then determine what type of a progression a^2, b^2, c^2 are in. (1985)

11.7 In any triangle ABC, prove that $\angle A = 2\angle B$ if and only if $a^2 - b^2 = bc$. (See Exercise (24.76(b)) as a sequel to this.)

11.8 In a triangle ABC, AD is the altitude from A. Given $b > c$, $\angle C = 23^\circ$ and $AD = \dfrac{abc}{b^2 - c^2}$, find $\angle B$. (1994)

11.9 A circle is inscribed in an equilateral triangle of side a. Find the area of a square inscribed in this circle. (1994)

11.10 If two sides of a triangle ABC and the included angle are given by $a = (1 + \sqrt{3})$cm, $b = 2$cm and $C = 60^\circ$ find the other two angles and the third side. (1979)

11.11 A triangle ABC has sides $AB = AC = 5$ cm and $BC = 6$ cm. Triangle $A'B'C'$ is the reflection of the triangle ABC in a line parallel to AB placed at a distance 2 cm from AB, outside the triangle ABC. Triangle $A''B''C''$ is the reflection of the triangle $A'B'C'$ in a line parallel to $B'C'$ placed at a distance 2 cm from $B'C'$ outside the triangle $A'B'C'$. Find the distance between A and A''. (1978)

11.12 Find the area of a triangle two of whose angles are 30° and 45° and the included side is $(\sqrt{3} + 1)$ cms. (1988)

11.13 ABC is a triangle inscribed in a circle of radius r. If $AB = AC$ and h is the altitude from A to BC, then the triangle ABC has perimeter $P = 2(\sqrt{2hr - h^2} + \sqrt{2hr})$. Find the area A of the triangle and also $\lim\limits_{h\to 0} \dfrac{A}{P^3}$. (1989)

11.14 If in a triangle ABC, $\dfrac{2\cos A}{a} + \dfrac{\cos B}{b} + \dfrac{2\cos C}{c} = \dfrac{a}{bc} + \dfrac{b}{ca}$, find $\angle A$. (1993)

11.15 AD and AE are the altitude and the internal angle bisector through A of a triangle ABC. P, Q are points on AB, AC respectively, such that $\angle AQP = \angle ABC$. Complete the following relations : (1980)

(i) $\dfrac{BD}{BA} = \dfrac{AB}{...}$, given that $\angle A$ is a right angle

(ii) $\dfrac{\text{area of } \Delta APQ}{\text{area of } \Delta ABC} = \dfrac{(....)}{AC^2}$ (nothing else being given)

(iii) $\angle DAE = \frac{1}{2}[(....) - \angle C]$, given that $\angle B > \angle C$.

11.16 Which of the following sets of conditions on a triangle are feasible (i.e., such that there exists a triangle ABC satisfying them simultaneously)? (1986)

(A) $b\sin A = a,\ A < \frac{\pi}{2}$ (B) $b\sin A > a,\ A > \frac{\pi}{2}$

(C) $b\sin A > a,\ A < \frac{\pi}{2}$ (D) $b\sin A < a,\ A < \frac{\pi}{2},\ b > a$

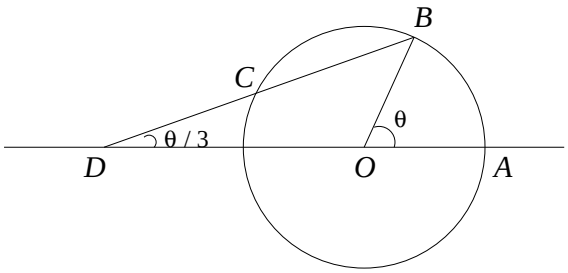
(E) $b\sin A < a,\ A > \frac{\pi}{2},\ b = a$

11.17 If in triangle PQR, $\angle R = \frac{\pi}{2}$ and $\tan\frac{P}{2}$ and $\tan\frac{Q}{2}$ are roots of a quadratic $ax^2 + bx + c = 0$ then which of the following statements is true? (1999)

(A) $a + b = c$ (B) $b + c = a$ (C) $c + a = b$ (D) $a = c$

11.18 Construct a triangle with base 9 cm, altitude 4 cm, the ratio of the other two sides being 2 : 1. (1979)

11.29 In the accompanying figure, A, B, C are points on a circle with centre O and D is the point where the lines BC and OA meet. If CD equals the radius of the circle, prove that $\angle BDO = \frac{1}{3}\angle BOA$. Can this give a straight edge and compass construction for trisecting a given angle BOA?



11.20 Suppose P is a point outside a straight line L and we want to draw the perpendicular, say PQ, from P to L using only a straight edge and compass. Can we do so by drawing a circle centred at P which just touches L

and letting Q be the point of contact of this circle with the line L? Give a valid construction for the desired perpendicular.

11.21 Let L be a straight line passing through the centre of a circle C. Show how to draw a chord of C of a given length and perpendicular to L.

11.22 Give a construction for a rectangle with a given base and having area equal to that of a given rectangle. (Because of this construction, if we are given or are able to construct line segments of lengths a and b, say, then we can construct a line segment of length $\frac{a}{b}$.)

11.23 Show how to construct a square whose area equals that of a given rectangle. What significance does such a construction have?

11.24 Using the preceding exercises (or otherwise) and the fact that $\cos 72^\circ = \frac{1}{\sqrt{5}+1}$, give a straight edge and compass construction for drawing a regular pentagon. (The problem of constructing a regular n-gon has an interesting history. It can be shown that a construction is possible for $n = 2, 3, 4, 5, 6, 8, 10, 12, 15$ and 16, for example but that no construction is possible for $n = 7, 9, 11, 13, 14$. Interestingly, a construction *is* possible for $n = 17$. So it was a challenging problem to characterise those n for which a regular n-gon can be constructed with only a straight edge and compass. The final answer (due to Gauss) is that this is the case if and only if $n = 2^r p_1 p_2 \dots p_k$, where $p_1, p_2, \dots, p_k$ are distinct primes of the form $2^{2^t} + 1$. Such primes are called **Fermat primes**. Note that 3, 5, 17 and 257 are Fermat primes. They correspond to $t = 0, 1, 2$ and 3 respectively. It can also be shown that the next case, viz., $2^{2^4} + 1$, i.e., $2^{16} + 1$ is a prime. But Euler proved that $2^{2^5} + 1$ is not a prime and it is still not known whether there are infinitely many such primes.)

*11.25 Give a construction using only a compass (i.e., without a ruler) to bisect a given line segment.

11.26 Which of the numbers $\sin 15^\circ, \cos 15^\circ, \sin 15^\circ \cos 15^\circ, \sin 15^\circ \cos 75^\circ$ is /are rational? (1998)

11.27 Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a circle of unit radius. Find the product of the lengths of the line segments A_0A_1, A_0A_2 and A_0A_4. (1998)

11.28 A cyclic quadrilateral whose sides are a, b, c, d is such that a circle can also be inscribed in it. Prove that the tangents from the vertices of the quadrilateral to the inscribed circle are of lengths $\frac{ab}{s}, \frac{bc}{s}, \frac{ca}{s}$ and $\frac{da}{s}$, where $2s = a + b + c + d$.

11.29 The diagonals of a quadrilateral inscribed in a circle of radius 65 cm are of lengths 78 cm and 50 cm and inclined to each other at an angle $\sin^{-1}\frac{3}{5}$. Show that the sides of the quadrilateral are $11\sqrt{26}, 5\sqrt{26}, 5\sqrt{26}$ and $19\sqrt{26}$ cms.

11.30 $ABCDEF$ is a regular hexagon. A circle is drawn touching AB and AF and a circle of equal radius touches this circle and the sides CD and DE. A third circle touches this circle and also the side EF. Prove that its radius is $\dfrac{6+5\sqrt{3}}{52}$ times a side of the hexagon.

11.31 A is a vertex of a regular n-gon inscribed in a circle of radius a centred at O. P is the mid-point of OA. Prove that the sum of the fourth powers of the distances of the vertices of the polygon from P is $\dfrac{33n}{16}a^4$.

11.32 The right angle C of a right-angled triangle ABC is divided into n equal parts by lines which meet the hypotenuse AB in points $P_1, P_2, \ldots, P_{n-1}$. Show that $\dfrac{1}{CP_1}+\dfrac{1}{CP_2}+\ldots+\dfrac{1}{CP_{n-1}}=\dfrac{a+b}{2ab}\left(\cot\dfrac{\pi}{4n}-1\right)$, where $BC=a$ and $CA=b$. What happens when $n=2$?

11.33 Suppose l, m, n are real numbers satisfying $lm+mn+nl=1$. Prove that

$$\frac{1}{mn(1+l^2)}+\frac{1}{nl(1+m^2)}+\frac{1}{lm(1+n^2)}=\frac{2}{lmn\sqrt{1+l^2}\sqrt{1+m^2}\sqrt{1+n^2}}.$$

11.34 If the equations $x\cos\alpha+y\cos\beta=c$ and $x^2-2xy\cos\gamma+y^2=c^2$ have a unique solution, then prove that one of the four angles $\alpha+\beta+\gamma$, $\alpha-\beta+\gamma$, $-\alpha+\beta+\gamma$ and $-\alpha-\beta+\gamma$ is an odd multiple of π.

11.35 If P, Q, R are the feet of the perpendiculars drawn from the centroid to the sides of a triangle ABC, prove that the area of the triangle PQR is $\frac{1}{18}(a^2+b^2+c^2)\sin A\sin B\sin C$.

11.36 Let x, y, z be the lengths of the chords of the incircle of a triangle ABC intercepted by the altitudes of ABC. Prove that

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}-\frac{1}{4r^2}=\frac{1}{8rR\cos A\cos B\cos C}$$

11.37 Suppose a, b, c, x, y, z are all rational and $a+b+c=0=x+y+z$. Prove that the area of the triangle with sides $\sqrt{a^2+x^2}$, $\sqrt{b^2+y^2}$ and $\sqrt{c^2+z^2}$ is rational and equals $\frac{1}{2}(-bcx^2-cay^2-abz^2)^{1/2}$.

11.38 Let P be a point on the arc BC of the circumcircle of an equilateral triangle ABC. Prove that $AP=BP+PC$. Can the problem be done with complex numbers (e.g. by taking A, B, C as $1, \omega, \omega^2$ respectively)?

11.39 Prove the following inequalities using trigonometric substitutions.

(i) $\sqrt{abc}+\sqrt{(1-a)(1-b)(1-c)}<1$ for all $a, b, c \in (0,1)$

(ii) $\sqrt{ab}+\sqrt{cd}\le\sqrt{(a+d)(b+c)}$ for all positive a, b, c, d

(iii) $2|(x+y)(1-xy)|\le(1+x^2)(1+y^2)$ for all real x, y.

Chapter 12

HEIGHTS AND DISTANCES

How do you find the width of a river without actually crossing it and how do you measure the height of a tree without climbing it? These are among the most fascinating applications of geometry and trigonometry as far as the layman is concerned. Although the mathematics needed in these problems is relatively elementary, such problems of 'heights and distances' have their own charm. Their very wording is a welcome relief from the maze of formulas and expressions which predominate most other problems. After solving a problem of this type one gets the feeling that one has done something 'really useful' which you do not get after showing that a certain determinant vanishes even though the latter may be far more challenging and the solution far more elegant.

The most vital thing in attacking the problems of heights and distances is to correctly understand and visualise the data. A good diagram is half the solution. Assigning suitable variables to the quantities involved, translating the data into a system of equations in these unknowns and finally solving this system are relatively routine affairs. However, as the problem in Comment No. 4 shows, sometimes a few facts from geometry can drastically simplify the solution while Comment No. 10 gives an example of a problem where some geometry is needed even to formulate the problem. Comment No. 11 and 12 deal with problems involving the motion of an object under gravity.

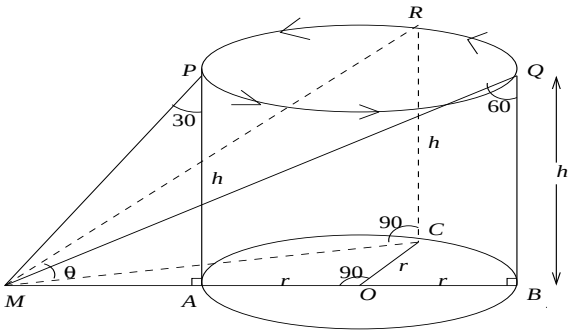
Problems about heights and distances are not asked at JEE every year possibly because they tend to be somewhat repetitious. It is not easy to come up with a novel and yet simple problem. But when asked, the problems of heights and distances are among the relatively straightforward problems.

Main Problem : A bird flies in a circle on a horizontal plane. An observer stands at a point on the ground. Suppose 60° and 30° are the maximum and the minimum angles of elevation of the bird and they occur when the bird is at points P and Q respectively on its path. Let θ be the angle of elevation of the bird when it is at a point on the arc of the circle exactly mid-way between P and Q. Then the numerical value of $\tan^2\theta$ is ..... . (JEE 1998)
(Assume that the observer is not inside the vertical projection of the path of the bird.)

First Hint: Let A and B be the vertical projections of the points P and Q on the ground. Show that the point, say M, where the observer is standing must be on the line AB.

Second Hint: Let R be a point on the path of the bird exactly mid-way between P and Q and C its vertical projection on the ground. Let O be the centre of the vertical projection of the path of the bird. Prove that $\angle MOC$ is a right angle.

Solution: Let h be the height above ground of the plane of the flight of the bird. The vertical projection of the path of the bird is a circle (as the path is a circle in a parallel plane). Let O be the centre of this circle and let A, B, C, M be the points as described in the hints. Then $PA = QB = RC = h$. From the data, A and B must be the closest and the farthest points of the projected circle from M. So AB is a diameter of the projected circle and M lies on it outside the circle (as given in the last part of the problem). It is given further that



$\frac{h}{AM} = \tan 60° = \sqrt{3}$ and $\frac{h}{BM} = \tan 30° = \frac{1}{\sqrt{3}}$. Letting r be the radius of the projected circle, we have $h = \sqrt{3}(OM - r) = \frac{1}{\sqrt{3}}(OM + r)$.

Solving simultaneously, $OM = \dfrac{h}{2}(\sqrt{3} + \dfrac{1}{\sqrt{3}}) = \dfrac{2h}{\sqrt{3}}$ and $r = \dfrac{h}{2}(\sqrt{3} - \dfrac{1}{\sqrt{3}}) = \dfrac{h}{\sqrt{3}}$.
Now as R is mid-way between P and Q in the original path, C is mid-way between A and B in the projected path. Hence OC is perpendicular to the diameter AB. So $\tan^2\theta = \dfrac{h^2}{CM^2} = \dfrac{h^2}{OM^2 + r^2} = \dfrac{3h^2}{h^2(4+1)} = \dfrac{3}{5}$. (The calculations can be simplified a little by observing that the answer to the problem depends

only on the relative proportions of the various distances and heights. So we are free to choose the units. A convenient choice is to take $AM = 1$. Then h comes out as $\tan 60° = \sqrt{3}$. As a result, $MB = h \cot 30° = \sqrt{3} \times \sqrt{3} = 3$. This gives $AB = MB - MA = 3 - 1 = 2$. So $OM = 2$ and hence $OC = \sqrt{1+4} = \sqrt{5}$. This gives $\tan\theta = \dfrac{h}{OC} = \dfrac{\sqrt{3}}{\sqrt{5}}$.)

Comment No. 1:

Drawing a reasonably good diagram is a must in all problems of heights and distances. Often the problem involves visualisation of three-dimensional data. A correct diagram not only tells the examiner that the candidate has correctly understood the problem, but also makes him somewhat less demanding in justifications which would normally be required. For example, in the solution above, it was claimed that AB is a diameter of the projected circle and further that the point M lies on the line AB. If the problem were a problem of geometry, this would require some proof. (One such proof is by letting the line MO cut the circle at points A' and B' with A' lying between M and B'. Then all points of the circle lie between the two parallel tangents, one at A' and the other at B'. Hence A' and B' are the points of the circle which are, respectively, closest to and farthest from M. So A' must coincide with A and B' with B.) But in a problem of heights and distances, geometry is only a tool and so such elaborate justifications are hardly expected. The diagram makes it all clear.

It is not, however, necessary that the diagram be artistic or drawn to scale. (It is, for example, unnecessary, that an angle whose measure is 30 degrees be shown as such.) In any case, this is not always possible when a two-dimensional diagram is drawn to illustrate a three dimensional situation. The diagram should reveal your understanding of the problem and be an aid to understanding the solution you give. Labelling the diagram is very important. Even if it is obvious sometimes which point is P and which is Q, do not force the examiner to figure this out and thereby invite his displeasure.

Comment No. 2:

Problems of heights and distances are real life problems and so certain common sense assumptions can be made unless inconsistent with the data of the problem. For example, the earth is assumed to be planar which is reasonable because in the given problem a very small portion of the surface of the earth is involved. If the problem were from astronomy involving, say, the earth, the sun and the moon, then one would have to assume these bodies as spheres which could degenerate into points if the distances involved were very large as compared to the radii of these heavenly bodies. Similarly, in the problem above, nothing is given about the height of the observer and so it is taken to be zero. (Actually, in the present problem it really does not matter, since we could then take the vertical projection of the path of the bird on the horizontal plane passing through the observer instead of taking the vertical projection on the ground.) In those problems where the observer is given to be, say, a man

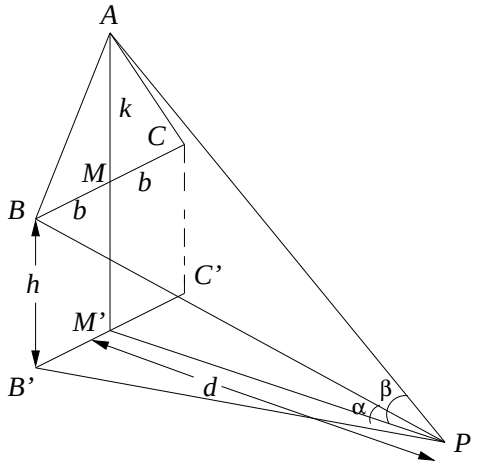
of height 180 cm, it is to be assumed that 180 cm is not the full height of the man but the height of his eyes above the ground! (And, for those who love hair splitting, by an 'eye' here we mean the centre of the 'eye'!) When the problem involves towers, poles or trees you may safely assume that they are vertical unless the contrary is specified (for example, if the problem deals with the tower of Pisa!). By the way, towers rarely appear anywhere else in mathematics, just as urns rarely appear anywhere besides probability problems!

Comment No. 3:

Instead of specifying heights of vertical objects (such as towers, trees or poles) in terms of the angles of elevation of their tops, as seen from a point on the ground, one can specify the angle subtended at the point on the ground by these objects (also called, less accurately, the angle subtended by their tops). This is the language used in the following problem.

A signpost in the form of an isosceles triangle is mounted on a pole of height h fixed to the ground. The base BC of the triangle is parallel to the ground. A man standing on the ground at a distance d from the signpost finds that the top vertex A of the triangle subtends an angle β and either of the other two vertices subtends the same angle α at his feet. Find the area of the triangle. (JEE 1988)

Once again, a good diagram is half the solution. Let M be the mid-point of BC and let B', C' and M' be the vertical projections of B, C and M respectively to the ground. Then M' is the mid-point of $B'C'$ and also the foot of the pole. Let P be the point on the ground where the man is standing. The fact that B and C subtend equal angles at P implies that PM' is at right angles to $B'C'$. (A justification of this may be omitted. In fact, even an explicit verbal statement of this fact is unnecessary if the diagram makes it abundantly clear. But it is better to give it in order to be on the safer side.) So $d = PM'$ and $h = BB' = MM' = CC'$. Let $AM = k$ and $BC = 2b$. Then $BM = B'M' = b$ and so $B'P = \sqrt{b^2 + d^2}$.



The area of the triangle is kb and so we would be through if we can find the values of k and b (in terms of the given parameters, viz., d, h, α and β). For this we need a system of two equations in k and b. Such a system is easily obtained from the fact that AM' and BB' subtend angles β and α respectively at the point P. Thus we have $\tan\alpha = \dfrac{BB'}{B'P} = \dfrac{h}{\sqrt{b^2 + d^2}}$ and $\tan\beta = \dfrac{AM'}{M'P} = \dfrac{h + k}{d}$,

which yield $k = d\tan\beta - h$ and $d^2 + b^2 = h^2\cot^2\alpha$ or $b = \sqrt{h^2\cot^2\alpha - d^2}$. As said before, the area of the triangle ABC equals kb i.e., $(d\tan\beta - h)\sqrt{h^2\cot^2\alpha - d^2}$.

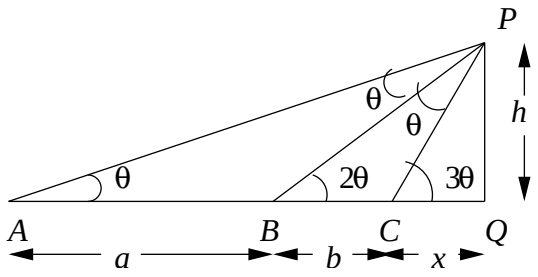
Comment No. 4:

Both the problems above involved the angles of elevation of different points in space from a common point on the ground. There are also problems which involve the angles of elevation of the same point as seen from different points on the ground. The simplest case is the one where these points and the point in space lie in the same vertical plane. This requires that the points on the ground be collinear, as in the following problem.

A balloon is observed simultaneously from three points A, B and C on a straight road directly beneath it. The angular elevation at B is twice that at A and the angular elevation at C is thrice that at A. If the distance between A and B is a and the distance between B and C is b, find the height of the balloon in terms of a and b. (JEE 1979)

Let P be the point where the balloon is located and Q the projection of P onto the ground. As the road is directly beneath P, the points P, A, B, C and Q all lie in the same vertical plane, with A, B, C and Q lying on a horizontal line.

Let h be the height of the balloon, θ be the angle of elevation of the balloon as seen from A. Then the angles of elevation from B and C are 2θ and 3θ respectively. Let x be the distance between C and Q.



We have three equations from the data, viz.,

$$h = (a + b + x)\tan\theta \tag{1}$$

$$h = (b + x)\tan 2\theta \tag{2}$$

$$h = x\tan 3\theta \tag{3}$$

By eliminating θ and x from this system we shall get an equation involving a, b and h, which can then be solved to express h in terms of a and b. Thus the problem is now reduced purely to a problem in algebra. One way to do it would be to express $\tan 2\theta$ and $\tan 3\theta$ in terms of $\tan\theta$ (for which standard formulas are available) and then to substitute $\tan\theta = \dfrac{h}{a + b + x}$ from (1). This will eliminate θ and will give us two equations each expressing h in terms of a, b

and x. Specifically,

$$h = (b+x)\frac{2\dfrac{h}{a+b+x}}{1-(\dfrac{h}{a+b+x})^2} \text{ and } h = x\frac{3\dfrac{h}{a+b+x}-(\dfrac{h}{a+b+x})^3}{1-3(\dfrac{h}{a+b+x})^2}$$

Obviously, eliminating x from this system will be cumbersome. So instead of following the mechanical approach of converting the data to a system of equations, it is a good idea to look for some geometric short cuts. One such shortcut is to observe that in the triangle PAB, $\angle APB = 2\theta - \theta = \theta$. So the triangle is isosceles, giving $PB = AB = a$. So, from triangle PBQ, we have $h = a\sin 2\theta$. Thus we have already expressed h in terms of a and θ. To express it in terms of a and b we need an equation involving a, b and θ. For this, observe that $\angle BPC = 3\theta - 2\theta = \theta$. So, applying the sine rule to the triangle PBC we have $\dfrac{b}{\sin\theta} = \dfrac{a}{\sin(\pi-3\theta)} = \dfrac{a}{\sin 3\theta}$. Writing $\sin 3\theta$ as $3\sin\theta - 4\sin^3\theta$ and cancelling $\sin\theta$ (which is obviously non-zero) we get $a = b(3-4\sin^2\theta) = b(3-2(1-\cos 2\theta)) = b(1+2\cos 2\theta)$. This gives the value of $\cos 2\theta$ as $\dfrac{a-b}{2b}$ and hence of $\sin 2\theta$ as $\dfrac{\sqrt{4b^2-(a-b)^2}}{2b}$ i.e., as $\dfrac{\sqrt{3b^2-a^2+2ab}}{2b}$. So, finally, $h = a\sin 2\theta = \dfrac{a\sqrt{3b^2-a^2+2ab}}{2b}$.

Note the advantage of the geometric approach over the algebraic approach. In the algebraic approach we introduced an auxiliary variable x (besides θ) and then tried to eliminate it which proved to be cumbersome. In the geometric approach, we bypassed x and got h directly in terms of a and θ. It may be argued that this worked because the angles of elevations at A, B, C were related in a particularly simple manner. But then, the ability to spot and apply such simplifying features in a given problem is an asset and sometimes problems are designed precisely to test this ability.

A special case of this problem is the following problem.

A man observes a tower of height h from a point A on the ground. He moves a distance d towards the foot of the tower and finds that the angle of elevation has doubled. He moves further a distance of $\frac{3}{4}d$ in the same direction and finds that the angle of elevation is three times that at A. Prove that $36h^2 = 35d^2$. (JEE 1986)

Evidently, this is a special case of the last problem with $a = d$ and $b = \frac{3}{4}d$. Substituting these values in the answer to the last problem, we get $h = \dfrac{d\sqrt{3(\frac{3}{4})^2 - 1 + 2\times\frac{3}{4}}}{2\times\frac{3}{4}}$. Upon squaring and simplifying we get $36h^2 = 35d^2$ as desired. But the data of the problem is unrealistic. In solving the last problem we saw that $h = a\sin 2\theta$. In the present problem this means $\dfrac{\sqrt{35}}{6}d = d\sin 2\theta$. This gives $\theta = \frac{1}{2}\sin^{-1}\dfrac{\sqrt{35}}{6}$ which is a little above $40°$. But that would mean

that the angle of elevation at C is more than 120 degrees! In other words, in going from B to C, the man has passed Q, the foot of the tower, and then after reaching C he has looked at the tower without turning back but instead bending his neck backwards! Mistakes like this can happen when a special case of a problem is designed by indiscriminately assigning special values to some of the parameters.

Comment No. 5:

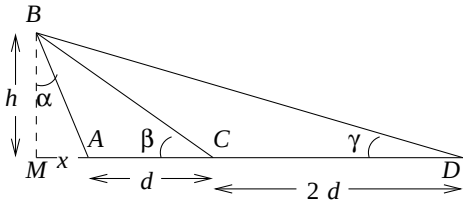
The data in the following problem (JEE 1994) is of a similar spirit as the last two problems. However, one of the angles of elevation has been replaced by making the tower lean!

A tower AB leans towards west making an angle α with the vertical. The angular elevation of B, the topmost point of the tower, is β as observed from a point C due east of A at a distance d from A. If the angular elevation of B from a point D due east of C at a distance $2d$ from C is γ, prove that

$$2\tan\alpha = 3\cot\beta - \cot\gamma \tag{4}$$

The diagram is very similar to the one above except that since it is customary to show east as pointing to the right rather than to the left, we take the mirror image and change the labels of various points and angles. However, in the last two problems we had to find the height of the tower using a certain given relationship among the various angles of elevation. In the present problem, on the other hand, we have to establish a certain relationship among the various angles of elevation.

Let h be the vertical height of the tower. (This is not the same as its length as the tower is leaning.) Let M be the vertical projection of B, the top of the tower, on the ground and let x denote the distance between M and A, the foot of the tower. We then have a system of three equations, viz.,



$$x = h\tan\alpha, \quad x + d = h\cot\beta \ \text{ and } \ x + 3d = h\cot\gamma \tag{5}$$

This is a system of three equations and the desired relationship (4) can be obtained if we can eliminate h, d and x from it. Normally, we can eliminate only two quantities from three equations. But the present system is homogeneous and so three quantities can be eliminated. The elimination itself is extremely easy and left as an exercise.

When this problem was asked in the 1994 JEE, it stated that the point C was due west of A (instead of due east). Naturally, this changes the problem drastically and then (4) can no longer hold. (Interestingly, the Hindi version did say that C was due east of A.) The JEE question papers are set and printed

with great care and so mistakes like this are very rare. But, needless to say, when they do occur, such mistakes are very confusing to the candidates as they are forced to check and recheck their work to see if *they* have committed any mistake. The practical advice given at the end of Comment No. 14 in Chapter 10 applies here as well. If you are sure of yourself, do not spend much time on one particular problem. If you think there is a mistake in it, say so.

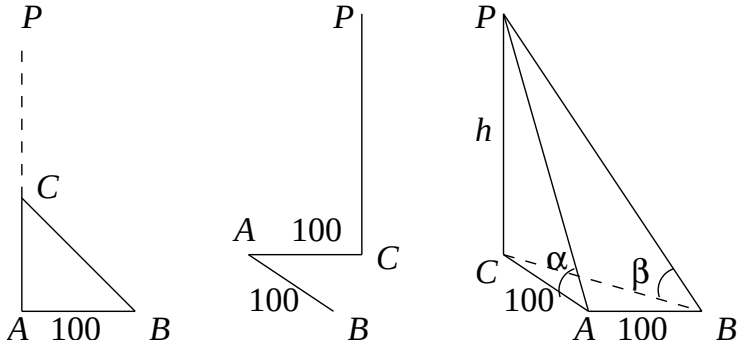
Comment No. 6:

Let us now see some problems where the point observed does not lie in the same vertical plane as the points from which it is observed. In such problems the data is three-dimensional and not planar (as was also the case with the Main Problem and the problem about the signpost in Comment No. 3 above). The diagrams have to be drawn on a plane, viz., the plane of the paper. So it takes a certain skill to draw a revealing diagram. This skill does *not* require artistic ability. If you have correctly visualised the data and know a few standard conventions, then with a little practice you can easily master it.

As a simple problem of this type we have the following.

A tower is observed from two stations A and B, where B is East of A at a distance of 100 meters. The tower is due North of A and due Northwest of B. The angles of elevations of the tower from A and B are complementary. Find the height of the tower. (JEE 1978)

Let P and C denote, respectively, the top and the foot of the tower. From the data, CAB is an isosceles, right-angled triangle. But if we draw it as such as we are in trouble. Let us see why. It is a universally accepted convention that in a map East points horizontally forward and North points vertically upwards. So the triangle CAB will have to be drawn as in Figure (a) below.



(a) (b) (c)

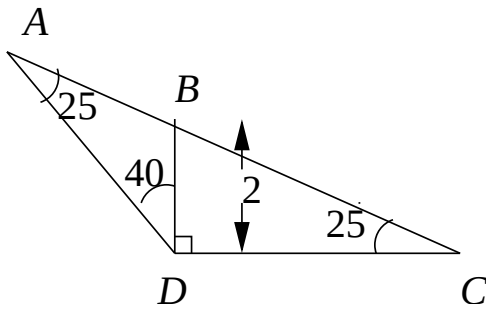
But then since the tower itself is vertical, it is also a standard practice to show it by a vertical line. This would look very confusing because the points A, C and P would appear collinear which is not the case. So somewhere we have to make a compromise. A standard compromise is to retain the horizontality of East and the verticality of the tower but draw North by a slanting line. So the triangle CAB can be shown either as an acute-angled triangle as in (b)

or as an obtuse-angled triangle as in (c). The latter is preferable because it clearly shows that the side BC is the longest side of the triangle ABC and in the present problem the lengths of the sides are relevant.

Having the diagram as in (c), the problem itself is easy. As usual, let h denote the height of the tower and let α, β be the angles of elevation from A and B respectively. We are given that $AB = 100$. Hence $AC = 100$ and $BC = 100\sqrt{2}$. Further, α and β are given to be complementary. So we have $\frac{h}{100} = \tan\alpha = \cot\beta = \frac{100\sqrt{2}}{h}$ which gives $h^2 = (100)^2\sqrt{2}$ and hence $h = 100(2)^{1/4}$ meters. This is approximately 118.9207 meters. But such a reduction is generally not expected, except when either calculating devices such as tables of logarithms are allowed or when some numerical values are given specifically as in the following problem. (JEE 1983)

Four ships A, B, C and D are at sea in the following relative position: B is on the straight line segment from A to C, B is due North of D and D is due West of C. The distance between B and D is 2 km, $\angle BDA = 40°, \angle BCD = 25°$. What is the distance between A and D? [Take $\cot 25°$ as 2.1445.]

This problem shows that not every problem in heights and distances must involve a tower or a pole! This is purely a problem in a plane (viz., the plane of the sea) and once the data is translated as in the figure, it is a problem about solving a triangle. The data implies that ABD is a triangle in which $BD = 2$, $\angle ABD = 90° + 25° = 115°$ and $\angle DAB = 180° - (115° + 40°) = 25°$.

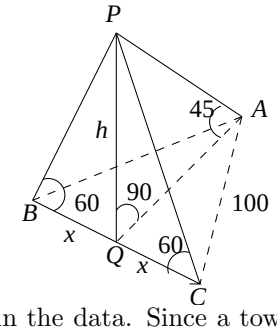


We have to find the length of the side AD. One way to do so is to apply the sine rule to the triangle ABD to get $\frac{AD}{\sin 115°} = \frac{BD}{\sin 25°}$ and hence $AD = 2\cot 25°$. (In the present problem, there is an easier way to see this since the triangle ADC comes out to be isosceles.) Using the given numerical value of $\cot 25°$, we get $AD = 4.289$ km. (In the original JEE 1983 problem, instead of $\cot 25°$, the value of $\sin 25°$ was given as 0.423. The value of $\cot 25°$ can, of course, be found from this as $\frac{\sqrt{1-(0.423)^2}}{0.423}$. But computing this without log tables or calculators (which were rare in 1983) is laborious. It is hard to say whether specifying $\sin 25°$ instead of $\cot 25°$ was intentional or a mistake.)

Comment No. 7:

Here is a problem where the height of a tower is to be found from its angles of elevation at three non-collinear points.

ABC is a triangular park with $AB = AC = 100$ m. A television tower stands at the mid-point of BC. The angles of elevation of the top of the tower at A, B, C are $45°, 60°, 60°$ respectively. Find the height of the tower. (JEE 1989)



There is some redundancy in the data. Since a tower is to be assumed to be vertical unless stated otherwise, the fact that it is equidistant from B and C automatically implies that the angles of elevation at B and C are equal. So it would have sufficed to specify only one of them. Let P and Q be the top and the foot of the tower. Then Q is the mid-point of the side BC. Let h and $2x$ denote, respectively, the height of the tower and the length of the side BC (in meters). Then we have a system of two equations in two unknowns, h and x, viz.,

$$h = \sqrt{(100)^2 - x^2}\,\tan 45° = \sqrt{10000 - x^2} \text{ and } h = x \tan 60° = \sqrt{3}\, x.$$

Putting $x = \frac{h}{\sqrt{3}}$ into the first equation and squaring we get $h^2 = 10000 - \frac{h^2}{3}$ giving $h^2 = 7500$ and hence, finally, $h = 50\sqrt{3}$ meters.

Comment No. 8:

By now, the general strategy in solving problems of heights and distances should be clear. Even though the problem asks to find the value of only one quantity, such as the height of the tower, we introduce one or more auxiliary variables so that the data can be translated into a system of equations involving the variable of our interest and the auxiliary variables. The choice of the auxiliary variables can often be made in more than one ways. In the present problem, for example, we could have taken the vertical angle A as a variable instead of the length of the side BC. Sometimes a judicious choice of the auxiliary variables simplifies the calculations. In the present problem, for example, we have used x to denote half the length rather than the full length of the side BC because it is this half length which comes into the picture. Had we let x be the length of the side BC, we would have to deal with $\frac{x}{2}$ instead of x. Mathematically, this makes little difference. But when speed is crucial, even a small simplification can reduce your chances of a numerical slip.

However, it is not a good idea to introduce too many auxiliary variables since this makes the resulting system of equations complicated. Sometimes considerations of symmetry can drastically reduce the number of auxiliary variables. Here is a good illustration. (JEE 1980)

PQ is a vertical tower, with P as its foot and Q as its top. A, B, C are three points in the horizontal plane through P. The angles of elevation of Q from A, B, C are equal and each is equal to θ. The sides of the triangle are

a, b, c and the area of the triangle is Δ. Show that the height of the tower is

$$\frac{abc\tan\theta}{4\Delta}.$$

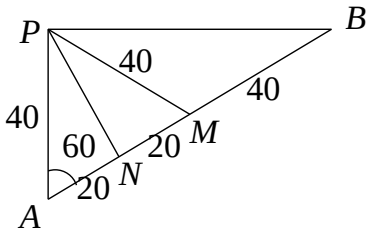
We skip the diagram since it is similar to that in the last problem and also because by now you have seen a fair number of diagrams to be able to draw one on your own. Let h be the height of the tower. In order to make use of the angles of elevations at A, B, C we would need three auxiliary variables denoting their distances from P, the foot of the tower. But this is unnecessary. The fact that all these angles of elevation are equal implies that A, B, C are equidistant from P. Put differently, this means that P is the circumcentre of the triangle ABC. Further, if R denotes its circumradius then we get $\tan\theta = \frac{h}{R}$. So the problem is now reduced to showing that $abc = 4R\Delta$, which follows easily by noting, for example, that $c = 2R\sin C$ and $ab\sin C = 2\Delta$.

Comment No. 9:

The last problem was essentially a problem about the triangle ABC. When the common angle of elevation of the tower from the three vertices A, B, C is known, finding its height is as good as finding the circumradius of the triangle ABC. And the problem essentially asks you to express the circumradius in terms of the sides and the area of the triangle. Here is another problem of a similar spirit.

A vertical tower stands at a point P. Points A and B are located to the South and East of P respectively. M is the mid-point of AB, PAM is an equilateral triangle and N is the foot of the perpendicular from P on AB. Let AN be 20 meters and the angle of elevation of the top of the tower at N be $\tan^{-1}(2)$. Determine the height of the tower and the angles of elevation of the top of the tower at A and B. (JEE 1990)

This problem, too, is more about the right-angled triangle APB. Since the angle of elevation of the top of the tower at the point N is given to be $\tan^{-1}(2)$, the height, say h, of the tower is twice the distance PN. So the first part of the problem really reduces to finding the length of PN which is given to be an altitude of the triangle APB. So we concentrate more on this triangle and suppress the tower from the figure so that the figure now represents a planar object and hence can be drawn more realistically.



Since the triangle PAM is equilateral, $\angle PAM = 60°$. Further, M is the mid-point of the hypotenuse AB. So $PA = PM = AM = MB$. The perpendicular from P to AB then is also a median of the equilateral triangle

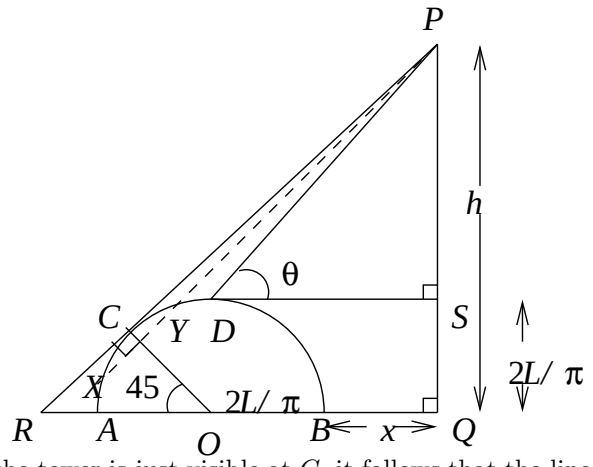
PAM. Hence $AN = \frac{1}{2}AM$. Thus we get $PA = AM = 2AN = 40$ and hence $PN = \sqrt{(40)^2 - (20)^2} = 20\sqrt{3}$. As noted before this means that the height h of the tower is $40\sqrt{3}$ meters. Now to know the angles of elevations at A and B we merely have to know the distances PA and PB. The former we already know as 40. The latter is then $40 \tan 60^\circ = 40\sqrt{3}$. So the angle of elevation of the top of the tower at A is $\tan^{-1}(\frac{h}{PA}) = \tan^{-1}(\frac{40\sqrt{3}}{40}) = 60^\circ$. Similarly, the angle of elevation at B equals $\tan^{-1}(\frac{h}{PB}) = \tan^{-1}(\frac{40\sqrt{3}}{40\sqrt{3}}) = 45^\circ$.

Comment No. 10:

In all the problems so far, the location of the point from which an object such as a tower is viewed was given. Sometimes, we have to first determine the point from which the object is visible and then consider the angle of elevation. The following problem is of this type.

A semicircular arch AB of length $2L$ and a vertical tower PQ are situated in the same vertical plane. The feet A and B of the arch and the base Q of the tower are at the same horizontal level, with B between A and Q. A man at A finds the tower hidden from his view due to the arch. He starts crawling up the arch and just sees the topmost point P of the tower after covering a distance $\frac{L}{2}$ along the arch. He crawls further to the topmost point of the arch and notes the angle of elevation of P to be θ. Compute the height of the tower in terms of L and θ. (JEE 1997*)

From the data, the radius of the arch is $\frac{2L}{\pi}$. Let O be its centre, D its topmost point and C the point on it from which the top of the tower becomes visible. Since the arc length AC is given to be $\frac{L}{2}$, $\angle AOC = 45^\circ$. Also $\angle AOD = 90^\circ$. That the tower is not visible from any point, say X, on the portion of the arch from A to C means that the line joining X to P meets the arch again at a point Y above X.



Since the top of the tower is just visible at C, it follows that the line PC touches the arch at C. Let this line cut the line AB at R. The tangent to the arch at D is horizontal. Let it meet the tower at S. Let h be the height of the tower. To express h in terms of L and θ, we need to introduce some auxiliary variables. We already know that OA, OB, OC and OD equal $\dfrac{2L}{\pi}$ each. Hence SQ also equals $\dfrac{2L}{\pi}$. We let BQ be x. Then the fact that the angle of elevation

of the top as seen from D is θ gives us

$$\tan\theta = \frac{PS}{DS} = \frac{PS}{OQ} = \frac{PQ - SQ}{OB + BQ} = \frac{h - \frac{2L}{\pi}}{x + \frac{2L}{\pi}} \tag{6}$$

To get rid of x, we need one more equation. As RCO is a right angled triangle and $\angle ROC = 45°$, we see that $\angle CRO$ also equals $45°$. This gives $OR = \sqrt{2}OC = \frac{2\sqrt{2}L}{\pi}$. Further, RQP is also a right-angled, isosceles triangle and hence $RQ = PQ$. In terms of x and h this means

$$h = PQ = RQ = RO + OB + BQ = \frac{2\sqrt{2}L}{\pi} + \frac{2L}{\pi} + x \tag{7}$$

From (7), $x + \frac{2L}{\pi} = h - \frac{2\sqrt{2}L}{\pi}$. Putting this into (6) and solving for h we get the height of the tower as $h = \frac{2L}{\pi}\frac{\sqrt{2}\tan\theta - 1}{\tan\theta - 1}$.

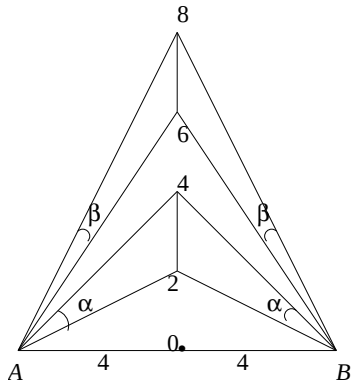
Comment No. 11:

The Main Problem dealt with a moving object. However, the location of the various points in its path was specified directly. In some problems, the motion is specified by some equation, called the **equation of motion** of the object. Typically, this is a differential equation involving the velocity and/or the acceleration of the object and we have to solve it first to determine the position of the relevant points in the path of the object. Consequently, such problems belong more properly to solutions of differential equations. However, when the differential equation is relatively easy to solve, it is the trigonometrical part which is more dominating. Here is an example of such a problem.

A 2 meter long object is fired vertically upwards from the mid-point of two locations A and B, 8 meters apart. The speed of the object after t seconds is given by $\frac{ds}{dt} = (2t+1)$ meters per second. Let α and β be the angles subtended by the object at A and B, respectively, after one and two seconds. Find the value of $\cos(\alpha - \beta)$. (JEE 1987)

It is not clear from the problem whether the object itself is vertical in its motion. We assume this to be the case. Also the problem does not explicitly say that A and B are in the same horizontal plane. This has to be inferred from the fact that the angles subtended by the object at A and B are equal, not just once but twice during the motion. (A rigorous justification of this claim will require some geometric argument which can be skipped in a problem of heights and distances as noted above.) Because of these assumptions, when the

bottom of the object is at a height h above A and B, its top is at a height $h+2$ (all in meters). The general solution of the equation $\frac{ds}{dt} = 2t+1$ is $s = t^2+t+c$, where the constant c is determined by the position at $t = 0$. For the point at the bottom of the object, $s = 0$ when $t = 0$ and so $c = 0$. Thus at $t = 1$, the bottom and the top of the object are at heights 2 and 4 meters above the ground respectively, while at $t = 2$, they are at heights 6 and 8 respectively.



From here onwards, the problem is purely a trigonometric one. From the data, we have

$$\alpha = \tan^{-1}\frac{4}{4} - \tan^{-1}\frac{2}{4} = \frac{\pi}{4} - \tan^{-1}\frac{1}{2} \tag{8}$$

$$\text{and} \quad \beta = \tan^{-1}\frac{8}{4} - \tan^{-1}\frac{6}{4} = \tan^{-1}2 - \tan^{-1}\frac{3}{2} \tag{9}$$

We are asked to find $\cos(\alpha-\beta)$. This can be done by computing $\cos\alpha$, $\cos\beta$, $\sin\alpha$ and $\sin\beta$ from (8) and (9). But that would be cumbersome. A better idea is to use (8) and (9) to first calculate $\tan\alpha$ and $\tan\beta$ and thereby to find $\tan(\alpha-\beta)$ which would eventually give $\cos(\alpha-\beta)$. From (8) and (9) we immediately get $\tan\alpha = \dfrac{1-\frac{1}{2}}{1+1\times\frac{1}{2}} = \dfrac{1}{3}$ and $\tan\beta = \dfrac{2-\frac{3}{2}}{1+2\times\frac{3}{2}} = \dfrac{1}{8}$. Hence $\tan(\alpha-\beta) = \dfrac{\frac{1}{3}-\frac{1}{8}}{1+\frac{1}{3}\times\frac{1}{8}} = \dfrac{1}{5}$. So, $\cos(\alpha-\beta) = \dfrac{5}{\sqrt{25+1}} = \dfrac{5}{\sqrt{26}}$.

Comment No. 12:

One of the simplest and yet important examples of motion is the motion of a projectile. This is studied in physics. But the treatment is purely mathematical and classically forms a part of a branch of mathematics called dynamics. In most of the problems about projectiles all we need is that if a particle is projected at an angle α with the horizontal with an initial speed u, then at time t, the vertical and the horizontal distances it travels from the starting point are given respectively by

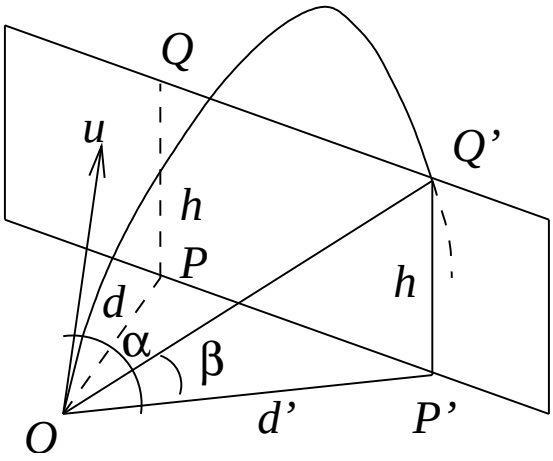
$$x = x(t) = u\cos\alpha\, t \quad \text{and} \quad y = y(t) = u\sin\alpha\, t - \frac{1}{2}gt^2 \tag{10}$$

where g is the acceleration due to gravity. As a typical application of these formulas we do the following problem whose solution will also require the maximisation of a trigonometric function.

A man is standing on the ground at a distance d from a vertical wall of height h. He wants to fire a shot that will go over the wall. Find the minimum

speed with which such a shot must be fired and the direction in which it must be fired with this minimum speed.

It is intuitively clear that for the minimum possible speed, the shot will just touch the wall at its top and go over it (or 'graze' it as it is sometimes described). For this, the shot must be fired in a plane perpendicular to the wall. If it is fired (with the same speed and the same initial angle with the horizontal) in some other vertical plane then it will take longer to reach the plane of the wall and by that time the shot would descend further down vertically so that it will hit the wall and not go over it. Clear as this point is, we do not have to *assume* it in our solution. Our reasoning will, in fact, prove it rigorously. We assume that the wall is of negligible width. If not, we add its width to d, the perpendicular distance between the man and the wall. (In effect, this means that we are measuring the distance of the man from the backside of the wall. We do so to avoid the possibility where the shot clears the front side of the wall but lands on the top of the wall and hence fails to go over it. Similarly, if the man's height is to be taken into account, we subtract it from h, the height of the wall, because the height that really matters in the problem is not the height of the top of the wall above the ground, but the height above the point from where the shot is to be fired.) Let O be the point at which the man is standing, P be the foot of the perpendicular from O to the wall and Q the point on the top of the wall directly above P. Then OPQ is a right-angled triangle with $OP = d$ and $PQ = h$.



Now suppose that a shot is fired from O, not necessarily in the vertical plane containing O, P and Q but possibly in some other vertical plane which meets the wall in a vertical segment $P'Q'$ with P' on the ground as shown in the figure above. Let $d' = OP'$. Then $OP'Q'$ is also a right angled triangle with $P'Q' = PQ = h$. However $d' > d$. We shall show later that if the shot is fired with the minimum possible speed, then P' must coincide with P and hence $d' = d$. Let u be the initial speed of the shot and α be the angle its initial

direction makes with OP'. Then the motion of the shot is in the vertical plane containing O, P' and Q' as shown in the figure. Call $\angle Q'OP'$ as β.

From (10) above, the shot will reach the plane of the wall when $t = \dfrac{d'}{u\cos\alpha}$. At this time the shot will be at a height $u\sin\alpha\dfrac{d'}{u\cos\alpha} - \frac{1}{2}g(\dfrac{d'}{u\cos\alpha})^2$ above the ground. The shot will clear the wall if and only if this height is at least h, i.e., if and only if

$$h \leq d'\tan\alpha - \frac{1}{2}g\frac{d'^2}{u^2\cos^2\alpha} \tag{11}$$

The rest of the solution is going to be a series of manipulations with this inequality. Multiplying both its sides by $2u^2\cos^2\alpha$, we can rewrite it as

$$gd'^2 \leq 2u^2(d'\sin\alpha\cos\alpha - h\cos^2\alpha) \tag{12}$$

For a fixed d', the minimum possible value of u which will satisfy (12) will occur when the factor $2(d'\sin\alpha\cos\alpha - h\cos^2\alpha)$ is maximum. So now the problem reduces to finding the maximum value of a trigonometric expression. As mentioned in Chapter 10, Comment No. 11, this can often be done without using the methods of calculus by rewriting the expression as the sine or cosine of some angle. In the present case, $2(d'\sin\alpha\cos\alpha - h\cos^2\alpha)$ can be rewritten as $d'\sin 2\alpha - h\cos 2\alpha - h$. Since $\sin\beta = \dfrac{h}{\sqrt{h^2+d'^2}}$ and $\cos\beta = \dfrac{d'}{\sqrt{h^2+d'^2}}$, the coefficient of u^2 in the R.H.S. of (12) further reduces to $\sqrt{h^2+d'^2}(\sin(2\alpha-\beta) - \sin\beta)$. (Note, incidentally, that if $2\alpha - \beta < \beta$, then this coefficient is negative and so (12) can never hold since its L.H.S. is positive. In other words, for the shot to clear the wall we must have $\alpha \geq \beta$. This is, of course, obvious by sheer common sense too. But it is not as strong as we need.)

Thus we see that for a fixed plane of motion (and hence for a fixed d' and a fixed β), the minimum speed with which the shot must be fired to go over the wall will occur when $2\alpha - \beta = \frac{\pi}{2}$, i.e., when $\alpha = \dfrac{\pi}{4} + \dfrac{\beta}{2}$. Moreover, for this value of α, equality holds in (12). The resulting equation can be rewritten as $gd'^2 = u^2\sqrt{h^2+d'^2}(1-\sin\beta)$. As $\sin\beta = \dfrac{h}{\sqrt{h^2+d'^2}}$, this further simplifies to

$$u^2 = g(h + \sqrt{h^2+d'^2}) \tag{13}$$

This equation gives the minimum value of the initial speed with which the shot must be fired in a vertical plane, viz., the vertical plane containing O and P'. As this plane varies, so does d' and hence the minimum speed of the shot also varies. As g and h are fixed, it is clear from (13) that u will be minimised when d' is minimised. But as we already know, the minimum value of d' is d, the perpendicular distance between the man and the wall. This gives a mathematical foundation to our intuitive prediction that for the minimum possible initial speed of the shot to clear the wall, the shot must be fired in the vertical plane containing O and P. This minimum speed is $\sqrt{g(h+\sqrt{h^2+d^2})}$.

As for the initial angle α, we already know that it equals $\frac{\pi}{4} + \frac{\beta}{2}$, where β now equals $\tan^{-1}\frac{h}{d}$.

EXERCISES

12.1 From the top of a light-house 60 meters high with its base at sea level, the angle of depression of a boat is 15°. Find the distance of the boat from the foot of the light-house. (1983)

12.2 The angles of elevation of the top of a vertical tower from two points A and B on the ground are α and β respectively. The points A and B are at a distance d apart and the segment AB subtends an angle γ at the foot of the tower. Express the height of the tower in terms of d, α, β and γ. (1982)

12.3 AB is a vertical pole. The end A is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion CB subtends an angle β at P. If $AP = nAB$, then show that $\tan\beta = \frac{n}{2n^2+1}$. (1980)

12.4 A ladder rests against a vertical wall at an angle α to the horizontal. Its foot is pulled away through a distance a so that it slides a distance b down the wall making an angle β with the horizontal. Show that $a = b\tan\frac{\alpha+\beta}{2}$. (1985)

12.5 An observer at O notices that the angle of elevation of the top of a tower is 30°. The line joining O to the base of the tower makes an angle of $\tan^{-1}\frac{1}{\sqrt{2}}$ with the North and is inclined Eastwards. The observer travels a distance of 300 meters towards the North to a point A and finds the tower to his East. The angle of elevation of the top of the tower at A is $\phi°$. Find ϕ and the height of the tower. (1993)

12.6 A tower of slant height a leans due North and subtends angles ϕ_1, ϕ_2 at two points on a road running Northwest from its base. The distance between the points is b. Prove that if α is the angle the tower makes with the vertical then

$$\sin\alpha = \frac{\sqrt{2}\left[a^2\sin^2(\phi_1-\phi_2) - b^2\sin^2\phi_1\sin^2\phi_2\right]^{1/2}}{a\sin(\phi_1-\phi_2)}$$

12.7 A tripod has three equal legs each making an angle α with the ground, their feet forming an equilateral triangle. Prove that the angle between any two of the legs is $\cos^{-1}\frac{1-3\cos 2\alpha}{4}$.

12.8 If a and b are the shortest distances from A and B to a straight road on which P and Q lie, $\alpha, \alpha_1, \alpha_2$ are the angles subtended at A by PQ, PB, QB

respectively and β, β_1, β_2 are the angles subtended at B by PQ, PA, QA respectively, then prove that

$$\frac{a}{b} = \frac{\sin\alpha \sin\beta_1 \sin\beta_2}{\sin\beta \sin\alpha_1 \sin\alpha_2}$$

*12.9 A triangle ABC is placed on a plane inclined at an angle θ with a horizontal plane so that the height of each vertex above the horizontal plane is equal to the perpendicular from that vertex to the opposite side of the triangle. Prove that $\cos 2\theta = 4(\cos A + \cos B + \cos C) - 5$.

12.10 In the problem in Comment No. 12, suppose that the shot is fired in the vertical plane through O and P with an initial speed which is higher than the minimum speed needed, say at an initial speed $\lambda\sqrt{g(h + \sqrt{h^2 + d^2})}$, where $\lambda > 1$. Show that for the shot to clear the wall, the angle α which its initial direction makes with the horizontal must satisfy

$$\tan^{-1}\frac{\sec\beta + \tan\beta}{\sec\phi + \tan\phi} \leq \tan\alpha \leq \tan^{-1}\frac{\sec\beta + \tan\beta}{\sec\phi - \tan\phi}$$

where $\beta = \tan^{-1}\dfrac{h}{d}$ and $\phi = \cos^{-1}\dfrac{1}{\lambda^2}$.

12.11 Two particles are projected simultaneously from a point A so as to pass through another point B, the speed of projection in each case being V. If α, α' are the angles of projection, prove that the particles will pass through B at times separated by the interval

$$\frac{2V}{g}\frac{\sin\frac{1}{2}(\alpha - \alpha')}{\cos\frac{1}{2}(\alpha + \alpha')}$$

12.12 A regular hexagon stands with one side on the ground. A particle is projected so as to just graze the four upper corners. Show that the ratio of the speed of the particle on reaching the ground to its least speed is $\sqrt{31} : \sqrt{3}$.

12.13 Find the greatest range of a particle, projected with a given speed u, on a given inclined plane which makes an angle β with the horizontal.

Chapter 13

MAXIMA, MINIMA AND CONCAVITY

In the chapters so far there were only marginal references to calculus. In Chapter 5, we showed how derivatives and integrals can be used sometimes to prove binomial identities. In Chapter 9, in Comment No. 2 we mentioned that the slope of the tangent to a curve can be obtained by differentiation while in Comment No. 14 we solved a problem in which the centre of a circle was given as a certain limit.

In the present chapter we begin to use calculus in full swing. Logically, before doing so we should study the more basic concepts of limits, continuity and derivatives. But we postpone them to Chapter 15 and assume a working knowledge of them here. Thus, we begin with the problems of maxima and minima. For one thing, these problems are among the most popular problems of calculus. The existence and uniqueness of a derivative has little appeal to a layman. But show him how to make the most economical tin can to hold a given volume and he is impressed!

The use of derivatives in maxima/minima problems comes from the fact that the sign of the derivative of a function tells us whether the function is increasing or decreasing. Not surprisingly, derivatives are also a powerful tool in proving certain inequalities as will be shown in Comment No. 11. The sign of the second derivative of a function also has an interesting significance in terms of what is called the concavity of its graph. This again leads to certain important inequalities. In particular, we shall prove the Jensen's inequality mentioned in Chapter 6, Comment No. 6.

The problems of finding the maxima/minima appear almost every year in the JEE, often in combination with other topics such as areas. Once the function to be maximised/minimised is set up properly, the rest of the work is routine. So these questions are among the 'sure shot' questions.

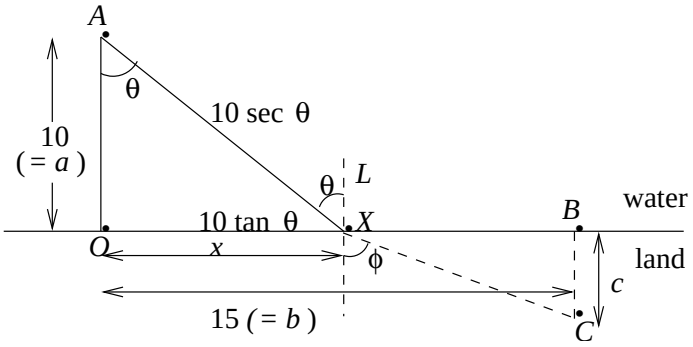
Main Problem : A swimmer who can swim half as fast in the sea as he can run on its shore is located at a point A which is at a distance of 10 meters inside the sea from a point O on a straight shore and wants to reach a point B on the shore at a distance 15 meters from O. If d meters is the distance he must swim in order to reach B as quickly as possible, then d^2, expressed as a rational number in the reduced form is ..... . (JEE 1983, modified)

First Hint: Draw a diagram. Let X be the point on the shore the swimmer reaches by swimming and let θ be the angle $\angle OAX$. Express the time taken by the swimmer as a function of θ.

Second Hint: Minimise this function.

Solution: The distance swum is $10\sec\theta$ while the distance run on the shore (from X to B) is $15 - 10\tan\theta$. Let the (maximum) speeds of swimming and running be, respectively, u and $2u$ meters per second. Then the total time taken to reach B is $f(\theta)$, where

$$f(\theta) = \frac{10\sec\theta}{u} + \frac{15 - 10\tan\theta}{2u} \tag{1}$$



This function is to be minimised over the interval $0 \le \theta \le \alpha$, where $\alpha = \tan^{-1}(\frac{15}{10}) = \tan^{-1}\frac{3}{2}$. We have,

$$f'(\theta) = \frac{10\tan\theta\sec\theta}{u} - \frac{10\sec^2\theta}{2u} \tag{2}$$

which vanishes when $\sin\theta = \frac{1}{2}$, i.e., when $\theta = \frac{\pi}{6}$. Note that $\frac{\pi}{6}$ lies in the interval $[0, \alpha]$ and further that $f'(\theta) < 0$ for $\theta < \frac{\pi}{6}$ while $f'(\theta) > 0$ for $\theta > \frac{\pi}{6}$. Hence $f(\theta)$ is decreasing on $[0, \frac{\pi}{6}]$ and increasing on $[\frac{\pi}{6}, \alpha]$. Therefore, $f(\theta)$ attains its minimum on $[0, \alpha]$ at $\theta = \frac{\pi}{6}$. So d = the distance swum (in meters) in the quickest path $= 10\sec(\frac{\pi}{6}) = 10\frac{2}{\sqrt{3}}$. Hence $d^2 = \frac{400}{3}$.

Comment No. 1:

Like problems on heights and distances, the problems of maxima and minima are often posed as real-life problems. When this happens, certain common sense

assumptions are implicit. For example, in the Main Problem we assume that the shore (i.e., the dividing line between the sea and the land) is a straight line. Also we ignore the fact that the water very near the shore is too shallow to swim! (If necessary, we can shift the shore into the sea so that such shallow water areas are considered to belong to the land rather than to the sea). Also in the solution above it is assumed that the swimmer will swim and run in straight lines. The justification for this is, of course, that when the speed is uniform in all directions, the quickest path between any two points will also be the shortest path and between any two given points the straight line is the shortest path. This requires a proof and the proof is not all that trivial. But here again, we take this as an intuitively clear fact.

Comment No. 2:

The time taken can also be expressed as a function of x, the distance (in meters) between O and X. It works out to be $\dfrac{\sqrt{x^2+100}}{u} + \dfrac{15-x}{2u}$ for $0 \leq x \leq 15$. The answer can also be obtained from this. But because of the radical sign, the calculations are a little more complicated and hence more prone to errors than in the case when it is expressed as a function of θ as done here. This illustrates how a particular choice of variables can be more convenient sometimes. The general tendency is to take various lengths as the primary variables and then to express angles in terms of them. In the solution above, it is the other way round. (There is, of course, no golden rule to tell beforehand which notation will be more convenient in a given problem. But the possibility should be kept in mind.)

Comment No. 3:

The motion of the swimmer consists of two parts : the motion in sea and the motion on the shore. The first part can be done most quickly by swimming straight from A to O. And, once at O, the second part can be done most quickly by running straight from O to B. It is therefore tempting to think that this is also the answer to the problem, much the same way as the quickest journey for a traveller is to reach the airport by the fastest possible vehicle and then to take the fastest flight to his destination. The catch is that unlike in the case of the traveller, the two parts of the swimmer's motion are *not* independent of each other. The first part determines the length and hence also the duration of the second part. (The approach in which an optimisation problem is broken into several parts and each part is optimised separately is called the **greedy approach**. Usually it does not work. The fallacy encountered here is analogous to that in trying to maximise/minimise a sum of two functions by trying to maximise/minimise each summand separately and adding the results. For example, each of the two functions $\sin\theta$ and $\cos\theta$ has maximum 1 and minimum -1 on the interval $[0, 2\pi]$. But the maximum of the sum function $\sin\theta + \cos\theta$ over the same interval is not 2. Nor is its minimum -2. These values are, respectively, $\sqrt{2}$ and $-\sqrt{2}$ as can be seen by differentiating or by writing $\sin\theta + \cos\theta$ as $\sqrt{2}\,\sin(\theta + \frac{\pi}{4}$.)

Comment No. 4:

More generally, we can replace the numbers 10 and 15 in the Main Problem by a and b respectively. Then $\alpha = \tan^{-1}\frac{b}{a}$. Also we assume that the ratio of the maximum speed of swimming to that of running is some positive real number λ. (In the Main Problem $\lambda = \frac{1}{2}$.) If $\lambda \geq 1$ (which is not likely to happen for a human swimmer but can happen for certain aquatic mammals), then $f'(\theta) < 0$ for all $\theta \in [0, \alpha]$. Thus $f(\theta)$ is decreasing throughout on $[0, \alpha]$ and hence the minimum occurs at the end-point α. That is, the quickest path for the swimmer is to swim straight from A to B. This is also the case if $0 < \lambda < 1$ and $\sin^{-1}\lambda \geq \alpha$ which reduces to $b \leq \dfrac{\lambda a}{\sqrt{1-\lambda^2}}$. In all other cases, the quickest path is given by $\theta = \sin^{-1}\lambda$.

Comment No. 5:

As a still further generalisation, suppose that the destination of the swimmer is not the point B on the shore but the point C on land at a distance c from the shore as shown in the figure in the solution to the Main problem. In that case, θ is also the angle the line AX makes with the normal line, say L to the shore at the point X. Let ϕ be the angle between L and the line XC. (Thus $\phi = \frac{\pi}{2}$ when the destination is on the shore as in the Main Problem.) Then $XC = c\sec\phi$, where $c\tan\phi = b - a\tan\theta$. Again, letting λv and v be the speeds in water and on land respectively, the time taken to go from A to C via X comes out to be

$$g(\theta) = \frac{a\sec\theta}{\lambda v} + \frac{\sqrt{c^2 + (b - a\tan\theta)^2}}{v} \tag{3}$$

As λ and v are fixed, a little computation, which we omit, shows that $g'(\theta) = 0$ when

$$\sin\theta = \frac{\lambda(b - a\tan\theta)}{\sqrt{c^2 + (b - a\tan\theta)^2}} \tag{4}$$

This equation is not easy to solve for θ. However, since $b - a\tan\theta = c\tan\phi$, it can be rewritten as

$$\sin\theta = \lambda\sin\phi \tag{5}$$

and in this form it gives some valuable information if we change the setting of the problem. Let us replace the swimmer by light and the sea and the land by two media, say, air and glass respectively. Then, the angles θ and ϕ become what are called the angle of incidence and the angle of refraction, respectively, for the light ray originating at A and falling at X. If we assume the law of nature that a light ray originating at A and reaching C will follow the quickest path (or the path of least resistance as it is often called) then it must pass through a point X on the boundary between the two media at which the angles of incidence and refraction are related to each other by (5). Here C is fixed and X varies. But we can take a fixed X and hence a fixed θ. In that case, the

ray originating at A and falling at X will change its direction after entering the second medium and the angle ϕ which the new direction makes with the normal is given by (5). It is clear that if $\lambda > 1$ (which is expressed by saying that the second medium is optically thicker than the first) then $\phi < \theta$, i.e., the ray will deflect towards the normal while if $\lambda < 1$, then it will deflect away from the normal. This is called the law of **refraction** in a qualitative form. Equation (5) is valid even if the boundary between the two media is not a straight line but some other curve, provided in that case that L is taken as the normal to the curve at the point of incidence.

Comment No. 6:

In its most general form, a maximisation problem involves a real-valued function (often called the **objective function**), say f, defined on a set, say S and the goal is to find a point $x_0 \in S$ such that $f(x_0) \geq f(x)$ for all $x \in S$. Such a point x_0 need not always exist (e.g., take $S = (0, 1)$ and $f(x) = \frac{1}{x}$ for $x \in S$). When it does, we say f attains its maximum on S at x_0 or that x_0 is a point of maximum of f on S. The value $f(x_0)$ is called the maximum value or simply the maximum of f on S. By an abuse of language sometimes we say that x_0 is a maximum of f on S (just as in school examinations we use the word 'highest' to mean either the highest score in a class or the student who obtains it). Sometimes the maximum is attained at more than one point, e.g., the function $f(x) = \cos x$ defined on the set $\mathbb{R}$ attains its maximum (which is 1) at an infinite number of points, viz., at all even multiples of π. Each of these points is a point of maximum. But the maximum value, which is 1, is always unique when it exists.

Analogous terminology is defined for the problem of minimisation. The two concepts are actually related. The minimum value of a function f is simply the negative of the maximum value of the function $-f$. As a result, once we prove a theorem for maximisation, we automatically get the corresponding theorem for minimisation. The terms '**optimum**' and '**extremum**' are often used to mean either a maximum or a minimum, depending on the context. The meaning of the latter is obvious. The former, which has come into vogue relatively recently, comes from what one opts for, i.e., what is desirable in a particular context.

We encounter optimisation problems even in real life. For awarding a prize we often need to find the top ranker. This is a maximisation problem where the objective function is the score of a student. The Travelling Salesman Problem, on the other hand, is a minimisation problem where the set S is the set of all possible circular tours and the objective function f assigns to each tour the time it takes. In both these problems, the set S is a finite set and so the existence of the optimum solution is not a problem. Moreover, if nothing else works, then the optimum can always be obtained by a straight comparison of all possible values. But this can be very time-consuming when the cardinality of the set S is large. (See the comments in Exercise (6.52)). In such cases we try to narrow down the search by 'short listing' the likely candidates for an optimum. This has to be done by methods tailored for a particular type of a

problem. We shall not go into it since these types of optimisation problems are beyond our scope. We remark, however, that these problems are not as trivial as they may appear at first sight because of the finiteness of the set S. In the Travelling Salesman Problem, for example, it is tempting to think that in the shortest tour, the closest pair of cities must be visited successively. But in Exercise (13.1) we shall give an example where this is not the case.

Comment No. 7:

When the domain set S of the objective function f is infinite, an optimum may not always exist. For example, as we saw above, the function $f(x) = \frac{1}{x}$ on the set $S = (0, 1)$ is not bounded above, i.e., it assumes arbitrarily large values. So there is no maximum value for this function. (It is tempting to call ∞ as the maximum of f here. Some old books do follow this convention. But it is better to avoid it since ∞ is not a real number as we saw in Chapter 6, Comment No. 20.) What about the minimum of f over $S = (0, 1)$. Certainly, the function f is bounded below by 1. In fact, it takes values which are arbitrarily close to 1 and greater than 1. In precise terms this means that for every $\epsilon > 0$, there exists some $x \in S$ for which $1 + \epsilon > f(x) > 1$. (For example, take x to be any real number between $\frac{1}{1+\epsilon}$ and 1.) Note, however, that there is no point x_0 in S at which f attains the value 1. The only point at which this could happen is 1 which is not in the open interval, since, by definition, an open interval[1] does not contain its end-points. So this function does not attain its minimum on S. If we take S to be the semi-open interval, then the function $f(x) = \frac{1}{x}$ does attain its minimum on S at the end-point 1. Of course, it still does not have a maximum on S.

Even when a function f has an optimum on an infinite set, there is, in general no method for finding it because straight comparison is clearly ruled out as it will never end. In some cases, the optimum can be located by some explicit construction, for example by showing that the points of maxima/minima must be the roots of some equation. The problems we shall tackle will fall under this category. Of course, this opens up a new question of how to find the roots of this equation. When there is no explicit construction for an optimum, one usually resorts to approximate optima. Sometimes the limit of a sequence of approximate optima gives an optimum of the original problem.

But all these problems are beyond our scope. We shall confine ourselves to a very special case of an optimisation problem where the set S is an interval of

[1] For the sake of record, an **open interval** (a, b) is a set of the form $\{x : a < x < b\}$, where a and b are some real numbers with $a < b$. The **closed interval** $[a, b]$ is the set $\{x : a \leq x \leq b\}$ while the **semi-open intervals** $[a, b)$ and $(a, b]$ are, respectively, the sets $\{x : a \leq x < b\}$ and $\{x : a < x \leq b\}$. These intervals are bounded and have two end-points or boundary points, viz., a and b. An end-point of an interval belongs to that interval if it is closed at that end. Sometimes we also consider unbounded open intervals of the form (a, ∞) defined as the set $\{x : a < x\}$ and of the form $(-\infty, a)$ defined as the set $\{x : x < a\}$, where a is a real number. The unbounded closed intervals $[a, \infty)$ and $(-\infty, a]$ are defined analogously with $\leq$ replacing $<$. The unbounded intervals have only one end-point each. We may also consider the entire real line as an interval $(-\infty, \infty)$. Note that it has no end-point and hence can be treated either as an open interval or as a closed interval.

real numbers (which may be open, closed or semi-open and also either bounded or unbounded) and the function f is continuous on S. Sometimes this interval S and the function f are given to us explicitly. Sometimes, as in the case of the Main Problem, we have to set these up ourselves. Even in this highly special case, the maxima and minima of f need not exist as we already saw where $S = (0, 1)$ and $f(x) = \frac{1}{x}$ which is continuous on S. (It may be noted that f is not continuous at 0. But 0 is not a point of the set $(0, 1)$.)

As far as the existence of extrema is concerned, the situation is pleasant when the interval is closed and bounded, as the following theorem shows.

Theorem 1: Every continuous, real valued function on a closed and bounded interval has a maximum and a minimum (in that interval).

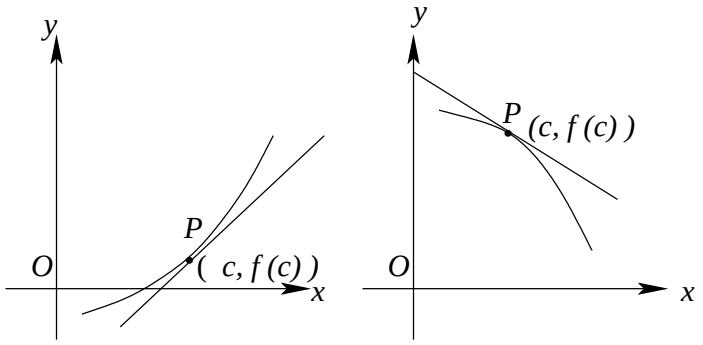
The proof of this theorem runs into the same basic properties of real numbers which we need in proving, for example, that every real number has a real cube root. We postpone such proofs to Chapter 16 (see Comment No. 5 there for a proof of this one). The theorem is not of much practical use, because it only asserts the existence of a maximum and also a minimum without giving any method for finding it, rather like saying that there is an eight-feet tall person without telling you his identity or any clue about his whereabouts. Still, the very existence of a maximum and minimum is theoretically important because it assures you that you are not wasting your time in trying to find something that simply does not exist, rather like waiting for a bus that has already left without your knowledge.

Once a maximum or a minimum of a function $f(x)$ on an interval S is known to exist, considerations of differentiability help in narrowing the search down considerably. Suppose c is an interior point of the interval S, i.e., not an end-point. Then there exists some $\delta > 0$ such that the open interval[2] $(c-\delta, c+\delta)$ is contained in S. (The exact value of δ is unimportant for most purposes. In the present case any positive number smaller than both $c - a$ and $b - c$ will do, where a and b are the left and the right end-points of S. (In case S is unbounded either on the left or on the right, then simply drop the corresponding restriction on δ. For example if $S = [a, \infty)$ and $c > a$ take δ to be any positive number less than $c - a$.)

Now suppose f is differentiable at c i.e., $f'(c)$ exists. By definition, $f'(c) = \lim_{x \to c} \dfrac{f(x) - f(c)}{x - c}$. We have not yet defined such limits (although, in principle, they are analogous to the limits of sequences defined in Chapter 6, Comment No. 6). Suffice it to say here that if x is very close to c, then the ratio $\frac{f(x)-f(c)}{x-c}$ will be very close to $f'(c)$. Now, if two real numbers are very close to each other and one of them is 0, nothing can be said about the sign of the other. But if one of them is positive then the other number will also be positive if it is 'sufficiently' close to the former. In particular, if $f'(c)$ is positive, then so will be the ratio $\frac{f(x)-f(c)}{x-c}$, provided x is sufficiently close to c. Hence its

[2]This interval is popularly called the δ-**neighbourhood** of the point c. It is precisely the set $\{x : |x - c| < \delta\}$. Sometimes we want to exclude the point c from it. Then we get what is called the **deleted** δ-**neighbourhood** of the point c. This equals the set $\{x : 0 < |x-c| < \delta\}$.

numerator and denominator must be of the same sign. Verbally, if x is just to the right of c, then we shall have $f(x) > f(c)$, while if x is just to the left of c, then $f(x) < f(c)$. This is sometimes expressed by saying that the function f is increasing at the point c. Geometrically, the tangent to the graph of $y = f(x)$ at the point $(c, f(c))$ has positive slope as shown in Figure (a) below. An entirely analogous situation arises if $f'(c) < 0$ as illustrated in (b)

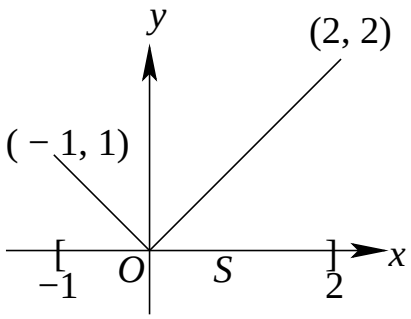


(a) $f'(c) > 0$ (b) $f'(c) < 0$

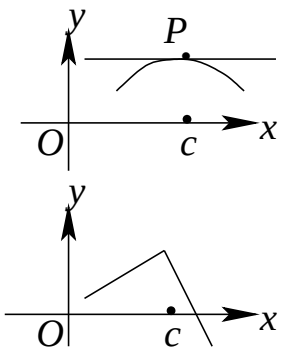
Note that in either case, there will be points on one side of c at which $f(x)$ will have a higher value than $f(c)$ while on the other side of c there will be points at which f has a lower value than at c. In particular we see that the function f can have neither a maximum nor a minimum at c. In effect, we have proved the theorem below.

Theorem 2: If a function $f(x)$ has either a maximum or a minimum at an interior point c of an interval S, and *if its derivative exists* at c, then $f'(c)$ must vanish.

Note that the theorem does *not* say that the function f is differentiable at c. For an actual example, take $f(x) = |x|$ and S to be the interval $[-1, 2]$. Then f attains its minimum on S at 0. But f is not differentiable at 0. Geometrically, the graph of the function $f(x) = |x|$ has a sharp corner or 'kink' at O and hence there is no well-defined tangent at O to the graph. (In fact, this is a standard example of a function which is continuous but not differentiable.) All that the theorem says is that in case f is differentiable at c, then $f'(c) = 0$. Note also that in this example, the function attains its maximum on S at the point 2. At this point f is differentiable and $f'(2) = 1 \neq 0$. This does not contradict the theorem, however, because here 2 is an end-point and not an interior point of $[-1, 2]$. (See Figure (c) below.)



(c) $f(x) = |x|$



(d) point of maximum

Intuitively the truth of this theorem is obvious. Suppose c is an interior point of the interval S and you are situated at the point, say $P = (c, f(c))$ on the graph of $y = f(x)$. Since c is an interior point, there are two directions you can move away from P, corresponding either to the forward or backward motion of x from c. Now suppose the function attains its maximum on S at c. Then P is like the crest of a wave or a summit and so no matter which direction you take you can only go downwards (see Figure (d) above). But for this to be possible, either there is no well defined tangent at P or else the tangent at P is horizontal. Translated in terms of derivatives, this means that either $f'(c)$ does not exist or else it vanishes. Similarly, if there is minimum at an interior point c then no matter which direction you take you can only move upwards from P. And for this to be possible, $f'(c)$ either fails to exist or else vanishes.

Note also that Theorem 2 gives only a necessary condition for a maximum to occur at an interior point. It does *not* say, for example, that in case $f'(c)$ vanishes, then the function f would have either a maximum or a minimum at c. The simplest example is to take $f(x) = x^3$ and $c = 0$. Here there is neither a maximum nor a minimum at c, even though $f'(c) = 0$. Such cases require further analysis based on the higher order derivatives of the function f at c. (See Theorem 8 of Chapter 16.)

Comment No. 8:

It is convenient to paraphrase Theorem 2 slightly by introducing a new term. We say that a point c is a **critical** point of a function f if either f is not differentiable at c or else it is differentiable at c and the derivative vanishes at c. Theorem 2 can now be worded succinctly to say that the maximum or minimum of a function f defined on an interval S can occur only at a critical point of f or at an end-point of S. Combined with Theorem 1, which asserts the existence of an extremum, we now have a workable procedure to find the extrema of a continuous function f on a closed and bounded interval, say $[a, b]$. All we have to do is to identify the critical points of f in $[a, b]$. Usually (although not always) the set, say C, of critical points is finite. Theorem 2 shortlists the extrema to the set $C \cup \{a, b\}$ which is also finite if C is finite. So the extrema

of f on $[a, b]$ can now be obtained by straight comparison. We illustrate this in the following problem.

A piece of wire of length L is to be divided into two parts, one of which is to be bent into a circle and the other into a square. Find the maximum and the minimum total area enclosed by the two pieces.

Let x and $L-x$ denote the lengths of the two segments. Here $0 \leq x \leq L$. The area enclosed by a circle of perimeter x is $\pi \left(\frac{x}{2\pi}\right)^2 = \frac{x^2}{4\pi}$ while that enclosed by a square of perimeter $L - x$ is $\frac{(L-x)^2}{16}$. So, the problem amounts to find the maximum as well as the minimum of the function

$$f(x) \quad = \quad \frac{x^2}{4\pi} + \frac{(L-x)^2}{16} \tag{6}$$

over the interval $[0, L]$.

Since f is continuous and the interval is closed and bounded, the maximum and the minimum exist by Theorem 1. Moreover, f is differentiable everywhere. Differentiating (1) and setting $f'(x) = 0$ gives $x = \frac{\pi L}{\pi+4}$ as the only critical point of f. This point lies in the interval $[0, L]$ since $0 < \frac{\pi}{\pi+4} < 1$. So, by Theorem 2 above, the extreme values of f must occur in the set consisting of the three points $f(0), f(L)$ and $f\left(\frac{\pi L}{\pi+4}\right)$. A direct calculation gives $f(0) = \frac{L^2}{16}, f(L) = \frac{L^2}{4\pi}$ and $f\left(\frac{\pi L}{\pi+4}\right) = \frac{L^2}{4(\pi+4)}$. A straight comparison (based on an approximate value of π) shows that f attains its minimum at $\frac{\pi L}{\pi+4}$ and its maximum at L. So $\frac{L^2}{4\pi}$ and $\frac{L^2}{4(\pi+4)}$ are the maximum and the minimum sums of areas enclosed by the circle and the square.

In this problem, the minimum occurred at an interior point while the maximum occurred at an end-point. In some problems the end-points may not exist or may not be given explicitly. But we can truncate the domain of the function f suitably to restrict it to a closed and bounded interval S. In essence we already did this in the Main Problem. In absence of any other information we assumed that the shore, i.e., the dividing boundary between the sea and the land was infinite in both the direction. This assumption is, of course, unrealistic. But if our interest is in minimising rather than maximising the time, then it is obvious that the point X can only lie between O and B. Accordingly, the variable θ, which could, in theory, assume any value in the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$, can be assumed to vary over the subset $[0, \alpha]$ (where $\alpha = \tan^{-1}\frac{3}{2}$), which is a closed and bounded interval. In the solution we saw that the only critical point of the function $f(\theta)$ defined by (1) was $\frac{\pi}{6}$. So we could as well have minimised $f(\theta)$ over $[0, \alpha]$ by straight comparison. The values $f(0), f(\frac{\pi}{6})$ and $f(\alpha)$ come out to be, $\frac{35}{2u}, \frac{30+15\sqrt{3}}{2\sqrt{3}u}$ and $\frac{5\sqrt{13}}{u}$ respectively. As we see by squaring, $\frac{35}{2u} < \frac{5\sqrt{13}}{u}$ and so the real contest for a minimum is between $f(\frac{\pi}{3}) = \frac{30+15\sqrt{3}}{2\sqrt{3}u}$ and $f(0) = \frac{35}{2u}$. Multiplying both the expressions by $\frac{2u}{5}$,

this reduces to a comparison between $\frac{6+3\sqrt{3}}{\sqrt{3}}$ and 7 and hence to a comparison between $6+3\sqrt{3}$ and $7\sqrt{3}$. Since $6 < 4\sqrt{3}$, as seen by squaring, we get that $f(\frac{\pi}{3}) < f(\alpha)$ and hence that f attains its minimum on S at $\frac{\pi}{3}$. (When you compare two expressions, say A and B, in your rough work it is both suggestive and efficient to do so as follows. Make two columns, one headed by A at the top and the other headed by B. Now go on reducing these expressions by a series of reductions, making sure that each reduction is order preserving. For example, adding or subtracting equal quantities from the two sides is all right. Multiplying both the sides by a positive quantity is permissible. Squaring is permissible if both the contestants are known to be positive. Through this series of reductions, ultimately reduce A and B to some expressions, say A' and B' respectively which are so simple that you can tell by inspection which of them is greater. You can then tell which of the original expressions is greater. For example, the comparison between $f(0)$ and $f(\frac{\pi}{3})$ can be made graphically as follows :

$\frac{35}{2u}$	$\frac{30+15\sqrt{3}}{2\sqrt{3}u}$
35	$\frac{30+15\sqrt{2}}{\sqrt{3}}$
7	$\frac{6+3\sqrt{3}}{\sqrt{3}}$
$7\sqrt{3}$	$6+3\sqrt{3}$
$4\sqrt{3}$	6
$(4\sqrt{3})^2$	36
48	36

Since in the last line the L.H.S. is bigger than the R.H.S., this is also the case in the first line. The procedure is reminiscent of a prolonged battle lasting over several generations, whose outcome is ultimately decided between the descendants of the original combatants!)

Comment No. 9:

Although inherently simple, the procedure above has certain disadvantages. First of all, as just illustrated, a 'straight' comparison may not be all that straight when the quantities to be compared are complicated expressions involving irrational numbers like π and $\sqrt{3}$. Secondly, in many problems, we are not interested in the extremum value *per se* but only in the point in the domain of the objective function at which it is attained. For example, in the Main Problem we merely wanted to find the quickest path for the swimmer and not the time the quickest path will take. So the work involved in computing this time (or in general the extreme value of the function) is really a waste.

Instead, in the solution to the Main Problem we used another method which is also based on derivatives but which is qualitatively different than that based on Theorem 2. Its underlying idea is extremely akin to common sense. Suppose that the prices of some commodity went on increasing till say, 1994, and thereafter they are falling every year. Then it is obvious that the commodity was dearest in 1994. An analogous statement holds if the prices were decreasing till 1994 and increasing afterwards. We already applied this reasoning in several

places in earlier chapters. In Comment No. 9 of Chapter 5, we saw that if n is an even integer, say $n = 2m$ then the binomial coefficient $\binom{n}{k}$ goes on increasing as k varies from 1 to m and thereafter decreases as k increases. So $\binom{n}{k}$ is maximum when $k = m$. Here the integer k is an example of what is called a **discrete variable**. It assumes only integer values. But we also applied this reasoning in Comment No. 11 of Chapter 3 to analyse the behaviour of the quadratic function $f(x) = ax^2 + bx + c$ of a variable x which is a **continuous variable** in that it assumes all real values. Equation (33) there shows that if $a > 0$ then $f(x)$ decreases as x gets closer to γ from the left but as x crosses γ and increases further, so does $f(x)$. So we concluded that $f(x)$ attained its minimum at γ. In Chapter 10 we found the extreme values of certain trigonometric functions by examining on which intervals they were increasing/decreasing.

Let us now apply this reasoning to find the extreme values of a general function f on an interval S. For the sake of definiteness we take this interval to be a closed and bounded interval $[a, b]$. Suitable modifications can be made when S is an interval of some other type. We say that a real valued function defined on S is **monotonically increasing** (or simply increasing) on S if whenever x and y are points of S with $x \leq y$, we have $f(x) \leq f(y)$. Note that even if $x < y$ we do not require that $f(x)$ be strictly less than $f(y)$. If this happens for all $x, y \in S$ we say that f is **strictly monotonically increasing** (or simply, strictly increasing) on S. The terms 'monotonically decreasing' and 'strictly monotonically decreasing' are defined analogously[3]. (Or, we may define f to be monotonically decreasing to mean that $-f$ is monotonically increasing.) Note that according to our definition, a constant function is both monotonically increasing and monotonically decreasing on every interval and it is the only function with this property. Of course, it is neither strictly increasing nor strictly decreasing. The function $f(x) = x$ is strictly increasing on the entire real line. The function $f(x) = x^2$ is strictly increasing on $[a, b]$ if $0 \leq a < b$ and strictly decreasing if $a < b \leq 0$. But if $a < 0 < b$, i.e., if 0 is an interior point of $[a, b]$, then f is neither increasing nor decreasing on the entire interval $[a, b]$. On the subinterval $[a, 0]$ it is strictly decreasing while on the subinterval $[0, b]$ it is strictly increasing. (A similar behaviour is shown by any quadratic whose leading coefficient is positive; see the graphs shown in Comment No. 11 of Chapter 3.)

Even though the definitions just made do not involve derivatives, it turns out that the most common method of testing whether a function is increasing/decreasing on an interval is by looking at the sign of its derivative. We already hinted at this in Comment No. 9 of Chapter 6. Also in the discussion preceding Theorem 2 above we saw that if $f'(c) > 0$, then f is increasing at c in the sense described there. It is natural to expect that if this holds at all points of an interval then the function should be increasing on that entire interval. This guess is quite correct. But the proof is not as immediate as one might expect.

[3]Some authors use the term 'monotonically increasing' to mean what we call as 'strictly increasing'. They use the term 'steadily increasing' to mean what we call monotonically increasing. The terms 'rising' and 'falling' or 'ascending' and 'descending' are also used especially to describe the graphs.

The precise statement is given below.

Theorem 3: Suppose a real-valued function f is continuous on a closed interval $[a, b]$ and differentiable at every point of the open interval (a, b). Then the following statements hold.

(i) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing on $[a, b]$.

(ii) If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly monotonically increasing on $[a, b]$.

(iii) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing on $[a, b]$.

(iv) If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly monotonically decreasing on $[a, b]$.

(v) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.

All these statements look very obvious and many people think that no proof is needed for them, especially for the last statement. Certainly, the converse of (v) is trivial and this may be one of the reasons for some people to believe that (v) is too obvious to need a proof. Even those who are careful enough not to confuse a statement with its converse are apt to think that (v) is obvious. Their argument is something like this. The derivative represents the rate of growth of a function. (Geometrically, it is the slope of the tangent to the graph of the function.) So if $f'(x) = 0$ for all $x \in (a, b)$, it means that the function is growing at the zero rate. But then it must be a constant. (Geometrically, the tangent at each point on the graph of the function is horizontal and so the graph must be a horizontal straight line.)

This argument is certainly plausible. But mathematically it is not a rigorous proof. A rigorous proof must be based strictly on the definition of a derivative and not on any other interpretations (such as the rate of growth) that we assign to it. Such interpretations are undoubtedly valuable. In fact, the applicability of mathematics to real life depends vitally on such interpretations. But such interpretations are like the Public Relation Officers of a company. The public knows them better than it knows the directors of the company and the PRO's can tell the public about the decisions taken by the directors. But the decision power lies with the directors and not with the PRO's! The PRO's can tell the public only what the directors have authorised them to tell. So if the argument given above, based on the rates of growth, is to be a valid mathematical proof, then it must be shown first of all using derivatives, that a function with a zero rate of growth is constant. But that, in essence, means we have to prove (v).

So, however obvious the statements above look, they need a proof. Statements (iii) and (iv) follow by applying (i) and (ii) respectively to $-f$ while (i) and (iii) together imply (v). So it suffices to prove only (i) and (ii). These follow immediately by applying a very celebrated result in calculus, called the **Lagrange Mean Value Theorem**. We shall derive it in Chapter 16 on theoretical calculus as a consequence of a basic property of real numbers, called completeness.

So we take Theorem 3 for granted and apply it to find the extrema of a function f on an interval $[a, b]$ by examining on which subintervals f is increasing/decreasing. Note that in Theorem 3, although the function f is required to be continuous at all points of $[a, b]$, including the end-points a and b, it is not required to be differentiable at the end-points. For example, we can apply Theorem 3 to a function of the form $f(x) = \sqrt{x}$ on an interval of the type $[0, b]$, where $b > 0$. As the simplest application of Theorem 3, if $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing on $[a, b]$ and so attains its maximum on $[a, b]$ at b and its minimum on $[a, b]$ at a. Here Theorem 2 would have been inconvenient since it would require us to consider the points where f' vanishes. As a less trivial application, consider the function $f(x) = |x|$ on the interval $[-1, 2]$. As noted before, this function is not differentiable at 0 and so we cannot apply Theorem 3 to the entire interval $[-1, 2]$. However, f is differentiable at all other points and so we can apply the theorem to the two sub-intervals $[-1, 0]$ and $[0, 2]$ separately, since they have 0 as an end-point and differentiability at an end-point is not required. On $[-1, 0]$, $f(x) = -x$ and so $f'(x) = -1 < 0$ for all $x \in [-1, 0]$. Hence by Part (iv) of Theorem 3, f is strictly decreasing on $[-1, 0]$. Similarly, on the interval $[0, 2]$, $f(x) = x$ and hence $f'(x) = 1 > 0$ for all $x \in [0, 2]$ whence f is strictly increasing on $[0, 2]$. But then f must attain its minimum on $[-1, 2]$ at 0. As for the maximum, -1 and 2 are the two candidates. Since $f(-1) = 1$ and $f(2) = 2$, a direct comparison gives 2 as the point of maximum of f on $[-1, 2]$. None of this is new to us. But see how effortlessly we got it using Theorem 3.

The solution to the Main Problem was also based on Theorem 3. The crucial point was that $f'(\theta) > 0$ for all $\theta < \frac{\pi}{6}$. Hence by (iv) of Theorem 3, f was strictly decreasing on $[0, \frac{\pi}{6}]$. Similarly, because of the positivity of the derivative, f was strictly increasing on $[\frac{\pi}{6}, \alpha]$. Therefore, f attained its minimum at $\frac{\pi}{6}$. (Note that, in the solution we did not mention Theorem 3 explicitly. This is typical of theorems that are used very frequently in a particular context.)

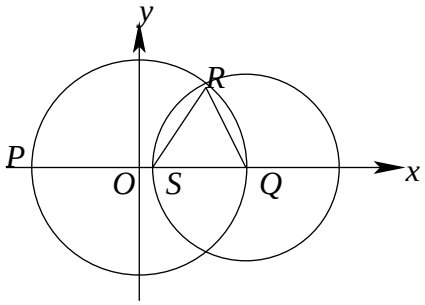
Comment No. 10:

Here is another problem which can be done very easily using Theorem 3.

The circle $x^2 + y^2 = 1$ cuts the x-axis at P and Q. Another circle with centre at Q and variable radius intersects the first circle at R above the x-axis and the line segment PQ at S. Find the maximum area of the triangle QSR. (JEE 1994)

Without loss of generality, we take P as $(-1, 0)$ and Q as $(1, 0)$. Let r be the radius of the variable circle. Since its centre is at $(1, 0)$, its equation is $(x - 1)^2 + y^2 = r^2$. Solving this simultaneously with $x^2 + y^2 = 1$ we get $R = (\frac{2 - r^2}{2}, \frac{r}{2}\sqrt{4 - r^2})$. Since $QS = r$, we get that the area of the triangle QRS equals $\frac{r^2}{4}\sqrt{4 - r^2}$. Call this as $f(r)$. We have to maximise

this as a function of r. The domain of f is not given. But evidently $r > 0$ and for $\sqrt{r^2-4}$ to be real, we must also have $r \leq 2$. So we take $[0,2]$ as the domain of f, even though the end point 0 corresponds to a degenerate circle and the end-point 2 gives a degenerate triangle QRS.



A direct computation gives $f'(r) = \dfrac{r(8-3r^2)}{4\sqrt{4-r^2}}$. So f' vanishes at $r = \frac{2\sqrt{2}}{\sqrt{3}}$ and at $r = 0$, which is an end-point of the interval $[0,2]$. Moreover, since the denominator is positive and the factor r in the numerator is also positive on $(0,2)$, we get, without doing any extra work, that $f'(r) > 0$ for $0 < r < \frac{2\sqrt{2}}{\sqrt{3}}$ and also that $f'(r) > 0$ for $r > \frac{2\sqrt{2}}{\sqrt{3}}$. By Theorem 3, f is increasing on $[0, \frac{2\sqrt{2}}{\sqrt{3}}]$ and decreasing on $[\frac{2\sqrt{2}}{\sqrt{3}}, 2]$. So the maximum area of the triangle occurs when $r = \frac{2\sqrt{2}}{\sqrt{3}}$. By direct calculation, its value is $\frac{4}{3\sqrt{3}}$ sq. units. (Of course, the present problem could have been done equally easily with Theorem 2 too, because at both the end points of the interval $[0,2]$, the function $f(r)$ vanishes. As there is only one critical point and the value of f at it is positive, it must be the point of maximum by Theorems 1 and 2 put together. Yet another illustration that when a variety of methods is available, rarely is one method superior all the time.)

Comment No. 11:

Not surprisingly, Theorem 3 is also a powerful tool in proving inequalities. To appreciate its power, let us first take an inequality which can also be proved without Theorem 3, and in fact, without calculus in any form. Suppose, for example, that we want to prove that $f(x) = x^3$ is a strictly monotonically increasing function of x. This can be done purely algebraically as follows. Suppose $x < y$. To show that $x^3 < y^3$, we factorise $y^3 - x^3$ as $(y-x)(y^2+yx+x^2)$. The first factor is positive by assumption. The second factor can be rewritten as $\frac{1}{2}(2y^2+2xy+2x^2)$ and hence as $\frac{1}{2}[x^2+y^2+(x+y)^2]$. As $x \neq y$, at least one of the three squares is positive. So both the factors of y^3-x^3 are positive, whence $y^3 - x^3 > 0$, i.e., $x^3 < y^3$ as desired. This is a simple but rather tricky proof. Now consider the function $f(x) = x^3$. Its derivative $3x^2$ is positive everywhere except at $x = 0$. So as a direct consequence of Part (ii) of Theorem 3, we see that $x^3 < y^3$ if either $0 \leq x < y$ or $x < y \leq 0$. The only case left is that $x < 0 < y$. This can be settled directly since a number and its cube always have the same sign. Alternatively, we can apply Theorem 3 separately to the intervals $[x,0]$ and $[0,y]$ and get $f(x) < f(0) < f(y)$ and hence $f(x) < f(y)$ (cf. Exercise (13.8)). Note that this proof will work equally easily for any odd power

of x. The algebraic proof, on the other hand, will become more complicated. (A clever way out is to factorise $y^n - x^n$ as $(y-x)(y^{n-1}+y^{n-2}x+\ldots+yx^{n-2}+x^{n-1})$ and first consider the case when x and y are of the same sign. But this is not as elegant as the argument based on the sign of nx^{n-1}.)

Theorem 3 is especially useful in proving inequalities of the form $f(x) \geq g(x)$, where f and g are two differentiable functions of x. The way it works is as follows. Usually, we can identify some point, say x_0 at which $f(x_0) = g(x_0)$. We now let $h(x) = f(x) - g(x)$. Then h is also a differentiable function of x and $h(x_0) = 0$. Suppose further that we are also able to show that $f'(x) \geq g'(x)$ for all $x \geq x_0$. Then $h'(x) \geq 0$ for all $x \geq x_0$ and so by Theorem 3, h is increasing on the interval $[x_0, x_1]$ for any $x_1 > x_0$. But $h(x_0) = 0$. So we must have $h(x_1) \geq 0$, i.e., $f(x_1) \geq g(x_1)$. The reasoning here can be given a real life interpretation through the interpretation of derivative as a speed. Think of x as time and $f(x)$ and $g(x)$ as the positions of two runners, say A and B respectively, at time x. Take x_0 as the moment when the race begins. At this moment, both the runners are on the starting line. Thereafter, if A runs faster than (or at least as fast as) B, then it is evident that A will always be ahead of B. (Or at least that he will not lag behind.) The reasoning above will also apply if instead of $f(x_0) = g(x_0)$ we had $f(x_0) > g(x_0)$ (and $f'(x) \geq g'(x)$ for $x \geq x_0$). In fact, in that case we would have $f(x) > g(x)$ for all $x \geq x_0$. (In the language of the race, if the faster runner also had an initial headstart then surely he will always be ahead thereafter.)

We apply this reasoning to make good a promise in Chapter 6. At the end of Comment No. 9 there, we remarked that even though x lies in the interval $[\sin x, \tan x]$ for $0 \leq x < \frac{\pi}{2}$, we cannot easily tell on which side of the midpoint x lies, i.e., we cannot tell by elementary trigonometric arguments how $\frac{1}{2}(\sin x + \tan x)$ compares with x. Using calculus, we can settle this question. In fact we shall prove a slightly stronger result (JEE 1990), viz., that

$$2\sin x + \tan x \geq 3x \quad \text{where } 0 \leq x < \frac{\pi}{2}. \tag{7}$$

(Add the inequality $-\sin x \geq -x$ to this, we shall get $\sin x + \tan x \geq 2x$.)

To prove (7) in the manner indicated above, we take $f(x) = 2\sin x + \tan x$ and $g(x) = 3x$. Then $f(0) = g(0)$. So we would be through if we can prove that for $x \in [0, \frac{\pi}{2})$, $f'(x) \geq g'(x)$, i.e.,

$$2\cos x + \sec^2 x \geq 3 \quad \text{for } x \in [0, \frac{\pi}{2}) \tag{8}$$

If we hastily use that $\sec^2 x \geq 1$, (8) would reduce to proving that $2\cos x \geq 2$. But this is not true for $x \neq 0$. So we cannot afford to be so generous as to replace $\sec^2 x$ by 1. Some rethinking can give us a clue. For the given range of x, $\cos x$ and $\sec x$ are positive and reciprocals of each other. Hence their sum is at least 2, by the A.M.-G.M. inequality. So we add and subtract $2\sec x$ to the L.H.S. of (8) and further completing the square write it as $2\cos x + 2\sec x + (\sec x - 1)^2 - 1$. As just observed, the sum of the first two terms is at least 4. The third term, being a square, is at least 0 and the last term is a constant, -1. So we get that

the L.H.S. of (8) is at least $4+0-1$ which is the R.H.S. of (8). This proves (8) and hence (7). (A slicker, albeit tricky, way to prove (8) is to write the L.H.S. as a sum of *three* terms, viz., as $\cos x + \cos x + \sec^2 x$ and apply the A.M.-G.M. inequality. See the end of Comment No. 6 of Chapter 6.) We shall revisit the inequality (7) in Chapter 16, Comment No. 14.

Comment No. 12:

Of course, no theorem should be resorted to blindly and Theorem 3 is no exception. Note, for example, that sometimes Theorem 3 may not be applicable. It depends on derivatives whereas the concept of monotonicity does not. Suppose, for example, that we are given that $h(x) = f(x) - (f(x))^2 + (f(x))^3$, where $f(x)$ is some function defined for all real values of x and we want to see if anything can be said about the monotonicity of h from that of f. (JEE 1998) Here we cannot begin by writing $h'(x)$ as $f'(x) - 2f(x)f'(x) + 3(f(x))^2 f'(x)$ because $f'(x)$ is not given to exist. So Theorem 3 is inapplicable to $h(x)$ directly. However, if we call $f(x)$ as y, then we can write $h(x)$ as $y - y^2 + y^3 = g(y)$, say. As a function of y, g is differentiable with $g'(y) = 1 - 2y + 3y^2 = (1-y)^2 + 2y^2 > 0$ for all y. So $g(y)$ is a strictly increasing function of y. Since $h(x) = g(f(x))$ it now follows that if f increases as x increases then h will also increase (although not necessarily strictly) and similarly, if f is decreasing then so is h.

Sometimes, there are situations where an inequality to be proved is obvious from other considerations and the method given in Comment No. 11 above based on Theorem 3 is of little or no help. As a glaring example, we have $\sin x < 2$ for all x. Here taking derivatives is not going to help because $\cos x$ is not necessarily less than 0. (It should be noted that a bigger function does not necessarily have a bigger derivative.) Similarly, in Chapter 6, Comment No. 9, we proved that $\cos(\sin\theta) \geq \sin(\cos\theta)$ for all $\theta \in [0, \frac{\pi}{2})$ by showing that $\cos\theta$ lies between the two. Taking derivatives will only complicate the inequality here.

Comment No. 13:

Apart from requiring the existence of derivatives, a more serious limitation of Theorem 3 is that the derivative of the function f must maintain its sign throughout on the interval (a, b). If f' changes sign at just one intermediate point, say c, then correspondingly, the function f changes its behaviour from increasing to decreasing or vice versa. Indeed, this was the very essence of applying Theorem 3 to problems of finding maxima and minima. But what if f' changes its sign at more than one point? In that case, Theorem 3 cannot be applied so directly to find the maximum/minimum of the function f on the entire interval $[a, b]$. Nevertheless, the reasoning still goes through to give what is called a **local extremum**. Just as a local champion is a champion in a restricted locality, a local maximum is a maximum over a small neighbourhood. In precise terms, we say that a function f defined over an interval has a local maximum at an interior point c of S if there exists some $\delta > 0$ such that the δ-neighbourhood, i.e., the interval $(c - \delta, c + \delta)$ is contained in S and further that for every x in this neighbourhood, we have $f(c) \geq f(x)$. A local minimum is defined analogously. The actual value of δ does not matter as long as it is

positive. In real-life we may give more respect to a local champion depending on the size of the locality over which he is the champion. But as far as local maxima/minima are concerned we do not do so. Of course, if a function f attains its maximum over the entire interval S at a point c and c is an interior point, then automatically c is also a local maximum. But the converse is false, as is to be expected and will soon be shown by examples.

To stress the difference between a maximum and a local maximum, the former is often called a **global maximum**. The terms **absolute maximum** and **relative maximum** are also used sometimes to mean a global and a local maximum respectively. Note that a global or absolute maximum can occur at an end-point as well. But a local or a relative maximum can occur only at an interior point. (Some authors do allow a local maximum/minimum to occur at an end-point. In that case the definition is slightly modified so as to require conditions like $f(c) \geq f(x)$ to hold for x only on one side of c, viz., the side which is a part of the set S.) Another point of difference between an absolute and a relative maximum is that even though an absolute maximum of a function f can sometimes occur at more than one point (e.g., the function $f(x) = \cos x$ is maximum at all even multiples of π), the values of the function f at all these points are equal. This need not be so for relative maxima as can be shown, for example, for the function $f(x) = \frac{1}{5}x^5 - \frac{3}{2}x^4 + \frac{11}{3}x^3 - 3x^2$ which has two local maxima, one at 0 and another at 2.

The reasoning given in Theorems 2 and 3 goes through to give us the following important result.

Theorem 4: Suppose f is a real-valued function defined on an interval S and c is an interior point of S. Then :

(i) if f has a local maximum/minimum at c, then c is a critical point of f, i.e., $f'(c)$ either fails to exist or vanishes.

(ii) if f is continuous at c and there exists some $\delta > 0$ such that $(c-\delta, c+\delta) \subset S$ and further $f'(x) \geq 0$ for all $x \in (c-\delta, c)$ and $f'(x) \leq 0$ for all $x \in (c, c+\delta)$, then f has a local maximum at c.

(iii) if f is continuous at c and there exists some $\delta > 0$ such that $(c-\delta, c+\delta) \subset S$ and further $f'(x) \leq 0$ for all $x \in (c-\delta, c)$ and $f'(x) \geq 0$ for all $x \in (c, c+\delta)$, then f has a local minimum at c.

Let us illustrate the concepts just defined and the procedure for obtaining them.

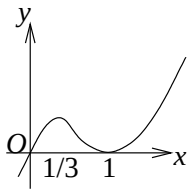
Find all maxima and minima of the function

$$y = x(x-1)^2, \;\; 0 \leq x \leq 2 \;\; \text{(JEE 1989)} \tag{9}$$

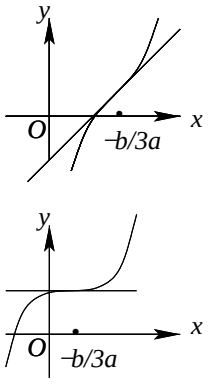
Call the given function as $f(x)$. Then $f(x) = x^3 - 2x^2 + x$ and so $f'(x) = 3x^2 - 4x + 1$ which factors as $(3x-1)(x-1)$. This gives $x = \frac{1}{3}$ and $x = 1$ as two critical points. Both lie in the interval $[0, 2]$. The factorisation of $f'(x)$ also shows that $f'(x) < 0$ for $\frac{1}{3} < x < 1$ and $f'(x) > 0$ when either $x < \frac{1}{3}$ or when $x > 1$. So by the theorem above, f has a relative or a local maximum

at $\frac{1}{3}$ and a relative or a local minimum at 1. The values of f at these points are $\frac{4}{27}$ and 0 respectively. To find the global maxima/minima over the interval $[0, 2]$, we have to also take into account the values of f at the end-points 0 and 2. Clearly, $f(0) = 0$ and $f(2) = 2$. A comparison of $f(0), f(\frac{1}{3}), f(1)$ and $f(2)$ shows that f has its global maximum on $[0, 2]$ at 2 while its minimum occurs at two places, viz., 0 and 1. (If instead of the closed interval $[0, 2]$ we consider the open interval $(0, 2)$ then the minimum is attained only at 1 while the maximum is not attained at all, even though there is a local maximum at $\frac{1}{3}$.)

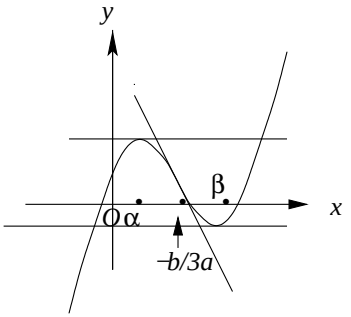
Although not asked in the problem, it is instructive to draw the graph of $f(x)$. We already know four points on the graph, viz., $(0, 0), (\frac{1}{3}, \frac{4}{27}), (1, 0)$ and $(2, 2)$. We also know that the tangents at the points $(\frac{1}{3}, \frac{4}{27})$ and $(1, 0)$ are horizontal and further that there is a local maximum at $(\frac{1}{3}, \frac{4}{27})$ and a local minimum at $(1, 0)$. Hence small portions of the graph near these points must look like a hat $(\frown)$ and an inverted hat $(\smile)$ respectively. Moreover, from the sign of $f'(x)$, we know that f is increasing on $(0, \frac{1}{3})$, decreasing on $(\frac{1}{3}, 1)$ and then again increasing on $(1, 2)$. This information is sufficient to give a reasonably good sketch of the graph as shown (without scale).



The same procedure can be applied to draw the graph of a cubic, say $y = f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$. Again, we may suppose, without loss of generality, that $a > 0$ as otherwise we first draw the graph of $-f(x)$ and then take its reflection. Here $f'(x) = 3ax^2 + 2bx + c$ which is a quadratic in x. Its discriminant is $4(b^2 - 3ac)$ and depending upon its sign, three cases arise (cf. Chapter 3, Comment No. 11). If $b^2 < 3ac$ then $f'(x) > 0$ for all x and so the graph of $y = f(x)$ is strictly rising on the entire real line. If $b^2 = 3ac$, then the derivative is positive except at one point where it is 0. So in this case also f is strictly increasing (cf. Exercise (13.8)). In both these cases the graphs are as in Figure (a) below.



(a) $a > 0, b^2 \leq 3ac$



(b) $a > 0, b^2 > 3ac$

In the remaining case where $b^2 - 3ac > 0$, f' has two real zeros, say

α and β with $\alpha < \beta$. With the same reasoning as given above for (9), there is a local maximum at α and a local minimum at β and the graph looks like that in Figure (b) above. (In both (a) and (b), something interesting happens at $x = -\frac{b}{3a}$, which we shall discuss later. Note that in (b) this is precisely the arithmetic mean of α and β.)

More generally if $f(x)$ is a polynomial of degree n then $f'(x)$ is a polynomial of degree $n-1$ and so there could be as many as $n-1$ points of local maxima/minima. It is easy to show that if these points are all distinct then they must alternate with each other. That is, between any two local maxima there is at least one local minimum and vice versa. This follows by factorising $f'(x)$ as $a(x-c_1)(x-c_2)\ldots(x-c_{n-1})$, where $a \neq 0$ and $c_1 < c_2 < \ldots < c_{n-1}$. There is a change of sign of $f'(x)$ every time x crosses one of the c's.

Comment No. 14:

So far we studied the maxima and minima of real-valued functions of one real variable. In real-life problems, we frequently encounter functions of two or more variables. For example, our diet consists of so many types of food items and therefore its nutritive value (and also its cost) is a function of what quantities of these food items we consume. So problems of optimising functions of several real variables are very important from a very practical point of view. But they are considerably more complicated and are, in fact, beyond the scope of JEE. Still, we briefly consider the simplest among them, viz., where we have to optimise a function, say $f(x, y)$, of two real variables x and y as the point (x, y) varies over some subset, say S, of the xy-plane.

The simplest among these are those where the set S is a part of some plane curve such as a straight line or an ellipse. In such a case the two variables x and y are related to each other by the equation of that curve and so effectively f is a function of just one real variable. We already encountered an instance of this in the problem done in Comment No. 8 above, where we broke a wire of length L into two parts. Let x and y denote the lengths of these two parts. Then the objective function is the sum of the two areas, viz., $\dfrac{x^2}{4\pi} + \dfrac{y^2}{16}$. However, here x and y always lie on the line $x + y = L$ and so we could express f as a function of just one variable x as in (6).

If instead of a straight line, the curve happens to be more complicated, such as a circle or a hyperbola, then we can use its parametric equations to express the objective function as a function of just one variable. We can then apply the methods we have studied above to optimise it. As an illustration, we have the following problem.

Find the point on the curve

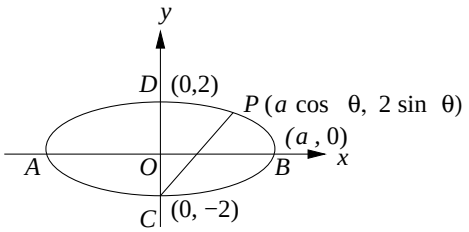
$$4x^2 + a^2y^2 = 4a^2, \quad (4 < a^2 < 8) \tag{10}$$

that is farthest from $(0, -2)$. (JEE 1987)

The objective function in this problem is the distance of a point (x, y) from $(0, -2)$, i.e., $\sqrt{x^2 + (y+2)^2}$. But since the distance is always non-negative, minimising it is the same as minimising its square, which is a little easier to

handle. (Such simplifying tricks, although mathematically not very significant, often save you a lot of time and risk of errors.) So we take $f(x,y) = x^2+(y+2)^2$ as the function to be minimised as the point (x,y) satisfies (10).

Rewriting (10) as $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$, we see that it is an ellipse. Its standard parametrisation is $x = a\cos\theta,\ y = 2\sin\theta$, where $0 \le \theta \le 2\pi$. Substituting these, our objective function becomes $g(\theta) = a^2\cos^2\theta+(2\sin\theta+2)^2$ and we have to maximise it over the interval $[0, 2\pi]$. Differentiating, $g'(\theta) = -2a^2\cos\theta\sin\theta+ 8(\sin\theta+1)\cos\theta$. Setting $g'(\theta) = 0$ we run into two possibilities : (i) $\cos\theta = 0$ and (ii) $a^2\sin\theta = 4\sin\theta + 4$, i.e., $\sin\theta = \dfrac{4}{a^2-4}$. The first possibility gives $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$. As a^2 is given to lie between 4 and 8, the ratio $\dfrac{4}{a^2-4}$ lies between 1 and ∞. So the second possibility has no solution. This gives $\frac{\pi}{2}$ and $\frac{3\pi}{2}$ as the only critical points of $g(\theta)$. We can now calculate the values of g at these critical points and also at the end points 0 and 2π and compare the four values to see where g attains its maximum. This can be done very easily geometrically. Let AB and CD be the major and minor axes of the ellipse. Then the point $(0,-2)$ is C, the lower end of the minor axis of the ellipse. The values $0, \frac{\pi}{2}, \frac{3\pi}{2}$ and 2π of the parameter θ correspond to the points B, D, C and B again on the ellipse. Clearly, $CD = 4$ while $CB = \sqrt{a^2+4}$. As we are given that $a^2 < 8$, $CD > CB$ and so the farthest point from C is D, i.e., the point $(0,2)$.



Note the role played by the restriction $4 < a^2 < 8$ in this problem. If $a^2 < 4$, then CD would be the major and not the minor axis of the ellipse. The point farthest from C would still be D. When $a^2 = 4$, (10) represents a circle and D is the point diametrically opposite to C. If a^2 were greater than 8, say $a^2 = 12$ then something interesting happens. In this case, $\dfrac{4}{a^2-4} = \frac{1}{2}$ and so the possibility (ii) in the solution above gives $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ as two more critical points of $g(\theta)$. Let the corresponding points on the ellipse be E and F. They are symmetrically located about the minor axis and now the farthest point from C is not D but either of E and F. As θ varies from 0 to 2π, the corresponding point, say P, on the ellipse starts moving counter-clockwise. Its distance from C goes on increasing till it reaches E. Thereafter it starts falling till D, then rises again till F, falls again till C (which is certainly the closest point to C) and then again rises till P comes back to B. In other words, the distance function, or equivalently, the function $g(\theta)$, has a local minimum at D (i.e., at $\theta = \frac{\pi}{2}$) and another local minimum at C (i.e., at $\theta = \frac{3\pi}{2}$). The latter is also the absolute minimum. The former is only a relative minimum. At E and F (i.e., when $\theta = \frac{\pi}{6}$ and $\frac{5\pi}{6}$), the distance function has a global as well as a local maximum. (We invite you to draw a suitable diagram to see this.)

There is a purely geometric way to do the problem above. For $r > 0$, let

C_r be the circle of radius r centred at C. The equation of C_r is

$$x^2 + (y+2)^2 = r^2 \tag{11}$$

As r varies so does C_r and hence its points of intersection with the ellipse given by (10). For small values of r, C_r cuts the ellipse at exactly two points while for very large values of r, the ellipse will lie entirely inside C_r and hence C_r will not intersect the ellipse at all. Now suppose Q is a point on the ellipse which is farthest from C and let M be the distance CQ. Then we claim that C_M, i.e., the circle of radius M with centre at C, touches the ellipse at Q, i.e., the two have a common tangent at Q. For, if this is not the case, then the ellipse will cut the circle C_M at the point Q. That means a small portion of the ellipse on one side of the point Q will lie inside the circle C_M while a small portion on the other side will lie outside C_M. But this would mean that on the ellipse there are points that are farther from C than Q is, contradicting that Q is a farthest point.

This important observation reduces the search for Q to a point (x_0, y_0) on the ellipse given by (10) at which the tangent is perpendicular to the line joining C to Q. This gives the slope of the tangent to the ellipse at Q as $-\dfrac{x_0}{y_0+2}$. But, on the other hand, it is well-known that the slope of the tangent to the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1$ at (x_0, y_0) is $-\dfrac{4x_0}{a^2 y_0}$. Equating the two slopes we get either $x_0 = 0$ or $y_0 = \dfrac{8}{a^2 - 4}$. If $4 < a^2 < 8$ this would force $y_0 > 2$ which is impossible if (10) is to hold. So $x_0 = 0$ is the only possibility which gives $(0, 2)$ as the farthest point. However, if $a^2 = 12$, say, as considered above, then $y_0 = 1$ is indeed a possibility. The corresponding values of x_0 are $\pm\frac{\sqrt{3}}{2}a$. And both these values give farthest points from C. In this case $x_0 = 0$ gives $y_0 = \pm 2$. At $(0, -2)$ the distance is smallest, while at $(0, 2)$ there is a local minimum, exactly the same answer we got earlier. (At a local minimum also, the circle passing through it must touch the ellipse, since nearby points on the ellipse on both the sides of a local minimum must lie outside the circle.)

This geometric method can be put in a general setting. Let $f(x, y) = x^2 + (y+2)^2$ and $h(x, y) = 4x^2 + a^2 y^2$. Then f is the function which we seek to minimise on the set S given by (10). We can rewrite (10) and (11) as

$$h(x, y) = k_1 \tag{12}$$

$$\text{and } f(x, y) = k_2 \tag{13}$$

respectively, where $k_1 = 4a^2$ and $k_2 = r^2$. Note that k_1 and k_2 are constants. Thus the ellipse is the set of points where $h(x, y)$ has a a certain constant value while the circle C_r is the set of points where the function $f(x, y)$ has a certain constant value. It is customary to express this by saying that the ellipse is a **level curve** of the function h while C_r is a level curve of the function h. By definition, a level curve[4] of a function of two variables is the set of those

[4]Depending on what the function measures, the level curves are sometimes given some

points where the function assumes a given constant function. As this constant changes we get different level curves of the same function. Usually the various level curves of the same function have some common feature. For example the level curves of the functions f and h above are concentric circles and ellipses respectively. (Some level curves may degenerate into a point or may even be empty. We usually exclude such degenerate level curves.)

The problem of finding a point on (10) which is farthest from $(0, -2)$ can be paraphrased as the problem of maximising the function $f(x, y)$ subject to the **constraint** (12) (which is the same as (10)). This type of optimisation is called a **constrained optimisation**. Sometimes there may be more than one constraints. Also in the problem above the constraint was in the form of an equality. But it is common to encounter problems in which at least some of the constraints are in the form of inequalities. In fact, many important real-life problems of optimisation belong to this category including those of linear programming mentioned in Chapter 6, Comment No. 12.

The geometric method given above can be generalised to say that if $Q = (x_0, y_0)$ is a point of a local extremum of a function $f(x, y)$ whose constraint is a level curve of some function $h(x, y)$, then at Q, the level curves of f and h touch each other, i.e., they have the same tangent. So the problem reduces to finding the tangents to level curves at given points on them. One way to do this is by casting the level curve in a parametric form. But if we do so, then we are missing the real strength of this method. For, in that case we might as well express the objective function $f(x, y)$ in terms of the parameter used to parametrise the level curve of the function h (as we did in the first solution to the problem above). The real beauty of the geometric method is that there is a way to find the tangent to a level curve *without* parametrising it. For example, one can resort to implicit differentiation to be studied in Chapter 15.

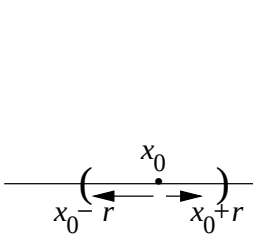
The geometric method is also applicable to constrained optimisation of functions of more than two variables, when the constraints are in the form of equalities. It is called the method of **Lagrange multipliers**. It will take us too far afield to discuss it. But to those who may be already knowing it, we shall illustrate its power in the next chapter on triangular optimisation.

Comment No. 15:

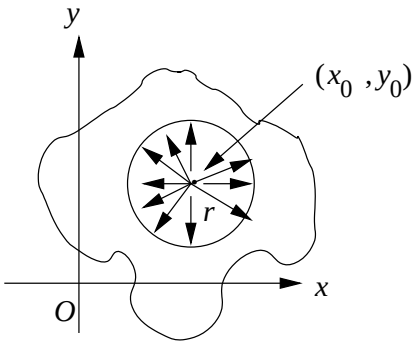
A level curve of a function of two variables generally arises as the boundary of some region in the plane. For example, the level curves (12) and (13) above are the boundaries of an elliptic region and a disc. We skip the formal definition of a region. Intuitively, a region is 'thick' and has a (positive) area while a curve is 'thin' and has no area (or zero area if you prefer). These are very crude descriptions, of course. The point to note is that a curve can be parametrised in terms of a single parameter. A region cannot. As a result, neither of the two methods studied in the last section is applicable when we have to optimise a

popular names. For example, in a topographical map, all points which are at a constant altitude above sea level are said to form a **contour**. Similarly, in a weather map, the level curves of the temperature function are called **isotherms** while those of barometric pressure are called **isobars**.

function $f(x, y)$ over a set S which is a plane region. This type of optimisation is beyond the JEE level. We remark, however, that the analogue of Theorem 2 holds for this type of optimisation also. To understand it we first need to know what is meant by an interior point of a set S. This is easy. In Comment No. 12 of Chapter 6, we defined the concept of an open disc of radius r centred at a point (x_0, y_0) as the set of all points (x, y) in the plane which are at a distance less than r from (x_0, y_0), i.e., the set $\{(x, y) : (x-x_0)^2+(y-y_0)^2 < r^2\}$. We say that a point (x_0, y_0) is an **interior point** of a set S in the plane if there exists some $r > 0$ such that the set $\{(x, y) : (x - x_0)^2 + (y - y_0)^2 < r^2\}$ is contained in S. Obviously, this is in direct analogy with the concept of an interior point of an interval. But there is an important difference. In the case of an interval, if c is an interior point, there are only two directions in which you can move starting from c, either the forward or the backward direction as shown in Figure (a) below. But in the case of an interior point of a subset of the plane, there are infinitely many directions in which you can move starting from that point as shown in Figure (b) below. Out of these infinitely many directions, there are two which are parallel to the positive and the negative x-axis and two others which are parallel to the positive and the negative y-axis. These four directions may be considered as the principal directions at (x_0, y_0), just as even though there are infinitely many directions, we regard East, West, North and South as the four principal directions.



(a) interior point in real line



(b) interior point in a plane

The essential idea in Theorem 2 was that if a function of one variable attains its maximum on a set S at an interior point c then no matter whether you move forward or backward from c, the value of f will decrease (or at least, remain constant). The analogous result for a function of two variables says that if its maximum is attained at an interior point, say (x_0, y_0), then no matter which of the infinitely directions you start moving from (x_0, y_0), the value of the function $f(x, y)$ must decrease. In particular this is the case if you move along the four principal directions which are parallel to the axes. As a result, in the case of a function of one variable, $f'(c)$ must vanish if it exists. The analogous assertion for functions of two variables involves the concept of what is called the **partial derivative** of $f(x, y)$ w.r.t. each of the two variables. In

essence, the partial derivative of a function w.r.t. x is obtained by treating it only as a function of that variable, holding all other variables fixed. Thus, for example, if $f(x,y) = x^2y^3$, then its partial derivative w.r.t. x denoted variously by $\frac{\partial f}{\partial x}$ or $f_x(x,y)$ or D_xf is $2xy^3$ while $\frac{\partial f}{\partial y}$ (or $f_y(x,y)$ or D_yf) is $3x^2y^2$. The analogue of Theorem 2 for functions of two variables says that if $f(x,y)$ attains its extremum at an interior point and the partial derivatives exist at that point then they must vanish. For example, the function $f(x,y) = x^2+3y^2$ is minimum at $(0,0)$ and indeed both the partial derivatives vanish there.

We shall not do problems of optimisation of a function of two variables over a region as they are beyond our scope. Nevertheless the idea that from a topmost point, no matter which direction you take you can only go downward appears in the solution of the following problem. The problem itself belongs to discrete mathematics because the variables involved are discrete and not continuous. But the key idea is the one just stated.

Suppose m and n are some positive integers. Assume that the mn squares of a rectangular board of size $m \times n$ are filled with real numbers in such a way that the entry in each square is the arithmetic mean of those which are adjacent to it either rowwise or columnwise. Prove that the numbers in all the squares must be the same. (JEE 1982)

The problem can be paraphrased in terms of an $m \times n$ matrix, say $\mathbf{A} = (a_{ij})$. If we rewrite a_{ij} as $f(i,j)$ we see that a matrix can be thought of as a function of two discrete variables, say, i and j. Although this is of little help in solving the problem, it inspires the idea that the problem can be looked at as a problem about a function of two variables. A square can have 2, 3 or 4 neighbours depending upon where it is located because there are four possible principal directions in which you can move from a given square. Since the entry in every square is the A.M. of those in its neighbours, some of these will have possibly higher entries and the others lower entries than the square with which we start. But if we start from a square with the highest entry in it then no matter which direction we take, we can only go downward. That means all the neighbours must have entries not exceeding the entry in that square. But in that case, the only way their average can be the entry in that square is that they are all equal to the entry in that square.

Once this key idea strikes us, the solution itself is easy. As there are only finitely many entries, there will be one which is largest among them all. Call it M and consider any square which has M in it. Then by what we said above, all its neighbours must also have M in them. We can now consider the neighbours of these neighbours. Then they must also have M in them. Continuing like this, all the squares will have M in them. (The argument is reminiscent of the spread of an infectious disease.)

Comment No. 16:

In special circumstances a function of two variables can be optimised over a region by geometric arguments. Suppose for example, that C_1 and C_2 are two plane curves and we want to find the shortest distance between them, i.e., a pair of points (P_0, Q_0) with P_0 on C_1 and Q_0 on C_2 such that for every pair of points

(P, Q) with P on C_1 and Q on C_2, $P_0Q_0 \leq PQ$. If the two curves intersect then trivially the shortest distance between them is 0 and we can take P_0 and Q_0 to be any point of intersection of C_1 and C_2. So the problem becomes interesting only when C_1 and C_2 have no point in common. We can parametrise C_1 and C_2 as, say,

$$C_1 \quad : \quad x = f_1(s),\ y = g_1(s) \tag{14}$$

$$\text{and } C_2 \quad : \quad x = f_2(t),\ y = g_2(t) \tag{15}$$

where the parameters s and t vary over some intervals, say S_1 and S_2 respectively. Let $S = S_1 \times S_2 = \{(s,t) : s \in S_1, t \in S_2\}$. Then the set S is a region in the st-plane. (It is a rather special type of region, called a rectangle with sides S_1 and S_2.) Define the real-valued function $d : S \longrightarrow \mathbb{R}$ by $d(s,t) = (f_1(s) - f_2(t))^2 + (g_1(s) - g_2(t))^2$. d is nothing but the square of the distance between the point on C_1 corresponding to s and the point on C_2 corresponding to t on C_2. So the problem of finding the shortest distance between C_1 and C_2 is equivalent to that of minimising the function d of two variables s and t over the region S in the st-plane. As remarked above, we are not in a position to tackle such problems in general. However, for the present problem, the same reasoning that was given in the second solution to the problem in Comment No. 14 above shows that if (P_0, Q_0) is the closest pair of points then the line joining P_0 and Q_0 is perpendicular to the tangent to C_1 at P_0 (as otherwise the circle with centre Q_0 and radius Q_0P_0 will contain some points of the curve C_1 inside it.) Similarly, the line joining P_0 and Q_0 is perpendicular to the tangent to C_2 at Q_0. Let s_0 and t_0 be the values of s and t that correspond to the points P_0 and Q_0 respectively. Because of the parametrisations (14) and (15), the slope of the tangent to C_1 at P_0 is $\dfrac{g_1'(s_0)}{f_1'(s_0)}$ while the slope of the tangent to C_2 at Q_0 is $\dfrac{g_2'(t_0)}{f_2'(t_0)}$. So the perpendicularity of these tangents to the line P_0Q_0 gives us the following system of two equations in two unknowns (viz., s_0 and t_0)

$$\frac{g_1'(s_0)}{f_1'(s_0)}\,\frac{g_2(t_0) - g_1(s_0)}{f_2(t_0) - f_1(s_0)} = -1 \tag{16}$$

$$\text{and} \quad \frac{g_2'(t_0)}{f_2'(t_0)}\,\frac{g_2(t_0) - g_1(s_0)}{f_2(t_0) - f_1(s_0)} = -1 \tag{17}$$

Here the functions f_1, f_2, g_1 and g_2 and their derivatives are known to us. So this system can, at least in theory, be solved to give values of s_0 and t_0. Usually, there will be more than one solution. (For example, if both the curves are bounded, e.g., an ellipse and a circle, then the farthest pair of points will also give rise to a solution. And sometimes there may be other solutions as well, as we saw in Comment No. 14 above for the case $a^2 = 12$.) In such cases the closest pair can be obtained by comparing the values of d for the various possible solutions of the system (16) and (17). But the calculations are generally too

involved and so methods taking advantage of the special features of the curves C_1 and C_2 work better in a given problem. (See Exercise (13.23) for instance.)

Sometimes the maxima/minima of a function $f(x, y)$ of two variables can be found by *ad-hoc* methods such as completion of squares. Yet another method is to fix one of the two variables, say y, and let $M(y)$ be the maximum (or the minimum) of $f(x, y)$ treated as a function of x, holding y as a constant. The maximum (or the minimum) of $M(y)$ as y varies will give the maximum (or the minimum) of $f(x, y)$. See Exercise (24.91) for an illustration. In principle, this is as natural as finding the tallest resident in a building by first finding the tallest resident on each floor and then finding the tallest among these tallest. But this method is seldom used because the calculations are often complicated. See Exercise (24.92) for optimisation of a linear function over a convex polygon.

Comment No. 17:

Note that so far we have not even once referred to the second derivative of the objective function in any problem. This may come as a surprise because the so-called 'second derivative test' is very popular (possibly because of the mechanical manner in which the derivatives can often be obtained) both with students and examiners. So, typically, in order to solve a maxima/minima problem, a student differentiates the objective function (which is an easy, almost mechanical task) and sets the derivative equal to 0. The resulting equation usually has only one solution. Even if there are any others, they can usually be ruled out on practical grounds, e.g., if such solutions would imply some box to have an infinite volume or some object to have a negative mass! The lone solution that is left is then subjected to the so-called **second derivative test**, i.e., the second derivative of the function to be optimized is found (another mechanical, although sometimes a bit laborious task) and evaluated at this point where the first derivative vanished. Depending upon whether the second derivative is positive or negative the point is declared as the point where the function has a minimum or maximum.

While this procedure is not totally baseless, the mechanical and sometimes hasty manner in which it is applied deserves some rebuttal. The last step especially, i.e., the second derivative test, is inadequate by itself. What is more, even in those problems where it works, it is dispensable and can be replaced by a careful study of the first derivative (which can usually be done with just a little more work than that involved in locating the zeros of the first derivative) as we have done in the solution of the Main Problem and also that of the problem in Comment No. 10. Before showing how these problems could also have been done using the second derivative test, let us first state, without proof, the result which establishes its validity.

Theorem 5: Suppose S is any interval (closed, open or semi-open) and f is differentiable on S. Suppose c is the only interior point of S at which $f^{'}$ vanishes. Then if $f^{''}(c) > 0, f$ attains its (global) minimum on S at c while if $f^{''}(c) < 0$, then f attains its (global) maximum on S at c.

Let us apply this theorem to some of the optimisation problems we already solved without the use of second derivative. In the Main Problem we saw that

$\frac{\pi}{6}$ was the only critical point of $f(\theta) = \frac{10\sec\theta}{u} + \frac{15-10tan\theta}{2u}$ in the interval $[0, \alpha]$, where $\alpha = \tan^{-1}\frac{3}{2}$. We already computed $f'(\theta)$ in (2). Further differentiation gives $f''(\theta) = \frac{10\sec^3\theta+10\tan^2\theta\sec\theta}{u} - \frac{10\sec^2\theta\tan\theta}{u} = \frac{10\sec\theta(\sec^2\theta+\tan^2\theta-\tan\theta\sec\theta)}{u}$. As we are interested only in the sign and not the actual value of $f''(\frac{\pi}{3})$, we observe that the denominator and the first factor of the numerator are positive for all $\theta \in (0, \frac{\pi}{2})$. The positivity of the second factor follows by rewriting it as $(\sec\theta - \tan\theta)^2 + \tan\theta\sec\theta$ and noting again that both $\tan\theta$ and $\sec\theta$ are positive on $(0, \frac{\pi}{2})$. In particular, $f''(\frac{\pi}{6}) > 0$. As it is the only critical point, by Theorem 5, f attains its minimum at $\frac{\pi}{3}$.

Similarly, in the problem solved in Comment No. 10, we had $f'(r) = \frac{r(8-3r^2)}{4\sqrt{4-r^2}}$ which gives $f''(r) = \frac{\sqrt{4-r^2}(8-6r^2) + \frac{r^2(8-3r^2)}{\sqrt{4-r^2}}}{4(4-r^2)}$. The only critical point of f in $(0, 2)$ was $\frac{2\sqrt{2}}{\sqrt{3}}$. At this point, $r^2 = \frac{8}{3}$ and hence $8 - 6r^2 = -8 < 0$. The second term in the numerator vanishes. As $4 - r^2 > 0$, we see that $f''(\frac{2\sqrt{2}}{\sqrt{3}}) < 0$, even without computing it explicitly. Hence by Theorem 5, $f(r)$ is maximum when $r = \frac{2\sqrt{2}}{\sqrt{3}}$. (We could have saved some time by not computing the second term in the numerator of $f''(r)$, because no matter what it is, we know that it will have $(8 - 3r^2)$ as a factor and hence will vanish when $r = \frac{2\sqrt{2}}{3}$. Such alertness pays off in time-constrained examinations. More generally, this trick is applicable when the first derivative, say $f'(x)$, is of the form $\frac{u(x)}{v(x)}$. If $f'(x_0) = 0$, then $u(x_0) = 0$ and the sign of $f''(x_0)$ depends solely on $u'(x_0)v(x_0)$.)

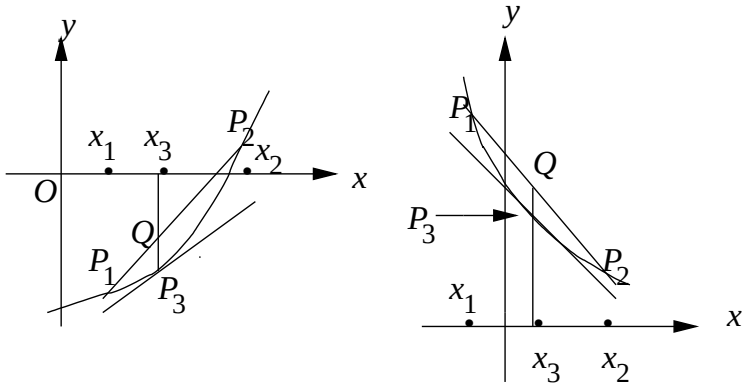
Strictly speaking, whenever we apply Theorem 5, it should be stressed that there is only one interior point where the derivative vanishes. This is often skipped. But the theorem may fail without this hypothesis. Let us revisit the problem done in Comment No. 13, of finding all maxima and minima of the function $f(x) = x(x-1)^2$ over the interval $[0, 2]$. We showed that there were two critical points, viz., $\frac{1}{3}$ and 1. Both are interior points of $[0, 1]$. Here $f''(x) = 6x - 4$. Then $f''(\frac{1}{3}) = -2 < 0$ while $f''(1) = 2 > 0$. But the function does not have a global maximum at $\frac{1}{3}$. It does have a global minimum at 1, but this will not be so if we change the domain to, say, $[-1, 2]$. In that case both the global maximum and minimum will occur at the end-points of $[-1, 2]$ and neither is captured by Theorem 5. What is captured by Theorem 5 is not the global but the local maximum and minimum. So Theorem 5 is really speaking a theorem about *local* extrema of a function. It is just that when there is only one local extremum and no other interior critical point, this lone local extremum becomes the global extremum, rather like an unopposed contestant being declared the champion.

Thus we may restate Theorem 5 to say that if f is differentiable everywhere on S, c is an interior point of S, $f'(c) = 0$ and $f''(c)$ exists, then there is a local maximum at c if $f''(c) < 0$ and a local minimum at c if $f''(c) > 0$. Even in this new form, Theorem 5 is always dispensable and sometimes less applicable than an argument based on Theorem 3. If $f'(c) = 0$ and $f''(c) > 0$

then f' is increasing at c, i.e., in some δ-neighbourhood of c we shall have $f'(x) > f'(c)$ and hence $f'(x) > 0$ for all $x \in (c, c+\delta)$ and similarly $f'(x) < 0$ for all $x \in (c-\delta, c)$. But then we could as well apply Theorem 3 to conclude that f has a relative minimum at c. A similar remark holds for a relative maximum. On the other hand there are situations where Theorem 3 can be applied but not the second derivative test. This can happen for two reasons. First, $f''(c)$ may not exist, i.e., f may not be twice differentiable at c, even though it is differentiable everywhere. An example of such a function will be given in Chapter 15, Comment No. 13. Secondly, the second derivative test is silent as to what happens if $f''(c) = 0$. As a simple example, take the function $f(x) = x^4$. Evidently, it has a local minimum, in fact a global minimum at 0. However we cannot tell this from the second derivative because $f''(x) = 12x^2$ vanishes at 0. But we can apply Theorem 3. Since $f'(x) = 4x^3$ which is positive for $x > 0$ and negative for $x < 0$, we get that f is decreasing on $(-\infty, 0)$ and increasing on $(0, \infty)$ and hence has a global minimum at 0. We shall take this up again in Chapter 16, Comment No. 13.

Comment No. 18:

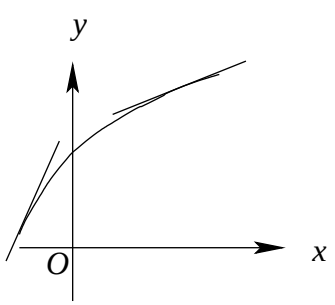
Thus, despite its popularity, the second derivative is really not such a good tool for finding maxima and minima. But there is something else for which the second derivative plays a very crucial role. It is analogous to the role of the first derivative in testing on which intervals the graph of a (differentiable) function, say $f(x)$, is rising and falling. Suppose now that the function $f(x)$ is twice differentiable at every interior point of an interval S. Let x_1 and x_2 be any two interior points of S with $x_1 < x_2$. Let $y_1 = f(x_1)$ and $y_2 = f(x_3)$ and let $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$ be the corresponding points on the graph of $y = f(x)$. Then geometrically, $f'(x_1)$ and $f'(x_2)$ are the slopes of the tangents to this graph at the points P_1 and P_2 respectively.



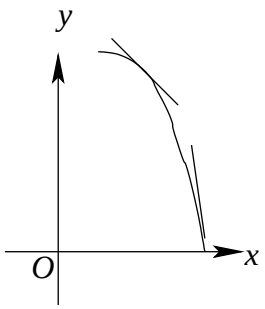
(a) concave upward (b) concave upward

Now suppose that $f''(x) \geq 0$ for all $x \in (x_1, x_2)$. Then by Theorem 3 (applied to f' and not to f), $f'(x)$ increases on the interval $[x_1, x_2]$. Geometrically this means that the slope of the tangent to the curve $y = f(x)$ goes on increasing

as x increases from x_1 to x_2. In other words the tangent bends upwards. (This should not be confused with the rise or fall of the graph, which depends on the sign of the first derivative. The tangent can bend upward regardless of whether the function is increasing as in Figure (a) above or decreasing as in (b) above. Of course in (b), the slope of the tangent is negative. Still it is increasing and the graph is bending upward. As an actual example, consider $f(x) = x^2$ on the interval $[-2, -1]$.) We express this by saying that the function $f(x)$ (or rather its graph) is **concave upwards** on the interval $[x_1, x_2]$.



(c) concave downward



(d) concave downward

Similarly, if $f''(x) \leq 0$ for all $x \in (x_1, x_2)$ then $f'(x)$ decreases as x moves from x_1 to x_2. Geometrically, this means that the tangent is bending downwards. Again this can happen independently of whether f itself is rising as in Figure (c) or falling as in (d). In either case, we say that the function f is **concave downwards** on the interval $[x_1, x_2]$. Evidently, f is concave downwards if and only if $-f$ is concave upwards. The function $f(x) = x^3$ is concave upwards on any interval contained in $[0, \infty)$ and concave downwards on any interval contained in $(-\infty, 0]$.

Let us now see what effect concavity has on the values of the function f at a point, say, x_3 in between x_1 and x_2, First suppose that f is concave upward on $[x_1, x_2]$ as in Figure (a) or (b). Let P_3 be the point (x_3, y_3), where $y_3 = f(x_3)$. Since the tangent is bending upwards as you move from P_1 to P_2, it is intuitively clear that the portion of the graph of $y = f(x)$ between the points P_1 and P_2 lies on or below the chord P_1P_2. In particular, the point P_3 lies at or below the point Q, where this chord meets the vertical line $x = x_3$. The equation of the chord $P_1P_2$is

$$y = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \tag{18}$$

Since the point $P_3 = (x_3, y_3)$ lies on or below this chord, we must have $y_3 \leq y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x_3 - x_1)$. Putting $y_i = f(x_i)$ for $i = 1, 2, 3$, this gives

$$f(x_3) \leq f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x_3 - x_1) \tag{19}$$

For applications, it is convenient to paraphrase this inequality slightly. Let $\lambda_1 = \dfrac{x_2 - x_3}{x_2 - x_1}$ and $\lambda_2 = \dfrac{x_3 - x_1}{x_2 - x_1}$. Since $x_1 \leq x_3 \leq x_2$, λ_1 and λ_2 are both non-negative. Moreover they always add up to 1. A direct computation shows that $x_3 = \lambda_1 x_1 + \lambda_2 x_2$. Geometrically, this means that x_3 divides the interval $[x_1, x_2]$ in the ratio $\lambda_2 : \lambda_1$ (and hence also that the point Q divides the chord P_1P_2 in the ratio $\lambda_2 : \lambda_1$). Now (19) can be rewritten as

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2) \tag{20}$$

which not only looks neater than (19) but has the advantage that it is completely symmetric w.r.t. x_1 and x_2 and hence remains valid even if $x_2 < x_1$. Note that our derivation of (20) began with (18) where we appealed to intuition. A rigorous proof can be given using the Lagrange Mean Value Theorem mentioned earlier in the comments following Theorem 3. We shall do so in Chapter 16.

If f is concave downwards, then we have to reverse the inequality in (20). It can be proved by noting that for such a function, the graph lies on or above the chord. Alternately, simply replace f by $-f$ and use (20), applied to $-f$.

An expression of the form $\lambda_1 x_1 + \lambda_2 x_2$, where λ_1, λ_2 are non-negative and add up to 1, is called a **convex combination**[5] of x_1 and x_2. More generally, we can define a convex combination of any finite number of points. Verbally, (20) says that if a function f is concave upward, then its value at a convex combination of two points cannot exceed the corresponding convex combination of its values at the two points. In this form (20) can be generalised and the result is an important inequality, called **Jensen's inequality** which we mentioned in Chapter 6.

Theorem 6: Suppose $f(x)$ is a twice differentiable real-valued function on an interval S and $f''(x) \geq 0$ at every interior point of S. Then for every positive integer n, for all $x_1, x_2, \ldots, x_n \in S$ and for all non-negative real numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ with $\lambda_1 + \lambda_2 + \ldots + \lambda_n = 1$, we have

$$f(\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_n x_n) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2) + \ldots + \lambda_n f(x_n) \tag{21}$$

If $f''(x) \leq 0$ for all interior points of S, then (21) holds with the inequality sign reversed.

Proof: We apply induction on the integer n. For $n = 1$, λ_1 must equal 1 and so (21) holds trivially, with equality. The case $n = 2$ is precisely the inequality (20). For the inductive step, suppose that the result is true for $n = k$. To prove that it holds for $n = k + 1$, suppose $x_1, x_2, \ldots, x_{k+1} \in S$ and non-negative real numbers $\lambda_1, \lambda_2, \ldots, \lambda_{k+1}$ adding up to 1 are given. If $\lambda_{k+1} = 1$, all other λ's are zero and there is nothing to prove. Otherwise, for

[5]The terms 'concave' and 'convex' are geometric in origin. But we shall not define them in their generality. Let us only say that a set is called **convex** if it has the property that whenever two points lie in it so does the entire line segment joining them. The inside of a triangle, circle and an ellipse are convex sets. An **annulus**, i.e., the portion between two concentric circles is not convex. More generally, no set with a 'hole' in it can be convex.

$i = 1, 2, \ldots, k$ set $\mu_i = \frac{\lambda_i}{1-\lambda_{k+1}}$ and write $f(\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_k x_k + \lambda_{k+1} x_{k+1})$ as $f((1-\lambda_{k+1})(\mu_1 x_1 + \mu_2 x_2 + \ldots + \mu_k x_k) + \lambda_{k+1} x_{k+1})$. Now apply the case $n = 2$ to get that this is at most $(1-\lambda_{k+1}) f(\mu_1 x_1 + \mu_2 x_2 + \ldots + \mu_k x_k) + \lambda_{k+1} f(x_{k+1})$. Putting the two together we get

$$\begin{aligned} f(\lambda_1 x_1 &+ \ldots + \lambda_k x_k + \lambda_{k+1} x_{k+1}) \\ &\leq (1 - \lambda_{k+1}) f(\mu_1 x_1 + \ldots + \mu_k x_k) + \lambda_{k+1} f(x_{k+1}). \end{aligned} \tag{22}$$

Since the λ's are non-negative, so are the μ's. Further, by very construction, $\mu_1 + \mu_2 + \ldots + \mu_k = 1$. So we can apply the truth of the result for the case $n = k$ (with the μ's replacing the λ's) to get that

$$f(\mu_1 x_1 + \mu_2 x_2 + \ldots + \mu_k x_k) \leq \mu_1 f(x_1) + \mu_2 f(x_2) + \ldots + \mu_k f(x_k) \tag{23}$$

If we put (23) into (22) and observe that for $i = 1, 2, \ldots, k$, $(1 - \lambda_{k+1})\mu_i = \lambda_i$, we get (21). The second assertion follows by applying (21) to $-f$.

An especially interesting case of (21) occurs when all the λ's are equal, each then being equal to $\frac{1}{n}$. In that case, under the hypothesis that $f''(x) \geq 0$ for all interior points of S we get that for any $x_1, x_2, \ldots, x_n \in S$,

$$f\left(\frac{x_1 + x_2 + \ldots + x_n}{n}\right) \leq \frac{f(x_1) + f(x_2) + \ldots + f(x_n)}{n} \tag{24}$$

In the next chapter, we shall see that by choosing the function f suitably, (24) can be made to work wonders. In particular, we shall derive the A.M.-G.M. inequality from (24).

We saw earlier that when f' changes sign as x passes through an interior point c, the rising/falling behaviour of f changes and so there is a local extremum at c. Similarly, when $f''(x)$ changes sign as x passes through an interior point c, there is a change of concavity, either from upward to downward or vice versa. Such a point c is called a point of **inflection** of the function f. In the graph of $y = f(x)$, something strange happens at the point $(c, f(c))$. Normally, the tangent to a curve lies entirely on one side of a small portion of the curve containing the point of contact. But at a point of inflection, the tangent must bend upwards on one side and downwards on the other. The only way this can happen is that the tangent must cross the curve at such a point. You can see this easily from the graph of $y = x^3$ for which 0 is a point of inflection. More generally, for the cubic $y = ax^3 + bx^2 + cx + d$, there is a point of inflection at $x = -\frac{b}{3a}$ as shown in Figures (a) and (b) of Comment No. 13.

EXERCISES

13.1 Suppose ABC is an isosceles triangle, with $\angle A = 90°$. Let M be the midpoint of BC and suppose P is a point on the median AM. Consider the Travelling Salesman Problem with the four cities A, B, C and P in which the time taken to go from one city to another is proportional to the shortest distance between them. Show that P and A are the closest cities, but that if $PM < (\frac{9-4\sqrt{2}}{14})BC$, then the shortest tour is $P - C - A - B - P$.

13.2 A window is in the form of a rectangle surmounted by a semi-circle. The rectangle is of clear glass while the semi-circle is of coloured glass which transmits only half as much light per sq.cm. as clear glass does. If the total perimeter is fixed, find the proportions of the sides of the rectangular part of the window which will admit the maximum light. (JAT 1979)

13.3 (a) Let (h, k) be a fixed point, where $h > 0, k > 0$. A straight line passing through this point cuts the positive directions of the coordinate axes at the points P and Q. Find the minimum area of the triangle OPQ, O being the origin. (1995)

(b) A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive coordinate axes at the points P and Q. Find the absolute minimum value of $OP + OQ$ as L varies, where O is the origin. (2002)

13.4 A point P is given on the circumference of a circle of radius r and centre O. Chords QR are parallel to the tangent at P. Maximise : (i) the area of the triangle PQR (ii) the inradius of the triangle OQR. (1990)

13.5 Find the coordinates of the point on the curve $y = \dfrac{x}{1+x^2}$, where the tangent to the curve has the greatest slope. (1984)

13.6 Find the maximum or the minimum value of the function $y = \frac{a^2}{x} + \frac{b^2}{a-x},\ a > 0,\ b > 0$ in the interval $(0, a)$. (JAT 1980)

13.7 Let $A(p^2, -p), B(q^2, q), C(r^2, -r)$ be the vertices of the triangle ABC. A parallelogram $AFDE$ is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Using calculus, show that the maximum area of such a parallelogram is $\frac{1}{4}(p+q)(q+r)(p-r)$. (1986)

13.8 Show that all the conclusions of Theorem 3 remain true even if the interval $[a, b]$ contains a finite number of critical points of f (there being no relaxation of the continuity requirement). Thus, for example, in order to show that a function is strictly increasing on $[a, b]$, it is all right even if its derivative is 0 at a finite number of points, as long as it is positive at all other points of (a, b). Hence show that the function $\sin x + x$ is strictly increasing over the entire real line.

13.9 Let f and g be increasing and decreasing functions respectively from $[0, \infty)$ to $[0, \infty)$. Let $h(x) = f(g(x))$. If $h(0) = 0$, determine which, if any, of the following statements is (are) correct. (1987)

(A) $h(x) - h(1)$ is always zero (B) $h(x) - h(1)$ is always negative
(C) $h(x) - h(1)$ is always positive (D) $h(x) - h(1)$ is strictly increasing.

13.10 Let

$$f(x) = \begin{cases} -x^3 + \frac{(b^3-b^2+b-1)}{(b^2+3b+2)}, & 0 \le x < 1 \\ 2x - 3 & 1 \le x \le 3. \end{cases}$$

Find all possible real values of b such that $f(x)$ has the smallest value at $x = 1$. (1993)

13.11 Let $f(x)$ be a quadratic expression which is positive for all real x. If $g(x) = f(x) + f'(x) + f''(x)$, find out which of the following statements is true for all real x. (1990)

(A) $g(x) < 0$ (B) $g(x) > 0$ (C) $g(x) = 0$ (D) $g(x) \geq 0$.

13.12 If $f(x) = \dfrac{x^2 - 1}{x^2 + 1}$ for all real x, determine which of the following statements is (are) true about the minimum value of f. (1998)

(A) It does not exist because f is unbounded (B) It equals 1
(C) It is not attained even though f is bounded (D) It equals -1.

13.13 Let $P(x) = a_0 + a_1x^2 + a_2x^4 + \ldots + a_nx^{2n}$ be a polynomial in a real variable x with $0 < a_0 < a_1 < a_2 < \ldots < a_n$. Determine which, if any, of the following statements is (are) true about the function $P(x)$. (1986)

(A) it has neither a maximum nor a minimum
(B) it has only one maximum
(C) it has only one minimum
(D) it has only one maximum and only one minimum.

13.14 Determine the intervals on which the function $y = 2x^2 - \ln|x|$ $(x \neq 0)$, is monotonically increasing and monotonically decreasing. (1983)

13.15 Determine whether each of the functions $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$ is increasing/decreasing in the interval $(0, 1]$. (1997)

13.16 (a) Identify the set of all x for which $\ln(1 + x) \leq x$. (1987)

(b) Determine which of $\cos(\ln\theta)$ and $\ln(\cos\theta)$, is larger for $e^{-\pi/2} < \theta < \frac{\pi}{2}$. (1983)

(c) Prove that $(1 + x)\ln(1 + x) + (1 - x)\ln(1 - x) > 0$ and hence that $(1 + x)^{1+x}(1 - x)^{1-x} > 1$ for $0 < x < 1$.

(d) Show that $1 + x\ln(x + \sqrt{1 + x^2}) \geq \sqrt{1 + x^2}$ for all $x \geq 0$. (1983)

(e) Find all linear functions which map $[-1, 1]$ onto $[0, 2]$. (1989)

13.17 (a) Use the function $f(x) = x^{1/x}$, $x > 0$ to determine the bigger of the two numbers e^π and π^e. (1981)

(b) Determine for which positive integer n, the expression $n^{1/n}$ is maximum. [*Hint:* Even though here the variable n is discrete, the expression $x^{1/x}$ makes sense for all positive reals (which includes all positive integers). This is in sharp contrast with the situation in Comment No. 9, where a binomial coefficient such as $\binom{m}{n}$ makes no sense if n is a continuous variable. See Exercise (22.23) for another example where calculus based methods cannot be used for maximising a function of a discrete variable.]

13.18 (i) If $y = a\ln x + bx^2 + x$ has its extremum values at $x = -1$ and $x = 2$, determine a and b. (1983)

(ii) Determine the points of maxima and minima of the function $f(x) = \frac{1}{8}\ln x - bx + x^2$, $x > 0$, where $b \geq 0$ is a constant. (1996)

13.19 Let $f(x) = \sin^3 x + \lambda \sin^2 x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. Find all values of λ for which $f(x)$ has exactly one minimum and exactly one maximum. (1985)

13.20 Find the shortest distance of the point $(0, c)$ from the parabola $y = x^2$. (1982)

13.21 Find the coordinates of all the points P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, for which the area of the triangle PON is maximum, where O denotes the origin and N, the foot of the perpendicular from O to the tangent at P. (1999)

13.22 ABC is an equilateral triangle with side a. Two particles start simultaneously from A and B and move towards B and C in straight lines with uniform speeds u and v respectively. Find the shortest distance between them.

13.23 Suppose C_1 and C_2 are the parabolas $x^2 = y-1$ and $y^2 = x-1$ respectively, Let P be a point on C_1 and Q be a point on C_2. Let P_1 and Q_1 be the reflections of the points P and Q respectively w.r.t. the line $y = x$. Prove that the point P_1 lies on C_2 and the point Q_1 lies on C_1 and $PQ \geq \min\{PP_1, QQ_1\}$. Hence or otherwise determine points P_0 and Q_0 on C_1 and C_2 respectively such that for all pairs (P, Q), where P lies on the parabola C_1 and Q lies on the parabola C_2, $P_0Q_0 \leq PQ$. (2000)
Can the problem be done by the method in of Comment No. 16, i.e. by observing that the line P_0Q_0 is a common normal to C_1 and C_2?

13.24 (a) If $f(x)$ is concave upwards on an interval S and $a, b, c \in S$, with $a < b < c$, prove that $f(a - b + c) \leq f(a) - f(b) + f(c)$.

(b) Which functions are both concave upwards and concave downwards?

13.25 Prove that if c is a point of inflection of $f(x)$ and $f''(c)$ exists then $f''(c) = 0$. Hence find the points of inflection of the sine and the cosine functions. Prove that 0 is not a point of inflection of $f(x) = x^4$ even though $f''(0) = 0$. Give a sufficient condition (in terms of $f'''(c)$) for a point c to be a point of inflection of the function f.

13.26 Give an alternate solution to the problem solved in Comment No. 12 of Chapter 10. [*Hint :* The problem was already reduced to finding $f(S)$ where $S = (-\infty, -1) \cup (0, \infty)$ and $f(x) = \dfrac{x^2 + x}{x - 1}$. Note that $f(1)$ is undefined. From the sign of $f'(x)$ conclude that f is increasing on $(-\infty, -1)$ and also on $(0, 1)$ and further that under f the image of each of these intervals is $(-\infty, 0)$. As for $(1, \infty)$, f is decreasing till $1 + \sqrt{2}$ and increasing thereafter. Calculation of $f(1 + \sqrt{2})$ completes the solution.]

Chapter 14

TRIGONOMETRIC OPTIMISATION

The title is somewhat misleading. If interpreted literally, it would mean the optimisation of trigonometric functions such as the sines and the cosines. We have already studied such optimisation problems in the last chapter. However, we mostly confined ourselves to the case of functions of one variable. In Comment No. 14, we handled the optimisation of a function of two variables subject to some equality constraint, because, in effect, such a constraint reduces the number of variables by one.

In this chapter, we shall consider some special constrained optimisation problems of functions of three variables. Typically, in these problems the three variables are the angles of a triangle ABC (so that the equality constraint satisfied by them is $A+B+C=\pi$) and the objective function is expressible as a function of these angles. The Main Problem is of this type. A clever use of some standard inequalities coupled with some trigonometric identities about triangles gives the solution. Optimisation of this type is also called 'triangular optimisation'. As a consequence we get many characterisations of equilateral triangles (Comment No. 8). Later we show how this sometimes leads to a solution of a system of equations where there are more unknowns than equations (Comments 9 and 10). In Comment No. 12, we consider triangular optimisation where the objective function is a function of the sides of a triangle. The relationship between the two types of triangular optimisations is studied in Comments 17 and 18.

Problems of triangular optimisation do not appear every year in the JEE, possibly because they tend to be rather repetitious, since in the vast majority of cases, the optimum is attained when the triangle is equilateral. A short question may be asked to test awareness of this fact (without proof).

Main Problem : If A, B, C are angles of a triangle, then the maximum value of $\cos A + \cos B + \cos C$, expressed as a reduced rational is ..... . (JEE 1984)

First Hint: Prove that the maximum must occur for an equilateral triangle.
Second Hint: For a fixed value of one of the three angles, show that the maximum must occur when the other two are equal.

Solution: Here the three angles A, B, C are constrained by the requirement that they are non-negative (in fact, positive, if we want to exclude degenerate triangles) and add up to π. Let S be the set of all ordered triples of the form (A, B, C) which satisfy these constraints, i.e.,

$$S = \{(A, B, C) = A \geq 0, B \geq 0, C \geq 0 \text{ and } A + B + C = \pi\}$$

and denote $\cos A + \cos B + \cos C$ by $f(A, B, C)$. Then the problem amounts to finding the maximum value of the function f over the set S.

Now suppose $(A, B, C) \in S$. We claim that unless $A = B$, we can find some $(A', B', C') \in S$ such that $f(A', B', C') > f(A, B, C)$. Indeed we let $C' = C$ and $A' = B' = \frac{A+B}{2}$. Note that $A' + B' = A + B$ and hence $(A', B', C') \in S$. Note that $\cos(\frac{A+B}{2}) = \cos(\frac{A'+B'}{2}) \neq 0$ (except in the degenerate case where $A = B = \frac{\pi}{2}$). Further, $\cos(\frac{A'-B'}{2}) = \cos 0 = 1 > \cos(\frac{A-B}{2})$. This gives,

$$\begin{aligned} f(A, B, C) &= \cos A + \cos B + \cos C \\ &= 2\cos(\tfrac{A+B}{2})\cos(\tfrac{A-B}{2}) + \cos C \\ &< 2\cos(\tfrac{A'+B'}{2})\cos(\tfrac{A'-B'}{2}) + \cos C' \\ &= \cos A' + \cos B' + \cos C' \\ &= f(A', B', C') \end{aligned}$$

By a similar reasoning, unless $B = C$, we can find some $(A'', B'', C'') \in S$ such that $f(A, B, C) < f(A'', B'', C'')$. A similar assertion holds if $C \neq A$. It then follows that when $f(A, B, C)$ is maximum, A, B, C must be all equal and hence each equals $\frac{\pi}{3}$ (since $A + B + C = \pi$). But $f(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$. So the maximum value of $\cos A + \cos B + \cos C$ is $\frac{3}{2}$.

Comment No. 1:

The problem here is an example of what is called **constrained optimisation**. Here 'optimisation' means either a maximisation or a minimisation, depending upon the context. The function to be optimised is called the **objective function**. (In the present problem, optimisation means maximisation and the objective function is the function $f(A, B, C) = \cos A + \cos B + \cos C$ of three real variables, A, B and C. The requirements which A, B, C have to satisfy (viz., that they are all non-negative and add up to π) are called **constraints**. The set S is called the **feasible set** and every member of it is called a **feasible solution**. A member of the feasible set at which the objective function attains its maximum (or minimum, as the case may be) is called an **optimal solution**. In our present problem, $(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3})$ is the optimal solution. A problem is called

infeasible if its feasible set is empty i.e., if there are no feasible solutions. Here the word 'solution' does not mean a solution to the main problem (which consists of optimising the objective function over the feasible set), but only to the subsidiary problem of determining if the constraints are mutually consistent. A trivial example of an infeasible problem is where the constraints are, say, that A, B, C are all non-negative and add up to $-\pi$. In fact, the main problem begins only after its feasible set is known to be non-empty. (For the given problem, this is trivially true since the angles of any triangle give a feasible solution. Note that here we are also allowing degenerate triangles, i.e., those where one or more of the angles is 0. This may seem absurd. But it is necessary to include them because sometimes an optimal solution can occur at a degenerate triangle. In fact, if, instead of maximising $\cos A + \cos B + \cos C$, we consider the problem of minimising it then it can be shown that the miminum occurs when the triangle has one angle 0 and the other two right angles or one angle π and the other two 0 each.)

We shall use the term **triangular optimisation** for an optimisation problem where the function to be optimised is a function of three variables, which are either the three angles or the three sides of a triangle. If we denote the sides of a triangle ABC by a, b, c as usual, then the obvious constraints they have to satisfy are that they are all non-negative and that the sum of any two of them exceeds the third. But if these are the only constraints then the objective function, say $g(a, b, c)$ will usually have neither a maximum nor a minimum, because a, b, c can take arbitrarily large or arbitrarily small values. So usually there is some other constraint which a, b, c have to satisfy. For example, the problem of finding the maximum area of a triangle of a given perimeter, say L, can be posed as a triangular optimisation problem if we recall the well-known Heron's formula for the area of a triangle with sides a, b and c, viz., $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$. Thus the problem is to maximise the function $g(a, b, c) = \sqrt{s(s-a)(s-b)(s-c)}$ subject to the constraints that (i) a, b, c are all positive, (ii) the sum of any two of them exceeds the third and finally (iii) $a + b + c = L$. (For various methods of solving this problem see Comment No. 12 below and onwards.)

All this fancy terminology does not, of course, solve the given problem. Nor is it expected from the candidates. It is given here simply because it makes the presentation look slightly neater and because it helps in discussing other problems of a similar spirit (see below). As far as the actual solution to the given problem is concerned, we might as well drop the first paragraph and reformulate the rest of the solution without using the symbols S and f.

Comment No. 2:

In a more conventional form the problem could have been to prove that in any triangle ABC, $\cos A + \cos B + \cos C \leq \frac{3}{2}$ with equality holding if and only if the triangle is equilateral. The essential idea of the solution above was to show that if some two of the three angles are unequal then the objective function can be increased (without affecting the third angle) and hence the optimal solution is provided by the equilateral triangle. There are numerous problems where

some inequality is to be proved for a triangle with the added stipulation that equality holds if and only if the triangle is equilateral. Many of these problems can also be done by invoking suitable trigonometric identities and then applying inequalities such as the A.M.−G.M. inequality.

As a typical example, let us prove that in any acute-angled triangle ABC, $\tan A + \tan B + \tan C \geq 3\sqrt{3}$ with equality holding if and only if the triangle is equilateral. (JEE 1998) The technique used in our solution can also be used here. In fact, if $A \neq B$ then with the notations introduced there we get another acute-angled triangle $A'B'C'$ for which $\tan A' + \tan B' + \tan C' < \tan A + \tan B + \tan C$. Indeed,

$$\begin{aligned}
\tan A + \tan B + \tan C &= \frac{\sin(A+B)}{\cos A \cos B} + \tan C \\
&= \frac{2\sin(A+B)}{\cos(A+B) + \cos(A-B)} + \tan C \\
&= \frac{2\sin(A'+B')}{\cos(A'+B') + \cos(A-B)} + \tan C' \\
&> \frac{2\sin(A'+B')}{\cos(A'+B') + 1} + \tan C' \\
&= \frac{2\sin(A'+B')}{\cos(A'+B') + \cos(A'-B')} + \tan C' \\
&= \tan A' + \tan B' + \tan C'
\end{aligned}$$

(Note that the assumption of acuteness of the triangle was needed to ensure that $\cos A \cos B$ and hence $\cos(A'+B') + \cos(A-B)$ is positive. For otherwise, even though $\cos(A'+B') + \cos(A-B) < \cos(A'+B') + 1$ (which is always positive), we cannot say that $\dfrac{1}{\cos(A'+B') + \cos(A-B)} > \dfrac{1}{\cos(A'+B')+1}$.) It now follows by exactly the same argument as in the solution to the Main Problem that $\tan A + \tan B + \tan C$ is minimum when and only when $A = B = C = \frac{\pi}{3}$. Obviously this minimum value is $3\sqrt{3}$.

But there is another solution to this problem based on the well-known trigonometric identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. For brevity, denote $\tan A, \tan B, \tan C$ by x, y, z respectively. As ABC is acute-angled, x, y, z are all positive and the A.M.−G.M. inequality which says $x+y+z \geq 3(xyz)^{\frac{1}{3}}$ can be applied. In view of the identity just mentioned, we get $x+y+z \geq 3(x+y+z)^{\frac{1}{3}}$. Taking cubes of both the sides and cancelling $x+y+z$ (which is positive) this gives $(x+y+z)^2 \geq 27$. Taking square root we get the desired inequality. If equality is to hold, then it must also hold in the A.M.−G.M. inequality, which can happen if and only if $x = y = z$ i.e., if and only if the triangle is equilateral.

Obviously, the second solution is simpler. Still this approach requires some caution. First of all, there are so many trigonometric identities that there is no unanimity as to which ones among them are standard enough to be assumed without proof. In an examination, therefore, the safe thing from a practical point of view is to at least briefly indicate a proof of the identity that is being used. For example, in the problem just solved, it is a good idea to mention that the identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ is proved by writing

$$\tan C = \tan(\pi - (A+B)) = -\tan(A+B) = \frac{\tan A + \tan B}{\tan A \tan B - 1}.$$

A more serious limitation of this approach is that it is not always easy to pick up the identity that will do the trick for a given problem. Also, in some cases, even though the trigonometrical formula used is a very standard one, the subsequent work may be rather tricky. In fact, the Main Problem falls under this category, With standard notations, the cosine formula gives

$$\begin{aligned}\cos A + \cos B + \cos C &= \frac{b^2+c^2-a^2}{2bc} + \frac{c^2+a^2-b^2}{2ca} + \frac{a^2+b^2-c^2}{2ab} \\ &= \frac{ab^2+ac^2+bc^2+ba^2+ca^2+cb^2-a^3-b^3-c^3}{2abc}\end{aligned}$$

So the problem of showing that $\cos A+\cos B+\cos C \leq \frac{3}{2}$ is equivalent to proving the inequality

$$ab^2+ac^2+bc^2+ba^2+ca^2+cb^2 \leq a^3+b^3+c^3+3abc \tag{1}$$

for any three positive real numbers a, b, c with the property that the sum of any two of them is bigger than the third. (This condition can actually be dropped, see Exercise (14.1).) But this inequality cannot be derived readily from some standard inequality such as the A.M.−G.M. inequality. In fact, the proof of (1) is rather tricky. It is based on the well-known factorisation of $a^3+b^3+c^3-3abc$ as $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$. Note that the second factor can be expressed as $\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$ and hence, being a sum of squares, is non-negative. (In fact, as was shown in Chapter 6, this is one of the ways of proving the A.M.−G.M. inequality for three positive real numbers, say, x, y, z. If we denote their cube roots by a, b, c respectively and apply the factorisation just given, we get $a^3+b^3+c^3 \geq 3abc$, i.e., $x+y+z \geq 3(xyz)^{\frac{1}{3}}$ with equality holding if and only if $a=b=c$ which is the same as $x=y=z$. In other words, what we are doing to prove (1) is more basic than the A.M.−G.M. inequality. This makes it unlikely that (1) will pop out as an application of the A.M.−G.M. inequality.)

To tackle (1) using this factorisation, we subtract $6abc$ from both the sides and multiply throughout by 2. As we have just mentioned, the R.H.S. reduces to $(a+b+c)[(a-b)^2+(b-c)^2+(c-a)^2]$. The L.H.S., on the other hand, becomes, $2(ab^2+ac^2+bc^2+ba^2+ca^2+cb^2)-12abc$. By regrouping the terms, this comes out as $2[a(b-c)^2+b(c-a)^2+c(a-b)^2]$. Thus (1) is equivalent to proving

$$2[a(b-c)^2+b(c-a)^2+c(a-b)^2] \leq (a+b+c)[(a-b)^2+(b-c)^2+(c-a)^2]$$

or, bringing all terms on one side, to proving

$$0 \leq (b+c-a)(b-c)^2+(a-b+c)(c-a)^2+(a+b-c)(a-b)^2 \tag{2}$$

But this is true! Each of the three terms of the R.H.S. is non-negative, being the product of two factors of which the first is given to be positive and the

second is a square. Thus we have proved (1) and hence shown that for any triangle ABC, $\cos A + \cos B + \cos C \leq \frac{3}{2}$. Further, equality holds if and only if it holds in (1) and hence in (2) also. But a sum of non-negative terms is 0 if and only if each term is 0. Since the factors $(a+b-c), (b+c-a)$ and $(c+a-b)$ are all positive, the factors $(a-b)^2, (b-c)^2$ and $(c-a)^2$ must all vanish, forcing $a = b = c$ i.e., the triangle ABC to be equilateral.

What if somebody is unable to think of the algebraic manipulations needed in reducing (1) to (2)? Then it is very unlikely that he will be able to prove (1). In fact, if we multiply the three expressions $(a + b - c), (b + c - a)$ and $(c + a - b)$, their product is positive since each of them is positive. If we expand the product we get

$$ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2 > a^3 + b^3 + c^3 + 2abc \tag{3}$$

The left hand sides of the inequalities (1) and (3) are equal. But the R.H.S. of (1) (which we want to prove) is only slightly greater than the R.H.S. of (3) (which is known to be true). This suggests that proving (1) will not be so easy. So, especially when time is limited, there is no point in spending much time along this line.

Comment No. 3:

The problem of maximising $\cos A + \cos B + \cos C$ subject to the constraints $A \geq 0, B \geq 0, C \geq 0$ and $A+B+C = \pi$ can be done if instead of the A.M.$-$G.M. inequality we use a stronger inequality, called **Jensen's inequality** which was already proved as Theorem 6 of the last chapter. For our purpose, the special case given in (24) of the last chapter would suffice. But we need a 'strict' version of it. So we state it in this new form. In Exercises (14.6) and (14.7) we shall indicate how the proof of Theorem 6 of the last chapter can be modified to derive the stricter version. We also change the notation slightly because we are now using f to denote the objective function of an optimisation problem.

Theorem 1: Suppose $h(x)$ is a twice differentiable, real-valued function on an interval $[a, b]$ and that $h''(x) > 0$ for all $a < x < b$. Then for every positive integer m and for all points $x_1, x_2, \ldots, x_m$ in $[a, b]$, we have

$$h\left(\frac{x_1 + x_2 + \ldots + x_m}{m}\right) \leq \frac{h(x_1) + h(x_2) + \ldots + h(x_m)}{m}\,.$$

Moreover, equality holds if and only if $x_1 = x_2 = \ldots = x_m$. A similar result holds if $h''(x) < 0$ for all $a < x < b$ except that the inequality sign is reversed.

Verbally stated, the inequality says that the value assumed by the function h at the arithmetic mean of a given set of points in the interval $[a, b]$ cannot exceed the arithmetic mean of the values assumed by h at these points. More compactly, the value at a mean is at most the mean of values if h'' is positive on the open interval (a, b) and the value at a mean is at least the mean of values if h'' is negative on it. (Note that h'' is allowed to vanish at one or both the end-points of the interval $[a, b]$.)

It is easy to derive the A.M.−G.M. inequality as a special case of the Jensen's inequality. Suppose $y_1, y_2, \ldots, y_m$ are positive real numbers. Let $x_i = \ln y_i$ for $i = 1, 2, \ldots, m$, where ln denotes the natural logarithm, i.e., logarithm w.r.t. the base e (so that for $i = 1, 2, \ldots, m$ we have $y_i = e^{x_i}$). Now take $[a, b]$ to be any interval containing all the numbers $x_1, x_2, \ldots, x_m$. The function $h(x) = e^x$ satisfies the conditions in the inequality since $h''(x) = e^x$ is positive everywhere. So by Jensen's inequality, we have $e^{\frac{x_1+x_2+\ldots+x_m}{m}} \leq \dfrac{e^{x_1} + e^{x_2} + \ldots + e^{x_m}}{m}$. Since $e^{x_i} = y_i$ for $i = 1, 2, \ldots, m$, this means $(y_1 y_2 \ldots y_m)^{\frac{1}{m}} \leq \dfrac{y_1 + y_2 + \ldots + y_m}{m}$. This is exactly the A.M.−G.M. inequality. Equality holds in it if and only if all x_i's are equal, which is the same thing as saying that all y_i's are equal.

Inasmuch as Jensen's inequality is stronger than the A.M.−G.M. inequality, it is to be expected that certain problems which cannot be done with the A.M.-G.M. inequality may be amenable to the use of Jensen's inequality. This indeed turns out to be the case. In fact, the problem of maximising $\cos A + \cos B + \cos C$ for a triangle, for which we abandoned the former approach can be tackled with Jensen's inequality. Here A, B, C all lie in the interval $[0, \pi]$ and the constraint that they add up to π is equivalent to saying that their arithmetic mean $\frac{A+B+C}{3}$ equals $\frac{\pi}{3}$. So we try to apply Jensen's inequality with $m = 3$. Which function should we take for $h(x)$? It is tempting to take $h(x) = \cos x$. But the trouble is that its second derivative, viz., $-\cos x$ is neither positive throughout nor negative throughout for $0 < x < \pi$. If the triangle ABC were given to be acute-angled then A, B, C would all lie in the interval $[0, \frac{\pi}{2}]$ and on this interval $h''(x)$ is negative everywhere (except at the end-point $\frac{\pi}{2}$) so that a straight application of the Jensen's inequality (with sign reversed as in the last part of it) would give $\cos(\dfrac{\pi}{3}) \geq \dfrac{\cos A + \cos B + \cos C}{3}$ with equality holding if and only if $A = B = C$. This is, of course, what was to be proved since $\cos(\frac{\pi}{3}) = \frac{1}{2}$.

But as the triangle ABC is not given to be acute-angled, we need to modify this solution a little. For this we first note that in any triangle ABC (whether acute-angled or not), $\sin\frac{A}{2}, \sin\frac{B}{2}$ and $\sin\frac{C}{2}$ are always positive. Hence,

$$\begin{aligned} \cos A + \cos B &= 2\cos(\tfrac{A+B}{2})\cos(\tfrac{A-B}{2}) \\ &= 2\sin\tfrac{C}{2}\cos(\tfrac{A-B}{2}) \\ &\leq 2\sin\tfrac{C}{2} \end{aligned}$$

Writing similar inequalities for $\cos B + \cos C$ and for $\cos C + \cos A$ and adding, we get

$$\cos A + \cos B + \cos C \leq \sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} \tag{4}$$

Instead of applying Jensen's inequality to the function $h(x) = \cos x$ (over the interval $[0, \pi]$) we apply it to the function $g(x) = \sin\frac{x}{2}$. Now there is no difficulty because $g''(x) = -\frac{1}{4}\sin\frac{x}{2} < 0$ for all $0 < x < \pi$. So applying the

inequality with $m = 3$ and noting that $g(\frac{A+B+C}{3}) = g(\frac{\pi}{3}) = \sin\frac{\pi}{6} = \frac{1}{2}$ we get

$$\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} \leq \frac{3}{2} \tag{5}$$

with equality holding only for an equilateral triangle. Combining (4) and (5) we now see that for any triangle (whether acute angled or not),

$$\cos A + \cos B + \cos C \leq \frac{3}{2}$$

Further, if equality holds, then in view of (4), equality must also hold in (5) and hence the triangle must be equilateral. The converse is trivial.

The other optimisation problem we encountered, viz., that of minimising $\tan A + \tan B + \tan C$ for an acute angled triangle ABC can also be done easily using Jensen's inequality. Here A, B, C all lie in the interval $[0, \pi/2)$ and we take $h(x) = \tan x$. Then $h''(x) = 2\sec^2 x \tan x > 0$ for all $0 < x < \pi/2$ and so Jensen's inequality gives

$$\tan A + \tan B + \tan C \geq 3\tan(\frac{A+B+C}{3}) = 3\tan\frac{\pi}{3} = 3\sqrt{3}$$

with equality holding if and only if $A = B = C$. (Here, strictly speaking, the function $h(x)$ is not defined at the end-point $\frac{\pi}{2}$. But that does not matter. In a more rigorous form, the inequality can be modified to take care of such anomalous behaviour at the end-points.)

Thus we see that Jensen's inequality works marvelously. But there is a serious limitation to its use. In the form in which we have stated it, it can be applied only to such optimisation problems about triangles where the objective function $f(A, B, C)$ is of the form $h(A)+h(B)+h(C)$ for some function $h(x)$ of a single real variable x. For example, Jensen's inequality cannot be applied directly for optimising $\cos A \cos B \cos C$ or $\tan A \tan B \tan C$. In the latter case, because of the trigonometric identity mentioned earlier we can write $\tan A \tan B \tan C$ as $\tan A+\tan B+\tan C$ which is in a suitable form for applying Jensen's inequality. But a similar conversion is not so easy for $\cos A \cos B \cos C$. (One can, of course, pull some other tricks. For example, when ABC is acute-angled, instead of maximising $\cos A \cos B \cos C$ one can maximise its logarithm which equals $\ln\cos A + \ln\cos B + \ln\cos C$ to which Jensen's inequality can be applied. But such tricks are of an *ad hoc* nature. They may not be always available.)

Jensen's inequality can also be used to give easier proofs of certain other trigonometric inequalities whose direct proofs are either difficult or clumsy. For example, applying Jensen's inequality to the function $h(x) = \sin x$ on the interval $[0, \pi]$ one gets the following result. (JEE 1997)

If n is a positive integer and $0 < A_i < \pi$ for $i = 1, 2, \ldots, n$, then

$$\sin A_1 + \sin A_2 + \ldots + \sin A_n \leq n\sin\left(\frac{A_1 + A_2 + \ldots + A_n}{n}\right)$$

A couple of other applications of Jensen's inequality will be given in the exercises.

Comment No. 4:

Jensen's inequality is based on a relatively elementary part of calculus, involving functions of only one variable and their derivatives. If more advanced methods based on the calculus of functions of several variables are allowed, then there is a very popular and powerful technique, called the method of **Lagrange's multipliers**, which gives an almost instantaneous solution to nearly all problems of triangular optimisations. It will take us too far to discuss the method. However, for those who already know it, we indicate how it works in the present problem, namely, that of maximising the function $f(A, B, C) = \cos A + \cos B + \cos C$. Here the three variables A, B, C have to satisfy certain constraints laid down earlier. Let us ignore the non-negativity requirements and concentrate on the remaining constraint, viz., $A + B + C = \pi$. Denote $A + B + C$ by $g(A, B, C)$. Then both f and g are real-valued functions of three real variables, viz., A, B and C. Suppose (A_0, B_0, C_0) is an optimal solution. Then the method of Lagrange's multipliers says that except when (A_0, B_0, C_0) is a degenerate feasible solution, the three partial derivatives[1] f_A, f_B, f_C of f at (A_0, B_0, C_0) are equal multiples of the corresponding partial derivatives g_A, g_B, g_C of g. In other words, there exists a real number λ (called a multiplier, whence the name of the method) such that the following system of three equations holds :

$$\begin{aligned} f_A(A_0, B_0, C_0) &= \lambda g_A(A_0, B_0, C_0) \\ f_B(A_0, B_0, C_0) &= \lambda g_B(A_0, B_0, C_0) \\ f_C(A_0, B_0, C_0) &= \lambda g_C(A_0, B_0, C_0) \end{aligned}$$

These equations, along with the constraint equation, viz., $g(A_0, B_0, C_0) = \pi$ give a system of four equations in the four unknowns A_0, B_0, C_0 and λ. Solving it, we are able to identify the optimal solution (A_0, B_0, C_0).

In our present problem, the partial derivatives of $f(A, B, C) = \cos A + \cos B + \cos C$ with respect to A, B and C are, respectively, $-\sin A, -\sin B$ and $-\sin C$. In the case of $g(A, B, C) = A+B+C$, each one of them is 1 everywhere. So the system we have to solve is :

$$\begin{aligned} -\cos A_0 &= \lambda \\ -\cos B_0 &= \lambda \\ -\cos C_0 &= \lambda \\ \text{and} \quad A_0 + B_0 + C_0 &= \pi \end{aligned}$$

This system is extremely easy to solve. The first three equations imply $A_0 = B_0 = C_0$. Along with the last equation this means each of them is $\frac{\pi}{3}$. Thus we get $(\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3})$ as the optimal solution. In other words, the triangle has to be equilateral. (Note that we have not solved the system completely, because

[1] If f is a function of several variables of which x is one, then its **partial derivative** w.r.t. the variable x, denoted by f_x or by $\dfrac{\partial f}{\partial x}$, is obtained by differentiating f w.r.t. x as if all other variables were constants. Thus, for example, $\dfrac{\partial}{\partial x}(2x^2y) = 4xy$ while $\dfrac{\partial}{\partial y}(2x^2y) = 2x^2$.

we have not determined the value of λ. Nor is it needed for our purpose. If one wants, it can of course be found easily as $-\cos\frac{\pi}{3}$, i.e., $-\frac{1}{2}$.)

The other two optimisation problems that we encountered (viz., that of minimising $\tan A + \tan B + \tan C$ for an acute-angled triangle ABC and that of maximising $\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2}$ for any triangle ABC can also be handled by the method of Lagrange multipliers. In fact, the equality constraint in all these problems is the same, viz., $g(A, B, C) = A + B + C = \pi$. The objective function $f(A, B, C)$ is, of course, different for different problems. But as long as it is symmetric in the three variables A, B and C, the system of equations will always give the equilateral triangle as the optimal solution. The exception, of course, is when an optimal solution is degenerate. For example, as remarked earlier, the minimum value of $\cos A + \cos B + \cos C$ occurs for two types of degenerate triangles,viz., those where one of the three angles is 0 and the other two are $\frac{\pi}{2}$ each and those where two angles are 0 and the third is π.

Comment No. 5:

It is now time to review together the four general methods we have used for triangular optimisation.

(i) **Trigonometric Method:** As just mentioned, the essence of this method is the observation that the cosine of an angle is at most 1 and that it equals 1 only when the angle is 0. This fact is applied to the difference between two of the angles A, B and C, holding the third angle fixed, to show that unless those two angles are equal, the objective function can be increased (or decreased as the case may be). Consequently, at an optimal solution, these two angles must be equal. If the objective function is symmetric (as has been the case in all the problems of triangular optimisation we considered so far), then every two of A, B and C must be equal to each other and hence the triangle ABC must be equilateral.

This method is elementary and easy to apply. Even when the objective function is only partially symmetric, i.e., symmetric in two but not in all the three variables, it can be applied to those two variables, holding the third variable fixed. Suppose for example, that we want to maximise $f(A, B, C) = \cos A + 2\cos B + \cos C$. This is symmetric in A and C. So by the same reasoning as given for maximising $\cos A + \cos B + \cos C$, at an optimal solution we must have $A = C$. Then $B = \pi - 2A$ which makes f effectively a function of just one variable, viz., $\cos A + 2\cos(\pi - 2A) + \cos A$ which equals $2\cos A - 4\cos^2 A + 2$. This can be maximised as a quadratic in $\cos A$ either by completing the square or using calculus. The maximum occurs when $A = \cos^{-1}\frac{1}{4}$. Thus, the maximum value of $\cos A + 2\cos B + \cos C$ for a triangle ABC is $\frac{9}{4}$. The method, of course, fails if the function is not even partially symmetric. This is not surprising. Basically, in a triangular optimisation problem, we are dealing with a function $f(A, B, C)$ of three variables. Because of the constraint $A + B + C = \pi$, any one of the variables can be expressed in terms of the other two. This effectively makes f a function of two variables. Optimisation of functions of several

variables requires advanced methods. It is only when f satisfies some other condition such as partial symmetry that we can hope to reduce the number of variables further so that elementary methods can be applied.

(ii) **Algebraic Method:** The essence of this method is to reduce the optimisation problem to some inequality (such as (1) which arose while maximising $\cos A + \cos B + \cos C$) using suitable trigonometric formulas or identities. The inequality is then established using some standard inequality such as the A.M.−G.M. inequality, or sometimes, by doing some more basic work (as was the case for proving (1)). The fundamental ideas are very simple, viz., (a) the square of any real number is non-negative and is 0 only when that number is 0 and (b) the sum of two or more non-negative numbers is non-negative and vanishes if and only if each of the terms is 0. When this method works, it works elegantly. But it is not always easy to come up with the right algebraic manipulations. (Sometimes, certain simplifying substitutions have to be used, see Exercise (24.93).) Still, it is an elementary method and deserves to be tried (but without spending too much time!).

(iii) **Jensen's Inequality:** This is a relatively advanced method. It is directly applicable when the objective function is, or can be recast, in a certain form viz., $h(A)+h(B)+h(C)$, where h is a function of one variable whose second derivative maintains the same sign over a suitable interval. But even when h fails to be so, the method can sometimes be applied with a suitable conversion of the problem, as in (4) above.

(iv) **Lagrange's Multipliers:** This is a highly advanced method based on the calculus of functions of several variables. It is applicable to all types of objective functions, not just those that are symmetric or partially symmetric. When applied to triangular optimisation problems with symmetric objective functions, the optimal solution is either degenerate or an equilateral triangle.

Naturally, for a particular given problem, some of these methods may work better than others. The method of Lagrange's multipliers is the surest but the most mechanical of all the four. The algebraic method is artistic and sometimes gives the answer very fast as was the case in the problem of minimising $\tan A + \tan B + \tan C$. Jensen's inequality also works fast once you are able to cast the objective function in a certain form. Such a recasting may involve some ingenuity sometimes. The trouble is that both these methods work only in the case of an optimisation problem where the objective function is symmetric. And in such cases, the method of Lagrange's multipliers makes a mincemeat of the problem. From an examination point of view, this is a boon if a question about triangular optimisation is asked in a 'fill in the blanks' or multiple choice form, where you don't have to show any reasoning. If the objective function is symmetric, then the optimal solution is either degenerate or an equilateral triangle. But degenerate triangles are often excluded from the very definition of

a triangle because of the requirement that the three vertices of a triangle must be distinct and non-collinear and, in any case, such absurdities are unlikely to be asked in an examination! So it is a safe bet to simply *assume* that the optimal solution is an equilateral triangle and proceed with the further work (namely, calculating the value of the objective function for an equilateral triangle). This saves you a lot of time.

Comment No. 6:

And it is precisely for this reason that problems of triangular optimisations of symmetric functions are unsuitable for being asked in the 'fill in the blank' or multiple choice form. Even when asked in a conventional form (where a candidate has to give reasoning), they invite some controversy unless the use of the more advanced methods (viz., those based on Jensen's inequality and on Lagrange's multipliers) is specifically allowed or specifically disallowed. For an examination held at a very low level (such as the tenth standard in Indian schools), hardly any candidates are exposed to the advanced methods and no controversy need arise. The same is the case for an examination at an advanced level (e.g., B. Sc.) because in that case *all* candidates are expected to be familiar with the advanced methods. But trouble arises at an examination which is at an intermediate level, where some but not all candidates know these methods. And once again, in absence of clear instructions, there is always some class of candidates that stands to suffer. If these methods are disallowed but this is not made known beforehand to the candidates then those who used them, despite being able to do the problem by the elementary methods, stand to lose. On the other hand, if these methods are allowed but this is not a public knowledge, then those candidates who know them but who are not sure of their permissibility are forced to try the elementary methods and thereby waste their precious time since, as a rule, the more elementary methods are also more time-consuming.

Comment No. 7:

Such controversies are, of course, irrelevant from a purely academic point of view. In fact, from an academic point of view, a very exciting development takes place when a problem can be done easily by reducing it to one form but not so easily (or sometimes not at all) through some other reduction. *We simply pose the other reduction as a new problem!* As a good illustration, we have already shown that $\cos A + \cos B + \cos C \leq \frac{3}{2}$. We could do this quite easily by either the trigonometric method or the use of Jensen's inequality, and almost instantaneously, by the method of Lagrange multipliers. But when we tried to do it by the algebraic method, we saw that the problem was equivalent to proving the inequality (1). We did prove (1), but only after some rather tricky manipulations. We, in fact, remarked that unless one could think of these manipulations, it is advisable not to spend too much time on it.

Now comes an interesting possibility. Instead of looking at (1) as a possible way to solve the problem of showing that $\cos A + \cos B + \cos C \leq \frac{3}{2}$, we turn the tables around. That is, we solve this problem by some other method and derive (1) as a corollary. Coupled with (3), this gives rise to a new problem, viz.,

"If a, b, c are positive real numbers such that the sum of any two of them exceeds the third, then show that

$$a^3 + b^3 + c^3 + 2abc < ab^2 + ac^2 + bc^2 + ba^2 + ca^2 + cb^2 \leq a^3 + b^3 + c^3 + 3abc$$"

We have already solved this problem. The first inequality, which is precisely the same as (3), follows by expanding the product $(a + b - c)(b + c - a)(c + a - b)$ which is positive (since all its factors are positive). It is the second inequality in this problem (which is the same as (1)) that is really challenging. We proved it above by using some tricky manipulations. But a more artistic approach is to begin by observing that (1) is equivalent to proving that $\cos A + \cos B + \cos C \leq \frac{3}{2}$, where ABC is a triangle whose sides are the given real numbers a, b and c and then proving the latter result trigonometrically. (That such a triangle is possible follows from the data that the sum of any two of these numbers exceeds the third.) In fact, this makes the problem interesting. Although it appears to be a purely algebraic problem, the solution is trigonometric. (See Comment No. 11 of Chapter 11 for more applications of trigonometry to algebra.) Because it is given that the sum of any two of the numbers a, b, c exceeds the third, there is an implicit hint in the problem that methods from trigonometry may be applicable. This is an important characteristic of a good problem. There is a built-in hint to what methods may be useful in solving it. And the purpose is to see who can take the hint. We can, of course, spoil the problem by masking the hint. For example, we may change the hypothesis to something like, "Let a, b, c be real numbers with $0 < a \leq b \leq c < a + b$." This, of course, implies that the sum of any two of a, b and c exceeds the third. So our solution still applies. But now the beauty of the problem is gone!

There are dozens of problems of triangular optimisations where the objective function is symmetric and therefore the optimal solution is an equilateral triangle. Each one of them gives rise to an inequality about three positive real numbers, with the restriction that the sum of any two of them exceeds the third. For example, using the formula $\sin\frac{A}{2} = \sqrt{\frac{(s-a)(s-b)}{bc}}$ (where $s = \frac{1}{2}(a+b+c)$) and its analogues, we can convert (5) to an inequality

$$\begin{aligned} \sqrt{a(a^2 - b^2 - c^2 + 2bc)} \quad &+ \quad \sqrt{b(b^2 - c^2 - a^2 + 2ca)} \\ &+ \quad \sqrt{c(c^2 - a^2 - b^2 + 2ab)} \leq 3\sqrt{abc} \end{aligned} \tag{6}$$

It would indeed be a challenge to prove this directly. But anybody who has a little perceptivity will be reminded of the cosine formula by the expressions under the radical signs. It is then very easy to reduce (6) to (5) and prove the latter trigonometrically.

There is a noteworthy point here for the good student. Merely solving a given problem somehow is not the end of it although, of course, this may be so from an examination point of view. When a student *studies* a problem, there is often so much to learn from it. Quite often, there are more than one method of solving it, some of them being applicable to other problems. (We

already saw many instances of this.) So, a comparative study of the relative merits and demerits of these methods commands merit. But a good student can occasionally go beyond this, by *creating* a new problem. If you can devise a good problem, it is a thrilling experience. Just as one original solution beats a dozen borrowed ones, one original problem beats a dozen original solutions. If you have a like-minded friend of comparable ability then the two of you can develop a healthy rivalry by devising and posing such problems to each other. Even for those who study strictly for examinations, this habit pays off. When you have the habit of generating problems yourself, you are less likely to be baffled when you see a new problem in the examination, as your chances of relating it to some known problem are much higher. In fact, in some cases you can read the paper-setter's mind by correctly figuring out what must have gone into it when he designed that problem. And, occasionally, you can beat him by designing a better problem!

Comment No. 8:

A particular by-product of problems of triangular optimisation deserves a special mention. Whenever the optimal solution to an optimisation problem is unique, we get a characterisation of it in terms of the value attained at it by the objective function. For example, the answer to the problem of maximising $\cos A + \cos B + \cos C$ for a triangle ABC automatically includes a characterisation of an equilateral triangle, viz., that a triangle ABC is equilateral if and only if $\cos A + \cos B + \cos C = \frac{3}{2}$. Similarly, because we know that among all acute-angled triangles ABC, $\tan A + \tan B + \tan C$ is minimum when ABC is equilateral, we get a characterisation of an equilateral triangle among all acute-angled triangles, viz., an acute-angled triangle ABC is equilateral if and only if $\tan A + \tan B + \tan C = 3\sqrt{3}$. (Actually, this characterisation is valid among *all* triangles, not just the acute-angled ones. To see this, note first that for a right-angled triangle ABC, the expression $\tan A + \tan B + \tan C$ is meaningless and hence cannot possibly equal $3\sqrt{3}$. In the remaining case, viz., when ABC is obtuse-angled, exactly one of $\tan A, \tan B$ and $\tan C$ is negative while the other two are positive and so the product $\tan A \tan B \tan C$ is negative. Because of the identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$, this means that for an obtuse-angled triangle ABC, $\tan A + \tan B + \tan C$ is negative and hence cannot equal $3\sqrt{3}$.)

Sometimes the given problem may not ask for the optimisation part but only for the associated characterisation of an equilateral triangle. For example, 'Prove that a triangle ABC is equilateral if and only if $\cos A + \cos B + \cos C = \frac{3}{2}$' or 'Prove that a triangle ABC is equilateral if and only if $\tan A + \tan B + \tan C = 3\sqrt{3}$.' In nearly all such problems, the direct implication is extremely easy. All we have to do is to substitute $A = B = C = \frac{\pi}{3}$ and evaluate the given expression. It is the converse that is really challenging. We can, of course, convert the problem to an optimisation problem and get the answer as we have already done for both these problems. But since the interest is only in the characterisation part, sometimes some easier methods are possible because now the advantage is that we are already given some condition about the triangle in

the form of an equality. Suppose for example that we are given that

$$\cos A + \cos B + \cos C = \frac{3}{2} \tag{7}$$

and we want to prove that the triangle ABC is equilateral.

We rewrite (7) as $(1 - \cos A) - (\cos B + \cos C) + \frac{1}{2} = 0$. The trick now is to write the L.H.S. as a quadratic in $\sin \frac{A}{2}$ using various trigonometric identities. We already converted $\cos B + \cos C$ to $2 \sin \frac{A}{2} \cos \frac{B-C}{2}$. Since $1 - \cos A = 2 \sin^2 \frac{A}{2}$, we get

$$2 \sin^2 \frac{A}{2} - 2 \cos \frac{B-C}{2} \sin \frac{A}{2} + \frac{1}{2} = 0 \tag{8}$$

This quadratic equation (in $\sin \frac{A}{2}$) has a real root (namely the value of the sine of half the angle A of the given triangle ABC). So its discriminant must be non-negative, i.e.,

$$\cos^2 \frac{B-C}{2} \geq 1 \tag{9}$$

But this is possible only when $\cos \frac{B-C}{2} = \pm 1$, i.e., only when $B - C = 0$ or $B - C = \pm\pi$. As B and C are angles of a triangle, the latter possibility is ruled out. So we must have $B = C$. By a similar reasoning, $A = B$. Hence the triangle ABC must be equilateral.

Note that with essentially the same amount of work we could have shown that $\cos A + \cos B + \cos C$ can never exceed $\frac{3}{2}$. For if it did for some triangle ABC, then its value will be of the form $\frac{3}{2} + \lambda$ for some $\lambda > 0$. In that case if we replace the R.H.S. of (7) by $\frac{3}{2} + \lambda$ and carry out the same steps as above, then instead of (9) we shall get,

$$\cos^2 \frac{B-C}{2} \geq 1 + 2\lambda \tag{10}$$

This is impossible if $\lambda > 0$ and so we get a contradiction. Note that the essential idea in this method is the same as that in our very first solution to the problem of maximising $\cos A + \cos B + \cos C$. We carried out some trigonometric manipulations to get $\cos \frac{B-C}{2}$ (or something similar) into the picture and then used the fact that the cosine of an angle cannot exceed 1 and that it equals 1 only when the angle is 0. (Of course, the angle could be any multiple of 2π. But for angles like $\frac{B-C}{2}$, the intervals in which B and C have to lie precludes any other multiples of 2π.)

Comment No. 9:

What is so remarkable about such characterisations of equilateral triangles is that the condition which the triangle ABC is given to satisfy is in the form of a single equation involving the three angles A, B and C. In addition, A, B, C always satisfy one equation, viz., $A + B + C = \pi$. From these two equations, we are expected to prove that $A = \frac{\pi}{3}$, $B = \frac{\pi}{3}$ and $C = \frac{\pi}{3}$. Now,

normally, two equations are *not* enough to determine the values of three unknowns uniquely. So the problem may seem impossible. And it usually is. For example, one of the characterisations of an equilateral triangles above was that $\tan A + \tan B + \tan C = 3\sqrt{3}$. If we replace $3\sqrt{3}$ by some bigger number, say 10, and consider the equation $\tan A + \tan B + \tan C = 10$, then this single equation (along with $A + B + C = \pi$) does not determine the angles uniquely. Indeed, consider the identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. For brevity, denote $\tan A, \tan B, \tan C$, respectively, by p, q, r. Then we have a system of two equations : (i) $p + q + r = pqr$ and (ii) $p + q + r = 10$. For a fixed $p \neq 0$, q and r satisfy the two equations $q + r = 10 - p$ and $qr = \frac{10}{p}$. Hence q and r are the two roots of the quadratic $x^2 + (p - 10)x + \frac{10}{p} = 0$. The fact that this quadratic has real roots does put some restriction on p, viz., $(p - 10)^2 \geq \frac{40}{p}$. But this does not determine it uniquely. In fact, this inequality is satisfied by all sufficiently large values of p. So there are infinitely many values which p can take and for each such value there will be one triangle ABC (actually two triangles if we interchange the angles B and C) which will satisfy the equation $\tan A + \tan B + \tan C = 10$. In other words, this equation (coupled with $A + B + C = \pi$, of course) has infinitely many solutions and hence cannot characterise the angles of any triangle.

A similar analysis holds for the characterisation of an equilateral triangle as a triangle of the form ABC for which $\cos A + \cos B + \cos C = \frac{3}{2}$. If we replace $\frac{3}{2}$ by something slightly smaller, say $\frac{5}{4}$, then the equation $\cos A + \cos B + \cos C = \frac{5}{4}$ has infinitely many solutions.

So, what is so special about the equations $\tan A + \tan B + \tan C = 3\sqrt{3}$ and $\cos A + \cos B + \cos C = \frac{3}{2}$ that makes it possible to determine A, B, C uniquely from any one of them? The answer lies in the fact that the values $3\sqrt{3}$ and $\frac{3}{2}$ are the respective optimal values of the objective functions represented by the left hand sides of these equations. Even if we ignore the optimisational part, it is not hard to fathom the reason how each of these equations single-handedly (along with the general equation $A + B + C = \pi$) determines that the triangle is equilateral. Let us do this for (7). We already converted it to a quadratic in $\sin \frac{A}{2}$, viz., (8). Normally, the next step would be to solve this quadratic and thereby express $\sin \frac{A}{2}$ in terms of B and C. But instead, we went for the existence rather than the actual value of a real root, and, applying the discriminant criterion, got (9). If we go into the logic behind the discriminant criterion, we get an alternate way of solving the problem. Multiplying throughout by 2 and completing the square, we can rewrite (8) as

$$\left(2\sin\frac{A}{2} - \cos\frac{B-C}{2}\right)^2 + \left(1 - \cos^2\frac{B-C}{2}\right) = 0 \qquad (11)$$

This is a single equation. But it has the force of two. The L.H.S. is a sum of two terms. The first term can never be negative as it is the square of a real number. The second term also cannot be negative since the cosine function can never be numerically bigger than 1. Therefore, the sum of these two terms vanishes if and only if each one of them vanishes. Thus from (11), we get a system of two

equations:

$$2\sin\frac{A}{2} - \cos\frac{B-C}{2} = 0 \tag{12}$$

and

$$\cos\frac{B-C}{2} = \pm 1 \tag{13}$$

As we have seen before, (13) implies $B = C$. From this and (12), we get $\sin\frac{A}{2} = \frac{1}{2}$ whence $A = \frac{\pi}{3}$ or $\frac{2\pi}{3}$. The second possibility is ruled out since $0 < A < \pi$. Hence ultimately, $A = \frac{\pi}{3}$. Further, $A + B + C = \pi$ and $B = C$ together imply that $B = C = \frac{\pi}{3}$.

More generally, if $x_1, x_2, \ldots, x_n$ are quantities which can never be negative (for whatever reason), then the single equation $x_1+x_2+\ldots+x_n = 0$ is equivalent to a system of n equations, viz., $x_1 = 0$, $x_2 = 0$, $\ldots$, $x_n = 0$. We in fact used this fact earlier (with $n = 3$) when we showed that equality could hold in (2) if and only if $a = b = c$.

Comment No. 10:

The general rule to keep in mind is that when you have fewer equations than unknowns, see if you can somehow get an equation which does the job of several equations. This general principle can, in fact, be applied to the characterisation of many other types of triangles. Consider, for example, the following problem.

Suppose ABC is a triangle in which $\cos A\cos B + \sin A\sin B\sin C = 1$. Prove that $a : b : c = 1 : 1 : \sqrt{2}$. (JEE 1986)

Superficially, the conclusion to be reached involves the sides of the triangle. But since only their ratios are involved, we can translate it in terms of the angles. $a : b : c = 1 : 1 : \sqrt{2}$ is equivalent to saying that $a = b$ and $a^2 + b^2 = c^2$, which, in turn, amounts to saying that the triangle is isosceles and right-angled at C. Thus we have to prove that $C = \frac{\pi}{2}$ and $A = B$ ($= \frac{\pi}{4}$ each). (This, by the way, is a very helpful habit. See if you can recast the problem i.e., its hypothesis or its conclusion in various forms. Although mere recasting does not solve the problem, it often inspires you to try certain methods. For instance, now that we have reduced the problem solely in terms of the angles, we can try to solve it by finding the values of $\sin C, \sin A, \cos A$, etc.)

Now, coming to the problem itself, we are given only one equation (besides the general equation $A+B+C = \pi$). So, somehow we have to cast this equation in a form in which it is equivalent to two separate equations. This can be done by adding and subtracting $\sin A\sin B$, to get, $\cos(A-B)+\sin A\sin B(\sin C-1) = 1$ or,

$$(1 - \cos(A - B)) + \sin A\sin B(1 - \sin C) = 0 \tag{14}$$

The expressions $1 - \cos(A - B)$ and $1 - \sin C$ are always non-negative. Also since A, B lie in the interval $(0, \pi)$, $\sin A, \sin B$ are positive. Therefore (14) is equivalent to a system of two equations, viz., $\cos(A - B) = 1$ and $\sin C = 1$. These two equations imply, respectively, that $A = B$ and $C = \frac{\pi}{2}$ as desired.

A slightly different approach, involving essentially the same work, is to show that the desired type of a triangle is the optimal solution to an optimisation

problem whose objective function is $f(A,B,C) = \cos A \cos B + \sin A \sin B \sin C$. Since $\sin A \sin B$ is positive, and $\sin C \leq 1$, we have $f(A,B,C) \leq \cos A \cos B + \sin A \sin B$ with equality holding if and only if $\sin C = 1$ i.e., if and only if, $C = \frac{\pi}{2}$. Now, $\cos A \cos B + \sin A \sin B = \cos(A-B) \leq 1$ with equality holding if and only if $A = B$. Putting these two inequalities together, $f(A,B,C) \leq 1$ with equality holding if and only if C is a right angle and $A = B$. Thus the desired triangle is the optimal solution to the problem of maximising $f(A,B,C)$.

Comment No. 11:

We saw that problems of triangular optimisation give as a by-product, characterisations of certain types of triangles. There is yet another type of problems which one can create from optimisation problems. For example, we saw that for an acute-angled triangle ABC, the function $f(A,B,C) = \tan A + \tan C + \tan C$ attains its minimum at an equilateral triangle and that this minimum is $3\sqrt{3}$. Now let k be a positive real number and consider the equation $\tan A + \tan B + \tan C = k$. Then for $0 < k < 3\sqrt{3}$, this equation has no solution. For $k = 3\sqrt{3}$, it has a unique solution, while for $k > 3\sqrt{3}$ it has more than one (in fact, infinitely many) solutions. (This can be proved by the same argument by which we showed in Comment No. 9 that the equation $\tan A + \tan B + \tan C = 10$ has infinitely many solutions. There was nothing special about the number 10. Any number greater than $3\sqrt{3}$ would have worked.) Thus we see that the only positive real number k for which the equation $\tan A + \tan B + \tan C = k$ has a unique (that means one and only one) solution is $k = 3\sqrt{3}$.

Along similar lines, we could ask the following problems:

(i) Find a value of k for which the equation

$$\cos A + \cos B + \cos C = k$$

for a triangle ABC has a unique solution.

(ii) Find a value of k for which the equation

$$\cos A \cos B + \sin A \sin B \cos C = k$$

for a triangle ABC has a unique solution.

These problems are nothing but disguised versions of the corresponding optimisation problems. The answers to them are, respectively, $\frac{3}{2}$ and 1. Try these problems on a friend and see if he/she can do them. Both the problems may appear slightly clumsy. But be prepared for clumsy problems occasionally. There are some reputed examinations where problems are generally not repeated. Because of several constraints (such as having to cover the entire syllabus uniformly) the paper-setters are not in a position to come up with fresh, good problems every year. So sometimes an old or a well-known problem is asked after giving it a new garb. And in this process, the old problem is likely to lose its inherent beauty (for example, an implicit hint it may carry in its very formulation). So be prepared for clumsy looking problems. And once again, the

best way to prepare for such problems is to design some problems yourself by modifying old problems!

Optimisation means either maximisation or minimisation. When a problem is to be solved by converting it to an optimisation problem, how do you tell if the objective function should be maximised or minimised? The answer is simple. Look at the degenerate cases. These are the cases where one of the angles is very large and the other two very small or those where one of the angles is very small and the other two are close to right angles. The general rule of thumb is that whenever the maximum of an objective function occurs at a non-degenerate triangle, its minimum will occur at a degenerate one (which, in effect, means that the minimum does not exist, since we do not consider such degenerate triangles as genuine triangles) and vice versa. Consider for example, the function $f(A, B, C) = \cos A + \cos B + \cos C$. As ABC tends to a degenerate triangle, $f(A, B, C)$ tends to 1. For a non-degenerate triangle which is close to these degenerate cases, the value of f is slightly higher than 1. So the interesting problem would be to maximise and not minimise $f(A, B, C)$. As we have already seen, the maximum occurs at an equilateral triangle. Similarly, if we take $f(A, B, C) = \cos A \cos B + \sin A \sin B \sin C$ we see that by taking B and C very small, the function can be made to approach -1 from above. So once again, it is the maximisation of f which is really interesting and, as we already know, the optimal solution is a triangle which is isosceles and right-angled at C.

In the case of the function $f(A, B, C) = \tan A \tan B \tan C$, if we confine ourselves only to acute-angled triangles then as you approach the degenerate case, one of the angles tends to 0 and the other two to $\frac{\pi}{2}$ each. So, $f(A, B, C)$ which also equals $\tan A + \tan B + \tan C$ tends to ∞. Hence in this case, one should look for the minimum and not the maximum of f. For a right triangle, f is undefined. For an obtuse-angled triangle ABC, $f(A, B, C)$ is negative. It approaches 0 as you approach one type of degeneracy (where one of the angles tends to π and the other two to 0). But as you approach the other type of degeneracy (where one of the angles tends to 0 and the other two to $\frac{\pi}{2}$ each, but one from above and the other from below) $f(A, B, C)$ tends to $-\infty$. Thus for an obtuse-angled triangle, neither the maximum nor the minimum of f exists.

Comment No. 12:

In all the problems of triangular approximation discussed so far in this chapter, the objective function was a function of the three angles of a triangle. Let us now consider a problem of triangular optimisation where the objective function is a function of the lengths of the three sides of a triangle.

"A piece of wire of length 10cm is to be bent so as to form the boundary of a triangle. Find the maximum area of the triangle so formed."

Let a, b, c denote the lengths (in centimeters) of the sides of the triangle formed by bending the wire. Then its area Δ (measured in square centimeters) is given by

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

where s, the semiperimeter, equals $\frac{1}{2}(a+b+c)$. In this problem $s = 5$. Maximis-

ing Δ is equivalent to maximising Δ^2. So the problem amounts to maximising the function $g(a,b,c) = 5(5-a)(5-b)(5-c)$ subject to the constraints that (i) a, b, c are positive, (ii) the sum of any two of them exceeds the third and (iii) $a+b+c = 10$. Notice that because of (ii), $5-a = \frac{1}{2}(b+c-a) > 0$. Similarly, $5-b$ and $5-c$ are positive. By applying the A.M.−G.M. inequality for three positive real numbers, we see that the product $(5-a)(5-b)(5-c)$ is maximum when $5-a = 5-b = 5-c$, i.e., when the triangle is equilateral. This makes $a = b = c = \frac{10}{3}$ and $\Delta^2 = \frac{625}{27}$.

More generally, instead of 10, we could consider the problem of maximising the area Δ of a triangle ABC with a given perimeter, say L. If one wants, one can convert this problem to a triangular optimisation problem where the objective function is a function of the angles A, B and C and apply any of the four methods discussed above for the solution to the Main Problem. The conversion is easy. If we denote by R the circumradius of the triangle ABC, then we have $a = 2R\sin A, b = 2R\sin B$ and $c = 2R\sin C$. This gives $L = a+b+c = 2R(\sin A+\sin B+\sin C)$ and hence R in terms of L, viz., $R = \frac{L}{2(\sin A+\sin B+\sin C)}$. Since Δ, the area of the triangle ABC, equals $2R^2 \sin A \sin B \sin C$, we get

$$\Delta = \frac{L^2 \sin A \sin B \sin C}{2(\sin A + \sin B + \sin C)^2} \tag{15}$$

Since $\frac{L^2}{2}$ is a (positive) constant, we might as well forget it and thus the original problem is now reduced to maximising the function

$$f(A,B,C) = \frac{\sin A \sin B \sin C}{(\sin A + \sin B + \sin C)^2} \tag{16}$$

where A, B, C are the angles of a triangle.

Even though the reduction itself was not difficult, one can see that the R.H.S. of (16) is more complicated than any of the examples discussed so far. Also the objective function is not amenable to the application of Jensen's inequality. The method of Lagrange's multipliers will work but the computations of the partial derivatives will be horrendous. So instead of maximising it directly, we first apply the A.M.-G.M. inequality to the numerator and get

$$\begin{aligned} f(A,B,C) = \frac{\sin A \sin B \sin C}{(\sin A + \sin B + \sin C)^2} &\leq \frac{(\frac{\sin A+\sin B+\sin C}{3})^3}{(\sin A + \sin B + \sin C)^2} \\ &= \frac{1}{27}(\sin A + \sin B + \sin C) \end{aligned} \tag{17}$$

with equality holding if and only if $\sin A = \sin B = \sin C$, i.e., if and only if the triangle is equilateral. Now the problem of maximising $\sin A + \sin B + \sin C$ is very similar to the Main Problem and so we omit the details. The optimal solution is $A = B = C = \frac{\pi}{3}$ and the optimal value is $\frac{3\sqrt{3}}{2}$. So the maximum value of $f(A,B,C)$ is $\frac{\sqrt{3}}{18}$. Recalling the factor $\frac{L^2}{2}$ we finally get the maximum value of Δ as $\dfrac{\sqrt{3}L^2}{36}$.

Comment No. 13:

In theory, every problem of triangular optimisation where the objective function is given as a function of the three sides of the triangle, say ABC, can be converted to an optimisation problem where the objective function is a function of the angles of the triangle, using the same method as above, viz., through the formulas $a = 2R\sin A, b = 2R\sin B$ and $c = 2R\sin C$. These values are substituted in the equality constraint which a, b, c have to satisfy. (In the present problem, this constraint is $a + b + c = L$, where L is a given constant.) This gives the circumradius R in terms of $\sin A, \sin B$ and $\sin C$. Now, if we replace a, b, c by $2R\sin A$ etc. in the objective function, say $g(a, b, c)$, we get it in terms of $\sin A, \sin B, \sin C$ and R. Since R has already been expressed in terms of $\sin A, \sin B$ and $\sin C$, we thus ultimately get a triangular optimisation problem where the objective function is a function of the angles of the triangles.

The other way conversion is also possible, but not unique, because the three angles of a triangle do not determine it uniquely. They determine it upto similarity. That is to say, any two triangles with given angles are similar to each other. They have the same shapes but not necessarily the same size. This does not matter since the solution to such optimisation problems gives only the shape of the triangle. However, if we want to convert such optimisation problems (where the objective function is expressed in terms of the angles A, B and C), to those where it is expressed as a function of the sides a, b and c, then in order to define the triangle ABC uniquely we have to impose some arbitrary condition on the sides. For example, we may require that the first side a is of a unit length, or that the perimeter $a + b + c$ is of a fixed length. A condition of this type is called a **normalisation condition**. Such a condition does not restrict the essence of the problem. It merely makes the problem specific. (It is analogous in spirit to choosing the coordinate system suitably so that the equation of a given curve such as an ellipse appears in the standard form.) It really does not matter which normalisation condition one puts and depending upon the problem, one choice may be more convenient than some other. A commonly used normalisation condition is to let R, the circumradius, be of unit length. This allows us to represent the coordinates of the vertices of the triangle in the form $(\cos\alpha, \sin\alpha)$, $(\cos\beta, \sin\beta)$, and $(\cos\gamma, \sin\gamma)$, for some α, β and γ.

Comment No. 14:

Because of these back-and-forth conversions of problems of triangular optimisations, if one wants, one can confine oneself entirely to problems of one of the two types. But this is not a good idea. Conversion is a tool and its use is optional, rather than mandatory. Whether to apply it or not in a particular problem will depend upon that particular problem. Basically, the choice depends on the simplicity of the constraint and of the objective function. In our present problem (viz., that of maximising the area of a triangle of a given perimeter L), the constraint, when expressed in terms of the sides is $a + b + c = L$. For triangular optimisation of a function of the three angles, the constraint is, of course, always the same, viz., $A + B + C = \pi$. Thus both these constraints are of the same type and equally simple. So there is little point in choosing one over the

other. What about the objective function, viz., the area Δ? There are many expressions available for Δ. For example, $\Delta = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C$, or, in a symmetric form $\Delta = 2R^2\sin B\sin C\sin C$. If these were the only available expressions for the area, then perhaps it would be a good idea to transform the given problem. But the fact is that the area Δ can also be expressed directly in terms of the sides a, b, c by the well-known **Heron's formula**

$$\Delta \quad = \quad \sqrt{s(s-a)(s-b)(s-c)} \tag{18}$$

where s is the semi-perimeter $\frac{1}{2}(a+b+c)$.

This looks like a formidable formula and would appear even more so if we express each of the four factors under the radical sign directly in terms of a, b and c. For, in that case it would look as

$$\Delta \quad = \quad \frac{1}{4}\sqrt{(a+b+c)(a+b-c)(a-b+c)(b+c-a)} \tag{19}$$

But in our problem s, equals $\frac{L}{2}$ which is a constant. So, for our purpose, (18) is a very convenient formula. True, there is the ugly radical sign. But we can get rid of it easily because obviously, maximising area is equivalent to maximising the square of the area. (This is a simple but useful trick which allows us sometimes to get rid of the messy radicals. It is useful especially when optimising some distance since the distance is often expressed as a square root of something. When the objective function is a sum of two or more radicals, the trick, of course, fails.) So we try to maximise the function $g(a,b,c) = s(s-a)(s-b)(s-c)$, where $s = \frac{L}{2}$ and a, b, c satisfy the constraints that they are all non-negative, that the sum of any two of them exceeds the third and finally that $a+b+c = L$.

This can be done by any of the four techniques we applied earlier for optimising a function of the three angles of a triangle. The first method is to hold one of the three variables fixed and then show that unless the values of the other two variables are equal, the objective function can be increased. For example, fix c. Then $(s-c)$ is also fixed. Also $(s-a)+(s-b)$ is fixed since it equals c. Note further that $(s-a), (s-b)$ and $(s-c)$ are all positive. (This is where we need that the sum of any two of a, b and c exceeds the third.) It follows from the A.M.$-$G.M. inequality for two real numbers that the product $(s-a)(s-b)$ and hence also the product $(s-a)(s-b)(s-c)$ is maximum when $s-a = s-b$, i.e., when $a = b$. By a similar reasoning, at an optimal solution $b = c$. Thus the area is maximised when the triangle is equilateral. (The maximum value of the area in terms of the given perimeter L comes out as $\sqrt{\dfrac{(a+b+c)abc}{16}} = \dfrac{\sqrt{3}L^2}{36}$.)

More directly, as we did in the special case above with $L = 10$, we could apply the A.M.$-$G.M. inequality to the product $(s-a)(s-b)(s-c)$ itself. The arithmetic mean of $s-a, s-b$ and $s-c$ is $\frac{3s-(a+b+c)}{3} = \frac{s}{3}$ which is a constant. It follows that their product is maximised when $s-a = s-b = s-c$, i.e., when $a = b = c = \frac{L}{3}$.

As for the more advanced methods of triangular optimisation, Jensen's inequality cannot be applied to the objective function $(s-a)(s-b)(s-c)$

directly. But as noted in Comment No. 3 above, taking logarithms will do the trick. There is no difficulty in taking the logarithms since all three numbers $s-a, s-b$ and $s-c$ are positive. We consider the function $h(x) = \ln x$. Then $h''(x) = -\frac{1}{x^2} < 0$ for all $0 < x < L$. So a straight application of Jensen's inequality gives

$$\ln(s-a) + \ln(s-b) + \ln(s-c) \leq 3\ln(\frac{s-a+s-b+s-c}{3}) = 3\ln\frac{s}{3}$$

with equality holding if and only if $s-a = s-b = s-c$, which is equivalent to $a = b = c$. Taking exponentials of both the sides this gives

$$(s-a)(s-b)(s-c) \leq (\frac{s}{3})^3 = \frac{s^3}{27}$$

Multiplying throughout by s, taking square roots and noting that $s = \frac{L}{2}$, we get that the area of a triangle with perimeter L is at most $\frac{\sqrt{3}L^2}{36}$ with equality holding only for an equilateral triangle.

Finally, the method of Lagrange's multipliers can also be applied very routinely. In fact, since the objective function as well as the constraint $a+b+c = L$ is symmetric in a, b and c, the system of equations in the four unknowns a, b, c and λ that we get from the partial derivatives will also be symmetric in a, b, c (although not necessarily between all the four unknowns). So, in any non-degenerate solution of it, the values of a, b, c will be equal. (The value of the fourth unknown λ may be different. But we are not interested in it *per se.*) Hence the optimal solution has to be an equilateral triangle.

Comment No. 15:

In the problem just solved we maximised the area of a triangle that can be formed by bending a piece of wire of a given length L. Note that there is no minimum to this problem since one of the pieces can be arbitrarily close to $L/2$ in which case the altitude on it and hence the area can be made arbitrarily small. Or we can take one of the three pieces close to 0 in length. (If we allow degenerate triangles, then the minimum area is 0. It is attained when one segment is $L/2$ or 0.) Sometimes we are confronted with a different but related optimisation problem. Suppose we want to construct a triangle which encloses a given area, say Δ. If the wire needed to form its boundaries is expensive, we would obviously like to minimise its perimeter. So here we have the problem of minimising the perimeter of a triangle, subject to the constraint that its area is a given constant, say Δ. Mathematically, we want to minimise the function $f(a, b, c) = a + b + c$ of three variables subject to the constraints that (i) a, b, c are all non-negative, (ii) the sum of any two of them exceeds the third and (iii) $s(s-a)(s-b)(s-c) = \Delta^2$. Note that here the semi-perimeter s is not a constant. It is a function of a, b, c, viz., $s = \frac{a+b+c}{2}$. When expressed solely in terms of a, b, c, the third constraint takes the form

$$(a+b+c)(a+b-c)(a-b+c)(b+c-a) \quad = \quad 16\Delta^2 \qquad (20)$$

This time it is silly to ask for a maximum since by taking the height small and base large, one can create a triangle of a given area whose perimeter is as large as one likes.

Let us now see which of our four standard methods (listed in Comment No. 7 above) are applicable to this problem. Because of the symmetry of the constraint (20) and of the objective function $a + b + c$, once again the method of Lagrange multipliers makes a mincemeat of the problem and gives the equilateral triangle as the answer. Since its area is the given constant Δ, each side has length $2\left(\frac{\Delta}{\sqrt{3}}\right)^{\frac{1}{2}}$ and hence the minimum possible perimeter of a triangle with area Δ equals $6\left(\frac{\Delta}{\sqrt{3}}\right)^{\frac{1}{2}}$.

Jensen's inequality is inapplicable. The objective function, viz., $a+b+c$ is of the form required by Jensen's inequality. But the equality constraint on a, b, c now is (20) which is not in the right form. If at all we want to apply Jensen's inequality, we shall first have to replace (20) by an equivalent constraint which can be cast in the form

$$h(a) + h(b) + h(c) \quad = \quad \text{a constant, say, } \lambda \tag{21}$$

where h is some function of one variable. We shall then have to call $h(a)$, $h(b), h(c)$ as x, y, z (say) respectively and express a, b, c as $h^{-1}(a), h^{-1}(b)$, $h^{-1}(c)$ respectively, where h^{-1} is the inverse function of h. We can then apply Jensen's inequality to this inverse function. Obviously this is a very complicated task. In fact for problems of triangular optimisations where the objective function is given in terms of the sides, Jensen's inequality is rarely applicable. The problem above was an exception since the constraint happened to be in the form $a + b + c =$ a constant.

Coming to the algebraic method, we are given that a, b, c are positive real numbers which satisfy (20) and we have to show that the perimeter of the triangle with a, b, c as sides cannot be less than that of an equilateral triangle of area Δ. This amounts to showing that

$$a + b + c \quad \geq \quad 6\left(\frac{\Delta}{\sqrt{3}}\right)^{\frac{1}{2}} \tag{22}$$

with equality holding if and only if $a = b = c$. For this, we apply the A.M.$-$G.M. inequality to the last three factors of the L.H.S. of (19). Since their sum is $a + b + c$, we get that

$$(a + b + c)(a + b - c)(a - b + c)(b + c - a) \quad \leq \quad \frac{(a + b + c)^4}{27} \tag{23}$$

with equality holding if and only if $a + b - c = a - b + c = b + c - a$, which is equivalent to $a = b = c$. (22) now follows immediately by putting (20) and (23) together and taking the fourth roots of both the sides of the inequality. Further, equality in (22) can hold if and only if it holds in (23), which, as noted above, happens if and only if $a = b = c$.

(Here we applied the A.M.−G.M. inequality for the product of three positive real numbers, viz., $a+b-c, a-b+c$ and $b+c-a$. If, instead, we apply it to the products of these numbers taken two at a time, we get $\sqrt{(a+b-c)(a-b+c)} \leq \frac{1}{2}[(a+b-c)+(a-b+c)] = a$ and two similar inequalities. Multiplying these three inequalities together and substituting in (19) we get the result that $\Delta \leq \frac{1}{4}\sqrt{(a+b+c)abc}$ with equality holding if and only if $a=b=c$ (JEE 2001). Alternately, we could expand the product $(a+b-c)(a-b+c)(b+c-a)$ and use the inequality (1) in Comment No. 2 above.)

Finally, let us apply the technique of holding one of the three variables fixed and showing that unless the other two are equal, we can improve the objective function. So assume a is fixed. Let ABC be a triangle with area Δ and base BC equal to a. We claim that unless ABC is isosceles (with $AB=AC$), we can find another triangle $A'BC$ with the same base and the same area for which the perimeter is smaller, i.e., $A'B + A'C < AB + AC$.

We can paraphrase this slightly. Let $h = \frac{2\Delta}{a}$. Then h is the altitude of any triangle with base a and area Δ. Since Δ and a are constants, so is h. Now fix a segment BC of length a and draw two lines L and L' parallel to the line BC at a distance h from it. Then any triangle with base BC and height h must have its third vertex either on L or on L'. By symmetry, we may suppose it lies on L. So our problem is reduced to showing that among all triangles with BC as a base and the third vertex lying on L, the isosceles triangle has the smallest perimeter.

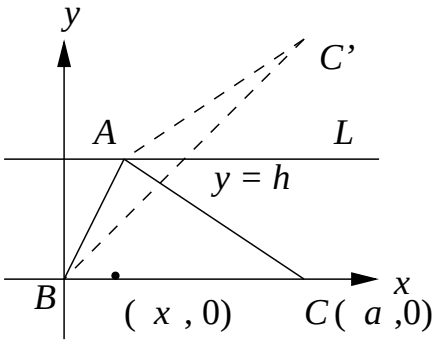
This itself is an interesting problem and can be tackled in several ways. The most straightforward method is based on calculus. We take B as the origin and BC as the x-axis, as shown in Figure (a) below. Then $C=(a,0)$ and the equation of the line L can be taken as $y=h$. A typical point A on L has the form (x,h) for some real number x. Let $f(x) = AB + AC = \sqrt{x^2+h^2} + \sqrt{(x-a)^2+h^2}$. We have to show that $f(x)$ is minimum when $AB=AC$, i.e., when $x=\frac{a}{2}$. To do this we differentiate $f(x)$ to get

$$\begin{aligned} f'(x) &= \frac{2x}{\sqrt{x^2+h^2}} + \frac{2x-2a}{\sqrt{(x-a)^2+h^2}} \\ &= \frac{2x\sqrt{(x-a)^2+h^2} + 2(x-a)\sqrt{x^2+h^2}}{\sqrt{x^2+h^2}\sqrt{(x-a)^2+h^2}} \end{aligned}$$

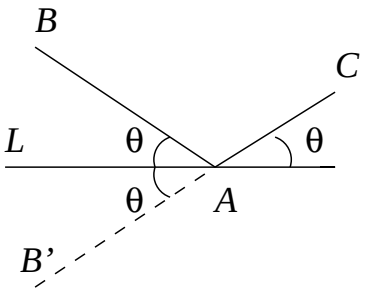
For this to vanish, we must have $x\sqrt{(x-a)^2+h^2} = -(x-a)\sqrt{x^2+h^2}$. After squaring and simplifying, this gives $x^2 = (x-a)^2$, i.e., $x = \pm(x-a)$. As the $+$ sign gives no solution we must have $x=-(x-a)$, i.e., $x=\frac{a}{2}$. As this is the only critical point of f and $f(x)$ tends to ∞ as $x \longrightarrow \pm\infty$, $f(x)$ has a minimum at $x=\frac{a}{2}$ as desired.

For a trigonometric solution, let A be any point on the line L and consider the triangle ABC. Since the altitude through A has length h we have $h(\cot B + \cot C) = a$ and $AB + AC = h(\text{cosec } B + \text{cosec } C)$. Thus the problem amounts to minimising $\text{cosec } B + \text{cosec } C$ subject to the constraint that $\cot B + \cot C$ is a constant, say, k. A direct, trigonometric solution of

this does not seem easy. We could, of course, write cosec B + cosec C as $\sqrt{1+x^2}+\sqrt{1+(k-x)^2}$, where $x = \cot B$ and then minimise this as a function of x. But then the proof is not substantially different from the one above. But there is a way out if we note that the problem is equivalent to proving that $(b+c)^2 \geq a^2+4h^2$, i.e., to showing that $(b+c)^2 - a^2 \geq \frac{16\Delta^2}{a^2}$. This follows from (20) and the inequality $(a+b-c)(a-b+c) = a^2-(b-c)^2 \leq a^2$.



(a)



(b)

There is also an elegant, purely geometric way to minimise $AB + AC$ as the point A moves along the line L. Let C' be the reflection of the point C in the line L. Then $AC = AC'$ and so $AB + AC$ is the same as $BA + AC'$. Now, the points B and C' are fixed while A moves on the line L. So the sum $BA + AC'$ will be minimised when it is the straight line distance between B and C'. For this to happen, B, A and C' must be collinear, or in other words, A must be the point of intersection of the line L and the line BC'. Since B and C are at equal distances from L, this can happen when and only when the triangle ABC is isosceles with $AB = AC$. (The major part of this argument is applicable even when B and C are not equidistant from L, i.e., even when BC is not parallel to L. But in that case the triangle ABC for which the perimeter is minimum will no longer be isosceles.) The key idea in this geometric method of minimisation is the triangle inequality, viz. that the sum of the two sides of a triangle is at least the third side. The dual form of the triangle inequality, viz. that the difference between the two sides is at most the third side can also be used to find a maximum sometimes, see Exercise (14.17).

Comment No. 16:

Summing up, we could do the problem of minimising the perimeter of a triangle of a given area by all standard methods, except by the use of Jensen's inequality. The last method, in fact, led to a new optimisation problem which was of interest in its own right, namely that of minimising the perimeter of a triangle of a given height on a given base. This is typical of mathematics. The search for a solution to one problem leads to another problem on the way. (And, in some cases, this new problem is more exciting than the original one.) That is why, even after you have solved a given problem by one method, it is a good

idea to keep trying other methods. In an examination, the solution which takes the least time is the best and it is pointless to try others. But while studying, it pays to do so since what you learn through this exercise may be useful elsewhere. In the present problem, for example, the elegant geometric solution that we saw above (based on the simple idea of the reflection of a point in a straight line) can easily be modified to do the following problem.

"Suppose two localities are situated on the same side of a river flowing in a straight line. A pumping station is to be constructed at a point on the river and pipe lines are to be laid from it to serve the localities. Where should the station be located so as to minimise the total length of the pipe lines?"

In fact, the same idea can be the key to success in some other problems which are not optimisation problems. Here is an example.

Suppose B and C are any two points on the same side of a line L. Show how to find a point A on the line L such that AB and AC are equally inclined to the line L. (JEE 1980)

For an easy solution, let B' be the reflection of B in L as shown in Figure (b) above. Then AB and AB' are equally inclined to L and so the given condition is equivalent to saying that AB' and AC are equally inclined to L. But B' and C are on opposite sides of L. So the only way for this to happen is for B', A, C to be collinear. Thus we get A as the point of intersection of the line L with the line $B'C$.

This problem is typical of those problems of geometric constructions where once you hit upon the key idea, the problem becomes trivial. But then how do you hit upon the key idea? There is no single answer to this. Apart from your innate abilities, sometimes luck also plays a role. But you can supplement these by your own resourcefulness. And your best resource is not a bunch of important formulas or a huge collection of solved problems. These things undoubtedly have their value. But your real resource is the ideas you have gathered. With a bunch of formulas, you can do a problem where one of those formulas fits in. With a collection of solved problems, you can do an analogous problem. But with a repertoire of ideas, you are your own master. If you have studied solutions of some problems for the sake of the ideas that they contain, then chances are that even a remote resemblance can trigger your imagination. That's exactly what happened in the case of the geometric construction asked above. The very fact that the problem deals with a pair of points lying on the same side of a line (and nothing else) can inspire you to think of some other problem where a similar piece of data was given, for example the problem of laying the pipe lines given above. Even though it was a very different type of a problem, viz., an optimisation problem, the key idea was the same.

Comment No. 17:

We now present yet another and a somewhat novel solution to the original problem we have been doing, viz., that of minimising the perimeter of a triangle having a given area Δ. It is novel because it is based on the peculiar relationship between this problem and the earlier problem. This relationship, which we already introduced in Chapter 6 (see Comment No. 5 there), is known as

duality. In the earlier problem, we maximised the area of a triangle having a fixed perimeter. There, the objective function was the area while the equality constraint was in terms of the perimeter function. In the problem we are now doing, things are exactly the other way. Here the constraint is that the area has a given value. But the objective function is the perimeter. (Note, however, that in the earlier problem, optimisation meant maximisation; in the present problem it means minimisation.)

We shall not go into a formal definition of duality. But basically, two problems are said to constitute a pair of (mutually) dual problems when their objective and constraint functions are interchanged. If we think of the two functions as the cost (or the resources needed) and the yield (or the profit) of some process, then mutually dual problems represent two different ways of improvement. And we can cite a number of examples. Thus, a factory can improve its performance in two ways. Either it can increase its production with the same number of workers or it can keep its production at the same level but lay off some workers. The strategies needed for these two optimisations are not quite the same and therefore we cannot expect the optimal solution to a problem to automatically give an optimal solution to its dual problem. Sometimes, however, the nature of the two objective functions is such that we can get the optimal solution to an optimisation problem from the optimal solution to the dual problem. This is exactly what happens in the present case.

Let $f(a, b, c)$ and $g(a, b, c)$ denote, respectively, the area and the perimeter of a triangle with sides a, b and c. We have already solved the problem of maximising $f(a, b, c)$ subject to a constraint of the form $g(a, b, c) = L$ and we know that the answer is that the maximum is attained for an equilateral triangle, i.e., when $a = b = c = \frac{L}{3}$. Suppose we now want to minimise $g(a, b, c)$ subject to a constraint of the form $f(a, b, c) = \Delta$. We claim that this happens when the triangle is equilateral, i.e., when $a = b = c = 2\left(\frac{\Delta}{\sqrt{3}}\right)^{\frac{1}{2}}$. We have already proved this, of course, by a variety of methods. But now we want to derive it as a consequence of the solution of the earlier problem. The basic idea is very simple. It says that if two triangles are similar and the ratio of the corresponding sides is λ, then the ratio of their perimeters is λ while the ratio of their areas is λ^2. In symbols, $g(\lambda a, \lambda b, \lambda c) = \lambda g(a, b, c)$, while $f(\lambda a, \lambda b, \lambda c) = \lambda^2 f(a, b, c)$ for all a, b, c.

So, suppose ABC is a triangle with sides a, b, c such that $f(a, b, c) = \Delta$. Let $L = g(a, b, c)$ and let $L_0 = 6\left(\frac{\Delta}{\sqrt{3}}\right)^{\frac{1}{2}}$. We have to prove that $L \geq L_0$ with equality holding if and only if ABC is equilateral. Suppose on the contrary that $L < L_0$. Then we shall get a contradiction as follows. Let $\lambda = \frac{L_0}{L}$. Then λ is a real number greater than 1. Now let $A'B'C'$ be a triangle with sides $\lambda a, \lambda b$ and λc. Then $A'B'C'$ is similar to ABC. By our observation, $A'B'C'$ has perimeter equal to $g(\lambda a, \lambda b, \lambda c) = \lambda g(a, b, c) = \lambda L = L_0$ and area equal to $f(\lambda a, \lambda b, \lambda c) = \lambda^2 f(a, b, c) = \lambda^2 \Delta$ which is greater than Δ since $\lambda > 1$. But this is a contradiction since because of the previous problem, the maximum possible area for a triangle with perimeter L_0 is Δ. So we must have $\lambda \leq 1$. Further,

if $\lambda = 1$ then $A'B'C'$ has the maximum possible area among all triangles with perimeter L_0. So $A'B'C'$ is equilateral. But then so is ABC, since ABC is similar (and, in fact, now congruent since $\lambda = 1$) to $A'B'C'$.

The trick adopted here could, of course, also be applied in the other direction. That is, having (independently) found the triangle with minimum perimeter for a given area, we could have used the method above to find a triangle of maximum area for a given perimeter. Which way the trick should be applied depends upon which of the pair of mutually dual problems is easier to solve independently. In our case, we saw that it was slightly easier to maximise the area for a given perimeter than to minimise the perimeter for a given area because the constraint involved in the former was of a simpler nature than in the latter. That is why, Jensen's inequality could be applied to it but not to the dual.

As another example of duality, we considered above the problem of minimising the perimeter of a triangle with a given base and a given height and found that the answer was an isosceles triangle. The dual problem would be to maximise the height for a given base and a given perimeter. The answer is, again, an isosceles triangle. We leave it to you to prove this both directly and using duality.

The concept of duality is not encountered very frequently in mathematics upto the Junior College level. There is, however, a class of certain special optimisation problems, called **linear programming** where it plays a very important role. These problems have many down-to-earth applications. They tell you, for example, how you should invest your money so as to maximise the interest you will get, how you should plan the schedule of your travel so as to minimise the time you spend in waiting and so on.

Comment No. 18:

We remarked earlier that normally a solution to an optimisation problem cannot be expected to easily give a solution to its dual optimisation problem. But in our case, this happened with surprising ease. It is instructive to fathom the reason for this pleasant surprise. Let us paraphrase both the problems in terms of triangular optimisation problems where the objective function is expressed as a function of the angles A, B, C rather than the sides a, b, c of a triangle. There is an advantage in doing so. When the constraints are given in terms of the sides, they can be of very different types for different problems. (In the present case, for example, for the earlier problem, it was relatively simple, viz., $a + b + c = L$. But, for the dual problem it was the more complicated (19) or (20).) But when translated in terms of angular optimisation, all problems have a common constraint viz., $A + B + C = \pi$. This brings all the problems of triangular optimisation to a common denominator, so to speak. For then, we can compare them simply in terms of their objective functions.

We have already converted the problem of maximising the area subject to a fixed perimeter. It amounts to maximising the function $f(A, B, C)$ given by (16) of the three angles A, B, C of a triangle. Let us now carry out a similar translation of the dual problem, viz., that of finding the minimum perimeter

for a triangle of a given area, say, Δ. This translation is done along the same lines as before. Instead of minimising the perimeter L we minimise its square L^2. Now, (15) already gives a relationship between Δ and L^2. If we use it to express L^2 in terms of Δ and $\sin A,\ \sin B,\ \sin C$ and then omit the constant factor 2Δ we see that the problem of minimising L^2 subject to a constant area is equivalent to that of minimising the function

$$h(A,B,C) \quad = \quad \frac{(\sin A + \sin B + \sin C)^2}{\sin A \sin B \sin C} \tag{24}$$

where A, B, C are the angles of a triangle.

As we see immediately from (16) and (24), the objective functions of the two problems (or rather, of their translations) are precisely the reciprocals of each other. So maximising the former is equivalent to minimising the latter and moreover the optimal solutions for the two will be the same. No wonder then that the solution to the problem of maximising the area subject to a given perimeter gave so easily the solution to the dual problem of minimising the perimeter subject to a given area. In effect, the two problems amount to the same!

EXERCISES

14.1 Prove that the inequality $(a+b-c)(b+c-a)(c+a-b)-abc \leq 0$ (which is a reformulation of (1)) holds for *any* three positive real numbers a, b, c with equality holding only if $a = b = c$. (1986)

14.2 If AB is a diameter of a circle and C is a point on the circle find when the area of ABC is maximum. (1983)

14.3 If the sides of a triangle subtend angles α, β, γ at the circumcentre, find the minimum value of the arithmetic mean of $\cos(\alpha + \frac{\pi}{2}), \cos(\beta + \frac{\pi}{2})$ and $\cos(\gamma + \frac{\pi}{2})$. (1987)

14.4 If a, b, c represent the sides of a triangle, prove that $3(ab + bc + ca) \leq (a + b + c)^2 \leq 4(ab + bc + ca)$. When are the first two expressions equal? (1979)

14.5 If $A > 0, B > 0$ and $A + B = \frac{\pi}{3}$, find the maximum value of $\tan A \tan B$.

14.6 Suppose h is differentiable on $[a, b]$ and $h''(x) > 0$ for all $x \in (a, b)$. Show that for any $x_1, x_2 \in [a, b]$ and for any positive real numbers λ_1, λ_2 with $\lambda_1 + \lambda_2 = 1$, the inequality $h(\lambda_1 x_1 + \lambda_2 x_2) < \lambda_1 h(x_1) + \lambda_2 h(x_2)$ holds true. (This is the same as (20) of the last chapter except that the inequality is strict. Modify the proof accordingly. What happens if $h'' < 0$ on (a, b)?)

14.7 Using the last exercise prove the assertion about when equality holds in Theorem 1. [*Hint:* The argument is of the same spirit as the proof of Theorem 1 in Chapter 6. If equality holds and $x_1 < x_2$ (say), let $x_1' =$

$x_2' = \frac{1}{2}(x_1 + x_2)$ and $x_i' = x_i$ for $i > 2$. Then $h(x_1') = h(x_2') < \frac{1}{2}(h(x_1) + h(x_2))$, whence $\sum_{i=1}^{m} h(x_i') < \sum_{i=1}^{m} h(x_i) = mh(\frac{x_1+\ldots+x_m}{m})$ by assumption. The contradiction comes by observing that $h(\frac{x_1+\ldots+x_m}{m})$ is the same as $h(\frac{x_1'+\ldots+x_m'}{m})$ which is at most $\frac{1}{m}\sum_{i=1}^{m} h(x_i')$.]

14.8 If a, b, m, n are all positive and $a \neq b$ prove that $(ma + nb)^{m+n} > (m + n)^{m+n} a^m b^n$. [*Hint:* Take logarithms and apply Exercise (14.6). The case where m, n are integers was covered in Exercise (6.13)(b).]

14.9 For every positive integer n, prove that $(n + \frac{1}{n})^{2n} > (n-1)^{n-1}(n+1)^{n+1}$.

14.10 Using Exercise (13.16) (c), prove that for any two distinct positive numbers a and b, $a^a b^b > (\frac{a+b}{2})^{a+b}$. Can this be derived from Exercise (14.6)?

14.11 Prove as many characterisations of equilateral triangles as you can (e.g., $\cot A + \cot B + \cot C = \sqrt{3}$, or, $\cot^2 A + \cot^2 B + \cot^2 C = 1$, or $R = 2r$, or, $a^2 + b^2 + c^2 = 4\sqrt{3}\Delta$ (IMO 1961).)

14.12 If $a, b, c \in (0, 1)$ and $ab+bc+ca = 1$, prove that $\frac{a}{1-a^2} + \frac{b}{1-b^2} + \frac{c}{1-c^2} \geq \frac{3\sqrt{3}}{2}$.

14.13 If h is the altitude through the vertex A of a triangle ABC, prove that $(b + c)^2 \geq a^2 + 4h^2$ with equality holding only for $b = c$.

14.14 Given two lines L_1, L_2 and points P_1, P_2 all in the same plane, construct (i) a circle which touches both L_1 and L_2 and passes through P_1 and (ii) a circle which passes through P_1 and P_2 and touches L_1.

*14.15 Given two intersecting lines L_1 and L_2 and a point P in their plane, show how to draw a line L through P which, along with L_1 and L_2, makes a triangle having (i) a given perimeter p, (ii) the least possible perimeter.

*14.16 Given a triangle ABC find the point P in it for which the sum $PA+PB+PC$ is minimum. (You will have to consider separately the case when one of the angles of the triangle exceeds 120 degrees.)

*14.17 Maximise the function $f(x) = |\sqrt{x^4 - 3x^2 - 6x + 13} - \sqrt{x^4 - x^2 + 1}\,|$ for $x \in \mathbb{R}$, using the dual form of the triangle inequality. (Anand Kumar)

*14.18 Let O be a point inside a triangle ABC and let D, E, F be the feet of the perpendiculars from O to the sides BC, CA, AB respectively. Prove that $OA + OB + OC \geq 2(OD + OE + OF)$ with equality holding only for an equilateral triangle. [*Hint* : First prove that $OA \sin A \geq EF$ and then take the projection of EF on BC. This result is known as the **Erdös–Mordell inequality** because it was posed as a problem by Erdös and solved by Mordell. Erdös, in addition to his own profound research, was also famous for posing a large number of problems in various areas of mathematics and offering cash rewards for their solutions.)

Chapter 15

LIMITS, CONTINUITY AND DERIVATIVES

We already used derivatives in Chapter 13 for finding the maxima and minima of functions. Derivatives are certain types of limits, Specifically, they are limits of the incrementary ratios of the form $\frac{\Delta y}{\Delta x}$, where the variable y is a given function of x. (Traditionally, x is called the independent and y the dependent variable. Also sometimes x is called the 'argument' and y the 'value' of the function.) The limit is taken as the increment in x, i.e., Δx tends to 0. Obviously, this limit cannot be evaluated by simply setting Δx equal to 0. The problem of finding derivatives will be considered in Comment No. 4.

In this chapter we also consider problems of geometric applications of derivatives such as finding the tangents and normals to curves which are not necessarily given as graphs of functions of the form $y = f(x)$. Two standard methods for doing so will be pointed out in Comments 2 and 3 respectively.

We shall also consider problems of finding limits of real-valued functions defined in a deleted neighbourhood of a point. Although in principle, the definition of such limits is not very different from that of the limits of sequences studied in Chapter 6, the techniques needed differ. (For example, induction will rarely work as the independent variable is continuous and not discrete.)

The concept of a limit naturally leads to that of continuity of a function. We shall do a few problems about testing the continuity and differentiability of a given function. Some theoretically important properties of continuous and differentiable functions are postponed to the next chapter on theoretical calculus.

Problems of testing continuity and differentiability appear almost every year in the JEE. Problems of finding derivatives *per se* are rarely asked recently. But problems of the geometric applications of derivatives are asked occasionally. (The Main Problem is of this type.) Sometimes the solution requires us to solve a differential equation. Such problems will be taken up in Chapter 19.

Main Problem : The curve $y = ax^3 + bx^2 + cx + 5$, touches the x-axis at $P(-2, 0)$ and cuts the y-axis at a point Q, where its gradient is 3. Then the value of $2a - 4b + 5c$ equals ..... (JEE 1994)

First Hint : Use the data to get a system of three equations in the three unknowns a, b and c.

Second Hint : Observe that if $f(x) = ax^3 + bx^2 + cx + 5$, then $f(-2) = 0$ and $f(0) = 5$.

Third Hint : The data implies that $f'(-2) = 0$ and $f'(0) = 3$.

Solution : Let $f(x)$ be as in the second hint. Then the curve cuts the y-axis at $(0, f(0))$, i.e., at $(0, 5)$. This is the point Q. The gradient of a curve at any point on it is simply the slope of the tangent at that point. So the second piece of data implies $f'(0) = 3$. Since $f'(x) = 3ax^2 + 2bx + c$, this gives one equation in a, b, c, viz.,

$$c = 3 \tag{1}$$

We need two more equations to determine a, b, c. The fact that $P(-2, 0)$ lies on the curve gives us one equation, viz., that $f(-2) = 0$. But we are also given that the x-axis touches the curve at P. This is another way of saying that the slope of the tangent at P is 0, or equivalently, $f'(-2) = 0$. Thus, the first piece of data gives us :

$$-8a + 4b - 2c + 5 = 0 \tag{2}$$

$$\text{and} \qquad 12a - 4b + c = 0 \tag{3}$$

In view of (1), these reduce, respectively, to $-8a + 4b = 1$ and $12a - 4b = -3$. Adding, $4a = -2$ and hence $a = -\frac{1}{2}$ which further gives $b = -\frac{3}{4}$. Since c is already known to be 3, this gives the value of $2a - 4b + 5c$ as 17.

Comment No. 1:

In this problem, the curve was given the form $y = f(x)$ and this made it easier to find the slope of the tangent by merely differentiating $f(x)$. A curve in a plane can also be specified parametrically by expressing both x and y as functions of some parameter, say t. Indeed, as we saw in Comment No. 8 of Chapter 9, this is how the locus of a point moving in a plane is often obtained. (A curve of the form $y = f(x)$ can also be looked at as a parametric curve by taking x itself as a parameter. Similarly, a curve of the form $x = g(y)$ can be parametrised as $x = g(y); y = y$, where y is a parameter.) So suppose $x = u(t);\ y = v(t)$ are the parametric equations of a curve. If we want we can eliminate t and get a single equation in x and y (as we have done many times in problems of finding a locus). We can then apply implicit differentiation (to be studied shortly) to get the slope of the tangent at a given point, say $P = (u(t_0), v(t_0))$ on the curve.

But this is generally not a good idea, because there is a much better way. We simply take the ratio $\dfrac{u'(t_0)}{v'(t_0)}$. Then in general[1] this ratio represents the slope of the tangent to the curve at the point $P = (u(t_0), v(t_0))$. (A good way to remember this is to write $u'(t)$ as dx/dt and $v'(t)$ as dy/dt and then 'cancel' dt to get the ratio equal to dy/dx. However, this is not a rigorous proof, because in the expression dx/dt, neither dx nor dt has an independent meaning. It is only the entire expression dx/dt which has a meaning, viz., the derivative of x w.r.t. t. The derivative is, to be sure, a *limit* of a ratio. But that does not mean it is a ratio itself. Just because something has been denoted like a ratio does not make it a ratio[2] in much the same way as a person named Goldsmith need not be a goldsmith by profession, even though the family name may have originated because some of his ancestors were goldsmiths by profession!

We use the formula for the slope of the tangent (and hence that of the normal) to a parametrised curve in the solution to the following problem (JEE 1983). Note that the elimination of the parameter is not easy and so implicit differentiation cannot be resorted to.

Find which, if any, of the following statements is true about the normal to the curve $x = a(\cos\theta + \theta\sin\theta),\ y = a(\sin\theta - \theta\cos\theta)$, at any point '$\theta$' on it.

(A) it makes a constant angle with the x-axis,

(B) it passes through the origin,

(C) it is at a constant distance form the origin.

Here by the point 'θ' one really means the corresponding point, say $P(\theta) = (a(\cos\theta + \theta\sin\theta), a(\sin\theta - \theta\sin\theta))$. The slope of the tangent to this curve at this point is $\dfrac{dy/d\theta}{dx/d\theta} = \dfrac{\theta\sin\theta}{\theta\cos\theta} = \tan\theta$ and hence the slope of the normal at $P(\theta)$ equals $-\cot\theta$. As this expression varies with θ, we see that statement (A) above is not true. To test the truth of (B) or (C), we first write the equation

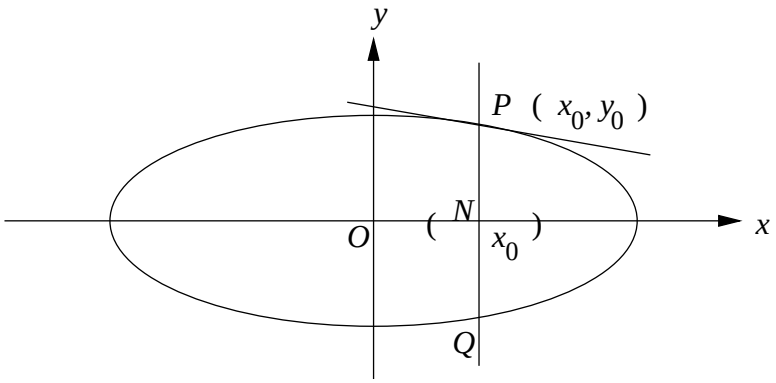
[1]This ratio makes sense only when the denominator $v'(t_0)$ is non-zero. In case it is 0 but the numerator $u'(t_0)$ is non-zero, the tangent at P is vertical, as for example, is the case with the parabola $x = at^2, y = 2at$ at the point $(0, 0)$. The trouble arises when both $u'(t_0)$ and $v'(t_0)$ vanish. Sometimes, even in such cases the curve may have a well-defined tangent at P. For example, instead of the usual parametrisation of a straight line $y = 2x$, suppose we parametrise it as $x = t^3, y = 2t^3$. The curve obviously has a tangent of slope 2 at every point. But at $t = 0$, both $\frac{dx}{dt}$ and $\frac{dy}{dt}$ vanish. Anomalies like this can be avoided by resorting to a standard, intrinsic parameter, called the **arc length parameter**, which basically represents the length of a portion of a curve from some fixed point of reference on it. For example, the parametrisation $x = x_0 + r\cos\theta, y = y_0 + r\sin\theta$ of a straight line considered in Comment No. 17 of Chapter 10 is an arc length parametrisation. But the arc length parametrisation is generally not easy to obtain. An easier way out is to take the limit of the ratio $\dfrac{u'(t)}{v'(t)}$ as $t \to t_0$. If this limit exists or is $\pm\infty$, then the curve has a tangent at P.

[2]The expressions dx and dt can each be given a meaning by defining what are called **differentials**. Then it turns out that the derivative is a genuine ratio of two differentials. It is then called, quite appropriately, a **differential quotient**. A mathematically rigorous treatment of differentials is far from easy. But they are very convenient, especially in solving differential equations as we shall see in Chapter 19.

of the normal at $P(\theta)$ as $y - a(\sin\theta - \theta\cos\theta) = -\cot\theta(x - a(\cos\theta + \theta\sin\theta))$, which reduces to $x\cos\theta + y\sin\theta = a$. The distance of the origin from it is $\dfrac{|a|}{\sqrt{\cos^2\theta + \sin^2\theta}} = |a|$. Hence (C) holds. (For (B) to hold a must vanish. But in that case the curve degenerates into a point and has no well defined normal.)

Comment No. 2:

In the problem just done, the curve was given parametrically and converting it to an implicit form was theoretically possible but practically complicated. In many problems the curve is already given in an implicit form, i.e., in the form $\phi(x, y) = 0$, where ϕ is some function of x and y. Indeed, this is the case with the familiar curves such as straight lines and conics. At the end of Comment No. 2 of Chapter 9, we already remarked that the slope of the tangent to an ellipse can be obtained using differentiation. More precisely, the kind of differentiation needed here is called **implicit differentiation**, which is somewhat a misnomer because it is the function and not the process of differentiation that is implicit here. Instead of applying it to a function of the form $y = f(x)$ we are applying it to a relation of the form $\phi(x, y) = 0$. Whether every such relation really defines y as a function or not is an important question. The answer is 'no' in general. As a simple counter-example, suppose $\phi(x, y) = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - 1$. Then the relation $\phi(x, y) = 0$ represents the ellipse, say $C = \{(x, y) : \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1\}$. Then since every vertical line of the form $x = x_0$ for $-a < x_0 < a$, cuts C in two distinct points, say, $P = (x_0, \frac{b}{a}\sqrt{a^2 - x_0^2})$ and $Q = (x_0, -\frac{b}{a}\sqrt{a^2 - x_0^2})$ as shown in the figure below, it is clear that C cannot be the graph of



a function of the form $y = f(x)$, because one of the cardinal requirements of a function is that it be single valued.

Suppose, however, that our interest is only in finding the slope of the tangent to the curve C at the point P on it. Then we can do so in a number of ways. We can parametrise the ellipse as $x = a\cos\theta, y = b\sin\theta$ and proceed as in the problem above. But it is not always easy to parametrise an arbitrary curve which is given implicitly. Another method is to solve the equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

as a quadratic in y. This gives $y = \pm\frac{b}{a}\sqrt{a^2 - x^2}$. As P lies above the x-axis, so do all points in a small neighbourhood of P. Therefore we choose the positive sign and write $y = \frac{b}{a}\sqrt{a^2 - x^2}$. Note that this does not give us the entire ellipse, but only the upper half of it. However, since our interest is only in finding the tangent at P, this half is sufficient for our purpose. Now that we have expressed the relevant portion of the curve C in the form $y = f(x)$, we can differentiate it to get the slope of the tangent at P as

$$\left(\frac{dy}{dx}\right)_{x=x_0} = f'(x_0) = -\frac{bx_0}{a\sqrt{a^2 - x_0^2}} \tag{4}$$

What we have done here is to express the relation $\phi(x, y) = 0$ *locally* in an explicit form as $y = f(x)$. The trouble with this approach is that it is not always easy to solve the given relation explicitly. In the present instance it was possible because it was a quadratic. But in general it is impossible.

There is an important result called the **implicit function theorem** which asserts that if $P = (x_0, y_0)$ is a point on the curve defined by a relation of the form $\phi(x, y) = 0$, then under certain rather general conditions on the function ϕ, there exists a small neighbourhood of the point in which we can explicitly define y as a function of x, say $y = f(x)$. The proof (and, indeed, even a complete statement of this theorem) is beyond our scope. Note further, that the theorem does *not* give any formula, nor even a method, for the function $y = f(x)$. All it says is that there exists a differentiable function $y = f(x)$ in a neighbourhood, say N, of the point x_0 such that for all $x \in N$ we have $\phi(x, f(x)) = 0$. In our present example, this means

$$\frac{x^2}{a^2} + \frac{[f(x)]^2}{b^2} = 1 \tag{5}$$

We can now differentiate both the sides and get

$$\frac{2x}{a^2} + \frac{2f(x)f'(x)}{b^2} = 0 \tag{6}$$

which is valid for all $x \in N$. In particular, putting $x = x_0$ and noting that $f(x_0) = y_0$, this gives

$$f'(x_0) = \frac{-x_0 b^2}{a^2 y_0} \tag{7}$$

which is the same as (4) since $y_0 = \frac{b}{a}\sqrt{a^2 - x_0^2}$. The crucial difference is that (4) was obtained by explicitly writing $f(x)$ as $\frac{b}{a}\sqrt{a^2 - x^2}$ while in (7) we merely used the existence of the function $f(x)$. Once this is understood, it is clear that we need not even mention the function $f(x)$ in the notation. That is, both in (5) and (6), we could have just as well retained y as such (instead of replacing it by $f(x)$). Instead of $f'(x)$ we write $\frac{dy}{dx}$ (or y'). This, in essence, is the method of **implicit differentiation**. It can be applied to find the equation of tangent to many familiar curves such as ellipses, hyperbolas, parabolas and many others.

In fact it is often applied routinely without being consciously aware that it is based on a highly non-trivial theorem.

Here is a problem where implicit differentiation is the best method.

Find the point(s) on the curve $y^3 + 3x^2 = 12y$, where the tangent is vertical. (JEE 2002)

Differentiating the given relation implicitly, $3y^2y' + 6x = 12y'$. This gives $y' = \frac{2x}{4-y^2}$. For the tangent to be vertical, the denominator must vanish which gives $y = \pm 2$ and correspondingly, $3x^2 = 12y - y^3 = \pm 16$. As x^2 cannot be negative, the value $y = -2$ is discarded. The other value, $y = 2$, gives $(\pm\frac{4}{\sqrt{3}}, 2)$ as the two points on the curve where the tangent is vertical. (We also note that at these points y' is not of the form $0/0$ as $2x \neq 0$.)

Implicit differentiation can also be applied to find derivatives of higher orders. If $y = f(x)$, and n is a positive integer, then the n-th derivative of y w.r.t. x is defined inductively as the derivative of the $(n-1)$-th derivative. It is denoted variously by $f^{(n)}(x)$, $y^{(n)}$, $D^n y$, $\frac{d^n y}{dx^n}$ or by $\frac{d^n}{dx^n}y$. Each notation is particularly convenient in certain contexts. But we shall not go into it. When n is small, $f^{(n)}(x)$ is more customarily denoted by putting an appropriate number of dashes, thus $f^{(2)}(x) = f''(x), f^{(3)}(x) = f'''(x)$ etc. Similarly, $y^{(3)} = y'''$. (Sometimes for the sake of notational uniformity, we also consider the 'zero-th derivative' $f^{(0)}(x)$. This simply means the function $f(x)$ itself.) We already saw in Chapter 13 what geometric significance the second derivatives have.

Here is a problem (JEE 1988) which involves the third derivative of an implicitly defined function.

If $y^2 = P(x)$, a polynomial of degree 3, which of the following expressions equals $2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right)$?

(A) $P'''(x) + P'(x)$ (B) $P''(x)P'''(x)$
(C) $P(x)P'''(x)$ (D) a constant

Differentiating $y^2 = P(x)$ implicitly w.r.t. x we get $2yy' = P'(x)$. Differentiating again, $2yy'' + 2(y')^2 = P''(x)$ or $2yy'' = P''(x) - 2(y')^2$. Multiplying throughout by y^2 this gives $2y^3y'' = y^2P''(x) - 2(yy')^2 = P(x)P''(x) - \frac{1}{2}(P'(x))^2$. One more differentiation gives $2\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right) = P(x)P'''(x) + P'(x)P''(x) - P'(x)P''(x) = P(x)P'''(x)$. Hence (C) is the correct answer. (Note that nowhere was it used that $P(x)$ is a polynomial of degree 3. It can be shown that a polynomial of degree n is characterised by the condition that its n-th derivative is a (non-zero) constant. The wording of the problem makes it tempting to think that this fact will probably be needed in the solution. But that turns out to be a false clue.)

Comment No. 3:

Even though implicit differentiation is the simplest method of finding the tangent (and the normal) at a given point on a curve, if we want to do something more with these tangents and the normals, other methods such as parametrising the curve may be preferable. Here is an illustration.

Suppose that the normals drawn at three different points on the parabola $y^2 = 4x$ pass through the point (h, k). Show that $h > 2$. (JEE 1981)

Here we first have to find the condition that the normal at a point P on the parabola $y^2 = 4x$ passes through (h, k). And then we have to show that in case there are three distinct points satisfying this condition, then $h > 2$. Obviously the condition will have to be in the form of an equation in some parameter and the problem then amounts to showing that $h > 2$ is a necessary condition for this equation to have three distinct roots. This suggests that we should represent the points on the parabola $y^2 = 4x$ in terms of a parameter. Here we can take y itself as a parameter. However, it is preferable to take the standard parametrisation of the parabola, viz., $x = t^2$, $y = 2t$ because then one can use the readymade formulas for the equation of the normal at a point $(t^2, 2t)$. (Mathematically, this makes little difference. But it saves some time and the possibility of errors.) Then the slope of the tangent at $(t^2, 2t)$ is $\frac{1}{t}$ and hence that of the normal is $-t$. So the equation of the normal at $(t^2, 2t)$ is $y - 2t = -t(x - t^2)$. This normal will pass through (h, k) if and only if $k - 2t = -t(h - t^2)$ which can be rewritten as $t^3 + (2 - h)t - k = 0$ or as $f(t) = 0$, where $f(t)$ is the cubic polynomial $f(t) = t^3 + (2 - h)t - k$. Here the leading coefficient is positive. So, as we saw in Chapter 13, Comment No. 13, the graph of $f(t)$ will either have a local maximum followed by a local minimum or will be strictly increasing depending upon whether the derivative $f'(t)$ has distinct real roots or not. In the present problem, it is given that $f(t)$ has three distinct zeros. So only the first possibility holds, i.e., the derivative $f'(t) = 3t^2 + (2 - h)$ has two distinct real roots. So, its discriminant, viz., $12(h - 2)$ is positive, i.e., $h > 2$ as was to be proved. (We stress again that the reasoning in classifying the graphs of a cubic was based on the Lagrange Mean Value theorem, which we shall study in the next chapter.)

Comment No. 4:

Finding the derivative of a given function is a relatively routine task (in sharp contrast with the task of finding antiderivatives which often demands considerable ingenuity, as will be seen in Chapter 17). There are four basic rules which express the derivatives of a sum, a difference, a product and a quotient of two functions in terms of the derivatives of the two functions. These rules are very standard and so we omit their explicit statements. In fact we have already used some of them. There is also the **chain rule**, which in essence gives the derivative of the composite of a function. This rule, too, is needed so frequently, that it is often used without an explicit mention. The usual statement of the chain rule is that if a function $f(x)$ is differentiable at a point x_0 and another function $g(x)$ is differentiable at the point $g(x_0)$, then the function $g(f(x))$ is differentiable at x_0 and has derivative $g'(f(x_0))f'(x_0)$ at x_0. In this form the statement looks rather complicated. But if we write the derivatives as differential quotients and introduce the symbols $y = f(x)$ and $z = g(y)$, then the chain rule can be written as $\dfrac{dz}{dx} = \dfrac{dz}{dy}\dfrac{dy}{dx}$ or more precisely as $\left(\dfrac{dz}{dx}\right)_{x=x_0} = \left(\dfrac{dz}{dy}\right)_{y=f(x_0)} \left(\dfrac{dy}{dx}\right)_{x=x_0}$.

If we interpret the derivative as related rate (i.e., $\frac{dy}{dx}$ as the rate at which y increases w.r.t. x), then in effect all that the chain rule says is that if A is running at the speed of 10 km/hour and B is running twice as fast as A, then B is running at the speed of 20 km/hour.

It is interesting to note that in order to apply the chain rule, we do not need the functions *per se* but only their derivatives. Suppose, for example (JEE 1979), that we are given that $f'(x) = \sin(x^2)$ and $y = f(x^2 + 1)$. Here $f(x)$ is not given. Nor do we need it. For, by the chain rule, $\frac{dy}{dx} = f'(x^2+1)\frac{d}{dx}(x^2+1) = \sin((x^2 + 1)^2) \times (2x) = 2x \sin(x^4 + 2x^2 + 1)$.

A particular case of the chain rule deserves an explicit mention. If the two functions f and g are inverses of each other, then writing y for $f(x)$, we have $x = g(y)$. Hence $1 = \frac{dx}{dx} = \frac{dx}{dy}\frac{dy}{dx} = g'(y)f'(x)$. Verbally, the derivative of the inverse function is the reciprocal of the derivative of the original function. This is also obvious geometrically, because if we sketch the graphs of $y = f(x)$ and $y = g(x)$ with the same axes, then they are mirror images of each other w.r.t. the line $y = x$. (To see this, draw the graphs of, say, $y = x^2$ and $y = \sqrt{x}$ for $x \geq 0$. These two functions are inverses of each other. See Exercise (13.23) for another such pair.) So the tangent to the first graph at a point (x_0, y_0) will be the mirror image of the tangent to the second graph at the point (y_0, x_0). But then the slopes of the two tangents are reciprocals of each other as the angles they make with the x-axis are complementary.

All these rules are parasitical in that they only serve to express the derivative of a function in terms of the derivatives of simpler functions. Before applying these rules we must first find some derivatives directly from the definition (or, *ab initio* as it is often called). The three basic derivatives of this type are :

$$\frac{d}{dx}x^n = nx^{n-1} \tag{8}$$

$$\frac{d}{dx}\sin x = \cos x \tag{9}$$

$$\frac{d}{dx}e^x = e^x \tag{10}$$

(8) is first proved for the case when n is a non-negative integer using either the binomial theorem or induction on n (along with the product rule for derivatives). It is then extended to negative integral exponents by writing x^{-n} as $\frac{1}{x^n}$ and applying the quotient rule. The case where n is of the form $\frac{1}{q}$ (where q is a positive integer), is covered by observing that $y = x^q$ is the inverse function of $y = x^{1/q}$ and applying the rule for the derivative of the inverse function mentioned above. The case $n = \frac{p}{q}$ then follows by writing x^n as $(x^{1/q})^p$ and applying the chain rule. To extend (8) to the case where the exponent n is irrational is beyond our scope.

The proofs of (9) and (10) reduce ultimately to showing that both the limits $\lim_{h\to 0}\frac{\sin h}{h}$ and $\lim_{h\to 0}\frac{e^h - 1}{h}$ equal 1. The former is a consequence of the inequality $\sin h < h < \tan h$ (for small positive values of h) mentioned in Chapter

6, Comment No. 9. The second limit is derived by writing e^h as $\lim_{n\to\infty}(1+\frac{h}{n})^n$. We omit the details which may be found in standard texts. Using (9) and the chain rule, we get the derivative of $\cos x$. All we have to do is to write $\cos x$ as $\sin(\frac{\pi}{2}-x)$ to get $\frac{d}{dx}\cos x = -\cos(\frac{\pi}{2}-x) = -\sin x$. Having known the derivatives of $\sin x$ and $\cos x$ we can now find the derivatives of other trigonometric functions such as $\tan x$ etc. We can then go on and apply the rule for the derivative of an inverse function to find the derivatives of the inverse trigonometric functions. For example, suppose $y = \sin^{-1} x$. Then $x = \sin y$ which gives $\frac{dx}{dy} = \cos y$ which equals $\pm\sqrt{1-\sin^2 y}$, i.e., $\pm\sqrt{1-x^2}$. As $\sin^{-1}$ is a strictly increasing function we choose the positive sign. Thus we get $\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$. The derivatives of the other inverse trigonometric functions are obtained analogously.

Similarly, since $\ln x$ is the inverse function of the exponential function, (10) coupled with the rule for differentiation of the inverse function gives the derivative[3] of $\ln x$ as $\frac{1}{x}$. Finally, for any $a > 0$, we write a^x as $e^{x\ln a}$. From (10) and the chain rule it then follows that $\frac{d}{dx}a^x = (\ln a)a^x$.

As a direct application of the basic formulas and the various rules, suppose (JEE 1981) we want to find $\frac{dy}{dx}$, where $y = e^{x\sin(x^3)} + (\tan x)^x$. We first write the second term as $e^{x\ln\tan x}$. (Obviously, this is not defined when $\tan x$ is negative or 0. But in problems like this, it is not expected that you identify the points where the derivative exists unless the question specifically asks you to do so.) The calculation of $\frac{dy}{dx}$ begins by applying (10) and the chain rule and proceeds as follows :

$$\begin{aligned}
\frac{dy}{dx} &= e^{x\sin(x^3)}\frac{d}{dx}(x\sin(x^3)) + (\tan x)^x\frac{d}{dx}(x\ln\tan x)\\
&= e^{x\sin(x^3)}(1\times\sin(x^3) + x\times 3x^2\cos(x^3))\\
&\quad +(\tan x)^x(1\times\ln\tan x + x\times\frac{1}{\tan x}\frac{d}{dx}\tan x)\\
&= e^{x\sin(x^3)}(\sin(x^3)+\cos(x^3)\frac{d}{dx}(x^3)) + (\tan x)^x(\ln\tan x + \frac{x\sec^2 x}{\tan x})
\end{aligned}$$

After gaining some practice, it is hardly necessary to show all the intermediate steps in the calculation. In fact, it is not difficult to write down the final answer directly. Sometimes it is a good idea to simplify the given function before differentiating it. For example, it is foolish to find $\frac{d}{d\theta}(\sin^2\theta+\cos^2\theta)$ by differentiating each term separately. Since the function is identically equal to 1, the derivative is 0 everywhere. (A problem where the simplification is so glaringly obvious is rarely asked. So one should keep looking for a more subtle possibility of a simplification.)

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[3] An alternate approach is to first define $\ln x$ as $\int_1^x \frac{1}{t}dt$. It then follows from the fundamental theorem of calculus (to be studied in Chapter 17) that $\frac{d}{dx}\ln x = \frac{1}{x}$. One then defines the exponential function as the inverse function of $\ln x$ and then applies the formula for the derivative of the inverse function to conclude that $\frac{d}{dx}e^x = e^x$.

Sometimes we want to find $\frac{dy}{dz}$, where y and z are functions of some other variable, say x. Here it is assumed tacitly that z as a function of x is invertible. For then one can express x in terms of z. By the chain rule, the desired derivative $\frac{dy}{dz}$ equals $\frac{dy}{dx}\frac{dx}{dz}$. To find the latter, it is not necessary to express x explicitly in terms of z. From the rule about the derivatives of inverse functions, we know that $\frac{dx}{dz} = \frac{1}{dz/dx}$ and this is sufficient to find $\frac{dy}{dz}$. As an example, suppose we want to differentiate $y = \log(\cos(a^{3x+2}))$ with respect to $z = a^{\log x}$. (JAT 1980) (Here log is to be taken as the natural logarithm. Nowadays, it is more customary to denote it by ln.) A direct calculation gives $\dfrac{dy}{dx} = \dfrac{-\sin(a^{3x+2})3\log a\, a^{3x+2}}{\cos(a^{3x+2})}$ and $\dfrac{dz}{dx} = \log a\, a^{\log x}\dfrac{1}{x}$. Hence $\dfrac{dy}{dz} = \dfrac{dy/dx}{dz/dx} = \dfrac{-3xa^{3x+2}\tan(a^{3x+2})}{a^{\log x}}$.

When y and z are given as functions of some variable x and we are asked to find $\frac{dy}{dz}$, we are at liberty to change the variable x to some other variable, say u, with a suitable substitution, because such a change does not affect $\frac{dy}{dz}$. For, $\frac{dy}{dz}$ also equals $\frac{dy/du}{dz/du}$. If the substitution is chosen judiciously, this ratio may be simpler to find than the ratio $\frac{dy/dx}{dz/dx}$. Suppose, for example, that we want to find the derivative of $\sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$ w.r.t. $\sqrt{1-x^2}$ at $x = \frac{1}{2}$. (JEE 1984) These functions are rather complicated to differentiate w.r.t. x. However, let us put $x = \cos\theta$. Then $2x^2 - 1 = 2\cos^2\theta - 1 = \cos 2\theta$ and so $\sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$ is simply 2θ. (Actually, the choice of the substitution $x = \cos\theta$ was dictated by looking at the expression $\sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$.) Similarly, $\sqrt{1-x^2}$ is simply $\sin\theta$. So the derivative of $\sec^{-1}\left(\dfrac{1}{2x^2-1}\right)$ w.r.t. $\sqrt{1-x^2}$ is the same as that of 2θ w.r.t. $\sin\theta$. This equals the ratio $\dfrac{\frac{d}{d\theta}2\theta}{\frac{d}{d\theta}\sin\theta} = \frac{2}{\cos\theta}$ which is simply $\frac{2}{x}$. Hence the desired derivative is $\frac{2}{1/2} = 4$.

Comment No. 5:

Derivatives are limits of a special type. So it is not surprising that certain limits can be evaluated by recognising them as derivatives either directly or with suitable manipulations. As an example of the former, suppose we want to evaluate the limit $\lim\limits_{h\to 0}\dfrac{(a+h)^2\sin(a+h) - a^2\sin a}{h}$. (JEE 1980) We note that this is nothing but the derivative of the function $f(x) = x^2\sin x$ at $x = a$. Hence it equals $2a\sin a + a^2\cos a$.

As an example where a given limit can be related to a derivative after some manipulation, consider $\lim\limits_{x\to 1}(1-x)\tan\dfrac{\pi x}{2}$. (JEE 1984) Here $\tan\dfrac{\pi x}{2}$ tends to $-\infty$ as $x \to 1$ from the right and to ∞ as $x \to 1$ from the left. The other factor, viz., $(1-x)$ tends to 0. So this limit is of the form $0\times\infty$ which is an indeterminate form, (see Exercise (6.50)). However, if we take the reciprocal,

we get $\dfrac{\cot\frac{\pi x}{2}}{1-x}$. Since $\cot\frac{\pi}{2}=0$, this is essentially the incrementary ratio of the function $\cot\frac{\pi x}{2}$ at $x=1$ except that the denominator is $1-x$ and not $x-1$. This can be easily corrected. Since the derivative of $\cot x$ is $-\text{cosec}^2 x$, we have $\dfrac{d}{dx}\cot\dfrac{\pi x}{2}=-\dfrac{\pi}{2}\text{cosec}^2\dfrac{\pi x}{2}$ and so $\lim\limits_{x\to 1}\dfrac{\cot\frac{\pi x}{2}}{1-x}=\dfrac{\pi}{2}\text{cosec}^2\dfrac{\pi}{2}=\dfrac{\pi}{2}$. The given limit equals the reciprocal of this, i.e., $\dfrac{2}{\pi}$.

Sometimes we encounter a limit of the form $\lim\limits_{x\to a}\dfrac{u(x)-v(x)}{x-a}$, where the functions u and v are such that $u(a)=v(a)$. In such a case we can rewrite the numerator as $(u(x)-u(a))-(v(x)-v(a))$. It then follows that the given limit is $u'(a)-v'(a)$, provided, of course, that both u and v are differentiable at a. For example, $\lim\limits_{x\to 0}\dfrac{\cos x-e^x}{x}$ equals $-\sin 0-e^0$, i.e., -1.

Comment No. 6:

In the examples just considered, the limits were either already in the form $\frac{0}{0}$ (i.e., where both the numerator and the denominator tend to 0) or could be reduced to it by some means (e.g., by taking reciprocals). It should be noted that not every limit of the form $\frac{0}{0}$ can be evaluated by recognising it as a derivative. Consider for example the limit $\lim\limits_{x\to 0}\dfrac{x2^x-x}{1-\cos x}$. (JEE 1980) Here the denomiator is not of the form $x-a$ and so the limit, as it stands, is not a derivative of any function. There is a way to express it in terms of several other derivatives, but it is not so obvious. We first express $1-\cos x$ as $2\sin^2\frac{x}{2}$. Now, $\lim\limits_{x\to 0}\dfrac{\sin\frac{x}{2}}{\frac{x}{2}}$ can be treated as the derivative of $\sin y$ at $y=0$ if we put $y=\frac{x}{2}$. Thus, by (9) this limit equals $\cos 0=1$. (Actually, it is not quite correct to say that we are using (9) here, because as remarked before, the derivation of (9) is based on the fact that $\lim\limits_{h\to 0}\dfrac{\sin h}{h}=1$. So we could have as well used this fact directly, bypassing (9).) Now in the original limit we divide both the numerator and the denominator of the given ratio by $\frac{x^2}{2}$ and write it as $\dfrac{2\left(\frac{2^x-1}{x}\right)}{\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2}$. As just remarked, the limit of the denominator as $x\to 0$ is the square of a derivative while that of the second factor in the numerator is the derivative of 2^x at $x=0$. The latter equals $\ln 2\times 2^0$ i.e., $\ln 2$ and so the given limit equals $\frac{2\ln 2}{(1)^2}$ i.e., $2\ln 2$.

The essence of the argument above was that for small non-zero values of x, $1-\cos x$ was comparable to $\sin^2 x$ and hence to x^2 in the sense that even though both $1-\cos x$ and x^2 tend to 0 as $x\to 0$, their ratio tends to a finite non-zero limit. This limit equals $\frac{1}{2}$. For many purposes the actual value of this limit is not as important as the knowledge[4] that it is finite and non-zero because this knowledge tells us how rapidly a given expression tends to 0. In

[4] As a real life analogy, in many situations the knowledge of the marital status of a person matters more than the identity of the spouse!

somewhat technical terms we say that for small x, $1-\cos x$ is of the same **order of magnitude** as x^2. This means, for example, that if the first seven places of decimal in the expansion of x are 0, then in $1-\cos x$, the first fourteen places of decimal will be 0. (Of course, if the ratio of the limits were greater than 1, this will not be quite true. Still, it is enough to give a qualitative idea as to how small $1-\cos x$ is.)

The basic idea introduced here is not entirely new. In Chapter 3, Comment No. 1, we defined the concept of the order or the multiplicity of a root of a polynomial. Suppose α is a root of order r of a polynomial $p(x)$. Then we can write $p(x)$ as $(x-\alpha)^r q(x)$, where $q(x)$ is a polynomial. Note that $q(\alpha) \neq 0$ as otherwise $(x-\alpha)$ would be a factor of $q(x)$ and thereby $(x-\alpha)^{r+1}$ will also divide $p(x)$, contradicting the definition of multiplicity of a root. Note that for $x \neq \alpha$, $q(x) = \frac{p(x)}{(x-\alpha)^r}$. So $\lim\limits_{x\to\alpha} \dfrac{p(x)}{(x-\alpha)^r} = \lim\limits_{x\to\alpha} q(x) = q(\alpha) \neq 0$. (Here we are using the continuity of the polynomial function $q(x)$ at α. We shall systematically study continuity later in this chapter.) In other words, for x close to α, $p(x)$ is of the same order of magnitude as $(x-\alpha)^r$. So the concept of the order of magnitude is a generalisation of that of multiplicity of a root. The function $f(x) = 1-\cos x$ has a zero at 0. But $f(x)$ is not a polynomial and so there is no question of factoring out any power of $(x-0)$ from it. Still, the fact that $\frac{f(x)}{(x-0)^2}$ tends to a finite non-zero limit gives us the same information. So we say that $1-\cos x$ has a zero of order (or multiplicity) 2 at 0. Similarly, $\sin x$ and $\tan x$ have zeros of order 1 each at 0 while $\tan x - 1$ has a zero of order 1 at $\frac{\pi}{4}$. Evidently, if $f(x)$ and $g(x)$ have a common zero, say α, of multiplicities m and n respectively, then α is a zero of order $m+n$ of the product function $f(x)g(x)$. As for the quotient $\frac{f(x)}{g(x)}$, it is easy to show that if $m > n$ then it has a zero of order $m-n$ at α; if $m = n$ then the ratio tends to a finite non-zero limit, while if $m < n$ then its limit does not exist (except as $\pm\infty$). It is also true that α is a zero of $f(x) \pm g(x)$. But very little can be said about its order. For example, take $f(x) = x$ and $g(x) = x - \frac{1}{2}x^n$, where n is a positive integer. Then both f and g have a zero of order 1 each at 0. But $f(x) - g(x) = \frac{1}{2}x^n$ has a zero of order n. As n can be chosen at will, nothing can be said about the order of the zero of $f-g$ in terms of its orders w.r.t. f and g.

Considerations of this type enable us to answer, just by inspection, some questions about limits where the information sought is of a qualitative type. Suppose for example, that we want to determine the integer n for which $\lim\limits_{x\to 0} \dfrac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number. (JEE 2002) Note that we are not asked to find the value of the limit. So the problem is of a qualitative type. As we already saw, $\cos x - 1$ is comparable to x^2 near $x = 0$. As for the second factor of the numerator, viz., $(\cos x - e^x)$, we saw at the end of the last Comment that $\frac{\cos x - e^x}{x}$ tends to a finite non-zero limit as $x \to 0$. So in order that the given limit have a finite non-zero value, the integer n must equal $2+1$, i.e., 3.

Comment No. 7:

We now move on to a more difficult class of limits. These limits are in the form $\frac{0}{0}$ or can be easily cast into it. But neither do they represent the derivative of some function, nor can they be related to derivatives as we did in the problems above. For example, consider $\lim_{x\to 0} \frac{1}{\sin x} - \frac{1}{x}$. It is easy to convert this to the $\frac{0}{0}$ form by writing $\frac{1}{\sin x} - \frac{1}{x}$ as $\frac{x-\sin x}{x \sin x}$. Here both the numerator and the denominator have 0 as a zero. The order of it for the denominator is 2. But as just remarked above, very little can be said about the numerator. So the fate of the ratio as x tends to 0 remains undecidable by the methods of the last Comment.

In such cases, there are three standard methods. Arguing as in the problem done in the last Comment, we get that $\lim_{x\to 0} \frac{x - \sin x}{x \sin x}$ is the same as $\lim_{x\to 0} \frac{x - \sin x}{x^2}$. We can now expand $\sin x$ as a power series in x as mentioned in the footnote of Chapter 11, Comment No. 1. Specifically, $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. If we do so, we get that the ratio $\frac{x - \sin x}{x^2}$ equals $-\frac{x}{6} + \frac{x^3}{5!} - \ldots$. As x tends to 0, each term tends to 0 and so the given limit equals 0.

This method is not mathematically wrong. But before we can apply it we have to do a lot of spadework. First, we have to define what a power series is. So far we have not even defined infinite series. Then we have to show that the power series above indeed has $\sin x$ as its sum. Finally, we have to justify that we can take the limits term by term to get the limit of the power series. Rigorous proofs of these assertions are far from easy and so the method is really beyond our scope. We mention it because power series were very popular and routinely used in old textbooks, without bothering much about rigour. Students studying from such books often fail to understand why they are not used any more at elementary levels. The reason is that an indiscriminate use of power series (and, in general, of any infinite series) can lead to disasters.

The second method of evaluating a limit like $\lim_{x\to 0} \frac{x-\sin x}{x^2}$ is based on the observation that we do not need the entire power series of $\sin x$ for this purpose. It is enough if we can show that $\sin x$ is a sum of x and something which tends to 0 more rapidly than the denominator x^2. This approach requires the use of a theorem called **Taylor's theorem** which is an extension of the Lagrange's Mean Value Theorem. We shall consider it in the next chapter.

The third method, which we shall follow here is based on what is called **l'Hôpital's rule**. There are various formulations of this rule. The simplest (and probably the most popular) formulation is as follows. Suppose that $f(x), g(x)$ are two functions which are differentiable at a point c. If $f(c)$ and $g(c)$ are both 0, then the limit $\lim_{x\to c} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$. Suppose, further that $g'(c) \neq 0$. Then l'Hôpital's rule says that $\lim_{x\to 0} \frac{f(x)}{g(x)}$ equals $\frac{f'(c)}{g'(c)}$. The proof follows by rewriting $\frac{f(x)}{g(x)}$ as $\frac{(f(x)-f(c))/(x-c)}{(g(x)-g(c))/(x-c)}$ and taking the limit of the numerator and the denominator as x tends to c.

In fact the proof is so trivial that it may appear that we can as well do without l'Hôpital's rule. And this indeed is the case when the rule is applied in the form stated above. For example, the limit $\lim_{x\to 1} \dfrac{1-x}{\cot \frac{\pi x}{2}}$ can be evaluated using l'Hôpital's rule since the derivative of the denominator at $x = 1$ is $-\text{cosec}^2 \frac{\pi}{2} \times \frac{\pi}{2} = -\frac{\pi}{2}$ while that of the numerator is -1. So by l'Hôpital's rule, the limit equals $\frac{2}{\pi}$. But we had anyway found this limit earlier by recognising it as the reciprocal of a derivative (except for the minus sign). Not surprisingly, the form of l'Hôpital's rule as given above is called its weak form.

The real strength of l'Hôpital's rule becomes apparent when it is applied in its stronger form. This happens when not only the functions $f(x)$ and $g(x)$ but also their derivatives $f'(x)$ and $g'(x)$ vanish at c. In such a case, assuming that f and g are twice differentiable at c, we can apply l'Hôpital's rule (the weak form) to the ratio $\frac{f'(x)}{g'(x)}$ to get that its limit as $x \to c$ equals $\frac{f''(c)}{g''(c)}$ provided, of course, that $g''(c) \neq 0$. Now to conclude from this that the ratio $\frac{f(x)}{g(x)}$ also tends to the same limit (viz., $\frac{f''(c)}{g''(c)}$) is non-trivial. It requires the use of what is called the Cauchy's Mean Value Theorem, which is a generalisation of the Lagrange's Mean Value Theorem and will be taken up in the next chapter.

Let us try this strong form of l'Hôpital's rule for the limit $\lim_{x\to 0} \dfrac{x2^x - x}{1 - \cos x}$ which we already found as $2 \ln 2$ by a rather tricky argument. With the stronger form of l'Hôpital's rule, the job can be done much more easily. The derivative of the numerator is $2^x - x \ln 2\ 2^x - 1$ while that of the denominator is $\sin x$. Both of these vanish at 0. So we take the second derivatives which are, respectively, $2 \ln 2\ 2^x + x(\ln 2)^2 2^x$ and $\cos x$. At $x = 0$, their values are $2 \ln 2$ and 1 respectively. As the latter is non-zero, we get the original limit equal to $\frac{2 \ln 2}{1}$ which is the same answer as before.

To convincingly prove the superiority of the l'Hôpital's rule, let us apply it to evaluate $\lim_{x\to 0} \frac{x - \sin x}{x^2}$ with which we are still fumbling. The first derivative of the denominator vanishes at 0. The second derivative is identically 2 and so the stronger form of the rule will be needed. The first derivative of the numerator is $1 - \cos x$ which indeed vanishes at $x = 0$ as it ought to for the rule to apply. The second derivative, viz. $\sin x$, also vanishes at 0. So the desired limit is 0. (If we call the limit as L and write $x = 2y$ then $\frac{x - \sin x}{x^2} = \frac{2y - \sin 2y}{4y^2} = \frac{y - \sin y}{2y^2} + \frac{\sin y \sin^2(y/2)}{y^2}$. As x tends to 0, so does y. Hence the limit of the first term is $L/2$ while that of the second is easily seen to be 0. This gives an equation $L = L/2 + 0$, whence $L = 0$. It would thus appear that we can bypass the l'Hôpital's rule here too. But this derivation presupposes that L exists and only calculates its value. The l'Hôpital's rule, on the other hand, proves the existence of the limit *and also* gives its value.)

If $\frac{f''(c)}{g''(c)}$ is also of the form $\frac{0}{0}$ we try the third derivatives. We keep repeating this process till we reach a higher order derivative of g which does not vanish at c. (There is however, no guarantee that this process will always terminate. There are non-zero functions all whose derivatives vanish at some

point. But they are very weird. We omit them.) For example, suppose that in the problem just solved, instead of x^2 we had x^3 in the denominator. Then one more differentiation would give us $\lim_{x\to 0} \dfrac{x-\sin x}{x^3} = \dfrac{1}{6}$. (Here, too, putting $x = 2y$ will give us an equation for the limit, viz., $L = L/4 + 1/8$ and hence $L = 1/6$. But, as commented above, this method is not logically sound. A reasoning based on a wrong presumption of the existence of a limit can lead to disaters. See Exercise (15.27).)

Comment No. 8:

The limits to which we applied the l'Hôpital's rule so far were of the form $\frac{0}{0}$ or $0 \times \infty$ or $\infty - \infty$. The rule can also be applied to limits of the form $\frac{\infty}{\infty}$ if we first convert them to the $\frac{0}{0}$ form by taking reciprocals (i.e., by writing $\frac{f(x)}{g(x)}$ as $\frac{\frac{1}{g(x)}}{\frac{1}{f(x)}}$). These forms are examples of what are called **indeterminate forms** because in such cases the limit cannot be determined by using the usual rules for finding the limit of a quotient, a difference or a product. There is yet another form where the rule can be applied. It is called the 1^∞ form and arises when we want to take the limit of an exponential expression of the form $(f(x))^{g(x)}$, where the base tends to 1 while the exponent tends to ∞ as x tends to c (say). (The limit of the sequence $\{(1+\frac{1}{n})^n\}$ is also of the form 1^∞. Of course, here the variable n is discrete and is tending to ∞ rather than to a finite value. But the essential idea is the same.) In such cases the standard method is to take the logarithm of the expression, i.e., to take $g(x)\ln f(x)$ which is of the form $\infty \times 0$. The limit of this logarithm can often be evaluated using the techniques we have studied so far. Suppose $\lim_{x\to c} g(x)\ln f(x) = L$. Then taking back the exponentials, the original limit viz., $\lim_{x\to} f(x)^{g(x)}$ equals e^L. Here we are using a result about limits to the effect that the limit of a logarithm is the logarithm of the limit. This is a consequence of the continuity of the log function. (See Theorem 1 in the next chapter.)

As an illustration of this procedure let us find $\lim_{x\to 0}\{\tan(\frac{\pi}{4}+x)\}^{1/x}$ (JEE 1993). We first take logarithms and find $\lim_{x\to 0} \dfrac{\ln(\tan(\frac{\pi}{4}+x))}{x}$. Recognising this as a derivative we see that its value is the value of $\dfrac{\sec^2(\frac{\pi}{4}+x)}{\tan(\frac{\pi}{4}+x)}$ at $x = 0$. A direct substitution gives this value as 2. So the original limit is e^2.

The applicability of l'Hôpital's rule (and other techniques for finding limits) can be further enhanced by suitable substitutions. Suppose, for example, that we want to evaluate $\lim_{x\to\infty}\left(\dfrac{x+6}{x+1}\right)^{x+4}$. (JEE 1990) Rewriting $\dfrac{x+6}{x+1}$ as $\dfrac{1+\frac{6}{x}}{1+\frac{1}{x}}$ we see that it tends to 1 as $x \to \infty$. So the given limit is in the indeterminate form 1^∞ as in the last problem. But the difference is that now we are

taking limit as x tends to ∞. In such cases, the standard substitution is to put $y = \frac{1}{x}$, express the given function in terms of y and then take the limit as $y \to 0$. (Actually, it suffices to consider only the positive values of y, i.e., to consider the limit as y tends to 0 from the right, or from above as it is called.) In the present problem $x = \frac{1}{y}$ changes $\left(\frac{x+6}{x+1}\right)^{x+4}$ to $\left(\frac{1+6y}{1+y}\right)^{\frac{1+4y}{y}}$ whose natural logarithm is $\frac{1+4y}{y}(\ln(1+6y) - \ln(1+y))$. Rewriting this as $(1+4y)\frac{\ln(1+6y)-\ln(1+y)}{y}$, we see that the first factor tends to 1 as y tends to 0. As for the second factor, applying l'Hôpital's rule, we see that its limit as $y \to 0$ equals the value of $\frac{6}{1+6y} - \frac{1}{1+y}$ at $y = 0$, i.e., 5. Hence the original limit equals e^5.

Comment No. 9:

Despite the importance of 1'Hôpital's rule in evaluating limits, it is not such a golden tool as it appears. Its applicability is restricted, first of all, by the differentiability requirement. For example, the function $\sqrt{x}$ is not differentiable at 0 and so the rule cannot be applied to a limit like $\lim\limits_{x\to 0^+} \dfrac{\sin(\sqrt{x})}{\sqrt{x}}$,, whereas a mere substitution $\sqrt{x} = y$ would give the answer. It is also to be noted that for a limit like $\lim\limits_{x\to 0} \frac{\sin x}{x}$, l'Hôpital's rule gets undeserved credit because to apply it you need the derivative of the sine function and to find the latter you need to evaluate $\lim\limits_{x\to 0} \frac{\sin x}{x}$ anyway !

A good example of a bad use of l'Hôpital's rule is given by $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\tan x}{\sec x}$ which is in the $\dfrac{\infty}{\infty}$ form. If we notice that $\sec x$ is a factor of $\tan x$, then cancelling it, this limit is simply $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sin x}{1}$ which trivially equals 1, by continuity of the sine function. Or, as remarked above, we can convert the given limit to $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\cos x}{\cot x}$ and apply l'Hôpitals's rule. But if we try to apply l'Hôpital's rule directly to $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\tan x}{\sec x}$ the problem reduces to finding $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sec^2 x}{\tan x \sec x}$. Now $\sec x$ is an obvious common factor. If we ignore it, the subsequent derivatives will get horrendous. Even if we cancel $\sec x$ now, it is not of much help. For, we would then have to find $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sec x}{\tan x}$. Another application of l'Hôpital rule would reduce this to $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sec x \tan x}{\sec^2 x}$. Cancelling sec x at this stage would get us back to the original problem! And this cycle would repeat endlessly.

Finally, l'Hôpital's rule may give wrong results if applied to limits which are not in the indeterminate forms. A simple counter-example is $\lim\limits_{x\to 0} \dfrac{x+1}{x+2}$ which is $\dfrac{1}{2}$. But l'Hôpital's rule would give it as $\lim\limits_{x\to 0} \dfrac{1}{1} = 1$.

So, like any other tool in mathematics, l'Hôpital's rule is to be applied with discretion.

Comment No. 10:

The limits we considered so far were related to derivatives in some form or the other. Let us now study limits in general. Logically, the definition and the basic properties of limits should precede limits. But we have chosen to reverse the order for two reasons. First, we believe you already have a working knowledge of the former. Secondly, if we begin with a general limit of the form $\lim_{x\to c} f(x)$, then despite all emphatic warnings, there is a tendency to confuse it with $f(c)$, i.e., with the value of $f(x)$ at $x = c$. To some extent this is natural, because for most of the functions we encounter such as the polynomials, trigonometric and exponential functions, the two are indeed equal. And once this wrong notion (viz., that $\lim_{x\to c} f(x)$ is conceptually the same as $f(c)$) sets in the mind, it is difficult to remove it. If we start with derivatives, which are limits of the form $\lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, this danger is automatically precluded because in this case the expression is not even defined at $h = 0$. Also, from a practical point of view, the derivatives are probably the most important types of limits.

We already defined in Chapter 6, Comment No. 20 what is meant by saying that $\lim_{n\to\infty} a_n = L$, where $\{a_n\}_{n=1}^{\infty}$ is a sequence of real numbers and L is also a real number. The definition of $\lim_{x\to c} f(x) = L$ is basically similar, except for two points of differences. First, the variable n in the case of a sequence is a discrete variable while now we are dealing with a variable x which is continuous, in that it assumes values in some open interval containing the point c. Secondly, n was tending to ∞, whereas here the variable x is approaching c. The crucial point is that n can never be actually equal to ∞. In fact, as stressed in Chapter 6, ∞ is not even a real number, let alone a positive integer. Here, on the other hand, there is nothing to automatically preclude x from assuming the value c.

The most critical point to note is that as far as the limit of $f(x)$ as x tends to c is concerned, the value of f at points near c matters a lot, but its value at c makes absolutely no difference. In fact, it makes no difference whether f is defined at c or not. In order for $\lim_{x\to c} f(x)$ to make sense, it is perfectly all right if we start with a function which is defined in some neighbourhood, say, $(c-\delta_0, c+\delta_0)$ of c from which the point c itself has been deleted, i.e., on the set $(c-\delta_0, c+\delta_0) - \{c\}$. Quite appropriately, this set is called the **deleted** δ_0-neighbourhood of c. Note that this is precisely the set $\{x : 0 < |x-c| < \delta_0\}$, where both the inequalities are strict. Geometrically, it is the set of all points whose distance from c is less than δ, not counting the point c itself[5].

Lest this sounds paradoxical, here is a real life analogy to highlight the conceptual difference between $f(c)$ and $\lim_{x\to c} f(x)$. Think of $f(x)$ as the income of a person x. Now suppose c is a young man in whom I am interested as a

[5]Actually, this is consistent with the layman's understanding of a neighbourhood. When a person talks of his neighbours, he naturally excludes himself. Ask anyone which planet in the solar system is closest to Earth. Most persons will think that the answer is either Venus or Mars and will try to refer to a book an astronomy. Very few people will give the correct answer, viz., the Earth itself!. Indeed, it takes a certain weirdness of mind to call c itself as a neighbour of c, just as it looks weird to say that a straight line is parallel to itself.

possible match for my daughter. Naturally, one of the inquiries I would want to make is how much he earns. It is insulting, and in any case, not of much avail to ask him directly. Instead, I find out the financial status of persons who are close to him. The closer these persons are to c, the better idea I get about his financial status. The picture that I gather this way is like the limit of $f(x)$ as x approaches c, i.e., it corresponds to $\lim_{x\to c} f(x)$. Sometimes this picture may be misleading, i.e., $f(c)$ may differ considerably from $\lim_{x\to c} f(x)$. This can happen, for example, if c is a poor person who lives in a rich locality (possibly on borrowed money) or a person who lives very simply in a hutment but has a huge fortune tucked in solid securities. Of course, normally people do live in neighbourhoods commensurate with their incomes. In such cases, $f(c)$ will coincide with $\lim_{x\to c} f(x)$. The mathematical equivalent of this is that the income function is continuous at the point c. We shall study this important concept soon.

With this preamble, we are now ready for a formal definition of the statement $\lim_{x\to c} f(x) = L$. The motivation is completely analogous to that behind Definition 1 in Chapter 6. Intuitively, we want to say that $f(x)$ is very close to L, i.e., that the difference $|f(x) - L|$ is very small, if x is sufficiently close to c but not equal to c i.e., if the difference $|x - c|$ is small but not 0. The standards of closeness are specified by positive real numbers. And the latter depends on the former. This gives the following definition.

Definition 1: We say that $f(x)$ tends to L as $x \to c$ or that L is a **limit** of $f(x)$ as x tends to c, and write $\lim_{x\to c} f(x) = L$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - c| < \delta \tag{11}$$

Note that we are not allowing x to equal c. But we are allowing $f(x)$ to equal L. Indeed if $f(x)$ is the constant function with value L, then this is the case for all values of x. It follows that in such a case no matter what c is, we have $\lim_{x\to c} f(x) = L$. The definition above can be verified directly in this case by taking δ to be any positive real number we want (e.g., $\delta = 1$). In other words, for the limit of a constant function, δ is independent of c and also of ϵ. As a less trivial example, we prove that for any $c \in \mathbb{R}$, $\lim_{x\to c} x = c$. Indeed, given any $\epsilon > 0$, we take $\delta = \epsilon$. Then (11) holds. Note that here the choice of δ depended only on ϵ and not on c. But in general it could depend on both. For example, take $f(x) = x^2$ and $c = 3$. We claim that $\lim_{x\to 3} x^2 = 9\ (= 3^2)$. Let $\epsilon > 0$ be given. Then we have to find a $\delta > 0$ such that

$$9 - \epsilon < x^2 < 9 + \epsilon \tag{12}$$

$$\text{whenever } 3 - \delta < x < 3 + \delta,\ x \neq 3 \tag{13}$$

We may suppose ϵ is so small that $9 - \epsilon > 0$, i.e., $\epsilon < 9$. In that case (12) is

equivalent to

$$\sqrt{9-\epsilon} < x < \sqrt{9+\epsilon} \tag{14}$$

Now (13) will imply (14) (and hence (12)) if and only if the interval $(3-\delta, 3+\delta)$ is contained in the interval $(\sqrt{9-\epsilon}, \sqrt{9+\epsilon})$. Note that the latter interval contains the point 3 but is not centred at 3. So the largest δ that will work for a given ϵ (assumed to be less than 9) is the smaller of the two numbers $3-\sqrt{9-\epsilon}$ and $\sqrt{9+\epsilon}-3$. Rationalising both the numbers and comparing their denominators, we see that the second number is smaller. So we can take $\delta = \sqrt{9+\epsilon}-3$ or any positive number smaller than it.

Thus we have shown that $\lim\limits_{x\to 3} x^2 = 9$. By a similar argument we can show that $\lim\limits_{x\to 5} x^2 = 25$. However, for the same ϵ, the δ that will now work will be at most $\sqrt{25+\epsilon}-5$. Again by rationalising, it is easy to show that this number is smaller than $\sqrt{9+\epsilon}-3$. In other words, a δ that worked for $c=3$ may not work for $c=5$ even though the ϵ is the same.

In the examples given above, for a given $\epsilon > 0$, we found the best (i.e., the largest) δ which will answer the condition (12) in the definition of a limit. But in general, this is not easy if the function $f(x)$ involved is a complicated one. Nor is it necessary. All that the definition requires is that for every $\epsilon > 0$, there exists *some* $\delta > 0$ such that (11) holds. Obviously, when a particular δ works for a given ϵ, so will any smaller (but positive) δ and unless the problem specifically asks us to find the best δ that will work, we are at liberty to use other simplifications and come up with *some* δ that will work. Suppose, for example, that we want to verify the definition of $\lim\limits_{x\to 3} x^2 = 9$ for $\epsilon = 0.01$. We already found above the best δ that will work, viz., $\delta = \sqrt{9.01}-3$. But if we are not so ambitious, there is a simpler way. Factoring x^2-9, we have $|x^2-9| = |x-3|\ |x+3|$. The first factor is less than δ which we are yet to choose. Now, by the triangle inequality, $|x+3| = |(x-3)+6| \le |x-3|+6 < \delta+6$. Hence we get that

$$|x^2-9| \ < \ \delta(\delta+6) \ \text{ whenever } |x-3| < \delta. \tag{15}$$

So we would be through if we can find some $\delta > 0$ for which

$$\delta(\delta+6) < 0.01 \tag{16}$$

We could do this by considering the roots of the quadratic $\delta^2+6\delta-0.01$. But since we are not after the largest δ that will work, we can take a short cut. To begin with we may suppose that $\delta \le 1$. For if a larger δ works, then $\delta = 1$ will surely work. Now, since $\delta \le 1$, we have $\delta+6 \le 7$. So the L.H.S. of the inequality (16) is at most 7δ. Therefore, if we take $\delta = \frac{0.01}{7} \approx 0.00142857$ then (16) will be satisfied. Note that this is less than the best δ, viz., $\sqrt{9.01}-3 \approx 0.001662$. The simplicity gained may not mean much in this problem because here the function $f(x) = x^2$ is relatively uncomplicated. But if it is a cubic polynomial,

say $f(x) = x^3 - 2x^2 + 5x + 7$, then finding the best δ would have been very difficult.

Comment No. 11:

In fact, in most cases where we have to show that a particular function tends to a particular limit, a δ which answers the definition of a limit is rarely found explicitly in terms of ϵ. All that is done is to show that some suitable δ *exists*. In other words, a limit is rarely evaluated, or verified, direct from the definition. This is typical of mathematics. A few basic things are derived directly from the definition and then the rest are built up from these basic ones. This is what we did with derivatives and also with limits of sequences. In fact the basic properties which express the limit of one function in terms of others are so similar to those listed in Theorem 3 of Chapter 6, that it is hardly necessary even to state them explicitly. Typically, they say that the limit of a sum is the sum of limits. (The very first property (i) asserts the uniqueness of limits.) We have already used them several times (except possibly the sandwich theorem which we shall soon use. A property which we used but which was not listed in Theorem 3 was that the limit of a log is the log of the limit and similarly the limit of sine is the sine of the limit. These are consequences of the continuity of the log function and the sine function which we shall study soon.)

We already showed above that the limit of a constant function is the constant everywhere and also that $\lim_{x\to c} x = c$ for all $c \in \mathbb{R}$. Starting from these building blocks and using the rule for the product and the sum of two functions we get that $\lim_{x\to c} x^n = c^n$ for every $c \in \mathbb{R}$ and for every integer $n \geq 0$ and hence that for any polynomial $p(x)$, $\lim_{x\to c} p(x) = p(c)$ for every $c \in \mathbb{R}$. Using the quotient rule, the result extends to a rational function. That is, if $p(x)$ and $q(x)$ are two polynomials then $\lim_{x\to c} \dfrac{p(x)}{q(x)} = \dfrac{p(c)}{q(c)}$ for all $c \in \mathbb{R}$ for which $q(c) \neq 0$. In case c is a root of $q(x)$, the limit may still exist if $p(c)$ also vanishes, i.e., if c is a root of $p(x)$ as well. As remarked in Comment No. 6 above, in such a case, the fate of $\lim_{x\to c} \dfrac{p(x)}{q(x)}$ depends on the orders of c as roots of $p(x)$ and $q(x)$.

In fact, as far as calculations of limits of rational functions are concerned, these are the only interesting cases. The other limits are dull in that $\lim_{x\to c} f(x)$ always equals $f(c)$ and, as mentioned earlier, gives the wrong impression that conceptually the two are the same.

A simple but useful result is that if $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$. (This is the analogue of Exercise (6.40)(b).) The proof is based on the inequality $||f(x)|-|L|| \leq |f(x)-L|$ which is a form of the triangle inequality. The converse is true if $L = 0$ for, in that case, $f(x) \to 0$ is equivalent to $|f(x)| \to 0$ since $|f(x) - 0| = ||f(x)| - 0|$. But in general the converse is false. A simple counter-example is the function $f(x)$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ -1 & \text{if } x \text{ is irrational} \end{cases}$$

Here $|f(x)|$ is the constant function 1 and so for every $c \in \mathbb{R}$, $\lim_{x\to c} |f(x)| = 1$. But we claim that $\lim_{x\to c} f(x)$ does not exist for any value of c. For suppose $\lim_{x\to c} f(x) = L$. Take $\epsilon = 1$. Then by the definition of a limit, there exists some $\delta > 0$ such that for all x in the deleted δ-neighbourhood of c, $|f(x)-L| < 1$. Now, no matter how small δ is, this deleted neighbourhood will contain infinitely many rational numbers and also infinitely many irrational numbers. (Here we are using the fact that the set of rational numbers as well as the set of irrational numbers is dense. See Exercise (6.46).) So, in this deleted δ-neighbourhood, there will be points x_1 and x_2 such that $f(x_1) = 1$ and $f(x_2) = -1$. But this would mean that $|1 - L| < 1$ and $|-1 - L| < 1$ and hence, by the triangle inequality, that $2 = |1 - (-1)| = |(1 - L) - (-1 - L)| \leq |1 - L| + |-1 - L| < 1 + 1 = 2$, a contradiction. So $\lim_{x\to c} f(x)$ does not exist.

It is instructive to paraphrase the argument above a little informally. Fix any real number c. Then since the set of rationals as well as the set of irrationals are dense, regardless of whether c itself is rational or not, it can be 'approached' both through the set of rationals and also through the set of irrationals. As far as the former is concerned, the function $f(x)$ above is the constant function 1. So its limit as x tends to c is also 1. On the other hand, on the set of irrationals, the function is like the constant function -1. So $f(x)$ approaches -1 as x tends to c through the irrationals. But that would mean that $f(x)$ has two possible limits as x tends to c, contradicting the uniqueness of the limit. (It is no coincidence that the proof of the uniqueness of limits is also got by obtaining a contradiction based on the triangle inequality. In fact, we invite you to give this proof yourself.)

In the example just given, the function $f(x)$ was defined by one formula on the set of rationals and by another formula on its complement, viz., the set of irrationals. Examples like this, where a function is defined by two different formulas, may appear weird. But one such case does occur in real life. Take, for example, the income tax function $f(x)$ which gives the tax due if your taxable income is x. Usually, there is no tax upto a certain limit, say x_0, and after that the tax is proportional to the income above this limit. Thus, in this case, we have $f(x) = 0$ for $x < x_0$ and $f(x) = \alpha(x - x_0)$ for $x \geq x_0$, where α is some constant. (In a more realistic situation, the tax rate is higher for higher income slabs. In such a case, the function $f(x)$ will have a different formula for, say $x > x_1$, where $x_1 > x_0$ and so on.)

There are also other examples where the formula for a function is different on the two sides of some point. For example, the function $f(x) = |x - 3|$ equals $x - 3$ for $x > 3$ and $3 - x$ for $x < 3$. (For $x = 3$, either formula will work.) When a function $f(x)$ is defined by one formula for all $x < c$ (or, for x 'below' c, or to the left of c) and by another formula for $x > c$, the consideration of $\lim_{x\to c} f(x)$ splits into two natural parts, one the **limit from the right** (or from above) and the other, the **limit from the left** (or from below), denoted respectively by $\lim_{x\to c^+} f(x)$ and $\lim_{x\to c^-} f(x)$. The definitions are analogous to that of $\lim_{x\to c} f(x)$ except that in the case of the former we require (11) to hold only

for $0 < x - c < \delta$ (or equivalently, for $c < x < c + \delta$) while in case of the latter we require it only for $c - \delta < x < c$. Evidently, for any function defined in a deleted neighbourhood of a point c, $\lim\limits_{x \to c} f(x)$ exists if and only if both the limits $\lim\limits_{x \to c^+} f(x)$ and $\lim\limits_{x \to c^-} f(x)$ exist and are equal to each other.

As an illustration let us consider $\lim\limits_{x \to 1} \dfrac{\sqrt{1 - \cos 2(x-1)}}{x-1}$. (JEE 1998) The expression under the radical sign equals $2\sin^2(x-1)$. But we must not hastily equate its square root with $\sqrt{2}\sin(x-1)$, because near the point 1, the latter expression is negative on the left side of 1. So the given function equals $\sqrt{2}\,\dfrac{\sin(x-1)}{x-1}$ for $x > 1$ and $-\sqrt{2}\,\dfrac{\sin(x-1)}{x-1}$ for $x < 1$. Accordingly, here the right handed limit is $\sqrt{2}$ while the left handed limit is $-\sqrt{2}$. As these two are unequal, the given limit does not exist.

Another standard example of a function which suddenly changes colour is the greatest integer function $f(x) = [x]$, which we already encountered in Comment No. 2 of Chapter 4 and also in Chapter 6, Comment No. 10. If c is not an integer, then there exists a sufficiently small deleted neighbourhood of c in which $[x] = [c]$. So in such a case, $\lim\limits_{x \to c}[x] = [c]$. Suppose however, that c is an integer. Then for x slightly greater than c, $[x] = [c]$ and so $\lim\limits_{x \to c^+}[x] = [c]$. However, if x is slightly to the left of c, then $[x] = [c]-1$ and so $\lim\limits_{x \to c^-}[x] = [c] - 1$.

Sometimes, the nature of the function is such that it is defined only on one side of a point c. For example, $f(x) = \sqrt{x+3}$ makes sense only for $x \geq -3$. In such a case, only one of the two limits is defined. (In the present example, the right handed limit at -3 is easily seen to be 0 but the left handed limit at -3 is undefined.) Sometimes, the domain of a function is expressly given as a closed interval, say $[a, b]$. In such a case, the left handed limit at a and the right handed limit at b are undefined.

Since derivatives are special cases of limits, it is not surprising that like one-sided limits, we can also talk of the left and right handed derivatives of a function $f(x)$ defined in a neighbourhood of a point c. These are defined as $\lim\limits_{h \to 0^-} \dfrac{f(c+h) - f(c)}{h}$ and $\lim\limits_{h \to 0^+} \dfrac{f(c+h) - f(c)}{h}$ respectively. The latter is denoted by $f'_+(c)$ (or by $f'(c+)$) and the former by $f'_-(c)$ (or by $f'(c-)$). It is clear that f is differentiable at c if and only if $f'_+(c)$ and $f'_-(c)$ both exist and are equal. Thus for example, the function $f(x) = |x-1|$ is right differentiable at 1 with $f'_+(1) = \lim\limits_{h \to 0^+} \dfrac{|1+h-1| - |1|}{h} = \lim\limits_{h \to 0^+} \dfrac{h}{h} = 1$. By a similar calculation, $f'_-(1) = -1$. Hence f is not differentiable at 1. However, at every $c \neq 1$, it is differentiable with $f'(c) = 1$ if $c > 1$ and $f'(c) = -1$ if $c < 1$. This follows because given any $c \neq 1$, we can find a sufficiently small neighbourhood, say N of c, such that N lies completely on one side of 1. In that case, throughout N, $f(x)$ equals either $x - 1$ or $1 - x$. Like any other limits, in finding derivatives, what matters is what happens near a point, i.e., in some neighbourhood (which could be as small as we like) of that point. What happens outside that neighbourhood

is entirely irrelevant. Attributes and properties which depend only on what happens in a neighbourhood of a point are called **local** in character, a term which we already encountered when we talked about local maxima and minima in Chapter 13. Thus a limit is a local concept. Hence so are other attributes based on limits such as differentiability and continuity. An integral of a function over an interval, or its absolute maximum (or minimum) on an interval, on the other hand, are global properties.

When a function is defined only on one side of the point, only one of the two derivatives makes sense. In such cases, the word 'differentiable' is often used in a restricted sense. For example, we just showed that the function $f(x) = |x-1|$ is differentiable everywhere except at 1. But if the domain of the function is given to be only, say the interval $[1, 4]$, then we often say that f is differentiable on it, because here the differentiability at the end-point 1 simply means the existence of $f'_+(1)$.

Comment No. 12:

As stressed before, $\lim_{x\to c} f(x)$ is conceptually quite different from $f(c)$. Examples can be given when one of them is defined but not the other and also where both are defined but are not equal. However, for most 'well-behaved' functions such as polynomials the two are equal. The equality of $f(c)$ with $\lim_{x\to c} f(x)$ is given a special name. When this happens, we say that the function $f(x)$ is **continuous** at the point c. Intuitively, this means that the function does not suddenly change at c (like a very rich man living in a poor neighbourhood). In other words, the value at c is what you would expect looking at the values of $f(x)$ for points in a deleted neighbourhood of c. Continuous functions are very important in real life applications because the things that we encounter in real life are generally continuous. This is the content of the saying that Nature abhors sudden change. Even when the change appears sudden, in reality it is not so. For example, when a cricket ball is hit by a bat we think that its speed (and direction) has changed suddenly in a moment. But when viewed microscopically, the change takes place gradually over a very short interval of time. Similarly, when we turn an electric switch on, the current in the wire does not instantaneously change from 0 to some steady state value. The change occurs continuously over a short interval of time. For this reason, although it is very easy to give examples of discontinuous functions (by taking any continuous function and changing the value of $f(c)$ arbitrarily, without changing the value of $f(x)$ at nearby points), such examples often appear very artificial.

A function which is continuous at every point of a set S is said to be continuous on the set S. The reasoning given above in the last Comment shows that all rational functions are continuous at all points except at the roots of the polynomial in the denominator. The basic theorems about limits translate into corresponding theorems about continuity. Suppose for example that $h(x) = f(x) + g(x)$ for all x in some neighbourhood of a point c (including c itself). Suppose f and g are continuous at c. Then $\lim_{x\to c} h(x) = \lim_{x\to c} f(x) + \lim_{x\to c} g(x) = f(c) + g(c) = h(c)$ which shows that the sum of two continuous functions is continuous.

The same goes for the difference, the product and the quotient of two continuous functions. Also from the result about the limit of the absolute value of a function, it follows that continuity of $f(x)$ implies that of $|f(x)|$.

As with limits, we can split hair and talk of the left and right continuity of a function at a point. We skip the obvious definitions. The work we did above shows that the function $f(x) = [x]$ is continuous at every c which is not an integer. But if c is an integer, then it is only right continuous at c but not left continuous. Similarly, $\sqrt{x}$ is right continuous at 0.

A function $f(x)$ can be discontinuous at a point c for a variety of reasons. If $\lim\limits_{x\to c} f(x)$ exists but differs from $f(c)$ the discontinuity is called **simple** or **removable**, because such a discontinuity can be removed by redefining $f(c)$ as $\lim\limits_{x\to c} f(x)$. A more serious type of discontinuity is one where both $\lim\limits_{x\to c^-} f(x)$ and $\lim\limits_{x\to c^+} f(x)$ exist but are unequal. Such a discontinuity is called, quite appropriately, a **jump discontinuity** because intuitively, in such a case the function jumps up or down (depending which of the two limits is larger) as x passes through c. Such discontinuities can occur typically when the formula for the function changes suddenly at c. For example, the function $f(x) = [x]$ just considered has a jump discontinuity at every integer. Still more serious discontinuities are those where at least one of the two limits fails to exist. We already had an example of such a function, viz., the function $f(x)$ which equals 1 if x is a rational and -1 if x is irrational. Another example will be given soon.

It is interesting to note that differentiability of a function $f(x)$ at a point $\alpha \in \mathbb{R}$ can be characterised in terms of a removable discontinuity of an associated function $g(x)$ defined by $g(x) = \frac{f(x)-f(\alpha)}{x-\alpha}$. This function is defined at all points where $f(x)$ is defined, except at α. By definition, differentiability of $f(x)$ at α is equivalent to saying that $\lim\limits_{x\to\alpha} g(x)$ exists. But this is the same thing as saying that $g(x)$ has a removable discontinuity at α. If we define $g(\alpha) = f'(c)$ then g becomes continuous at α. Thus we have proved that a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ is differentiable at α if and only if there is a function $g : \mathbb{R} \longrightarrow \mathbb{R}$ which is continuous at α and satisfies $f(x) - f(a) = g(x)(x-\alpha)$ for all $x \in \mathbb{R}$. (JEE 2001) Rewriting $f(x)$ as $f(\alpha) + g(x)(x-\alpha)$, we see that if g is continuous at α, so is f. In other words, differentiability of f at a point α implies its continuity at α. The converse is false. The standard counter-example is the function $f(x) = |x|$. It is continuous everywhere. But it is not differentiable at 0. Another example will be given in Comment No. 13.

Just as we have the chain rule for the derivative of a composite function, there is an analogous result that the composite of two continuous functions is also a continuous function. As with derivatives, this rule is very useful in proving continuity of complicated functions from simpler functions. For example, suppose we know that the sine function is continuous everywhere. Then the functions $\sin(x^2)$ and $\cos x$ are continuous everywhere, the latter using the identity $\cos x = \sin(\frac{\pi}{2} - x)$.

It is also true that the inverse function of a continuous real-valued function on an interval is continuous. But the proof is not so direct. It is

based on the fact that if such a function is invertible then it has to be strictly monotonically increasing or strictly monotonically decreasing. We shall take this up in the next chapter, Comment No. 6. As an application, the continuity of the logarithm follows from that of the exponential function while that of the inverse trigonometric functions follows from the continuity of the trigonometric functions.

Of course, to start with, the continuity of some basic functions such as the sine and the exponential functions has to be proved independently. This requires us to use certain inequalities and the Sandwich Theorem. We illustrate this for the function $f(x) = \sin x$. To prove its continuity at a point, say c, is equivalent to showing that $|\sin(c+h) - \sin c| \to 0$ as $h \to 0$. Let us rewrite the expression as $|\sin c\,(\cos h - 1) + \cos c\ \sin h|$ and hence as $|2\sin c\ \sin^2 \frac{h}{2} + \cos c\ \sin h|$. By triangle inequality, this is at most $2|\sin c|\sin^2 \frac{h}{2} + |\cos c||\sin h|$. For $0 < h < \frac{\pi}{2}$, this is at most $2|\sin c|\frac{h^2}{2} + |\cos c|h$. Here $|\sin c|$ and $|\cos c|$ are fixed numbers. Thus we have proved that for $0 < h < \frac{\pi}{2}$,

$$0 \ \leq\ |\sin(c+h) - \sin c| \ \leq 2\,|\sin c|\frac{h^2}{2} + |\cos c|h \tag{17}$$

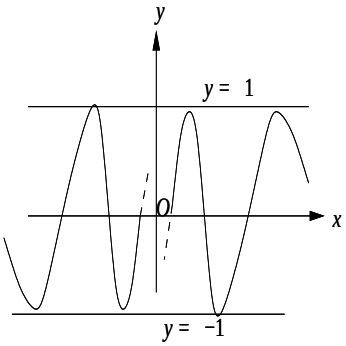
As $h \to 0^+$, the first and the last terms tend to 0. Therefore the Sandwich Theorem implies that the middle term also tends to 0 and this establishes the right continuity of the sine function at c. The left continuity is established similarly. (Or one may simply note that for $h < 0$, $\sin h = -\sin(-h)$ and apply the right continuity.) Put together, we see that the sine function is continuous everywhere.

Comment No. 13:

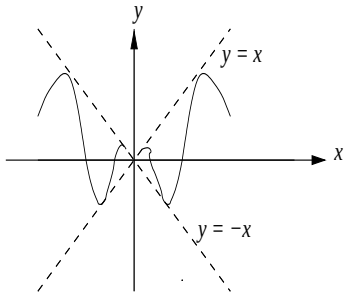
As another classic application of the Sandwich Theorem, we study three related functions f, g, h defined by $f(x) = \sin\frac{1}{x}$, $g(x) = x\sin\frac{1}{x}$ and $h(x) = x^2\sin\frac{1}{x}$ for $x \neq 0$. We define $f(0) = g(0) = h(0) = 0$. For $x \neq 0$ all the three functions are continuous and have derivatives of all orders. But at 0, they behave very differently as we will show.

We note first that $\lim\limits_{x\to 0}\sin(\frac{1}{x})$ does not exist. The best way to see this is to draw the graph of $y = f(x) = \sin\frac{1}{x}$. The sine function is periodic with period 2π. So $f(x_1) = f(x_2)$ whenever x_1 and x_2 are two numbers whose reciprocals differ by a multiple of 2π. As $x \to 0$, the reciprocal of x changes more and more rapidly. As a result, the function $f(x)$ keeps on repeating its values over shorter and shorter intervals. Put differently, the frequency of oscillation of $f(x)$ increases as x moves close to 0. For example, all points of the form $\frac{1}{n\pi}$, where n is an integer go to 0, while those of the form $\frac{2}{(4n+1)\pi}$ are taken to 1 by f. As a result, the graph looks like that in Figure (a) below. Note that here $f(-x) = -f(x)$ for all x. A function with this property is called an **odd function**. The graph of any such function is symmetric about the origin because if (x, y) lies on it so does $(-x, -y)$ which is the reflection of (x, y) w.r.t. the origin. (Illustrate this yourself by drawing the graphs of a few simple odd functions such as $x, x^3, \sin x, \tan x$.)

It is clear from the graph that as $x \to 0$, $\sin\frac{1}{x}$ tends to no definite limit. For a rigorous proof, suppose, if possible that $\lim_{x\to 0} \sin\dfrac{1}{x} = L$. Take $\epsilon = \frac{1}{2}$. Then there exists $\delta > 0$ such that if N denotes the deleted δ-neighbourhood $(-\delta, \delta) - \{0\}$, then $|\sin\frac{1}{x} - L| < \frac{1}{2}$ for all $x \in N$. Now, no matter how small δ is, both the points $\frac{1}{n\pi}$ and $\frac{2}{(4n+1)\pi}$ will lie in N if n is chosen sufficiently large in magnitude (depending on δ, of course). But then we would have $|0 - L| < \frac{1}{2}$ on the one hand and $|1 - L| < \frac{1}{2}$ on the other, whence the triangle inequality will give $|0 - 1| < \frac{1}{2} + \frac{1}{2}$. Hence $1 < 1$, a contradiction. Actually the same reasoning shows that neither the left handed nor the right handed limit of $f(x)$ exists at 0. In particular the function is not continuous at 0.



(a) $y = \sin\frac{1}{x}$



(b) $y = x\sin\frac{1}{x}$

In striking contrast, we shall prove that $\lim_{x\to 0} x\sin\dfrac{1}{x}$ exists and equals 0. We can see this from the graph. But the trouble is that in order to draw a good graph, we need to know this limit beforehand. In other words, a graph is a good way to visually see the behaviour of a function if it is drawn for you by someone else! When you have to draw it yourself, you first have to study the function carefully. In the present case, the periodicity of the sine function again means that the graph will oscillate up and down with increasing frequency as x gets close to 0. But there is a difference. Because of the factor x, the amplitude of the oscillations will go on decreasing to 0 as x tends to 0. We can say this neatly using the Sandwich Theorem. For any $x > 0$ we have $-1 \le \sin\frac{1}{x} \le 1$ and hence,

$$-x \le x\sin\frac{1}{x} \le x \tag{18}$$

Geometrically, this means that the graph of $y = x\sin\frac{1}{x}$ lies between the straight lines $y = -x$ and $y = x$ for $x > 0$. It does meet these lines at points of the form $(\frac{2}{(4n\pm1)\pi}, \pm\frac{2}{(4n\pm1)\pi})$. As both $-x$ and x tend to 0 as $x \to 0^+$, the Sandwich Theorem and (18) give that $x\sin\frac{1}{x} \to 0$ as $x \to 0^+$. For $x < 0$, the same argument applies except that the inequality (18) is reversed. Or one can note

that $g(-x) = g(x)$. A function with this property is called an **even function**. This means that the graph of $y = x \sin \frac{1}{x}$ is symmetric about the y-axis. It is shown in Figure (b) above.

Since $\lim_{x\to 0^+} g(x)$, $\lim_{x\to 0^-} g(x)$ both equal 0 and $g(0) = 0$ by definition, we see that the function $g(x)$ is continuous at 0 and hence everywhere. We claim, however, that it is not differentiable at 0. (JAT 1979) Note that at any point c other than 0, we can apply the chain rule to get $g'(c) = \sin \frac{1}{c} - \frac{1}{c^2} \cos \frac{1}{c}$. But at 0, the chain rule is useless since the function $y = \frac{1}{x}$ is not differentiable at 0 (in fact, not even defined at 0). So to decide whether $g'(0)$ exists or not, we shall have to proceed strictly from the definition of a derivative of a function at a point. For $x \neq 0$, we have $\dfrac{g(x) - g(0)}{x} = \sin \frac{1}{x}$. As just proved, $\lim_{x\to 0} \sin \dfrac{1}{x}$ does not exist and so we conclude that $g(x)$ is not differentiable at 0.

Finally, let us consider the third function defined by $h(x) = x^2 \sin \frac{1}{x}$ for $x \neq 0$ and $h(0) = 0$. We leave it to you to sketch its graph which is qualitatively similar to the graphs of g except that it lies entirely between the two parabolas $y = -x^2$ and $y = x^2$ and it is not symmetric about the y-axis. (Instead, it is symmetric about the origin as is the case with the graph of f.) Note that $h(x)$ is the product of the function $y = x$ and the function $g(x)$. As both these are continuous everywhere, so is the function $h(x)$. As for its differentiability, we have $h'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$ for all $x \neq 0$. But this is of no use[6] to find $h'(0)$. So again, we proceed strictly from the definition. Here $\frac{h(x)-h(0)}{x} = x \sin \frac{1}{x}$ which does tend to 0 as x tends to 0. So we see that $h(x)$ is differentiable at 0, with $h'(0) = 0$. Note however, that h' is not continuous at 0. So h is not continuously differentiable at 0. As one can predict, things will be better if instead of x^2 we have x^3. This indeed is the case. For, as x tends to 0, $\frac{d}{dx}(x^3 \sin \frac{1}{x}) = 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x}$ also tends to 0. However, here the first derivative is not differentiable at 0. With $x^4 \sin \frac{1}{x}$, this will be so, i.e., the function is twice differentiable, but the second derivative will not be continuous at 0. And so on.

Comment No. 14:

So far we defined limits of the form $\lim_{x\to c} f(x)$, where c is a real number. We can also define limits of the form $\lim_{x\to\infty} f(x)$ or of the form $\lim_{x\to -\infty} f(x)$. For the former to make sense the function must be defined for all sufficiently large values of x, say for all $x > x_0$ for some real number x_0. When this is the case, the definition of $\lim_{x\to\infty} f(x) = L$ is that for every $\epsilon > 0$, there exists some R (depending on ϵ) such that for all $x > R$, $|f(x) - L| < \epsilon$. An equivalent approach

[6] Some people have a tendency to find $h'(0)$ by taking the limit of $h'(x)$ as x tends to 0. It can be shown as an application of the Mean Value Theorem that if this limit exists then it indeed equals $h'(0)$ (see Exercise (16.11)). In the present example, though, $\lim_{x\to 0} h'(x)$ does not exist because although $2x \sin \frac{1}{x}$ tends to 0 as just proved, the second term $-\cos \frac{1}{x}$ has no limit as x approaches 0. The argument resembles that given for the non-existence of $\lim_{x\to 0} \sin \dfrac{1}{x}$.

is to put $y = \dfrac{1}{x}$ and require that $\lim\limits_{y\to 0^+} f(\dfrac{1}{y}) = L$. The other limit $\lim\limits_{x\to -\infty} f(x)$ is defined analogously. Both types of limits have properties analogous to those of the limits of the type $\lim\limits_{x\to c} f(x)$. But there is no such thing as continuity[7] or differentiability at ∞ or at $-\infty$.

As examples, suppose $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials of degree m and n respectively. Since $q(x)$ vanishes at most at a finite number of points, $f(x)$ is certainly defined for all sufficiently large x. If $m < n$ then $\lim\limits_{x\to\infty} = 0$. If $m = n$ then this limit equals the ratio $\dfrac{a_m}{b_n}$, where a_m and b_n are the leading coefficients of $p(x)$ and $q(x)$ respectively. This follows by dividing both the numerator and the denominator by x^n. Specifically, suppose $p(x) = a_0 + a_1x + a_2x^2 + \ldots a_mx^m$ and $q(x) = b_0 + b_1x + \ldots + b_nx^n$ with $a_m \neq 0$ and $b_n \neq 0$. Then

$$f(x) = \frac{a_mx^m + a_{m-1}x^{m-1} + \ldots + a_1x + a_0}{b_nx^n + b_{n-1}x^{n-1} + \ldots + b_1x + b_0} \tag{19}$$

$$= \frac{a_m}{b_n}\frac{1}{x^{n-m}} \frac{1 + \dfrac{a_{m-1}}{a_m}\dfrac{1}{x} + \ldots + \dfrac{a_1}{a_m}\dfrac{1}{x^{m-1}} + \dfrac{a_0}{a_m}\dfrac{1}{x^m}}{1 + \dfrac{b_{n-1}}{b_n}\dfrac{1}{x} + \ldots + \dfrac{b_1}{b_n}\dfrac{1}{x^{n-1}} + \dfrac{b_0}{b_n}\dfrac{1}{x^n}} \tag{20}$$

The desired assertions follow by noting that $\frac{1}{x}, \frac{1}{x^2}$ etc all tend to 0 as x tends to ∞. In fact, after a little practice, it is hardly necessary to write down (20) elaborately, when we are interested only in the answer which can be obtained just by looking at (19). We reason like this, for large x, the dominating term in the numerator is a_mx^m. All other terms are insignificant compared to it. Similarly, for large x, the only term in the denominator that matters is b_nx^n. Hence $\lim\limits_{x\to\infty} \dfrac{p(x)}{q(x)}$ is the same as $\lim\limits_{x\to\infty} \dfrac{a_mx^m}{b_nx^n}$, which in turn, is the same as $\dfrac{a_m}{b_n} \lim\limits_{x\to\infty} \dfrac{1}{x^{n-m}}$ for $n \geq m$.

Indeed this type of a reasoning applies even when the expression involves some other functions. Suppose, for example, that we want to evaluate $\lim\limits_{x\to\infty} \sqrt{\dfrac{x - \sin x}{x + \cos^2 x}}$. (JEE 1979) Note that neither $\lim\limits_{x\to\infty} \sin x$ nor $\lim\limits_{x\to\infty} \cos^2 x$ exists. But, both these functions are bounded. So, for large x they are insignificant as compared to x. Hence dropping them, the given limit is the same as $\lim\limits_{x\to\infty} \sqrt{\dfrac{x}{x}}$ which is obviously 1. The argument can, of course, be formalised by dividing both the numerator and the denominator by x and then applying the Sandwich Theorem to show that $\dfrac{\sin x}{x}$ and $\dfrac{\cos^2 x}{x}$ both tend to 0 as x tends to ∞. And it is indeed a good idea to indulge in such formalisation till you outgrow the need for it.

[7] It is sometimes customary to say that a function $f(x)$ is continuous at ∞ if $\lim\limits_{x\to\infty} f(x)$ exists, for one can then define $f(\infty)$ to be this limit. But this is only a manner of speaking.

Comment No. 15:

When we are asked to identify all the points where a given function is continuous, differentiable, twice differentiable etc. we first have to identify the troublesome points. These are the points where the function is either undefined because some parts of it are undefined for certain values of the argument (e.g., some denominator becoming 0, some expression under a radical sign becoming negative etc.) or suddenly changes colours either explicitly (for example, when in the problem itself the function is defined by different formulas on different sets) or implicitly (because the function involves some other functions such as $|x|, [x]$ which suddenly change colours). If these trouble spots are isolated then usually things are nice on the portions of the real line between two consecutive 'bad' points.

As a straightforward illustration, suppose we want to identify the set of points where the function $f(x) = \begin{cases} (x-1)^2 \sin \frac{1}{(x-1)} - |x| & \text{if } x \neq 1 \\ -1 & \text{if } x = 1 \end{cases}$ is not differentiable. (JEE 1981)

Here we have to consider the point $x = 1$ separately because the function has a different definition at it than at nearby points. Similarly, $|x|$ has a different definition for $x < 0$ than for $x > 0$. So 0 is also a possible troublemaker. At other points there is no problem about existence of derivatives. The function considerably resembles the function $h(x)$ we studied in Comment No. 13 above. In fact we can rewrite $f(x)$ as $h(x-1) - |x|$. As $h(x)$ was differentiable at $x = 0$, by the chain rule we get that $h(x-1)$ is differentiable at $x = 1$. The other term, viz. $|x|$ is differentiable at $x = 1$ anyway. Thus we see that f is differentiable at $x = 1$. As for the point 0, the term $(x-1)^2 \sin \frac{1}{(x-1)}$ causes no problem. But the second term $-|x|$ fails to be differentiable (even though it is continuous). Hence $f(x)$ cannot be differentiable at $x = 0$. Summing up, the set of points where the function $f(x)$ is not differentiable is the singleton set $\{0\}$.

As another illustration (JEE 1998), consider the function $f(x) = \min\{x, x^2\}$, for every real number x. Here, for every x, $f(x)$ is the smaller of the two numbers x and x^2 and we first have to determine which possibility holds where. Clearly, for $x < 0$, $x < x^2$ since the latter is positive. If $x \geq 0$ then we have $x^2 \leq x$ for $x \leq 1$. But for $x > 1$, $x < x^2$. So the function $f(x)$ is given by

$$f(x) = \begin{cases} x & , \text{ if } x \leq 0 \\ x^2 & , \text{ if } 0 < x < 1 \\ x & , \text{ if } 1 \leq x \end{cases} .$$

Here the troublemakers are the points 0 and 1. However, since both x and x^2 tend to 0 as $x \to 0$ and both tend to 1 as $x \to 1$, $f(x)$ is continuous at both these points and hence on the entire real line. (Another method is to use Exercise (6.5)(a) to write $f(x)$ as $\frac{1}{2}(x + x^2 - |x - x^2|)$ and apply the continuity of the absolute value function. In fact this reasoning shows that the minimum of any two continuous functions is also a continuous function. A similar result holds for the maximum.) As for differentiability, $f'(0-)$ equals the value of $\frac{d}{dx}x$ at $x = 0$ since $f(x) = x$ for $x \leq 0$. Hence $f'(0-) = 1$. By a similar reasoning,

$f'(1+) = 1$. To find $f'(0+)$ and $f'(1-)$ we have to take the derivative of x^2 at 0 and 1. They equal 0 and 2 respectively. So $f(x)$ is differentiable at neither 0 nor 1. It is differentiable at all other points. (Although it is not asked here, it is a good idea to at least mentally draw a graph of $f(x)$. Here it is the straight line $y = x$, except that the portion between $x = 0$ and $x = 1$ is an arc of the parabola $y = x^2$. From the graph, the answer can be written down by inspection.)

Of course, as with other problems where a function is given by some formula, one should always be on the alert to see if the formula can be simplified before subjecting it to any analysis hastily. Such alertness can sometimes pay rich dividends. As a good illustration (JEE 1981), consider the function $f(x) = \dfrac{\tan(\pi[x-\pi])}{1+[x]^2}$. Let us first find the troublesome points in the conventional manner. Here the denominator is positive everywhere and so we do not have to worry about its vanishing. In fact, it is constant over intervals of the form $(n, n+1)$, where n is an integer. At integral values of x, the denominator is discontinuous. As a result it cannot be differentiable there either. So we now confine ourselves entirely to intervals of the form $(n, n+1)$, where n is an integer. Over every such interval, the behaviour of $f(x)$ will depend entirely on the numerator, viz., $\tan(\pi[x-\pi])$. The tangent function has derivatives of all orders at all points except at odd multiples of $\frac{\pi}{2}$. So now we have to exclude values of x for which $\pi[x-\pi]$ is an odd multiple of $\frac{\pi}{2}$. But regardless of what x is, the expression $[x-\pi]$ is always a whole number and so the product $\pi[x-\pi]$ can never be an odd multiple of $\frac{\pi}{2}$. But actually, more is true. Since $\pi[x-\pi]$ is an integral multiple of π, its tangent is always 0. So regardless of what the denominator is, the function has the value 0 everywhere. Being a constant function, it has derivatives of all orders everywhere! Note that the denominator, considered separately, has points of discontinuity viz., all integers. But the identical vanishing of the numerator makes the ratio continuous even at these points. Had we taken a moment out to reflect and observe that $\pi[x-\pi]$ is always an integral multiple of π, the answer would have come instantaneously.

It can be argued that if we had, say, $\tan(x[x-\pi])$ in the numerator, then this kind of a tricky simplification would not be valid and conventional analysis would have to be resorted to. But then this is the very characteristic of examinations intended to test qualities rather than mere knowledge. It is one thing to know various formulas and methods. It is quite another to be able to figure out which one to use where.

A word of caution is also necessary regarding the use of standard theorems which assert continuity and differentiability of a complicated function in terms of those of simpler functions. The hypotheses of these theorems are only sufficient conditions which will guarantee the conclusion. But sometimes the conclusion may hold even when these conditions are not met. For example, the product of two continuous functions is continuous. But the product of a discontinuous function with a continuous function can also be continuous sometimes, a trivial example being when the other function is identically 0. In fact, even the product of two discontinuous functions can be continuous, for example, when they are reciprocals of each other. (See also Exercise (15.20).) Similarly, the function

$|x|$ is not differentiable at 0. So we cannot apply the chain rule to test the differentiability of the function $\cos(|x|)$ at $x = 0$. But the function $\cos(|x|)$ *is* differentiable at 0, because since the cosine function is even, we can write $\cos(|x|)$ as $\cos x$ for all x and *now* apply the chain rule. (Note, however, that $f(x) = |\cos x|$ is non-differentiable at points where $\cos x = 0$, i.e., when x is an odd multiple of $\frac{\pi}{2}$. At every such point, $f'_+ = 1$ while $f'_- = -1$. This is also clear from the graph of $y = |\cos x|$ which consists of a chain of arches of base width π and height 1 each, placed one after the other.)

EXERCISES

15.1 Find the equation of the normal to the curve $x^2 = 4y$ which passes through the point $(1, 2)$. (1984)

15.2 Find the centre of the circle passing through the point $(0, 1)$ and touching the curve $y = x^2$ at $(2, 4)$. (1983)

15.3 If the line $ax + by + c = 0$ is normal to the curve $xy = 1$, which of the following statements is (are) true? (1986)

(A) $a > 0, b > 0$ (B) $a > 0, b < 0$ (C) $a < 0, b > 0$ (D) $a < 0, b < 0$

15.4 Which normal to the curve $y = x^2$ forms the shortest chord? (1992)

15.5 (a) Tangent at a point P_1 (other than $(0, 0)$) on the curve $y = x^3$ meets the curve again at P_2. The tangent at P_2 meets the curve at P_3 and so on. Show that the abscissas of $P_1, P_2, P_3, \ldots, P_n$, form a G.P. Also find the ratio

$$[\text{area } (\Delta\ P_1P_2P_3)]/[\text{area } \Delta\ (P_2P_3P_4)]. \qquad (1993)$$

(b) Let α and β be the intercepts with the axes made by the tangent at a point (x_0, y_0) on the curve $x^3 + y^3 - 3axy = 0$. Prove that $x_0y_0 = \dfrac{2a^2\alpha\beta}{\alpha\beta - a^2}$.

(c) Prove that there are just two straight lines which are both tangent and normal to the curve $y^2 = x^3$ and that the x-coordinates of the points where either of them touches the curve and cuts it at right angles are $\frac{8}{9}$ and $\frac{2}{9}$ respectively. Show these lines in a rough sketch of the curve.

15.6 (a) Find the equations of the straight lines perpendicular to $x+y-2 = 0$ and touching the ellipse $x^2 + 8y^2 - 2x - 32y + 25 = 0$. (JAT 1980)

(b) Find all the tangents to the curve $y = \cos(x + y)$, $-2\pi \leq x \leq 2\pi$, that are parallel to the line $x + 2y = 0$. (1985)

(c) Find the points on the curve $y^3 - 3xy + 2 = 0$ where the tangent is horizontal and also the points where the tangent is vertical. (1994)

(d) Find the equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$. (2002)

15.7 (a) If $y = \dfrac{5x}{\sqrt[3]{(1-x)^2}} + \cos^2(2x+1)$, find $\dfrac{dy}{dx}$. (1980)

(b) If $xe^{xy} = y + \sin^2 x$, find $\dfrac{dy}{dx}$ at $x = 0$. (1996)

(c) Find the equation of the normal to the curve $y = (1+x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$. (1993)

15.8 If $x + |y| = 2y$, which of the following statements is (are) true about y as a function of x? (1984)

(A) it is defined for all real x (B) it is continuous at $x = 0$
(C) it is differentiable for all x (D) $\frac{dy}{dx} = \frac{1}{3}$ for $x < 0$

15.9 Suppose $f(x) = (x+1)^2$ for $x \geq -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ w.r.t. the line $y = x$, find $g(x)$. (2002)

15.10 Let $D(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$, where f_1, f_2, g_1, g_2 are differentiable functions of x. Show by an example that in general $D'(x)$ is different from the determinant $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$. Prove that the correct value of $D'(x)$ is the sum $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$. State and prove a similar result for a 3×3 determinant of functions of x. What happens if all the functions except those in a particular row (or column) are constants?

15.11 (a) Find the derivative with respect to x of the function $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \sin^{-1}(\dfrac{2x}{1+x^2})$ at $x = \frac{\pi}{4}$. (1984)

(b) Find the derivative of $f(x) = \log_x(\ln x)$ at $x = e$. (1985)

(c) Suppose $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$. If $|p(x)| \leq |e^{x-1} - 1|$ for all $x \geq 0$, prove that $|a_1 + 2a_2 + \ldots + na_n| \leq 1$. (2000)

(d) If $f_r(x), g_r(x), h_r(x)$, $r = 1, 2, 3$, are polynomials in x such that $f_r(a) = g_r(a) = h_r(a)$, $r = 1, 2, 3$ and

$$F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix},$$

find the value of $F'(x)$ at $x = a$. (1985)

(e) Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Find the value of $\dfrac{d^3}{dx^3}(f(x))$ at $x = 0$. (1997*)

15.12 (a) If $x = \sec\theta - \cos\theta$ and $y = \sec^n\theta - \cos^n\theta$ (where n is a positive integer), then show that $(x^2+4)\left(\frac{dy}{dx}\right)^2 = n^2(y^2+4)$. (1989)

(b) Let $y = (1-x)^{-\alpha}e^{-\alpha x}$, $(x \neq 1)$, where α is a constant. Prove that for every positive integer n,
$$(1-x)y^{(n+1)} - (n+\alpha x)y^{(n)} - n\alpha y^{(n-1)} = 0.$$ (JAT 1980)

(c) If $(a+bx)e^{y/x} = x$, then prove that $x^3\dfrac{d^2y}{dx^2} = \left(x\dfrac{dy}{dx} - y\right)^2$. (1983)

(d) If $y = e^x\tan^{-1}x$, prove that $(1+x^2)y'' - 2(1-x+x^2)y' + (1-x)^2y = 0$

(e) Suppose u and v are functions of x that have derivatives of all orders. For every positive integer n, prove that $\dfrac{d^n}{dx^n}(uv) = \sum\limits_{k=0}^{n}\binom{n}{k}\dfrac{d^ku}{dx^k}\dfrac{d^{n-k}v}{dx^{n-k}}$. (This is known as **Leibnitz rule**. Note the formal resemblance with the binomial theorem. Not surprisingly, a proof based on induction can be given if you use the right binomial identities.)

(f) Prove that $\dfrac{d^n}{dx^n}\left(\dfrac{\ln x}{x}\right) = \dfrac{(-1)^n n!}{x^{n+1}}(\ln x - H_n)$, where H_n is the n-th harmonic number $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$.

(g) Suppose y_1, s and t are functions of x which are twice differentiable. Let $y_2 = sy_1$ and $y_3 = ty_1$. Prove that
$$\begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = y_1^3(s't'' - s''t').$$

15.13 (a) Suppose $F(x) = f_1(x)f_2(x)\ldots f_n(x)$, where n is a positive integer and every f_i is differentiable and non-zero at a point x_0. Prove that
$$\frac{F'(x_0)}{F(x_0)} = \frac{f_1'(x_0)}{f_1(x_0)} + \frac{f_2'(x_0)}{f_2(x_0)} + \ldots + \frac{f_n'(x_0)}{f_n(x_0)}.$$
(This is sometimes called **logarithmic differentiation**, because an easy proof can be given by taking logarithms and then differentiating. But this proof is incomplete since it is not given that the functions are positive. An inductive proof can be given using the product rule.)

(b) Let $F(x) = f(x)g(x)h(x)$ for all real x, where $f(x), g(x)$ and $h(x)$ are differentiable functions. Suppose at some point x_0, $F'(x_0) = 21F(x_0), f'(x_0) = 4f(x_0), g'(x_0) = -7g(x_0)$ and $h'(x_0) = kh(x_0)$. Find the value of k. (1997*)

(c) If $y = \dfrac{ax^2}{(x-a)(x-b)(x-c)} + \dfrac{bx}{(x-b)(x-c)} + \dfrac{c}{x-c} + 1$, prove that $\dfrac{y'}{y} = \dfrac{1}{x}\left(\dfrac{a}{a-x} + \dfrac{b}{b-x} + \dfrac{c}{c-x}\right)$. (1998)

(d) Find the derivative of $\sin(x^2+1)$ w.r.t. x from first principles. (1978)

15.14 (a) If $G(x) = -\sqrt{25 - x^2}$, then find $\lim_{x\to 1} \dfrac{G(x) - G(1)}{x - 1}$. (1983)

(b) If $f(a) = 2, f'(a) = 1, g(a) = -1, g'(a) = 2$, find the value of $\lim_{x\to a} \dfrac{g(x)f(a) - g(a)f(x)}{x - a}$. (1983)

(c) The function $\frac{\ln(1+ax)-\ln(1+bx)}{x}$ is not defined at $x = 0$. Find the value which should be assigned to f at $x = 0$ so that it is continuous at $x = 0$. (1983)

(d) Evaluate $\lim_{x\to 0} \dfrac{2^x - 1}{\sqrt{1+x} - 1}$. (1982)

(e) If $f(9) = 9, f'(9) = 4$, find $\lim_{x\to 9} \dfrac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$. (1988)

(f) Evaluate $\lim_{x\to a} \dfrac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$, $(a \neq 0)$ (1978)

15.15 Find the following limits using l'Hôpital's rule or otherwise.

(a) $\lim_{x\to 1} \dfrac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$ (See Exercise (5.14)(c) for an application of this limit.)

(b) $\lim_{h\to 0} \dfrac{\ln(1+2h) - 2\ln(1+h)}{h^2}$ (1997*)

(c) $\lim_{x\to 0} \dfrac{x\tan 2x - 2x\tan x}{(1-\cos 2x)^2}$ (1999) (d) $\lim_{x\to 0} \left(\dfrac{1+5x^2}{1+3x^2}\right)^{\frac{1}{x^2}}$. (1996)

(e) $\lim_{x\to 0} \left(\dfrac{f(1+x)}{f(1)}\right)^{1/x}$, it being given that $f(1) = 3$ and $f'(1) = 6$, (2002)

(f) $\lim_{x\to\infty} a^x \sin(\dfrac{b}{a^x})$ where $a > 1$, (JAT 1979)

(g) $\lim_{x\to 0}(\dfrac{1}{x^2} - \dfrac{1}{\tan^2 x})$ (h) $\lim_{x\to -\infty} \left(\dfrac{x^4 \sin(\frac{1}{x}) + x^2}{(1+|x|^3)}\right)$ (1987)

(i) $\lim_{x\to 0} f(x)$, where $f(x) = \begin{cases} \dfrac{\sin[x]}{[x]} & , [x] \neq 0 \\ 0 & , [x] = 0 \end{cases}$ (1985)

(j) $\lim_{x\to 0} g(f(x))$, where

$$f(x) = \begin{cases} \sin x & , x \neq n\pi,\ n = 0, \pm 1, \pm 2, \pm 3, \ldots \\ 2 & , \text{otherwise} \end{cases}$$

$$\text{and } g(x) = \begin{cases} x^2 + 1 & , x \neq 0, 2 \\ 4 & , x = 0 \\ 5 & , x = 2 \end{cases}$$

15.16 (a) Let $f(x) = \begin{cases} (x^3+x^2-16x+20)/(x-2)^2 & \text{if } x \neq 2 \\ k & \text{if } x = 2 \end{cases}$
If $f(x)$ is continuous for all x, find k. (1981)

(b) Find a and b such that $\lim\limits_{x\to 0}\left(\dfrac{a\cot x}{x}+\dfrac{b}{x^2}\right)=\dfrac{1}{3}$. (JAT 1980)

(c) Let $g(x)$ be a polynomial of degree one and $f(x)$ be defined by

$$f(x) = \begin{cases} g(x) & , x \leq 0 \\ \left[\frac{(1+x)}{(2+x)}\right]^{1/x} & , x > 0. \end{cases}$$

Find the continuous function $f(x)$ satisfying $f'(1) = f(-1)$. (1987)

(d) Find the values of a and b so that the function

$$f(x) = \begin{cases} x + a\sqrt{2}\sin x, & 0 \leq x < \pi/4, \\ 2x\cot x + b, & \pi/4 \leq x \leq \pi/2, \\ a\cos 2x - b\sin x, & \pi/2 < x \leq \pi \end{cases}$$

is continuous for $0 \leq x \leq \pi$. (1989)

(e) Let $f(x) = \begin{cases} \dfrac{1-\cos 4x}{x^2}\ , & x<0 \\ a\ , & x=0\ . \\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4} & x>0 \end{cases}$ Determine the value of a, if possible, so that the function is continuous at $x = 0$. (1990)

(f) Let $f(x) = \begin{cases} (1+|\sin x|)^{a/|\sin x|} & , -\frac{\pi}{6} < x < 0\ , \\ b & , x = 0\ , \\ e^{\tan 2x/\tan 3x} & , 0 < x < \frac{\pi}{6} \end{cases}$. Determine a and b such that f is continuous at $x = 0$. (1994)

(g) Let $f(x) = \begin{cases} x+a & \text{if } x<0 \\ |x-1| & \text{if } x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x+1 & \text{if } x<0 \\ (x-1)^2+b & \text{if } x \geq 0 \end{cases}$, where a and b are non-negative real numbers. Determine the composite function $g \circ f$. If $(g \circ f)(x)$ is continuous for all real x, determine the values of a and b. Further, for these values of a and b, is $g \circ f$ differentiable at $x = 0$? Justify your answer. (2002)

15.17 Find a $\delta > 0$ such that $|x^3 - 7^3| < 0.01$ whenever $|x - 7| < \delta$. (JAT 1980)

15.18 If $\lim\limits_{x\to a} f(x)g(x)$ exists, must both $\lim\limits_{x\to a} f(x)$ and $\lim\limits_{x\to a} g(x)$ exist? (1981)

15.19 Let $f(x) = \begin{cases} 1 & \text{when } x \text{ is rational} \\ 0 & \text{when } x \text{ is irrational.} \end{cases}$
Show that f is discontinuous everywhere. (JAT 1979)

15.20 (i) Let $f(x)$ be a continuous function and $g(x)$ be a discontinuous function. Prove that $f(x) + g(x)$ is a discontinuous function. (1987)

(ii) Suppose $f(x)$ is discontinuous and $g(x)$ is continuous at a point c. If $g(c) \neq 0$, show that $f(x)g(x)$ is discontinuous at c. Show by examples that when $g(c) = 0$, nothing can be said in general.

(iii) Give an example where both $f(x), g(x)$ are discontinuous at a point c but $f(x) + g(x)$ and $f(x)g(x)$ are continuous at c.

15.21 (a) Determine if the following functions are continuous and/or differentiable at $x = 0$.

(i) $f(x) = x(\sqrt{x} - \sqrt{x+1})$ (1985)

(ii) $f(x) = [\tan^2 x]$ (1993)

(iii) $f(x) = \begin{cases} xe^{-(\frac{1}{|x|}+\frac{1}{x})}, & x \neq 0 \\ 0, & x = 0. \end{cases}$ (1997*)

(b) Let $f(x) = \begin{cases} \frac{1}{x}[e^{2x}\cos x - 2\sin x - 1], & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$. Find $f'(0)$. (JAT 1980)

(c) For the function $f(x) = \begin{cases} \dfrac{x}{1+e^{\frac{1}{x}}}, & x \neq 0, \\ 0, & x = 0, \end{cases}$ evaluate the derivative from the right $f'(0+)$ and the derivative from the left $f'(0-)$. (1983)

(d) Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2, \\ 3-x, & 2 < x \leq 3. \end{cases}$ Determine the form of $g(x) = f(f(x))$ and hence find the points of discontinuity of g, if any. (1983)

(e) If $f(x) = |x-2|$ and $g(x) = f[f(x)]$, find $g'(x)$ for $x > 20$. (1990)

(f) Suppose $f(x) = \dfrac{x-1}{2x^2-7x+5}$ for $x \neq 1$ and $f(1) = -\frac{1}{3}$. Find $f'(1)$. (1979)

15.22 Identify the sets of points in the domains where the following functions are continuous and also the sets of points where they are differentiable.

(a) $f(x) = \dfrac{x}{(1+|x|)}$ (1987) (b) $f(x) = 1 + |\sin x|$ (1986)

(c) $f(x) = [x \sin \pi x]$ (1986) (d) $f(x) = [x] \sin\left(\frac{\pi}{[x+1]}\right)$ (1996)

(e) $g(x) = f(|x|) + |f(x)|$, where $f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases}$ (1986)

(f) $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \dfrac{x^2}{4} - \dfrac{3x}{2} + \dfrac{13}{4}, & x < 1 \end{cases}$ (1988)

(g) $f(x) = \begin{cases} 1-x, & x < 1 \\ (1-x)(2-x), & 1 \leq x \leq 2 \\ 3-x, & x > 2. \end{cases}$ (1997)

(h) $f(x) = [x]^2 - [x^2]$ (1999)

(i) $f(x) = \begin{cases} \tan^{-1} x & \text{if } |x| \le 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$ (2002)

(j) $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$. (1999)

(k) $g(x) = \begin{cases} \max\{f(t) : 0 \le t \le x\}, & 0 \le x \le 1 \\ 3 - x, & 1 < x \le 2 \end{cases}$
where $f(x) = x^3 - x^2 + x + 1$ (1985)

15.23 (a) Draw a graph of the function $y = [x] + |1 - x|, \quad -1 \le x \le 3$. Determine the points, if any, where this function is not differentiable. (1989)

(b) Let $f(x) = x|x|$. Find the set of points where $f(x)$ is twice differentiable. (1992)

(c) Let $f(x) = \begin{cases} xe^{ax}, & x \le 0 \\ x + ax^2 - x^3, & x > 0 \end{cases}$, where a is a positive constant. Find the interval in which $f'(x)$ is increasing. (1996)

15.24 Let $f(x) = \begin{cases} x^2/2 & , \ 0 \le x < 1, \\ 2x^2 - 3x + \frac{3}{2} & , \ 1 \le x \le 2. \end{cases}$ Discuss the continuity of f, f' and f'' on $[0, 2]$. (1983)

15.25 (a) If $f(x) = \frac{1}{2}x - 1$, determine which of the following statements are true on the interval $[0, \pi]$. (1989)

(A) $\tan[f(x)]$ and $1/f(x)$ are both continuous
(B) $\tan[f(x)]$ and $1/f(x)$ are both discontinuous
(C) $\tan[f(x)]$ and $f^{-1}(x)$ are both continuous
(D) $\tan[f(x)]$ is continuous but $1/f(x)$ is not.

(b) For the following functions, find their inverses or else show that they do not exist.

(i) $f(x) = 3x - 5$ (1998)
(ii) $f(x) = 2^{x(x-1)}$ for $x \ge 1$ (1999)
(iii) $f(x) = (a - x^n)^{1/n}$, where $a > 0$ and n is a positive integer. (1983)

15.26 If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \le x \le 2 \\ 37 - x, & 2 < x \le 3 \end{cases}$, determine which of the following statements is/are true. (1993)

(A) $f(x)$ is increasing on $[-1, 2]$, (B) $f(x)$ is continuous on $[-1, 3]$,
(C) $f'(2)$ does not exist, (D) $f(x)$ has the maximum value at $x = 2$.

15.27 Let $a_n = (-1)^n$. Then $a_n + a_{n+1} = 0$ for every n. Let L be the limit of the sequence $\{a_n\}$. Then L also equals $\lim_{n\to\infty} a_{n+1}$. Hence $2L = L + L = \lim_{n\to\infty} (a_n + a_{n+1}) = 0$. So $L = 0$. Similarly, $L^2 = \lim_{n\to\infty} a_n a_{n+1} = -1$. So, L is not even a real number! What goes wrong?

Chapter 16

THEORETICAL CALCULUS

So far we studied the basic concepts of calculus (or, more precisely, one half of calculus, called the differential calculus) and also applied them, for example, to find the maxima and minima of various functions. The crucial tool in doing so was Theorem 3 of Chapter 13. We remarked that although it appeared obvious, its rigorous proof was far from trivial and required the use of Lagrange's Mean Value Theorem. The proof of this theorem, in turn, vitally uses an important property of the real number system, called its **completeness**. Completeness is also needed crucially in proving some properties of continuous functions and also in proving the existence of such things as logarithms and square roots.

In this chapter we fill most of these gaps. We begin with an intuitive understanding of completeness in Comment No. 1. In Comment No. 2 we give a formulation of completeness which is probably the most palpable. Using it we derive important properties of continuous functions and thereafter the various Mean Value Theorems. This part of calculus is often called **analysis**. It provides the theoretical foundation underlying its applications.

The essence of differentiability of a function at a point is its approximability by a linear function in a neighbourhood of that point. We shall study this and some problems of error estimation based on this. As is to be expected, higher order derivatives give stronger approximations. This leads to a well known theorem, called Taylor's theorem which we shall state without proof.

Questions based on theoretical calculus are comparatively infrequent at the JEE. (Mean value theorems were removed from the JEE syllabus. But they are reintroduced from 2003 onwards.) The concepts involved and the skills needed here are qualitatively different than at most other JEE topics. As a result, even those doing well in JEE sometimes face difficulties with this material later. Why not familiarise yourself with it ahead of time?

Main Problem : Let $\mathbb{R}$ be the set of real numbers and $f : \mathbb{R} \longrightarrow \mathbb{R}$ be such that for all x and y in $\mathbb{R}$, $|f(x) - f(y)| \leq |x - y|^3$. If $f(10) = 100$, the value of $f(20)$ is ..... . (JEE 1988, modified)

First Hint: Show that f is differentiable everywhere.

Second Hint: Show that f' vanishes identically.

Third Hint: Show that f is a constant function.

Solution : Fix any $x \in \mathbb{R}$. Then for every $h \neq 0$ we have $|f(x+h)-f(x)| \leq |h|^3$ and hence

$$0 \leq \left| \frac{f(x+h) - f(x)}{h} \right| \leq h^2 \tag{1}$$

By the Sandwich Theorem we get that $f'(x) = 0$. So, f' is identically 0. Therefore by Theorem 3 in Chapter 13, $f(x)$ is a constant function. As it is given that $f(10) = 100$, it follows that $f(20)$ also equals 100.

Comment No. 1:

Although the solution is extremely short and simple, the theorems needed in it are far from trivial. The Sandwich Theorem is, of course, quite elementary. But as stressed in Chapter 13, Theorem 3 there is one of those results which look quite obvious, but whose rigorous proofs are very deep. The reason is that they are based on a very fundamental property of the real number system, called its completeness. In essence, completeness asserts the existence of certain real numbers which we normally take for granted. That is why many results based on completeness look obvious but are not so.

Before we formally define completeness and derive Theorems such as Theorem 3 of Chapter 13 from it, let us illustrate what completeness does. In Chapter 4, Comment No. 21, we proved that $\sqrt{2}$ is not a rational number. In making this statement, we are presupposing that there is a real number, denoted by $\sqrt{2}$, and defined by the requirements that it is positive and its square is 2. And the proof given there shows that this number cannot be rational. But how do we know that such a real number indeed exists? We never bothered about such questions before. But until we know that the answer is in the affirmative, all that we proved in Chapter 4, Comment No. 21 means that there is no rational number whose square is 2, which is not quite the same thing as saying that $\sqrt{2}$ is not rational, much the same way as saying that Mr. X is not married to a Muslim is not quite the same as saying that Mrs. X is not a Muslim. The former statement holds even when Mr. X is unmarried and hence is not as strong as the latter. (In Exercise (6.48), we gave a construction for a sequence of rational numbers which converges to $\sqrt{2}$. So we may define $\sqrt{2}$ as the limit of this sequence. But this is not of much help either, because how do we know that the sequence constructed there is convergent?)

More generally, in Comment No. 6 of Chapter 2, we gave an intuitive argument to define the q-th root of a real number $a > 1$, where q is a positive integer. The idea was that as x increases starting from $x = 1$, so does x^q. For values of x near 1, $x^q < a$, while for $x > a$, $x^q > a$. So somewhere between 1 and a there is a real number, say u, at which u^q must equal exactly a. This again looks very obvious, in fact, as obvious as saying that if a person was 2 feet tall when he was 3 years old and is a six feeter today, then at some time in between, he must have been exactly five feet tall. But analogies prove nothing. Real-life analogies can make something easier to understand. They cannot be a substitute for a mathematical proof.

To appreciate the point involved here, suppose our concept of a number is limited only to rational numbers. This was indeed the case a long time ago and continued to be so for ages since the set of rationals is large enough to permit all four basic arithmetical operations. Now, within the rational number system, if you take the squares of positive rationals, it divides the set of positive rationals into two parts, those whose squares are less than 2 and those whose squares are bigger than 2. But there is no rational which sits right at the 'boundary' in the sense that its square will be exactly 2. It is as if we start running in our territory and suddenly find ourselves in the enemy camp with nothing to indicate that you are crossing the border. If this can happen within the rational number system, logically there is nothing to suppose that things will necessarily be better if we have real numbers instead of rationals.

Similarly, having defined the power a^x (where $a > 0$) for all rational exponents x, when we want to extend the definition to the case where x is irrational, we are in trouble. We would like a^x to be some real number which is bigger than all numbers of the form a^u, where u is a rational number less than x and, at the same time, smaller than every real number of the form a^v, where v is a rational number greater than x. Using the fact that the set of rationals is dense (see Exercise (6.46)), it is not hard to show that there cannot be more than one such number. In other words, if it exists, it is unique. But how do we know that it exists in the first place?

Then again, there are many things which are defined as limits of one type or the other. For example, one popular way to define the number e is as the limit of the sequence $a_n = (1 + \frac{1}{n})^n$ as $n \to \infty$. Here every term of the sequence is rational. Moreover, it is easy (and standard) to show that it is strictly monotonically increasing i.e., that $a_n < a_{n+1}$ for every $n \geq 1$. Still, its terms do not grow much for large n. In fact, it can be shown that $a_n < 3$ for all n. So intuitively the terms are heading to some definite number. Still, it can be shown that they are not heading to any rational number. Given any rational number q, logically, one of the following two things will happen. Either some term (and hence all subsequent terms) of the sequence will exceed q, or all the terms will be less than or possibly equal to q. In the latter event, it can be shown (this part is not so trivial) that we can always find another rational $r < q$ which is also bigger than every a_n. Put differently, every rational q will be either too small or too big to be the limit of the sequence $a_n = (1 + \frac{1}{n})^n$. In other words, even though every term of it is rational, it does not converge

to anything in the rational number system. (By the way, the terms of this sequence have a real-life interpretation. Simple arithmetic shows that if you invest one rupee in a bank which allows interest at the fabulous rate of 100% per anum, then $(1 + \frac{1}{n})^n$ is the amount you stand to get at the end of one year if the interest is compounded n times annually at equal rests. As $n \to \infty$, the situation approaches what is called continuous compounding, a concept which we shall study in Chapter 19. Intuitively, it means your money is growing all the time and every moment the interest you have earned is added to your deposit. So if you are living in the world of rational numbers, then under continuous compounding you will get nothing at the end of the year, because the sequence $\{(1 + \frac{1}{n})^n\}$ does not converge to any rational, much as we would like it to do!)

The recurring theme in all these examples is that in the rational number system, there are many 'gaps'. Of course, these gaps are not the kind of gaps that you see in the set of all integers (often denoted by $\mathbb{Z}$). Between any two distinct integers, there is a minimum gap of one unit. Because of these gaps, we cannot always perform division in the set of integers. For example, if we want to divide 10 rupees equally among 3 persons, we know that 3 is too small an answer and 4 is too big. The correct answer lies somewhere in the gap between 3 and 4. In fact, if we fill all such gaps, we get precisely the rational number system (generally denoted by $\mathbb{Q}$) starting from the integers. In this system we can perform division (except by 0) without any problem. In the system $\mathbb{Z}$ we can do the other three basic arithmetic operations, viz., addition, subtraction and multiplication. In $\mathbb{Q}$, we can, of course, do all these. But we can also do division. So the rational number system is complete as far as basic arithmetic is concerned. (And, to a large extent, this is the reason why for a long time, the concept of a number stopped at rationals. Even today, the layman's understanding of a number rarely goes beyond $\mathbb{Q}$. Even when he needs some irrational numbers such as π for practical purposes, he routinely equates them with some rationals, such as $\frac{22}{7}$ or 3.1416.)

Unlike for integers, for the rational numbers, there is no such thing as two consecutive rational numbers, because between any two rationals we can find another rational (for example, their arithmetic mean) and hence infinitely many rationals. So if you picture the rational numbers as points on a straight lines, you won't see any gaps even with the best microscope. Still there are gaps in the sense that certain numbers which should have been there (such as the limits of some sequences of rationals or the square roots of some rationals) are not there. Intuitively, the real number system $\mathbb{R}$ is obtained by filling all these gaps in $\mathbb{Q}$, in the same spirit as $\mathbb{Q}$ was obtained from $\mathbb{Z}$ by filling the gaps necessary for carrying out division.

The construction of real numbers from rationals is well beyond our scope. But suffice it to say that just as after extending $\mathbb{Z}$ to $\mathbb{Q}$, we could perform division with no problem, except where it was inherently impossible (because of the vanishing of the denominator), after we extend $\mathbb{Q}$ to $\mathbb{R}$ we no longer encounter the difficulties that we had in $\mathbb{Q}$. In other words, in $\mathbb{R}$ every sequence will converge unless it is inherently impossible for it to converge (e.g., a sequence whose terms are not bounded, or a sequence whose terms are

alternately 0 and 1). The same holds for limits of the type $\lim_{x \to c} f(x)$ where $f(x)$ is a real-valued function of x. Put differently, the real number system is **complete** as far as existence of limits is concerned. In it, the limits which should exist, indeed exist. (There are, however, other types of inadequacies in $\mathbb{R}$. For example, there is no square root of -1 in it. This leads to complex numbers as we saw in Chapter 2.)

Naturally, the real number system is the ideal playground for any games based on the concept of a limit. The definition of a limit makes sense even in the world of rational numbers. All we have to do is to restrict the epsilons and deltas to rational numbers. But such a definition will be of little interest if the limits which ought to exist don't exist. Completeness of the real numbers is just the thing that is needed to make the game interesting. The limiting concept being the very heart of calculus, calculus in $\mathbb{Q}$ is as dull as a boxing match between two weaklings in which both the players are defensive and neither has much of a chance or will to attack the other. Calculus in $\mathbb{R}$, on the other hand, is like a match in which both the players are strong and aggressive and neither of them misses a single opportunity to attack the other!

Comment No. 2:

Any serious study of calculus must take into account the completeness of the real number system. There are various formulations of completeness. They are, of course, logically equivalent to each other. And each is more convenient to apply in a particular context than others. We give here a formulation which is intuitively clear and easy to understand since it does not require us to define any new terms. What is more, this formulation will also easily imply all the important theorems we have been mentioning.

We remarked above that the sequence $a_n = (1 + \frac{1}{n})^n$ is monotonically increasing and bounded above (by 3) and so we expect it to converge, but that it does not converge within $\mathbb{Q}$ the rational number system. More generally, it stands to reason that any sequence of real numbers which is monotonically increasing and bounded above ought to converge to some real number. This is a consequence of completeness of $\mathbb{R}$. What is more interesting, completeness (in any other formulation) can be derived from it. In other words, it is not just a consequence of completeness, but is as strong as completeness. So we choose to define completeness in this form. The standard definitions are somewhat different. But we believe that this property of real numbers is obvious enough to be taken as an axiom.

Axiom : The real number system $\mathbb{R}$ has the property that every monotonically increasing sequence of real numbers which is bounded above (by some real number) is convergent in $\mathbb{R}$, i.e. has some real number as its limit.

An axiom is a statement whose truth is taken for granted. In every branch of mathematics, we start from a few axioms and then go on proving theorems which are based either directly on these axioms or indirectly based on them (that is, they are based on theorems which have first been deduced from the axioms).

For example, one of the axioms of geometry is that through two distinct points one and only one line can be drawn. So, axioms are the foundations on which we construct the theorems. We remark that we are assuming the statement above as axiom because we are not constructing real numbers. If we do, then the statement above can be *proved* from the definition of real numbers. So it is then no longer necessary to assume it as an axiom. But we do not aim at this degree of completeness. (Note that we have only listed the axiom which is equivalent to completeness. In addition, there are many other properties of real numbers which we use all the time, for example, the commutativity of their addition and multiplication. We are not listing them here.)

Most elementary texts on calculus simply *assume* the various consequences of completeness (such as Theorems 1 and 3 in Chapter 13) without proof. We follow a via media in which we take the convergence of a monotonically increasing, bounded sequence as axiom and derive the rest from it. The proofs will give a good glimpse of the kind of arguments needed in theoretical calculus. They usually involve very little computation. Instead, they require logical deduction of one statement from another.

For example, note that an immediate consequence of the axiom is that every monotonically decreasing sequence of real numbers which is bounded below has a limit. For, if $\{x_n\}$ is such a sequence, then the sequence $\{-x_n\}$ is a monotonically increasing sequence. Moreover, if M is a lower bound for $\{x_n\}$ (i.e., $x_n \geq M$ for every n) then $-M$ is an upper bound for $\{-x_n\}$. So by the Axiom, there exists some real number L such that $-x_n \to L$ as $n \to \infty$. But then $x_n \to -L$ as $n \to \infty$. Hence the sequence $\{x_n\}$ has a limit within $\mathbb{R}$.

It is often useful to observe that if a monotonically increasing sequence $\{a_n\}$ has an upper bound, say b and converges to a limit L (whose existence is asserted by the Axiom above), then

$$a_n \leq L \leq b \text{ for every } n = 1, 2, 3, \ldots \tag{2}$$

The first part of this inequality is proved by a *reductio ad absurdum* argument (i.e., by the method of contradiction). For suppose, if possible, that $L < a_m$ for some integer m. Take ϵ to be any positive number smaller than $a_m - L$. (For the sake of definiteness we may take it to be $\frac{1}{2}(a_m - L)$.) Then for all $n \geq m$, we have $a_n \geq a_m > L + \epsilon$. But that means at most a finite number of terms of the sequence $\{a_n\}$ lie in the interval $(L - \epsilon, L + \epsilon)$, contradicting that $a_n \to L$ as $n \to \infty$. The second inequality in (2) follows by applying Theorem 3(v) of Chapter 6 to the sequences $\{a_n\}$ and the constant sequence $\{b\}$.

Similarly, if a monotonically decreasing sequence $\{b_n\}$ is bounded below by a (say) and converges to L^*, then we have

$$a \leq L^* \leq b_n \text{ for every } n = 1, 2, \ldots \tag{3}$$

Let us apply this reasoning to the sequence $b_n = \frac{1}{n}$. Trivially, this is a monotonically decreasing sequence of non-negative terms. So its limit, say L, exists and is non-negative. We claim that $L = 0$. For otherwise, $L > 0$ and then the reciprocal sequence $a_n = \dfrac{1}{b_n}$ will converge to the real number $\dfrac{1}{L} = M$

(say). But then so would the sequence $\{a_{n+1}\}$. A direct computation gives $a_{n+1} = a_n + 1$. So, taking limits of both the sides we would get $M = M + 1$, a contradiction.

Thus, using completeness, we have given a rigorous proof of the fact that $\frac{1}{n}$ tends to 0 as $n \to \infty$. We often take this for granted. But it is not quite so. In fact, this assertion is equivalent to the **Archimedian property** mentioned in a footnote of Chapter 6, Comment No. 20. (An equivalent formulation is that there is no real number which is bigger than every positive integer, or, in other words, that the set of positive integers is not bounded above as a subset[1] of $\mathbb{R}$.)

Comment No. 3:

As the first non-trivial application of completeness to study the behaviour of continuous functions, we now prove a well-known property of continuous functions, called the **Intermediate Value Property** (or IVP for short). We remarked a while ago that if a six feeter today was 2 feet tall when he was 3 years old then we automatically conclude that at some time or the other he must have been exactly 5 feet tall. The particular figures 6,2,3 and 5 are irrelevant here, as is the fact that $6 = 2 \times 3$ and $5 = 2 + 3$ (although there are some Number Theory buffs[2] who are quick to notice such relationships, however irrelevant they be to the matter at hand). Instead of 5 we could have had any number between or 'intermediate' to 2 (the height at the earlier time) and 6 (the height now). What matters is not the numbers but the fact that the height is a continuous function of time. If it were not, then it is possible, at least in theory, that a person who was 4 feet tall became 6 feet instantaneously. But we know, or rather assume, that Mother Nature cannot do such a thing. (It is also true that the height is a monotonically increasing function. But that is not relevant here. If instead of height we had some other function, say temperature, which could go up and down, even then we would say that between two moments when the temperature was 2° and 6° it must have been exactly 5° at least once, provided it changed continuously with time.)

The Intermediate Value Property asserts that if $f(x)$ is continuous over an interval $[a, b]$ and m is a number which is intermediate to $f(a)$ and $f(b)$, (i.e., $f(a) \leq m \leq f(b)$ or $f(b) \leq m \leq f(a)$ depending upon which of $f(a)$ and $f(b)$ is greater), then there exists some point c in $[a, b]$ at which $f(c) = m$. We already

[1] The stress here is on the fact that $\mathbb{N}$ is not bounded above *in the real number system*. As a subset of $\mathbb{Q}$, it is trivial to show that $\mathbb{N}$ has no upper bound. For, if $\frac{p}{q}$ were such an upper bound, where p and q are positive integers, then we immediately get a contradiction because the positive integer $p + 1$ is surely greater than $\frac{p}{q}$. In order to show that $\mathbb{N}$ has no upper bound in $\mathbb{R}$ either, it is tempting to try an easy proof based on the denseness of $\mathbb{Q}$ as follows. Suppose a real number x is an upper bound on $\mathbb{N}$. Then there is at least one rational, say y, between x and $x + 1$. So, y would be an upper bound of $\mathbb{N}$ in $\mathbb{Q}$, a contradiction. But this proof is not valid, because the proof of denseness of rationals is based on the Archimedian property in the first place, see Exercise (6.46). Indeed the two are equivalent.

[2] According to a famous anecdote about the great Indian mathematician Ramanujan, when his friend once casually mentioned that the cab he travelled in was numbered 1729, Ramanujan excitedly quipped,"But that's the smallest integer which can be written as the sum of two cubes in more than one way". He was right since $1729 = 12^3 + 1 = 10^3 + 9^3$.

mentioned this property in Exercise (3.15)(a). But the proof given there applies only for quadratic functions. Before proving it for any continuous function, let us prove a result about continuous functions which is of an independent interest.

Theorem 1: Suppose $f(x)$ is a function defined in some neighbourhood N of a point c and continuous at c. Suppose $\{x_n\}$ is a sequence which converges to c. Then the sequence $\{f(x_n)\}$ converges to $f(c)$.

Proof: We first have to find out if the sequence $f(x_n)$ is really defined, i.e., whether x_n lies in the domain of f. Actually, this need not be so for all n. Still, we are given that $x_n \to c$ as $n \to \infty$. Since N is a neighbourhood of c, there exists some $\delta_0 > 0$ such that the interval $(c - \delta_0, c + \delta_0)$ is contained in N. By definition of convergence, there exists some n_0 such that for all $n \geq n_0$, $x_n \in (c - \delta_0, c + \delta_0) \subset N$. Since f is defined at all points of N, we see that $f(x_n)$ is defined at least for all $n \geq n_0$. Since the convergence of a sequence is unaffected by its first few terms, we may as well suppose that $f(x_n)$ is defined for *all* n.

We now have to show that $f(x_n) \to f(c)$ as $n \to \infty$. We do so by directly showing that the definition is satisfied. Let $\epsilon > 0$ be given. By continuity of f at c, there exists some $\delta > 0$ such that

$$|f(x) - f(c)| < \epsilon \text{ whenever } |x - c| < \delta \tag{4}$$

We assume $\delta < \delta_0$ for otherwise we can always replace δ by any positive number smaller than δ_0 and (4) will still hold. Now since $\{x_n\}$ is given to converge to c, there exists some m ($\geq n_0$) such that

$$|x_n - c| < \delta \text{ whenever } n \geq m \tag{5}$$

Putting (5) and (4) together, we have

$$|f(x_n) - f(c)| < \epsilon \text{ whenever } n \geq m \tag{6}$$

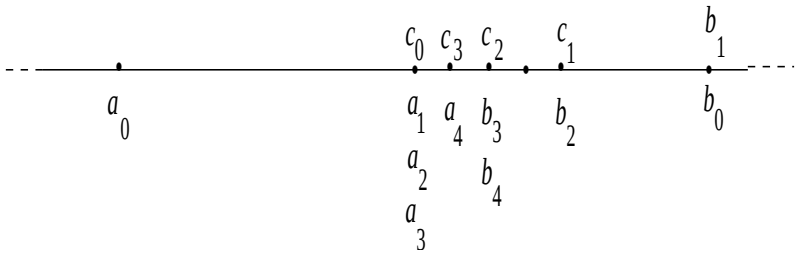
In other words, we have shown that the condition in the definition of $\lim_{n\to\infty} f(x_n) = f(c)$ holds true for the ϵ that was given. As ϵ was arbitrary, we have established that $f(x_n) \to f(c)$ as $n \to \infty$. Thus the proof of the theorem is complete.

This proof is quite different from the kind of proofs that you need for trigonometric identities or for, say, proving the concurrency of three lines. There are no computations involved here. Just a cool logical deduction of one statement from its previous ones. But one has to be careful about the language so as to ensure that what is said is exactly what is meant. For example, in a statement like, 'for every $\epsilon > 0$, there exists $\delta > 0$, such that ...' if you instead say 'there exists a $\delta > 0$ such that for every $\epsilon > 0$, ...' the whole meaning is changed drastically. In the former statement, the δ was allowed to change as ϵ changed. In the latter, the same δ is claimed to work for every ϵ. The difference is rather like the difference between the statements, 'for every man there is a woman who loves him' and the statement, 'there is a woman who loves every man'.

We remark that the converse of the result above is also true. That is, if we are given that for every sequence $\{x_n\}$ which converges to c, the sequence $\{f(x_n)\}$ converges to $f(c)$, then the function f is continuous at c. Read together with the converse, the Theorem above characterises continuity of a function in terms of convergence of sequences. Verbally, the two together say that continuity of a function amounts to preservation of limits of sequences, i.e., the value of a function at the limit of a sequence is the limit of the image sequence. It is this characterisation which allows us to transform one sequence to another, a more convenient one, while finding limits. Suppose, for example, that we want to find $\lim_{n\to\infty} a_n$, where the terms a_n are all positive. Sometimes we may be in a better position to find $\lim_{n\to c} \ln a_n$. Suppose this limit is L. Then by the theorem above, the limit of the original sequence $\{a_n\}$ is e^L. Here we are using the fact that the exponential function is continuous at L. As a quick illustration, let us calculate $\lim_{n\to\infty}(1+\frac{1}{n})^n$. We already know that the answer is e. But let us also get it from the method above. We have $\ln\left[(1+\frac{1}{n})^n\right] = n\ln(1+\frac{1}{n}) = \frac{\ln(1+h)}{h}$, where $h = \frac{1}{n}$. Now as $n \to \infty$, $h \to 0^+$ as we proved from completeness. But we already know the limit of the ratio $\frac{\ln(1+h)}{h}$ as h tends to 0 through *all* real values (not necessarily values of the form $\frac{1}{n}$, where n is a positive integer). Writing the ratio as $\frac{\ln(1+h)-\ln 1}{h}$, this limit is the derivative of $\ln(1+x)$ at $x = 0$. Hence its value is $\frac{1}{1+0} = 1$. Hence $\lim_{n\to\infty} n\ln(1+\frac{1}{n}) = 1$. By the theorem above and the continuity of the exponential function, we get that the original limit, viz., $\lim_{n\to\infty}(1+\frac{1}{n})^n$ equals e^1 i.e., e.

But, interesting as the converse is, its proof is a little clumsy as it is based on a *reductio ad absurdum* argument. Moreover, we shall need only the theorem above and not its converse. So we now return to the proof of the Intermediate Value Property. We are given a function f which is continuous at every point of the interval $[a, b]$. We are given that either $f(a) \le m \le f(b)$ or $f(b) \le m \le f(a)$. Without loss of generality we may suppose that the first possibility holds. For otherwise we replace m by $-m$ and f by $-f$ which is also continuous. If $f(a) = m$ or $f(b) = m$, then we are through because then we can take c as a and b respectively. Hence assume $f(a) < m < f(b)$. In such a case, to find a point c such that $f(c) = m$ we shall use a technique called **successive bisection** in which we go on dividing intervals into two parts. The basic idea is simple and akin to common sense. Suppose we want to find out the house of some friend knowing only that it is somewhere on a 1 km long road. If it is at either of the two ends of the road, well and good. Otherwise, instead of starting from one end and going on asking at every house, we inquire at a point half-way on the road. If the friend happens to live there we are lucky. Otherwise we ask in which half of the road he lives and proceed only $\frac{1}{4}$ km in that direction. We inquire again and if we do not find him there, we go forward or backward, but only $\frac{1}{8}$ km this time. In this process either we shall find the friend or at least we shall come within a given distance from him.

Now to carry out an analogous procedure for the hunt of a point c in $[a, b]$ at which the function f has value m, we first set up a notation. Call a as a_0 and b as b_0. Let $c_0 = \dfrac{a_0 + b_0}{2}$. If $f(c_0) = m$, we are done. Otherwise either $f(c_0) < m$ or $f(c_0) > m$. In the first case, call c_0 as a_1 and b_0 as b_1. In the second case, call a_0 as a_1 and c_0 as b_1. In either case we have $f(a_1) < m < f(b_1)$. We now subdivide $[a_1, b_1]$ into two equal subintervals and repeat the argument. We repeat this process if necessary. (The figure below shows a situation, where $a_1 = a_2 = a_3 = c_0, b_1 = b_0, b_2 = c_1, b_3 = c_2, a_4 = c_3$ and $b_4 = b_3$.)



In this process, either we shall reach some c_r (viz., the mid-point of $[a_r, b_r]$) such that $f(c_r) = m$ or else we shall get an infinite sequence of nested intervals $[a_n, b_n]$ such that

$$\left.\begin{array}{rll} & \text{(i)} & [a_n, b_n] \subset [a_{n-1}, b_{n-1}] \text{ for every } n = 1, 2, \ldots \\ & \text{(ii)} & b_n - a_n = \dfrac{b_0 - a_0}{2^n} \text{ for all } n = 0, 1, 2, \ldots \\ \text{and} & \text{(iii)} & f(a_n) < m < f(b_n) \text{ for all } n = 0, 1, 2, \ldots \end{array}\right\} \tag{7}$$

From (i) it is clear that the sequence $\{a_n\}$ is monotonically increasing. Moreover b_1 is an upper bound for it. So by the first part of our Axiom about completeness, the sequence $\{a_n\}$ converges, say to some number c. Similarly, the sequence $\{b_n\}$ is monotonically decreasing, bounded below (by a_1) and hence convergent, say to c^*. Because of (2) and (3) and the fact that $a_n \leq b_n$ for every n, we have

$$a \leq a_n \leq c \leq c^* \leq b_n \leq b \text{ for every } n = 1, 2, 3, \ldots \tag{8}$$

In particular it follows that both c and c^* lie in the interval $[a, b]$. So far we have used only (i) of (7). We now claim that because of (ii) in (7), c and c^* are equal. We already showed that $\frac{1}{n} \to 0$ as $n \to \infty$. Now for every positive integer n, we have $n < 2^n$ and hence $0 < \frac{1}{2^n} < \frac{1}{n}$. Therefore by the Sandwich Theorem, $\frac{1}{2^n}$ and hence $\frac{b-a}{2^n}$ tend to 0 as n tends to ∞. So, by (ii) of (7), we get that $(b_n - a_n) \to 0$ as $n \to \infty$. On the other hand, since $\{a_n\}$ and $\{b_n\}$ converge to c and c^* respectively, we have $b_n - a_n \to c^* - c$ as n tends to ∞. As limits are unique, we get $c^* - c = 0$, i.e., $c = c^*$.

We now put (iii) of (7) to use along with Theorem 1 above. Since $\{a_n\}$ converges to c and $f(x)$ is continuous at c, we get that $f(a_n) \to f(c)$ as

$n \to \infty$. But since $f(a_n) < m$ for every n, we must have (again using Part (v) of Theorem 3 in Chapter 6) $f(c) \leq m$. But we also have $b_n \to c$ as $n \to \infty$. So again by Theorem 1 above, $f(b_n) \to f(c)$ as $n \to \infty$. But then because of (iii) of (7), we must have $m \leq f(c)$. As we have already proved $f(c) \leq m$, it follows that $f(c)$ must exactly equal m. The proof of the Intermediate Value Property is now complete.

There is a slight paraphrase of the IVP (Intermediate Value Property) which is worth noting. Suppose I is an interval (not necessarily closed or bounded) and $J = f(I)$ is the image of I under a function f which is continuous at all points of I (and possibly at some other points too). We claim that the set J is an interval of real numbers. For if not[3], then there will be three points, say α, β and γ, such that $\alpha < \beta < \gamma$ and α, γ are in J but $\beta \notin J$. Since $\alpha, \beta \in f(I)$, there exist $a, b \in I$ such that $f(a) = \alpha$ and $f(b) = \gamma$. Consider the interval $[a, b]$ (or the interval $[b, a]$ in case $a > b$). Since $a, b \in I$, we have $[a, b] \subset I$. By IVP, there exists some $c \in [a, b] \subset I$ such that $f(c) = \beta$. But this means $\beta \in f(I)$, a contradiction. So we may reformulate the IVP by saying that every continuous function maps intervals onto intervals.

Comment No. 4:

We now study some applications of the Intermediate Value Property. First we make good a promise about the existence of square roots, and more generally, of the q-th roots of real numbers for any integer $q > 1$. Consider the function $f(x) = x^q$. It is continuous at all points and $f(0) = 0$. Moreover, $f(x) \to \infty$ as $x \to \infty$. So given any positive real number, say α, we can find some $R > 0$ such that $f(R) > \alpha$. If we apply IVP (Intermediate Value Property) to the function f over the interval $[0, R]$ we immediately get that there is some u in $[0, R]$ such that $f(u) = \alpha$. This means $u^q = \alpha$. In other words, u is a q-th root of α. Thus we have shown that every positive real number has at least one positive q-th root. Actually, this q-th root is unique since it can be shown that the function $f(x) = x^q$ is strictly increasing on $(0, \infty)$. (See Chapter 13, Section 11 for details.) If q is even then $-u$ is another q-th root of α. Similarly by studying the behaviour of the function $f(x) = x^q$ on the interval $(-\infty, 0]$ and applying IVP to a suitable closed subinterval of it we get that for odd q, every negative real number has a negative q-th root (which is also unique).

As another application, we give an alternate proof of the result that every odd degree polynomial with real coefficients has at least one real root. In Chapter 3, Comment No. 2, we derived this as a consequence of the Fundamental Theorem of Algebra. But the latter is a very deep result whose proof is well beyond our scope. So it is desirable to have another, more elementary proof. One such proof is given by the IVP. Denote the polynomial by $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$, where $a_n \neq 0$. Then we can rewrite

[3] We are using here that a set J having the property that whenever any two points are in J so is every point between the two, is an interval. This is not hard to prove using other versions of completeness. Here we may simply take it as the *definition* of an interval. It is clear that all the intervals of the various types such as (a, b), $[a, \infty)$ etc. are intervals in this sense.

$f(x)$ as $a_n x^n(1+\frac{a_{n-1}}{a_n}\frac{1}{x}+\ldots+\frac{a_1}{a_n}\frac{1}{x^{n-1}}+\frac{a_0}{a_n}\frac{1}{x^n})$. The expression in the parentheses tends to 1 as $x \to \pm\infty$. We are assuming n to be odd. So $x^n \to \infty$ as $x \to \infty$ and $x^n \to -\infty$ as $x \to -\infty$. Assuming first that a_n is positive, we have $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$. So given any real number R, there are points a and b such that $f(a) < R$ and $f(b) > R$. By the IVP, there exists a point c such that $f(c) = R$. This shows not only that f has a zero, but that it is an onto function. If $a_n < 0$, the argument is similar. (Or, as usual, one can replace $f(x)$ by $-f(x)$ and apply the earlier part.) The analogous property of real polynomials of even degrees, given at the end of Comment No. 4, Chapter 3, can also be proved by applying the IVP to the intervals $[-R, 0]$ and $[0, R]$ for a sufficiently large R.

Sometimes the IVP can be applied in a negative manner, i.e., instead of applying it to prove the existence of a point at which the given function assumes an intermediate value, we apply it to the contra-positive, i.e., to prove the non-existence of a function which is barred from attaining certain values. Suppose, for example, that $f(x)$ is a continuous function defined for $1 \leq x \leq 3$. We are further given that $f(x)$ takes rational values for all x and that $f(2) = 10$ and are asked to find out the value of $f(1.5)$. (JEE 1997) With no other information about $f(x)$ (such as that it satisfies some differential equation), it may seem an impossible task to determine any formula for $f(x)$. But look at the condition that $f(x)$ takes only rational values. Now between any two (unequal) real numbers there is at least one irrational number. (This follows from the denseness of the set of all irrational numbers. See Exercise (6.46).) Since $f(x)$ is given to be continuous everywhere in $[1, 3]$, the IVP would then imply that $f(x)$ is identically constant. For, if $f(a) \neq f(b)$ for some $a, b \in [1, 3]$, then let α be any irrational between $f(a)$ and $f(b)$ and the function will have to assume α as a value at some point between a and b, contradicting the hypothesis that f takes only rational values. Since f is constant and $f(2) = 10$ we must also have $f(1.5) = 10$. The crucial point in this argument was that the set of irrational numbers is dense. So more generally, any continuous function which never assumes any value in a dense set must be a constant over any interval. (It is also easy to see this from the alternate formulation of the IVP given above, because if the image interval contains no point of a dense set, it must degenerate to a single point.)

Comment No. 5:

While studying the applications of IVP, one must not forget its limitations as well. First, it does not say that the point c at which f attains the given intermediate value m is unique. And, indeed, it need not be so, as can be seen by simple examples (e.g., take $f(x) = \cos x$ on $[0, 3\pi]$ with $m = \frac{1}{2}$.) A more serious limitation is that even when such a point c is unique, the proof above gives no explicit construction for obtaining it in a finite number of steps. The proof does give us an infinite sequence which will converge to c. But that is only a poor consolation. Using it we can come as close to c as we want. But that is not quite the same thing as hitting upon c exactly.

Let us illustrate this with an example. In Figure (d) of Comment No. 8 in Chapter 11, we gave a straight edge and compass construction to find a line, say L, through a given point, say D, lying between two rays, say R_1 and R_2 (meeting at A) such that the portion of L intercepted between R_1 and R_2 is divided at D in the ratio $1:1$. Now, if we merely want to prove the existence of such a line L, we can do so very easily using the Intermediate Value Property. In fact, a layman will do it just by common sense, without realising that he is really using IVP. First draw through D a line parallel to R_1 and let it cut R_2 at the point M. Now let X be a variable point on the ray R_2 which is farther from A than M is. Then the line through X and D will cut the ray R_1 at some point Y (say). Consider the ratio $\frac{XD}{DY}$. This ratio is undefined when $X = M$ since in that case the line XD is parallel to R_1. For X close to M, the point Y is very far out on the ray R_1 and so the ratio $\frac{XD}{DY}$ is close to 0. As X moves on R_2 away from M, this ratio keeps increasing. In fact as X tends to infinity on the ray R_2, the ratio tends to infinity. So, somewhere along the way, the ratio must be exactly 1. In essence, what is used here is the IVP. Let x be the distance between M and X. Using the sine rule, it is easy to express the lengths of the segments XD and DY and hence the ratio $\frac{XD}{DY}$ in terms of x. When so done, it will be a continuous function of x, defined for all $x > 0$. The reasoning given above is equivalent to saying that $f(x) \to 0$ as $x \to 0^+$ while $f(x) \to \infty$ as $x \to \infty$. So by the IVP, there is some value, say c at which $f(c)$ equals 1. Let C be the point on R_2 at a distance c from M. Then the line through C and D is the desired line L.

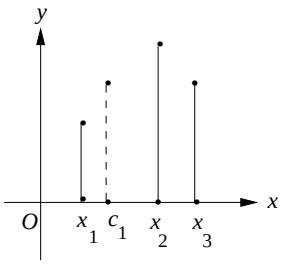
The point to note is that this reasoning does not give an explicit construction for the desired line L. The best we can do is to fix two points, say X_1 and X_2 on R_2, one very close to M and another very far so that the ratio $\frac{X_1D}{DY_1}$ is less than 1 and the ratio $\frac{X_2D}{DY_2}$ exceeds 1. Then let X_3 be the mid-point of the segment X_1X_2 and consider the ratio $\frac{X_3D}{DY_3}$. If it equals 1, we are lucky. Otherwise, depending upon whether it is smaller or bigger than 1, we move half-way towards X_2 or X_1 and continue. In essence we are repeating the proof of the IVP. But this process may never end. On this background, we can appreciate the explicit construction given in Comment No. 8 of Chapter 11. In fact, as remarked there, for some problems, no geometric construction is possible. For example, there is no construction to draw a line through D for which the intercept XY will be exactly of a given length, say, λ. But the existence of such a line can be proved using the IVP. Let $g(x)$ be the length of the segment XY, expressed as a function of x. Then $g(x)$ is continuous and tends to ∞ as $x \to 0+$ and also as $x \to \infty$. So, as long as λ is more than the minimum value of $g(x)$, the IVP will give at least two lines through D which will have intercepts of lengths λ each between R_1 and R_2.

The IVP was also inherent in the solution to the problem at the beginning of Comment No. 2 of Chapter 9 where we wanted to find the number of values of c for which the line $y = 4x + c$ touches the curve $\frac{x^2}{4} + y^2 = 1$. The intercept of such a line will be either the empty set or a segment (which could degenerate to a point). The length of this segment is a continuous function of the parameter

c. So its range is an interval of $\mathbb{R}$ having 0 as an endpoint. This means there are chords of arbitrarily small lengths parallel to the given line. The limiting position of such chords will give a tangent parallel to $y = 4x$. As remarked in Comment No. 6 of Chapter 9, IVP (for functions of two variables) also sometimes simplifies sign determination. But its discussion is beyond our scope.

Comment No. 6:

As a theoretically important application of IVP, we now prove that the inverse function, if any, of a continuous function, say f, is continuous. In order that the inverse function exists, it is first of all necessary that the function f be one-to-one and onto. So suppose f is a continuous function which maps an interval $[a, b]$ bijectively onto an interval $[c, d]$. We first claim that either f is strictly increasing or else it is strictly decreasing on $[a, b]$. As f is given to be one-to-one, strict monotonicity is the same as monotonicity. So it suffices to show that f is either monotonically increasing or monotonically decreasing on the entire interval $[a, b]$. If neither holds, then there exist $x_1 < x_2 < x_3$ in $[a, b]$ such that either $f(x_1) < f(x_2) > f(x_3)$ or $f(x_1) > f(x_2) < f(x_3)$. In the first case, we must have either $f(x_1) < f(x_3)$ or $f(x_1) > f(x_3)$ since equality of the two is precluded by the assumption that f is one-to-one. If $f(x_1) < f(x_3)$, then $f(x_3)$ lies between $f(x_1)$ and $f(x_2)$. So applying the IVP to the interval $[x_1, x_2]$, we get some $c_1 \in (x_1, x_2)$ such that $f(c_1) = f(x_3)$ as shown in the figure. Obviously, $c_1 \neq x_3$. But this contradicts that the function f is one-to-one. On the other hand, if $f(x_1) > f(x_3)$, then $f(x_1)$ lies between $f(x_2)$ and $f(x_3)$ and we apply the IVP to the interval $[x_2, x_3]$ to get a contradiction. Thus no matter how $f(x_1)$ and $f(x_3)$ are related to each other, we get a contradiction in the case $f(x_1) < f(x_2) > f(x_3)$.



We still have to handle the case where $f(x_1) > f(x_2) < f(x_3)$. Again, we consider two subcases depending upon which of $f(x_1)$ and $f(x_3)$ is greater. Applying the IVP to an appropriate subinterval gives a contradiction in each subcase.

Now for continuity of f^{-1}, we assume f is strictly increasing, as otherwise we replace f by $-f$. First let $y_0 \in (c, d)$. Then $y_0 = f(x_0)$ for a unique $x_0 \in (a, b)$. Let $\epsilon > 0$ be given. Choose $0 < \epsilon_1 < \epsilon$ such that $[x_0 - \epsilon_1, x_0 + \epsilon_1] \subset (a, b)$. As f is increasing and continuous, the image of the interval $[x_0 - \epsilon_1, x_0 + \epsilon_1]$ will be the interval $[f(x_0 - \epsilon_1), f(x_0 + \epsilon_1)]$, which contains $f(x_0)$, i.e., y_0, in its interior (although not necessarily at its centre). Let us now define δ by $\delta = \min\{f(x_0) - f(x_0 - \epsilon_1),\ f(x_0 + \epsilon_1) - f(x_0)\}$. Then $\delta > 0$ and moreover, $f^{-1}((y_0 - \delta, y_0 + \delta)) \subset f^{-1}(f([x_0 - \epsilon_1, x_0 + \epsilon_1])) = [f^{-1}(y_0) - \epsilon_1, f^{-1}(y_0) + \epsilon_1]$ as f is one-to-one. But $[f^{-1}(y_0) - \epsilon_1, f^{-1}(y_0) + \epsilon_1] \subset (f^{-1}(y_0) - \epsilon, f^{-1}(y_0) + \epsilon)$. Thus we have shown that f^{-1} is continuous at every interior point y_0 of the interval $[c, d]$. Proof of continuity of f^{-1} at the end-points c, d is similar.

We remark that in this argument the domain of f was an interval. It does

not matter much whether this interval is closed or open, bounded or unbounded. But if we replace it by, say, the union of two intervals, then the theorem may be false. For example, let S be the union of the closed interval $[-\frac{\pi}{2}, 0]$ and the open interval $(2\pi, 2\pi + \frac{\pi}{2})$. Then the function $f(x) = \sin x$ is continuous and strictly monotonically increasing on S and maps it onto the semi-open interval $[-1, 1)$. The inverse function f^{-1} maps $[-1, 1]$ onto $[-\frac{\pi}{2}, 0]$ and $(0, 1)$ onto $(2\pi, 2\pi + \frac{\pi}{2})$. The inverse function is continuous everywhere but not at 0. More precisely, it is left continuous but not right continuous at 0.

Comment No. 7:

A few more applications of the Intermediate Value Property will be given in the exercises.We now turn to another important consequence of continuity which we stated as Theorem 1 in Chapter 13, It asserts that every continuous, real-valued function on a closed and bounded interval has a maximum and a minimum. Like the IVP, this is also an existence theorem in the sense that it merely proves the existence of the desired point by expressing it as the limit of some sequence whose convergence is guaranteed by completeness of the real line. Not surprisingly, the same technique of successive bisection is used in its proof. As a result, this time we give the proof a little tersely than that of the Intermediate Value Property.

So, suppose that $f(x)$ is continuous at every point of an interval $[a, b]$. It suffices to prove that f attains its maximum on $[a, b]$. The case of the minimum can be deduced by taking the maximum of $-f$ as usual.

We first show that the function $f(x)$ is bounded above on $[a, b]$. That is, we claim that there exists some M such that $f(x) \leq M$ for all $x \in [a, b]$. We proceed by contradiction. That is, assume f is not bounded above on $[a, b]$. Divide the interval $[a, b]$ into two halves. Then either f is not bounded above in the left half or it is not bounded above in the right half. For, if it were bounded above in both with M_1 and M_2 as upper bounds (say) respectively, then $\max\{M_1, M_2\}$ will be an upper bound on $f(x)$ for the entire interval $[a, b]$, contradicting the hypothesis. Call $[a, b]$ as $[a_0, b_0]$ and let $[a_1, b_1]$ be that half on which f is not bounded above. (In case f is unbounded above on both the halves, make a choice. For example, always take the left half.) We now repeat this argument for $[a_1, b_1]$ and get $[a_2, b_2]$ as one of its halves on which f is not bounded above. Continuing in this manner we get, exactly as in the proof of the Intermediate Value Property, an infinite sequence of intervals $[a_n, b_n]$ for $n = 0, 1, 2, \ldots$ such that

$$\left.\begin{array}{rll} & \text{(i)} & [a_n, b_n] \subset [a_{n-1}, b_{n-1}] \text{ for every } n = 1, 2, \ldots \\ & \text{(ii)} & b_n - a_n = \dfrac{b_0 - a_0}{2^n} \text{ for all } n = 0, 1, 2, \ldots \\ \text{and} & \text{(iii)} & f \text{ is not bounded above on } [a_n, b_n] \\ & & \text{for all } n = 0, 1, 2, \ldots \end{array}\right\} \tag{9}$$

Now, exactly as in the proof of the IVP above, the sequences $\{a_n\}$ and $\{b_n\}$ converge to a common limit, say c in $[a, b]$. More precisely, a_n tends to c from below and b_n tends to c from above. We are given that $f(x)$ is continuous

everywhere and hence, in particular at c. Let us first assume that c is an interior point of $[a, b]$, i.e., $a < c < b$. Fix some $\delta_0 > 0$ such that $(c - \delta_0, c + \delta_0) \subset [a, b]$. Now take $\epsilon = 1$ in the definition of continuity of $f(x)$ at c. Then there exists some positive δ (which may be assumed to be smaller than δ_0), such that

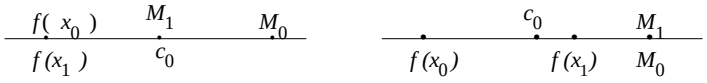
$$f(c) - 1 < f(x) < f(c) + 1 \text{ for all } x \in (c - \delta, c + \delta) \tag{10}$$

In particular, this means that $f(x)$ is bounded above (by $f(c) + 1$) on the interval $(c - \delta, c + \delta)$. Since $a_n \to c^-$ and $b_n \to c^+$ as $n \to \infty$, and $\delta > 0$, we see that for all sufficiently large values of the integer n, $c - \delta < a_n \leq c$ and $c \leq b_n < c + \delta$. Put together this means that for all large n, the interval $[a_n, b_n]$ is contained in the interval $(c - \delta, c + \delta)$. Since $f(x)$ is bounded above on the latter, so it is on the subinterval $[a_n, b_n]$. But this contradicts (iii) of (9) directly. If the point c is an end-point of $[a, b]$ then the argument is even simpler, since we need only one of the two sequences $\{a_n\}$ and $\{b_n\}$ (the former if $c = b$ and the latter if $c = a$) to get a contradiction. In all cases, we get a contradiction because of our original assumption that f is not bounded above on the interval $[a, b]$. Thus we have shown that there exists some M such that $f(x) \leq M$ for all $x \in [a, b]$. In fact, we have shown that *every* continuous real-valued function on $[a, b]$ is bounded above. We shall use this general result later in the course of the proof.

Although we have so far shown that f has an upper bound on $[a, b]$, we still have to show that f attains its maximum on $[a, b]$ at some point, i.e., there is some x^* in $[a, b]$ such that $f(x) \leq f(x^*)$ for all $x \in [a, b]$. Note that this does not follow automatically from the fact that M is an upper bound for f on $[a, b]$. If there is some point, say x^* in $[a, b]$ such that $f(x^*) = M$, then of course, we would be through. But the catch is that such a point need not exist. The fact that M is an upper bound only means that for every point $x \in [a, b]$, the inequality $f(x) \leq M$ holds. But the inequality may be strict everywhere. (For example, 100 is an upper bound for the sine function on any interval. But there is no point at which the sine function has value 100.)

This example also suggests what may be a way out. The upper bound M on f may be extravagantly large. If we can replace it by a smaller number, say M' which is also an upper bound, then the chances are better that there will be some point x^* in $[a, b]$ at which it is assumed as a value. If not, we replace M' by a still smaller upper bound. Continuing like this, we should either get a desired maximum or else an infinite sequence to which we can apply completeness.

Now to the details. We already have an upper bound M on $f(x)$ for $x \in [a, b]$. We rename it as M_0. Now take an arbitrary point x_0 in $[a, b]$. Then $f(x_0) \leq M_0$. If luckily equality holds, then we can take $x^* = x_0$ as a point of maximum of f on $[a, b]$. Suppose, however, that $f(x_0) < M_0$. Let c_0 be the A.M. of the two, i.e., $c_0 = \frac{1}{2}(f(x_0) + M_0)$. Then $f(x_0) < c_0 < M_0$. Now there are two possibilities. Either c_0 is also an upper bound for $f(x)$ on $[a, b]$ or else it is not. If the first possibility holds, we call x_0 as x_1 and c_0 as M_1 as shown in Figure (i) below. If the second possibility holds, then there is some $x_1 \in [a, b]$ such that $f(x_1) > c_0$. We choose any such x_1 and call M_0 as M_1 as shown in Figure (ii) below.



(i) (ii)

In either case we have x_1 as a point of $[a, b]$ and M_1 as an upper bound of $f(x)$ on $[a, b]$. Moreover the length of the interval $[f(x_1), M_1]$ is half of the length of $[f(x_0), m_0]$ in the first case (i.e., where $c_0 = M_1$), while in the second case it is even smaller since $f(x_1) > c_0$.

We now repeat the argument with x_1 and M_1. If $f(x_1) = M_1$ we take $x^* = x_1$ as a point of maximum of f on $[a, b]$. Otherwise we construct $x_2 \in [a, b]$ and M_2 such that M_2 is an upper bound and the interval $[f(x_2), M_2]$ is contained in the interval $[f(x_1), M_1]$ and has at most half the length as the latter. Continuing further, either we shall find x^* at some stage or else we shall get a sequence of points $\{x_n\}$ in $[a, b]$ and a monotonically decreasing sequence $\{M_n\}$ of upper bounds for $f(x)$ on $[a, b]$ such that the sequence $\{f(x_n)\}$ is monotonically increasing. (Note that we are *not* saying that the sequence $\{x_n\}$ is monotonically increasing. In fact we know nothing about the sequence $\{x_n\}$ other than that it lies in the interval $[a, b]$.) Moreover, because of the construction, $0 \leq (M_n - f(x_n)) \leq \frac{M_0 - f(x_0)}{2^n}$ and hence $M_n - f(x_n) \to 0$ as $n \to \infty$. Once again, by completeness, M_n and $f(x_n)$ approach a common limit, say M^*. From this we leave it to you to check that M^* is also an upper bound on $f(x)$ for $a \leq x \leq b$. We claim that this upper bound really gives us a point of maximum, i.e., we can now find some $x^* \in [a, b]$ such that $f(x^*) = M^*$. For, suppose this is not the case. Then for every $x \in [a, b]$ we have $f(x) < M^*$. We can then define a continuous real-valued function g on $[a, b]$ by $g(x) = \frac{1}{M^* - f(x)}$. Since the sequence $\{f(x_n)\}$ converges to M^* from below, by taking $x = x_n$ for a sufficiently large n, we can make $M^* - f(x_n)$ arbitrarily small and hence $g(x)$ arbitrarily large. But this contradicts what we proved above, viz., that *every* continuous, real-valued function on $[a, b]$ is bounded above.

Thus we have shown that every continuous real-valued function on a closed and bounded interval attains its maximum and similarly its minimum. Some parts of the proof could have been done a little more tersely had we used more conventional versions of completeness. But that would have required us to introduce certain new terms and to prove some results about them first. The version of completeness we have given in the form of the Axiom above is probably the easiest to accept as obvious.

Comment No. 8:

The theorem just proved was stated as Theorem 1 in Chapter 13, Comment No. 7. Theorem 2 there was elementary and its proof required little more than the definition of a derivative. It asserted that if a function $f(x)$ assumes its maximum or minimum at an interior point, say c, and is differentiable at this point, then $f'(c)$ must vanish. We now put these two theorems together. The former guarantees the existence of an extremum while the latter tells us what happens to the derivative at the extremum point. What results is a chain of

related theorems, the most frequently used among them being the Lagrange's Mean Value Theorem. The hypotheses of these theorems typically involve functions that are continuous on a closed and bounded interval, say $[a, b]$, and which are differentiable at every point of the open interval (a, b). We need continuity on the closed interval because without it the function need not have an extremum (e.g., the function $f(x) = \frac{1}{x}$ is continuous on the open interval $(0, 1)$ but has neither a maximum nor a minimum). However, as far as differentiability is concerned only the interior points matter and so there is no need to assume right/left differentiability at the end-points. Thus, for example, the theorems we shall prove will be applicable to the function $f(x) = \sqrt{x}$ on the interval $[0, 1]$, even though it is not (right) differentiable at the end-point 0. (For another such example, see Comment No. 13 of the last chapter.)

The proofs we shall give are all very standard. But they are generally not included in texts at the JEE level. So we give them here.

The forerunner in the series is the following theorem, called **Rolle's Theorem** which is proved by merely putting together Theorems 1 and 2 of Chapter 13.

Theorem 2: Suppose f is a real-valued function which is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b). (In other words, $f'(c)$ exists at every interior point. No assumption is made regarding differentiability at an end point.) Assume further that $f(a) = f(b) = 0$. Then there exists some $c \in (a, b)$ such that $f'(c) = 0$.

Proof : If f is identically 0 on $[a, b]$ then f' vanishes identically on (a, b) and so we may take c to be any point of (a, b). If f is not identically 0 on $[a, b]$ then there is some $x_0 \in (a, b)$ such that $f(x_0) \neq 0$. So either $f(x_0) > 0$ or $f(x_0) < 0$. Without loss of generality, assume that the first possibility holds. (Otherwise, as usual, replace f by $-f$.) Now, by Theorem 1 of Chapter 13, there exists some point c in $[a, b]$ such that $f(c) \geq f(x)$ for all $x \in [a, b]$. In particular $f(c) \geq f(x_0) > 0$. So c must be an interior point of $[a, b]$, since $f(a) = f(b) = 0$. In other words f attains its maximum on $[a, b]$ at the point $c \in (a, b)$. By assumption, $f'(c)$ exists. So by Theorem 2 of Chapter 13, $f'(c) = 0$.

As a direct application of Rolle's theorem we get that in the interval $(1, 3)$ there is at least one value of x which satisfies $x \ln x = 3 - x$. For, we take $f(x) = (x - 3) \ln x$ which is differentiable for all $x > 0$ and hence in particular on $[1, 3]$ (which automatically implies continuity on $[1, 3]$). Clearly, $f(1) = 0 = f(3)$. So by Rolle's theorem $f'(x) = 0$ for some $x \in (1, 3)$. But $f'(x) = \frac{x-3}{x} + \ln x$. Equating this with 0 gives $x \ln x = 3 - x$ as desired. (JAT 1980)

Rolle's theorem gives us another method to show that a given function, say $g(x)$, has at least one zero in a given interval. Sometimes this may work where the IVP is not very convenient to apply. Suppose, for example that we are given that $a + b + c = 0$ and are asked to show that the quadratic $3ax^2 + 2bx + c = 0$ has at least one root in $[0, 1]$. (JEE 1983) One way to do this would be to consider the function $g(x) = 3ax^2 + 2bx + c$. If we could show

that $g(0)$ and $g(1)$ are of opposite signs, then the IVP will imply that $g(x)$ has at least one zero in $[0,1]$. Now, $g(0) = c$ while $g(1) = 3a+2b+c = 2a+b$ (since $a+b+c = 0$). But without further data, there is nothing to suppose that c and $2a+b$ are of opposite signs. In fact, one of the limitations of the IVP is that in case a function changes sign twice in an interval, then its values at both the end-points will be of the same sign and hence the IVP will give us nothing, unless we split the interval suitably and apply the IVP to the subintervals.

But if we recognise $3ax^2+2bx+c$ as $f'(x)$, where $f(x) = ax^3+bx^2+cx$, we see that $f(0) = f(1) = 0$ by hypothesis. So by Rolle's theorem, $3ax^2+2bx+c$ vanishes at least once in $[0,1]$.

A geometric paraphrase of Rolle's theorem is that if f satisfies the conditions in the theorem then there is some point on the graph of the function at which the tangent is horizontal. Verbally, Rolle's theorem says that if the continuity and differentiability requirements are satisfied then between any two zeros of a function f lies at least one zero of f'. Now suppose a, b, c are three distinct zeros of a function f with $a < b < c$ (say), then by Rolle's theorem, there are at least two zeros of f', one, say u in (a,b) and another, say v in (b,c). Clearly $u < v$ and hence they are distinct. So we conclude that if a function has 3 distinct zeros then its derivative has at least 2 distinct zeros. Sometimes the nature of f' precludes it from having two distinct zeros. Then we get that the original function cannot have three distinct zeros. For example, suppose $f(x) = ax^3+bx^2+cx+d$. Then $f'(x) = 3ax^2+2bx+c$ is a quadratic polynomial with discriminant $4(b^2-3ac)$. If $b^2 < 3ac$ then f' has no real zeros and hence the original cubic cannot have more than one real root. But by the Intermediate Value Property, it has at least one real root. So, if $b^2 < 3ac$, then the cubic ax^3+bx^2+cx+d has exactly one real root. (We hasten to add here, that $b^2 > 3ac$ is only a necessary condition for the cubic to have three real roots. That it is not sufficient is shown by the cubic $f(x) = x^3-3x+3$.)

More generally, if a real polynomial $p(x)$ has r distinct real roots, then $p'(x)$ has at least $r-1$ distinct real roots, whence $p''(x)$ has at least $r-2$ distinct real roots and so on. Actually, this still holds even if the roots are not distinct. But then we have to count them according to their multiplicities and the argument gets a little complicated. (See Exercise (16.19)).

Comment No. 9:

By applying Rolle's theorem to a judiciously chosen function, we get a variety of theorems. They are essentially of the same spirit as Rolle's theorem but are sometimes more convenient to apply than Rolle's theorem. By far, the most important among them is the Lagrange's Mean Value Theorem. But occasionally we need something stronger than that. So we begin with the following theorem, from which everything we need will follow. (Many authors prefer to go directly from Rolle's theorem to Lagrange's theorem. A direct proof of Lagrange's theorem for a function $f(x)$ on an interval $[a,b]$ can be obtained by applying Rolle's theorem to the function $\phi(x) = (b-a)f(x) - (f(b)-f(a))x + af(b) - bf(a)$.)

Theorem 3: Suppose f, g, h are three functions which are continuous on $[a,b]$

and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$\begin{vmatrix} f'(c) & g'(c) & h'(c) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0 \tag{11}$$

Proof: Define a function $\phi : [a, b] \longrightarrow \mathbb{R}$ by

$$\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix}$$

When expanded, $\phi(x)$ is a linear combination of $f(x), g(x)$ and $h(x)$. So ϕ is continuous on $[a, b]$ and differentiable on (a, b). Also by the properties of the determinants, $\phi(a) = 0 = \phi(b)$. Hence by Rolle's theorem, there exists some $c \in (a, b)$ such that $\phi'(c) = 0$. Since the entries in all except one row of the determinant are constants, by Exercise (15.10) $\phi'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \end{vmatrix} = 0.$ So (11) holds.

By choosing the functions f, g and h suitably, we can derive from this theorem certain important results. Suppose, for example, that we take h to be the constant function 1. Then no matter what c is, $h'(c) = 0$. So expanding the determinant in (11) we get

$$g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a)) \tag{12}$$

If we further assume that g' never vanishes in (a, b), then it follows that $g(a) \neq g(b)$, as otherwise we get a contradiction by applying Rolle's theorem to the function $g(x) - g(a)$ on the interval $[a, b]$. We can then divide (12) by $g'(c)(g(b) - g(a))$ to get the following result, called **Cauchy's Mean Value Theorem**.

Theorem 4: Suppose f and g are continuous on $[a, b]$ and differentiable on (a, b) and $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists some $c \in (a, b)$ such that $\dfrac{f'(c)}{g'(c)} = \dfrac{f(b) - f(a)}{g(b) - g(a)}$.

Cauchy's MVT is just the tool we need to prove the so-called strong form of l'Hôpital's rule mentioned in Comment No.7 of the last chapter. The precise statement is as follows.

Theorem 5: Suppose that the functions f and g are continuous in a neighbourhood N of a point c and are differentiable in the deleted neighbourhood $N - \{c\}$ of c. Assume further that $f(c) = 0 = g(c)$ but that g does not vanish in $N - \{c\}$ and $\lim\limits_{x \to c} \dfrac{f'(x)}{g'(x)} = L$. Then $\lim\limits_{x \to c} \dfrac{f(x)}{g(x)} = L$.

Proof : We shall first show that $\lim_{x\to c^+} \frac{f(x)}{g(x)} = L$. Let $x \in N$ and $x > c$. Since $g(x) \neq 0$ while $g(c) = 0$, we can apply Cauchy's MVT to get some $c(x)$ (depending on x), in the interval (c, x) such that

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f'(c(x))}{g'(c(x))} \tag{13}$$

As $x \to c^+, c(x) \to c^+$. So $\lim_{x\to c^+} \frac{f'(c(x))}{g'(c(x))}$ is the same as $\lim_{x\to c^+} \frac{f'(x)}{g'(x)}$. (To see this, merely observe that for $\delta > 0, c < x < c + \delta$ implies $c < c(x) < c + \delta$.) But the latter equals L. So by (13) $\lim_{x\to c^+} \frac{f(x)}{g(x)} = L$. A similar argument works for the left-handed limit, $\lim_{x\to c^-} \frac{f(x)}{g(x)}$, except that, we apply Cauchy's MVT to the interval $[x, c]$ rather than to $[c, x]$. Since both limits equal L, the proof is complete.

Comment No. 10:

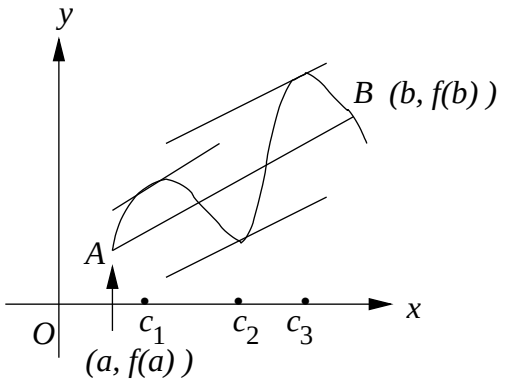
The stage is now set for the **Lagrange's Mean Value Theorem**.

Theorem 6: Suppose $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists some c such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Proof: This follows at once by taking $g(x) = x$ in Cauchy's MVT. (A direct proof was already indicated before the statement of Theorem 3.)

Although only a special case of Cauchy's MVT, the Lagrange's MVT is applied so frequently, that when one simply says 'Mean Value Theorem (or MVT)' without qualification, it means Lagrange's Mean Value Theorem. Before discussing any such applications, a few comments are in order. The term 'mean value' may give the wrong impression that the number c (whose existence is asserted by the theorem) is the arithmetic mean of a and b (i.e., $\frac{a+b}{2}$) or some other mean such as the geometric or the harmonic mean. That this is not the case can be seen from simple examples like $f(x) = x^3$ and $[a, b] = [0, 1]$. Here $f'(x) = 3x^2$ and $\frac{f(b)-f(a)}{b-a} = 1$. So the only point c satisfying the conclusion of the MVT is $c = \frac{1}{\sqrt{3}}$. There is, in fact, no general formula to express c in terms of a, b and the function f. The term 'mean' denotes here the average rate of change. As we move from a to b the net change in the function is $f(b) - f(a)$ (which could be positive, negative or zero). So $\frac{f(b)-f(a)}{b-a}$ is the average or the mean rate of change over the interval $[a, b]$. Lagrange's theorem asserts that under the conditions specified, there exists at least one c such that the instantaneous rate of change at c equals the average rate of change over $[a, b]$. Naturally, this point c depends as much on the function f as the points a and b. If we take a smaller interval, say, $[a_1, b_1]$, contained in $[a, b]$ then the MVT would give a point $c_1 \in (a_1, b_1)$ such that $f'(c_1) = \frac{f(b_1) - f(a_1)}{b_1 - a_1}$. But in general $c_1 \neq c$.

Geometrically, if we draw the graph of $y = f(x)$ for $a \leq x \leq b$, then $\dfrac{f(b)-f(a)}{b-a}$ is precisely the slope of the chord joining the end-points $A = (a, f(a))$ and $B = (b, f(b))$. Lagrange's theorem asserts the existence of a point on the graph at which the tangent is parallel to the chord AB. As seen from the figure, there may be several such points. In other words, the mean value c given by the MVT need not be unique. In fact examples can be given where the function f is not a constant but has infinitely many mean values. The exercises will also give examples to show that the conditions in the hypothesis are vital.



As a straightforward verification of Lagrange's MVT (JAT 1979), consider the function $f(x) = \sqrt{x-1}$ for $1 \leq x \leq 3$. Note that the function is not defined for $x < 1$. However, the function $\sqrt{x-1}$ is continuous from the right at $x = 1$. As it is continuous at all $x \in (1, 3]$, the first part of the hypothesis of Lagrange's MVT is satisfied. As for the second part, the function is *not* differentiable (i.e., not right differentiable) at 1. But this does not matter because the hypothesis requires differentiability only on the open interval $(1, 3)$. For $x \in (1, 3)$, we indeed have $f'(x) = \frac{1}{2\sqrt{x-1}}$. Now, $\frac{f(3)-f(1)}{3-1} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$. So a point $c \in (1, 3)$ will answer the conclusion of Lagrange's MVT if and only if $\frac{1}{2\sqrt{c-1}} = \frac{1}{\sqrt{2}}$. Solving, this gives $c = \frac{3}{2}$ as the 'mean value'.

Comment No. 11:

With the Lagrange's MVT at our disposal, we can now give rigorous proofs of some statements where we relied only on intuition earlier. Take for example, Theorem 3 of Chapter 13 in which we linked the increasing/decreasing behaviour of a function over an interval with the sign of its derivative. We can now derive all the five statements there at a stroke. Suppose x_1, x_2 are any two points in the domain interval with $x_1 < x_2$. By Lagrange's MVT, there exists $c \in (x_1, x_2)$ such that $f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1}$. Let us rewrite this as

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) \tag{14}$$

As the second factor on the R.H.S. is positive, the sign of the L.H.S. is decided by $f'(c)$. For example, if $f'(c) < 0$ then $f(x_1) < f(x_2)$. Hence f is strictly decreasing. All other statements follow equally easily. A similar reasoning works to establish parts (ii) and (iii) of Theorem 4 of Chapter 13.

Another place where we appealed to intuition to derive conclusions about the behaviour of a function from its derivatives was when we considered

the concavity upward or downward of a function $y = f(x)$ in Comment No. 18 of Chapter 13. Specifically, there we started with a function $y = f(x)$ which is twice differentiable on an interval S and two interior points x_1 and x_2 of S with $x_1 < x_2$. We claimed that if the second derivative $f''(x)$ is non-negative for all $x \in (x_1, x_2)$, then the graph of the function $y = f(x)$ bends upwards and hence for any point $x_3 \in (x_1, x_2)$, the point $(x_3, f(x_3))$ on the graph lies on or below the chord joining the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the graph. Algebraically, this is equivalent to saying that

$$f(x_3) \leq f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x_3 - x_1) \tag{15}$$

(This is the same as (19) in Chapter 13.)

We can now give a rigorous proof of this inequality which is based on Lagrange's MVT without appealing to geometric intuition. First we rewrite (15) in a slightly different form. Let $\lambda = \frac{x_3 - x_1}{x_2 - x_1}$. Since $x_1 < x_3 < x_2$, we have $0 < \lambda < 1$. By a direct calculation, (15) is equivalent to

$$f(x_3) \leq (1 - \lambda) f(x_1) + \lambda f(x_2) \tag{16}$$

Now, to prove (16) we first apply Lagrange's MVT to the intervals $[x_1, x_3]$ and $[x_3, x_2]$ to get points $c_1 \in (x_1, x_3)$ and $c_2 \in (x_3, x_2)$ such that

$$f(x_3) - f(x_1) = f'(c_1)(x_3 - x_1) = f'(c_1)\lambda(x_2 - x_1) \tag{17}$$

$$\text{and } f(x_2) - f(x_3) = f'(c_2)(x_2 - x_3) = f'(c_2)(1 - \lambda)(x_2 - x_1) \tag{18}$$

Since $c_1 < x_3$ and $x_3 < c_2$, we have $c_1 < c_2$. We are assuming that $f''(x) \geq 0$ throughout (x_1, x_3). So by Lagrange's MVT applied to f', we have

$$f'(c_1) \leq f'(c_2) \tag{19}$$

If we multiply the inequality (17) by $(1 - \lambda)$ and the inequality (18) by λ (both of which are positive) and use (19), we get

$$(1 - \lambda)(f(x_3) - f(x_1)) \leq \lambda(f(x_2) - f(x_3)) \tag{20}$$

which gives (16) and hence (15) as desired. Note further that if $f''(x) > 0$ for all $x \in (x_1, x_2)$ then strict inequality holds in (19) and hence also in (20). Correspondingly, strict inequality will also hold in (16) and hence in (15). Thus, in such a case, the function $y = f(x)$ is strictly concave upward.

Comment No. 12:

Important as these applications are, they do not use Lagrange's MVT in its full strength. As we saw, the conclusion of Lagrange's MVT can be written as Equation (14) above. But in the applications so far, we used it only to decide if the L.H.S. was positive or negative. Put differently, our concern was more with the sign of $f'(c)$ than with its magnitude. This is rather like looking at,

say, only the gender of a person and ignoring his/her other qualities. While this may be the right thing to do for certain purposes, for many other purposes it is the other attributes such as education, physical fitness and income that may be more relevant.

As an example of an application of Lagrange's MVT where not the sign but the magnitude of $f'(c)$ matters, take $f(x) = \sin x$, $x_1 = 0$ and $x_2 = \theta$ in (14). Then it says that for every $\theta > 0$, there is some $c \in (0, \theta)$ such that $\sin\theta = \cos c\ \theta$. Now no matter what c is, $\cos c \leq 1$. Therefore we get that for all $\theta > 0$, $\sin\theta \leq \theta$. This may not sound very impressive because we know this inequality already and in fact, this inequality is needed in finding the derivative of the sine function. But the point to note is that the same method can be applied to *any* function, say $f(x)$ and any interval over which the hypothesis of MVT holds. We fix a and take b as $a + h$, where we allow h to be negative as well, in which case, Lagrange's theorem is to be applied to the interval $[b, a]$ rather than to the interval $[a, b]$. Then (14) translates as

$$f(a + h) = f(a) + hf'(c) \tag{21}$$

for some $c \in (a, a + h)$.

Let us interpret this in a novel way. By continuity of f, $f(a+h)$ tends to $f(a)$ as $h \to 0$. So for small values of $|h|$, we may take $f(a)$ as an approximation to $f(a + h)$. This approximation is called the **zero-th order approximation** or the constant approximation to f at a. Thus, in the R.H.S. of (21), the first term is the approximate value of $f(a+h)$ under the zero-th order approximation while the second term is, by definition, the **error** of the approximation. It tells you how far the approximate value (here $f(a)$) is from the exact value (viz., $f(a + h)$).

With this interpretation, Lagrange's MVT gives us a formula for the error in the 0-th order approximation. At first sight it may appear that this formula is of little use because, as stressed after the statement of Theorem 5 above, in general there is no formula to write down the mean value c in terms of a, b and f. But this hardly detracts from the merit of the MVT. In fact, it is very rare that you will get an exact formula for the error, for the simple reason that if you have such a formula, then it is as good as having the exact answer and then there is no need to go for an approximation in the first place! It is rather like this. Suppose your watch is not showing the correct time. But if you know exactly how fast or how slow it is, then effectively your watch *is* showing the correct time!

When we go for approximations, the thing we look for is not an exact formula for the error, but an assurance that the error is not more than some number. When this number is smaller than (or possibly equal to) the degree of inaccuracy we are prepared to tolerate, the approximation is as good as the exact answer for practical purposes. For example, if we want the approximation to be correct upto three places of decimals, it means we must ensure that the error is not more than 10^{-3}. An approximate answer along with an estimation of the error (i.e., an upper bound on its absolute value) is the next best thing to an exact answer.

When viewed in this new light, (21) is a powerful tool for estimating the error caused by the 0-th order approximation. In many cases it is not hard to find some upper bound, say M, on $|f'(x)|$ for $x \in (a, b)$. Indeed, many times this can be done by inspection. (Note that since our interest is only in an estimation of the error and not computing it exactly, there is little point in trying to find the best upper bound on $|f'(x)|$ if doing so means spending a lot of effort. Avoid, for example, finding the maximum of $|f'(x)|$ by the methods given in Chapter 13. What we want here is any upper bound which can be obtained with little or no effort as long as it is not extravagantly large.) Then (21) says that the error involved in approximating $f(a + h)$ by $f(a)$ is at most $M|h|$ in magnitude.

Suppose for example, we want to estimate $\sqrt{2}$. Take $f(x)$ as $\sqrt{x}$. Then $f'(x) = \frac{1}{2\sqrt{x}}$ for $x > 0$. Taking $a = 1$ and $h = 1$ we get $\sqrt{2} - 1 = \frac{1}{2\sqrt{c}}$ for some $c \in (1, 2)$. Clearly, $\frac{1}{2\sqrt{c}} < \frac{1}{2}$ for all $c > 1$ and hence, in particular, for all $c \in (0, 1)$. So applying (21) and taking $M = 0.5$ we get that $\sqrt{2} \approx 1$ with an error less than 0.5. This is, of course, a very crude approximation. But we can improve upon it. The reason we took $a = 1$ was that we know the exact value of $f(1)$, i.e., of $\sqrt{1}$. Things will improve if we retain this feature and can move a closer to 2. One such value is 2.25 since $\sqrt{2.25} = 1.5$. So now we take $a = 2.25$ and $h = -0.25$. Then by (21), $\sqrt{2} - 1.5 = -0.25\frac{1}{2\sqrt{c}}$ for some $c \in (2, 2.25)$. We again take $M = 0.5$. ($\frac{1}{2\sqrt{2}}$ would be a better upper bound on $\frac{1}{2\sqrt{c}}$ for $c \in (2, 2.25)$. But to compute it we need to know $\sqrt{2}$ in the first place! So we settle for $M = 0.5$.) We now get $\sqrt{2} \approx 1.5$ with an error less than 0.125. If we can move a still closer to 2, (e.g., $a = 1.96$ with $\sqrt{1.96} = 1.4$), we would get still better an approximation. Obviously, there are limitations to this method. In effect, here we are searching for the exact value of $\sqrt{2}$ by creating a sequence $1, 1.5, 1.4, \ldots$ which will converge to $\sqrt{2}$. For each term of the sequence we are applying the zero-th order approximation and estimating the error using the MVT. In case the error estimate is not satisfactory, we are moving to the next term of the sequence. This is not a very efficient procedure.

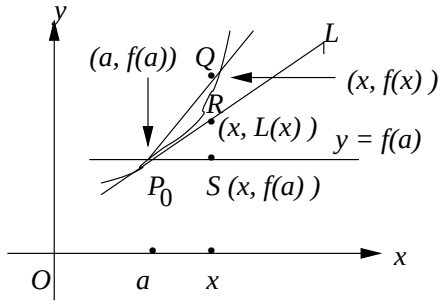
Comment No. 13:

A radically superior approach is to see if (21) itself can be improved upon. Naturally, inasmuch as (21) was based on Lagrange's MVT, such an improvement may be possible only if the function $f(x)$ satisfies stronger conditions than those needed in Lagrange's MVT. Note that the zero-th order approximation is based only on the continuity of $f(x)$ at the point a while its error estimation is based on the assumption that f be continuous on the interval $[a, a + h]$ and differentiable on the interval $(a, a + h)$. As proved in Comment No. 12 of the last chapter, differentiability is a stronger concept than continuity. Let us see what this means in terms of approximation of $f(a + h)$. To say that f is continuous at a point a simply means that for numerically small values of h, $f(a + h)$ is close to $f(a)$, i.e., that the difference $|f(a + h) - f(a)|$ is small. But differentiability at a says more. It says that for small non-zero values of h,

the ratio $\frac{f(a+h)-f(a)-hf'(a)}{h}$ and hence also the ratio $\frac{|f(a+h)-f(a)-hf'(a)|}{|h|}$ nearly equals 0. This means that not only is the numerator small, but it is of a smaller order of magnitude than $|h|$.

As a real-life analogy, suppose you spot a moving car at a point P on a road at some moment and you want to figure out where it would be ten seconds later. In the absence of any other information, all you can say is that it will be near P. But suppose you also know that its speed was, say 60 km/hr at the moment it was spotted. Then you can estimate its position ten seconds later more accurately as at a point Q further along the road at a distance $\frac{1}{6}$ km from P. This answer need not be exact since the speed of the car could change in ten seconds. But it is certainly better than the earlier answer. And, if the time interval were shorter, say, only one second, then the approximate answer would indeed be very close to the exact answer.

Returning to mathematical terminology, $f(a) + hf'(a)$ will be a better approximation to $f(a+h)$ than $f(a)$ is. It is called the **first order approximation** or the **linear approximation** to f at a. To understand the latter term, call $a+h$ as x. Then $f(a)+hf'(a)$ is the same as $f(a)+f'(a)(x-a)$. Call this expression as $L(x)$. Then it is a linear (i.e., a first degree) polynomial in x. The graph of $y = L(x)$ is a straight line passing through the point $(a, f(a))$ $(= P_0)$ (say) and having slope $f'(a)$. But this is exactly the tangent to the graph of $y = f(x)$ at the point P_0 on it.



Geometrically, the zeroth order approximation to $f(x)$ at a approximates the portion of the graph of $y = f(x)$ near the point P_0 by the horizontal straight line $y = f(a)$, while the first order approximation approximates it by a straight line which is the tangent to the graph at P_0. Obviously, this is a better approximation because not only are the functions $f(x)$ and $L(x)$ equal at a (each having value $f(a)$), but even their derivatives are equal.

The next in line is the second order or a quadratic approximation to $f(x)$ near the point a. We would like this to be a quadratic expression in x, say $Q(x) = Ax^2 + Bx + C$, where A, B, C are some constants. We would like not only $Q(x)$ to agree with $f(x)$ at $x = a$, but also to have the same first and second derivatives there as $f(x)$. In other words we want $Q(a) = f(a), Q'(a) = f'(a)$ and also $Q''(a) = f''(a)$. These three conditions determine A, B, C uniquely and then it is easy to show (see Exercise (16.24)) that $Q(x)$ must equal $f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2$. A necessary and sufficient condition for the quadratic approximation to exist is that $f''(a)$ should exist (i.e., that f be twice differentiable at a).

Naturally, we expect the quadratic approximation to be even better than the linear approximation, which itself is better than the zero degree or

constant approximation. (In terms of the analogy of the moving car given above, the second order approximation is arrived at after knowing not only the position and the speed of the car at the moment it was spotted but its acceleration as well at that moment.) To see this by a numerical example, let us go back to finding an approximate value of $\sqrt{2}$ by taking $f(x)$ as $\sqrt{x}$, a as 1 and h also as 1. The zeroth order approximation gives $\sqrt{2} \approx 1$ as we already saw. Since $f'(a) = \frac{1}{2\sqrt{1}} = \frac{1}{2} = 0.5$ the first order approximation gives $\sqrt{2} \approx 1 + (0.5) \times 1 = 1.5$ which is better than the zeroth order approximation. Further since $f''(x) = -\frac{1}{4x\sqrt{x}}$, we have $f''(a) = -\frac{1}{4} = -0.25$. So the second order approximation gives $\sqrt{2} \approx 1 + 0.5 - \frac{1}{8} \times 1^2 = 1.375$ which is still better, the exact value of $\sqrt{2}$ being 1.414213....

More generally, for any positive integer n, if the function $f(x)$ has derivatives of order upto n at a, then we have the following n-th order approximation to $f(a+h)$.

$$\begin{aligned} f(a+h) &\approx \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(a) h^k \\ &= f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \ldots + \frac{f^{(n)}(a)}{n!}h^n \end{aligned} \tag{22}$$

We now need an estimation of the error in this approximation. For $n = 0$ this task was done by (21) which is essentially, a reformulation of Lagrange's MVT. For a general n there is an analogous theorem called **Taylor's theorem** which does the job. We state it without proof. A proof can be given by repeated applications of Rolle's theorem and will be indicated in the exercises (Exercises (16.25) and (16.26)).

Theorem 7: Suppose n is a positive integer and $f : [a, b] \longrightarrow \mathbb{R}$ is a function for which the $(n-1)^{\text{th}}$ derivative $f^{(n-1)}(x)$ exists and is continuous at every point of $[a, b]$, while the nth derivative $f^{(n)}(x)$ exists for all x in (a, b). Then there exists $c \in (a, b)$ (depending on a, b, f and n) such that

$$\begin{aligned} f(b) &= f(a) + f'(a)(b-a) + \tfrac{1}{2}f''(a)(b-a)^2 + \cdots \\ &\quad + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \frac{f^{(n)}(c)}{n!}(b-a)^n \end{aligned} \tag{23}$$

If we write $b = a + h$ and omit the last term on the R.H.S. then we get precisely the $(n-1)$-th order approximation to $f(a+h)$ at a. The last term, rewritten as $\dfrac{f^{(n)}(c)}{n!}h^n$ is the error term. (This term is also popularly called the **remainder**.) If M is any upper bound on $|f(n)(x)|$ for $x \in (a, b)$, then the error is at most $\frac{M}{n!}|h|^n$. As pointed out in Exercise (6.52), $n!$ is a very large number even for a relatively small n. So the Taylor approximation is a very good approximation and is used heavily in practical problems. As a good illustration, let us take $f(x) = \sin x$, $a = 0$ and $h = \theta$. The successive derivatives of the sine function recur with a cycle of 4 and so their values at 0 (starting from the

zero-th derivative) are $0, 1, 0, -1, 0, 1, 0, -1, 0, 1, \ldots$. Hence for an odd n, say for $n = 2r + 1$, (23) becomes

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \ldots + (-1)^{r-1}\frac{\theta^{2r-1}}{(2r-1)!} + \frac{(-1)^r \cos c}{(2r+1)!}\theta^{2r+1} \tag{24}$$

for some $c \in (0, \theta)$. Now no matter what c is, $|\cos c| \leq 1$ and so the error is at most $\dfrac{|\theta|^{2r+1}}{(2r+1)!}$ in magnitude. Let us use this to find an approximate value of $\sin 34°$. We could, of course, draw a right angled triangle with one angle equal to 34 degrees. But to measure its sides we would need highly sophisticated instruments to get even a modest accuracy (say upto three places of decimals). But with (24) the job is easy. Since $34° = \dfrac{34}{180} \times \pi$ radians $\simeq 0.5934$ radians, we apply (24) with $r = 3$ and $\theta = 0.5934$. Since $\theta^7 < (0.6)^7 < 0.0285$ and $\dfrac{1}{7!} = \dfrac{1}{840} < .0012$, we see from (24) (and the estimate for the remainder in the Taylor's theorem) that $\theta - \dfrac{\theta^3}{6} + \dfrac{\theta^5}{120}$ will give an accuracy of upto four places of decimals. With a hand calculator, for $\theta = 0.5934$, this value comes out to be 0.55918811542... The value of $\sin 34°$ obtained from the tables is 0.559192.

We can similarly get Taylor approximations of $\cos\theta$ and of e^x at the point 0. We can of course, take the approximations with any real number a. But for $a = 0$, the successive derivatives of these functions are particularly easy to calculate at 0 and so it is convenient to take the approximations at 0. By an easy calculation, we get

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \ldots + \frac{(-1)^r\theta^{2r}}{(2r)!} + \frac{(-1)^{r+1}\sin\alpha}{(2r+2)!}\theta^{2r+2} \tag{25}$$

$$\text{and} \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^{n-1}}{(n-1)!} + \frac{e^\beta}{n!}x^n \tag{26}$$

for some $\alpha \in (0, \theta)$ and $\beta \in (0, x)$.

In (24) and (25), it can be shown that for a fixed θ the remainder tends to 0 as n tends to ∞. Similarly, in (26) for a fixed x, e^β is at most $e^{|x|}$ and hence the remainder tends to 0, too. This leads to the power series for the trigonometric and the exponential functions. They were very popular at one time and old books on trigonometry use them frequently. But a lot of care is necessary in doing so. For example, in Comment No. 7 of the last chapter, we showed how it is tempting to evaluate a limit like $\lim\limits_{x\to 0} \dfrac{x - \sin x}{x^2}$ using the power series for $\sin x$. Similarly, from the power series of $\sin\theta$ and of $\cos\theta$, it is tempting to think that the relations $\frac{d}{d\theta}\sin\theta = \cos\theta$ and $\frac{d}{d\theta}\cos\theta = -\sin\theta$ follow if we differentiate the power series term-by-term. This indeed, can be justified, but only after a lot of work, which is beyond our scope.

However, as remarked in Comment No. 7 of the last chapter, many times we do not need the entire power series. We can often get by with the

Taylor approximation. This can be done quite rigorously assuming Theorem 6 (whose proof is not significantly more complicated than that of the Lagrange's MVT). Indeed, if we apply (24) with $n = 3$ (i.e., with $r = 1$), then we have $\sin x = x - \dfrac{\cos c_x}{6}x^3$ for some $c_x \in (0, x)$. The number c_x may change as x changes. But no matter what it is, the absolute value of the remainder is at most $\dfrac{|x|^3}{6}$. So we have

$$-\frac{|x|}{6} \leq \frac{x - \sin x}{x^2} \leq \frac{|x|}{6} \tag{27}$$

By the Sandwich theorem, the middle term tends to 0 as $x \to 0$. By a similar reasoning, expanding $\sin x$ to one more term (i.e., using (24) with $n = 5$ instead of $n = 3$) and using essentially the same reasoning, we get $\lim\limits_{x\to 0} \dfrac{x - \sin x}{x^3} = \dfrac{1}{6}$. Earlier, we had to use the l'Hôpital's rule to evaluate such limits.

Similarly, at the end of Comment No. 6 of the last chapter, we did the problem of finding the integer n for which x^n is comparable to $(\cos x - 1)(\cos x - e^x)$ in the sense that $\dfrac{(\cos x - 1)(\cos x - e^x)}{x^n}$ tends to a finite non-zero limit as $x \to 0$. (JEE 2002) We can do this easily using power series. For, the numerator is the product $(-\dfrac{x^2}{2!} + \dfrac{x^4}{4!} + \ldots)(-x - x^2 - \dfrac{x^3}{3!} + \ldots)$. When expanded, the significant term will be x^3 and to match it, n will have to be 3. Here we are assuming that two power series can be multiplied the way two polynomials are multiplied, i.e., by multiplying the various terms of the first factor by those of the second and then grouping together the 'like' terms. Once again, this needs a justification which is beyond our scope. So, for those of us who are shy of using such 'shoddy' methods, we can avoid them by using, instead, the Taylor expansions of $\cos x$ and e^x to write the two factors of the numerator, respectively, as $(-\dfrac{x^2}{2!} + a_x x^3)$ and $(-x + b_x x^2)$, where a_x, b_x remain bounded as $x \to 0$. Now we can expand the numerator 'cleanly' as $-\dfrac{1}{2}x^3 - (a_x + \dfrac{b_x}{2})x^4 + a_x b_x x^5$ and see that its ratio with x^3 will tend to a finite non-zero limit as $x \to 0$.

Comment No. 14:

Taylor's theorem also enables us to settle some delicate inequalities. At the end of Comment No. 9 of Chapter 6, we mentioned that for small positive values of x, we cannot tell by elementary methods which of $\sin x + \tan x$ and $2x$ is bigger, because while x is bigger than $\sin x$, it is less than $\tan x$. In Comment No. 11 of Chapter 13 we settled this question by proving a stronger inequality, viz.,

$$2\sin x + \tan x \geq 3x \quad \text{for } 0 \leq x < \frac{\pi}{2} \tag{28}$$

We proved this by comparing the derivatives of both the sides. But the proof was rather tricky. A more systematic way to prove the inequality, at least for

small values of x, is by taking the Taylor expansions of both the sides at 0. For the R.H.S., it is simply $3x$. Call the L.H.S. as $f(x)$. Then $f'(x) = 2\cos x + \sec^2 x$ and so $f'(0) = 3$. Hence $f(x) \approx 3x$. But this is no good because the error could be either positive or negative and depending on that $f(x)$ could be either smaller or bigger than $3x$. So we try higher order approximations. We leave it to you to compute the successive derivatives of f at 0. You will find that the derivatives of order 2,3 and 4 all vanish at 0. So even the fourth order approximation will not work. But the fifth derivative $f^{(v)}(0)$ comes out to be 18. So by Theorem 6, $f(x) \approx 3x + \frac{18}{120}x^5$ with an error term which is numerically at most of order x^6. Now we can say that for sufficiently small positive values of x, the L.H.S. of (28) decidedly beats the R.H.S. We leave it to you (Exercise (16.27)) to check, along similar lines, which of $3\sin x + \tan x$ and $4x$ is bigger for small positive values of x.

Comment No. 15:

The essence of the method above was to go on taking successive derivatives of a given function at a point, say a, till you encounter a non-zero derivative, because then you get a significant term in the Taylor expansion compared to which the error is negligible in magnitude. The sign of the expression is then decided by the sign of this derivative. (In the problem above, in effect we applied this reasoning to the function $2\sin x + \tan x - 3x$ which vanished at 0 along with its first four derivatives.)

An analogous strategy can be tried to determine if a function $f(x)$ has a local maximum/minimum at an interior point, say c, in its domain. If $f'(c) \neq 0$ then there is neither a local maximum nor a local minimum at c as we already saw in Chapter 13. This is also evident from the Taylor expansion near c, which gives $f(x) - f(c) \approx (x - c)f'(c)$ with an error which is insignificant. Here the sign of $f(x) - f(c)$ depends upon whether $x - c$ is positive or negative. But if $f'(c) = 0$, then the error becomes significant and we have to go for the second order approximation, viz., $f(x) - f(c) \approx \frac{1}{2}f''(c)(x - c)^2$ plus an error term. If $f''(c) \neq 0$, then the error is insignificant and the sign of $f(x) - f(c)$ is the same no matter on which side of c x lies, as long as it is sufficiently close to c. But if $f''(c) = 0$, then the error becomes significant and we have to look at $f'''(c)$. Continuing in this line we get the following theorem.

Theorem 8: Suppose $f(x)$ has derivatives of all orders in a neighbourhood of a point c and n is the smallest positive integer such that $f^{(n)}(c) \neq 0$. Then

(i) if n is odd, f has neither a local minimum nor a local maximum at c

(ii) if n is even, then f has a local minimum or a local maximum at c according as $f^{(n)}(c) > 0$ or $f^{(n)}(c) < 0$.

We leave you to prove and illustrate this by simple examples such as $f(x) = x^n$. But the essential idea can be illustrated by a real-life analogy. The successive derivatives at a point are like a descending hierarchy of bosses in an office. When $f'(c) \neq 0$, it is like the top boss being present. Then he rules and

the other bosses are insignificant. But when he is absent, the next in line takes charge and dictates the terms! And this goes on.

EXERCISES

16.1 Suppose S is a bounded solid. Prove that there is some horizontal plane which divides S into two solids of equal volume. [*Hint* : Use IVP.]

16.2 Let C be the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and d be any number with $0 < d < 2b$. Let L be a straight line. Prove that there exists a chord of length d of C which is parallel to L. (Do not actually find it. Simply prove its existence.)

16.3 Suppose $f : [a, b] \longrightarrow [a, b]$ is a continuous function. Prove that there is some point $x_0 \in [a, b]$ such that $f(x_0) = x_0$. What does this say about the graph of $y = f(x)$? [*Hint* : Apply the IVP to the function $g(x) = f(x) - x$. A point such as x_0 here is called a **fixed point** of f and the result is expressed by saying that the interval $[a, b]$ has the fixed point property. Do $\mathbb{R}$ and $(0, 1)$ have fixed point property?]

16.4 Suppose $f : [0, 1] \longrightarrow \mathbb{R}$ is continuous and $f(0) = f(1)$. Prove that there exists $c \in [0, \frac{1}{2}]$ such that $f(c) = f(c + \frac{1}{2})$.

*16.5 Given a point P on an ellipse C, prove that there exist points Q and R on C such that the triangle PQR is similar to a given triangle.

16.6 Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be defined by $f(x) = 2x + \sin x$. Determine if f is one-to-one and also if it is onto. (2002)

16.7 Let $f(x) = |x|$ and $[a, b] = [-1, 1]$. Prove that Lagrange's theorem does not hold, i.e., there is no $c \in (-1, 1)$ such that $f'(c) = \frac{f(1) - f(-1)}{2}$. What goes wrong?

16.8 Prove that Lagrange's theorem also fails for $[0, 1]$ if f is defined by $f(x) = 0$ for $x \neq 1$ and $f(1) = 1$.

16.9 Let $f(x) = x \sin \frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. (See Comment No. 13 of Chapter 15.) Prove that in the interval $[0, \frac{1}{\pi}]$, there are infinitely many values of x, where f' vanishes. All these are 'mean values' in the sense of the MVT.

*16.10 Suppose f is continuous on $[a, b]$ and differentiable on (a, b). Assume further that $f(b) - f(a) = b - a$. Prove that for every positive integer n, there exist distinct points $c_1, c_2, \ldots, c_n$ in (a, b) such that $f'(c_1) + f'(c_2) + \cdots + f'(c_n) = n$.

16.11 Suppose f is continuous on $[a, b]$ and differentiable on (a, b). Assume that $\lim_{x \to a^+} f'(x)$ exists and equals L. Prove that the right derivative of f at a,

i.e., $f'_+(a)$ also exists and equals L. [*Hint*: For every $x \in (a,b)$, there exists some $c_x \in (a,x)$ such that $\frac{f(x)-f(a)}{x-a} = f'(c_x)$. As $x \to a^+, c_x \to a^+$.]

16.12 Prove that a derived function, i.e., a function which is the derivative of some other function cannot have a simple discontinuity. (See the footnote in Chapter 15, Comment No. 13.)

16.13 Without solving the equation, show that between $x = -2$ and $x = -1$, there lies one and only one real root of $x^4 + 3x + 1 = 0$. (JAT 1979)

16.14 Which, if any, of the following statements can be concluded from the MVT for any x ? (JAT 1980)

(A) $|\tan^{-1} x| \geq |x|$ (B) $|\tan^{-1} x| \leq |x|$ (C) $|\tan^{-1} x| \geq 2|x|$.

16.15 Suppose $f(x), g(x)$ are differentiable, $f'(x) = g(x)$ and $g'(x) = -f(x)$ for all x. Let $h(x) = (f(x))^2 + (g(x))^2$. If $h(5) = 11$, find $h(10)$. (1982)

16.16 For all x in $[0,1]$, let the second derivative $f''(x)$ of a function $f(x)$ exist and satisfy $|f''(x)| \leq 1$. If $f(0) = f(1)$, then show that $|f'(x)| < 1$ for all x in $[0,1]$. (1981)

16.17 (a) Show that the hypothesis $f(a) = f(b) = 0$ in Rolle's theorem can be replaced by the weaker one $f(a) = f(b)$.

(b) Give a direct proof of Cauchy's MVT by considering the function $\psi(x) = f(x) - \frac{f(b)-f(a)}{g(b)-g(a)} g(x)$. (1982)

(c) Can you get Cauchy's MVT by applying Lagrange's MVT separately to the numerator and the denominator and then taking the ratio?

(d) Show that $(343.01)^{1/3} - 7 < 7 - (342.99)^{1/3}$ by applying MVT to the function $f(x) = x^{1/3}$. [*Hint*: Note that $f'(x)$ is strictly decreasing. An algebraic proof by cubing both the sides is messy.]

16.18 Suppose $p(x)$ is a polynomial with real coefficients and $a < b$. Assume further that $p(a)$ and $p(b)$ are both non-zero. Prove that the interval $[a,b]$ contains an even (possibly none) or an odd number of roots of $p(x)$ depending upon whether $p(a)p(b)$ is positive or negative. (Here multiple roots are to be counted according to their multiplicities.)

16.19 Prove that if an interval (a,b) contains k roots of a real polynomial then it contains at least $k-1$ roots of $p'(x)$, all roots being counted according to their multiplicities. [*Hint:* First observe that if α is a root of multiplicity $r > 1$ of $p(x)$, then it is a root of multiplicity $r-1$ of $p'(x)$.]

16.20 (a) Prove, both with and without calculus, that if $(n-1)a_1^2 - 2na_2 < 0$ then the roots of the polynomial $p(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_{n-1}x + a_n$ cannot be all real.

(b) Let $p_m(x) = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^m}{m!}$. By induction on n prove that $p_{2n-1}(x)$ has exactly one real root while $p_{2n}(x) > 0$ for all real x.

16.21 Let α be a repeated root of a quadratic equation $f(x) = 0$ and $A(x), B(x)$ and $C(x)$ be polynomials of degrees 3, 4 and 5 respectively. Show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by $f(x)$, where the dash denotes the derivative. (1984)

16.22 What error estimate do you get by Lagrange's MVT if you approximate $\sqrt{2}$ as 1.4 ?

16.23 Suppose that $f(x)$ is continuous at a point a. Prove that it is differentiable at a if and only if there exists some linear function $L(x)$ (i.e., a function of the form $L(x) = Ax+B$ for some constants A and B) such that $\frac{|f(x)-L(x)|}{|x-a|}$ tends to 0 as x tends to a. (Verbally, a function is differentiable at a point, if and only if it is continuous at that point and can be approximated by a linear function near that point.) What happens if f is not given to be continuous at a?

16.24 Verify that if $Q(x)$ is a quadratic polynomial in x such that $Q(a) = \alpha, Q'(a) = \beta$ and $Q''(a) = \gamma$, where α, β and γ are some given real numbers, then $Q(x)$ must be $\alpha + \beta(x-a) + \frac{1}{2}\gamma(x-a)^2$.

16.25 (a) Suppose n is a positive integer, g is a function whose $(n-1)^{\text{th}}$ derivative, $g^{(n-1)}$, is continuous on $[a, b]$ and differentiable on (a, b). Assume further that $g(b) = 0, g(a) = 0, g'(a) = 0, \ldots, g^{(n-1)}(a) = 0$. Prove that there is some $c \in (a, b)$ such that $g^{(n)}(c) = 0$.

(b) Suppose g'' is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$. Assume that g vanishes at ± 1 and also at some $c \in (-1, 1)$ and $g'(0) = g''(0) = 0$. Prove that g''' has at least one zero in $(-1, 1)$.

16.26 Prove Taylor's theorem. [*Hint* : Let $g(x) = f(x) - f(a) - f'(x)(x-a) - \cdots - \frac{f^{(n-1)}(a)}{(n-1)!}(x-a)^n - K(x-a)^n$ where K is such that $g(b) = 0$. Apply (a) of the last exercise to g.]

16.27 Decide which of $3\sin x + \tan x$ and $4x$ is bigger for small positive values of x. Deduce that although the point x lies in the interval $[\sin x, \tan x]$ to the left of the point which divides it in the ratio $1 : 2$, it lies to the right of the point which divides this interval in the ratio $1 : 3$. Similarly decide which of $|\tan^{-1} x|$ and $|x|$ is bigger for any x. (JAT 1980)

16.28 Prove Theorem 8. Using it, determine whether at $x = 0$, $f(x) = x^5 - 5x^4 + 5x^3$ is maximum/minimum. Is it is increasing/decreasing at $x = 0$? (JAT 1980)

16.29 Prove that in the Main Problem, the function f will be constant if, instead of assuming that $|f(x) - f(y)| \leq |x-y|^3$, we assume that $|f(x) - f(y)| \leq |x-y|^2$ for all x, y (which is a weaker hypothesis if $|x-y| < 1$). Will it still hold if we only assume $|f(x) - f(y)| \leq |x-y|$ for all x, y ?

Chapter 17

AREAS AND ANTIDERIVATIVES

Traditionally, calculus is divided into two halves, the differential calculus (where we primarily study derivatives and their applications such as maxima and minima) and the integral calculus, where we study integration. Analogy with the real-life usage of the term is not misleading. In national integration, the various individuals or communities contribute what they have, so as to build a nation with a collective identity. An integral is obtained by putting together the contributions (measured as the values of some real valued function) of individual points of the domain.

In this chapter we study integrals and their applications to finding the areas of plane figures and also to finding the limits of certain sums. The Fundamental Theorem of Calculus establishes a link between derivatives and integrals. This link is used so heavily that it gives the convenient but conceptually wrong impression that integration is just the reverse of differentiation. Because of this linkage, the problem of finding an antiderivative of a given function becomes very important. There is no golden method for doing so. Some standard 'techniques of integration' are covered here with illustrations. But we do not give elaborate tables of formulas as these are generally available in standard texts on calculus. At any rate, finding the antiderivative is more of an art and an individual's skill and experience matter more than dozens of formulas crammed in the head. Evaluation of definite integrals will be taken up in the next chapter.

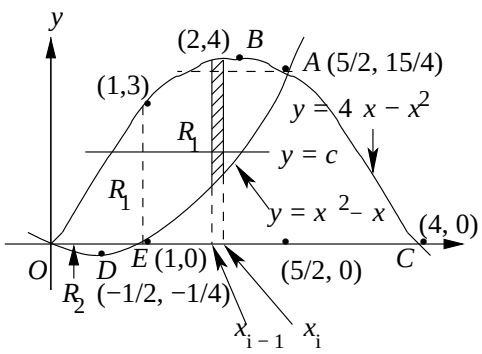
Problems on integration are asked in the JEE every year. Those where an area is to be evaluated are usually straightforward once you sketch the region correctly. Similarly, those where you have to apply integrals to evaluate the limits of certain sums are also easy once you cast the sum in a suitable form. Problems asking for antiderivatives require practice and skill. Those based on the second form of the fundamental theorem demand conceptual understanding.

Main Problem : The ratio in which the x-axis divides the area bounded by the parabolas $y = 4x - x^2$ and $y = x^2 - x$, when expressed as a reduced rational, is ..... . (JEE 1994)

First Hint : Identify the points of the intersection of the two parabolas and sketch the region.

Second Hint : Compute the areas above and below the x-axis by integration.

Solution : To find the points of intersection of the parabolas, we have to solve their equations simultaneously. Adding the two equations gives $y = \frac{3}{2}x$. Putting this into the second equation gives the quadratic $x^2 - \frac{5}{2}x = 0$ with roots $x = 0$ and $x = \frac{5}{2}$. The two corresponding points of intersection are $(0, 0)$ and $(\frac{5}{2}, \frac{15}{4})$. Completing squares, the first parabola is downward with $(2, 4)$ as its top point. It cuts the x-axis at $(0, 0)$ and at $(4, 0)$. Similarly, the second parabola is an upward parabola with $(\frac{1}{2}, -\frac{1}{4})$ as its bottom point and cuts the x-axis at $(0, 0)$ and $(1, 0)$. The region, say R, bounded by the two parabolas is as shown in the sketch, where the points $(\frac{5}{2}, \frac{15}{4})$, $(2, 4)$, $(4, 0)$, $(\frac{1}{2}, -\frac{1}{4})$ and $(1, 0)$ are denoted by A, B, C, D and E respectively.



Let R_1 and R_2 be the portions of R above and below the x-axis, respectively and let A_1 and A_2 denote their areas. A_2 is easy to calculate because R_2 is bounded above by the x-axis and below by the curve $y = x^2 - x$ throughout the interval $0 \le x \le 1$. So, by direct integration,

$$A_2 = \text{ area of } R_2 = \int_0^1 x - x^2 dx = \frac{x^2}{2} - \frac{x^3}{3}\Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \qquad (1)$$

However, to compute A_1 by this method, we have to be careful. Although R_1 is bounded above by $y = 4x - x^2$ for all $x \in [0, \frac{5}{2}]$, the lower boundary is not the same for all x. For $0 \le x \le 1$, it is the x-axis while for $1 \le x \le \frac{5}{2}$, the lower boundary is the parabola $y = x^2 - x$. So A_1 is the sum of the two integrals

$$A_1 = \text{ area of } R_1 \quad = \quad \int_0^1 (4x - x^2)dx + \int_1^{\frac{5}{2}} (4x - x^2 - x^2 + x)dx$$

$$\begin{aligned} &= \int_0^1 (4x - x^2)dx + \int_1^{\frac{5}{2}} (5x - 2x^2)dx \\ &= (2x^2 - \frac{x^3}{3})\Big|_0^1 + (\frac{5x^2}{2} - \frac{2x^3}{3})\Big|_1^{\frac{5}{2}} \\ &= 2 - \frac{1}{3} + \frac{125}{8} - \frac{125}{12} - \frac{5}{2} + \frac{2}{3} \\ &= \frac{121}{24} \end{aligned} \tag{2}$$

From (1) and (2), the ratio of the two areas is $\frac{121/24}{1/6} = \frac{121}{4}$. (The problem does not specify the order in which the two areas are to be taken while forming their ratio. So $\frac{4}{121}$ is also a valid answer.)

Comment No. 1:

Although not specifically asked in the question, it is always a good idea to sketch the regions whose area is to be found. As with the diagrams involved in problems about heights and distances, the sketch need not be very accurate (i.e., to scale) or artistic. But it must reveal the vital features, for example, exactly which curves form the various portions of the boundaries of the regions involved, the points of intersection and especially those points where some crucial change takes place, e.g., where the equation of the boundary curve changes (e.g., the point $(1, 0)$ in the Main Problem where the lower boundary of the region R_1 changes its equation).

Such sketches not only help you to correctly set up the integrals to express the various areas, but sometimes they enable you to take short cuts. For example, in the Main Problem, we found the area of R_1 by splitting it into two parts with the vertical line $x = 1$. But there is an easier way out. Just as in complementary counting (Chapter 1, Comment No. 1) we find the cardinality of a set by subtracting the cardinality of its complement from that of the ambient set, the area of R_1 can also be obtained by finding the area of R and subtracting the area of R_2 from it. (It can be argued that the regions R_1 and R_2 here are not complementary, because they are not disjoint, since the portion of the x-axis between the points O and E is common to them. But this common portion is a thin line segment which has zero area. So, as far as areas are concerned, we may take R_1 and R_2 as disjoint. This convention will be followed throughout. Whenever two regions overlap only on a curve, they will be treated as non-overlapping.) This approach is advantageous because the area of R_2 is already calculated, while that of R is easier to find than R_1 because unlike R_1, the lower boundary of R is given by $y = x^2 - x$ throughout $0 \le x \le \frac{5}{2}$. So we get,

$$\text{area of } R = \int_0^{\frac{5}{2}} (5x - 2x^2)dx = (\frac{5}{4}x^2 - \frac{2}{3}x^3)\Big|_0^{\frac{5}{2}} = \frac{125}{8} - \frac{125}{12} = \frac{125}{24} \tag{3}$$

and hence

$$A_1 = \text{ area of } R - A_2 = \frac{125}{24} - \frac{1}{6} = \frac{121}{24} \tag{4}$$

which is the same as (2) but is easier to obtain. It saves you both time and the risk of numerical slips.

Comment No. 2:

It is instructive to understand the reasoning that underlies the method of calculating areas as integrals. The method given in the solution above was based on what is called **vertical slicing**, in which we slice the region R_1 into a large number of thin strips by vertical lines, i.e., lines of the form $x = x_i$, where $x_0, x_1, x_2, \ldots, x_n$ (say) are some real numbers lying in the interval $[0, \frac{5}{2}]$ with $x_0 = 0$, $x_n = \frac{5}{2}$ and $x_0 < x_1 < x_2 < \ldots < x_{i-1} < x_i < \ldots < x_{n-1} < x_n$. (In technical terms, we call any such collection $(x_0, x_1, \ldots, x_{n-1}, x_n)$, a **partition** of the interval $[0, \frac{5}{2}]$ and the point x_i the i-th **node** of the partition. Let $f(x_i)$ be the length of the portion of the line $y = x_i$ intercepted by the region R_1. From the figure above in the solution to the Main Problem, we have

$$f(x_i) = \begin{cases} 4x_i - x_i^2, & \text{if } 0 \le x_i \le 1 \\ 5x_i - 2x_i^2, & \text{if } 1 \le x_i \le \frac{5}{2} \end{cases} \tag{5}$$

For $i = 1, 2, \ldots, n$, denote $x_i - x_{i-1}$ by Δx_i. We emphasise that this is just a convenient notation and denotes a single real number and not the product of the two real numbers Δ and x_i. Now, the i-th slice of R_1 is precisely the portion of R_1 between the adjacent vertical lines $x = x_{i-1}$ and $x = x_i$. If Δx_i is small, this portion can be taken approximately as a rectangle with width Δx_i and height $f(x_i)$. (See the shaded strip in the figure above. Of course, we might as well take $f(x_{i-1})$ instead of $f(x_i)$ as the height of this rectangle. It makes little difference because neither answer is exact, since the strip is not a perfect rectangle anyway.)

The exact area of the region R_1 is the sum of the areas of these strips. So its approximate area is the sum of their approximate areas. In other words,

$$A_1 = \text{ area of } R_1 \approx \sum_{i=1}^{n} f(x_i)\Delta x_i \tag{6}$$

Moreover, we expect that the smaller the widths Δx_i, the less will be the error of this approximation. So the exact answer would come if we take the limit of the sum in the R.H.S. of (6) as each Δx_i tends to 0. Note that as each Δx_i tends to 0, the number of strips, i.e., the integer n has to tend to infinity. So the limit here is a peculiar type of limit in which the number of summands tends to ∞ but individually each summand tends to 0. Such a limit is therefore analogous to the indeterminate form $\infty \times 0$. It is called the **(definite) integral** of the function $f(x)$ (defined by (5) above) over the interval $[0, \frac{5}{2}]$ and denoted by $\int_0^{\frac{5}{2}} f(x)dx$. The function $f(x)$ is called the **integrand**, a term coined in an obvious analogy with the term 'summand' and giving a clear hint that integration is a refined form or a limiting case of summation. A sum such as that in the R.H.S. of (6) is called a **Riemann sum** of the function $f(x)$ for the partition $(x_0, x_1, \ldots, x_i, \ldots x_n)$

in honour of the mathematician who studied them first. Definite integrals are therefore also called **Riemann integrals**. Note that it is not required that the nodes be equally spaced, or in other words, that all Δx_i's be equal. The largest of the Δx_i's is called the **mesh** or the **norm** of the partition. Clearly to say that each Δx_i tends to 0 is equivalent to saying that the mesh of the partition tends to 0. So we may say that a Riemann integral is the limit of a Riemann sum as the mesh of the partition of the underlying interval tends to 0.

By a similar (and in fact, somewhat simpler) reasoning, if we slice the region R_2 into vertical strips, then its area A_2 is given by the definite integral $\int_0^1 x - x^2 dx$ as in (1) above.

The notations for a definite integral also owe their origins to summation. The integral of a function $f(x)$ over an interval $[a, b]$ is denoted by $\int_a^b f(x)dx$. The integral sign $\int$ is an elongated 'S' which was formerly used to denote summation (instead of the Σ which is more common now). Also the part $f(x)dx$ corresponds to $f(x_i)\Delta x_i$ which is a typical term in the Riemann sum. However, even though the latter is a product of two real numbers, viz., $f(x_i)$ and Δx_i, $f(x)dx$ is a single symbol. It is not the product of $f(x)$ and dx. In case this sounds paradoxical, recall from Comment No. 1 of Chapter 15 that even though a derivative $\frac{dy}{dx}$ is the limit of a ratio, (specifically, of the ratio $\frac{\Delta y}{\Delta x}$), the derivative itself is not a ratio! However, in informal terms, it is customary to say, for example, that $f(x)dx$ is the area of a strip of height $f(x)$ and 'elemental' width dx. Just as treating a derivative as a ratio of two differentials dy and dx gains convenience and causes little harm if done with care, with integrals also, this informal splitting of $f(x)dx$ is often convenient. In fact this is even reflected in the notation. Thus, for example, when $f(x)$ happens to be of the form $\frac{1}{g(x)}$, then instead of writing $\int_a^b \frac{1}{g(x)}dx$, we write $\int_a^b \frac{dx}{g(x)}$. Similarly we often write $\int_a^b \frac{f(x)\,dx}{g(x)}$ instead of $\int_a^b \frac{f(x)}{g(x)}dx$.

Another point of analogy between summation and integration is that in a sum of the form $\sum_{i=1}^{n} a_i$, the variable i is only an index variable and replacing it with any other variable does not alter the sum. Thus, the sum above can as well be denoted by $\sum_{j=1}^{n} a_j$ or by $\sum_{r=1}^{n} a_r$ etc. Similarly, in the case of an integral $\int_a^b f(x)dx$, the variable x is called the **variable of integration**. The integral depends solely on the function f and the 'upper' and the 'lower' limits b and a respectively. As long as they remain unchanged, we may denote the same integral by $\int_a^b f(y)dy$ or by $\int_a^b f(u)du$ etc. In fact, some authors prefer

to dispense with the variable of integration altogether and write the integral simply as $\int_a^b f$. But we will stick to the more traditional notation.

In defining $\int_a^b f(x)dx$, it is tacitly assumed that $b > a$ as otherwise the interval $[a, b]$ would make no sense. But sometimes we have consider integrals of the form $\int_a^b f(x)dx$ when $b \leq a$. If $a = b$ then the interval $[a, b]$ degenerates to a single point, a (which is the same as b). In this case, there is only one partition of the interval $[a, b]$. There is only one Riemann sum and it is 0. So in this case $\int_a^b f(x)dx = 0$ no matter what the function $f(x)$ is. When $b < a$, the interval $[a, b]$ (which, by definition, is the set $\{x \in I\!R : a \leq x \leq b\}$) becomes the empty set. However, in this case the interval $[b, a]$ makes sense and we *define* $\int_a^b f(x)dx$ to be $-\int_b^a f(x)dx$.

As with many other types of limits, a definite integral is very rarely evaluated by actually computing all its Riemann sums. The most common method of evaluating definite integrals is based on what is called the **fundamental theorem of calculus**. There are two versions of it. The **first form**, which is useful in evaluating definite integrals is as follows. A proof can be given using Lagrange's Mean Value Theorem. But it is far too lengthy and we omit it. An intuitive justification will be given in Comment No. 9 below.

Theorem 1: Suppose $f(x)$ is continuous on an interval $[a, b]$ and there is a function $F(x)$ such that $F'(x) = f(x)$ for all $x \in [a, b]$. Then $\int_a^b f(x)dx$ exists and equals $F(b) - F(a)$. Put succinctly,

$$\int_a^b F'(x)dx = F(b) - F(a) \tag{7}$$

A function $F(x)$ with the property that $F'(x) = f(x)$ is called an **antiderivative** of $f(x)$. In applying this theorem, after you find an antiderivative, say $F(x)$, of the integrand, it is customary to write $F(x)\Big|_a^b$ (or, more precisely, $F(x)\Big|_{x=a}^{x=b}$) instead of $F(b)-F(a)$. We already applied this theorem in evaluating the integrals in (1) and (2) above. Note that in the case of (2), we had to split the region R_1 into two parts which resulted in a corresponding splitting of the integral $\int_0^{\frac{5}{2}} f(x)dx$ (where $f(x)$ is as in (5)) into the two integrals $\int_0^1 f(x)dx$ and $\int_1^{\frac{5}{2}} f(x)dx$. This happened because the formula for $f(x)$ was different in the subinterval $[1, \frac{5}{2}]$ than in the subinterval $[0, 1]$. Consequently, $f(x)$ had very different antiderivatives in the two subintervals. In effect, what we have used here is the following property of integrals, called the **additivity property**. It

says that for any function[1] $f(x)$ and real numbers $a < c < b$, we have

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

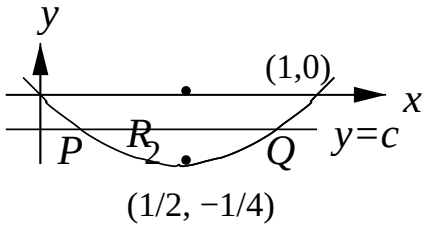
.

Note that because of our convention that $\int_a^b f(x)dx$ means $\int_b^a f(x)dx$ in case $b < a$, (7) holds even when $a < b$. It also holds if $a = b$ since both the sides are 0 in this case. Similarly, the additivity property above holds for any a, b, c no matter in which order they lie. (Of course, we assume that they all lie in an interval in which $f(x)$ is defined. But the order need not necessarily be $a < c < b$.)

Comment No. 3:

Returning to the Main problem, we now show how the areas of R_2 and R_1 can be found by a different type of slicing. So far we found both the areas by the method of vertical slicing, i.e., by slicing with lines parallel to the y-axis. But in the plane, the y-axis is like any other line. It is only when a particular frame of coordinates is fixed that the y-axis acquires any special importance. But the area of a region is a geometric attribute of it. The equations of its boundary curves may change as we change the frame of reference. But its area must remain the same, much the same way as the duration of the life span of a person must remain the same even though the years of his birth and death may vary depending upon whether we follow the Christian calendar or some other calendar. So, for finding the area of a region, we are at liberty to slice it by lines parallel to any direction of our choice. In particular, we can slice the region horizontally, i.e., by lines parallel to the x-axis.

Let us first do this for the region R_2. From the figure it is clear that R_2 lies entirely between the two horizontal lines $y = -\frac{1}{4}$ and $y = 0$. In other words, for any point (x, y) in R_2, we have $-\frac{1}{4} \le y \le 0$. So we slice R_2 by lines of the form $y = c$, where $-\frac{1}{4} \le c \le 0$. Denote by $g(c)$ the length of the portion of the line $y = c$ intercepted by the region R_2. As shown in the figure, this intercept is the line segment PQ, where P and Q are the points of intersection of the line $y = c$ with the parabola $y = x^2 - x$.



[1] When we say 'any' function, we are tacitly assuming here that the function is such that the limit of the Riemann sums indeed exists, as otherwise the integral would not even be defined. Such functions are called **(Riemann) integrable**. There do exist functions which are not integrable. But they are weird and we shall not go into them. It turns out that all continuous functions are Riemann integrable. As our interest in integrals is more practical than theoretical, we may safely confine ourselves to continuous functions while talking about integrals.

To find these points of intersection, we have to solve the quadratic equation $c = x^2 - x$. The two roots are $\frac{1}{2} \pm \frac{1}{2}\sqrt{4c+1}$. Clearly, the smaller root gives P as $(\frac{1}{2} - \frac{1}{2}\sqrt{4c+1}, c)$ while the larger root gives Q as $(\frac{1}{2} + \frac{1}{2}\sqrt{4c+1}, c)$. Hence the length of the segment PQ is $\sqrt{4c+1}$. Call this $g(c)$.

If we now slice R_2 by lines of the form $y = y_i$, where $(y_0, y_1, \ldots, y_i, \ldots, y_m)$ is a partition of the interval $[-\frac{1}{4}, 0]$, then arguing exactly as in (6) above, the approximate area of R_2 will be $\sum_{i=1}^{m} g(y_i)\Delta y_i$ and its exact area will equal the definite integral $\int_{-\frac{1}{4}}^{0} g(y)dy$. Thus,

$$A_2 = \text{ area of } R_2 = \int_{-\frac{1}{4}}^{0} g(y)dy = \int_{-\frac{1}{4}}^{0} \sqrt{4y+1}dy \tag{8}$$

This integral is not as easy to compute as the one in (1). But not too difficult either. An antiderivative of $\sqrt{4y+1}$ is $\frac{1}{6}(4y+1)^{3/2}$ as can be seen by direct differentiation. Later we shall study systematic methods for obtaining antiderivatives. So by Theorem 1, we get

$$A_2 = (\frac{1}{6}(4y+1)^{3/2})\Big|_{-\frac{1}{4}}^{0} = \frac{1}{6} - 0 = \frac{1}{6} \tag{9}$$

which is the same answer as (1).

To apply this method for the region R_1, we first have to determine the values between which y varies in the region R_1. From the figure in the solution to the Main Problem, these values are 0 and 4. Next, for every c between 0 and 4, we have to find the length of the portion of the line $y = c$ intercepted by the region R_1. R_1 is bounded on the left by the parabola $y = 4x - x^2$ for all $y \in [0, 4]$. However, on the right it is bounded by the parabola $y = x^2 - x$ if $0 \leq y \leq \frac{15}{4}$, and by the parabola $y = 4x - x^2$ if $\frac{15}{4} \leq y \leq 4$. As a result, the left end-point of the intercept by R_1 on the line $y = c$ for all $c \in [0, 4]$ is obtained by solving the quadratic $4x - x^2 = c$ and taking the smaller root. It gives the point $(2 - \sqrt{4-c}, c)$ But the right end point changes depending upon whether $0 \leq c \leq \frac{15}{4}$ or $\frac{15}{4} \leq c \leq 4$. In case of the former, it is obtained by taking the larger root of the quadratic $x^2 - x = c$. It gives the point $(\frac{1}{2} + \frac{1}{2}\sqrt{4c+1}, c)$ as the right end of the intercept of $y = c$ by R_1. Hence the length of this segment PQ is $\frac{1}{2}\sqrt{4c+1} + \sqrt{4-c} - \frac{3}{2}$. Call this $h(c)$. However, for $\frac{15}{4} \leq c \leq 4$, the right end point is given by the larger root of the quadratic $4x - x^2 = c$ which is $2 + \sqrt{4-c}$. Hence for $c \in [\frac{15}{4}, 1]$, we have $h(c) = 2\sqrt{4-c}$.

If we now slice R_1 by lines of the form $y = y_i$, where $(y_0, y_1, \ldots, y_i, \ldots, y_m)$ is a partition of the interval $[0, 4]$, then arguing exactly as in (6) above, the approximate area of R_1 will be $\sum_{i=1}^{m} h(y_i)\Delta y_i$ and its exact area will equal the definite integral $\int_{0}^{4} h(y)dy$. Since the formula for $h(y)$ changes at $y = \frac{15}{4}$, this

integral will have to be split into two integrals before it can be evaluated. The splitting gives

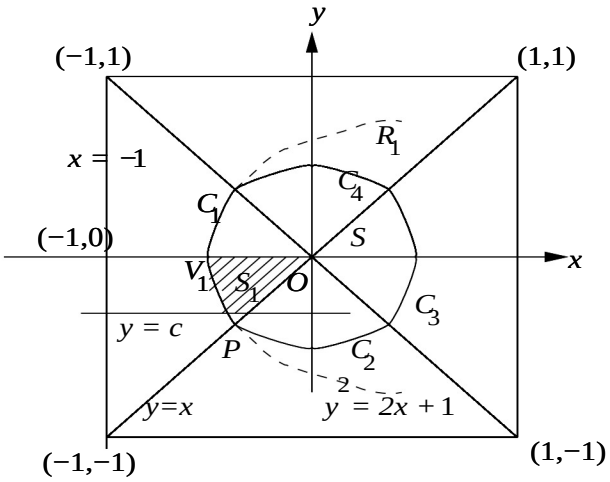
$$A_1 = \int_0^{\frac{15}{4}} \left(\frac{1}{2}\sqrt{4y+1} + \sqrt{4-y} - \frac{3}{2} \right) dy + \int_{\frac{15}{4}}^{4} 2\sqrt{4-y}dy \qquad (10)$$

We leave it to you to check that these integrals come out to be, respectively, $\frac{117}{24}$ and $\frac{1}{6}$. Adding, we get the same answer as (2). Admittedly, the calculations involved here are far more complicated than in the vertical slicing. But that is exactly the point we want to make. In theory, every slicing is as good as any other. But for a particular problem a particular slicing may be more convenient. In the present problem, both the parabolas were vertical (i.e., had vertical axes). If they were, instead, $x = 4y - y^2$ and $x = y^2 - y$, then the horizontal slicing would be more convenient to find the area of the region bounded by the two. Even as the problem stands, if we want the area of the portion of R between the horizontal lines $y = 0$ and $y = \frac{15}{4}$, then as just shown, the first integral in (10) would give the answer. But to do the problem by vertical slicing, we would have to divide the interval $[0, \frac{5}{2}]$ into *three* parts, viz., $[0, 1], [1, \frac{3}{2}]$ and $[\frac{3}{2}, \frac{5}{2}]$ and accordingly write the area as the sum of three integrals.

Comment No. 4:

In some problems the boundaries of the region whose area is to be found are not specified by explicit equations. Instead, they have to be obtained first from the data. Here is a good illustration of such a problem. (JEE 1995)

Consider a square with vertices at $(1, 1), (-1, 1), (-1, -1)$ and $(1, -1)$. Let S be the region consisting of all points inside the square which are nearer to the origin than to any edge. Sketch the region S and find its area.



We first identify the region S. The centre of the square is at $O = (0, 0)$.

One of the sides of the square has equation $x = -1$. Let C_1 be the set of points which are equidistant from the point O and from this side. Then C_1 is a parabola with focus at O and directrix $x = -1$. Its vertex, say V_1, is at $(-\frac{1}{2}, 0)$. The latus rectum is 2 (being twice the distance between the directrix and the focus). This gives the equation of the parabola C_1 as $y^2 = 2(x + \frac{1}{2}) = 2x + 1$. (This can also be obtained directly by equating the distance of a point (x, y) from O, viz., $\sqrt{x^2 + y^2}$ with $|x + 1|$, the distance from the line $x = -1$.) This parabola divides the plane into two regions. The region, say R_1, to the right of this parabola consists of points which are closer to the origin than to the line $x = -1$ while its complement consists of those points which are closer to the direcrix than to the focus. (The common boundary of these two regions is precisely the parabola C_1.)

Similarly, the other three sides of the given square determine three regions, say R_2, R_3 and R_4, bounded by parabolas C_2, C_3 and C_4 (say), respectively. The equations of these parabolas can be written down directly or be derived from that of C_1, since all of them are the reflections of C_1 in suitable lines. For example, C_2 is the reflection of C_1 in the line $y = x$ and hence the equation of C_2 is $x^2 = 2y + 1$. The given region S is the intersection of these four regions. It is sketched in the figure above.

By symmetry, S is the union of eight mutually congruent, non-overlapping pieces. One of these pieces, S_1, is shown by shading in the figure. Its boundary has three parts. One of them is the portion of the parabola C_1 between the vertex V_1 and the point P, which is the point of intersection of the parabolas C_1 and C_2. The other two portions of the boundary of S_1 are the straight line segments PO and OV_1. So the area of S will be 8 times that of S_1. To find the latter we first identify the point P as the point of intersection of the parabolas $y^2 = 2x + 1$ and $x^2 = 2y + 1$. By symmetry, P lies on the line $y = x$ and this makes it easier to identify it by solving the quadratic $x^2 = 2x + 1$ which gives $x = 1 \pm \sqrt{2}$. We have to take the negative sign as P is to the left of the y-axis. This gives P as $(1 - \sqrt{2}, 1 - \sqrt{2})$.

Finally, to find the area of S_1 it is clear that horizontal slicing is better than the verical one because with horizontal slicing, we do not have to split the region S_1. The region S_1 lies between the lines $y = 1 - \sqrt{2}$ and $y = 0$. For any $c \in [1 - \sqrt{2}, 0]$, the line $y = c$ cuts the parabola $y^2 = 2x + 1$ at $(\frac{c^2-1}{2}, c)$ and the line OP (whose equation is $y = x$) at the point (c, c). Hence the length of the intercept by S_1 of the line $y = c$ is $c - \frac{c^2-1}{2}$ for all $c \in [1 - \sqrt{2}, 0]$. Therefore,

$$\begin{aligned}
\text{area of } S_1 &= \int_{1-\sqrt{2}}^{0} (y - \frac{y^2 - 1}{2}) dy \\
&= \left(\frac{y^2}{2} - \frac{y^3}{6} + \frac{y}{2} \right) \Big|_{1-\sqrt{2}}^{0} \\
&= -\frac{(1 - \sqrt{2})^2}{2} + \frac{(1 - \sqrt{2})^3}{6} - \frac{1 - \sqrt{2}}{2} \\
&= \frac{(1 - \sqrt{2})^3 - 3(1 - \sqrt{2})^2 - 3(1 - \sqrt{2})}{6} = \frac{4\sqrt{2} - 5}{6} \qquad (11)
\end{aligned}$$

Multiplying this by 8, the area of $S = \dfrac{16\sqrt{2} - 20}{3}$ sq. units. (It is generally not expected that an answer like this be simplified further. In particular, there is no need to convert it into a decimal form. Generally, in answers to numerical problems of integration, only elementary simplifications such as adding fractions, cancelling common factors, expanding powers of surds and grouping the like terms together are expected. If the answer involves some irrational numbers such as π, e or some radicals or some logarithms, leave them as they are. If the answer involves an expression like $\sin^{-1}(\frac{1}{2})$, then it is expected that you write it as $\frac{\pi}{6}$. But, for an expression like $\sin^{-1}(\frac{3}{5})$ which has no such alternate expression, you may leave it as it is. In examinations in applied subjects such as engineering, it is often important to reduce the final answer to a standard decimal form, because there you often have to do something practical with the answer, such as designing a beam which will take a given load. In examinations in science subjects such as physics and chemistry such a simplification is desirable. But to decide whether it is mandatory, it is best to follow the instructions. In an examination intended to test mathematical skills, such a conversion is not needed unless specifically asked.)

Comment No. 5:

We do not need integrals to compute the areas of simple figures such as triangles and parallelograms. In fact, in the Main Problem of Chapter 9 and in Comment No. 5 of it we computed several areas of this type. (See also Exercises (9.19) and (9.59).) More generally, we can calculate the area of any polygonal figure without calculus. Indeed, the area of a rectangle *has to be* obtained without calculus. Given a rectangle with height h and base b (say), we can choose the axes along two of its sides so that the rectangle is precisely the region below the graph of the constant function $f(x) = h$ defined for $0 \leq x \leq b$. Hence its area equals the integral $\int_0^b h dx$ which is simply bh. It is, therefore, tempting to conclude that the area of a rectangle can be found using integrals. But this is logically fallacious, because when we expressed the area of a region as a certain integral, the derivation was based on an equation like (5). But (5) was obtained by assuming the formula for the area of a rectangle in the first place!

However, for other geometric figures such as triangles and trapezia, we have a choice. We can find their areas purely geometrically by constructing rectangles whose areas match those of the given figures, as is shown in elementary geometry books or we can apply integrals. The former is, of course, preferable. However, just for curiosity, we shall show in the exercises how the standard formula for the area of a triangle with vertices at $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) can be derived using integrals.

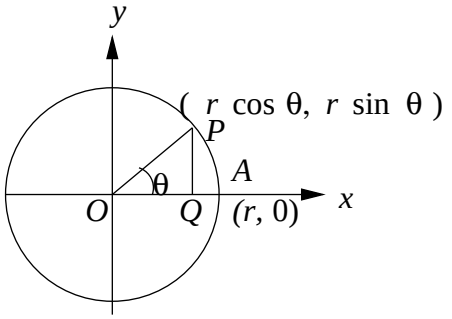
It should be noted, however, that the area of a circular disc cannot be found without integrals. Most people learn a formula for it in schools (where they also pick up the wrong terminology 'area of a circle' instead of the correct phrase 'area bounded by a circle' and still more fussily, 'area of the region bounded by a circle') long before they study calculus and hence are under the

impression that it is an 'elementary' result. More generally, they also think that the formula for the area of a sector of angle θ of a circle of radius r (viz., $\frac{1}{2}\theta r^2$) is an obvious one. When they cannot prove it, they try to circumvent the difficulty by saying that this is the very definition of the measure of an angle. (And, in particular, the number π is defined as the ratio of the area of a circle of radius r to the area of a square with side r.) This approach is perfectly feasible. But then the price one has to pay is that one has to give a proof that the arc length of this sector is $r\theta$ (and, in particular, a proof that the perimeter of a circle of radius r is $2\pi r$). So somewhere you have to do some hard work. You cannot, take *both* the formulas as definitions of the measure of an angle. You can take either one of them as the definition and then have to prove the other as a theorem. Trying to pocket both the formulas as definitions would be like trying to eat a cake and have it!

The standard practice is the other way round. That is, we *define* the measure of an angle in terms of the arc length of a sector (specifically, as the ratio of the arc length to the radius of the sector). Having defined an angle this way, we prove the inequality $\sin\theta < \theta < \tan\theta$ (for small positive values of θ) as mentioned in Chapter 6, Comment No. 9. One then goes on to find the derivatives of the trigonometric and then of the inverse trigonometric functions. With all this spadework, the formula for the area of a sector with angle θ at the center and radius r can be obtained using calculus as follows.

For simplicity assume θ to be acute. (Otherwise divide θ into two or more equal parts and add the areas of the corresponding subsectors.) Take the equation of the circle as $x^2 + y^2 = r^2$ and let A and P be the points $(r, 0)$ and $(r\cos\theta, r\sin\theta)$ respectively. Draw the perpendicular PQ from P to OA. Then $Q = (r\cos\theta, 0)$. Clearly, the area of the sector is the sum of the area of the triangle OPQ and the area of the portion between the x-axis and the arc PA of the upper half of the circle, with equation $y = \sqrt{r^2 - x^2}$.

Now the area of the triangle is $\frac{1}{2}r^2\sin\theta\cos\theta$. The second area is the integral $\int_{r\cos\theta}^{r} \sqrt{r^2 - x^2}dx$. For the moment we borrow an antiderivative of the integrand. It is $\frac{x}{2}\sqrt{r^2 - x^2} + \frac{r^2}{2}\sin^{-1}(\frac{x}{r})$ as can be verified by directly computing its derivative. Using Theorem 1, we now get

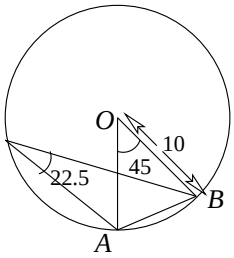


$$\begin{aligned}
\text{area of sector} &= \int_{r\cos\theta}^{r} \sqrt{r^2 - x^2}dx \\
&= \frac{1}{2}r^2\sin\theta\cos\theta + \left(\frac{x}{2}\sqrt{r^2 - x^2} + \frac{r^2}{2}\sin^{-1}\frac{x}{r}\right)\Big|_{r\cos\theta}^{r} \\
&= \frac{1}{2}r^2\sin\theta\cos\theta + \frac{\pi}{4}r^2 - \frac{r^2}{2}\sin\theta\cos\theta - \frac{r^2}{2}\sin^{-1}(\cos\theta)
\end{aligned}$$

$$= \frac{\pi}{4}r^2 - \frac{r^2}{2}\sin^{-1}(\sin(\frac{\pi}{2}-\theta)) = \frac{\pi}{4}r^2 - \frac{r^2}{2}(\frac{\pi}{2}-\theta)$$
$$= \frac{1}{2}r^2\theta \qquad (12)$$

Taking $\theta = 2\pi$ we get the area of a circular disc of radius r as πr^2. A similar integration gives the area bounded by an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Cutting it into 4 equal quarters, the area is $4\int_0^a \frac{b}{a}\sqrt{a^2-x^2}dx$, which comes out to be simply, πab, i.e. π times the product of the semi-major and the semi-minor axes of the ellipse. (Interestingly, if $b = a$, then the ellipse reduces to a circle of radius a and the area becomes πa^2 as is to be expected.)

Of course, formula (12) is so standard that in some problems it can be applied without proof. Suppose, for example, that we want to find the area of the smaller part of of a disc of radius 10 cm cut off by a chord AB which subtends an angle of $22\frac{1}{2}^\circ$ at the circumference. (JEE 1979) Let O denote the centre of the disc. Then by a well-known result in geometry, $\angle AOB = 45^\circ = \frac{\pi}{4}$. So by (12) above, the area of the sector between the radii OA and OB is $\frac{\pi}{8} \times 100$ sq. cms. The area of the triangle OAB, on the other hand, is $\frac{1}{2}r^2\sin\frac{\pi}{4} = \frac{1}{2\sqrt{2}} \times 100$ sq. cms. Subtracting the two we get $(\frac{\pi}{8} - \frac{1}{2\sqrt{2}}) \times 100$ sq. cms. as the area of the smaller portion of the disc cut off by the chord AB.



Comment No. 6:

A few more problems about finding areas will be given in the exercises. Sometimes a problem of finding area is combined with some other problems. Suppose for example, that we want to find all the possible values of $b > 0$, so that the area of the bounded region enclosed between the parabolas $y = x - bx^2$ and $y = \frac{x^2}{b}$ is maximum. (JEE 1997*) We skip the diagram which is qualitatively similar to that in the solution to the Main Problem since here also one of the parabolas is vertically downward and the other vertically upward. To find the points of intersection of the two parabolas we solve the quadratic $x - bx^2 = \frac{x^2}{b}$. $x = 0$ is an obvious root. The other root then, is $\frac{b}{b^2+1}$. For values of x between the two roots, the expression $\frac{x^2}{b} + bx^2 - x$ is negative. Hence $\frac{x^2}{b} < x - bx^2$. So the area between the two parabolas is given by the integral $\int_0^{b/(b^2+1)} (x - bx^2 - \frac{x^2}{b})dx = \int_0^{b/(b^2+1)} (x - \frac{b^2+1}{b}x^2)dx$. Writing c for $\frac{b}{b^2+1}$, this integral comes out to be $\frac{c^2}{2} - \frac{c^2}{3} = \frac{c^2}{6}$. This is maximum whenever c is maximum (since $c \geq 0$). But $c = \frac{b}{b^2+1}$. So c is maximum when $\frac{1}{c}$ is minimum, i.e., $b + \frac{1}{b}$ is minimum. By the A.M.-G.M. inequality (or using calculus, if you prefer), this happens when $b = 1$.

This problem is a good example of how ideas from various topics have be combined to get the answer. It takes a little familiarity with coordinate geometry to recognise the shapes of the parabolas. Without such recognition, it is difficult to even start the solution. To find the points of intersection and to set up the integral for the area between the parabolas, we need elements of the theory of quadratic equations. The area, of course, comes by integration, which is the heart of the problem. Finally, to maximise the area we could use derivatives. But a standard application of the A.M.-G.M. inequality gave the answer effortlessly.

Comment No. 7:

Sometimes the problem of finding an area is asked in a disguised form. Let $f(x)$ be continuous and positive on an interval $[a, b]$ and let R be the region below the graph of f, i.e., the region bounded above by the curve $y = f(x)$, below by the x-axis, on the left by the line $x = a$ and on the right by the line $x = b$. Let A be the area of R. Then $A = \int_a^b f(x)dx$. But in arriving at this formula we used the fact that A is approximately the sum $\sum_{i=1}^{n} f(x_i)\Delta x_i$, where $(x_0, x_1, \ldots, x_i, \ldots, x_n)$ is a partition of the interval $[a, b]$ into n subintervals. (See (6) above except that there the region R_1 was of a slightly different type.) We call such sums as Riemann sums. And the integral is the limit of such sums as the mesh of the partition tends to 0.

We now consider a special type of a Riemann sum in which all the subintervals of $[a, b]$ are of the same length. If the number of subintervals is n, then evidently $\Delta x_i = \frac{b-a}{n}$ for every $i = 1, 2, \ldots, n$. Further $x_i = a + \frac{i(b-a)}{n}$ for $i = 0, 1, 2, \ldots, n$. So for this partition the Riemann sum is simply the sum $\frac{(b-a)}{n} \sum_{i=1}^{n} f(a + \frac{i(b-a)}{n})$. Now if all the Riemann sums tend to the limit A, then these *particular* Riemann sums must also tend to the same limit A. When all n subintervals are of equal lengths, the mesh is simply $\frac{b-a}{n}$ and to say that the mesh tends to 0 is equivalent to saying that n tends to ∞. It follows that if we can recognise a given sum as a Riemann sum of a suitable function over a suitable interval partitioned into equal parts, then its limit as the number of these parts tends to 0 will be the Riemann integral of that function over that interval.

Let us apply this reasoning to evaluate $\lim_{n\to\infty} a_n$, where $a_n = \sum_{i=1}^{n} \frac{1}{n+i} = \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{2n}$. In Exercise (6.40)(d) we pointed out that this limit is not 0, even though each individual term $\frac{1}{n+i}$ tends to 0 as n tends to ∞ for $i = 1, 2, \ldots, n$. We can now evaluate the limit by first rewriting a_n as $\frac{1}{n} \sum_{i=1}^{n} \frac{1}{1 + \frac{i}{n}}$ and hence as $\frac{2-1}{n} \sum_{i=1}^{n} f(1 + \frac{i}{n})$, where $f(x) = \frac{1}{x}$. In this form a_n is the Riemann sum of the function $f(x) = \frac{1}{x}$ for the interval $[1, 2]$ partitioned

into n equal parts. So by what we have said above, $\lim_{n\to\infty} a_n = \int_1^2 \frac{dx}{x} = \ln x\Big|_1^2 = \ln 2 - \ln 1 = \ln 2$.

By a similar reasoning, $\lim_{n\to\infty}(\frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{6n}) = \ln 6$ as the given sum is the Riemann sum of $f(x) = \frac{1}{x}$ over the interval $[1, 6]$ partitioned into $5n$ equal parts. (JEE 1981)

Comment No. 8:

The reason the area of a region can be expressed as a definite integral is basically that an area is an **additive attribute**. This means when a region is divided into two or more mutually non-overlapping subregions, its area is the sum of the area of these subregions. We encounter many other additive attributes like mass and volume in real life. Not surprisingly, integrals are also useful to give formulas for such quantities. We illustrate this in the case of the mass of a wire. Suppose we have a piece of wire of length L. If its linear mass density (i.e., mass per unit length) is uniform, say ρ throughout, then its mass is simply the product ρL. (This corresponds to the area of the portion below the graph of a constant function. This portion is a rectangle and its area is simply the product of the (constant) height and the length of the base interval.) Suppose, however, that the linear density is not uniform. Fix one end of the wire and let $f(x)$ denote the linear density at a point on the wire x units away from the fixed end. The variable takes values in the interval $[0, L]$. A partition, say $(x_0, x_1, \ldots, x_i, \ldots, x_n)$ of the interval $[0, L]$ induces a corresponding physical partition of the wire into n pieces. As mass is additive, the mass of the wire is the sum of the masses of the pieces. Now the mass of the i-th piece is approximately $f(x_i)\Delta x_i$ and hence the mass of the wire is approximately $\sum_{i=1}^{n} f(x_i)\Delta x_i$. This is exactly a Riemann sum of the function $f(x)$ for the partition $(x_0, x_1, \ldots, x_n)$ of the interval $[0, L]$. As the mesh of the partition tends to 0, it tends to $\int_0^L f(x)dx$. This is, therefore, the mass of the wire.

Exactly the same reasoning applies to find the resistance of a wire, the amount of heat to raise its temperature by one degree and many other things which are additive in nature. This explains why the definite integrals are so important in physical applications.

Comment No. 9:

In fact, if we apply the reasoning here to take the integral of the speed function, we see why the Fundamental theorem of calculus is so natural. Consider the motion of a particle on a straight line. Let t denote the time and s the position of the particle. Here s denotes the algebraic distance of the position of the particle, measured from some fixed point, say P_0, on the line. That is, we take s to be positive on one side of P_0 and negative on the other. Then the motion is specified by expressing s as a function of t, say $s = F(t)$. Clearly, its derivative $\frac{ds}{dt} = F'(t)$ is the speed of the particle at time t. It is positive when

the particle is moving forward and negative when it is moving backward. (Draw your own diagram to see this.)

Now let $[a, b]$ be a time interval. During this time interval the particle could have changed its direction any number of times and might even have stood still for some part. But no matter what the motion was during this time interval, the net displacement[2] over this time interval is $F(b) - F(a)$ for the simple reason that $F(a)$ and $F(b)$ denote, respectively, the initial and the final positions of the particle. We now calculate the net displacement in another way. Denote $F'(t)$ by $f(t)$. This is the speed of the particle at time t and we assume that it is a continuous function of t. If the motion is uniform, $f(t)$ is a constant, say v, and in that case the distance moved over the time interval $[a, b]$ is simply the product $v(b - a)$. But, in general $f(t)$ varies with t. In such a case we can find the total displacement over $[a, b]$ as follows. Take a partition, say $(t_0, t_1, \ldots, t_i, \ldots, t_n)$ of the interval $[a, b]$ and consider its i-th subinterval $[(t_{i-1}, t_i]$. Its length is $\Delta_i = (t_i - t_{i-1})$. Now if this interval is small (i.e., if Δt_i is small), then it is reasonable to assume, because of continuity of the function $f(t)$, that the speed does not vary much over this subinterval and hence that the displacement over it is approximately $f(t_i)\Delta_i$. The total displacement over the interval $[a, b]$, then, is approximately $\sum_{i=1}^{n} f(t_i)\Delta_i$. Moreover, as the mesh of the partition tends to 0, this approximation will be more accurate. So the exact value of the net displacement over the time interval $[a, b]$ would be the limit of these sums. But, by definition, the limit of such sums is the definite integral $\int_a^b f(t)dt$.

Thus we have found the total displacement of the particle over the time interval from $t = a$ to $t = b$ in two ways. The easier way, it was simply $F(b) - F(a)$. The harder way, it came to be $\int_a^b f(t)dt$ where $f(t) = F'(t)$. But whether you compute it the easier way or the harder way, the two answers must agree. Equating the two we get precisely Theorem 1 above except that the variable x is replaced by the variable t.

Comment No. 10:

We hasten to add here, that the argument given above is not a rigorous proof of the Fundamental Theorem of Calculus. It is based on a physical interpretation of the derivative, as the speed. As remarked after the statement of Theorem 3 in Chapter 13, arguments based on such interpretations, no matter how plausible they look, are not a substitute for a mathematical proof. As mentioned earlier, a rigorous proof of Theorem 1 is non-trivial.

Nevertheless, the argument above shows that the two basic concepts of calculus, viz., the derivatives and the (definite) integrals are intimately related. In fact, as mentioned before, the most common way of evaluating an integral $\int_a^b f(x)dx$ is by first finding an antiderivative, say $F(x)$, of the integrand $f(x)$.

[2]We could also say the net distance travelled. But by convention, the distance travelled is taken to be always positive. So the word 'displacement' is preferred. If a bus goes from one town to another and comes back, its net displacement is 0. But it will be awkward to say that the distance travelled by it is 0, since that would give the impression that it never moved!

Once this is obtained, there is hardly anything left to be done except to evaluate the antiderivative F at the end points b and a and take the difference. This procedure is followed so ubiquitously and routinely, that the layman can hardly be blamed if he thinks that integrals are the same thing as antiderivatives. This is also reflected in the terminology. Antiderivatives were (and to some extent are still) popularly called **indefinite integrals** and written using the integral sign without the upper and the lower limits. Thus for example, we write $\int x^2 dx = \frac{x^3}{3}$. Mathematically, this is no different than saying that $\frac{x^3}{3}$ is an antiderivative of x^2, i.e., $\frac{d}{dx}(\frac{x^3}{3}) = x^2$. This is purely a statement about derivatives and not about integrals. But because of Theorem 1, this knowledge enables us to evaluate a definite integral of the form $\int_a^b x^2 dx$ as $\frac{1}{3}(b^3 - a^3)$. Because of this close relationship (which is established by Theorem 1) between antiderivatives and definite integrals, the former began to be called 'indefinite integrals' much the same way that a girl Miss X who marries a man Mr. Y is subsequently called Mrs. X. (To stretch the simile further, the marriage is solemnised by Theorem 1!)

Another unintended misuse of the Fundamental Theorem of Calculus is to believe that integration is just the reverse of differentiation. While this may be justified on practical grounds at an elementary level (at which a rigorous treatment of definite integrals is often skipped), to think so permanently is quite wrong. The difference between a definite integral $\int_a^b f(x)dx$ and an indefinite integral $\int f(x)dx$ is not just the superficial difference that the upper and the lower limits are missing in the latter. The difference between them is conceptual. The former is defined as the limit of a sum of a certain form. To understand it you need to know nothing about derivatives. The latter, on the other hand, is the opposite of derivatives.

Comment No. 11:

There is another way to interpret the statement that integration is the opposite of differentiation. Suppose we fix some a and consider the interval from a to a variable x. We can then rewrite Equation (7) as

$$\int_a^x F'(t)dt = F(x) - F(a) \tag{13}$$

Note that we have changed the variable of integration from x to t, because now we are using x to denote the upper limit of integration. As stressed just before stating Theorem 1, a change in the variable of integration does not change the integral. Logically, there is nothing wrong in writing $\int_a^x f(x)dx$ as long as we understand the double role of the variable x, one as the variable upper limit of integration and another as the variable of integration. But it is better to avoid such double roles since they are likely to be confused with each other.

Verbally, Equation (13) says that if we start with a function $F(x)$,

differentiate it to get $F'(x)$ and then integrate the derivative $F'(x)$, then we get back the original function $F(x)$ except for a constant (viz., $F(a)$). To illustrate this with a particular example, take $F(x) = x^3$ and $a = 2$. Then $F'(x) = 3x^2$, so $F'(t) = 3t^2$. Hence for any x we have, $\int_2^x 3t^2 dt = t^3\Big|_2^x = x^3 - 8$ which differs from $F(x)$ just by a constant 8.

Here we first differentiated and then integrated and (almost) got back the original function. Let us see what happens if we reverse the procedure. That is, we now start with some function $f(t)$. For simplicity, we assume that it is continuous for all t. Fix some a and consider the variable interval $[a, x]$. Let us define a new function $g(x)$ by

$$g(x) = \int_a^x f(t)dt \tag{14}$$

(Note that $g(x)$ is defined even for $x < a$ because of our convention that in such a case $\int_a^x$ means $-\int_x^a f(t)dt$.) Quite appropriately the function $g(x)$ is called a **function defined by an integral**, because that's exactly what it is! For example, suppose we take $f(t)$ to be $2 + t$ and $a = 1$. Then $g(x) = \int_1^x 2 + tdt = (2t + \frac{t^2}{2})\Big|_1^x = 2x + \frac{x^2}{2} - \frac{5}{2}$. If we differentiate this function, we see that $g'(x) = 2+x$ which is exactly $f(x)$, the function we started with. Note further, that this time we got back the original function on the nose (i.e., without the addition or subtraction of any constant). You may wonder if this was a coincidence. Try it with a few other functions such as $f(t) = \cos t, f(t) = e^t$ etc. and you will be convinced that it is not so. The following theorem, called the **second form** of the fundamental theorem of calculus should then come as no surprise.

Theorem 2: Suppose $f(x)$ is continuous in an interval I and a is a fixed point of I. For $x \in I$ define $g(x)$ by (14). Then $g'(x) = f(x)$ for all $x \in I$. In symbols, for all $x \in I$, we have

$$\frac{d}{dx}\left(\int_a^x f(t)dt\right) = f(x) \tag{15}$$

The point a plays very little role in this theorem. If instead of a we took some other point, say c in I, then the two functions $\int_a^x f(t)dt$ and $\int_c^x f(t)dt$ differ by a constant (viz., $\int_a^c f(t)dt$) and so they have the same derivatives. Like Theorem 1, we skip the proof of Theorem 2. A plausible argument, can, however, be given by taking $f(t)$ to be the speed of a moving particle at time t. In that case $\int_a^x f(t)dt$ is simply the position of the particle at time x relative to its position at $t = a$. So the derivative of this function ought to give you the speed, i.e., the original function f. But once again, this is not a mathematical proof.

Theorems 1 and 2 taken together mean that integration and differentiation are opposite processes no matter in which order they are performed. They are like oxidation and reduction or hydrolysis and ester formation or, to give simpler analogies, like dressing and undressing (except that you cannot undress before you dress).

Thus in this sense the two theorems are complementary to each other and together constitute the Fundamental Theorem of Calculus. But they differ considerably in their usefulness. As we have already seen, the first form of the Fundamental Theorem is used heavily in evaluating antiderivatives. The second form is mostly of a theoretical interest. In essence it asserts that every continuous function f *has* an antiderivative, viz., the one given by (14). This knowledge, in itself, is of little use unless you can identify an antiderivative in what is called a closed form, i.e., in terms of some familiar functions such as the rational functions, roots, the exponential and the trigonometric functions and their inverses. For example, in (12) above we could calculate the area of a sector only because we could identify an antiderivative of the function $\sqrt{r^2 - x^2}$ in a closed form. Theorem (2) also gives us an antiderivative of $\sqrt{r^2 - x^2}$, e.g., $g(x) = \int_0^x \sqrt{r^2 - t^2}dt$. But if we try to apply Theorem 1 with this antiderivative, we shall get

$$\begin{aligned} \int_{r\cos\theta}^{r} \sqrt{r^2 - x^2}dx &= g(r) - g(r\cos\theta) \\ &= \int_0^r \sqrt{r^2 - t^2}dt - \int_0^{r\cos\theta} \sqrt{r^2 - t^2}dt \qquad (16) \end{aligned}$$

But now we are stuck as we can evaluate neither of these two integrals without knowing an antiderivative of the integrand in a closed form. Because of the additivity property of the integrals we can rewrite the R.H.S. of (16) as a single integral, viz., $\int_{r\cos\theta}^{r} \sqrt{r^2 - t^2}dt$. But that takes us back to exactly where we started, viz., the integral $\int_{r\cos\theta}^{r} \sqrt{r^2 - x^2}dx$. All we have gained is that the variable of integration has been changed from x to t.

Thus, in effect, when it comes to actually finding an antiderivative of a given function, Theorem 2 is just about as evasive as someone who says that his daughter is married to his son-in-law! The only useful inference from such a statement is that the person who makes it has a daughter and that the daughter has a husband. Similarly, the high point of Theorem 2 is that every continuous function *has* an antiderivative and moreover that the antiderivative can be expressed as a definite integral. If we have some independent means of evaluating definite integrals (other than those based on Theorem 1) then this knowledge may be of some practical use. There are many functions such as $\sin(x^2)$ or $e^{\cos x}$ whose antiderivatives cannot be obtained in closed forms. The mere knowledge that such functions have antiderivatives is of little practical use.

Comment No. 12:

However, even as it stands, Theorem 2 is useful in identifying a function from its integral. Suppose, for example, that we are given, either directly or indirectly, that some (continuous) function $f(x)$ has the property that for all positive values of a and b, $\int_a^b f(x)dx$ equals, $\ln(\frac{b}{a})$. Writing $\ln(\frac{b}{a})$ as $\ln b - \ln a$ and replacing b by x, we get that

$$\int_a^x f(t)dt = \ln x - \ln a \tag{17}$$

for all $x > 0$. Differentiating both the sides w.r.t. x and using (15) we get $f(x) = \frac{1}{x}$ for all $x > 0$.

As a more direct application of Theorem 2, suppose we are given that $f(x)$ is the integral of $\dfrac{2\sin x - \sin 2x}{x^3}$, for $x \neq 0$ and are asked to find $\lim\limits_{x\to 0} f'(x)$. Note that f is not given to be defined at 0. But this does not matter since we are not asked to find $f'(0)$ but $\lim\limits_{x\to 0} f'(x)$ for which it is enough to know $f'(x)$ for all x in some deleted neighbourhood of 0. Note also that $f(x)$ is not given explicitly either. We can of course, try to find it by finding an antiderivative of $\dfrac{2\sin x - \sin 2x}{x^3}$. But this is difficult. Nor is it needed. By Theorem 2, even without knowing $f(x)$ explicitly, we know its derivative $f'(x)$. It is simply the given function $\dfrac{2\sin x - \sin 2x}{x^3}$. So the problem is, in disguise, the problem of evaluating $\lim\limits_{x\to 0} \dfrac{2\sin x - \sin 2x}{x^3}$. We have evaluated such limits using l'Hôpital's rule in Chapter 15, Comment No. 7 or using the Taylor approximation of the numerator (Chapter 16, Comment No. 13) near 0. Either way, the answer comes out to be 1. (JEE 1979)

For applications like this, it is convenient to cast Theorem 2 in a more general form. In the integral $\int_a^x f(t)dt$, the lower limit of integration is constant while the upper one, although variable, is a very simple function of x. We can more generally consider integrals of the form $\int_a^{v(x)} f(t)dt$, where $v(x)$ is some differentiable function of x taking values in the interval on which the function $f(t)$ is defined (as otherwise the integral would not make sense). Call $v(x)$ as y. Then $\int_a^{v(x)} f(t)dt$ equals $\int_a^y f(t)dt$, which, as a function of y is differentiable and has derivative $f(y)$ w.r.t. y, by Theorem 2. This derivative is the same as $f(v(x))$. But then, by the chain rule, if we multiply it by $\frac{dy}{dx}$, we get its derivative w.r.t. x. Put together, we have

$$\frac{d}{dx}\int_a^{v(x)} f(t)dt = f(v(x))v'(x) \tag{18}$$

Lest this appears too abstract, we invite you to verify its truth by taking, say,

$f(t) = \cos t$ and $v(x) = x^2$. Still more generally, we can consider integrals of the form $\int_{u(x)}^{v(x)} f(t)dt$, where $u(x)$ and $v(x)$ are some differentiable functions of x taking values in the interval on which the function $f(t)$ is defined. Fix any a in this interval and write $\int_{u(x)}^{v(x)} f(t)dt$ as the difference $\int_{a}^{v(x)} f(t)dt - \int_{a}^{u(x)} f(t)dt$. Applying (18) to each term we get

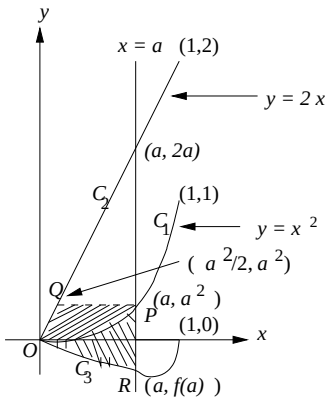
$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)dt = f(v(x))v'(x) - f(u(x))u'(x) \tag{19}$$

As an application of (18) we evaluate $\lim_{x\to 0} \dfrac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$. As we have done before in Chapter 15, the factor $\sin x$ in the denominator can be replaced by x since $\frac{\sin x}{x}$ tends to 1 as x tends to 0. (If you are still hesitant about such replacements, rewrite the quotient as $\dfrac{g(x)}{x^2}\dfrac{x}{\sin x}$, where $g(x) = \int_0^{x^2} \cos t^2 dt$ and observe that the second factor tends to 1 as $x \to 1$.) So the problem reduces to finding the limit of $\dfrac{g(x)}{x^2}$ as x tends to 0. This is of the $\dfrac{0}{0}$ form. So we apply l'Hôpital's rule and try to evaluate $\lim_{x\to 0} \dfrac{g'(x)}{2x}$. Now, by (18), $g'(x) = 2x\cos(x^4)$. Hence for $x \neq 0$, $\frac{g'(x)}{2x}$ is simply $\cos(x^4)$. This tends to 1 as x tends to 0. Therefore the given limit is also 1 by l'Hôpital's rule. (JEE 1997*)

Comment No. 13:

Earlier we did several problems of computing the areas of regions determined by some given functions and in doing so we used Theorem 1. Sometimes we are given some information about the area of a region determined by some (unknown) function and we are asked to find this function. In such cases, we need Theorem 2. Here is an illustration of such a problem. (JEE 1998)

Let C_1 and C_2 be the graphs of the functions $y = x^2$ and $y = 2x$, $0 \le x \le 1$, respectively. Let C_3 be the graph of a function $y = f(x)$, $0 \le x \le 1$, $f(0) = 0$. For a point P on C_1, let the lines through P parallel to the axes, meet C_2 and C_3 at Q and R respectively (see figure). If for every position of P (on C_1), the areas of the shaded regions OPQ and ORP are equal, determine the function $f(x)$.



Referring to the figure above, let $P = (a, a^2)$ for some a, $0 \leq a \leq 1$. Then $Q = (\frac{a^2}{2}, a^2)$ and $R = (a, f(a))$. The area of the region ORP equals $\int_0^a x^2 - f(x)dx$. However, for the region OPQ it is better to find its area by horizontal rather than by vertical slicing. It comes out as $\int_0^{a^2} \sqrt{y} - \frac{y}{2}dy$. Equating the two areas and integrating, $\frac{2}{3}a^3 - \frac{a^4}{4} = \frac{a^3}{3} - \int_0^a f(x)dx$, or with a change of notation, $\int_0^x f(t)dt = \frac{x^4}{4} - \frac{x^3}{3}$. Therefore, by Theorem 2, $f(x) = \frac{d}{dx}(\frac{x^4}{4} - \frac{x^3}{3}) = x^3 - x^2$.

Comment No. 14:

We remarked that Theorem 1 is the most standard way of evaluating definite integrals. There are some exceptions. We shall take some of them up in the next chapter. But by and large, whenever we want to evaluate an integral $\int_a^b f(x)dx$, unless there is something special about the integrand that will give a quick answer (examples of such situations will be given later), one generally tries to first find an antiderivative $F(x)$ of $f(x)$ and then apply Theorem 1.

Of course, this raises the question : how does one find an antiderivative of a given function? As it often happens in mathematics, (and in real life too), when a process is easy its reverse process is usually difficult. Going downhill is fun, going uphill is tiring. Mixing two things is easy, separating them out is difficult and sometimes impossible. In mathematics too, multiplying two polynomials is a routine job. But factorising the product can take a very long time. Similarly, finding the derivative of a given function is easy. But finding an antiderivative is difficult. There is, in fact, no set procedure which will always work. A few general principles can be laid down. But ultimately it is the ingenuity and the experience of an individual that steal the show.

Let us note, first of all, that antiderivatives are not unique. If $F(x)$ is an antiderivative of a function $f(x)$ so is $F(x) + c$, where c is any constant. But we can show that over any given interval, any two antiderivatives of the same function must differ by a constant. This follows as an application of Lagrange's MVT. Indeed suppose $F(x)$ and $G(x)$ are two antiderivatives of the same function $f(x)$ of an interval, say I. Let $H(x) = F(x) - G(x)$. Then H' vanishes identically on I. Hence by Part (v) of Theorem 3 of Chapter 13, $H(x)$ is a constant function on I. (However, if we replace I by a union of two disjoint intervals, then $H(x)$ will be constant on each one of them, but not necessarily the same constant for both.)

Put differently, an antiderivative of a function on an interval is unique upto a constant. Therefore, after obtaining any one antiderivative of a function, it is customary to add a constant (almost invariably denoted by c or C) to it, called the **constant of integration**. Thus we write $\int x^2 dx = \frac{x^3}{3} + c$ etc. As

far as evaluation of a definite integral, say $\int_a^b f(x)dx$ is concerned, this constant hardly matters, because when we take $F(b) - F(a)$ it will cancel out. But when we use antiderivatives for some other purposes, such as solving differential equations, it plays a vital role.

The simplest antiderivatives are those which are obtained by simply remembering certain basic derivatives. All we have to do is to rewrite the formulas. For example, instead of saying $\frac{d}{dx}\sin x = \cos x$ we now say $\int \cos x dx = \sin x + c$. Similarly, for $\alpha \neq 0$, the formula $\frac{d}{dx}x^\alpha = \alpha x^{\alpha-1}$ can be rewritten, after dividing by α as $\int x^{\alpha-1}dx = \frac{1}{\alpha}x^\alpha + c$ and hence further as $\int x^\beta dx = \frac{x^{\beta+1}}{\beta+1} + c$ which is valid for $\beta \neq -1$. For $\beta = -1$ this formula breaks down. However we have $\ln x$ as an antiderivative of $\frac{1}{x}$ for $x > 0$ and $\ln(-x)$ as an antiderivative for $x < 0$. The two can be combined together to say that $\int \frac{dx}{x} = \ln|x| + c$ for all $x \neq 0$. But it is generally all right to merely say that $\int \frac{dx}{x} = \ln x + c$. Collectively we call all such antiderivatives as **integrals of the standard type**. For example, $\int \frac{dx}{x^2+1}$ which equals $\tan^{-1} x$ is an integral of the standard type because the only proof we have for it is to use the standard formula for the derivative of $\tan^{-1} x$.

Another simplifying feature comes from the linearity of derivatives. If $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$ respectively, then for any constants a and b, $aF(x)+bG(x)$ is an antiderivative of $af(x)+bg(x)$. This can, of course, be extended, by induction, to the linear combination of any (finite) number of functions. Note, in particular, that we can now find antiderivatives of any polynomial functions. We in fact used them in the solution to the Main Problem. Although very elementary, an ingenious use of the antiderivatives of polynomials can simplify the solution to a problem. Here is a good example.

A cubic $f(x)$ vanishes at $x = -2$ and has relative minimum/maximum at $x = -1$ and $x = \frac{1}{3}$. If $\int_{-1}^{1} f(x)dx = 14/3$, find the cubic $f(x)$. (JEE 1992)

The most straightforward way to do the problem would be to take $f(x)$ as $a_0 + a_1x + a_2x^2 + a_3x^3$ and use the data to get a system of four equations in the four unknowns a_0, a_1, a_2 and a_3 and then to solve it. But we can simplify the work considerably by first looking at the derivative $f'(x)$. Since $f(x)$ is a cubic, $f'(x)$ is a polynomial of degree 2. The second part of the data implies that it must have -1 and $\frac{1}{3}$ as roots. So $f'(x)$ must be of the form $k(x+1)(x-\frac{1}{3})$ for some constant k. Expanding, $f'(x) = k(x^2 + \frac{2}{3}x - \frac{1}{3})$. Taking antiderivatives, $f(x)$ must be of the form $\frac{kx^3}{3} + \frac{kx^2}{3} - \frac{kx}{3} + c$ for some constant c. Now we have only two constants, viz., k and c. We can determine them from the other pieces of data. The condition $f(-2) = 0$ gives, by direct substitution, $c = \frac{2k}{3}$.

We need one more equation. It is given by the fact that $\int_{-1}^{1} f(x)dx = \frac{14}{3}$. Again taking antiderivatives, we have

$$\begin{aligned}\int_{-1}^{1} f(x)dx &= \int_{-1}^{1}(\frac{kx^3}{3}+\frac{kx^2}{3}-\frac{kx}{3}+\frac{2k}{3})dx \\ &= \frac{kx^4}{12}+\frac{kx^3}{9}+\frac{kx^2}{6}+\frac{2kx}{3}\Big|_{-1}^{1} \\ &= 0+\frac{2k}{9}+0+\frac{4k}{3}=\frac{14k}{9} \qquad (20)\end{aligned}$$

Since we are given that this integral equals $\frac{14}{3}$ we get $14k = 42$, i.e., $k = 3$. The cubic $f(x)$ is now identified as $x^3 + x^2 - x + 2$. (Some further short cuts are still possible. For example, the interval $[-1, 1]$ is symmetric about the origin. As we shall see in Comment No. 12 of the next chapter, the integral of any odd function over such intervals is 0. So we could have as well omitted the terms of degree 3 and 1 when we took the integral of $f(x)$ over $[-1, 1]$.)

Comment No. 15:

Before we go on with the task of finding antiderivatives of more complicated functions, it is worth mentioning here a simple device which extends the applicability of the formulas we have learnt so far, and more generally, of any such formulas. Although only a special case of the rule of substitution which we shall study presently, it is well worth a special mention. To illustrate, we already know that $\int x^2dx = \frac{x^3}{3}$. Now suppose we want to find $\int(ax+b)^2dx$, where a and b are some constants. (We assume further that $a \neq 0$ as otherwise the problem is trivial.) One way to do this would be to expand $(ax+b)^2$ as $a^2x^2+2abx+b^2$. Integrating term by term and adding, we get $a^2\frac{x^3}{3}+abx^2+b^2x+c$ as an antiderivative of $(ax+b)^2$, where c is the constant of integration. This expression is not particularly elegant. But we can rewrite it as $a^2\frac{x^3}{3}+abx^2+b^2x+\frac{b^3}{a}+(c-\frac{b^3}{a})$ and thence as $\frac{1}{a}(ax+b)^3+c'$, where $c' = (c-\frac{b^3}{a})$. Now, as a, b and c are constants, so is c'. And the values assumed by c are the same as those assumed by c'. So we might as well call c' as c. (This is expressed by saying that $-\frac{b^3}{a}$ is 'absorbed' in c. In integration such replacements are made routinely.)

Thus we see that $\frac{1}{a}\frac{(ax+b)^3}{3}+c$ is also an antiderivative of $(ax+b)^2$. We could, of course, *verify* this by differentiation. But we have now *obtained* it. Similarly, obtain an antiderivative of $(ax+b)^3$ using the binomial theorem to expand it first. It will come out to be $\frac{1}{a}\frac{(ax+b)^4}{4}+c$. These examples lead us to guess that, in general, the following result would be true.

Theorem 3: If $F(x)$ is an antiderivative of $f(x)$, then for any constants a and b (with $a \neq 0$), $\frac{1}{a}F(ax+b)$ is an antiderivative of $f(ax+b)$.

It is trivial to prove this by a simple application of the chain rule since

$\frac{d}{dx}F(ax+b) = F'(ax+b)\frac{d}{dx}(ax+b) = af(ax+b)$. Dividing by a gives the result. The result, in fact, would not be worth stating were it not for the fact that it saves us an unnecessary strain on memory. Some students have the habit of remembering formulas like

$$\int \frac{1}{ax+b}dx = \frac{\ln(ax+b)}{a} \tag{21}$$

In view of what is said above, this is a consequence of the basic formula $\int \frac{1}{x}dx = \ln x$. Similarly, one of the standard formulas is $\int \frac{dx}{1+x^2} = \tan^{-1} x$. Once this is remembered, it is quite unnecessary to remember a formula for $\int \frac{dx}{x^2+a^2}$. We can rewrite $\frac{1}{x^2+a^2}$ as $\frac{1}{a^2}\frac{1}{1+(\frac{x}{a})^2}$. Then the theorem above implies that $\frac{1}{a}\tan^{-1}(\frac{x}{a})$ is an antiderivative of $\frac{1}{x^2+a^2}$. In fact, more generally, we have

$$\int \frac{dx}{(ax+b)^2+c^2} = \frac{1}{ac}\tan^{-1}\left(\frac{ax+b}{c}\right) \tag{22}$$

Good memory can, of course, save you some precious time. But if not aided with reasoning, a heavily taxed memory is prone to errors and can get you into trouble just as a heavily used chequebook unaided by sufficient bank balance can get you into jail!

As another interesting application of the theorem above, we have $\int \sin(ax+b)dx = -\frac{1}{a}\cos(ax+b)$ and $\int \cos(ax+b)dx = \frac{1}{a}\sin(ax+b)$ which follow, respectively, from the antiderivatives of $\sin x$ and $\cos x$. In particular we now have antiderivatives of $\sin nx$ and $\cos nx$ for any positive integer n. This is important because powers of $\sin x$ and $\cos x$ can be expressed in terms of the sines and cosines of multiples of x. (This is essentially, the converse of Exercise (7.21)(d).) For example, from $\cos^4 x = \frac{1}{4}(1+\cos 2x)^2 = \frac{1}{4} + \frac{1}{2}\cos 2x + \frac{1}{8}(1+\cos 4x) = \frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$, we get

$$\int \cos^4 x dx = \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c \tag{23}$$

Similarly $\sin 3x = 3\sin x - 4\sin^3 x$ gives $\int \sin^3 x dx = -\frac{3}{4}\cos x + \frac{1}{12}\cos 3x$. (For odd powers, there is an easier way, see (32) below.)

Comment No. 16:

Formulas (21) and (22) (with $a = 1$) are two of the three prototypes of a method of integrating a rational function, i.e., a function of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are two polynomials with real coefficients. Without loss of generality, we may suppose that the degree of $f(x)$ is less than that of $g(x)$. For otherwise, we can use the euclidean algorithm for polynomials mentioned in Comment No. 8 of Chapter 2 to write $\frac{f(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}$, where $q(x)$ (the

quotient) and $r(x)$ (the remainder) are polynomials with deg $(r(x)) < \deg(g(x))$. Here $q(x)$ being a polynomial, its integral is very easy to find and so the problem reduces to that of integrating the other term $\dfrac{r(x)}{g(x)}$ which is a rational function with the degree of the numerator less than that of the denominator.

So we suppose we want to integrate a rational function $\dfrac{f(x)}{g(x)}$ with deg $(f(x)) <$ deg $(g(x))$. We may further suppose that $g(x)$ is monic, as otherwise we can always take out its leading coefficient, say a and write $\dfrac{f(x)}{g(x)}$ as $\dfrac{1}{a}\dfrac{f(x)}{h(x)}$, where $h(x)$ is monic and has the same degree (and the same roots) as $g(x)$. We now recall from Chapter 3, Comment No. 2, that as a consequence of the Fundamental Theorem of Algebra, every real polynomial factors as the product of real factors some of which are linear (i.e., of the form $ax+b$) while the others are irreducible quadratics (i.e., of the form $ux^2 + vx + w$, with $v^2 < 4uw$, or equivalently, of the form $(ax+b)^2+c^2$ for some $c \neq 0$). If further, the polynomial is monic, then we may suppose that these factors are of the form $(x + b)$ or of the form $(x + b)^2 + c^2$ respectively. We apply this to the polynomial $g(x)$ in the denominator of our rational function $\dfrac{f(x)}{g(x)}$. So, let us assume, with a slight change of notation that

$$g(x) = (x - a_1)\dots(x - a_k) \times ((x + b_1)^2 + c_1^2))\dots((x + b_r)^2 + c_r^2)) \qquad (24)$$

where the quadratic factors are all irreducible. Here k and r are non-negative integers with $k + 2r =$ deg $(g(x))$. For the moment we assume that the linear as well as the quadratic factors are all distinct. The case of multiple factors is rather complicated and will be taken up later. With our assumptions it can be shown that for a given $f(x)$ (with degree lower than that of $g(x)$), there exist unique real numbers $A_1, A_2, \dots, A_k, B_1, \dots, B_r, C_1, \dots, C_r$ such that

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - a_1} + \dots + \frac{A_k}{x - a_k} + \frac{B_1x + C_1}{(x + b_1)^2 + c_1^2} + \dots + \frac{B_rx + C_r}{(x + b_r)^2 + c_r^2} \qquad (25)$$

The terms on the right are known as **partial fractions** of the rational function $\dfrac{f(x)}{g(x)}$. It is beyond our scope to prove the existence of such a decomposition for every rational function, where the degree of the denominator exceeds that of the numerator. Nor is it important for our purpose as we shall be interested only in particular problems, where the existence of a decomposition into partial fractions will follow by actually finding it. Once we have the decomposition (25), it is easy to find an antiderivative for $\dfrac{f(x)}{g(x)}$ by integrating each partial fraction separately. For terms of the form $\dfrac{A_i}{x - a_i}$ we can apply (21) directly. For partial fractions of the form $\dfrac{B_ix + C_i}{(x + b_i)^2 + c_i^2}$ if $B_i = 0$, then we can apply (22) directly. However, if $B_i \neq 0$, we first rewrite the numerator as

$B_i(x+b_i)+(C_i-B_ib_i)$. This gives the following splitting of the integral of the corresponding partial fraction

$$\int \frac{B_ix+C_i}{(x+b_i)^2+c_i^2}dx = \int \frac{B_i(x+b_i)}{(x+b_i)^2+c_i^2}dx + \int \frac{C_i-B_ib_i}{(x+b_i)^2+c_i^2}dx \tag{26}$$

The second term on the right can be evaluated using (22). For the first term we borrow a formula

$$\int \frac{x\,dx}{x^2+c^2} = \frac{1}{2}\ln(x^2+c^2) \tag{27}$$

which is easily verified by differentiation. (Like the other antiderivatives which we have borrowed readymade so far, later we shall see a way of *arriving at* (27).)

This is the third prototype we need for the method of partial fractions. Combining it with Theorem 3, we now have

$$\int \frac{x+b}{(x+b)^2+c^2}dx = \frac{1}{2}\ln((x+b)^2+c^2) \tag{28}$$

The success of this method depends on two things, first on our ability to factorise $g(x)$ completely as in (24) and secondly, on our ability to obtain the constants $A_1,\ldots,A_k,B_1,\ldots,B_r,C_1,\ldots,C_r$. The first part is very difficult and there is in general no method for doing it for arbitrary polynomials. However, polynomials of small degrees can often be factored by trial. Or sometimes the problem itself gives $g(x)$ in a factorised form. Once a factorisation is obtained, one way to determine the constants $A_1,\ldots,C_r$ is to bring the terms on the R.H.S. of (25) to a common denominator $g(x)$ and add. The numerator will then be a polynomial with degree at most $k+2r-1$ in which the coefficients will be some linear combinations of the constants $A_1,\ldots,C_r$. Comparing these with the corresponding coefficients of $f(x)$ we get a system of $2r+k$ linear equations in as many unknowns. Solving this system, we get the values of $A_1,\ldots,C_r$. This straightforward method is called, quite appropriately, the **method of undetermined coefficients**.

As a simple illustration, let us find $\int \frac{dx}{x^2-a^2}$. Here the denominator factors as a product of linear factors, viz., $(x-a)(x+a)$. So the partial fraction resolution of the integrand is of the form $\frac{A}{x-a}+\frac{B}{x+a}$, where A and B are to be determined. To this end, add the terms to get $\frac{(A+B)x+(A-B)a}{x^2-a^2}$. Equating this with $\frac{1}{x^2-a^2}$ we see that $(A+B)x+(A-B)a$ must identically equal 1. This is possible only when $A+B=0$ and $(A-B)a=1$. Solving, $A=\frac{1}{2a}$ and $B=-\frac{1}{2a}$. So we have $\frac{1}{x^2-a^2}=\frac{1}{2a}(\frac{1}{x-a}-\frac{1}{x+a})$. Applying (21), we get $\int \frac{dx}{x^2-a^2}=\frac{1}{2a}(\ln(x-a)-\ln(x+a))+c$ which can also be written as $\frac{1}{2a}\ln(\frac{x-a}{x+a})+c$.

After a little experience, the partial fractions of a function like $\dfrac{1}{(x+a)(x+b)}$ can be written down by inspection. Also, for an expression like $\dfrac{2x+5}{x^2-1}$, instead of rushing in with the method of undetermined coefficients, it is better to pause for a moment and split the numerator as $2(x+1)+3$. Then the given expression becomes $\dfrac{2}{x-1}+\dfrac{3}{x^2-1}$. The second term can be resolved by inspection as $\dfrac{3/2}{x-1}-\dfrac{3/2}{x+1}$ as just mentioned. So once again, no method should be resorted to blindly. Always be alert for short cuts.

In fact, because of our assumption that in (24) there are no multiple factors, there is always a slicker way to find the coefficients $A_1, A_2, \ldots, A_k$. We illustrate this for A_1. Since $(x-a_1)$ is a factor of $g(x)$, we can write $g(x)$as $(x-a_1)h_1(x)$. (In fact, $h_1(x)$ is just the product of all the factors in (24) excluding $(x-a_1)$.) Note that $h_1(a_1) \neq 0$, as otherwise $(x-a_1)$ would be a multiple factor of $g(x)$. Now we multiply both the sides of (24) by $(x-a_1)$ and take limits as $x \to a_1$. Then the limit of the first term on the R.H.S. will equal A_1 while that of every other term will equal 0. As for the L.H.S., we can cancel the factor $(x-a_1)$ from the denominator. Then its limit will be simply the value of $\dfrac{f(x)}{h_1(x)}$ at $x=a_1$. So, equating the limits of both the sides, we get that $A_1 = \dfrac{f(a_1)}{h(a_1)}$. Let us verify this for the partial fraction resolution of $\dfrac{1}{x^2-a^2}$ which we just obtained by the method of undetermined coefficients. Once again we let $\dfrac{1}{x^2-a^2} = \dfrac{A}{x-a}+\dfrac{B}{x+a}$. But this time, to find A we merely multiply $\dfrac{1}{x^2-a^2}$ by $(x-a)$ to get it as $\dfrac{1}{x+a}$. Setting $x=a$ in this gives A as $\dfrac{1}{2a}$. Similarly, B is obtained by setting $x=-a$ in $\dfrac{(x+a)}{x^2-a^2}$, i.e., in $\dfrac{1}{x-a}$. This way we get A and B with a lot less effort than before.

For the coefficients $B_1, \ldots, B_r$ and $C_1, \ldots, C_r$ in (25), there are no such short cuts unless we are prepared to factor the irreducible quadratics using complex numbers. Still, the fact that we can at least find the A_i's quickly, simplifies the computation of the B's and the C's. As an example, let us find $\int \dfrac{7x^3+6x^2-3x+2}{x^4-1}dx$. As the first step we factorise the denominator and write

$$\frac{7x^3+6x^2-3x+2}{x^4-1} = \frac{A}{x-1}+\frac{B}{x+1}+\frac{Cx+D}{x^2+1} \tag{29}$$

To find A, B, C and D by the method of undetermined coefficients will involve a system of 4 equations in 4 unknowns. But, if we use the method above we quickly identify A as the value of $\dfrac{7x^3+6x^2-3x+2}{(x^2+1)(x+1)}$ at $x=1$, i.e., as 3. Similarly, B comes out as -1. If we substitute these into (29) and add the terms on the R.H.S., its numerator comes out to be $(Cx+D)(x^2-1)+(2x+4)(x^2+1)$.

Expanding and equating with $7x^3+6x^2-3x+2$ gives a system of four equations in the two unknowns C and D, because the constant terms and the coefficients of x, x^2 and x^3 must all match. We need to take only two of these. Equating the constant terms, we get $-D+4=2$, i.e., $D=2$, while equating the coefficients of x^3 we get $C+2=7$ so that $C=5$. (We could have as well taken either of the other two equations. But it is clear from the expression that D cannot appear in the coefficient of x^3 while C cannot appear in the constant term. So it is better to take these two 'pure' equations instead of the other two 'mixed' ones. It is noteworthy that even in such minor matters, a little alertness can pay off in terms of time and accuracy.) Thus we have,

$$\begin{aligned}\int \frac{7x^3+6x^2-3x+2}{x^4-1}dx &= \int \frac{3\,dx}{x-1} - \int \frac{dx}{x+1} + \int \frac{5x+2}{x^2+1}dx \\ &= 3\ln|x-1| - \ln|x+1| \\ &\quad +\frac{5}{2}\ln(x^2+1) + 2\tan^{-1}x + c \qquad (30)\end{aligned}$$

Summing up, integrating a rational function by the method of partial fractions is a rather laborious job. But, assuming you can factorise the denominator, it is a sure method. Also, as we shall soon see, certain other integrals can be converted to those of rational functions. Hence one should be prepared to use partial fractions if need be.

Comment No. 17:

We now study an important technique of integration, called the **method of substitution**. This technique is extremely versatile and powerful. It will not only enable us to *derive* all the antiderivatives which we so far only borrowed ready made and verified, it will also enable us to integrate rational functions in which the denominator has repeated factors. In addition, it will enable us to obtain many other antiderivatives. In fact, there is no end to the applicability of the method of substitution. Coming up with the right substitution that will do the trick in a given problem can be a thrilling experience sometimes, comparable to hitting just the right construction that unfolds a pure geometry problem or solving some puzzle. So the technique of substitution to find antiderivatives is important mathematically, from an examination point of view and also as intellectual fun.

The essential idea in the method of substitution is to replace the variable x in an expression like $\int f(x)dx$ by an expression of the form $g(u)$, where u is some other variable and $x=g(u)$ is some (differentiable) function of u. As a result, $f(x)$ equals the composite function $f(g(u))$. Now suppose, $F(x)$ is an antiderivative of $f(x)$. Then, by definition, $\dfrac{d}{dx}(F(x)) = f(x)$. Now consider the composite function $F(g(u))$. This is a function of u and by the chain rule we have $\dfrac{d}{du}(F(g(u)) = F'(g(u))\dfrac{dx}{du} = f(g(u))g'(u)$. Hence, again by definition, $F(g(u))$ is an antiderivative of $f(g(u))g'(u)$. Let us rewrite this using the integral sign

as

$$\int f(x)dx = \int f(g(u))g'(u)du = \int f(g(u))\frac{dx}{du}du \tag{31}$$

This is the formal statement of the **rule of substitution** for indefinite integrals. In essence it is no different than a paraphrase of the chain rule for derivatives and, therefore, hardly a deep result. But it is a very useful result because if the function $g(u)$ is chosen judiciously, then the second term in (30) may be much easier to find than the first term. As a result, we get the answer in terms of u. We must convert this back in terms of x to write an antiderivative of the original function.

As a good illustration, in Comment No. 5 above, while finding the area of a sector of a circle we simply borrowed an antiderivative of the function $f(x) = \sqrt{r^2 - x^2}$. But now we can *arrive* at it by a simple substitution, $x = r\sin u$. (That this is a valid substitution follows from the fact that in the original problem the variable x could only assume values between $-r$ and r, and indeed only between 0 and r as the sector was assumed to be acute angled. But such questions of validity of the substitution are generally skipped at elementary levels. The rationale taken is that the substitution is used only as a tool and as long as the antiderivative obtained using it works, why bother about the sanctity of the tool?) So here $g(u) = r\sin u$ and hence $g'(u) = r\cos u$. $\sqrt{r^2 - x^2}$ becomes $r\cos u$. So $f(g(u))g'(u) = r^2\cos^2 u$. To integrate this w.r.t. u, we use a method similar to that used in (23). We write $\cos^2 u$ as $\frac{1}{2} + \frac{1}{2}\cos(2u)$. Then $\int r^2\cos^2 u du = \frac{r^2 u}{2} + \frac{r^2}{4}\sin(2u)$. To convert this back in terms of x we put $u = \sin^{-1}(\frac{x}{r})$ and $\sin 2u = 2\sin u\cos u = \dfrac{2x\sqrt{r^2 - x^2}}{r^2}$. This gives us an antiderivative of $\sqrt{r^2 - x^2}$.

After gaining some practice it is hardly necessary to elaborately introduce the functions f and g. We simply say, 'Put $x = r\sin u$'. Also instead of saying $\dfrac{dx}{du} = r\cos u$, it is quite all right to split the differentials and write '$dx = d(r\sin u) = r\cos u\ du$'. The work above is then customarily written in the following form which everybody understands.

$$\begin{aligned}
\int \sqrt{r^2 - x^2}\, dx &= \int \sqrt{r^2 - r^2\sin^2 u}\, d(r\sin u) \quad (\text{with } x = r\sin u) \\
&= \int r^2\cos^2 u\ du \\
&= \int \frac{r^2}{2}(1 + \cos 2u)\ du \\
&= \frac{r^2}{2}(u + \frac{1}{2}\sin 2u) + c \\
&= \frac{r^2}{2}\sin^{-1}(\frac{x}{r}) + \frac{r^2}{2}\sin u\cos u + c \\
&= \frac{r^2}{2}\sin^{-1}(\frac{x}{r}) + \frac{1}{2}x\sqrt{r^2 - x^2} + c
\end{aligned}$$

and, as usual, some of the intermediate steps may be skipped with practice.

Of course, the question that still remains is how in a problem like this, one thinks of the substitution $x = r \sin u$ in the first place. To this there is no mathematical answer. But generally one takes a cue from standard trigonometric and other identities so as to get rid of the radicals. In the present problem the substitution $x = r \cos u$ could also have worked. If the integrand involved, say $\sqrt{x^2 - r^2}$, then the right thing to try would be $x = r \sec u$, based on the identity $\tan^2 u + 1 = \sec^2 u$. (Another possibility would be to put $x = r \cosh u$ and use the identity about hyperbolic sines and cosines, viz., $\sinh^2 u + 1 = \cosh^2 u$.)

Comment No. 18:

In a majority of problems the substitution needed is not of the form $x = g(u)$ as described above, but of the opposite type, i.e., where instead of expressing x as a function of u, u is expressed as a function of x, say $u = h(x)$. Of course, in this case we may take the inverse function and write $x = h^{-1}(x)$. Whether the function h is invertible is one of those questions that are not considered when using the substitution as a tool. So the two types of substitutions are not radically different. But in a particular problem, one type may be easier to conceive than the other. For example, the substitution $x = r \sin u$ used in the problem above could also have been expressed in this new form by $u = \sin^{-1}(\frac{x}{r})$ which we used anyway while converting the answer back in terms of x. But looking at the expression $\sqrt{r^2 - x^2}$, it is a little easier to think of putting x as $r \sin u$ rather than u as $\sin^{-1}(\frac{x}{r})$. In some problems the situation is the other way round.

In Theorem 3 in Comment No. 15 above, we already encountered this type of substitution. There the function was $h(x) = ax + b$ which was invertible since $a \neq 0$. The same proof given there would apply, more generally, to show that if $F(x)$ is an antiderivative of $f(x)$, then for any differentiable function $h(x)$ of x, $\frac{d}{dx}F(h(x)) = F'(h(x))h'(x) = f(h(x))h'(x)$. It follows that $F(h(x))$ is an antiderivative of $f(h(x))h'(x)$,i.e.,

$$\int f(h(x))h'(x)\, dx = F(h(x)) \tag{32}$$

In other words, if we can cast the integrand in a form $f(h(x))h'(x)$ for some suitable function $h(x)$ of x, then the problem of finding the antiderivative reduces to that of finding an antiderivative of the function $f(u)$ w.r.t. u. For example, $\sin^3 x = (\cos^2 x - 1)(-\sin x)$ shows that the substitution $u = \cos x$ will work for $\int \sin^3 x\, dx$. Sometimes some recasting may be needed for (32) to apply. For example, we cannot apply it to $\int \dfrac{dx}{(5 - \sin x)\cos x}$ as it is. But if we multiply both the numerator and the denominator by $\cos x$, then substitution $u = \sin x$ will work after writing $\cos^2 x$ in the denominator as $1 - \sin^2 x$.

As a special, but important case (besides Theorem 3) we have

$$\int \frac{h'(x)}{h(x)} dx = \ln h(x) + c \tag{33}$$

which follows by putting $u = h(x)$ which transforms the L.H.S. to $\int \frac{du}{u}$, a standard integral. We already obtained (27) as a special case of this. Other important antiderivatives we get this way are of the basic trigonometric functions. From the derivatives of $\sin x$ and $\cos x$, we know that $-\tan x = \dfrac{-\sin x}{\cos x}$ is in a form to which (33) can be applied with $h(x) = \cos x$, giving $\int -\tan x \; dx = \ln \cos x$. $\int \cot x \; dx$ is handled similarly. Thus,

$$\int \tan x \; dx \quad = \quad -\ln \cos x = \ln \sec x \tag{34}$$

$$\text{and} \quad \int \cot x \; dx \quad = \quad \ln \sin x = -\ln \; \text{cosec}\, x \tag{35}$$

An antiderivative of $\sec x$ is obtained by writing it as $\dfrac{\sec^2 x + \tan x \sec x}{\tan x + \sec x}$ to which (33) can be applied. A similar manipulation holds for $\int \; \text{cosec}\, x \; dx$. These results are, of course, very standard. But we have presented them as illustrations of (33) which is a special case of (32).

Returning to the more basic formula (32), let us apply it to find $\displaystyle\int \frac{x}{1+x^4} dx$. (JEE 1978) This could be done by partial fractions. But the factorisation of the denominator is cumbersome, viz., $(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$. Instead, we take advantage of the fact that the denominator contains no odd degree terms and hence can be expressed as a function of x^2. As for the numerator, there is only one factor x, which together with dx is simply $\frac{1}{2}d(x^2)$. So the substitution $u = x^2$ reduces the problem to finding $\frac{1}{2}\int \dfrac{du}{1+u^2}$, which, being of a standard type, equals $\frac{1}{2}\tan^{-1} u$. Hence the original antiderivative is $\frac{1}{2}\tan^{-1}(x^2)$.

Comment No. 19:

Formula (32) also allows us to apply the method of partial fractions in the general case, i.e., without the assumption that the denominator has no repeated factors. The full statement of the general situation will be notationally complicated. But we illustrate how the case of multiple factors is handled. Assume as before that we have a rational function $\dfrac{f(x)}{g(x)}$, where the degree of the denominator exceeds that of the numerator. Suppose a_1 is a real root of $g(x)$ of multiplicity m. This means that factor $(x-a_1)^m$ appears in the factorisation of $g(x)$ instead of just the factor $(x-a_1)$ as in (24). Consequently, instead of the first term on the R.H.S. of (25) (viz., $\dfrac{A_1}{x-a_1}$) we now have a sum of m terms of the type

$$\frac{D_1}{x-a_1} + \frac{D_2}{(x-a_1)^2} + \ldots + \frac{D_m}{(x-a_1)^m} \tag{36}$$

where $D_1, D_2, \ldots, D_m$ are some real numbers. Just as there is a short cut to find the number A_1 in (25), there is an easy method to find D_m by taking

$\lim_{x\to a_1} \dfrac{(x-a_1)^m f(x)}{g(x)}$. The other constants $D_{m-1}, D_{m-2}, \ldots, D_2$ and D_1, can be determined by taking the successive derivatives of this ratio at a_1. But the calculations are a little more involved. The method of undetermined coefficients (applied to these D's along with the constants corresponding to all other factors of $g(x)$) is, of course, available.

The situation about repeated quadratic factors is similar. Suppose the factor $(x+b_1)^2 + c_1^2$ (where $c_1 \neq 0$) appears with multiplicity m. Then in the partial fraction decomposition of $\dfrac{f(x)}{g(x)}$, instead of the term $\dfrac{B_1 x + C_1}{(x+b_1)^2 + c_1^2}$ we now have a battery, viz.,

$$\frac{E_1 x + F_1}{(x+b_1)^2 + c_1^2} + \frac{E_2 x + F_2}{((x+b_1)^2 + c_1^2)^2} + \ldots + \frac{E_m x + F_m}{((x+b_1)^2 + c_1^2)^m} \tag{37}$$

where $E_1, \ldots, E_m$ and $F_1, \ldots, F_m$ are some real numbers.

Thus, to find an antiderivative of a rational function using partial fractions, it is enough if we can integrate each of the three prototypes $\dfrac{1}{(x-a)^i}$, $\dfrac{x}{((x+b)^2+c^2)^i}$ and $\dfrac{1}{((x+b)^2+c^2)^i}$, where a, b, c are real, $c \neq 0$ and i is an integer greater than 1. (The case $i = 1$ corresponds to a simple factor.) The first one comes easily using the substitution $u = x - a$. Its integral is $\dfrac{-1}{(i-1)(x-a)^{i-1}}$. As for $\displaystyle\int \frac{x\,dx}{((x+b)^2+c^2)^i}$, we split it as

$$\int \frac{x\,dx}{((x+b)^2+c^2)^i} = \frac{1}{2}\int \frac{2(x+b)\,dx}{((x+b)^2+c^2)^i} - b\int \frac{dx}{((x+b)^2+c^2)^i} \tag{38}$$

The first term on the R.H.S. can be found with the substitution $u = (x+b)^2 + c^2$. The second term (which is also the third prototype listed above) can be handled with the substitution $x + b = c\tan u$ which gives $(x+b)^2 + c^2 = \sec^2 u$ and $dx = c\sec^2 u\ du$. This transforms the integral $\displaystyle\int \frac{dx}{((x+b)^2+c^2)^i}$ into $\int \cos^{2i-2} u\ du$ which can be handled in the manner indicated above for (23), i.e., by expressing $\cos^{2i-2} u$ as a linear combination of $1, \cos 2u, \cos 4u, \ldots, \cos((2i-2)u)$. There is little point in writing down all these formulas elaborately. Instead we illustrate the method by finding $\displaystyle\int \frac{x^3+3x+2}{(x^2+1)^2(x+1)}dx$. (JEE 1999) Here the denominator has $(x^2+1)^2$ as the only multiple factor. Instead of applying the method of undetermined coefficients to resolve into partial fractions, let us try to rewrite the numerator so that the terms may have some common factors with the denominator. By inspection, one such break-up is $x^3 + 3x + 2 = (x^3 + x) + (2x + 2) = x(x^2+1) + 2(x+1)$. Hence we have

$$\int \frac{x^3+3x+2}{(x^2+1)^2(x+1)}dx = \int \frac{x\,dx}{(x^2+1)(x+1)} + \int \frac{2\,dx}{(x^2+1)^2} \tag{39}$$

As just noted, the substitution $x = \tan u$ converts the second integral to $\int 2\cos^2 u \; du$, i.e., to $\int(1 + \cos 2u) \; du$ which equals $u + \frac{1}{2}\sin 2u$ i.e., $u + \sin u \cos u$, or $\tan^{-1} x + \dfrac{x}{x^2+1}$. As for the first integral on the R.H.S. of (39), the denominator has no repeated factors. Let $\dfrac{x}{(x^2+1)(x+1)} = \dfrac{A}{x+1} + \dfrac{Bx+C}{x^2+1}$. Then A is the value of $\dfrac{x}{x^2+1}$ at $x = -1$. Hence $A = -\frac{1}{2}$. Add the terms on the R.H.S. and equate the numerator with x. From the coefficient of x^2 we get $-\frac{1}{2} + B = 0$. Hence $B = \frac{1}{2}$. Also equating the constant term with 0, we get $C = \frac{1}{2}$. So, $\dfrac{x}{(x^2+1)(x+1)} = \dfrac{1}{2}(\dfrac{-1}{x+1} + \dfrac{x}{x^2+1} + \dfrac{1}{x^2+1})$. Integrating both the sides w.r.t. x, we get $\displaystyle\int \frac{x\,dx}{(x^2+1)(x+1)}$ as the sum $-\frac{1}{2}\ln(x+1) + \frac{1}{4}\ln(x^2+1) + \frac{1}{2}\tan^{-1} x$. So from (39), the given indefinite integral equals $-\dfrac{1}{2}\ln(x+1) + \dfrac{1}{4}\ln(x^2+1) + \dfrac{3}{2}\tan^{-1} x + \dfrac{x}{x^2+1} + c$. In problems like this it is a good idea to check that the answer is correct by differentiating it. We leave this verification to you.

Comment No. 20:

We emphasize again that just because we have a rational function (with the denominator already factored or easily factorisable) to integrate, it does not mean that we should mechanically resolve it into partial fractions. Alternate methods should always be explored. As a good illustration, let us find $\displaystyle\int \frac{x^2\,dx}{(a+bx)^2} dx$. (JEE 1979) Here the degree of the numerator equals that of the denominator and so before we can resolve into partial fractions we would have to apply the division algorithm. Thereafter we shall have to deal with the multiple factor in the denominator. But there is a way to bypass all this if we look at the expression $\dfrac{x^2}{(a+bx)^2}$ a little closely. It cannot be easily split because even though the numerator has just one term, the denominator is a sum of terms. (We assume here that $b \neq 0$ as otherwise the problem would be trivial.) If it were the other way round, that is, if the denominator had just one term, then we could split the numerator and accordingly the ratio too. But this can be arranged with a change of variable, i.e., with a suitable substitution! Simply put $u = a + bx$. Then $x = \frac{u-a}{b}$ and $dx = \frac{1}{b}du$. So the given integral transforms into $\displaystyle\int \frac{(u-a)^2\,du}{b^3u^2}$ which can be evaluated by splitting the integrand as $\dfrac{1}{b^3}(1 - \dfrac{2a}{u} + \dfrac{a^2}{u^2})$. The desired antiderivative is, therefore, $\dfrac{1}{b^3}(u - 2a\ln u - \dfrac{a^2}{u}) + c$. To put it back in terms of the original variable x, all we have to do now is to put $u = a + bx$. This part is absolutely mechanical and whether to insist upon it in a particular examination is a matter of policy. To play it safe, it is better to do the conversion. Also in the interest of complete-

ness, one must deal with the case $b = 0$, howsoever trivial it is. If $b = 0$ then $a \neq 0$ and the integrand is simply $\dfrac{x^2}{a^2}$. So an antiderivative is $\dfrac{x^3}{3a^2} + c$.

Comment No. 21:

As mentioned before, certain other integrands can be converted to rational functions using suitable substitutions. One standard substitution of this type is $u = \tan\frac{x}{2}$. Because of the identities $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2} = \dfrac{2\tan\frac{x}{2}}{\sec^2\frac{x}{2}} = \dfrac{2u}{1+u^2}$ and $\cos x = \cos^2\frac{x}{2} - \sin^2\frac{x}{2} = \dfrac{1-\tan^2\frac{x}{2}}{\sec^2\frac{x}{2}} = \dfrac{1-u^2}{1+u^2}$ we can convert any algebraic expression in the trigonometric functions of x into a rational function in u. Note that $\frac{1}{2}\sec^2\frac{x}{2}\,dx = du$ and hence $dx = \dfrac{2\,du}{1+u^2}$.

Let us evaluate $\int \dfrac{\sin x}{\sin x - \cos x}dx$ using this substitution. (JEE 1978) With the formulas just given the integral transforms into $\displaystyle\int \frac{4u\,du}{(1+u^2)(2u+u^2-1)}$. The denominator factorises as $(u+1+\sqrt{2})(u+1-\sqrt{2})(u^2+1)$. We leave it to you to find the partial fraction resolution of the integrand. It comes out as

$$\frac{4u}{(u^2+2u-1)(u^2+1)} = \frac{1/2}{u+1+\sqrt{2}} + \frac{1/2}{u+1-\sqrt{2}} + \frac{-u+1}{u^2+1} \tag{40}$$

Integrating term by term and combining the logarithmic terms together,

$$\int \frac{4u\,du}{(u^2+2u-1)(u^2+1)} = \frac{1}{2}\ln\left(\frac{u^2+2u-1}{u^2+1}\right) + \tan^{-1}u + c$$

Recalling that $u = \tan\frac{x}{2}$ and $\dfrac{2u}{u^2+1} = \sin x$ etc. we get

$$\int \frac{\sin x}{\sin x - \cos x}dx = \frac{1}{2}\ln(\sin x - \cos x) + \frac{x}{2} + c \tag{41}$$

This was a straightforward calculation. But, once again, it is not the best method! With a clever trigonometric manipulation, the problem can be done without *any* substitution. The idea is to note that the denominator $\sin x - \cos x$ is a factor of $\sin^2 x - \cos^2 x$ which can be written as a single term, viz., $-\cos 2x$. So multiplying both the numerator and the denominator by $\sin x + \cos x$ and with further simplification we get

$$\begin{aligned}
\frac{\sin x}{\sin x - \cos x} &= \frac{\sin^2 x + \sin x\cos x}{\sin^2 x - \cos^2 x} \\
&= \frac{1}{2}\frac{1 - \cos 2x + \sin 2x}{-\cos 2x} \\
&= \frac{1}{2}(1 - \tan 2x - \sec 2x)
\end{aligned}$$

Term by term integration gives

$$\int \frac{\sin x\, dx}{\sin x - \cos x} = \frac{x}{2} - \frac{1}{4}\ln(\sec 2x) - \frac{1}{4}\ln(\tan 2x + \sec 2x) + c \qquad (42)$$

which is an equally valid answer. If we combine the logarithmic terms in (42) together we get $-\frac{1}{4}\ln\frac{\cos 2x}{\tan 2x + \sec 2x} = \frac{1}{4}\ln\frac{\cos^2 2x}{1+\sin 2x}$. If we write $1 + \sin 2x$ as $(\sin x + \cos x)^2$, then this further reduces to $\frac{1}{2}\ln\frac{\cos 2x}{\sin x + \cos x}$. Writing the numerator as $\cos^2 x - \sin^2 x$ this finally becomes $\frac{1}{2}\ln(\cos x - \sin x)$. This does not quite tally with (41) where we have $\frac{1}{2}\ln(\sin x - \cos x)$. This happens because in taking the various square roots, we have not bothered about the signs. At any rate, if we keep in mind that $\int \frac{dt}{t} = \ln|t|$ no matter whether t is positive or negative, we can replace $\ln(\sin x - \cos x)$ by $\ln|\sin x - \cos x|$ in (41). A similar change in (42) shows that (42) is the same answer as (41) (which is, however, not mandatory, since the same function can have two antiderivatives differing by a constant).

Even if you cannot think of the (rather tricky) manipulation needed to get (42), everything is not lost if you can think of some other substitution. In finding $\int \frac{x^2}{(a+bx)^2}dx$ in Comment No. 20, the substitution was motivated by the desire to reduce the denominator to a single term. A similar consideration applies to $\frac{\sin x}{\sin x - \cos x}$. We can write $\sin x - \cos x$ as $\sqrt{2}\sin(x - \frac{\pi}{4})$, which is a single term. So we put $u = x - \frac{\pi}{4}$ which gives $dx = du$ and reduces $\int \frac{\sin x}{\sin x - \cos x}dx$ to $\frac{1}{\sqrt{2}}\int \frac{\sin(u + \frac{\pi}{4})}{\sin u}du$. Expanding the numerator this comes to be $\frac{1}{2}\int(1 + \cot u)\, du$, i.e., $\frac{u}{2} + \frac{1}{2}\ln\sin u + c$. Putting back $u = x - \frac{\pi}{4}$, the antiderivative becomes $\frac{x}{2} - \frac{\pi}{8} + \frac{1}{2}\ln(\sin x - \cos x) - \frac{1}{4}\ln 2 + c$ which differs from (42) by a constant. But that is all right.

Whenever a problem can be done by several methods, it is a good idea to compare them in terms of their generality. In the present case, the second method is the least general. It worked only because the denominator $\sin x - \cos x$ was a factor of $\cos 2x$. The third method, on the other hand, will work if, we instead had any linear combination of $\sin x$ and $\cos x$ in the denominator. But it will not work if we had a combination of $\sin x$ and $\cos 2x$ etc. The first method is the most general. But the price you pay is that after the problem is reduced to that of integrating a rational function, it may not be easy to factorise the denominator completely. For example, try it for an integral like $\int \frac{\sin x\, dx}{\cos x + \cos 2x}$. But if we recognise that the denominator can be expressed in terms of $\cos x$ while $\sin x\, dx = d(\cos x)$ except for a sign, we see that the substitution $u = \cos x$ will work.

Comment No. 22:

Summing up, there is no golden method which is the best in all situations. Nor is there any general method to tell beforehand which substitution will work

in a given problem. When the integrand can be cast in the form $f(h(x))h'(x)$, (32) applies. When there is no obvious way to do this, the idea is that the substitution should get rid of the trouble makers such as a radical or a sum of two terms in the denominator. We already had examples of both.

Sometimes the substitution that will do the trick does not come naturally in one shot. That is, we may try a substitution which will not quite solve the problem but nevertheless will bring it to a form from which another substitution will lead to a solution. We illustrate this with a couple of examples. First, suppose we want to find $\int \frac{dx}{x^2(x^4+1)^{3/4}}$. (JEE 1984) To get rid of the ugly surd in the denominator we put $x = \sqrt{\tan u}$ so that $dx = \frac{\sec^2 u}{2\sqrt{\tan u}}du$. The integral now reduces to $\int \frac{\sec^2 u\ du}{2\sqrt{\tan u}\ \tan u\ \sec u\ \sqrt{\sec u}}$ which still involves radicals but is somewhat better than the original function in that it does not involve a root of a sum. In fact, on simplification, this integral is simply $\int \frac{\cos u\ du}{2\sin u\sqrt{\sin u}}$. So the substitution $\sin u = v$ will now work. It converts the integral to $\frac{1}{2}\int (v)^{-\frac{3}{2}}dv$ which equals $-v^{-\frac{1}{2}} + c = \frac{-1}{\sqrt{v}} + c$. In terms of u this equals $\frac{-1}{\sqrt{\sin u}} + c$ and in terms of the original variable x, it equals $-\frac{(x^4+1)^{1/4}}{x} + c$. (A slicker, albeit somewhat trickier, solution is to write the denominator of the integrand as $x^5(1+x^{-4})^{3/4}$ and then put $u = 1+x^{-4}$ to transform the integral to $-\int \frac{du}{4u^{3/4}}$.)

As our second example of this type (and as our last example of the method of substitution), we find $\int (x^{3m} + x^{2m} + x^m)(2x^{2m} + 3x^m + 6)^{1/m}dx$ for $x > 0$, where m is a natural number. (JEE 2002) As all the terms are various powers of x^m, the first substitution that comes naturally to the mind is $u = x^m$. With this substitution $mx^{m-1}dx = du$, which on multiplication by x becomes $mx^m dx = x\ du = u^{1/m}du$. Writing the first factor of the given integrand as $(x^{2m} + x^m + 1)x^m$, the given integral now becomes $\frac{1}{m}\int (u^2+u+1)(2u^2+3u+6)^{1/m}u^{1/m}du$ which can be further rewritten as $\frac{1}{m}\int (u^2+u+1)(2u^3+3u^2+6u)^{1/m}du$. But now we observe that u^2+u+1 is the derivative of $2u^3+3u^2+6u$ except for a coefficient. So as our second substitution, we put $v = 2u^3+3u^2+6u$ so that $dv = 6(u^2+u+1)\ du$. The integral now becomes $\frac{1}{6m}\int v^{1/m}dv = \frac{1}{6(m+1)}v^{1+\frac{1}{m}} + c$. In terms of the original variable, this equals $\frac{1}{6(m+1)}(2x^{3m} + 3x^{2m} + 6x^m)^{1+\frac{1}{m}} + c$.

Comment No. 23:

Even though the technique of substitution is so powerful, in essence it is

just a restatement of the chain rule for derivatives. We now give a technique of integration which is based on another frequently used rule about derivatives, viz., the product rule. Let us state it in the language of antiderivatives. Suppose $F(x)$ and $G(x)$ are antiderivatives of two continuous functions $f(x)$ and $g(x)$ respectively. Then $F'(x) = f(x)$ and $G'(x) = g(x)$. Now, by the product rule for derivatives $\dfrac{d}{dx}(F(x)G(x)) = f(x)G(x) + F(x)g(x)$. Let us rewrite this as

$$F(x)g(x) = \frac{d}{dx}(F(x)G(x)) - F'(x)G(x) \tag{43}$$

This is hardly new or profound. But let us see what we get if we take the antiderivatives of both the sides. Of course, the antiderivatives may differ by a constant. So we get

$$\int F(x)g(x)dx = F(x)G(x) - \int F'(x)G(x)dx + c \tag{44}$$

where c is a constant. A verbal paraphrase of this result is instructive. Suppose we want to integrate a function which is given as a product of two functions, $F(x)g(x)$. Sometimes it so happens that we cannot easily integrate this product but we can integrate it partly in the sense that we can find an antiderivative $G(x)$ for the factor $g(x)$. Then the formula above reduces the problem of integrating $F(x)g(x)$ to that of integrating the product $F'(x)G(x)$. If the latter can be solved, then (44) gives a formula for an antiderivative of $F(x)g(x)$. As an example, suppose we want to find $\int x\cos x\ dx$. Take $F(x)$ as x and $g(x)$ as $\cos x$. Then we already know an antiderivative of $g(x)$, viz., $\sin x$. We call it $G(x)$. Clearly $F'(x) = 1$. So by (44), $\int x\cos x\ dx = x\sin x - \int 1.\sin x\ dx$. As we already know the last integral as $-\cos x$, we get

$$\int x\cos x\ dx = x\sin x + \cos x + c \tag{45}$$

whose correctness can be verified by differentiating the R.H.S.

This method of finding antiderivatives is known as **integration by parts** because it begins by integrating a 'part' or a factor, viz., $g(x)$ of the integrand. The functions $F(x)$ and $g(x)$ are sometimes called the first and the second functions respectively. With a little practice, it is, of course, quite unnecessary to identify them explicitly or to give symbols to them. (Nor is it necessary that the 'first' function should always be written as the first factor of the integrand. The nomenclature is based on the role played by the factor and not on its position.) Note that although in theory either of the two factors of the integrand can be taken as the first function, in practice this may be of little value if we are not able to compute the integral that arises from it. For example, in the integral $\int x\cos x\ dx$ just considered, we know an antiderivative of the factor x too, viz., $\dfrac{x^2}{2}$. But if we take x as the second function and apply integration by parts, then instead of (45) we shall get

$$\int x\cos x\ dx \quad = \quad \frac{x^2}{2}\cos x - \int \frac{x^2}{2}\frac{d}{dx}(\cos x)\ dx$$

$$= \frac{x^2}{2}\cos x + \int \frac{x^2}{2}\sin x\ dx \tag{46}$$

which is true but useless since we now have to evaluate an integral which is more complicated than the one with which we started! (Here we are assuming that our sole goal is to find an antiderivative. Sometimes we are in a situation where we cannot evaluate an integral exactly anyway and have to settle for an approximation. In that case converting that integral to a more complicated one which we cannot evaluate either, may not always be a bad idea if this more complicated integral can be estimated more accurately than the original integral. See Comment No. 9 of the next chapter and also Exercise (18.31).)

Sometimes the integral $\int F'(x)G(x)\ dx$ that appears in the R.H.S. of (44) again has to be evaluated by parts. If we do so taking $G(x)$ as the first and $F'(x)$ as the second function, then we get back to where we started, i.e., at having to find $\int F(x)g(x)\ dx$. This is, of course, quite silly. But if we can integrate $F'(x)G(x)$ by taking $G(x)$ as the second function (i.e., by taking an antiderivative, say $H(x)$, of it) then the problem reduces to finding $\int F''(x)H(x)dx$. This process can continue.

As an illustration, we evaluate $\int_0^{\pi/2} x^2\cos x\ dx$. (JAT 1980) This is a definite integral. But if we first find an indefinite integral, then Theorem 1 can be applied to find the definite integral. To find the indefinite integral, we integrate by parts taking x^2 as the first function and then again applying integration by parts to the integral that results as shown below.

$$\begin{aligned}
\int x^2\cos x\ dx &= x^2\sin x - \int 2x\sin x\ dx \\
&= x^2\sin x - \int 2x\frac{d}{dx}(-\cos x)\ dx \\
&= x^2\sin x + 2x\cos x + \int \frac{d}{dx}(2x)(-\cos x)\ dx \\
&= x^2\sin x + 2x\cos x - \int 2\cos x\ dx \\
&= x^2\sin x + 2x\cos x - 2\sin x + c
\end{aligned} \tag{47}$$

After a little practice, some of the intermediate steps may be omitted. But care has to be taken about the signs of the terms. From (47) and Theorem 1, we have

$$\int_0^{\pi/2} x^2\cos x\ dx = x^2\sin x + 2x\cos x - 2\sin x\Big|_0^{\pi/2} = \frac{\pi^2}{4} - 2 \tag{48}$$

Instead of obtaining the antiderivative first and then using Theorem 1, the two steps can be combined if one uses the following version of (44) which is valid for definite integrals over an interval $[a, b]$.

$$\int_a^b F(x)g(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b F'(x)g(x)dx \tag{49}$$

In that case we would be putting the upper and lower limits on all integrals in the derivation of (47).

Sometimes the integrand is not given as a product of two functions. We can still apply integration by parts if we take the constant 1 as the second function. As a simple illustration, consider $\int \ln x \; dx$. We write the integrand as $\ln x \times 1 = \ln x \times \dfrac{d}{dx}x$. Then integration by parts gives

$$\int \ln x dx = x \ln x - \int x \frac{d}{dx}(\ln x) dx = x \ln x - x + c \tag{50}$$

Similarly, we can evaluate $\int \tan^{-1} x \; dx$, $\int \sin^{-1} x \; dx$ etc.

Comment No. 24:

The fact that the exponential function e^x is its own derivative and hence also its own integral has certain interesting implications in integration by parts when this function is one of the two factors of the integrand. Suppose, for example, that we want to find $\int e^{ax} \sin bx \; dx$, where a and b are some (non-zero) real numbers. Since we already know $\frac{1}{a}e^{ax}$ as an antiderivative of e^{ax}, integration by parts gives

$$\int e^{ax} \sin bx \; dx = \frac{1}{a}e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos bx \; dx \tag{51}$$

Applying integration by parts again to the second term on the right, we have

$$\begin{aligned} \int e^{ax} \sin bx \; dx &= \frac{1}{a}e^{ax} \sin bx - \frac{b}{a}(\frac{1}{a}e^{ax} \cos bx + \frac{b}{a} \int e^{ax} \sin bx \; dx) \\ &= \frac{1}{a}e^{ax} \sin bx - \frac{b}{a^2}e^{ax} \cos bx - \frac{b^2}{a^2} \int e^{ax} \sin bx \; dx \end{aligned} \tag{52}$$

This may seem like an exercise in futility, because now we have to find the same integral with which we started in the first place! More applications of integrating by parts would continue the cycle endlessly. But if we look carefully, we see that the last term of the R.H.S. of (52) is not quite the same as its L.H.S. In fact, if we transport it to L.H.S. then with a little further manipulation we get

$$\int e^{ax} \sin bx \; dx = \frac{e^{ax}}{a^2 + b^2}(a \sin bx - b \cos bx) + c \tag{53}$$

In an entirely analogous manner we have

$$\int e^{ax} \cos bx \; dx = \frac{e^{ax}}{a^2 + b^2}(a \cos bx + b \sin bx) + c \tag{54}$$

(which also follows by putting (53) into (51)). The interesting part of the derivation of (53) was that normally we try to evaluate something unfamiliar by relating it, through a sequence of manipulations, to something we already

know. But in (52) we ended up with something that was almost but not quite equal to what we started with and the difference between the two enabled us to get an equation for the thing we were after. We shall see more instances of this rather unexpected situation in the next chapter. (See also Exercise (17.24).)

Another interesting consequence of the relation $\frac{d}{dx}e^x = e^x$ in integration by parts occurs when we want to integrate a function of the form $e^x(f(x)+f'(x))$, where $f(x)$ is some differentiable function of x. It is most natural to try to do this by separately integrating $e^x f(x)$ and $e^x f'(x)$. For the latter, integration by parts is tempting with $f(x)$ as the second function. Thus we get $\int e^x f'(x)dx = e^x f(x) - \int e^x f(x)dx$. So we again have to evaluate $\int e^x f(x)dx$. But note that now it occurs with a minus sign and hence without doing any further work, it simply cancels with $\int e^x f(x)dx$ which was a part of the original integral. The result can be put as

$$\int e^x(f(x) + f'(x)dx = e^x f(x) \tag{55}$$

which can also be verified by differentiation. The interesting part is that in some situations we may not be able to find either $\int e^x f(x)dx$ or $\int e^x f'(x)dx$ individually. But because of (55), we can easily find their sum. In fact in doing so, we have to do virtually nothing once we recognise that in the given integrand, the coefficient of e^x was the sum of a function and its derivative.

As an illustration, let us find $\int \frac{(x-1)e^x}{(x+1)^3}dx$. (JEE 1983) This integral is not in any of the standard forms. Nor can it be tackled by substitution. Integration by parts is the only go. But for this we need to begin either with an antiderivative of e^x or that of the other factor $\frac{(x-1)}{(x+1)^3}$. The former approach is obviously simpler. But if we take e^x as the second function then later we would have to integrate $e^x \frac{d}{dx}\left(\frac{(x-1)}{(x+1)^3}\right)$ which will be even more complicated than the original problem. The other approach is to take $\frac{(x-1)}{(x+1)^3}$ as the second function. For this we shall have to first find an antiderivative of $\frac{(x-1)}{(x+1)^3}$. This can indeed be done, for example, using partial fractions. But the antiderivative will quite likely involve terms containing logarithms and afterwards we would have to integrate e^x times these logarithms, which is not going to be easy either. But the dilemma can be solved if we can recognise $\frac{(x-1)}{(x+1)^3}$ as the sum of a function, say $f(x)$, and its derivative. Once this idea strikes us, actually finding the function $f(x)$ is a matter of inspection. The factor $(x+1)^3$ in the denominator of $f(x) + f'(x)$ suggests that $f(x)$ will have $(x+1)^2$ in its denominator. So we try $f(x) = \frac{1}{(x+1)^2}$ as our simplest choice. Then $f'(x) = \frac{-2}{(x+1)^3}$. When added to $\frac{1}{(x+1)^2}$ this does give $\frac{(x-1)}{(x+1)^3}$. So by (55), we get

$$\int \frac{(x-1)e^x}{(x+1)^3}dx = \frac{e^x}{(x+1)^2} + c.$$

Comment No. 25:

The exercises will contain more problems about finding antiderivatives. It should be noted that as with derivatives, it often pays to first simplify the given integrand. Indeed sometimes such a simplification may be the focal point of the problem and once you get it the part of finding an antiderivative itself may be easy. As a good illustration, suppose we want to find $\int (e^{\log x} + \sin x)\cos x \; dx$. (JEE 1981) Here the exponential function appears only superficially, since $e^{\log x}$ is simply x. (As the base of the logarithm is not specified, we assume it is e. Nowadays, it is more common to denote the natural logarithms by ln rather than by log.) So the given integral is simply the sum of the two integrals $\int x\cos x \; dx$ and $\int \sin x \cos x \; dx$. The first one was already evaluated above in (45). The second one, again by a trigonometric simplification, is $\frac{1}{2}\int \sin 2x \; dx$ and hence equals $-\frac{1}{4}\cos 2x$. Adding, the given integral is $x\sin x + \cos x - \frac{1}{4}\cos 2x + c$.

We remark that despite all the techniques at our disposal, not every function has an antiderivative that can be expressed in a closed form. Even such simple functions as $\sin(x^2)$ or e^{x^2} belong to this category. For functions involving radicals, it is more of an exception than a rule to be able to find their antiderivatives in a closed form. Suppose, for example, that $f(\theta) = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$, where a and b are some positive real numbers. If luckily $a = b$, then $f(\theta)$ is a constant function and we can find its antiderivative. But if $a \neq b$, then try as we might we cannot come up with a function $F(\theta)$ such that $F'(\theta) = f(\theta)$. By Theorem 2, an antiderivative definitely exists. Indeed $F(\theta) = \int_0^\theta \sqrt{a^2\sin^2 u + b^2\cos^2 u}du$ is an antiderivative. But it is not in a closed form. It is of a theoretical and little practical importance. It is not hard to show that the definite integral $\int\limits_0^{2\pi} \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}d\theta$ is the perimeter of an ellipse with equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$. (The case $a = b$ gives $2\pi a$ as the perimeter of a circle of radius a.) The fact that the integrand has no antiderivative in a closed form does not, of course, mean that the ellipse has no perimeter! It simply means that we cannot evaluate it exactly. We mention here that the integral $\int \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}d\theta$ belongs to an important class of integrals called **elliptic integrals**. The name comes from the fact that the integral above is the perimeter of an ellipse. But in fact such integrals also come in many other contexts and are very important in applications.

When a definite integral cannot be evaluated exactly one has to resort to evaluating it approximately. This is beyond our scope. But we mention the general idea, which is based on the fact that a definite integral is the limit of a Riemann sum. Let $I = \int\limits_a^b f(x)dx$. Let $(x_0, x_1, \ldots, x_n)$ be a partition of $[a, b]$ into n equal parts. Then $\Delta x_i = \frac{b-a}{n}$ for each i. Denote this by h. Let

$y_i = f(x_i)$ for $i = 1, 2, \ldots, n$. Denote by R_n and L_n the sums $h\sum_{i=1}^{n} y_i$ and $h\sum_{i=1}^{n} y_{i-1}$ respectively and let $T_n = \frac{1}{2}(R_n + L_n)$. (Here '$R$' stands for 'right' since R_n is obtained by summing the values of f at the right end points of the subintervals. Similarly, 'L' stands for 'left'. The 'T' stands for 'trapezium', because T_n can be interpreted as the sum of the areas of certain trapezia which approximate the area below the graph of $f(x)$ on $[a, b]$.)

When n is large, both R_n and L_n approximate I. But T_n gives a much better approximation, because, being the arithmetic mean of R_n and L_n, it tends to cancel the errors of the two. As a concrete example, let $I = \int_0^1 \frac{dx}{1+x^2}$. Then the exact value of I is $\tan^{-1} 1 = \frac{\pi}{4} \approx 0.7853981\cdots$. We leave it to you to check with a hand calculator that $R_{10} \approx 0.7599815\cdots$, $L_{10} \approx 0.8099815\cdots$ and $T_{10} \approx 0.7849815\cdots$.

When n is even, say $n = 2m$, a far better approximation to I is given by the sum, say S_n, which is defined as $\frac{h}{3}\sum_{i=1}^{m}(y_{2i-2} + 4y_{2i-1} + y_{2i})$. We leave it to you to check that in the example above $S_{10} \approx 0.7853981\cdots$ which is almost but not quite the exact value $\frac{\pi}{4}$ as the latter is irrational.

Approximations to I by T_n and S_n are called, respectively the **trapezoidal rule** and the **Simpson's rule**.

EXERCISES

17.1 If R is a triangle with vertices at $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3), prove that its area equals half the absolute value of the determinant

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

[*Hint* : Without loss of generality assume $x_1 \le x_2 \le x_3$. As the area is unchanged under a shifting of the origin, (x_2, y_2) may be taken as $(0, 0)$. Carry out the integration over $[x_1, 0]$ and $[0, x_3]$ and add.]

17.2 Let $O(0, 0), A(2, 0)$ and $B(1, \frac{1}{\sqrt{3}})$ be the vertices of a triangle. Let R be the region consisting of all those points P inside ΔOAB which satisfy

$$d(P, OA) \le \min\{d(P, OB), d(P, AB)\},$$

where d denotes the distance from the point to the corresponding line. Sketch the region R and find its area. (1997*)

17.3 Find the areas of the regions with the following specifications (a * indicates that a sketch is also asked) :

(i) bounded by the curves $y = x^2$ and $x^2 + y^2 = 1$ in the first quadrant, (JAT 1979)

(ii) bounded by the parabola $x^2 = 4ay + 4a^2$ and the line $3x + 4y = 0$, (JAT 1980)

(iii) bounded by the curve $x^2 = 4y$ and the straight line $x = 4y - 2$, (1981)

(iv) bounded by the x-axis, the curve $y = |x \sin \pi x|$ and the lines $x = -1$ and $x = \frac{3}{2}$, (1982)

(v) bounded by the x-axis and the curves defined by $y = \tan x,\ -\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$ and $y = \cot x,\ \frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$, (1984)

(vi)* bounded by the curves $y = \sqrt{5 - x^2}$ and $y = |x - 1|$, (1985)

(vii) bounded by the curves $x^2 + y^2 = 4,\ x^2 = -\sqrt{2}y$ and $x = y$, (1986)

(viii) bounded by the curves $x^2 + y^2 = 25,\ 4y = |4 - x^2|$ and $x = 0$, *above* the x-axis, (1987)

(ix) bounded by the curve $y = \tan x$, the tangent drawn to it at $x = \frac{\pi}{4}$ and the x-axis, (1988)

(x) bounded by the curve $y = x(x - 1)^2$, the y-axis and the line $y = 2$, (1989)

(xi) bounded by the curves $y = ex \ln x$ and $y = \dfrac{\ln x}{ex}$, where $\ln e = 1$, (1990)

(xii)* bounded by the curves $y = x^2$ and $y = 2/(1 + x^2)$, (1992)

(xiii) lying in the third quadrant, on the left of the line $8x + 1 = 0$ and bounded by the curves $x = -2y^2$ and $y = f(x)$, where $f(x)$ is a continuous function given by

$$f(x) = \begin{cases} 2x, & |x| \leq 1 \\ x^2 + ax + b, & |x| > 1 \end{cases} . \quad (1999)$$

(xiv) bounded by the curves $y = f(x)$, the x-axis, $x = 0$ and $x = 1$, where $f(x) =$ Maximum $\{x^2, (1 - x)^2, 2x(1 - x)\}$, for $0 \leq x \leq 1$, (1997)

(xv) bounded by the curves $y = x^2,\ \ y = |2 - x^2|$ and $y = 2$ and lying to the right of the line $x = 1$. (2002)

17.4 Find the area bounded by the x-axis, part of the curve $y = 1 + \frac{8}{x^2}$ and the ordinates at $x = 2$ and $x = 4$. If the ordinate at $x = a$ divides the area into two equal parts, find a. (1983)

17.5 Find all values of m for which the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals $\frac{9}{2}$. (1999)

17.6 Let $g(x)$ be a function defined on $[-1, 1]$. If the area of the equilateral triangle with two of its vertices at $(0, 0)$ and $[x, g(x)]$ is $\sqrt{3}/4$, determine which of the following can be $g(x)$. (1989)

(A) $g(x) = \pm\sqrt{1 - x^2}$ (C) $g(x) = -\sqrt{1 - x^2}$

(B) $g(x) = \sqrt{1 - x^2}$ (D) $g(x) = \sqrt{1 + x^2}$

17.7 Prove that for any positive integer k (and $x \neq n\pi$ for any integer n), the sum $2[\cos x + \cos 3x + \ldots + \cos(2k-1)x]$ equals $\dfrac{\sin 2kx}{\sin x}$. Hence prove that $\int\limits_0^{\pi/2} \sin 2kx \cot x = \frac{\pi}{2}$. (Strictly speaking, the integrand is undefined at $x = 0$. But the discontinuity is removable. Such difficulties will be handled in Comment No. 15 of the next chapter.) (1990)

17.8 (a) Let $b \neq 0$ and for $j = 0, 1, 2, \ldots, n$, let S_j be the area of the region bounded by the y-axis and the curve $xe^{ay} = \sin by$, $\dfrac{j\pi}{b} \leq y \leq \dfrac{(j+1)\pi}{b}$. Show that $S_0, S_1, S_2, \ldots, S_n$ are in geometric progression. Also find their sum for $a = -1$ and $b = \pi$. (2001)

(b) Show that the area of a circle of radius r can also be found by taking $\lim\limits_{n\to\infty} A_n$, where A_n is the area of a regular n-gon inscribed in the circle. How will you modify this method to obtain the area of a sector?

17.9 Evaluate the following limits :

(a) $\lim\limits_{n\to\infty} \dfrac{1}{n} \sum\limits_{r=1}^{n} \sin(\dfrac{\pi r}{n})$ (b) $\lim\limits_{n\to\infty} \sum\limits_{r=1}^{n} \dfrac{1}{\sqrt{2nr - r^2}}$ (JAT 1980)

(c) $\lim\limits_{n\to\infty} \dfrac{1}{n} \ln \dfrac{(2n)!}{n^n n!}$ (d) $\lim\limits_{n\to\infty} \dfrac{1}{n} \sum\limits_{r=1}^{2n} \dfrac{r}{\sqrt{n^2 + r^2}}$ (1997)

17.10 For $n > 1$, let $S_n = \sum\limits_{k=1}^{n} \sqrt{k}$ and $T_n = \sum\limits_{k=1}^{n} \frac{1}{\sqrt{k}}$. Evaluate S_{10000} approximately as an integral and obtain an upper bound on the error. Also show that $2\sqrt{n} - 2 < T_n < 2\sqrt{n} - 1$. Hence find $[T_{10000}]$, the integral part of T_{10000}.

17.11 Show that there is a unique value of the positive integer k for which the limit $\lim\limits_{n\to\infty} \dfrac{1}{n^k}[1^{10} + 2^{10} + \ldots + (n-1)^{10} + n^{10}]$ is finite and non-zero. Find the limit for this value of k. What happens for other values of k? (See also the end of Comment No. 2, Chapter 2.)

17.12 A straight 2 kilometer road runs from A to B. The population density at a point x k.m. away from A is $\rho(x)$ while the per capita income of people living at that point is $\sigma(x)$. Express (i) the total population, (ii) the total income and (iii) the average per capita income for the road in terms of definite integrals.

17.13 Suppose a solid in space lies between the planes $z = a$ and $z = b$, where $a < b$. Asume that for every $c \in [a, b]$, the area of cross-section of the solid by the plane $z = c$ is $A(c)$. Show that the volume of the solid is $\int_a^b A(z)dz$.

17.14 Using the last exercise find volumes of (i) a hemisphere of radius r and (ii) a right circular cone of height h and base radius r.

17.15 Show that $\int_0^{n\pi+v} |\sin x| dx = 2n + 1 - \cos v$, when n is a positive integer and $0 \le v < \pi$. (1994)

17.16 (a) Investigate for maxima and minima the function

$$\int_1^x [2(t-1)(t-2)^3 + 3(t-1)^2(t-2)^2] dt. \quad (1988)$$

(b) Let $f(x) = \int_{-1}^x t(e^t - 1)(t-1)(t-2)^2(t-3)^5 dt$. Find the point(s) at which $f(x)$ has a local minimum. (1999)

17.17 (a) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a differentiable function with $f(1) = 4$ and $f'(1) = 10$. Find the value of $\lim_{x\to 1} \int_4^{f(x)} \frac{2t}{x-1} dt$. (1990, modified))

(b) If $f(x)$ is continuous, find $\frac{d}{dx} \int_0^x f(u) \sin(x+u) du$.

17.18 Find (i) $\lim_{x\to x_0} \frac{1}{x^2 - x_0^2} \int_{x_0}^x \sin(t^2) dt$ (ii) $\lim_{x\to 0} \frac{1}{x^6} \int_0^{x^2} \frac{t^2 dt}{t^6 + 1}$.

17.19 If the variables x and y are related by the equation

$$x = \int_0^y \frac{1}{\sqrt{1+4t^2}}\, dt$$

prove that $\frac{d^2y}{dx^2}$ is proportional to y. (Thomas)

17.20 If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B\ln(9e^{2x} - 4) + C$, determine A, B and C. (1990)

17.21 Let $a + b = 4$, where $a < 2$ and let $g(x)$ be a differentiable function. If $\frac{dg}{dx} > 0$ for all x, prove that $\int_0^a g(x)dx + \int_0^b g(x)dx$ increases as $(b-a)$ increases. (1997)

17.22 (a) If $\int_0^x f(t)dt = x + \int_x^1 tf(t)dt$, find the value of $f(1)$. (1998)

(b) Let $f(x) = \int_1^x \sqrt{2-t^2} dt$. Find the real roots of the equation $x^2 - f'(x) = 0$. (2002)

17.23 Suppose $f(x)$ is a function satisfying the following conditions

(a) $f(0) = 2,\ f(1) = 1$,

(b) f has a minimum value at $\frac{5}{2}$, and

(c) for all x,

$$f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$$

where a, b are some constants.

Determine the constants a, b and the function $f(x)$. (1998)

17.24 Try to find $\int \sec^3\theta d\theta$ by parts. You get the same term on the right hand side, but with a minus sign. Bring it to the left and then evaluate $\int \sec^3\theta d\theta$. Using it obtain $\int \sqrt{a^2+x^2}dx$, a being a constant.

17.25 (i) Find $\int \sin^2 x dx$ by a method similar to that in the last exercise and also by writing $\sin^2 x$ in terms of $\cos 2x$. Why can't $\int \sin(x^2)dx$ be evaluated using integration by parts?

(ii) Analogously, find $\int \sin x \cos x dx$ by parts in two different ways. Also find it by writing the integrand as $\frac{1}{2}\sin 2x$. You get three different answers. Is anything wrong?

17.26 (a) Suppose $f(x) = u(x) + iv(x)$ is a complex-valued function of a real variable, where $u(x)$ and $v(x)$ are real-valued functions of x. Define $f'(x)$ to be $u'(x) + iv'(x)$. (In other words, a complex-valued function of a real variable is differentiated by differentiating its real and imaginary parts separately.) Prove that the basic rules for the derivatives of sums, differences, products and quotients remain valid for complex-valued functions too. Prove also that the chain rule holds true for a complex valued function of the form $f(g(t))$, where $g(t)$ is a real valued function of a real variable t.

(b) Prove that $\dfrac{d}{dx}e^{ibx} = ibe^{ibx}$ for every real b and hence that $\dfrac{d}{dx}e^{cx} = ce^{cx}$ for every complex number c.

(c) Using (b) obtain (54) and (53) together as the real and imaginary parts of $\int e^{(a+ib)x}dx$. (This is an alternate derivation which does not need integration by parts. It is an interesting, although not a very profound, application of complex exponentials and in particular of the Euler formula.)

17.27 Evaluate the following antiderivatives :

(i) $\int \sqrt{1+\sin(\frac{1}{2}x)}\ dx$ (1980) (ii) $\int \dfrac{x^2}{\sqrt{1-x}}\ dx$ (1980)

(iii) $\int \sqrt{\dfrac{1-\sqrt{x}}{1+\sqrt{x}}}\,dx$ (1985) (iv) $\int \left[\dfrac{(\cos 2x)^{1/2}}{\sin x}\right] dx$ (1987)

(v) $\int(\sqrt{\tan x}+\sqrt{\cot x})\,dx$ (1989) (vi) $\int \dfrac{(x+1)}{x(1+xe^x)^2}\,dx$ (1996)

(vii) $\int \dfrac{\sin^{-1}\sqrt{x}-\cos^{-1}\sqrt{x}}{\sin^{-1}\sqrt{x}+\cos^{-1}\sqrt{x}}\,dx$ (1986) (viii) $\int \left(\dfrac{1-\sqrt{x}}{1+\sqrt{x}}\right)^{1/2}\dfrac{dx}{x}$ (1997*)

(ix) $\int \left(\dfrac{1}{\sqrt[3]{x}+\sqrt[4]{x}}+\dfrac{\ln(1+\sqrt[6]{x}\,)}{\sqrt[3]{x}+\sqrt{x}}\right) dx$ (1992)

(x) $\int \cos 2\theta \ln(\dfrac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta})\,d\theta$ (1994)

(xi) $\int \sin^{-1}(\dfrac{2x+2}{\sqrt{4x^2+8x+13}})\,dx$ (2001)

(xii) $\int (x+1)e^x \ln x\,dx$

17.28 Just as (49) is a version of integration by parts for definite integrals, prove the following version of the rule of substitution (also called the change of variable formula) for definite integrals.

Let $f:[a,b]\longrightarrow \mathbb{R}$ be continuous. Suppose $h:[c,d]\longrightarrow[a,b]$ is a continuously differentiable function with $h(c)=a$ and $h(d)=b$. Then

$$\int_a^b f(x)dx=\int_c^d f(h(t))h'(t)dt=\int_c^d f(x)\frac{dx}{dt}dt$$

17.29 Evaluate the following definite integrals :

(i) $\int\limits_0^{\frac{1}{2}} \dfrac{x\sin^{-1}x}{\sqrt{1-x^2}}\,dx$ (1984)

(ii) $\int\limits_{-2}^{2} |1-x^2|\,dx$ (1989)

(iii) $\int_2^3 \dfrac{2x^5+x^4-2x^3+2x^2+1}{(x^2+1)(x^4-1)}\,dx$ (1993)

(iv) $\int_1^{e^{37}} \dfrac{\pi\sin(\pi\ln x)}{x}dx$ (1997)

17.30 Let $\dfrac{d}{dx}F(x)=\dfrac{e^{\sin x}}{x}$. Find a possible value of k for which

$$\int_1^4 \frac{2e^{\sin x^2}}{x}dx=F(k)-F(1). \qquad (1997)$$

17.31 Evaluate $\ln 2$ approximately by computing T_{10} and S_{10} for the definite integral $\int\limits_0^1 \dfrac{dx}{1+x}$. (The exact value is $0.6931471\cdots$.)

Chapter 18

DEFINITE INTEGRALS

We already defined and evaluated definite integrals in the last chapter, and even applied them to find the areas of given regions. But the evaluation was based invariably on the Fundamental Theorem of Calculus. As a result, those problems were, in effect, problems on finding antiderivatives, for which we studied a number of standard techniques.

While the Fundamental Theorem of Calculus remains the most standard way of evaluating a definite integral $\int_a^b f(x)dx$, there are situations where it cannot be applied directly in the form we have stated. This can happen for a number of reasons. For example, the function $f(x)$ may not be continuous at all points of the interval $[a, b]$. Or it may not have an antiderivative in a closed form. Yet another possibility is that it has an antiderivative in a closed form but that may not be sufficient to give us the answer in a desired form as in the Main problem.

Sometimes a definite integral can still be evaluated in such situations using some other methods. We study a few such methods in this chapter. The general idea is to relate such an integral to another, similar integral using some special features of the function such as its symmetry or its periodicity or some integer parameter associated with it. This relationship can sometimes give the answer directly. Or, as in the Main Problem, this new integral may further have to be related to yet another integral and this process continued till we get an integral which we can evaluate directly. This latter type of a relationship is called a **reduction formula** (a term which is also used in a different context, viz. the binomial coefficients).

Cases where the integrand has discontinuities in the interval $[a, b]$ are handled by first taking the integral over subintervals which do not contain any points of discontinuity. This is discussed briefly in Comment No. 15 and 16.

Problems of evaluation of definite integrals using techniques other than merely the Fundamental Theorem of Calculus are asked almost every year in the JEE.

Main Problem : Let I be the value of the integral

$$\int_0^1 x^{70}(1-x)^{30}dx$$

Then the number $\binom{100}{30}I$, when expressed as a reduced rational, equals ..... .

First Hint: More generally, try to evaluate $I_{m,n} = \int_0^1 x^m(1-x)^n dx$ for non-negative integers m, n.

Second Hint: Express $I_{m,n}$ in terms of $I_{m+1,n-1}$ for $n > 0$.

Third Hint: Find $I_{m,0}$.

Solution: For $n > 0$, integration by parts gives

$$\begin{aligned} I_{m,n} &= \int_0^1 x^m(1-x)^n dx \\ &= \frac{1}{m+1}x^{m+1}(1-x)^n\Big|_0^1 + \frac{n}{m+1}\int_0^1 x^{m+1}(1-x)^{n-1}dx \\ &= \frac{n}{m+1}I_{m+1,n-1} \end{aligned} \tag{1}$$

Repeated applications of this formula (with decreasing values of the second parameter n) give,

$$\begin{aligned} I_{m,n} &= \frac{n}{m+1}I_{m+1,n-1} \\ &= \frac{n}{m+1}\frac{n-1}{m+2}I_{m+2,n-2} \\ &= \cdots \\ &= \frac{n!}{(m+1)(m+2)\ldots(m+n)}I_{m+n,0} \\ &= \frac{1}{\binom{m+n}{n}}I_{m+n,0} \end{aligned} \tag{2}$$

By a direct calculation,

$$I_{m+n,0} = \int_0^1 x^{m+n}dx = \frac{1}{m+n+1}$$

.

So, $\binom{m+n}{n}I_{m,n} = \frac{1}{m+n+1}$. In particular, $\binom{100}{30}\int_0^1 x^{70}(1-x)^{30} = \frac{1}{101}$.

Comment No. 1:

Obviously, the parameters 70 and 30 in the given problem have no special significance. So the problem really is to evaluate $I_{m,n}$ for general values of m and n. The direct approach would be to expand $(1-x)^n$ by the binomial theorem as $\sum_{i=0}^{n}(-1)^i\binom{n}{i}x^i$. The integral $I_{m,n}$ would then come out to be the sum $\sum_{i=0}^{n}(-1)^i\frac{1}{m+i+1}$. The trouble is that there is no easy way to evaluate this sum in a closed form. Of course, if the value of n is small, say 3 or 4, then we could evaluate it simply by brute force. But for $n = 30$ this is clearly impracticable. So we abandon this approach and look for alternate methods.

A clue is provided by the number $\binom{100}{30}$ appearing in the statement of the problem. It suggests that in general, the binomial coefficient $\binom{m+n}{n}$ must be involved somewhere in the evaluation of $I_{m,n}$. This much anybody can guess. A perceptive person can go a little deeper. The binomial coefficients involve factorials, which are defined recursively, (i.e., to know $k!$, you first have to know $(k-1)!$ and then multiply it by k). So perhaps, in this problem, too, the key to success lies in relating the integral $\int_0^1 x^m(1-x)^n dx$ with a similar integral but with lower values of at least one of the parameters. Once this idea is conceived, it is easy to integrate by parts and come up with the relation (1).

The observation that $I_{m,n}$ can be evaluated easily if n is small not only shows that there is some progress in going from $I_{m,n}$ to $I_{m+1,n-1}$ but also suggests that in order to make further progress we should apply (1) again and again till the value of n is reduces to, say 1 or 2. The ultimate is, of course, $n = 0$. This is how we get (2). The rest of the work is routine. (Some people tend to think that m and n must be positive integers. They may prefer to apply (1) till n is reduced to 1, but not to 0. In that case, instead of (2), they will end up with

$$I_{m,n} = \frac{n!}{(m+1)(m+2)\ldots(m+n-1)} I_{m+n-1,1}$$

which is equally good, but evaluation of $\int_0^1 x^{m+n-1}(1-x)dx$ is slightly more time-consuming than that of $\int_0^1 x^{m+n}dx$. This is a minor point but deserves to be noted. Because of the impossibility of dividing by 0, many people have a dread of letting a parameter take the value 0. And, sometimes, such a value may, in fact, be inadmissible. But it should not be supposed that it is *always* disallowed. In fact, sometimes it is very convenient, especially when the parameter occurs as an exponent, as in the present problem. Similarly, in the method of mathematical induction, sometimes the statement to be proved may make sense for $n = 0$ (possibly with special conventions such as $0! = 1$) and in that case, the induction may very well begin with it instead of the normal starting point, viz., $n = 1$.)

Comment No. 2:

A slight reformulation of the solution above is noteworthy. When you cannot solve a problem in the general case, it is a good idea to do a few special, simple cases of it, look for some pattern in the answers, guess the answer to the general case and finally prove it. In doing the special cases, however, try to be as general as you can. If, for example, the problem involves several parameters (as in the present problem), then instead of giving particular values to all of them, see if you can get by with particular values for only some of them. In the present problem there are two parameters, viz., m and n. Let us keep m as it is and give small values to n. By direct calculations, for $n = 0, 1$ and 2, the expression $\binom{m+n}{n} I_{m,n}$ comes out to be, respectively, $\frac{1}{m+1}, \frac{1}{m+2}$ and $\frac{1}{m+3}$. This suggests that in the general case the answer is probably $\frac{1}{m+n+1}$. Of course, one has to prove this guess. And that can be done by induction on n. (Note that there is no induction on the other parameter m. So the statement $P(n)$ to be proved by induction on n has to be taken as 'For *every* non-negative integer m, the expression $\binom{m+n}{n} I_{m,n}$ equals $\frac{1}{m+n+1}$.') We can start the induction with $n = 0$. The proof of the inductive step is essentially the work involved in (1). Indeed, if we multiply both the sides of (1) by $\binom{m+n}{n}$ we get

$$\begin{aligned}
\binom{m+n}{n} I_{m,n} &= \binom{m+n}{n} \frac{n}{m+1} I_{m+1,n-1} \\
&= \frac{(m+n)(m+n-1)\dots(m+1)}{n!} \frac{n}{m+1} I_{m+1,n-1} \\
&= \frac{(m+n)(m+n-1)\dots(m+2)}{(n-1)!} I_{m+1,n-1} \\
&= \binom{m+n}{n-1} I_{m+1,n-1} \\
&= \binom{(m+1)+(n-1)}{n-1} I_{m+1,n-1}
\end{aligned}$$

We now apply the inductive hypothesis to the right hand side and get that it equals $\frac{1}{m+n+1}$, thereby completing the proof of the inductive step. (Note that even though the index m has been raised by 1, there is no difficulty in applying the inductive hypothesis because the induction is not on m but on n whose value has indeed been lowered.)

Basically, the work involved in the solution above and in the proof by induction is the same since both of them rely crucially on (1). But there is a subtle difference. In the earlier solution, we applied (1) repeatedly and *arrived* at the answer. In induction, on the other hand, we first *guessed* the answer and then *verified* it.

Comment No. 3:

There is also a very different and somewhat obscure way of evaluating $\binom{m+n}{n} I_{m,n}$, i.e., of evaluating $\int_0^1 \binom{m+n}{n} x^m (1-x)^n dx$. It begins by recognising that the integrand is one of the terms in the binomial expansion of $[x + (1 - x)]^{m+n}$, or equivalently, if we replace the first term x by tx, where t is some

parameter independent of x, then the integrand is the coefficient of t^m in the expansion of $[tx+(1-x)]^{m+n}$. Now consider the integral $\int_0^1 [tx+(1-x)]^{m+n}dx$. Here the integration is w.r.t. x, treating t as a constant. The value of this integral will depend on the value of t. Denote it by $f(t)$. We can evaluate this integral in two ways, viz., directly and also by writing the integrand as a sum (using the binomial theorem) and integrating each term separately. The key idea behind the solution is to equate the two values of $f(t)$ obtained this way.

For a direct evaluation of the integral (JEE 1981), we write $tx+(1-x)$ as $(t-1)x+1$ and get

$$\begin{aligned} f(t) &= \int_0^1 [(t-1)x+1]^{m+n}dx \\ &= \frac{1}{(m+n+1)(t-1)}[(t-1)x+1]^{m+n+1}\Big|_0^1 \\ &= \frac{t^{m+n+1}-1}{(m+n+1)(t-1)} \\ &= \frac{1}{m+n+1}(t^{m+n}+t^{m+n-1}+\ldots+t^2+t+1) \end{aligned}$$

(Strictly speaking this derivation is valid only for $t \neq 1$. But the equality holds for $t=1$ also as both the sides come out to be 1.)

On the other hand, evaluating the integral by first expanding the integrand we get,

$$\begin{aligned} f(t) &= \int_0^1 [tx+(1-x)]^{m+n}dx \\ &= \int_0^1 \sum_{i=0}^{m+n} \binom{m+n}{i} x^i(1-x)^{m+n-i}t^i dx \\ &= \sum_{i=0}^{m+n} \binom{m+n}{i} \left(\int_0^1 x^i(1-x)^{m+n-i}dx\right) t^i \\ &= \sum_{i=0}^{m+n} \binom{m+n}{i} I_{i,m+n-i}t^i \end{aligned}$$

Both the expressions for $f(t)$ are polynomials (of degree $m+n$) in t. As they are equal for all values of t, the coefficients of like powers of t on the two sides must be equal to each other. (A similar reasoning was used in the solution to a problem in Comment No. 8 of Chapter 5 and also underlies the method of undetermined coefficients for obtaining the partial fractions of a rational function.) In particular, equating the coefficients of t^m in both the expressions for $f(t)$ we get the desired result. (The strange thing about the technique adopted here is the double role played by the parameter t. As far as integration w.r.t. x is concerned, it is a constant. At the same time we also treated it as a real variable and considered a function of it, viz., $f(t)$. In effect, we are varying the parameter. For this reason, this technique is called the **variation of parameter**. It is a powerful technique. Note that the given

problem makes no reference to the parameter t, or to any other parameter for that matter. We have devised it for our convenience.)

Comment No. 4:

As an interesting application of the formula for $I_{m,n}$, we can evaluate the alternating sum of the reciprocals of the binomial coefficients with a given upper index, i.e., a sum of the form $\sum_{i=0}^{n} \frac{(-1)^i}{\binom{n}{i}}$ for a fixed (non-negative) integer n. Denote this sum by S_n. Thus $S_0 = 1, S_1 = 1 - 1 = 0, S_2 = 1 - \frac{1}{2} + 1 = \frac{3}{2}, S_3 = 0, S_4 = \frac{5}{3}$ and so on. Because of the symmetry relation, viz., $\binom{n}{i} = \binom{n}{n-i}$, it is easy to see that $S_n = 0$ if n is odd. For in this case each reciprocal appears twice with opposite signs in the sum. However, for even n, it is not easy to evaluate S_n directly, because although there are numerous identities involving the binomial coefficients, there are hardly any involving their reciprocals. With a slight change of notation, the formula proved above expresses $\dfrac{1}{\binom{n}{k}}$ as $(n+1)I_{k,n-k}$ and thereby allows the desired sum to be expressed as an integral. Indeed,

$$\begin{aligned} S_n &= \sum_{k=0}^{n} (n+1)(-1)^k I_{k,n-k} \\ &= (n+1)\sum_{k=0}^{n} \int_0^1 (-x)^k (1-x)^{n-k} dx \\ &= (n+1)\int_0^1 \sum_{k=0}^{n} (-x)^k (1-x)^{n-k} dx \end{aligned}$$

Here the integrand is a sum of a geometric progression with first term $(1-x)^n$ and common ratio $\frac{-x}{1-x}$. This gives

$$\begin{aligned} S_n &= (n+1)\int_0^1 \frac{(1-x)^n \left[1 - (\frac{-x}{1-x})^{n+1}\right]}{1 + \frac{x}{1-x}} dx \\ &= (n+1)\int_0^1 (1-x)^{n+1} + (-1)^{n+2} x^{n+1} dx \\ &= (n+1)\left[\frac{1}{n+2} + \frac{(-1)^{n+2}}{n+2}\right] \\ &= \begin{cases} 0 & \text{if } n \text{ is odd} \\ \dfrac{2n+2}{n+2} & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

Comment No. 5:

For a fixed value of m, the integral $I_{m,n} = \displaystyle\int_0^1 x^m (1-x)^n dx$ depends only upon the other parameter, viz., n. Formula (1) serves to express $I_{m,n}$ in terms of $I_{m+1,n-1}$, where the value of the second parameter is reduced by 1. Repeated applications of this formula led to (2) where the integral $I_{m+n,0}$ could

be evaluated directly. This technique is fairly common in evaluating certain types of definite integrals. Suppose we have a family of related functions. It happens sometimes that the integral of a typical member of this family cannot be evaluated directly, but can be easily expressed in terms of the integral of some other member of the family. Often the family is indexed by an integer parameter, say $n \geq 0$. A formula which expresses (or 'reduces') the integral of the nth function in terms of that of the $(n-1)^{\text{th}}$ (or lower indexed) function is called a **reduction formula.** The integrals of the functions corresponding to the lower values of a parameter n (say $n = 0$ or 1) can often be evaluated directly. Repeated application of the reduction formula then gives the value of the integral of the nth function for a general n.

As an instructive example, consider the family of functions $\sin^n x, n = 0, 1, 2, \ldots$ (where by $\sin^0 x$ we mean the constant function 1). We can then prove the following reduction formula.

$$\int_0^{\pi/2} \sin^n x dx = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} x dx, \quad \text{for} \quad n \geq 2 \tag{3}$$

This can be proved very easily by writing $\sin^n x$ as $\sin^{n-1} x \frac{d}{dx}(-\cos x)$. Integration by parts gives

$$\begin{aligned} \int_0^{\pi/2} \sin^n x dx &= -\sin^{n-1} x \cos x \Big|_0^{\pi/2} + \int_0^{\pi/2} (n-1) \sin^{n-2} x \cos^2 x dx \\ &= (n-1) \int_0^{\pi/2} \sin^{n-2} x dx - (n-1) \int_0^{\pi/2} \sin^n x dx \end{aligned} \tag{4}$$

Bringing the last term on the left gives (3)

The integrals $\int_0^{\pi/2} \sin^0 x dx$ and $\int_0^{\pi/2} \sin x dx$ can be calculated directly and come out to be $\frac{\pi}{2}$ and 1 respectively. Hence repeated applications of the reduction formula give

$$\int_0^{\pi/2} \sin^{2m} x dx = \frac{(2m)!}{[2^m m!]^2} \frac{\pi}{2} \tag{5}$$

and

$$\int_0^{\pi/2} \sin^{2m+1} x dx = \frac{[2^m m!]^2}{(2m+1)!}. \tag{6}$$

Formulas (5) and (6) enable us to derive, after some more work on the integrals $\int_0^{\pi/2} \sin^n x dx$, a very interesting formula, due to **Wallis** which expresses π as an infinite product. More precisely stated, the formula says,

$$\lim_{n \to \infty} \frac{2.2}{1.3} \cdot \frac{4.4}{3.5} \cdot \frac{6.6}{5.7} \cdots \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{\pi}{2}.$$

See Exercise (18.5) for a proof of this formula. Note the role of the limits of integration, viz., 0 and $\frac{\pi}{2}$ in deriving the reduction formula (3). The reasoning in obtaining (4) would be valid if we replace 0 and $\frac{\pi}{2}$ by any two real numbers, say a and b, except that the first term may not vanish. Instead, it will be some number, say, c_n depending on n. We would still get a reduction formula for $\int_a^b \sin^n x dx$. But it will not be easy to apply it to get a closed form expression for $\int_a^b \sin^n x dx$ (i.e., an expression which gives its value directly in terms of a, b and n, instead of just relating it to the value of a similar integral).

Comment No. 6:

If we go back to the Main Problem, the substitution $x = \sin^2\theta$ converts the integral $I_{m,n} = \int_0^1 x^m(1-x)^n dx$ into the integral $2\int_0^{\pi/2} \sin^{2m+1}\theta\cos^{2n+1}\theta d\theta$. This suggests an alternate way of solving the problem by obtaining a reduction formula for an integral of this type, keeping in mind that we now have two integer parameters, viz., m and n, and not just one.

More generally, we could get a reduction formula for the integral, say $J_{p,q} = \int_0^{\pi/2} \sin^p\theta\cos^q\theta d\theta$, where the non-negative integers p and q are not necessarily odd. The method is again based on integration by parts. Writing $\sin^p\theta\cos^q\theta$ as $\sin^p\theta\cos^{q-1}\theta\frac{d}{d\theta}(\sin\theta)$ and keeping in mind that $\cos^{q-1}(\frac{\pi}{2}) = 0$ for $q \geq 2$, we get

$$\begin{aligned} J_{p,q} &= \int_0^{\pi/2} \sin^p\theta\cos^q\theta d\theta = \int_0^{\pi/2} (\sin^p\theta\cos^{q-1}\theta)\frac{d}{d\theta}(\sin\theta)d\theta \\ &= (\sin^{p+1}\theta\cos^{q-1}\theta)\Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{d}{d\theta}(\sin^p\theta\cos^{q-1}\theta)\sin\theta d\theta \\ &= -p\int_0^{\pi/2} \sin^p\theta\cos^q\theta d\theta + (q-1)\int_0^{\pi/2} \sin^{p+2}\theta\cos^{q-2}\theta d\theta \end{aligned}$$

Writing $\sin^{p+2}\theta$ in the last integrand as $\sin^p\theta - \sin^p\theta\cos^2\theta$, we get the reduction formula

$$J_{p,q} = \frac{q-1}{p+q} J_{p,q-2} \tag{7}$$

which is valid for $q \geq 2$. The values of $J_{p,1}$ and $J_{p,0}$ have to be calculated. The first one is the integral $\int_0^{\pi/2} \sin^p\theta\cos\theta d\theta$ which easily seen to be $\frac{1}{p+1}\sin^{p+1}\theta\Big|_0^{\pi/2} = \frac{1}{p+1}$. The second one, viz., $J_{p,0}$, is precisely the integral $\int_0^{\pi/2} \sin^p\theta d\theta$ whose value can be found from (5) or (6) depending upon whether p is even or odd. We can now apply (7) repeatedly to evaluate $J_{p,q}$ for any p and q. In particular, taking

$p = 2m+1$ and $q = 2n+1$, since q is odd we shall end up with $J_{2m+1,1}$ which equals $\frac{1}{2m+2}$ as noted above. Thus we get

$$\begin{aligned} J_{2m+1,2n+1} &= \frac{2n}{2m+2n+2} J_{2m+1,2n-1} \\ &= \frac{n}{m+n+1}\frac{n-1}{m+n} J_{2m+1,2n-3} \\ &\cdots \\ &= \frac{n(n-1)(n-2)\ldots.2.1}{2(m+n+1)(m+n)\ldots(m+2)(m+1)} \end{aligned} \tag{8}$$

The R.H.S. of (8) is $\frac{1}{2(m+n+1)\binom{m+n}{n}}$. But $I_{m,n} = 2J_{2m+1,2n+1}$ as we observed already. So (8) gives another solution to the Main Problem.

Comment No. 7:

An analogous problem is the following (JAT 1980):

For $n \geq 0$, let $I_n = \int_0^{\pi/2} \theta \sin^n \theta d\theta$. Show that for $n \geq 2$,

$$I_n = \frac{n-1}{n} I_{n-2} + \frac{1}{n^2} \tag{9}$$

This follows by taking essentially the same steps as those in proving (3). The integrals $I_0 = \int_0^{\pi/2} \theta d\theta$ and $I_1 = \int_0^{\pi/2} \theta \sin\theta d\theta$ come out to be $\frac{\pi^2}{4}$ and 1 respectively, by direct calculation, the latter requiring integration by parts. We can now apply (9) repeatedly and get $I_2, I_4, I_6, \ldots$ and also $I_3, I_5, I_7, \ldots$. Thus we can evaluate I_n for any given value of n. However, because of the term $\frac{1}{n^2}$ in (9), we cannot get formulas analogous to (5) and (6). (The situation in (3) was better because there I_n was of the form $g(n)I_{n-2}$, where $g(n) = \frac{n-1}{n}$. In (9), on the other hand, I_n is of the form $g(n)I_{n-2} + h(n)$, where $h(n) = \frac{1}{n^2}$. This extra term $h(n)$ makes it impossible to express I_n directly in terms of n. As noted earlier, a similar complication would arise for integrals of the type $\int_a^b \sin^n x dx$ if the numbers a and b are allowed to take values other than 0 and $\pi/2$.)

Comment No. 8:

In the last problem, it was easy to obtain the reduction formula for I_n but thereafter, it was not easy to obtain a simple, closed form expression for I_n. The situation in the next problem is exactly opposite. Here, the expression for the value of the integral comes out to be very simple. But the reduction formula is rather tricky.

Let $I_m = \int_0^{\pi} \frac{1-\cos mx}{1-\cos x} dx$. Use mathematical induction to prove that

$$I_m = m\pi \tag{10}$$

for $m = 0, 1, 2, \ldots$. (JEE 1995)

In view of the utter simplicity of (10), it is tempting to think that the right way to prove it is by proving a relation of the type

$$I_m = I_{m-1} + \pi \tag{11}$$

which reduces to proving

$$\int_0^\pi \frac{\cos(m-1)x - \cos mx}{1-\cos x}dx = \pi \tag{12}$$

The integrand here can be simplified to $\frac{\sin(\frac{2m-1}{2}x)}{\sin\frac{x}{2}}$. Putting $u = x/2$ this becomes $\frac{\sin((2m-1)u)}{\sin u}$ and (12) is equivalent to proving that $J_m = \pi/2$ where $J_m = \int\limits_0^{\pi/2} \frac{\sin((2m-1)u)}{\sin u}du$ for $m \geq 1$. This can be done by obtaining a reduction formula for J_m. Indeed, we have $J_{m+1} - J_m = \int\limits_0^{\pi/2} \frac{2\cos 2mu \sin u}{\sin u}du$ which comes out to be simply $\frac{1}{m}\sin 2mu\Big|_0^{\pi/2} = 0$. Thus $J_{m+1} = J_m$ for all m. By direct calculation, J_1 comes out as $\pi/2$. So, $J_m = \pi/2$ for all m which proves (12).

Although simple, this solution is a little surprising, because on the face of it, the L.H.S. of (12) would depend on m and it is not easy to strike that it is a constant independent of m. There is another way which bypasses (12) if, instead of $\cos(m-1)x$, we consider $\cos(m-2)x$. The difference $\cos(m-2)x - \cos mx$ equals $2\sin(m-1)x\sin x$. This is still not of much help because the problem deals with the cosines and not with the sines of integrals multiples of x. But what if we *add* $\cos(m-2)x$ and $\cos mx$? Then we get $2\cos(m-1)x\cos x$. The key to the solution lies in rewriting this slightly as

$$(1-\cos(m-2)x) + (1-\cos mx) = 2 - 2\cos(m-1)x\cos x$$

Adding and subtracting $2\cos(m-1)x$ from the R.H.S. and dividing throughout by $1-\cos x$ gives

$$\frac{1-\cos(m-2)x}{1-\cos x} + \frac{1-\cos mx}{1-\cos x} = 2\frac{1-\cos(m-1)x}{1-\cos x} + 2\cos(m-1)x \tag{13}$$

The fractions appearing in (13) are nothing but the integrands of I_{m-2}, I_m and I_{m-1}. So we integrate both the sides w.r.t. x from 0 to π. Noting that for $m > 1$, $\int\limits_0^\pi \cos(m-1)xdx = \frac{1}{m-1}\sin(m-1)x\Big|_0^\pi = 0$, we have

$$I_{m-2} + I_m = 2I_{m-1} \tag{14}$$

which is valid for all $m \geq 2$. Note that (14) expresses I_m not wholly in terms of I_{m-1} or wholly in terms of I_{m-2} but as a linear combination of both, i.e.,

as $2I_{m-1} - I_{m-2}$. So this reduction formula is of a somewhat different nature than the earlier ones we have encountered. Whenever such formulas are to be applied in a proof by induction, the induction has to begin by verifying the truth for the first *two* values of the parameter m and not just the first one as is normally done. This is so because normally in a proof by induction, the truth of a statement $P(k)$ is derived from that of the statement $P(k-1)$. In the present problem, however, the truth of $P(k)$ will require that of both $P(k-1)$ and of $P(k-2)$. (An analogous situation was encountered in the problem done in Comment No. 11 of Chapter 4.)

Once this point is understood, the proof itself is a triviality. The cases $m = 0$ and $m = 1$ are true because in them the integrand is identically 0 and 1 respectively. As for the inductive step, if $I_{k-1} = (k-1)\pi$ and $I_{k-2} = (k-2)\pi$ are both true, then from (14) we must have $I_k = k\pi$, thereby completing the inductive step.

This problem is yet another illustration of how sometimes an approach (viz., (11)) which suggests itself encounters an obstacle. However, if we analyse the failure then one can sometimes get a clue to remedy it (as was done through (13)). Actually, the two approaches are not radically different, because if we rewrite (14) as $I_m - I_{m-1} = I_{m-1} - I_{m-2}$, it is simply $2J_m = 2J_{m-1}$.

Comment No. 9:

Even though we may not always be able to obtain a closed form expression for an integral after having obtained a reduction formula for it, the reduction formula may sometimes enable us to give some estimate about the value of the integral. This is the case in the following problem.

Let A_n be the area bounded by the curve $y = (\tan x)^n$ and the lines $x = 0$, $y = 0$ and $x = \frac{\pi}{4}$. Prove that for $n > 2$,

$$A_n + A_{n-2} = \frac{1}{n-1} \tag{15}$$

Deduce that $\frac{1}{2n+2} < A_n < \frac{1}{2n-2}$. (JEE 1996)

Evidently, A_n is the integral $\int_0^{\pi/4} \tan^n x dx$. For $n > 2$, writing the integrand as $\tan^{n-2} x(\sec^2 x - 1)$, we get $A_n = \int_0^{\pi/4} \tan^{n-2} x \sec^2 x dx + A_{n-2}$. The substitution $u = \tan x$ converts the integral to $\int_0^1 u^{n-2} du$ whose value is $\frac{1}{n-1}$. This gives (15). Repeated applications of (15) would give

$$\begin{aligned} A_n &= -A_{n-2} + \frac{1}{n-1} \\ &= A_{n-4} + \frac{1}{n-1} - \frac{1}{n-3} \\ &= -A_{n-6} + \frac{1}{n-1} - \frac{1}{n-3} + \frac{1}{n-5} \\ &= \ldots \end{aligned}$$

which will go on till we get A_2 if n is even and A_1 if n is odd. Both these integrals can be calculated easily. But the trouble is that we would also get a series (with alternating positive and negative terms) for which there is no closed form formula for the sum. So we cannot evaluate A_n explicitly as a function of n. Nevertheless, the second part of the problem can be done directly from (15) and the elementary observation that since $0 < \tan x < 1$ for all $x \in (0, \frac{\pi}{4})$, we have that $\tan^k x < \tan^{k-1} x$ for all $x \in (0, \frac{\pi}{4})$ and for every positive integer k. Integrating, we get $A_k < A_{k-1}$ and hence

$$A_k < A_{k-2} \tag{16}$$

for every $k > 2$. (Here we are implicitly using the fact that if $f(x)$ and $g(x)$ are any two functions defined on an interval $[a, b]$ such that $f(x) < g(x)$ for all $x \in (a, b)$ then $\int_a^b f(x)dx < \int_a^b g(x)dx$. This certainly looks obvious. But a formal proof will be given after the solution of the present problem.)

From (16) we get $A_n + A_{n-2} > 2A_n$. Combining this with (15) gives $\frac{1}{n-1} > 2A_n$, i.e., $A_n < \frac{1}{2n-2}$. To get the lower bound on A_n, we apply (16) with $k = n + 2$ to get, by a similar reasoning, that $2A_n > A_{n+2} + A_n$. By (15) again (applied with n replaced by $n + 2$) we get $2A_n > \frac{1}{n+1}$, i.e., $\frac{1}{2n+2} < A_n$. This completes the solution. Thus we have shown that although we cannot evaluate A_n exactly, it lies somewhere in the interval $[\frac{1}{2n+2}, \frac{1}{2n-2}]$. The length of this interval is $\frac{1}{(n^2-1)}$, which is quite small (much smaller than $\frac{C}{n}$ for any constant C) for large n. Hence for large n, any point in this interval, e.g., $\frac{1}{2n}$ can be taken as an approximate value of A_n with a guarantee that the error cannot exceed $\frac{1}{n^2-1}$.

Comment No. 10:

We now prove the assertion that was used implicitly in the solution above. None of the results we proved so far about integrals involved inequalities. Actually, for indefinite integrals, the concept of an inequality does not make sense, because the indefinite integral of a function is not a single function but a whole bunch of functions any two of which differ by a constant. As a result, from the inequality of two functions we cannot conclude much about the inequality of their antiderivatives. For example, we have $\cos x \leq 1$ for all x. If we take their antiderivatives as $\sin x$ and x respectively, then we do have $\sin x \leq x$ for all $x > 0$. But $\sin x > x$ if $x < 0$. If we take $\sin x + 100$ as an antiderivative of $\cos x$ then the inequality $\sin x + 100 > x$ holds even for some positive values of x. (Certainly it holds for all $x \leq 99$.)

For definite integrals the situation is different. Suppose $f(x)$ and $g(x)$ are two (functions) defined on the same interval $[a, b]$. Then their definite integrals over $[a, b]$, i.e., $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ are some real numbers and so it makes sense to compare them. Intuitively we expect that the larger of the two functions would have a larger integral. The precise statement is as follows.

Theorem 1: Suppose $f(x)$ and $g(x)$ are continuous on an interval $[a, b]$, where $a < b$. If

$$f(x) \leq g(x) \text{ for all } x \in (a, b) \tag{17}$$

then

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx \ . \tag{18}$$

Moreover, if strict inequality holds in (17) for at least one point in (a, b) then it holds in (18).

Proof: Let $h(x) = g(x) - f(x)$ for $x \in [a, b]$. Then by (17), $h(x) \geq 0$ for all $x \in (a, b)$. Now let $H(x) = \int_a^x h(t)dt$ for all $x \in [a, b]$. Then by the second form of the Fundamental Theorem of Calculus (Theorem 2 of the last chapter), $H'(x) = h(x)$ for all $x \in [a, b]$. Thus we have $H'(x) \geq 0$ for all $x \in (a, b)$. Therefore by Lagrange's Mean Value Theorem, $H(b) \geq H(a)$. But $H(a) = 0$. So we get $H(b) \geq 0$. But $H(b) = \int_a^b g(x) - f(x)dx = \int_a^b g(x)dx - \int_a^b f(x)dx$. This proves (18).

In the proof of the second part, we shall use this first part. Assume there is some $c \in (a, b)$ such that $f(c) < g(c)$. Then $h(c) > 0$. Note that h is continuous at c. Taking $\epsilon = \frac{1}{2}h(c)$ in the definition of the statement $\lim_{x \to c} h(x) = h(c)$, we get a $\delta > 0$ such that the interval $(c - \delta, c + \delta)$ is contained in $[a, b]$ and further, for all $x \in (c - \delta, c + \delta)$, $h(c) - \frac{1}{2}h(c) < h(x) < h(c) + \frac{1}{2}h(c)$, which, in particular means

$$h(x) \geq \frac{1}{2}h(c) \text{ for all } x \in (c - \delta, c + \delta) \tag{19}$$

We now decompose the interval $[a, b]$ into three subintervals, viz., $[a, c - \delta]$, $[c - \delta, c + \delta]$ and $[c + \delta, b]$. Because of the additivity property of integrals, we have

$$\int_a^b h(x)dx = \int_a^{c-\delta} h(x)dx + \int_{c-\delta}^{c+\delta} h(x)dx + \int_{c+\delta}^b h(x)dx \tag{20}$$

By the first part of the Theorem, all the three integrals on the R.H.S. are non-negative since $h(x) \geq 0$ for all $x \in [a, b]$. But about the middle one we can say something stronger, because of (19), viz.,

$$\int_{c-\delta}^{c+\delta} h(x)dx \geq \int_{c-\delta}^{c+\delta} \frac{1}{2}h(c)dx = h(c)\delta \tag{21}$$

So from (20) and (21), $\int_a^b h(x)dx \geq h(c)\delta$. But this means $\int_a^b g(x)dx \geq \int_a^b f(x)dx + \frac{1}{2}h(c)\delta$. But $\frac{1}{2}h(c)\delta > 0$. So strict inequality holds in (18). This completes the proof of the second part.

We have given the proof using the Fundamental Theorem of Calculus. An alternate proof is possible using the definition of a definite integral as the limit of Riemann sums. For, in the proof of the first part, the non-negativity of $h(x)$ throughout $[a, b]$ implies that all Riemann sums are non-negative and hence that their limit is also non-negative. Similarly in the second part if the partition is sufficiently fine then its Riemann sum can be easily shown to be at least $h(c)\delta$. This proof is, in fact, superior because it stays closer to the true identity of an integral, viz., as the limit of a sum rather than its alternate identity as an antiderivative.

As an immediate corollary of this theorem, we get the following estimates on the value of a definite integral $\int_a^b f(x)dx$ in terms of upper and lower bounds on the values of the integrand $f(x)$ on the interval $[a, b]$.

Corollary 2: Suppose $f(x)$ is continuous on $[a, b]$ and m and M are lower and upper bounds on the values of $f(x)$ on $[a, b]$. That is,

$$m \ \leq \ f(x) \ \leq \ M \ \text{ for all } x \in [a, b] \tag{22}$$

Then

$$m(b-a) \ \leq \ \int_a^b f(x)dx \ \leq \ M(b-a) \tag{23}$$

Moreover if strict inequality holds even for one point for either of the two inequalities in (22) then the corresponding inequality in (23) is strict.

If we divide (23) throughout by $b - a$ (which is positive), we get

$$m \ \leq \ \frac{\int_a^b f(x)dx}{b-a} \ \leq \ M \tag{24}$$

Although this is hardly different from (23), it is worth stating because of the significance of the middle term. It is called the **average value** or the **mean value** of the function $f(x)$ over the interval $[a, b]$. This name comes from the fact that it is the height of the rectangle on the base $[a, b]$ whose area equals the area below the graph of $y = f(x)$ over the same base. Of course, depending upon what $f(x)$ stands for, it could have many other interpretations. For example, if $f(x)$ is the linear mass density of a piece of wire of length L at a point x units away from one of its ends (see Comment No. 8 of the last chapter) then the integral $\int_0^L f(x)dx$ is the mass of the wire and $\int_0^L f(x)dx/L$ is its average linear mass density.

When the difference $M - m$ is large, it means the function $f(x)$ can vary a lot over the interval $[a, b]$. In that case the inequalities (23) and (24) given by this corollary are rather crude because there may be some parts of the interval $[a, b]$, where $f(x)$ is considerably bigger than m or considerably smaller than M. In that case the value of the integral may be considerably bigger than $m(b-a)$ or

considerably smaller than $M(b-a)$. For example, the integral $\int_0^\pi \sin x dx$ equals 2 and so the average value of $\sin x$ over the interval $[0, \pi]$ is $\frac{2}{\pi}$ (≈ 0.63662) as we see by direct computation. But the inequality (24) would only give that it lies between 0 and 1 since the minimum and the maximum values of $\sin x$ for $0 \leq x \leq \pi$ are 0 and 1 respectively.

This is to be expected, because in spirit, all that (24) says is that the average rainfall of a country lies somewhere between the rainfalls at its driest and wettest spots! For a vast country like India with considerable variation of rainfall, this is a useless piece of information. To get a better picture we must divide the country into a large number of small regions and consider the lower and upper bounds on their rainfalls. Analogously, a much sharper inequality than (23) is obtained if we take a partition, say $(x_0, x_1, \ldots, x_{n-1}, x_n)$ of $[a, b]$. Suppose for $i = 1, 2, \ldots, n$, m_i and M_i are, respectively, some lower and upper bounds on the values of $f(x)$ in the i-th subinterval $[x_{i-1}, x_i]$. Then by (23) applied to this subinterval, we have

$$m_i \Delta x_i \ \leq \ \int_{x_{i-1}}^{x_i} f(x)dx \ \leq \ M_i \Delta_i \tag{25}$$

If we add the middle terms for $i = 1$ to n, we get $\int_a^b f(x)dx$. So we get

$$\sum_{i=1}^{n} m_i \Delta x_i \ \leq \ \int_a^b f(x)dx \ \leq \ \sum_{i=1}^{n} M_i \Delta x_i \tag{26}$$

Note that unless $f(x)$ is constant, strict inequality holds. In general (26) is a much better estimate than (23) because continuity of $f(x)$ means that if the mesh of the partition $(x_0, x_1, \ldots, x_n)$ is small then the function $f(x)$ does not vary much over $[x_{i-1}, x_i]$ for any i and hence that the numbers m_i and M_i can be chosen to be close to each other. As a result, the upper and the lower bounds on the integral do not differ much from each other. These bounds are often called the **upper** and the **lower Riemann sums** of $f(x)$ for the given partition. Every Riemann sum for the same partition lies between them and they all tend to the integral as the mesh of the partition tends to 0. When $f(x)$ is a monotonic function, these sums are very easy to write down since M_i and m_i are simply the values at the end points of the i-th subinterval. In particular, taking $f(x) = \frac{1}{1++x+x^2}$ and $[a, b]$ as $[0, 1]$, (26) shows that $\sum_{k=1}^{n} \frac{n}{n^2+kn+k^2} < \frac{\pi}{3\sqrt{3}} < \sum_{k=0}^{n-1} \frac{n}{n^2+kn+k^2}$ for every n, (JEE 2008).

However, even as its stands, the inequality (23) can give a negative type of information. For example, let I be the integral $\int_0^1 (1 + e^{-x^2})dx$. As remarked in the last chapter, we cannot obtain an antiderivative of e^{-x^2} in a closed form and so we cannot evaluate this integral exactly. Suppose, however, that we

are merely asked to decide if this integral I can equal any of the values $-1, 2$ and $1 + e^{-1}$. (JEE 1981) Note that here the integrand decreases strictly as x increases from 0 to 1. So its maximum and minimum values over $[0, 1]$ are, respectively, 2 and $1 + e^{-1}$ and we can apply (23) with $M = 2$, $m = 1 + e^{-1}$, $a = 0$ and $b = 1$. Moreover strict inequality will hold since there are values of x in $[0, 1]$, where $1 + e^{-x^2}$ lies strictly between m and M. But this means $1 + e^{-1} < I < 2$. Hence I cannot equal any of the three given values.

Comment No. 11:

The applicability of Theorem 1 (and hence its consequences listed above) is limited by the fact that given two functions $f(x)$ and $g(x)$ over an interval $[a, b]$, it generally does not happen that one of them is bigger than the other at *all* points of $[a, b]$. Typically, at some points of $[a, b]$, we have $f(x) \geq g(x)$ while at the remaining points the opposite is true. Equivalently, the difference function $h(x) = f(x) - g(x)$ will, in general, change its sign at least once. In such cases, in absence of any further information it is impossible to tell how the integrals $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ are related to each other (or equivalently, whether the integral $\int_a^b h(x)dx$ is positive or negative). In fact the two integrals may even be equal if the parts, where $f(x) \geq g(x)$ 'balance' those where the opposite holds. For example, $\int_0^1 2xdx$ and $\int_0^1 3x^2dx$ both equal 1. But the integrands are equal only at $x = 0$ and at $x = \frac{2}{3}$. For $x \in (0, \frac{2}{3})$, $2x$ dominates $3x^2$ while for $x \in (\frac{2}{3}, 1]$, $3x^2$ dominates $2x$.

But once again, Theorem 1 can be applied in a negative manner as in the following problem. (JEE 1981)

Let a, b, c be non-zero real numbers such that

$$\int_0^1 (1 + \cos^8 x)(ax^2 + bx + c)dx = \int_0^2 (1 + \cos^8 x)(ax^2 + bx + c)dx \qquad (27)$$

Determine which of the following statements is true about the quadratic equation $ax^2 + bx + c = 0$:

(a) It has no root in $(0, 2)$ (b) it has at least one root in $(0, 2)$

(c) it has a double root in $(0, 2)$ (d) it has complex roots.

Note that the integrands on the two sides are identical, but the interval of integration, viz., $[0, 1]$, in the L.H.S. is contained in that in the R.H.S., viz., $[0, 2]$. If we split the latter into two subintervals $[0, 1]$ and $[1, 2]$, and use the additivity property of the definite integral, then the integral on the R.H.S. is a sum of two integrals of which one part cancels with the L.H.S. So (27) can be replaced by

$$\int_1^2 (1 + \cos^8 x)(ax^2 + bx + c)dx = 0 \qquad (28)$$

Here the integrand, say $f(x)$, is the product of two factors of which the first factor is positive everywhere. So the sign of $f(x)$ is determined by that of

the quadratic expression $ax^2 + bx + c$. If the quadratic has either no roots or a double root in $(1, 2)$ then it will maintain its sign throughout the interval i.e., will be positive throughout or negative throughout except for vanishing at one or two points. But in that case the same will be true of the integrand $(1+\cos^8 x)(ax^2+bx+c)$. The second part of Theorem 1 would then imply that the integral $\int_1^2 (1 + \cos^8 x)(ax^2 + bx + c)dx$ would be either positive or negative. But that would contradict (28). So we conclude that the quadratic equation $ax^2 + bx + c = 0$ has at least one root in $(1, 2)$ and hence at least one root in $(0, 2)$. So out of the given statements (b) is correct.

The framing of the question deserves some comments. First, instead of giving the hypothesis in the form (28) (which is really relevant to the problem) it is given in a twisted form as (27). Evidently the idea here is to test the knowledge of the additivity property of definite integrals. Secondly, the integrand is such that its antiderivative can be found in a closed form if we first identify an antiderivative of $1 + \cos^8 x$ and then integrate by parts. But the work involved will be cumbersome. Indeed, this itself is a hint that some alternate approach should be tried. This point should always be kept in mind in tackling problems about definite integrals. When the integrand is such that an antiderivative can be found rather easily, it is best to find it. When it is impossible to find it, obviously some other techniques have to be applied to deal with the definite integral. (A few such techniques will be studied presently.) But when the integrand is such that an antiderivative *can be* obtained but only with considerable work, it a matter of judgement. Sometimes the credit allocated to the question can be a guide. If it is substantial, then probably the hard work is indeed expected. But for a short question expected to be answered in a minute or two, it is an indication that some clever bit of reasoning rather than brute force computation will do the trick.

Comment No. 12:

We now study a few instances where some peculiar property of the integrand enables us to find its integral over certain intervals without finding an antiderivative. In fact we already mentioned one such property. In Chapter 15, Comment No. 13, we defined the concept of an odd function. To recall, a function $f(x)$ is said to be odd if for every x in its domain, $-x$ is also in the domain and moreover $f(-x) = -f(x)$. We saw that the graph of such a function is symmetric w.r.t. the origin, i.e., whenever a point (x, y) is on it, so is the point $(-x, -y)$, which is the reflection of (x, y) in the origin. The name 'odd function' probably comes from the fact that the function $f(x) = x^n$ is odd if and only if the integer n is odd. There are other examples of odd functions such as $\sin x, \tan x, x \sin^2 x$ and $\sin^{-1} x$ etc. Less obvious examples are $\sinh x$ $(= \frac{1}{2}(e^x - e^{-x}))$ and the function $y = \ln \frac{1+x}{1-x}$. These are special cases of the fact that for any function $g(x)$ defined on an interval containing 0 in its centre, the function $g(x) - g(-x)$ is always an odd function.

In the solution to the problem done at the end of Comment No. 14 of the last chapter, we remarked that if $f(x)$ is odd and defined over an interval

of the form $[-a, a]$ then $\int_{-a}^{a} f(x)dx = 0$. This can be seen from the symmetry property of the graph of $f(x)$ just mentioned. But it can also be done using the additivity property of definite integrals. Let I, I_1 and I_2 denote the integrals $\int_{-a}^{a} f(x)dx$, $\int_{-a}^{0} f(x)dx$ and $\int_{0}^{a} f(x)dx$ respectively. Then $I = I_1 + I_2$. We claim that I_1 and I_2 cancel each other. For, with a change of variable $u = -x$, $I_1 = \int_{-a}^{0} f(x)dx = -\int_{a}^{0} f(-u)du = \int_{a}^{0} f(u)du$, since $f(-u) = -f(u)$. If we interchange the upper and lower limits, this further becomes $-\int_{0}^{a} f(u)du$ which is identical to $-I_2$ since x and u are just 'dummy' variables of integration. So, we have proved that $I_1 = -I_2$. Adding we get $I = 0$.

An entirely similar argument shows that if $f(x)$ is an even function (i.e., if $f(-x) = f(x)$ for all x in the domain of f), then the integrals I_1 and I_2 are equal to each other. (This is also obvious geometrically since now the graph is symmetric about the y-axis. Illustrate this to yourself by drawing the graphs of some familiar even functions such as $1, x^2, \cos x$.) We record the two results together.

$$\int_{-a}^{a} f(x)dx = \begin{cases} 0 & \text{if } f \text{ is an odd function} \\ 2\int_{0}^{a} f(x)dx & \text{if } f \text{ is an even function} \end{cases} \tag{29}$$

Note that this is a very special type of a result in that it is applicable only because the interval of integration is symmetric about 0. For a general interval of the form $[a, b]$, there is no similar formula. But when the interval *is* symmetric about 0 then (29) is a convenient tool. For even functions it reduces the problem to finding the integral over the right half of the interval. For an odd function, we have to do absolutely nothing. The answer is 0 without finding any antiderivative of $f(x)$.

A problem asking you outrightly to integrate an odd function over an interval of the form $[-a, a]$ is unlikely to be asked, except possibly when it takes a while to recognise that the integrand is an odd function. But integration of an odd function could occur as a part of some other problem. Suppose for example, that we are asked to evaluate $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx$. (JEE 1997) Here the denominator is an even function while the numerator is a sum of an odd function, viz., $2x$ and an even function $2x \sin x$. So we can split the integral as the sum of two integrals, one of an odd function $\frac{2x}{1+\cos^2 x}$ and the other of an even function $\frac{2x \sin x}{1+\cos^2 x}$. By (29) the first integral vanishes while the second one equals $2\int_{0}^{\pi} \frac{2x \sin x}{1+\cos^2 x}dx$. This integral cannot be evaluated by finding an antiderivative in a closed form. We shall take up its evaluation a little later.

Comment No. 13:

The relationships $f(-x) = -f(x)$ and $f(-x) = f(x)$ satisfied by odd and even functions respectively, are instances of what are more generally called as **functional equations**. In essence, a functional equation is an equation which relates the values assumed by a function at two or more points which are themselves related in a particular manner. The definition of an odd function can be paraphrased to say that it is a function $f(x)$ which satisfies the functional relation $f(x) + f(y) = 0$ whenever $x + y = 0$. Of course, this does not identify the function uniquely as there are many odd functions. Sometimes with some additional information, a function satisfying a given functional equation can be identified uniquely. We shall take up such problems in Chapter 20. For the moment we are only interested in studying what implications such functional equations have on the definite integrals of these functions. Equation (29) above is a result of this type.

Periodicity of a function is yet another property which can be stated as a functional equation. By definition, a real number T is called a **period** of a function $f(x)$ if whenever x is in the domain of f, so is $x + T$ and further $f(x + T) = f(x)$. If T is a period, so are all integral multiples of T (including negative ones). So generally we consider only positive periods. The least positive period, say T^* of a function is often called *the* period of it. It is not hard to show (see Exercise (18.7)) that the only periods of f in such a case are the multiples of T^*. We already know that 2π is the period of $\sin x$ and $\cos x$ while π is the period of $\tan x, |\sin x|, \sin^2 x$ and $\cos^2 x$. It is easily seen that if T is a period of $f(x)$ then, for every non-zero real number α, $\frac{T}{\alpha}$ is a period of $f(\alpha x)$.

In essence, if T is a (positive) period of a function $f(x)$, then its values repeat after an interval of length T. Naturally, anything which depends upon these values will also recur after the same period. For example if a function $f(x)$ with period T is continuous at a point c, it will be so at all points of the form $c \pm nT,\ n = 1, 2, 3, \ldots$. In particular, the derivative of a periodic function is periodic. The local maxima, local minima, the points of inflection will all recur with periodicity T. Since there are many processes in real life (such as temperature at a place) which show similar recurring behaviour, it is hardly surprising that periodic functions are important in real-life applications. Another reason is that uniform circular motion is tremendously important in modern life. The electricity we get is generated by a uniformly rotating armature. The position of a point moving in a plane is specified by expressing its x and y coordinates as functions of time, say t. If the motion is uniform circular then both these functions are periodic.

As for integrals, it is obvious that if $f(x)$ is periodic with a period T, then for all a, b we must have

$$\int_a^b f(x)dx = \int_{a+T}^{b+T} f(x)dx = \int_{a-T}^{b-T} f(x)dx \tag{30}$$

This follows because a partition, say $(x_0, x_1, \ldots, x_n)$ of $[a, b]$ induces the partitions $[x_0 \pm T, x_1 \pm T, \ldots, x_n \pm T)$ of the intervals $[a \pm T, b \pm T]$ and vice versa.

The corresponding Riemann sums of $f(x)$ on all the three intervals are equal by the periodicity of $f(x)$ and hence so are their limits. But that's precisely what (30) says.

Repeated applications of (30) give that for every integer n (positive, negative or 0),

$$\int_{a+nT}^{b+nT} f(x)dx = \int_a^b f(x)dx \tag{31}$$

Both (30) and (31) are immediate consequences of periodicity and are obvious. As an application of (31), we prove that the integral of a periodic function of (positive) period T is the same over *all* intervals of length T. This is not immediately obvious from (31). But an intuitive argument can be given as follows. A periodic function is like a bus ride along a circular route. The integral is what you get in one round trip. Then no matter where you board the bus, as long as you complete one round and return to the same spot, your net gain ought to be the same!

The result can be expressed more formally by saying that if $f : \mathbb{R} \longrightarrow \mathbb{R}$ is periodic with a period T, then for any two real numbers a and b

$$\int_a^{a+T} f(x)dx \; = \int_b^{b+T} f(x)dx \tag{32}$$

or in other words that the integral $\int\limits_a^{a+T} f(x)dx$ is independent of a. (JEE 1984)

To prove this it suffices to fix one of the intervals arbitrarily as, say, $[0, T]$ and then show that for every $a \in \mathbb{R}$,

$$\int\limits_a^{a+T} f(x)dx = \int_0^T f(x)dx \tag{33}$$

We shall do so by writing each side as a sum of two quantities and showing that equal quantities appear on each side. Consider the semi-open interval $[a, a+T)$. Since it is of length T, it will contain exactly one integral multiple, say kT of T. (The value of k will depend upon a and will be negative if $a \leq -T$.) In other words,

$$a \;\leq\; kT \;<\; a + T \tag{34}$$

We decompose the interval $[a, a + T]$ into two subintervals, $[a, kT]$ and $[kT, a + T]$. (The first interval could degenerate to a single point if $a = kT$.) By the additivity property of definite integrals, we have

$$\int_a^{a+T} f(x)dx = \int_a^{kT} f(x)dx + \int_{kT}^{a+T} f(x)dx \tag{35}$$

Also from (34) we have $0 < a + T - kT \leq T$. So breaking the interval $[0, T]$ at the point $a + T - kT$ we get

$$\int_0^T f(x)dx = \int_0^{a+T-kT} f(x)dx + \int_{a+T-kT}^T f(x)dx \tag{36}$$

Since $k-1$ and k are integers, by (31) the first term in the R.H.S. of (35) equals the second term in the R.H.S. of (36) while the second term of the R.H.S. of (35) equals the first term of the R.H.S. of (36). This proves (33).

The proof just given only assumed that the function $f(x)$, besides being periodic, is integrable on every interval (i.e., its Riemann integral exists). It is not given here that $f(x)$ is continuous. If this were the case then we could apply the second form of the Fundamental Theorem of Calculus and get a slicker proof. Indeed, define a function $F : \mathbb{R} \longrightarrow \mathbb{R}$ by $F(a) = \int_a^{a+T} f(x)dx$. Here a is a variable and both the upper and the lower limits of the integral are functions of a. So by (19) of the last chapter, we have $F'(a) = \dfrac{d}{da} \int_a^{a+T} f(x)dx = f(a+T)\dfrac{d}{da}(a+T) - f(a)\dfrac{d}{da}(a) = f(a+T) - f(a)$. But by periodicity of f (with T as a period), this is identically 0. So F is a constant function. Hence $F(a) = F(b)$ for all a, b. That is, (32) holds.

We remark again that even though the second proof is more elegant, the first proof is superior not only because it is more general but also because it stays closer to the true identity of a definite integral as the limit of Riemann sums.

As an application of (32) we can calculate $\int_0^{\pi} \sin^2 x dx$. Call this integral I. The function $\sin^2 x$ has π as a period (in fact, as its smallest period). So by (32), I also equals $\int_{\pi/2}^{3\pi/2} \sin^2 x dx$. The substitution $u = x - \frac{\pi}{2}$ converts this latter integral to $\int_0^{\pi} \cos^2 u du$ which, by a change of the dummy variable of integration, is the same as $\int_0^{\pi} \cos^2 x dx$. Therefore the integrals $\int_0^{\pi} \sin^2 x dx$ and $\int_0^{\pi} \cos^2 x dx$ both equal I. Adding, $2I = \int_0^{\pi} (\sin^2 x + \cos^2 x)dx$ which obviously equals π. So the given integral $\int_0^{\pi} \sin^2 x dx$ equals $\frac{\pi}{2}$. Of course, we could have as well evaluated this integral by finding an antiderivative of the integrand. This can be done by writing $\sin^2 x$ as $\frac{1}{2}(1 - \cos 2x)$ so that $\int \sin^2 x dx = \frac{x}{2} + \sin x \cos x + c$. In fact, this approach is superior because it will give us the value of $\int_a^b \sin^2 x dx$ for *any* a and b, whereas the earlier solution applies only when $b - a$ is a multiple of π, the period. But we just wanted to illustrate a method. There can be situations

where the antiderivative is very cumbersome to find but some alternate method gives the answer (see Exercise (18.15)(xv) for a similar problem).

Comment No. 14:

The crucial part of the alternate calculation of the integral $I = \int_0^\pi \sin^2 x dx$ just given was that using some special property of the integrand and the interval (viz., that its length coincided with a period of the integrand) we could relate the given integral I to some other integral J over the same interval. (Specifically, J was the integral $\int_0^\pi \cos^2 x dx$ and it happened to be equal to I.) Neither I nor J was so easy to compute individually. But the sum $I + J$ was ridiculously easy to compute. This gave another relationship between I and J. Combined with the earlier relationship (viz., $I = J$) we could get the values of I and J. Even though our interest was only in I we needed J as a sort of a catalyst to find I.

We now study another important illustration of this technique. The preceding example was based on (32) where the two integrals were equal because of the fact that the integrand was a periodic function with period T, which was also the length of the interval of integration. A real life analogy was that in a round tour which keep on repeating itself in cycles, no matter where you begin it, you get the same thing in one full round. Our second illustration can also be motivated by a real-life example. Suppose we have a piece of wire of length L with two ends, say A and B. A point P on the wire can be specified by specifying its distance (along the wire) from either of the two ends. In fact, if the distance of P from A is x, then its distance from B is $L - x$. Now let $f(x)$ be the linear mass density of the wire at a point P which is at a distance x from A. Then as we saw in Comment No. 8 of the last chapter, the mass of the wire is the integral, $\int_0^L f(x)dx$. If, instead, we measure the distance from the other end B of the wire, then the mass would be the integral $\int_0^L f(L-x)dx$. But obviously, the mass of the wire is the same no matter how we calculate it. So the two integrals must be the same, i.e., we must have

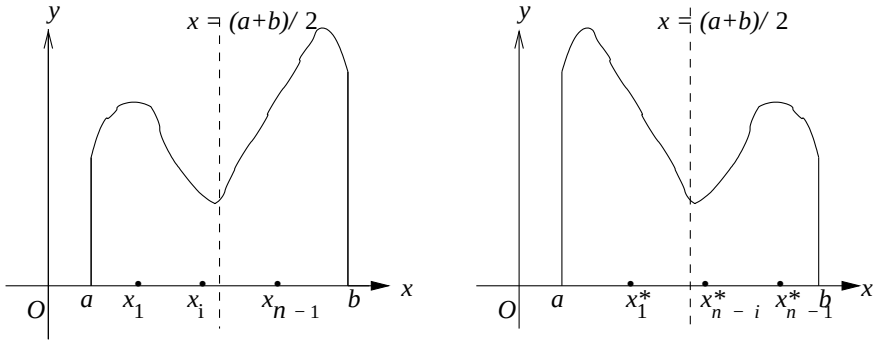
$$\int_0^L f(x)dx = \int_0^L f(L-x)dx \tag{37}$$

The argument here can be generalised for any definite integral over any interval. The result is

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx \tag{38}$$

This can be proved geometrically by observing that the graphs of $y = f(x)$ and $y = f(a + b - x)$ over the intervals are mirror images of each other in the vertical line passing through the centre of the interval, i.e., the line $x = \frac{a+b}{2}$ as shown in the figure above. Therefore the areas below the two graphs are also

reflections of each other. But areas are invariant under a reflection. This gives (38). The argument can be made more rigorous (and less dependent on geometric intuition) by observing that every partition, say $(x_0, x_1, \ldots, x_i, \ldots, x_n)$ of the interval $[a, b]$ gives rise to another partition $(x_0^*, x_1^*, \ldots, x_i^*, \ldots, x_n^*)$ of $[a, b]$, where $x_i^* = a+b-x_{n-i}$ for $i = 0, 1, \ldots, n$. The lengths of the subintervals of the first partition are the same as those of the subintervals of the second partition but in a reverse order. Using this, one can equate their Riemann sums and (38) would follow by taking limits as the mesh of the partition tends to 0.



In this proof all that we assumed was that the function $f(x)$ was integrable over $[a, b]$. Frequently, it is also continuous and in that case a slicker proof of (38) can be given by putting $u = a+b-x$ in the integral on the R.H.S. which converts it to $-\int_b^a f(u)du$, i.e., to $\int_a^b f(u)du$ which is precisely the L.H.S.

Evidently, (37) is a special case of (38). In most applications, this special case is sufficient. The way (38) (or (37)) is applied is as follows. Call the integrals on the two sides as I and J. Then $I = J$. Also $I+J = \int_a^b (f(x)+f(a+b-x))dx$. Sometimes the nature of the function $f(x)$ (and of the numbers a and b) is such that the sum $f(x) + f(a + b - x)$ has a particularly simple expression. For example, suppose $f(x) = \sin^2 x$ and the interval is $[0, \frac{\pi}{2}]$. Then $f(a + b - x) = f(\frac{\pi}{2} - x) = \cos^2 x$ and so $f(x) + f(a + b - x)$ is a constant! In such cases, it is easier to find $I + J$ than to find either I or J individually. Since we already know $I = J$, we can determine I once we know the value of $I + J$.

As a typical illustration of this procedure, let us complete the evaluation of the integral $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx$ (JEE 1997) which we began at the end of Comment No. 12. There we showed that this integral equals $4\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$. Call the latter integral as I. Since $\sin(\pi-x) = \sin x$ and $\cos(\pi-x) = -\cos x$, by (37) I also equals $\int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}dx$. Adding the two, we get $2I = \pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$. We can evaluate this by finding an antiderivative of the integrand. (Earlier we were stuck because of the troublemaker x in the numerator.) Using the

substitution $u = \cos x$, the indefinite integral $\int \dfrac{\sin x}{1+\cos^2 x}dx$ comes out to be $-\tan^{-1}(\cos x) + c$ and so $\int\limits_0^{\pi} \dfrac{\sin x}{1+\cos^2 x}dx$ equals $\tan^{-1}(\cos 0) - \tan^{-1}(\cos \pi) = \tan^{-1} 1 - \tan^{-1}(-1) = \pi$. Hence $I = \frac{\pi^2}{2}$. The original integral is 4 times this, i.e., $2\pi^2$.

Equation (37) can also be applied to simplify a given integral. For example, consider the definite integral $\displaystyle\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x}dx$. (JEE 1983) Here if we write $\sin 2x$ as $2\sin x \cos x$ we see that the integrand is a rational function of $\sin x$ and $\cos x$. Therefore, as remarked in Comment No. 21 of the last chapter, the substitution $u = \tan \frac{x}{2}$ will convert the integral to one where we have to integrate a rational function of u w.r.t. u, for which the method of partial fractions is available. However, because of the product $\sin x \cos x$ in one of the terms of the denominator, the resulting rational function in u will have a fourth degree polynomial in the denominator and hence will probably not be easy to factorise. So we try another approach. If we do carry out the substitution $u \tan \frac{x}{2}$, the indefinite integral $\displaystyle\int \frac{\sin x + \cos x}{9 + 16\sin 2x}dx$ reduces to the indefinite integral $\displaystyle\int \frac{2(2u + 1 - u^2)}{9(u^2+1)^2 + 64u(1-u^2)}du$, the very sight of which is formidable. It is an asset to be able to see such dead ends before you have wasted a lot of time. This is where trying alternate approaches helps. If the problem was really meant to be done this laborious way, then the problem could as well have been posed simply as a problem of finding an indefinite integral, because once you find it, evaluating the definite integral from it is merely a matter of substituting the values of the upper and the lower limits of integration into it. The very fact that the problem is posed as a definite integral is an indication that it is intended to be done by some other method. Of course, this is not a hard and fast rule. The right thing to do is to keep the mind open to all possible approaches and first see how easy or difficult each one is going to be without actually carrying out a lot of work.

So let us try to apply (37). Call the given integral I. Since $\sin(\frac{\pi}{4} - x) = \dfrac{1}{\sqrt{2}}(\cos x - \sin x)$, $\cos(\frac{\pi}{4} - x) = \dfrac{1}{\sqrt{2}}(\cos x + \sin x)$ and $\sin 2(\frac{\pi}{4} - x) = \cos 2x$, we get $I = \displaystyle\int_0^{\pi/4} \frac{\sqrt{2}\cos x\ dx}{9 + 16\cos 2x}$. This is more manageable than the original expression for I because now we can write the denominator entirely in terms of $\sin x$ and so the term $\cos x\ dx$ in the numerator suggests that the substitution $u = \sin x$ will work. We leave it to you to check that this substitution transforms I to the definite integral $\displaystyle\int_0^{1/\sqrt{2}} \frac{\sqrt{2}\ du}{25 - 32u^2}$. True, this still needs partial fractions. But it is much simpler than the earlier approach. Using the formula $\displaystyle\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c$ which we derived in Comment No. 16 of the last chapter, we

now have
$I = -\frac{\sqrt{2}}{32}\int_0^{1/\sqrt{2}} \frac{du}{u^2 - (\frac{5}{4\sqrt{2}})^2} = -\frac{\sqrt{2}}{32}\frac{2\sqrt{2}}{5}\ln|\frac{4\sqrt{2}u-5}{4\sqrt{2}u+5}|\Big|_0^{1/\sqrt{2}}$ which comes out to be $= -\frac{1}{40}(\ln\frac{1}{9} - \ln 1) = \frac{\ln 3}{20}$.

As an example where (38) (rather than (37)) is needed, consider the integral $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$. (JEE 1994) Again call this I. Then by (38) I also equals $\int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}}dx$. Hence $2I = \int_2^3 1\ dx = 1$. So $I = \frac{1}{2}$.

Formulas (37) and (38), coupled with some special properties of other functions such as the trigonometric ones, sometimes allow us to relate integrals over different intervals as well. Here is an example. (JEE 1990) Suppose we want to prove that

$$\int_0^{\pi/2} f(\sin 2x)\sin x\ dx = \sqrt{2}\int_0^{\pi/4} f(\cos 2x)\cos x\ dx \tag{39}$$

Denote the integral on the L.H.S. by I. There is little point in applying (37) to I since that would only give $I = \int_0^{\pi/2} f(\sin 2x)\cos x\ dx$ which is difficult to relate to the R.H.S. However, the fact that the integral on the R.H.S. is over the interval $[0, \frac{\pi}{4}]$ suggests that one possible line of attack is to split I as $I_1 + I_2$, where

$$I_1 = \int_0^{\pi/4} f(\sin 2x)\sin x\ dx \tag{40}$$

$$\text{and}\quad I_2 = \int_{\pi/4}^{\pi/2} f(\sin 2x)\sin x\ dx \tag{41}$$

We apply (37) to I_1. As for I_2 we use the substitution $u = x - \frac{\pi}{4}$ to get

$$I_1 = \frac{1}{\sqrt{2}}\int_0^{\pi/4} f(\cos 2x)(\cos x - \sin x)\ dx \tag{42}$$

$$\begin{aligned}\text{and}\quad I_2 &= \int_0^{\pi/4} f(\cos 2u)\sin(\frac{\pi}{4}+u)\ du \\ &= \frac{1}{\sqrt{2}}\int_0^{\pi/4} f(\cos 2u)(\sin u + \cos u)\ du \\ &= \frac{1}{\sqrt{2}}\int_0^{\pi/4} f(\cos 2x)(\sin x + \cos x)\ dx \qquad (43)\\ &\quad \text{(by a change of the dummy variable)}\end{aligned}$$

Adding (42) and (43) we get $I = \sqrt{2}\int_0^{\pi/4} f(\cos 2x)\cos x\ dx$ which is precisely the R.H.S. of (39).

Comment No. 15:

The exercises will give more examples of integrals that can be evaluated using (37), and more generally, (38). (The list deliberately includes a few where finding antiderivatives is the best method! This point should be kept in mind. In a book, the chapter where a given problem occurs can give some hint regarding the method to be used. In an examination, no such clue exists generally and one has to first figure out where the problem belongs, much the same way that when you study clinical pathology, you study diseases of one system of the body at a time; but when you practise medicine, you first have to figure out which part of the patient's body is really affected and external symptoms can sometimes be misleading in this respect!)

In all the examples of definite integrals we considered so far, the integrand was continuous at all points of the interval of integration. This enabled us to apply the Fundamental Theorem of Calculus. Even where this theorem could not be applied (because of the failure of the integrand to have an antiderivative in a closed form), we did not have to worry about the existence of the integral. Although we never proved it, it can be shown that every continuous function is integrable.

Occasionally we have to integrate functions which have points of discontinuity in the interval of integration. We shall not go into the question of the existence of the definite integral in such cases. We assume that we have a function $f(x)$ defined on an interval and that it has only a finite number of points of discontinuity, say $c_1, c_2, \ldots, c_k$, with $a \leq c_1 < c_2 < \ldots c_k \leq b$. For notational uniformity, call a as c_0 and b as c_{k+1}. In such a case, it can be shown that if f is bounded on $[a, b]$ then it is integrable on $[a, b]$, i.e., the limit of its Riemann sums exists as the mesh of the partition tends to 0. Moreover, its integral over $[a, b]$ is simply the sum of the integrals over the subintervals into which the interval $[a, b]$ is split by the points $c_1, c_2, \ldots, c_k$. In symbols,

$$\int_a^b f(x)\,dx = \int_{c_0}^{c_1} f(x)\,dx + \int_{c_1}^{c_2} f(x)\,dx + \ldots + \int_{c_k}^{c_{k+1}} f(x)\,dx \qquad (44)$$

Earlier we have used a similar splitting for evaluating some areas because the region had to be split into two (or more) subregions to find its area. But that was necessitated not by a discontinuity of the integrand but because it suddenly changed a formula at some intermediate point. (See for example, (5) of the last chapter.)

A typical term in the R.H.S. of (44) is of the form $\int\limits_{c_{i-1}}^{c_i} f(x)dx$. Note that now the function f has no discontinuity in the open interval (c_{i-1}, c_i). If further, the right-handed limit at c_{i-1} and the left handed limit at c_i both exist then by redefining $f(c_{i-1})$ and $f(c_i)$ (if necessary) so as to coincide with these limits, we may effectively suppose that f is continuous at the end points c_{i-1} and c_i too. In that case, the integral $\int\limits_{c_{i-1}}^{c_i} f(x)dx$ can be evaluated with any of the methods we have used so far. Adding all these $(k+1)$ integrals, we get the

value of the integral $\int_a^b f(x)\,dx$ by (44).

We illustrate this procedure with several examples. First consider the integral, say $I = \displaystyle\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}\,dx$. (JEE 1983) Here the integrand is continuous in the open interval $(0, \frac{\pi}{2})$. But since $\tan x$ is discontinuous at $\frac{\pi}{2}$ while $\cot x$ is discontinuous at 0, the integrand has discontinuities at the end points, viz., 0 and $\frac{\pi}{2}$. But it is easily seen that the integrand tends to 1 as $x \to 0^+$ while it tends to 0 as $x \to \frac{\pi}{2}^-$. So we might as well carry out the integration as if these discontinuities did not exist. Now, we cannot evaluate I by directly finding an antiderivative of the integrand. However, since $\tan(\frac{\pi}{2} - x) = \cot x$ and $\cot(\frac{\pi}{2} - x) = \tan x$, if we apply (37) we see that I also equals the integral $\displaystyle\int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx$. Adding the two integrals, $2I = \displaystyle\int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$. So $I = \frac{\pi}{4}$. (In Exercise (17.7) we had a situation where the integrand had a removable discontinuity only at the left end point. But there the integral was obtained by finding an antiderivative in a closed form.)

In this example, we did not have to split the interval of integration since both the discontinuities occurred at the end points. As an example where the integrand is discontinuous at an interior point, consider the integral $I = \displaystyle\int_0^{1.5} [x^2]\,dx$ where [ ] denotes the greatest integer function. (JEE 1988) (Although this notation is fairly standard, square brackets are also often used simply to enclose complicated expressions so as to avoid confusion, especially where ordinary parentheses are already in use, as for example, in the expression $\ln\,[\sqrt{(1-x)} + \sqrt{(1+x)}]$. One then has to decide from the context whether they are really meant to convey the integral part.)

Here the integrand is discontinuous at those points where x^2 is a positive integer. Considering that $(1.5)^2 = 2.25$ lies between 2 and 3, in the interval $[0, 1]$ the integrand is discontinuous only at $x = 1$ and $x = \sqrt{2}$. (We ignore 0 as it is an end point, and at 0, $[x^2]$ is right continuous anyway.) So the given integral splits into three integrals, one over $[0, 1]$, one over $[1, \sqrt{2}]$ and one over $[\sqrt{2}, 1.5]$. Over each of these intervals, the integrand is constant except possibly for the end points, the values of these constants being $0, 1$ and 2. So by (44), the given integral equals $0\times(1-0)+1\times(\sqrt{2}-1)+2\times(1.5-\sqrt{2}) = \sqrt{2}-1+3-2\sqrt{2} = 2-\sqrt{2}$ ≈ 0.5858. (Normally, the last conversion of the answer is not expected. But in the present problem it is given because the data was in a decimal form.)

Comment No. 16:

We now consider a situation where the integrand $f(x)$ is continuous on the open interval (a, b) but has more serious types of discontinuities at the end points than those considered in the examples above. These arise when either the limit $\lim\limits_{x\to a^+} f(x)$ and/or the limit $\lim\limits_{x\to b^-} f(x)$ does not exist. We consider only the first case here. The discussion of the second case is entirely similar. (If neither of the two limits exists, we insert a point c and consider the two integrals $\displaystyle\int_a^c f(x)\,dx$

and $\int_c^b f(x)\ dx$ separately and add. This way, we have to deal with only one difficulty at a time.)

So we assume $f(x)$ is continuous on the semi-open interval $(a, b]$. Then for all sufficiently small positive values of δ (specifically, for all $0 < \delta < b - a$), the point $a + \delta$ is in (a, b) and moreover the function $f(x)$ is continuous on the closed interval $[a + \delta, b]$. So the integral $\int_{a+\delta}^b f(x)dx$ is defined. Naturally, its value will depend on δ. Denote it by $I(\delta)$. As δ gets smaller and smaller, the interval $[a + \delta, \delta]$ captures more and more of the the interval $[a, b]$ and so the most reasonable definition of $\int_a^b f(x)dx$ will be $\lim_{\delta\to 0^+} I(\delta)$. We indeed adopt this definition. Moreover, it can be shown that in case $f(x)$ is continuous at the point a too, so that $\int_a^b f(x)dx$ is already defined, then as $\delta \to 0^+$, $I(\delta)$ indeed approaches $\int_a^b f(x)dx$. So the new definition is consistent with the old one in case the latter makes sense. But there are situations where the old definition does not make sense and still the new one does.

Consider, for example, the indefinite integral $\int_0^1 \ln x\ dx$. (JAT 1979) Here the integrand tends to $-\infty$ as $x \to 0^+$. So it is not continuous on the interval $[0, 1]$. But it is continuous on $[\delta, 1]$ for every $\delta \in (0, 1)$. So the integral $I(\delta) = \int_\delta^1 \ln x\ dx$ is defined. Indeed we can evaluate it explicitly. In (50) of the last chapter we integrated by parts and found $x \ln x - x$ as an antiderivative of $\ln x$. Hence $I(\delta) = (x \ln x - x)\Big|_\delta^1 = \delta - 1 - \delta \ln \delta$. So to find the given integral, we have to take the limit of this expression as $\delta \to 0^+$. The limit of the last term is of the form $0 \times (-\infty)$, which is an indeterminate form. To find it we put $y = \frac{1}{\delta}$ and take the limit as $y \to \infty$. So, $\lim_{\delta\to 0^+} \delta \ln \delta = \lim_{y\to\infty} \dfrac{-\ln y}{y}$. The latter can be evaluated by l'Hôpitals's rule as $\lim_{y\to\infty} \dfrac{-1/y}{1} = 0$. So $\delta \ln \delta \to 0$ as $\delta \to 0^+$ and hence we have $\lim_{\delta\to 0^+} I(\delta) = 0 - 1 - 0 = -1$. Hence according to our new definition, the integral $\int_0^1 \ln x\ dx$ has value -1.

The integral $\int_0^1 \ln x\ dx$ is an example of what is called an **improper integral**. More generally, the term is applied to any integral $\int_a^b f(x)\ dx$ in which the integrand $f(x)$ tends to $\pm\infty$ as x approaches either of the end points a and b. The study of such integrals is beyond our scope. However, see Exercise (18.16) for a couple of examples.

EXERCISES

18.1 If $I_n = \int_0^{\pi/2} x^n \sin x dx$, then prove that

$$I_n + n(n-1)I_{n-2} = n(\frac{\pi}{2})^{n-1}$$

for $n \geq 2$. Deduce that $I_3 = \frac{3}{4}\pi^2 - 6$.

18.2 Obtain reduction formulas for the indefinite integrals of the following functions.

(a) $\int x^m \cos x dx$ (b) $\int x^m e^x dx$
(c) $\int (\ln x)^m dx$ (d) $\int x^a (\ln x)^m dx \quad (a \neq -1)$

18.3 Determine a positive integer $n \leq 5$, such that the integral $\int_0^1 e^x (x-1)^n dx$ equals $16 - 6e$. (1992)

18.4 Letting $I_n = \int_0^{\pi/2} \sin^n x dx$, prove that for all $m \geq 0$

$$1 \leq \frac{I_{2m}}{I_{2m+1}} \leq 1 + \frac{1}{2m}$$

[*Hint*: Compare the integrands of I_{2m+1}, I_{2m} and I_{2m-1} and use the reduction formula.]

18.5 Using (5), (6) and the last exercise, prove Wallis formula.

18.6 (a) Prove that a linear combination of two even functions is even and that of two odd functions is odd. What can be said about the product, the quotient and the composite of two functions in terms of these two functions being even/odd?

(b) Prove that the only function which is both even and odd is the identically zero function.

(c) Prove that for every function $f(x)$, the function $\frac{1}{2}(f(x) + f(-x))$ is even and the function $f(x) - f(-x)$ is odd. Hence show that every function can be expressed uniquely as the sum of an even function and an odd function. (This is of the same spirit as Part (v) of Exercise (2.34).)

(d) Prove that the derivative of a (differentiable) even function is odd and vice versa. Interpret this result geometrically. (1983, 1987)

(e) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ and $g : \mathbb{R} \longrightarrow \mathbb{R}$ be continuous functions. Find the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [f(x) + f(-x)][g(x) - g(-x)]dx$. (1990)

18.7 (a) Prove that the sum and the difference of any two periods of a function $f(x)$ are also periods of it.

(b) Let T^* be the smallest positive period of a function $f(x)$. Prove that the only possible periods of $f(x)$ are multiples of T^*. [*Hint:* Given a period T, find an integer m such that $mT^* \le T < (m+1)T^*$. Apply (a).]

18.8 (a) Let $T > 0$ be a fixed real number. Suppose f is a continuous function such that for all $x \in \mathbb{R}$, $f(x+T) = f(x)$. Let $I = \int_0^T f(x)\,dx$. Express the value of $\int_3^{3+3T} f(2x)\,dx$ in terms of I. (2002)

(b) If $g(x) = \int_0^x \cos^4 t\,dt$, determine which of the following expressions equals $g(x+\pi)$. (1997)

(A) $g(x)+g(\pi)$ (B) $g(x)-g(\pi)$ (C) $g(x)g(\pi)$ (D) $\dfrac{g(x)}{g(\pi)}$.

18.9 For $x > 0$, let $f(x) = \int_1^x \dfrac{\ln t}{1+t}\,dt$. Determine the function $f(x) + f(\frac{1}{x})$ and prove that $f(e) + f(\frac{1}{e}) = \frac{1}{2}$. (Here $\ln t = \log_e t$.) (2000)

18.10 Prove that $\int_0^{\pi} xf(\sin x)\,dx = \dfrac{\pi}{2}\int_0^{\pi} f(\sin x)\,dx$. (1982)

18.11 Prove that $\int_0^1 \tan^{-1}\left(\dfrac{1}{1-x+x^2}\right)dx = 2\int_0^1 \tan^{-1} x\,dx$. Hence or otherwise, evaluate the integral $\int_0^1 \tan^{-1}(1-x+x^2)\,dx$. (1998)

18.12 Must the value of $\int_0^{2a} [f(x)/\{f(x)+f(2a-x)\}]dx$ equal a ? (1988)

18.13 Let f be a positive function. Let $I_1 = \int_{1-k}^{k} xf(x(1-x))\,dx$ and $I_2 = \int_{1-k}^{k} f(x(1-x))\,dx$ where $2k-1 > 0$. Determine which of the following expressions is the value of $\frac{I_1}{I_2}$. (1997*)

(a) 2 (b) k (c) $\frac{1}{2}$ (d) 1.

18.14 If f and g are continuous functions on $[0,a]$ satisfying the relations $f(x) = f(a-x)$ and $g(x)+g(a-x) = 2$, then show that $\int_0^a f(x)g(x)\,dx = \int_0^a f(x)\,dx$. (1989)

18.15 Evaluate the following definite integrals :

(i) $\int_0^{\pi} e^{\cos^2 x} \cos^3(2n+1)x \, dx$ (1985) (ii) $\int_0^{\pi/2} \frac{x \sin x \cos x}{\cos^4 x + \sin^4 x} \, dx$ (1985)

(iii) $\int_{\pi/4}^{3\pi/4} \frac{\phi}{1+\sin\phi} \, d\phi$ (1993) (iv) $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ (1993)

(v) $\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x} + e^{-\cos x}} \, dx$ (1999) (vi) $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ (1999)

(vii) $\int_0^{\pi/4} \ln(1+\tan x) \, dx$ (1997*) (viii) $\int_{-1}^{1} x - [x] \, dx$ (1998)

(ix) $\int_{-1/2}^{1/2} \left([x] + \ln \frac{1+x}{1-x} \right) dx$ (2002) (x) $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2\sin x] \, dx$ (1999)

(xi) $\int_0^{\pi} \frac{x \, dx}{1+\cos\alpha \sin x}$, $0 < \alpha < \pi$ (1986)

(xii) $\int_0^{\pi/2} f(x) \, dx$, where $f(x) = \begin{vmatrix} \sec x & \cos x & \sec^2 x + \cot x \operatorname{cosec} x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$ (1987)

(xiii) $\int_0^{1} \log \left[\sqrt{(1-x)} + \sqrt{(1+x)} \right] dx$ (1988)

(xiv) $\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \left(\frac{x^4}{1-x^4} \right) \cos^{-1} \left(\frac{2x}{1+x^2} \right) dx$ (1995)

(xv) $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} \, dx$, where $n > 0$ (1996)

18.16* Assuming that (37) is valid for improper integrals too, and using the symmetry of $\sin x$ about $x = \frac{\pi}{2}$, show that $\int_0^{\pi/2} \ln(\sin x) \, dx = -\frac{\pi}{2} \ln 2$. Using this, evaluate $\int_0^{\pi/2} \log(\tan x + \cot x) \, dx$. (JAT 1979)

18.17 Using $\int_0^{1} \ln x \, dx = -1$, find $\lim_{n\to\infty} \frac{(n!)^{1/n}}{n}$. (JAT 1979)

18.18 For large n, show that $\int_0^{1} \frac{nx^{n-1} dx}{1+x^2}$ nearly equals $\frac{1}{2}$.

18.19 For every positive integer n, prove that

$$\sum_{k=0}^{n^2} (-1)^k \frac{\binom{n^2}{k}}{\binom{n+k}{k}} = \frac{1}{n+1}$$

Chapter 19

DIFFERENTIAL EQUATIONS

Differential equations easily constitute one of the most important parts of mathematics from the point of view of applications to real life. The primary reason is that the various laws of physics are often in the form of equations involving rates of change of one quantity w.r.t. another, e.g., speed, acceleration, the rate of linear expansion of a wire w.r.t. temperature, the electrical current (which is the rate at which charge flows) and so on. Since the mathematical equivalent of a rate is a derivative, differential equations arise very naturally in real life and methods for solving them acquire paramount importance.

The theory of differential equations is too rich to be contained even in a single book, let alone a single chapter. We present here only a glimpse of one corner (but an important one, nevertheless) of it. The Main Problem is a simple but typical application to fluid flow. This is followed by other problems involving rates of growth of both continuous and discrete quantities (Comment No. 5 and 6). In Comment No. 7, we study some generalities about differential equations. The subsequent sections deal with various elementary methods of solving first order equations. Applications of differential equations to determine classes of plane curves are taken up in Comment No. 16 onwards. They include the problem of finding orthogonal trajectories. At the end there is a brief mention of singular solutions.

Differential equations have been introduced in the JEE syllabus relatively recently. Generally only elementary questions are asked. But because differential equations are not included at full steam, sometimes the problems appear tricky or obscure in the absence of the relevant supplementary material.

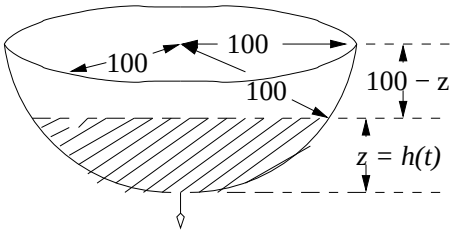
Main Problem : A hemispherical tank of radius 1 meter is initially full of water and has an outlet of 12 cm^2 cross-sectional area at its bottom. The outlet is opened at some instant. The flow through the outlet is according to the law $v(t) = \lambda\sqrt{h(t)}$, where t is the time measured in seconds, $v(t)$ is the velocity of the flow (in cm/sec) at time t, $h(t)$ is the height (in centimeters) of the water level above the outlet at time t and λ is some constant. If T is the time (in seconds) it takes to empty the tank, then $\frac{\lambda T}{\pi}$, expressed as a reduced rational equals ..... . (JEE 2001, modified)

First Hint: Find the rate at which the volume of water in the tank is changing.

Second Hint: Write a differential equation for $h(t)$.

Third Hint: Solve the differential equation with suitable initial conditions.

Solution: The problem is ambiguous because if a hole is cut at the bottom of a hemisphere, it reduces its height and what is left is not a hemisphere but a frustum of a hemisphere. (This difficulty would not arise if the base of the tank were a flat surface as in the case of a box-shaped tank or a barrel.) This will complicate the calculations. As the tank is *given* to be hemispherical despite the hole, we assume that the decrease in height because of the hole is negligible. (Or, we suppose that the 'hole' consists of a spherical cap made of some porous membrane through which water can flow.)



Now let $V(t)$ be the volume of the water in the tank at time t. Then the rate at which V is depleted is the product of the area of the hole and the velocity of the water flowing through. So,

$$\frac{dV}{dt} = -12v(t) = -12\lambda\sqrt{h(t)} \tag{1}$$

Since the area of the cross-section of the sphere at a height z cm above the hole is $\pi(\sqrt{(100)^2 - (100-z)^2})^2$ i.e., $\pi(200z - z^2)$ cm^2, by Exercise (17.13) $V(t)$ is given by

$$V = V(t) = \int_0^{h(t)} \pi(200z - z^2)dz \tag{2}$$

Hence from the fundamental theorem of calculus (Theorem 2 of Chapter 17),

$$\frac{dV}{dt} = \pi(200h(t) - h(t)^2)h'(t) \tag{3}$$

From (1) and (3) and writing h for $h(t)$, we get

$$\pi(200h - h^2)\frac{dh}{dt} = -12\lambda\sqrt{h} \quad (4)$$

$$\text{or} \quad \pi(200h^{\frac{1}{2}} - h^{\frac{3}{2}})dh = -12\lambda\, dt \quad (5)$$

Solving,

$$\pi(\frac{400}{3}h^{\frac{3}{2}} - \frac{2}{5}h^{\frac{5}{2}}) = -12\lambda t + c \quad (6)$$

where c is some constant. We are given that

$$h = 100 \text{ when } t = 0 \quad (7)$$

This gives the value of c as $\pi(\frac{400}{3}100^{\frac{3}{2}} - \frac{2}{5}100^{\frac{5}{2}}) = \pi\frac{14}{15}10^5$. Putting this into (6), we have

$$\pi(\frac{400}{3}h^{\frac{3}{2}} - \frac{2}{5}h^{\frac{5}{2}}) = -12\lambda t + \pi\frac{14}{15}10^5 \quad (8)$$

As T is the time when the tank is emptied, we have $h = 0$ when $t = T$. This gives $\frac{\lambda T}{\pi} = \frac{14}{180}10^5 = \frac{70000}{9}$.

Comment No. 1:

The form of the fundamental theorem of calculus used in going from (2) to (3) is its second form (Theorem 2 of Chapter 17). Its use can be avoided by evaluating the integral in (2) (using the first form of the fundamental theorem of calculus, Theorem 1 of Chapter 17) to get

$$V(t) = \pi\left(100z^2 - \frac{1}{3}z^3\right)\Big|_0^{h(t)} = \pi\left(100(h(t))^2 - \frac{1}{3}(h(t))^3\right)$$

and then differentiating w.r.t. t to get (3). This is, of course, a waste of time. But it is not substantial. In fact, for many persons, it is less than the time it would take to see how to avoid the waste! If, instead of a hemisphere, the tank were in the shape of the boundary of a solid whose cross-sectional area by a horizontal plane at a height z from the bottom is a complicated function of z, say, $e^z\sin(\frac{\pi z}{200})$ instead of the simple integrand in (2), then evaluating the integral and then differentiating it would indeed be a substantial waste of time, so that the use of the second fundamental theorem becomes desirable (although still not mandatory). And if the cross-sectional area is some function like e^{z^2} whose antiderivative cannot be found in a closed form, then the first fundamental theorem is of no help and the use of the second fundamental theorem becomes mandatory. Whenever a student studies a solution (either his own or somebody else's) to some problem, it is a healthy habit to see if the problem, and indeed every step of the solution, could be done in some other way. Although for that particular problem the alternate methods may not be of much advantage, there

may be some variations of the problem where they may be superior and, in some cases, indispensable.

Comment No. 2:

The constant λ equals $0.6\sqrt{2g}$, where the constant 0.6 is the contraction factor because of the fact that the cross-section of the outflowing stream is smaller than that of the orifice and g is the acceleration due to gravity. In the cgs system its value is usually taken as 980 cm/sec^2. In a more conventional problem, you could be asked to find the time, say T, taken to empty the tank. From the work above, T comes out as $\frac{70000\pi}{9\lambda}$ i.e., $\frac{70000\pi}{5.4\sqrt{2g}}$ seconds. In examinations where calculational aids such as logarithm tables, slide rules or calculators are allowed, this can be computed as 970.7285 seconds. Otherwise it can be left as it is. (See the general remarks made about numerical problems in Chapter 1, Comment No. 2, Chapter 12, Comment No. 6 and Chapter 17, Comment No. 4.)

Comment No. 3:

The same method can be applied to do a more general problem. Suppose that the tank is in the form of a frustum of a hemisphere of radius R. Assume that the base has radius r_0 and contains an outlet of cross-sectional area A. Assume that the depth of the water in the tank is d at the time of opening the outlet. Then by duplicating the reasoning in the solution to the Main Problem, the time T, taken to empty the tank, is given by

$$T = \frac{\pi\left(4Rh_0d^{\frac{1}{2}} + \frac{4R}{3}d^{\frac{3}{2}} - 2h_0^2d^{\frac{1}{2}} - \frac{2}{5}d^{\frac{5}{2}} - \frac{4}{3}h_0d^{\frac{3}{2}}\right)}{0.6A\sqrt{2g}} \tag{9}$$

where $h_0 = R - \sqrt{R^2 - r_0^2}$. (This is the height of the base of the tank above the lowest point of the hemisphere.) To do the original problem without neglecting the loss of height because of the hole at the bottom, we now put $R = 100$ cm, $A = 12$ cm^2, $r_0 = \sqrt{\frac{12}{\pi}}$ cm, $h_0 = 100 - \sqrt{10000 - \frac{12}{\pi}}$cm and $d = 100 - h_0 = \sqrt{10000 - \frac{12}{\pi}}$ cm to get $T = 971.0596$ seconds which is slightly *more* than the previous answer, viz., 970.7285 seconds. This may come as a surprise because now there is slightly less water in the tank. The explanation is that now its bottom is slightly raised and hence there is a decrease in the height of the water level above the bottom and consequently also a decrease in the velocity of flow.

Comment No. 4:

This problem illustrates a typical, simple application of differential equations. We want to express the height h of the water level as a function of the time t. For this we first obtained a differential equation, (4), for its derivative, viz., $\frac{dh}{dt}$ and then solved it. The latter part was rather easy in this problem. But we had to do some work to get the differential equation in the first place because the problem does not give $\frac{dh}{dt}$ directly. Because we know the geometry

of the tank, we can express the volume V at a time t in terms of the height h at time t, and thereby, using the chain rule, we can express $\frac{dV}{dt}$ in terms of h and $\frac{dh}{dt}$. So we can get $\frac{dh}{dt}$ if we can get hold of $\frac{dV}{dt}$. But the problem does not specify $\frac{dV}{dt}$ either! So, to find it we had to use the velocity $v(t)$ at time t, which the problem does specify.

In some problems, formation of the differential equation is easy because the data directly says something about the rate at which the desired quantity is changing. Solving the differential equation is, of course, not always easy. Moreover, sometimes some more work is involved after solving the differential equation. Here is an illustration of such a problem. (JEE 1997*)

Water is being drained out from two reservoirs A and B. At any moment, the rate of flow of water from each reservoir is proportional to the quantity of water in that reservoir at that moment. Initially, the reservoir A has twice as much water as the reservoir B. An hour later, A has 1.5 times as much water as B. Find out when the two reservoirs will contain equal quantities of water.

Let t denote the time (in hours). Let $x = x(t)$ and $y = y(t)$ be the quantities of water in A and B respectively at time t. Then we have

$$\frac{dx}{dt} = \alpha x \quad \text{and} \quad \frac{dy}{dt} = \beta y \tag{10}$$

where α and β are some constants. Writing these equations as $dx = \alpha dt$ and $dy = \beta dt$, their solutions are $x = ae^{\alpha t}$ and $y = be^{\beta t}$, where a, b are some constants. Putting $t = 0$ we see that a, b are the initial quantities of water in the reservoirs. They are not given. But it is given that $a = 2b$. Also putting $t = 1$ gives $ae^{\alpha} = \frac{3}{2}be^{\beta}$. Combining these two equations, $4e^{\alpha} = 3e^{\beta}$. Taking logarithms, we get

$$\alpha - \beta = \ln 3 - \ln 4 \tag{11}$$

Let t^* be the time when $x(t^*) = y(t^*)$. Then $ae^{\alpha t^*} = be^{\beta t^*}$. Again, combining this with $a = 2b$ and taking logarithms we get

$$(\alpha - \beta)t^* = \ln \frac{1}{2} \tag{12}$$

From (10) and (11), $t^* = \frac{\ln 2}{\ln 4 - \ln 3}$. So this is the time when the two reservoirs will have equal quantities of water. Approximate numerical value of t^* can be found if the values of $\ln 2$ and $\ln 3$ are given (since we would then also know $\ln 4$ which equals $2 \ln 2$). But in a mathematical problem where the focus is on the differential equations, this subsequent work is not very crucial.

It is important to note that the actual values of the constants a, b, α, β cannot be determined from the data because to determine them we would need four equations, while the data gives only two. But the individual values of a and b are not needed, since the problem deals only with the proportion of the quantities in the two reservoir. It is very important to realise this, as otherwise you are likely to foolishly spend your precious time in trying to find something which is neither possible nor needed. (A more mature way to begin the solution

would be to choose the units of measurement of the quantities of water in such a way that initially the two reservoirs hold 2 and 1 units.) Similarly, we cannot determine α and β individually. But that does not matter either since the answer depends only on the difference $\alpha - \beta$ which we can find from the first part of the data. The standard strategy in solving a problem is to introduce some variables relevant to the problem, to translate the data of the problem into a system of equations in those variables and finally to solve it so as to get the values of the variables we introduced. The present problem is a sort of an exception to this.

Comment No. 5:

The situation in the last problem where the rate of change of some quantity is proportional to the quantity present at that time is rather commonly encountered in nature. It is called the **law of continuous growth** (or **decay**, in case there is a decrease, rather than an increase, as in the last problem). When other things are equal, it certainly stands to reason that the growth is proportional to the quantity on which it takes place. (It is to be noted that we are talking of the absolute rate of growth here. The relative, or the so-called percentage rate of growth is constant. In symbols, if x is the quantity at time t, then $\frac{dx}{dt}$ is proportional to x. The ratio $\frac{dx/dt}{x}$ is a constant.) Many living organisms such as bacterial colonies are seen to obey the law of continuous growth. The loss of mass of radioactive elements also obeys it, except that here the rate of growth is negative. Sometimes some artificial variables such as the money invested in a bank are made to grow continuously. If a bank allows simple interest at the rate of say, r per cent per annum, then 1 rupee will give a total of $1+\frac{r}{100}$ rupees at the end of one year. But if the interest is compounded every six months, the total will be slightly more, viz., $(1 + \frac{r}{200})^2$, i.e., $1 + \frac{r}{100} + \frac{r^2}{40{,}000}$ rupees. More generally, compounding n times a year (at equal rests) will give a total of $(1 + \frac{r}{100n})^n$ rupees at the end of one year. If the interest is compounded continuously, then the total at the end of one year will be $\lim_{n\longrightarrow\infty}(1 + \dfrac{r}{100n})^n$. It can be shown that this limit is $e^{\frac{r}{100}}$. We can also get the same answer by writing the differential equation $\frac{dx}{dt} = \frac{r}{100}x$, where $x = x(t)$ is the total at time t. With the initial condition $x(0) = 1$, the solution is $x(t) = e^{\frac{rt}{100}}$, so that $x(1) = e^{\frac{r}{100}}$. For this reason, law of continuous growth is also sometimes called the **law of continuous compounding**.

Strictly speaking, such a law is applicable only to quantities which are continuous in nature, i.e., which can assume any real multiple of some basic unit as a possible value. This includes all the quantities of classical mechanics such as length, area, volume, mass, energy, momentum, time and so on. On the other hand, variables which take only whole multiples of some basic unit as possible values are called **discrete variables.** Examples of such variables are the number of balls in a box, the number of times a particular thing is done and many variables that we encounter in problems of permutations and combinations. The population of a country is also a discrete variable. For such variables the concept of a derivative w.r.t. time does not make sense and hence

they cannot obey the law of continuous growth. However, when the basic unit is very small and the multiples involved are large, then such variables can be treated approximately as if they are continuous variables. Thus, the number of persons in a small community cannot be treated as a continuous variable. But the population of a large country can. When so done, the population often satisfies the law of continuous growth. The following problem is based on this idea.

The (continuous) rates of growth of the populations of two countries, A and B, are 3% per annum and 4% per annum respectively. At the beginning of the year 1950, the population of A was twice that of B. Find when the two countries will have equal populations, given that $\ln 2 \simeq 0.6931471$.

For a solution, measure time t (in years) starting from the beginning of 1950. Let $x = f(t), y = g(t)$ be the population of A, B respectively at time t. The data is $\frac{dx}{dt} = 0.03x$, $\frac{dy}{dt} = 0.04y$ and $f(0) = 2g(0)$. Solving, $x = f(t) = f(0)e^{0.03t}$ and $y = g(t) = g(0)e^{0.04t}$. So $\frac{x}{y} = \frac{f(t)}{g(t)} = 2e^{-0.01t}$. The populations will be equal when $e^{0.01t} = 2$, i.e., when $t = 100 \ln 2 \simeq 69.31471$, i.e., sometime during 2019, (to be more precise, sometime during April, 2019).

Comment No. 6:

There are situations where one of the variables is continuous and the other discrete. Here is an example of such a problem.

The food deficit of a country is 10 per cent. Its population grows continuously at the rate of 3% p.a. while every year its annual food production is 4% more than that of the last year. Assuming that the per capita food requirement remains the same, find out after how many years the country will be self-sufficient in food. (JEE 2000, modified)

Here the population is to be taken as a continuous variable and as a function of time t which is also a continuous variable. But the food is not produced continuously. It is produced in seasons and so the food produced in the nth year will be a function of n which is a discrete variable. Once this difference is clearly understood, the solution itself is not difficult. Let $x = x(t) =$ population at time t and $y_n =$ annual food production in the nth year (with y_0 denoting the food production at the start). Here y_0 is like the principal invested in a bank which gives simple interest at the rate of 4 p.c.p.a. and compounding is done at the end of each year. But x_0 is like the principal invested in a bank account which gives continuously compounded interest at the rate of 3 p.c.p.a. Choose suitable units of population and food so that $x(0) = 1$ and $y_0 = 0.9$. Then $x(t) = e^{0.03t}$ for $t \geq 0$, while $y_n = 1.04y_{n-1}$ for all integers $n \geq 1$. By induction on n, it follows easily that for every $n \geq 0$, $y_n = (0.9)(1.04)^n$. For self-sufficiency, $y_n \geq x(n)$, i.e., $(1.04)^n \geq \frac{10}{9}e^{.03n}$. Thus n is the smallest integer $\geq \frac{\ln 10 - \ln 9}{\ln(1.04) - 0.03} \simeq 11.426504$. So $n = 12$. In other words, the country will be self-sufficient in food at the end of the twelfth year.

Comment No. 7:

Before giving more applications of differential equations it is worthwhile to pause for a while and consider some generalities about them. Simply stated, a differential equation is an equation involving the independent and the dependent variables of a function and also its derivatives. Thus, if a variable y is a function of a variable x, then a typical differential equation will be of the form

$$\phi(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \ldots, \frac{d^ry}{dx^r}) = 0 \tag{13}$$

where r is a positive integer and ϕ is a function of $r+2$ variables. Naturally we assume that this function ϕ involves the highest derivative $\frac{d^ry}{dx^r}$, i.e., not independent of it, as otherwise we may replace r by a lower value. The integer r then is well-defined and called the **order** of the differential equation. (The restriction is analogous to the requirement that the leading coefficient of a polynomial must be non-zero, as otherwise its degree will not be well defined). For notational simplicity we denote the derivatives by $y', y'', \ldots, y^{(r)}$. Then $e^{y''}\cos y' - \frac{\sin(y-y')}{y'''x} = 0$ is a differential equation of order 3 while $y'' = 0$ is differential equation (or d.e., a commonly used short form) of order 2. (Note that it does not matter that this equation does not involve the first order derivative y'. What matters is that the highest order derivative must be involved.)

Frequently, the function ϕ in (13) is a polynomial in $r+2$ variables. In that case its degree in the last variable (viz., $\frac{d^ry}{dx^r}$ or $y^{(r)}$) is called the **degree** of the differential equation. For example, $2 + xy^3 + xy'^8 - yy'' + 27xy''^3 = 0$ is a d.e. of order 2 and degree 3. Note that what matters is the degree of the topmost order derivative (in the present case y''). It makes no difference that some lower order derivative appears with a higher degree.

Differential equations of orders upto 2 arise very frequently in physical applications because so many basic concepts of physics can be expressed mathematically as either the first or the second order derivatives. The simplest examples are velocity and acceleration. The laws of motion are, therefore, expressible as certain differential equations where the independent variable is the time. It is a common practice to denote derivatives w.r.t. time by dots rather than dashes. For example, motion in a free fall under gravity is given by the second order equation

$$y'' = g \text{ or } \ddot{y} = g \tag{14}$$

where g denotes the acceleration due to gravity. (The motion of a projectile which we considered in Comment No. 12 of Chapter 12 comes under this, except that there we also had a horizontal component of the motion which was uniform, i.e., had no acceleration.)

Similarly, a **simple harmonic motion** is a motion in a straight line in which the acceleration is directed to a fixed point (called the center) in the

line and proportional to the displacement as measured from the centre. If we denote the displacement by x, the equation of a simple harmonic motion is

$$\ddot{x} = -\alpha x \tag{15}$$

where α is a positive constant. (For a solution, see Exercise (19.13).)

Similarly, the rate of growth (or 'decay' as it is called when it is negative) of a quantity is simply its derivative w.r.t. time. We already saw examples of differential equations arising from such problems. Another important source of differential equations is the various curves in a plane or in space. Since the slope of the tangent to a plane curve of the form $y = f(x)$ is simply the derivative $\dfrac{dy}{dx}$, it is hardly surprising that properties of curves involving tangents and related concepts such as normals lead to first order differential equations. We shall consider a few such equations later.

Just as in algebra we often consider not just a single equation but a system of simultaneous equations involving more than one variables, with differential equations also we encounter situations where there is only one independent variable, say x, but more than one dependent variables, say y and z and instead of a single d.e., we have system of d.e.'s in which the derivatives of y and z are entangled with each other. Equation (10) above is a simple example of a system of two d.e.'s. But it is not a typical example, because the two variables x and y are not linked together. So it is merely a collection of two separate d.e.'s put together. If, instead, we had a system like, say,

$$\frac{dx}{dt} = \alpha y \quad \text{and} \quad \frac{dy}{dt} = \beta x$$

then the two variables are linked together. Naturally solving such a system is more challenging than solving (10).

The way we apply differential equations to solve problems is, in principle, no different than the way algebra is applied. In algebra, we introduce one or more variables (or 'unknowns' as they are often called) and convert the data to an equation (or a system of equations) in these unknowns. And finally, we solve this equation (or the system) to get a solution to the original problem. In differential equations we do the same. The only difference is that is that in algebra, these equations are algebraic while in differential equations, they involve derivatives.

But this makes a lot of difference both in the nature of the solutions and the methods for obtaining the solutions. Let us first see what is meant by a solution of a differential equation in its general form (13). Ideally, this is a function of the form $y = f(x)$ defined on some interval, say I, of the real line such that if in (13) we replace x by any point, say $x_0 \in I$ and $y, y', y'', \ldots, y^{(r)}$ by $f(x_0), f'(x_0), f''(x_0), \ldots, f^{(r)}(x_0)$ respectively, then (13) holds. The point is that the solution is a *function* of the independent variable x and not just a single number (or sometimes a set of numbers) as in the case of algebraic or trigonometric equations. For example, one can easily verify by direct substitution that if we put $\alpha = \lambda^2$ in (15) (which is permissible since α is positive), then both

$x = \cos \lambda t$ and $x = 10 \sin \lambda t$ are solutions of it. It is often not very important (and, in many cases, not possible either) to obtain a solution of a d.e. in the form of $y = f(x)$. In such a case, it is all right if we can obtain it only implicitly in the form of a relation $g(x, y) = 0$, where g is a function of two variables. For in that case one can appeal to the implicit function theorem mentioned in Chapter 15, Comment No.2. For example the relation $x^2 + y^2 - 5 = 0$ is a solution of the d.e. $x + yy' = 0$ as one sees by implicit differentiation. Parts (a) to (d) of Exercise (15.12) are in essence, verifications that the relations given there are solutions of the d.e.'s appearing as conclusions of those exercises.

But, as we have emphasized in Comment No. 14 of Chapter 17, it is one thing to *verify* something, it is quite another to *obtain* it. When it comes to solutions of differential equations, the difference between the two is even more pronounced. In Chapter 3 we remarked that there is no golden method to solve even an algebraic equation, much less an equation involving combinations of exponential or trigonometric equations (e.g., an equation like $e^x - \tan^2 x + x^8 = 0$. Similarly, in Chapter 17 we remarked that we cannot obtain antiderivatives of many functions (e.g., $\sin(x^2)$) in a closed form. But these problems are nothing when compared with the problem of solving a differential equation. Not only is there no method to solve the general differential equation (13), even for highly special cases of it (such as a differential equation of order 1 or 2), there are no general methods to obtain the solution in an exact form. But then, that is precisely why solving a differential equation is a challenge that demands a good deal of skill and perceptivity (unlike, for example, solving a system of linear equations, which can be done quite mechanically).

Another important point to note is that the solution of a d.e. is not unique. We can easily illustrate this. We saw that $x^2 + y^2 = 5$ is a solution of the d.e. $x + yy' = 0$. But so is the relation $x^2 + y^2 = 39$, and, more generally, the relation $x^2 + y^2 = c$, where c is a positive constant. (If $c = 0$, the relation degenerates into a single point $(0, 0)$ while for $c < 0$, it represents the empty set. In such cases the concept of the derivative y' makes no sense.) Moreover, using Lagrange's Mean Value Theorem we can show that these are the only solutions. For, put $u = x^2 + y^2$. Then $u' = x + yy'$ and so the d.e. $x + yy' = 0$ is equivalent to the d.e. $u' = 0$. But then by Theorem 3 of Chapter 13 (which looks obvious but needs Lagrange's MVT for a proof) u is constant over every interval I. So for this reason we say that $x^2 + y^2 = c$ is the **general solution** of the d.e. $x + yy' = 0$. A solution obtained by giving a particular value to the constant c is called a **particular solution**. (The situation is analogous to the solutions of trigonometric equations considered in Equations (1) to (3) of Chapter 10. The difference, however, is that the variable n there was a discrete variable taking only integral values. Here the variable c (or, rather the 'parameter' c as it is preferably called, since it is awkward to call a constant a variable!) is a continuous variable taking values in some interval of real numbers, which in the present case is the interval $(0, \infty)$.)

Similarly, in addition to the solutions of (15) we gave earlier, we see that

for any two values of the constants c_1 and c_2, the function

$$x = c_1 \cos \lambda t + c_2 \sin \lambda t \tag{16}$$

is a solution of (15). It can again be shown, using Lagrange's MVT that this is the general solution of (13) but the proof is not so direct. Similarly

$$y = \frac{1}{2} g t^2 + c_1 t + c_2 \tag{17}$$

where c_1 and c_2 are arbitrary constants is the general solution of (14).

It is not a coincidence that in all these examples, the number of arbitrary constants occurring in the general solution was the same as the order of the equation. It can be shown (although the proof is not so easy) that the general solution of the differential equation (13) involves r constants, say $c_1, c_2, \ldots, c_r$ which can take values in some intervals, say $I_1, I_2, \ldots, I_r$ respectively, independently of each other. As a result, these constants are said to be mutually independent and arbitrary. A solution obtained by giving these constants some particular values (withing their respective intervals of variation) is called a **particular solution** of (13).

In a given problem, after we have obtained the differential equation (from the general physical or geometric aspects of the data of the problem), the standard procedure is to first find its general solution. Then we use the specific pieces of data of the problem to write a system of r equations in the r unknowns $c_1, c_2, \ldots, c_r$. Solving this system we ultimately obtain a particular solution of the differential equation. For example, in the solution to the Main Problem, we first obtained the differential equation (4) from the physical data of the problem. Then we solved it to get (6) as its general solution. Since the d.e. was of order 1, there was only one constant, viz., c, in the general solution. To find its value, we had to use (7) which was given as a part of the data. This finally gave us (8) as the particular solution of the d.e. (4). (The solution of the problem itself, may involve further work, of course. But as far as the d.e. part of it is concerned, the solution is over the moment we obtain a particular solution.)

As in the Main Problem, the independent variable in many physical problems (especially those where some motion is involved), is the time, say t. And the particular data of the problem consists of what happens 'initially' i.e., at $t = 0$. So these conditions (which enable us to obtain a particular solution from the general solution) are often called the **initial conditions** and the term is also used figuratively even in those situations where the independent variable does not stand for time. For example, suppose we want a solution of $x + yy' = 0$ which satisfies the 'initial condition' that the point $(2, 3)$ lie on the curve represented by the solution. We already saw that the general solution of this equation is $x^2 + y^2 = c$, where c is a (positive) constant. Geometrically, this gives us a family of concentric circles all centred at the origin. Out of this family, the circle that satisfies the initial condition will have $c = 2^2 + 3^2 = 13$. So $x^2 + y^2 = 13$ is the desired particular solution.

Comment No. 8:

With these generalities about the nature of the solution of a d.e., let us now get down to the task of finding the general solution of a given differential equation (and more generally, of a system of equations). As mentioned earlier this is a very challenging task. But we shall confine ourselves to a very special case of it. First, we shall deal only with a single d.e. and not a system of d.e.'s. Secondly, we shall confine ourselves only to equations of order 1, i.e., to equations of the form $\phi(x, y, y') = 0$, where ϕ is a given function of three variables. Using the implicit function theorem mentioned earlier (or rather, an extension of it), it can be shown that every such equation can be expressed in the form

$$y' = \frac{dy}{dx} = \psi(x, y) \tag{18}$$

where $\psi(x, y)$ is a (known) function of two variables x and y. For example the d.e. $x + yy' = 0$ which we have considered several times can be written as $y' = \frac{x}{y}$. Such a recasting may not necessarily make the equation easier to solve. But it helps in conceptual understanding, because there is a geometric way of looking at (17). A solution of $\phi(x, y) = 0$ will consist of a family of curves in the plane. Let $P_0 = (x_0, y_0)$ be a fixed point in the plane. Then there is a unique member, say C_0 of this family which passes through P_0. Equation (18) does not immediately tell us what this curve C_0 will be. That we shall come to know only after finding the general solution of (18) and then the particular solution which satisfies the initial condition that it passes through P_0. But what (18) does tell us, even before solving (18) is that whatever this curve C_0 may be, its slope at the point P_0 is $\psi(x_0, y_0)$. So if we draw a small line segment with slope $\psi(x_0, y_0)$, it will approximately look like the curve C_0 in a neighbourhood of P_0. If we sketch several such segments in a plane at various points of it, we get a qualitative idea of what the solution curves will look like. And, in absence of an exact solution, this gives a method for finding an approximate solution. We omit the details of this procedure except to remark that it is similar to the picture of the lines of magnetic force that you get by putting iron filings on a piece of paper near a magnet and tapping the paper gently.

So we confine ourselves only to differential equations which are or can be reduced to the form (18). Even this very special case is not always amenable to an exact solution because the nature of the function $\psi(x, y)$ can be quite complicated. We shall content ourselves by solving (18) in some special cases, starting from the simplest case and then considering a few relatively more complicated cases.

Comment No. 9:

The simplest case is where the function $\psi(x, y)$ is free of one of the two variables. Suppose it does not involve y at all. Then effectively, if is a function of just one variable x, sat $f(x)$ and in this case the problem of solving (18) is simply the problem of finding an antiderivative of the function $f(x)$. We already addressed this problem in Chapter 17. If the variable x is absent in

$\psi(x,y)$, then it can be written as $\frac{dy}{dx} = g(y)$. Clearly we may assume $g(y)$ does not vanish identically, as otherwise $y = c$ is the general solution of (18). Taking reciprocals of both the sides we get $\frac{dx}{dy} = \frac{1}{g(y)}$. We have, in effect, reversed the roles of x and y here. In other words we are expressing x as a function of y here. We can now find an antiderivative of $\frac{1}{g(y)}$ w.r.t. y and that gives us the general solution. The differential equation (4) which we encountered in the solution to the Main Problem is of this type, if we divide both the sides by $\pi(200h - h^2)$. The equation (5) is obtained by cancelling the factor $\sqrt{h}$ from both the sides and then splitting the differentials. This is done very commonly in solving differential equations and it proves to be very convenient. In fact, we could as well have written (18) as $dy = \psi(x,y)dx$. Those who do not like to split differentials can, of course, bypass (5) and instead write,

$$\pi(200h^{1/2} - h^{3/2})\frac{dh}{dt} = -12\lambda \tag{19}$$

then apply the chain rule to the L.H.S. to rewrite it as $\frac{d}{dt}(\pi(\frac{400}{3}h^{3/2} - \frac{2}{5}h^{5/2}))$ and then integrate both the sides w.r.t. t. The work involved is exactly the same. So the superiority of (19) over (5) is puritanical rather than practical. From now onwards, we shall split the differentials without hesitation.

Comment No. 10:

The next in line is the case where the function $\psi(x,y)$ appearing in the R.H.S. of (18) can be expressed as a product of the form $f(x)g(y)$, where f and g are functions of one variable only. In this case we let $h(y) = \dfrac{1}{g(y)}$ and after splitting the differentials write (18) as

$$h(y)dy = f(x)dx \tag{20}$$

which can be solved by simply integrating both the sides. For example, we saw that $x^2 + y^2 = c$ is the general solution of the d.e. $x + yy' = 0$. Now we can *obtain* it as follows. First rewrite the equation as $yy' = -x$ and then split the differentials to recast it as $ydy = -xdx$ which is in the form (20). Integrating both the sides we have $\int ydy = -\int xdx$, i.e., $\dfrac{y^2}{2} + c_1 = -\dfrac{x^2}{2} + c_2$, which simplifies to $x^2 + y^2 = 2(c_2 - c_1)$. Here superficially there are two constants c_1 and c_2. But any value assumed by the expression $2(c_2 - c_1)$ can also be assumed by a single constant c. So the general solution can as well be written as $x^2 + y^2 = c$, where c is a constant.

The process of converting a given d.e. to the form (20) is called **separation of variables**. Once an equation is reduced to the form (20) it is as good as solved, because then all that is left is finding antiderivatives. Sometimes the functions $h(y)$ and $f(x)$ are such that their antiderivatives cannot be found in a closed form. Even then we say that $\int h(y)dy = \int f(x)dx + c$ is the general solution of (20), because as far as the d.e. part of the problem is concerned, it is over.

Comment No. 11:

It is instructive to recast (20) in the language of what are called differentials. We mentioned them in a footnote in Comment No. 1 of Chapter 15 but did not use them since a mathematically precise definition of a differential is far from easy. But they are very convenient in solving differential equations and so we briefly study them. Suppose u is a function of a variable x, say $u = f(x)$. Then the differential du is *defined*[1] as the expression $f'(x)dx$. Thus for example, $d(x^3) = 3x^2dx$, $d(\sin x) = \cos x dx$ etc. In reality, we are not defining any new concept here. We are merely introducing a new notation which will prove to be very convenient in solving differential equations. In other words to say $d(\sin x) = \cos x dx$ is not conceptually different than saying that $\frac{d}{dx}\sin x = \cos x$. But the advantage gained is that the equality of two derivatives can now be stated in terms of differentials, without an overt reference to a particular independent variable. Take for example, the rule for the derivative of a sum. Suppose u and v are two variables. Then so is their sum $u + v$. Instead of saying that the derivative of $u + v$ is the sum of the derivative of u and that of v we can say

$$d(u + v) = du + dv \tag{21}$$

where we have not mentioned the independent variable at all. In fact once we accept that a differential has an independent meaning of its own then (21) is valid regardless of what the independent variable is. If it is x, then (21) translates as $\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$. If the independent variable is t, then it translates as $\frac{d(u+v)}{dt} = \frac{du}{dt} + \frac{dv}{dt}$ and so on. Similarly the product and the quotient rules for derivatives can be paraphrased, respectively, as

$$d(uv) = udv + vdu \tag{22}$$

$$\text{and} \quad d(\frac{u}{v}) = \frac{vdu - udv}{v^2} \tag{23}$$

As an illustration of the use of (23), since $\tan x = \frac{\sin x}{\cos x}$, we have $d(\tan x) = d(\frac{\sin x}{\cos x}) = \frac{\cos x d(\sin x) - \sin x d(\cos x)}{\cos^2 x} = \frac{\cos^2 x dx + \sin^2 dx}{\cos^2 x} = \frac{dx}{\cos^2 x} = \sec^2 dx$. We emphasize again that the work done is hardly different than that in differentiating $\tan x$ using the quotient rule in the usual form. It is, in fact, difficult to appreciate the convenience gained by the use of differentials until we solve d.e.'s as we do now.

[1] The main objection to this definition is that we have not really defined what is meant by the expression $f'(x)dx$ which we are treating as if it were a product of two things, one the derivative $f'(x)$ which is a perfectly well-defined entity, and the other the differential dx. Moreover, du is not defined intrinsically in terms of u, but in terms of something extraneous, viz., the differential dx and the function $f(x)$. If simultaneously u is a function of some other variable, say $u = g(y)$, then du will be defined as $g'(y)dy$. Considerable work is needed to show that these two definitions coincide.

We have a theorem (which we already used) that if the derivative of a function vanishes identically, then the function must be a constant. In the language of differentials this translates as "If $du = 0$ then $u \equiv c$ for some constant c". Let us revisit some of the equations we have solved earlier in this new light. Take the equation $x + y\dfrac{dy}{dx} = 0$. We solved it above by separation of variables to get $x^2 + y^2 = c$ as its general solution. Instead, we can first write the equation in the form of differentials as $xdx + ydy = 0$. Now we recognise the L.H.S. as $d(\frac{1}{2}(x^2+y^2))$. So the solution is $\frac{1}{2}(x^2+y^2) = c$. Calling $2c$ as c, this is the same as $x^2 + y^2 = c$. Similarly, we can rewrite (20) as $h(y)dy - f(x)dx = 0$ and hence as $d(H(y) - F(x)) = 0$ where $H(y)$ and $F(x)$ are antiderivatives of $h(y)$ and $f(x)$ respectively. Hence the general solution of (20) is $H(y)-F(x) = c$, which we could also get by integrating both the sides of (20).

So once again it might appear that we have gained nothing by using the language of differentials. But the point is that there are many differential equations which are not in the separate variables form but which can be cast (often after some manipulations) in the form $du = 0$ for some differential du. As an example, consider the differential equation

$$\frac{dy}{dx} = \frac{y - 2x^3}{x + 2yx^2} \tag{24}$$

Here the variables cannot be separated as the equation stands. However let us rewrite it using differentials as

$$(2x^3 - y)dx + (2yx^2 + x)dy = 0 \tag{25}$$

which is still not in the form $du = 0$ for any variable u. But if we group together xdy and ydx then we can hardly fail to notice that because of (23), $xdy - ydx$ is very close to being $d(\dfrac{y}{x})$, except that we shall have to divide it by x^2 first. So let us divide (25) by x^2 and rewrite it as

$$2xdx + 2ydy + d(\frac{y}{x}) = 0 \tag{26}$$

Here the first two terms can be recognised as the differentials of x^2 and y^2 respectively. So (26) can be written as

$$d(x^2 + y^2 + \frac{y}{x}) = 0 \tag{27}$$

Therefore $x^2 + y^2 + \dfrac{y}{x} = c$ is the general solution of (24). That it is indeed so can, of course, be verified by direct differentiation. But once again, art lies in obtaining the solution. Verifying it is pure labour.

A differential equation which is in the form $du = 0$ for some function $u(x, y)$ of two variables is said to be **exact**. Note that this is a property of a particular form and not of the intrinsic equation itself. For example, we would obviously treat (25) and (26) as the same equations. But (25) is not exact while (26) is.

More generally, consider a differential equation in the form

$$M(x,y)dx + N(x,y)dy = 0 \tag{28}$$

where M and N are functions of two variables x and y. All the d.e.'s of first order we have solved so far can be written in this form. So we may very well consider (28) as the general form for a d.e. of first order. There is an easy way to tell if this equation is exact or not. It requires the concept of partial derivatives which we touched upon only briefly when we considered the method of Lagrange multipliers for trigonometric optimisation in Chapter 14. In essence, the partial derivative w.r.t. x is obtained by differentiation w.r.t. x in which all other variables are treated as constants. The criterion for exactness of the equation (28) above then says that it is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \text{ i. e. } M_y = N_x \tag{29}$$

A proof of this criterion is beyond our scope. But we can illustrate its applicability. First consider (25). Here $M = 2x^3 - y$ and so $M_y = -1$. But $N = 2yx^2 + x$ and so $N_x = 4xy + 1$. So (25) is not exact. On the other hand, if we divide it throughout by x^2, then M changes to $2x - \frac{y}{x^2}$ and hence M_y becomes $-\frac{1}{x^2}$. Similarly, N changes to $2y + \frac{1}{x}$ and hence $N_x = -\frac{1}{x^2}$. So again by (29) the new equation, viz.,

$$(2x - \frac{y}{x^2})dx + (2y + \frac{1}{x})dy = 0 \tag{30}$$

(which is essentially the same as (26) after a regrouping of the terms) is exact.

Formula (29) only gives a criterion to tell if the d.e. (28) is exact or not, i.e., whether it can be expressed in the form $du = 0$ or not. Once it is so expressed, its solution is immediate, viz., $u = c$. Unfortunately, even if (29) tells us that a particular equation is exact, it does not, by itself, tell us *how to* express it in the form $du = 0$ for some suitable function[2] u of two variables. There is a systematic method for obtaining this u. But it is beyond our scope. In simple problems this can usually be done by inspection, after a regrouping of the terms if necessary. For example, we just saw that (30) above is exact. If we regroup the terms together, we get (26) and from there (27) which is in the form $du = 0$.

Summing up, once a d.e. is cast in a form which is exact, things are fairly rosy. (The case where the variables can be separated can also be considered as coming under this because if, in (28), M is a function of x only and N is a function of y only then the condition of exactness in (29) is trivially satisfied.) But what if the equation, in the form it is given, is not exact? In that case, one usually tries to multiply it throughout by some factor, say $F = F(x,y)$ such

[2] A function u with this property is called a **potential function**. The name obviously comes from applications to physics. But we shall not go into it.

that the new equation $FMdx + FNdy$ is exact. Such a factor F is called an **integrating factor**. (The term 'integration' is used for finding antiderivatives and hence more generally, for solving differential equations.) For example, (25) is not exact. But (30), which is obtained from it by multiplying by $\frac{1}{x^2}$ is exact. Hence $\frac{1}{x^2}$ is an integrating factor for (25).

The million dollar question now is how to find an integrating factor for a general first order d.e. given in the general form (28). There can be no easy answer to this, because if there were, then the problem of solving at least the first order differential equations would be a simple task. Various formulas for obtaining integrating factors are known. But they are beyond our scope. Moreover all of them are applicable only in special circumstances. There is no golden formula that will give an integrating factor in every case. This is an art and not a drill.

Comment No. 12:

Nevertheless, we consider one important class of differential equations called **linear differential equations** of first order which can be solved by finding an easy integrating factor. These equations are special cases of (18) where the function $\psi(x, y)$ is of the form $f(x)y + g(x)$, where $f(x)$ and $g(x)$ are functions of one variable. When either $f(x)$ or $g(x)$ is the zero function, this equation reduces to the separate variable form which we already know how to tackle. The justification of the name 'linear' is beyond our scope. For notational convenience, it is customary to replace $f(x)$ by $-f(x)$ and rewrite a typical linear first order equation in the form

$$\frac{dy}{dx} + f(x)y = g(x) \tag{31}$$

We can also rewrite this as

$$(f(x)y - g(x))dx + dy = 0 \tag{32}$$

which is in the standard form (28) with $M = f(x)y - g(x)$ and $N = 1$. Clearly, $N_x = 0$ while $M_y = f(x)$. So this equation is never exact except in the trivial case where $f(x)$ vanishes identically.

But it is easy to obtain an integrating factor for (31) and then to solve it using the fact that the exponential function is its own derivative. Indeed, let us try an integrating factor of the form $H(x)$, where H is a function of one variable. Multiplying (32) throughout by $H(x)$ gives

$$[H(x)(f(x)y - g(x))]dx + H(x)dy = 0 \tag{33}$$

By (29) this will be exact if and only if $\frac{\partial}{\partial x}(H(x)) = \frac{\partial}{\partial y}[H(x)f(x)y - H(x)g(x)]$. Since $H(x)$ is a function of x only, $\frac{\partial}{\partial x}(H(x))$ is the same as $H'(x)$

(the ordinary derivative of $H(x)$ w.r.t. x) and so the condition for exactness reduces to

$$H'(x) = H(x)f(x) \tag{34}$$

We can rewrite this as

$$f(x) = \frac{H'(x)}{H(x)} \tag{35}$$

Suppose now that $F(x)$ is any antiderivative of the function $f(x)$. Then we can integrate both the sides of (34) and get $\ln(H(x)) = F(x)$ or equivalently $H(x) = e^{F(x)}$. (Here we are using Formula (33) of Chapter 17. We are ignoring the constant of integration, because all we want is *some* integrating factor $H(x)$. If we do take into account the constant of integration, we shall get $H(x)$ as $ke^{F(x)}$ for some (positive) constant k.) Of course, we could have also *started* by taking $H(x)$ as $e^{F(x)}$ and then *verified* that it is an integrating factor for (33). But it is always more satisfying to *arrive at* something than to borrow it as readymade.

As commented above, after having obtained an integrating factor, in general we still have to do some work to get the equation in the form $du = 0$ from which we can read out the solution instantaneously. But in the present case this is very easy. To make it still simpler, let us multiply the original equation (31) (rather than its version (32)) by the integrating factor $e^{F(x)}$ we just found. Then writing $F'(x)$ for $f(x)$ it becomes

$$e^{F(x)}\frac{dy}{dx} + e^{F(x)}F'(x)y = e^{F(x)}g(x) \tag{36}$$

The L.H.S. is simply the derivative of the product $e^{F(x)}y$ w.r.t. x. So (36) can be rewritten as

$$\frac{d}{dx}(e^{F(x)}y) = e^{F(x)}g(x) \tag{37}$$

If we integrate both the sides w.r.t. x and divide throughout by $e^{F(x)}$ (which is never 0), we get

$$y = e^{-F(x)}\left[\int e^{F(x)}g(x)dx + c\right] \tag{38}$$

where c is an arbitrary constant. So finally, this is the general solution of the equation (31) with which we started. Note that there are two integrations to be carried out here. First, we integrate $f(x)$, i.e., the coefficient of y in (31) to get $F(x)$. Then we need to integrate the product $e^{F(x)}g(x)$ w.r.t. x.

In applications, we often have to solve a linear equation (31) with a given initial condition, say $y = y_0$ when $x = x_0$. (Together, it is often called an **initial value problem**.) One way to do this is to ignore the initial condition till we get the general solution (38) and *then* use it to find the value of c. Or, we can incorporate the initial condition in (38) itself by writing definite integrals

instead of indefinite integrals. Without loss of generality, we suppose that the antiderivative $F(x)$ of $f(x)$ is so chosen that $F(x_0) = 0$. (Otherwise simply subtract a constant from it.) After getting (37), we take the definite integrals of both the sides over the interval $[x_0, x]$. Since x is used here to denote one of the limits of integration, we call the variable of integration t rather than x. Then we have

$$\int_{x_0}^{x} \frac{d}{dt}(e^{F(t)}y)\ dt = \int_{x_0}^{x} e^{F(t)}g(t)\ dt \tag{39}$$

By the Fundamental Theorem of Calculus (first form), the L.H.S. is simply $e^{F(x)}y - e^{F(x_0)}y_0 = e^{F(x)}y - y_0$ since $F(x_0) = 0$. So the particular solution is given by

$$y = e^{-F(x)}\left[y_0 + \int_{x_0}^{x} e^{F(t)}g(t)\ dt\right] \tag{40}$$

As a simple illustration, let us find a particular solution of

$$\frac{dy}{dx} + (\tan x)\ y = x^2 \cos x \tag{41}$$

which satisfies the initial condition $y(\pi) = 10$. Let us first do the problem by finding the general solution of (41). This is in the form (31) with $f(x) = \tan x$. Hence an integrating factor is $e^{\int \tan x dx} = e^{\ln|\sec x|} = \pm \sec x$. It does not matter which sign we choose, since both the signs will give us integrating factors. We take $\sec x$ as the integrating factor. We can now get the general solution from (38). But instead of remembering the clumsy formula (38), it is often a good idea to multiply the given equation by the integrating factor and then work afresh. If we do so in the present case we get $\frac{d}{dx}(\sec x\ y) = x^2$. Integrating both the sides we get $(\sec x)y = \frac{x^3}{3} + c$ or $y = \frac{x^3 \cos x}{3} + c\cos x$ as the general solution of (41). We are further given that $y = 10$ when $x = \pi$. This determines c as $-10 - \frac{\pi^3}{3}$. So the desired particular solution is $y = (\frac{x^3 - \pi^3}{3} - 10)\cos x$.

Alternately, we can get the answer by using (40) (instead of (38)) with $x_0 = \pi$ and $y_0 = 10$. However, we must first ensure that the antiderivative $F(x)$ which is $\ln\sec x$ in the present problem is so chosen that it vanishes at $x = \pi$. But since $\sec\pi = -1$, $\ln\sec\pi$ is undefined. So we should have taken $F(x)$ as $\ln(-\sec x)$ which is defined and vanishes when $x = -\pi$. Since $F(x) = \ln(-\sec x), e^{-F(x)} = -\cos x$ and so from (40), we get $y = -\cos x\ (10 + \int_{\pi}^{x} -\sec t\ \ t^2\cos t dt) = -\cos x\ (10 + \frac{\pi^3 - x^3}{3})$ which is exactly the same answer as before.

This example is probably not very convincing because it may be argued that the integrating factor $\sec x$ for (41) could as well have been found simply by inspection. This is indeed a valid point and only underscores the maxim

that no single formula should be adhered to blindly. Other possibilities should always be kept open. The reason the integrating factor in the present problem comes out to be so simple is that the nature of the function $f(x)$ here is such that its antiderivative $F(x)$ is in the form of a logarithm. In such cases, the exponential $e^{F(x)}$ assumes a particularly simple form. But obviously this is not the case in all problems. In fact, in a majority of the problems that arise in applications, we are lucky to get the antiderivative $F(x)$ in a closed form. And even when this is so, the second antiderivative that we have to find in (38), viz., $\int e^{F(x)}g(x)dx$ can rarely be expressed in a closed form. In such cases it is difficult to apply (38). However the situation can be salvaged somewhat through (40) by resorting to approximate integration which was briefly mentioned at the end of Chapter 17.

Comment No. 13:

The fact that the integrating factor $e^{F(x)}$ of a linear d.e. is always positive is also helpful in proving certain inequalities through differential equations. We already saw in Comment No. 11 of Chapter 13 how certain inequalities of functions can be proved through the inequalities of their derivatives. In comment No. 10 of the last chapter, we saw how an inequality of integrands leads to an inequality of the definite integrals. Although there is no such thing as a differential inequality, we illustrate here how the ideas that are used in the solution of a differential equation (and specifically, that of the integrating factor for a linear d.e.) can lead to a result about inequalities.

Let $f(x), x \geq 0$, be a continuous function, and let $F(x) = \int_0^x f(t)dt,\ x \geq 0$. If for some $c > 0$, $f(x) \leq cF(x)$ for all $x \geq 0$, then show that $f(x) = 0$ for all $x \geq 0$. (JEE 2001)

By the fundamental theorem of calculus (the second form), we have $F'(x) = f(x)$ for all $x \geq 0$. This enables us to rewrite the inequality in the hypothesis as

$$F'(x) \quad \leq \quad cF(x) \ \text{ for all } x \geq 0 \tag{42}$$

It is tempting to divide both the sides by $F(x)$ and rewrite this as $\dfrac{F'(x)}{F(x)} \leq c$. One can now take the definite integral of both the sides over the interval $[0, x]$. Using Formula (33) in Chapter 17 and Theorem 1 of Chapter 18, this would give $\ln F(t)\Big|_0^x \leq cx$ for all $x \geq 0$. But the trouble is that $F(0) = 0$ and so we cannot divide by $F(x)$ in (42). In fact, if we look at the assertion of the problem carefully, it is equivalent to proving that $F(x) = 0$ for all $x \geq 0$. So dividing by $F(x)$ is not going to work even if we exclude the point 0 from the interval of integration.

Let us, however, rewrite (42) as

$$F'(x) - cF(x) \quad \leq \quad 0,\ x \geq 0 \tag{43}$$

If instead of an inequality, we had an equality sign, then (43) would be a differential equation for $F(x)$. Moreover, it would be a linear differential equation with integrating factor $e^{\int -c\,dx} = e^{-cx}$. So even though (43) is not a d.e. we can still multiply the L.H.S. by e^{-cx} and express it as a derivative, viz., $\dfrac{d}{dx}(e^{-x}F(x))$. Note that the exponential function is always positive. So the inequality will be preserved when multiplied by it. Thus from (43) we get

$$\frac{d}{dx}(e^{-cx}F(x)) \quad \leq \quad 0, \text{ for all } x \geq 0 \tag{44}$$

It is now easy to finish the solution. By (44) and Theorem 3 of Chapter 13, the function $e^{-cx}F(x)$ is monotonically decreasing on the interval $[0, \infty)$. In particular this means that for every $x \geq 0$, $e^{-cx}F(x) \leq e^0F(0) = 0$ since $F(0) = 0$. As the factor e^{-x} is positive for all x, this gives $F(x) \leq 0$ for all $x \geq 0$. But since $F(x) = \int\limits_0^x f(t)dt$ and $f(t) \geq 0$ for all $t \geq 0$, we also have by Theorem 1 of Chapter 18 that $F(x) \geq 0$ for all $x \geq 0$. Putting the two inequalities together we must have $F(x) = 0$ for all $x \geq 0$. But then its derivative, which equals $f(x)$ must also vanish for all ≥ 0.

This problem is remarkable in that although it is not a problem about differential equations, the solution was based on ideas used in solving certain types of differential equations. We encountered a similar situation at the end of Comment No. 16 of Chapter 14 where, even though the problem asked for a geometric construction, the key to it was based on a technique used in trigonometric optimisation.

Comment No. 14:

Returning to the problem of solving a general differential equation of first order, we now present a technique which converts a given differential equation to another. The technique itself is not new. In Chapter 17, we saw how a clever substitution can work wonders in obtaining antiderivatives. The same is true of differential equations. We begin with a simple illustration (JEE 1996). Suppose we want to find the equation of the curve passing through the origin, in the form $y = f(x)$, which satisfies the differential equation

$$\frac{dy}{dx} = \sin(10x + 6y) \tag{45}$$

This equation, as it stands, is not of any type we know how to solve. Expanding $\sin(10x + 6y)$ as $\sin 10x \cos 6y + \cos 10x \sin 6y$ is not of much help either. But let us put $10x + 6y = v$. We can now express either y in terms of x and v or x in terms of y and v. We choose the former and write $y = \dfrac{v - 10x}{6}$ whence $\dfrac{dy}{dx} = \dfrac{1}{6}(\dfrac{dv}{dx} - 10)$. With these substitutions, (45) becomes

$$\frac{dv}{dx} = 6\sin v + 10 \tag{46}$$

which can be easily solved since the R.H.S. is a function of v only. Rewriting (46) as $\dfrac{dv}{6\sin v + 10} = dx$ we see that now the problem is reduced to that of finding an antiderivative of $\frac{1}{6\sin v+10}$. For this, we use the standard substitution $t = \tan\frac{v}{2}$ (given in Comment No. 21 of Chapter 17). With this substitution, we get

$$\begin{aligned}
\int \frac{dv}{6\sin v + 10} &= \int \frac{2\,dt}{12t + 10(t^2+1)} = \int \frac{dt}{5t^2 + 6t + 5} \\
&= \int \frac{dt}{(\sqrt{5}\,t + \frac{3}{\sqrt{5}})^2 + \frac{16}{5}} \\
&= \frac{1}{4}\tan^{-1}\left(\frac{(\sqrt{5}\,t + \frac{3}{\sqrt{5}})\sqrt{5}}{4}\right) \\
&= \frac{1}{4}\tan^{-1}\left(\frac{5t+3}{4}\right) \\
&= \frac{1}{4}\tan^{-1}\left(\frac{5\tan(5x+3y)+3}{4}\right) \qquad (47)
\end{aligned}$$

The general solution of (45) is now

$$\frac{1}{4}\tan^{-1}\left(\frac{5\tan(5x+3y)\ +3}{4}\right) = x + c \qquad (48)$$

where c is an arbitrary constant. For each value of c, this is a curve in the xy-plane. We are given that the curve passes through the origin. This means (48) is satisfied if we put $x = 0$ and $y = 0$ in it. This determines c as $\frac{1}{4}\tan^{-1}(\frac{3}{4})$. But the answer is required to be in the form $y = f(x)$. For this, after putting this value of c in (48), we multiply both the sides by 4 and take tangents to get

$$5\tan(5x+3y)\ +3 = 4\tan(4x + \tan^{-1}\frac{3}{4}) \qquad (49)$$

from which we get finally $y = \dfrac{1}{3}\tan^{-1}\left[\dfrac{4\tan(4x+\tan^{-1}\frac{3}{4}) - 3}{5}\right] - \dfrac{5}{3}x$ as the answer to the given problem. (It will be seen that the differential equation part in the problem was relatively minor as compared with other work such as finding antiderivatives and the algebraic manipulations. This is to be expected because, at the JEE level, only rudimentary d.e.'s can be asked.)

We did this problem by expressing the dependent variable y in terms of the new variable v we introduced by $v = 10x + 6y$. Let us see what would have happened had we chosen to express x in terms of v and y by the relation $x = \frac{1}{10}(v - 6y)$. To transform (45), we use differentials to write $dx = \frac{1}{10}(dv - 6dy)$. Putting this into (45) we get $\frac{dy}{dx} = \frac{10dy}{dv-6dy} = \sin v$ which simplifies to

$$dy = \frac{\sin v}{10 + 6\sin v}dv \qquad (50)$$

which is also in the separate variables form. However, to solve it we would have to evaluate $\int \frac{\sin v}{10+6\sin v}dv$. This is analogous to the antiderivative $\int \frac{dv}{6\sin v+10}$ we had to find in the earlier approach, but a little more complicated. So you can see that in some problems replacing y may be more convenient than replacing x and vice versa. There are also situations where both the variables need to be replaced. See Exercise (19.17) for a simple illustration of this.

Comment No. 15:

There is an important class of problems where the substitution $v = \frac{y}{x}$ will work. This is the class of those equations in which the function $\psi(x, y)$ appearing in the R.H.S. of (18) is a ratio of two homogeneous polynomials in x and y of equal degree. Here by 'homogeneous' we mean that every monomial in it has the same total degree (see Comment No. 9 of Chapter 2 for definitions). For example, the method will apply if $\psi(x, y) = \frac{x^2y-4y^3}{x^3-x^2y}$ but not when $\psi(x, y) = \frac{x^2y-4y^3}{x^3-x^2y^2}$. To illustrate how it works for the former, suppose we have to solve

$$\frac{dy}{dx} = \frac{x^2y - 4y^3}{x^3 - x^2y} \tag{51}$$

Now put $v = \dfrac{y}{x}$, i.e., $y = vx$. Then $dy = vdx + xdv$ and so the L.H.S. equals $v + x\frac{dv}{dx}$. As for the R.H.S., putting $y = vx$ changes $\frac{x^2y-4y^3}{x^3-x^2y}$ to $\frac{v-4v^3}{1-v}$. Note that this is solely in terms of v. (In fact this is precisely where we need the hypothesis that the numerator and the denominator are homogeneous polynomials of the same degree.) The d.e. (51) thus changes to

$$x\frac{dv}{dx} = \frac{v - 4v^3}{1 - v} - v \tag{52}$$

which can be solved by separating the variables.

Comment No. 16:

As it happens with problems about finding antiderivatives (and more generally, with many other problems in mathematics), sometimes several methods may be available to solve a particular d.e. and a judicious choice pays off. Consider, for example, the d.e.

$$\frac{dy}{dx} = (y - 1)\cot x \tag{53}$$

We can bring the term $y\cot x$ to the left and solve this as a linear d.e. But if we put $v = y - 1$ then the d.e. changes to $\frac{dv}{dx} - v\cot x = 0$ which is also linear but easier to solve since the R.H.S. is now 0. But a still better way is to divide by $y - 1$ and separate the variables! The exercises will provide a consolidated list of d.e.'s where you have to decide which is the best method for each.

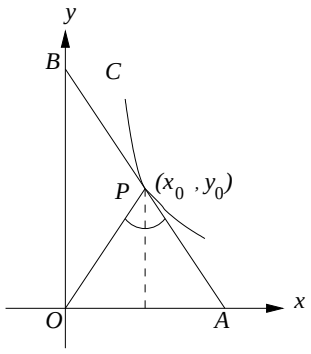
We now turn to the geometric applications of differential equations. As mentioned earlier, the foremost source of such applications lies in the fact that the slope of the tangent to the graph of a function $y = f(x)$ is given by the first

derivative $\dfrac{dy}{dx}$. The second order derivative $\dfrac{d^2y}{dx^2}$ also figures in the study of such geometric attributes as the **curvature**, which is a measure of what a layman would call as a 'bend' in a curve. But that is beyond our scope. So we confine ourselves to problems involving tangents and related concepts such as normals.

As a simple but typical illustration, suppose that a curve C has the property that if the tangent drawn at any point P on C meets the co-ordinate axes at A and B, then P is the mid-point of AB. It is further given that the curve passes through the point $(1, 1)$ and we are asked to find the equation of the curve. (JEE 1998)

Let $P = (x_0, y_0)$ be a point on C. Let the tangent to C at P meet the axes at A and B respectively. Since P is the midpoint of AB, we have $A = (2x_0, 0)$ and $B = (0, 2y_0)$. Hence the slope of AB is $-\frac{y_0}{x_0}$. But as this line touches C at P, its slope also equals the value of $\dfrac{dy}{dx}$ at (x_0, y_0). Hence

$$\left(\frac{dy}{dx}\right)_{x=x_0} = -\frac{y_0}{x_0} \tag{54}$$



(This is also obvious from the diagram if we observe that P being the midpoint of the hypotenuse of a right angled triangle AOB, the triangle OPA is isosceles. So the lines PO and PA are equally inclined to the vertical line at P. Hence their slopes are negatives of each other.)

In (54), P_0 was *any* point on C. So we can as well replace x_0 and y_0 by x and y respectively to get

$$\frac{dy}{dx} = -\frac{y}{x} \tag{55}$$

As the curve C is given to pass through the point $(1, 1)$, its equation will be a particular solution of (55) which is satisfied if we put $x = 1$ and $y = 1$. To obtain it we first solve (55) by separating the variables. The general solution is $\ln y + \ln x = c$ or equivalently, $xy = k$, where c and k are arbitrary constants. (An alternate derivation is to write (55) as $xdy + ydx = 0$, i.e., as $d(xy) = 0$ which gives $xy = k$ for some constant k.) The fact that $(1, 1)$ lies on the curve determines k as 1. So the equation of the curve C is $xy = 1$.

Comment No. 17:

In the problem just solved, the initial condition appeared only at the end. If we drop it, then we see that curves of the form $xy = k$ for various values of the constant k represent the various solutions of the differential equation (55). Every member of this family is a rectangular hyperbola with asymptotes along the axes. (When $k = 0$, this hyperbola degenerates into a pair of straight lines, viz., the two axes. In this case (55) holds for all points on the x-axis except the

origin. For points on the y-axis, it holds in the sense that the reciprocals of both the sides are 0 and hence equal. Consideration of such degeneracies is itself an interesting branch of the theory of differential equations. But it is beyond our scope.)

In the language introduced in Comment No. 8 of Chapter 9, the family of all hyperbolas of the form $xy = k$ is a one-parameter family of curves. As another example, we already saw that the general solution of the d.e. $x+yy' = 0$ is $x^2 + y^2 = c$, where c is a (positive) constant. This is also a one-parameter family of circles centred at the origin.

It is not a coincidence that the general solution of a first order d.e. represents a one parameter family of plane curves. It is a consequence of the fact that the general solution involves one arbitrary constant. More generally, the general solution of a differential equation of order r will involve r mutually independent arbitrary constants and hence will represent an r-parameter family of curves. As noted in Comment No. 11 of Chapter 9, such a family has r degrees of freedom. This means that you need to be given r (independent) pieces of data to determine any member of it. This corresponds to the fact that a particular solution of a d.e. of order r is determined by r initial conditions. We cannot give many examples of this type as we are confining ourselves only to first order d.e.'s. But consider a simple second order d.e.

$$\frac{d^2y}{dx^2} = g \tag{56}$$

(which is same as (14) except for the notation). One integration gives $\frac{dy}{dx} = gx + c_1$, where c_1 is a constant. One more integration gives $y = \frac{1}{2}gx^2 + c_1x + c_2$ as the general solution. This represents a 2-parameter family of parabolas.

Conversely, every r-parameter family gives rise to a differential equation of order r. All we have to do is to write the equation of a typical member of that family. It will involve r constants, say $c_1, c_2, \ldots, c_r$. We now differentiate this equation r times w.r.t. x so as to get r new equations. Thus in all we have a system of $r+1$ equations involving $x, y, y', y'', \ldots, y^{(r)}, c_1, c_2, \ldots, c_r$. The desired d.e. of order r can be obtained by eliminating $c_1, c_2, \ldots, c_r$ from these $r + 1$ equations. (The ease with which such an elimination can be done may vary. In some problems, some of the constants may drop out right during differentiation. In some others, the elimination may be quite challenging.)

Before applying this procedure, it is important to ensure that the constants $c_1, c_2, \ldots, c_r$ are independent and cannot be replaced by fewer constants. Suppose for example, that we are given (JEE 1998) that the general solution of a d.e. is $y = (C_1 + C_2)\cos(x + C_3) - C_4e^{x+C_5}$, where C_1, C_2, C_3, C_4, C_5 are arbitrary constants. Here superficially there are 5 constants and so it is tempting to think that the order of the d.e. is 5. However, C_1 and C_2 appear only in the sum $C_1 + C_2$. Any solution that results by giving values, say a_1 and a_2 to C_1 and C_2 respectively can also be obtained by treating $C_1 + C_2$ as a single constant and assigning the value $a_1 + a_2$ to it. So these two constants can be replaced by a single one (which could as well be denoted by C_1 or by

C_2). Similarly, $C_4e^{x+C_5}$ equals $C_4e^{C_5}e^x$ which shows that C_4 and C_5 can be replaced by a single constant, say C_6. (If, however, either C_4 or C_5 appeared again somewhere else, then we would have to be wary before making such a replacement.) Summing up, even though there are superficially 5 constants, in reality there are only 3 and so the d.e. representing the given family of solutions has order 3 and not 5.

Let us obtain the differential equation of the 1-parameter family of hyperbolas $xy = k$ where k is a constant. Here the very first differentiation eliminates k and so without doing any further work we get $xy' + y = 0$ (which is the same as (55)) as the d.e. of the family. The d.e. of the family of all circles centred at the origin is obtained equally immediately. As a less trivial example, consider the family of all straight lines passing through a given point, say $(2, 1)$. A typical member of this family is

$$y - 1 = m(x - 2) \tag{57}$$

where m is a constant, representing the slope of the line. Differentiating (57), we get $\frac{dy}{dx} = m$. Putting this back into (57) gives

$$y - 1 = \frac{dy}{dx}(x - 2) \tag{58}$$

as the d.e. of the family of all straight lines passing through the point $(2, 1)$. It is easy to solve (58) and get back (57).

As an example of the d.e. obtained from a 2-parameter family of curves, consider the family of all circles of radii 1 in the plane. A typical member of this family is of the form

$$(x - a)^2 + (y - b)^2 = 1 \tag{59}$$

where a and b are arbitrary constants. Differentiating (59) twice we get

$$(x - a) + (y - b)y' = 0 \tag{60}$$

$$\text{and} \quad 1 + (y - b)y'' + y'^2 = 0 \tag{61}$$

From (61), $y - b = -\dfrac{1 + y'^2}{y''}$. Putting this into (60) gives $x - a = \dfrac{y'(1 + y'^2)}{y''}$. Putting these into (59) gives, upon simplification

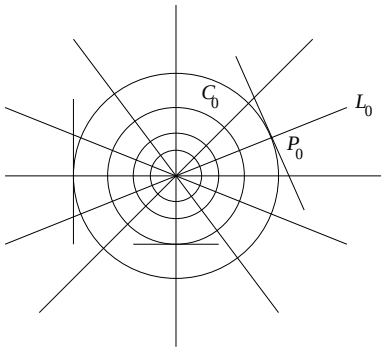
$$(1 + y'^2)^3 = (y'')^2 \tag{62}$$

as the differential equation of the 2-parameter family of all unit circles in the plane. Similarly one can obtain the d.e. of the 3-parameter family of all circles in the plane, the 2-parameter family of all circles with their centres on the x-axis and so on. Once you write the equation of a typical member of the family, the rest of the work is routine, in sharp contrast with the problem in reverse, i.e., the problem of identifying the family of curves represented by a given d.e.

Comment No. 18:

Here is a standard application of the the d.e. representing a 1-parameter family of plane curves. For physical applications, given such a family, it is often important to find another family of curves whose members are at right angles everywhere to those of the original family. That is, given a point P, there is a unique member of the first family, say C_1 passing through P (because it is a one parameter family) and a unique member, say C_2, of the second family passing through P. We want their tangents at P to be at right angles. When this relationship holds between two families of curves, they are said to be **orthogonal trajectories** of each other. As a very simple example, the family of all horizontal lines and that of all vertical lines are orthogonal trajectories of each other. More generally the same is true if one family consists of all lines parallel to a given line and the other, of all lines perpendicular to it.

Sometimes the requirement of orthogonality is satisfied at all except a few points called **singularities**. Still the two families are called orthogonal trajectories of each other. As an example, consider the family of all concentric circles centred at the origin and that of all straight lines passing through the origin. Let $P = (x_0, y_0)$ be any point in the plane other than the origin. Then through P there is a unique circle, say C_0, centred at the origin, viz., the circle $x^2 + y^2 = x_0^2 + y_0^2$. Also there is a unique line, say L_0 passing through P and O, viz., the line $x_0y = y_0x$. It is easily seen that C_0 and L_0 are always orthogonal to each other. At the origin, the circle degenerates into a point and orthogonality makes no sense. So here there is one singularity, viz., the origin.



While it is not difficult to *verify* that two given families are orthogonal to each other, it is generally not easy to simply *guess* the family of orthogonal trajectories of a given family. This job can be done by differential equations. Let C_1 and C_2 be the members of the first and the second family passing through a point $P(x_0, y_0)$. Then the slopes of the tangents at P to C_1 and C_2 are negative reciprocals of each other. This means that the value of $\frac{dy}{dx}$ for the first curve at P is the value of $-\frac{dx}{dy}$ for the second curve at P. In other words, if we know the differential equation of the first family, then we can get the d.e. of the second family simply by replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$. Solving this new d.e. will identify the second family.

Let us illustrate this procedure in the case of the family of concentric circles centred at the origin. As we have already seen, a typical member of this 1-parameter family is $x^2 + y^2 = c$ and so the d.e. of this family is $x + y\frac{dy}{dx} = 0$. Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ we get

$$x - y\frac{dx}{dy} = 0 \tag{63}$$

as the d.e. of the family of orthogonal trajectories. This can be easily solved by separating variables and integrating. The general solution is $\ln|y| = \ln|x| + c$, or, equivalently[3], $y = kx$ for some constant k. This is precisely the family of all straight lines through the origin. Of course, we already know this answer. But now we have arrived at it systematically. A few simple problems of finding orthogonal trajectories will be given in the exercises.

Comment No. 19:

The one-to-one correspondence between parametrised families of plane curves and differential equations implies that if we start with a parametrised family of curves, form its d.e. and solve it, we should get back the same family that we started with. But sometimes something curious happens. Consider, for example, the family of all circles of radii 1 with centres lying on the x-axis. This is a 1-parameter family. A typical member of it has an equation of the form

$$(x-a)^2 + y^2 = 1 \tag{64}$$

where a is an arbitrary constant. Differentiating (63) gives $x - a = -yy'$. Putting this into (64) again, we get

$$y^2(1+y'^2) = 1 \tag{65}$$

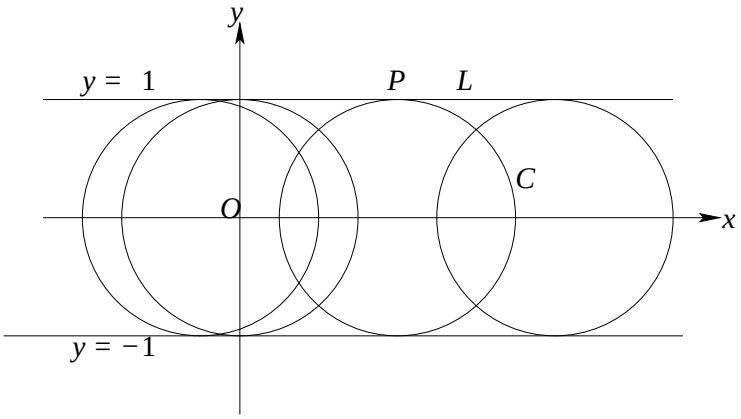
as the differential equation of the family (64).

Let us now see what happens if we solve (65). Rewriting it as $\frac{dy}{dx} = \pm\frac{\sqrt{1-y^2}}{y}$ and separating the variables we get $\sqrt{1-y^2} = c \pm x$ as the general solution. Curves given by this are of the form $(c+x)^2 + y^2 = 1$ or of the form $(c-x)^2 + y^2 = 1$. Together, they constitute the family of all circles of radii 1 with centres lying on the x-axis. So we got back the 1-parameter family of circles that we started with. But the strange thing is that (64) has two more solutions which are not of this form. They are given by $y = 1$ and $y = -1$. For, in either case y' vanishes identically and so (65) is satisfied!

The mysterious appearance of these two extra solutions of (65) can be explained geometrically. In the figure below a few members of the family of curves (64) are sketched. The two solutions $y = \pm 1$ represent two horizontal lines. Let L be the line $y = 1$, P be any point on L and C be the (unique) member of the family (64) passing through P. Then C is a circle of radius 1 which has P as its topmost point. Hence the value of $\frac{dy}{dx}$ at P as a point on the circle C, is 0. But this is also the value of $\frac{dy}{dx}$ at P as a point of L since the line L has slope 0 at all points. Thus we see that the line L, although not a member of the family represented by (64), has the property that at every point P on it, it shares with C not only the coordinates x and y of P (as happens when any two curves intersect) but also the slope $\frac{dy}{dx}$. So any equation which involves only

[3]Actually, we only get $|y| = |kx|$ and hence $y = \pm kx$. By considerations of continuity for functions of two variables it can be shown that the same sign must prevail at all points on each ray starting at the origin. Such fine justifications, although important in theory, are often omitted in solutions of actual problems.

x, y and $\frac{dy}{dx}$ and which is satisfied by every member of the family represented by (64) will also be satisfied by the line L. Since (65) is one such equation, it is hardly surprising that $y = 1$ is a solution of it. An identical explanation holds for the line $y = -1$, except that a typical point on it is the lowest point of the circle passing through it.



A solution which satisfies a differential equation but is not a member of the family of curves represented by it is called a **singular solution**, because it cannot be obtained by giving any value to the arbitrary constants in the general solution. Geometrically, the singular solution of the d.e. obtained from a 1-parameter family of curves represents a curve which has the property that at every point on it, it touches a member of the original family of curves. (Sometimes, such a curve is called an **envelope** of that family.) Figuratively, a singular solution is like an old faithful servant who gets along so well with every member of the family that, as a result, although not a member of the family, enjoys the same social status! Not all d.e.'s have singular solutions. There are systematic methods for obtaining them (in case they exist) but we shall not go into them. As in the example just given, sometimes a singular solution can be identified just by inspection.

EXERCISES

19.1 A spherical rain drop evaporates at a rate proportional to its surface area at any instant t. Find the differential equation giving the rate of change of the radius of the rain drop. (1997*)

19.2 Which of the following functions is a solution of the differential equation $(\frac{dy}{dx})^2 - x\frac{dy}{dx} + y = 0$? (1999)

(A) $y = x$ (B) $y = 2x$ (C) $y = 2x - 4$ (D) $y = 2x^2 - 4$.

19.3 Suppose a radio-active substance obeys the law of decay $\frac{dx}{dt} = -\lambda x$, where $\lambda > 0$ is a constant. Prove that the time taken for any quantity of the

substance to reduce to half its size is a constant, independent of the quantity. (This constant is called the **half-life** of that substance and can vary considerably from substance to substance.)

19.4 Suppose the ratio of the radio-active carbon isotope (C_{14}) to the non-radio-active carbon (C_{12}) is the same in all living organisms. Let this ratio be α. After death, C_{14} decays exponentially, while C_{12} remains intact. Suppose β is the ratio of C_{14} to C_{12} in a fossil, today. If the half-life of C_{14} is T, show that the 'age' of the fossil is $\dfrac{(\ln\alpha - \ln\beta)\,T}{\ln 2}$. (This method is popularly known as **carbon dating**.)

19.5 Use Taylor's Theorem (Theorem 6 in Chapter 16) to show that if $|x|$ is small then $\ln(1+x) \approx x$.

19.6 One of the rules of thumb adopted in banking circles is the **seventy percent rule** which says that if interest at the rate of α p.c.p.a. is compounded annually, then the principal will be doubled in about $\dfrac{70}{\alpha}$ years. What is the mathematical basis for this rule? Show that the rule is valid for small values of α, e.g., $\alpha = 8$ or 10 but that it fails if α is large, e.g., $\alpha = 70$. (Of course, no bank gives interest at such a fabulous rate!)

19.7 Let $u(x)$ and $v(x)$ satisfy the differential equations $\dfrac{du}{dx} + p(x)u = f(x)$ and $\dfrac{dv}{dx} + p(x)v = g(x)$, where $p(x), f(x)$ and $g(x)$ are continuous functions. If $u(x_1) > v(x_1)$ for some x_1 and $f(x) > g(x)$ for all $x > x_1$, prove that any point (x, y), where $x > x_1$, does not satisfy the equations $y = u(x)$ and $y = v(x)$. (1997)

19.8 Let $y = f(x)$ be a curve passing through $(1, 1)$ such that the triangle formed by the coordinate axes and the tangent at any point of the curve lies in the first quadrant and has area 2. Form the differential equation and determine all such possible curves. (1995)

19.9 A curve $y = f(x)$ passes through the point $P(1, 1)$. The normal to the curve at P is $a(y-1) + (x-1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, determine the equation of the curve. Also obtain the area bounded by the y-axis, the curve and the normal to the curve at P. (1996)

19.10 A curve passing through the point $(1, 1)$ has the property that the perpendicular distance of the origin from the normal at any point P on the curve is equal to the distance of P from the x-axis. Determine the equation of the curve. (1999)

19.11 (a) A normal is drawn at a point $P(x, y)$ of a curve. It meets the x-axis at Q. If PQ is of constant length k, then show that the differen-

tial equation describing such curves is $y\dfrac{dy}{dx} = \pm\sqrt{k^2 - y^2}$. Find the equation of such a curve passing through $(0, k)$. (1994)

(b) Prove that the differential equation in (a) has a singular solution.

19.12 Find the order and the degree of the differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where c is a positive parameter. (1999)

19.13 Although we have not studied any methods for solving second order differential equations, some equations can be solved by elementary techniques. For example, by integrating twice w.r.t. x, we solved (56). Similarly, to solve (15) multiply both the sides by $\dot{x}$. Solve the resulting equation to get a d.e. for $\dot{x}$ and then finally solve this new equation.

19.14 Identify the orthogonal trajectories of the following families of curves:

(i) the family of all rectangular hyperbolas of the form $xy = k$

(ii) the family of all circles passing through the origin and cetres on the x-axis

(iii) the family of all parabolas with centre at the origin and foci on the x-axis

(iv) the family of all curves with equations of the form $e^x \cos y = c$, where c is a constant.

19.15 Find the differential equation of the family of all circles of radii 1 with centres on the circle $x^2+y^2 = 4$. Without solving this differential equation, identify two singular solutions of it.

19.16 Solve the following first order differential equations:

(i) $y' - x^{-1}y = x^{-1}y^2$ (ii) $xy' - 2y = x^4$

(iii) $(2x + e^y)dx + xe^y dy = 0$ (iv) $ye^{x/y}dx + (y - xe^{x/y})dy = 0$

(v) $(x + y)^2 y' = 1$ (vi) $(x^2 + y^2)dx + xydy = 0$

(vii) $y' = xy + x^3y^3$ [*Hint:* Put $v = 1/y^2$.] (viii) $(xy + x^3y^3)y' = 1$.

*(ix) $y' = e^x + e^y + 1$. [*Hint:* Multiply by e^{-y} and put $u = e^{-y} + 1$.]

19.17 Solve the differential equation $(a_1x + b_1y + c_1)dx = (a_2x + b_2y + c_2)dy$, where $a_1, a_2, b_1, b_2, c_1, c_2$ are constants. [*Hint :* The case $c_1 = c_2 = 0$ is handled by Comment No. 15. Reduce the general case to this one with the substitutions $x = u + h$ and $y = v + k$, where h, k are suitably chosen.]

19.18 Solve the following second order d.e.'s by first treating them as first order d.e.'s in y'.

(i) $xy'' + y' = (y')^2$ (ii) $yy'' - y'^2 = 0$ (iii) $y'' + (1 + y^{-1})(y')^2 = 0$.

Chapter 20

FUNCTIONAL EQUATIONS AND RELATIONS

The concept of a functional equation was already introduced in Chapter 18, Comment No. 13. In essence it is an equation that relates the values assumed by a function at several points which are themselves related in a particular manner. Even and odd functions, as well as periodic functions are examples of functions that satisfy certain functional equations and we already studied some of their properties.

The problem we undertake in this chapter is to solve a functional equation, that is, to identify the function from a functional relation which it is given to satisfy. In principle, this is analogous to solving a differential equation, where we try to find the function from an equation satisfied by its derivatives. But the techniques needed are different. Unlike differential equations, there are no standard procedures to classify or to solve functional equations. An *ad hoc* type of reasoning is often needed. Sometimes a functional equation leads to a differential equation whose solution then gives a solution of the functional equation.

There is a brief discussion of binary relations on a set. Although the theory developed here only scratches the surface, it introduces some commonly used terminology such as an equivalence relation.

Solutions of functional equations are not asked every year at the JEE. But functions satisfying special functional equations such as the even and the odd functions and the periodic functions appear in many problems. The topic of binary relations is not specifically included in the JEE syllabus anymore. So direct questions on them are rarely asked in recent years.

Main Problem: A function $f : \mathbb{R} \longrightarrow \mathbb{R}$ satisfies the equation

$$f(x+y) = f(x)f(y) \tag{1}$$

for all x, y in $\mathbb{R}$ and $f(x) \neq 0$ for any x in $\mathbb{R}$. Let the function be differentiable at 0 and $f'(0) = 2$. Then the numerical value of $f(\ln 10)$ is .... . (JEE 1990, modified)

First Hint: Show that $f(0) = 1$.

Second Hint: Show that $f'(x) = 2f(x)$ for all x in $\mathbb{R}$.

Third Hint: Show that $f(x) = e^{2x}$ for all $x \in R$.

Solution : Let $a = f(0)$. Then by putting $x = y = 0$ in (1), we have $a = a^2$. Hence $a = 0$ or $a = 1$. As it is given that $f(x)$ is always non-zero, we must have $f(0) = 1$. Now consider any $x \in \mathbb{R}$. For any $h \neq 0$, we have $f(x+h) = f(x)f(h)$ by (1). Since $f(0) = 1$, we get

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= \frac{f(x)f(h) - f(x)}{h} \\ &= f(x)\frac{f(h) - f(0)}{h} \end{aligned} \tag{2}$$

As $h \to 0$ (with x fixed), the ratio $\frac{f(h)-f(0)}{h}$ tends to $f'(0)$ which is given to exist and equal 2. So by (2), $f'(x)$ exists and equals $2f(x)$ at every $x \in \mathbb{R}$. Put differently, $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx} = 2y \tag{3}$$

which can be easily solved by separating the variables. The general solution is $y = ke^{2x}$ for some constant k. The initial condition $f(0) = 1$ determines k as 1. So we get $f(x) = e^{2x}$ for all x. Putting $x = \ln 10$, $f(\ln 10) = e^{2 \ln 10} = e^{\ln 100} = 100$.

Comment No. 1:

It is not necessary that we should be given that $f(x) \neq 0$ for every $x \in \mathbb{R}$. It is enough if we assume that f is not identically 0, i.e., that there exists at least one value of x for which $f(x)$ is non-zero. For, if c is any such value, then taking $x = c$ and $y = 0$ in (1), we have $f(c) = f(c+0) = f(c)f(0)$ which implies $f(0) = 1$ since $f(c) \neq 0$. The rest of the argument goes through unchanged.

Comment No. 2:

Equation (1) is a typical example of what is called a **functional equation**. It expresses a relationship among the values of the function at points in the domain which are themselves related under a certain relationship. For example,

verbally, (1) says that whenever we have three real numbers such that one of them is the sum of the other two, then the value of f at this real number is the product of its values at those two real numbers. Or, more compactly, the value at a sum is the product of the values. This property does not uniquely determine the function $f(x)$. However, with some additional information we may be able to narrow down the scope of $f(x)$. For example, the argument above would have worked equally well had we merely been given that $f'(0)$ exists (without being given that it equals 2). In that case, denoting $f'(0)$ by α, instead of (3) we would have had $\frac{dy}{dx} = \alpha$ and finally, $f(x) = e^{\alpha x}$ for all $x \in \mathbb{R}$. In fact we can call this as the **general solution** of (1) (assuming, of course, differentiability at 0). A particular value of α would give us a particular solution as in the Main Problem. (Exercise (4.41) also dealt with a functional equation.)

The procedure used to solve (1) is noteworthy. As a starter, we gave some special values to the variables x and y to get $f(0)$. This was then combined with the additional information that $f'(0) = 2$ to get the differential equation (3). Finally, the d.e. was solved. So here differential equations were used as a tool.

Comment No. 3:

We now study a functional equation which is of the same spirit as (1). Suppose $f(x)$ satisfies

$$f(x + y) = f(x) + f(y) \tag{4}$$

for all $x, y \in \mathbb{R}$. (A function satisfying this condition is often said to be an **additive function**.) Suppose further that we are given that $f'(0)$ exists and equals c. Then we can determine $f(x)$ by taking steps that are similar to those taken for (1). First we show that $f(0) = 0$ by putting $x = 0 = y$ in (4). Then we show that for every $x \in \mathbb{R}$, $f'(x) = 0$ by writing $\frac{f(x+h)-f(x)}{h}$ as $\frac{f(h)}{h}$. Finally, we solve the differential equation $y' = c$ to get $y = cx + k$ for some constant k. The fact that $f(0) = 0$ yields $k = 0$. Hence we have proved that $f(x) = cx$ for all $x \in \mathbb{R}$. (Or we could let $g(x) = e^{f(x)}$. Then (4) implies that $g(x)$ satisfies the functional equation (1). Moreover, by the chain rule, $g'(0) = e^c$. So by the generalisation of the Main Problem considered above, we get $g(x) = e^c x$ for all $x \in \mathbb{R}$, which implies $f(x) = cx$ for all $x \in \mathbb{R}$. This argument is not substantially different from the one above but looks slicker because we are avoiding duplication of work.)

Comment No. 4:

We just solved (4) under the additional hypothesis of differentiability of $f(x)$ at 0. It is interesting, that we get exactly the same solution under a weaker additional hypothesis that $f(x)$ is merely continuous at 0. As is to be expected, now the problem is more challenging.

We first prove that any function which satisfies (4) and is continuous at 0 is continuous everywhere (JEE 1981). This is equivalent to saying that for every fixed $x \in \mathbb{R}$, $f(x + h)$ tends to $f(x)$ as $h \to 0$. This is easy because $f(x + h) = f(x) + f(h)$ by (1) and since $f(0) = 0$ (as already proved), by

continuity of $f(x)$ at 0, $\lim_{h\to 0} f(h) = f(0) = 0$.

Having proved that $f(x)$ is continuous everywhere, we now prove that there is some constant c such that $f(x) = cx$ for all $x \in \mathbb{R}$. In Comment No. 4 we derived this assuming that $f'(x) = c$. Now we are not given this number c. Nor are we given that f is differentiable at 0. So we cannot begin by letting c be $f'(0)$. How, then, do we identify this magic number c? The answer employs a clever trick in mathematics. If at all such a c exists so that $f(x) = cx$ for all $x \in \mathbb{R}$, then, in particular, we must have $f(1) = c$. So we have no choice but to let $c = f(1)$. We indeed do so.

Now comes the task of showing that

$$f(x) = cx \tag{5}$$

for *all* $x \in \mathbb{R}$. So far we have proved $f(0) = 0$ and so (5) holds for $x = 0$. Also, $f(1) = c$ by definition, and so (5) holds for $x = 1$ too. From these two values, we have to extend the truth of (5) to all real values of x in stages. That is, first we prove that (5) holds for all positive integers, then for all integers, then for all rationals and finally for all real numbers. (In principle, this is also how we gradually extend the definition of the power a^x, where a is a fixed positive real number.)

Note first that because of (4), if (5) holds for say, $x = x_1$ and for $x = x_2$ then it also holds for $x = x_1 \pm x_2$. As a result, since (5) holds true for $x = 1$, it holds for $x = 1 + 1$, then for $x = 2 + 1$ and hence, by induction, for all positive integers n. Thereafter taking $x_1 = 0$ and $x_2 = n$, it holds for $x = 0 - n = -n$. Hence it holds for all negative integers also. Next we claim that (5) holds true for every rational value of x, say $x = \frac{m}{n}$, where m and n are integers and n is positive. Let $u = f(\frac{1}{n})$. Then $f(\frac{2}{n}) = f(\frac{1}{n}) + f(\frac{1}{n}) = u + u = 2u$ by (4). Similarly, $f(\frac{3}{n}) = f(\frac{2}{n}) + f(\frac{1}{n}) = 2u + u = 3u$. Continuing like this (or by induction), we have $f(\frac{k}{n}) = ku$ for every positive integer k. In particular, taking $k = n$, we have $nu = f(1) = c$. Hence $u = c\frac{1}{n}$. Thus we have proved that (5) holds for $x = \frac{1}{n}$. But then by the same argument as above, it holds for all integral multiples of $\frac{1}{n}$, and hence for $x = \frac{m}{n}$.

Thus we have proved that (5) holds for all rational values of x. We now show that it holds for all real values of x. Note that so far we have not used continuity of f. We need it precisely at this juncture. We shall also need the fact that the set of all rationals is dense in the set of real numbers. For, then we can apply Exercise (6.47). Given any real number α, fix a sequence $\{x_n\}$ of rational numbers converging to it. Then by continuity of f at α and Theorem 1 of Chapter 16, we have $f(\alpha) = \lim_{n\to\infty} f(x_n)$. As we have already proved that (5) holds for all rationals, we have $f(x_n) = cx_n$ for all n. So $f(\alpha) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} cx_n = c\alpha$ since $x_n \to \alpha$ as $n \to \infty$. Thus we have shown that (5) holds for every real α. This completes the proof that (5) holds for all $x \in \mathbb{R}$.

Note that continuity of f (which followed from its continuity at 0), was vitally used in the last phase of the argument. It can, in fact, be shown

by examples, that without continuity, (5) may fail. The examples are rather complicated and beyond our scope.

Comment No. 5:

We have seen that (1) and (4) are essentially the same functional equations. A few other related functional equations will be given in the exercises. As with trigonometric equations (see Comment No. 14 of Chapter 10), sometimes a problem about functional equations may turn out to be erroneous. Of course, this happens very rarely. Here is an example.

Let $f(x+y) = f(x)f(y)$ for for all x and y. Suppose $f(5) = 2$ and $f'(0) = 3$. Find $f'(5)$. (JEE 1981)

It is not given here that $f(x) \neq 0$ for every $x \in \mathbb{R}$. However, from Comment No. 1 above, the fact that $f(5) = 2 \neq 0$ implies that $f(x) \neq 0$ for all $x \in \mathbb{R}$. We already solved this equation in the Main Problem, except that instead of $f'(0) = 2$, we now have $f'(0) = 3$. The crucial step in the solution was to get Equation (3) (which was another way of saying that $f'(x) = 2f(x)$ for all $x \in \mathbb{R}$). In the present problem, we shall instead get

$$f'(x) = 3f(x) \tag{6}$$

for all $x \in \mathbb{R}$. In particular, putting $x = 5$ we get $f'(5) = 3f(5) = 6$ since it is given that $f(5) = 2$. This may seem to complete the solution and indeed it does as far as the given question is concerned. But the point is that if we solve (6) as a differential equation (the way we solved (3)) with the initial condition $f(0) = 1$, then we get $f(x) = e^{3x}$ as the solution. So we should have $f(5) = e^{15}$. But we are given that $f(5) = 2$. This means that there is an inconsistency in the data and so the problem is a hypothetical one.

Comment No. 6:

Sometimes a functional equation may be combined with something else. Suppose for example, that we want to find the natural number a for which

$$\sum_{k=1}^{n} f(a+k) = 16(2^n - 1) \tag{7}$$

where the function f satisfies the relation

$$f(x+y) = f(x)f(y) \tag{8}$$

for all natural numbers x, y and further $f(1) = 2$. (JEE 1992)

(8) is identical to (1) but the difference is that (8) is given to hold only for positive integral values of x and y. In fact, the function f in this problem is a function of a discrete variable and so there is no such thing as its continuity or differentiability. The method which we applied for solving (1) cannot be applied here. Nor is it needed, because our task here is not to determine $f(x)$ for all real x. Our task is limited to determining $f(x)$ when x is a positive integer and for this the data is sufficient. Putting $x = 1 = y$ in (8) we get $f(2) = [f(1)]^2 = 2^2$.

Then again putting $x = 2, y = 1$ into (8) we get $f(3) = f(2)f(1) = 2^2 \times 2 = 2^3$. Continuing like this (or more formally, by induction), we get $f(x) = 2^x$ for every positive integer x.

Now that we know $f(x)$, (7) reduces to $\sum_{k=1}^{n} 2^{a+k} = 16(2^n - 1)$. The L.H.S. is the sum of a geometric progression with common ratio 2. So we get $2^{a+1}(2^n - 1) = 16(2^n - 1)$ which gives $2^{a+1} = 16 = 2^4$ and hence $a = 3$.

Comment No. 7:

In the functional equations we studied so far, there were three variables, say x, y, z and the functional equation amounted to saying that whenever these three variables were related in a particular manner, the values $f(x), f(y), f(z)$ were related in a particular manner. (For example, (1) amounts to saying that whenever $z = x + y$, we have $f(z) = f(x)f(y)$ while (4) amounts to saying that whenever $z = x + y$, we have $f(z) = f(x) + f(y)$. We now turn to functional equations in which there are only two variables, say x and y such that whenever they are related in a particular way, the values $f(x)$ and $f(y)$ are related in a particular way. Suppose, for example (JEE 1996), that for every $x \neq 0$,

$$af(x) + bf(\frac{1}{x}) = \frac{1}{x} - 5 \tag{9}$$

where $a \neq b$ and we are asked to find the value of the integral $\int_1^2 f(x)dx$. Here the functional equation (9) can be paraphrased as saying that whenever $xy = 1$, we have $af(x) + bf(y) = y - 5$. If we put $x = 1$ in (9), we get $(a + b)f(1) = -4$. From this we can get $f(1)$ if $a + b \neq 0$. But this is not given.

To solve an equation like (9), the trick is to recognise that the relationship between the two variables x and y, is of a reciprocal type. That is, if we interchange x and y, the same relationship will hold. Put differently, if we replace x by $\frac{1}{x}$ in (9), we get

$$af(\frac{1}{x}) + bf(x) = x - 5 \tag{10}$$

Now, for a fixed non-zero x, (9) and (10) together can be treated as a system of two linear equations in the two unknowns $f(x)$ and $f(\frac{1}{x})$. To solve this system, we multiply (9) by a, (10) by b and subtract to get

$$(a^2 - b^2)f(x) = \frac{a}{x} - 5a - bx + 5b \tag{11}$$

for all non-zero x.

The L.H.S. can be 0 when either $a - b = 0$ or $a + b = 0$. We are given that $a \neq b$. However $a + b$ may be 0. But in that case $b = -a \neq 0$ (as otherwise $a - b = 0$). So (11) becomes a quadratic equation in x which can have at most two solutions. But that contradicts the fact that (9) and hence (10) are valid for *all* non-zero x.

So we conclude that $a^2 - b^2 \neq 0$. Then from (11) we have $f(x) = \dfrac{a}{(a^2-b^2)x} - \dfrac{bx}{a^2-b^2} - \dfrac{5}{a+b}$ for all $x \neq 0$. Now a routine integration gives $\int_1^2 f(x)dx = \dfrac{2a\ln 2 - 10a + 7b}{2(a^2-b^2)}$.

Note the role played by the term $\frac{1}{x}$ on the R.H.S. of (9). If that term were absent and we had only a constant, say c, on the R.H.S. then instead of (11) we would get $(a^2-b^2)f(x) = c(a-b)$. If $a^2 \neq b^2$ this would mean that $f(x)$ is a constant. However, if $a^2 = b^2$ then there would be either no solution or infinitely many solutions, depending upon whether $c(a-b)$ is non-zero or zero. Such cases are degenerate. Barring them, a functional equation of the form (9) always has a unique solution unlike an equation like (1) or (4) which have infinitely many solutions. Note further, that unlike (1) or (4), (9) was solved by a purely algebraic argument, with no differential equations, and not even continuity considerations involved anywhere.

Comment No. 8:

As noted earlier the equation (9) gives a relationship between $f(x)$ and $f(y)$ whenever x and y are related by the equation $xy = 1$ (in a simpler language, whenever they are reciprocals of each other). The case where the relationship is of the form $x + y = 0$ is of an entirely similar nature. In such a case instead of (9) we have an equation like

$$af(x) + bf(-x) = g(x) \tag{12}$$

where a, b are some constants and $g(x)$ is some function of x. Working exactly as we did for (9) we get $af(-x)+bf(x) = g(-x)$ which along with (12) gives $(a^2 - b^2)f(x) = ag(x) - bg(-x)$ which determines $f(x)$ when (12) is not degenerate, i.e., when $a \neq \pm b$. In the degenerate case there will be no solution if $ag(x) - bg(-x)$ is not 0 while there will be infinitely many solutions if it is 0.

Ironically, it is precisely these degenerate cases of (12) which occur most frequently. Suppose for example that $a = b = 1$ and $g(x)$ vanishes identically. Then (12) can be written as

$$f(-x) = -f(x) \text{ for all } x \tag{13}$$

which is, of course, exactly the definition of an odd function. Putting $x = 0$ in (13) we get $f(0) = -f(0)$ and hence $f(0) = 0$. But that is just about all the progress we can make. Even if we are further given that $f'(0)$ exists, there is not much we can conclude from it. We cannot, for example, deduce that $f(x)$ is differentiable everywhere as we could do in the solution to the Main Problem. The trouble is that because of degeneracy, the functional equation (13) is not as strong as (1). You get nothing new by replacing x with $-x$ in it. As a result, there are many, many odd functions. In fact one way to construct such a function is to take absolutely any function $f(x)$ defined for $x \geq 0$, with the sole restriction that $f(0) = 0$. Then we define $f(x) = -f(-x)$ for $x < 0$. This new function satisfies (13) by the very construction.

Of course, if we *know* that f is differentiable everywhere (in addition to being odd), then by Exercise (18.6)(d), $f'(-x) = f'(x)$ for all $x \in \mathbb{R}$, or in other words, that f' is an even function. If further, f' is differentiable, i.e., f is twice differentiable, then f'' is odd (again by Exercise (18.6)(d)) and hence $f''(0) = 0$. Continuing like this, we can show that at 0, derivatives of f of even orders must vanish (provided they exist) and hence that the Taylor expansion of $f(x)$ at 0 consists of only odd powers of x. (This is probably the reason an odd function is called that way.)

Similarly, if we take $a = 1, b = -1$ and $g(x) = 0$ in (12) we get precisely the definition of an even function. The condition does not determine an even function uniquely. In fact, for an even function we cannot even determine the value at 0. But its derivative, in case it exists, is an odd function. (That the derivative need not exist everywhere is shown by taking $f(x) = |x|$ which is an even function but not differentiable at 0.) Hence $f'(0)$ must vanish in case it exists. Similarly, the derivatives of all odd orders must vanish at 0 and so the Taylor expansion at 0 will consist only of even powers of x.

Comment No. 9:

Basically, the reason an equation like (9) can be solved is that the reciprocal of a reciprocal gives you back the original number. Similarly, (12) can be solved for $f(x)$ (except in degenerate cases) because $-(-x) = x$ for all $x \in \mathbb{R}$. The root cause of this, of course, is that -1 is a number whose square equals 1. Put differently, it is a square root of unity, a term we introduced in Chapter 2 and used heavily in Chapter 7. In the real number system there are no other roots of unity (except for 1, of course). As a result, we cannot have other functional equations analogous to (12) in the real number system. But with complex numbers the situation is quite different. For every positive integer $n \geq 2$, $\alpha = e^{\frac{2\pi}{n}i}$ is a primitive n-th root of unity. If we take a non-zero complex number z and go on multiplying it by α we get the complex numbers $z, z\alpha, z\alpha^2, z\alpha^3, \ldots$ which recur in a cycle of length n. It follows that if we have a functional equation relating the values of a function f at some of these points, then by applying it with z replaced by $z\alpha$ then by $z\alpha^2$, and so on, we shall get a system of linear equations whose solution will determine $f(z)$ except in non-degenerate cases. We illustrate this for $n = 3$. As usual we use ω to denote the complex cube root of unity $e^{\frac{2\pi}{3}i}$.

Suppose $f(z)$ is a possibly complex-valued function of a complex number z which satisfies a functional equation of the form

$$af(z) + bf(\omega^2 z) = g(z) \tag{14}$$

for all complex values of z, where a, b are some fixed complex numbers and $g(z)$ is some function of z. Replacing z by ωz and then by $\omega^2 z$ we get the following two equations

$$af(\omega z) + bf(z) = g(\omega z) \tag{15}$$

$$\text{and} \quad af(\omega^2 z) + b(\omega z) = g(\omega^2 z) \tag{16}$$

(14), (15) and (16) together form a linear system of three equations in the three unknowns $f(z), f(\omega z)$ and $f(\omega^2 z)$ which can be expressed in a matrix form as

$$\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix} \begin{bmatrix} f(z) \\ f(\omega z) \\ f(\omega^2 z) \end{bmatrix} = \begin{bmatrix} g(z) \\ g(\omega z) \\ g(\omega^2 z) \end{bmatrix} \tag{17}$$

By a direct computation, the determinant of the coefficient matrix is $a^3 + b^3$. So when $a^3 + b^3 \neq 0$, the functional equation (14) has a unique solution.

Comment No. 10:

Periodicity of a function is another property which can be expressed as a functional equation, viz., $f(x + T) = f(x)$ for all x. This can be paraphrased to say that whenever x and y are related by $y - x = T$, a fixed constant, $f(x)$ and $f(y)$ are related by the condition that they are equal. More generally we can consider a functional equation of the form

$$af(x) + bf(x + T) = g(x) \tag{18}$$

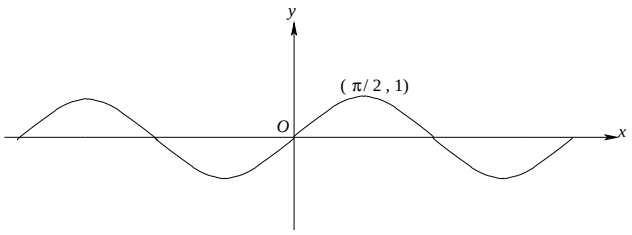
where a, b, T are constants and $g(x)$ is a function of x. This looks very similar to (12) or (14). But here the method given above for solving a functional equation like (12) or (14) will not work. If we replace x by $x + T$, we get $af(x + T) + bf(x + 2T) = g(x + T)$. Replacing x by $x + T$ again will give $af(x + 2T) + bf(x + 3T) = g(x + 2T)$. But no matter how many times we do this, x will never appear again in the sequence $x, x + T, x + 2T, x + 3T, \ldots$ (except in the trivial case where $T = 0$).

That is why a relation like (18) cannot determine the function $f(x)$ uniquely. In particular we see that for every $T > 0$, there are infinitely many functions with T as a period. We are absolutely free to define $f(x)$ any way we please for values of x in some semi-open interval, say $[0, T)$. Thereafter the freedom is over. Given any $x \in \mathbb{R}$, we subtract the largest integral multiple of T from it to write it as $x = mT + y$ for a unique integer m (which could be negative) and a unique $y \in [0, T)$. Then because of periodicity $f(x)$ has to equal $f(y)$.

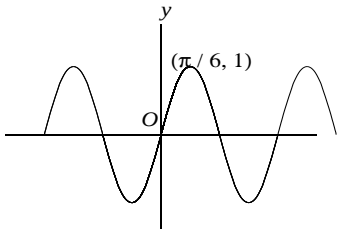
Periodicity does help, however, in proving certain global properties of a function. For example, as we proved in Comment No. 13 of Chapter 18, the value of the integral of a periodic function with period T is the same on every interval of length T. The exercises will give a few more results about periodicity. Periodicity of a function also helps in drawing its graph because once we have drawn it over an interval, say $a, a + T]$ of length T, we merely have to draw its replicas over the intervals $[a + T, a + 2T]$, $[a + 2T, a + 3T]$ and so on and also on intervals $[a - T, a]$, $[a - 2T, a - T]$ and so on. As a concrete illustration let us draw the graph of $y = \sin(3x + \frac{\pi}{4})$. (JEE 1978, modified) We can, of course, apply the general methods for drawing the graph of any function, such as first studying where the graph is increasing/decreasing. But in the present case, the graph of the sine and the cosine functions are very standard ones and

so if we could relate the given function to either of these two, the work will be simplified. One way to do so is to expand $\sin(3x + \frac{\pi}{4})$ as $\frac{1}{\sqrt{2}}(\sin 3x + \cos 3x)$. But this is not of much help, since drawing the graph of a sum of two functions from the graphs of the two functions is not a very easy thing to do.

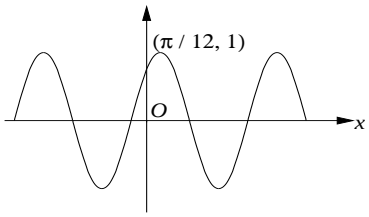
A far better idea is to start with the very standard graph of the sine function, shown in Figure (a) below and see how it can be modified to get the graph of the given function. Let us first rewrite $3x + \frac{\pi}{4}$ as $3(x + \frac{\pi}{12})$ and observe that the sine function is periodic with period 2π. Therefore the function $\sin 3x$ is periodic with period $\frac{2\pi}{3}$. Hence it is easy to draw the graph of $y = \sin 3x$ from that of $\sin x$. We simply compress the latter horizontally by a factor of 3 as shown in the Figure (b) below. Even though this graph is not asked, it is a good idea to draw it as an auxiliary graph. For, once we draw it, the graph of $y = \sin 3(x + \frac{\pi}{12})$ is obtained simply by shifting the graph of $y = \sin 3x$ to the left by $\frac{\pi}{12}$ units. It is shown in Figure (c).



(a) $y = \sin x$



(b) $y = \sin 3x$



(c) $y = \sin(3x + \frac{\pi}{4})$

Comment No. 11:

We now consider functional equations of a different type. They arise through compositions of two functions. Suppose the function h is the composite $g \circ f$ of two functions, i.e., $h(x) = g(f(x))$ for all x. Given the functions f and g, finding h is merely a matter of substitution. In fact, this is as straightforward as multiplying two integers. But we now consider the reverse problem, that is, given a function h we want to find two functions f and g such that $h = g \circ f$. This is analogous to the problem of factoring a given integer, say n, as a product of two integers, say $n = uv$. There is, of course, a trivial answer to the problem. Simply take $u = n$ and $v = 1$ or vice versa! But if we want to factorise n in a

non-trivial way, that can be quite difficult. And sometimes there need not be any solution. (This happens when the integer n is a prime.)

The situation with the problem of expressing a given function h as a composite $g \circ f$ is somewhat similar. There is always a trivial solution. For example, either take $f(x) = x$ for all x and $g = h$ or vice versa. But a non-trivial factorisation is not generally easy to obtain. In some cases, the answer (or at least one possible answer) is obvious. For example, suppose $h(x) = \sin(x^2)$ for $x \in \mathbb{R}$. Here one obvious solution is to take $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. But one can also take $f(x) = x^4$ and $g(x) = \sin(\sqrt{x})$. (Here g is defined only for $x \geq 0$. That does not matter while forming the composite $g \circ f$ since the range of f is the set of all non-negative real numbers. If we require that g be defined on all real numbers, we may take it to be $g(x) = \sin(\sqrt{|x|})$.)

Summing up, the problem of solving an equation of the form

$$g(f(x)) = h(x) \tag{19}$$

where h is a known function but both f and g are unknown functions is not a very interesting one. However, some special cases of this problem are interesting. One such case is where the function f is also given (in addition to the function h). In this case if a solution exists, we say that the function h can be factored[1] through the function f, because we can now write $h = g \circ f$.

For a given h and f whether the equation (19) has a solution or not depends on the relationship between f and h. If, for example, f is an even function, then no matter what g is, the composite $g \circ f$ is an even function. It follows that if f is even and h is not, then (19) cannot be solved for g. More generally, if x_1 and x_2 are two points such that $f(x_1) = f(x_2)$ but $h(x_1) \neq h(x_2)$ then (19) can have no solution.

Another important special case of (19) arises when the function h is the identity function $h(x) = x$ for all x. In that case the functions f and g have to be the inverses[2] of each other. Exercise (15.25)(b) deals with a few problems of finding inverses of functions.

We now consider another special case of (19) in which we further require that f and g be the same functions. Thus the problem now amounts to finding, for a given function h, another function f such that the composite $f \circ f$ equals h. This composite is often denoted[3] by f^2. With this new notation the problem

[1] It is important to note that the word 'factoring' here is applied here w.r.t. composition and not w.r.t. the ordinary product of two functions. That is, one must not confuse $g \circ f$ with gf which is defined simply as $(gf)(x) = g(x)f(x)$ for all x. Clearly, fg and gf are always equal. But $g \circ f$ is in general, different from $f \circ g$ as one can see by taking $f(x) = \sin x$ and $g(x) = x^2$, for example.

[2] More precisely, here we say that g is a **left inverse** of f and that f is a **right inverse** of g. The term 'inverse' usually implies a two sided inverse, i.e., we want both the composites $g \circ f$ and $f \circ g$ to be the identity functions.

[3] This is a dangerous practice because $(f^2)(x)$ can easily be confused with $(f(x))^2$. But if one is careful, then the notation is convenient when we have to apply the same function f again and again. More generally, in this new notation, f^3 means $f \circ f \circ f$. By induction, f^n is defined as the composite $f \circ (f^{n-1})$. This process is often called **iteration**, which simply means repetition, with f^n being called the n-th iterate of f. It is an important technique in finding

now asks whether the given function h has a 'square root'. Suppose for example that h is the identity function $h(x) = x$ for all x. Then the problem has two easy solutions. We can define $f(x) = x$ or $f(x) = -x$ for all $x \in \mathbb{R}$. In either case we see that f^2 i.e., $f \circ f$ is precisely h.

What if we take the function $h(x) = -x$ for all $x \in \mathbb{R}$? We claim that h has no continuous square root, i.e., that there exists no continuous function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that for every $x \in \mathbb{R}$, $f(f(x)) = -x$. Suppose such a function does exist. Then this function must be one-to-one, for $f(x_1) = f(x_2)$ would imply $f(f(x_1)) = f(f(x_2))$, i.e., $-x_1 = -x_2$ and hence $x_1 = x_2$. But as an application of the Intermediate Value Property, it follows that every one-to-one continuous function from $\mathbb{R}$ to $\mathbb{R}$ is either strictly increasing or strictly decreasing on $\mathbb{R}$ (see Comment No. 6 of Chapter 16). So this function f is either strictly increasing or strictly decreasing. In either case, the composite $f \circ f$ is strictly increasing. But the function $h(x) = -x$ is strictly decreasing. So $f \circ f$ cannot equal h for any continuous function f. In other words, h has no continuous square root.

We remark that if we do not insist on continuity, then the function h above *has* a square root, i.e., we can find a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that $f(f(x)) = -x$ for every $x \in \mathbb{R}$. But the construction of such a function is complicated and beyond our scope. (In spirit, it is similar to the construction of a solution of (4) which is not of the form (5).) If instead of real numbers, we work with complex numbers, then the function $h(z) = -z$ does have a square root. Simply define $f(z) = iz$ for any complex number z.

Comment No. 12:

We conclude with a brief discussion of binary relations on a set. By definition, a **binary relation** (or simply a **relation**) on a set S is simply a subset, say R, of the cartesian product $S \times S$, i.e., the set of all ordered pairs of the form (x, y), where x, y are two (not necessarily distinct) elements of S. If a and b are two elements of S such that the pair $(a, b) \in R$, then we say a is related to b under the relation R, or simply that it is R-related to b and express this in symbols by aRb. (We could, of course, use $(a, b) \in R$. But aRb is quite common.) If a

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approximate solutions to problems where exact solutions are impossible. We start with some point x_0 and then consider the sequence of iterates $\{x_0, x_1, x_2, x_3, \ldots\}$, where each term is obtained by applying f to the preceding term. This is a recursive sequence and in many cases, if the function f and the starting point x_0 are chosen properly, it converges to a solution of the given problem. In such a case, even if we are unable to evaluate this limit (which is the exact answer to the problem), we can calculate x_n for a large n and take that as an approximate solution. We shall not study iterative procedures of this type. However, we invite you to take a hand calculator, x_0 as 1.5, $f(x) = x - \dfrac{x^3 - 2}{3x^2}$ and calculate the first few iterates. You will find that they are $x_1 = 1.2962963, x_2 = 1.2609322, x_3 = 1.2599219, x_4 = 1.2599210, x_5 = 1.2599210, \ldots$ which converge very rapidly to the cube root of 2. This method of finding cube roots approximately is an instance of a more general method called the **Newton−Raphson method** for solving equations. The essential idea is that given a function $g(x)$ and a real number α we define a new function $f(x)$ by $f(x) = x - \dfrac{g(x) - \alpha}{g'(x)}$. Then if we start with a 'suitable' x_0, the iterates of it under powers of f will generally converge to a solution of the equation $g(x) = \alpha$. But it is beyond our scope to prove this.

is not R-related to b, we write $a \not R b$. For example if S is the set of all positive integers and $R = \{(x, y) \in S \times S : x \text{ divides } y\}$, then $5R10$, $5R5$ but $10 \not R 5$.

This definition of a binary relation may sound artificial, because in real life we use the term 'relation' to mean some sort of a statement which involves two things (often human beings) at a time. Thus familiar examples of relations in real life are 'A and B live together', 'A loves B' or 'this summer is worse than the last one'. In all such cases, we can identify the relationship with the set of all pairs of those things or persons who have the stated relationship between them. For example, we may let S be the set of all human beings and L the subset $\{(x, y) \in S \times S : x \text{ loves } y\}$. Then the statement 'A loves B' is equivalent to saying that the pair (A,B) belongs to this subset L or equivalently, 'ALB'.

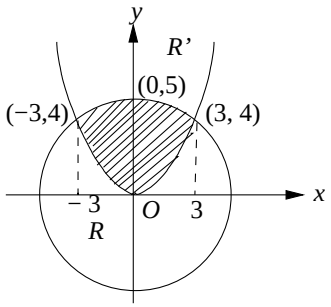
Every function, say f, from a set S to itself[4] defines a binary relation on the set S, viz., the subset $\{(x, y) \in S \times S : y = f(x)\}$. (This subset is often called the graph of the function, because that's what it is when the set S is the real line or some subset of it.) Note, however, that because of the requirement that a function be defined at all points of the domain set and also be single valued, not every binary relation on a set arises out of some function. Still, in analogy with the terminology for a function, given any binary relation R on a set S we define its **domain** and **range** to be the subset $\{x \in S :$ there is at least one $y \in S$ such that $(x, y) \in R\}$ and the subset $\{y \in S :$ there is at least one $x \in S$ such that $(x, y) \in R\}$ respectively. Put differently, the domain of R is precisely the image of the subset R under the first projection function from $S \times S$ to S (i.e., the function which takes a point (x, y) to the point x) and similarly the range of R is the image of R under the second projection.

For example (JEE 1979), let $S = \mathbb{R}$, the set of all real numbers and consider the two relations on $\mathbb{R}$ defined by

$$R = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R},\ x^2 + y^2 \leq 25\} \tag{20}$$

$$\text{and } R' = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R},\ y \geq \frac{4}{9}x^2\} \tag{21}$$

and we are asked to find the domain and the range of $R \cap R'$. The easiest way to do so is to sketch R and R' as subsets of the cartesian plane. Clearly, R is the set of all points in the closed disc of radius 5 centred at the origin while R' is the set of all points that lie on or above the parabola $y = \frac{4}{9}x^2$. Their intersection is the shaded region shown in the figure.



[4]More generally, just as we can consider a function from one set, say X, to another set, say Y, we can also consider a relation between elements of two different sets. In fact we already did so under the name of an incidence relation in Comment No. 10 of Chapter 1. However, unless the two sets are equal, concepts such as symmetry or transitivity do not make sense for such relations.

The domain of $R \cap R'$ is the projection of this region to the x-axis. To find it we first identify the points of intersection of the circle $x^2 + y^2 = 25$ and the parabola $y = \frac{4}{9}x^2$. By solving appropriate quadratics (or even by trial), these points are $(-3, 4)$ and $(3, 4)$. So the domain of $R \cap R'$ is the interval $[-3, 3]$. The range of $R \cap R'$ is the projection of the shaded region on the y-axis. By inspection, it is the interval $[0, 5]$. Note that $R \cap R'$ is not a function for two reasons. First, its domain is not the entire real line $\mathbb{R}$ but only the subset $[-3, 3]$ of it. Even if we ignore this point (as some authors do), note that for every $c \in (-3, 3)$, the vertical line $x = c$ cuts the region $R \cap R'$ in more than two points (in fact, in an interval of length $\sqrt{25 - c^2} - \frac{4}{9}c^2$). So the relation $R \cap R'$ cannot be a function.

In fact, no subset of the cartesian plane that contains an interior point can define a function from a subset of $\mathbb{R}$ to $\mathbb{R}$. However, if R happens to be a set of the form $R = \{(x, y) : \phi(x, y) = 0\}$, where $\phi(x, y)$ is a real-valued function of two real variables x and y, then in general, R represents some plane curve. Even then it need not be a function. (For example, if $\phi(x, y) = x^2 + y^2 - 25$, then R is a circle which cannot be the graph of any function since some vertical lines cut it into more than one point.) However, as mentioned in Comment No. 2 of Chapter 15, under certain rather general conditions on the function ϕ, the relation $\phi(x, y) = 0$ defines y as a function of x locally, i.e., in a neighbourhood of a given point.

Note that in the definition of a binary relation R on a set S, there is no restriction on R other than that it be a subset of $S \times S$. As a result, the number of binary relations on a given set is very large. When the set S is infinite there are infinitely many binary relations on it. Even when S is a finite set with n elements (say), the number of binary relations on S is 2^{n^2} which is a very large number even for relatively small values of n. Of these, we are generally interested only in those relations which satisfy some additional properties. By far the most important ones among such properties are reflexivity, symmetry and transitivity. We do not define them here[5] as they are generally defined and illustrated in standard texts. We remark however, that when the set S is the real line (or some subset of it), the conditions of reflexivity and symmetry can be easily characterised geometrically. Let L be the straight line $y = x$. Then a binary relation R on a subset S of $\mathbb{R}$ is reflexive if and only if it contains all points of the line L that lie in the set $S \times S$, i.e., if and only if $(S \times S) \cap L \subset R$. Similarly, the binary relation R is symmetric if and only if it is symmetric w.r.t. the line L, i.e., it coincides with its own reflection in the line L.

When the set S is finite, a binary relation R on it can also be expressed by a matrix with only 1 and 0 as possible entries. Specifically, list the elements

[5] Just for the sake of record, a binary relation R on a set S is called **reflexive** if for every $x \in S$, we have xRx. Verbally, every element is related to itself under R. This terminology evidently comes from the reflexive pronouns that you study in grammar. A relation R is called **symmetric** if for all $x, y \in S$, 'xRy' implies 'yRx'. The use of this term is obvious. Finally, R is called **transitive** if for all x, y, z in S, whenever we have xRy and yRz we also have xRz. In informal terms, whenever you can go (or 'transit') from x to y and also from y to z under R, you can also transit from x to z directly under R.

of S as $x_1, x_2, \ldots, x_n$ in an arbitrary but fixed manner. Now form an $n \times n$ matrix A whose (i,j)-th entry a_{ij} is 1 if $x_i R x_j$ and 0 if $x_i \not R x_j$. Clearly this matrix determines the relation uniquely. Some of the basic properties of the relation R can be paraphrased in terms of this matrix A. This will be given in Exercise (20.16).

A binary relation R which is reflexive, symmetric and transitive is called an **equivalence relation**. It is a generalisation of the equality relation for it is trivial to check that the relation R defined by 'xRy if and only if $x = y$' is an equivalence relation. Many times we encounter situations where two things are not equal on the nose, but we would nevertheless like to treat them as equal. This condition can be expressed conveniently in terms of a suitable equivalence relation. For example, a shoe designer treats two men as equal if they have the same shoe size. In formal terms we can define a binary relation R on the set, say M, of all men by declaring that for $x, y \in M$, xRy if and only if x and y have the same shoe size. Clearly R is an equivalence relation. Note that, in essence, the shoemaker is classifying all men according to their shoe sizes. Two men are in the same class if and only if they are related to each other under the equivalence relation just defined. Similarly, we can classify human beings according to their heights, religions, economic status and so on. In each case, the classification arises out of a suitably defined equivalence relation. Note that in every case, the classes are mutually disjoint. That is, two classes are either equal (and hence have everything in common) or else they are completely disjoint (and hence have nothing in common). There is no such thing as two classes partially overlapping with each other. Such a classification of a set is also called a **partition** or a **decomposition** of the set.

There is, in fact, a one-to-one correspondence between the various decompositions of a set S and the various equivalence relations on it. Suppose R is an equivalence relation on S. For each $x \in R$, denote by $[x]$ (or, more precisely, by $[x]_R$) the **equivalence class** of S containing the element x. This is defined as the set $\{a \in S : xRa\}$, i.e., as the set of all elements of S that are related to x under R. It is easy to show that for any two $x, y \in S$, the equivalence classes $[x]$ and $[y]$ are either equal or mutually disjoint depending upon whether xRy or $x \not R y$. This way we get a decomposition of S into R-equivalence classes. These equivalence classes are often given different names in different contexts.

As an instructive example, let S be the set of all integers, m a fixed positive integer, and define a binary relation R on S by xRy to mean that m divides $x - y$, i.e., that there exists some integer u such that $mu = x - y$. We check that this is an equivalence relation. First, for every integer x, we have $m \times 0 = x - x$ and hence xRx. Thus the relation R is reflexive. Secondly, if xRy, then $x - y = mu$ for some integer m. But then $-u$ is also an integer and $y - x = m \times (-u)$ which proves that R is symmetric. Finally, for transitivity, if x, y, z, u, v are integers such that $x - y = mu$ and $y - z = mv$, then $x - z = mw$ where $w = u + v$. But since u, v are integers, so is w. This shows that whenever xRy and yRz hold, so does xRz. So R is indeed an equivalence relation. We already encountered it in Comment No. 16 of Chapter 4. What we called as

residue classes modulo m there are nothing but the equivalence classes of the relation R just defined.

As another example of an equivalence relation (JEE 1980), let L be the set of of all straight lines in the euclidean plane. Then parallelism is an equivalence relation on L. Here we regard two identical lines as parallel to each other. (In case this sounds awkward, define two lines to be parallel to each other if and only if they make equal angles with a given (directed) line in the plane.) The equivalence classes of this relation correspond to various directions, or equivalently, to various slopes. (In the case of a direction, we usually want to distinguish between two opposite directions. In that case instead of lines, we consider half-lines or rays.)

The concept of an equivalence relation and of an equivalence class is very convenient for clarification of ideas. For example, in elementary texts, a vector is often defined as a directed line segment. At the same time, two directed line segments which have the same direction and equal lengths are considered to represent the same vector. A beginner often finds it difficult to reconcile the two. This difficulty can be overcome by first considering the set of all directed line segments and then defining an equivalence relation on this set by declaring that two line segments are related to each other if and only if they have the same direction and equal lengths. It is easily verified that this is an equivalence relation. A vector can now be defined as an equivalence class under this relation. Mathematically this is not a great achievement. But it clears the conceptual difficulty.

As another application, we mentioned in Comment No. 8 of Chapter 1 that certain counting problems can be paraphrased as counting the numbers of equivalence classes w.r.t. suitably defined equivalence relations. Again this does not necessarily simplify the problem. But it tells us exactly what it is that we are counting. Otherwise there is a good chance that we may err by counting the same thing more than once.

The exercises will give more examples of equivalence relations and their associated decompositions. We conclude the discussion of binary relations on a set by a brief mention of another important class of relations. First we need to define a property called **anti-symmetry**. This is not merely the absence of symmetry, but the very opposite of symmetry. Symmetry means that whenever xRy holds, yRx also holds. Lack of symmetry simply means that *sometimes* we may have xRy but not yRx. Antisymmetry says that the two possibilities can never coexist, except in a trivial case. Formally, we say that a binary relation R on a set S is antisymmetric if for all $x, y \in S$, whenever xRy and yRx both hold, we must have $x = y$. As standard examples, suppose $S = \mathbb{R}$, the set of all real numbers. If we define xRy to mean that $x \leq y$, then R is an antisymmetric relation, because $x \leq y$ and $y \leq x$ together imply that $x = y$. As another example, suppose S is the power set of some set X, i.e., the set of all subsets of X including the empty set and the entire set X. Let $A, B \in S$. Then A, B are subsets of X and we define R by letting 'ARB' mean that $A \subset B$, i.e., A is contained in B. Then it is easily seen that R is antisymmetric.

Antisymmetry is the very essence of what is called an order relation on a set S. An equivalence relation on a set tries to equate two elements which are unequal but which we would like to treat as equal for a particular purpose. An order relation, on the other hand, strives to compare elements of S two at a time and to declare that one of them is smaller or prior (in some sense) to the other. Formally, a binary relation R on a set S is called a **partial order** if it is reflexive, transitive and anti-symmetric. The relations defined by '$\leq$' and '$\subset$' in the two examples just given are easily seen to be partial orders. The word 'partial' stresses that the relation acts only partially. That is, it may happen that there are two elements x and y which are not comparable under R. That is to say, it may happen that neither xRy nor yRx holds true. If at least one of the two possibilities must hold, we say that R satisfies the **law of dichotomy**. In the examples given above the usual order relation on $\mathbb{R}$ satisfies the law of dichotomy. But the relation '$\subset$' on the power set of a set does not, in general. For, given two subsets, say A and B of a set X, it may happen that neither is contained in the other. A partial order in which the law of dichotomy holds is called a **total order**, because every two elements are comparable under it.

EXERCISES

20.1 Show that if the functional equation (1) holds and $f(x) \neq 0$ for at least one value of x, then $f(x) > 0$ for all $x \in \mathbb{R}$. As a result, $\ln f(x)$ is defined. Using this and the solution of (5), give an alternate (although essentially similar) solution of (1) and hence of the Main Problem.

20.2 In Comment No. 4, we proved that if $f(x)$ satisfies (4) and is given to be continuous at 0, then it is continuous everywhere. Prove that, more generally, if we are given that $f(x)$ is continuous at *any one* point $x_0 \in \mathbb{R}$, then it is continuous everywhere.

20.3 Given a function f satisfying the following conditions for every x and y

(i) $f(x+y) = f(x)f(y)$ and (ii) $f(x) = 1 + xg(x)$, where $\lim_{x\to 0} g(x) = 1$,

find $f'(x)$ if it exists. (JAT 1979)

20.4 Find all continuous functions defined from the set of positive real numbers to itself which satisfy the functional equation $f(xy) = f(x)f(y)$ for all positive x and y.

20.5 Find all continuous real-valued functions defined on the set of positive real numbers which satisfy the condition that $f(xy) = f(x) + f(y)$ for all positive x, y.

20.6 Let $f(\frac{x+y}{2}) = \frac{f(x)+f(y)}{2}$ for all real x and y.

(a) Prove that if $f(x)$ is continuous at 0, then it is continuous everywhere.

(b) If $f'(0)$ exists and equals -1 and $f(0) = 1$, find $f(2)$. (1995)

20.7 Which, if any, of the following functions is periodic? (1983)

(A) $f(x) = x - [x]$ (B) $f(x) = \sin\dfrac{1}{x}$ (C) $x\cos x$

20.8 (a) Let $f(x)$ be a periodic function. Prove that either $f(x)$ has a smallest positive period or else it has arbitrarily small positive periods (that is, given any $\epsilon > 0$, there is some period T such that $0 < T < \epsilon$). [*Hint:* If there is no smallest positive period, get a strictly decreasing sequence of positive periods. Apply completeness and then use Exercise (18.7)(a).]

(b) Suppose $f(x)$ is continuous everywhere and periodic. Prove that unless $f(x)$ is constant, it has a smallest positive period. [*Hint:* Otherwise, use (a) above to show that for every $c \in \mathbb{R}$ and for every positive integer n, there exists some $x_n \in (c, c + \frac{1}{n})$ such that $f(x_n) = f(0)$. Now apply Theorem 1 of Chapter 16 to show that $f(c) = f(0)$.]

(c) Suppose $f(x)$ and $g(x)$ are periodic functions with periods T_1 and T_2 respectively. Show by an example that $f(x) + g(x)$ need not be periodic. Prove, however, that if the ratio T_1/T_2 is rational, then the function $f(x) + g(x)$ is periodic.

20.9 Draw the graph of $y = \frac{1}{\sqrt{2}}(\sin x + \cos x)$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$. (1979)

20.10 Given functions f and h, prove that there exists a function g such that $g \circ f = h$ if and only if $h(x_1) = h(x_2)$ whenever $f(x_1) = f(x_2)$. (This gives a necessary and sufficient condition for (19) to have a solution for g when f and h are given.)

20.11 If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin\sqrt{x})^2$, determine which of the following statements is/are true. (1998)

(A) $f(x) = \sin^2 x,\ g(x) = \sqrt{x}$ (B) $f(x) = \sin x,\ g(x) = |x|$

(C) $f(x) = x^2,\ g(x) = \sin\sqrt{x}$ (D) f and g cannot be determined.

20.12 Determine which, if any, of the following statements is correct for the function $f(x) = |x - 1|$. (1983)

(A) $f(x^2) = (f(x))^2$ (B) $f(x+y) = f(x) + f(y)$ (C) $f(|x|) = |f(x)|$.

20.13 If $y = f(x) = \frac{x+2}{x-1}$, determine which of the following statements is/are correct. (1984)

(A) $x = f(y)$ (C) y increases with x for $x < 1$

(B) $f(1) = 3$ (D) f is a rational function of x.

20.14 Instead of a single functional equation we can also have a system of functional equations involving more than one function. For example, suppose $f(x)$ and $g(x)$ are two functions which simultaneously satisfy the following three conditions, which interlink $f(x)$ and $g(x)$:

(i) $f(x+y) = f(x)g(y) + g(x)f(y)$ for all x, y

(ii) $g(x+y) = g(x)g(y) - f(x)f(y)$ for all x, y

(iii) $f'(0) = 1$, $g'(0) = 0$

Show that $f(x) = \sin x$ and $g(x) = \cos x$ for all x.

20.15 Suppose R_1 and R_2 are two binary relations on a set S. Is it true that the domain of $R_1 \cap R_2$ is the intersection of the domains of R_1 and R_2? What about the range of $R_1 \cap R_2$?

20.16 Let $S = \{x_1, x_2, \ldots, x_n\}$ be a finite set with n elements and R a binary relation on S. Let $A = (a_{ij})$ be the $n \times n$ matrix in which $a_{ij} = 1$ or 0 according as $x_i R x_j$ or $x_i \not R x_j$. Prove that :

(a) R is reflexive if and only if all the diagonal entries a_{ii} are 1 each

(b) R is symmetric if and only if the matrix A is symmetric

(c) for any $i, j \in \{1, 2, \ldots, n\}$, the (i, j)-th entry of the matrix A^2 is the number of k's such that $x_i R x_k$ and $x_k R x_j$ (intuitively, every such x_k is like an intermediary between x_i and x_j under the relation R)

(d) R is transitive if and only if whenever an entry of A^2 is non-zero, the corresponding entry of A is also non-zero.

20.17 Using parts (a) and (b) of the last exercise, find how many binary relations on a set with n elements are (i) symmetric, (ii) reflexive, (iii) both reflexive and symmetric and (iv) neither reflexive nor symmetric. (There is no easy way to count the number of transitive relations on a finite set.)

20.18 Let $f : S \longrightarrow T$ be any function. For $x, y \in S$ define xRy to mean that $f(x) = f(y)$. Prove that R is an equivalence relation on S. Identify the equivalence classes. Use this result to give an alternate proof that parallelism is an equivalence relation on the set of all straight lines in the euclidean plane. (Here every line is to be considered parallel to itself.)

20.19 Define a binary relation R on the set of complex numbers by $z_1 R z_2$ if and only if there exists some non-zero real number α such that $z_1 - \alpha z_2 = 0$.

(a) Prove that R is an equivalence relation. (1982)

(b) Identify the equivalence classes.

(c) What would be the answer if we further require $\alpha > 0$ in the definition?

20.20 Repeat the last exercise for the relation R defined by $z_1 R z_2$ if and only if $z_1 - z_2$ is purely imaginary.

20.21 Let S be the set of all ordered pairs of the form (a, b) where a, b are positive integers. For $(a, b), (c, d) \in S$, define $(a, b)R(c, d)$ to mean that $a+d = b+c$. Prove that R is an equivalence relation. What do the equivalence classes correspond to? Do a similar problem if S is the set of all ordered pairs (a, b) of integers with $b \neq 0$ and R is defined by $(a, b)R(c, d)$ if $ad = bc$.

20.22 For two integers x and y write $x|y$ to mean that x divides y. Prove that on the set of positive integers this defines a partial order but that on the set of all integers, antisymmetry fails. Does the law of dichotomy hold for this relation?

Chapter 21

VECTORS

Vectors are studied both in physics and in mathematics and the difference in the two approaches can be confusing at the beginning. In physics, a vector is a quantity which has both magnitude and direction. Thus, velocity, acceleration and momentum are vectors while speed, mass and energy are not. (Quantities of the latter type are often called 'scalars' to distinguish them from vectors.) In mathematics, on the other hand, a vector is a directed line segment and a scalar is simply a real number. It would be better if the physicists say that a vector is a physical quantity which can be *represented* by a directed line segment. As if to add to the confusion, even in mathematics, there is yet another approach to vectors in which a vector is defined as an ordered pair (or an ordered triple) of real numbers. The first approach is geometric and the second algebraic. The interplay between the two is brought out fully in Comment No.s 4 and 5.

The basic operations with vectors are their linear combinations, the dot product and the cross product (which is a peculiarity of three-dimensional vectors). In addition, there is the scalar triple product and the vector triple product. We assume familiarity with their definitions and elementary properties. The concepts of linear dependence, basis, an orthogonal basis and an orthonormal basis are defined in Comment No. 13. Applications of vectors to geometry, which were begun in Chapter 8, are continued in Comment No. 16 and 17. In Comment No. 18, we introduce you to an important technique in mathematics, called the perturbation technique. Exercise (21.25) gives an introduction to quaternions. In essence, they are mixtures of scalars and vectors just as complex numbers are mixtures of real numbers and purely imaginary numbers.

Short, numerical questions on vectors are asked every year in the JEE. The long questions include proving some identity or some geometric result using vector methods.

Main Problem : Suppose **a, b, c** are three vectors of unit lengths, every two of which are inclined to each other at the angle $\cos^{-1}\frac{3}{5}$. If $\mathbf{a}\times\mathbf{b} = p\mathbf{a}+q\mathbf{b}+r\mathbf{c}$, where p, q, r are scalars, then q^2, expressed as a rational number in the reduced form, equals ..... . (JEE 1997*, modified)

First Hint: Let $\cos\theta = \frac{3}{5}$. Without loss of generality assume $\mathbf{a} = \mathbf{i}, \mathbf{b} = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}$ and $\mathbf{c} = \cos\theta\mathbf{i} + \lambda\mathbf{j} + \mu\mathbf{k}$, where $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ is a right-handed system of mutually perpendicular unit vectors and λ, μ are some scalars.

Second Hint: λ and μ can be taken respectively to be $\sin\theta\cos\phi$ and $\sin\theta\sin\phi$, where $\cos\phi = \dfrac{\cos\theta}{1+\cos\theta} = \dfrac{3}{8}$.

Third Hint: Write a system of three equations in the three unknowns p, q, r and solve.

Solution: To justify the first hint, take $\mathbf{i} = \mathbf{a}$, $\mathbf{j}$ to be a unit vector perpendicular to $\mathbf{i}$ in the plane spanned by the vectors $\mathbf{a}$ and $\mathbf{b}$ and $\mathbf{k} = \mathbf{i}\times\mathbf{j}$. Since $\mathbf{c}$ is a unit vector and $\mathbf{c}\cdot\mathbf{a} = \cos\theta$, $\mathbf{c}$ must be of the form $\cos\theta\mathbf{i} + \lambda\mathbf{j} + \mu\mathbf{k}$ with $\cos^2\theta + \lambda^2 + \mu^2 = 1$. Since $\lambda^2 + \mu^2 = \sin^2\theta$, we may assume λ, μ are of the form $\sin\theta\cos\phi$ and $\sin\theta\sin\phi$ for some angle ϕ. $\cos\theta = \mathbf{b}\cdot\mathbf{c} = \cos^2\theta + \sin^2\theta\cos\phi$ gives the value of $\cos\phi$ as $\frac{3}{8}$ and hence also of $\sin\phi$ as $\pm\dfrac{\sqrt{55}}{8}$. Finally, $\mathbf{a}\times\mathbf{b}$ equals $\sin\theta\mathbf{k}$ while $p\mathbf{a}+q\mathbf{b}+r\mathbf{c} = (p+q\cos\theta+r\cos\theta)\mathbf{i}+(q\sin\theta+r\sin\theta\cos\phi)\mathbf{j}+r\sin\theta\sin\phi\mathbf{k}$. Matching the corresponding components of these two vectors, we get:

$$\begin{aligned} r\sin\theta\sin\phi &= \sin\theta \\ q\sin\theta + r\sin\theta\cos\phi &= 0 \\ p + q\cos\theta + r\cos\theta &= 0. \end{aligned}$$

The first equation gives $r = \operatorname{cosec}\phi$. Putting this in the second equation, we get $q = -\cot\phi$. So $q^2 = \cot^2\phi = \frac{9}{55}$.

Comment No. 1:

Instead of obtaining a system of three equations by equating the coefficients of **i, j** and **k** as was done above, we can also get such a system from the equation $\mathbf{a}\times\mathbf{b} = p\mathbf{a} + q\mathbf{b} + r\mathbf{c}$ by taking the dot products of both the sides with the vectors **a**, **b** and **c**. Here $\mathbf{a}\cdot(\mathbf{a}\times\mathbf{b})$ and $\mathbf{b}\cdot(\mathbf{a}\times\mathbf{b})$ both vanish. The third scalar triple product (or the 'box product'), viz., $\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})$ equals $\sin^2\theta\sin\phi$. So we get

$$\begin{aligned} p + (q+r)\cos\theta &= 0 \\ (p+r)\cos\theta + q &= 0 \\ \text{and}\quad (p+q)\cos\theta + r &= \sin^2\theta\sin\phi\ . \end{aligned}$$

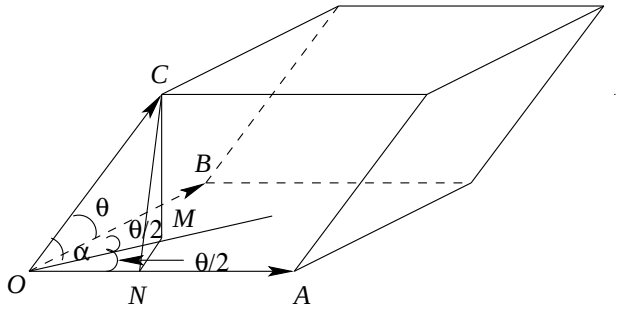
From the first two equations, $p = q$ (which is also obvious from the data which is symmetric about **a** and **b**). Replacing p by q in the second and the third equation and solving we get $r = \operatorname{cosec}\phi$ and $q = -\cot\phi$. But the calculations are a little more complicated than those for the earlier system of equations. However, if we

convert this system to a numerical one using the values of $\cos\theta, \sin\theta$ and $\sin\phi$ (and the fact $p = q$) which we already know, then it reduces to a system of two equations, viz. $8q + 3r = 0$ and $6q + 5r = \pm\frac{2\sqrt{55}}{5}$ which can be solved quite easily for q to get $q = \pm\frac{3}{\sqrt{55}}$ as before.

Comment No. 2:

There is also a purely geometric way of getting the box product $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$. Represent the vectors **a**, **b** and **c** by three directed line segments of unit lengths $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively. Then $\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})$ is the volume of the parallelopiped with sides OA, OB, OC. To find it, let M be the foot of the perpendicular drawn from C to the plane spanned by the lines OA and OB. Then $CM = \sin\alpha$ where $\alpha = \angle COM$ and the volume of the parallelopiped is $\sin\theta\sin\alpha$.

To find α, note first that since OC is equally inclined to OA and to OB, its projection OM bisects $\angle BOA$ which equals θ. Let N be the foot of the perpendicular from M to OA. Then $ON = OM\cos(\frac{\theta}{2}) = OC\cos\alpha\cos(\frac{\theta}{2}) = \cos\alpha\cos(\frac{\theta}{2})$.



But CN is also perpendicular to OA and so $ON = OC\cos\theta = \cos\theta$. This gives $\cos\alpha = \dfrac{\cos\theta}{\cos(\theta/2)}$. We are given that $\cos\theta = \frac{3}{5}$. From this we easily get $\cos(\dfrac{\theta}{2}) = \sqrt{\dfrac{1+\cos\theta}{2}} = \dfrac{2}{\sqrt{5}}$. This gives $\cos\alpha = \dfrac{3/5}{2/\sqrt{5}} = \dfrac{3}{2\sqrt{5}}$ and hence $\sin\alpha = \dfrac{\sqrt{11}}{2\sqrt{5}}$. So ultimately, $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \sin\alpha\sin\theta = \dfrac{2\sqrt{55}}{25}$, which is the same value as before. The rest of the work is as in the last comment. In the general case, $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$ comes out to be $(1 - \cos\theta)\sqrt{1 + 2\cos\theta}$. But once again, the trigonometrical work is rather substantial. So in this problem it is much easier to do a particular numerical case than the general one.

Comment No. 3:

Here is an analogous problem.

Suppose four distinct vectors of equal lengths are equally inclined to each other. If θ is the common angle between every two of them, find $\cos\theta$

Denote the vectors by **a, b, c, d** and, without loss of generality assume that they are all of length 1 each. By symmetry, **d** must be of the form $p(\mathbf{a} + \mathbf{b} + \mathbf{c})$ for some scalar p. Since $\mathbf{a} \cdot \mathbf{d} = \cos\theta$, we get,

$$\cos\theta = p(1 + 2\cos\theta)$$

But $\mathbf{d}$ being a unit vector, we also have $\mathbf{d} \cdot \mathbf{d} = 1$, which yields

$$p^2(3 + 6\cos\theta) = 1$$

Eliminating p we get $3\cos^2\theta - 2\cos\theta - 1 = 0$. Solving, $\cos\theta$ equals either 1 or $\frac{-1}{3}$. The first possibility implies that all the four vectors are equal, which is not the case. So $\cos\theta = \frac{-1}{3}$.

An alternate solution is possible using the work in the solution to the Main Problem. Take $\mathbf{a} = \mathbf{i}$ and $\mathbf{b} = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}$. Then $\mathbf{c}$ and $\mathbf{d}$ are of the form

$$\begin{aligned} \mathbf{c} &= \cos\theta\mathbf{i} + \sin\theta\cos\alpha\mathbf{j} + \sin\theta\sin\alpha\mathbf{k} \quad \text{and} \\ \mathbf{d} &= \cos\theta\mathbf{i} + \sin\theta\cos\beta\mathbf{j} + \sin\theta\sin\beta\mathbf{k} \end{aligned}$$

for some angles α and β. Here $\cos\alpha$ and $\cos\beta$ both equal $\frac{\cos\theta}{1+\cos\theta}$. So $\alpha = \pm\beta$. If we take $\alpha = \beta$, then $\mathbf{c}$ and $\mathbf{d}$ would be the same vectors, which is not the case. So we must have $\beta = -\alpha$. Then $\sin\beta = -\sin\alpha$. With this substitution, $\mathbf{c} \cdot \mathbf{d} = \cos\theta$ gives

$$\begin{aligned} \cos^2\theta + \sin^2\theta\cos^2\alpha - \sin^2\theta\sin^2\alpha &= \cos\theta \\ \text{or} \quad \cos^2\theta + (1 - \cos^2\theta)(2\cos^2\alpha - 1) &= \cos\theta \end{aligned}$$

Substituting $\cos\alpha = \frac{\cos\theta}{1+\cos\theta}$ and writing x for $\cos\theta$, this gives an equation for x which simplifies to $3x^2 - 2x - 1 = 0$. Of the two solutions, viz., $x = 1$ and $x = \frac{-1}{3}$, the first is rejected as before.

Obviously the first solution is better. Normally it is an asset to be able to recognise the similarity of a given problem to some other problem whose solution is already known. Such a recognition makes you more comfortable and sometimes also saves duplication of work. But one should be open to the possibility that some entirely different approach may work better for the given problem. The present problem is an instance of this. A lot of work in the solution to the Main Problem was spent in finding the components of the vectors in terms of a suitably chosen right-handed system of basis vectors. The present problem, on the other hand, merely asks for the common angle between the vectors. So it is not surprising that a short cut is possible.

This problem has an interesting geometric application. If A, B, C, D are the vertices of a regular tetrahedron with centre O, then the angle θ above is the angle between any two of the lines joining the centre to the vertices. Its approximate value is $109°28'$. This has an application in chemistry for the tetrahedral model of a methane molecule, where the carbon atom is at the centre and the hydrogen atoms at the four vertices. While this knowledge is not of much help in doing the problem mathematically, it can sometimes alert you if you have made a numerical mistake and are way off the correct answer (for example if your answer is positive). And, in case a student knows the exact answer from chemistry, he can simply fill the blank with it. In a 'fill-in-the-blank' question, nobody questions how you arrived at the answer.

Comment No. 4:

The solutions given to the Main Problem represent the two standard methods to handle vectors. The first method is purely algebraic. In this method, everything is resolved along a fixed right-handed[1] system $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ of unit vectors. (As in the case of the solution to the Main Problem, this system can often be chosen as per our convenience.) Once this system is fixed, everything is done in terms of the components of the vectors along **i, j** and **k**. Suppose, for example, that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$ are three vectors and λ is a real number (often called a scalar so as to distinguish it from a vector). Then the following formulas express the basic concepts involving them.

$$||\mathbf{a}|| = \sqrt{a_1^2 + a_2^2 + a_3^2} \text{ (length of a vector)} \tag{1}$$

$$\mathbf{a} + \mathbf{b} = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j} + (a_3 + b_3)\mathbf{k} \text{ (sum of vectors)} \tag{2}$$

$$\lambda\mathbf{a} = (\lambda a_1)\mathbf{i} + (\lambda a_2)\mathbf{j} + (\lambda a_3)\mathbf{k} \text{ (scalar multiplication)} \tag{3}$$

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \text{ (the dot product)} \tag{4}$$

$$\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \text{ (cross product)} \tag{5}$$

$$(\mathbf{a}\ \mathbf{b}\ \mathbf{c}) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \text{ (scalar triple product)} \tag{6}$$

The length of a vector is also called its **magnitude** or **absolute value** (or, sometimes, **norm**) and is sometimes denoted by single rather than double bars. The sum of two vectors is also known as their **resultant**, especially when they represent forces acting concurrently. The dot product is also called the **scalar product** (and, sometimes, the **inner product**) while the cross product is also called the **vector product**. The scalar triple product is also often denoted by $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$ and called the **box product** (because of its geometric meaning given below). There is an alternate formulation of the cross product $\mathbf{a} \times \mathbf{b}$ as a determinant, viz.,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \tag{7}$$

which makes it easier to remember it. It should be noted that the first row of this determinant consists of vectors while the other two rows have ordinary real numbers as entries. Such a 'hybrid' determinant is only symbolic and must not be indiscriminately subjected to the laws that hold for ordinary determinants. For example, if we interchange the second and the third rows, then the sign of the determinant changes and this corresponds to saying that $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$ (a property often called the **anti-commutativity** of the cross product). But this cannot be taken as a *proof* until we have shown that the rule of interchange of

[1] See Comment No. 13 below for an explanation of this peculiar term.

rows is valid for such mixed determinants also. Till then, the only acceptable proof is using (5) (which is the 'official' version of (7)). The determinant that appears in (6) is, however, a genuine determinant and we are free to subject it to any of the laws for determinants. For example, we see at once that $(\mathbf{a}\ \mathbf{b}\ \mathbf{c}) = -(\mathbf{a}\ \mathbf{c}\ \mathbf{b}) = (\mathbf{b}\ \mathbf{c}\ \mathbf{a})$ etc.

In effect, in the algebraic approach, we are identifying a vector $\mathbf{a}$ with an ordered triple of the form (a_1, a_2, a_3). Formulas (1) to (6) above can then be taken as the *definitions* of the corresponding concepts. One of the advantages of this approach is that it generalises easily to n-dimensional vectors for any n (except for the cross product which is a peculiarity of three-dimensional space).

At the other extreme, there is the purely geometric approach. In this approach a vector is identified with a directed line segment. The basic concepts of length, direction and the angle between two directions are taken as intuitively clear and are not defined. The other concepts are then defined in terms of these. For example, the addition of two vectors is defined by the well-known parallelogram law (see Chapter 2, Comment No. 16). The dot product is defined in terms of the angle between the vectors. Specifically, suppose $\mathbf{a}$ and $\mathbf{b}$ are two vectors. If at least one of them is the zero vector, we define $\mathbf{a} \cdot \mathbf{b}$ to be 0. Otherwise, let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. Then we define $\mathbf{a} \cdot \mathbf{b}$ by

$$\mathbf{a} \cdot \mathbf{b} = ||\mathbf{a}||\ ||\mathbf{b}|| \cos\theta \tag{8}$$

which can be considered to apply even in the case, where one of the vectors is $\mathbf{0}$, because in such a case even though θ is undefined, one of the other two factors of the R.H.S. is 0 and hence we can take the R.H.S. to vanish. Similarly, the cross product $\mathbf{a} \times \mathbf{b}$ is defined as the vector with length $||\mathbf{a}||||\mathbf{b}|| \sin\theta$ having direction perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ so that $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b}$ forms a right handed system. Note that $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$. It is immediate that for any two vectors $||\mathbf{a}||^2||\mathbf{b}||^2 = (\mathbf{a} \cdot \mathbf{b})^2 + ||\mathbf{a} \times \mathbf{b}||^2$.

Since $|\cos\theta| \leq 1$ for any θ, an immediate consequence of (9) is that for any two vectors $\mathbf{a}$ and $\mathbf{b}$,

$$|\mathbf{a} \cdot \mathbf{b}| \leq ||\mathbf{a}||\ ||\mathbf{b}|| \tag{9}$$

with equality holding if and only if one of the two vectors is a scalar multiple of the other.

It is interesting to note that in the algebraic approach things are turned around. That is, in the algebraic approach, (8) is taken as the *definition* of the angle between two (non-zero) vectors $\mathbf{a}$ and $\mathbf{b}$. In other words, the angle between $\mathbf{a}$ and $\mathbf{b}$ is *defined* as $\cos^{-1} \frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}||\ ||\mathbf{b}||}$. Of course, for this definition to make sense it must first be shown that the expression $\frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}||\ ||\mathbf{b}||}$ does not exceed 1 in absolute value (which is what (9) says). In other words, we must first establish (9) using the algebraic definitions of length and the dot product given in (1) and (4) above respectively. This amounts to showing that

$$a_1b_1 + a_2b_2 + a_3b_3 \leq \sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}\ .$$

But this is exactly the Cauchy−Schwarz inequality (for the case $n = 3$) given in Chapter 6. Summing up, in the algebraic approach, we first define the length and the dot product by (1) and (4), then prove (9) using the Cauchy−Schwarz inequality and then finally define the angle between two vectors as given by (8). In the geometric approach, we take the length and the angle as intuitively clear terms, then define the dot product using (8) and finally derive (9) from (8).

The algebraic approach is not substantially different from three-dimensional coordinate geometry. Its major advantage is that the powerful machinery of algebra becomes available to prove results about vectors. Take, for example, what is known as the **bilinearity**[2] of the dot product. Simply stated this means that for any three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ and for any two scalars λ and μ, the following relations hold:

$$(\lambda\mathbf{a} + \mu\mathbf{b}) \cdot \mathbf{c} = \lambda(\mathbf{a} \cdot \mathbf{c}) + \mu(\mathbf{b} \cdot \mathbf{c}) \tag{10}$$

$$\text{and} \quad \mathbf{a} \cdot (\lambda\mathbf{b} + \mu\mathbf{c}) = \lambda(\mathbf{a} \cdot \mathbf{b}) + \mu(\mathbf{a} \cdot \mathbf{c}) \tag{11}$$

These follow very easily from (2), (3) and (4) above. But if we want to establish (10) geometrically, then we shall first have to draw a parallelogram with sides represented by the vectors $\lambda\mathbf{a}$ and $\mu\mathbf{b}$ (which itself runs into several cases depending upon the signs of λ and μ) and then consider the angles made by the direction of $\mathbf{c}$ with those of the sides and the diagonal of this parallelogram. This becomes a messy task. A similar comment holds for (11). Similarly, the algebraic proof of the cyclicity of the scalar triple product, i.e., the relation

$$(\mathbf{a}\ \mathbf{b}\ \mathbf{c}) = (\mathbf{b}\ \mathbf{c}\ \mathbf{a}) = (\mathbf{c}\ \mathbf{a}\ \mathbf{b}) \tag{12}$$

is an easy consequence of the properties of determinants as noted above. But a geometric proof is far from easy. In fact the result is not obvious at all. It is only when we interpret the triple product $(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$ as the volume of a box (formally called a parallelopiped[3]) with sides represented by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, that we get the desired equality. Similarly, proving certain results about the cross product requires its interpretation as the area (or more precisely the oriented area) of a parallelogram having the two vectors as adjacent sides.

Analogous to the scalar triple product, there is the **vector triple product**, say $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ of three vectors. It is possible (although not so trivial) to show (cf. Exercise (21.7 (b))) that

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \tag{13}$$

a relation which is needed frequently in simplifying expressions. Again the algebraic proof is straightforward (albeit a little tedious) while the geometric

[2]The prefix 'bi' means 'two'. The dot product is a function of two variables or 'arguments' as they are sometimes called. Equation (10) says that if the second variable is held fixed then the dot product is linear as a function of the first variable. Similarly, (11) says that the dot product is linear in the second argument.

[3]The correct spelling is actually, 'parallelepiped'. But nowadays, 'parallelopiped' is more common.

proof is not so easy because the vector triple product has no easy geometric significance as the scalar triple product. Note also that in general $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq \mathbf{b} \times (\mathbf{c} \times \mathbf{a})$. In fact the two sides have hardly anything in common. However, we have the following well-known identity which is proved by applying (13) to each term (and using commutativity of the dot product, i.e., $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ etc.).

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0} \tag{14}$$

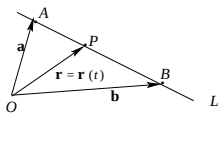
Note that we cannot take either the dot or the cross product of a scalar and a vector. We can only take a scalar multiple of a vector. Thus, for example, if $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are three vectors then the expressions $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ and $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ are meaningful (representing, respectively, the scalar triple product $(\mathbf{u}\ \mathbf{v}\ \mathbf{w})$ and the multiple of the vector $\mathbf{w}$ by the scalar $\mathbf{u} \cdot \mathbf{v}$), but neither $(\mathbf{u} \cdot \mathbf{v}) \cdot \mathbf{w}$ nor $\mathbf{u} \times (\mathbf{v} \cdot \mathbf{w})$ is a meaningful expression. (JEE 1998) Care has to be exercised to avoid writing such meaningless expressions unwittingly[4]. Note also that the scalar 0 is quite different from the vector $\mathbf{0}$ (called the zero vector or sometimes, the **null vector**). It is a standard practice to denote vectors by writing small arrows on top (e.g., $\vec{a}, \vec{A}$ etc.). If a vector happens to be a unit vector, i.e., a vector of length 1, then it is customary to stress this fact by putting a 'hat' rather that an arrow on the top (e.g., $\hat{u}, \hat{i}, \hat{j}$ etc.) In printing, vectors which are denoted by single symbols (such as **a, b, c, u, v, w, 0**) are often shown in bold face. But when a vector is specified as the directed segment from a point, say A to another point, say B, it is more customary to denote it by $\overrightarrow{AB}$.

Comment No. 5:

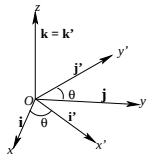
A link between the algebraic and the geometric approach is established through the concept of what is called the **position vector** of a point, say P, in space w.r.t. a given fixed point, say O in space. By definition, this is the vector $\overrightarrow{OP}$. If Q is any other point then the vector $\overrightarrow{PQ}$ is simply the vector $\overrightarrow{OQ} - \overrightarrow{OP}$. The choice of the initial point O is at our disposal and in applications, a particular choice may simplify the calculations considerably. If O is taken as the origin w.r.t. a fixed (right-handed) system of rectangular cartesian coordinates in space, and in this system the coordinates of P are x, y, z (say) then the position vector $\overrightarrow{OP}$ of the point P is simply the vector $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, where $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are unit vectors along the coordinate axes. (See Figure (a) below.) It is customary to denote this vector by $\mathbf{r}$ especially when the point P is moving in space and we want to find its locus or study various aspects of its motion such as its velocity and acceleration. The coordinates x, y, z of P are precisely the components of the vector $\mathbf{r}$ along the three unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$. (More precisely, the components of $\mathbf{r}$ are the vectors $x\mathbf{i}, y\mathbf{j}$ and $z\mathbf{k}$ rather than the

[4]On the other hand, the meaninglessness of certain expressions can also be an advantage as it sometimes obviates the need to put parentheses. For example, the expression $\mathbf{u} \cdot \mathbf{v} \times \mathbf{w}$ can only mean $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ as the other possible insertion of parentheses gives $(\mathbf{u} \cdot \mathbf{v}) \times \mathbf{w}$ which is meaningless. Similarly, $\mathbf{u} \cdot \mathbf{v}\ \mathbf{w}$ can only mean $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$. Note, however, that in the case of the vector triple product, the parentheses are mandatory since both the expressions $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ and $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ are meaningful but unequal.

scalars x, y, z, which are called projections of $\mathbf{r}$.) The motion of P is specified completely by expressing x, y, z as functions of some parameter t (usually, but not always, the time t). Instead, we can express $\mathbf{r}$ as a single (vector valued) function of the parameter t. We then get what is called a **vector equation** of the path of P.



(a) vector equation of a line



(b) rotation of coordinate axes

Suppose, for example, that P is moving on a straight line, say L. Fix any two (distinct) points, say A and B on L (see Figure (a) again). Denote the position vectors of A and B by $\mathbf{a}$ and $\mathbf{b}$ respectively. Then a vector equation of the line L is given by

$$\mathbf{r} = \mathbf{r}(t) = (1-t)\mathbf{a} + t\mathbf{b} \tag{15}$$

Note that $\mathbf{r}(0) = \mathbf{a}$, which simply means that P is at A when $t = 0$. Similarly, P is at B when $t = 1$. Values of the parameter t between 0 and 1 give points on the segment $\overline{AB}$. More specifically, if $0 < t < 1$, then the point $\mathbf{r}(t)$ divides the segment AB in the ratio $t : (1-t)$. This follows by calculating $||\mathbf{r}(t) - \mathbf{a}||$ and $||\mathbf{b} - \mathbf{r}(t)||$ which come out to be $t||\mathbf{b} - \mathbf{a}||$ and $(1-t)||\mathbf{b} - \mathbf{a}||$ respectively. With a slight change of notation, we get that the point which divides the segment AB in the ratio $\lambda : \mu$, where λ, μ are positive real numbers, has position vector $\mu\mathbf{a} + \lambda\mathbf{b}$. This result, often called the **section formula** will be needed later. In particular the mid-point of AB has position vector $\frac{1}{2}(\mathbf{a} + \mathbf{b})$.

Equation (15) gives a necessary and sufficient condition for collinearity of three points in space in terms of their position vectors. Specifically, it says that points with position vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are collinear if and only if there exists some t such that $\mathbf{c} = (1-t)\mathbf{a} + t\mathbf{b}$. In particular we see that for any two vectors $\mathbf{a}$ and $\mathbf{b}$ and any real number k, the points with position vectors $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$ and $\mathbf{a} + k\mathbf{b}$ are collinear because we can write $\mathbf{a} + k\mathbf{b}$ as $(1-t)(\mathbf{a} + \mathbf{b}) + t(\mathbf{a} - \mathbf{b})$, where $t = \frac{1-k}{2}$. (JEE 1984)

Vector equations can also be written easily for curves like circles and ellipses in space. For an ellipse in the xy-plane in the standard form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the vector equation is $\mathbf{r}(\theta) = a\cos\theta\mathbf{i} + b\sin\theta\mathbf{j}$. Here we can replace $\mathbf{u}, \mathbf{v}$ by any pair of mutually perpendicular unit vectors in space. Then we get the vector equation of an ellipse in the plane spanned by $\mathbf{u}, \mathbf{v}$, whose centre is at the origin and whose axes lie along the vectors $\mathbf{u}, \mathbf{v}$ and have lengths $2a, 2b$ respectively. (See Exercise (21.28)(e) for an illustration.)

Comment No. 6:

It should be noted that the representation of a vector in terms of coordinates is subject to a particular frame of reference. If we change the frame of reference

then the same vector will have a different representation. Such a change can always be effected in two parts, viz.,

(i) **parallel translation:** Here the origin is shifted to some point, say O', but the new axes are parallel to the corresponding old axes. As a result, the vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ remain the same. In this case it is very easy to relate the vectors OP and $O'P$. Let $\mathbf{v}$ be the vector $\overrightarrow{OO'}$ (i.e., the position vector of the new origin w.r.t. the old one. Then $\mathbf{r} = \mathbf{r}' + \mathbf{v}$. As a result, everything is 'shifted' by the vector $\mathbf{v}$. In terms of components, this can be rewritten as $x = x' + v_1$, $y = y' + v_2$ and $z = z' + v_3$, where v_1, v_2, v_3 are the coordinates of O' in the old frame of reference. Obviously, a change of this type is very easy to handle.

(ii) **rotation of axes:** In this type of a change, the origin remains where it is but the directions of the axes change. This case is far more complicated (and interesting) than parallel translation. We shall not go into it completely as that would take us too far afield. But the special case when the z axis remains as it is can be handled easily. Let $\mathbf{i}', \mathbf{j}'$ and $\mathbf{k}'$ be the unit vectors along the new axes. Then $\mathbf{k}' = \mathbf{k}$. As $\mathbf{i}'$ and $\mathbf{j}'$ are perpendicular to $\mathbf{k}'$ and hence to $\mathbf{k}$, they must lie in the plane spanned by $\mathbf{i}$ and $\mathbf{j}$. Moreover, if θ is the angle $\mathbf{i}'$ makes with $\mathbf{i}$ then it is also the angle $\mathbf{j}'$ makes with $\mathbf{j}$ (see Figure (b) above). So $\mathbf{j}'$ makes an angle $\theta + \frac{\pi}{2}$ with $\mathbf{i}$. By a reasoning similar to the initial part of the solution to the Main Problem, we have

$$\mathbf{i}' = \cos\theta\mathbf{i} + \sin\theta\mathbf{j} \quad \text{and} \quad \mathbf{j}' = -\sin\theta\mathbf{i} + \cos\theta\mathbf{j} \tag{16}$$

It is very easy to 'invert' this system of equations, i.e., to express $\mathbf{i}$ and $\mathbf{j}$ in terms of $\mathbf{i}'$ and $\mathbf{j}'$. The answer comes as

$$\mathbf{i} = \cos\theta\mathbf{i}' - \sin\theta\mathbf{j}' \quad \text{and} \quad \mathbf{j} = \sin\theta\mathbf{i}' + \cos\theta\mathbf{j}' \tag{17}$$

These equations allow us to convert the old coordinates into new ones and vice versa. Thus, we have,

$$\begin{aligned} x'\mathbf{i}' + y'\mathbf{j}' + z'\mathbf{k}' &= x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \\ &= x(\cos\theta\mathbf{i}' - \sin\theta\mathbf{j}') + y(\sin\theta\mathbf{i}' + \cos\theta\mathbf{j}') + z\mathbf{k}' \\ &= (x\cos\theta + y\sin\theta)\mathbf{i}' + (-x\sin\theta + y\cos\theta)\mathbf{j}' + z\mathbf{k}' \end{aligned}$$

which gives (JEE 1983)

$$x' = x\cos\theta + y\sin\theta, \quad y' = -x\sin\theta + y\cos\theta \quad \text{and} \quad z' = z \tag{18}$$

It is interesting to note which expressions are invariant (a formal term for what a layman would call 'unchanging') under a change of the frame of reference. Take for example, the expression $x + y + z$ i.e., the sum of the

components of the vector $\overrightarrow{OP}$ along the vectors **i**, **j** and **k**. If we rotate the axes in the manner above then the sum of the components of $\overrightarrow{OP}$ along the vectors $\mathbf{i}', \mathbf{j}'$ and $\mathbf{k}'$ will be $x' + y' + z'$ where x', y' and z' are given by (18). It is easily seen that $x' + y' + z'$ is, in general, different from $x + y + z$. In other words, the sum of the components is not invariant under a rotation of the axes. In striking contrast, a little computation shows that $x'^2 + y'^2 + z'^2 = x^2 + y^2 + z^2$. In other words, the sum of the squares of the components of the vector $\overrightarrow{OP}$ along the basic vectors is invariant, even though the components themselves are not. This may appear puzzling. But there is an explanation. The expression $x^2 + y^2 + z^2$ has a geometric significance. It is precisely the square of the length of the vector $\overrightarrow{OP}$. So it is an intrinsic geometric attribute of the vector $\overrightarrow{OP}$. Expressions for it may change depending upon which frame of reference we have but the length itself cannot change. A similar comment holds for other concepts such as the dot or the cross products, which have purely geometric definitions. Suppose, for example, that **a** and **b** are two vectors. Let $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ be their representations in the 'old' system and let $\mathbf{a} = a'_1\mathbf{i}' + a'_2\mathbf{j}' + a'_3\mathbf{k}'$ and $\mathbf{b} = b'_1\mathbf{i}' + b'_2\mathbf{j}' + b'_3\mathbf{k}'$ be their representations in the new system obtained by rotating the axes around the z-axis by an angle θ. Then using (18) we can express a'_1, a'_2, a'_3 in terms of a_1, a_2, a_3 (and θ) and similarly express b'_1, b'_2, b'_3 in terms of b_1, b_2, b_3. A straightforward computation gives that $a_1b_1 + a_2b_2 + a_3b_3 = a'_1b'_1 + a'_2b'_2 + a'_3b'_3$. This could have been predicted because both the sides are expressions for the dot product $\mathbf{a} \cdot \mathbf{b}$ of the vectors **a** and **b**. The dot product can be defined in a purely geometric manner, without introducing any frame of reference and hence will be invariant under a change of coordinate frames. (This is very analogous to saying that a rose will smell sweet by any other name!)

The invariance of the geometric attributes of vectors allows us the freedom to choose the coordinate frame as per our convenience. We already exploited this freedom in the first solution to the Main Problem, where we took the vector **i** to be **a** and the vector **j** to lie in the plane spanned by the vectors **a** and **b**.

Comment No. 7:

The equivalence of the two approaches to vectors permits us to use whichever one is more convenient in a particular problem. The difference between the two is essentially the difference between coordinate geometry and pure geometry. As remarked in Chapter 8, a solution using pure geometry is elegant but often elusive while a solution using coordinates is surer, albeit sometimes tedious. This difference is even more pronounced when we deal with vectors and as a result we encounter problems where neither approach is satisfactory. We illustrate this with a problem.

Let **a**, **b**, **c** be vectors of length 3, 4 and 5 respectively. Let **a** be perpendicular to $\mathbf{b} + \mathbf{c}$, **b** to $\mathbf{c} + \mathbf{a}$ and **c** to $\mathbf{a} + \mathbf{b}$. Find the length of the vector $\mathbf{a} + \mathbf{b} + \mathbf{c}$. (JEE 1981)

For a geometric solution, we would have to begin with a diagram. Note

that it is *not* given that the vectors are mutually perpendicular. (This will turn out to be true but it is not obvious merely from the data.) So drawing an accurate diagram is not easy. This rules out a geometric approach to the problem. For an algebraic solution, we shall have to begin by writing $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ in the form $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, where ($\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$) is a right-handed system of mutually perpendicular unit vectors. The data of the problem then translates into equations of the form $a_1^2 + a_2^2 + a_3^2 = 9$ and $a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) = 0$. There are six such equations and from them we have to determine the value of the expression $\sqrt{(a_1 + b_1 + c_1)^2 + (a_2 + b_2 + c_2)^2 + (a_3 + b_3 + c_3)^2}$. While this can be done, the calculations would look very lengthy. A little simplification is possible by choosing $\mathbf{a} = 3\mathbf{i}$. This is equivalent to taking $a_1 = 3, a_2 = 0$ and $a_3 = 0$. But even with these simplifications, the work is lengthy.

For problems of this and many other types, there is a third method which works best. It is called the **vector method**! In this method, we first prove a few basic rules about vectors. These rules can be proved using either the algebraic or the geometric approach. But once these rules are proved, all the simplifications are made using only these rules and without directly appealing either to the coordinates or to geometry. We do not list all these basic rules. In fact some of them (e.g., the commutativity of the addition of vectors which simply says that for any two vectors $\mathbf{a}$ and $\mathbf{b}$, we have $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$) are so obvious that they are used without an explicit mention. In the present problem we shall need two basic rules about the dot product. One of them is what is called its **symmetry**, which says that for any two vectors $\mathbf{a}$ and $\mathbf{b}$, we have $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$. The other rule is the bilinearity of the dot product stated in equations (10) and (11) above. Because of these two rules, we can manipulate the dot products of vectors much the same way as we manipulate the products of ordinary real numbers. In particular, repeated application of these two rules gives

$$(\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot (\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{b} + \mathbf{c} \cdot \mathbf{c} + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) \quad (19)$$

We are given that $\mathbf{a}$ is perpendicular to $\mathbf{b} + \mathbf{c}$ which gives $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = 0$, which, by bilinearity and symmetry means $\mathbf{a} \cdot \mathbf{b} + \mathbf{c} \cdot \mathbf{a} = 0$. The other two perpendicularity conditions give us similar equations. If we put these into (19) and further note that for any vector $\mathbf{v}$, $\mathbf{v} \cdot \mathbf{v}$ is simply $||\mathbf{v}||^2$, then (19) gives

$$||\mathbf{a} + \mathbf{b} + \mathbf{c}||^2 = ||\mathbf{a}||^2 + ||\mathbf{b}||^2 + ||\mathbf{c}||^2 \quad (20)$$

To finish the solution we now merely note that the R.H.S. is given to be $9 + 16 + 25 = 50$. So $||\mathbf{a} + \mathbf{b} + \mathbf{c}|| = \sqrt{50} = 5\sqrt{2}$. (Note incidentally, that we have also proved along the way that $\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = 0$, which along with the perpendicularity of $\mathbf{a}$ and $\mathbf{b} + \mathbf{c}$ implies $\mathbf{b} \cdot \mathbf{c} = 0$, i.e., $\mathbf{b} \perp \mathbf{c}$. Similarly, $\mathbf{a} \perp \mathbf{c}$ and $\mathbf{a} \perp \mathbf{b}$. But this is not given in the statement of the problem.)

Comment No. 8:

Like the dot product, the cross product is also bilinear (i.e., equations (10) and (11) hold if we replace the dot product throughout by the cross product).

However, it is not symmetric. Instead, it is what is called **skew-symmetric**, i.e., for any two vectors $\mathbf{a}$ and $\mathbf{b}$, we have $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$. Also $|\mathbf{a} \times \mathbf{b}|$ equals the area of the parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. Equivalently, it is twice the area of the triangle two of whose sides are represented by the vectors $\mathbf{a}$ and $\mathbf{b}$. (The third side will be the vector $\mathbf{b} - \mathbf{a}$ or $\mathbf{a} - \mathbf{b}$ depending in which direction it is taken.

These simple facts put together give an easy solution to the following problem (JEE 1987) using vector methods. A direct algebraic solution will be too complicated while a purely geometric solution seems impossible.

If A, B, C, D are any four points in space prove that

$$\left| \overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD} \right| = 4 \text{ (area of triangle } ABC) \qquad (21)$$

We first express the given vectors in terms of the position vectors of the various points w.r.t. some fixed point in space. It will simplify the calculations to take this point as D because then the vectors $\overrightarrow{AD}$ etc., can be denoted by single symbols. (In the present problem the gain is not all that substantial. Still one should be on the lookout for such simplifying choices.) So let us denote the vectors $\overrightarrow{DA}$, $\overrightarrow{DB}$ and $\overrightarrow{DC}$ by $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. Then $\overrightarrow{AB} = \overrightarrow{DB} - \overrightarrow{DA} = \mathbf{b} - \mathbf{a}$ and so $\overrightarrow{AB} \times \overrightarrow{CD} = (\mathbf{b} - \mathbf{a}) \times (-\mathbf{c}) = (\mathbf{a} - \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} - \mathbf{b} \times \mathbf{c}$. Rewriting the other two cross products in a similar manner, the L.H.S. of (21) becomes $|\mathbf{a} \times \mathbf{c} - \mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} - \mathbf{c} \times \mathbf{a} + \mathbf{c} \times \mathbf{b} - \mathbf{a} \times \mathbf{b}|$. Applying skew-symmetry of the cross product to the last three terms, this becomes $2|\mathbf{a} \times \mathbf{c} - \mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a}|$. The R.H.S. is twice the area of the parallelogram with sides AB and AC and hence equals $2|\ \overrightarrow{AB} \times \overrightarrow{AC}\ |$, i.e., $2|(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})|$. From the bilinearity of the cross product and the fact that $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, this further reduces to $2|(\mathbf{b} \times \mathbf{c}) - (\mathbf{a} \times \mathbf{c}) - (\mathbf{b} \times \mathbf{a})|$. The proof is completed by observing that for any vector $\mathbf{v}$, $||\mathbf{v}|| = ||-\mathbf{v}||$.

Comment No. 9:

The scalar triple product is a function of three variables and it is linear in each of the three arguments. (This follows from the bilinearity of the dot and the cross products.) As a result we can expand the scalar triple product of a complicated form into simpler scalar triple products. The cyclic symmetry of the scalar triple product (relation (12)) is also frequently needed. Like the cross product, the scalar triple product is also skew-symmetric, i.e., interchanging any two vectors in it reverses its sign. These simple facts give a quick solution to the following problem. (JEE 1980)

Prove that

$$(\mathbf{a} + \mathbf{b}) \cdot [(\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})] = \mathbf{2}\ (\mathbf{a}\ \mathbf{b}\ \mathbf{c})$$

For a solution, we first apply bilinearity of the cross product to reduce the bracketed expression to $\mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{a}$. Next we apply the bilinearity of the dot product to get the L.H.S. as the sum of six box products, viz.,

(**a b c**), (**a b a**), (**a c a**), (**b b c**), (**b b a**) and (**b c a**). Because of cyclic symmetry, the last one equals the first while the remaining four box products vanish because in each of them two entries are equal. (This corresponds to the vanishing of a determinant in which two rows are identical). So we get the result. (It is not necessary to elaborately mention the bilinearity of the cross and the dot products. After a little practice, you can expand the L.H.S. mechanically as the sum of eight box products six of which vanish.) In an entirely analogous manner we can show that $(\mathbf{a}-\mathbf{b})\cdot(\mathbf{b}-\mathbf{c})\times(\mathbf{c}-\mathbf{a})$ vanishes. (JEE 1989)

Comment No. 10:

Algebraically, the scalar triple product is a certain determinant while geometrically it is the volume of a certain parallelopiped. This dual role enables us to prove certain results which are not so easy to prove directly. Consider, for example, a determinant, say $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$. Let $A = \sqrt{a_1^2+a_2^2+a_3^2}$, $B = \sqrt{b_1^2+b_2^2+b_3^2}$ and $C = \sqrt{c_1^2+c_2^2+c_3^2}$. We claim $|\Delta| \le ABC$. A direct algebraic proof of this is possible but a little complicated. But geometrically, $|\Delta|$ is nothing but the volume of the parallelopiped with sides along the vectors $\mathbf{a} = a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i}+c_2\mathbf{j}+c_3\mathbf{k}$ and A, B, C are simply the lengths of these vectors. If we take the parallelogram spanned by the vectors **a** and **b** as the base of the parallelopiped then the base area is $|\mathbf{a}\times\mathbf{b}|$ which is at most AB. The height cannot exceed $||\mathbf{c}|| = C$. So the volume is at most ABC and this implies the desired inequality, viz., that $|\Delta| \le ABC$.

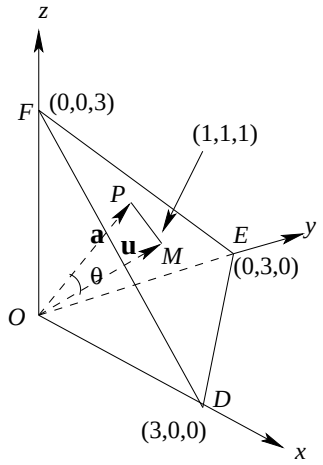
As a consequence, we can do the following problem. (JEE 2002)

Let V be the volume of the paralleopiped formed by the vectors $\mathbf{a} = a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i}+c_2\mathbf{j}+c_3\mathbf{k}$. If a_r, b_r, c_r, where $r = 1, 2, 3$, are non-negative real numbers and $\sum_{r=1}^{3}(a_r+b_r+c_r) = 3L$, show that $V \le L^3$.

We first paraphrase the given condition. Let $P = a_1+a_2+a_3$, $Q = b_1+b_2+b_3$ and $R = c_1+c_2+c_3$. Then we are given that the arithmetic mean of P, Q, R is L. By the A.M.−G.M. inequality, it follows that $PQR \le L^3$. So we shall be through if we can show that $V \le PQR$. We already proved above that $V \le ABC$, where A, B, C are the lengths of the vectors **a**, **b**, **c** respectively. So it suffices to show that $ABC \le PQR$, which, in turn, would follow if we can show separately that $A \le P$, $B \le Q$ and $C \le R$. (As with many other problems about inequalities, what we are giving here is one particular line of thinking. We are replacing the inequality to be proved by progressively stronger inequalities and hoping that we can prove these stronger inequalities. But these implications are not reversible and there is always a danger that we are not able to prove the stronger inequalities and still the original inequality is true. If that happens, we have to abandon the line of thinking we have followed and try some other. Fortunately, this does not happen in the present problem.) Since A and P are both non-negative, proving $A \le P$ is equivalent to proving that $A^2 \le P^2$, i.e., that $a_1^2+a_2^2+a_3^2 \le (a_1+a_2+a_3)^2$. But this follows by directly expanding

the square and using the hypothesis that a_1, a_2, a_3 are non-negative. Similarly, we get $B \leq Q$ and $C \leq R$. As already observed, this implies the result.

The solution above was based on the inequality $|(\mathbf{a}\ \mathbf{b}\ \mathbf{c})| \leq ||\mathbf{a}||\ ||\mathbf{b}||\ ||\mathbf{c}||$ which we proved by interpreting the scalar triple product geometrically. It should not, however, be supposed that the geometric approach is always superior. Indeed, in the second part of the solution above, we needed the inequalities $A \leq P$ etc., which we proved algebraically. If we try to prove them geometrically, we would first have to interpret A and P geometrically. We already know that A is the length of the vector $\mathbf{a}$. As for $P = a_1 + a_2 + a_3$, we can recognise it as the dot product $\mathbf{a} \cdot \mathbf{u}$, where $\mathbf{u}$ is the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$. Similarly we get $Q = \mathbf{b} \cdot \mathbf{u}$ and $R = \mathbf{c} \cdot \mathbf{u}$. The trouble is that this interpretation does not readily imply that $||\mathbf{a}|| \leq \mathbf{a} \cdot \mathbf{u}$. The most standard inequality about the dot product, viz., (9) is of no use here since it will give an inequality which is the other way, viz., $|\mathbf{a} \cdot \mathbf{u}| \leq ||\mathbf{a}||\ ||\mathbf{u}||$, i.e., $|P| \leq \sqrt{3}A$. It is, in fact, not true in general that $||\mathbf{a}|| \leq \mathbf{a} \cdot \mathbf{u}$. To prove it, we shall crucially need that a_1, a_2, a_3 are all non-negative, or in other words, that the vector $\mathbf{a} = \mathbf{a_1 i} + \mathbf{a_2 j} + \mathbf{a_3 k}$ is in the first orthant[5]. The vector $\mathbf{u}$ makes an angle $\cos^{-1} \frac{1}{\sqrt{3}}$ with each of the vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$. We first show that this is the largest angle any vector in the first orthant can make with $\mathbf{u}$. To see this let θ denote the angle between $\mathbf{a}$ and $\mathbf{u}$. Let D, E, F and M be respectively the points $(3,0,0)$, $(0,3,0)$, $(0,0,3)$ and $(1,1,1)$. All these points lie in the first orthant and on the plane whose equation is $x + y + z = 3$. Also DEF is an equilateral triangle with centre M. The vector $\overrightarrow{OM}$ is simply the vector $\mathbf{u}$. Let P be the point on the plane $x + y + z = 3$ such that $\overrightarrow{OP}$ is parallel to the vector $\mathbf{a}$.



Note that since $\mathbf{a}$ is in the first orthant, P must lie in the triangle DEF. Clearly, $\tan\theta = \frac{MP}{OM}$ will be maximum when MP is maximum and this will be the case when P is at any of the vertices D, E or F. For any of these points, $\tan\theta = \frac{DM}{OM} = \frac{\sqrt{4+1+1}}{\sqrt{3}} = \sqrt{2}$ which gives $\cos\theta = \frac{1}{\sqrt{3}}$.

It *now* follows (using (8)) that for any vector $\mathbf{a}$ in the first orthant, $\mathbf{a} \cdot \mathbf{u} \geq ||\mathbf{a}||\ ||\mathbf{u}|| \dfrac{1}{\sqrt{3}} = ||\mathbf{a}||$, i.e., $P \geq A$ as desired. Obviously, this proof is more complicated than its algebraic counterpart which consists merely of expanding $(a_1 + a_2 + a_3)^2$ and observing that it is at least $a_1^2 + a_2^2 + a_3^2$ since

[5] Just as a plane is divided into four quadrants depending upon the signs of the two coordinates, a space is divided into 8 ($=2^3$) orthants. In the first orthant, all the three coordinates are non-negative.

$2(a_1a_2 + a_2a_3 + a_3a_1) \geq 0$.

So the thing to keep in mind is that neither the geometric nor the algebraic approach is always superior to the other. Deciding which will work better where, calls for a judgement and (as in this problem) sometimes the answer is different for different parts of the same problem.

An advantage of the algebraic approach is the ability to choose the frame of reference $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ suitably as we did in the solution to the Main Problem. Here is a problem where a certain inequality about a scalar triple product can be proved more comfortably if we use this freedom.

Let $\vec{u}$ and $\vec{v}$ be unit vectors. If $\vec{w}$ is a vector such that $\vec{w} + (\vec{w} \times \vec{u}) = \vec{v}$, then prove that $|(\vec{u} \times \vec{v}) \cdot \vec{w}| \leq \frac{1}{2}$ and that the equality holds if and only if $\vec{u}$ is perpendicular to $\vec{v}$. (JEE 1999)

Let us first give a solution by vector methods. Since $\vec{w}$ and $\vec{w} \times \vec{u}$ are at right angles to each other, the equation $\vec{w} + (\vec{w} \times \vec{u}) = \vec{v}$ gives

$$||\vec{w}||^2 + ||\vec{w} \times \vec{u}||^2 = ||\vec{v}||^2 \tag{22}$$

Let α be the angle between $\vec{w}$ and $\vec{u}$. Then since $\vec{u}, \vec{v}$ are unit vectors, (22) implies $||\vec{w}||^2(1 + \sin^2 \alpha) = 1$. Since $\sin^2 \alpha \leq 1$ we further get

$$||\vec{w}||^2 \geq \frac{1}{2} \text{ with equality if and only if } \alpha = \pm\frac{\pi}{2} \tag{23}$$

Also, taking the dot products of the two sides of the equation $\vec{w} + (\vec{w} \times \vec{u}) = \vec{v}$ with the vectors $\vec{u}, \vec{v}$ and $\vec{w}$ we get, respectively,

$$\vec{u} \cdot \vec{w} = \vec{u} \cdot \vec{v} \tag{24}$$

$$\vec{v} \cdot \vec{w} + (\vec{v}\ \vec{w}\ \vec{u}) = ||\vec{v}||^2 = 1 \tag{25}$$

$$\text{and} \quad ||\vec{w}||^2 = \vec{v} \cdot \vec{w} \tag{26}$$

From (25) and (26) we get $(\vec{v}\ \vec{w}\ \vec{u}) = 1 - ||\vec{w}||^2$. Further, since $||\vec{v}|| = 1$, from (22) we have $1 - ||\vec{w}||^2 \geq 0$ and so we must have $|(\vec{v}\ \vec{w}\ \vec{u})| = 1 - ||\vec{w}||^2$ which combined with (23) gives $|(\vec{v}\ \vec{w}\ \vec{u})| \leq \frac{1}{2}$ with equality holding if and only if $\vec{u}$ and $\vec{w}$ are perpendicular to each other. But in view of (24) this is equivalent to saying that $\vec{u}$ and $\vec{v}$ are perpendicular to each other. To finish the solution we need only note that $|(\vec{u} \times \vec{v}) \cdot \vec{w}| = |(\vec{v}\ \vec{w}\ \vec{u})|$.

To get an algebraic solution, we take $\vec{i} = \vec{u}$ and $\vec{j}$ to be a vector perpendicular to $\vec{i}$ in the plane spanned by $\vec{u}$ and $\vec{v}$. Then $\vec{v} = \cos\theta\vec{i} + \sin\theta\vec{j}$, where θ is the angle between $\vec{u}$ and $\vec{v}$. Now let $\vec{k} = \vec{i} \times \vec{j}$ and suppose $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$. Then the equation $\vec{w} + (\vec{w} \times \vec{u}) = \vec{v}$ reduces to $w_1\vec{i} + (w_2 + w_3)\vec{j} + (w_3 - w_2)\vec{k} = \cos\theta\vec{i} + \sin\theta\vec{j}$. Equating coefficients of $\vec{i}, \vec{j}, \vec{k}$, we get $w_2 = w_3$ and further $w_2 = w_3 = \frac{1}{2}\sin\theta$ and $w_1 = \cos\theta$. Hence $|(\vec{u} \times \vec{v}) \cdot \vec{w}| = |(\sin\theta)\vec{k} \cdot \vec{w}| = |w_3 \sin\theta| = \frac{1}{2}\sin^2\theta$ which is clearly $\leq \frac{1}{2}$ with equality holding if and only if $\theta = \pm\frac{\pi}{2}$.

Here the algebraic proof is not only shorter than the one by vector methods, it is far more straightforward. In an examination, one should try both

the approaches (except, of course, when the problem specifically says that a certain approach is to be followed) and then pick the one that is more suitable.

Comment No. 11:

Like the scalar triple product, the vector triple product is also a function of three vector variables and is linear in each of the three arguments. However, unlike the scalar triple product, the vector triple product is not cyclically symmetric. Nor does it have any succinct geometric interpretation. This puts serious restrictions on its maneuverability. The best way to handle a vector triple product is to use formula (13) above. We illustrate this with a couple of examples.

If $\vec{A}, \vec{B}$ and $\vec{C}$ are vectors such that $|\vec{B}| = |\vec{C}|$, prove that

$$\Big[(\vec{A}+\vec{B}) \times (\vec{A}+\vec{C})\Big] \times (\vec{B} \times \vec{C}) \cdot (\vec{B}+\vec{C}) = 0. \quad \text{(JEE 1997)} \tag{27}$$

Let us first work with the vector $\Big[(\vec{A}+\vec{B}) \times (\vec{A}+\vec{C})\Big] \times (\vec{B} \times \vec{C}) = \vec{U}$ (say). The expression in the brackets is the vector $\vec{B} \times \vec{A} + \vec{A} \times \vec{C} + \vec{B} \times \vec{C}$. Taking the cross product of each term with $\vec{B} \times \vec{C}$ and applying (13) (and properties of the scalar triple product) we get $\vec{U} = (\vec{B}\ \vec{A}\ \vec{C})\vec{B} - (\vec{B}\ \vec{A}\ \vec{B})\vec{C} + (\vec{A}\ \vec{C}\ \vec{C})\vec{B} - (\vec{A}\ \vec{C}\ \vec{B})\vec{C} + (\vec{B}\ \vec{C}\ \vec{C})\vec{B} - (\vec{B}\ \vec{C}\ \vec{B})\vec{C}$. Four of these scalar triple products vanish and we are left with $\vec{U} = (\vec{B}\ \vec{A}\ \vec{C})\vec{B} - (\vec{A}\ \vec{C}\ \vec{B})\vec{C} = (\vec{B}\ \vec{A}\ \vec{C})(\vec{B}-\vec{C})$ again by the properties of the scalar triple products. Let us write this as $\lambda(\vec{B}-\vec{C})$, where $\lambda = (\vec{B}\ \vec{A}\ \vec{C})$. Now, the L.H.S. of (27) is simply $\vec{U} \cdot (\vec{B}+\vec{C}) = \lambda(\vec{B}-\vec{C}) \cdot (\vec{B}+\vec{C})$. Using bilinearity (and symmetry) of the dot product, this reduces to $\lambda(\vec{B} \cdot \vec{B} - \vec{C} \cdot \vec{C}) = \lambda(|\vec{B}|^2 - |\vec{C}|^2) = 0$ since $|\vec{B}| = |\vec{C}|$.

Comment No. 12:

Here is another problem involving vector triple products. (JEE 1994)

For any four vectors $\mathbf{a},\ \mathbf{b},\ \mathbf{c}, \mathbf{d}$ show that the vector

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) + (\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c})$$

is parallel to $\mathbf{a}$.

Call the given vector as $\mathbf{u}$. Note that the expression for $\mathbf{u}$ is cyclically symmetric in $\mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ (but not w. r. t. $\mathbf{a}$). In order to show that $\mathbf{u}$ is parallel to $\mathbf{a}$ we want to express $\mathbf{u}$ as a multiple of $\mathbf{a}$. If we apply (13) to each term in the expression for $\mathbf{u}$ we shall get $\mathbf{u}$ as a linear combination of $\mathbf{c},\ \mathbf{d}$ and $\mathbf{b}$ which is not what we want. So let us rewrite the first term as $-(\mathbf{c} \times \mathbf{d}) \times (\mathbf{a} \times \mathbf{b})$ and then apply (13) to get

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = -(\mathbf{c} \times \mathbf{d}) \times (\mathbf{a} \times \mathbf{b}) &= [(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{a}]\ \mathbf{b} - [(\mathbf{c} \times \mathbf{d}) \cdot \mathbf{b}]\ \mathbf{a} \\ &= (\mathbf{a}\ \mathbf{c}\ \mathbf{d})\ \mathbf{b} - (\mathbf{b}\ \mathbf{c}\ \mathbf{d})\ \mathbf{a} \end{aligned} \tag{28}$$

Because of cyclic symmetry, we can immediately write down similar expressions for the other two terms, viz., $(\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b})$ and $(\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c})$. But when

we add the three together, the terms involving $\mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ will not cancel as we would like them to do. To overcome this difficulty we expand the second term in a manner similar to the first to get

$$\begin{aligned}(\mathbf{a}\times\mathbf{c})\times(\mathbf{d}\times\mathbf{b}) = -(\mathbf{d}\times\mathbf{b})\times(\mathbf{a}\times\mathbf{c}) &= [(\mathbf{d}\times\mathbf{b})\cdot\mathbf{a}]\,\mathbf{c} - [(\mathbf{d}\times\mathbf{b})\cdot\mathbf{c}]\,\mathbf{a} \\ &= (\mathbf{a}\,\mathbf{d}\,\mathbf{b})\,\mathbf{c} - (\mathbf{c}\,\mathbf{d}\,\mathbf{b})\,\mathbf{a} \qquad (29)\end{aligned}$$

However, we expand the third term, viz., $(\mathbf{a}\times\mathbf{d})\times(\mathbf{b}\times\mathbf{c})$ differently. We *do not* rewrite it as $-(\mathbf{b}\times\mathbf{c})\times(\mathbf{a}\times\mathbf{d})$ as we did with the first two terms. Instead we take it as it is and expand using (13) to get

$$\begin{aligned}(\mathbf{a}\times\mathbf{d})\times(\mathbf{b}\times\mathbf{c}) &= [(\mathbf{a}\times\mathbf{d})\cdot\mathbf{c}]\,\mathbf{b} - [(\mathbf{a}\times\mathbf{d})\cdot\mathbf{b}]\,\mathbf{c} \\ &= (\mathbf{c}\,\mathbf{a}\,\mathbf{d})\,\mathbf{b} - (\mathbf{b}\,\mathbf{a}\,\mathbf{d})\,\mathbf{c} \qquad (30)\end{aligned}$$

We now add (28) to (30). Since $(\mathbf{a}\ \mathbf{c}\ \mathbf{d}) = -(\mathbf{c}\ \mathbf{a}\ \mathbf{d})$ and $(\mathbf{a}\ \mathbf{d}\ \mathbf{b}) = (\mathbf{b}\ \mathbf{a}\ \mathbf{d})$ the coefficients of $\mathbf{b}$ and $\mathbf{c}$ cancel out and we get $\mathbf{u}$ as a scalar multiple of $\mathbf{a}$ as desired.

Normally whenever there is cyclic symmetry, all the terms are expanded in the same manner. We have seen many instances of this in problems dealing with triangles. The present problem is exceptional in this respect (and hence somewhat tricky). There is indeed a way to do this problem in a cyclically symmetric manner using the concept of resolution of a vector along a basis, which we shall study shortly. (See Comment No. 18 for the alternate solution.)

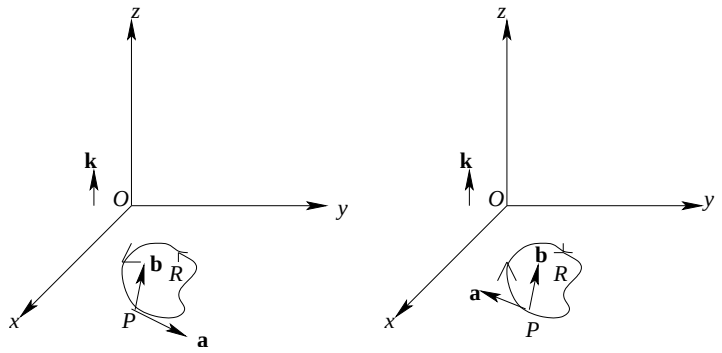
Comment No. 13:

The essence of the algebraic approach to vectors is to express every vector in terms of the standard basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. In the vector method, however, we often choose a basis which is more convenient for a particular problem. The situation is rather like this. When we give dates according to, say, the Christian calendar, we are telling how many years after the birth of Christ a particular event occurred. This may seem ridiculous if the event has nothing to do with Christ. But the practice acquires sanctity because of its universal acceptance. But in a particular context, some other reference point of time may be more relevant. Suppose for example, that some person was born in 1986 A.D. and graduated from college in 2002 A.D. In his biography, it will be far more logical to state that he graduated when he was 16. In fact, this will drive home his unusual prodigy more forcefully, with the reader spared of having to do any arithmetic!

The fundamental idea behind a basis is that of linear independence. Formally, a collection of vectors $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$ is called **linearly dependent** if there exist scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$ not all 0, such that $\lambda_\mathbf{1}\mathbf{v_1}+\lambda_\mathbf{2}\mathbf{v_2}+\ldots+\lambda_\mathbf{k}\mathbf{v_k} = \mathbf{0}$. It is easy to show that this is equivalent to saying that at least one member of the collection can be expressed as a linear combination of the others. A set which is not linearly dependent is called linearly independent. If $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$ is such a set then for any real numbers $\alpha_1, \alpha_2, \ldots, \alpha_k, \beta_1, \beta_2, \ldots, \beta_k$, equality of $\sum_{i=1}^{k} \alpha_i \mathbf{v}_i$

and $\sum_{i=1}^{k} \beta_i \mathbf{v}_i$ implies that $\alpha_i = \beta_i$ for every $i = 1, 2, \ldots, k$. In other words, a single vector equation is equivalent to k scalar equations. This is the reason why the concept is so important. Put differently, this says that the components of a vector along the $\mathbf{v_i}$'s are uniquely defined, provided the vector lies in their linear span.

It can be shown that in a three-dimensional vector space every four (or more) vectors are linearly dependent. Two vectors are linearly dependent if and only if one of them is a scalar multiple of the other. If they are linearly independent, then they form a basis for all vectors in the plane spanned by them. For three vectors, say $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$, they are linearly dependent if and only if they are coplanar which is equivalent to saying that their box product $(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$ vanishes. If they are linearly independent then they form a basis for the entire vector space. If every two of them are at right angles then they are automatically linearly independent and are said to constitute an **orthogonal basis**. If further they are all of length 1, then the basis is said to be **orthonormal**.



(a) Counter-clockwise traversal (b) Clockwise traversal

A basis $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is said to be **right-handed** or **left-handed** according as the scalar triple product $(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$ is positive or negative. This tallies with the physical definition, that is $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is right-handed if and only if whenever the index and the middle fingers of the right right hand point in the directions of $\mathbf{a}$ and $\mathbf{b}$ respectively, the thumb points in the direction of the vector $\mathbf{c}$. Note that this depends only on the cyclic order of the three vectors. Thus, if $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is a right-handed system, then so are $(\mathbf{b}, \mathbf{c}, \mathbf{a})$ and $(\mathbf{c}, \mathbf{a}, \mathbf{b})$, but the system $(\mathbf{a}, \mathbf{c}, \mathbf{b})$ is left-handed. Note further that if we take the negative of any one of the three vectors, e.g., $(\mathbf{a}, -\mathbf{b}, \mathbf{c})$ or $(\mathbf{a}, \mathbf{b}, -\mathbf{c})$, then we get a left-handed system. Thus the two types of systems are mirror images of each other, or in other words, they have opposite orientations. In fact, it is possible to give a rigorous definition of clockwise or counter-clockwise orientation of the boundary of a plane region, say R, in terms of right- or left-handed systems. Specifically, suppose that you are at a point P while traversing the boundary. Let $\mathbf{a}$ be a vector in the forward direction of motion at P. Let $\mathbf{b}$ be any vector at P which points into the region R. Then the boundary is being traversed counter-clockwise or clockwise

according as the system $(\mathbf{a}, \mathbf{b}, \mathbf{k})$ is right-handed or left-handed. (See the figure above. It is, of course, assumed here that $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ is a right-handed system to begin with.)

The advantage which an orthogonal basis has over an ordinary one is as follows. Suppose $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is a basis. Then for every vector $\mathbf{u}$, there exist unique scalars α, β and γ such that $\mathbf{u} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$. To determine them, we take the dot product of $\mathbf{u}$ with $\mathbf{a}, \mathbf{b}, \mathbf{c}$ to get

$$\left.\begin{array}{rcl} ||\mathbf{a}||^2\alpha + (\mathbf{a}\cdot\mathbf{b})\beta + (\mathbf{a}\cdot\mathbf{c})\gamma &=& \mathbf{a}\cdot\mathbf{u} \\ (\mathbf{b}\cdot\mathbf{a})\alpha + ||\mathbf{b}||^2\beta + (\mathbf{b}\cdot\mathbf{c})\gamma &=& \mathbf{b}\cdot\mathbf{u} \\ (\mathbf{c}\cdot\mathbf{a})\alpha + (\mathbf{c}\cdot\mathbf{b})\beta + ||\mathbf{c}||^2\gamma &=& \mathbf{c}\cdot\mathbf{u} \end{array}\right\} \tag{31}$$

This is a system of three linear equations in three unknowns. It can be shown (cf. Exercise (21.7)(e)) that the determinant of its coefficient matrix is simply $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^2$ which is non-zero as the three vectors are linearly independent. So the values of α, β, γ can be determined from it. The calculations are straightforward, but a little cumbersome. If, however, the basis $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is orthogonal then the determination is immediate since all non-diagonal entries are zero.

If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are given in terms of the standard basis as, say, $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, and $\mathbf{u}$ is also given as $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$, then taking dot products with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ we get the following system of three equations in the three unknowns α, β and γ.

$$\left.\begin{array}{rcl} a_1\alpha + b_1\beta + c_1\gamma &=& u_1 \\ a_2\alpha + b_2\beta + c_2\gamma &=& u_2 \\ a_3\alpha + b_3\beta + c_3\gamma &=& u_3 \end{array}\right\} \tag{32}$$

Here the determinant of the coefficient matrix is $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$ and so solving this system, which is a little easier to write down than (31), also gives the values of α, β, γ.

When we have two linearly independent vectors, say $\mathbf{a}$ and $\mathbf{b}$, the vector $\mathbf{a} \times \mathbf{b}$ is a non-zero vector perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. So, $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$ is a basis. For want of a better name, we call such a basis as **semi-orthogonal**, because out of the three pairs of basis vectors, two are mutually orthogonal. As a result, when a vector, say $\mathbf{u}$, is expressed as $\alpha\mathbf{a} + \beta\mathbf{b} + \gamma(\mathbf{a} \times \mathbf{b})$ the coefficients α, β, γ can be computed rather easily although not as easily as for an orthogonal basis. Putting $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ in (31), we see that the last equation immediately determines γ. Moreover, the first two equations do not involve γ. So, determining α and β is easier too than for an arbitrary basis.

In physical applications, sometimes we do not have a full basis but just a single non-zero vector, say $\mathbf{a}$ and we want to know the components of a given vector $\mathbf{u}$ along and perpendicular to $\mathbf{a}$. To find these, let $\mathbf{b}$ be the orthogonal projection of $\mathbf{u}$ on the plane perpendicular to $\mathbf{a}$. Then the vectors $\mathbf{a}, \mathbf{b}, \mathbf{u}$ are coplanar and we can write $\mathbf{u}$ as $\alpha\mathbf{a} + \beta\mathbf{b}$. Since $\mathbf{a}$ is perpendicular to $\mathbf{b}$, taking dot product with $\mathbf{a}$ gives $\alpha = \dfrac{\mathbf{a}\cdot\mathbf{u}}{||\mathbf{a}||^2}$. So the component of $\mathbf{u}$ along $\mathbf{a}$ is $\alpha\mathbf{a} = \dfrac{\mathbf{a}\cdot\mathbf{u}}{||\mathbf{a}||^2}\mathbf{a}$ while the component perpendicular to $\mathbf{a}$ is $\beta\mathbf{b} = \mathbf{u} - \alpha\mathbf{a} = \mathbf{u} - \dfrac{\mathbf{a}\cdot\mathbf{u}}{||\mathbf{a}||^2}\mathbf{a}$. (JEE

1988) Note that we have not found β and $\mathbf{b}$ separately. Nor is it asked. In case $\mathbf{a}$ is a unit vector, the scalar α equals $\mathbf{u}\cdot\mathbf{a}$ and is also often called the **orthogonal projection** (or simply the projection) of the vector $\mathbf{u}$ along (or in the direction of) $\mathbf{a}$. (Occasionally, the vector $(\mathbf{u}\cdot\mathbf{a})\mathbf{a}$ is taken as the projection. In that case it is the same as the component. But again, sometimes components are taken as scalars. One should be wary of which convention is followed.) The term 'projection' is also used sometimes when $\mathbf{a}$ is not a unit vector. But in that case $\mathbf{a}$ has to be first replaced by the unit vector, say $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{||\mathbf{a}||}$ and then the orthogonal projection of $\mathbf{u}$ along $\mathbf{a}$ is not $\mathbf{u}\cdot\mathbf{a}$ but $\dfrac{\mathbf{u}\cdot\mathbf{a}}{||\mathbf{a}||}$.

Comment No. 14:

With this background, we are ready to tackle problems where either the problem itself or its solution involves linearly dependent/independent sets of vectors. As a starter, suppose we are given two mutually perpendicular non-zero vectors $\mathbf{A}$ and $\mathbf{B}$ and a non-zero scalar c and we have to find the vector $\mathbf{X}$ which satisfies the equations $\mathbf{A}\cdot\mathbf{X} = c$ and $\mathbf{A}\times\mathbf{X} = \mathbf{B}$. (JEE 1983) In the algebraic approach we would write $\mathbf{X}$ as $x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}$ and try to determine the unknowns x_1, x_2, x_3 by writing a system of three equations in them. Let $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{B} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$. The vector equation $\mathbf{A}\times\mathbf{X} = \mathbf{B}$ is equivalent to a system of three equations of the form $a_2x_3 - a_3x_2 = b_1$. But these equations are not independent of each other and so in reality it gives only two equations. (See Exercise (21.7)(c) for more details.) However, the condition $\mathbf{A}\cdot\mathbf{X} = c$ gives one more equation, viz., $a_1x_1 + a_2x_2 + a_3x_3 = c$. Thus together we have three equations in three unknowns. But solving them is cumbersome. The vector method, however, provides a simple solution. Since $\mathbf{A}\perp\mathbf{B}$, the system $(\mathbf{A}, \mathbf{B}, \mathbf{A}\times\mathbf{B})$ is a right-handed orthogonal basis. The condition $\mathbf{A}\times\mathbf{X} = \mathbf{B}$ implies that $\mathbf{X}$ is perpendicular to $\mathbf{B}$ and is therefore, a linear combination of the other two basis elements, say, $\mathbf{X} = \alpha\mathbf{A} + \beta(\mathbf{A}\times\mathbf{B})$. The given equation $\mathbf{A}\cdot\mathbf{X} = c$ determines α as $\dfrac{c}{||\mathbf{A}||^2}$. Also $\mathbf{A}\times\mathbf{X} = \beta\mathbf{A}\times(\mathbf{A}\times\mathbf{B}) = \beta(\mathbf{A}\cdot\mathbf{B})\mathbf{A} - \beta(\mathbf{A}\cdot\mathbf{A})\mathbf{B} = -\beta||\mathbf{A}||^2\mathbf{B}$ using (13) and the fact that $\mathbf{A}\perp\mathbf{B}$. So the equation $\mathbf{A}\times\mathbf{X} = \mathbf{B}$ implies $\beta||\mathbf{A}||^2 = -1$ giving $\beta = \dfrac{-1}{||\mathbf{A}||^2}$. Thus $\mathbf{X} = \dfrac{c}{||\mathbf{A}||^2}\mathbf{A} + \dfrac{-1}{||\mathbf{A}||^2}(\mathbf{A}\times\mathbf{B})$. (The problem *can* be solved algebraically if we exercise the freedom to choose the vectors $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$. As it is given that $\mathbf{A}\perp\mathbf{B}$ we can choose $\mathbf{i}, \mathbf{j}, \mathbf{k}$ along $\mathbf{A}, \mathbf{B}, \mathbf{A}\times\mathbf{B}$ respectively. In that case $\mathbf{A} = |\mathbf{A}|\mathbf{i}$, $\mathbf{B} = |\mathbf{B}|\mathbf{j}$ and $\mathbf{A}\times\mathbf{B} = |\mathbf{A}||\mathbf{B}|\mathbf{k}$. Let $\mathbf{X} = x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}$. Then the equations $\mathbf{A}\cdot\mathbf{X} = c$ and $\mathbf{A}\times\mathbf{X} = \mathbf{B}$ translate into $|\mathbf{A}|x_1 = c$, $|\mathbf{A}|x_2 = 0$ and $-|\mathbf{A}|x_3 = |\mathbf{B}|$. This system is very easy to solve and gives the same answer as above. Actually, this approach is not substantially different from the one above. There we invoked the freedom to choose the basis. Now we are invoking the freedom to choose the vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.)

However, the freedom to choose the vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ is not available in all the problems. Suppose, for example, that we are given that $\vec{A} = 2\vec{i} + \vec{k}, \vec{B} =$

$\vec{i}+\vec{j}+\vec{k}$ and $\vec{C}=4\vec{i}-3\vec{j}+7\vec{k}$ and we want to determine a vector $\vec{R}$ such that $\vec{R}\times\vec{B}=\vec{C}\times B$ and $\vec{R}\cdot\vec{A}=0$. (JEE 1990) Here the very statement of the problem involves $\vec{i},\vec{j}$ and $\vec{k}$ and so we cannot choose them as per our liking. Once again, it is not a good idea to write down $\vec{R}$ as (say) $x\vec{i}+y\vec{j}+z\vec{k}$ and rush to get a system of three equations in the three unknowns x,y,z. Instead, let us pause and rewrite the first condition on $\vec{R}$ as $(\vec{R}-\vec{C})\times\vec{B}=\vec{0}$. This can happen only if $\vec{R}-\vec{C}$ is a multiple of $\vec{B}$, say $\lambda\vec{B}$, for some real number λ. This means $\vec{R}=\lambda\vec{B}+\vec{C}$. Now there is only one unknown, viz., λ and to find it we use the condition $\vec{R}\cdot\vec{A}=0$. We have $(\lambda\vec{B}+\vec{C})\cdot\vec{A}=0$ and hence $\lambda=-\dfrac{\vec{C}\cdot\vec{A}}{\vec{B}\cdot\vec{A}}=-\dfrac{15}{3}=-5$. Hence $\vec{R}=-5\vec{B}+\vec{C}=-\vec{i}-8\vec{j}+2\vec{k}$.

In the same vein, suppose we want to find a unit vector $\vec{u}$ coplanar with the vectors $\vec{i}+\vec{j}+2\vec{k}$ and $\vec{i}+2\vec{j}+\vec{k}$ and perpendicular to the vector $\vec{i}+\vec{j}+\vec{k}$. (JEE 1992) The coplanarity condition implies $\vec{u}=\lambda(\vec{i}+\vec{j}+2\vec{k})+\mu(\vec{i}+2\vec{j}+\vec{k})=(\lambda+\mu)\vec{i}+(\lambda+2\mu)\vec{j}+(2\lambda+\mu)\vec{k}$ for some λ and μ. Perpendicularity with $\vec{i}+\vec{j}+\vec{k}$ implies $4\lambda+4\mu=0$ i.e., $\mu=-\lambda$. So $\vec{u}=-\lambda\vec{j}+\lambda\vec{k}$. Since $\vec{u}$ is a unit vector, we further get $2\lambda^2=1$ i.e., $\lambda=\pm\frac{1}{\sqrt{2}}$. So $\vec{u}=\pm\frac{1}{\sqrt{2}}(\vec{j}-\vec{k})$. Had we started with $\vec{u}=u_1\vec{i}+u_2\vec{j}+u_3\vec{k}$ and written down a system of three equations for u_1,u_2 and u_3 we would have gotten the same answer. But the work would have been more clumsy.

As a problem about orthogonal projections, suppose $\vec{b}=4\vec{i}+3\vec{j}$ and $\vec{c}$ are two vectors perpendicular to each other in the xy-plane. We want to find all vectors in the same plane having projections 1 and 2 along $\vec{b}$ and $\vec{c}$ respectively. (JEE 1987) We first convert $\vec{b}$ to a unit vector by writing $\vec{b}=5\hat{b}$, where $\hat{b}=\dfrac{4}{5}\vec{i}+\dfrac{3}{5}\vec{j}$. Then in the xy-plane, the unit vectors $\hat{c}$ which are perpendicular to $\vec{b}$ are $\pm(\dfrac{3}{5}\vec{i}-\dfrac{4}{5}\vec{j})$. So the vectors having projections 1 and 2 along **b** and **c** are $\hat{b}+2\hat{c}=(\dfrac{4}{5}\vec{i}+\dfrac{3}{5}\vec{j})\pm 2(\dfrac{3}{5}\vec{i}-\dfrac{4}{5}\vec{j})$. They come out as $2\vec{i}-\vec{j}$ and $-\dfrac{2}{5}\vec{i}+\dfrac{11}{5}\vec{j}$.

Comment No. 15:

Linear dependence of three vectors (which is equivalent to their coplanarity) can be easily characterised in terms of their scalar triple product, which, in turn, can be expressed as a determinant. Not surprisingly, some problems about linear dependence/independence of three vectors often reduce to problems about determinants. As an example, suppose that the vectors $a\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}$ are coplanar, where a,b,c are all different from 1. We want to find the value of the expression $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$. (JEE 1987) The coplanarity condition means that $\Delta=0$, where Δ is the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$. By expansion, $\Delta=abc-a-b-c+2$. Also by direct expansion, the given expression equals $\dfrac{3-2a-2b-2c+ab+bc+ca}{(1-a)(1-b)(1-c)}$ and further $\dfrac{3-2a-2b-2c+ab+bc+ca}{1+ab+bc+ca-a-b-c-abc}$. Using $\Delta=0$, i.e., $abc=a+b+c-2$, the denominator equals the numerator.

So the expression equals 1.

It was remarked that the cross product of two vectors can be expressed as a symbolic determinant (see (7)). Here is a problem about such determinants. (JEE 1989)

If the vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar, prove that

$$\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = \vec{0} \tag{33}$$

Because of coplanarity, the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are linearly dependent. Hence there exist scalars, α, β, γ not all 0, such that

$$\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \tag{34}$$

Taking dot products of both the sides with $\vec{a}$ and $\vec{b}$ gives, respectively,

$$\alpha(\vec{a}\cdot\vec{a}) + \beta(\vec{a}\cdot\vec{b}) + \gamma(\vec{a}\cdot\vec{c}) = 0 \tag{35}$$

$$\text{and} \quad \alpha(\vec{b}\cdot\vec{a}) + \beta(\vec{b}\cdot\vec{b}) + \gamma(\vec{b}\cdot\vec{c}) = 0 \tag{36}$$

Taking dot product with $\vec{c}$ will give one more equation. But since $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent, it will convey no new information. Anyway, these two equations along with the original vector equation (34) are sufficient for our purpose. It is tempting to say that (33) follows by eliminating α, β and γ from (34), (35) and (36). This reasoning is basically not incorrect. The only catch is that as it stands, (34) is a vector equation. We can replace it by three separate equations obtained by resolving $\vec{a}, \vec{b}, \vec{c}$ along a fixed basis, say as $\vec{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\vec{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\vec{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$. Then (34) is equivalent to a system of three equations of the form $\alpha a_1 + \beta b_1 + \gamma c_1 = 0$. These equations, coupled with (35) and (36) give us $\begin{vmatrix} a_1 & b_1 & c_1 \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = 0$, $\begin{vmatrix} a_2 & b_2 & c_2 \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = 0$ and $\begin{vmatrix} a_3 & b_3 & c_3 \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = 0$. Multiplying these by $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ respectively and adding gives (33). Alternatively, we imitate the reasoning behind this determinant criterion. Specifically, if (35) and (36) hold then α, β, γ are proportional to the respective 2×2 determinants of coefficients. In other words, there exists some k such that

$$\alpha = k[(\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{c}) - (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{b})] \tag{37}$$

$$\beta = k[(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{a}) - (\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{c})] \tag{38}$$

$$\text{and } \gamma = k[(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{a})] \tag{39}$$

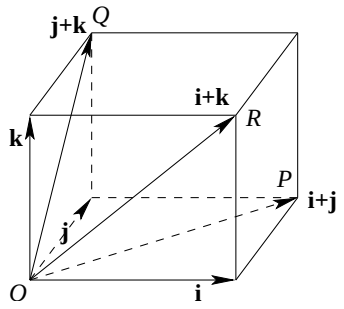
Since α, β, γ are not all 0, we have $k \neq 0$. So if we substitute (37) to (39) into (34) and divide by k we get the expanded version of (33).

Comment No. 16:

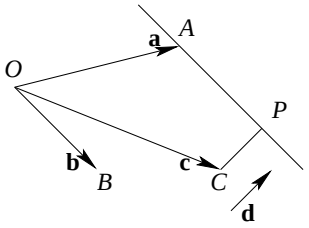
We now study some applications of vector methods to geometry. We study two kinds of problems, viz., those in which the data and/or the conclusion involves a vector and thereafter those where neither the data nor the conclusion refers to any vector but vectors are introduced as means.

As a typical problem of the first type, suppose three vectors of length $a, 2a$ and $3a$ meet at a point O. These vectors are directed outwards along the diagonals OP, OQ and OR of the three faces of a cube meeting at O. We want to find their resultant in the form $\mathbf{r} = x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$ and to calculate its magnitude. (JAT 1980)

Without loss of generality we may take the cube as the unit cube with one vertex at O and the remaining seven vertices at the points with position vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}, \mathbf{i}+\mathbf{j}, \mathbf{j}+\mathbf{k}, \mathbf{i}+\mathbf{k}$ and $\mathbf{i}+\mathbf{j}+\mathbf{k}$. (See Figure (a) below.) We take P, Q, R as the points with position vectors $\mathbf{i}+\mathbf{j}, \mathbf{j}+\mathbf{k}$ and $\mathbf{i}+\mathbf{k}$. (The problem does not specify this. By permutations, five other choices are possible and so the answer to the problem is not unique.) Now, a unit vector in the direction of OP is $\frac{1}{\sqrt{2}}(\mathbf{i}+\mathbf{j})$. So the vector of length a along OP is $\frac{a}{\sqrt{2}}(\mathbf{i}+\mathbf{j})$. Similarly, the other two vectors meeting at O are $\frac{2a}{\sqrt{2}}(\mathbf{j}+\mathbf{k})$ and $\frac{3a}{\sqrt{2}}(\mathbf{i}+\mathbf{k})$. Hence their resultant (which is just another name for the vector sum) is $\frac{a}{\sqrt{2}}((\mathbf{i}+\mathbf{j})+2(\mathbf{j}+\mathbf{k})+3(\mathbf{i}+\mathbf{k}) = \frac{4a}{\sqrt{2}}\mathbf{i}+\frac{3a}{\sqrt{2}}\mathbf{j}+\frac{5a}{\sqrt{2}}\mathbf{k}$.



(a)



(b)

The following problem deals with vectors in a plane. (JAT 1980)

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be the position vectors of points A, B, C in a plane. A straight line is drawn through A parallel to $\mathbf{b}$. Let P be the foot of the perpendicular from C onto this straight line. Let $\mathbf{d}$ be a vector perpendicular to $\mathbf{b}$. Find the vector equation of PC and the length of this perpendicular.

We first draw a diagram as in (b) above. Since $\mathbf{d}$ is perpendicular to $\mathbf{b}$ which is parallel to the line AP, we get that $\mathbf{d}$ is parallel to the line PC. So the vector equation of the line PC is $\mathbf{c}+t\mathbf{d}$, where t is a real parameter. Let t_0

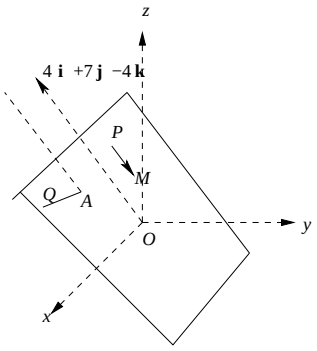
be the value of this parameter for which we get the point P, i.e., the position vector of P is $\mathbf{c} + t_0\mathbf{d}$. But since AP is parallel to $\mathbf{b}$ we also have that the position vector of P is $\mathbf{a} + s\mathbf{b}$ for some scalar s. So we get a vector equation

$$\mathbf{c} + t_0\mathbf{d} = \mathbf{a} + s\mathbf{b} \tag{40}$$

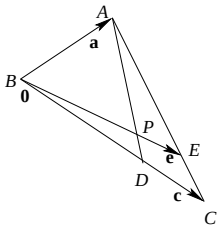
which is equivalent to a system of two scalar equations which can be solved to get the values of the unknowns t_0 and s. But our interest is only in t_0 which can be obtained by simply taking the dot product of both the sides with $\mathbf{d}$. Since $\mathbf{b} \cdot \mathbf{d} = 0$ we get $\mathbf{c} \cdot \mathbf{d} + t_0||\mathbf{d}||^2 = \mathbf{a} \cdot \mathbf{d}$ giving $t_0 = \dfrac{\mathbf{a} \cdot \mathbf{d} - \mathbf{c} \cdot \mathbf{d}}{||\mathbf{d}||^2}$. The length of the perpendicular PC is $||t_0\mathbf{d}||$ which is $\dfrac{|\mathbf{a} \cdot \mathbf{d} - \mathbf{c} \cdot \mathbf{d}|}{||\mathbf{d}||}$.

In three-dimensional coordinate geometry, a single linear equation in the coordinates (i.e., an equation of the form $ax + by + cz + d = 0$, where a, b, c are not all 0) represents a plane. A plane can be specified in various ways, e.g., by three points on it, or by a point and a line in it or by a pair of lines in it. In vector methods, the most convenient way to specify a plane is by the 'point, normal' form where we are given a point in the plane and a vector perpendicular to it (often called a normal to the plane). The following problem is based on this idea.

Find the equation of the plane passing through the point $(3, -2, 1)$ and perpendicular to the vector $4\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$. Let PM be the perpendicular from the point $P\ (1, 2, -1)$ to this plane. Find the length of PM. (JAT 1980)



(c)



(d)

Let $A = (3, -2, 1)$ and $Q = (x, y, z)$ be a general point. Then (see Figure (c) above) Q lies in the given plane if and only if the vector $\overrightarrow{AQ} = (x - 3)\mathbf{i} + (y + 2)\mathbf{j} + (z - 1)\mathbf{k}$ is perpendicular to the vector $4\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$, i.e., if and only if $4(x - 3) + 7(y + 2) - 4(z - 1) = 0$, or, $4x + 7y - 4z + 6 = 0$. So this is the equation of the plane. To find M, note that the line PM is parallel to the vector $4\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$ and hence every point on it has position vector of the form $(\mathbf{i} + 2\mathbf{j} - \mathbf{k}) + t(4\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}) = (4t + 1)\mathbf{i} + (7t + 2)\mathbf{j} + (-4t - 1)\mathbf{k}$ for some real number t. For this point to lie on the given plane, we must have

$4(4t+1)+7(7t+2)+4(4t+1)+6=0$, which gives $81t+28=0$, i.e., $t=-\dfrac{28}{81}$. This gives $\overrightarrow{PM}=-\dfrac{28}{81}(4\mathbf{i}+7\mathbf{j}-4\mathbf{k})$. Hence the length of the perpendicular PM is $\dfrac{28}{81}||(4\mathbf{i}+7\mathbf{j}-4\mathbf{k})||=\dfrac{28}{81}\times 9=\dfrac{28}{9}$.

When a plane is given in some other form, its equation can often be obtained by first converting it to a 'point, normal' form. For example, suppose we are given the coordinates of three (non-collinear) points, say, A, B and C in a plane. Then the vector $\overrightarrow{AB}\times\overrightarrow{AC}$ is normal to it. See Exercises (21.15), (21.16) and (21.28) for more applications of vectors to solid coordinate geometry problems involving lines and planes.

Comment No. 17:

We now turn to applications of vector methods to purely geometric problems where neither the hypothesis nor the conclusion contains any vectors but they are used as means. In Chapter 8 (see Comments 5 to 9 in it) we saw that vectors come in handy where the problem involves perpendiculars. Vectors can also be used to represent the angles between given directions (as was done in Comment No. 3 above where the problem was, in essence, a purely geometric problem). Another concept which is very convenient is the section formula mentioned after equation (15). Here is a typical application of it.

In a triangle ABC, D and E are points on BC and AC respectively such that $BD=2DC$ and $AE=3EC$. Let P be the point of intersection of AD and BE. Find BP/PE by vector methods. (JEE 1993)

We first take the position vectors of the various points involved. The choice of the origin is at our disposal and since our goal is to find the ratio BP/PE, it is convenient to take the origin at B. So let $\mathbf{a},\mathbf{c},\mathbf{d},\mathbf{e}$ and $\mathbf{p}$ be the position vectors of A, C, D, E and P w.r.t. B (see Figure (d) above). Because of the section formula we have

$$\mathbf{d}=\frac{2}{3}\mathbf{c} \text{ and } \mathbf{e}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{c} \tag{41}$$

Now let the ratio BP/PE be $\lambda:1$. Then again by the section formula and (41),

$$\mathbf{p}=\frac{\lambda}{\lambda+1}\mathbf{e}=\frac{\lambda}{4(\lambda+1)}(\mathbf{a}+3\mathbf{c}) \tag{42}$$

We need to get another expression for $\mathbf{p}$ in terms of $\mathbf{a}$ and $\mathbf{c}$ so that equating the two expressions for $\mathbf{p}$ we shall get a system of equations in which λ will be one of the unknowns. For this purpose we are forced to introduce another auxiliary unknown, say μ, equal to the ratio AP/PD. Then again by the section formula,

$$\mathbf{p}=\frac{1}{\mu+1}\mathbf{a}+\frac{\mu}{\mu+1}\mathbf{d}=\frac{1}{\mu+1}\mathbf{a}+\frac{2\mu}{3(\mu+1)}\mathbf{c} \tag{43}$$

Since the points A, B, C are non-collinear, the vectors $\mathbf{a}$ and $\mathbf{c}$ are linearly independent. So equating the coefficients of $\mathbf{a}$ and $\mathbf{c}$ in (42) and (43) we get

$$\frac{\lambda}{4(\lambda+1)} = \frac{1}{\mu+1} \quad \text{and} \quad \frac{3\lambda}{4(\lambda+1)} = \frac{2\mu}{3(\mu+1)} \tag{44}$$

which together give $\mu = \frac{9}{2}$ and $\lambda = \frac{8}{3}$. So $BP : PE = 8 : 3$. (For a slicker solution, eliminating $\mathbf{c}$ from the two equations in (41) gives $8\mathbf{e} = 2\mathbf{a} + 9\mathbf{d}$, or equivalently, $\frac{8}{11}\mathbf{e} = \frac{2}{11}\mathbf{a} + \frac{9}{11}\mathbf{d}$. The L.H.S. is a point on the segment BE while the R.H.S. is a point on the segment AD. So, both the sides equal $\mathbf{p}$ which implies that P divides BE in the ration $8 : 3$. The problem can also be done by applying the well-known **Menelaus theorem** in pure geometry to the triangle ADC whose sides are cut by the line BPE. The product of the the three ratios $BP/PE, AE/AC$ and CD/DB equals 1 which gives $\lambda \times \frac{3}{4} \times \frac{1}{2} = 1$. Turning the tables around, one can prove Menelaus theorem using the section formula. Using the section formula we also get a quick proof of the concurrency of the three medians of a triangle as will be pointed out in Exercise (21.18).)

Comment No. 18:

Finally, we make good a promise given at the end of Comment No. 12 that the problem done there, viz., that of showing that the vector

$$\mathbf{u} = (\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) + (\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c}) \tag{45}$$

is parallel to $\mathbf{a}$ can also be done in a manner which is cyclically symmetric w.r.t. $\mathbf{b}, \mathbf{c}, \mathbf{d}$ (unlike the proof given earlier). Let us first suppose that the vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are not coplanar. Then they form a basis and so we can write

$$\mathbf{a} = \alpha\mathbf{b} + \beta\mathbf{c} + \gamma\mathbf{d} \tag{46}$$

for some scalars α, β, γ. Taking dot products of both the sides with $\mathbf{c} \times \mathbf{d}$, $\mathbf{d} \times \mathbf{b}$ and $\mathbf{b} \times \mathbf{c}$ (and noting that $(\mathbf{c}\ \mathbf{d}\ \mathbf{b}) = (\mathbf{d}\ \mathbf{b}\ \mathbf{c}) = (\mathbf{b}\ \mathbf{c}\ \mathbf{d})$) we get, respectively

$$(\mathbf{a}\ \mathbf{c}\ \mathbf{d}) = \alpha(\mathbf{b}\ \mathbf{c}\ \mathbf{d}), \quad (\mathbf{a}\ \mathbf{d}\ \mathbf{b}) = \beta(\mathbf{b}\ \mathbf{c}\ \mathbf{d}), \quad \text{and} \quad (\mathbf{a}\ \mathbf{b}\ \mathbf{c}) = \gamma(\mathbf{b}\ \mathbf{c}\ \mathbf{d}) \tag{47}$$

We now duplicate equation (28) obtained above and also write its two cyclic analogues. Combining them with (47) we get

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = (\mathbf{a}\ \mathbf{c}\ \mathbf{d})\mathbf{b} - (\mathbf{b}\ \mathbf{c}\ \mathbf{d})\mathbf{a} = (\mathbf{b}\ \mathbf{c}\ \mathbf{d})(\alpha\mathbf{b} - \mathbf{a}) \tag{48}$$

$$(\mathbf{a} \times \mathbf{c}) \times (\mathbf{d} \times \mathbf{b}) = (\mathbf{a}\ \mathbf{d}\ \mathbf{b})\mathbf{c} - (\mathbf{b}\ \mathbf{c}\ \mathbf{d})\mathbf{a} = (\mathbf{b}\ \mathbf{c}\ \mathbf{d})(\beta\mathbf{c} - \mathbf{a}) \tag{49}$$

$$(\mathbf{a} \times \mathbf{d}) \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a}\ \mathbf{b}\ \mathbf{c})\mathbf{d} - (\mathbf{b}\ \mathbf{c}\ \mathbf{d})\mathbf{a} = (\mathbf{b}\ \mathbf{c}\ \mathbf{d})(\gamma\mathbf{d} - \mathbf{a}) \tag{50}$$

Adding, we get $\mathbf{u} = (\mathbf{b}\ \mathbf{c}\ \mathbf{d})(\alpha\mathbf{b} + \beta\mathbf{c} + \gamma\mathbf{d} - 3\mathbf{a})$. By (46) this equals $-2(\mathbf{b}\ \mathbf{c}\ \mathbf{d})\mathbf{a}$ which is evidently parallel to $\mathbf{a}$.

Note that we have crucially used the hypothesis that the vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are not coplanar. Let us now see how we can modify the solution if this condition is not satisfied, i.e., if $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar. If they are not only coplanar but collinear, then all the three cross products $\mathbf{b} \times \mathbf{c}$, $\mathbf{c} \times \mathbf{d}$ and $\mathbf{d} \times \mathbf{b}$ vanish and so

$\mathbf{u} = \mathbf{0}$ which is parallel to $\mathbf{a}$ (in a degenerate sense, because the null vector has no fixed direction and can be considered to be parallel to any vector). So let us suppose that $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar but not collinear. In that case they span a plane. Resolve the vector $\mathbf{a}$ along and perpendicular to this plane, i.e., write $\mathbf{a} = \mathbf{v} + \mathbf{w}$, where $\mathbf{v}$ is in the plane spanned by $\mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ and $\mathbf{w}$ is perpendicular to it. Then $(\mathbf{v}\ \mathbf{c}\ \mathbf{d}) = 0$ as $\mathbf{v}, \mathbf{c}, \mathbf{d}$ are coplanar and so $(\mathbf{a}\ \mathbf{c}\ \mathbf{d}) = (\mathbf{v}\ \mathbf{c}\ \mathbf{d}) + (\mathbf{w}\ \mathbf{c}\ \mathbf{d}) = (\mathbf{w}\ \mathbf{c}\ \mathbf{d})$. Similarly, $(\mathbf{a}\ \mathbf{d}\ \mathbf{b}) = (\mathbf{w}\ \mathbf{d}\ \mathbf{b})$ and $(\mathbf{a}\ \mathbf{b}\ \mathbf{c}) = (\mathbf{w}\ \mathbf{b}\ \mathbf{c})$. Now, the first equalities in (48) to (50) are still true. Because of what we have just said, and because $(\mathbf{b}\ \mathbf{c}\ \mathbf{d})$ is now 0, we get

$$\mathbf{u} = (\mathbf{w}\ \mathbf{c}\ \mathbf{d})\mathbf{b} + (\mathbf{w}\ \mathbf{d}\ \mathbf{b})\mathbf{c} + (\mathbf{w}\ \mathbf{b}\ \mathbf{c})\mathbf{d} \tag{51}$$

From this it can be shown that $\mathbf{u} = \mathbf{0}$. We leave this as an exercise (Exercise (21.22)). It then follows that even in this case, $\mathbf{u}$ is parallel to $\mathbf{a}$ (again in a degenerate sense).

There is yet another technique by which we can prove the result in the degenerate case where $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar, after we have proved it for the case where they are not coplanar. It is called a **perturbation technique**. In fact it is more to acquaint you with this technique that we are presenting this alternate solution. In layman's language, a perturbation means a slight change. A general definition of a perturbation is beyond our scope. But an intuitive understanding of it is sufficient. A quality which is unchanged under a small perturbation is said to be **stable**. For example, suppose ABC is a triangle. Let us move the vertices A, B, C slightly to get another triangle, say, $A'B'C'$. If ABC is equilateral then in general $A'B'C'$ will not be so. But if ABC is not equilateral and the perturbation is sufficiently small, then $A'B'C'$ cannot be equilateral either. So being equilateral is an unstable property while not being equilateral is a stable property. Choosing smaller and smaller perturbations, we can express a given equilateral triangle as the limit of non-equilateral triangles.

We do something similar about coplanarity of the vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ in the problem above. If we perturb these vectors slightly, the new vectors will, in general, not be coplanar. In fact it suffices to perturb just one vector. Again we assume that $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar but not collinear. Let $\mathbf{p}$ be a unit vector perpendicular to the plane spanned by $\mathbf{b}, \mathbf{c}, \mathbf{d}$. Then the vectors $\mathbf{p}, \mathbf{c}, \mathbf{d}$ are not coplanar and so $(\mathbf{p}\ \mathbf{c}\ \mathbf{d}) \neq 0$. Now let t be a non-zero real number and let $\mathbf{b}'$ be the vector $\mathbf{b} + t\mathbf{p}$. Then $(\mathbf{b}'\ \mathbf{c}\ \mathbf{d}) = (\mathbf{b}\ \mathbf{c}\ \mathbf{d}) + t(\mathbf{p}\ \mathbf{c}\ \mathbf{d}) = t(\mathbf{p}\ \mathbf{c}\ \mathbf{d}) \neq 0$ and so $\mathbf{b}', \mathbf{c}, \mathbf{d}$ are not coplanar. Let $\mathbf{u}'$ be the vector obtained by replacing $\mathbf{b}$ by $\mathbf{b}'$ in (45). Then because we have already done the case of non-coplanar vectors, we have

$$\mathbf{u}' = -2(\mathbf{b}'\ \mathbf{c}\ \mathbf{d})\mathbf{a} \tag{52}$$

We now take the limits of both the sides as t tends to 0. Since $\mathbf{b}' = \mathbf{b} + t\mathbf{p}$, it is evident that $\mathbf{b}'$ tends to $\mathbf{b}$ as t tends to 0. It can then be shown that as t tends to 0, $\mathbf{u}'$ tends to $\mathbf{u}$ and $(\mathbf{b}'\ \mathbf{c}\ \mathbf{d})$ tends to $(\mathbf{b}\ \mathbf{c}\ \mathbf{d})$. We shall not prove this. What is needed basically is that the dot product and the cross product are continuous functions of their arguments. The limiting case of (52) is $\mathbf{u} = -2(\mathbf{b}\ \mathbf{c}\ \mathbf{d})\mathbf{a}$ which gives the result even in the 'degenerate' case where $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar. (In fact, we have shown that in this case, $\mathbf{u} = \mathbf{0}$.)

In general, in problems where the associated functions are continuous, perturbation techniques can be applied to study degenerate cases by approximating them by non-degenerate cases. Their study is beyond our scope. However, here is a problem where a simple property of continuous functions gives the solution.

Let $\vec{A}(t) = f_1(t)\hat{i} + f_2(t)\hat{j}$ and $\vec{B}(t) = g_1(t)\hat{i} + g_2(t)\hat{j}$, $t \in [0,1]$, where f_1, f_2, g_1, g_2 are continuous functions. If $\vec{A}(t)$ and $\vec{B}(t)$ are non-zero vectors for all t and $\vec{A}(0) = 2\hat{i} + 3\hat{j}, \vec{A}(1) = 6\hat{i} + 2\hat{j}, \vec{B}(0) = 3\hat{i} + 2\hat{j}$ and $\vec{B}(1) = 2\hat{i} + 6\hat{j}$, then show that $\vec{A}(t)$ and $\vec{B}(t)$ are parallel for some t. (JEE 2001)

For a solution, note that $\vec{A}(t) \times \vec{B}(t) = h(t)\vec{k}$ where $h(t) = f_1(t)g_2(t) - f_2(t)g_1(t)$ is a continuous function of t. From the data we have $h(0) = 2 \times 2 - 3 \times 3 = -5 < 0$. Also, $h(1) = 6 \times 6 - 2 \times 2 = 32 > 0$. Hence by the Intermediate Value Property of continuous functions, $h(t) = 0$ for some $t \in [0,1]$. But that means $\vec{A}(t) \times \vec{B}(t) = \vec{0}$. Since $\vec{A}(t)$ and $\vec{B}(t)$ are given to be non-zero, this implies that they are parallel.

EXERCISES

21.1 (a) How many vectors of unit length are perpendicular to the vectors $\vec{a} = (1,1,0)$ and $\vec{b} = (0,1,1)$? (1987)

(b) Let the vectors $2\mathbf{i} - \mathbf{k}$ and $\mathbf{j} + 2\mathbf{k}$ be the adjacent sides of a triangle. Find the area of the triangle. (JAT 1979)

(c) Find a unit vector perpendicular to the plane determined by the points $P\ (1,-1,2)$, $Q\ (2,0,-1)$ and $R\ (0,2,1)$. (1983, 1994)

(d) Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 10\vec{a} + 2\vec{b}$, and $\overrightarrow{OC} = \vec{b}$, where O, A and C are non-collinear points. Let p denote the area of the quadrilateral $OABC$ and let q denote the area of the parallelogram with OA and OC as adjacent sides. If $p = kq$, find the value of k. (1997)

(e) Let $A_1, A_2, \ldots, A_n$ be the vertices of a regular n-gon with centre O. Let P be any point. Prove that

(i) $\overrightarrow{PA_1} + \overrightarrow{PA_2} + \ldots + \overrightarrow{PA_n} = -n\ \overrightarrow{OP}$

(ii) $\overrightarrow{OA_1} \times \overrightarrow{OA_2} + \overrightarrow{OA_2} \times \overrightarrow{OA_3} + \ldots + \overrightarrow{OA_{n-1}} \times \overrightarrow{OA_n} = (n-1)\overrightarrow{OA_1} \times \overrightarrow{OA_2}$ (1982)

21.2 (a) For three vectors $\vec{u}, \vec{v}, \vec{w}$, which of the following expressions is not equal to any of the remaining three? (1998)

(A) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (B) $(\vec{v} \times \vec{w}) \cdot \vec{u}$ (C) $\vec{v} \cdot (\vec{u} \times \vec{w})$ (D) $(\vec{u} \times \vec{v}) \cdot \vec{w}$

(b) If $\mathbf{X} \cdot \mathbf{A} = \mathbf{X} \cdot \mathbf{B} = \mathbf{X} \cdot \mathbf{C} = 0$ for some non-zero vector $\mathbf{X}$, determine the value of $[\mathbf{A}\ \mathbf{B}\ \mathbf{C}]$. (JAT 1980, 1983)

(c) Find the area of the triangle with vertices $A\ (1,-1,2)$, $B\ (2,1,-1)$ and $C\ (3,-1,2)$. (1983)

(d) Find the volume of the parallelopiped whose sides are given by the vectors $2\mathbf{i} - 3\mathbf{j}, \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $3\mathbf{i} - \mathbf{k}$. (1983)

(e) If $\vec{A}, \vec{B}, \vec{C}$ are three non-coplanar vectors, find the value of the expression $\dfrac{\vec{A}\cdot\vec{B}\times\vec{C}}{\vec{C}\cdot\vec{A}\times\vec{B}} + \dfrac{\vec{B}\cdot\vec{A}\times\vec{C}}{\vec{C}\cdot\vec{A}\times\vec{B}}$. (1985)

(f) Using (13) show that $(\vec{A}\times\vec{B})\cdot(\vec{B}\times\vec{C})\times(\vec{C}\times\vec{A}) = (\vec{A}\cdot\vec{B}\times\vec{C})^2$ (JAT 1979)

(g) Let $\vec{V} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{W} = \vec{i} + 3\vec{k}$. If $\vec{U}$ is a unit vector, then find the maximum value of the scalar triple product $[\vec{U}\ \vec{V}\ \vec{W}]$. (2002)

(h) Determine which, if any, of the following expressions equals the scalar $\vec{A}\cdot(\vec{B}+\vec{C})\times(\vec{A}+\vec{B}+\vec{C})$. (1981)

(a) 0 (b) $[\vec{A}\ \vec{B}\ \vec{C}] + [\vec{B}\ \vec{C}\ \vec{A}]$ (c) $[\vec{A}\ \vec{B}\ \vec{C}]$.

21.3 (a) If a, b, c are distinct, prove that the vectors $(1, a, a^2), (1, b, b^2), (1, c, c^2)$ are linearly independent.

(b) If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and the vectors $A = (1, a, a^2),\ B = (1, b, b^2),$ $C = (1, c, c^2)$ are non-coplanar, find the value of abc. (1985)

21.4 Find 3-dimensional vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ satisfying $\vec{v}_1\cdot\vec{v}_1 = 4,\ \vec{v}_1\cdot\vec{v}_2 = -2,\ \vec{v}_1\cdot\vec{v}_3 = 6,\ \vec{v}_2\cdot\vec{v}_2 = 2,\ \vec{v}_2\cdot\vec{v}_3 = -5$ and $\vec{v}_3\cdot\vec{v}_3 = 29$. (2001)

21.5 (a) If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other, then find the angle between $\vec{a}$ and $\vec{b}$. (2002)

(b) Let $\vec{A}, \vec{B}$ and $\vec{C}$ be unit vectors. Suppose that $\vec{A}\cdot\vec{B} = \vec{A}\cdot\vec{C} = 0$ and that the angle between $\vec{B}$ and $\vec{C}$ is $\frac{\pi}{6}$. Is $\vec{A} = \pm 2(\vec{B}\times\vec{C})$? (1981)

(c) Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\mathbf{a}\times(\mathbf{a}\times\mathbf{c}) + \mathbf{b} = \mathbf{0}$, find the acute angle between $\mathbf{a}$ and $\mathbf{c}$. (1997*)

(d) A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. Determine the angle between the vector $\vec{a}$ and the vector $\hat{i} - 2\hat{j} + 2\hat{k}$. (1996)

(e) Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, then find the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$. (1986)

(f) Let $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} + \vec{j}$. If $\vec{c}$ is a vector such that $\vec{a}\cdot\vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 30°, find the value of $|(\vec{a}\times\vec{b})\times\vec{c}|$. (1999)

21.6 A vector $\vec{a}$ has components $2p$ and 1 w.r.t. a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter-clockwise sense. If, w.r.t. the new system, $\vec{a}$ has components $p+1$ and 1, determine the possible values of p. (1986)

21.7 (a) If $\vec{C} = \vec{A} \times \vec{B}, \vec{B} = \vec{A} \times \vec{C}$ and $\vec{A} \neq \vec{0}$, then prove that $\vec{B} = \vec{C} = \vec{0}$. (JAT 1979)

(b) Prove (13) by vector methods.

(c) Let $\mathbf{A}, \mathbf{B}$ be vectors with $\mathbf{A} \neq \mathbf{0}$. Prove that the equation $\mathbf{A} \times \mathbf{X} = \mathbf{B}$ has a solution if and only if $\mathbf{A} \perp \mathbf{B}$ and that when this is the case, every solution is of the form $\mathbf{U} + \lambda\mathbf{A}$, where $\mathbf{U}$ is a fixed vector and λ is arbitrary. Do the problem both algebraically and by vector methods.

(d) If $\vec{A} = (1,1,1)$, $\vec{C} = (0,1,-1)$ are given vectors, find a vector $\vec{B}$ satisfying the equations $\vec{A} \times \vec{B} = \vec{C}$ and $\vec{A} \cdot \vec{B} = 3$. (1985)

(e) For any three vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$, prove that

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^2 = \begin{vmatrix} \mathbf{a}\cdot\mathbf{a} & \mathbf{a}\cdot\mathbf{b} & \mathbf{a}\cdot\mathbf{c} \\ \mathbf{b}\cdot\mathbf{a} & \mathbf{b}\cdot\mathbf{b} & \mathbf{b}\cdot\mathbf{c} \\ \mathbf{c}\cdot\mathbf{a} & \mathbf{c}\cdot\mathbf{b} & \mathbf{c}\cdot\mathbf{c} \end{vmatrix}$$

[*Hint*: Use (13) and the fact that for any two vectors $\mathbf{u}$ and $\mathbf{v}$, $(\mathbf{u}\cdot\mathbf{v})^2 = ||\mathbf{u}||^2||\mathbf{v}||^2 - ||\mathbf{u}\times\mathbf{v}||^2$.]

(f) Let $\mathbf{a} \neq \mathbf{0}$. If $\mathbf{a}\cdot\mathbf{u} = \mathbf{a}\cdot\mathbf{v}$ and $\mathbf{a}\times\mathbf{u} = \mathbf{a}\times\mathbf{v}$, prove that $\mathbf{u} = \mathbf{v}$. Show by examples that neither equation implies the equality by itself. (In the language of Exercise 4.12 (a), neither the dot nor the cross product satisfies the cancellation law. But the two together do.)

(g) Prove that if $\mathbf{a}\cdot\mathbf{u} = \mathbf{a}\cdot\mathbf{v}, \mathbf{b}\cdot\mathbf{u} = \mathbf{b}\cdot\mathbf{v}$ and $\mathbf{c}\cdot\mathbf{u} = \mathbf{c}\cdot\mathbf{v}$ for three linearly independent vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ then $\mathbf{u} = \mathbf{v}$. (This makes it easy to establish the equality of two vectors.)

21.8 Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors and $\vec{p}, \vec{q}, \vec{r}$ be the vectors defined by the relations $\vec{p} = \dfrac{\vec{b}\times\vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]}$, $\vec{q} = \dfrac{\vec{c}\times\vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]}$, $\vec{r} = \dfrac{\vec{a}\times\vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]}$. Then find the numerical value of the expression $(\vec{a}+\vec{b})\cdot\vec{p} + (\vec{b}+\vec{c})\cdot\vec{q} + (\vec{c}+\vec{a})\cdot\vec{r}$. (1988)

21.9 Solve the equation $\mathbf{p}\times((\mathbf{x}-\mathbf{q})\times\mathbf{p}) + \mathbf{q}\times((\mathbf{x}-\mathbf{r})\times\mathbf{q}) + \mathbf{r}\times((\mathbf{x}-\mathbf{p})\times\mathbf{r}) = \mathbf{0}$ for $\mathbf{x}$ where $\mathbf{p}, \mathbf{q}, \mathbf{r}$ are three mutually perpendicular vectors of the same magnitude. (1997*)

21.10 If $\vec{b}, \vec{c}$ are any two non-collinear unit vectors and $\vec{a}$ is any vector, simplify the expression $(\vec{a}\cdot\vec{b})\,\vec{b} + (\vec{a}\cdot\vec{c})\,\vec{c} + \frac{\vec{a}\cdot(\vec{b}\times\vec{c})}{|\vec{b}\times\vec{c}|}(\vec{b}\times\vec{c})$. (1996)

21.11 Let $\vec{a}$ and $\vec{b}$ be two non-collinear unit vectors. If $\vec{u} = \vec{a} - (\vec{a}\cdot\vec{b})\vec{b}$ and $\vec{v} = \vec{a}\times\vec{b}$, which of the following expressions is(are) equal to $|\vec{v}|$? (1999)

(A) $|\vec{u}|$ (B) $|\vec{u}| + |\vec{u}\cdot\vec{a}|$ (C) $|\vec{u}| + |\vec{u}\cdot\vec{b}|$ (D) $|\vec{u}| + \vec{u}\cdot(\vec{a}+\vec{b})$.

21.12 (a) Determine the value of a for which the points with position vectors $60\mathbf{i}+3\mathbf{j}, 40\mathbf{i}-8\mathbf{j}$ and $a\mathbf{i}-52\mathbf{j}$ are collinear. (1983)

(b) If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=4\hat{i}+\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+\alpha\hat{j}+\beta\hat{k}$ are linearly dependent vectors and $|\vec{c}|=\sqrt{3}$, determine all possible values of α and β. (1998)

(c) Let a, b, c be distinct non-negative real numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ lie in a plane, find how c is related to a and b. (1993)

(d) A, B, C, D are points with position vectors $3\hat{i}-2\hat{j}-\hat{k}$, $2\hat{i}+3\hat{j}-4\hat{k}$, $-\hat{i}+2\hat{j}+2\hat{k}$ and $4\hat{i}+5\hat{j}+\lambda\hat{k}$ respectively. If the points A, B, C, D lie on a plane, find the value of λ. (1986)

21.13 (a) Let $\vec{a}=2\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$. Find a unit vector $\vec{c}$ which is coplanar with $\vec{a}$ and $\vec{b}$ and is perpendicular to $\vec{a}$. (1999)

(b) Let $\vec{a}=2\hat{i}-\hat{j}+\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-2\hat{k}$. Find which of the following vectors is (are) in the plane of $\vec{b}$ and $\vec{c}$ and has (have) projection on of magnitude $\sqrt{(2/3)}$ on $\vec{a}$. (1993)

(A) $2\hat{i}+3\hat{j}-3\hat{k}$ (B) $2\hat{i}+3\hat{j}+3\hat{k}$ (C) $-2\hat{i}-\hat{j}+5\hat{k}$ (D) $2\hat{i}+\hat{j}+5\hat{k}$.

21.14 The position vectors of the vertices A, B, C of a tetrahedron $ABCD$ are $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}$ and $3\hat{i}$ respectively. The altitude from vertex D to the opposite face ABC meets the median line through A of the triangle ABC at a point E. If the length of the side AD is 4 and the volume of the tetrahedron is $\frac{2\sqrt{2}}{3}$, find the position vector of the point E for all its possible positions. (1996)

21.15 A, B, C, D are the points $(1,-2,-1), (4,0,-3), (1,2,-1)$ and $(2,-4,-5)$ respectively. Find the shortest distance between the lines AB and CD by vector methods. (JAT 1979)

21.16 Two straight lines whose equations are $6x+4y-5z=4, x-5y+2z=12$ and $\frac{x-9}{2}=\frac{y+4}{-1}=\frac{z-5}{1}$ are given. Find the angle between them and also the equation of the plane containing them. (JAT 1980)

21.17 $OACB$ is a parallelogram, D is the mid-point of OA. Using vector methods, prove that BD and CO intersect in the same ratio. Determine this ratio. (1988)

21.18 Using the section formula prove that the medians of any triangle are concurrent and that the point of concurrence divides each median in the ratio $2:1$.

21.19 Show, by vector methods, that the angular bisectors of a triangle are concurrent and find an expression for the position vector of the point of concurrency in terms of the position vectors of the vertices. (2001)

21.20 In a triangle OAB, E is the mid-point of OB and D is a point on AB such that $AD : DB = 2 : 1$. If OD and AE intersect at P, determine the ratio $OP : PD$ using vector methods. (1989)

21.21 Prove by vectors methods that the point of intersection of the diagonals of a trapezium lies on the line passing through the mid-points of the parallel sides. (You may assume that the trapezium is not a parallelogram.) (1998)

21.22 Suppose three vectors $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are coplanar but not collinear and $\mathbf{w}$ is a vector perpendicular to the plane spanned by them. Prove that $(\mathbf{w}\ \mathbf{c}\ \mathbf{d})\mathbf{b} + (\mathbf{w}\ \mathbf{d}\ \mathbf{b})\mathbf{c} + (\mathbf{w}\ \mathbf{b}\ \mathbf{c})\mathbf{d} = \mathbf{0}$. (Hint : Without loss of generality, assume that $\mathbf{b}, \mathbf{c}, \mathbf{d}$ are unit vectors in the xy-plane).

21.23 For two lines in a plane, prove that being parallel is an unstable property. Similarly, for three lines in a plane, prove that their being concurrent is an unstable property. (The basic idea here is the same as that in Exercise (1.36), where we defined what is meant by a set of lines in a plane to be in general position.)

21.24 For any two vectors $\mathbf{u}$ and $\mathbf{v}$, prove that

(a) $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$. When does equality hold? (Predictably, this result is called the **triangle inequality** for vectors.)

(b) $||\mathbf{u} - \mathbf{v}|| \geq |\ ||\mathbf{u}|| - ||\mathbf{v}||\ |$. (This is an equivalent, although less frequently used, form of the triangle inequality.)

21.25 A complex number $x+iy$ can be identified with a pair of real numbers, viz., (x, y). When so done, the addition and the multiplication of the complex numbers take the form $(a, b) + (c, d) = (a + b, c + d)$ and $(a, b)(c, d) = (ac - bd, ad + bc)$. A real number x can be identified with a complex number whose imaginary part is 0, i.e., with the number $(x, 0)$. Similarly, a purely imaginary number iy corresponds to the pair $(0, y)$ whose real part is 0. (See Chapter 2, Comment No. 19 for details.)

Analogously, we define a **quaternion** as an ordered pair of the form $(u_0, \mathbf{u})$, where u_0 is a real number and $\mathbf{u}$ is a (three dimensional) vector. For brevity, let us denote such a quaternion simply by u. We may call u_0 and $\mathbf{u}$ as the scalar and the vector components of u respectively. A real number can be identified with a quaternion whose vector component is $\mathbf{0}$, i.e., with a quaternion of the form $(a, \mathbf{0})$. Similarly a vector can be identified as a quaternion whose scalar component is 0. Thus the system of quaternions includes both real numbers and three-dimensional vectors.

Given two such quaternions $u = (u_0, \mathbf{u})$ and $v = (v_0, \mathbf{v})$ we define their addition $u + v$ in the expected manner, i.e., as the quaternion $(u_0 + v_0, \mathbf{u} + \mathbf{v})$. However, their product uv is defined by

$$uv = (u_0, \mathbf{u})(v_0, \mathbf{v}) = (u_0 v_0 - \mathbf{u} \cdot \mathbf{v}, u_0\mathbf{v} + v_0\mathbf{u} + \mathbf{u} \times \mathbf{v})$$

Note the formal resemblance with multiplication of complex numbers. Prove that:

(a) the quaternionic addition and multiplication are both associative and that the former is distributive over the latter (i.e., for any three quaternions u, v, w, $u(v+w) = uv + uw$).

(b) the quaternionic addition is commutative but the multiplication is not so. However, a scalar quaternion, say $u = (u_0, \mathbf{0})$ commutes with every quaternion, i.e., $uv = vu$ for every quaternion v.

(c) the zero quaternion $0 = (0, \mathbf{0})$ and the unit quaternion $1 = (1, \mathbf{0})$ are identities of addition and multiplication respectively, i.e., $u + 0 = u$ and $u1 = u = 1u$ for every quaternion u.

(d) for every quaternion u, the quaternion $-u = (-u_0, -\mathbf{u})$ is the additive inverse of u, i.e., $u + (-u) = 0$.

(Most of these are ritualistic verifications except possibly for the associativity of multiplication and distributivity of addition over multiplication. With all these properties we say technically that the quaternions form a ring. It is equally customary to denote a quaternion u by a formal sum of the form $u_0 + u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and then define multiplication by rules like $\mathbf{i} \times \mathbf{j} = \mathbf{k} = -(\mathbf{j} \times \mathbf{i})$. But then the formula for the complete multiplication as well as the verification of the properties becomes lengthy.)

21.26 Given a quaternion $u = (u_0, \mathbf{u})$ define the **norm** $N(u)$ and the **conjugate** $\bar{u}$ of u to be the real number $\sqrt{u_0^2 + ||\mathbf{u}||^2}$ and the quaternion $(u_0, -\mathbf{u})$ respectively. (Again note the formal resemblance with the absolute value and the conjugate of a complex number.) Prove that

(a) for any two quaternions u and v, $\overline{u+v} = \bar{u} + \bar{v}$ and $\overline{uv} = \bar{v}\bar{u}$ (and not $\bar{u}\bar{v}$ as one might expect)

(b) $N(u) = \sqrt{u\bar{u}} = \sqrt{\bar{u}u}$ for every quaternion u

(c) $N(u) = 0$ if and only if $u = 0$, the zero quaternion

(d) the norm is multiplicative, i.e., for any two quaternions u and v, we have $N(uv) = N(u)N(v)$

(e) every non-zero quaternion u has a multiplicative inverse, i.e., a quaternion v such that $uv = vu = 1$. (As a result, the quaternions form what is called a division ring, basically a ring in which division is possible. The quaternions are important in number theory and also in mechanics. Using quaternions, one can, for example, prove the celebrated four square theorem of Lagrange that every positive integer can be expressed as a sum of four squares. The role of quaternions in theorems like this is not very surprising since if we write a quaternion $u = (u_0, \mathbf{u})$ as $u_0 + u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$, then $N(u)$ is simply $\sqrt{u_0^2 + u_1^2 + u_3^2 + u_3^2}$.)

21.27 For any two (three-dimensional) vectors $\mathbf{u}$ and $\mathbf{v}$, prove that $(1 + |\mathbf{u}|^2)(1 + |\mathbf{v}|^2) = (1 - \mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} + \mathbf{v} + (\mathbf{u} \times \mathbf{v})|^2$. (This follows from (d) of the last exercise. But a direct proof is easier.) (1998)

21.28 Solid coordinate geometry is generally not studied as a separate topic at the JEE level. Analogy with plane coordinate geometry and the use of vectors often suffices as we already saw in Comment No. 16. An important difference is that in the plane, the direction of a line is specified by a single number, viz. its slope. In the space, this is not possible. Instead, we talk of the **direction ratios** of a line L, defined as the components of any vector $\mathbf{u}$ parallel to L. They are not unique. If $\mathbf{u}$ is a unit vector, then they are unique upto a $\pm$ sign and are called the **direction cosines** of the line.

(a) If α, β, γ are the angles L makes with the axes, then prove that the direction cosines of L are $(\cos\alpha, \cos\beta, \cos\gamma)$ (which justifies the name). Show that any two of these angles determine the third.

(b) If the direction ratios of two lines are (l_1, m_1, n_1) and (l_2, m_2, n_2) respectively, find an expression for the angle between their diections.

(c) Obtain an expression for the perpendicular distance of a point $P_1 = (x_1, y_1, z_1)$ from (i) a plane $ax + by + cz + d = 0$ (ii) a line L passing through a point $P_0 = (x_0, y_0, z_0)$ and having direction cosines (l, m, n).

(d) Using vector methods, show that the segments joining the vertices of a tetrahedron and the centroids of the opposite faces are concurrent and further that they are divided in equal ratios by this point of concurrence. (This is in direct analogy with the concurrence of the medians of a triangle.)

*(e) If $OABC$ is a tetrahedron, prove that its four altitudes are concurrent if and only if $OA^2+BC^2 = OB^2+CA^2 = OC^2+AB^2$. [*Hint:* Taking the origin at O, the normals to the faces are $\mathbf{b}\times\mathbf{c}, \mathbf{c}\times\mathbf{a}, \mathbf{a}\times\mathbf{b}$ and $\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}+\mathbf{a}\times\mathbf{b}$. Here the analogy with a triangle fails. So one should be wary in generalising from plane to solid geometry.]

21.29 Let $\vec{u} = \vec{i} + \vec{j} + 3\vec{k}$, $\vec{v} = 3\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{w} = -4\vec{i} + 5\vec{j}$. Determine the values of λ for which the equation $x\vec{u} + y\vec{v} + z\vec{w} = \lambda(x\vec{i} + y\vec{j} + z\vec{w})$ has a non-trivial solution, i.e., a solution where not all of x, y, z are zero. (1982)

(Translated in terms of linear equations, the problem asks to find those values of λ for which there exists a non-trivial solution to the system

$$\mathbf{A}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \end{bmatrix} \tag{53}$$

where $\mathbf{A}$ is the 3×3 matrix $\begin{bmatrix} 1 & 3 & -4 \\ 1 & -3 & 5 \\ 3 & 1 & 0 \end{bmatrix}$. Such values are called **eigenvalues** of the matrix $\mathbf{A}$ and are very important in applications. It is easy to show that the eigenvalues are the same as the characteristic roots of the matrix $\mathbf{A}$ as defined in Exercise (2.37).)

Chapter 22

FINITISTIC PROBABILITY

The uncertainties of life (the only certain thing about which is death) make probability a fascinating subject even for a layman. Thus a farmer wants to know about the chances of timely rain, a gambler wants to know his chances of winning the next round and an insurance company is interested in the probability that an insured person will live for the next five years or so. (The words 'chance' or 'chances' or 'likelihood' are also used, if somewhat colloquially, instead of 'probability' which is rather bookish. The phrase 'odds against' also conveys probability. If the odds against an event are $a : b$, it means the probability of its occurrence is $\frac{a}{a+b}$. Notations also vary. The probability of an event A is denoted variously by $P(A), Pr(A)$ or by $P_r(A)$.)

We already saw in Chapter 1 how counting problems can be posed as probability problems under the assumption that the set of all possible cases is finite and that all these cases are equally likely. The situation where the set of all possibilities (or the 'sample space' as it is formally called) is finite but all its elements are not necessarily equally likely falls under the present chapter. We study certain standard techniques such as the conditional and complementary probabilities, the conjunction and disjunction of events, Venn diagrams, tree diagrams. The concept of independence of events is introduced in Comment No. 7. In Comment No. 13 we study an important special type of probability problems where a certain experiment or 'trial' (which could be a toss of a coin or a rolling of a die) is carried out in succession, independently of each other. The calculations of probabilities in such problem involve the binomial coefficients and so they are called binomial probabilities. In Comment No. 14 we do a problem where a probability is found using a recurrence relation.

Probability problems are asked every year in JEE. If you have clear thinking, they are quick scoring questions as the other computational work is often minor.

Main Problem : Two players A and B play a chess match which consists of a series of games. Whoever first wins two (not necessarily consecutive) games wins the match. Suppose the probabilities of A's winning, losing and drawing a game against B are, respectively, 0.5, 0.3 and 0.2. Then the probability that A wins the match at the end of the tenth game (but not earlier), written in the standard decimal notation, is ..... .

First Hint: The outcome of the tenth game is a win for A. Consider the possible outcomes of the first nine games.

Second Hint: Divide the possible outcomes of the first nine games into two classes depending upon B's winning a game.

Solution: The probability that A wins the tenth game is 0.5. To win the match with the tenth game A has to win exactly one out of the first nine games. The remaining 8 are either all draws or exactly one of them is won by B and the remaining 7 are draws. The probability of the first event is $\binom{9}{1}(0.5)(0.2)^8 = 1152 \times 10^{-8}$ while that of the second event is $\binom{9}{1}(0.5)\binom{8}{1}(0.3)(0.2)^7 = 13824 \times 10^{-8}$. Hence the desired probability is $(0.5) \times (1152 + 13824) \times 10^{-8} = 7488 \times 10^{-8} = 7.488 \times 10^{-5}$.

Comment No. 1:

The numbers $0.5, 0.3$ and 0.2 can be replaced with any three positive real numbers a, b, c adding up to 1. Also 10 can be replaced by any positive integer n. In that case the probability, say p_n, of A's winning the match with the n-th game (but not earlier) comes out to be

$$(n-1)a^2c^{n-3}[c + (n-2)b]$$

Comment No. 2:

The next problem (JEE 1983) is also of a similar spirit as the Main Problem but differs in one essential respect.

Cards are drawn one-by-one at random from a well-shuffled full pack of 52 cards until two aces are obtained for the first time. If N is the number of cards required to be drawn, then show that

$$Pr\{N = n\} = \frac{(n-1)(52-n)(51-n)}{50 \times 49 \times 17 \times 13}, \quad \text{where } 2 \leq n \leq 50 \tag{1}$$

(The expression on the L.H.S. is to be read as 'the probability that N equals n'.)

In problems such as this, where something (such as a card or a ball) is drawn repeatedly from a set, it makes a vital difference whether the draws are made with or without replacements. In case the card is replaced after each draw, then each outcome is independent of the previous ones as was the case in

the problem above and so the probability of a particular outcome (in this case an ace showing) is the same at every draw. Otherwise, this probability depends on the outcomes of the earlier draws and that changes the very nature of the problem.

In the present problem, it is not explicitly given whether the cards are drawn with or without replacement. But this can be inferred from the last part of the problem. It is given that N can assume only 2 to 50 as possible values. If the cards were drawn with replacements, then N could take any value greater than 1. So we assume that they are drawn without replacements. Now fix an integer n, $2 \leq n \leq 50$. In order that the second ace comes at the n-th draw it is necessary that in the first $(n-1)$ draws exactly one ace shows. This is equivalent to choosing $n-1$ cards from the 52 cards with the selection containing exactly one ace. As there are 4 aces and 48 other cards, the number of such selections is $4 \times \binom{48}{n-2}$. Hence the probability that exactly one ace comes during the first $n-1$ draws is the ratio $\dfrac{4 \times \binom{48}{n-2}}{\binom{52}{n-1}}$. (Note that we have ignored the order in which the $n-1$ cards are drawn. If we take it into account then each selection would correspond to $(n-1)!$ sequences of draws. But then both the numerator and the denominator of the ratio above will get multiplied by the same factor, viz., $(n-1)!$ so that the probability will be unaffected.) After these $n-1$ draws, $52-(n-1)$, i.e., $53-n$ cards are left, of which exactly 3 are aces. So the probability that an ace will show at the n-th draw is $\frac{3}{53-n}$. So the desired probability is the product of the fractions $\dfrac{4 \times \binom{48}{n-2}}{\binom{52}{n-1}}$ and $\frac{3}{53-n}$. Writing out the binomial coefficients in terms of factorials and cancelling common factors gives the answer.

The first part, viz., that of finding the probability of drawing exactly one ace in the first $n-1$ draws could also have been done in a manner which is analogous to the solution to the last problem. If we arrange the cards in a row in the order in which they are drawn, then in all there are ${}^{52}P_{n-1}$ i.e., $\frac{52!}{(52-(n-1))!}$, i.e., $\frac{52!}{(53-n)!}$ permutations. The lone ace can be any one of the four aces and can occur at any one of these $n-1$ places. The remaining $n-2$ places have to be filled by non-aces of which there are 48. This can be done in ${}^{48}P_{n-2}$, i.e., $\frac{48!}{(50-n)!}$ ways. So the number of favourable permutations is $\frac{4(n-1)48!}{(50-n)!}$ and the probability of the first event, viz., getting exactly one ace in the first $n-1$ draws is $\dfrac{4(n-1)48!(53-n)!}{(50-n)!52!}$ which is the same as the earlier value, viz., $\dfrac{4 \times \binom{48}{n-2}}{\binom{52}{n-1}}$.

Comment No. 3:

In both the problems so far, we had to find the probability of some event, say A, occurring on the n-th round. But in the first problem, this probability was independent of what happened on the first $n-1$ rounds while in the second one, the occurrence of the event A presupposed that a certain other event, say B, occurred earlier (namely that of exactly one ace showing in the first $n-1$

draws). In the solution given above, the number $\frac{3}{53-n}$ was not the probability of the event A. It was what is called the **conditional probability** of A, it being given that B has occurred (the probability of which was found to be $\frac{4 \times \binom{48}{n-2}}{\binom{52}{n-1}}$). The product of these two probabilities gave the probability of the event A. This is known as the **law of conditional probability**. To state it in a compact form, certain standard symbols are used. The probability of an event A is usually denoted by $P(A)$. The (conditional) probability of occurrence of A, it being given that B has occurred, is denoted by $P(A|B)$ (read as 'probability of A, given B'). The law of conditional probability can then be stated as

$$P(A) = P(B)P(A|B) \tag{2}$$

We emphasise that here we are assuming that the nature of the events A and B is such that occurrence of A presupposes that of B, i.e., A is a 'sub-event' of B in a sense. Later on we shall give a general version of the law of conditional probability which is applicable without any such restriction.

(2) is hardly a profound result. In fact, what it says is sheer common sense. But it is very useful. When we cannot find $P(A)$ easily, we often have to use some suitable event B as a 'stepping stone', i.e., we find $P(B)$, $P(A|B)$ and apply (2). Sometimes this 'auxiliary' event B is indicated by the problem itself. Sometimes we may have to invent it ourselves and this may call for some ingenuity. (An example of such a problem will be given later.) It is rather like this. Suppose we want to locate some town in the map of India. This may not be easy. But, if we know or can find out the state to which it belongs then we can first locate that state in the map of India and then locate the town in the magnified map of that state. In more complicated problems, finding $P(B)$ itself may require finding $P(C)$ and $P(B|C)$ for some suitable event C. (In terms of our analogy, this is like locating a village by first locating its state, then its district and then finally the village.

Comment No. 4:

A few other equally simple but useful results deserve to be stated. One is about what is called the **complementary probability** of an event, say A, i.e., the probability that it will *not* occur. Obviously, this probability is $1 - P(A)$. It is customary to say that this is the probability of the complementary event of A, which is denoted generally by A', or by $\bar{A}$, or by A^c (or $c(A)$). The interpretation of the complementary event A' is simply the logical negation of A, although in a particular context we can give it a positive meaning. For example, if an urn contains white, black and red balls and A is the event that 'a ball drawn from the urn is white', then A' is the event that 'a ball drawn from the urn is not white'. But if it is known that the urn contains only white and black balls then A' can be stated as 'a ball drawn from the urn is black'. When $P(A)$ is denoted by the symbol p, it is a standard practice to denote the complementary probability $1 - p$ by q.

There are situations where the complementary probability is easier to find. Suppose, for example, that we want the probability, say p, that in a

class of 50 students at least two students have their birthdays on the same day (ignoring leap year, and assuming that every day of the year is equally likely to be a person's birthday). Here the total number of possible cases is 365^{50}. Counting the number of cases in which the event occurs runs into a large number of cases depending upon the frequencies of occurrences of particular days. But the complementary event is that every two persons have distinct birthdays. Each favourable case corresponds to a permutation of length 50 of 365 days. The number of such permutations is the product $365 \times 364 \times 363 \times \ldots \times 317 \times 316$ i.e., the product $365^{50} \prod_{i=1}^{50}(1 - \frac{i-1}{365})$. So the desired probability is $1 - \prod_{i=1}^{50}(1 - \frac{i-1}{365})$. (In an examination in mathematics, a number like this is not expected to be calculated. But just for curiosity, with a FORTRAN program, it comes out to be more than 97%. So the event is almost a certainty.)

Comment No. 5:

Two other elementary concepts are the **disjunction** and the **conjunction** of two events, say A and B, denoted respectively, by $A \cup B$ (or occasionally by $A + B$) and by $A \cap B$ (or by $A.B$ or simply AB). Simply stated, $A \cup B$ is the event which occurs when (and only when) at least one of A and B occurs while $A \cap B$ is the event which occurs when (and only when) both A and B occur. As the notations suggest, the disjunction and conjunction of events are intimately related with the union and intersection of subsets respectively. Indeed let S be a set of all possible cases relevant to some problem. (Formally, such a set is called a **sample space**.) For example, in the problem above involving a pack of cards, for a fixed positive integer n, we could take S as the set of all permutations of length n of the 52 cards in the pack. We can then identify an event relevant to that problem with the set of those cases (in the sample space) in which it holds true. (This subset associated with an event is sometimes called the **truth set** of that event.) For example, in the solution above, the event A corresponds to the set of those permutations (in the set S) in which an ace occurs exactly twice; once at the end and once earlier while the event B corresponds to those permutations (in S) in which there is exactly one ace in the first $n - 1$ places (there being no restriction on the card in the n-th place).

It is clear that the truth set of the disjunction of any two events A and B is precisely the union of the truth set of A and the truth set of B. Similarly the truth set of the conjunction $A \cap B$ is the intersection of the truth sets of A and B. (This, indeed, is the reason why the same symbols, viz., $\cup$ and $\cap$ are used for both.)

The sample space S is usually so chosen that every element in it is equally likely. Phrases like 'a typical element' or 'randomly chosen' are often used to stress this. In the case of a coin or a die or a pack of cards, the adjectives 'fair', 'unbiased' or 'well-shuffled' etc., are used to convey that in the sample space of all possible outcomes, every element is equally likely. But even when such phrases are not explicitly mentioned, they are to be understood from the

context unless the contrary is explicitly stated. As an example of a sample space where all the elements do not have the same probability, suppose a biased coin which has a probability 0.4 of showing a head is tossed thrice. Here the sample space, say S, consists of 8 ordered triplets of the form (x, y, z) where each entry is either an H (for a head) or a T (for a tail). Here the triplet (H, H, H) occurs with probability $(0.4)^3 = 0.064$, but the triplets (H, T, T), (T, H, T) and (T, T, H) occur with probability 0.144 each. It should also be noted that the choice of the sample space relevant to a problem is not unique. Consider, for example, a problem where a pair of fair dice is rolled. Here, we can take as the sample space the set S of all ordered pairs of the form (x, y), where x and y denote the figures on the first and the second die respectively. All elements have the same probability, viz., $\frac{1}{36}$. But if the problem deals with the total score (as is often the case), then for the sample space we may as well take the set, say T, of the integers from 2 to 12. But now the probabilities of their occurrences are not all equal. For example, the total score $i + j$ will equal 5 in precisely 4 cases, viz., when $i = 1, 2, 3$ or 4 and correspondingly, $j = 4, 3, 2$ or 1. So the probability of obtaining a sum 5 is $\frac{4}{36}$, i.e., $\frac{1}{9}$. By a similar reasoning, the probability of getting 7 as a sum is $\frac{6}{36} = \frac{1}{6}$. (This calculation comes up so frequently in problems where a pair of fair dice is rolled, that it can be taken as a standard result. No justification needs to be given when it is used only as a means to do some bigger problem.)

When all the elements of the sample space are equally likely, probability problems are essentially reduced to problems of counting. Recall that for any (finite) set X, we denote the number of elements in it (also called the **cardinality** of X) by $|X|$. (The notations $\#(X)$ and $n(X)$ are also used sometimes.) With this notation, the probability of an event A (which is now taken to be a subset of the sample space S) is simply the ratio $\frac{|A|}{|S|}$. This is the classic definition of probability of an event as the ratio of the number of favourable cases to the total number of cases. Using this correspondence, elementary results about cardinalities of subsets of a space can be translated into elementary laws of probability. For example, the law of complementary probability is a consequence of the fact that the truth set of the complementary event A' is simply the complement (in the sample space S) of the truth set of A. If we divide the equation $|S - A| = |S| - |A|$ throughout by $|S|$, we get $P(A') = 1 - P(A)$. The law of conditional probability (2) given above follows by writing $\frac{|A|}{|S|}$ as $\frac{|A|}{|B|}\frac{|B|}{|S|}$ and recognising the first factor on the R.H.S. as the conditional probability $P(A|B)$, i.e., the probability of A given B. Here it is implicit that A is a subset of B. But even when this is not the case, $P(A|B)$ is the same as $P(A \cap B\ |B)$ and since $A \cap B$ is always a subset of B, we have the general **law of conditional probability** (also called **Bayes theorem**)

$$P(A \cap B) = P(B)P(A|B) \tag{3}$$

which is valid for any two events A and B (in the same sample space, of course).

(The very concept of a conditional probability is sometimes confusing to a beginner. The confusion is best illustrated by the following problem. Suppose

a box contains m biased and n unbiased coins, the probability of showing a head for a biased coin being p. A coin is drawn at random. Naturally, its probability of being biased is $\frac{m}{m+n}$. Suppose, however, that the coin drawn is tossed and a head shows up. Then the probability of the coin drawn being biased is no longer $\frac{m}{m+n}$. This is confusing to a beginner who is most likely to argue, "The nature of the coin is decided the moment it is drawn from the box. Whether it shows a head or a tail when tossed has no bearing on the type of that coin. So how can this knowledge change the probability of the coin drawn being biased?" The answer lies in the fact that the two probabilities here are conceptually different. Let A be the event that the coin drawn is biased and B be the event that the drawn coin shows a head when tossed. Then the first probability, when nothing is known about the outcome of the toss, is $P(A)$, the probability of the event A. And this is clearly $\frac{m}{m+n}$. The second probability, on the other hand, is no longer $P(A)$ but the conditional probability $P(A|B)$, i.e., the probability of the occurrence of A, *it being given that the event B has occurred*. By (3), this equals the ratio $\frac{P(A\cap B)}{P(B)}$ and there is no reason to suppose that it will always equal $P(A)$. In the present problem, we have not yet computed the probabilities of the events $A \cap B$ and B. We shall do so later (at the end of Comment No. 11). But the discussion should clarify the distinction between the probability of an event and its conditional probability, given some other event. As a real-life analogy, suppose there are as many men in a town as there are women and it is known that one of them has committed a certain murder. Then in the absence of any other information, the probability of the murderer being a man is $\frac{1}{2}$. Suppose, however, that there is some evidence to show that the murderer is a smoker. Now, if the percentage of smokers among men is more than that in women, then this additional information makes it more likely that the murderer is a man. This is, in fact, the conditional probability that the murderer is a man given that he/she is a smoker.)

The well-known result that for any two subsets A and B of S

$$|A \cup B| = |A| + |B| - |A \cap B| \tag{4}$$

(which is a special case of the principle of inclusion and exclusion) translates into

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \tag{5}$$

An especially noteworthy case of (5) occurs when the events A and B are **mutually exclusive**, i.e., can never hold simultaneously. In that case $P(A \cap B) = 0$ and hence we have $P(A \cup B) = P(A) + P(B)$. More generally, if the events $B_1, B_2, \ldots, B_k$ are mutually exclusive then

$$P(B_1 \cup B_2 \cup \ldots \cup B_k) = P(B_1) + P(B_2) + \ldots + P(B_k) \tag{6}$$

This relation is very useful when we cannot find the probability of a complicated event B directly. We then decompose it into mutually exclusive events whose probabilities can be determined relatively easily. So this is an instance of the

well-known strategy of 'divide and rule'! Actually, we can go a little further. Suppose in this situation that A is any other event. Then the event $A \cap (B_1 \cup B_2 \cup \ldots \cup B_k)$ is the disjunction of the mutually exclusive events $A \cap B_1$, $A \cap B_2$, $\ldots$, $A \cap B_k$ and so $P(A \cap (B_1 \cup B_2 \cup \ldots \cup B_k)) = \sum_{i=1}^{k} P(A \cap B_i)$. For each $i = 1, 2, \ldots, k$, we can use (3) and write $P(A \cap B_i)$ as $P(B_i)P(A|B_i)$. Hence,

$$P(A \cap (B_1 \cup B_2 \cup \ldots \cup B_k)) = \sum_{i=1}^{k} P(B_i)P(A|B_i) \tag{7}$$

A collection of events is called **exhaustive** if in any situation at least one of them must occur. For example, for any event A, the pair $\{A, A'\}$ is exhaustive. (We could, of course, replace A' by any event whose truth set is bigger than that of A'.) Very frequently, we run into a situation where some events, say $B_1, B_2, \ldots, B_k$ are mutually exclusive and exhaustive. In that case, the L.H.S. of (7) is simply $P(A)$ and we get

$$P(A) = \sum_{i=1}^{k} P(B_i)P(A|B_i) \tag{8}$$

We emphasize that the basic laws above (viz., (3) and (5) to (8)) remain true even without the assumption that all the elements of the sample space S are equally likely. But the simple-minded derivations given above will not work because now we can no longer equate $P(A)$ with the ratio $\frac{|A|}{|S|}$. Instead, we shall now have to consider a non-negative real-valued function f on the sample space S such that for each $x \in S$, $f(x)$ is the probability of the occurrence of the element x. In that case, for a subset A of S, $P(A)$ will equal $\sum_{x \in A} f(x)$, i.e., the sum of the probabilities of occurrences of the various elements of the subset A. (The function $f : S \longrightarrow \mathbb{R}$ is sometimes called a **weight function** because, for every $x \in S$, $f(x)$ is the (relative) weight or importance we attach to x. Obviously, the total weight $P(S)$, i.e., $\sum_{x \in S} f(x)$ has to equal 1.) If we take this approach then all the basic laws can be proved. We omit the proofs because they are routine and, at any rate, our interest is not so much in proving the laws but in applying them.

The concepts of conditional probability and of a sample space are also useful in resolving certain situations which can be confusing to a beginner. Suppose, for example, that a box contains m white and n black balls and that we draw two balls from the box one after the other. Let A be the event that the second ball is white. If the balls are drawn with replacement, then, of course, $P(A) = \frac{m}{m+n}$. Suppose, however, that they are drawn without replacement. Then a layman is likely to say that $P(A) = \frac{m-1}{m+n-1}$ if the first ball is white and that $P(A) = \frac{m}{m+n-1}$ if the first ball is black. Obviously there is something wrong here since $P(A)$ is a fixed number and its value should not depend on our knowledge of the outcome of the first ball. The correct way to express these

statements about $P(A)$ is using the concept of conditional probability. Let B be the event that the first ball is white. Then $P(B) = \frac{m}{m+n}$ while $P(B') = \frac{n}{m+n}$. The probabilities obtained above, viz., $\frac{m-1}{m+n-1}$ and $\frac{m}{m+n-1}$ are, precisely speaking, the *conditional probabilities* $P(A|B)$ and $P(A|B')$ respectively. Now there is nothing wrong if they are unequal, since they represent two different things. Moreover, neither of them is equal to $P(A)$. To find $P(A)$ we can use (8) (in a very special case) and get $P(A) = P(B)P(A|B) + P(B')P(A|B') = \frac{m(m-1)}{(m+n)(m+n-1)} + \frac{nm}{(m+n)(m+n-1)}$ which comes out to be $\frac{m}{m+n}$.

Note also that $P(A)$ is the same as $P(B)$. In fact, it can be shown with a little more calculation, that if we draw one more ball (without replacement) then its probability of being white is also $\frac{m}{m+n}$. (We are, of course, assuming here that there are at least three balls to begin with, i.e., $m + n \geq 3$.) This may seem like a coincidence, but it is not so. Using suitable sample spaces, let us prove for example, that $P(A) = P(B)$. Number the balls from 1 to $m + n$ in an arbitrary but fixed manner. We now have $m + n$ distinct balls of which m are white and n black. The sample space, say S, consists of all ordered pairs of the form (x, y), where x and y are two distinct balls. Then $|S| = (m + n)(m + n - 1)$. Moreover, since the balls are drawn in a random order, all elements of S are equally likely. Now let F and G be the subsets $\{(x, y) \in S : x \text{ is white}\}$ and $\{(x, y) \in S : y \text{ is white}\}$, respectively. Then F and G are nothing but the favourable sets for the events B and A respectively and so $P(A) = \frac{|G|}{|S|}$ while $P(B) = \frac{|F|}{|S|}$. Now, consider the bijection $\theta : S \longrightarrow S$ defined by interchanging x and y, i.e., $\theta(x, y) = (y, x)$. Then $\theta(F) = G$, and so we have $|F| = |G|$. Hence it is not a coincidence that $P(A) = P(B)$ although their direct calculations differed considerably in their length. More generally, the argument we have given shows that for any $r \leq m+n$, the probability that the r-th ball is white is the same, viz., $\frac{m}{m+n}$. (JEE 1980) In other words, whether the balls are drawn with or without replacements, the probability that a ball in a particular draw is white is the same. The *conditional probability* of its being white is of course different and will depend considerably on the knowledge of the outcomes of the earlier draws in case the balls are drawn without replacements. The same reasoning applies if we have balls of more than two colours.

Even when elements of a sample space S are not equally likely, if we can find a bijection, say, $\theta : S \longrightarrow S$ which is weight-preserving, i.e. $f(\theta(x)) = f(x)$ for all $x \in S$ (f being the weight function) and which maps the truth set of an event A onto that of an event B, then we have $P(A) = P(B)$, because both the probabilities are sums with the same summands, possibly in a different order.

Comment No. 6:

When we apply (3) (or (7)) to find probability $P(A)$ of an event A, by finding the probabilities $P(B_i)$ and then the conditional probabilities $P(A|B_i)$, the events B_i's are of our choice and generally it is obvious what these auxiliary events B_i's should be. In some problems, this is not so. The obvious choice does not work; at least it does not work easily and so we have to insert one more auxiliary event. Here is a good example of such a problem.

Eight players $P_1, P_2, \ldots, P_8$ play a knock-out tournament. It is known that whenever the players P_i and P_j play, the player P_i will win if $i < j$. Assuming that the players are paired at random in each round, what is the probability that the player P_4 reaches the final? (JEE 1999)

Here there is no uncertainty about the outcome of any match. The player P_1 will reach the final (and in fact will win it) no matter how the players are paired at each round. However, whether any other player reaches the final or not will depend upon the pairing. Reaching the final is equivalent to winning in both the first and the second round. Let A be the event that the player P_4 wins the second round (which presupposes that he also wins the first round). We want to find $P(A)$. The most natural approach to try is to let B be the event that P_4 wins the first round, find $P(B)$, then find the conditional probability $P(A|B)$ and finally apply (3). This approach begins well, because it is easy to find $P(B)$. The straightforward way will be to count all possible ways of pairing 8 players and then take those in which P_4 is paired with P_i for some $i > 4$. By elementary counting techniques (see the discussion about equation (12) in Chapter 1), the former can be shown to be $\frac{8!}{2^4 \times 4!} = 7 \times 5 \times 3 \times 1 = 105$. For counting the latter, divide the set of pairings into four mutually disjoint classes depending upon which of P_5, P_6, P_7, P_8 is paired with P_4. In each case, the remaining 6 players can be paired in $5 \times 3 \times 1 = 15$ ways. So the total number of favourable cases (for the event B, i.e., for P_4 to win the first round) is 60. Hence $P(B) = \frac{60}{105} = \frac{4}{7}$. But it is unnecessary to go through all this trouble if we look at the first round differently. In any pairing of the 8 players, P_4 will be paired with one of the remaining 7 players and each of these 7 players is equally likely to be P_4's opponent in the first round. Now, P_4 wins the first round if and only if this opponent is one of the 4 players P_5, P_6, P_7 and P_8. So the probability of P_4's winning the first round is simply $\frac{4}{7}$. This is a good illustration of how a clever choice of a sample space can simplify the calculation of probability. In the first approach, the sample space consisted of all possible 105 ways of pairing the 8 players. In the second approach, it consisted of the set of all players other than P_4.

So, whichever approach we follow, we get $P(B) = \frac{4}{7}$. We now have to find the conditional probability $P(A|B)$, i.e., the probability of P_4's winning the second round, given that he has already won the first round. Now, unfortunately, there is no slick way to calculate this. If we know who are the winners (besides P_4) of the first round, then of course, we can immediately tell the chances of P_4's winning the second round exactly the same way we calculated his chance of winning the first round. But then we would have to make so many cases depending upon the outcome of the first round and calculate P_4's probability of winning the second round in each case separately since this probability is not the same in all cases. (If, for example, all of P_1, P_2, P_3 won the first round, besides P_4, then P_4 has no chance of winning the second round. In other cases, he has some hopes.)

The way out here is to introduce a suitable intermediate event C and calculate the conditional probabilities $P(C|B)$ and $P(A|B \cap C)$ instead of the conditional probability $P(A|B)$. Note that no matter how the players are paired

at the first round, at least two of P_1, P_2 and P_3 are sure to enter the second round. We are assuming that P_4 is also in the second round. So at most one of the four players P_5, P_6, P_7 and P_8 enters the second round. Moreover, this player has to be different from the player, say P_i, with whom P_4 was paired in the first round. And the only way this can happen is if two of the three players of the set $\{P_5, P_6, P_7, P_8\} - \{P_i\}$ were paired with each other. So we let C be the event that in the first round, two of the players from P_5 to P_8 are paired together. It is easy to find the conditional probability $P(C|B)$ now by finding its complementary probability $P(C'|B)$. Indeed, given B, let P_i be the player paired with P_4 in the first round, where $5 \leq i \leq 8$. Then the remaining 6 players can be paired off in 15 ways and the only way C can fail is if the three players P_1, P_2, P_3 are paired with the three players of the set $\{P_5, P_6, P_7, P_8\} - \{P_i\}$. This can happen in 3! i.e., in 6 possible ways. So, $P(C'|B) = \frac{6}{15} = \frac{2}{5}$ and hence $P(C|B) = 1 - \frac{2}{5} = \frac{3}{5}$. Multiplying, we get $P(B \cap C) = P(B)P(C|B) = \frac{4}{7} \times \frac{3}{5} = \frac{12}{35}$.

To complete the solution, we still have to calculate $P(A|B \cap C)$. But this is easy. If B and C both hold, it means that in the second round, besides P_4, there will be a player P_j for some j, $5 \leq j \leq 8$. The two other players will be from P_1, P_2 and P_3. It is now clear that P_4 will win the second round if and only if he is paired with P_j. And the probability of this happening is $\frac{1}{3}$, following either of the two methods we used to calculate $P(B)$. So by the law of conditional probability once more, we get $P(A) = P(B \cap C)P(A|B \cap C) = \frac{12}{35} \times \frac{1}{3} = \frac{4}{35}$ as the answer finally. We could, of course, give the same argument without explicitly introducing the intermediate event C. In that the case, having known that P_4 has won the first round, we would have to divide the possible outcomes of the first round into two categories, viz., those in which some player lower than P_4 also wins the first round and those in which there is no such player. In the case of each of the outcomes in the first category, the probability that P_4 wins the second round is $\frac{1}{3}$ while in the case of any outcome of the second category, it is 0. This would also give the solution and the calculations would be basically the same. But the introduction of the intermediate event C makes the solution look neater. Moreover, the moment you divide those outcomes of the first round in which P_4 is a winner into two categories, in effect, you are introducing the event C in your mind, without possibly naming it. Whether you introduce it on paper or not, does not change the crux of the solution.

We mention that there is an unusually short solution to this problem. It hinges on the idea of selecting the sample space ingeniously. At the end of any knock-out tournament, the original set of players gets divided into two mutually disjoint subsets, the 'champion's pool' and the 'runner-up's pool'. The champion's pool consists of the champion and all other players who are beaten by him directly or indirectly till the final is played. The runner-up's pool is its complement, and consists of the runner-up and all players beaten by him directly or indirectly before the final. In the present problem P_1 is the champion. If P_4 reaches the third round then he will be necessarily the runner-up and moreover he will have to be the best player in his pool. Now in P_4's pool, there are three players besides himself. These can be chosen from the 7 players in $\binom{7}{3} = 35$ ways. In order that P_4 is the runner-up, it is necessary and sufficient that all

these three players are from the players P_5 to P_8 and this choice can be made in $\binom{4}{3} = 4$ ways. So the desired probability is $\frac{4}{35}$.

In terms of elegance, this solution matches Andre's solution to the problem of counting the number of balanced arrangements of pairs of parentheses or the pure geometry solution to the Main Problem of Chapter 10 (about finding how many sides a regular polygon has, given an equation about the lengths of some of its diagonals) or the combinatorial proof of the identity (2) in the solution to the Main Problem of Chapter 5. Whether in pure geometry or in probability problems, a miraculous short-cut is sometimes available if you focus upon the key idea. With such a short solution, a skeptic might question whether it is really a valid solution or whether, despite some flaw in the reasoning, the figure $\frac{4}{35}$ has come up fortuitously. To see the validity of the solution, let us compare the sample spaces in the two solutions. In the 'pedestrian' solution, the sample space corresponded directly to the pairings we made at each round. To be more precise, there the sample space, say X, consisted of all ordered pairs of the form (E, F). Here E is a decomposition of the set of eight players into 4 (mutually disjoint) subsets of two elements each. For example, $E = \{\{P_1, P_6\}, \{P_4, P_3\}, \{P_2, P_8\}, \{P_7, P_5\}\}$ is one such typical decomposition. Note that the order of the four subsets is immaterial. Also the two elements in each pair can be written in either order. Hence the same E can be written in $4!2^4 = 384$ ways. Indeed, this is one of the ways to show that the total number of such decompositions is $\frac{8!}{4!2^4} = 105$. Now, for a given E, the second entry F consists of a decomposition of the set of the winners in E into two subsets of two elements each. For example, with E as above, the set of winners in it is the set $\{P_1, P_3, P_2, P_5\}$ and F can be $\{\{P_1, P_2\}, \{P_3, P_5\}\}$, or $\{\{P_1, P_3\}, \{P_2, P_5\}\}$ or $\{\{P_1, P_5\}, \{P_3, P_2\}\}$. In fact, it is clear that since there are four winners in every decomposition E, the second entry F has 3 possibilities. So the sample space X consists of $105 \times 3 = 315$ elements. All these elements are equally likely.

Let us now consider the sample space, say Y, in the short solution to the problem. This consists of all decompositions of the set of players into two mutually disjoint subsets (the two 'pools') each having 4 elements. In any such decomposition, exactly one of the subsets contains the player P_1. And if we know this, then we know the decomposition. So, we might as well let Y consist of all subsets of four players of which P_1 is one. Clearly, the total number of such subsets is $\binom{7}{3} = 35$. So $|Y| = 35$. Here also all 35 elements are equally likely.

We now define a function $f : X \longrightarrow Y$ as follows. Take a typical element, say (E, F) of the sample space X. Then E amounts to a pairing of the 8 players in the first round and F amounts to a pairing of the winners in E. In the first round, P_1 defeats the player he is paired with. In the second round, he directly defeats one of the winners of the first round. But his opponent in the second round has already defeated one player in the first round. Clubbing all these three players with P_1 gives an element of the set Y and we define $f((E, F))$ to be this element. Verbally, $f((E, F))$ is nothing but the 'champion's pool' described above and so we may call f as the 'champion pool function'. For example, let E be as above, viz., $E = \{\{P_1, P_6\}, \{P_4, P_3\}, \{P_2, P_8\}, \{P_7, P_5\}\}$.

Then the winners in E are P_1, P_2, P_3 and P_5. We already saw that here F can take three possible values, viz., $\{\{P_1, P_2\}, \{P_3, P_5\}\}$, or $\{\{P_1, P_3\}, \{P_2, P_5\}\}$ or $\{\{P_1, P_5\}, \{P_3, P_2\}\}$. If we take the first possibility, i.e., if we let $F = \{\{P_1, P_2\}, \{P_3, P_5\}\}$, then we get $f((E, F)) = \{P_1, P_6, P_2, P_8\}$, because in the first round P_1 defeats P_6 and in the second round he defeats P_2 who has defeated P_8 in the first round. Similarly, if we let $F = \{\{P_1, P_3\}, \{P_2, P_5\}\}$ (with no change in E), then $f((E, F)) = \{P_1, P_6, P_3, P_4\}$, while setting $F = \{\{P_1, P_5\}, \{P_3, P_2\}\}$ would give $f((E, F)) = \{P_1, P_6, P_5, P_7\}$.

Thus f is a function from a set with 315 elements to a set with only 35 elements. Obviously, f cannot be a one-to-one function. In fact, we contend that under f, every element of Y has exactly 9 preimages (or, in other words, f is a 'nine-to-one' function). A typical element of Y is of the form $\{P_1, P_i, P_j, P_k\}$, where i, j, k are three distinct integers between 2 to 8. Now if this subset is a champion's pool, then exactly one of the three players P_i, P_j and P_k was beaten by P_1 directly in the first round. Call him M. M can be chosen in 3 ways. Call the remaining two players as L and N with L having a higher rating (i.e., a lower subscript) than N. Then in the second round, P_1 defeats L while in the first round, L defeats N. This means that in the first round, $\{P_1, M\}$ and $\{L, N\}$ were two pairs of the corresponding decomposition, say E, of the set of 8 players. The remaining 4 players can be paired off among themselves in 3 ways. Depending on the pairing, two of these players, say P and Q, will be the winners in E. They both enter the second round. But in the second round, P_1 is already paired with L. So P and Q must be paired with each other. In other words, the decomposition F must equal $\{\{P_1, L\}, \{P, Q\}\}$. So, once E is determined, we have no further choice in F. Summing up, for each possible choice of M (from the set $\{P_i, P_j, P_k\}$), we could build up a decomposition E in 3 different ways and this decomposition then uniquely determined F. It follows that for a given element $\{P_1, P_i, P_j, P_k\}$ of Y, there are exactly 9 elements $(E, F) \in X$ such that $f((E, F)) = \{P_1, P_i, P_j, P_k\}$. In other words, every element of the sample space Y has exactly 9 preimages in the sample space X under the function f. Each of these 9 elements has a weight (or probability of occurrence) equal to $\frac{1}{315}$. So, together the weight of 9 elements in the preimage is $\frac{9}{315} = \frac{1}{35}$. But this is exactly the probability of occurrence of each element of the sample space Y. So, the two solutions are basically the same. In the slow solution, we find the desired probability by directly adding the probabilities of the favourable cases in the sample space X. In the slick solution, we first group these favourable cases into bunches of size 9 each and then add the favourable bunches. But just see what a tremendous saving of work is achieved because of this clever bunching.

We have given the solution to this problem rather extensively, because it is a good drill in sharpening one's thinking process. If you thoroughly master both the solutions and also the proof of their equivalence, the probability of your successfully attacking other probability problems will, most probably increase!

Comment No. 7:

An important relationship between two events, say A and B occurs when there is no relationship between them! That is, the occurrence of each of them

is completely independent of the occurrence or otherwise of the other. For example, if a coin is tossed twice and A is the event that a head shows on the first round while B is the event that a head shows on the second, then obviously A and B are mutually independent. Mathematically, A is **independent** of B if the probability of the occurrence of A is unaffected by the occurrence of B. In other words, $P(A)$ is the same as the conditional probability $P(A|B)$, viz., the probability of A given B. In view of (3) this is equivalent to saying that $P(A \cap B) = P(A)P(B)$ which is often taken as the definition of mutual independence of two events.

Here is a simple problem based directly on this definition.

A and B are two independent events. The probability that both A and B occur is $\frac{1}{6}$ and the probability that neither of them occurs is $\frac{1}{3}$. Find the probability of the occurrence of A. (JEE 1984)

The first part of the numerical data means $P(A \cap B) = \frac{1}{6}$. Taking complementary probability, the second part means $P(A \cup B) = \frac{2}{3}$. Using (5) and the independence of A and B, these two together imply (i) $P(A)P(B) = \frac{1}{6}$ and (ii) $P(A) + P(B) - P(A)P(B) = \frac{2}{3}$. We thus have a system of two equations in two unknowns, $P(A)$ and $P(B)$. Solving, we get $P(A) = \frac{1}{2}$ or $\frac{1}{3}$ (and correspondingly, $P(B) = \frac{1}{3}$ or $\frac{1}{2}$). The data is insufficient to determine which of these two possibilities holds.

It is easy to show that if A and B are independent, then so are A and B'. This follows by writing $P(A \cap B') = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B')$. Here is a simple problem based on this.

Let A and B be two events such that $P(A) = 0.3$ and $P(A \cup \bar{B}) = 0.8$. If A and B are independent, find $P(B)$. (JEE 1990)

Independence of A and B implies that of A and $\bar{B}$. So, $0.8 = P(A \cup \bar{B}) = P(A) + P(\bar{B}) - P(A \cap \bar{B}) = P(A) + P(\bar{B}) - P(A)P(\bar{B})$. Putting $P(A) = 0.3$ and solving we get $P(\bar{B}) = \frac{5}{7}$. Hence $P(B) = 1 - P(\bar{B}) = \frac{2}{7}$.

Comment No. 8:

More generally, given n events, say $A_1, A_2, \ldots, A_n$ (where $n \geq 2$), we say that they are **independent**, or more precisely, **collectively independent**, if for every set of indices $j_1, j_2, \ldots, j_m$ with $1 \leq j_1 < j_2 < \ldots < j_m \leq n$, we have $P(A_{j_1} \cap A_{j_2} \cap \ldots \cap A_{j_m}) = P(A_{j_1})P(A_{j_2}) \ldots P(A_{j_m})$. This, in particular implies that they are pairwise independent, i.e. that every two of them are mutually independent as we see by taking $m = 2$. But the converse is false as the following problem shows.

A lot contains 50 defective and 50 non-defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as (i) $A = \{$ the first bulb is defective$\}$, (ii) $B = \{$the second bulb is non-defective$\}$ and (iii) $C = \{$the two bulbs are both defective or both non-defective$\}$. Determine (a) whether A, B, C are pairwise independent, and (b) whether A, B, C are independent. (JEE 1992)

The two bulbs can, in all, be drawn in $100 \times 100 = 10,000$ equally likely ways. A holds true in 50×100, i.e., in 5,000 of these cases. B holds true in $100 \times 50 = 5,000$ cases also. C holds true in $(50 \times 50) + (50 \times 50) = 5,000$

cases. So $P(A) = P(B) = P(C) = \frac{1}{2}$. Also A and B both hold true in $50 \times 50 = 2,500$ cases. So $P(A \cap B) = \frac{1}{4} = P(A)P(B)$. Hence A and B are mutually independent. For A and C to hold together, both the bulbs must be defective and this can happen in 2,500 cases. So $P(A \cap C) = \frac{1}{4} = P(A)P(C)$, proving the independence of A and C. By a similar reasoning, B and C are independent. But A, B and C can never hold simultaneously. Hence $P(A \cap B \cap C) = 0 \neq P(A)P(B)P(C)$. So they are not (collectively) independent.

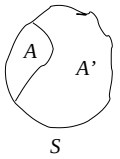
Note that in the solution above, we really did not *prove* the independence of A and B. In fact when we said that the two bulbs can be drawn in 10,000 *equally likely* ways, we implicitly *assumed* that the outcomes of the two draws are independent of each other. This is also the case with events in many real-life problems. Their independence has simply to be assumed as a matter of common sense. This holds good, for example, where the events represent the outcomes of objects drawn with replacements, identical dice rolled together, a coin flipped in succession and so on. This can sometimes lead to simple but somewhat tricky problems. Suppose, for example (JEE 1998), that a fair coin is tossed repeatedly. If a tail appears on the first four tosses, we tend to think that on the fifth toss, a head is far more likely to show than a tail. (If a head is taken to symbolise success and a tail a failure, then this is, in fact, the psychology of gamblers. After all, fate cannot be so cruel as to punish anyone five times in a row!) But mathematically, the outcome of the fifth toss is completely independent of those on the first four tosses and so the probability of a head showing on the fifth toss is still only $\frac{1}{2}$ and no higher. Note that the probability, say p_1, which is asked here is the conditional probability that a head shows on the fifth toss, it being given that all the first four tosses show tails. This probability is $\frac{1}{2}$ as we just saw. The apparent paradox here arises if we confuse p_1 with the probability, say p_2, of at least one head showing in 5 tosses. p_2 can be found very easily by finding its complementary probability. Since all the five tosses are independent, the probability of getting a tail on all is $(\frac{1}{2})^5 = \frac{1}{32}$. So, $p_2 = \frac{31}{32}$ which is nearly equal to one. Similar paradoxes are also the gist of some popular jokes. For example a surgeon assures a worried patient, "The operation I am going to perform on you has a 10% chance of success. All my nine patients so far have died. You are the tenth and so sure to be cured!" Jokes apart, this example shows that philosophically, the concept of independence of events is a highly elusive one and often borders on an article of faith. For example, in the problem above the rationalist assumption is that the five tosses of the coin are collectively independent. But gamblers think otherwise. And, for believers in astrology, the outcome of your matrimonial search is dependent upon the position of Mars at the time of your birth!

Leaving this debate aside, we shall continue to take the rationalist view, viz., as long as there is no demonstrable link between two events in real life, they are independent. Many simple problems in probability can be done almost mentally if the collective independence of events is kept in mind. For example, if three faces of a fair die are yellow, two red and one blue and the die is tossed thrice then the probability that the colours yellow, red and blue appear on the first, second and the third tosses respectively is simply $\frac{1}{36}$, which is the product

of $\frac{3}{6}$, $\frac{2}{6}$ and $\frac{1}{6}$, these three numbers being the probabilities of three collectively independent events corresponding to the three tosses. (JEE 1992)

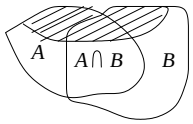
Comment No. 9:

In set theory, it is customary to show the intersections, unions and complements of subsets of a set, say S by drawing appropriate Venn diagrams. In fact, as we saw in Chapter 1, they are very helpful in counting problems. Problems in probability are intimately related to counting problems. Indeed, when the sample space is such that all its elements are equally likely, they are essentially counting problems in disguise. So it is hardly surprising that Venn diagrams also serve to give very vivid interpretations of the various elementary laws of probability. Although such reformulations are hardly profound, they serve the pedagogical purpose of making these laws crystal clear. For example, in the figure below, (a), (b) and (c) illustrate, respectively, the law for the complementary probability, the law for probability of the disjunction of two events and the law for conditional probability.



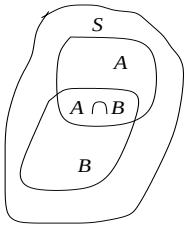
$$P(A') = \frac{|S|-|A|}{|S|} = 1 - P(A)$$

(a) Probability of complementary event



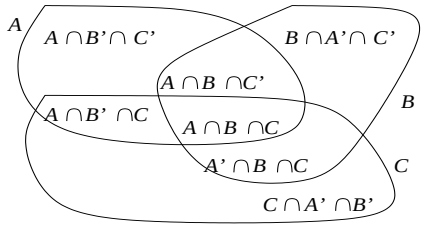
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(b) Probability of disjunction of two events



$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{P(A \cap B)}{P(B)}$$

(c) Conditional probability



(d) Disjunction of three events

Simple questions about probabilities of various combinations of events can sometimes be answered by simply looking at the appropriate Venn diagram. Here is an easy example.

Suppose A, B are two events. Express, in terms of $P(A), P(B)$ and $P(A \cap B)$, the probability that exactly one of A and B occurs. (JEE 1984)

Let C be the event that occurs when (and only when) exactly one of A and B occurs. From the diagram (b) above, it is clear that $A \cup B$ is the disjunction of three mutually exclusive events, viz., $A \cap B'$, $A' \cap B$ and $A \cap B$. So $P(A \cup B) = P(A \cap B') + P(A' \cap B) + P(A \cap B)$. But the disjunction of $A \cap B'$ and $A' \cap B$ is precisely the event C. So, $P(C) = P(A \cap B') + P(A' \cap B) = P(A \cup B) - P(A \cap B)$, which by (5) equals $P(A) + P(B) - 2P(A \cap B)$.

As a less elementary example, consider the following problem.

In a certain city only two newspapers A and B are published. It is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. It is also known that 30% of those who read A but not B look into advertisements and 40% of those who read B but not A look into advertisements while 50% of those who read both A and B look into advertisements. What percentage of population reads an advertisement? (JEE 1984)

Let S be the set of all persons in the city. Use the same symbols A and B to denote the readerships of A and B respectively. These are subsets of S. We are given that (i) $\frac{|A|}{|S|} = \frac{25}{100}$, (ii) $\frac{|B|}{|S|} = \frac{20}{100}$ and (iii) $\frac{|A \cap B|}{|S|} = \frac{8}{100}$. Now let C be the subset of S consisting of those persons who read an advertisement. We are given that (iv) $\frac{|A \cap B' \cap C|}{|A \cap B'|} = \frac{30}{100}$, (v) $\frac{|A' \cap B \cap C|}{|A' \cap B|} = \frac{40}{100}$ and finally, (vi) $\frac{|A \cap B \cap C|}{|A \cap B|} = \frac{50}{100}$. And we are asked to find $\frac{|C|}{|S|}$. This can be done most vividly by shading certain portions of the sets $A \cap B'$, $A' \cap B$ and $A \cap B$ in the diagram (b) above, the shaded portions corresponding to those persons who look at advertisements. (Such diagrams are only symbolic. They are generally not drawn to scale. So it is not necessary that the areas of the various regions be proportional to the cardinalities of the subsets of S which they represent.) Clearly, C is the union of the three mutually disjoint subsets, $A \cap B' \cap C$, $A' \cap B \cap C$ and $A \cap B \cap C$. So we have

$$|C| = |A \cap B' \cap C| + |A' \cap B \cap C| + |A \cap B \cap C| \tag{9}$$

We also have $|A| = |A \cap B| + |A \cap B'|$ and hence $|A \cap B'| = |A| - |A \cap B| = \frac{25-8}{100}|S| = \frac{17}{100}|S|$. Multiplying this by (iv) gives $|A \cap B' \cap C| = \frac{30}{100}\frac{17}{100}|S|$. Similarly, $|A' \cap B|$ comes out as $\frac{20-8}{100}|S| = \frac{12}{100}|S|$ and combined with (v) this gives $|A' \cap B \cap C| = \frac{40}{100}\frac{12}{100}|S|$. Finally, from (iii) and (vi), $|A \cap B \cap C| = \frac{50}{100}\frac{8}{100}|S|$. Substituting these three values in (9) gives $|C| = (\frac{510+480+400}{10,000})|S| = \frac{1390}{10,000}|S|$. In other words, 13.90% of the population reads an advertisement.

Although the problem was stated in terms of percentages, it can be easily recast in terms of probabilities. For example, the first statement of the data is equivalent to saying that a person chosen at random from the city reads newspaper A is $\frac{25}{100}$, i.e., $\frac{1}{4}$. Similarly, the fourth statement says that the probability that a person who reads only A looks at the advertisements is $\frac{2}{5}$. The problem then asks to find the probability that a person at random from the city reads an advertisement. It is in fact, often helpful to go the other way round, that is recast a probability problem as a problem about percentages. Although such a conversion is hardly profound, it enables one to visualise the problem better.

Venn diagrams also help in visualising implications. Suppose A and B are events such that occurrence of A always implies that of B. In other words,

whenever A occurs, B must also occur. If we identify the events with their corresponding truth sets, then this is equivalent to saying that A is a subset of B. In particular this means that $P(A) \leq P(B)$. The converse is false as can again be seen from a Venn diagram. Just because a region A has a smaller size than B, there is no reason to suppose that it is a subset of B. In fact, it may even be disjoint from B. And even if $A \cap B$ is not the empty set, it cannot be necessarily concluded that either $A \subset B$ or $B \subset A$. So nothing can be said as to the occurrence of which of these events implies that of the other. (JEE 1998)

The Venn diagram (b) above showed only two events. A similar, but slightly more cluttered diagram can be drawn when there are three (or more) events as shown in (d). We illustrate its uses in the following problem.

For the three events A, B and C, P(exactly one of A and B occurs) = P(exactly one of B and C occurs) = P(exactly one of C and A occurs) = p and P(all the three events occur simultaneously) = p^2, where $0 < p < \frac{1}{2}$. Find the probability that at least one of the three events occurs. (JEE 1996)

For notational brevity, let us denote the conjunction $A \cap B$ by simply AB etc. (which is one of the standard notations for a conjunction anyway). The data then says that (i) $P(AB' \cup A'B) = p$, (ii) $P(BC' \cup B'C) = p$, (iii) $P(CA' \cup C'A) = p$ and (iv) $P(ABC) = p^2$. We have to find $P(A \cup B \cup C)$. We can write $A \cup B \cup C$ as the disjunction of seven mutually exclusive events as shown in (d), where we write the symbols a, b, c, x, y, z and w to denote the probabilities of the events represented by the respective regions. Thus, $a = P(AB'C'), x = P(A'BC), w = P(ABC)$ etc. From the figure it is clear that $AB' \cup A'B$ is the disjunction of the mutually exclusive events $AB'C', AB'C, A'BC$ and $A'BC'$. So (i) amounts to saying that (i)$a + x + b + y = p$. Similarly, (ii) and (iii) can be replaced by (ii)$b + y + c + z = p$ and (iii)$c + z + a + x = p$. Solving, $a + x = b + y = c + z = \frac{p}{2}$. By (iv), $w = p^2$. So the desired probability, which equals $a + b + c + x + y + x + w$ is $p^2 + \frac{3p}{2} = \frac{2p^2+3p}{2}$.

Note that the data is insufficient to determine the values of x, y, z, a, b, c individually, since we have only three equations in 6 unknowns. Nor are these individual values needed. In fact, as the solution shows, we could treat $a+x, b+y$ and $c + z$ as single variables. So this is one of those problems where the data appears insufficient initially but later turns out to be sufficient. And the credit goes to the Venn diagram. Note also that the hypothesis $0 < p < \frac{1}{2}$ was never used in the solution. It is, in fact, a *consequence* of the solution, because if $p > \frac{1}{2}$, then $p^2 + \frac{3p}{2}$ would exceed 1, which is a contradiction since the probability of an event can never exceed 1. This also suggests that the problem could have been posed as a consistency problem. Suppose, for example, that in a city three newspapers A, B and C are published and that a surveyor reports that the percentages of the population who read (i) exactly one out of A and B, (ii) exactly one out of B and C, (iii) exactly one out of C and A and (iv) all three newspapers, are, respectively, p, q, r and s. Then by exactly the same work as above we get that the percentage of the population which read at least one of the three newspapers is $\frac{p+q+r+2s}{2}$. So if the figures p, q, r, s are such that $p + q + r + 2s > 200$ we can conclude that the data is inconsistent and hence

the survey is a fake one!

The fundamental idea behind inconsistency is that the probability, say p, of an event must always lie between 0 and 1 (both included). Using Venn diagrams one may be able to write an equation for p and using algebra it may be possible to solve it. But if all the solutions lie outside the interval $[0, 1]$, then there is an inconsistency in the data. As a simple example (JEE 1982), suppose A and B are events with $P(A) = 0.5$, $P(B) = 0.9$ and $P(A \cap B) = 0.2$. Then by (5), $P(A \cup B) = 0.5 + 0.9 - 0.2 = 1.2 > 1$. So the data is inconsistent.

Comment No. 10:

It is perhaps also fair here to point out a possible misuse of Venn diagrams. It was remarked above that the Venn diagrams are only a visual aid to thinking and are generally not drawn to scale. In other words, if the sample space S is shown by a region and an event by a subregion, say A of S, then it is not necessary that the ratio of the area of A to that of S be equal to the probability of that event. If we inadvertently make this assumption, we can sometimes get wrong answers. We illustrate this with a famous problem which often generates a lot of controversy.

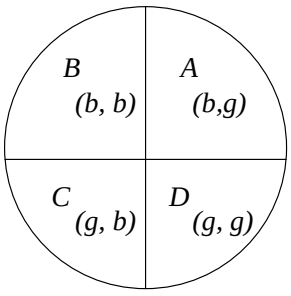
In a township, every family has two children and the probability of a child being a boy is $\frac{1}{2}$. All the boys and their fathers assemble at a meeting.

(i) If a boy present at this meeting is picked at random, what is the probability that his sibling (i.e., the other child in his family) is also a boy?

(ii) If a man present at this meeting is picked at random, what is the probability that he has two sons?

These questions could be paraphrased in terms of conditional probabilities. Thus (i) is equivalent to asking (iii) given that one child in a family is a boy, what is the probability that the other child is also a boy? while (ii) amounts to asking (iv) given that a man has a son, what is the probability that he has two sons?

Many people think that the answer to both (i) and (ii) is the same and that it is $\frac{1}{3}$. In an attempt to justify it, they draw a nice Venn diagram in which the township is represented by a disc which is divided into four equal quadrants, say, A, B, C and D as shown below. Each quadrant is labelled by a symbol which indicates the sexes of the two children in a family belonging to that quadrant. For example, the label (b, g) for the quadrant A indicates that A consists of all those families in which the first child is a boy and the second a girl. Because of the hypothesis that the probability of a child being a boy is $\frac{1}{2}$, all the four types of families are equally likely and so it is a perfectly legitimate Venn diagram.



Now, to answer the question (i), a boy in the township can come only from a family belonging to either A, B or C. So the sample space now is no longer the entire disc but only three quarters of it, viz., the union $A \cup B \cup C$. Of this, the favourable set consists of only the quadrant B and so the desired probability is $\frac{1}{3}$. The flaw in this reasoning is that it is based on the implicit assumption that when you take $A \cup B \cup C$ as the sample space, every element in it is equally likely. But this assumption is wrong. It is true that the number of *families* in each of the quadrants A, B and C is the same. But as far as the *boys* in the township are concerned, half of them come from the quadrant B and one fourth each from A and C. So, if we take the region $A \cup B \cup C$ as the sample space for the set of boys, then the probability that a boy at random will come from A, B and C is, respectively, $\frac{1}{4}, \frac{1}{2}$ and $\frac{1}{4}$. His sibling is also a boy if and only if he comes from B and the probability of this is $\frac{1}{2}$. So the correct answer to the question (i) (and hence also to (iii)) is $\frac{1}{2}$ and *not* $\frac{1}{3}$. Actually, the correct answer to (iii) could have been arrived at even without a Venn diagram merely on the basis of independence of events. In absence of any information to the contrary, we have to assume that the sexes of the children in a family are independent of each other. So, even if it is known that one of the children is a boy, the probability of the other child being a boy is unaffected by it and hence is still $\frac{1}{2}$. (This is analogous to the problem in Comment No. 8 regarding the toss of a fair coin five times in a row.)

However, as far as question (ii) above is concerned, the argument given above is quite valid. Only three fourths of the men in the township are present at the meeting and exactly one third of those at the meeting come from each of A, B and C. Those who have two sons come from B only. So their proportion at the meeting is $\frac{1}{3}$ which is also the answer to (ii) (and hence also to (iv)).

Comment No. 11:

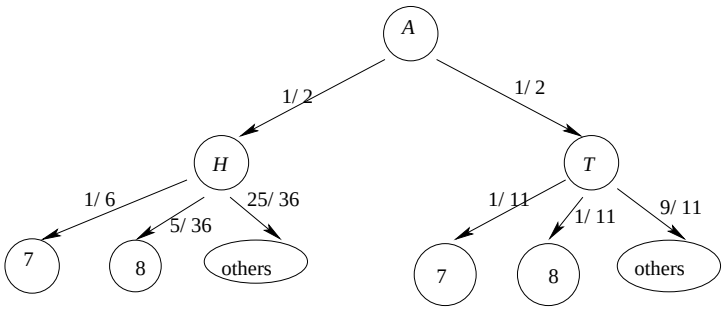
Another limitation on the use of Venn diagrams is that all the events must be shown in terms of subsets of a common sample space, say S. When the same problem involves events of very different types such as flipping a coin and drawing a card, it becomes rather artificial to express them in terms of a common sample space. The natural sample space for the first event is a set $\{H, T\}$ with just two elements, while the pack of cards itself is the right choice for a sample space for the second event. If we want to handle both the problems with a common sample space S, then we would have to let S consist of all ordered pairs of the form (x, y), where x is either H or T and y is a card in the pack. The event that an ace is drawn will then correspond not to the subset of the four aces, but to the subset of S consisting of all ordered pairs of the form (x, y), where x is unrestricted (i.e., can be either H or T) but y is an ace. Obviously, this is clumsy.

In such problems, there is another type of diagram called **tree diagrams** which is more convenient. The name is in analogy with natural trees where at each node, several branches shoot out. So, in a tree diagram for a probability problem, there are certain points or **nodes** which represent events. The various possibilities depending on the outcome of an event are represented by branches

or arrows shooting out from its corresponding node. However, unlike in a natural tree, these branches are often shown as heading downward, as in a family tree. (Sometimes they are also shown as heading towards the right.) To make the tree diagram more revealing, sometimes each branch is labelled by a number which gives its probability. Also, the nodes are labelled so as to indicate the outcome of the previous event leading into it. Sometimes, some more information about the event is written alongside the node representing it. At the lowest level there are 'terminal nodes' (also called 'leaves' sometimes) from which no branches shoot out. They represent the final outcomes whose probabilities are to be found. We illustrate the use of such tree diagrams with the following problem.

An unbiased coin is tossed. If the result is a head, a pair of unbiased dice is rolled and the number obtained by adding the numbers on the two faces is noted. If the result is a tail, a card from a well-shuffled pack of eleven cards numbered $2, 3, \ldots, 12$ is picked and the number on the card is noted. What is the probability that the noted number is either 7 or 8? (JEE 1994)

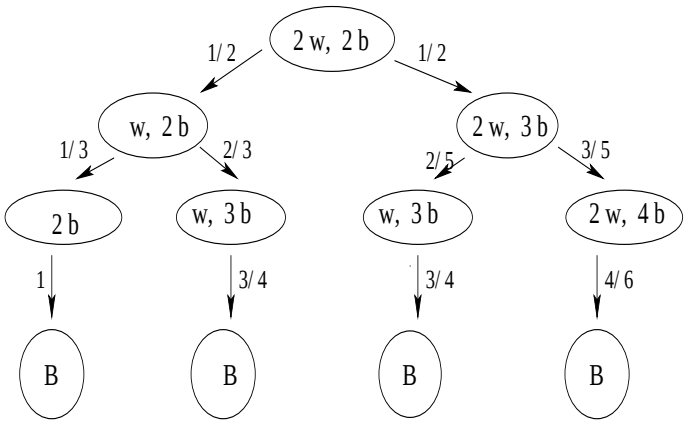
For the solution, we draw the following diagram. The topmost node A corresponds to the toss of a coin. The left branch represents a head and the right one a tail. As the coin is fair, both are labelled with $\frac{1}{2}$. The node H (for a head) at the end of the left branch corresponds to rolling a pair of unbiased dice, while the right edge terminates into a node T (for a tail) corresponding to the drawing of a card. In general, the outcomes of these two events may have nothing in common. But in the present problem each of them has the same possible outcomes, viz., the integers from 2 to 12. However, the probabilities of the outcomes are different. For the rolling of dice, we already remarked in Comment No. 5 above how to calculate the probabilities of the various outcomes. Since our interest is only in two of these outcomes, viz., 7 and 8, we group together the remaining 9 outcomes as 'others' and show only three edges from H leading to the terminal nodes labelled 7, 8 and 'others'. Their probabilities are, respectively, $\frac{1}{6}, \frac{5}{36}$ and $1 - \frac{1}{6} - \frac{5}{36}$, i.e., $\frac{25}{36}$. For the edges originating at the node T all the 11 outcomes are equally likely. Again we show only three edges with probabilities $\frac{1}{11}$, $\frac{1}{11}$ and $\frac{9}{11}$.



The desired probability is now obtained by considering all possible paths originating at A and terminating at any of the four terminal nodes, two of which are marked with 7 and two with 8. There are four such paths. For

each one of them, we multiply the labels of the edges along that path to get the probability of reaching the destination along that path. Then we add these four products to get the desired probability. In the present case, the answer is $\frac{1}{2}(\frac{1}{6}+\frac{5}{36})+\frac{1}{2}(\frac{1}{11}+\frac{1}{11})=\frac{1}{2}(\frac{11}{36}+\frac{2}{11})=\frac{193}{792}$.

As another example, suppose an urn contains 2 white and 2 black balls. A ball is drawn at random. If it is white, it is not replaced into the urn. Otherwise it is replaced with another ball of the same colour. The process is repeated. Find the probability that the third ball drawn is black. (JEE 1987)



The tree diagram is shown above. Each node (except the terminal nodes) is labelled to indicate the contents of the urn at the time of drawing the ball. This makes it easier to mark the edges issuing from that node with their respective probabilities. There are two edges going out from each (non-terminal) node. As we want the colour of the third ball drawn, the terminal nodes will be 3 levels below the topmost node and there will be eight of them. To avoid unnecessary cluttering in the diagram, we show only 4 of them, labelled B, as we want only the probability that the third ball will be black.

The desired probability can now simply be read off from the tree diagram as $\frac{1}{2}(\frac{1}{3}\times 1+\frac{2}{3}\times\frac{3}{4})+\frac{1}{2}(\frac{2}{5}\times\frac{3}{4}+\frac{3}{5}\times\frac{2}{3})$ which comes out to be $\frac{1}{2}(\frac{1}{3}+\frac{1}{2}+\frac{3}{10}+\frac{2}{5})=\frac{23}{30}$.

As yet another example, we complete the solution of the problem introduced above while commenting on conditional probability after Equation (3). Here we have a box containing m biased and n unbiased coins, the probability of a biased coin showing a head being p. It is given further that a coin drawn at random showed a head when tossed. We want to find the probability that the selected coin is biased. We introduced two events, A and B, the first being that the drawn coin is biased and the second being that it shows a head when tossed. We calculated $P(A)$ as $\frac{m}{m+n}$ but postponed the calculation of $P(B)$ and of $P(A\cap B)$. The latter is the probability that the drawn coin is biased and shows a head when tossed. Since the probability of a biased coin showing a head is given to be p, we have $P(A\cap B)=\frac{mp}{m+n}$. To find $P(B)$ we need a tree diagram, which we leave to the reader to draw. (After some practice, in simple

problems one may as well draw such diagrams only mentally. But it is always a good idea to draw them, at least roughly, on a piece of paper.) Here from the initial node there are two branches, one leading to the coin being biased and the other to its being unbiased. Their probabilities are $\frac{m}{m+n}$ and $\frac{n}{m+n}$ respectively. From each of these two nodes there are two branches, one leading to a head and the other to a tail. We are interested only in the branches leading to heads. Their probabilities are p (for the biased coin) and $\frac{1}{2}$ (for the unbiased coin). So, $P(B) = \frac{mp}{m+n} + \frac{n}{2(m+n)} = \frac{2mp+n}{2(m+n)}$. As noted earlier, the desired probability is the ratio of $P(A \cap B)$ to $P(B)$, which comes out to be $\frac{2mp}{2mp+n}$. It is instructive to compare this with $\frac{m}{m+n}$, which is the probability of the coin being biased, without having any information as to the outcome of its toss. If $p = \frac{1}{2}$, then the two are equal. This is to be expected, because in this case, there is hardly any difference between an unbiased and a biased coin. It is just that m of the $m+n$ coins are called biased and the remaining unbiased. However, if $p < \frac{1}{2}$, then $\frac{2mp}{2mp+n} < \frac{m}{m+n}$, i.e., $P(A|B) < P(A)$. This means that the additional knowledge that the first toss is a head makes it less likely that the coin is biased. This is, of course, also obvious by common sense. Similarly, if $p > \frac{1}{2}$, then the knowledge that the toss is a head improves the chances that the coin drawn is a biased one.

Comment No. 12:

The tree diagrams are akin to how a person's estate goes on splitting with the various shares going to his descendants. The terminal nodes are like persons who have no issues and who are, therefore, the ultimate beneficiaries. These terminal nodes constitute the sample space and the probability or the 'weight' of each node is the portion of the estate he inherits, which is the product of the probabilities of the edges of the path leading from the initial node to that particular terminal node. This analogy makes it a little easier to develop familiarity with the tree diagrams. When applied to probability problems, these diagrams aid the thought process by ensuring that no possibility is missed out and that no two distinct possibilities are inadvertently mixed. The tree diagrams are, of course, only a visual representation of a certain thought process. After a little practice, it is not necessary to draw them every time. In fact, it is impracticable to draw them in full in those problems where there is a long chain of events (e.g., flipping a coin ten times). At any rate, if the thought process is clear, one can get the answer even without the tree diagrams as we illustrate in the problem below.

A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 and the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till all defective articles are found. What is the probability that the testing procedure ends at the twelfth testing? (JEE 1986)

Here the catch is that it is not given beforehand whether the lot contains 2 or 3 defective articles. Therefore even if 2 defective articles are found, we cannot be sure that the testing procedure is over, until either one more defective article

is found or until all articles have been tested. So, the testing procedure will end at the twelfth testing if and only if the twelfth article is defective and 2 out of the earlier 11 articles are also defective. The probability of this happening is 0 if, to start with the lot has only 2 defective articles, the probability of which is $\frac{2}{5}$. The other possibility, viz., that there are 3 defective articles has probability $\frac{3}{5}$. Assuming that we are in this case, the rest of the solution is analogous to that of the problem above in Comment No. 5 regarding drawing aces from a pack of cards. We let the sample space S consist of all permutations of length 12 of the 20 articles. Then $|S| = {}^{20}P_{12} = \frac{20!}{8!}$. All these permutations are equally likely as the articles are drawn randomly. Let A be the set of the 'favourable' permutations, i.e., those permutations (in S), where the last article is defective and two of the first 11 articles are defective. The last defective article can be any of the 3 defective ones. The places for the remaining 2 defective articles can be chosen in $\binom{11}{2}$, i.e., in 55 ways, and each choice can be filled in 2 ways by the 2 defective articles. The remaining 9 places in the permutation can be filled by the 17 non-defective articles in ${}^{17}P_9$, i.e., in $\frac{17!}{8!}$ ways. Multiplying, we get $|A| = 3 \times 55 \times 2 \times \frac{17!}{8!} = 330 \times \frac{17!}{8!}$. So, $P(A) = \frac{|A|}{|S|} = 330 \times \frac{17!}{8!}\frac{8!}{20!} = \frac{330}{18\times19\times20} = \frac{11}{228}$.

Summing up, the probability that the testing is over at the twelfth round is 0 if the first possibility holds (viz., that the lot contains 2 defective articles) while it is $\frac{11}{228}$ if the second possibility holds (viz., that the lot contains 3 defective articles). As the probabilities of these two possibilities are given to be $\frac{2}{5}$ and $\frac{3}{5}$ respectively, the answer to the problem is $0 \times \frac{2}{5} + \frac{11}{228} \times \frac{3}{5} = \frac{33}{1140}$. Note that we did not draw any tree diagram. Such a diagram is actually inherent in the thought process, for example, the initial branching of the problem into two possibilities depending upon whether the lot contains 2 or 3 defective articles. It is not necessary that the diagram be drawn explicitly on a piece of paper. It is a good idea to draw it as a part of the rough work. But in probability problems, it is not an essential part of the final solution, the way a diagram is a must in trigonometric problems of heights and distances. In fact, in a numerical problem like this, the moment an examiner sees the correct answer, viz., $\frac{33}{1140}$, he is likely to ignore any lacunas in its presentation. And he can hardly be blamed for it, because, after all, it is the thought process that really matters. And in the present problem, it is simply impossible for anybody to get the weird answer $\frac{33}{1140}$ without going through the correct thought process!

A tree diagram is also implicit in the solution to the following problem.

Suppose an anti-aircraft gun can take a maximum of four shots at an enemy plane moving away from it. The probabilities of hitting the plane at the first, the second, the third and the fourth shot are 0.4, 0.3, 0.2 and 0.1 respectively. Find the probability that the the aircraft is shot down. (JEE 1981)

Here it is easier to find the complementary probability, i.e., the probability, say p, that all the shots fail. We (mentally) draw a tree with four nodes corresponding to the four shots. At each node there are two edges. The edge corresponding to success of the shot terminates into a terminal node as no more shots are fired. The edge corresponding to the failure of the shot leads to the node corresponding to the next shot, if any, or to the terminal node correspond-

ing to the failure of all four shots. This is the node we want and the path from the initial node to it has four edges with probabilities 0.6, 0.7, 0.8 and 0.9 respectively. (These are obtained as complementary probabilities of the success probabilities of the shots.) So $p = 6 \times 7 \times 8 \times 9 \times 10^{-4} = \frac{3,024}{10,000}$. Hence the desired probability equals $1 - \frac{3,024}{10,000} = \frac{6,976}{10,000}$.

There is an alternate way to do this problem. Note that the four shots represent four independent events. Whether the second shot will be fired or not does depend on the outcome of the first shot. But if fired, its outcome is independent of the outcome of the first shot. The same holds for the third and the fourth shots. So, as far as probabilities are concerned, we might suppose that all the four shots are fired regardless of the outcomes of the earlier shots if any. (If we want, we may regard the shots fired after a successful shot as dummy shots.) Now the sample space, say X, consists of 16 sequences of the form (x, y, z, w), where each entry is either an S (for success) or an F (for failure). However, these 16 elements do not have equal probabilities. For example the sequence (S, F, S, S) has probability $\frac{4\times7\times2\times1}{10,000} = \frac{56}{10,000}$. But the sequence (F, S, S, S) has probability $\frac{6\times3\times2\times1}{10,000} = \frac{36}{10,000}$. There is only one unfavourable sequence, viz., (F, F, F, F) whose probability is $\frac{6\times7\times8\times9}{10,000} = \frac{3,024}{10,000}$. Subtracting this from 1 gives the same answer as before.

Although the second solution is quite correct mathematically, some people are not easily convinced of its validity. To compare the two solutions, the sample space, say Y, in the tree approach consists of only the 5 terminal nodes. We can represent them with sequences in which a string of F's is followed by a single S if any. In other words, Y is the 5-element set $Y = \{(S), (F, S), (F, F, S), (F, F, F, S), (F, F, F, F)\}$. The probabilities of these elements are, respectively, $\frac{4}{10}$, $\frac{18}{100}$, $\frac{84}{1000}$, $\frac{336}{10,000}$ and $\frac{3,024}{10,000}$. Now define a function $f : X \longrightarrow Y$ as follows. A typical element of X is a sequence of length 4 with entries F or S. We scan this sequence from the left and terminate it as soon as we encounter an S. Thus, for example, $f((F, S, F, S)) = (F, S)$. $f((F, F, F, F)) = (F, F, F, F)$. Note that under f, all the four sequences $(F, S, F, F), (F, S, F, S)$, (F, S, S, F) and (F, S, S, S) are mapped onto the same element of Y, viz., (F, S). Further, the sum of the probabilities of these four elements of the sample space X is the same as the probability of the single element (F, S) of the sample space Y. This is actually true for every element of Y. Its probability is precisely the sum of the probabilities of all the elements in its inverse image under f. Put differently, whether we do the problem with X as the sample space or with Y as the sample space, we are ultimately adding the same probabilities. In the former, we add them directly. In the latter we first bunch them together (under the function f) and add the bunches together. (This is very similar to the comparison of the two solutions to the problem discussed in Comment No. 6 above, except that there all the elements of the sample spaces were equally likely.)

Comment No. 13:

In many problems, we encounter a sequence of identical but independent events (sometimes called 'trials' or 'experiments') such as a series of matches

played, a coin tossed in succession (or, what amounts to the same, several such coins tossed simultaneously) and our interest is only in the number of trials a particular outcome occurs and not in which ones of the trials it occurs. Suppose, for example, that a biased coin, which has a probability p of showing a head, is tossed 5 times and we want the probability, say p_3 that a head shows exactly thrice. Here it does not matter *which* three of the tosses give a head. Such a problem can, of course, be done with a tree diagram. But there is a much better way. Let us first find the probability, say α, that the first three tosses are heads (and the last two tails). As the tosses are independent of each other, $\alpha = p^3(1-p)^2$, or, p^3q^2, following the standard practice of writing q for $1-p$. But this would also be the probability that the i-th, j-th and the k-th tosses are heads and the remaining two are tails, where i, j, k are any three integers such that $1 \leq i < j < k \leq 5$. The choice of such a triplet can be made in $\binom{5}{3}$ i.e., in 10 ways. In effect, we have expressed the desired event (viz., that of exactly 3 heads showing) as a disjunction of 10 mutually exclusive events, each of which has probability p^3q^2. So, the desired probability p_3 equals $10p^3q^2$.

More generally, the probability of getting (exactly) r heads in n tosses is $\binom{n}{r}p^rq^{n-r}$. As this is also one of the terms in the binomial expansion of $(p+q)^n$, this probability is often called the **binomial probability**. Actually the resemblance is not coincidental. When we expand $(p+q)^n$ by multiplying the product of the n factors $(p+q)(p+q)\ldots(p+q)$, the term p^rq^{n-r} occurs $\binom{n}{r}$ times, every occurrence corresponding to choosing the term p from r of these factors and the term q from the remaining $n-r$ factors.

We already used binomial probability in the solution to our Main Problem in an implicit manner when we considered the outcome of the first 9 games. More precisely there we used what is called the trinomial probability, because there were three possible outcomes in each game. Here are a couple of problems where binomial probability is the central idea.

As a very straightforward example, suppose a man takes a step forward with probability 0.4 and backwards with probability 0.6. We are asked to find the probability that at the end of eleven steps he is one step away from the starting point. (JEE 1987) Here the event is the disjunction of two events (i) the man takes exactly 6 steps forward and 5 backwards (in whatever order) and (ii) he takes 5 steps forward and 6 backwards. The probabilities of these two events are, respectively, $\binom{11}{6}p^6q^5$ and $\binom{11}{5}p^5q^6$, where $p = 0.4$ and $q = 0.6$. As these two events are mutually exclusive, the desired probability is the sum of these two. It comes out to be $\binom{11}{5}p^5q^5$, or numerically, $462 \times (0.24)^5$.

Here is a less elementary problem. (JEE 1989)

Suppose the probability for A to win a game against B is 0.4. If A has an option of playing either a "best of 3 games" or a "best of 5 games" match against B, which option should he choose so that the probability of his winning the match is higher? (No game is a draw.)

Let p_1 and p_2 be, respectively, the probabilities of A's winning a 'best of 3' match and a 'best of 5' match. The problem does not make it clear how 'best of 3' is decided. In one of the methods, if the same player wins the first two games then the third game is not played. In another method, all the three games are

played anyway. It does not really make any difference which method is followed as far as probability is concerned, because the last game may be treated as a dummy game if necessary. (See the remarks at the end of Comment No. 12 above regarding the equivalence of the two solutions to the problem about the anti-aircraft gun.) Thus we assume that in a 'best of 3' match, 3 games are played. For A to win the match, he must win 2 or 3 of these games. The probability of winning exactly 2 out of 3 games is $\binom{3}{2}(0.4)^2(0.6) = \frac{36}{125}$ while that of winning all three is $(0.4)^3 = \frac{8}{25}$. Adding the two, $p_1 = \frac{44}{125}$. The calculation of p_2 is similar. Adding $\binom{5}{3}(0.4)^3(0.6)^2$, $\binom{5}{4}(0.4)^4(0.6)$ and $(0.4)^5$ we get $p_2 = \frac{720+240+32}{3,125} = \frac{992}{3,125}$. Since $p_1 = \frac{44}{125} = \frac{1,100}{3,125}$, we have $p_1 > p_2$ and so A should choose the 'best of 3 games' option.

In this problem, the probability p was given. In some problems this has to be first found from some other information. Here is an illustration.

In a multiple choice question there are four alternative answers, of which one or more are correct. A candidate will get marks in the question only if he correctly ticks all the correct answers. The candidate decides to tick the answers at random. If he is allowed upto three chances to answer the question, find the probability that he will get marks in the question. (JEE 1985)

Let X be the set of all four alternative answers and let C be the set of the correct answers. Then C is given to be a non-empty subset of X. In all, X has $2^4 = 16$ subsets and hence 15 non-empty subsets. In absence of any other information, we must assume that all these 15 subsets are equally likely. Now let A be the set of the alternatives ticked by the candidate. Then A is also a non-empty subset of X. The assumption that the candidate is marking the alternatives at random amounts to saying that he is choosing one non-empty subset A of X at random. The candidate will get marks if and only if A is the same as C. So the probability of this happening is $\frac{1}{15}$. Call this as p. This is the probability of getting marks at one attempt.

From this point onwards, the problem becomes ambiguous. The candidate is allowed upto three chances to answer the question. But it is not given how many of these he will exercise. This may seem like a silly question. But if each attempt is going to cost him some fee, it is quite possible that he may opt to quit after one or two attempts. As nothing is mentioned by way of this possibility, we go by common sense and assume that the candidate exercises all three chances. Even then, if we go strictly by the text of the problem, his answers at every attempt will be given randomly. In other words, all the three attempts are independent of each other and the probability of success at each is the same, viz., $p = \frac{1}{15}$. In that case, the probability of at least one success is found most easily by finding the complementary probability, i. e. the probability of failure on all three attempts, which is $(\frac{14}{15})^3$. Hence the probability that the candidate gets marks in the question is $1 - (\frac{14}{15})^3 = \frac{631}{3,375}$.

Let us, however, again let common sense prevail. Then evidently the candidate will choose three different non-empty subsets, say A_1, A_2 and A_3 at the three attempts. In other words, he will choose a permutation of length 3 of the 15 possible options. There are $15 \times 14 \times 13$ such permutations. The number

of such permutations in which none of the A_i equals C (the correct choice) is $14\times13\times12$. So the probability of failure is $\frac{14\times13\times12}{15\times14\times13}=\frac{4}{5}$. Hence the probability of getting marks is $1-\frac{4}{5}$, i.e., $\frac{1}{5}$ which is slightly higher than the earlier answer, viz., $\frac{631}{3,375}$.

The statement of the problem is silent whether the candidate is told after each attempt whether his attempt was successful or not as is the case in some interactive examinations (such as interviews). Here, if the candidate knows that he is unsuccessful at an attempt, we tend to think that this additional information will improve his chances at the next attempt. But this does not make any difference, because even without this information, the candidate will choose a different option anyway to improve his chances. The problem then becomes analogous to that of the anti-aircraft gun at the end of Comment No. 12. With a tree approach, the probability of getting marks in an interactive examination is $\frac{1}{15}+\frac{14}{15}\frac{1}{14}+\frac{14}{15}\frac{13}{14}\frac{1}{13}$ which is $\frac{1}{5}$, exactly the same as before.

In the following problem we have to find p from the binomial probability.

One hundred identical coins, each with probability p of showing a head, are tossed once. If $0<p<1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, find p. (JEE 1988)

Here it is given that $\binom{100}{50}p^{50}q^{50}=\binom{100}{51}p^{51}q^{49}$, where, as usual, $q=1-p$. Cancelling the common factors this reduces to $\frac{q}{50}=\frac{p}{51}$, i.e., $51(1-p)=50p$, which gives $p=\frac{51}{101}$.

As the final example of binomial probability, let us consider what is called a **ruin problem**. In this problem a player A plays a game against a machine. At each round he deposits one rupee in a slot and then flips a coin which has a probability p of showing a head. If a head shows, he gets back the rupee he deposited and one more rupee from the machine. If a tail shows, he loses his rupee. The player is said to be ruined if he is left with no money. If A starts with 10 rupees, find the probability that he is ruined by the 14-th round or earlier.

For the solution, note first that since the player starts with an even number of rupees, he cannot be ruined at the end of an odd-numbered round. So we consider separately the probabilities of his ruin on the 10th, the 12th and the 14th round. He will be ruined at the end of the tenth round if and only if he gets a tail on each of the first 10 rounds. The probability of this is q^{10}, where, as usual, $q=1-p$. To be ruined at the 12-th round (but not earlier) he must have exactly 1 head in the first 10 rounds, and a tail on the 11-th and the 12-th round. The probability of this is $\binom{10}{1}pq^{11}$. A ruin on the 14-th round (but not earlier) can ocur in two ways : (i) 2 heads in the first 10 rounds and all tails in the next 4 rounds, or (ii) 1 head in the first 10 rounds, 1 in the next two and no head in the 13th and 14th rounds. The probabilities of (i) and (ii) are, respectively, $\binom{10}{2}p^2q^8\times q^4=45p^2q^{12}$ and $10pq^9\times2pq\times q^2=20p^2q^{12}$. Adding all possibilities, the probability of a ruin on the 14-th round or earlier is $q^{10}(1+10pq+65p^2q^2)$. In the case of a fair coin, this becomes $\frac{121}{16,384}$. After the 14-th round, the calculations become more complicated, because at each even-numbered round, the ruin can occur in many different ways.

We caution that the method of binomial probability *does not* apply when the trials are dependent, as for example, when a person is taking successive tests or playing games in a row and the probability of his success at a particular test or game depends on his performance on the earlier ones. In such cases, the tree diagrams (or rather, the thought process behind them) is the only go. See Problem 5 in the Main Paper in the Educative Commentary on JEE 2003 Mathematics Papers (available on the author's personal home page) for an example of such a problem.

Comment No. 14:

Suppose a trial is conducted a number of times independently and we want the probability of an event which is specified by the number of successful trials. Then the binomial probability is the right tool as we saw above. For example, suppose a coin with p as the probability of showing a head is tossed 10 times in a row. If the event is that exactly three heads show, then its probability is $\binom{10}{3}p^3q^7$. If the event A is, say, that at least three heads and at least four tails show, then we have to first express the event as the disjunction of several events, viz., (i) 6 heads, 4 tails (ii) 5 heads, 5 tails, (iii) 4 heads, 6 tails and (iv) 3 heads, 7 tails, find the binomial probabilities in each case and add the four answers to get $P(A)$. If the restriction imposed by the event is regarding the absolute positions of successful and unsuccessful trials, then we get the answer directly from the independence of the trials, without having to use binomial probability. For example, with the same coin as above the probability that a head will show on the first, the third and the seventh toss and a tail will show on the fourth and the eighth tosses (there being no restriction on the other tosses) then the answer is simply p^3q^2.

Sometimes, however, we encounter problems where the restriction is neither on the number of successful trials nor on their absolute positions, but on their *relative* positions, for example, a restriction to the effect that no three heads should appear consecutively, or that every head should be preceded at least with two tails and so on. Such problems tend to be fairly complicated, even for the relatively simple case of an unbiased coin, where $p = q = \frac{1}{2}$. In such problems, it is very difficult to come up with the exact number of favourable cases. And so techniques such as induction have to be used. We illustrate this in the following problem. (JEE 2000)

A coin has probability p of showing a head when tossed. For a positive integer n, let p_n be the probability that when the coin is tossed n times, no two (or more) consecutive tosses show heads. Prove that $p_1 = 1, p_2 = 1 - p^2$ and that for $n \geq 3$,

$$p_n = (1-p)p_{n-1} + p(1-p)p_{n-2}\ . \tag{10}$$

Prove further by induction on n, that for all $n \geq 1$,

$$p_n = A\alpha^n + B\beta^n \tag{11}$$

where α and β are the roots of the quadratic equation $x^2 - (1-p)x - p(1-p) = 0$

and

$$A = \frac{p^2 + \beta - 1}{\alpha\beta - \alpha^2} \quad \text{and} \quad B = \frac{p^2 + \alpha - 1}{\alpha\beta - \beta^2} \tag{12}$$

The assertion $p_1 = 1$ holds trivially because when there is only one toss, there is no question of two consecutive heads appearing. That $p_2 = 1-p^2$ follows by considering the complementary probability, viz., the probability that both the tosses are heads. Now suppose $n \geq 3$. We have a chain of n independent events, viz., n tosses of the coin and the outcome will be any of 2^n possible sequences of length n of the form $(x_1, x_2, \ldots, x_n)$ with each $x_i = H$ (for a head) or T (for a tail). (Note that these 2^n sequences are not equally likely unless $p = \frac{1}{2}$, i.e., unless the coin is unbiased.) Let us call such a sequence as successful if no two consecutive entries are H. The first entry x_1 can be H (with probability p) or T (with probability $1 - p$). If it is T, then the sequence $(x_1, x_2, x_3, \ldots, x_n)$ will be a successful sequence of length n if and only if the sequence $(x_2, x_3, \ldots, x_n)$ is a successful sequence of length $n-1$, the probability of which is p_{n-1}. So the probability of success with $x_1 = T$ is $(1 - p)p_{n-1}$.

Suppose, however, that $x_1 = H$. If x_2 also equals H then we already have two consecutive heads and cannot have success. However, if $x_2 = T$, then we shall have success if and only if the sequence $(x_3, x_4, \ldots, x_n)$ is a successful sequence of length $n - 2$ and the probability of this is p_{n-2}. So the probability of success with the first toss H is $p(1 - p)p_{n-2}$. Adding these two probabilities gives (10).

Now to prove (11) by induction, the cases $n = 1$ and $n = 2$ amount to showing, respectively, that

$$1 = A\alpha + B\beta = \frac{p^2 + \beta - 1}{\beta - \alpha} + \frac{p^2 + \alpha - 1}{\alpha - \beta} \tag{13}$$

$$\text{and} \quad 1 - p^2 = A\alpha^2 + B\beta^2 = \frac{p^2\alpha + \alpha\beta - \alpha}{\beta - \alpha} + \frac{p^2\beta + \alpha\beta - \beta}{\alpha - \beta} \tag{14}$$

both of which follow by direct easy computations. For the inductive step, we have to use (10). Note, however, that (10) expresses p_n not in terms of p_{n-1} alone, but in terms of p_{n-1} and p_{n-2}. So, we cannot expect the truth of (11) for a given value of n (say, $n = k$) to follow from that for $n = k - 1$. We have to assume (11) true for $n = k - 1$ and for $n = k - 2$ also. (This is also the reason why the induction had to begin with the verification of the first two cases and not just the first one case as is normally the case in a proof by induction.)

Now, assuming that (11) holds for $n = k - 1$ and for $n = k - 2$ and substituting into the R.H.S. of (10) (with n replaced by k), we get

$$p_k = (1 - p)(A\alpha^{k-1} + B\beta^{k-}) + p(1 - p)(A\alpha^{k-2} + B\beta^{k-2}) \tag{15}$$

But since α and β are the roots of the quadratic equation $x^2-(1-p)x-p(1-p) = 0$, we have $1 - p = \alpha + \beta$ and $p(1 - p) = -\alpha\beta$. If we substitute these in the R.H.S. of (15) it reduces to $A\alpha^k + B\beta^k$. So we have proved (11) for $n = k$ thereby completing the induction.

Note that the last part of the problem (viz., proving (11)) is exactly of the same spirit as the problem solved in Comment No. 12 of Chapter 4. In effect we are asked to verify that (11) is a solution of the recurrence relation (10), satisfying the initial conditions $p_1 = 1$ and $p_2 = 1 - p^2$. The only difference is that here we first obtained the recurrence relation from some problem while in the earlier problem it was given to us. Recurrence relations arise naturally in many connections. They can be thought of as the discrete analogues of differential equations. There is a general theory of solving recurrence relations such as (10). And not surprisingly, it is analogous in certain respects to the theory of solving differential equations. If we use it then we can *arrive at* (11), instead of merely *verifying* it by induction as we did above. But we shall not go into it.

EXERCISES

22.1 Balls are drawn one-by-one without replacement from a box containing 2 black, 4 white and 3 red balls till all the balls are drawn. Find the probability that the balls drawn are in the order 2 black, 4 white and 3 red. (1978)

22.2 Two fair dice are tossed. Let x be the event that the first die shows an even number and y be the event that the second die shows an odd number. Are the two events x and y (i) independent, (ii) mutually exclusive ? (1979)

22.3 An objective type test paper consists of 10 questions with 4 choices for each question. If a student selects his answers randomly for each question, what is the probability that he gets at most 9 correct answers? If two students were to answer the same paper independently, what is the probability of all their answers being identical *and* wrong? (JAT 1979)

22.4 (a) Two events A and B have probabilities 0.25 and 0.50 respectively. The probability that both A and B occur simultaneously is 0.14. Find the probability that neither A nor B occurs. (1980)

(b) The probability that at least one of the events A and B occurs is 0.6. If A and B occur simultaneously with probability 0.2, find $P(\bar{A}) + P(\bar{B})$. (1987)

(c) For two given events A and B which of the following statements is(are) true about $P(A \cap B)$? (1988)

(A) it is not less than $P(A) + P(B) - 1$

(B) it is not greater than $P(A) + P(B)$

(C) it is equal to $P(A) + P(B) - P(A \cup B)$

(D) it is equal to $P(A) + P(B) + P(A \cup B)$.

(d) Let E and F be two independent events. The probability that both E and F happen is $\frac{1}{12}$ and the probability that neither E nor F happens is $\frac{1}{2}$. Determine which of the following statements *may be* true. (1993)

(A) $P(E)=\frac{1}{3}, P(F)=\frac{1}{4}$ (B) $P(E)=\frac{1}{2}, P(F)=\frac{1}{6}$
(C) $P(E)=\frac{1}{6}, P(F)=\frac{1}{2}$ (D) $P(E)=\frac{1}{4}, P(F)=\frac{1}{3}$.

(e) If E and F are independent events such that $0 < P(E) < 1$ and $0 < P(F) < 1$, which of the following statements are true? (1989)

(A) E and F are mutually exclusive

(B) E and F^c (the complement of the event F) are independent

(C) E^c and F^c are independent

(D) $P(E|F) + P(E^c|F) = 1$.

(f) If two events A and B are such that $P(A^c) = 0.3, P(B) = 0.4$ and $P(AB^c) = 0.5$, find $P[B|(A \cup B^c)]$. (1994)

(g) If $\bar{E}$ and $\bar{F}$ are the complementary events of the events E and F respectively, and if $0 < P(F) < 1$, which of the following statements is(are) true? (1998)

(A) $P(E|F) + P(\bar{E}|F) = 1$ (B) $P(E|F) + P(E|\bar{F}) = 1$
(C) $P(\bar{E}|F) + P(E|\bar{F}) = 1$ (D) $P(E|\bar{F}) + P(\bar{E}|\bar{F}) = 1$.

(h) If X and Y are two sets, which, if any, of the following equals $X \cap (X \cup Y)^c$? (1979)

(A) X (B) Y (C) $\emptyset$.

(i) If the probability for A to fail in an examination is 0.2 and that for B is 0.3, is it true that the probability that either A or B fails is 0.5? (1989)

22.5 Find a necessary and sufficient relation between $P(A)$ and $P(B)$ in order that $P(A \cup B) = P(A \cap B)$. (1985)

22.6 (a) An investigator interviewed 100 students to determine their preferences for the three drinks : milk (M), coffee (C) and tea (T). He reported the following: 10 students had all the three drinks, M, C and T ; 20 had M and C ; 30 had C and T ; 25 had M and T ; 12 had M only ; 5 had C only ; and 8 had T only. Using a Venn diagram find how many did not take any of the three drinks. (1978)

(b) A, B, C are events such that $P(A) = 0.3, P(B) = 0.4, P(C) = 0.8$, $P(AB) = 0.08, P(AC) = 0.28$ and $P(ABC) = 0.09$. If further $P(A \cup B \cup C) \geq 0.75$, then show that $P(BC)$ lies in the interval $0.23 \leq x \leq 0.48$. (1983)

22.7 The probability that an event A happens in one trial of an experiment is 0.4. Three independent trials of the experiment are performed. Find the probability that the event A happens at least once. (1980)

22.8 (a) If letters of the word 'ASSASSIN' are arranged at random, find the probability that no two S's are together. (1983)

(b) Seven white balls and three black balls are randomly placed in a row. Find the probability that no two black balls are placed adjacently. (1998)

(c) If 7 white, 6 black and 1 red balls are arranged randomly in a row, what is the probability that no two balls of the same colour are adjacent? What if we have 6 white, 6 black and 1 red balls?

22.9 (a) Three identical dice are rolled. Find the probability that the same number will appear on each of them. (1984)

(b) An unbiased die with faces marked 1, 2, 3, 4, 5 and 6 is rolled four times. Find the probability that out of the four face values obtained, the minimum face value is not less than 2 and the maximum face value is not greater than 5. (1993)

(c) Fifteen coupons are numbered $1, 2, \ldots, 15$, respectively. Seven coupons are selected at random one at a time with replacement. Find the probability that the largest number appearing on a selected coupon is 9. (1983)

(d) Three numbers are chosen at random without replacement from the set $\{1, 2, \ldots, 10\}$. Find the probability that the minimum of the chosen numbers is 3 or their maximum is 7.

22.10 For a biased die the probabilities for the different faces to turn up are:

Face	1	2	3	4	5	6
Probability	0.1	0.32	0.21	0.15	0.05	0.17

The die is tossed and you are told that either face 1 or face 2 has turned up. Find the probability that it is face 1. (1981)

22.11 A box contains 100 tickets numbered $1, 2, \ldots, 100$. Two tickets are chosen at random. It is given that the maximum number on the two chosen tickets is not more than 10. Find the probability that the minimum number on them is 5. (1985)

22.12 If $\frac{1+3p}{3}, \frac{1-p}{4}, \frac{1-2p}{2}$ are probabilities of three mutually exclusive events then identify the set of possible values of p. (1986)

22.13 A student appears for tests I, II and III. The student is successful if he passes either in tests I and II or in tests I and III. The probabilities of the student passing in tests I, II and III are p, q and $\frac{1}{2}$ respectively. If the probability that the student is successful is $\frac{1}{2}$, which of the following possibilities can hold? (1986)

(A) $p = q = 1$ (B) $p = q = \frac{1}{2}$ (C) $p = 1, q = 0$ (D) $p = 1, q = \frac{1}{2}$.

22.14 The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has a 75% chance of passing in at least one, a 50% chance of passing in at least two, and a 40% chance of passing in exactly two. Which of the following relations are true? (1999)

(A) $p+m+c = \frac{19}{20}$ (B) $p+m+c = \frac{27}{20}$ (C) $pmc = \frac{1}{10}$ (D) $pmc = \frac{1}{4}$.

22.15 A box contains 2 fifty paise coins, 5 twentyfive paise coins and and a certain fixed number N (≥ 2) of ten and five paise coins. Five coins are taken out of the box at random. Find the probability that the total value of these coins is less than one rupee and fifty paise. (1988)

22.16 (a) Three distinct integers are selected at random from the set of integers from 1 to 20. Find the probability that the numbers selected are in an A.P.

(b) If p and q are chosen randomly from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, with replacement, determine the probability that the roots of the equation $x^2 + px + q = 0$ are real. (1997)

22.17 If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, find the probability that 2 white and 1 black ball will be selected. (1998)

22.18 An urn contains m white and n black balls. A ball is drawn at random and is put back into the urn along with k additional balls of the same colour as that of the ball drawn. A ball is again drawn at random. What is the probability that the ball drawn now is white? (2001)

22.19 A box contains N coins, m of which are fair and the rest are biased. The probability of getting a head when a fair coin is tossed is $\frac{1}{2}$, while it is $\frac{2}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and is tossed twice. The first time it shows head and the second time it shows tail. What is the probability that the coin drawn is fair? (2002)

22.20 If a coin which has probability $\frac{3}{4}$ of showing a head is tossed n times then show that the probability that no three heads appear consecutively is $(\frac{3}{4})^{n+1}(2 + \alpha^{n+1} + \beta^{n+1})$ where α and β are the roots of the equation $3x^2 + 2x + 1 = 0$.

22.21 Urn A contains 6 red and 4 black balls and urn B contains 4 red and 6 black balls. One ball is drawn at random from urn A and placed in urn B. Then one ball is drawn at random from urn B and placed in urn A. If one ball is now drawn at random from urn A, find the probability that it is found to be red. (1988)

22.22 There are four machines and it is known that exactly two of them are faulty. They are tested, one by one, in a random order till the faulty machines are identified. Find the probability that only two tests are needed. (1998)

22.23 One hundred persons stand in a queue and buy tickets one-by-one. When a ticket is given, the birthday of the person is also noted. A free ticket is given to the first person, if any, whose birthday is already recorded. Ignoring leap year and assuming that all birthdays are equally likely, find which person in the queue has the maximum chance of getting a free ticket.

Chapter 23

INFINITISTIC PROBABILITY

When the sample space is infinite, the naive definition of probability as the ratio of the number of favourable cases to the total number of cases breaks down completely. Even the relatively more general definition as the sum of the probabilities of the favourable cases becomes inapplicable. Problems of this type come under 'infinitistic probability'. Like other things involving infinity, the right way to attack them is by approximating them by a sequence of problems of finitistic probability studied in the last chapter. The Main Problem is of this type where the sample space consists of the possible outcomes of an infinite sequence of trials. The answer naturally leads us to the definition of the sum of an infinite series. We do this in Comment No. 2. We then extend some of the problems of finitistic probabilities to those of infinitistic probability. Especially interesting is a ruin game between two players. We also study the concept of an average or expected value.

An essentially different type of infinistic probability arises when the sample space is an interval of real numbers. Such problems are beyond JEE. But we give a very brief exposure to them because so many real-life problems can be cast in this form.

Problems of infinitistic probability are asked very rarely in the JEE. Still, they are a natural sequel to finitistic probability and well deserve at least a cursory look. Moreover, they cover some very interesting problems that deal with outcomes of games which could, in theory, go on for infinitely many rounds till some player wins them. (The Main Problem involves one such game.) Such problems cannot be handled with finitistic probability. What is more important, a 'game' need not always be interpreted literally. The same logic applies in many serious real life problems wherever there is an element of uncertainty, e.g. in deciding the premium for an insurance. So, the topic is worth a study.

Main Problem : In a game, three players A, B and C, toss a coin cyclically in that order (that is, $A, B, C, A, B, C, A, B, \ldots$) till a head shows and the player who gets that head is said to win the game. Suppose the coin has probability 0.4 of a head showing. Then the probability that B wins the game is ..... . (JEE 1998)

First Hint: Let α, β and γ be the probabilities of wins for A, B, C respectively. Obtain a system of three equations in α, β, γ.

Second Hint: Prove that $\beta = \frac{6}{10}\alpha$.

Third Hint: Express γ in terms of α.

Solution: Let A, B, C also denote the events that the corresponding players win the game. Then certainly these three events are mutually exclusive. We further claim that they are exhaustive. For, the complementary event of $A \cup B \cup C$ is $A' \cap B' \cap C'$, i.e., the event that none of the three players wins the game. But the only way this can happen is that the coin would go on being tossed without a head ever showing. For every positive integer n, let p_n be the probability that the coin is tossed n times with no head showing. Then $p_n = (\frac{3}{5})^n$. Since $|\frac{3}{5}| < 1$, we have $p_n \to 0$ as $n \to \infty$. So, $P(A' \cap B' \cap C') = \lim_{n\to\infty} p_n = 0$. But this means that $P(A \cup B \cup C) = 1 - P(A' \cap B' \cap C') = 1$, i.e., that the events A, B, C are exhaustive. So if we let α, β and γ be as in the hint, we have

$$\alpha + \beta + \gamma = 1 \tag{1}$$

We need two more equations to get the values of α, β and γ. We are letting α be the probability that the first player wins the game. In order that the second player B win, it is necessary, first of all, that the first toss be a tail. The probability of this is 0.6. After the first toss shows a tail, the coin will be flipped in the order $B, C, A, B, C, A, B, \ldots$ until a head shows. So it is as if the same game starts again, except now the first player is B and not A. So the probability that B will win this new game is α. Thus we get

$$\beta = \frac{6}{10}\alpha \tag{2}$$

By a similar reasoning, in order for C to win, the first two tosses have to be tails, the probability of this being $\frac{36}{100}$. Thereafter, a new game starts with C as the first player. This gives

$$\gamma = \frac{36}{100}\alpha \tag{3}$$

Substituting (2) and (3) in (1) and solving, we get $\alpha = \frac{100}{196} = \frac{25}{49}$. From (2) we then get $\beta = \frac{15}{49}$ as B's probability of winning.

Comment No. 1:

More generally, if the coin had a probability p of showing a head, then (1) would remain unaffected. However, instead of (2) and (3), we would have, respectively, $\beta = q\alpha$ and $\gamma = q^2\alpha$, where $q = 1 - p$. Solving, we would get

$$\alpha = \frac{1}{1+q+q^2}\ , \quad \beta = \frac{q}{1+q+q^2} \ \text{ and } \ \gamma = \frac{q^2}{1+q+q^2} \tag{4}$$

Comment No. 2:

The justification needed for (1) was twofold. First we had to show that the events A, B, C are mutually exclusive. This was easy. But we also had to show that they are exhaustive. In ordinary terms, this means that at least one of them must occur. But this actually was not true in the strict sense. For, it is theoretically possible that we could go on flipping a coin forever, with no head showing. Put differently, the event $A' \cap B' \cap C'$ is not a logical impossibility. What we have done in the proof above is to show that this event is *improbable*, i.e., its probability is 0. It is not easy to accept that an event which is logically possible has probability 0. It defies our intuition which is usually based on probability problems where only a finite number of events is involved as was the case with all the problems in the last section.

When a problem of this type, where a possibly infinite sequence of events is involved, is asked at the level of the Joint Entrance Examination, completely rigorous solutions are not expected. For example, in the solution given above, it is quite all right to assume it as intuitively clear that the events A, B and C are exhaustive and thereby write down (1) without any justification. However, for the sake of completeness, we include here a rather lengthy discussion of how a rigorous definition of probability can be given in a problem like this. There are other ways to solve the problem. One such method, based on infinite series will be given in Comment No. 3. Those who are not interested in the theoretical justification we are going to present here, may simply skip it and go straight to Comment No. 3.

Probability problems where the sample space is infinite, are qualitatively different from those where it is finite. The present problem is of this type. In this problem, the sample space S consists of all infinite sequences of the form $(x_1, x_2, \ldots, x_n, x_{n+1}, \ldots)$, where each entry is either an H (for a head) or a T (for a tail). (These being the only two possible values for each entry, such sequences are called **binary sequences**, in analogy with the binary representation of integers where only two symbols, viz., 0 and 1 are needed.) Suppose now that A is an event, treated as a subset of S. Then the simple-minded definition of probability of A, viz., $P(A) = \frac{|A|}{|S|}$ breaks down because $|S|$ is infinite. This is not such a serious problem because even when the sample space is finite, this definition does not apply if all the elements of S are not equally likely. In that case, as we saw in Comment No. 5 of the last chapter, we can consider a non-negative real-valued function (often called a 'weight function') $f : S \longrightarrow \mathbb{R}$ with the property that $\sum_{x \in S} f(x) = 1$ and define $P(A)$ as the sum $\sum_{x \in A} f(x)$. When the

sample space S is infinite, even this alternate definition makes no sense. And *that* is the real trouble. First of all, for this definition to make sense, we would have to consider infinite sums since the subset A of S could be infinite. A more serious difficulty, however, is in defining the weight function $f : S \longrightarrow \mathbb{R}$. Suppose for example, that S is the sample space consisting of all binary sequences obtained by flipping an unbiased coin an infinite number of times. Then it stands to reason that all elements of S should have the same weight, say α which should be a non-negative real number. The trouble is that if α is positive then the sum $\sum_{x \in S} f(x)$ would be infinite (as the set S is infinite) and hence cannot be equal to 1 (which is a prime requirement of all probability problems). So we have no choice but to declare that $f(x) = 0$ for every $x \in S$ or verbally, that every single element of S has weight (or probability) 0. But still, the total weight of the set S is not 0, in fact it is 1. This looks paradoxical.

Fortunately, this paradox is not so hard to swallow because there are other real-life situations where a similar paradox arises. As the simplest example, take an interval, say $[a, b]$. It consists of infinitely many real numbers. Individually, every single real number has length 0. But the whole interval has a positive length, viz., $b - a$. Similarly, a piece of wire consists of infinitely many tiny particles of matter. Individually they have no mass (or a mass 0 each). But put together, the piece of wire has a positive mass.

A rigorous treatment of probability when the sample space is infinite requires what is called **measure theory** which is well beyond the level of the Joint Entrance Examination. (We only mention here that lengths, areas, volumes and mass are some of the examples of measures.) However, a working definition when the sample space S consists of infinite binary sequences as in the example above can be given as follows. In fact, more generally we consider the situation of k-ary sequences, where k is some fixed integer ≥ 2. (The cases $k = 2$ and $k = 3$ are called, respectively, binary and ternary sequences.) So let Y be a set with k elements, say, $Y = \{y_1, y_2, \ldots, y_k\}$. We let S be the set of all possible infinite sequences $(x_1, x_2, \ldots, x_n, x_{n+1}, \ldots)$, where every entry is some element of the set Y. We assume we are given some weight function $f : Y \longrightarrow \mathbb{R}$ which takes only non-negative values and further has the property that the sum of the weights of all elements of Y is 1, i.e., $\sum_{i=1}^{k} f(y_i) = 1$. Or, if we denote the weights $f(y_1), f(y_2), \ldots, f(y_k)$ by $w_1, w_2, \ldots, w_k$ respectively, then we assume that each w_i is non-negative and $w_1 + w_2 + \ldots + w_k = 1$. (In the case of a sample space arising from a coin tossed infinitely often, the integer k equals 2, y_1 and y_2 can be taken to be H and T respectively, where H stands for a head and T for a tail. If the coin is unbiased then $w_1 = w_2 = \frac{1}{2}$. But in general, $w_1 = p$ and $w_2 = 1 - p = q$, where p is the probability of a head showing. In the case of a problem involving an infinite number of tosses of a fair die, Y consists of the six figures 1 to 6 marked on the faces of the die and each w_i equals $\frac{1}{6}$.)

So, we assume we have some fixed set Y (with k elements) and a fixed weight function $f : Y \longrightarrow \mathbb{R}$. Let $Y^2 = Y \times Y$ be the set of all ordered pairs

of elements of the set Y. Then Y^2 has k^2 elements of the form (y_i, y_j), where each i and j can take any value from 1 to k. Now define a weight function $f_2 : Y^2 \longrightarrow \mathbb{R}$ as follows. Take a typical element (y_i, y_j) of Y^2. Then we define its weight $f_2(y_i, y_j)$ to be simply the product of the weights of its entries, i.e., as $f(y_i)f(y_j)$ (which we are also denoting by $w_i w_j$). Surely, the function f_2 so defined takes only non-negative values. But to ensure that it is really a weight function we must verify that the weight of the entire set Y^2 is indeed 1. But this is easy. For, the total weight of Y^2 equals the sum of all possible products of the form $w_i w_j$, i. e. the double sum $\sum_{i=1}^{k}\sum_{j=1}^{k} w_i w_j$. If we fix the index i, then the sum $\sum_{j=1}^{k} w_i w_j$ equals w_i since $w_1 + w_2 + \ldots + w_k = 1$. So it is clear that this double sum equals $w_1 + w_2 + \ldots + w_k$ which again equals 1. Hence the total weight of Y^2 is 1 as desired. (An equivalent, although essentially similar, method is to expand the product $(w_1 + w_2 + \ldots + w_k)^2$ which equals 1 and to observe that in the expansion every term is equal to the weight of the corresponding element of the set Y^2.)

We can generalise this construction to any positive integer n. Keeping Y and the weight function $f : Y \longrightarrow \mathbb{R}$ as before let us denote by Y^n the set of all sequences of length n with entries coming from the set Y. (To keep conformity with this notation, since S is the set of all infinite sequences with entries coming from the set Y, S is also sometimes denoted by Y^∞. Also, for $n = 1$, we often identify Y^1 with Y, because a 1-tuple is just one element of Y.) A typical element of Y^n is of the form $(x_1, x_2, \ldots, x_n)$, where each x_i is some element of the set Y. (Such an element is also often called an ordered n-tuple of elements of Y.) For each $i = 1, 2, \ldots, n$, $f(x_i)$ is the weight of the element x_i of Y. We now define a weight function $f_n : Y^n \longrightarrow \mathbb{R}$ by taking the product of the weights of the entries. In symbols, if $(x_1, x_2, \ldots, x_n)$ is a typical element of Y^n, then we define $f_n((x_1, x_2, \ldots, x_n)) = f(x_1)f(x_2)\ldots f(x_n)$. Suppose for example, that $n = 4$ and the set Y is the set $\{H, T\}$ with H for a head and T for a tail on the toss of a coin. Assume $f(H) = p$ and $f(T) = 1 - p = q$. Then Y^4 consists of all ordered quadruples. For the function $f_4 : Y^4 \longrightarrow \mathbb{R}$ we have, $f_4((H, H, T, H)) = p^3 q$ while $f_4((T, T, T, T)) = q^4$. If the coin is unbiased then f_4 is a constant function with value $\frac{1}{16}$ at every element of the set Y^4.

Note that since $|Y| = k$, the set Y^n has k^n elements, because every entry in an ordered n-tuple $(x_1, x_2, \ldots, x_n)$ can be chosen in k different ways, independently of each other. Further, if we add the weights of all these k^n elements, we always get 1. This is a consequence of the fact that $w_1 + w_2 + \ldots + w_k = 1$ which implies that the power $(w_1 + w_2 + \ldots + w_k)^n$ also equals 1. If we expand this power, it consists of k^n terms, each of which is a product of n factors of the form w_i. So each term corresponds to the weight of an element of the set Y^n.

Because of the weight function $f_n : Y^n \longrightarrow \mathbb{R}$, for any subset, say

B of Y^n, we can define its probability $P(B)$ as the sum $\sum_B f_n(x_1, x_2, \ldots, x_n)$, where the sum ranges over all elements of the set B. This is exactly the kind of probability that we encounter in problems where a coin is tossed n times or n dice are rolled together. So this is really nothing new to us. But now comes the main question. We could generalise from Y^2 to Y^n. Can we generalise from Y^n to Y^∞ in a similar manner? The answer is 'no' because for each n, the set Y^n is finite, while the set Y^∞ (which we have been calling as S) is infinite and as we already saw, on an infinite set no weight function can work satisfactorily.

This really poses a dilemma. But there is a way out. It is suggested by the manner in which we define the sum of an infinite series, say, $\sum_{i=1}^{\infty} a_i$ or, when written out in full, the sum $a_1 + a_2 + \ldots + a_n + a_{n+1} + \ldots$. Obviously, this cannot be obtained by actually adding an infinite number of terms for there is no such process. It can only be defined as the limit of finite sums. The most standard approach is to let S_n be the sum of the first n terms, i.e., the sum $a_1 + a_2 + \ldots + a_n$, or, $\sum_{i=1}^{n} a_i$. S_n is called the n-th **partial sum** of the original infinite sum $\sum_{i=1}^{\infty} a_i$, because it is only a part of it. As n grows larger and larger, the partial sum S_n captures more and more terms of the original sum $\sum_{i=1}^{\infty} a_i$. So it is reasonable to expect that as n tends to infinity, S_n should approach the infinite sum $\sum_{i=1}^{\infty} a_i$. This is actually a meaningless statement until we have defined what we mean by $\sum_{i=1}^{\infty} a_i$. So we kill two birds with one stone if we simply *define* $\sum_{i=1}^{\infty} a_i$ as the limit of S_n as n tends to infinity. Or in symbols, we define $\sum_{i=1}^{\infty} a_i$ as $\lim_{n\to\infty} S_n$ provided this limit exists (which is not always the case). For example, in the case of a geometric series $1 + r + r^2 + \ldots + r^n + r^{n+1} + \ldots$ (where we assume $r \neq 1$), the partial sum $S_n = 1 + r + r^2 + \ldots + r^{n-1} = \frac{1-r^n}{1-r}$. If $|r| < 1$, then by Theorem 4 in Chapter 6, $r^n \to 0$ as $n \to \infty$ and so in this case, $\sum_{i=0}^{\infty} r^i = \lim_{n\to\infty} \frac{1-r^n}{1-r} = \frac{1}{1-r}$. But if $|r| > 1$ then $\lim_{n\to\infty} S_n$ does not exist and so the infinite sum makes no sense.

The formula for the sum of an infinite geometric series is needed frequently in problems of infinitistic probability. So we record it explicitly for future use.

$$\sum_{i=0}^{\infty} ax^i = \frac{a}{1-x} \text{ whenever } |x| < 1 \tag{5}$$

We shall follow a somewhat similar approach to define $P(A)$ for a subset A of the set $S = Y^\infty$, where Y is a finite set (with k elements) and a weight function $f : Y \longrightarrow \mathbb{R}$ is given. The n-th partial sum of an infinite series $a_1 + a_2 + \ldots + a_n + a_{n+1}, \ldots$ is obtained by 'truncating' it at the n-th term, i.e., by dropping all the terms of the series after the n-th term. Let us see how this concept of 'truncation' works for an infinite sequence. A typical element of the set Y^∞ (which we have been calling as S) is an infinite sequence of terms coming from the set Y. If we truncate it at the n-th term, then we would get a sequence of length n with entries from the set Y, i.e., an element of the set Y^n. This gives us a function say g_n, called the n-th truncation function from S to Y^n. Specifically, take a typical element, say $(x_1, x_2, \ldots, x_{n-1}, x_n, x_{n+1}, \ldots)$ of S. Then $g_n(x_1, x_2, \ldots, x_{n-1}, x_n, x_{n+1}, \ldots) = (x_1, x_2, \ldots, x_{n-1}, x_n)$. Again, taking the example above, if $Y = \{H, T\}$ and $n = 4$ then the 4-th truncation of, say, $(H, T, T, H, H, H, T, T, H, \ldots)$, i.e., $g_4(H, T, T, H, H, H, T, T, H, \ldots)$ equals (H, T, T, H).

Now given a subset A of S, let $A_n = g_n(A)$. That is, A_n is the image of the subset A under the truncation function g_n. Then A_n is a subset of the set Y^n. We can give A_n a geometric interpretation. The n-th truncation function $g_n : S \longrightarrow Y^n$ is rather like taking the shadow of S on Y^n. So, for a subset A of S, the set $A_n = g_n(A)$ is like its shadow in Y^n. Of course, in taking this shadow we are completely ignoring so many entries in the original infinite sequences and so the shadow A_n cannot give a completely accurate picture of the original subset A of S. But just as the n-th partial sum of an infinite series captures more and more terms of the original infinite sum as n grows, similarly, as n grows, the shadow A_n of A (or, in other words, the n-th truncation of A) will give a more and more accurate picture of A. Notice that A_n is a subset of a finite set Y^n on which a weight function f_n has already been defined. So we know how to define $P(A_n)$. We have not defined $P(A)$ yet. But in view of what we have said, for large values of n, $P(A_n)$ should be very close to $P(A)$. So the most reasonable definition of $P(A)$ is the limit of the sequence $\{P(g_n(A)\}$ as n tends to ∞, or in symbols,

$$P(A) = \lim_{n \to \infty} P(g_n(A)) \tag{6}$$

provided, of course, that this limit exists.

To illustrate what is happening, we once again take the example where $Y = \{H, T\}$ with $f(H) = p$ and $f(T) = q = 1 - p$. Suppose A is the set of those (infinite) binary sequences whose second and fifth terms are H and whose fourth term is T. For notational simplicity, denote $P(g_n(A))$ by $p_n(A)$. Then (6) can be rewritten as $P(A) = \lim_{n \to \infty} p_n(A)$. Let us find $p_n(A)$ for various values of n. For $n = 1$, the set A_1 obtained by truncating various elements of A right after the first entry is the entire set Y, because in A we have sequences whose first entry is H and also sequences whose first entry is T. So $p_1(A) = f_1(H) + f_1(T) = f(H) + f(T) = p + q = 1$. However, for $n = 2$, we have that A_2 consists of only those ordered pairs whose first entry is unrestricted but whose second entry is H. In other words, $A_2 = \{(H, H), (T, H)\}$ and so

$p_2(A) = f_2(H,H) + f_2(T,H) = p^2 + qp = (p+q)p = p$. By a similar reasoning, A_3 consists of 4 ordered triples whose second entry is H, with no restriction on the other two entries. So, $p_3(A) = f_3(H,H,H) + f_3(T,H,H) + f_3(H,H,T) + f_3(T,H,T) = p^3 + qp^2 + p^2q + pq^2$ which comes out to be simply p. For $n = 4$, A_4 consists of all binary sequences of length 4 whose second term is H and fourth term is T. There are 4 such sequences. They are obtained by writing a term T at the end of the four members of the set A_3. From this (or directly), we get $p_4(A) = pq$. By a similar reasoning, $p_5(A) = p^2q$. In fact it is easy to see that for all $n \geq 5$, we have $p_n(A) = p^2q$. This is because after $n = 5$ there are no restrictions on the terms of the sequences in A. So, the set $A_n = g_n(A)$ will consist of all binary sequences of length n whose second and fifth terms are H and fourth term is T. There are 2^{n-3} sequences of this type in all. And the sum of their weights is $(p+q)p(p+q)qp(p+q)^{n-5}$ which is simply p^2q.

Summing up, in this illustration we see that after $n = 5$, the sequence $\{p_n(A)\}$ is constant. Therefore its limit as n approaches ∞ is also the same constant, viz., p^2q. Thus we have shown that $P(A) = p^2q$. We can now legitimately say that if a coin with p as the probability of showing a head is tossed an infinite number of times then the probability that a head will show on the second and the fifth toss and a tail will show on the fourth toss is p^2q.

As another example, in our Main Problem, the event $A' \cap B' \cap C'$ consists of just one sequence, viz., the sequence whose every term is T. The n-th truncation of $A' \cap B' \cap C'$ therefore consists of just one element, viz., the sequence of length n with all tails. The probability of this is q^n. So in our new notation $p_n(A' \cap B' \cap C') = q^n$. Since $0 < q < 1$, this tends to 0 as n tends to ∞. So, according to the definition (6), $P(A' \cap B' \cap C') = 0$. Verbally, this amounts to saying that if a coin is tossed infinitely often, then no matter how biased it is, we are sure to get at least one head, except, of course, when it is so hopelessly biased as to have $p = 0$. More generally, it can be shown that for any fixed positive integer r, the probability of getting at least r heads in infinitely many tosses is 1. Indeed, call this event as A. Then for every $n \geq r$, its n-th truncation A_n is the event that a head shows at least r times in n tosses of the coin. Taking binomial probability, $P(A_n) = \sum_{i=r}^{n} \binom{n}{i} p^i q^{n-i}$, or equivalently, $1 - P(A_n) = \sum_{i=0}^{r-1} \binom{n}{i} p^i q^{n-i}$. This is a finite sum and so to take its limit as n tends to ∞, we take the limits of each of the terms and add. The i-th term is $\binom{n}{i} p^i q^{n-i} = \dfrac{n(n-1)\ldots(n-i+1)}{i!q^i} p^i q^n$ which is at most $\dfrac{p^i}{i!q^i} n^i q^n$. Here for a fixed i, $\dfrac{p^i}{i!q^i}$ is a constant and it can be shown that $\lim_{n\to\infty} n^i q^n = 0$ (assuming again, that $q < 1$). So, $\lim_{n\to\infty} 1 - P_n(A) = 0$ and hence $\lim_{n\to\infty} P_n(A) = 1$. Put verbally, if you are prepared to wait long enough you are sure to get r heads, except that how long is long enough is not specified!

The definition (6) is not, however, a golden tool which can be applied right away to solve probability problems dealing with events in the sample space

S. In fact, before we can apply the definition (6) to solve problems, we first have to solve the problems created by (6)! Note, first of all, that for some subsets the limit in the R.H.S. of (6) may not exist. In that case such subsets do not have any probability! This may sound absurd and indeed examples of such 'pathological' subsets are not easy to construct. Nevertheless, they do exist. We may ignore them because for events that we encounter in probability problems the limit in (6) generally does exist and so there is no difficulty in defining their probabilities. Still, we have to prove all familiar laws about probability. For example, suppose A and B are two mutually exclusive events. Then we have to prove that $P(A \cup B) = P(A) + P(B)$. Results like this are easy to prove when the sample space is finite. However, with (6) as the definition, this is not as obvious as it looks. A natural approach to try would be to show that for every n, $P(g_n(A \cup B)) = P(g_n(A)) + P(g_n(B))$ and then take limits of both the sides as n tends to ∞. Fortunately, the sets $g_n(A \cup B)$ and $g_n(A) \cup g_n(B)$ are equal. But the trouble is that even though A and B are mutually disjoint subsets of S, their truncations $g_n(A)$ and $g_n(B)$ may overlap and so we cannot conclude that $P(g_n(A) \cup g_n(B)) = P(g_n(A)) + P(g_n(B))$ and hence that $P(g_n(A \cup B)) = P(g_n(A)) + P(g_n(B))$. A proof can still be salvaged. But it is not easy. Even the very special case of complementary probability, viz., showing that for any event A, $P(A) + P(A') = 1$ is fraught with difficulties.

We are hardly in a position to resolve these difficulties. So we shall simply assume that the basic laws of probability (as, for example, those laid down in (3) and (5) to (8) of the last section) continue to hold even for infinitistic probabilities, i.e., where the sample space S is infinite. We are, in fact, interested only in a special case where the sample space S consists of infinite k-ary sequences (usually with $k = 2$) whose entries come from a finite set (with k elements) on which some weight function has been given. The definition (6) given above can then be summarised in a simple manner. Take an event, say A, pertaining to the occurrence of an infinite number of trials. For every positive integer n, let A_n be the 'truncated' event, obtained by restricting ourselves only to the first n trials. Find the probability, $P(A_n)$ of this truncated event, which can be done by the various methods studied earlier since A_n is an event in a finite sample space. Then take the limit of $P(A_n)$ as n tends to ∞. This limit is the probability of the event A.

Comment No. 3:

Let us now go back to the Main Problem and find the probability β, of B's winning the game directly using the definition (6) (applied to the event B, of course). We shall work the problem with any value of p and put $p = 0.4$ at the end. Here, for a positive integer n, the truncated event B_n is that B wins the game in n tosses of the coin. This can be further expressed as a disjunction of a number of mutually exclusive events. Note that B can win only on the 2nd, the 5th, the 8th, ... and, in general on the $3r + 2$-th toss, where r is some non-negative integer. To win at the end of the $3r + 2$-th toss, the first $2r + 1$ tosses must be tails and the $3r + 2$-th toss must be a head. The probability of this happening is $q^{3r+1}p$. So, the probability of B's winning in the first n

rounds is the sum of these probabilities. Thus we get,

$$P(B_n) = \sum_{r=0}^{t_n} q^{3r+1}p \tag{7}$$

where the upper index t_n of summation is the largest integer m such that $3m + 2 \leq n$. (In other words, t_n is the integral part of the real number $\frac{n-2}{3}$, a standard notation for which is $[\frac{n-2}{3}]$. The R.H.S. of (6) is the sum of a geometric progression with the first term pq and common ratio q^3. We can evaluate it in a closed form as $\dfrac{pq(1-q^{3(t_n+1)})}{1-q^3}$. But this is not very important, because our ultimate interest in not so much in (7) *per se* as it is in the limits of the two sides of (7) as n tends to ∞. The L.H.S. tends to $P(B)$ i.e., to β. As for the R.H.S., as n tends to ∞ so does the integer t_n. So the limit of the R.H.S. of (7) is nothing but the infinite geometric sum with the first term pq and common ratio q^3. As $0 \leq q < 1$, we have $|q^3| < 1$. So (5) is applicable and we get that β, i.e., the probability of B's winning the game equals $\frac{pq}{1-q^3}$ which simplifies to $\frac{q}{1+q+q^2}$ since $p = 1 - q$. Now we put $p = 0.4$ to get $q = 0.6$ and hence $\beta = \frac{60}{196} = \frac{15}{49}$, the same answer as before.

The procedure adopted here was to first find the probability of the truncated event B_n, as in (7), and then to take its limit as $n \to \infty$. After some practice, this procedure becomes so ritualistic that it can be omitted. Thus instead of going through (7), it is all right if we write directly

$$P(B) = \sum_{r=0}^{\infty} q^{3r+1}p \tag{8}$$

and then evaluate the sum using (5) as we did.

A few more problems based on the application of (5) will be given in the exercises.

Comment No. 4:

An essentially similar (and in fact, a little simpler) problem is the following:

A pair of fair dice is rolled together till a sum of either 5 or 7 is obtained. Find the probability that 5 comes before 7. (JEE 1989)

When two fair dice are rolled independently, the probabilities that the total score is 5 and 7 are, respectively, $\frac{1}{9}$ and $\frac{1}{6}$. So, the desired probability, say p, is clearly the sum $\sum_{n=1}^{\infty} p_n$, where, for every positive integer n, p_n is the probability that 5 appears at the end of the n-th rolling with neither 5 nor 7 occurring at any earlier rolling. The probability that neither 5 nor 7 occurs on a given rolling is $1 - \frac{1}{9} - \frac{1}{6} = \frac{13}{18}$. Hence $p_n = (\frac{13}{18})^{n-1}\frac{1}{9}$. The series $\sum_{n=1}^{\infty} p_n$ is therefore the geometric series $\displaystyle\sum_{n=1}^{\infty}(\frac{13}{18})^{n-1}\frac{1}{9}$. Calling $n-1$ as our new index

variable r, this is the same as $\sum_{r=0}^{\infty}(\frac{13}{18})^r\frac{1}{9}$. (A change of index like this is a simple but useful technique in reducing a given sum to a standard form where it is easier to manipulate. The index variable is, after all, a dummy variable and the sum is independent of what symbol we use for it. So, we may denote r by n as well. But it is better to do so only after a little practice, because initially it is quite puzzling to call $n-1$ as n.) We are now in a position to apply (5) and get the value of this infinite sum as $\frac{1}{9}\frac{1}{1-\frac{13}{18}}=\frac{2}{5}$. This is, therefore, the required probability. (Note that by the same reasoning, the probability that 7 shows before 5 is $\frac{3}{5}$. The sum of these two is precisely 1. This is not surprising since if the dice are rolled indefinitely till a sum of 5 or 7 is obtained, then obtaining 7 is precisely the complementary event of obtaining 5. It is, in fact, intuitively clear, even without infinite series, that if the choice is between the sums 5 and 7, then the chances of getting them first before the other are proportional to their probabilities, viz., to $\frac{1}{9}$ and $\frac{1}{6}$, because we may disregard all other rollings (i.e., where neither 5 nor 7 shows) as dummy ones. A rigorous argument requires infinite series as shown above. But if you are sure of your intuition, and the question is an objective type one, you can save a lot of time by appealing to your intuition.)

Comment No. 5:

We can now convert some of the problems of finitistic probability to similar problems of infinitistic probability. Consider for example, the chess match between two players, A and B in the Main Problem of the last chapter, or more precisely, its generalisation in Comment No. 1 after its solution. Thus, let the positive numbers a, b and c denote, respectively, the probabilities of A's winning, losing and drawing a game against B. The rules of the match are that whoever first wins two (not necessarily consecutive) games wins the match. For a positive integer n, let p_n be the probability that A wins the match at the end of the n-th game (but not earlier). Then evidently, $\sum_{i=1}^{n} p_i$ is the probability that A wins the match on or before the n-th game. But this is precisely the n-th partial sum of the infinite series $\sum_{i=1}^{\infty} p_i$. Replacing the index variable i with n, this series is the same as $\sum_{n=1}^{\infty} p_n$. So, $\sum_{n=1}^{\infty} p_n$ will give the probability, say p, that A wins the match (sometime or the other). We have already calculated p_n as $(n-1)a^2c^{n-3}[c+(n-2)b]$. So the desired probability p is now the sum of the infinite series $\sum_{n=1}^{\infty}(n-1)a^2c^{n-3}[c+(n-2)b]$.

The trouble is that, unlike the series we have encountered in the problems so far, the present one is not a geometric series. But it is related to a geometric

series. Putting $a = 1$ in (5) we get

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots + x^n + \ldots \tag{9}$$

Here both the sides are functions of the variable x. The L.H.S. is a single function of x which is defined for all $x \neq 1$. The R.H.S. is an infinite sum of terms. Each of these terms is a function of x defined for all real values of x. But the sum is meaningful only for $|x| < 1$. The familiar summation formula for a derivative, viz., that the derivative of a sum of functions is equal to the sum of their derivatives is valid for the sum of a *finite* number of functions. It is *not* unconditionally true for an infinite sum of functions. However, for an infinite series like that in (9) it can be shown that it is valid. Put simply, we can get the derivative of the R.H.S. of (9) by 'term-by-term differentiation' exactly as we would for a finite sum. In fact we can do this term-by-term differentiation again and again, any (finite) number of times. The justification of this fact is non-trivial and well beyond our scope. So we shall simply take it for granted.

Differentiating (9) term-by-term, we get $\sum_{n=1}^{\infty} nx^{n-1} = \frac{1}{(1-x)^2}$. Differentiating again, $\sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}$. (These identities[1] are valid only for $|x| < 1$.) Using these identities, the infinite sum $\sum_{n=1}^{\infty}(n-1)c^{n-2}$ comes out to be $\frac{1}{(1-c)^2}$, while $\sum_{n=1}^{\infty}(n-1)(n-2)c^{n-3}$ equals $\frac{2}{(1-c)^3}$. Substituting these,

$$\begin{aligned} p &= \sum_{n=1}^{\infty} p_n = \sum_{n=1}^{\infty} a^2(n-1)c^{n-2} + \sum_{n=1}^{\infty} a^2 b(n-1)(n-2)c^{n-3} \\ &= \frac{a^2}{(1-c)^2} + \frac{2a^2b}{(1-c)^3} = \frac{a^3 + 3a^2b}{(a+b)^3} \quad (\text{since } 1 - c = a + b). \end{aligned} \tag{10}$$

Interchanging a and b we get the probability, say q, of B's winning the match sometime or the other. It comes out to be $\frac{b^3+3ab^2}{(a+b)^3}$. Note that $p + q = 1$ which means that if the match is played long enough then either A or B is sure to win eventually.

It is interesting to note that if our interest was solely in finding p, i.e., the probability of A's winning the match sometime or the other, then an alternate derivation of (10) is possible using the same intuitive reasoning as was used in the solution to the last problem regarding the occurrence of a sum 5 before the

[1] The first identity can be proved without differentiation by recognising the partial sum, say $T_n = 1 + 2x + 3x^2 + \ldots + nx^{n-1}$, as the sum of an arithmetico-geometric progression, encountered in Comment No. 6 of Chapter 5. Following the method there, T_n comes out to be $\frac{1-x^n}{(1-x)^2} - \frac{nx^n}{(1-x)}$. $\lim_{n\to\infty} T_n$ can now be found using the result of Exercise (6.51). A similar approach is also possible for the second indentity but is more complicated.

first occurrence of a 7 when a pair of dice are rolled a number of times till either 5 or 7 is obtained. Treating the drawn games as dummy ones, the probability that A will register the first win is $\frac{a}{a+b}$, while the probability that B will register the first win is $\frac{b}{a+b}$. Now suppose A has won one game. Thereafter, if he wins a game again before B, then he wins the match and the probability of this happening is $\frac{a^2}{(a+b)^2}$. However, after A's first win, it is possible that B registers a win first. The probability of this is $\frac{ab}{(a+b)^2}$. When this happens, in order to win the match, A must record the next win. So the probability that A wins the match in this manner is $\frac{aba}{(a+b)^3}$. There is only one more way A can win the match, viz., B registers the first win and the two wins after that are registered by A. By a similar reasoning as above, the probability of this case is $\frac{ba^2}{(a+b)^3}$. As these are the three mutually exclusive possibilities for A to win the match, his total probability p of winning the match is $\frac{a^2}{(a+b)^2} + \frac{2a^2b}{(a+b)^3}$ which gives the same answer as (10).

An even shorter argument is that once we discard the draws as dummy games, the probabilities of the players of winning a particular game have to be proportionately multiplied so as to add up to 1. That is, instead of a and b, we must now assume them to be $\frac{a}{a+b}$ and $\frac{b}{a+b}$ respectively. But now winning the match is the same as winning it in a best of three matches encountered in one of the problems in Comment No. 13 of the last chapter. So, by the reasoning given there, A's probability of winning is $\frac{3a^2b}{(a+b)^3} + \frac{a^3}{(a+b)^3}$ which is the same answer as before.

Comment No. 6:

Let us now convert the problem discussed in Comment No. 14 of the last chapter. There we found the probability p_n that in n tosses of a coin (with probability p of showing a head), no two (or more) heads appear consecutively. In view of (6), $\lim_{n\to\infty} p_n$ will give the probability that no two consecutive tosses will show a head if the coin is tossed an infinite number of times. To find this limit we use (11) of the last chapter, viz., $p_n = A\alpha^n + B\beta^n$, where A and B are some constants and α and β are the roots of the quadratic equation $x^2 - (1-p)x - p(1-p) = 0$. We can take α as $\dfrac{1-p+\sqrt{1+2p-3p^2}}{2}$ and β as $\dfrac{1-p-\sqrt{1+2p-3p^2}}{2}$. As we are assuming that $0 < p < 1$, it is easy to show that $-1 < \beta < 0 < \alpha < 1$. Hence both α^n and β^n tend to 0 as n tends to ∞. So, no matter what the constants A and B are, we get that $\lim_{n\to\infty} p_n = 0$. Hence the complementary probability, viz., that of at least two heads appearing consecutively is 1. If the appearance of two consecutive heads is considered as some sort of a victory, then verbally this result means that if you wait long enough, you are sure to win.

We remarked that in the last chapter equation (10) was a recurrence relation for the sequence $\{p_n\}$, while (11) was its solution. Using (11), we just showed that $\lim_{n\to\infty} p_n = 0$. It is interesting to note that assuming that $\lim_{n\to\infty} p_n$

exists, its value can be found directly from (10), i.e., without having to solve it. Indeed, call this limit as L. Then letting n approach ∞ in the recurrence relation (10), we get $L = (1-p)L + p(1-p)L$, i.e., $p^2 L = 0$. Again, because of our assumption that $0 < p < 1$, $p^2 \neq 0$ and hence we must have $L = 0$. The price we pay for the simplicity of this argument is that the existence of the limit of the sequence $\{p_n\}$ has to be known beforehand. There is an easy justification for this based on the fact that $\{p_n\}$ is a monotonically decreasing sequence, i.e., $p_{n+1} \leq p_n$ for all n. But we shall not go into it. Justifications of this type are not expected at the JEE level.

Actually, if our sole interest is in L, i.e., $\lim\limits_{n\to\infty} p_n$, and not in p_n *per se*, then we can do even better. We just saw that we could dispense with (11) and get the answer directly from (10). But actually, we can dispense with (10) as well. Indeed, here L is the probability that no two consecutive heads appear in an infinite sequence of tosses. Recall that the key idea in obtaining the recurrence relation (10) in the last section was to consider the possible outcomes of the first two tosses. Using the same idea we can directly get an equation for L (again, assuming that it exists). Here L is the probability that no two consecutive tosses show heads in an infinite sequence of tosses. The possible outcomes of the first toss are T with probability q and p with probability p. In the former, the probability that there are no two consecutive H's in subsequent tosses is L again. The latter case is divided into two subcases depending upon the outcome of the second toss. If it is H, then we already have two consecutive H's. If it is T, then the probability that there be no two consecutive heads subsequently is L again. Putting it all together we get the equation

$$L = qL + pqL = (1-p^2)L \tag{11}$$

whose only solution is $L = 0$ since $p^2 < 1$.

This alternate method for obtaining L directly (i.e., without first finding the probability p_n, and even without finding a recurrence relation for the sequence $\{p_n\}$) is very important because there are situations where an equation like (11) can be written down for the probability in an infinistic problem, but where it is not easy to write down a recurrence relation for the probabilities in the corresponding finitistic problem, or situations where, even if such a recurrence relation can be written down, it is not easy to solve it explicitly. We discuss below one such problem which is a modified (and a more realistic) version of the ruin game discussed in Comment No. 13 of the last chapter, because here the player plays not against a machine but against another player.

Comment No. 7:

Suppose two players A and B play a game in which a coin with probability p of a head showing is tossed at each round. Initially, A and B have with them a and b rupees respectively. At each round, if a head shows then A gets one rupee from B. But if a tail shows, he gives B one rupee. A player is said to be ruined when he runs out of money and then the game stops the first time this happens. Find the probability that A is ruined.

The numbers a and b are often called the initial capitals of the two players. Note that this is an infinitistic probability problem because theoretically, the game can go on indefinitely without either player being ruined. This can happen, for example, when a and b are at least 2 each and the coin shows alternately a head and a tail (or a tail and a head). The problem asks for the probability that A will be ruined first. One way to do the problem would be to try to find, for each positive integer n, the probability, say p_n, that A is ruined at the end of the n-th round or earlier and then to take the limit $\lim_{n\to\infty} p_n$. Obviously, p_n will depend upon p, the probability of a head, for the given coin. But in addition, it will also depend on the parameters a and b. So we better denote it by $p_n(a, b)$. But we can exercise some economy of notation. As the game goes on, the total amount of money the two players have together remains a constant, viz., $a + b$. Let us call this the total capital and denote it by c. Then we can drop it from the notation. At any time if k denotes the amount A has then $c - k$ is the amount which B has. Initially, $k = a$. The game goes on as long as $0 < k < c$. It ends when either k becomes 0 (with A ruined) or k becomes c (with B ruined). So these two values are sort of the 'boundaries' which the variable k must stay within for the game to go on. With this change of notation, let us denote by $p_n(k)$ the probability that in a game in which A starts with k rupees (and B with $c - k$ rupees) A is ruined on or before the n-th round. Our problem is to find $p_n(a)$ for a given n and a.

So assume we have a situation where A has k rupees, and B has $c - k$ rupees. Now, if the first toss is a head then after it A will have $k + 1$ rupees. If $k = c - 1$, the player B is ruined at this stage. Otherwise the game will go on as if a new game has started with A having $k + 1$ rupees and the probability that in this game A will be ruined on or before the $(n - 1)$-th round is $p_{n-1}(k + 1)$. A similar reasoning applies if the first toss is a tail. Combining these two possibilities, we get a recurrence relation

$$p_n(k) = pp_{n-1}(k + 1) + (1 - p)p_{n-1}(k - 1) \tag{12}$$

which is valid for $1 < k < c - 1$. If $k = 1$, then the second term on the R.H.S. is meaningless because in that case A is already ruined. So we may set $p_{n-1}(0) = 1$ and then (12) remains valid for $k = 1$ also. Similarly, by setting $p_{n-1}(c) = 0$ (because that amounts to B's ruin and hence to the impossibility of A's ruin), we can make (12) valid for $k = c - 1$ too. We therefore set

$$p_n(0) = 1 \text{ and } p_n(c) = 0 \tag{13}$$

for all $n \geq 1$. Then (12) remains valid for all $n \geq 2$ and for all k with $1 \leq k \leq c - 1$. An equation like (13) is said to specify the **boundary conditions** for the recurrence relation (12), because as we saw above, 0 and c are the boundaries within which k has to stay. Just as a differential equation has infinitely many solutions but the boundary conditions determine one particular solution uniquely, a recurrence relation such as (12) also has infinitely many solutions but the boundary conditions (13) determine one particular solution uniquely. (The analogy can be taken further. Just as in the case of a differential

equation, the number of boundary conditions needed equals the order of the differential equation, the same is true for recurrence relations. Actually, the analogy between differential equations and recurrence relations runs deeper than this. But we shall not go into it.)

Note that (12) is a recurrence relation with two integer parameters, viz., n and k. In this respect it resembles the reduction formula (1) we derived in the solution to the Main Problem of Chapter 17 for the integral $\int_0^1 x^m(1-x)^n dx$ which also had two integer parameters, viz., m and n. (The resemblance will be clearer if we write $p_{n,k}$ instead of $p_n(k)$.) But there, repeated applications of the reduction formula eventually led us to an explicit expression for this integral. The trouble in the present problem is that there are two terms on the R.H.S. of (12). Repeated applications of (12) will go on increasing the number of terms still more and is unlikely to give a handy closed form expression for $p_n(k)$. There is, in fact, no easy way to solve (12). Fortunately, that does not matter because our interest is not in $p_n(k)$ (which represents the probability of A's ruin in at most n rounds if he starts with a capital of k rupees) *per se*. For our purpose, this probability is only a means and not the goal. Our goal is merely to find the *limit* of $p_n(k)$ as n tends to ∞. This limit will, in general depend on k and so let us call it L_k. Just as we started with equation (10) in the last section and by letting n tend to ∞ on both the sides, obtained (11) (in the present section), if we let n tend to ∞ on both the sides of Equation (12) above, we get $L_k = pL_{k+1} + (1-p)L_{k-1}$, or, with a change of indexing,

$$pL_k = L_{k-1} - (1-p)L_{k-2} \tag{14}$$

which is valid for all $k \geq 2$. Similarly, the 'boundary conditions' in (13) turn into

$$L_0 = 1 \text{ and } L_c = 1 \tag{15}$$

Note that (14) itself is a recurrence relation for the sequence $\{L_k\}$. This is a finite sequence since the integer variable k can take only the values between 0 and c (which is the sum of the initial capitals of the two players). The method of solving a recurrence relation like (15) is very similar to that of solving the recurrence relation (10) in the last section. There, we did not actually solve the relation but only verified that (11) was its solution. The solution (11) was expressed in terms of the powers of the roots of the quadratic equation $x^2 - (p-1)x - p(1-p) = 0$. In the present problem too, the general solution of (14) depends crucially on the roots of the quadratic equation $px^2 - x + (1-p) = 0$. Note that the roots are 1 and $\frac{1-p}{p}$. These are distinct unless $p = \frac{1}{2}$. In the case of distinct roots, the solution of (14) is of the form

$$L_k = A(1)^k + B(\frac{1-p}{p})^k = A + B(\frac{1-p}{p})^k \tag{16}$$

where the constants A and B can be determined from (15) which gives a system

of two equations, viz.,

$$A + B = 1 \text{ and } A + B(\frac{1-p}{p})^c = 0 \tag{17}$$

Solving this system and substituting in (16) gives an explicit formula for L_k in terms of p and c, viz.,

$$L_k = \frac{p^k(1-p)^c - p^c(1-p)^k}{p^k((1-p)^c - p^c)} \tag{18}$$

which is valid for $p \neq \frac{1}{2}$, i.e., for a biased coin. For an unbiased coin, i.e., for $p = \frac{1}{2}$, the form of the general solution is slightly different from (16). We omit the details but only remark that in that case L_k comes out to be simply $\frac{c-k}{c}$. It is interesting to note that this is also the limit of the R.H.S. of (18) as $p \to \frac{1}{2}$, as can be shown, for example, by rewriting the R.H.S. as $\dfrac{\lambda^c - \lambda^k}{\lambda^c - 1}$, where $\lambda = \frac{1-p}{p}$ and taking limit as $\lambda \to 1$. (To evaluate this latter limit, either apply the L'Hôpital's rule or write the numerator as $(\lambda^c - 1) - (\lambda^k - 1)$ and extract a factor $\lambda - 1$ from both the numerator and the denominator.)

Going back to the original problem now, the probability that A will be ruined is obtained by putting $k = a$ and $c = a + b$ in (18). It comes out to be $\dfrac{(1-p)^{a+b} - (1-p)^a p^b}{(1-p)^{a+b} - p^{a+b}}$ provided $p \neq \frac{1}{2}$ and $\frac{b}{a+b}$ if $p = \frac{1}{2}$. The latter stands to our intuition because if the coin is unbiased, the players' relative chances of ruin should be inversely proportional to their initial capitals. But the case of a biased coin is far from easy to predict by mere intuition. In the same game, B's probability of ruin can be obtained by a similar calculation. But that is hardly necessary. We merely interchange a with b and p with $1 - p$ in the answers obtained above. It is interesting to note that the probabilities of the ruins of the two players always add up to 1 as is easily verified by a direct calculation. In other words, if the game is played long enough then it is sure to end. This again may sound intuitively obvious. But one should be wary of letting one's intuition run too wild. For example, if we take the 'average' number of rounds the game will go on, then it can be shown (as we shall see later) that in the case of an unbiased coin, the game will go on for ab rounds on the average. This means that if A starts with just 1 rupee and his adversary B with 1000 rupees as their initial capitals, then on an average it will take 1000 rounds for the game to end. There is, of course, a high probability (viz., 50%) that A will be ruined right at the first round. So we tend to think that even if he is lucky at the first round, he will be able to pull on for only a few more rounds. So here our intuition fails miserably.

The problem above deals with a form of gambling. But as it happens many times in mathematics, the same reasoning applies to a totally different problem. Suppose a particle is constrained to move along a straight line. In one unit of time it moves one step forward or one step backward, the probabilities of these being p and $1 - p$ respectively, where $0 < p < 1$ is fixed. Initially, the

particle is at a point a unit steps to the right of a reference point O. C is a point at a distance c units to the right of O, where c is some positive integer. We assume $0 < a < c$. Suppose at O and C there are two 'absorbing barriers'. That is, if the particle hits either of these, its motion stops. It is clear that the probability that the particle will hit the barrier at O sometime or the other is precisely the probability that in the game above the player A will be ruined. (We take a and $c - a$ as the initial capitals of A and B respectively.) Similarly, the probability that the particle will be absorbed by the barrier at C is the probability that B will be ruined.

This type of a motion is called a **random walk**. It is quite different from the type of motion we encounter in classical mechanics which is governed by Newton's laws of motion. The branch of mathematics in which such random walks are studied is called **statistical mechanics**.

Modifications of the game above (and hence also of a random walk) are possible where at each round there is a possibility of a draw in which case neither player gives the other anything (which, in a random walk means the particle remains stationary for that particular unit of time). In that case there will be three probabilities α, β and γ (adding to 1) corresponding to A's getting one rupee (from B), losing one rupee (to B) and no change, respectively. We leave it as an exercise to find the analogues of (12), (14) and (18) for this modified game.

Comment No. 8:

In dealing with the ruin game above, we mentioned the concept of the 'average' duration of the game. Let us formalise it now. The concept of an average is, of course, familiar right from school. Suppose, for example, that there are n students in a class and that in some examination, n_1 of them get a score x_1 each, n_2 get a score of x_2 each, ..., n_k get a score of x_k each with $n_1+n_2+\ldots+n_k = n$. Then the average score, say A, is simply $\frac{n_1x_1+n_2x_2+\ldots+n_kx_k}{n}$ which can also be written as $\sum_{i=1}^{k} \frac{n_i}{n} x_i$. The latter formulation can further be interpreted in terms of probability. For each $i = 1, 2, \ldots, k$, $\frac{n_i}{n}$ is simply the probability, say p_i, that the score of a student equals x_i. So the average A could also have been defined as $\sum_{i=1}^{k} p_i x_i$. Here the possible scores are $x_1, x_2, \ldots, x_k$ and the probability or the 'expectation' that the score will be equal to x_i is p_i. For this reason, the average is also often called the **expected value**.

The advantage in defining the average in terms of probabilities is that it can be generalised even when the number of possible values assumed is infinite. Suppose, for example, we have a situation where there is an infinite sequence of (real) values, say $\{x_i\}_{i=1}^{\infty}$. Let us suppose that the value x_i is assumed with probability p_i, for $i = 1, 2, \ldots, n, n+1, \ldots$. (This is also expressed more compactly by writing $p_i = P(x = x_i)$, where x is a variable ranging over the set of possible values.) Then we must have $\sum_{i=1}^{\infty} p_i = 1$. (The sequence $\{p_i\}_{i=1}^{\infty}$ is often called the **probability distribution** of a **discrete random variable**,

say x, which assumes the sequence $\{x_i\}_{i=1}^{\infty}$ as its possible values. These are technical terms and it is not necessary to know them in doing problems. But their use simplifies the expression of ideas.) The average or the expected value of x is now defined as the infinite sum $\sum_{i=1}^{\infty} p_i x_i$. There are examples where this series is not convergent. But we shall assume that in all the problems we are dealing with, this series is convergent. Trivially, this series is convergent when only finitely many of the probabilities p_i are non-zero because in that case it is essentially a finite sum. This is always so in the case of problems where the sample space is finite. That is why calculation of average in the case of a finitistic probability problem is a simple arithmetical problem. For example, suppose two fair dice are rolled together and we want to find the average value of the total score. (JAT 1979) Here there are 11 possible values, viz., the integers from 2 to 11. Their probabilities are respectively $\frac{1}{36}, \frac{2}{36}, \ldots, \frac{6}{36}, \frac{5}{36}, \ldots, \frac{1}{36}$. By a direct calculation, the average total score comes out to be 7.

Things are, however, quite different for problems of infinitistic probability. As a typical illustration, we do the following problem in which averages for both types of problems are to be found.

Suppose a coin has probability p of a head showing when tossed.

(i) Find the average number of times a head will show in n tosses of the coin, where n is a fixed positive integer.

(ii) Find the average number of times the coin is tossed till a head shows for the first time.

The first part is a problem about finitistic probability. Let x denote the number of heads in n tosses of the coin. Then x is a discrete random variable which assumes the integers from 0 to n as its possible values. Moreover, for $r = 0, 1, \ldots, n$, $P(x = r)$ is the binomial probability $\binom{n}{r} p^r q^{n-r}$ (where $q = 1 - p$ as usual). So the average, say A_n, is the sum $\sum_{r=0}^{n} \binom{n}{r} p^r q^{n-r} r$. To evaluate this sum first rewrite it as $q^n \sum_{r=0}^{n} \binom{n}{r} r(\frac{p}{q})^r$. Now by the binomial theorem we have $(1 + x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r$ which is true for all x. If we differentiate both the sides w.r.t. x and multiply by x, we get $\sum_{r=0}^{n} \binom{n}{r} r x^r = nx(1 + x)^{n-1}$. This is true for all values of x. In particular, putting $x = \frac{p}{q}$, we get $\sum_{r=0}^{n} \binom{n}{r} r(\frac{p}{q})^r = n\frac{p}{q}(1+\frac{p}{q})^{n-1}$ which reduces to $\frac{np}{q^n}$ since $p + q = 1$. So A_n is simply pn. This is consistent with our intuition because if we toss the coin n times, then the fraction of tosses showing a head would be p. This, indeed, is the intuitive meaning of probability.

Part (ii) of the problem comes under infinitistic probability. Here the number of tosses till the first head appears is a discrete random variable, say x, which takes all positive integers as possible values. Since $x = r$ if and only if the first $r - 1$ tosses show a tail and the r-th toss shows a head, we get

$p_r = P(x = r) = q^{r-1}p$ for every $r = 1, 2, \ldots$. Hence the expected number of tosses, say A, equals the infinite sum $\sum_{r=1}^{\infty} rq^{r-1}p$. We already evaluated this sum above in Comment No. 5 as $\frac{p}{(1-q)^2}$. This is simply $\frac{1}{p}$. Again, this is intuitively obvious at least when p is of the form $\frac{1}{k}$ for some positive integer k. For, in that case the chance of a success (i.e., getting a head) is '1 out of k'. So for a success we expect k attempts would be needed on the average.

There is an ingenious way to get the answer to (ii) without summing an infinite series, and in fact, without even having to calculate p_n for each n. It is based on the idea of what is called the weighted average of averages. To see what it means suppose that a country S is divided into 10 states, say, $S_1, S_2, \ldots, S_{10}$. Let $A_1, A_2, \ldots, A_{10}$ be the average per capita incomes of these states respectively. Denote by A the average per capita income of the whole country. Then it would be a mistake to equate A with $\frac{1}{10}(A_1 + A_2 + \ldots + A_{10})$. In other words, the average for the whole country is generally different from the average of the averages for the states. To calculate A from the A_i's, each state S_i must be given a 'weightage' proportional to its population, say n_i. Then letting $n = n_1 + n_2 + \ldots + n_k$ be the total population, the correct value of A is $\frac{n_1A_1 + n_2A_2 + \ldots + n_kA_k}{n}$ which is the same as $p_1A_1 + p_2A_2 + \ldots + p_kA_k$, where, for $i = 1, 2, \ldots, k$, $p_i = \frac{n_i}{n}$ is the probability that a person from the country picked at random belongs to the state S_i. In other words, A is the 'weighted' average of the averages $A_1, A_2, \ldots, A_k$, the 'weights' (or the relative importance) being the respective probabilities. In fact this formulation makes sense and remains valid even for infinitistic probability. We state this formally as the **law of weighted average of averages**.

Suppose S is a (possibly infinite) sample space which is the disjoint union of subsets $S_1, S_2, \ldots, S_k$. Let g be a real-valued function defined on S and let $A, A_1, A_2, \ldots, A_k$ be the averages of g on the sets $S, S_1, S_2, \ldots, S_k$ respectively. Then $A = p_1A_1 + p_2A_2 + \ldots + p_kA_k$, where, for $i = 1, 2, \ldots, k$, p_i is the probability that an element of S (picked randomly) belongs to the subset S_i.

As an application of this law for finitistic probability, let us again calculate the average total score when a pair of fair dice is rolled. Here we take the sample space S to consist of all 36 equally likely ordered pairs of the form (x, y), where x and y are the figures that appear on the first and the second die respectively. We take g to be the function (of two variables) defined by $g(x, y) = x + y$, which gives the total score. Earlier we calculated the average value of the total score as 7 by finding the probability of each of the 11 possible scores (from 2 to 12). We can now do the problem more efficiently as follows. Divide the set S into six subsets depending upon the figure on the second die. In symbols, for $i = 1, 2, \ldots, 6$, let S_i be the subset $\{(x, y) \in S : y = i\}$. Each S_i has 6 elements corresponding to the 6 values of the first variable x from 1 to 6. The average A_i of g on the subset S_i can be calculated easily because the values of g on elements of S_i are simply $1 + i, 2 + i, \ldots, 6 + i$, each equally likely. Since the average of the 6 integers from 1 to 6 is $\frac{7}{2}$, we get $A_i = \frac{7}{2} + i$ for $i = 1, 2, \ldots, 6$. Also, since each

S_i has 6 elements and all 36 elements of S are equally likely, we have $p_i = \frac{1}{6}$ for every $i = 1, 2, \ldots, 6$. So by the law of weighted average of averages, A, i.e., the average total score for the entire set S is simply $\frac{1}{6}\sum_{i=1}^{6}(\frac{7}{2} + i)$. This sum comes out to be $\frac{7}{2} + \frac{1+2+3+4+5+6}{6}$ which is 7, the same answer as before.

As a far more convincing example of the power of this law, we do part (ii) of the problem above, viz., that of finding the average number, say A, of times a coin has to be tossed to get the first head. Here the sample space S consists of all infinite binary sequences with entries H (for a head) and T (for a tail). The real-valued function g on S is defined as follows. Let $(x_1, x_2, \ldots, x_n, x_{n+1}, \ldots)$ be a typical element of S. Then we define $g(x_1, x_2, \ldots, x_n, x_{n+1}, \ldots)$ to be the least integer i such that $x_i = H$. Note that in this case we must also have $x_1 = x_2 = \ldots = x_{i-1} = T$. (The function g is undefined if the sequence consists of all tails. But as we have seen earlier, the probability of this is 0 and so the calculation of the averages is unaffected by it.) The function g assumes the value n with probability $p_n = q^{n-1}p$ and evaluating the sum $\sum_{n=1}^{\infty} p_n n$, we found above that $A = \frac{1}{p}$. But there is a better way. We divide the sample space S into two mutually disjoint subsets S_1 and S_2, where S_1 consists of all infinite binary sequences whose first entry is H and S_2 consists of those whose first entry is T. Since the probability that the first toss is a head is p, clearly the probabilities p_1 and p_2 for a sequence in S to lie in S_1 and S_2 are, respectively, p and q. Hence by the law of weighted average of averages, we have $A = pA_1 + qA_2$, where A_1 and A_2 are the averages of the function g on the subsets S_1 and S_2 respectively. Finding A_1 is very easy, because for every sequence in S_1, the value of g is 1. So g is a constant function on S_1 and hence its average A_1 is simply this constant value, viz., 1. Now let us turn to S_2. By definition, for every sequence in S_2, the first toss is a tail. After this, it is as if we have freshly begun to toss the coin and the average number of tosses (not counting the first toss which has resulted in a tail) to get a head is A by assumption. If we include the first 'wasted' toss, then we get the average value of g on S_2. In short, $A_2 = A + 1$. Hence we get an equation

$$A = pA_1 + qA_2 = p \times 1 + q \times (A + 1) = p + (1 - p)(A + 1) \tag{19}$$

solving which we get $A = \frac{1}{p}$, the same answer as before.

Using a similar reasoning, we can find the average number of tosses till a head shows twice (not necessarily in succession). Denote the average by E this time. If the first toss is H we need one more head and because of the problem just solved this would take $\frac{1}{p}$ tosses on the average. On the other hand, if the first toss is a tail, we still have to wait E more rounds on the average. Combining, we get

$$E = p(1 + \frac{1}{p}) + q(E + 1) = 2 + (1 - p)E \tag{20}$$

solving which we get $E = \frac{2}{p}$, which again tallies with our intuition. Suppose,

however, that we want the average number of tosses, say B, till two heads appear *consecutively*. Then we divide the sample space into three mutually disjoint subsets, viz., those sequences for which the first toss is a T, those for which the first toss is H but the second toss is T and those for which the first two tosses are both heads. Their probabilities are q, pq and p^2 respectively. For sequences of the first type, we have 'wasted' one toss and need B more tosses on the average. So the average for the first subset is $B+1$. By a similar reasoning, for the second set it is $B+2$. For the third, since two heads have already appeared, regardless of what happens later, the function g (giving the number of tosses till two heads appear consecutively) is constant with value 2. So this is also the average for the third subset. Adding and applying the law of weighed average of averages, we get

$$B = q(1+B) + pq(B+2) + 2p^2 \tag{21}$$

Putting $q = 1-p$ and solving we get $B = \frac{1+p}{p^2}$. For an unbiased coin this value is 6. We leave it to you to show by a similar reasoning that for an unbiased coin, three consecutive heads will require 14 tosses on the average. As a practical advice, suppose you go to a casino and want to play a game where each toss of an unbiased coin costs you one rupee and you win a reward if you get a head thrice in a row. If you are just 'averagely' lucky, then don't play the game unless the reward is at least 14 rupees!.

Notice that in these solutions, we did not calculate the probability with which the function g assumes a particular value. For example, we did not find the probability, say p_n, that two heads will occur for the first time at the end of the n-th toss. This is the same as the probability that the n-th toss and exactly one of the earlier tosses is a head. From this it follows that $p_n = (n-1)p^2q^{n-2}$. We could have then found the average E above by summing the series $\sum_{n=1}^{\infty} n(n-1)p^2q^{n-2}$. For this we would need the same reasoning that was used in deriving (10) above. However, if we let p_n be the probability that two *consecutive* heads appear for the first time at the end of the n-th toss, then calculating p_n is much more complicated. It can be done by writing down a recurrence relation considering the possible outcomes of the first two tosses, viz., that for all $n \geq 3$,

$$p_n = qp_{n-1} + pqp_{n-2} \tag{22}$$

with the initial conditions $p_1 = 0$ and $p_2 = p^2$. This is exactly the same as (10) in the last chapter. But the initial conditions are different. So the solution will be similar but not quite the same. It can be shown by induction that if α and β are the roots of the quadratic equation $x^2 - qx - pq = 0$, then for all $n \geq 1$,

$$p_n = C\alpha^{n-1} + D\beta^{n-1} \quad \text{where } C = \frac{p^2}{(\alpha-\beta)} \text{ and } D = \frac{p^2}{(\beta-\alpha)} \tag{23}$$

The identities $\sum_{n=1}^{\infty} n\alpha^{n-1} = \frac{1}{(1-\alpha)^2}$ and $\sum_{n=1}^{\infty} n\beta^{n-1} = \frac{1}{(1-\beta)^2}$ (which are valid

since in Comment No. 6 above it was proved that $|\alpha| < 1$ and $|\beta| < 1$) can now be used to compute B, the average cost of winning. The answer comes out to be $\frac{p^2(2-\alpha-\beta)}{(1-\alpha)^2(1-\beta)^2}$ which is a symmetric function of the roots α and β and can be shown to equal $\frac{1+p}{p^2}$ even without identifying α and β explicitly. A similar appoach is possible for three consecutive tosses but is more complicated because now the analogous recurrence relation will be

$$p_n = qp_{n-1} + pqp_{n-2} + p^2qp_{n-3} \tag{24}$$

for $n \geq 4$ with the intitial conditions $p_1 = p_2 = 0$, $p_3 = p^3$ and so the formula for p_n will involve the roots of the cubic equation $x^3 - qx^2 - pqx - p^2q = 0$.

There is a method which is intermediate to the two methods above, that is, we do not explicitly solve the recurrence relation (22) or (24). But we multiply both its sides by n and sum them over for all permissible values of n (i.e., $n \geq 3$ for (22) and $n \geq 4$ for (24)). Thus, in the case of (22) we get

$$\sum_{n=3}^{\infty} np_n = q\sum_{n=3}^{\infty} np_{n-1} + pq\sum_{n=3}^{\infty} np_{n-2} \tag{25}$$

With a little manipulation of the sums on the R.H.S. this can be rewritten as

$$\sum_{n=3}^{\infty} np_n = q\sum_{n=2}^{\infty} np_n + q\sum_{n=2}^{\infty} p_n + pq\sum_{n=1}^{\infty} np_n + 2pq\sum_{n=1}^{\infty} p_n \tag{26}$$

By definition, the average B equals the sum $\sum_{n=1}^{\infty} np_n$. The L.H.S. of (26) is almost this except for the two missing terms, viz., p_1 and $2p_2$. Keeping in mind that $p_1 = 0$ and $p_2 = p^2$, the L.H.S. is simply $B - 2p^2$. Similarly, the first and the third series on the R.H.S. equal B each. As for the remaining two infinite series, $\sum_{n=1}^{\infty} p_n$ is the probability that two consecutive heads will occur at least once. Hence it equals 1, because the complementary probability that no two consecutive heads occur is 0 as we saw above in (11). Hence the last sum is 1 and so is the second sum since the only missing term, viz., p_1 is 0. Now that we have found expressions for the values of all the five infinite series in (26) we get an equation

$$B - 2p^2 = qB + q + pqB + 2pq \tag{27}$$

which is exactly the same as (21). This is not surprising because (27) is based on (22) and the reasoning which went into deriving (22) is basically the same as that in deriving (21), viz., to classify according to the outcomes of the first two tosses. But in deriving (21) we could completely bypass infinite series using the law of weighted average of averages. We leave it to you to do a similar manipulation to the recurrence relation (24) and show that the average number of tosses for the first-time occurrence of three consecutive heads is $\dfrac{1+p+p^2}{p^3}$. As already noted, for a fair coin, this comes out to be 14.

Finally, we apply the law of weighted average of averages to find the average duration of the ruin game given at the start of comment No. 7 above. Suppose the total capital is c. For each integer k with $0 \leq k \leq c$, denote by D_k the average duration of the game in which initially the player A has k rupees (and hence B has $c-k$ rupees). Evidently, the game does not even start if k is either 0 or c and so we have

$$D_0 = 0 \quad \text{and} \quad D_c = 0 \tag{28}$$

which are like 'boundary conditions'. The sample space S again consists of infinite binary sequences and splitting it into two subsets depending upon the outcome of the first toss of the coin, we get, using the law of weighted average of averages,

$$D_k = p(D_{k+1}+1) + q(D_{k-1}+1) = pD_{k+1} + (1-p)D_{k-1} + 1 \tag{29}$$

which is valid for $0 < k < c$. With a change of indexing, this can be rewritten as

$$pD_k = D_{k-1} - (1-p)D_{k-2} - 1 \tag{30}$$

This is highly analogous to (14) except for the last term -1. So, not surprisingly, the nature of the solution changes crucially depending upon whether $p = \frac{1}{2}$ or not, i.e., depending upon whether the coin is biased or not. We skip the derivation of the solution. But we mention the final answer.

For $p \neq \frac{1}{2}$, we set $q = 1-p$ as usual. Then the average duration of the game when A starts with a rupees (and B with $c-a$ rupees) to begin with is given by

$$D_a = \frac{a}{q-p} - \frac{c}{q-p}\frac{1-(q/p)^a}{1-(q/p)^c} \tag{31}$$

while for $p = \frac{1}{2}$,

$$D_a = a(c-a) = ab \tag{32}$$

where again, (32) can be treated as a limiting case of (31) as $p \to \frac{1}{2}$ (i.e., as $\frac{q}{p} \to 1$).

Comment No. 9:

In the problems studied so far in this section, the sample space was infinite because it comprised the possible outcomes of an infinite sequence of a finitistic process (such as flipping a coin or rolling a pair of dice). There are other problems where the sample space is infinite because it consists of an interval or a region in a plane or in space. Such problems are really beyond the level of the Joint Entrance Examination. Still, a few simple ones can be done by simple geometric arguments and so deserve a brief mention.

As a start, suppose the sample space S consists of an interval, say $[a, b]$ of real numbers. Given a subset A of S, we would like to define $P(A)$, i.e.,

the probability that an element x of S picked at random, belongs to the subset A. Since we want $P(S) = 1$, intuitively $P(A)$ should be the ratio of the 'size' of A to that of S. The trouble, of course, is to define what one means by 'size'. Cardinality is not the right measure of size here since all intervals of real numbers have infinite cardinalities. There is, in fact, no satisfactory definition of the 'size' applicable to all subsets of S. In case the subset A happens to be a subinterval, say $[c, d]$ of S, then its length is a natural way to express size. So, for a point x from the interval $[a, b]$, the probability that x lies in $[c, d]$ should be defined as $\frac{d-c}{b-a}$. This is actually a fairly satisfactory definition of probability. Note that with this definition, $P(S)$ indeed comes out to be 1, as it should. Further, for any single point, say y in S, we can think of the singleton set $\{y\}$ as a degenerate interval of length 0. When so done the probability of each single point comes out to be 0. This is consistent with the observation that in an infinite sample space, a single point has no weight. Note also that it makes no difference whether we consider closed intervals or open intervals or semi-open intervals. $P((c, d)), P([c, d)), P((c, d])$ are all equal to $P([c, d])$, i.e., to $\frac{d-c}{b-a}$. This is again consistent with the fact that these intervals differ from each other by only one or two points and a finite number of points has no weight.

How do we define the size of a subset A of S which is not an interval? If A is the union of two (or more) mutually disjoint subintervals then we define the size of A as the sum of the lengths of these subintervals. The trouble is that not every subset of S is of this form. Defining the size (or the 'measure') of such subsets is a non-trivial question. We shall not go into it, because we are only going to present a glimpse of the probability problems of this type and the sets that we encounter in them will be intervals or finite union of intervals.

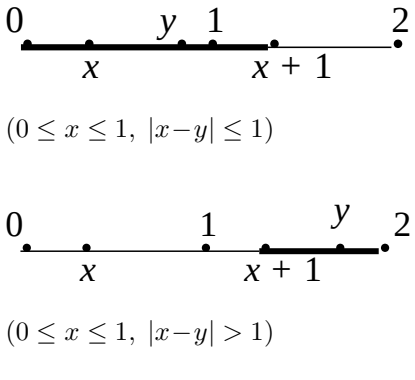
By analogy, for a subset of a plane, its area is the right measure of its size. Thus if the sample space S is a subset of the plane, then for any subset A of S, $P(A)$ will be the ratio of the area of A to that of S (assuming the latter to be a finite, non-zero real number). Again, a rigorous definition of the area of a plane figure is far from trivial. But we shall work with our intuitive understanding of it. Similarly, for a figure in space probability will be defined in terms of volumes.

With this understanding, we solve a simple problem as an illustrative example.

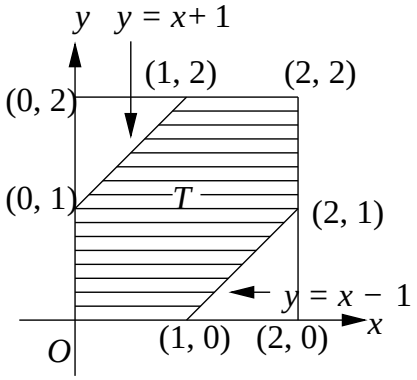
Suppose two points are picked at random from an interval of length 2. Find the probability that the distance between them is at most 1.

Call the points picked as x and y and let $d = d(x, y) = |x - y|$ be the distance between them. Then d can take any possible value in the interval $[0, 2]$ and the desired probability, say p, is the probability that d lies in the interval $[0, 1]$. So it is tempting to think that p is simply the ratio of the lengths of these two intervals, viz., $\frac{1}{2}$. But this is not the correct answer and it is instructive to see the reason. Without loss of generality, take the original interval to be $[0, 2]$. Again, by symmetry around the mid-point of the interval, suppose x lies in the left half of the interval, i.e., $0 \leq x \leq 1$. Then in order that $|x - y| \leq 1$, y can lie anywhere in the interval $[0, x + 1]$ whose length is $x + 1$ which is always greater than or equal to 1, and in fact close to 2 if x is close to the center. On the other

hand, for $|x - y| > 1$ to hold, y must lie in the semi-open interval $(x + 1, 2]$ (see Figure (a) below) whose length is $1 - x$. So if y is chosen at random from the interval $[0, 2]$ it is far more likely that $d \leq 1$ than $d > 1$. Put differently, smaller values of d are more likely than the large ones and so it is not correct to conclude that $p = \frac{1}{2}$ (in which the implicit assumption is that all values of d are equally likely).

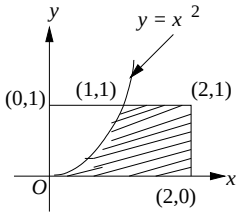


(a)

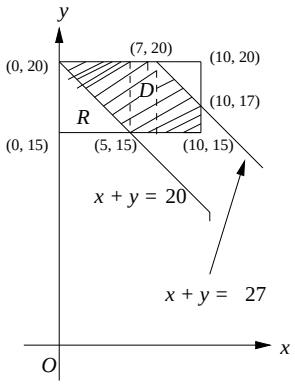


(b)

The correct approach to the problem is as follows. Picking two points x and y randomly from the interval $[0, 2]$ is equivalent to picking a single point (x, y) randomly from the square S shown in Figure (b) above. So we take S as our sample space. Now, the condition $|x - y| \leq 1$ is satisfied if and only if the point (x, y) lies in the shaded trapezium, say T, bounded by the lines $y = x + 1$ and $y = x - 1$. (We may call T as the 'favourable region' for the event whose probability we are seeking.) Therefore p, i.e., the probability that $|x - y| \leq 1$ is the ratio of the area of T to that of the square S. By elementary calculation this ratio is $\frac{3}{4}$.



(c)



(d)

By a similar reasoning, if points x, y are chosen randomly from the intervals $[0, 2]$ and $[0, 1]$ respectively, then the probability that $y \leq x^2$ is $\frac{2}{3}$. Here the sample space is the rectangle shown in Figure (c) above and the 'favourable region' is shown by shading. It is the portion of the rectangle lying below the parabola $y = x^2$. Its area is found using calculus.

In both these problems the points x and y were chosen independently of each other and that is why we could take the sample space to be the cartesian product of the intervals over which they ranged. Contrast this with the next problem where the two variables are not independent.

A piece of wire of length $4L$ is bent at random to form a rectangle. Find the probability that its area is at most $\frac{L^2}{4}$.

Let x and y be the length and the breadth of the rectangle. Then we have $x + y = 2L$. So x and y are not independent of each other. In fact, either one determines the other completely. So here there is only one real variable. We choose it to be x. Then x takes values in the interval $[0, 2L]$. (Here we allow the breadth to possibly exceed the length. If we don't then x will be restricted to the interval $[L, 2L]$. But that will not affect the answer.) The area of the rectangle is xy i.e., $x(2L - x)$. If this is to be at most $\frac{L^2}{4}$, then x must not lie in between the two roots of the quadratic equation $x^2 - 2Lx + \frac{L^2}{4} = 0$. In other words, x must lie in the union of the two intervals $[0, \frac{2-\sqrt{3}}{2}L]$ and $[\frac{2+\sqrt{3}}{2}L, 2L]$. The sum of the lengths of these two intervals is $(2 - \sqrt{3})L$. So the desired probability is $\frac{2-\sqrt{3}}{2}$.

In all these problems whenever a variable, say x, was picked at random from an interval, say $[a, b]$, we assumed that it varied uniformly over that interval. That is, points from one part of the interval were as likely to be selected as points from any other part (of the same size). In many real-life problems this is not so. For example, in the first problem we considered, suppose that the interval of length 2 represents a 2-kilometer long straight road in some city and that the points represent persons living on that road. It is quite possible that some parts of the road are more densely populated than some others. In that case, a person living in such parts has a higher chance of getting selected and so the answer to the problem will no longer be $\frac{3}{4}$. To find the correct answer in such a case, we have to be given what is called a **probability density function**. In essence, the value of this function at a point x_0 tells you how likely it is for a point to belong to a neighbourhood of x_0. But we shall not go into problems of this type. We shall confine ourselves only to those situations where the point chosen varies uniformly over a given interval. (This amounts to saying that the probability density function is a constant function.)

Here is a very realistic problem to be done with this assumption.

A commuter leaves home at a time which varies uniformly from 8:00 to 8:10 a.m. The time it takes him to reach the station varies uniformly (and independently of his starting time from home) between 15 to 20 minutes. From the station there are two trains, one departing at 8:20 a.m. and the other at 8:27 a.m. Find the probability that the commuter misses the first train but makes the second.

Here we have two independent variables, say x and y, which can be taken to vary uniformly over the intervals $[0, 10]$ and $[15, 20]$ respectively. So the sample space is the rectangle R shown in Figure (d) above. Its area is 50 units. The 'favourable region', say D, of R lying between the lines $x + y = 20$ and $x + y = 27$ has area 33 units. So the desired probability is $33/50 = 0.66$.

The problems just considered differ from the earlier problems on probability in one important respect. In the earlier problems the variables involved (e.g., the number of tosses of a coin) were **discrete** in that they assumed only integer values. In the present problems, the variables range over entire intervals and can assume any possible values in these intervals. They are, therefore, called **continuous variables**. We made this distinction in Comment No. 5 of Chapter 19 in connection with continuous compounding. There we observed that if A is the amount you get by compounding continuously, then there are two ways of finding A. One is by writing a differential equation for the continuous rate of growth and then solving it. Another method is to take $\lim_{n\to\infty} A_n$, where, for every positive integer n, A_n is the amount that you get by compounding n times a year (at equally spaced rests). In a nutshell, continuous compounding is the limit of discrete compounding.

A similar approach is also possible for problems of continuous probability. We illustrate this for the first such problem we did, viz., that of finding the probability, say p, that the distance between two points x and y chosen randomly from the interval $[0, 2]$ be at most 1. Let us divide the interval $[0, 2]$ into n equal subintervals by points of the form $\frac{2i}{n}$, where i is an integer ranging from 0 to n. Let S_n be the set of these $n + 1$ points. Let p_n be the probability that two points picked at random from the set S_n be at most 1 apart from each other. As n becomes larger and larger, the set S_n resembles the interval $[0, 2]$ more and more. So we expect that if we take the limit of p_n as n tends to infinity, then we should get p. This indeed turns out to be the case as we show now.

Let us first calculate p_n explicitly. We are not requiring x and y to be distinct. So, the pair (x, y) can be formed in $(n + 1)^2$ ways. The assumption that in the original problem x and y range uniformly over the entire interval $[0, 2]$ forces us to assume that these $(n + 1)^2$ pairs are equally likely. Now let us count the number, say f_n, of the 'favourable' pairs among these. Let $x = \frac{2i}{n}$ and $y = \frac{2j}{n}$. Then $|x - y| = \frac{2|i-j|}{n}$. We have to make two cases depending upon whether n is even or odd, because depending on that the point 1 is or is not in the set S_n. First assume that n is odd, say $n = 2m - 1$, where m is a positive integer. By symmetry, we may calculate the number of favourable pairs $(\frac{2i}{n}, \frac{2j}{n})$ in which $0 \le x < 1$, i.e., in which $0 \le i < m$ and multiply the count by 2. From Figure (a) above, it is clear that if $0 \le x \le 1$, then $|x - y| \le 1$ if and only if $0 \le y \le x + 1$. Hence for every i with $0 \le i \le m - 1$, $(\frac{2i}{n}, \frac{2j}{n})$ is a favourable pair if and only if $0 \le j \le i + m - 1$. So for each such i, there are $i + m$ favourable pairs. Hence the number of favourable pairs in which $0 \le x < 1$ is the sum $\sum_{i=0}^{m-1} i + m$ which, after a little calculation comes out to be $\dfrac{3m^2 - m}{2}$. As already observed, f_n is twice this. So we have shown that for $n = 2m - 1$,

$f_n = 3m^2 - m$ and hence $p_n = \dfrac{f_n}{(n+1)^2} = \dfrac{3m^2 - m}{4m^2}$. This tends to $\frac{3}{4}$ as $m \to \infty$. Thus we have shown that as n tends to infinity through odd values, the discrete probability p_n indeed tends to $\frac{3}{4}$, which we already know as the answer to the problem.

The case where n is even, say $n = 2m$, is handled similarly. Again by symmetry, the number of favourable pairs in which $0 \leq x < 1$ is the same as those in which $1 < x \leq 2$. However, this time we also have to consider the case $x = 1$ (i.e., $i = m$) for which every value of y will give a favourable pair. We omit the details but remark that f_n, i.e., $f_{2m} = 3m^2 + 3m + 1$ and hence for $n = 2m$, $p_n = \dfrac{3m^2 + 3m + 1}{4m^2 + 4m + 1}$. So $p_n \to \frac{3}{4}$ as n tends to ∞ through even values. As we have already shown this to be the case as $n \to \infty$ through odd values, we can now say that $\lim\limits_{n \to \infty} p_n = \dfrac{3}{4}$. In other words, the probability for the continuous problem is the limit of the sequence of probabilities for the corresponding discrete problem, exactly the way continuous compounding is the limit of compounding n times a year as $n \to \infty$.

Comment No. 10:

The essence of the problems done in the last comment was that when we have two mutually independent continuous random variables x and y varying over some intervals, say $[a, b]$ and $[c, d]$ respectively, we can replace them by a single random variable (x, y) varying uniformly over the rectangle with sides $[a, b]$ and $[c, d]$. And the area of a subset of this rectangle provided the right measure to calculate probability. The situation is analogous when we have three pairwise continuous random variables. We replace them by a single variable varying over a rectangular box and take the volume as the appropriate measure. In fact, the method can be applied even when we have more than three (but finitely many) random variables provided we define the concept of a higher dimensional volume (or **hypervolume** as it is sometimes called). This is indeed possible to do inductively. But we shall not go into it.

EXERCISES

23.1 Let $x = \sum\limits_{n=0}^{\infty} \cos^{2n}\phi$, $y = \sum\limits_{n=0}^{\infty} \sin^{2n}\phi$ and $z = \sum\limits_{n=0}^{\infty} \cos^{2n}\phi \sin^{2n}\phi$, where $0 < \phi < \frac{\pi}{2}$. Determine which of the following relationships holds. (1993)

(A) $xyz = xz + y$ (B) $xyz = xy + z$
(C) $xyz = x + y + z$ (D) $xyz = yz + x$.

23.2 Find the rational number which equals the number $2.\overline{357}$ with recurring decimals. (1983)

23.3 Find out at what time between 4.00 p.m. and 5.00 p.m. the hour and the minute hands of a clock coincide. Do the problem by simple arithmetic

and also using infinite series. (For the latter, go on adding the time it takes the hour hand to reach the former position of the minute hand).

23.4 Two mathematicians A and B, 1 km. apart on a straight road start walking towards each other with constant speeds of 5 km./hour and 4 km./hour respectively. A fly, which flies at a constant speed of 20 km./hour is originally on A's head. It flies towards B, and after touching B's head, immediately reverses its direction and flies towards A, then again towards B and so on. Find how much total distance the fly covers till A and B meet. Do the problem with geometric series.

23.5 Do the last exercise with plain arithmetic. (According to a popular anecdote about the great mathematician **von Neumann**, at a social party he was asked this question. He answered it correctly in just a few seconds. The person who asked it said, somewhat defeatedly, "Well, I'm glad you got it the easy way. Many people try to do this with infinite series." von Neumann was puzzled and said, "But that's exactly the way I did it.")

23.6 Suppose we have n types of objects with an unlimited supply of objects of each type. For $r \geq 0$, let a_r be the number of ways of choosing r out of these n types of objects. (Here objects of the same type are indistinguishable from each other.) Prove that a_r is the coefficient of x^r in the product of the n infinite series each equalling $1 + x + x^2 + x^3 + \ldots = \sum_{k=0}^{\infty} x^k$.

23.7 In the last exercise, assume that the binomial theorem is valid for the expansion of $(1-x)^{-n}$, with the understanding that for every integer $r \geq 0$, $\binom{-n}{r}$ means the expression $\frac{(-n)(-n-1)\ldots(-n-r+1)}{r!}$. Then show that $a_r = \binom{n+r-1}{r}$ for $r \geq 0$. (This gives an alternate proof of the formula for the number of selections with repetitions. A combinatorial proof was given in Chapter 1, Comment No. 5. Instead of assuming the validity of the binomial theorem to expand $(1+x)^{-n}$ as a power series in x one can start with (9) and differentiate both the sides $n-1$ times, assuming that the R.H.S. can be differentiated term-by-term.)

23.8 Prove that the number of selections of n objects from $3n$ objects in which n are alike and the rest unlike is the coefficient of x^n in $(1+x)^{2n}(1-x)^{-1}$.

23.9 If n is a positive integer, prove that the coefficients of x^n in the expansions of $\dfrac{(1+x)^n}{1-x}$ and $\frac{(1+x)^{2n+1}}{1-x}$ are 2^n and 4^n respectively.

23.10 Using the last exercise, prove that for every positive integer n,
(i) $\sum_{r=0}^{n} 2^r \binom{2n-r}{n} = 4^n$ (ii) $\sum_{r=0}^{n-1} \frac{1}{2^r}\binom{n+r-1}{r} = 2^{n-1}$ (iii) $\sum_{k=0}^{n} \binom{n+k}{n}\frac{1}{2^k} = 2^n$.
(The last sum was evaluated earlier using (45) of Chapter 5.)

23.11 (a) Let S be a set with n elements. What is the average number of elements in a subset of S, taken at random from all possible 2^n subsets?

(b) In (a), suppose $S = \{1, 2, \ldots, n\}$. For a positive integer $r \leq n$ let $P_r(S)$ be the set of all r-subsets of S (i.e., subsets having r elements each). For $A \in P_r(S)$, let $f(A)$ be the least element of A. Prove that the average value of the function f is $\frac{n+1}{r+1}$. (Tomescu)

23.12 Cards are drawn randomly one after another from an ordinary pack. What is the average number of draws till the first ace shows if the draws are (i) with replacements (ii) without replacements?

23.13 A bag contains tickets marked from 1 to n^2+1 with r tickets marked with r for $1 \leq r \leq n^2+1$. A person who picks a ticket marked with a perfect square m^2 (for some $1 \leq m \leq n$) gets a reward of m rupees. Show that the average reward is $\dfrac{n^2(n+1)^2}{2(n^2+1)(n^2+2)}$ rupees.

23.14 For the problem in Exercise (22.22), find the average number of tests needed till both the faulty machines are identified. More generally, suppose n machines are tested in a random order, it being known beforehand that exactly m of them are faulty. Write a recurrence relation for the average number of tests needed till all the faulty machines are identified.

23.15 (a) Suppose that in the game in the Main Problem, the coin has probability p of showing a head. Assume further that each player goes on putting one rupee in a pool every time he flips a coin and the winner takes all the money put in the pool by then. Find the expected net gains of A, B and C and show that they add up to 0. (The game is therefore called a **zero sum game.**) Which player has the maximum expected net gain? (The game is said to **poised** in the favour of such a player.)

(b) In a gambling machine, a player wagers 1 rupee and chooses a number, say x, from 1 to 6 as his lucky number. A die is cast thrice. If an x shows, then the player gets back his rupee and one bonus rupee for each occurrence of x. Otherwise he loses his wagered rupee. If you are 'averagely' lucky, would you play the game?

23.16 A person A has two friends B and C. B calls A on the phone every morning at a time which varies uniformly between 8 : 00 and 8 : 30 and unless the phone is busy at that time, the call goes on for ten minutes or till 8 : 30 whichever is earlier. If C also calls A at a time which varies uniformly over 8 : 00 to 8 : 30, find the probability that the phone is busy when she calls. (Assume there are no other callers and also that A makes no calls during this time.)

23.17 If two real numbers, say x and y are picked at random from an interval, say, $[a, b]$, prove that the probability that they will be equal is 0, while the probabilities $Pr\{x < y\}$ and $Pr\{x > y\}$ are $\frac{1}{2}$ each. (It is in this sense that equality of two real numbers is an accident while their inequality is a

rule. Geometrically, if two points are taken at random on a line segment then the probability that they will coincide is 0.)

23.18 Suppose two lines L_1 and L_2 are drawn at random in a plane. Prove that the probability that they will be parallel to each other is 0. [*Hint:* Consider the angles made by these lines with a fixed direction and apply the last exercise. Verbally, for two lines in a plane, parallelism is an accident. By taking more than two random variables, the result can be generalised to say that given any finite number of lines in a plane, the probability that at least two of them are parallel to each other is 0. The condition for concurrency of three lines in a plane can be expressed as a certain equality, specifically, the equality of a certain determinant with the number 0. So it can be shown by a similar argument, that given a finite set of lines at random in a plane, the probability that at least three of them are concurrent is 0. This tallies with the concept of a set of lines 'in general position' as defined in Exercise (1.36). See also Exercise (21.23).]

23.19 Although the study of infinite series is beyond our scope, here is an illustration of how they have fascinated mathematicians for long. The infinite sum $\sum_{n=1}^{\infty} \frac{1}{n}$ is not finite even though its n-th term tends to 0 as n tends to ∞. This is equivalent to saying that the n-th harmonic number H_n tends to ∞ as $n \to \infty$, see Exercise (6.52). In sharp contrast, the infinite sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is finite. But the trouble in evaluating it is that unlike in the case of a geometric series, there is no closed form expression for the partial sums of the series of reciprocals of squares. That is, there is no formula for the sum $1 + \frac{1}{4} + \frac{1}{9} + \ldots + \frac{1}{n^2}$. Surprisingly, the infinite sum $\sum_{n=1}^{\infty} \frac{1}{n^2}$ *can* be evaluated. We assume that the power series expansion of $\sin x$ is

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \ .$$

(i) Suppose $f(x) = 1 - \dfrac{x}{3!} + \dfrac{x^2}{5!} - \dfrac{x^3}{7!} + \dfrac{x^4}{9!} + \cdots + \dfrac{(-1)^n x^n}{(2n+1)!} + \cdots$. Show that the zeros of f are of the form $(n\pi)^2$, where n is a positive integer. [*Hint* : Clearly f has no negative zeros. For $x > 0$, show that $f(x) = \frac{\sin y}{y}$, where $y^2 = x$.]

(ii) Assuming that the result of Exercise (3.5)(a) is valid for power series (i.e., for 'polynomials of infinite degree') and applying it to the function $f(x)$ in (i), prove that $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. (An assumption like this is, of course, unwarranted and can lead to disasters. But Euler made it anyway and 'proved' this beautiful result. A logically rigorous proof can be given using theorems about what are called **Fourier series**. But that is beyond our scope.)

Chapter 24

MISCELLANEOUS TIPS AND REVIEW

The past twentythree chapters have exposed you to a wide range of topics from counting techniques to infinitistic probability and along the way, many other important topics such as algebra, number theory, geometry, trigonometry, calculus, differential equations and vectors. At the end of each chapter, there was a fairly long list of exercises with varying degrees of difficulty.

In this last chapter we are in a different ball game. When you try to do an exercise from a given chapter, the very location of it tells you what methods will be needed in the solution. In an examination, there is usually no such hint. Problems are given (often in a random order) and it is for you to first figure out which area they belong to and then what methods are likely to give the solutions. The first guess may not turn out to be the right one and may have to be revised and even totally abandoned later on. After the problem is correctly diagnosed and the method decided upon, there comes the task of actual computation. This is often routine. But sometimes there is room for ingenious short cuts. And finally there is the question of presenting the solution.

Although it is not the purpose of this book to coach you for JEE, in this chapter we give you a few tips of a general nature for these four steps, because there is something to learn here too. The illustrations given will also serve as a review of the earlier chapters. It must, however, be kept in mind that there is no mechanical procedure guaranteed to succeed in every problem. No matter how many tips are given, the decision to apply which tip where and how is ultimately yours. And, for this there is no tip!

We also indicate what to do in the case of an erroneous question. Such errors are extremely rare in the JEE question papers. Still one should be prepared for their accidental occurrence. The exercises vary considerably over the topics and their degree of difficulty. A few of them introduce some standard theorems.

Main Problem : Suppose f is an even function defined on the interval $(-5, 5)$. Then the sum of the squares of the four real values of x satisfying the equation $f(x) = f\left(\dfrac{x+1}{x+2}\right)$ is ..... (JEE 1996, modified)

First Hint: Identify the four values as the roots of two quadratic equations.

Solution: As $f(x) = f(-x)$ for all $x \in (-5, 5)$, the equality $f(x) = f(\dfrac{x+1}{x+2})$ can occur when $x = \pm\dfrac{x+1}{x+2}$. The first possibility gives the quadratic equation $x^2 + 2x = x + 1$, i.e., $x^2 + x - 1 = 0$ gives $x = \dfrac{-1 \pm \sqrt{5}}{2}$. Similarly, the second possibility leads to $x^2 + 3x + 1 = 0$ which gives $x = \dfrac{-3 \pm \sqrt{5}}{2}$. All these four values do lie in the interval $(-5, 5)$ as can be checked easily.

The sum of the squares of $\dfrac{-1 \pm \sqrt{5}}{2}$ is $\dfrac{1+5}{2} = 3$ while the sum of the squares of $\dfrac{-3 \pm \sqrt{5}}{2}$ is $\dfrac{9+5}{2} = 7$. Hence the desired answer is 10.

Comment No. 1:

The sum of the squares of the roots of a quadratic (and more generally of any polynomial equation) can also be found without actually solving it, by expressing it in terms of the elementary symmetric functions of the roots as we saw in Chapter 3, Comment No. 4. In the present problem, however, the saving is not substantial. If instead of $\frac{x+1}{x+2}$ we had a complicated rational function of x, this alternate approach would be mandatory. However, if we have to take only those roots that lie in a given interval, then there is no easy way to tell which ones they are, short of actually finding them. This sort of a verification, although very easy in the present problem can be bypassed. The question already tells us that there are four possible values. If the question had not specified how many possible values of x we shall get, or if the question did specify the number of values we are expected to get and the actual number we got was more than this, then, of course, we would have to eliminate some of them and take only those that lie in the given interval $(-5, 5)$.

Comment No. 2:

Although from a mathematical point of view, the Main Problem was a very simple problem, it is a good illustration of the kind of approach and strategy needed in an examination, especially, in an examination where time is precious. The problem deals with an even function. Normally, when a problem involves some concept, one should be in a position to recollect at least some of the more important results about it and to use them as and when needed. At the same time, one should also keep oneself open to the possibility that these results

might not be needed and that the key to the problem may lie somewhere else. For example, in the present problem no properties of an even function (such as the symmetry of its graph or the fact that its derivative is an odd function) are needed. All that is needed is the definition of an even function. Thereafter, the problem is essentially a problem about quadratic equations. This is, in fact, the correct 'diagnosis' of the problem. And as in clinical practice, correct diagnosis is half the solution.

Sometimes the lack of information is a big help in correct diagnosis since it restricts the possibilities to the relevant ones. In the present problem, for example, suppose instead of an abstract function $f(x)$ some actual even function, say $\cos x$ or $\cosh x$ was given without explicitly telling you that it was an even function. That could complicate the diagnosis because this particular function may have many other properties besides being an even function. And chances are that by focusing on them you may miss the really relevant one, viz., that it is an even function. (A real-life analogy would be that a patient who talks too much can mislead the physician and put him in a worse position than a veterinarian!)

As another example, consider the Main Problem in Chapter 16 where you are given $f(10)$ and some condition on $f(x)$ and are asked to find the value of $f(20)$. Here evidently, 20 has no particular significance and therefore in reality the problem asks you to determine $f(x)$ for all x. Once this idea strikes you, you can think of various ways of doing it. Trying to get hold of a differential equation satisfied by $f(x)$ is one possibility. And the inequality given does precisely that.

When the problem asks something which is not normally possible, it is a sign that its focus is somewhere else. For example, two equations in three unknowns are generally not enough to determine them uniquely. But as we saw in Comment No. 9 of Chapter 14, a system of two equations like $A+B+C=\pi$ and $\tan A+\tan B+\tan C=3\sqrt{3}$ in the three unknowns A, B, C has a unique solution. Here the problem is really that of trigonometric optimisation. (See also Comment No. 10 of Chapter 10.)

In multiple choice type questions, the given answers can provide a clue to the focus of the problem. Indeed, this can be taken as a characteristic of a well-designed multiple choice question. (Regrettably, not all multiple choice questions are of this type. In fact many of them are just conventional questions artificially posed in the multiple choice format.) Consider for example Exercise (15.3) (JEE 1986), where you are given that the line $ax+by+c=0$ is normal to the curve $xy=1$. The alternatives given involve the signs of a and b. Since the slope of a straight line is determined by the coefficients of x and y in its equation, it follows that the focus of the question is on the sign of the slope of the normal to the curve $xy=1$. Once this is realised, the answer comes without much calculation because the curve is a well-known one (viz., a rectangular hyperbola with the axes as the asymptotes) and so even a mental sketch of the curve tells you that every normal to it has positive slope.

Comment No. 3:

The ability to quickly get to the focus of the problem is really an asset. It

can be sharpened with practice and reasoning. But in essence it is an innate quality. A few general guidelines can, however, be given. We list them here with the illustrations coming mostly from problems already solved (or given as exercises) in the earlier chapters.

(i) Carefully see what is asked in the given problem. It may happen that the method that comes most naturally to the mind not only gives what is asked but also something more. In that case the extra gain is of little value and the time spent for it is really a waste as far as that particular problem is concerned. For example, when a problem asks only the number of solutions to something, it is useless to try to actually identify them. There may be a short cut to find the number of solutions without actually solving the equation. We saw an instance of this at the beginning of Comment No. 2 of Chapter 9. Similarly when we are only asked to find the order of a certain differential equation that would result from a given family of curves, there is no need to find that d.e. In the same vein, while proving the concurrency of three lines there is no need to actually find their common point as we saw in Chapter 8. Instead, see if there are easier ways to prove concurrency. Similarly, if the problem merely asks you to identify the type of curve represented by a given locus (as in Exercise (9.15)(b)), there is no need to spend time in finding the equation of the locus. When a definite integral is to be evaluated, the right way is not always through finding antiderivatives.

We caution you, however, against thinking that this reasoning is applicable everywhere. There are situations, where, for example, finding the number of solutions takes almost the same amount of work as determining which ones they are. An example of this will be given at the end of Comment No. 6 below. This only illustrates that no tip should be adhered to blindly. The golden rule is to keep your mind open to various possibilities but not be swayed by any.

Another type of problems where the goal can be reached more elegantly by not taking the most straightforward path, arise in coordinate geometry, where you have to prove a certain relationship between certain points that are not given explicitly. Suppose for example, that you are asked to prove that the tangents at some two points P and Q on an ellipse are perpendicular to each other. If the points P and Q are given explicitly, then the best way to do so is to compute the slopes, say m_1 and m_2, of the tangents at these points and show that their product is -1. But suppose that P and Q are given as the points of intersection of some line L with the ellipse (as in Exercise (10.30)). (JEE 1997) In such a case, identifying the two points P and Q individually is a wasted labour. Our interest is only in showing that $m_1 m_2 = -1$ and there may be a way to do this without computing either m_1 or m_2 explicitly, for example by first finding a quadratic whose roots are m_1 and m_2 (or closely related to them). Exercise (9.61) is an even more glaring example of this.

(ii) An especially noteworthy case of problems where we have to draw the line between what is asked and what appears to be the most natural way to do it, is problems which ask for the existence of something. In real life if you want to prove the statement 'There is a man who is eight feet tall', the best proof would be to actually produce such a man and in case that is not possible, then at least to give his whereabouts. In mathematics, on the other hand, there is a subtle distinction between asserting the existence of something and actually producing it. As we saw in Chapter 16, completeness of the real number system guarantees the existence of so many things e.g., limits of certain sequences, points where a continuous function attains an intermediate value, points where it attains its extrema and so on. But it never gives an explicit construction for any of these things. All that the proof gives is that these points arise as limits of certain sequences. So in problems based on any of these or similar theorems, you can rarely prove the existence by an actual construction. See, for example, the problem done at the end of Comment No. 18 of Chapter 21. If the functions f_1, f_2, g_1, g_2 were given explicitly then the desired value of t can be obtained by solving some equations. But when all that is given is that these four functions are continuous, any attempt to find the desired t explicitly is doomed to fail. It is very important to realise such inevitable failures by carefully drawing the line between what is asked and what is attempted.

Sometimes the proof of existence is given by a contradiction. For example, in Comment No. 16 of Chapter 6, we studied several applications of the pigeon hole principle. In these problems we want to prove the existence of two (or more) things which share some common property. The skill lies in paraphrasing this as saying that there exist two distinct points at which a suitably defined function takes the same value. If this is not the case, then the function constructed would be one-to-one and the contradiction comes because of the fact that there is no room for a one-to-one function from a set to a set of smaller cardinality. Note that the pigeon hole principle does not give you explicitly two points at which the function assumes the same value. It merely asserts the existence of such a pair using cardinality considerations. So again the focus is on proving existence and not on actually finding.

(iii) Analogy with familiar problems is also an important tool in correct diagnosis both in clinical practice and in mathematics. This is where experience really counts. If you have already done a problem very similar to the one asked, then of course, you have a field day. But even when this is not so, see if there is some feature of the given problem which rings a bell. For example, in the Main Problem of Chapter 11, it was given that three circles touch each other externally and that their common tangents are concurrent and further that their common point is equidistant from each point of contact. As it stands, it is unlikely that you have already done a problem with this rather weird hypothesis. But give it a thought and

analyse it. Let O be the point where the common tangents meet and let P, Q, R be the points of contacts of the circles, two at a time. Then these points lie on the lines joining the centres of the circles taken two at a time. Moreover, OP, OQ, OR are perpendicular to these lines and the data $OP = OQ = OR$ now implies that the perpendicular distance of O from all the three lines joining the centres of the circles is the same. But this is exactly the property of the incentre of a triangle. So the point O must be the incentre of the triangle with vertices at the centres of the three circles. Once this idea strikes you, you can of course change the label O to the more familiar I for the incentre. You have correctly diagnosed the problem.

As another example, consider the problem solved at the end of Comment No. 5 of Chapter 6, in which we are given that $a_1, a_2, \ldots, a_n$ are positive real numbers whose product is a fixed number c and are asked to minimise the sum $a_1 + a_2 + \ldots + a_{n-1} + 2a_n$. The catch here is that the last term is $2a_n$. If it were a_n, the answer would come immediately from the all too familiar A.M.$-$G.M. inequality. But why don't we take $2a_n$ as a single entity, say b_n? For notational uniformity we can rename the remaining a_i's as b_i's if we want. The problem now asks us to minimise $b_1 + b_2 + \ldots + b_{n-1} + b_n$ subject to the constraint that $b_1 b_2 \ldots b_{n-1} \frac{b_n}{2} = c$ which can be rewritten as $b_1 \ldots b_{n-1} b_n = 2c$. This is a familiar problem.

Sometimes even a remote similarity can give you a clue like a lightning. See the end of Comment No. 16 of Chapter 14 for an excellent illustration of this. The problem solved in Comment No. 13 of Chapter 13 is yet another illustration, where even though the data is in the form of an inequality, its resemblance with a first order linear differential equation provides the key, viz., multiplying both the sides of the inequality by the integrating factor.

(iv) Sometimes, the data of the problem is given in a twisted form and some untwisting is necessary before you can get to the focal point of the problem. A common instance of this arises in coordinate geometry where the slope of a line is given indirectly by specifying that it is parallel or perpendicular to a given line. In trigonometry, if ABC is a triangle, $\sin A$ is sometimes twisted as $\cos(\frac{B+C-A}{2})$. As a less trivial example, suppose it is given that $\int_0^1 (1 + \cos^8 x)(ax^2 + bx + c)dx$ equals $\int_0^2 (1 + \cos^8 x)(ax^2 + bx + c)dx$. This is an unnecessarily clumsy way of saying that $\int_1^2 (1 + \cos^8 x)(ax^2 + bx + c)dx$ vanishes. There is not much one can immediately conclude from the equality of two integrals over different intervals. But the vanishing of a single integral over an interval forces one to conclude that the integrand must have changed its sign within that interval and this is indeed the key to the solution (given in Comment No. 11 of Chapter 18).

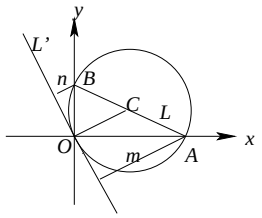
As another example, we saw at the end of Comment No. 5 in Chapter

11 that there are many conditions which are equivalent to the three sides of a triangle being in an A.P. A problem where this condition is to form a part of the hypothesis can be twisted, for example, by giving, instead, that its altitudes are in H.P. Similarly, in Exercise (4.9)(b) (JEE 1998), where we are asked to find the number of divisors of 240 which are of the form $4n+2$, $(n \geq 0)$ a direct count is too time-consuming. But if we note that $4n+2$ divides 240 if and only if $2n+1$ divides 120, the work is reduced by half because now we only have to find the number of odd divisors of 120. Actually, the progress made is more than half. For, write 120 as 15×8. Since every odd integer is relatively prime to 8, it follows (cf. the end of Comment No. 17, Chapter 4), that any odd divisor of 120 must divide 15. So we now only have to find the number of divisors of 15, or $d(15)$ in the notation of Exercise (4.37). This can be done by inspection.

In Chapter 9 we saw instances of how the essence of a problem in coordinate geometry may sometimes lie in a simple geometric fact. As another example, consider the following problem. (Tata-McGraw Hill Series, *Course in Mathematics for IIT-JEE*, p. 834 (modified)).

A line L meets the coordinate axes in A and B. A circle is circumscribed about the triangle OAB. Let L' be the tangent to this circle at O. If m and n are the distances of L' from the points A and B, express the diameter of the circle in terms of m and n.

It is obviously better to start by taking A as $(a, 0)$ and B as $(0, b)$. (Equivalently, we are taking the equation of L in the two intercepts form as $\frac{x}{a} + \frac{y}{b} = 1$.) Then the equation of the circle with AB as a diameter is $(x-a)x+y(y-b) = 0$, i.e. $x^2+y^2-ax-by = 0$. (See the derivation of Equation (21) in Chapter 9.) Then the equation of L' is $ax+by = 0$. Hence we get $m = \frac{a^2}{\sqrt{a^2+b^2}}$ and $n = \frac{b^2}{\sqrt{a^2+b^2}}$. Adding, we see that the diameter (viz., $\sqrt{a^2+b^2}$) equals $m+n$. In this mechanical solution, little use was made of the tangency of L' except to find its equation. Had L' been some other line with a given equation, we could still express a, b and hence the diameter in terms of m and n.



But if we pause for a moment and focus on the essence of the problem, we can see that most of the data is superfluous. The only relevant part of it is that AB is a diameter of a circle and m, n are the lengths of the perpendiculars from A, B to the tangent at some point O on the circle. Let C be the mid-point of AB. A simple diagram shows that these perpendiculars are the parallel sides of a trapezium and hence their sum is simply twice the perpendicular distance of the mid-point C from the tangent at O. But the latter is nothing but the radius of the circle. So, without finding m and n individually, $m+n$ equals the diameter of the circle! We have now done justice to the problem by correctly recognising the role of the special feature that the line through O touches the circle.

In these examples, the given problem was reduced to another one which was qualitatively similar. But sometimes the twist may be such that it shifts the focus of the problem from one area of mathematics to another. For example, suppose we want to evaluate the integral $\int_0^{2\pi} [\sin x + \cos x] dx$ where [ ] denotes the integral part. Here the integrand is piecewise constant and discontinuous whenever $\sin x + \cos x$ equals $0, 1$ or -1. Solving the corresponding trigonometric equations and splitting the interval at $\pi/2, 3\pi/4, \pi, 3\pi/2$ and $7\pi/4$ gives the integral as $-\pi$. So, in this problem the focus is not on integration but on solving trigonometric equations. Similarly, in evaluating $(1+\omega)^{100}$ where ω is a complex cube root of unity, the focus is not on the binomial theorem but on knowing that $1+\omega = -\omega^2$ which gives the answer as $\omega^{200} = \omega^{198}\omega^2 = \omega^2$.

Untwisting of the data may also be helpful when a problem looks quite difficult if attempted by the most straightforward method which does not recognise the special features of the problem but which becomes quite simple when it is paraphrased using these special features. Take for example, Exercise (1.1)(c) (JEE 1985), where we are asked to find the number of onto functions from a set A with n elements to itself. The general problem of finding the number of onto functions from one set to another is quite difficult and the answer cannot be given in an easy form. In the given problem, however, the function is from A to itself. As the set A is finite, such a function is onto if and only if it is one-to-one. Once this key idea strikes us, the answer is simple, viz., $n!$. Similarly, in Exercise (2.28) (JEE 1981), we consider 3×3 determinants with only 0 and 1 as entries and let B and C be, respectively, the sets of those determinants with values 1 and -1 respectively. To compare the cardinalities of the sets B and C, it is not necessary to count them separately. In fact, this would be rather time-consuming since there are 2^9 possible 3×3 determinants with 0 and 1 as entries. But a direct bijection from B to C is obtained if we simply interchange any two fixed rows, say the first and the second. So B and C have the same numbers of elements. Here the problem is superficially a counting problem but the correct focus is on properties of determinants.

(v) In problems involving a particular positive integer, that particular integer may or may not have any special role. For example, consider Exercise (4.21), where you have to show that if 91 divides $n^5 - 5$ then it also divides $n - 31$. Here the integers 91, 31 and 5 have definite roles to play. The result will not hold true if we replace them by arbitrary integers. But in the Main Problem of Chapter 2, where you are asked to find the sum of the products of the integers from 1 to 10 taken three at a time, the integer 10 has no particular significance. It could as well be replaced by any positive integer n. In fact, doing so is a good idea for two reasons. First it reduces the possibility of numerical mistakes. (It is less likely that in a hurry you will write n^5 when you meant n^6 than you will write

100000 when you meant 1000000.) More importantly, doing so opens the door for other methods such as induction or recurrence relations. (For the particular problem referred here, the solution given in Chapter 2 is the best. But the problem can also be done by writing down a recurrence relation for S_n.)

Similarly, in some computational problems certain quantities which could assume any real values (with some restriction such as they are to be non-negative) appear with specific numerical values. Sometimes this may simplify the problem as we saw in Comments No. 1 and 2 of Chapter 21. Similarly, the first problem in Comment No. 1 of Chapter 9 dealt with common tangents of the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. As remarked there, finding a common tangent of two such curves is a rather laborious task in general. But, for these two particular curves, the job can be done simply by inspection.

On the other hand, in many numerical problems the particular figures play little role and sometimes they can even be a nuisance. For example, in the problem done in Comment No. 6 of Chapter No. 11, the length of the median was given as $\frac{1}{\sqrt{11 - 6\sqrt{3}}}$. Having to carry this clumsy figure through the solution is not only time consuming but prone to numerical mistakes. In such cases, it is best to replace it, as we did, by a variable y, get the answer entirely in terms of y and *then* put back the given value of y. Similarly, in many probability problems, the particular values of the probabilities given have very little role to play and could as well be replaced by some real numbers a, b, c etc. lying in the interval $[0, 1]$. Sometimes the very familiarity with certain standard notations makes it easier to work out a problem using them rather than one which uses their particular values. For example, it is much easier to deal with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ than, say, with the ellipse $2x^2 + 5y^2 = 6$, even though as an algebraic expression, the latter is certainly simpler.

While it is not always easy to tell beforehand if in a given problem, the particular values of the variables are a boon or a bane or neither, while diagnosing the problem, one should keep in mind the possibility that there *may be* some purpose behind the particular figures and in case there appears to be none, whether it would be a good idea to replace them by variables.

(vi) In some problems, on the other hand, it pays to specialise rather than to generalise. This is especially true for problems containing a parameter, say n, which can assume positive integers as values, where you have to guess the answer and prove it by induction. Working out a few special cases with lower values of n often enables you to see some pattern. Further experimentation can then give you the key idea (see for example, Comment No. 1 of Chapter 4). Similarly, in the solution to the last problem in Comment No. 16 of Chapter 5, the crucial idea was the symmetry of the

coefficients in the expansion of $(1+x+x^2)^n$. This may not be so noticeable for $n = 1$. But if you take $n = 2$, you will hardly miss it.

Similarly, in the case of a problem dealing with a polygon with n sides, see first what it reduces to in the familiar cases of a triangle or a quadrilateral, which can be worked out 'by hand'. Even among polygons with the same number of sides, say 4, it sometimes pays to consider special cases. For example, if you are asked to prove a result about cyclic quadrilaterals, see what happens in the case of a rectangle, which is a special type of a cyclic quadrilateral. This may sometimes give you a clue to the general case. Even when it does not, it can give you an idea about the degree of difficulty of the problem. And, sometimes (although not very frequently), by some clever trick you may even be able to reduce the general case to the special one as we saw at the end of Comment No. 11 in Chapter 8, where a problem about a pair of concentric ellipses was solved by converting it to a similar problem about a pair of concentric circles.

Sometimes it pays to first generalise and then specialise (to a case other than the one asked). Suppose, for example, that you are given four positive real numbers a, b, c, d and are asked to prove

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4 \qquad (*)$$

The most straightforward approach would be to expand the L.H.S. But that would be horrendous. Instead, the cyclic form of the L.H.S. suggests that there is probably nothing special about 4 here and that for any $n \geq 1$ and for any positive real numbers $x_1, x_2, \ldots, x_n$, we should be able to prove that

$$\frac{x_1}{x_2} + \frac{x_2}{x_3} + \ldots + \frac{x_{n-1}}{x_n} + \frac{x_n}{x_1} \geq n \qquad (**)$$

In a way, we are making the problem harder. But this is actually a boon because now the temptation to elaborately add the terms of the L.H.S. is eliminated as impossible. Instead, we specialise to $n = 2$ (since it is the first non-trivial case of (**)). Then we see that by letting $x = \frac{x_1}{x_2}$, (**) reduces to

$$x + \frac{1}{x} \geq 2 \qquad (***)$$

But this is a very familiar result which follows immediately from the A.M.−G.M. inequality (see Chapter 6, Comment No. 6). So, quite possibly, the A.M.−G.M. inequality is also the key to success for (**) and hence for (*) as well. This indeed is the case, because if we divide both the sides of (**) by n, the L.H.S. is simply the A.M. of the n numbers $\frac{x_1}{x_2}, \frac{x_2}{x_3}, \ldots, \frac{x_{n-1}}{x_n}$ and $\frac{x_n}{x_1}$ while the R.H.S. is their G.M., since their product is 1. Thus we have proved (**) and hence the original (*).

It can be argued that we could have as well divided both the sides of (*) by 4 and applied the A.M.−G.M. inequality. But the question is : How does one think of it? Once the idea of using the A.M.−G.M. inequality strikes you, there is hardly anything left in the problem. (In fact, then you also realise that (**) would remain true even if the denominators were *any* permutation of the numerators, not necessarily a cyclic one.) But unless you make it a practice to try to apply the A.M.−G.M. inequality in every problem involving inequalities, its applicability is not likely to occur to you from (*). Nor is it likely to strike you for (**). But (***) is a known special case of (**) proved using the A.M.−G.M. inequality. So *now* it is very natural to try the same for (**) and hence for (*). (Incidentally, this also highlights the importance of remembering not just the results but their proofs too. If you remember only (***) but have forgotten that it is a consequence of the A.M.−G.M. inequality, you are unlikely to get any clue for proving (**) or (*).) Of course, we could as well have skipped (**) and gone directly from (*) to (***) if we are motivated by a desire to look for a problem which is similar to some problem we already know. But the similarity between (*) and (***) is not so directly visible. On the other hand, it is quite natural to generalise from (*) to (**) and equally natural to specialise from (**) to (***).

(vii) In some problems the data comes from real-life situations and the first task is to convert the problem into a mathematical one. We had instances of this in Chapters 1, 4, 12, 13, 19 and 22. In the problems we considered, such a conversion was relatively standard and so the focal point of the problem was easy to locate. But in the exercises we shall give a couple of problems where the very task of converting them into mathematical problems is a challenge. There is a branch of mathematics, called **mathematical modelling**, devoted to this. But we shall not go into it.

(viii) In situations where a real-life problem is to be converted into a mathematical one, certain simplifying assumptions are allowed even though they may be contrary to real life. For example, the swimmer in the Main Problem of Chapter 13 is assumed to be able to swim even where the water is too shallow near the shore! (If necessary we can change the definition of a shore to exclude from the sea that portion of it where the water is too shallow to swim.) Similarly, we are to ignore the effect of waves or currents on the motion as the data of the problem is silent about these things. In the same vein, in problems of heights and distances in trigonometry, the surface of the earth is assumed to be flat and not spherical (except when the problem specifies it to be so). Also when a man of height h (say) stands on the ground and the angle of elevation of some pole as seen by the man is given, it is understood that h is not the full height of the man but only the distance between his eye and the ground.

Admittedly, there is no hard and fast rule to indicate which assumptions are allowed and which are not in the case of real-life problems. In a

combinatorial problem where balls are to be put into boxes, unless the capacity of a box is specified (or implied), it is to be assumed that it is big enough to hold any number of balls. On the other hand, in a problem where guests are to be seated on chairs, it is to be assumed that no two guests can occupy the same chair at the same time! Common sense is the only guide here. The thing to keep in mind is that in a serious examination designed to test the mathematical ability of a student, there is no room for a frivolous display of originality of thinking or for pedantic correctness. In the second problem of Comment No. 12 of Chapter 22, for example, where the probability of a gun hitting a target is asked, the correct answer is not 0 (as the gun does not actually hit the target), but rather the probability that a shot fired from the gun hits the target. (Such questions are sometimes asked in some quiz shows. But that is done mostly for fun. Stories also abound that such questions are asked at the IAS interviews to test the presence of mind of the candidate. According to one such popular story, a candidate was asked how a dumb person would request that he be given a pair of scissors. After the answer was given (by miming), he was immediately asked how a blind person would do the same thing. A candidate who lacks alertness is likely to ignore the obvious answer 'verbally'. While such questions may occur in anecdotes and may serve some purpose in oral examinations, they have no place in serious written examinations.)

(ix) Even in a purely mathematical problem, certain assumptions are implicit so as to avoid trivial exceptions or degeneracies. For example if the problem deals with the Euler line of a triangle (i.e., the line passing through the orthocentre, the circumcentre and the centroid of the triangle), then it is to be assumed that the triangle is not equilateral as otherwise all these three points would coincide and the Euler line would not be defined. Similarly, when we are dealing with an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a and b are some variable parameters, it is to be understood that when $a = b$, what we have is a circle and not an ellipse. (Equivalently, for such problems, a circle is to be considered as a special case of an ellipse.)

(x) Occasionally, there are situations where the data of a problem can be interpreted in more than one way. This may happen because of a lack of uniformity in certain conventions. For example, some persons treat a square as a special case of a rhombus while some don't and depending on it, the answer to a problem may change. In a problem where the faces of a cube are to be coloured using only three out of a given set of colours, the phrase 'only three' can mean 'at most three' or 'exactly three' (i.e., three colours are used and each one is used at least once). In some cases, an ambiguity may also arise because the data is insufficient as for example, in the problem done at the beginning of Comment No. 16 of Chapter 21. Similarly two circles may be given to touch each other without specifying whether they do so externally or internally. Or the problem may refer to an

angle bisector of a triangle without being specific whether it is the internal or the external angle bisector. If the subsequent data is consistent with only one of the two interpretations, then obviously the other interpretation can be dropped.

(xi) Sometimes the problem makes sense with either of the two interpretations but the amount of work involved is not the same. (See the Main Problem in Chapter 19 for an example of such a problem.) In such problems no matter what policy is adopted subsequently in evaluation, there is always some class of the candidates that suffers. If, for example, only the hard option is given full credit then those who took the soft option (which may look natural in some way) despite being able to do the hard one, stand to lose. If on the other hand, the soft option is also given full credit, then those who chose the hard option have wasted their precious time (which can be quite costly in a competitive examination).

There is very little that a candidate can do in such a case. If time is not a problem, then the safest thing to do is to solve the problem with all possible interpretations. But in doing so, care must be taken to point out right at the beginning that the candidate intends to do the problem with more than one interpretation. Otherwise, if the first interpretation is not the one the examiner wants, he may simply stop reading further and there is a danger that the subsequent interpretation (the one that is wanted) may not get examined at all. Strictly speaking, the examiners are supposed to read every line written by the candidate. But they are, after all, human beings. They are overworked. And moreover, once an answer goes astray there is very little chance that it will cure itself subsequently. So they can hardly be blamed for not reading further once they see that the problem has been wrongly interpreted.

If time is short, see if some clue is provided by the credit allotted to the question. If it is disproportionately high as compared with the work involved in the softer option, then probably the paper-setters had the harder option in mind. In any case, no matter which option a candidate takes, it is always a good idea, to mention at the beginning of the solution that the problem could have been interpreted some other way but that the candidate is sticking to a particular interpretation. Even if the discarded way happens to be the one the examiners want, the candidate may get some credit (or at least some sympathy) for his awareness of the other possibility. If, in addition, the candidate also writes briefly how the problem with the alternate interpretation can be tackled or how the solution he has given can be modified to handle the alternate interpretation, that further enhances his chances of getting some partial credit. (These considerations are, of course, inapplicable for a multiple choice type question. But the usual practice in the case of a multiple choice question containing an ambiguity is to give as correct any answer which answers it correctly with at least one interpretation.)

Comment No. 4:

After a problem is correctly diagnosed, the next task is, of course, to identify the methods that are likely to give the answer. This will naturally depend upon the problem itself and frequently there is a choice. For example, in the Main Problem, after the problem was reduced to a couple of quadratic equations, we could either solve them or express the sums of their squares in terms of the elementary symmetric functions of the roots. Similarly, if the problem requires us to prove some binomial identity, we have a variety of methods given in Chapter 5. Then again, for problems of geometry, we can either apply pure geometry or coordinates or complex numbers or vectors or sometimes even trigonometry as we have illustrated in Chapters 8 through 11 and 21. Similarly, there is a wide variety of methods for evaluating integrals or solving differential equations.

We have already discussed the merits and demerits of the various methods in the respective chapters. So here we give only a few general guidelines.

(i) Draw a preliminary diagram wherever possible, whether the problem specifically asks or not. These diagrams are often necessary just to understand the problem and can inspire ideas. Even in problems such as counting or probability problems, the Venn diagrams are often helpful. Similarly, when you are dealing with an A.P., think of the terms as located on a straight line at equally spaced points. When you want to think of a sequence converging to some point, draw a dot on the line and let the tip of your pen hop towards that point. Whenever you encounter an inequality of the form $a \leq x \leq b$, where a, b are real numbers, immediately picture it as



When you are dealing with functions, especially several functions at a time, including their composites, draw a diagram in which the sets are shown as blocks and functions as arrows. (See for example, the diagram illustrating Theorem 2 in Comment No. 7 of Chapter 3.) If you have made a habit of drawing such diagrams, it hardly takes any time to draw them in your rough work in the examination. Whatever time you do spend in drawing them is amply rewarded by the ideas that they inspire.

It is wrong to think that diagrams are needed only where the data is of a geometric or of a physical nature or where there is some physical motion to simulate the concepts involved (such as the concept of a function, which can be thought of as a transition from one set to another). Diagrams can also be used to simulate a thought process. The tree diagrams that we introduced in Comment No. 11 of Chapter 22 are an excellent example of this. More generally, when you are doing a problem which requires you to break the reasoning into several cases, you can help yourself by drawing a rough diagram. You do not have to show this diagram in your fair work.

But it is very beneficial to draw it to think of the right solution in the first place.

We illustrate this with a solution to Exercise (22.17). (JEE 1998) Here we are given three boxes holding white and 1 black, 2 white and 2 black, and 1 white and 3 black balls respectively. This is a physical data and you can picture it by actually drawing the boxes and the balls therein. (It is of course, unnecessary, that you draw a coloured diagram in case the balls are coloured! In fact, instead of showing the balls by tiny discs, it is enough to simply put appropriate symbols to indicate this. For example, as shown in the diagram below, we can put 3W 1B inside the first box to indicate that it contains 3 white balls and 1 black ball.)

3 W 1 B	2 W 2 B	1 W 3 B	
W	W	B	3/4 x 1/2 x 3/4 = 9/32
W	B	W	3/4 x 1/2 x 1/4 = 3/32
B	W	W	1/4 x 1/2 x 1/4 = 1/32

But the real use of diagrams comes later. The problem further tells you that one ball is picked at random from each box and asks you to find the probability that 2 white and 1 black balls are picked. Clearly this falls into three cases depending upon which box a black ball is picked from. We can picture these three cases by writing underneath each box the colour of the ball picked from that box as shown above. If you do this, then computing the probability of that particular colour combination becomes very vivid. It is shown by taking the products of the appropriate probabilities. It is now trivial to add the three numbers and get the answer.

When you see the 'polished' solution to this problem (such as the one in the Answers to Exercises at the back), it will rarely be accompanied by such a diagram. This often gives the impression that the problem can be done without the diagram. But that only means that you are drawing the diagram mentally instead of on a piece of paper. (To acquire the habit of drawing such diagrams, it is a good idea to draw them when you are reading a solution given by somebody else. That will not only enable you to understand the solution better, it will teach you how to think in tackling similar problems.)

(ii) Give a try to as many approaches as you can think of, without spending too much time on any particular one (except, of course, when one of them quickly leads to a solution). Do not disregard the straightforward approach. It is a common misconception that every problem has some hitch to it. Many problems don't. For example, the problems of finding loci, maxima and minima and areas can often be done by a straightforward

method. Also as remarked in Comment No. 20 of Chapter 2, sometimes it is not a bad idea to evaluate a 3×3 determinant by direct expansion, instead of spending time looking for some tricky row operations that will give the value less laboriously.

This is certainly not to suggest that the brute force approach is the best one. In many problems it is not and a good problem is one whose very formulation gives you an idea of which of the several alternative approaches is most suited for that problem. For example, the equation of a straight line can be written down in many different ways. If the problem involves the intercepts made by the line on the axes then the intercepts form is the best choice as was the case in Exercise (10.24). Similarly, Exercise (10.33) (JEE 1993) involves the distances of various points of a line from a fixed point $a(-5, -4)$ on that line. This suggests that it may be advantageous to parametrise the line in terms of the algebraic distance of a point P from A. The problem can be done by taking the equation of the line in the usual form as $y + 4 = m(x + 5)$. But this is laborious.

Similarly, there are many different formulas for the area of a triangle and although in theory any one of them can be used, in a particular problem, one of them may be more directly applicable than the other, depending on the data. In problems of finding antiderivatives, several different substitutions may look promising but some of them may prove to be too laborious later on (e.g., Exercise (17.27)(iii)).

It often happens that the standard approach to handle a particular piece of data in general is not the best one while handling the same piece of data in a special context. Suppose, for example, that you are given that x, y, z are in A.P. There are various ways to handle this. The most direct is to let d be the common difference and put $y = x+d$ and $z = x+2d$. But in some problems (for example, when you are dealing with the product xyz) it is better to take the three numbers as $y - d, y$ and $y + d$. In many problems, the common difference does not come into the picture and merely writing $2y = x + z$ (or, equivalently, the middle term as the A.M. of the other two terms) serves the purpose.

So these are the three standard ways of handling the data that x, y, z are in A.P. when nothing else is known about them. But suppose you are given further that x, y, z are the roots of a cubic. Then the sum $x + y + z$ is known to you and equating it with $3y$ will tell you one of the roots of the cubic. Similarly, if x, y, z stand for the angles of a triangle, then the hypothesis that x, y, z are in A.P. is equivalent to saying that $y = 60°$. Here the special context enables you to use the hypothesis more effectively.

As a more instructive example, (see Exercise (11.6)) suppose you are given that in a triangle ABC, $\cot A, \cot B, \cot C$ are in A.P. and are asked to find what type of a progression a^2, b^2, c^2 are in. (JEE 1985) Here hastily writing $2 \cot B = \cot A + \cot C$ will be a mistake. Instead, see how the hypothesis can be paraphrased in terms of the sides of the triangle using any

of the standard properties of a triangle. Since the sides are proportional to the sines of the opposite angles, we can replace the denominators of $\cot A, \cot B, \cot C$ by a, b, c respectively so as to get that $\frac{\cos A}{a}, \frac{\cos B}{b}, \frac{\cos C}{c}$ are in A.P. Clearing the denominators (i.e., multiplying every term by the constant abc), we further get that $bc\cos A, ca\cos B, ab\cos C$ are in A.P. Multiply by 2 and use the cosine formula to further get that $b^2+c^2-a^2, c^2+a^2-b^2, a^2+b^2-c^2$ are in A.P. We can *now* apply the fact that the middle term is the A.M. of the other two to clinch the answer. (Or, we can continue the reduction by subtracting $a^2+b^2+c^2$ from each term to get that $-2a^2, -2b^2, -2c^2$ are in A.P. and finally multiply each term by $-\frac{1}{2}$.) The key idea here is the linearity of arithmetic progressions (that is, the fact that numbers in an A.P. remain so when subjected to a linear change).

Just as the data and the context, i.e., the additional information as to what the quantities represent, have a bearing on the choice of the correct approach, so can the subsequent work to be done in the problem. Suppose for example, that we are looking for the minimum of the function $f(x) = \dfrac{x^2-1}{x^2+1}$ as in Exercise (13.12). Here hastily applying the quotient rule to take the derivative is a mistake. Instead if you rewrite $f(x)$ as $1-\dfrac{2}{x^2+1}$, then since the first term is a constant and there is only one more term, you can see, even without derivatives, that the function increases as x increases numerically. So here, a little algebraic manipulation so as to split the numerator suitably was the right approach. Suppose, however, that you need to find the minimum of the function $f(x) = \dfrac{(4x^2+1)^3}{16x^4}$ which arises during the solution to Exercise (15.4), where the goal is to find the normal to $y = x^2$ that forms the shortest chord. Here, it is *not* a good idea to expand the numerator and thereby express $f(x)$ as a sum of four terms. The reason is that later on we shall have to solve the equation $f'(x) = 0$ for which we shall have to factor $f'(x)$. Writing $f(x)$ as a sum of four terms will give $f'(x)$ as a sum of three terms (since one of the summands of $f(x)$ is a constant). But this is not of much help in factorisation of $f'(x)$. Instead, if we leave the cube in the numerator as it is and find $f'(x)$ by the quotient rule for the derivatives, then the numerator of $f'(x)$ will have $(4x^2+1)^2$ as a readymade factor and cancelling it (since it can never vanish), the calculation will be drastically simplified.

So the general rule that can be laid down is to keep in mind the nature and the context of the data and also the purpose of a particular part of a problem while selecting the right approach.

(iii) As far as possible, avoid an approach that will entail working with expressions involving too many radicals or minus signs since they increase your chances of slips of hand. For example, when you have a choice to work with the cotangent of an angle and the tangent of its complementary

angle, take the latter, because many but not all formulas about the cotangent involve negative signs and that puts a strain on you. In coordinate geometry, where the actual location of a given point does not matter, take it to lie in the first quadrant, as we did in the alternate solution to the first problem of Comment No. 6 in Chapter 9.

(iv) Try to assess the type and the amount of work you will have to do in the approach you are trying. Ask yourself whether you are fully utilising the data. If not, chances are that you are missing something. Read the question again and make sure you understand it correctly. (True, there are some exceptions to this. For example, in the problem solved at the end of Comment No. 2 of Chapter 15, the solution never used the part of the hypothesis that $P(x)$ was a polynomial of degree 3. But such cases are very rare.)

(v) Be prepared to discard a particular line of thinking if you find that it will involve something which is inherently impossible, difficult or time-consuming. For example, in the Main Problem in Chapter 10, it was tempting to convert the data as a cubic equation in $\cos\theta$. But unless the cubic has an obvious solution, there is no easy way to solve it. Similarly, for the problem in Comment No. 8 of Chapter 18, the initial attempt of proving (10) there through proving (11) had to be abandoned.

(vi) Sometimes the line you are following will, if successful, give something more than the problem asks. If so, chances are that you are aiming too high. For example, in the Main Problem of Chapter 11, we are given the inradius r as 4 units. We are asked to determine the ratio $\frac{r_1r_2r_3}{r_1+r_2+r_3}$. It is very tempting to try to do so by determining r_1, r_2, r_3. But, as remarked in Comment No. 2 there, the single piece of information about the length of the inradius is insufficient to determine r_1, r_2, r_3 individually. So this approach is destined to fail. The sooner you realise this the less time you waste. (A situation like this should be carefully distinguished from the one where the approach you are following is not inherently impossible but will take more time than what is asked. For example, in the problem at the beginning of Comment No. 2 of Chapter 9, we were asked to find the number of values of the parameter c that have a certain property. Here actually finding these values was not inherently impossible. But our interest was only in their number for which a short cut was possible.)

(vii) If none of the approaches you can think of appears likely to succeed, see if you can modify any one of them or can think of a new one. If you can do so quickly, well and good. Otherwise, you are probably missing something and merely trying the same things over and over again is not likely to help. In such cases, it is better to abandon the problem temporarily and return to it later with a fresh mind than spend more of your precious time on it. Just exactly at what stage you want to leave the problem will depend, naturally, on the credit allotted to the problem and on your ability to do the remaining problems.

(viii) Finally, after you have decided which line you want to take, before actually starting the work, make sure once again that you have understood the problem correctly and also, in case the problem has some numerical data, that you have taken it down correctly. This deserves to be stressed because in the elation that comes with the knowledge that the problem is within your reach, there is a danger that you may make a hasty and an incorrect start. For example, you may take the given point as $(2, 3)$ when in reality it is $(3, 2)$. In that case all the work that you do, even if it is correct and consistent with your initial assumption, may go down the drain if it is an objective question, where the examiner looks only at your final answer. Even when the question is of the conventional type, where you have to show your work, most examinations do not have uniform policies as to what partial credit should be given in such cases. It is therefore best to play safe by ensuring that you are not boarding the wrong train. This vigilance costs you nothing in terms of time or energy since you are exercising it before you have put in any work. But it can protect you against a lot of waste.

Comment No. 5:

After you have diagnosed the problem and decided what approach will work, comes the question of actually doing the computations. In fact, in some problems this is the predominant part. For example, if you are asked to prove a trigonometric identity, the focus of the problem is not a secret. Nor is the method of attack a hidden one. It has to consist of a series of algebraic manipulations using some standard trigonometric formulas. Similarly, when you are asked to count the number of elements in a given set (say, a set of certain permutations), the difficulty is not in understanding what is to be done but in deciding how to do the counting. In such problems the brute force method (such as fully expanding both the sides of the identity to be proved) is usually not the right one. Once again, there is no golden rule to find the right method. So all we can do is give a consolidated list of some standard techniques already encountered at various places in the earlier chapters.

(i) **Divide and Rule:** We already encountered this technique in Chapter 1 in counting problems and also in Chapter 22 in probability problems. It is also useful in problems of finding areas as we saw in the Main Problem of Chapter 17. Divide and rule is also the motivation behind resolving a rational function into partial fractions. In these problems the technique was used to split the thing to be counted (or measured) into several parts whose cardinalities or areas can be found easily. But sometimes it can also be used to prove an identity of the form $A = B$. It may not be easy to do so directly. But suppose we are able to split A into two parts, say A_1 and A_2 and similarly B into two parts, say B_1 and B_2 in such a way that we can easily show that $A_1 = B_1$ and $A_2 = B_2$. Then it follows that $A = B$. This is how we proved an important property of the definite integral of a periodic function (Equation (32) in Chapter 18). At the

end of Comment No. 6 of Chapter 6 we illustrated how the power of the A.M.−G.M. inequality be enhanced by first splitting one or more terms.

In applying 'divide and rule' it is very important to ensure that the parts are mutually disjoint (or, in the case of probability problems, that the events into which the original event is broken are mutually exclusive). If this is not the case, then we have to keep track of the overlaps using the principle of inclusion and exclusion discussed in Comment No. 9 of Chapter 1. This principle itself can be considered basically as a case of 'divide and rule', except that here we are also regrouping some of the subsets by taking their unions and also adding and subtracting certain cardinalities. Both these techniques will be studied shortly. (For the case of two subsets, say A and B, we first divide their union $A \cup B$ into three mutually disjoint subsets, viz., $A \cap B'$, $B \cap A'$ and $A \cap B$, where $'$ denotes complementation w.r.t. the ambient set. Grouping together $|A \cap B'|$ and $|A \cap B|$ gives $|A|$ while grouping together $|A' \cap B|$ and $|A \cap B|$ gives $|B|$. But if we want to do both these groupings, we need to add and subtract $|A \cap B|$.)

Divide and rule can also be applied multiplicatively instead of additively. That is, instead of writing the given expression as the sum of suitable parts, we factorise it suitably and then work with the individual factors which are easier to handle. In problems asking you to show that some integer is divisible by some integer n, it is often preferable to take the prime power factorisation of n and prove divisibility separately by each of these prime powers. (As examples, see Exercises (4.21) and (4.22).)

A question naturally arises as to what is the right way to divide. The answer of course depends on the specifics of a particular problem. Sometimes there is a clear hint. For example, in the calculation of areas by integration, a splitting is mandatory at a point where the integrand changes its formula. In other situations, a split is indicated by identifying a part which is easier to count or to measure or to show equal to something else. Where symmetry is involved, a splitting which respects the symmetry is the right one. Suppose for example, that H is the orthocentre and DEF is the **pedal triangle** of a triangle ABC (i.e., D, E, F are the feet of the altitudes of triangle ABC). Then if we want to split the pedal triangle, the most natural splitting will be into the three triangles, EHF, FHD and DHE.

Where 'divide and rule' is to be applied multiplicatively, standard factorisations such as that of a positive integer into prime powers, that of a difference of two squares (or cubes etc.) are useful. Sometimes these may have to be combined with other facts. For example, in some problems writing $\cos 2x$ as $(\cos x + \sin x)(\cos x - \sin x)$ will do the trick. At the end of Comment No. 12 of Chapter 7, we worked out a problem where factorising x^3+x^2+x into complex factors (viz., as $x(x-\omega)(x-\omega^2)$) was the key to the solution. And, at the end of the same chapter, we did one trigonometric

simplification by factorising 1 as $(\sin^2 2A + \cos^2 2A)(\sin^2 2B + \cos^2 2B)$, probably the most uncanny example of 'divide and rule'!

(ii) **Grouping Together:** This is the opposite of 'Divide and Rule'. In the latter we split something into parts. In grouping the idea is to combine together several parts so that their troublesome parts cancel each other. The idea is very simple and sometimes we use it in day-to-day life too. Suppose for example, that I buy 1.5 liters of cow milk sold at the rate of Rs. 14.50 per liter and 2.5 liters of whole milk sold at Rs. 21.50 per liter. The total cost can be found by simple arithmetic. But the jaggy fractions make it messy. But here is a way out. Simply add 14.50 and 21.50 to get Rs. 36 as the cost of one liter each of the two types of milk together. This is a whole number and so the cost of the fractional 1.5 liters of milk of each type is simply 36×1.5 which is 54 by a mental calculation. Add to this the cost Rs. 21.50 of one liter of whole milk. The total cost is therefore $54 + 21.50$ i.e., Rs. 75.50. Try this with a milkman. He will be impressed! The essence of this trick is to add the two fractions 14.50 and 21.50 together to get a whole number, 36, which is easier to handle.

Grouping can also be used to simplify products. We multiply together factors which are individually complicated but whose product is something easier to handle. We saw many instances of both additive grouping and multiplicative grouping in Chapter 7. The alternate solution to the Main Problem given in Comment No. 1 there was based on a clever grouping of factors.

The basic idea behind the change of order of summation is also a regrouping of terms. Suppose we have a double sum of the form, say $S = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}$ (which could, for example, represent the sum of all entries in an array with m rows and n columns). To evaluate the sum as it is, we first have to evaluate, for each fixed $i = 1, 2, \ldots, m$, the sum $\sum_{j=1}^{n} a_{ij}$ whose value, say S_i, will in general depend on i. Then the double sum S is the sum of these S_i's, i.e., $S = \sum_{i=1}^{m} S_i$. (In geometric terms, we are first adding all entries in each row and then adding these row sums to get the value of the double sum.) Now, in some problems, it may not be easy to get a succinct expression for S_i or even if it is so, it may not be easy to find $\sum_{i=1}^{m} S_i$. In such cases it may sometimes pay off to change the order of summation. That is, we first group together all terms in a given column and then add these column sums. Thus, S now equals $\sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij}$ which may be easier to find. Exercise (5.23) is an excellent example of this. (JEE 1985)

It deserves to be stressed that the technique of grouping together can be applied only when the number of terms (or factors) is finite. As we

have never dealt with infinite sums (except the geometric series in Chapter 23, Comment No. 2) and infinite products (except for a casual reference in the form of the Wallis formula in Comment No. 5 of Chapter 18), this restriction does not matter to us. But just to illustrate how things can go wrong if we apply groupings indiscriminately to infinite sums, consider the infinite sum $1-1+1-1+1-1+1-1+\ldots$. If we group together the terms two at a time starting from the first term we get

$$(1-1)+(1-1)+(1-1)+\ldots$$

from which we can conclude that the sum is 0. But if we group together the terms two at a time, starting from the second term, we get

$$1-(1-1)-(1-1)-(1-1)-\ldots$$

from which, the same sum ought to equal 1. So we get $0=1$. This only shows that extreme care is necessary in extending the manipulations which we do routinely for finite sums to infinite sums. The formula for change of order of summation also breaks down as will be shown in the exercises.

In fact, many things which we simply take for granted can go wrong when infinity comes in. Take for example, the pigeon hole principle in Comment No. 16 and 17 of Chapter 6. Its essential idea is that there cannot be a bijection from a (finite) set to a set of smaller cardinality. In particular, there cannot be a bijection between a set and a proper subset of it. (In classical language 'a part cannot be equal to the whole'.) But if the set is infinite, such a bijection can exist. For example, let $\mathbb{Z}$ be the set of all integers and let E denote the subset of all even integers. Then the function $f:\mathbb{Z}\longrightarrow E$ defined by $f(x)=2x$ is a bijection, even though E is a proper subset of $\mathbb{Z}$. (See Exercises (24. 8) and (24.15) too.)

Frequently, the technique of grouping is combined with splitting. That is, we split two (or more) things into several parts each and then group together a part of one with a part of the other. In effect, this is what was done above to find the cost of the milk. Sometimes the splitting is such that the parts cancel each other and saves us the redundant work of calculating each one of them. This happens typically, when some area is evaluated by splitting it into several parts. For example, in the solution to Exercise (17.3)(xv), the area came out to be the sum $\int_0^1(\sqrt{2+y}-\sqrt{2-y}\,)dy+\int_1^2(\sqrt{2+y}-\sqrt{y}\,)dy$. If we evaluate each integral separately by finding an antiderivative of the integrand and evaluating it at the upper and the lower limits of integration, we see that in both the integrals, the value of the antiderivative of $\sqrt{2+y}$ (viz., $\frac{2}{3}(2+y)^{3/2}$) at $y=1$ is involved. But in the first integral, this value will appear with a positive sign while in the second one it will appear with a negative sign. Since they will cancel each other, it is foolish to spend time in finding this value. It is therefore better to split each integrand into two parts and combine the first parts of each. Then the area can be rewritten as $\int_0^2\sqrt{2+y}\,dy-\int_0^1\sqrt{2-y}\,dy-$

$\int_1^2 \sqrt{y}\, dy$. Now there is no need to evaluate the antiderivative of $\sqrt{y+2}$ at $y = 1$.

The 'split and regroup' technique is also the essence of summing certain types of series called **telescopic series**. Typically, in such a series, each term is broken into two parts so that one part of each term cancels with the other part of the next term. (This explains the peculiar name because in a telescope there are coaxial cylinders so that the outer diameter of each is the same as the inner diameter of the one in which it fits.) Suppose, for example, that we want to find a closed form expression for the sum $\sum_{k=1}^{n} \frac{1}{k^2+k}$. This is neither a geometric nor an arithmetic series. In fact, it does not come under any of the standard types of sequences for which closed form expressions are known for the sums of their terms. But let us rewrite $\frac{1}{k^2+k}$ as $\frac{1}{k(k+1)}$ and thence as $\frac{1}{k} - \frac{1}{k+1}$. The next term will then be $\frac{1}{k+1} - \frac{1}{k+2}$. Thus the two occurrences of $\frac{1}{k+1}$ cancel each other. As this happens for every pair of consecutive terms, we are only left with the first part of the first term and the second part of the last term, viz., with $1 - \frac{1}{n+1}$. So, this is the sum of the series. (The reasoning becomes a little more vivid if we write the series as $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \ldots + (\frac{1}{n-1} - \frac{1}{n}) + (\frac{1}{n} - \frac{1}{n+1})$.)

(iii) **Borrow:** This is what we do when we introduce an extra term or an extra factor in a given expression. Of course, as in real life, in mathematics too, the borrowing has to be made up by repayment. That leads to the 'add and subtract' or 'multiply and divide' technique which we use innumerably often. For example, we factorise $x^2 - y^2$ by adding and subtracting the term xy. We derive the product rule for derivatives by adding and subtracting $f(x)g(x_0)$ to $f(x)g(x) - f(x_0)g(x_0)$. Similarly when we rationalise an expression $\frac{1}{\sqrt{x}-\sqrt{y}}$ as $\frac{\sqrt{x}+\sqrt{y}}{x-y}$, we are multiplying and dividing by $\sqrt{x} + \sqrt{y}$. Similarly, when we rewrite a complex number $\frac{1}{x+iy}$ as $\frac{x-iy}{x^2+y^2}$ so as to have a real denominator we are multiplying and dividing by $x - iy$.

Indeed, so common is this technique that there is hardly any need to give more examples. It is the purpose of applying this technique that deserves some comment. When we add fractions, we apply this technique to each fraction so as bring them all to a common denominator. The same thing applies when we add several rational functions. However, sometimes the data of the problem may be such that common denominators can be ensured in an unusual manner. Take for example, Exercise (2.2) (JEE 1982), where we have to work with a sum of the form $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \ldots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$, it being given that $a_1, a_2, \ldots, a_n$ are in A.P. Here we can bring all the terms to a common denominator in the usual manner by multiplying all the denominators together. But that would not lead us anywhere. If, instead, we first rationalise each term then because of the hypothesis that $a_1, a_2, \ldots, a_n$ are

in A.P., all the denominators will be automatically equal and the solution will come by merely adding the numerators.

The technique of complementary counting (and hence also complementary probability) can also be looked at as an instance of borrowing. Suppose we want the cardinality $|A|$ of a set A. In complementary counting, we take some superset S of A and count $|S|$ and $|S-A|$ (which are easier to count than $|A|$). Here, in essence, we are adding and subtracting $|S-A|$ to $|A|$ to get $|A|$ as $|A|+|S-A|-|S-A|$ and then grouping the first two terms together. As illustrations, see Comment No. 4 of Chapter 22 and Comment No. 6 of Chapter 1, where complementary counting was combined with an ingeneous transform.) There is also a continuous version of complementary counting. We used it at the end of Comment No. 5 of Chapter 17 to find the area of the smaller portion of a disc cut off by a chord. (See also Exercise (13.7).) In a generalised sense, we are also resorting to borrowing when we calculate, say, the square of a number like 998 by writing it as $(1000-2)^2$ and evaluating it as $1000000-4000+4=996004$ instead of finding it directly by multiplying 998 with itself. This only shows that the same underlying idea can manifest itself in so many ways.

In proving inequalities, we often borrow an intermediate term. For example, if a, b, c, d are positive real numbers with $a < c$ and $b > d$ then proving that $\frac{a}{b} < \frac{c}{d}$ is simplified if we separately prove $\frac{a}{b} < \frac{c}{b}$ and $\frac{c}{b} < \frac{c}{d}$. (See Exercise (24.64) for a good illustration.) In proving inequalities of the form $f(m) < f(n)$ for positive integers $m < n$, we often prove $f(k) < f(k+1)$ for every k. In effect, here we are borrowing several intermediate terms. (See Comment No. 9, Chapter 6 for illustrations.)

(iv) **Transform:** It so happens sometimes that a problem looks rather complicated as it is. But when you transfer it to an equivalent problem, the latter may be easier to handle. The solution can then be translated back in terms of the original problem. As a classic example, we all know that it is much easier to add two numbers than to multiply them. Suppose we want to multiply two numbers, say x and y. We assume that both are positive as the sign can be added later on after we multiply their absolute values. Now we take their logarithms w.r.t. some fixed base, say 10 or e. Then $\log(xy) = \log x + \log y$. From the tables of logarithms[1] we read $\log x$ and $\log y$, add them (which is presumably easier than multiplication) and from the tables of anti-logarithms, read xy. This is the essence of the use of tables of logarithms. Nowadays they are replaced by calculators. But the basic properties of logarithms remain intact. Using the same basic property as the one mentioned here, we can convert a G.P. to an A.P. which is a little easier to handle. (See Comment No. 6 of Chapter 2.) Through logarithms (and their inverses, viz., exponentials, we can convert

[1]It can be argued that the tables give only approximate values of logarithms. But then so does even the most advanced computer today. The purpose here is to illustrate the method of transforms and not to go into the question of accuracy of the log tables.

a functional relation of one type into another, as we saw in Comment No. 3 of Chapter 20.

The entire subject of coordinate geometry discussed in Chapter 9 is an example of a transform technique, where we convert a problem in geometry to a problem in algebra. Vectors and trigonometry are yet two more tools to transform a problem in geometry (see Chapters 21 and 11 respectively). As we saw in Comment No. 12 and 13 of Chapter 8, sometimes problems of plane geometry can be transfromed most conveniently to problems of complex numbers, especially where rotations are involved. And, as we saw in Comment No. 17 of the same chapter, sometimes the tables can be turned around. That is, a problem about complex numbers can be transformed into a problem in geometry. In Chapter 7 we saw numerous examples of transforming certain trigonometric products and sums using the complex roots of unity. Exercise (11.33) is an example where a problem of algebra is solved by transforming it to a problem in trigonometry.

Within coordinate geometry itself, we can transform the origin by shifting and the axes by rotation so as to bring the equation of a given curve to a particularly simple form. In Comment No. 18 and 19 of Chapter 10, this enabled us to completely classify the curves represented by the general second degree equation in x and y.

A transform also sometimes enables you to apply a theorem which is not (directly) applicable to the original problem. For example, suppose p is an odd prime and we want to prove that $4(p-3)!+2$ is divisible by p (cf. Exercise (4.8)(i)). Let $u = 4(p-3)!+2$ and $v = (p-1)(p-2)v$. Since p is relatively prime to $(p-1)(p-2)$, p divides u if and only if it divides v. But the advantage in working with v is that it equals $4(p-1)! + 2p^2 - 6p + 4$ and now Wilson's theorem becomes applicable.

The technique of duality tranforms an optimisation problem of one type to that of another as we saw in Chapter 14, Comment No. 17 and 18. The fundamental theorem of calculus studied in Chapter 17 tranforms results about derivatives into results about definite integrals. (For example, the chain rule translates into the rule of integration by substitution while the rule for the derivative of a product of two functions translates into integration by parts.)

As just mentioned, a brilliant example of the transform technique in problems of counting occurs in Andre's solution to finding the number of balanced arrangements of n pairs of parentheses. After reducing the problem to that of counting the number of unbalanced arrangements, the key idea was to establish a bijection between the set of all unbalanced arrangements of n pairs of parentheses and the set of all arrangements of $n+1$ right and $n-1$ left parentheses. Equally elegant was the alternate solution to the problem of pairing off 8 players solved in Comment No. 6 of Chapter 22. As a less ingenious but more important application of transforms, we found the number of combinations with repetitions allowed

in Comment No. 5 of Chapter 1.

In both these examples, the tranformation that did the trick was in the form of a bijection between two sets. Usually, the less obvious the bijection, the more ingenious is the application. This was also the basic idea in combinatorial proofs of some combinatorial identities in Chapter 5 including the ones in its Main Problem.

Sometimes a transformation consists merely of a substitution. In Comment No. 17 to 22 of Chapter 17 we saw plenty of examples of finding antiderivatives using suitable substitutions. Comment No. 14 of Chapter 19 does the same for solving differential equations. As less ambitious but noteworthy examples, in Comment No. 1 of Chapter 10, we had to abandon the algebraic approach to solve the Main Problem because the cubic $8x^3 - 4x^2 - 4x + 1 = 0$ has no rational root. If we apply the rational root test directly to this cubic, then a possible rational root is of the form $\frac{p}{q}$, where p divides 1 and q divides 8. We may assume $p = 1$. But the possible values of q are $\pm 1, \pm 2, \pm 4$ and ± 8. So before we can conclude that the polynomial $8x^3 - 4x^2 - 4x + 1$ has no rational roots we shall have to dispose off 8 possibilities by verification. But as Exercise (10.12) shows, the simple substitution $y = 2x$ reduces the work drastically. The problem in Comment No. 10 of Chapter 5 illustrates a situation where among two equivalent substitutions, one is preferable to the other.

Similarly, in Comment No. 21 of Chapter 2, we evaluated the determinant $\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix}$ by first replacing the squares by single symbols. Retaining the squares would only complicate the expression and increase the chances of mistakes. So here again, a simple transform (viz., replacing a square by a symbol) simplified the matters. In some problems, things are the other way round. Suppose, for example, that we want to minimise some distance, say d. Typically, the formula for d is the square root of some rather complicated expression. In such cases it is better to get rid of the ugly radical sign by minimising d^2 rather than d as the two are equivalent. In Exercise (5.11), the binomial theorem could not be applied directly to the sum because the exponent $r - 1$ of (-3) did not match the lower index of the binomial coefficient ${}^{3n}C_{2r-1}$. But writing -3 as $(i\sqrt{3})^2$ and $(-3)^{r-1}$ as $\frac{1}{i\sqrt{3}}(i\sqrt{3})^{2r-1}$ does the trick.

Thus we see that a transform technique appears in many different contexts. The transform itself can be quite elementary or subtle. Ingenious use of it is an art. We remark that just as logarithms serve to convert products into sums, there are transform techniques available to convert a differential equation into an ordinary algebraic equation and thereby making it much easier to solve. They are called **Laplace transforms** and are beyond our scope.

(v) **Use symmetry:** Symmetry is a geometric concept and hence its role in

geometric problems is obvious. When things are symmetric about a line or a plane, many conclusions can be drawn without explicit justifications, since the justifications can be easily given using symmetry. We already exploited this fact in the solution to the last problem in Comment No. 11 of Chapter 8, and the first problem of Comment No. 3 of Chapter 11, where we used symmetry about an angle bisector. In the problem done in Comment No. 3 of Chapter 12, the data was symmetric about a vertical plane and this made the solution easier. Consideration of symmetry also helps in choosing suitable coordinate axes. For example, in the solution to the Main Problem of Chapter 8, by taking the internal angle bisector of the angle at A as the x-axis, the slopes of the sides AB and AC could be taken to be of the form m and $-m$.

Even when no geometry is involved directly, the concept of symmetry makes sense because of some of the conventions we follow. Take for example, the binomial coefficients $\binom{n}{r}$, where n is a positive integer and $0 \leq r \leq n$. Normally we represent the integers from 0 to n by equally spaced points on a straight line. As a result, the points r and $n-r$ are symmetrically located w.r.t. the centre $\frac{n}{2}$ which itself is an integer if n is even. That is why the combinatorial identity $\binom{n}{r} = \binom{n}{n-r}$ is called the symmetry relation. It says that terms that are symmetrically located w.r.t. the centre are equal. We used this identity in Chapter 5. Another interpretation to it is that in the binomial expansion of $(x+y)^n$, $\binom{n}{r}$ and $\binom{n}{n-r}$ are, respectively, the coefficients of $x^r y^{n-r}$ and $x^{n-r}y^r$. But the expression $(x+y)^n$ is symmetric between x and y in the sense that it is unaffected if we interchange x and y. Under this interchange, the terms $x^r y^{n-r}$ and $x^{n-r}y^r$ get interchanged. So their coefficients must be equal.

Symmetry of this type (i.e., symmetry w.r.t. the centre of a row) can inspire the key idea as we saw in the solution to the last problem of Comment No. 16 of Chapter 5. The problem dealt with the expansion of $(1+x+x^2)^n$ as, say, $a_0 + a_1 x + a_2 x^2 + \ldots + a_{2n} x^{2n}$. The middle term of the expansion is $a_n x^n$. It is not immediately obvious that the coefficients are symmetric about the centre. But this follows by writing $(1+x+x^2)^n$ as $x^n(\frac{1}{x}+1+x)^n$ and observing that $\frac{1}{x}+1+x$ is symmetric about the centre and hence any function of it will be unaffected if x is replaced by $\frac{1}{x}$. The solution to the problem depended crucially on the symmetry of the a_r's, i.e., on the relation $a_r = a_{2n-r}$.

More generally, as we defined in Comment No. 4 of Chapter 3, a function of more than two variables is said to be symmetric in all its arguments if the interchange of any two of them (leaving others unchanged) leaves its value unaffected. The case of three variables is especially interesting because trigonometric identities about triangles contain numerous examples of symmetric functions of three variables. For example, the area or the circumradius or the inradius of a triangle is a symmetric function of the lengths of its sides.

The symmetry considered so far was what is called **axial symmetry** which arises because there is an 'axis' (which is usually a line, but could be a single point or a plane), which has the property that the mid-point of the line segment joining two mutually symmetric points always lies on the axis. There is another kind of symmetry called **rotational symmetry** which arises when a geometric figure has the property that when it is rotated about a point (or a line sometimes) through a certain angle (and integral multiples of it), we get exactly the same figure. For example, an equilateral triangle has rotational symmetry, the angle of rotation being 120 degrees. More generally, a regular n-gon has rotational symmetry. This fact was crucially used in the pure geometry solution in Comment No. 2 of Chapter 10. In solid geometry, a regular tetrahedron has rotational symmetry about each of its altitudes. The solution to the problem in Comment No. 18 of Chapter 8 was based on this.

The analogue of rotational symmetry for functions of several variables is the concept of **cyclic symmetry**. It is usually applied only for a function of three variables. Specifically, a function of three variables, say x, y, z is said to be cyclically symmetric if whenever we replace x, y, z by y, z, x respectively, the function does not change. A cyclically symmetric function need not be symmetric as is shown, for example, by the function $x^2y + y^2z + z^2x$. For cyclically symmetric functions, it is customary to use the sigma notation such as $\sum x^2y$ which means that the other terms of the summation are obtained by replacing x, y, z by y, z, x and then by z, x, y respectively. The advantage of cyclical symmetry is that it saves the duplication (or rather triplication!) of work. For example, as noted in Comment No. 3 of Chapter 8, once we find the equation of one altitude of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3), we can write down the equations of the other altitudes instantaneously using cyclic symmetry. Cyclic symmetry also serves sometimes to keep a check on our calculations. Suppose, for example that we find the coordinates of the orthocentre of this triangle. If the expressions we get are not cyclically symmetric, then we have made a mistake somewhere.

Cyclic symmetry also helps in proving identities where the expressions involved are cyclically symmetric. This was amply illustrated in the problem worked out in Comment No. 13 of Chapter 7. (Note also the use of transform technique in it. The product $\sin(x-y)\sin(y-z)\sin(z-x)$ would be quite messy to expand by brute force. But the new variables A, B, C introduced there in a cyclical manner made it manageable. Indeed, this brings us to the next tip.)

(vi) **Choose efficient notation and units:** Many people tend to think that the notations used are a superficial part of mathematics, comparable to the name of a person. 'What is there in a name?' they ask. What difference does it make whether a triangle is called ABC or PQR? According to these persons, the only advantage a particular notation has is that consistent use of a particular notation breeds some familiarity which makes

things a little comfortable. For example, we consistently denote by a, b, c the lengths of the sides opposite to the vertices A, B, C respectively of a triangle ABC. If we denote them by some other symbols, say, D, x and ϕ respectively, then the cosine formula for $\cos A$ would be $\dfrac{x^2 + \phi^2 - D^2}{2x\phi}$ which nobody would recognise. Worse still, if we denote the sides opposite to A, B, C by b, a, c respectively (instead of the usual order (a, b, c)), then $\cos A$ would be $\dfrac{a^2 + c^2 - b^2}{2ca}$ which would be horribly confusing. Still, if we use it consistently, then over a passage of time, it will not look so awkward, just as 'Mumbai' for 'Bombay' and 'Chennai' for 'Madras' no longer look so awkward now.

These people do have a point. It is true that once a formula is proved, merely changing the labels of some of the symbols makes little difference as long as care is taken to ensure that the same symbol is not used to denote two different things at the same time. For example, it is perfectly all right to denote a triangle ABC by PQR, but in that case you better not denote its circumradius by R.

But things are quite different when you have a choice of notations while setting up the problem. Again, if your freedom is restricted only to the labels of the variables, then it does not mean much as it makes little difference if you denote a relevant angle by α or by β. But if you have the freedom to choose the variables as well, then by choosing them cleverly you can simplify the subsequent work. Just now we mentioned an instance of this from Comment No. 13 of Chapter 7, where by introducing the variables A, B, C in a cyclic manner, the product $\sin(x - y)\sin(y - z)\sin(z - x)$ could be tamed. It may be argued that here one of the variables is superfluous since $A+B+C = 0$. In the solution of the problem, later on we did, in fact, replace C by $-A - B$. But while introducing the variables it is preferable to do so without spoiling cyclic symmetry.

When setting up a real-life problem as a mathematical problem, we can often choose the quantities which we want to take as independent and then express the other quantities in terms of these basic quantities. For example, as remarked in Comment No. 2 of Chapter 13, in the case of the Main Problem there, we could express the time taken by the swimmer as a function of x, the distance which he swims parallel to the shore. In fact, for many persons this would be the most natural choice. But then the expression for the time taken will involve messy radicals. If, instead, we choose as our basic variable the angle θ which the direction of his motion in sea makes with the perpendicular from him to the shore (as we did in the solution given there), that simplifies the expression for the time taken and hence the subsequent work. (Of course we could as well have started with x as the independent variable and after encountering the ugly radical $\sqrt{a^2 + x^2}$ made the substitution $x = a\tan\theta$ to transform it to the neater $a\sec\theta$. But when you can, why not spend a moment and be wiser right

from the start?)

In problems of constrained optimisation, we have to optimise a function of two (or more) variables subject to some constraint. Frequently, the constraint can be incorporated right into the choice of the variables. Suppose for example, that we want to minimise the perimeter of a rectangle of a given area, say A. Denoting the sides of the rectangle by x and y, this amounts to minimising the function $f(x, y) = 2x + 2y$ subject to the constraint $xy = A$. But we can right at the start, take the sides of the rectangle as x and $\frac{A}{x}$ and consider the problem of minimising the function $2(x + \frac{A}{x})$ of the single variable x. A similar strategy was followed in solving the problem in Comment No. 8 of Chapter 13, where we had to cut a piece of wire into two parts.

In a problem where you are given some odd positive integer, say n, instead of working with n, it often pays to replace n by $2m + 1$ (or sometimes by $2m - 1$) and work with m instead of n. You then do not have to keep track of the hypothesis that n is odd. You have already incorporated it into a notation and made a beginning. As a more illuminating example of a problem where the data is incorporated into the choice of the notation, consider Exercise (6.11), where it was given that the real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in A.P. and we were asked to find the intervals in which β and γ lie. The normal tendency would be to write three equations for x_1, x_2, x_3 using elementary symmetric functions of the roots. These equations will involve β and γ. There is one more equation viz., $2x_2 = x_1 + x_3$ coming from the fact that the roots are in A.P. Eliminating x_1, x_2, x_3 from these four equations will give us a relationship between β and γ. But the work can be shortened considerably by taking the three roots in the form $x_2 - h, x_2$ and $x_2 + h$, where h is the common difference of the A.P. Since $x_1 + x_2 + x_3 = 1$ we further get $x_2 = \frac{1}{3}$. Hence the three roots of the cubic are $\frac{1}{3}, \frac{1}{3} - h$ and $\frac{1}{3} + h$, where we may suppose $h \geq 0$. Taking the product of the roots we get $\frac{1}{3}(\frac{1}{9} - h^2) = -\gamma$ from which the range of γ can be determined as the interval $[-\frac{1}{27}, \infty)$. Similarly, for β we have $\beta = x_1x_2 + x_2x_3 + x_3x_1 = \frac{1}{3} - h^2$, whence the range of β is the interval $(-\infty, \frac{1}{3}]$. The same strategy (viz., that of expressing three terms of an A.P. in terms of the middle term and the common difference) also simplifies the computational work in Exercise (2.3)(d). In a locus problem involving, say, the normal at a typical point on a conic, it is preferable to take the parametric equations of the conic to represent this typical point (e.g. Exercise (9.24)(xii)) as that reduces the number of variables from 2 to 1. Generally, the smaller the number of variables, the less number of equations you have to write and the simpler is the solution. But there is no hard and fast rule that one should always go for as few variables as possible. There are other considerations too, such as cyclic symmetry which one should not destroy as far as possible. In Comment No. 12 of Chapter 9, we solved a problem where it was easier and more natural to find the equation of a line by expressing it as a mem-

ber of a 2-parameter rather than of a 1-parameter family. Ultimately, it is a matter of judgement and once again, experience is a good guide.

What has been said about notations also applies to units. Sometimes a problem depends only on the relative proportions of the quantities involved. This gives us the freedom to set one of them equal to an arbitrary number of our liking. A judicious choice can simplify the calculations. The Main Problem of Chapter 12 is a good illustration of this. Similarly, if a problem in trigonometry involves only the shape and not the size of a triangle, then it is only the relative proportions of the sides that matters. So we can set one of the sides equal to 1 (or some other value). However, in the interest of symmetry, it may sometimes be preferable to set some other length associated with the triangle, say its circumradius R, equal to 1. A progression (whether arithmetic, geometric or harmonic) of positive numbers remains so if all the numbers are divided by the same (positive) number. So without loss of generality, one of the terms of the progression can be taken to be 1. This reduces the number of variables by 1 and thereby simplifies the calculations.

(vii) **Work backwards:** In complicated problems where you have to prove an equality of the form $A = B$, there may not be any easy way of going from A to B or vice versa. But it may be possible to show that both equal some third expression, say C. In such a case, you can first go from A to C and then backwards from C to B. Then you get a proof of the original identity to be proved. We already encountered an instance of this in Comment No. 14 of Chapter 7.

A similar remark applies when you are proving that some statement A implies some other statement B. When there is no easy way to do this directly, you try to replace B with a statement, say C, which is logically equivalent to B (i.e., B holds if and only if C holds). If you can prove that A implies C, then the problem is solved. We encountered this in the solution to the second part of the second problem of Comment No. 5, Chapter 11. Here you are reasoning 'forward' from A to C and then 'backward' from C to B.

To be able to apply this method effectively, you must know beforehand certain standard characterisations from geometry and trigonometry. The former include the various characterisations of parallelograms, rhombi, cyclic quadrilaterals, tangency to circles and so on. From trigonometry, the fact that the sides a, b, c of a triangle ABC are proportional to $\sin A, \sin B, \sin C$ allows you to convert problems involving the sides of a triangle to those involving its angles and *vice versa*. We used this convertibility in dualising problems of triangular optimisation (Comment No. 17 of Chapter 14) and also solving some problems such as the one in Comment No. 14 of Chapter 7. (See also Exercise (11.14).)

When we apply backward reasoning to prove that a statement A implies a statement B, by proving instead that A implies C, it is not

always necessary that the statement C should be logically equivalent to B. It is perfectly all right if the statement C is stronger than B, but it must never be weaker. This type of backward reasoning is often needed in proving inequalities. Suppose we are given some data and want to prove an inequality of the form $a \leq b$ from it. If this can be done directly, well and good. Otherwise we may often be able to find something, say c which is easily seen to be smaller than or at most equal to b. In that case, the inequality $a \leq c$ is stronger than the inequality $a \leq b$ which is asked. So if you can prove the former, you are through. Sometimes it is easier to find some c such that $a \leq c$. In that case, the inequality $a \leq b$ will follow if you can show $c \leq b$. This type of reasoning is especially needed when we want to verify the definition of a limit, say, $\lim_{x \to c} f(x) = L$ directly from the definition. Here we start with a positive real number ϵ (whose value may or may not be specified). Now we have to find *some* δ that will answer the definition of limit. So we would like to make $|f(x) - L|$ smaller than ϵ by taking $0 < |x - c| < \delta$. When the expression $|f(x) - L|$ is complicated, the standard practice is to replace it by something bigger, say $H(x)$. Since the inequality $H(x) < \epsilon$ is stronger than the inequality $|f(x) - L| < \epsilon$, it suffices to find some $\delta > 0$ for which $H(x) < \epsilon$ whenever $0 < |x - c| < \delta$. Of course, there may be some values of x which are outside the interval $(c - \delta, c + \delta)$ for which the inequality $H(x) < \epsilon$ fails but the weaker inequality $|f(x) - c| < \epsilon$ holds. This means that the δ we have found can possibly be enlarged, or in other words that it is not the best δ that will work for the given ϵ. But that is all right because our task was not to find the best δ that will work but only *some* δ that will work. (See the last part of Comment No. 10 of Chapter 15 for an actual example.)

(viii) **Look for Short cuts.** The tips given so far such as regrouping of terms or transforming a problem are often needed as key ideas to tackle the problem as a whole. Their use is almost mandatory because if you do not think of them, then the problem becomes impossible or horribly laborious. But, in addition, there are problems where short cuts of a somewhat *ad hoc* type are available for some of the intermediate steps. They are not mandatory. That is, even if you can't think of them, you will still be able to complete that step. But a little alertness can save you precious time and spare you the risk of numerical mistakes. For example, in Equation (29) of Chapter 17, we resolved the expression $\frac{7x^3+6x^2-3x+2}{x^4-1}$ into partial fractions by the method of undetermined coefficients. If done in the most straightforward manner, this would require us to solve a system of four equations in four unknowns. But as explained there, we can get the values of two of these unknowns very easily by another method. The problem is then reduced to solving a system of two equations in two unknowns. But that is not the last word. As remarked in the solution there, the choice of these two equations, if made cleverly, can make a mincemeat of the problem. Do look for such short cuts. As another example, if as a part

of the solution to some problem on vectors, you have to find the length of the vector $8\mathbf{i}+4\mathbf{j}+4\mathbf{k}$, then it is foolish to square the numbers 8, 4 and 4 and to add the squares. Instead, it is far better to take out the common factor 4 and get the length as $4|2\mathbf{i}+\mathbf{j}+\mathbf{k}| = 4\sqrt{6}$. (In essence you are using here the elementary fact $|\alpha\mathbf{v}| = |\alpha|\,|\mathbf{v}|$. Anybody can prove this, of course. But the real test comes when you apply it without being told to apply it.) So a little alertness pays here too.

When it comes to determinants, an ingenious application of the row and/or the column operations can simplify the work drastically. In fact some problems are intended to test precisely this. But sometimes as a part of some other problem (e.g., one involving linear dependence of vectors) you have to evaluate some determinant where there is not much room for such clever tricks and so the only course left is to expand the determinant. Mathematically, it does not matter with respect to which row or column you choose to expand. But sometimes one particular choice may reduce the risks of numerical mistakes. For example, the solution to Exercise (21.12) (d) ultimately reduces to evaluating the determinant $\begin{vmatrix} -1 & 5 & -3 \\ -4 & 4 & 3 \\ 1 & 7 & \lambda+1 \end{vmatrix}$. All the entries except one are some numbers. If you expand this determinant w.r.t. the first row (as is done commonly), the factor $\lambda+1$ will occur twice and increase the possibility of a numerical slip. But if you expand it w.r.t. the third row (or the third column), this factor occurs only once and the determinant comes out easily as $16(\lambda+1)+132$.

However, we hasten to caution you against trying to be the smart guy all the time. For many computations, there is no short cut, and even if there is, the time you spend in figuring it out may well exceed the time you will spend in doing the computation in a straightforward manner. Nor does the ability to see cheap short cuts necessarily signify mathematical acumen (unlike the ability to quickly see the focus of the whole problem). Ultimately, it is a question of the individual's bend of mind to decide how far out one should go to look for such short cuts. Still, the ability to spot them quickly is an asset and if used wisely, saves you time and also reduces numerical slips. After some practice, a clever student developes the ability to instinctively smell if there a room for a short cut, just as after some experience, a clever customs officer develops the ability to instinctively suspect a law breaking passenger.

Comment No. 6:

After you have finished the work and obtained the answer, before you start writing it as fair work, pause for a moment to ensure that you have made no numerical mistakes. In some problems, the only way to ensure this is to go over the calculations again. But in some other problems, there may be other methods. For example, if the problem asks you to solve some equations (whether algebraic, trigonometrical or differential ones) you can substitute the answer and

verify if it is indeed a solution. This will, of course, not reveal if you have missed any solutions. But it will at least tell you that the ones that you got are correct.

As remarked at the end of the solution to the Main Problem in Chapter 9, sometimes there are more than one ways to arrive at the answer and if you can afford the time, it is a good idea to see if they tally. Even if you cannot do the given problem by another method, you may be able to do some special cases of it and see if the answer tallies. For example, the problem may ask you to find the answer for a general positive integer n occurring in the statement of the problem. For lower values of n, you may be able to find the answer by hand. Do so and see if your general answer specialises to it. (In doing so, avoid degenerate cases, of course. For example, if the problem involves a polygon with n sides, the least value of n you can try is 3.) If a problem asks for the area of a cyclic quadrilateral, see what your answer reduces to in the case of a rectangle, whose area is already known.

In some problems, there may be some inherent hints that can alert you, for example if the area of a figure comes out to be negative or larger than the area of some rectangle containing it! Similarly, if the probability of some event comes out to be negative or greater than 1, it is a sure sign that something has gone wrong. Even when it comes out to lie between 0 and 1, some other check may be available. For example, if A and B are two events such that occurrence of A always implies that of B and in your answer, the values you get are such that $P(A) > P(B)$, evidently you have made a mistake.

In problems where the data is symmetric so is the answer as a general rule. For example, in Exercise (15.4) you have to identify the normals of the parabola $y = x^2$ that form the shortest chords. As the curve is symmetric about the y-axis, so will be such chords. So, if you begin the solution by considering a normal at a point (x_0, x_0^2) and then getting some equation for x_0, this equation must involve only even powers of x_0. If not, then you have made a slip somewhere. Checks like this alert you well in advance and heeding them can save some precious time and energy.

In non-numerical problems, if the answer is coming out to be surprisingly simple, take it as a warning. For example, if the area of a figure which is specified by some parameters is coming out to be independent of one of the parameters, then check again to see if you have inadvertently omitted some term containing that parameter. (Very rarely, there are exceptions. For example, the answer to Exercise (22.18) comes out to be independent of k, the number of additional balls. This is quite baffling at the outset and in fact, no explanation is known that will make you see intuitively why it is so.)

In some problems, considerations of dimension can help. Suppose for example, that the quantities a, b, c, r etc. represent certain lengths then an expression for area in terms of these quantities ought to only have terms of dimension 2 in it such as $\frac{\pi}{2}ab$ or $a\sqrt{r^2 + bc}$. If you get a term like abc, something has gone wrong. Although these considerations do not guard you against all types of mistakes (e.g., they will not tell you if the scalar factors such as $\frac{\pi}{2}$ are correct or not) they help you at least in some cases. And the time spent on them is like the money we spend on insurance.

In fact, in the case of numerical problems, this bit of wisdom applies not only to the entire solution but often to various intermediate steps as well. For example, in finding the area of a region specified by various curves, you often first have to find the points of intersections of various pairs of curves. For each such point, immediately verify by direct substitution of its coordinates that it indeed lies on both the curves under question. If not, you have made a numerical mistake. If you let such a mistake at an initial stage go unchecked, the subsequent work that you do regarding the evaluation of integrals may go down the drain, even though it is correct by itself in the sense that the antiderivatives you have found are correct but the points at which they are to be evaluated are wrongly identified. Again, while finding the antiderivatives themselves, if you have resolved some rational function into partial fractions, then before integrating them do ensure that the resolution is correct. This takes far less time than finding the resolution.

Similarly, if the problem requires you to begin by finding the cross product of two numerically given vectors, then before doing anything further with the cross product you obtained, ensure that it is indeed perpendicular to each of the two vectors. This can be done almost by inspection by taking the dot product. The few extra seconds you spend on it are well worth because a numerical mistake in taking the cross product may vitiate all your subsequent work.

As mentioned earlier, most examinations do not have a uniform policy as to which numerical mistakes are to be condoned. Until such a policy is made known to the candidates well in advance, there is always some class of students that suffers no matter which way the policy is. If, for example, the policy is to condone numerical mistakes, then those who spend their precious time in checking their answers stand to lose. On the other hand, if the policy is not to condone any mistakes but this is not made abundantly clear to the candidates, then those who have done a problem correctly except for a silly minor slip, are given a raw deal.

So once again, in the absence of any other information, the safe bet is to assume that *no* mistakes will be condoned. As remarked at the end of Comment No. 5 above, taking clever short cuts for the intermediate steps of a problem is desirable but not so terribly vital. But when it comes to ensuring that no numerical slips are committed, every single intermediate step matters and matters crucially. In fact, if you make a silly slip at some early stage and then complete the solution, you may have virtually wasted your time. As a result, in a competitive examination, you may be worse off than someone who simply did not touch that problem but instead spent his time on doing some other problem (correctly).

It is difficult to give an example of a single problem where all these tips will be applicable. Indeed, as mentioned earlier, art lies in deciding which trick to pull where. Still, as a sample, we do the following problem. (JEE 2001)

Find the number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in

the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$.

Obviously, the problem is a combination of determinants and trigonometric equations. Note that we are not asked to actually identify the solutions but only to find how many solutions there are. Obviously, when $\sin x = \cos x$, the determinant vanishes and this gives $x = \frac{\pi}{4}$ as a root. But there seems no easy way to determine if there are any others. So, without spending much time in this direction, let us get down to working out the solution methodically. First, to evaluate the determinant, let us increase our efficiency and reduce the chances of numerical mistakes by substituting a for $\sin x$ and b for $\cos x$. Then the determinant becomes $\begin{vmatrix} a & b & b \\ b & a & b \\ b & b & a \end{vmatrix}$. If we add the three rows we can take out a factor $a+2b$ and expand the rest of the determinant. But even if we miss this, the determinant is manageable enough for straight expansion and comes out as $a^3 + 2b^3 - 3ab^2$. If we put back $a = \sin x$ and $b = \cos x$ at this stage, and set the expression to 0, the resulting equation will look complicated. So, we apply the divide and rule technique to first factor the expression. Then the roots of the factors put together will give us all the roots. Obviously, it is preferable to first factorise $a^3 + 2b^3 - 3ab^2$ as it is and *then* substitute.

A direct factorisation of $a^3 + 2b^3 - 3ab^2$ is easy enough. But let us make it still easier by observing that right at the determinant stage, if $a = b$ then the determinant would vanish as all its rows would be equal. So $(a - b)$ must be a factor. Similarly adding the three rows of the determinant gives $a + 2b$ as a factor. If we are extra sharp we will notice that $(a - b)$ will be a double factor, because putting $a = b$ makes the first two rows equal and this is enough for the determinant to vanish. But putting $a = b$ also makes the second and the third rows equal, *independently of the equality of the first two rows* and hence $a - b$ is a factor again. (We cannot apply this reasoning to the first and the third row now because their equality follows from that of the first two rows and of the last two rows. So it cannot give us anything more. In particular, we cannot conclude that $(a-b)^3$ is a factor of $a^3+2b^3-3ab^2$, the value of the determinant.) In fact, once we know that $(a - b)^2$ and $a + 2b$ are factors of the determinant, there is no need to expand it. Since it is a polynomial of degree 3 in a and b, the other factor, if any, must be a constant and its presence does not affect the solution.

Whether we are extra sharp or just averagely sharp or not sharp at all, the factorisation of the determinant is the same, viz., $(a - b)^2(a + 2b)$. So, putting back $a = \sin x$ and $b = \cos x$, the roots of the given equation are those of $\sin x = \cos x$ and those of $\sin x = -2\cos x$. These can be rewritten as $\tan x = 1$ and $\tan x = -2$ respectively. In the given interval, viz., $[-\frac{\pi}{4}, \frac{\pi}{4}]$, the former has only one root while the latter has none. Hence the number of solutions of the given equation is 1. (Even though the problem asks only the number of roots, with the work we have put in, finding the roots was as easy as finding how many of them there were. But, as noted at the start, there is no obvious short cut to find only the latter directly.)

Comment No. 7:

So far we have discussed at fair length the three phases of solving a problem, viz., correctly diagnosing it, finding the line of attack that will work, and finally, actually doing the working. In an examination (and more generally, in any situation where you want to prove to someone else that you have solved the problem) there is the fourth phase, that of presenting the solution. This is usually a simple task which requires certain elementary discipline but little artistic or literary abilities. So after some practice, it could indeed become boring unless the problem you have solved is a particularly exciting or challenging one. Still, it is a skill one has to acquire, at least from an examination point of view. It is to be kept in mind that even though the candidate writes only his own answerbook, the examiner who evaluates it usually evaluates hundreds of other answerbooks and often in a severely limited time. This makes it vital that you present the solution so that the examiner can understand it easily. Things that look obvious to you may not be so to everyone. Or you may have given a justification somewhere earlier for a statement you make. But unless you help the examiner to locate it by numbering the earlier statement, he may not be convinced. If you simply say 'clearly' and it is not clear to the examiner, he may arguably think that you are bluffing.

There can never be total unanimity as to what degree of completeness of justification may be expected from a candidate. Surely, in an examination, where the time is severely limited, nobody can defend every step. So it becomes essentially a matter of judgement and can sometimes lead to controversies. Partly to avoid such controversies and partly to increase the speed of evaluation (so as to keep up with the ever growing number of candidates), there is a greater trend nowadays to ask multiple choice or 'fill-in-the-blank' type of questions. The rationale is that anybody who can come up with the correct answer has necessarily done the correct thinking.

While such an assumption is all right for some problems (such as the Main Problem in Chapter 2, see Comment No. 1 there), for some problems it might not be so. For example, as remarked at the end of Comment No. 1 above, the verification that all the four values $\frac{-1\pm\sqrt{5}}{2}$ and $\frac{-3\pm\sqrt{5}}{2}$ in the solution to the Main Problem lie in the interval $(-5,5)$ can be skipped because the problem already gives us that the interval $(-5,5)$ contains four points that satisfy the given condition. Cleverness of this sort borders on a 'killing instinct'. An even more glaring example where it pays is given by the following problem. (JEE 1996)

If $f(x)=\sin^2 x+\sin^2\left(x+\frac{\pi}{3}\right)+\cos x\cos\left(x+\frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right)=1$, then find $(g\circ f)(x)$.

Here nothing is given about the function g other than that its value at $\frac{5}{4}$ is 1. So there is nothing we can say about $g(f(x))$ unless $f(x)=\frac{5}{4}$. So an honest way to do the problem would be to show that $f(x)$ is a constant with value $\frac{5}{4}$ everywhere. This can be done by combining a little calculus with trigonometry. (A purely trigonometric proof is also possible.) Direct differen-

tiation gives $f'(x) = \sin 2x + \sin(2x + \frac{2\pi}{3}) - \sin(2x + \frac{\pi}{3})$ which simplifies to $2\sin(2x + \frac{\pi}{3})\cos\frac{\pi}{3} - \sin(2x + \frac{\pi}{3})$ which is identically 0. So $f(x)$ is a constant. By direct calculation, $f(0) = 0 + \frac{3}{2} + \frac{1}{2} = \frac{5}{4}$. So $f(x) = \frac{5}{4}$ for all x and therefore $(g \circ f)(x) = 1$ for all x.

But, a student with a killing instinct can avoid all this honest work. He can argue as follows. The only point at which the value of g is known is $\frac{5}{4}$. So, if at all the problem has a solution, then $f(x)$ must identically equal $\frac{5}{4}$ and hence $(g \circ f)(x)$ must equal 1 for all x. In a fill-in-the-blank type question, where no justification needs to be given, such a 'legal cheating' pays in terms of the precious time that is saved. While this can hardly be advocated academically, it is difficult to dispute the practical advantage of this approach. The only way to preclude it is to ask the question in the conventional form where the justification has to be given. As we already saw in Chapter 14, the answer to so many problems of triangular optimisation is an equilateral triangle, that in a question where no work is to be shown, it is a safe bet to get the answer by simply assuming the triangle to be equilateral. Posing such problems as multiple choice or fill-in-the-blank type questions fails to distinguish between those who do them honestly and those who cheat legally. Sometimes the saving of time is so substantial as to make one wonder whether the purpose of the problem is really to test mathematical ability or to test the killing instinct.

Comment No. 8:

Killing instinct also pays in multiple choice questions by helping you eliminate some of the possibilities quickly. We saw an instance of this in the first problem done in Comment No. 6 of Chapter 9. Here is an even more glaring example. (JEE 1979)

Let α be a fixed angle and A be the 2×2 matrix $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$. For every positive integer n we have to determine which of the following matrices equals A^n, i.e., the n-th power of A, which is also a 2×2 matrix obtained by multiplying A by itself again and again.

(a) $\begin{bmatrix} \cos^n\alpha & -\sin^n\alpha \\ \sin^n\alpha & \cos^n\alpha \end{bmatrix}$ (b) $\begin{bmatrix} \cos n\alpha & (-1)^n \sin n\alpha \\ (-1)^n \sin n\alpha & \cos n\alpha \end{bmatrix}$

(c) $\begin{bmatrix} \cos n\alpha & -\sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{bmatrix}$ (d) none of these.

In a conventional type of a problem where we are asked to find a succinct expression for A^n (rather than to choose from a given list), an honest attempt would begin by computing A^n for small values of n. For example, for $n = 2$ we have

$$\begin{aligned} A^2 &= \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\alpha - \sin^2\alpha & -2\sin\alpha\cos\alpha \\ 2\sin\alpha\cos\alpha & \cos^2\alpha - \sin^2\alpha \end{bmatrix} \end{aligned} \tag{1}$$

If we multiply this by A we shall get A^3. But the expression will be rather formidable. But if we recast A^2 as $\begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$ and *then* multiply it by A, then A^3 will, after some simplification, come out to be $\begin{bmatrix} \cos 3\alpha & -\sin 3\alpha \\ \sin 3\alpha & \cos 3\alpha \end{bmatrix}$. Thus we see a pattern and guess that A^n equals $\begin{bmatrix} \cos n\alpha & -\sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{bmatrix}$ for every positive integer n. It is not difficult to show by induction on n that this indeed is the case.

There are more efficient ways to arrive at the answer. But not everybody is likely to get them quickly. For example, in Comment No. 20 of Chapter 2, we saw that a matrix of the form $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$ can be identified with the complex number $x+iy$ and further that under this correspondence, matrix multiplication corresponds to the multiplication of complex numbers. As a result, the given matrix A corresponds to the complex number $\cos\alpha + i\sin\alpha$ which equals $e^{i\alpha}$ by the Euler formula. So the n-th power A^n corresponds to the complex number $(e^{i\alpha})^n$ which equals $e^{in\alpha} = \cos n\alpha + i\sin n\alpha$. But this complex number again corresponds to the matrix $\begin{bmatrix} \cos n\alpha & -\sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{bmatrix}$.

But as the problem stands, we merely have to select the correct alternative without showing any reasoning. Many times, the selection of the correct alternative is done most efficiently by eliminating the wrong ones. In the present problem, for example, the answer (b) is ruled out instantaneously as false even for $n = 1$. Answer (a) would be correct (or at least close to it) were matrices multiplied entry-wise. That is, the (i, j)-th entry of the product matrix is the product of the (i, j)-th entries of the two matrices. But this is certainly not the case. So it does not take a good student much time to realise that the choice (a) is cooked up by the paper-setters just to appear plausible. This narrows down the choice to (c) or (d). The examination strategists are generally quick to tip the students that in multiple choice questions, the 'none of these' alternative is very rarely the correct answer and further that it is usually added only as a safety device to save the embarrassment if even the intended answer turns out to be wrong either because of a misprint or because of a slip in calculation. So without doing any honest work, (c) is the correct choice! Even a candidate whose honesty is not marred by considerations of examination strategy will find it easier to *verify* (c) (by induction, for example) once it is suspected as the correct answer, than to *arrive at* it with no clue to its form.

In the example given above the choice was to be made from four answers. In a True/False question, there are only two possibilities and so the killing instinct has a greater chance of success. Suppose, for example, that the question asks you to decide whether it is true that for every integer $n > 1$, the inequality $(n!)^{\frac{1}{n}} < \frac{n+1}{2}$ is true. (See Exercise (6.15)(a), JEE 1981). This inequality can be derived as a consequence of the A.M.−G.M. inequality and a sincere student will certainly try to do so before concluding that it is true for all $n > 1$. But

a candidate who is willing to gamble may simply verify it by direct calculation for $n = 2$ and 3, say, and jump to the conclusion that it is true for all $n > 1$. In a conventional examination he will get no credit. But as the question stands, he gets full credit and some bonus in terms of the time he saves.

Another place where the gamblers have a field day is when the given alternatives are not logically independent of each other. Suppose for example, that whenever the alternative (A) holds, (B) must also hold. Now, if it is given that only one answer is correct, then obviously it cannot be (A), since in that case (B) would also be a correct answer. When all the alternatives are hierarchically ordered, this reasoning shows that the weakest among them is the right choice as we see in the following problem. (JEE 2001)

If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, which of the following numbers is never exceeded by $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$?

(A) 4 (B) 9 (C) 8 (D) 6

Here, without doing any work, the correct answer is (B), because if the given expression never exceeds 6 (say), then it can never exceed 9 either! To discourage such a cheap short cut, the question ought to have asked to select the *smallest* among the given choices that is never exceeded by $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$. In that case, some justification would have been needed before selecting 9 as the correct choice. One such justification is by actually giving three unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ for which the expression $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ equals 9. If we take three vectors that are inclined to each other at an angle of 120° we get such a triad.

(In a conventional form the problem would ask to maximise the value of the expression $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ as $\vec{a}, \vec{b}$ and $\vec{c}$ vary over the set of all unit vectors. But this is a rather tricky problem. (See Exercise (24.50) for a solution.) So in a problem like this, a sincere student has no chance. It can perhaps be argued that this is intentional, because after all, there is such a thing as 'intelligent gambling' and sometimes it is a desirable quality. Still it is hard to defend the question in the form it is posed. Perhaps the modification above, along with some figure higher than 9 thrown in as a possible choice would have punished the pure gamblers and rewarded the intelligent ones!)

In some multiple choice questions, even though the options are mutually exclusive, the fact that only one of them is correct provides a clue which unduly simplifies the work. Suppose for example, that we have to choose the acute angle of intersection of the curve C_1, $y = [|\sin x| + |\cos x|]$ and the curve C_2, $x^2 + y^2 = 5$ from the four alternatives : $\tan^{-1}(1/2), 0, \pi/2$ and $\tan^{-1}(2)$. Here $[\ ]$ is the greatest integer function and so the first curve consists of a number of horizontal line segments on or above the x-axis. Each such segment cuts the circle C_2 in two points symmetrically located about the y-axis. The absolute value of the slope of the tangent to the circle at either of these two points will give the desired angle since the tangent to C_1 at any point is horizontal. But, for two different lines above the x-axis, these tangents will have different slopes. As the problem has only one answer, C_1 ought to consist of just one horizontal line, say $y = c$. The value of c can be found by putting $x = 0$ (say)

which gives $c = 1$. The line $y = 1$ cuts C_2 at $(\pm 2, 1)$ and the desired angle is $\tan^{-1}(2)$. To do the problem honestly, one would have to prove that for every x, $|\sin x| + |\cos x| \in [1, 2)$. This is a consequence of the inequalities $1 \leq |\sin x| + |\cos x| \leq \sqrt{2}$ the first of which follows from the geometric interpretation of the expression as the sum of the two sides of a rightangled triangle with hypotenuse 1, while the other can be proved by maximising the function by any method, e.g. using calculus or by writing $|\sin x| + |\cos x|$ as $1.|\sin x| + 1.|\cos x|$ and applying the Cauchy Schwarz inequality or simply by observing that for every x, $|\sin x| + |\cos x| = |\sin x \pm \cos x| = \sqrt{2}\,|\sin(x \pm \frac{\pi}{4})|$.

In some multiple choice questions, the correct choice can also be determined simply by trying each possible choice one by one. Sometimes, this is, in fact, intended. For example, Exercise (19.2) asks which of the four given functions is a solution of differential equation $(\frac{dy}{dx})^2 - x\frac{dy}{dx} + y = 0$. Here it is not easy to *find* the solution and hence all that can be asked reasonably is to *verify* if a particular function is a solution. But sometimes a case-by-case verification provides an unintended sneak path. Suppose for example, that α, β are the roots of $x^2 - x + p = 0$ and γ, δ are the roots of $x^2 - 4x + q = 0$, where p and q are integers. We are further given that $\alpha, \beta, \gamma, \delta$ are in G.P. and are asked to determine (JEE 2001) which of the following pairs are the values of p, q respectively.

$$\text{(A)}\ \ -2, -32 \quad \text{(B)}\ \ -2, 3 \quad \text{(C)}\ \ -6, 3 \quad \text{(D)}\ \ -6, -32.$$

The honest approach here is to determine p and q from the data. Taking β, γ, δ as $\alpha r, \alpha r^2$ and αr^3 respectively, we get

$$\alpha(1 + r) = 1 \quad \text{and} \quad \alpha r^2(1 + r) = 4 \tag{2}$$

whence $r^2 = 4$ giving $r = \pm 2$. Similarly, taking products of the roots of each quadratic, we get

$$p = \alpha^2 r \quad \text{and} \quad q = \alpha^2 r^5 \tag{3}$$

After this we have a choice. The clean option is to put the values $r = \pm 2$ into (2) to get $\alpha = \frac{1}{3}$ or $\alpha = -1$. (3) then gives $p = \frac{2}{9}$ or -2. As it is given that p is an integer, we discard the first possibility, leaving us with $\alpha = -1$ and $r = -2$. Then from (3) again, q comes out as -32. So (A) is the correct choice. Here we found p and q entirely on our own. We made no use of the given options. We looked at them only at the end because the question asked which one of them was correct. A less puritanical approach is to conclude from (3) and $r^2 = 4$ that $q = 16p$. Now, without finding p and q, we look at the given options and see that this relation holds only in (A). So, if at all one of the choices is correct, it has to be (A). Even this approach is not all that bad as it does involve some thinking (for example, denoting the roots in terms of just two symbols, viz., α and r, making use of the fact that they are in G.P.).

The irony of the problem is that even this much thinking can be bypassed. We simply go on scanning the given options one by one. Naturally, we begin

with (A) (unless there is something to indicate that one of the remaining three options has a greater chance to be correct). As luck has it, the roots of the quadratics $x^2 - x - 2 = 0$ and $x^2 - 4x - 32 = 0$ can be found by inspection and they come out to be in G.P. So (A) is the correct option! It is doubtful if the paper-setters really intended to permit such a back door entry. But who are we to argue with success?

Comment No. 9:

Thus we see that even though the purpose of the so-called objective type questions is to relieve the candidates of the need to *write* the justification, sometimes it also relieves them of the need to *find* a justification, or the need to approach the problem in an honest manner. Fortunately (or unfortunately, depending upon your point of view) killing instinct does not work in all problems. And, academically, even when it works in a given problem, you should always be in a position to back it up with correct reasoning whether such a reasoning is asked in an examination or not. (That is why, in this book, most of the multiple choice questions taken from the JEE papers have been converted into a form where you have to find the answer. Advocating the use of cheap short cuts in a book like this will be the equivalent of a physician prescribing performance enhancing drugs to competitive sportsmen. This book is not meant for that sort of a thing.)

After you have solved the problem honestly, the kind and the extent of justification that is expected of you varies depending upon various factors such as the maturity of the intended reader, the time available and also the type of problem. In a research paper the justifications given are very terse as the readers are expected to be mature enough to fill the details. On the other hand, in a book such as this one, written with a view to teach new ideas to young readers, the justification includes rather too many details and also lots of comments meant to sharpen the reader. When you write in an examination, the person who examines your answerbook will, of course, be an experienced person and it is hardly expected that you educate him!. So the way solutions are written in this book tells you what is the right way to think and approach a problem but it does not tell you the right way to write in an examination! Also to be kept in mind is the fact that a research paper or a book are written patiently and are meant to be read again and again by various readers. In an examination, the writing is often done in a hurry. And once the examiner (and possibly some such authority as the Head Examiner, or the moderator or whatever) is through with it, it is virtually impossible that it will be read by anybody ever again. As a result and especially in a subject like mathematics, the requirements of grammar and diction that apply for a book are relaxed for writing in an examination. As long as the examiner can effortlessly understand what you are trying to say, he does not bother about other things. So it is perfectly all right if you write the mathematical contents of your steps and not much verbiage that you see in a book (and, in particular, in the present book). Certain abbreviations such as $\Longrightarrow$ (for 'implies'), iff (for 'if and only if'), WLOG (for 'without loss of generality'), $\therefore$ (for 'hence' or 'therefore') and $\because$ (for 'since')

which are rarely used in books can be used freely in an examination.

This book is not meant to coach you for JEE (or for any other particular examination for that matter). Still a few guidelines of a general type can be given. (We skip guidelines that advise you to write legibly, leaving wide margins and avoiding overwriting and to begin each new question on a new page! Not that these things don't matter. But we hope you have already learnt these elementary things from other sources. Here we concentrate more on the tips of an academic type, applicable especially to mathematics.)

(i) When there is a numerical answer to a problem (such as the probability of some event, the value of some integral, the equation of some locus), display it prominently by encircling or drawing a box around it. Sometimes the very nature of the problem is such that it is simply impossible to get the answer by bluffing. (For example, the Main Problem of Chapter 2, or the first problem illustrating the use of tree diagrams in Comment No. 11 of Chapter 22 or most of the definite integrals in Exercise (18.15).) In such cases, if your answer is correct, the examiner can presume that you have done the problem correctly. He is still duty bound to read the various steps of the solution. But because of the presumption he has made in your favour, he will look at your work only to ensure that you are not copying the answer from somebody else.

(ii) If the problem asks you to prove or disprove something (with justification) (or, in other words to decide if the given statement is true or false), for god's sake first make it abundantly clear which side you are on by writing words like 'True' (or 'Yes') or 'False' (or 'No'). This declaration must come right at the start. If it does not, it can become irritating to the examiner. After you pronounce your judgement, you can append it with a justification. The justification must consist of a short proof if you are claiming that the statement is true. Otherwise, it *must* consist of what is called a **counter-example**, i.e., an example where the statement fails to hold. Take for example, Exercise (9.1)(a) (JEE 1985), where you are asked to prove or disprove that if $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{vmatrix}$, then the two triangles with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and $(a_1, b_1), (a_2, b_2), (a_3, b_3)$ must be congruent. Here the answer is 'No' and the reason is that the equality of the determinants only tells you that the two triangles have equal areas and that does not necessarily mean that they are congruent. A student who writes this has undoubtedly solved the problem correctly. Still, his answer is technically incomplete until he actually gives a pair of non-congruent triangles where the two determinants are equals. One such pair is obtained, for example, by taking one triangle with vertices at $(0, 0), (1, 0)$ and $(0, 1)$ and the other with vertices at $(0, 0), (1, 0)$ and $(3, 1)$. Nor is it enough (or necessary) to tell what the correct statement would be. Suppose for example, that you are asked (JEE 1985) to prove

or disprove that if three complex numbers are in A.P. then they lie on a circle. Here the statement is 'False' and the correct reasoning is that the condition implies that the three complex numbers would be collinear and so in general they will not lie on a circle. But if you write 'No, the three points will lie on a straight line' you are really not answering the question as it is asked. Many examiners would probably give you full credit. But technically, your answer is incomplete without a counter-example, such as the complex numbers $0, 1, 2$. From an educative point of view it is certainly a good idea to know why the statement is false and how it can be rectified. But as far as the given question is concerned, a valid counter-example is the only thing that counts.

(iii) When you draw a diagram, whether because the question asks for it or on your own, be sure to label the relevant points. In your rough work this is not very vital because you are the only one to read it and you are also the one to have drawn it. But if you want the examiner to look at and appreciate it, do not *assume* that he knows what you have in mind. So no matter how 'obvious' you think it is, make it explicit what stands for what.

(iv) A similar remark applies for the notations you use. The given problem may itself involve some notations. Do not change them even if they be different from the ones you are more used to. That would cause tremendous confusion. If you introduce any new notations, be sure they do not clash with the notations given in the problem and among themselves. (For example, if a triangle is denoted by PQR, do not denote its circumradius by R, even though this is a very standard notation.) Most importantly, properly introduce all the symbols which you have added on your own unless they are very standard, such as a, b, c for the lengths of the sides opposite to the vertices A, B, C of a triangle. Moreover, when you define these variables, use a separate line for each new type of a variable. For example, you may say

$$\text{Let} \quad D, E, F \quad \text{be the mid-points of the sides } BC, CA \text{ and } AB \text{ respectively.}$$

and then, on a separate line,

$$\text{Let} \quad \Delta \;=\; \text{the area of } \Delta ABC.$$

Note also the gaps on either side of the symbols introduced. That makes them more vivid and easier to spot. In a textbook, this is often done in a running manner. We have also followed this practice. But the reader of a book can afford to spend more time with the book and read the same portion again and again if need arises. The examiner is in a hurry. It is very irritating for him to have to search for the notations. And if he does not understand your notations, he is unlikely to ever read your

paper again. Your only hope then is that your paper will be picked up by a Head Examiner who may have the patience to try to see what you really mean and do you justice. So why unnecessarily take chances?

(v) Show a similar consideration for the examiner when he reads the solution. Unlike in a book like this, use a separate paragraph for each piece of reasoning. Be explicit in your references. For example, instead of saying

Using sine rule for the triangle above,

it is far better to say

Using sine rule for the triangle DEF,

If you are using earlier parts of the solution as a justification for the subsequent ones, be sure that you number the earlier parts and refer to them by numbers, such as

Substituting (5) into (3) we get,

instead of forcing the examiner to figure out the justifications even though you may think that they are obvious.

(vi) What about justifications which are obvious to everyone? These arise when you use some very standard result such as the Pythagoras theorem or the cosine formula for a triangle. It would be too much if you have to mention the Pythagoras theorem every time you use it. And nobody expects you to do that. But the thing to remember is that you must justify its applicability. That is, suppose ABC is a triangle which is known to be (either from the hypothesis or from an earlier part of the proof) right-angled at B. Then while concluding that $AB^2 + BC^2 = AC^2$, it is always a good idea to cite this as a justification. The justification can be short and parenthetical, such as

$$AB^2 + BC^2 = AC^2 \quad (\text{since } \angle ABC = 90^\circ)$$

or

$$AB^2 + BC^2 = AC^2 \quad (\text{from (2)})$$

if in the statement numbered (2) you have shown that $\angle ABC$ is a right angle. When you write either of these two lines, the examiner will know that you are using Pythagoras theorem without your saying so. When you apply a result such as the sine rule which is valid for any triangle, the question of its applicability does not arise. In that case, no justification is needed.

(vii) The trouble arises when you use a result which you have not proved earlier in the solution but want to use as a 'well-known' result. Trigonometry

especially abounds in such results. There are numerous formulas and identities about triangles. It is very difficult to decide unanimously which of these can be assumed without proof. Everybody would agree that the sine rule is one such result. But can the identity

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C \tag{4}$$

for a triangle ABC be taken as standard? Some people would answer in the affirmative because this identity is very easy to prove (by writing $\tan C$ as $-\tan(A+B)$). But then can the same thing be said about

$$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1 \tag{5}$$

which follows from (4) immediately if we multiply both the sides by $\cot A \cot B \cot C$? If we assume this one too as standard, what about some consequences of it? Just exactly where do we stop? Similarly, the formulas for the cosines of the angles of a triangle in terms of its sides are certainly standard. But can the same thing be said about the cosines of half these angles?

Such questions become especially relevant when a particular problem can be done very effortlessly by pulling some formula or identity. Suppose, for example, that the problem asks the area of the pedal triangle of a triangle ABC. Now, formulas are available (see, for example, p. 239 of *Plane Trigonometry - Part I* by S. L. Loney) for expressing the sides and the angles of the pedal triangle in terms of A, B, C and a, b, c. If we use them (along with any of the formulas for the area of a triangle), the answer comes instantaneously. Whether such a solution qualifies for full credit becomes a matter of debate. The fact of the matter is that many candidates have spent several months of intense preparation, in which they may have solved dozens of problems using these formulas and identities. So in their limited world, such formulas may be very familiar and hence standard. But the examiner may be coming from a walk of life where most of these formulas are rarely used. So, for him they may not appear so 'well-known'.

The true solution to the problems that arise because of disparities like this would come only if the organisers of a particular examination declare either by prescribing some book or by explicit listing which results may be used without proof. (In pure geometry, thanks to the domination of Euclid for centuries, there is a general unanimity as to which results are standard.) Till then the only advice that can be given is that make a judgement depending upon the credit allotted to the question and the proportional time meant to be spent on the question. For example, if you are asked to prove (5) above and the proportionate time is less than a minute, you can use (4) as a standard result. But if you are allowed, say, 3 minutes, then it is excepted that you give a proof of (4) as well. And, finally, when in doubt, do add a justification, or, if you are hard pressed for time, at least say that the result you are quoting is a well-known one.

(viii) After you have presented your solution following the general guidelines above (including a prominent display of the answer in the case of a numerical problem) it is best to leave the problem right there. Again, this is where a book such as the present one differs from an examination. As has been emphasized again and again in this book, when you are learning from a problem, it is highly rewarding to go over the solution and see if you can give a better solution or if you can do a more general problem with the same method. Comments on the relative merits and demerits of various solutions have also been inserted liberally in this book so as to help you gain certain insights and increase your mathematical maturity, an invaluable quality.

In an examination, however, you are not teaching the examiner. In fact, he is an overworked human being who may get irritated if you try to do so. Nor is it going to help you, as in most examinations there is no provision for any bonus points for such editorial comments. The exceptions are those examinations where a very small number is to be selected from highly gifted contestants. In such examinations, remarks made by the candidates showing their quality of thinking can be used as tie-breakers. For examinations where a huge number of candidates appear, an occasional indication of an alternate solution or a remark to the effect that some part of the hypothesis was not needed in its full strength may be appreciated, because this does not happen frequently. But beyond this, avoid the temptation to show off. Your maturity is best revealed through the way you set up the problem and through a concise and yet precise solution. Using fancy expressions does not help. In fact, if such expressions are not matched with mathematical acumen, they appear quite ridiculous and sometimes an attempt to bluff. For example, if you want to show that an angle, say $\angle BIO$ is a right angle, give a proof. It is no use saying, 'From the geometry of the figure, it is manifest *a priori* that $\angle BIO$ is a right angle.'! (See, however, the remarks at the end of the next subsection.)

(ix) Some restraint is also called upon in explaining how you arrived at the answer. Academically, in the case of a thought-oriented problem, how you arrived at the answer is often far more important than the solution itself just as in a mystery novel, what led the detective to suspect the murderer is far more thrilling than who the murderer is. Socially, too, when you solve a challenging problem after a lot of thinking (often with many abortive attempts), it is a very exciting experience to tell a like-minded friend how you approached it and ultimately how the breakthrough came. But, once again, remember that in a conventional type examination, the examiner is interested only in the solution and not in your thought process. He wants to taste the pudding and not to know your recipe!

This point deserves some elaboration because the solutions given in this book are generally aimed at emphasizing the analytical thinking needed in arriving at the solution. In an examination, this thinking should

stay in the background and only the 'polished' solution need be given. In a book, occasionally some other exercises are referred to. The solution in an examination, on the other hand, has to be self-contained, using only very standard results. If you use a result which is not so standard, a brief justification of it is in order. This necessitates some changes of presentation. As a good illustration (JEE 1984), consider Exercise (16.21), where you have to prove that a certain polynomial given in the form of a determinant is divisible by a certain quadratic $f(x)$ which is given to have α as a repeated root. The solution given in the 'Answers to the Exercises' at the back tells you how the problem can be approached by paraphrasing the assertion to be proved. In an examination it is better to keep such a paraphrase in your mind and, instead, give the solution in the following polished form.

$$\text{Let } g(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}.$$

Setting $x = \alpha$ makes the first two rows equal. So $g(\alpha) = 0$. (1)

Since the second and the third rows of $g(x)$ consist only of constants,

$$g'(x) = \begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix} \qquad (2)$$

Putting $x = \alpha$ in this determinant, $g'(\alpha) = 0$. (3)

From (1) and (3), α is a root of $g(x)$ and also of $g'(x)$.

So α is a multiple root of $g(x)$.

Hence $g(x)$ is divisible by $(x - \alpha)^2$. (4)

It is given that $f(x)$ is a polynomial of degree 2 having α as a repeated root.

So, $f(x) = k(x - \alpha)^2$ for some (non-zero) constant k. (5)

From (4) and (5) it follows that $g(x)$ is divisible by $f(x)$.

Q. E. D.

This solution avoids an ostentatious display of the thought process. Still it is inherent in a subtle manner in the solution and a discerning

examiner can hardly miss it. For example, in step (1) a brief justification is given as to why the determinant vanishes on putting $x = \alpha$. But a similar justification for (3) is omitted. Note also the one line justification given for step (2) instead of appealing to Exercise (15.10). However, in step (4), once it is noted that α is a root of both $g(x)$ and $g'(x)$, no further justification is given as to how it follows that α is a multiple root of $g(x)$. This is a relatively standard fact and can be used without justification. Similarly, in the very last line, it is not shown how (4) and (5) imply that $f(x)$ divides $g(x)$. Technically, a justification is needed here because $f(x)$ is not quite the same as $(x - \alpha)^2$. But the justification is of a pedantic type. (That is, write $g(x) = h(x)(x - \alpha)^2$ for some polynomial $h(x)$. Let $q(x) = \frac{1}{k}h(x)$. Then $g(x) = q(x)f(x)$.) It is best to refrain from such trivial justifications. They cost you in terms of time and get you nothing in return. In fact, they can irritate the examiner. This may seem to contradict the remarks at the end of the last subsection. The point to keep in mind is that one should not attempt to fulfill the need of a justification by lofty language. But exactly which steps require further justifications is a matter of judgement. Admittedly, there is no unanimity on this. Considerations such as the level of the examination and the time allotted play a role here. In many cases, once you have indicated the quality of your thinking (for example, by giving the justification for (1) and (2) above), the examiner knows that you are capable of giving justifications when needed. He may also be impressed by the fact that you are careful enough to say that the constant k in Step (5) is non-zero. In that case, he will probably not hold it against you if you omit subsequent trivial justifications.

Comment No. 10:

The guidelines given above so far are applicable under normal circumstances when things are smooth, i.e., when the questions are clear (not necessarily easy) and convey exactly what the paper-setters have in mind. But occasionally certain pathological situations arise for various reasons. For example, there may be some ambiguity about the problem, or it may be outrightly wrong. Worse still, it may be technically correct but may not be solvable with the methods which the candidates are expected to know. Or it may be so solvable but the computational work it demands is unreasonably long for the credit it carries.

In a reputed examination like the JEE, such pathologies are extremely unlikely to occur as the question papers are prepared and printed with great care. But then that's precisely why, a mistake, when it does occur, is most devastating because the presumption is that there simply cannot be any mistake in the question and hence the mistake must lie in the candidate's work. A sincere candidate then often spends an inordinate time in checking and rechecking his calculations to see if he has committed any mistake. When he finds none, he is still hesitant to say that there is something wrong with the question. As a result, he is baffled and often demoralised that he cannot get the answer. It is

one thing not to be able to answer a question which is simply too difficult for a student. Even a good student is prepared for it. It is quite another to get stuck on a question which is well within one's reach. The time wasted and the consequent demoralisation adversely affect the candidate's performance in the other questions as well. So a word of advice is in order regarding what to do in such cases, however atypical they are, just as a driver's training is not complete without at least some rudimentary knowledge of what to do in the case of an accident.

(i) **Real-life Problems:** As remarked in Item (vii) of Comment No. 3 above, certain common sense assumptions are implicit in such problems. Sometimes this demands that you know the basic rules of some games such as chess. But in the well-designed examinations, care is taken to explain what is meant. (For example, what is meant by 'non-taking queens' as in Exercise (24.56).) So ambiguities are not likely to arise for this reason.

(ii) **Ambiguous Problems:** There can be ambiguities arising for other reasons such as those mentioned in Items (ix) or (x) of Comment No. 3 above. We have already advised what to do in such cases.

Yet another source of ambiguities is the use of a notation other than the one you are used to. Sometimes the paper-setters explain the notation used either in the question itself or in the general instructions given at the beginning of the question paper. The latter are prone to be skipped by students who believe that such instructions are of a generalistic type (such as 'use of unfair means is strictly prohibited') and want to rush to the 'relevant' part of the question paper.

Although most of the notations used in this book are, by now, quite standard, occasionally the given problem may involve a familiar concept but in an unfamiliar notation. For example, as remarked in a footnote in Comment No. 1 of Chapter 1, various notations are common to denote the complement of a subset. Chances are that you have been consistently exposed to one of them. In that case a problem stated with a different notation may be discomforting. But you can usually make out the intended meaning from the context by applying a little common sense. Suppose, for example (JEE 2001), that the problem asks you to identify which of the following subsets is the domain of definition of the function $f(x) = \dfrac{\log_2(x+3)}{x^2+3x+2}$.

(A) $\mathbb{R}\backslash\{-1,-2\}$ (B) $(-2,\infty)$

(C) $\mathbb{R}\backslash\{-1,-2,-3\}$ (D) $(-3,\infty)\backslash\{-1,-2\}$.

Here all the notations are standard except possibly the \ which we have mentioned but rarely used. In case you cannot remember what it means, look at the question. As the denominator of $f(x)$ vanishes at -1 and -2, these points are to be excluded from the domain. Since three of the

choices refer to subsets containing these 'bad' points, the \ must evidently mean their exclusion. So it must have been used to denote complements. (Incidentally, in order for the logarithm in the numerator to be defined we must also have $x > -3$. Hence (D) is the correct answer.)

In short, just because you see something unfamiliar, don't panic. It is no use asking the invigilator. Even if he knows, he is generally not authorised to tell you. But, with a little presence of mind, you can generally make out what is intended.

(iii) **Incorrect Problems:** What if a question is not ambiguous but outrightly wrong? For example, in the formulation of Exercise (9.37) (JEE 1993) as it originally appeared in the question paper of the 1993 JEE, because of a misprint of a sign, the equations of the circles were such that one of them was completely inside the other!. Similarly, in a trigonometry problem (JEE 1994) of heights and distances in the 1994 JEE, a point was given to lie due West of a tower, which ought to have been due East. (Interestingly, in the Hindi version there was no mistake!)

Sometimes there is an inherent inconsistency in the problem. We encountered (JEE 1981) an example of this situation in Comment No. 5 of Chapter 20, where the functional equation $f(x + y) = f(x)f(y)$ was given subject to the conditions $f(5) = 2$ and $f'(0) = 3$. Here the two conditions are mutually inconsistent. If we ignore this inconsistency then the problem can be solved as shown there. But since the data is vacuous, any answer is true, albeit vacuously, much the same way that you are free to say anything you please about a six-legged person. Similarly, Exercise (17.22)(a) can be 'solved' by differentiating both the sides of the given equation using the fundamental theorem of calculus to get that $f(x) = \frac{1}{x+1}$ and hence $f(1) = \frac{1}{2}$ as the answer. But the function $f(x) = \frac{1}{1+x}$ does not satisfy the given equation.

Sometimes the inconsistency is not with mathematics but with common sense. For example, at the end of Comment No. 4 of Chapter 12 we encountered a problem where a man walking towards a tower finally comes to a point, where the angle of elevation of the tower would be above 120°, forcing the man to bend his neck backwards to see the tower!

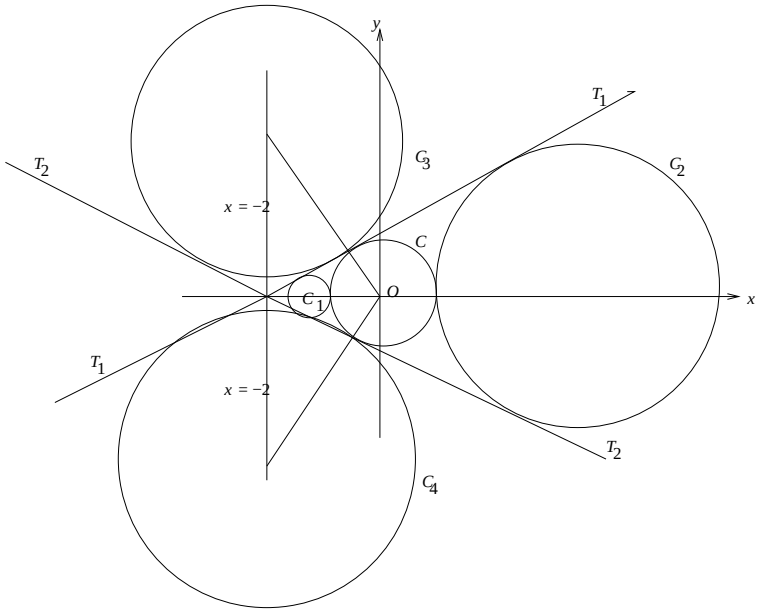
The only advice that can be given to a candidate in such cases is that he should go through his calculations to ensure that they are correct. Then he should read the question again carefully to see if he missed something vital. If possible, he should solve the problem by some other means. If he is convinced that he is right and the question is wrong, he should make a remark to that effect and then proceed with other questions without letting this question affect his mind. He should bother about it only if some time is left at the end after solving all problems within his reach.

(iv) **Unsolvable Problems:** Sometimes a problem is not incorrect mathematically. But the work needed for its solution is well beyond the level

of JEE. We already gave two examples of such problems in Comment No. 14 of Chapter 10 and remarked that the dilemma they pose to a sincere student is worse than that in the case of a problem which is outrightly wrong. All we can do is to repeat the advice given there, viz., not to spend too much time on such problems.

(v) **Laborious Problems:** Just as there is no commonly accepted norm of the difficulty of a problem, there is no unanimity as to what amount of computational work can be 'reasonably' expected from a candidate. But once in a while, probably because of a lapse in paper-setting, a problem may turn out to demand work which is qualitatively within the scope of the JEE syllabus but whose quantity is unreasonable by any standards as compared to the credit allotted for it. Here is an example of a problem posed for 10 points in a three hour examination with 200 points (so that, proportionately, a candidate gets nine minutes to solve the problem).

Let T_1, T_2 be the two tangents drawn from $(-2, 0)$ to the circle C : $x^2 + y^2 = 1$. Determine the circles touching C and having T_1, T_2 as their pairs of tangents. Further, find the equations of all possible common tangents to these circles, when taken two at a time. (JEE 1999)



As shown in the figure above, here there are two obvious circles, say, C_1 and C_2 with their centres lying on the x-axis which answer the description in the problem. If these were the only ones, then, along with C, in all there are three circles. So there are three pairs of circles out of these. For the pair C and C_1, finding the common tangents is very easy

and so is the case for the pair C and C_2. Comparatively the pair C_1 and C_2 takes more work because even though their direct common tangents (viz., T_1 and T_2) are already with us, we have to find the inverse common tangents too. Still, the total work involved is not unreasonable for a time limit of eight to nine minutes. And this is probably what the paper-setters had in mind.

The trouble, however, is that if one looks at the figure of the problem keenly, one can see that there are two more circles, say, C_3 and C_4 with their centres lying on the line $x = -2$ which also satisfy the description given in the problem, namely, that they touch the circle C and have T_1, T_2 as tangents. (If by a tangent from a point we mean only a ray originating at that point and not the entire line, then C_3 and C_4 are eliminated. But this is not a standard convention. Nor does the question make it clear.) So now we have *five* circles, viz., C, C_1, C_2, C_3 and C_4. When taken two at a time they give *ten* pairs (as opposed to three). Of these, at least six pairs (namely, those where C is not in the pair) are troublesome as opposed to only one with the earlier interpretation. So effectively, the degree of computational work needed has become six times as before. Now, by *any* standards nine minutes is far too inadequate. Thus we have a case of a problem which is technically not incorrect and not inherently difficult either, but whose complete solution would require unusually long and highly repetitious work.

The trouble once again, is that there is nothing to tell the good student exactly what he is expected to do to get the full credit for this question. Should he simply ignore the circles C_3 and C_4? Or should he consider them but find the common tangents only in the cases of one or two of the possible ten pairs of circles and leave the remaining cases with a comment that they can be handled similarly? Or is he expected to work out all the ten cases completely? In a competitive examination, awarding full credit to a candidate who does not even think of C_3 and C_4 would be a gross injustice to the perceptivity of a candidate who notices them.

Once again, all that can be advised is to make a judgement depending on the time allotted. When the work is repetitious, it is better to say so, handle a few cases and quit with a comment that the remaining cases can be handled similarly.

There is really no way to do justice to the bright and the perceptive students in such pathological cases except by making a provision for bonus points. At present, there are hardly any examinations with such a provision. So the only advice that can be given is of a general type, viz., "Do not spend too much time on any particular problem. Look at the credit allotted to the question and budget your time accordingly. And most importantly, let such anomalies not affect your performance in other problems." All one can hope is that such things do not happen. While setting a paper, the paper-setters' minds often get channelised and are occasionally prone to miss something that

does not come to surface immediately. This is quite human and also the reason that the author is usually a poor proof-reader for his own work. The situation could improve somewhat if after the paper is set, it is moderated by someone who tries to solve the problems with a fresh mind. Even as the things stand, such instances are extremely rare in the JEE as compared with several other examinations at comparable levels. (Because here we have collected together such instances spread over more than two decades, they may appear to be numerous. But the average comes out to be less than once in two years, which is quite commendable).

EXERCISES

24.1 Prove or disprove that if $f_1(x)$ and $f_2(x)$ are defined on domains D_1 and D_2 respectively, then $f_1(x) + f_2(x)$ is defined on $D_1 \cup D_2$. (1988)

24.2 Prove or disprove that if $x - r$ is a factor of a polynomial $f(x) = a_n x^n + \ldots + a_0$ repeated m times $(1 < m \leq n)$, then r is a root of $f'(x)$ repeated m times. (1983)

24.3 If $f : \mathbb{R} \longrightarrow \mathbb{R}$ is a function defined by $f(x) = x^2$, determine which, if any, of the following statements is true. (1979)

(a) f is injective but not surjective

(b) f is surjective but not injective

(c) f is bijective

24.4 Suppose $a_0, a_1, \ldots, a_n$ and α are real numbers. Prove that the function $y = e^{\alpha x}$ is a solution of the differential equation

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \ldots a_1 \frac{dy}{dx} + a_0 y = 0$$

if and only if α is a root of the polynomial $a_n \lambda^n + a_{n-1} \lambda^{n-1} + \ldots + a_1 \lambda + a_0 = 0$. (A differential equation of this type in which $a_n \neq 0$ is called a **linear differential equation** of order n. This polynomial is called the **characteristic polynomial** and its roots the **characteristic roots** of the linear differential equation. In Exercise (2.37), the same terms were used for square matrices. The two are actually related. But the relationship is far beyond our scope.)

24.5 If the bisector of the angle P of a triangle PQR meets QR in S, determine which, if any, of the following statements is true. (1979)

(A) $QS = SR$ (B) $QS : SR = PR : PQ$ (C) $QS : SR = PQ : PR$.

24.6 From a point O inside a triangle ABC, perpendiculars OD, OE, OF are drawn to the sides BC, CA, AB respectively. Prove that the perpendiculars from A, B, C to the sides EF, FD, DE of the triangle DEF are concurrent. (1978)

24.7 A quadrilateral $ABCD$ is inscribed in a circle S and A, B, C, D are the points of contact with S of another quadrilateral which is circumscribed about S. If this quadrilateral is also cyclic, prove that $AB^2 + CD^2 = BC^2 + AD^2$. (1978)

24.8 Suppose at a dance party that every boy dances with at least one girl but that no girl dances with every boy. Prove that there exist boys b, b' and girls g, g' at the party such that b dances with g and b' dances with g' but neither does b dance with g' nor b' with g. Paraphrase the problem in terms of subsets of the set of all girls at the party. (Putnam competition.)

24.9 Suppose we have a collection of 11 stones with the property that whenever we remove any one from it, the remaining ten stones can be grouped into two piles of equal total masses. Prove that all the stones are of equal mass. Paraphrase the problem in terms of an 11×11 matrix (or equivalently, in terms of a system of 11 equations in 11 unknowns). (Do not solve it. The solution depends on the concept of the rank of a matrix.)

24.10 Let A be the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. For every positive integer n, prove that $A^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, where F_n is the n-th Fibonacci number (starting from $F_0 = 0, F_1 = 1, F_2 = 1, F_3 = 2, \ldots$).

24.11 Is the binary relation $\cap$ defined on the set of complex numbers in Exercise (6.2) a partial order? a total order? (The relation was defined by saying $z_1 \cap z_2$ if $x_1 \leq x_2$ and $y_1 \leq y_2$, where $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.)

24.12 Given complex numbers a and b and a real number r ($> |a| + |b|$) identify the set of all complex numbers z which satisfy $|z - a| + |z - b| \leq r$ and show that this set is convex.

24.13 Suppose α and β are real numbers such that $\alpha^3 - 3\alpha^2 + 5\alpha = 17$ and $\beta^3 - 3\beta^2 + 5\beta = -11$. Determine $\alpha + \beta$. (Regional Mathematics Olympiad)

24.14 If $\alpha, \beta, \gamma, \delta \in (0, 2\pi)$ are four distinct solutions of the equation $\cos 2\theta + a\cos\theta + b\sin\theta + c = 0$, where a, b, c are some real constants, then prove that $\alpha + \beta + \gamma + \delta$ is a multiple of 2π.

24.15 For every ordered pair of positive integers, (m, n), define

$$a_{m,n} = \begin{cases} 0 & \text{if } n > m \\ -1 & \text{if } n = m \\ \frac{1}{2^{m-n}} & \text{if } m > n \end{cases}$$

Verify that for every fixed n, $\sum_{m=1}^{\infty} a_{m,n}$ is 0 and hence that the double sum $\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_{m,n}$ is also 0. However, for a fixed m show that the sum $\sum_{n=1}^{\infty} a_{m,n}$

equals $-\dfrac{1}{2^{m-1}}$ and hence that the double sum $\sum\limits_{m=1}^{\infty}\sum\limits_{n=1}^{\infty} a_{m,n}$ equals -2. In other words, the order of summation cannot be changed.

24.16 Suppose that the Sun rises and sets at 6:00 a.m. and 6:00 p.m. respectively at some place. A man starts working in the sun at 8 : 00 a.m. and at any time works at a rate proportional to the length of his shadow. He finishes half the work by noon, rests for two hours and resumes work. Find when the work will be complete.

24.17 Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. Find the range of $m(b)$ as b varies. (2001)

24.18 Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending a right angle at the centre. Then which of the following types of curves is the locus of the centroid of the triangle PAB as P moves on the circle? (2001)

(A) a parabola (B) a circle

(C) an ellipse (D) a pair of straight lines.

24.19 If in the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of the fifth and the sixth terms is zero, find the value of $\dfrac{a}{b}$ in terms of n. (2001)

24.20 Let T_n denote the number of triangles which can be formed using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, find n. (2001)

24.21 Find the number of **onto** functions from $\{1, 2, 3, 4\}$ to $\{1, 2\}$. (2001)

24.22 Let the positive numbers a, b, c, d be in A.P. Then determine which of the following statements is true about abc, abd, acd, bcd. (2001)

(A) They are in A.P. (B) They are in G.P.

(C) They are in H.P. (C) They are NOT in A.P./G.P./H.P.

24.23 If the sum of the first $2n$ terms of the A.P. 2, 5, 8, ..., is equal to the sum of the first n terms of the A.P. 57, 59, 61, ... , find n. (2001)

24.24 Let z_1 and z_2 be n-th roots of unity which subtend a right angle at the origin. Find which of the following forms n must have. (2001)

(A) $4k + 1$ (B) $4k + 2$ (C) $4k + 3$ (D) $4k$

24.25 The complex numbers z_1, z_2 and z_3 satisfy $\dfrac{z_1 - z_3}{z_2 - z_3} = \dfrac{1 - i\sqrt{3}}{2}$. Which of the following types of triangles has vertices z_1, z_2 and z_3 ? (2001)

(A) of area zero (B) right-angled isosceles

(C) equilateral (D) obtuse-angled isosceles

24.26 If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-\ldots\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-\ldots\right)=\frac{\pi}{2}$ for $0<|x|<\sqrt{2}$, find x. (2001)

24.27 Find the maximum value of $(\cos\alpha_1)(\cos\alpha_2)\ldots(\cos\alpha_n)$ under the restrictions $0\le\alpha_1,\alpha_2,\ldots,\alpha_n\le\frac{\pi}{2}$ and $(\cot\alpha_1)(\cot\alpha_2)\ldots(\cot\alpha_n)=1$. (2001)

24.28 If $\alpha+\beta=\frac{\pi}{2}$ and $\beta+\gamma=\alpha$, then which of the following expressions equals $\tan\alpha$? (2001)

(A) $2(\tan\beta+\tan\gamma)$ (B) $\tan\beta+\tan\gamma$

(C) $\tan\beta+2\tan\gamma$ (D) $2\tan\beta+\tan\gamma$

24.29 A man from the top of a 100 metros high tower sees a car moving towards the tower at an angle of depression of 30°. After some time, the angle of depression becomes 60°. Find the distance (in metros) travelled by the car during this time. (2001)

24.30 Find the number of integer values of m, for which the x-coordinate of the point of intersection of the lines $3x+4y=9$ and $y=mx+1$ is also an integer. (2001)

24.31 Find the equation of the directrix of the parabola $y^2+4y+4x+2=0$. (2001)

24.32 Let PQ and RS be the tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, find which of the following expressions equals $2r$. (2001)

(A) $\sqrt{PQ.RS}$ (B) $\frac{PQ+RS}{2}$

(C) $\frac{2PQ.RS}{PQ+RS}$ (D) $\sqrt{\frac{PQ^2+RS^2}{2}}$.

24.33 Find the area of the parallelogram formed by the lines $y=mx$, $y=mx+1$, $y=nx$ and $y=nx+1$. (2001)

24.34 Find the equation of the common tangent touching the circle $(x-3)^2+y^2=9$ and the parabola $y^2=4x$ above the x-axis. (2001)

24.35 If the triangle formed by the tangent to the curve $f(x)=x^2+bx-b$ at the point $(1,1)$ and the coordinate axes, lies in the first quadrant and has area 2, find the value of b. (2001)

24.36 If $f:[1,\infty)\longrightarrow[2,\infty)$ is given by $f(x)=x+\frac{1}{x}$, find $f^{-1}(x)$. (2001)

24.37 Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0. \end{cases}$ Find which of the following expressions equals $f(g(x))$ for all x. (2001)

(A) x (B) 1 (C) $f(x)$ (D) $g(x)$.

24.38 Find $\lim\limits_{x\to 0} \dfrac{\sin(\pi \cos^2 x)}{x^2}$. (2001)

24.39 Find the left-hand derivative of $f(x) = [x]\sin(\pi x)$ at $x = k$, k an integer. (2001)

24.40 Let $f : (0, \infty) \longrightarrow \mathbb{R}$ and $F(x) = \int\limits_0^x f(t)dt$. If $F(x^2) = x^2(1 + x)$, find $f(4)$. (2001)

24.41 Find the set of points where the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = \max\{x, x^3\}$ is NOT differentiable. (2001)

24.42 Which of the following statements is true about the function $f(x) = xe^{x(1-x)}$? (2001)

(A) $f(x)$ is increasing on $\left[-\dfrac{1}{2}, 1\right]$ (B) $f(x)$ is decreasing on $\mathbb{R}$

(C) $f(x)$ is increasing on $\mathbb{R}$ (D) $f(x)$ is decreasing on $\left[-\dfrac{1}{2}, 1\right]$.

24.43 Evaluate $\int\limits_{-\pi}^{\pi} \dfrac{\cos^2 x}{1 + a^x}dx$, $a > 0$. (2001)

24.44 Which of the following functions is differentiable at $x = 0$? (2001)

(A) $\cos(|x|) + |x|$ (B) $\cos(|x|) - |x|$

(C) $\sin(|x|) + |x|$ (D) $\sin(|x|) - |x|$

24.45 Find the number of solutions of $\log_4(x - 1) = \log_2(x - 3)$. (2001)

24.46 Let $f(x) = \dfrac{\alpha x}{x + 1}$, $x \neq -1$. Then find the value of α for which $f(f(x)) = x$. (2001)

24.47 Let $\vec{a} = \vec{i} - \vec{k}, \vec{b} = x\vec{i} + \vec{j} + (1 - x)\vec{k}$ and $\vec{c} = y\vec{i} + x\vec{j} + (1 + x - y)\vec{k}$. Which of the following does $[\vec{a}\ \vec{b}\ \vec{c}]$ depend on? (2001)

(A) only x (B) only y

(C) NEITHER x NOR y (D) both x and y

24.48 For a positive integer n, let H_n denote the n-th harmonic number defined as the sum $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$. Considering the area below the curve $y = \frac{1}{x}$, show that :

(i) $H_n - 1 < \ln n < H_{n-1}$ for all $n \geq 2$

(ii) the sequence $\{H_{n-1} - \ln n\}_{n=1}^{\infty}$ is monotonically increasing and bounded above by 1.

(From (ii) and the completeness axiom in Chapter 16, the sequence $H_{n-1} - \ln n$ and hence also the sequence $H_n - \ln n$ is convergent. For the limit, see Exercise (6.53).)

24.49 If m, n are positive integers, prove that $\dfrac{(2m)!(2n)!}{(m+n)!}$ is an integer.

24.50 (i) Let $\vec{a}, \vec{b}, \vec{c}$ be three unit vectors. Find a unit vector $\vec{d}$ which is equally inclined to each of $\vec{a}, \vec{b}, \vec{c}$.

(ii) For three unit vectors $\vec{a}, \vec{b}, \vec{c}$, show that the maximum possible value of $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ is 9. [*Hint:* Without loss of generality, take the vectors in the form $\sin\alpha\cos\theta_1\mathbf{i} + \sin\alpha\sin\theta_1\mathbf{j} + \cos\alpha\mathbf{k}$, $\sin\alpha\cos\theta_2\mathbf{i} + \sin\alpha\sin\theta_2\mathbf{j} + \cos\alpha\mathbf{k}$ and $\sin\alpha\cos\theta_3\mathbf{i} + \sin\alpha\sin\theta_3\mathbf{j} + \cos\alpha\mathbf{k}$ respectively.]

(iii) Give another solution to (ii) by rewriting $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ as $9 - |\vec{a} + \vec{b} + \vec{c}|^2$.

24.51 Three numbers are drawn from $\{1, 2, \ldots, p\}$ with replacements. Prove that the probability that their sum is $2p$ is $\dfrac{(p-1)(p+4)}{2p^3}$.

24.52 India plays two matches each with West Indies and Australia. In any match the probabilities of India getting points $0, 1$ and 2 are $0.45, 0.05$ and 0.50 respectively. Assuming that the outcomes are independent, find the probability of India getting at least 7 points. (1992)

24.53 Sixteen players $S_1, S_2, \ldots, S_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair a winner is decided on the basis of a game played between the players of the pair. Assume that all the players are of equal strength.

(a) Find the probability that the player S_1 is among the winners.

(b) Find the probability that exactly one of the two players S_1 and S_2 is among the eight winners. (1997*)

24.54 Numbers are selected at random, one at a time, from the two-digit numbers $00, 01, 02, \ldots, 99$ with replacement. An event E occurs if and only if the product of the two digits of a selected number is 18. If four numbers are selected, find the probability that the event E occurs at least 3 times. (1993)

24.55 Two players A and B play five games in a row. Assume that there are no draws and that the probability of A's winning a game is $\frac{4}{7}$. Find the probability that A wins three games in succession.

24.56 Two queens are placed at random on an 8×8 chess board. Find the probability that they are non-taking (i.e., that they do not lie in the same row or in the same column or on a line parallel to either of the diagonals).

24.57 Give an alternate solution to the problem solved in Comment No. 17 of Chapter 5 as follows. Let A be a set with n elements, say $x_1, x_2, \ldots, x_n$. Let the sample space consist of all ordered pairs (P, Q) of subsets of A. For each fixed $i = 1, 2, \ldots, n$ let E_i be the event that $x_i \notin P \cap Q$. Find $Pr(E_i)$.

24.58 Suppose AD is the internal angle bisector of a triangle ABC. If $\angle B = 2\angle C$ and $CD = AB$, prove that $\angle A = 72°$. (Regional Mathematics Olympiad)

24.59 Let BE and CF be the altitudes of an acute-angled triangle ABC and O its orthocentre. Suppose a line through O cuts AB at K and AC at L. Drop perpendiculars KM and LN from K, L to BE and CF respectively. Prove that FM is parallel to EN. (Regional Mathematics Olympiad)

24.60 From the vertices A, B, C of a triangle ABC, perpendiculars AD, BE, CF are drawn to any straight line. Show that the perpendiculars from D, E, F to BC, CA, AB respectively are concurrent.

24.61 Let A, B, C be collinear points in a plane, with B lying between A and C. Construct isosceles triangles APB and BQC on one side of the line AC and an isosceles triangle ARC on the other side of it such that each of the angles $\angle APB, \angle BQC$ and $\angle ARC$ equals 120°. Prove that PQR is an equilateral triangle. (Regional Mathematics Olympiad)

24.62 Prove that the product of the first 1000 even positive integers differs from the product of the first 1000 odd positive integers by a multiple of 2001. (Regional Mathematics Olympiad)

24.63 Find the number of positive integers x for which $\left[\frac{x}{99}\right] = \left[\frac{x}{101}\right]$. (Regional Mathematics Olympiad)

24.64 Prove that $\frac{x^2}{z} < \frac{x^2 + y^2 + z^2}{x + y + z} < \frac{z^2}{x}$ where x, y, z are positive real numbers with $x < y < z$. (1972)

24.65 If x, y, z are the sides of a triangle, prove that

$$|x^2(y-z)+y^2(z-x)+z^2(x-y)| < xyz.$$ (Regional Mathematics Olympiad)

24.66 If a, b, c are positive real numbers such that $abc = 1$, prove that :

(i) $a^{b+c}b^{c+a}c^{a+b} \leq 1$. (Indian National Mathematics Olympiad)

*(ii) $(a-1+\frac{1}{b})(b-1+\frac{1}{c})(c-1+\frac{1}{a}) \leq 1$ (IMO 2000)

24.67 Let n be an odd integer and A be a symmetric $n \times n$ matrix in which each of the integers 1 to n appears exactly once in each row. Prove that every such integer appears exactly once in its diagonal. (Regional Mathematics Olympiad)

24.68 Find all real values of a for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real. (Regional Mathematics Olympiad)

24.69 Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ which satisfy $f(x+y) = f(x)f(y)f(xy)$ for all $x, y \in \mathbb{R}$. (Indian National Mathematical Olympiad)

24.70 If $a_1, a_2, \ldots, a_n$ are all real and $A_1, A_2, \ldots, A_n$ are all positive integers, prove that the equation $\dfrac{A_1}{x-a_1} + \dfrac{A_2}{x-a_2} + \ldots + \dfrac{A_n}{x-a_n} = 0$ has all real roots.

24.71 In any triangle ABC, prove that

$$\begin{vmatrix} \cos(A+2B) & \cos B & \cos C \\ \cos A & \cos(B+2C) & \cos C \\ \cos A & \cos B & \cos(C+2A) \end{vmatrix} = 0.$$

24.72 The sides of a triangle are three consecutive integers and its inradius is 4 units. Find its circumradius. (Regional Mathematics Olympiad.)

24.73 Suppose A is a fifty-element subset of $\{1, 2, \ldots, 99, 100\}$ such that no two numbers from A add up to 100. Show that A contains a perfect square. (Regional Mathematics Olympiad)

24.74 In a plane convex quadrilateral of area 32, the sum of the lengths of one pair of the opposite sides and one diagonal is 16. Determine the possible length of the other diagonal. (K. N. Ranganathan)

24.75 There are doors numbered from 1 to 1000 in a row. At any time each door is either open or closed. An attendant visits them one by one in that order in each of the 1000 rounds he makes. In the k-th round he changes the state of those doors (from open to closed or vice versa as the case may be) whose numbers are multiples of k. Intitially all the doors are open. Find how many doors will be open at the end of the last round.

24.76 (a) An integer is called **squarefree** if it is not divisible by the square of any prime, or equivalently, if it is a product of distinct primes. Prove that every positive integer m can be written uniquely as u^2v where u, v are positive integers and v is squarefree. Prove further that if m divides n^2 where n is a positive integer, then uv divides n.

*(b) Find the least possible perimeter of a triangle ABC with integer sides in which $\angle C > 90^\circ$ and $\angle A = 2\angle B$. [*Hint:* See Exercise (11.7). After cancelling common factors, show that $b = u^2$ for some integer u. Show further that the interval $(3u^2, 4u^2)$ contains some perfect square.]

*24.77 Suppose cakes numbered 1 to n are arranged clockwise in a circle. A person starts from the first cake and keeps going clockwise cyclically, eating every alternate cake available. Let $f(n)$ be the number of the cake that is left out at the end. (For example, $f(10) = 4$ because the cakes that get eaten in the first round are numbered $1, 3, 5, 7, 9$, those eaten in the second round are $2, 6, 10$, and only cake numbered 8 gets eaten in the third round.) In the degenerate case $n = 1$, take $f(1)$ as 1. Find an easy expression for $f(n)$. Using your formula, compute $f(1000)$. [*Hint:* Reduce the problem to the case where n is odd. Then experiment.]

24.78 Take an $m \times n$ chessboard. Show that each diagonal of it passes through $m + n - d$ squares where d is the g.c.d. of m and n. (If a diagonal passes only through a vertex of a square, it is not considered to pass through that square.) [*Hint:* First consider the case where m, n are coprime.)

24.79 Two plane polygonal figures are said to be **equidecomposable** to each other if each one of them can be cut into a finite number of polygonal pieces which when rearranged (as in a jigsaw puzzle) give the other. For example, any two parallelograms with the same base and equal heights are equidecomposable as one sees easily. Trivially, equidecomposable figures have the same area. The converse, known as **Wallace-Bolyai-Gerwien theorem** is far from trivial. A proof can be given through the following steps. (Interestingly, its analogue for polyhedra is false. This was a famous question posed by a great mathematician **Hilbert**.)

(i) Equidecomposability is an equivalence relation.

(ii) The union of two disjoint squares is equidecomposable to a square. (Pythagoras theorem gives the size of such a square, and one of the classical proofs of it gives the desired decomposition.)

*(iii) Every rectangle is equidecomposable to a square. (Let $ABCD$ be a rectangle. By (ii) we may suppose $AB < AD < 2AB$. Take E on side BC so that AE is the geometric mean of AB and AD. Then the square on AE is a desired square.)

(iv) Every triangle and hence every polygonal figure is equidecomposable to a square. (Because of (i), this proves the Wallace-Bolyai-Gerwien theorem.)

24.80 Let X and Y be two sets with n and m elements respectively. Then the total number of functions from X to Y is m^n and it is easily seen that exactly ${}^mP_n$ of these functions are injective (for $m \geq n$). But there is no easy formula for the number of surjective functions from X to Y in case

$m \leq n$. (By Exercise (1.27), this number equals $m!S_{n,m}$ where $S_{n,m}$ is a Stirling number of the second kind. But this is not of much help since there is no easy formula for the Stirling numbers either.) Prove that the number of surjective functions from X to Y equals each of the following two summations:

(i) $\displaystyle\sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n$ (ii) $\displaystyle\sum_{\substack{(n_1,n_2,\ldots,n_m)\in\mathbb{N}^m \\ n_1+n_2+\ldots+n_m=n}} \frac{n!}{n_1!n_2!\ldots n_m!}$

where in (ii) the sum is taken over all ordered m-tuples of positive integers which add up to n (or equivalently, over all compositions of the integer n into m parts in the language of Exercise (1.25)(b)). [*Hint*: Let $Y = \{y_1, y_2, \ldots, y_m\}$. For (i) apply the principle of inclusion and exclusion. For $k = 1, 2, \ldots, m$, let A_k be the set of those functions from X to Y which do not assume the value y_k. For (ii), show that $\frac{n!}{n_1!n_2!\ldots n_m!}$ is the number of functions from X to Y for which the element y_k has exactly n_k preimages, for $k = 1, 2, \ldots, m$. An alternate expression for the sum in (ii) can be given if we assume the power series expansion of the exponential function e^x as $1 + x + \dfrac{x^2}{2!} + \ldots + \dfrac{x^r}{r!} + \ldots = \displaystyle\sum_{r=0}^{\infty}\frac{x^r}{r!}$. Consider the product $(e^x - 1)^m = (e^x - 1)(e^x - 1)\ldots(e^x - 1)$ (m times). Then for every composition $(n_1, n_2, \ldots, n_m)$ of n, the product of the terms $\dfrac{x^{n_1}}{n_1!}, \dfrac{x^{n_2}}{n_2!}, \ldots, \dfrac{x^{n_m}}{n_m!}$ is simply $\dfrac{x^n}{n_1!n_2!\ldots n_m!}$. It follows therefore that the sum in (ii) equals $n!$ times the coefficient of x^n when $(e^x - 1)^m$ is expanded as a power series.]

24.81 Suppose there are m roads going East-West and n roads going North-South in a city. There is a house at the intersection of every two roads. A person starts from a house at one of the corners of the city. In each move he can only go to an adjacent house along one of the roads. Under what condition on m and n will he be able to visit every house exactly once and return to where he started? (The problem can be paraphrased in terms of elementary graph theory. The graph involved here is called a **mesh** of size $m \times n$, with vertices at the houses and edges joining adjacent houses. The round path that is asked for is called a **Hamiltonian cycle** in this graph. Not every graph has such a cycle, a trivial counterexample being a graph with fewer edges than vertices. Many popular puzzles can be worded in terms of Hamiltonian cycles in suitable graphs. For example, the problem of moving a knight on a chess board exactly once through every square.)

24.82 The **digital sum** $a \oplus b$ of two non-negative integers a and b is defined as follows. Let $\displaystyle\sum_{i=0}^{m} a_i 2^i$ and $\displaystyle\sum_{i=0}^{m} b_i 2^i$ be their binary representations. We assume that at least one of the 'leading digits' a_m and b_m is non-zero (except when $a = b = 0$ in which case we take $m = 0$). Now let $c_i = 1$ or 0 according as $a_i + b_i$ is odd or even. Then $a \oplus b$ is defined to be the

integer $\sum_{i=0}^{m} c_i 2^i$. For example, $79 \oplus 50 = 125$ because in the binary system, $79 = 1001111$ and $50 = 0110010$. Put differently, the digital sum is the ordinary sum in the binary system but without carries. As a result, it shares some of the properties of the ordinary sum such as commutativity and associativity but differs sharply in some others. For example, $a \oplus b$ may be less than both a and b. Indeed, as an extreme case, we have $a \oplus a = 0$ for all non-negative integers! Given non-negative integers $a_1, a_2, \ldots, a_k$ with $a_1 \oplus a_2 \oplus \ldots \oplus a_k \neq 0$, prove that there exists an index r and a positive integer x such that $b_1 \oplus b_2 \oplus \ldots \oplus b_k = 0$ where $b_i = a_i$ for $i \neq r$ and $b_r = a_r - x$. In other words, a non-vanishing digital sum can be made to vanish by reducing one of its summands suitably, leaving the others intact. (This is trivial for $k = 2$, because $a_1 \oplus a_2 \neq 0$ gives $a_1 \neq a_2$. Reduce the larger of $\{a_1, a_2\}$ to the smaller. But the proof for $k = 3$ applies equally well to all $k > 3$. So you may assume $k = 3$.)

24.83 (a) Two persons A and B play a game, in which they alternately select integers from 1 to 10 and keep on totalling together the integers selected. The player who first reaches the figure 100 (exactly) wins. Prove that the player who plays first can ensure his victory. (Technically, this is called a **winning strategy** for the game. The game can also be played physically by removing matchsticks from a pile with 100 matchsticks originally, with each player having to take 1 to 10 sticks at a time. The player to empty the pile is the winner. In this form the game is called a **nim game**, the word 'nim' meaning 'take'.) [*Hint*: A player who reaches 89 first is sure to win the game.]

(b) What will be the winning strategy for a player if instead of 100 we have n matchsticks and each player has to take 1 to k sticks at a time where n, k are some positive integers?

(c) Suppose instead of one, we have several piles of matchsticks and at each turn a player can take any number of sticks from any one (but not more) of them, the player to take the last stick(s) being the winner. What is the winning strategy for this game? [*Hint*: The answer is easy for the case of two piles. Interpret it in terms of digital sums and apply Exercise (24.82) to handle the general case.]

*(d) As a truly non-trivial variation, suppose there are two piles with m and n matchsticks to begin with and in each move, a player can remove any number of sticks from either one or an equal number of sticks from both. Determine the winning strategy. (This game can be played on a (sufficiently big) chess board where a chip initially placed at (m, n) is to be moved to $(0, 0)$. In each move, a player can move it westward, southward or southwestward.)

24.84 (a) Two circles C_1 and C_2 of radii r_1 and r_2 respectively touch each other externally. A direct common tangent touches them at P_1 and P_2 respectively. Prove that $P_1P_2 = 2\sqrt{r_1 r_2}$.

(b) In (a), suppose a circle C of radius r touches both C_1 and C_2 externally and also touches one of their direct common tangents. Prove that $r = \dfrac{r_1 r_2}{(\sqrt{r_1} + \sqrt{r_2})^2}$.

*(c) Three circles C_1, C_2 and C_3 of radii r_1, r_2, r_3 respectively touch each other externally. A fourth circle C of radius r touches each one of these three circles externally. Prove that :

$$(\frac{1}{r} + \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3})^2 = 2(\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2})$$

[*Hint:* Let M_1, M_2, M_3 and M be the centres of the circles. Let α, β, γ be respectively the angles subtended at M by the segments M_2M_3, M_3M_1 and M_1M_2. The relation $\alpha + \beta + \gamma = 2\pi$ implies the identity $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2\cos\alpha\cos\beta\cos\gamma + 1$. Expressing these cosines in terms of r, r_1, r_2, r_3 gives the result after a good deal of algebraic manipulation, which is simplified considerably by working in terms of the elementary symmetric functions of r_1, r_2, r_3, viz., $\sigma_1 = r_1 + r_2 + r_3$, $\sigma_2 = r_1r_2 + r_2r_3 + r_3r_1$ and $\sigma_3 = r_1r_2r_3$. A proof using coordinates is also possible if we observe that M_3 and M lie on a hyperbola with foci at M_1 and M_2. This result is known as **Decartes' theorem** even though he stated it without proof. It can be modified when some of the circles touch internally rather than externally, by taking the radius of the outer circle as negative.

(d) It is customary to paraphrase the last result in the language of curvatures. Intuitively, the curvature of a curve at a point tells you how 'curved' the curve is at that point, i.e. the degree of its bend. It is intuitively obvious that the curvature of an ellipse is highest at the ends of its major axis and lowest at the ends of its minor axes, while in the case of a circle it is uniform at all points. A straight line should have curvature 0 everywhere since it does not bend at all. (For this reason, a layman often refuses to recognise a straight line as a curve!) A formal definition of curvature can be given in terms of the derivative of the angle the tangent makes with a fixed direction. For our purpose, it suffices to know that the curvature of a circle is inversely proportional to its radius. Denoting the curvatures of the four circles in (c) by $\kappa_1, \kappa_2, \kappa_3$ and κ (read as 'kappa'), Descartes' theorem says that $2(\kappa^2 + \kappa_1^2 + \kappa_2^2 + \kappa_3^2) = (\kappa + \kappa_1 + \kappa_2 + \kappa_3)^2$. Note that the relationship is symmetric in all the four variables even though our notation is not so. Note further that (b) comes as a special case if we treat a straight line as a circle with zero curvature everywhere.

24.85 If $x_1, x_2, \ldots, x_n$ are any real numbers, find the maximum value of

$$\sin x_1 \cos x_2 + \sin x_2 \cos x_3 + \ldots + \sin x_{n-1} \cos x_n + \sin x_n \cos x_1$$

24.86 A box contains m white, n black and k red balls. Balls are drawn from it, one by one, without replacement until all the balls left are of the same colour. Find the probability that the balls left are all red.

24.87 Sum the infinite series (i) $\sum_{k=1}^{\infty} \cot^{-1}(2k^2)$ (ii) $\sum_{k=1}^{\infty} \cot^{-1}(\frac{k^2}{2})$ and (iii) $\sum_{n=1}^{\infty} \dfrac{\tan(\theta/2^n)}{\cos(\theta/2^{n-1})}$ where θ is not a multiple of $\pi/2$ by a power of 2. (See Exercise (10.20) for the first two series.)

24.88 For a right-angled triangle with circumradius R and inradius r, prove that $R \geq (1+\sqrt{2})r$. When does equality hold?

24.89 Let $f(x) = x^3 + bx^2 + cx + d$. If $f(0)$ and $f(-1)$ are odd integers, prove that not all the zeros of $f(x)$ are integers. If, further, b, c are given to be integers, prove that none of the zeros of $f(x)$ can be an integer.

24.90 If the equation $(\tan^2\theta + 1)^2 + 4a(\tan^2\theta + 1)\tan\theta + 16\tan^2\theta = 0$ has four distinct roots in $(0, \pi/2)$, prove that $a \in (-5/2, -2)$.

24.91 Find the minimum values of the functions (i) $x^2 + 3y^2 - 6x - 2y$ and (ii) $x^2 + 2xy + 3y^2 - 6x - 2y$ of the real variables x, y.

24.92 (a) Let S be a (not necessarily regular) convex polygon with vertices $A_1, A_2, \ldots, A_n$. Prove that for every point P of S other than a vertex, there exists a line L passing through P such that S contains points on L on both the sides of P, or equivalently, $S \cap L$ contains a segment which is bisected at P. However, if $P = A_i$ for some i, then no such line exists. (Because of this property, for an arbitray convex subset S of a plane, a vertex is *defined* as a point of S, for which no such line can be drawn. It is then easy to see that for a disc, and more generally for a region bounded by an ellipse, the vertices are precisely the points on the boundary. Note that a closed half-plane is a convex set with no vertices.)

(b) Suppose S is a convex polygon and $f(x, y) = ax + by$ is a linear function. Let P be a point of S other than a vertex. Prove that there exist vertices V_1, V_2 of S such that $f(V_1) \leq f(P) \leq f(V_2)$.

(c) Using (b), show that f attains its maximum as well its minimum on S at some vertex of S. (Similar results hold for convex polytopes in higher dimensions mentioned in Comment No. 12 of Chapter 6. This is the starting point of an extremely useful branch of mathematics, called linear programming.)

24.93 (a) Given any three real numbers a, b, c, prove that there exist unique real numbers x, y, z such that $a = y + z, b = z + x$ and $c = x + y$. Prove further that x, y, z are positive if and only if a, b, c represent the sides of some triangle. Interpret x, y, z in terms of this triangle. (Because of this substitution, called **Ravi substritution**, any problem

involving the sides of a triangle can be converted to a problem involving three real numbers which are positive but unrestrained otherwise. This is especially convenient in proving certain inequalities.)

(b) Using Ravi substitution, prove that in any triangle ABC with area Δ, $2(ab+bc+ca)-a^2-b^2-c^2 \geq 4\sqrt{3}\Delta$ with equality holding only if ABC is equilateral. (This inequlaity is called the **Hadwiger-Finsler inequality**. It is stronger than saying $ab+bc+ca \geq 4\sqrt{3}\Delta$, which, in turn, is stronger than saying that $a^2+b^2+c^2 \geq 4\sqrt{3}\Delta$. The last inequality is called the **Weizenböck inequality** and was encountered in Exercise (14.11).)

24.94 (a) With usual notations, prove that in any triangle ABC, $abc = 4Rrs$ and $ab+bc+ca = s^2+4Rr+r^2$. Hence show that a triangle is uniquely determined by its perimeter, inradius and circumradius.

(b) Prove that a line which bisects both the perimeter and the area of a triangle must pass through its incentre. [*Hint:* Use the formula for the length of an angle bisector of a triangle quoted in Comment No. 10 of Chapter 11.]

24.95 (a) Let θ be a fixed angle between 0 and π. Show that the minimum of $\cot\theta_1+\cot\theta_2$ where θ_1, θ_2 are positive and $\theta_1+\theta_2=\theta$ occurs when $\theta_1=\theta_2=\frac{\theta}{2}$.

(b) In (a), suppose, more generally, that λ_1, λ_2 are any positive real numbers. Prove that the minimum of $\lambda_1\cot\theta_1+\lambda_2\cot\theta_2$ occurs when $\dfrac{\sin\theta_1}{\sin\theta_2}=\dfrac{\sqrt{\lambda_1}}{\sqrt{\lambda_2}}$.

(c) Let P be a variable point on the side BC of a triangle ABC and y, z be the prependicular distances of P from the sides CA, AB respectively. Show that the minimum of the sum $\dfrac{b}{y}+\dfrac{c}{z}$ occurs when OP is the angle bisector of the angle A.

(d) Let x, y, z be the perpendicular distances of a point P inside a triangle ABC from its sides BC, CA, AC respectively. Find when the sum $\dfrac{a}{x}+\dfrac{b}{y}+\dfrac{c}{z}$ is minimised.

24.96 (a) For any two triangles ABC and $A'B'C'$, prove that

$$a^2\cot A' + b^2\cot B' + c^2\cot C' \geq 4\Delta$$

with equality holding if and only if the two triangles are similar (with the given orders of vertices). (Here a, b, c and Δ are the sides and the area of the first triangle. Note that the Weizenböck inequality follows as a special case by taking $A'B'C'$ to be equilateral.)

(b) For any two triangles ABC and $A'B'C'$ with sides a, b, c, a', b', c' and areas Δ, Δ' respectively, prove that

$$a^2(b'^2 + c'^2 - a'^2) + b^2(c'^2 + a'^2 - b'^2) + c^2(a'^2 + b'^2 - c'^2) \geq 16\Delta\Delta'$$

with equality holding if and only if the two triangles are similar (in the given order of vertices). (This is simply a paraphrase of (a). But in this form, it is called **Pedoe's inequality**.)

24.97 Suppose $f(x)$ is a twice differentiable and $g(x)$ a continuous function on an interval $[a, b]$ and that they satisfy the differential equation $f''(x) + g(x)f'(x) - f(x) = 0$, for all $x \in [a, b]$. If f vanishes at both a and b, prove that it vanishes identically on $[a, b]$.

24.98 (a) Prove that any point in a plane is uniquely determined by its distances from any three non-collinear points in that plane. That is, if A, B, C are non-collinear and P, Q are such that $|PA| = |QA|, |PB| = |QB|$ and $|PC| = |QC|$ then P and Q must coincide with each other.

(b) A **congruence** or an **isometry** or a **rigid body motion** of a cartesian plane is a mapping, say T, of the plane into itself which preserves all distances, i.e. for every two points A, B in the plane, we have $d(T(P), T(Q)) = d(P, Q)$ where d denotes the usual euclidean metric. Suppose T is a congruence of the plane which fixes the origin. Prove that depending upon whether T preserves or reverses orientation, T is a rotation around the origin or a reflection in a line passing through the origin. [*Hint* : Let A, B, C be the images under T of the points $(0, 0), (1, 0)$ and $(0, 1)$. Construct an isometry S of the desired type which also takes $(0, 0), (1, 0)$ and $(0, 1)$ to A, B, C respectively. Then apply (a) to conclude that $T = S$.]

(c) If points of the plane are denoted by column vectors of the form $\begin{pmatrix} x \\ y \end{pmatrix}$, then show that in (b) T is given in terms of matrices by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

in the first case and by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & -\cos\alpha \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

in the second for some α. What is the geometric significance of α in the two cases? (Note that in each case, the column vectors of the matrix form an orthonormal basis for the plane vectors. For this reason, such matrices are called **orthogonal**. It can be shown that the determinant of every such matrix is ± 1. In case it is 1, the matrix is called **special orthogonal**. Of the two matrices here, only the first one is special orthogonal.)

(d) Prove that every isometry of the plane is either a rotation followed by a parallel translation or a reflection into a line followed by a parallel translation. (Actually, a stronger result is true. See (g) below.)

(e) If points of the plane are denoted by complex numbers then show that every isometry T of the plane is given by a function $f(z)$ which is of the form $f(z) = az + t$ or of the form $f(z) = a\overline{z} + t$ where a, t are complex numbers with $|a| = 1$. Deduce that every isometry of the plane is a bijection. Show further that in the first case, T is a translation if $a = 1$ while if $a \neq 1$, then it is a rotation of the plane around some point (not necessarily the origin) in it.

(f) Let L be a straight line in a plane. A **glide reflection** in L is the composite of the reflection in L followed by a translation by some vector parallel to L. Suppose the equation of the line L is given in the complex form as $\overline{b}z + b\overline{z} = c$ where c is a real number and b is a complex number with $|b| = 1$. (cf. Comment No. 16 in Chapter 8.) Show that a glide reflection in L is a function of the form $f(z) = bc - b^2\overline{z} + ibd$ for some real number d. Obtain an expression for it using 2×2 matrices in terms of β, c and d where β is such that $b = e^{i\beta}$.

(g) Using (e) and (f) show that every orientation preserving isometry of the plane is either a rotation or a parallel translation while every orientation reversing isometry is a glide reflection. (The catch is that in the last case, the translation involved is by a vector parallel to the line into which the reflection is taken, something which is not asserted by (d).)

(h) Prove that the composite of any two rotations of the plane is again a rotation or a translation. (This is not immediately obvious geometrically when the rotations are about two different points. But this follows easily from the classification of isometries in (g).) Similarly, prove that the composite of any two glide reflections is a rotation or a translation. (The study of isomeries of a plane is important both in physics and also in applied art because the symmetric patterns on a wall paper or a floor tiling etc. are related to isometries of the plane. Classification of isometries of space is also important in the study of crystal structures, but is more complicated. As expected, orthogonal matrices of order 3 play a crucial role in it.)

24.99 (a) Determine a three digit number which is twice the sum of the squares of the digits.

(b) Prove that there is only one digit which can simulataneously be the leading digit of 2^n and 5^n for some positive integer n. Find this digit.

24.100 Find all primes p for which $(p-1)! + 1$ has no prime factors besides p (which is always a factor by Wilson's theorem).

Appendix 1

More about Matrices

Matrices have been re-introduced into the JEE recently, after a long gap. The topic is vast enough to occupy whole books. And it will take a few years before its role and extent in the JEE are standardised. The official JEE websites give the follwing as the syllabus for matrices.

Matrices as a rectangular array of real numbers, equality of matrices, addition, multiplication by a scalar and product of matrices, transpose of a matrix, determinant of a square matrix of order up to three, inverse of a square matrix of order up to three, properties of these matrix operations, diagonal, symmetric and skew-symmetric matrices and their properties, solutions of simultaneous linear equations in two or three variables.

We already introduced most of these terms and many more, such as Hermitian and skew-Hermitian matrices, thier eigenvlues (also called characteristic roots) and trace. We did not explicitly define a diagonal matrix. But we did define diagonal entries of a square matrix in Exercise (2.34). A **diagonal matrix** is defined to be a square matrix all whose non-diagonal entries are zero. It is clear that the sum as well as the product of two diagonal matrices (of the same order) ia also a diagonal matrix.

It is not clear from the syllabus what degree of detail and depth is expected of the topics mentioned explicitly. But any serious study of matrices ought to include eigenvlues. As these are not even remotely mentioned, it appears that only elementary problems about matrices can be asked at the JEE. Another source of information would be the questions asked about matrices during the recent years. But apart from the small size of this data source, the difficulty is that some of these questions, although ostensibly based only on the topics listed in the syllabus, can be solved more easily if some advanced tools (not mentioned in the syllabus) are used. In some cases, the question would appear more natural in the context of certain concepts not introduced in the syllabus. There are also some questions where matrices are involved only superficially and the real question is only on properties of determinants. As for the use of matrices in the solutions of systems of linear equations, we have covered the major results in Chapter 3. But when there are only two or three variables, the system can often also be solved without matrices by direct elimination of variables.

In this appendix, we present illustrations of these various types of problems asked in the JEE from 2003 to 2004. As at many other places in the book, even when the original question is in a multiple choice format, we give it in a conventional form, unless there is some special reason to retain it in the multiple choice form.

We begin with a problem which is essentially a problem about determinants.

Problem 1: If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, find the value of α. (JEE 2004)

Solution:

It would be laborious to first calculate the matrix A^3 and then take its determinant. The key idea is to use the product formula for the determinant which says that the determinant of the product of two square matrices (of the same order) is the product of their determinants. Applying this twice in succession, we have $|A^3| = |A|^3$. So in the present problem, $|A| = 5$. But, by a direct computation, $|A| = \alpha^2 - 4$. So we get $\alpha^2 - 4 = 5$ giving $\alpha = \pm 3$.

The next problem can also be done with mere knowledge of some standard properties of determinants. Specifically, the properties that we shall need are :

(i) the determinant of the transpose of a (square) matrix is the same as the determinant of that matrix,

(ii) determinant of the product of two square matrices of the same order is the product of their determinants, and

(iii) if A is an $n \times n$ matrix and α is a real number then $\det(\alpha A) = \alpha^n \det(A)$.

We present this solution first. But after the solution, we shall see that there is more to the problem. The statement of the problem runs as follows.

Problem 2: If M is a 3×3 matrix with $M^T M = I$ (where I is the 3×3 identity matrix), and $\det(M) = 1$, then prove that $\det(M - I) = 0$. (JEE 2004.)

Solution: We are given a 3×3 matrix M with $M^T M = I$ and $\det(M) = 1$. From (i), we have $\det(M^T) = 1$. Therefore,

$$\begin{aligned} \det(M - I) &= \det(M^T)\det(M - I) \\ &= \det(M^T(M - I)) \quad \text{by (ii)} \\ &= \det(I - M^T) \\ &= \det(I - M^T)^T \quad \text{by (i) again} \\ &= \det(I - M) \\ &= (-1)^3 \det(M - I) \quad \text{by (iii)} \end{aligned} \tag{1}$$

Thus we have shown that $\det(M - I) = -\det(M - I)$, which implies $\det(M - I) = 0$ as was to be proved. Note that this argument will remain valid if instead of 3 we have any odd positive integer.

It is instructive to interpret the conclusion of the problem in a different way. The matrix $M - I$ is the matrix that results if we put $\lambda = 1$ in the matrix $M - \lambda I$ where λ is a variable. The determinant of this matrix is the characteristic polynomial, say $p(\lambda)$, of the matrix M (cf. Exercise (2.37). In this light, the problem says that if M is a square matrix of order 3 for which $M^T M = I$, then M has 1 as a characteristic root. Again, this holds even if the order is any odd positive integer.

We can go a little further. By Theorem 8 of Chapter 3, the assertion that $\det(M - I) = 0$ is equivalent to saying that there exists a non-zero solution to the equation

$$(M - I)\mathbf{x} = \mathbf{0} \tag{2}$$

which, in turn, means

$$M\mathbf{x} = \mathbf{x} \tag{3}$$

The problem does not specify if the entries of the matrix M are real or not. If they are complex, then the condition $M^TM = I$ does not mean much. But if they are real, then it means a lot. A (real) matrix A which satisfies the condition $A^TA = I$ is called **orthogonal**. The peculiar name comes from the concept of orthogonality, i.e. perpendicularity of vectors. To see this, suppose $A = (a_{ij})$ is a 3×3 (real) matrix with the property that $A^TA = I$. Writing this out fully, we get

$$A^TA = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{131} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4}$$

Now let us identify the columns of A with three dimensional vactors. Thus, for example, the first column of A, viz., $\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix}$ will be identified with the vector $a_{11}\mathbf{i}+a_{21}\mathbf{j}+a_{31}\mathbf{k}$. Let us denote this vector by $\mathbf{v}_1$. Similarly, identify the second and the third column of A with vectors $\mathbf{v}_2$ and $\mathbf{v}_3$.

Now if we carry out the multiplication of the two matrices in the middle term of (4), we see that the entry in the (i, j)-th place will be simply the dot product of the vectors $\mathbf{v}_i$ and $\mathbf{v}_j$. For example, the entry in the second row and the first column of the product matrix is $a_{12}a_{11} + a_{22}a_{21} + a_{32}a_{31}$ and this is precisely the dot product $\mathbf{v}_2 \cdot \mathbf{v}_1$. Doing this for all entries, we can rewrite (4) as

$$A^TA = \begin{pmatrix} \mathbf{v}_1 \cdot \mathbf{v}_1 & \mathbf{v}_1 \cdot \mathbf{v}_2 & \mathbf{v}_1 \cdot \mathbf{v}_3 \\ \mathbf{v}_2 \cdot \mathbf{v}_1 & \mathbf{v}_2 \cdot \mathbf{v}_2 & \mathbf{v}_2 \cdot \mathbf{v}_3 \\ \mathbf{v}_3 \cdot \mathbf{v}_1 & \mathbf{v}_3 \cdot \mathbf{v}_2 & \mathbf{v}_3 \cdot \mathbf{v}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{5}$$

If we equate the corresponding entries of the matrices we have got, we see that (5) simply says that the dot product $\mathbf{v}_i \cdot \mathbf{v}_j$ equals 1 if $i = j$ and 0 if $i \neq j$. In other words, every two distinct columns of the matrix A are orthogonal to each other when treated as vectors. This explains why such a matrix is called orthogonal. Another way to say that a matrix is orthogonal is to say that its column vectors form an orthonormal basis. As these concepts can be generalised for higher dimensional vectors too, we see that this paraphrase is valid for every n, and not just for $n = 3$.

Although matrix multiplication is not commutative in general, it can be shown that if A, B are square matrices of the same order, then the equations $AB = I$ and $BA = I$ imply each other. (Put differently, a left inverse of a square matrix is also its right inverse.) If we use this result, then the equality $M^TM = I$ implies $MM^T = I$, i.e. $(M^T)^TM^T = I$. In other words, whenever a square real matrix M is orthogonal, so is its transpose M^T. So, what we have proved above shows that the column vectors of M^T form an orthonormal basis.

But the column vectors of M^T are nothing but the row vectors of the original matrix M. Thus we have shown that if M is an orthogonal matrix then its column vectors, as well as its row vectors, form orthonormal bases.

If M is an orthogonal matrix, then $M^TM = I$. Taking determinants of both the sides we get $\det(M^TM) = \det(M^T)\det(M) = (\det M)^2 = 1$, whence $\det M = \pm 1$. If the + sign holds, then the matrix M is called **special orthogonal**. Returning to the problem above, what we have proved amounts to saying that every special orthogonal matrix of order 3 has 1 as an eigenvalue. Again, the proof also holds if we repalce 3 by any odd positive integer n. However, for $n = 3$, something more can be said. Let us take M as $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Denote the row vectors of the matrix M by $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ respectively. In other words, if we take M as above, then $\mathbf{u}_1 = a_{11}\mathbf{i} + a_{12}\mathbf{j} + a_{13}\mathbf{k}$ etc., where $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ is a fixed right-handed orthonormal basis. We already know that $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ is an orthonormal basis. We already know that for any orthogonal matrix M of order 3, $\det(M) = \pm 1$. It is further true that the + or − sign holds depending upon whether the orthonormal basis $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ is right-handed or left-handed. We are given that the first possibility holds. Thus the hypothesis of the problem can be paraphrased to say that M is a 3×3 matrix whose row vectors form a right-handed orthonormal basis.

The conclusion of the problem is that the determinant of the matrix $M - I$ is 0. If we write this matrix out, then we see that its row vectors are nothing but the vectors $\mathbf{u}_1 - \mathbf{i}, \mathbf{u}_2 - \mathbf{j}$ and $\mathbf{u}_3 - \mathbf{k}$. As the determinant equals the box product of its row vectors, its vanishing is equivalent to saying that the three vectors $\mathbf{u}_1 - \mathbf{i}, \mathbf{u}_2 - \mathbf{j}$ and $\mathbf{u}_3 - \mathbf{k}$ are linearly dependent. This can be further paraphrased to say that there exists a non-zero vector $\mathbf{v}$ which is perpendicular to all of them. Put differently, there exists a non-zero vector $\mathbf{v}$ such that

$$\mathbf{u}_1 \cdot \mathbf{v} = \mathbf{i} \cdot \mathbf{v} \tag{6}$$

$$\mathbf{u}_2 \cdot \mathbf{v} = \mathbf{j} \cdot \mathbf{v} \tag{7}$$

$$\text{and} \quad \mathbf{u}_3 \cdot \mathbf{v} = \mathbf{k} \cdot \mathbf{v} \tag{8}$$

Here $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ was a fixed right-handed system we started with. We could as well replace it with some other right-handed system, say $(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$. Then geometrically, this means that given any two right-handed systems $(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$ and $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$, there exists a direction (viz., the direction of the vector $\mathbf{v}$), which makes the same angles with $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ respectively as it does with the vectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ respectively. This is a purely geometric result. Thus we see that matrices have non-trivial applications to geometry. (A direct proof can be given using the concept of what are called direction cosines.) As already noted, the result can also be expressed in terms of the box product. Thus our little discovery can also be put as follows.

For any two right-handed systems $(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$ and $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$, the box product $[(\mathbf{u}_1 - \mathbf{w}_1)\ (\mathbf{u}_2 - \mathbf{w}_2)\ (\mathbf{u}_3 - \mathbf{w}_3)]$ vanishes.

A direct proof of this result is not so easy.

Equation (3) also acquires some significance when the entries of the matrix M are real. Let $\mathbb{R}^3$ be the set of all three dimensional column vectors. If $\mathbf{v} \in \mathbb{R}^3$, then $M\mathbf{v}$ is also an element of $\mathbb{R}^3$. This way we get a function, say f from $\mathbb{R}^3$ to $\mathbb{R}^3$, defined by $f(\mathbf{v}) = M\mathbf{v}$. This function is often called the **linear transformation** induced by M. The word 'linear' comes from the fact that such a function preserves a linear combination, i.e. for any two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, and for any real scalars α and β, we have

$$f(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha f(\mathbf{u}) + \beta f(\mathbf{v})$$

There is an intimate link between linear transformations and matrices. We shall not go into it. But we note that Equation (3) above can now be paraphrased to say that the function f has a non-zero fixed point. That is, the linear transformation induced by an orthogonal matrix of order 3 has at least one non-trivial fixed point. Again, this holds if 3 is replaced by any odd positive integer.

Orthogonal matrices seem to be favoured by the JEE paper-setters if we go by the number of questions about them, even though such matrices are not explicitly mentioned in the syllabus. Unfortunately, sometimes this leads to a hastily set question. The following is an example.

Problem 3: Let a, b, c be positive real numbers with $abc = 1$. Let

$$A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$$

If $A^T A = I$, where A^T is the transpose of A and I is the identity matrix, then determine the value of $a^3 + b^3 + c^3$. (JEE 2003)

Solution: Here the reference to the transpose A^T is redundant, because since A is symmetric, A^T equals A and we could as well have been given that $A^2 = I$. A direct computation gives

$$A^T A = A^2 = \begin{pmatrix} a^2+b^2+c^2 & ab+bc+ca & ab+bc+ca \\ ab+bc+ca & a^2+b^2+c^2 & ab+bc+ca \\ ab+bc+ca & ab+bc+ca & a^2+b^2+c^2 \end{pmatrix} \tag{9}$$

Equating this with I, the identity matrix, gives a system of two equations

$$a^2 + b^2 + c^2 = 1 \tag{10}$$

$$\text{and } ab + bc + ca = 0 \tag{11}$$

But since a, b, c are given to be positive, (11) can never hold. Hence the data of the problem is inconsistent. Let us, however, ignore this inconsistency by dropping the hypothesis that a, b, c are all positive. In that case, (10) and (11) together imply that $(a + b + c)^2 = a^2 + b^2 + c^2 = 1$. Therefore we have

$$a + b + c = \pm 1 \tag{12}$$

(If we assume that a, b, c are positive then we must discard the negative sign. However, as just noted such an assumption is not consistent with (11).)

We are given that $abc = 1$. We have thus got hold of the numerical values of the three elementary symmetric functions of a, b, c. We have to find the value of $a^3 + b^3 + c^3$ which is also a symmetric function of a, b, c. To do this, we use the well-known factorisation

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \tag{13}$$

From (10), (11), (12) and $abc = 1$, we thus get $a^3 + b^3 + c^3 = 3 \pm 1$, i.e. 4 or 2. (Again, if we continue to assume that a, b, c are positive, then we get only one answer, viz., $a^3 + b^3 + c^3 = 4$.)

The solution officially given on the internet along with the problem consists of first showing that $\det(A) = \pm 1$ by using the fact that $\det(A^T A) = (\det(A))^2 = \det(I) = 1$. By a direct computation, the determinant of A comes out to be $3abc - (a^3 + b^3 + c^3)$ which is factorised using (13). The fact that the last factor in the R.H.S. of (13) is always non-negative, coupled with the positivity of a, b, c then implies that $\det(A) = -1$. Since $abc = 1$, we then get $a^3 + b^3 + c^3$ as 4.

From this solution it appears that the problem was designed to test the knowledge of elementary properties of determinants, of the factorisation (13) and of inequalities (needed to show that the second factor of (13) is non-negative). However, the whole problem is hypothetical. Even if we drop the assumption that a, b, c are positive, it can be shown that there are no real numbers which satisfy (10), (11) and (12). For, if there were, then (depending upon which sign holds in (12)), they would have to be the roots of the cubic $x^3 - x^2 - 1 = 0$ or of the cubic $x^3 + x^2 - 1 = 0$. But it is easy to show (using a little calculus) that both these cubics have only one real root each. So if the matrix A given in the problem has the property that $A^T A = I$, then its entries will have to be complex.

The inconsistency of the data in the problem becomes even more apparent if we note that the matrix A is orthogonal. For, in that case, the column vectors of A are mutually orthogonal. But, to say that the entries of A are all positive is equivalent to saying that all its column vectors lie in the first orthant. (See Comment No. 10 of Chapter 21 for a definition.) But no two vectors in the same orthant can be orthogonal to each other unless they lie in the boundary planes of the orthant (in which case some of their entries will have to be 0).

Here is yet another JEE problem where orthogonal matrices figure (without the name, of course, since they are not explicitly listed in the syllabus).

Problem 4: If $P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$, find $P^T Q^{2005} P$. (JEE 2005)

Solution: Although the order of the matrix Q is only 2, nobody can compute the 2005-th power of a 2×2 matrix by brute force! There has to be some other way. The trick lies in recognising that the columns of the matrix P form an

orthonormal basis for $\mathbb{R}^2$ and hence that the matrix P is orthogonal, i.e.

$$P^T = P^{-1} \tag{14}$$

Equation (14) enables us to express the otherwise formidable power Q^{2005} quite succinctly. By definition, $Q = PAP^T$. So, $Q^2 = (PAP^T)(PAP^T)$, which, using the associativity of matrix multiplication, can be written as $PA(P^TP)AP^T$. But by (14), P^TP is simply the identity matrix I of order 2. So, Q^2 is the same as PA^2P^T. Multiplying by Q again, we get $Q^3 = QQ^2 = (PAP^T)(PA^2P^T) = PA(P^TP)A^2P^T = PAIA^2P^T = PA^3P^T$. Continuing in this manner (or, more formally, by induction on n), we can show that for every positive integer n,

$$Q^n = PA^nP^T \tag{15}$$

In particular, $Q^{2005} = PA^{2005}P^T$. Our interest lies in $P^TQ^{2005}P$. We have

$$\begin{aligned} P^TQ^{2005}P &= P^T(PA^{2005}P^T)P \\ &= (P^TP)A^{2005}(P^TP) \\ &= IA^{2005}I = A^{2005} \end{aligned} \tag{16}$$

We are not quite done yet. We still have to calculate the 2005-th power of the matrix A. Again, we begin with some experimentation. A direct calculation gives

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \tag{17}$$

$$\text{and} \quad A^3 = AA^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \tag{18}$$

There is a clear pattern now. By induction on n it is easy to show that

$$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \tag{19}$$

for every positive integer n.

So, finally, from (16) and (19), we get that $P^TQ^{2005}P = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$.

Some alternate derivations of (19) are possible. The first one starts by writing the matrix A as $I + B$ where I is the identity matrix and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Since the identity matrix I commutes with every matrix, we can apply the binomial theorem to this sum and get

$$\begin{aligned} A^n &= (I + B)^n \\ &= \sum_{k=0}^{n} \binom{n}{k} I^k B^{n-k} \end{aligned} \tag{20}$$

All powers of the identity matrix equal I itself. As for the powers of B, a direct computation gives that B^2 is the zero matrix. Hence all powers of B from B^2 onwards vanish. The 0-th power B^0 is to be taken as the identity matrix I. Therefore, the sum in (20) consists of only two terms, viz., $I^n + nB$. On expansion, this is simply the R.H.S. of (19).

This derivation of (19) was possible because the matrix A could be written in a particularly simple form. We now give a derivation which does not require any such special feature of the matrix A. It is based on a famous theorem about matrices, called **Cayley-Hamilton theorem**. To state it we need the concept of the characteristic polynomial, say $p(\lambda)$, of an $n \times n$ matrix A, defined as the determinant of the matrix $A - \lambda I$. Thus, for the matrix $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the characteristic polynomial comes out to be

$$p(\lambda) \quad = \quad \begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix} = \lambda^2 - 2\lambda + 1 \tag{21}$$

Now, in any polynomial $p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \ldots + a_{n-1}\lambda^{n-1} + a_n\lambda^n$, where $a_0, a_1, \ldots, a_n$ are some fixed real (or complex) numbers, if we replace the indeterminate λ by a real or a complex number, say α, then the expression will be another real or complex number, which we generally denote by $p(\alpha)$. This is too familiar to require much elaboration. What is not so commonly done is the replacement of the indeterminate by a square matrix, say B. We then get the expression $p(B) = a_0I + a_1B + a_2B^2 + \ldots + a_{n-1}B^{n-1} + a_nB^n$ where I is the identity matrix of the same order as B. Since all powers, sums and scalar multiples of square matrices of a given order are square matrices of the same order, the expression $p(B)$ is again a square matrix. (Note that it is vital that B is a square matrix. If not, it would be meaningless to talk of B^2 and of higher powers of B.)

Now suppose that $p(\lambda)$ is the characteristic polynomial of a square matrix A. What happens if we replace the indeterminate λ by the matrix A itself and form the matrix $p(A)$? The celebrated Cayley-Hamilton theorem says that the answer is surprisingly simple. In fact, it is the identically zero matrix of the appropriate order! In other words, $p(A) = O$. Stated verbally, every square matrix satisfies its own characteristic equation.

The proof of Cayley-Hamilton theorem is far from trivial and well beyond our scope. But we can verify it for square matrices of orders 1 and 2. The case of a matrix of order 1 is trivial because every such matrix can be identified with some number, say c and its characteristic polynomial is simply $c - \lambda$ which indeed has c as a root (in fact the only root). The next case is that of a 2×2 matrix, say $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then by a direct computation,

$$p(\lambda) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = \lambda^2 - (a+d)\lambda + ad - bc \tag{22}$$

which is a generalisation of (21). Also, $A^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ca+dc & bc+d^2 \end{bmatrix}$. A direct

substitution gives

$$\begin{aligned} p(A) &= A^2 - (a+d)A + (ad-bc)I \\ &= \begin{bmatrix} a^2+bc & ab+bd \\ ca+dc & bc+d^2 \end{bmatrix} - (a+d)\begin{bmatrix} a & b \\ c & d \end{bmatrix} + (ad-bc)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} a^2+bc-(a+d)a+(ad-bc) & ab+bd-(a+d)b+0 \\ ca+dc-(a+d)c+0 & bc+d^2-(a+d)d+(ad-bc) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned} \tag{23}$$

For matrices of order 3 or more, a direct verification of the Cayley-Hamilton theorem is too cumbersome. There is, in fact, no easy proof. (There is no dearth of proofs. But there is no proof whose simplicity will match that of the statement of the theorem.)

In essence, the Cayley-Hamilton theorem allows us to express the n-th power A^n of a matrix A (of order n), as a linear combination of the powers $I(= A^0), A, A^2, \ldots, A^{n-1}$, say as,

$$A^n = c_{n-1}A^{n-1} + c_{n-2}A^{n-2} + \ldots + c_1 A + c_0 I \tag{24}$$

Multiplying both the sides by A, we get

$$A^{n+1} = c_{n-1}A^n + c_{n-2}A^{n-1} + \ldots + c_1 A^2 + c_0 A \tag{25}$$

If we substitute (24) into (25) we get an expression for A^{n+1} as a linear combination of $I, A, A^2, \ldots, A^{n-1}$. Repeating this process, we see that all higher powers of A can be expressed as linear combinations of $I, A, A^2, \ldots, A^{n-1}$.

We are now ready to give an alternate derivation of (19) using Cayley-Hamilton theorem. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. We already calculated $p(\lambda)$ in (1). So, by Cayeley-Hamiltonian theorem, we have

$$A^2 = 2A - I \tag{26}$$

Therefore,

$$\begin{aligned} A^3 &= 2A^2 - A = 4A - 2I - A = 3A - 2I \\ \text{and } A^4 &= 3A^2 - 2A = 6A - 3I - 2A = 4A - 3I \end{aligned} \tag{27}$$

The pattern is now clear, viz.

$$A^n = nA - (n-1)I \tag{28}$$

which can be proved easily by induction on n, using (26). But if we calculate $nA-(n-1)I$, it comes out to be precisely $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$. Thus we get an alternate proof of (19).

Finally, a word about the figure 2005 occurring in the problem. It has no special significance as far as the problem is concerned. Instead of 2005, the question could have been asked with any positive integer and that would have made no difference in the method of solution. So one may wonder why the particular number 2005 is chosen. The answer lies in a tradition followed at the International Mathematical Olympiads, held annually. In every olympiad, one of the questions asked involves the number of the calendar year in which that particular olympiad is held. Usually, that particular number has no special significance. The other lower level olympiads such as the Indian National Mathematics Olympiad or the various Regional Mathematics Olympiads in India have also adopted this practice. It appears that this fashion has now caught on with JEE. Probably, some of the papersetters are fans of Mathematics Olympiads!

In this problem, we gave three proofs of (19), one of which used the Cayley-Hamiltonian theorem. Because of the peculiar feature of the matrix A, the other proofs were simple too. And so, in this problem, the use of the Cayley-Hamilton theorem may not seem to give any advantage. But there are problems where its use gives a substantially simpler solution. Here is one such problem (JEE 2005) posed in the multiple choice format.

Problem 5: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$. Suppose $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$ where I is the unit matrix. Then the values of c and d are

(A) $-6, -11$ (B) $6, 11$ (C) $-6, 11$ (D) $6, -11$

Solution: There are various ways to do this problem. The most straightforward method is to calculate both the sides of the given matrix equation

$$A^{-1} = \frac{1}{6}(A^2 + cA + dI) \tag{29}$$

separately so as to get an equality of two 3×3 matrices and then equate the corresponding pairs of entries.

There is a formula for computing the inverse of a (non-singular) $n \times n$ matrix, say A, in terms of the determinant of A and the determinants of the various minors of A. The (i, j)-th minor M_{ij} of A is defined as the submatrix of A obtained by deleting the i-th row and the j-th column of A. For example, in the given matrix A, $M_{23} = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$. It can be shown that the (i, j)-th entry of A^{-1} equals $(-1)^{i+j} \frac{\det M_{ji}}{\det A}$. (Note that we are taking M_{ji} and not M_{ij}.)

A direct proof of this result is not difficult. It is based on the expansion of a determinant by a particular row or column. But this method of finding the inverse is rarely applied in practice for matrices of order bigger than 3 because of the large number of determinants to be evaluated. For matrices of order 3, however, the situation is not so bad, because the minors are 2×2 matrices whose determinants can be written down by inspection. In the present problem this

becomes especially easy because the matrix has many zero entries. So, if we do this exercise, then it is not hard to show that

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix} \tag{30}$$

The R.H.S. of (29) is relatively less laborious to compute. A direct multiplication of A with itself gives

$$A^2 = A \times A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} \tag{31}$$

Multiplying both the sides of (29) by 6 we now get

$$\begin{aligned} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} + c\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} + d\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+c+d & 0 & 0 \\ 0 & -1+c+d & 5+c \\ 0 & -10-2c & 14+4c+d \end{bmatrix} \end{aligned} \tag{32}$$

Comparison of the corresponding entries of the two sides now gives us a host of 9 equations. Some of these are of the $0 = 0$ type. Out of the remaining five we can choose any two we like so as to get the values of c and d as quickly as possible. Equating the entries in the second row third column we get $-1 = 5 + c$ which gives $c = -6$. Equating the entries in the first row and first column, we get $6 = 1 + c + d = -5 + d$ which gives $d = 11$. Hence (C) is the correct answer. (The remaining three equations give us no new information about c and d, because they are consistent with these values of c and d. They better be. Otherwise it would mean that the matrix A^{-1} *cannot* be expressed in the form given in the statement of the question.)

Obviously, this solution is far too long to be the expected solution. We have, in fact, given it only so that on its background the brevity of other solutions is easier to appreciate. One such solution is to multiply both the sides of the matrix equation (29) by A (or rather, by $6A$) and get

$$6I = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = A^3 + cA^2 + dA \tag{33}$$

straightforward multiplication gives

$$\begin{aligned} A^3 &= A \times A^2 \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -11 & 19 \\ 0 & -38 & 46 \end{bmatrix} \tag{34}$$

Substituting (31) and (34) into (33), we get

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 1+c+d & 0 & 0 \\ 0 & -11-c+d & 19+5c+d \\ 0 & -38-10c-2d & 46+14c+4d \end{bmatrix} \tag{35}$$

As before, this is equivalent to a set of 9 equations in c and d of which four are redundant. Out of the remaining five, we can choose any two. The simplest choice is to take $1+c+d=6$ and $-11-c+d=6$. Subtracting the second from the first gives $c=-6$. Substitution into either one gives $d=11$, the same answer as before.

Although somewhat shorter than the first solution, this solution is still too long. Taking the cube of a 3×3 matrix is a torture. In the present problem, the fact that 4 out of the 9 entries of A are 0 does make the torture a little bearable. In fact, if we further note the location of these zero entries, it is possible to do the calculations a little more quickly. Let us introduce two matrices B and C by

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{36}$$

$$\text{and} \quad C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} \tag{37}$$

It is trivial to check that

$$A = B + C \tag{38}$$

If A, B, C stood for real (or complex) numbers then an equation like this, coupled with the binomial theorem will enable us to express the powers of A in terms of those of B and C. This is so because the multiplication of real numbers is commutative. This is not the case with matrix multiplication in general. That is why the binomial theorem is not used so frequently with matrices. But if the two matrices whose sum is taken commute with each other, then the binomial theorem does apply. In the present case, the matrices B and C do commute with each other. In fact, it is trivial to show by direct computation that

$$BC = CB = O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{39}$$

Here we are crucially using the fact not only four of the entries of the matrix A vanish, but also that these vanishing entries make the first row and the first column of A identically zero except for one single entry at their intersection.

Now, let m be a positive integer. The relations (29) and (39), along with the binomial theorem enable us to write the power A^m as $\sum_{k=0}^{m} \binom{m}{k} B^k C^{m-k}$. Note that for $k = 1, 2, \ldots, m-1$, the k-th term will contain the product BC and will therefore vanish. That leaves us only with two terms, so that we get

$$A^m = B^m + C^m \tag{40}$$

for every positive integer m. What makes this equation powerful is that the powers of B and C are very easy to calculate. In fact, one can check directly that $B^2 = B$ and hence that all powers of B equal B. The picture is not so rosy with the matrix C, but not entirely hopeless either. Let us consider a 2×2 matrix E defined by

$$E = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \tag{41}$$

Clearly, E is a 2×2 submatrix of the 3×3 matrix C. But it is much more than that. All other entries of C are 0. In fact, we can consider C as having been obtained from E by adding a 'dummy' row and an equally dummy column consisting of all zeros. The effect of these dummies is that if C is multiplied (whether on the right or on the left) by any 3×3 matrix, say F, the product will depend only on the 2×2 submatrix, say G, of F obtained by deleting the first row and the first column of F. It is as if we multiply the two matrices E and G and add a dummy row and a dummy column of zeros to get the product of C and F. So in this sense, E is not just a submatrix of C. We may, figuratively, say that it is the soul or the core of C.

In particular we see that the powers of C can be obtained from the corresponding powers of the submatrix E and then adding a dummy row and a dummy column consisting of all zeros. Being a 2×2 matrix, it is much easier to form the various powers of E. Indeed, we have

$$E^2 = \begin{bmatrix} -1 & 5 \\ -10 & 14 \end{bmatrix} \quad \text{and} \quad E^3 = \begin{bmatrix} -11 & 19 \\ -38 & 46 \end{bmatrix} \tag{42}$$

from which we get

$$C^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} \quad \text{and} \quad C^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -11 & 19 \\ 0 & -38 & 46 \end{bmatrix} \tag{43}$$

As we already know that $B^m = B$ for all m, (40) now provides less laborious derivations of (31) and (34). So, the second solution given above can be shortened. It is possible that the paper-setters originally designed the problem for the 2×2 matrix E and then extended it to the matrix A.

The first solution given above can also be shortened considerably by working in terms of the matrix E. The inverse of a 2×2 matrix can be written down almost by inspection because the minors are 1×1 matrices, i.e. mere numbers

so that no work is needed in finding their determinants. So, by the same method that we applied to calculate A^{-1}, but with much less work, we get

$$E^{-1} = \frac{1}{6}\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix} \tag{44}$$

If we rewrite (30) and (44) by taking the coefficient $\frac{1}{6}$ inside the matrices (i.e. by multiplying every entry by it), we see that A^{-1} is obtained from E^{-1} by adding one row and one column, their common entry being 1 and all entries 0.

The reason such a shortening of our earlier solutions was possible is the peculiar feature of the matrix A. The method is applicable to more general situations. Let $A = (a_{ij})$ be an $n \times n$ matrix. Assume that there is some positive integer $k < n$ such that (i) for every $i = 1, 2, \ldots, k$, we have $a_{ij} = 0$ for every $j > k$ and (ii) for every $i = k+1, k+2, \ldots, n$, we have $a_{ij} = 0$ for every $j \leq k$. Verbally, (i) means that in the first k rows of A, all the entries from the $k+1$-th column onwards vanish while (ii) means that in the las $n-k$ rows of A, all the entries in the first k columns vanish. These conditions can be expressed in a visual form.

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1k} & 0 & 0 & \ldots & 0 \\ a_{21} & a_{22} & \ldots & a_{2k} & 0 & 0 & \ldots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ a_{k1} & a_{k2} & \ldots & a_{kk} & 0 & 0 & \ldots & 0 \\ 0 & 0 & \ldots & 0 & a_{k+1,k+1} & a_{k+1,k+2} & \ldots & a_{k+1,n} \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \ldots & 0 & a_{n-1,k+1} & a_{n-1,k+2} & \ldots & a_{n-1,n} \\ 0 & 0 & \ldots & 0 & a_{n,k+1} & a_{n,k+2} & \ldots & a_{n,n} \end{bmatrix} \tag{45}$$

Let P be the $k \times k$ submatrix of A consisting of the first k rows and the first k columns. Similarly, let Q be the $(n-k) \times (n-k)$ submatrix consisting of the last $n-k$ rows and the last $n-k$ columns of A. Then it is customary to write A in a more compact form as

$$A = \begin{bmatrix} P & \vdots & O_1 \\ \ldots & \ldots & \ldots \\ O_2 & \vdots & Q \end{bmatrix} \tag{46}$$

where O_1 is a $k \times (n-k)$ matrix with all zero entries and O_2 is an $(n-k) \times k$ matrix with all zero entries. (Often O_1 and O_2 are denoted by the same symbol O.) The 3×3 matrix A given in the statement of the problem is a matrix of this type, with $k = 1$, $Q = E$, the matrix defined by (41) and P consisting of just one number, viz. 1.

The work we have done can be easily adapted to prove the following more general result.

Theorem: Suppose the matrix A can be written as in (46). Then for every positive integer m we have

$$A^m = \begin{bmatrix} P^m & \vdots & O_1 \\ \cdots & \cdots & \cdots \\ O_2 & \vdots & Q^m \end{bmatrix} \tag{47}$$

Moreover,

$$\det A = \det P \times \det Q \tag{48}$$

If, further, the matrices P and Q are invertible then so is A and A^{-1} is given by

$$A^{-1} = \begin{bmatrix} P^{-1} & \vdots & O_1 \\ \cdots & \cdots & \cdots \\ O_2 & \vdots & Q^{-1} \end{bmatrix} \tag{49}$$

Finally, we give a solution of this problem based on the Cayley-Hamilton theorem. We begin by calculating the characteristic polynomial $p(\lambda)$ of the matrix A given in the statement of the question.

$$\begin{aligned} p(\lambda) &= \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} \\ &= (1-\lambda)[(1-\lambda)(4-\lambda)+2] \\ &= (1-\lambda)(\lambda^2-5\lambda+6) \qquad (50) \\ &= -\lambda^3+6\lambda^2-11\lambda+6 \qquad (51) \end{aligned}$$

Therefore, by the Cayley-Hamilton theorem, we get $-A^3+6A^2-11A+6I = 0$, or equivalently,

$$A^3 - 6A^2 + 11A - 6I = O \tag{52}$$

On the other hand, (33) can be rewritten as

$$A^3 + cA^2 + dA - 6I = O \tag{53}$$

A direct comparison between these two equations gives $c = -6, d = 11$ at least as one possible solution. To prove that this is the only solution is not so easy. But in a multiple choice question, one need not bother about it. In fact, the present question gives an unfair advantage to those who know at least the statement of Cayley-Hamilton theorem. This theorem is not a part of the JEE syllabus. But considering that many students appear for JEE when they are in their first or even second year of a degree program, it is not unthinkable that some of them

know it. In a conventional examination where a candidate has to show his work, a solution based on an advanced technique such as the Cayley-Hamilton theorem can be specifically disallowed. But in an objective type test what matters is the answer and not how you arrived at it. The situation is analogous to the use of Lagrange's multipliers (see Comment No. 4 of Chapter 14) which often make a mincemeat of problems of trigonometric optimisation.

On the other hand, for those who do not know the Cayley-Hamilton theorem, the problem is far too laborious (even with the short cuts indicated) compared to the time available to solve it.

We caution the reader against using the Cayley-Hamilton theorem indiscriminately by analogy with polynomial equations of real or complex numbers. Suppose for example, that a polynomial $p(\lambda)$ factors as $q(\lambda)r(\lambda)$. If α is a real or complex root of $p(\lambda)$, then it must be a root of either $q(\lambda)$ or $r(\lambda)$. This is so because, if both $q(\alpha)$ and $r(\alpha)$ are non-zero numbers, then so is their product. With matrices, the situation is different. The product of two nonzero matrices can be the zero matrix as we see in (39). As a result, even if we have $p(A) = O$ where A is a square matrix, we cannot conclude that either $q(A) = O$ or $r(A) = O$. Indeed, (50) gives us a factorisation of the characteristic polynomial $p(\lambda)$ of the matrix A in our question. By Cayley-Hamilton theorem, we have $p(A) = O$. But neither $I = A$ nor is $A^2 - 5A + 6I = O$.

A few more problems involving matrix multiplication and powers of matrices will be given as exercises. We now turn to problems where matrices are applied to find solutions of systems of linear equations. As noted before, since the JEE syllabus confines itself only to the cases where the number of variables is 2 or 3, matrices can often be dispensed with. But our stress is on the methods. So, we begin with a problem where it is *not* a good idea to do it without matrices.

Problem 6: Let

$$A = \begin{bmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{bmatrix}, B = \begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix},$$

$$U = \begin{bmatrix} f \\ g \\ h \end{bmatrix}, V = \begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix} \text{ and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

If $AX = U$ has infinitely many solutions, prove that $BX = V$ has no unique solution. If further, $afd \neq 0$, then show that $BX = V$ has no solution.

Solution: Although matrices are involved in the statement of the problem, the problem itself is not so much about matrices as about the existence and uniqueness of solutions of systems of linear equations in which the number of variables equals the number of unknowns. Matrices are only a convenient tool. The basic result that is needed is that a system such as

$$AX = U$$

has a unique solution if $\Delta \neq 0$, where Δ is the determinant of the matrix A. When $\Delta = 0$, then the system has,

(i) no solution if at least one of Δ_1, Δ_2 and Δ_3 is non-zero

(ii) either no solution or else infinitely many solutions if $\Delta_1 = \Delta_2 = \Delta_3 = 0$,

where, for $i = 1, 2, 3$, Δ_i is the determinant of the matrix obtained by replacing the i-th column of A by the column vector U.

First we apply this result to the system

$$AX = U \tag{54}$$

which is given to have infinitely many solutions. As a result, $\Delta, \Delta_1, \Delta_2, \Delta_3$ all vanish, where

$$\Delta = \begin{vmatrix} a & 1 & 0 \\ 1 & b & d \\ 1 & b & c \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} f & 1 & 0 \\ g & b & d \\ h & b & c \end{vmatrix}, \tag{55}$$

$$\Delta_2 = \begin{vmatrix} a & f & 0 \\ 1 & g & d \\ 1 & h & c \end{vmatrix} \quad \text{and} \quad \Delta_3 = \begin{vmatrix} a & 1 & f \\ 1 & b & g \\ 1 & b & h \end{vmatrix} \tag{56}$$

We could expand these determinants right now. But we are not sure exactly which ones among them will be needed. So let us postpone this and consider the system

$$BX = V \tag{57}$$

about whose solutions we have to prove something. We apply the same criterion as above. However, this time let us denote the corresponding determinants by $\Delta', \Delta'_1, \Delta'_2, \Delta'_3$. That is,

$$\Delta' = \begin{vmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{vmatrix}, \quad \Delta'_1 = \begin{vmatrix} a^2 & 1 & 1 \\ 0 & d & c \\ 0 & g & h \end{vmatrix}, \tag{58}$$

$$\Delta'_2 = \begin{vmatrix} a & a^2 & 1 \\ 0 & 0 & c \\ f & 0 & h \end{vmatrix} \quad \text{and} \quad \Delta'_3 = \begin{vmatrix} a & 1 & a^2 \\ 0 & d & 0 \\ f & g & 0 \end{vmatrix} \tag{59}$$

The first part of the problem amounts to showing that if $\Delta, \Delta_1, \Delta_2, \Delta_3$ are all 0, then so is Δ'. For this it is hardly necessary to expand Δ'. From (56) and (58), we see that if we take the transpose of Δ_2 and interchange the second and the third rows, we get Δ'. Hence $\Delta' = -\Delta_2$. Since Δ_2 vanishes so does Δ'. Hence the system (57) can have no unique solution.

For the second part, we have to show that under the additional hypothesis $afd \neq 0$, (57) has no solution. From the criterion above, this amounts to showing that at least one of Δ'_1, Δ'_2 and Δ'_3 is non-zero. By direct expansions,

these determinants equal, respectively, $a^2(dh - gc), a^2cf$ and $-a^2df$. We are given that $afd \neq 0$. This means a, f, d are all non-zero. Hence $-a^2df$ is also non-zero. In other words, $\Delta'_3 \neq 0$. As $\Delta' = 0$, this means (57) has no solution.

Problems based on the criterion for the existence and uniqueness of solutions of systems of three linear equations in three unknowns are quite common at the JEE. But in most of these problems, the coefficients of the system are in terms of some single parameter (often denoted by λ) and you are asked to identify the values of this parameter for which the system has a unique solution, no solution etc. The present problem is a welcome departure from these stereotyped problems. Another nice feature of the problem is the use of the properties of determinants made in its solution (for example, that the determinant of the transpose of a matrix is the same as that of the original matrix). This saved a lot of computations. For example, we could show $\Delta' = 0$ without calculating it because we could relate it to Δ_2 which was given to vanish. Many students believe that the use of matrices in systems of equations is only to provide a compact notation. Such students feel more comfortable in solving systems of linear equations (especially when the system has only 2 or 3 equations) by writing them out fully and manipulating them. In the present problem, such an approach would be very complicated to say the least.

This problem dealt with the existence and the uniqueness of solutions of systems of equations rather than with the solutions *per se*. The next problem, on the other hand, asks you to actually solve a system and then do some computations with the solutions.

Problem 7: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$. Suppose U_1, U_2, U_3 are three column vectors such that

$$AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \; AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \text{ and } AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

and U is a 3×3 matrix whose first, second and third columns are U_1, U_2, U_3 respectively. Find the values of

(i) $|U|$

(ii) the sum of the elements of U^{-1}

(iii) $[3 \;\; 2 \;\; 0]\, U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$. (JEE 2006, modified.)

Solution: The straightforward way is to begin by identifying each of the column vectors U_1, U_2, U_3. Each requires us to solve a system of three equations in three unknowns. For example, suppose $U_1 = \begin{bmatrix} u_{11} \\ u_{21} \\ u_{31} \end{bmatrix}$. Then the equation

$AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is equivalent to the system

$$\begin{aligned} u_{11} &= 1 && (60) \\ 2u_{11} + u_{21} &= 0 && (61) \\ 3u_{11} + 2u_{21} + u_{31} &= 0 && (62) \end{aligned}$$

which can be solved by inspection to get

$$U_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \tag{63}$$

Similar calculations yield

$$U_2 = \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix} \text{ and } U_3 = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix} \tag{64}$$

Put together this gives the matrix U as

$$U = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \tag{65}$$

Now that we have identified the matrix U explicitly, we can answer any questions about it. Part (i) asks for its determinant $|U|$. A straightforward calculation gives

$$|U| = 1 \times (-1) + 2 \times (-7) + 2 \times 9 = 3 \tag{66}$$

A slightly better approach is to put together the three equations given in the statement of the problem to give the single matrix equation

$$AU = B \tag{67}$$

where A and B are the matrices given by

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix} \tag{68}$$

The matrix A is non-singular since its determinant is 1 which is non-zero. Therefore A^{-1} exists and so from (8) we get

$$U = A^{-1}B \tag{69}$$

This gives us an alternate method for calculating the matrix U. But the computations involved in finding A^{-1} are not substantially different from those in

solving the three equations (60), (61), (62) simultaneously. Moreover, we still have to multiply the two matrices A^{-1} and B. So there is not much saving in this approach.

But there is a slicker way to find $|U|$ even without finding U. Taking determinants of both the sides in (67),

$$|A||U| = |B| \tag{70}$$

So, if we could find $|A|$ and $|B|$, then we would know $|U|$ (provided, of course, that $|A| \neq 0$). This approach may seem seem foolish on the face of it because in order to find the determinant of one matrix, we now have to calculate the determinants of two matrices. But it has two advantages. First, it does not involve the computation of the matrix U and we only have to work with A and B which are explicitly given. Of course, evaluating a 3×3 determinant simply by expanding all terms is not a very exciting job either. But in this particular problem, this part too is drastically simplified because of certain particular features of the matrices A and B. Note that for the matrix B all the entries below the main diagonal are zero. A matrix with this property is called **upper triangular**. Similarly, the matrix A given in (68) is **lower triangular** because all entries above the main diagonal vanish. In either case, the determinant of the matrix has only one non-zero term, viz. the product of the diagonal elements. So the determinants of B and of A can be written down simply by inspection.

Applying this reasoning, we have $|A| = 1$ and $|B| = 3$. Putting these values into (70) gives $|U| = 3$. This solution is certainly more elegant in terms of its approach, which is based on the multiplicativity property of determinants (i.e. the fact that the determinant of a product of two matrices is the product of the determinants.) Surely, it is much easier to multiply determinants (which are some real numbers) than to multiply two 3×3 matrices which is a nasty job.

This completes the solution to (i). This part of the problem is excellent because it rewards those who look for elegant solutions before proceeding with brute force computations.

We now turn to (ii) viz. to find the sum of the elements of the matrix U^{-1}. Once again, the most straightforward approach would be to begin by calculating the matrix U^{-1} explicitly. If we have already identified U by (65) in the solution to the previous problem, then using the standard formula for the inverse in terms of the cofactors of the elements of a matrix, one can show, by sheer computation, that

$$U^{-1} = \frac{\text{adj } (U)}{|U|} = \frac{1}{3}\begin{bmatrix} -1 & -2 & 0 \\ -7 & -5 & -3 \\ 9 & 6 & 3 \end{bmatrix} \tag{71}$$

from which it is trivial to verify that the sum of the elements of U^{-1} is 0.

But the computations involved in finding U^{-1}, while inherently simple, are highly repetitious and prone to errors. So once again we look for some better method. Note that we are not asked to find the matrix U^{-1} *per se* but only the sum of its elements. Just as in Part (i) we were able to find the determinant of

the matrix U even without finding U explicitly, let us see if we can do something similar here.

We begin by an observation which is applicable to any square matrix $C = (c_{ij})$ of order 3 (or of any order n for that matter). It is well-known that the columns of C are precisely the product matrices $C \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $C \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $C \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ respectively. Adding these three, we get

$$C \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} c_{11} + c_{12} + c_{13} \\ c_{21} + c_{22} + c_{23} \\ c_{31} + c_{32} + c_{33} \end{bmatrix} \tag{72}$$

Verbally, multiplying a matrix on the right by the column vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ gives a column vector whose entries are the row sums of the matrix C. An entirely analogous argument shows that $[1\ 1\ 1]C$ is a row vector whose entries are the column sums of the matrix C.

What if we perform both the operations, i.e. premultiplication by the row vector $[1\ 1\ 1]$ and also post-multiplication by the column vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$? Then, as expected, we get a 1×1 matrix whose lone entry is the sum of all the entries of C. It is customary to denote a 1×1 matrix by its lone entry. We then have

$$[1\ 1\ 1]\, C \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \sum_{i=1}^{3} \sum_{j=1}^{3} c_{ij} \tag{73}$$

This simple formula allows us to recast our goal. We want the sum, say S, of the elements of the matrix U^{-1}. Because of (73) we can rewrite this as

$$S = [1\ 1\ 1]\, U^{-1} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \tag{74}$$

Of course, merely recasting a problem does not solve it! Indeed, if we have already calculated the matrix U^{-1}, then it is foolish to use (74). Instead, it is far better just to look at the entries of U^{-1} and add. The true worth of a formula like (74) comes when we have not explicitly calculated the matrix U^{-1} but can nevertheless identify it in some other way, e.g. as the product of some simpler matrices. This is indeed the case in the present problem. For, since both A and B are invertible, from Equation (67) we have $U = A^{-1}B$ and hence

$U^{-1} = B^{-1}A$. Putting this into (74), we now have

$$S = [1\ 1\ 1]\ B^{-1}A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \tag{75}$$

Here S, which is simply a real number, is expressed as a product of four matrices! Using associativity of matrix multiplication we are free to group any of these together as long as we do not change their order. In particular, we can rewrite S further as

$$S = PQ \tag{76}$$

where

$$P = [1\ 1\ 1]\ B^{-1} \tag{77}$$

$$\text{and} \quad Q = A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \tag{78}$$

We already know that Q is a column vector whose entries are the row sums of A. From (68), we have

$$Q = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} \tag{79}$$

Unfortunately, P cannot be computed so readily. We know it is a row vector whose entries are the column sums of the matrix B^{-1}. The trouble is that even though we know B from (58), we have not yet found its inverse. It may thus appear that we have not really gained anything by this approach over the earlier approach. The only difference is that instead of the matrix U we now have to find the inverse of the matrix B. But this is no small gain. First, we are given B whereas we have to *find* U (which so far we have escaped). Secondly, even if we compute U as in (65), computing its inverse is considerably more laborious than finding the inverse of B. This happens again because B is an upper triangular matrix. It is easy to show that the inverse of every such matrix (if it exists) is also upper triangular and further that its diagonal entries are precisely the reciprocals of the corresponding diagonal entries of the original matrix. Even if we do not know this result, we can find B^{-1} simply by solving some equations.

So, let $B^{-1} = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}$. Then $BB^{-1} = I$ gives

$$\begin{bmatrix} x_1 + 2x_2 + 2x_3 & y_1 + 2y_2 + 2y_3 & z_1 + 2z_2 + 2z_3 \\ 3x_2 + 3x_3 & 3y_2 + 3y_3 & 3z_2 + 3z_3 \\ x_3 & y_3 & z_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{80}$$

which is a system of 9 equations in 9 unknowns. A similar system for finding U^{-1} would be horrendous. But not the present one. For we immediately get $x_3 = 0, y_3 = 0, z_3 = 1$ and thereafter $x_2 = 0, y_2 = 1/3, z_2 = -1$ and finally the first row entries, viz. $x_1 = 1, y_1 = -2/3$ and $z_1 = 0$. Hence

$$B^{-1} = \begin{bmatrix} 1 & -2/3 & 0 \\ 0 & 1/3 & -1 \\ 0 & 0 & 1 \end{bmatrix} \tag{81}$$

(Note, incidentally, that B^{-1} is upper triangular as we had remarked it would have to be and, further also, that its diagonal entries are the reciprocals of the corresponding diagonal entries of B. In fact, those who know this already could have incorporated this information to reduce the number of unknowns in (80) from 9 to 3 and thereby simplify the calculations still further.)

Putting (81) into (77), we get

$$P = [1 \quad -1/3 \quad 0] \tag{82}$$

Finally, putting (76), (79) and (72) together, we have

$$S = PQ = [1 \quad -1/3 \quad 0] \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} = 0 \tag{83}$$

which completes the solution to Part (ii) of the problem.

As in Part (i), in (ii) too, the gap of labour needed between the elegant and the brute force solutions is remarkable. But the difference is that in Part (i) the key idea in the elegant solution was a familiar one, viz. the multiplicativity property of determinants. And once you hit it, hardly any numerical work had to be done. Comparatively, the key idea behind the elegant solution to Part (ii), viz. Equation (73), is not so frequently used and hence not easy to come up with. Moreover, you have to do some numerical work to get B^{-1} anyway. As a result, even though Part (ii) of the problem is also very good, it is a shade below Part (i). True elegance lies in using something usual in an unusual way.

Finally, we turn to Part (iii), viz. to find the value of $[3 \quad 2 \quad 0]\ U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$. Note that this expression is a product of three matrices of orders 1×3, 3×3 and 3×1. Hence it equals a 1×1 matrix. It is customary to identify such a matrix with its only entry and Part (iii) of the problem asks us to identify this entry.

In Part (ii), even if we had calculated the matrix U explicitly, there was considerable more work to do in the brute force method to find the sum of all the entries of U^{-1}. In the present part, on the other hand, if we have calculated U already, then the best way to get the answer is to carry out the multiplications. Thus,

$$[3 \ \ 2 \ \ 0]\ U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \quad = \quad [3 \ \ 2 \ \ 0] \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$

$$= [-1 \;\; 4 \;\; 4] \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$
$$= -3 + 8 = 5 \tag{84}$$

In the last two parts, we could avoid the computation of the matrix U by resorting to some elegant tricks. There seems to be no way to do so in the present part. The only thing we can possibly try is to write U as $A^{-1}B$ by (11). Then reasoning as in the elegant solution to Part (ii), the desired number, say T, equals

$$T = [3 \;\; 2 \;\; 0]\, U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = [3 \;\; 2 \;\; 0]\, A^{-1}B \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$
$$= LM \tag{85}$$

where

$$L = [3 \;\; 2 \;\; 0]\, A^{-1} \tag{86}$$
$$\text{and} \quad M = B \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \tag{87}$$

As we already know B by (68), computation of M is immediate.

$$M = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \\ 0 \end{bmatrix} \tag{88}$$

The computation of L is also equally easy. But first we must find A^{-1}. As A is a lower triangular matrix, its inverse can be found by a method which is analogous to the one we used in Part (ii) for finding the inverse of B which was upper triangular. In fact, we now use the comments made there after finding B^{-1} to start with a considerably simplified format for A^{-1}. Since we already know that A^{-1} is going to be lower triangular and also that its diagonal entries will be the reciprocals of the corresponding diagonal entries of A, right at the start, we take A^{-1} in a highly simplified form, viz. $\begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$ where we have only three unknowns, viz. a, b and c. Then the requirement $AA^{-1} = I$ spells out as

$$\begin{bmatrix} 1 & 0 & 0 \\ 2+a & 1 & 0 \\ 3+2a+b & 2+c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{89}$$

from which we get $a = -2, c = -2$ and $b = 1$. Therefore

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \tag{90}$$

We are now in a position to compute L using (86), giving

$$L \;=\; [3\;\;2\;\;0]\,A^{-1} = [3\;\;2\;\;0]\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = [-1\;\;2\;\;0] \qquad (91)$$

Putting (91) and (88) into (85), we finally get

$$T = LM \;=\; [-1\;\;2\;\;0]\begin{bmatrix} 7 \\ 6 \\ 0 \end{bmatrix} = -7 + 12 = 5 \qquad (92)$$

which is the same answer as before. Unlike in Parts (i) and (ii) of the problem, in the present part the difference between the two approaches (one with finding U and the other without finding it explicitly) is not very dramatic. The fact that the row vector and the column vector appearing in the expression are transposes of each other would have been helpful had the matrix U been symmetric. In particular, this would have been the case if A^{-1} equalled the transpose of B (or, equivalently, B^{-1} were the transpose of A). For, in that case, the expression asked would have been of the form $[3\;\;2\;\;0]\;B^tB\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ which means the row vector L above would have been the transpose of the column vector M. But in that case LM would have been simply the sum of the squares of the entries of M. Summing up, if B^{-1} (which we calculated in (81)) had come out to be the transpose of A (which is given to us in the preamble), then there would indeed have been a significant short cut to the answer. (It is possible that the data in the original problem was so designed as to make U symmetric, but some of the figures got changed in its retrieval.)

The simplifications in the second part of the problem were due to the fact that that the matrix U could be expressed as $A^{-1}B$ where the matrices A^{-1} and B are, respectively, lower and upper triangular. Such a factorisation of a square matrix is called a LU**- decomposition** of it. Here L stands for 'lower triangular' and U for 'upper triangular'. If the matrix is symmetric then the two factors can further be chosen to be the transposes of each other. It takes some work to find an LU-decomposition of a matrix. But once it is obtained, it considerably simplifies the solution of systems of linear equations.

We conclude the discussion of the role of matrices in systems of linear equations with a problem where the solutions of a system of linear equations are given to satisfy a certain condition and the problem asks what properties of the coefficient matrice are implied by these conditions.

Problem 8: A is a 3×3 matrix and $\vec{u}$ is a column vector. If $A\vec{u}$ and $\vec{u}$ are orthogonal for all real $\vec{u}$, then prove that A is skew-symmetric and singular. (JEE 2006.)

Solution: Let

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \quad \text{and} \quad \vec{u} \quad = \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{93}$$

Then by a direct computation,

$$A\vec{u} = \begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} \tag{94}$$

Orthogonality of $A\vec{u}$ and $\vec{u}$ means that their dot product vanishes, i.e.

$$x(a_1x + b_1y + c_1z) + y(a_2x + b_2y + c_2z) + z(a_3x + b_3y + c_3z) \quad = \quad 0$$

or, after simplification,

$$a_1x^2 + b_2y^2 + c_3z^2 + (a_2 + b_1)xy + (c_2 + b_3)yz + (c_1 + a_3)zx = 0 \tag{95}$$

If this is to hold for all x, y, z, we claim that the coefficient of every term must vanish. To see this, put $x = 1$ and $y = z = 0$. Then we get $a_1 = 0$. Similarly we get $b_2 = 0$ by putting $y = 1$, $x = z = 0$, and $c_3 = 0$ by putting $z = 1$, $x = y = 0$. Hence (95) now reduces to

$$(a_2 + b_1)xy + (c_2 + b_3)yz + (c_1 + a_3)zx = 0 \tag{96}$$

for all x, y, z. We now put $x = y = 1$ and $z = 0$ to get $a_2 + b_1 = 0$. In a similar manner we show that the coefficients $c_2 + b_3$ and $c_1 + a_3$ vanish.

Thus we have shown that $a_1 = b_2 = c_3 = 0$ and $a_2 = -b_1, c_2 = -b_3, a_3 = -c_1$. Therefore the matrix A looks like

$$A = \begin{bmatrix} 0 & b_1 & c_1 \\ -b_1 & 0 & c_2 \\ -c_1 & -c_2 & 0 \end{bmatrix} \tag{97}$$

It is immediate that A is skew-symmetric. Also, its determinant comes out to be 0. So it is singular too. (More generally, one can show that for odd n, every skew-symmetric matrix of order n is singular. For, if A is such a matrix, then by definition, A^t, i.e. the transpose of A, equals $-A$. Since every matrix has the same determinant as its transpose, we get that A and $-A$ have the same determinant. But $|-A| = (-1)^n|A| = -|A|$ as n is odd. So, $|A| = -|A|$, which means $|A| = 0$.)

If one wants, the first part (that of showing the skew-symmetry of A) can also be done more elegantly. The orthogonality of $A\vec{u}$ with $\vec{u}$ can be paraphrased to say that $\vec{u}\ {}^tA\vec{u} = 0$. Taking transposes of both the sides, $\vec{u}\ {}^tA^t\vec{u} = 0$ and hence $\vec{u}\ {}^tP\vec{u} = 0$ for all $\vec{u}$, where $P = A + A^t$. Symmetry of P then forces it to vanish. (The proof of this is analogous to the proof that all the coefficients in (95) vanish.)

Note that the L.H.S. of (95) is simply $\vec{u}\ {}^tA\vec{u}$ where A and $\vec{u}$ are as in (93). It is a homogeneous polynomial of degree 2 in the three variables x, y, z. Such expressions are called **quadratic forms**. They have many applications, including classification of conics. And matrices, especially the symmetric ones, play an invaluable role in their analysis.

EXERCISES

A1.1 Prove that the characteristic roots of a diagonal matrix are precisely its diagonal elements.

A1.2 Let $C = AB$ where A, B are square matrices of order n. Let S, T, U be the linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^n$ induced by A, B, C respectively. Prove that $U = S \circ T$, i.e. that U is the composite of S and T.

A1.3 In Problem 4, what is the linear transformation induced by the matrix P? What would be the matrix P^{2005}?

A1.4 Let A and B be 3×3 matrices of real numbers, where A is symmetric, B is skew-symmetric and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^t = (-1)^k AB$, where $(AB)^t$ is the transpose of the matrix AB, and k is an integer, then find the possible values of k. (2008)

A1.6 If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ are two matrices, then determine those values of α for which $A^2 = B$. (2003)

A1.6 Consider the system of equations

$$\begin{aligned} x - 2y + 3z &= -1 \\ -x + y - 2z &= k \\ x - 3y + 4z &= 1 \end{aligned}$$

and consider the following two statements:

STATEMENT - 1 : The system of equations has no solution for $k \neq 3$.

STATEMENT - 2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

Determine whether

(i) STATEMENT - 1 is true,

(ii) STATEMENT - 2 is true and

(iii) STATEMENT - 2 is a correct explanation of STATEMENT - 1.

(2008)

Appendix 2

Solid Coordinate Geometry

Traditionally, solid geometry is the study of the geometric properties of objects in the three dimensional space. It is a natural sequel to the plane geometry, which is the study of the geometric properties of objects in the two dimensional space. The basic ideas of the two are the same. But solid geometry is in general more complex. One of the difficulties is the inherent impossibility of drawing an accurate diagram of a solid object on a plane surface such as a paper or a blackboard. Some distortions are inevitable. For example, when you draw a picture of a right circular cylinder or a right circular cone, the boundary circles appear as ellipses rather than as circles. Certain standard tricks such as perspectives, shadings and dotted lines are often used to give the impression that the object sketched is a solid object. Sometimes cross sections of objects by certain planes are shown. It takes a certain cultivation to correctly interpret plane diagrams of solid figures and drawing such diagrams is a science in itself.

These things are beyond the scope of JEE. In the past, 'pure' solid geometry was included in the JEE syllabus along with what is called **mensuration**, i.e. the calculation of the areas and volumes of various space figures such as cones, cylinders, tetrahedra and spheres. In those days, calculus was not a part of the syllabus and so the formulas for the areas and volumes of standard figures were assumed without proof. Nowadays, calculus is included in the syllabus but mensuration is not. Also, just as the plane geometry now included in the JEE syllabus is using coordinates, the same holds for the solid geometry which is recently introduced after a long gap. Explicitly the syllabus is as follows:

"Direction cosines and direction ratios, equation of a straight line in space, equation of a plane, distance of a point from a plane."

We have already introduced direction cosines and direction ratios of a line in Exercise (21.28) of the Chapter 21 on vectors. Standard texts on solid coordinate geometry often try to do things without vectors. But as we have seen many times in Chapter 21, the use of vectors considerably simplifies both the formulas and their derivations. The cross product, which is a peculiarity of three dimensional vectors comes especially handy. In Exercise (21.15), for example, we showed how the shortest distance between two given lines can be found using the cross product. Similarly, the formula for the volume of a paralleopiped can be expressed most succinctly in terms of the scalar triple product of the three vectors representing the three edges meeting at any vertex of the parallelopiped. (Indeed, as we remarked, for this reason, the scalar triple product is popularly called the box product.)

In this appendix, we shall use vectors to derive some standard formulas in solid coordinate geometry. We shall illustrate their applications through problems, some of which are taken from the JEE questions of recent years (specifically, from 2003 to 2008).

We begin by obtaining a condition for two straight lines, say L_1 and L_2 in space to meet at a common point. In plane coordinate geometry, each line is represented by a (linear) equation and the problem amounts to finding the condition for these two equations to have a common solution. This is a purely algebraic problem. For the sake of the record, suppose the equation of the line L_i is

$$a_i x + b_i y + c_i = 0 \quad (i = 1, 2) \tag{1}$$

Then the two lines meet at exactly one point if the determinant $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is non-zero. If this deteminant vanishes, then the two lines are either identical or parallel, depending upon whether one equation is a multiple of the other or not (which, too, can be paraphrased in terms of the determinants $\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ and $\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$ if one wants).

In space, there are two complications. First, there is the additional possibility that the two lines L_1 and L_2 may not lie in the same plane. Such lines are said to form a **skew** pair. As examples, oposite edges of any tetrahedran always lie along skew lines. Similarly a diagonal of the base of a room and a diagonal of the ceiling are either parallel or skew. A road and an overbridge (considered as portions of straight lines) provide yet another example. Another complication is that in the space, a single linear equation in the coordinates represents a plane and not a straight line. We need two equations to specify a line. Each equation represents a plane and the line is the intersection of these two planes. These equations can be chosen in infinitely many ways because there are infinitely many planes containing a given line. So, these equations are somewhat extraneous as far as the properties of the line are concerned. As a result, in the space it is customary to specify a line by specifying a point on it and some vector parallel to it (or equivalently, the direction numbers of the line). If $\mathbf{a}$ is the position vector of a (fixed) point P_0 on the line L and $\mathbf{u}$ is a vector parallel to L, then the position vector of a typical point P on L is of the form

$$\mathbf{r} = \mathbf{r}(t) = \mathbf{a} + t\mathbf{u} \tag{2}$$

where t is a real parameter. For $t = 0$ we get the point P_0.

We can convert (2) to a system of three equations in the cartesian coordinates (x, y, z) of P by resolving both the sides along the three basic vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$. Letting $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$, we get

$$x = a_1 + tu_1, \;\; y = a_2 + tu_2, \;\; z = a_3 + tu_3 \tag{3}$$

These are precisely the parametric equations[2] analogous to the parametric equa-

[2]Traditionally, it is customary to rewrite these in what is called the **symmetric form** as $\dfrac{x - a_1}{u_1} = \dfrac{y - a_2}{u_2} = \dfrac{z - a_3}{u_3}$.

tions of a line in a plane. The difference is that in the plane they are used only sparingly (it being more common to express the line by a linear equation in x and y). But in higher dimentional spaces it is far more common and convenient to express a line in a parametric form.

So, suppose now that we have two lines L_1 and L_2 with vector equations

$$\mathbf{r} = \mathbf{a} + t\mathbf{u} \quad \text{and} \quad \mathbf{r} = \mathbf{b} + s\mathbf{v} \tag{4}$$

respectively. Let us find a necessary and sufficient condition for these two lines to have a common point. This can be obtained very succinctly by observing that the shortest distance between two lines is 0 if and only if they intersect. In the solution to Exercise (21.15), we saw that the shortest distance, say d, between the two lines above is given by the projection of the vector $\mathbf{a} - \mathbf{b}$ on the vector $\mathbf{u} \times \mathbf{v}$ (which is perpendicular to both the lines). That is,

$$d = \frac{|(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{u} \times \mathbf{v})|}{||\mathbf{u} \times \mathbf{v}||} \tag{5}$$

(It is assumed here that $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$ as otherwise $\mathbf{u}$ and $\mathbf{v}$ would be parallel. In that case, the two lines will be identical or parallel depending upon whether $\mathbf{a}-\mathbf{b}$ is a scalar multiple of $\mathbf{u}$ or not.) As the shortest distance is 0 for intersectiong lines, we get

$$(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{u} \times \mathbf{v}) = 0 \tag{6}$$

as the criterion for the lines to meet. An alternate derivation (without using the formula for the shortest distance between lines) is to note that the two lines meet if and only if the two equations in (4) have a common solution, i.e. if and only if there exist t, s such that

$$\mathbf{a} + t\mathbf{u} = \mathbf{b} + s\mathbf{v} \tag{7}$$

Taking the dot product of both the sides with the vector $\mathbf{u} \times \mathbf{v}$, we get

$$\mathbf{a} \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{b} \cdot (\mathbf{u} \times \mathbf{v}) \tag{8}$$

which is the same as (6). Note that this only gives a condition for the two lines to intersect, but not the point of intersection. To find the latter, we have to solve the vector equation

$$\mathbf{a} + t\mathbf{u} = \mathbf{b} + s\mathbf{v} \tag{9}$$

for s and t, assuming that (8) holds. This amounts to expressing the vector $\mathbf{a}-\mathbf{b}$ (which is already known to be perpendicular to $\mathbf{u} \times \mathbf{v}$) as a linear combination of $\mathbf{u}$ and $\mathbf{v}$. Here we can use the method given in Comment No. 13 of Chapter 21, after noting that $(\mathbf{u}, \mathbf{v}, \mathbf{u} \times \mathbf{v})$ is a semi-orthogonal basis.

Returning to the condition for intersection of L_1 and L_2, (6) can be paraphrased algebraically in terms of the components of the vectors $\mathbf{a}, \mathbf{b}, \mathbf{u}$ and $\mathbf{v}$.

suppose that the vectors involved are $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$. Then (6) is equivalent to

$$\begin{vmatrix} a_1 - b_1 & a_2 - b_2 & a_3 - b_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = 0 \tag{10}$$

From this we get an easy solution to the following problem (JEE 2004).

Problem 1: If the lines $\dfrac{x-1}{2} = \dfrac{y+1}{3} = \dfrac{z-1}{4}$ and $\dfrac{x-3}{1} = \dfrac{y-k}{2} = \dfrac{z}{1}$ intersect, then find the value of k.

Solution: All we have to do is to put the numerical values of the entries of the determinant above. Thus, k is given by the equation

$$\begin{vmatrix} 2 & k+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

A direct expansion of the determinant gives $-10 + 2(k+1) - 1 = 0$ from which $k = \frac{9}{2}$.

So far, we assumed that both the lines were given in the parametric form. When one or both of them are given as the intersection of two planes, one method is to first convert such a line into a parametric form. Suppose, for example, that the line L_1 is the intersection of the planes with equations

$$\left.\begin{array}{rrcl} & A_1x + B_1y + C_1z + D_1 & = & 0 \\ \text{and} & A_1'x + B_1'y + C_1'z + D_1' & = & 0 \end{array}\right\} \tag{11}$$

with L_2 still being given parametrically, sa in the second equation in (4). We can recast the system (10) into parametric form. For this we need to identify the vectors $\mathbf{a}$ and $\mathbf{u}$. The vector $\mathbf{u}$ is easier to identify. It is a vector parallel to the line L_1. The vectors $A_1\mathbf{i} + B_1\mathbf{j} + C_1\mathbf{k}$ and $A_1'\mathbf{i} + B_1'\mathbf{j} + C_1'\mathbf{k}$ are normal to the planes represented respectively by the first and the second equations of the system (10). As the line L_1 is perpendicular to both of these, their cross product gives the vector $\mathbf{u}$. So we can take

$$\mathbf{u} = (B_1C_1' - C_1B_1')\mathbf{i} + (C_1A_1' - A_1C_1')\mathbf{j} + (A_1B_1' - B_1A_1')\mathbf{k} \tag{12}$$

To get $\mathbf{a}$ we need to identify any one point on L_1, i.e. any one solution of the system (10). This can be done by inspection. We can give one of the three variables an arbitray value (usually 0) and solve the system for the remaining two variables. (See the first solution to Problem 7 below for an illustration of this procedure.) We may get different vectors $\mathbf{a}$ for different values. But that is all right since the corresponding points all lie on the line L_1. After getting L_1 in the parametric form, we can apply (4) to get the condition for the lines L_1 and L_2 to have a common point.

But there is an easier way. We simply substitute the parametric equations of L_2 into (10). Specifically, we put $x = b_1 + sv_1, y = b_2 + sv_2, z = b_3 + sv_3$ into (10) to get

$$\left.\begin{array}{rrcl} & A_1(b_1+sv_1) + B_1(b_2+sv_2) + C_1(b_3+sv_3) + D_1 & = & 0 \\ \text{and} & A_1'(b_1+sv_1) + B_1'(b_2+sv_2) + C_1'(b_3+sv_3) + D_1' & = & 0 \end{array}\right\} \quad (13)$$

The required condition can now be obtained by eliminating s from these two equations. In the determinant form the condition for L_1 and L_2 to intersect is

$$\begin{vmatrix} A_1b_1 + B_1b_2 + C_1b_3 + D_1 & A_1v_1 + B_1v_2 + C_1v_3 \\ A_1'b_1 + B_1'b_2 + C_1'b_3 + D_1' & A_1'v_1 + B_1'v_2 + C_1'v_3 \end{vmatrix} = 0 \quad (14)$$

It is hardly a good idea to remember this. It is far better to remember the reasoning and work from it fresh in a particular numerical problem. This is a healthy habit which also enables you to tackle other related problems. For example, let P_1 and P_1' be the planes representing the first and the second equations, respectively, of (11). Then the value of s which we get from the first equation in (13) will give us the point of intersection of the plane P_1 with the line L_2. Thus we now know how to find the point of intersection of a line (given parametrically) with a plane (given by a single equation). We see further that if the coefficient of s (viz. $A_1v_1 + B_1v_2 + C_1v_3$) is non-zero then there will be a unique point of intersection. But if it is 0, then depending upon whether the constant term (viz. $A_1b_1 + B_1b_2 + C_1b_3 + D_1$) vanishes or not the line L_2 will be contained in the plane P_1 or parallel to it. Note that the expression $A_1v_1 + B_1v_2 + C_1v_3$ can be written as $\mathbf{n}\cdot\mathbf{v}$ where the vector $\mathbf{n} = A_1\mathbf{i} + B_1\mathbf{j} + C_1\mathbf{k}$ is normal to the plane P_1. This is consistent with the fact that a line is parallel to (or contained in) a plane if and only if it is perpendicular to the normal to the plane.

So far, we studied the condition for two lines L_1 and L_2 to intersect when both were given parametrically or when one was given parametrically and the other as the intersection of two planes. What happens when both the lines are given as intersections of planes? Suppose, for example, that in addition to (11), the line L_2 is also given as the intersection of two planes, say,

$$\left.\begin{array}{rrcl} & A_2x + B_2y + C_2z + D_2 & = & 0 \\ \text{and} & A_2'x + B_2'y + C_2'z + D_2' & = & 0 \end{array}\right\} \quad (15)$$

Once again, we can recast both L_1 and L_2 in the prarametric form and then get the condition for them to intersect in terms of the 16 constants A_1 to D_2'. But we look for a direct derivation. Note that any point common to L_1 and L_2 will satisfy all the four equations, two each in (11) and (15). So, the two lines have a point in common, then these four equations in three unknowns x, yz are consistent. Eliminting them we get

$$\begin{vmatrix} A_1 & B_1 & C_1 & D_1 \\ A_1' & B_1' & C_1 & D_1' \\ A_2 & B_2 & C_2 & D_2 \\ A_2' & B_2' & C_2' & D_2' \end{vmatrix} = 0 \quad (16)$$

as the necessary and sufficient condition for the two lines to meet. This is a very quick test. Unfortunately, it does not readily tell what happens further. That is, if the four equations do have a common solution, then the intersection of the corresponding four planes has at least one point in it. But it could be bigger, e.g. a straight line, or even a plane (in the extreme case when all the four equations represent the same plane). Similarly, if the condition above fails, this can happen in a variety of cases. Two of the four planes could be parallel to each other. Or, the lines of intersections could form a parallel or a skew pair.

Resolving which possibility holds can be done by considering the determinants of various submatrices of the corfficient matrix above. But that runs into a rather large number of cases. Instead of doing it in general, we solve the following miniature problem (JEE 2007) where we consider the intersection of three planes (instead of four) all passing through the origin.

Problem 2: Consider the following linear equations

$$ax + by + cz = 0$$
$$bx + cy + az = 0$$
$$cx + ay + bz = 0$$

Column I lists some conditions on a, b, c while **Column II** lists some statements about what the equations represent. Match each condition with its implication(s).

	Column I		**Column II**
(A)	$a+b+c \neq 0$ and $a^2+b^2+c^2 = ab+bc+ca$	(p)	the equations represent planes meeting only at a single point
(B)	$a+b+c = 0$ and $a^2+b^2+c^2 \neq ab+bc+ca$	(q)	the equations represent the line $x = y = z$
(C)	$a+b+c \neq 0$ and $a^2+b^2+c^2 \neq ab+bc+ca$	(r)	the equations represent identical planes
(D)	$a+b+c = 0$ and $a^2+b^2+c^2 = ab+bc+ca$	(s)	the equations represent the whole of the three dimensional space

Solution: Each of the three given equation represents a plane passing through the origin in the three dimensional space. But this plane degenerates into the whole space when the coefficients a, b, c are all 0. Except for this case, the three planes will meet either at a single point or on a single line or they may represent identical planes (e.g. when $a = b = c \neq 0$). These four possibilities correspond to the entries in **Column II** and the problem asks you to decide which possibility holds when. The crucial observation is that the determinant

of the coefficients of the system has a well-known factorisation. Specifically,

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3$$

$$= -(a+b+c)(a^2+b^2+c^2-ab-bc-ca) \qquad (16)$$

As there are two factors on the R.H.S., logically there are four possibilities depending upon which of these factors vanishes. These four possibilities correspond to the entries in **Column I**. When both the factors are non-zero, so is the determinant. Hence the system has a unique solution, which geometrically means that the three planes meet only at the origin. Thus we see that (C) implies (p).

In all the other three cases the determinant vanishes, and so there will be some degeneracy. Note that the second factor in the R.H.S. of (1) can be written as

$$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2] \qquad (17)$$

As a, b, c are all real, the expression in the brackets vanishes when and only when $a = b = c$. So in (A) and (D), we are given that $a = b = c$. This makes the three equations identical. In fact, each is the equation $a(x+y+z) = 0$. Moreover in this case $a+b+c = 3a$. In (A) we have $a \neq 0$ and so all three equations represent the same plane, viz. $x+y+z = 0$. However, in (D), we have $a = 0$ and so every equation degenerates to $0x+0y+0z = 0$ which is an identity satisfied by all points of the three dimensional space. Therefore, (A) implies (r) while (D) implies (s).

(B) is the most interesting case. Here we have $a+b+c = 0$. As a result, any point which satisfies any two of the three equations will automatically satisfy the third one. So, we focus our attention to the solution set of any two of the equations, say the first two equations. Since a, b, c are not all equal, at least two of them are non-zero and so certainly, each equation represents a (non-degenerate) plane. We claim that the first two equations represent distinct planes. For otherwise, their coefficients will be proportional and as a result we shall have $b^2 = ac, c^2 = ab$ and $a^2 = bc$ and hence $a^2+b^2+c^2 = ab+bc+ca$, a contradiction to the second part of (B). Therefore in (B), the three equations together represent some straight line, say L in space. That this line is, in fact, the line $x = y = z$ follows any point which satisfies $x = y = z$ satisfies all three equations since $a+b+c = 0$. Therefore, the line L surely contains all points on the line $x = y = z$. Therefore the two lines must be the same.

Those who are familiar with a little bit of linear algebra and in particular with the concept of the rank of a matrix (introduced on p. 109) can do the problem more efficiently. Let A be the 3×3 matrix of the coefficients of the given system of the three linear equations in three unknowns x, y, z. That is,

$$A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \tag{18}$$

In (16) we computed the determinant of A. This matrix A defines what is called a linear transformation from the three dimensional vector space $\mathbb{R}^3$ to itself. The set of solutions of the system constitutes a certain subspace, say V, of the domain space $\mathbb{R}^3$. (This subspace is often called the **null space** of the linear transformation, because it consists of those points where the linear transformation assumes the value 0, i.e. the null value. The dimension of the null space is called the **nullity** of the linear transformation.) In the present case, since the null space V is a subspace of $\mathbb{R}^3$ its possible dimensions are $0, 1, 2$ and 3 and these correspond to the entries in **Column II** in that order. (In fact, this explains why the entries in **Column II** are ordered in a particular manner.

There is a standard theorem of linear algebra, often called the **rank nullity theorem** which says that the nullity, that is, the dimension of the solution space V is $n - r$ where n is the dimension of the domain vector space (in our problem $n = 3$) and r is the rank of the matrix A. The rank can be defined intuitively as the maximum number of independent or 'really different' equations in the system. (For example, if one of the equations is a linear combination of some others, then it is superfluous and does not contribute to the rank.) The truth of this theorem is intuitively obvious because every linear equation puts one constraint on the coordinates x, y, z and thereby reduces the dimension of the solution space by 1, but a superfluous equation does not decrease the solution space and hence does not reduce the dimension.

This theorem is of little use unless we have some way to determine the rank of a matrix. There is a standard method for this using what is called **row reduction** of the matrix. But it will take us too far afield to discuss it. Instead we resort to the definition given in the footnote on p. 109, according to which the rank r of A is the size of the largest square submatrix of A with a non-zero determinant. This definition is generally not very convenient because there are so many square submatrices of a given matrix and it is hardly practical to evaluate their determinants. But in the present case since the only possible values of r are 0,1, 2 and 3, such a determination is not so difficult.

Suppose the 3×3 matrix A is itself non-singular. By (16), this corresponds to the entry (C) in **Column I**. In this case the rank r equals 3 and so by the rank nullity theorem, the solution space V has dimension $3 - 3 = 0$. In other words it consists of a single point, viz. the origin. This is the possibility (p) in the second column. At the other extreme, we have a situation where the rank is 0. This would mean that the coefficients a, b, c are all 0 (because they can be equated with the determinants of the 1×1 submatrices of A.) In this case both the factors of the R.H.S. of (1) vanish and so we are in (D) of the first column. Here the solution space has dimension $3 - 0 = 3$ which corresponds to (s) in **Column II**.

The two remaining cases are $r = 1$ and $r = 2$. To say that A has rank 1 means that at least one of the coefficients a, b, c is non-zero but that the determinant of any 2×2 submatrix vanishes. By inspection, this is equivalent to saying that the expressions $a^2 - bc, b^2 - ac$ and $c^2 - ab$ all vanish. Hence $a^2 + b^2 + c^2 = ab + bc + ca$. But, as noted before, that forces $a = b = c$ and that puts us in (A) in **Column I**. Hence (A) implies (r).

Finally, suppose A has rank 2. This means A is singular but at least one of the expressions $a^2 - bc, b^2 - ac$ and $c^2 - ab$ is non-zero. This also implies that $a^2 + b^2 + c^2 \neq ab + bc + ca$ because otherwise we shall have $a = b = c$ which would mean all these three expressions vanish. Since the R.H.S. of (1) vanishes while the last factor does not, we must have $a + b + c = 0$. So we are in (B) in **Column I**. By the theorem quoted above, the solution space V has dimension 1. So, it is a straight line L. That this line is $x = y = z$ is proved the same way as in the earlier solution.

This is really a very good problem requiring knowledge of solutions of systems of linear equations, a little bit of solid coordinate geometry and some algebraic manipulations. Note that the fact that the coefficients are real was vitally used in both the solutions. Without it, the condition $a^2 + b^2 + c^2 = ab + bc + ca$ will not imply $a = b = c$. (The rank nullity theorem cited above also holds for complex vector spaces. But in the solution based on it we also need something more.)

It may be argued that those who are exposed to the rank nullity theorem (which is not a part of the JEE syllabus) will have an unfair advantage. But nothing can be done about it in an examination where all that matters is whether a candidate has got the right answer rather than how he has obtained it. In the past some candidates appeared for JEE more than twice while simultaneously studying more advanced mathematics. So the possibility of some of them being exposed to the rank nullity theorem was a real one. Now that a candidate can appear for JEE only twice, it is unlikely that he has been exposed to this theorem in a regular course work. And, if he has acquired it all on his own, why not reward him?

Although the complete resolution of all possibilities of the solution set of a system of linear equations runs into a large number of cases, in a particular numerical problem, where only one possibility is of interest, the work can be tailor made to cover only that possibility. Moreover, this can often be done in a variety of ways. The following problem (JEE 2003) is a good illustration.

Problem 3: Determine the value(s), of k for which the line

$$\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$$

lies completely in the plane $2x - 4y + z = 7$.

Solution: The line is given in the parametric form. Equating each ratio with r, a typical point on the line is of the form $(r + 4, r + 2, 2r + k)$. This will lie in the given plane if and only if $2(r + 4) - 4(r + 2) + (2r + k) = 7$. This gives $k = 7$.

Here the coefficient of r in the equation we got was 0. If it were non-zero, then there is no value of k for which the line will lie completely in the given plane. In that case, for every k the equation will give the point of intersection of the given line with the given plane. In fact, this shows that one cannot get the answer simply by checking whether the point $(4, 2, k)$ (which surely lies on the given line) lies in the given plane. Doing so does give $k = 7$. But it leaves open the possibility that the line may intersect the plane at $(4, 2, 7)$. However, after noting that the point $(4, 2, 7)$ is common to the given line and the given plane, there are other ways to determine if the line lies entirely in the plane. One method is to observe that a line lies completely in a plane as soon as any *two* points on it do so. So, if we can find some point on the line, other than $(4, 2, 7)$, which lies in the given plane, we are through. One such point is obtained by taking each ratio, i.e. r to be 1. This gives $(5, 3, 9)$ as another point on the line and it is easy to check directly that its coordinates satisfy the equation of the plane.

Yet another method is to write the given line as the intersection of any two planes, e.g. the planes $x - 4 = y - 2$ and the plane $2(y - 2) = z - 7$. These two equations along with the equation of the given plane give a non-homogeneous system of three linear equations in three unknowns x, y, z. We already know that it has one solution. To say that the line lies completely in the plane is equivalent to saying that the corresponding homogeneous system has infinitely many solutions. And this can be done by the determinant criterion applied in Q. 5 above. In fact, this method is better when the line is given as the intersection of two planes and not in the parametric form. Of course, in such a case, if one wants, one can first write the line in a parametric form.

One more method is provided by vectors. From the data, it is clear that the vector, say $\mathbf{u} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$, is parallel to the given line. Also, the vector, say $\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ (obtained from the equation of the plane), is normal to the given plane. Taking dot product, it is easily checked that $\mathbf{u}$ and $\mathbf{v}$ are perpendicular to each other. Hence the line is perpendicular to the normal to the plane. Therefore it is either in the plane or parallel to it. The second possibility is already excluded.

As compared with the treatment of a straight line in the three dimensional space, a plane is much easier to handle. Just as a single (non-degenerate) linear equation $ax + by + c = 0$ (with a, b not both 0) represents a straight line in the plane, in the space an equation of the form $Ax + By + Cz + D = 0$ where A, B, C are not all zero represents a plane. Just as any two distinct points determine a unique line passing through them, any three non-collinear points, say $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and (x_3, y_3, z_3) determine a unique plane containig them. The equation of this plane is given explictly in the determinant form by

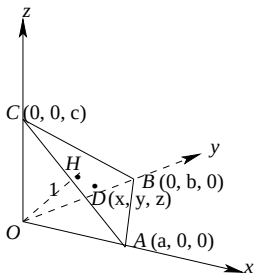
$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \tag{19}$$

The analogy extends further. For example, just as the vector $a\mathbf{i} + b\mathbf{j}$ is

perpendicular to the line $ax+by+c=0$, the vector $A\mathbf{i}+B\mathbf{j}+C\mathbf{k}$ is perpendicular to the plane $Ax+By+Cz+D=0$. The condition for parallelism of two planes and the formula for the distance between parallel lines and that for the distance of a point from a plane are straightforward analogues of the corresponding formulas in plane coordinate geometry. Here is a problem (JEE 2005) whose solution is based on such analogies.

Problem 4: A plane at a distance of 1 unit from the origin cuts the coordinate axes at A, B and C. If the centroid $D(x, y, z)$ of the triangle ABC satisfies the relation $\dfrac{1}{x^2}+\dfrac{1}{y^2}+\dfrac{1}{z^2}=k$, determine the value of k.

Solution: Analogous to the two intercepts formula for the equation of a straight line in a plane, there is the three intercepts formula for the equation of a plane in space. So, if we let $A=(a,0,0), B=(0,b,0)$ and $C=(0,0,c)$, then the equation of the plane is



$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \tag{20}$$

The formula for the perpendicular distance of a point from a plane is the exact analogue of the formula for the distance of a point from a straight line in a plane. As we are given that the perpendicular distance of the origin from the plane given by (20) is 1 unit, we get

$$\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=1 \tag{21}$$

The point $D=(x,y,z)$ is the centroid of the triangle ABC. This gives us three relations :

$$x=a/3, \quad y=b/3 \quad \text{and} \quad z=c/3 \tag{22}$$

Finally, we are given

$$\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=k \tag{23}$$

(22) and (23) together give

$$\frac{9}{a^2}+\frac{9}{b^2}+\frac{9}{c^2}=k \tag{24}$$

But by (21), $\dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}=1$. Hence from (24), we get $k=9$.

There are, however, some differences in the treatment of a line in a plane and that of a plane in space. In plane geometry we frequently talk of the slope of a line, defined in terms of the angle it makes with a fixed line (viz. the x-axis). In the space, we can define the slope of a plane, say P, as the tangent of the angle it makes with the xy-plane. But the trouble is that planes with equal slopes are not parallel to each other unless their lines of intersection with the xy-plane are parallel to each other. So, there is no formula for the angle between two planes in terms of their slopes. The difficulty can be overcome if we replace the slope of a plane by a vector normal to the plane. In the plane, if $ax + by + c = 0$ is a line, then the vector $a\mathbf{i} + b\mathbf{j}$ is perpendicular to the line, while the slope of the line is $-\frac{a}{b}$. As each can be obtained from the other, they convey the same information about the line. Specifically, each tells the direction of the line. Once the direction is known, any one point on the line determines it uniquely. In the space, if the equation $Ax + By + Cz + D = 0$ represents a plane, say P, then the vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is normal to the plane. This vector, along with any one point, say (x_0, y_0, z_0) on the plane determines it uniquely. In fact, the equation of P can as well be written as

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \tag{25}$$

But the slope of the plane P is $\dfrac{C}{\sqrt{A^2 + B^2}}$ and this does not determine the plane even if it is known to pass through a point (x_0, y_0, z_0).

As a result, in the three dimensional space, it is far more common to consider a normal vector to a plane than it is to consider a normal vector to a line in the plane and it is far less common to talk of the slope of a plane than it is to do so for a line in the plane. The normal vector to a plane can be read out instantaneously from its equation. Sometimes we first find the normal vector from the data and then use it to write the equation of the plane, as in the following problem (JEE 2008).

Problem 5: Consider the lines

$$L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2} \quad \text{and} \quad L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$$

Find the perpendicular distance of the point $(1, 1, 1)$ from the plane which passes through the point $(-1, -2, -1)$ and whose normal is perpendicular to the lines L_1 and L_2.

Solution: Call the desired plane as P and let $\mathbf{n}$ be a vector perpendicular to it. The vectors, say $\mathbf{u}_1$ and $\mathbf{u}_2$ parallel to L_1 and L_2 are already given to us, viz.

$$\mathbf{u}_1 = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k} \quad \text{and} \quad \mathbf{u}_2 = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$

As $\mathbf{n}$ is perpendicular to both L_1 and L_2, it is parallel to their cross product $\mathbf{u}_1 \times \mathbf{u}_2$. In fact, since the length of $\mathbf{n}$ does not matter, we may take $\mathbf{n}$ as $\mathbf{u}_1 \times \mathbf{u}_2$.

A direct computation gives

$$\mathbf{n} = \mathbf{u}_1 \times \mathbf{u}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -\mathbf{i} - 7\mathbf{j} + 5\mathbf{k} \tag{26}$$

This plane is to pass through the point $(-1, -2, -1)$. So, by (25), its equation is

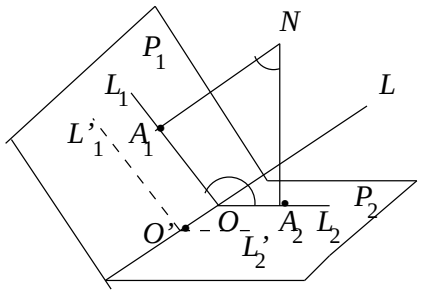
$$-(x+1) - 7(y+2) + 5(z+1) = 0 \tag{27}$$

i.e.

$$-x - 7y + 5z - 10 = 0 \tag{28}$$

The perpendicular distance of the point $(1, 1, 1)$ from this plane comes out as $\dfrac{-1-7+5-10}{1+49+25} = -\dfrac{13}{\sqrt{75}}$. The negative sign indicates that this point lies on the opposite side of the plane as the normal vector we have chosen. If we had taken an oppositely directed normal vector, we would have gotten $\dfrac{13}{\sqrt{75}}$. These are the algebraic distances. The problem asks for the geometric distance, which is always non-negative. So, we take the absolute value and get $\dfrac{13}{\sqrt{75}}$ as the answer.

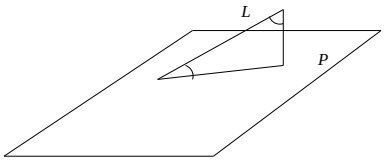
The normals to planes also provide a handy formula for the angle between two planes. The definition of an angle between two planes in space is much more complicated than that between two straight lines in a plane. Suppose the two planes are P_1 and P_2. If they do not intersect then they are parallel and the angle between them is 0. If they intersect, then let L be the line of their intersection. Take any point O on L. Through O, draw lines L_1, L_2 in the planes P_1, P_2 respectively, that are perpendicullar to L (see the figure). These lines are coplanar as they both lie in the plane through O which is perpendicular to L. The angle between these two lines is defined as the angle between the planes P_1 and P_2. (To show that this is well defined, one must show that it is independent of the point O on L which we start with. This is very easy, because if we replace O by another point O' on L, then the corresponding perpendiculars will be parallel to L_1 and L_2 respectively.)



Now let $\mathbf{n}_1, \mathbf{n}_2$ be vectors perpendiculars to the planes P_1 and P_2 respectively. Take any two points A_1 and A_2 (different from O) on the lines L_1 and L_2 respectively. Through A_1 and A_2 draw lines parallel to $\mathbf{n}_1$ and $\mathbf{n}_2$ respectively. Both these lines lie in the plane through O which is perpendicular to the line

L. So, they will intersect at some point, say N, in this plane. The quadrilateral OA_1NA_2 is a quadrilateral in this plane with the angles at A_1 and A_2 both right angles. So the other two angles are supplementary. Thus we see that the angle between two planes is supplementary to the angle between their normals. Actually there are two angles between any two intersecting lines, one acute and the other obtuse. So we may even say that the angle between two planes is the same as the angle between their normals with the understanding that both of them are acute or both obtuse. The advantage gained is that if we know the normal vectors to the two planes, then the angle between them can be found by a simple calculation.

The angle which a line makes with a plane is defined as the angle between that line and its vertical projection on that plane. By a reasoning similar to the above (and, in fact, a simpler one) it follows that the angle between a line and a plane is complementary to the (acute) angle between that line and the normal to the plane.



These simple observations enable us to solve the following problem. It also illustrates how the freedom to choose suitable coordinate system simplifies the work while tackling a problem from pure solid geometry.

Problem 6: Suppose two planes P and Q meet along a line L at an acute angle α. Let β be the acute angle which a line L' in the plane P makes with the plane Q. Prove that $\beta \leq \alpha$. Find when $\beta = \alpha$ and $\beta = 0$.

Solution: Without loss of generality, take Q to be the xy-plane, L to be the y-axis and L' to pass through O. The vector $\mathbf{k}$ is normal to the plane Q. Let $\mathbf{n}$ be a unit normal vector to the plane P. As the y-axis lies in P, $\mathbf{n}$ is perpendicular to the y-axis, or equivalently,

$$\mathbf{n} \cdot \mathbf{j} = 0 \tag{29}$$

We also have that $\mathbf{n}$ makes angle α with $\mathbf{k}$. Reversing the direction of $\mathbf{n}$ if necessary, we may suppose

$$\mathbf{n} \cdot \mathbf{k} = \cos\alpha \tag{30}$$

As $\mathbf{n}$ is a unit vector, (29) and (30) together give $\mathbf{n} = \pm\sin\alpha\mathbf{i} + \cos\alpha\mathbf{k}$. Once again without loss of generality we choose the negative sign so that the equation of the plane P comes in the succinct form

$$z = (\tan\alpha)x \tag{31}$$

Now let A be a point on the line L', other different from O. As L' lies completely inside the plane P, A will be of the form $(a, b, \tan\alpha)$. The projection of the vector $\overrightarrow{OA}$ on the plane Q (which we are taking as the xy-plane) is the vector

$\overrightarrow{OB}$ where B is the point $(a, b, 0)$. Then the angle β between the line L' and the plane Q is the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. So we get

$$\tan\beta = \pm\frac{a\tan\alpha}{\sqrt{a^2+b^2}} \tag{32}$$

where the choice of the sign depends on the sign of a. Taking a to be positive, we have $\tan\beta \leq \tan\alpha$ since $\frac{a}{\sqrt{a^2+b^2}} \leq 1$. This proves the first part. It is also clear from (32) that $\beta = \alpha$ if and only if $b = 0$, i.e. if and only if the line L' is perpendicular to the y-axis, which is the line L of the intersections of the planes P and Q. Finally, $\beta = 0$ if and only if $a = 0$, which, in turn, means that L' is parallel to L.

It is instructive to rewrite (32) slightly. We already know that the line OB is a line in the xy-plane. Let this line make an angle γ with the y-axis. Then $\sin\gamma = \dfrac{a}{\sqrt{a^2+b^2}}$. So, (32) can be rewritten as

$$\tan\beta = \sin\gamma\tan\alpha \tag{33}$$

This is a relation between three angles, the angle α between the two planes which meet along a line L, the angle β which a line L' in one of the planes makes with the other plane and the angle γ which the projection of this line L' on the other plane makes with the line L. Note that although we have used coordinates in deriving (33), the formula itself is purely trigonometrical.

In Comment No. 13 of Chapter 9, we solved a problem in plane coordinate geometry by consideting the one parameter family of lines passing through the point of intersection of two given lines. In space, the family of all straight lines passing though a given point is not a one parameter family, because if the point is $\mathbf{a}$ and $\mathbf{u}$ is a unit vector along a typical member of this family, say $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$, then all we have $u_1^2 + u_2^2 + u_3^2 = 1$ and so any two of the three variables u_1, u_2, u_3 determine the value of the third (except for the sign). But these two variables can be assigned values independently. So the family of all lines through a point is a two parameter family.

However, the family of all planes passing though a given line L (say) is a one parameter family. There are various ways to see this. Fix a plane P_0 containing L. Then any plane P containing L is determined by the angle it makes with P_0. In applications, the line L itself is specified as the intersection of two planes, say P_1 and P_2 with equations of the form $E_1 = 0$ and $E_2 = 0$ where E_1 and E_2 are linear expressions in x, y, z, i.e. expressions of the form $ax + by + cz + d$, where a, b, c, d are real numbers. If λ is a real parameter, then the equation $E_1 + \lambda E_2 = 0$ will always be a plane that will contain the line L. Moreover every plane containing L, can be written in this form, for some value of λ. The only exception is the plane P_2. If we want to avoid such exceptions, we take the equation of a typical member of this family as $(1-\lambda)E_1 + \lambda E_2 = 0$. Now both the planes P_1 and P_2 are included (with $\lambda = 0$ and $\lambda = 1$ respectively).

We put these ideas to use in the following problem (JEE 2005).

Problem 7: Find the equation of the plane containing the line $2x - y + z = 0,\ 3x + y + z = 5$ and at a distance $\frac{1}{\sqrt{6}}$ from the point $(2, 1, -1)$.

Solution: Call the given line as L. Then a point (x, y, z) lies on L if and only if its coordinates satisfy the following two equations:

$$2x - y + z - 3 = 0 \tag{34}$$

$$\text{and} \quad 3x + y + z - 5 = 0 \tag{35}$$

Note that we have recast the second equation slightly so as to make its R.H.S. vanish. The reason for this will be clear later. Each one of these two equations represents a plane and the line L is precisely the intersection of these two planes.

We are after a certain plane containing this line L. One way to tackle the problem would be to take the equation of the desired plane as

$$ax + by + cz + d = 0 \tag{36}$$

and determine the constants a, b, c, d. (Even though superficially there are four constants, what matters is only their relative proportions and so we need a system of three equations to determine them.)

One equation is provided by the condition that the point $(2, 1, -1)$ is at a distance $\frac{1}{\sqrt{6}}$ from the desired plane. This gives,

$$\frac{2a + b - c + d}{\sqrt{a^2 + b^2 + c^2}} = \pm\frac{1}{\sqrt{6}} \tag{37}$$

We need two more equations. These will come from the fact that every point on L lies in the plane. For this, we take the parametric equations of the line L. We first identify any one point, say P, on L by inspection. For example, by setting $y = 0$ in (34) and (35) we get $x = 2, z = -1$. So we may take P as $(2, 0, -1)$. We also need a vector, say $\mathbf{u}$ parallel to L. This is obtained by taking the cross product of the vectors perpendicular to the planes (34) and (35). Thus

$$\mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{vmatrix} = -2\mathbf{i} + \mathbf{j} - \mathbf{k} \tag{38}$$

So, the parametric equations of the line L are

$$x = 2 - 2t,\ y = t,\ z = 5t - 1 \tag{39}$$

Substituting (39) into (36) gives

$$(-2a + b + 5c)t + 2a - c + d = 0 \tag{40}$$

If this equation is to hold for all (real) values of t, then the coefficient of t as well as the constant term must vanish identically. Thus

$$-2a + b + 5c = 0 \tag{41}$$

$$\text{and} \quad 2a - c + d = 0 \tag{42}$$

We now solve these two simultaneously with (37). Putting $b = 2a - 5c$ and $d = c - 2a$ into (37) and squaring gives

$$6(2a - 5c)^2 = a^2 + (2a - 5c)^2 + c^2 \tag{43}$$

This is a homogeneous equation in a and c. As we are interested only in the relative proportions of the constants a, b, c, d we are free to set the value of one of them arbitrarily. Putting $c = 1$ and simplifying we get

$$19a^2 - 100a + 124 = 0 \tag{44}$$

which is a quadratic in a with roots $a = 2$ and $a = \frac{62}{19}$. We have already taken c as 1. From (41), we get $b = -1$ or $\frac{29}{19}$ epending upon whether $a = 2$ or $a = \frac{62}{19}$. Similarly, (42) gives the corresponding values of d as -3 and $-\frac{105}{19}$.

Thus, there are two possible planes satisfying the data and their equations are :

$$2x - y + z - 3 = 0 \tag{45}$$

$$\text{and} \qquad 62x + 29y + 19z - 105 = 0 \tag{46}$$

In this solution, we started with the general equation of a plane and determined the three (superficially four) constants in it. In effect, we considered the totality of all planes in the 3-dimensional space which is a 3-parameter family in the terminology of Comment No. 11 of Chapter 9. We then determined the values of these parameters from the data (viz. that the plane contains the given line L and is at a given distance from a given point).

We now present a more elegant way of doing the problem. The totality of all planes passing through a given line is a 1-parameter family. Let us call the expressions on the L.H.S. of (34) and (35) as E_1 and E_2 respectively. (This is the reason why we recast the equations so that the right hand sides are 0.) Now the equation of a typical plane containing the line L is of the form $(1 - \lambda)E_1 + \lambda E_2 = 0$, i.e.

$$\begin{aligned} (1 - \lambda)(2x - y + z - 3) + \lambda(3x + y + z - 5) &= 0 \\ \text{i.e.} \qquad (\lambda + 2)x + (2\lambda - 1)y + z + (-2\lambda - 3) &= 0 \end{aligned} \tag{47}$$

where λ is a (real) parameter. To determine its value, we equate the distance of the point $(2, 1, -1)$ from this plane with $\frac{1}{\sqrt{6}}$. This gives

$$\frac{2(\lambda + 2) + (2\lambda - 1) - 1 + (-2\lambda - 3)}{\sqrt{(\lambda + 2)^2 + (2\lambda - 1)^2 + 1}} = \frac{1}{\sqrt{6}} \tag{48}$$

which, after squaring both the sides and simplifying, reduces to a surprisingly simple quadratic in λ viz.

$$19\lambda^2 - 24\lambda = 0 \tag{49}$$

The roots of which are 0 and $\frac{24}{19}$. When these values of λ are substituted in (47) we get (45) and (46) as the two possible planes.

This second method is sometimes called the λ-method, because it is customary to denote the parameter by λ, although any other symbol is, of course, equally good. In this method the equation of the typical member of the family is often taken as $\lambda E_1 + E_2 = 0$ or as $E_1 + \lambda E_2 = 0$ instead of as $(1-\lambda)E_1 + \lambda E_2 = 0$ as we have done. This simplifies the calculations a little. But the danger is that in the first approach, there is no value of λ for which $\lambda E_1 + E_2$ equals E_1 while in the second, for no values of λ will $E_1 + \lambda E_2$ equal E_2. So if this plane is one of the answers, that will be missed out. In the prersent problem, for example, if we had taken the equation of a typical plane containing L as $\lambda E_1 + E_2 = 0$, then instead of (47) we would have got

$$(2\lambda + 3)x + (1 - \lambda)y - (1 + \lambda)z - 3\lambda - 5 = 0 \tag{50}$$

and instead of (48) we would have got

$$\frac{2\lambda + 3) + (1 - \lambda) - (\lambda + 1) - 3\lambda - 5}{\sqrt{(2\lambda + 3)^2 + (1 - \lambda)^2 - (1 + \lambda)^2}} = \frac{1}{\sqrt{6}} \tag{51}$$

which, after squaring and simplification becomes $24\lambda = -5$. This is not a quadratic but a linear equation in λ and has $-\dfrac{5}{24}$ as its only root. This value of λ does give (46) as an answer. But it misses out (45) which is also a valid answer. In fact, it is one of the two planes with which we started. So, if one follows this approach, one should, at the end, check if the plane missed out happens to be a solution. A better option is to parametrise the family of planes as $(1 - \lambda)E_1 + \lambda E_2 = 0$, which contains every plane containing the line L, without exception.

We now turn to locus problems. In plane coordinate geometry, the path traced by a moving point $P = (x, y)$ can be described parametrically as

$$x = f(t) \quad \text{and} \quad y = g(t) \tag{52}$$

where t is a parameter. If we eliminate t, we get a single equation in the variables x, y whose solution set is the locus of P. Conversely, the solution set of a single equation in x and y can often be parametrised so as to express it as the path traced by a moving point. Therefore these two approaches are interchangeable.

In the three dimesnional space, things are different. A single equation in x, y, z represents a surface such as a plane, a sphere, a cylinder or a cone. It is not what we would normally consider as a 'path' of a moving point. Nevertheless the term 'locus' is sometimes applied to it. The simplest surface is that represented by a linear equation in the coordiantes. This is a plane. Just as in plane geometry when the locus is a simple curve such as a straight line or a circle, it can often be identified by pure geometry, i.e. without coordinates (see e.g. Exercise (9.24)(viii), the same holds in solid geometry when the locus is a plane as we see in the following problem (JEE 2004).

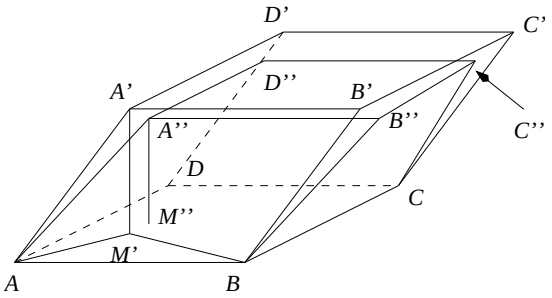
Problem 8: T is a parallelopiped in which A, B, C, D are vertices of one of the faces. The corresponding vertices of the opposite face are A', B', C', D'. T is

compressed to form another parallelopiped S with the face $ABCD$ remaining the same and A', B', C', D' shifted to A'', B'', C'', D'' respectively. If the volume of S is 90% that of T, prove that the locus of A'' is a plane.

Solution: This is one of those problems which take a while to understand but which are very simple once you understand them correctly. Let M' be the foot of the perpendicular from the vertex A' to the plane of the face $ABCD$. Then the volume, say V' of T is given by

$$V' = \text{area of } ABCD \times A'M' \tag{53}$$

When T is compressed to S, the vertices A, B, C, D remain fixed. But the vertices A', B', C', D' move to A'', B'', C'', D'' respectively. Let M'' be the foot of the perpendicular from A'' to the plane of the face $ABCD$. Since the volume of S is 90 % of V', we get



$$\frac{9}{10}V' = \text{area of } ABCD \times A''M'' \tag{54}$$

Dividing (2) by (1) and calling $A'M'$ as h, we get

$$A''M'' = \frac{9}{10}h \tag{55}$$

Here h is a constant. $A''M''$ is the (perpendicular) distance of the point A'' from the base plane $ABCD$. The points B'', C'', D'' are also at this height above the base plane. All these four points lie in a plane parallel to the base plane and at a height $\frac{9}{10}h$ above it. A'' is free to move anywhere on this plane. (Once the position of A'' is fixed, those of B'', C'' and D'' are automatically determined.) Hence the locus of A'' is a plane parallel to the plane of the face $ABCD$ and at a distance $\frac{9}{10}h$ from it, where h is the perpendicular distance between the faces $ABCD$ and $A'B'C'D'$.

In case you have difficulty in understanding a problem in solid geometry, an analogy with a problem in a plane often helps. In the present problem, for example, the plane analogue would be as follows. Suppose $ABB'A'$ is a parallelogram. Its vertices A', B' are moved to points A'', B'' respectively so that $ABB''A''$ is a parallelogram whose area is 90% of that of the parallelogram $ABCD$. Find the locus of A''. Since the area of a parallelogram with a fixed base is proportional to its height, we see that the vertex A'' has to move on a line parallel to AB. The key idea in the present problem is exactly same, except that instead of the area of a parallelogram, we are dealing with the volume of a parallelopiped with one face fixed.

(In the problem asked, the vertex A'' (and hence also the vertices B'', C'' and D'') could also lie on the other side of the face $ABCD$ at a distance $\frac{9}{10}h$ from it. In that case, the locus of A'' is not a single plane, but a union of two parallel planes lying on opposite sides of the plane of the face $ABCD$, each at a distance $\frac{9}{10}h$ from it. But in the statement of the problem, it is given that the original parallelopiped T is 'compressed' to the new parallelopiped S. Physically this means that the volume is gradually reduced from V' to $\frac{9}{10}V'$. In this process the moving vertices cannot possibly cross over to the other side of the plane $ABCD$. So this possibility is discarded.)

The solution given above was based on 'pure' solid geometry, because the gist of the problem was to find the locus of a point whose distance from a fixed plane is a constant and it is cleatr that this is a plane parallel to that plane. The problem can also be done using coordinates or vectors (in which case the volume of the parallelopiped will be the scalar triple product of the vectors representing the three edges meeting at one of the vertices). But neither approach will simplify the problem. In fact, they may only serve to complicate it. The essential idea is very simple, viz., that the volume of a paralleopiped is the product of the area of any face of it and the perpendicular distance between that face and its opposite face. There is no need to complicate it with coordinates or vectors. Indeed, this problem is a good test of a student's ability to grasp the essence of a problem without cluttering it with unnecessary gadgets.

Far more interesting are problems where the locus consists of a curve. To describe a 'curve' in space, there are two ways. The first is to express all the three coordinates of a point on as a function of a single parameter. That is, instead of (52) we have

$$x = f(t), \quad y = g(t) \quad \text{and} \quad z = h(t) \tag{56}$$

where t is a real parameter. We already saw how a straight line in space can be parametrised in this form. Another way to describe a curve in space is to write it as the intersection of two surfaces. These surfaces are not unique. For example, the same line can be expressed as the intersection of any two planes containing it. In fact, even the nature of the two surfaces can vary. For example, a circle can be expressed either as the intersection of a plane and a sphere, or of two spheres or of a right circular cylinder and a plane perpendicular to its axis or of a right circular cone and a plane perpendicular to its axis. Similarly, the nature of the curve depends not only on the two surfaces but also on their relative placement in space. For example, the intersection of a cone and a plane is a conic (by definition). But the nature of this conic depends on how the plane is situated w.r.t. the cone. (A complete classification of all cases was given at the end of Chapter 10.)

When a curve is expressed in the parametric form (56), it is easy to express it as the intersection of two surfaces. One way to do so is to eliminate t in the first two equations to get some relation in x, y, z, say

$$\phi(x, y, z) = 0 \tag{57}$$

Similarly, elimination of t in the second and the third equation will give

$$\psi(x, y, z) = 0 \tag{58}$$

Then the curve can be specified by these two simultaneous equations.

Conversely, suppose a curve C is given as the intersection of two surfaces, say S_1 and S_2 whose equations are given by (57) and (58) respectively and we want to express it in the parametric form (56). We already showed how to do this when both S_1 and S_2 are planes. But, in general, there is no easy way to do this. If one of the two equations, say (57), can be explicitly solved for one of the variables, say z, as

$$z = u(x, y) \tag{59}$$

then we can proceed as follows. (Note that the case where one of the equations is linear, i.e. when one of the surfaces is a plane is a special case of this.) Substituting this into (58) gives

$$\psi(x, y, u(x, y)) = 0 \tag{60}$$

Now, let C' be the vertical projection of the curve C onto the xy-plane, i.e. on the plane $z = 0$. In effect, C' is the intersection of the surface S_2 with the xy-plane. So, we may very well regard C' as a curve in the xy-plane. Its equation is given by

$$\psi(x, y, 0) = 0 \tag{61}$$

which is an equation in just two variables x and y. Suppose we are able to parametrise this plane curve C' in the form (52). Then we merely set $h(t) = u(f(t), g(t))$. Because of (60), for every t, $(f(t), g(t), h(t))$ is a point on the curve C and so (56) is a parametrisattion of the curve C. The success of this method depends on two factors, first, our ability to solve one of the two equations (56) and (57) explicitly for one of the variables, and the second, our ability to parametrise the projection of the curve (which is a two dimensional problem). We skip further illustrations of this procedure as they are beyond the JEE level. Nevertheless, here is a problem of parametrising a curve which is solved using vectors.

Problem 9: An ellipse centred at O with axes $2a$ and $2b$ lies in a plane which makes an angle α with the xy-plane. Suppose that α is also the angle its major axis makes with the xy-plane. Assume further that the vertical projection of its major axis on the xy-plane makes an angle β with the x-axis. Parametrise the ellipse.

Solution: Call the given ellipse as C and its vertical projection on xy-plane as C'. We already know a parametrisation of an ellipse in the standard form. More generally, let $\mathbf{u}$ and $\mathbf{v}$ be unit vectors along the major and minor axis of an ellipse centred at O. Then every point on C can be expressed uniquely

as $a\cos\theta\mathbf{u} + b\sin\theta\mathbf{v}$. So, if we let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and similarly, $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$, then a parametrisation of C is

$$\left.\begin{array}{rcl} x &=& au_1\cos\theta + bv_1\sin\theta \\ y &=& au_2\cos\theta + bv_2\sin\theta \\ z &=& au_3\cos\theta + bv_3\sin\theta \end{array}\right\} \tag{62}$$

Thus the problem is reduced to finding these vectors $\mathbf{u}$ and $\mathbf{v}$.

Let P be the plane of the ellipse C and L be the line of its intersection with the xy-plane. We are given that L passes through O and that P makes an angle α with the xy-plane. As in Problem 6, let L_1 and L_2 be the lines through O in the planes P and the xy-plane respectively, perpendicular to the line L of their intersection. Then the angle between L_1 and L_2 is α. But this is also the angle which the major axis of C makes with the xy-plane. So from the result proved in Problem 6, it follows that the projection of the major axis of the ellipse C lies along L_1 and its vertical projection on the xy-plane lies along L_2. This makes it easy to get hold of the unit vector $\mathbf{u}$. The line L_2 is given to make an angle β with the x-axis. So, a unit vector along L_2 is $\cos\beta\mathbf{i} + \sin\beta\mathbf{j}$. Thus we get that $u_1\mathbf{i} + u_2\mathbf{j}$ is a scalar multiple of $\cos\beta\mathbf{i} + \sin\beta\mathbf{j}$, i.e. that

$$u_1 = \lambda\cos\beta \quad \text{and} \quad u_2 = \lambda\sin\beta \tag{63}$$

for some scalar λ. Further, L_1 makes an angle α with L_2 and hence an angle $90^\circ - \alpha$ with the vector $\mathbf{k}$. This gives

$$u_3 = \sin\alpha \tag{64}$$

As $\mathbf{u}$ is a unit vector, (63) and (64) together give

$$u_1 = \cos\alpha\cos\beta,\ u_2 = \cos\alpha\sin\beta \ \text{ and } \ u_3 = \sin\alpha \tag{65}$$

To determine the unit vector $\mathbf{v}$ along the minor axis, we shall find two vectors perpendicular to it. Already, we have found one, viz. $\mathbf{u}$. We let the normal, say $\mathbf{n}$ to the plane P as the other vector perpendicular to $\mathbf{v}$. Of course we have to identify $\mathbf{n}$. For this again, we use $\mathbf{u} \perp \mathbf{n}$. Note also that $\mathbf{n}$ is perpendicular to the line L which is the intersection of P and the xy-plane. But we already know that the line L_2 in the xy-plane is perpendicular to L and $\cos\beta\mathbf{i} + \sin\beta\mathbf{j}$ is a unit vector along L_2. As L is also in the xy-plane, $\sin\beta\mathbf{i} - \cos\beta\mathbf{j}$ is a vector along L. As L lies in the plane P, we get $\mathbf{n}$ is perpendicular to $\sin\beta\mathbf{i} - \cos\beta\mathbf{j}$. So, we may take

$$\begin{aligned} \mathbf{n} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \sin\beta & -\cos\beta & 0 \\ \cos\alpha\cos\beta & \cos\alpha\sin\beta & \sin\alpha \end{vmatrix} \\ &= -\cos\beta\sin\alpha\mathbf{i} - \sin\beta\sin\alpha\mathbf{j} + \cos\alpha\mathbf{k} \end{aligned} \tag{66}$$

Now that we have found both $\mathbf{u}$ and $\mathbf{n}$, we get

$$\mathbf{v} = \mathbf{u}\times\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\alpha\cos\beta & \cos\alpha\sin\beta & \sin\alpha \\ -\cos\beta\sin\alpha & -\sin\beta\sin\alpha & \cos\alpha \end{vmatrix} = \sin\beta\mathbf{i} - \cos\beta\mathbf{j} \tag{67}$$

(Thus we see that $\mathbf{v}$ is parallel to the line L. As the point O is already on L, we get that the minor axis of the ellipse C lies in the xy-plane. Had we known this beforehand, we could have directly taken $\mathbf{v}$ as $\sin\beta\mathbf{i} - \cos\beta\mathbf{j}$.)

We now have

$$v_1 = \sin\beta,\ v_2 = -\cos\beta \text{ and } v_3 = 0 \tag{68}$$

Substitution from (65) and (68) into (62) completes the parametrisation of the ellipse C.

We conclude with a problem (JEE 2004) which is more like a puzzle. The knowledge of solid geometry needed in it is rudimentary. In particular, no coordinates are needed.

Problem 10: P_1 and P_2 are two planes passing through the origin. L_1 and L_2 are two lines passing through the origin such that L_1 is in P_1 but not in P_2 while L_2 is in P_2 but not in P_1. Show that there exist points A, B, C (none of them being the origin) and a suitable permutation A', B', C' of these points such that

(i) A is on L_1, B is on P_1 but not on L_1 and C is not in P_1, and

(ii) A' is on L_2, B' is on P_2 but not on L_2 and C' is not in P_2.

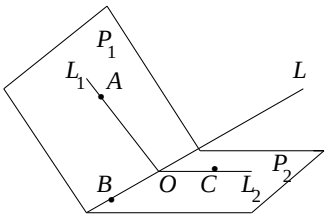
Solution: Let L be the line of intersection of P_1 and P_2. We are given that the origin 0 lies on L. Further the lines L_1, L_2 and L are concurrent at O. Let $\mathbb{R}^3$ be the euclidean (3-dimensional) space. Note that we have two chains of subsets here, each inclusion being strict.

$$\{O\} \subset L_1 \subset P_1 \subset \mathbb{R}^3 \tag{69}$$
$$\{O\} \subset L_2 \subset P_2 \subset \mathbb{R}^3 \tag{70}$$

Conditions (i) and (ii) can be stated as

$$A \in L_1 - \{O\}, B \in P_1 - L_1 \text{ and } C \in \mathbb{R}^3 - P_1 \tag{71}$$
$$A' \in L_2 - \{O\}, B' \in P_2 - L_2 \text{ and } C' \in \mathbb{R}^3 - P_2 \tag{72}$$

respectively. Further we want A', B', C' to be the same points as A, B, C but possibly in a different order. Now A is already in L_1 and hence in P_1 too. We claim that $A \notin P_2$. For, if A were in P_2, then it would lie on $P_1 \cap P_2$ which is the line L. As $A \in L_1$ already, this would mean that $A \in L_1 \cap L = \{O\}$, i.e. A would equal the origin O, a contradiction.



So we must have $A \notin P_2$. Therefore A will have to equal C' since A', B' lie in P_2 by (70) and (72). By a similar reasoning, we must have $A' = C$. Therefore,

by elimination, $B' = B$. This determines the permutation completely. With this knowledge, it is also easy to obtain the desired points. Since $B = B'$, this common point must lie on $P_1 \cap P_2$, i.e. on L. Then take A to be any point on L_1 other than O and C to be any point on L_2 other than O. This ensures $C \notin P_1$ and $C'(= A) \notin P_2$ as required.

One would hardly expect a permutation to figure in a problem in geometry. But a problem like this involves such a rare combination.

EXERCISES

A2.1 A plane passes through the point $(1, 1, 1)$ and is parallel to the lines whose direction ratios are $(1, 0, -1)$ and $(-1, 1, 0)$. If it cuts the coordinate axes at A, B, C respectively, find the volume of the tetrahedron $OABC$. (2004)

A2.2 Find the shortest distance between the lines L_1 and L_2 in Problem 5. (2008)

A2.3 Find the equation of the plane passing through the points $A(1, 2, 0)$, $B(5, 0, 1)$ and $C(4, 1, 1)$. Also determine the point Q such that the line segment joining the points $P(2, 1, 6)$ and Q is perpendicular to the plane and is bisected by it. (2003)

A2.4 Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - \hat{k}$. Find the set of all vectors in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\dfrac{1}{\sqrt{3}}$. (2006)

A2.5 Let $\vec{A}$ be a vector parallel to the line of intersection of the planes P_1 and P_2 through the origin. P_1 is parallel to the vectors $2\hat{j} + 3\hat{k}$ and $4\hat{j} - 3\hat{k}$ and P_2 is parallel to $\hat{j} - \hat{k}$ and $3\hat{i} + 3\hat{j}$. Find the angle between the vectors $\vec{A}$ and $2\hat{i} + \hat{j} - 2\hat{k}$. (2006)

A2.6 (i) A line is perpendicular to $x + 2y + 2z = 0$ and passes through $(0, 1, 0)$. Find the perpendicular distance of this line from the origin. (2006)

(ii) A plane passes through $(1, -2, 1)$ and is perpendicular to the two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$. Find the distance of the plane from the point $(1, 2, 2)$. (2006)

A2.7 Find the (acute) angle between any two diagonals of a cube.

A2.8 Let d be the perpendicular distance of a point (x_1, y_1, z_1) from a straight line L which passes through a point (a, b, c) and has direction cosines l, m, n. Prove that

$$d^2 = \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 - a \\ n & l \end{vmatrix}^2$$

ANSWERS TO EXERCISES

Generally, only the answers are provided here. In particular, no answer is provided where the problem asks you to prove something. However, for both types of problems, the key idea is given briefly in most cases. (No answer is given if the hint given with the question is sufficiently detailed or incisive or where the result comes by a straightforward calculation.)

Chapter 1

1. (a) 6 (b) Note that $|A| \leq |B| \leq |A|$. (c) $n!$ (A function from A to itself is onto if and only if it is one-to-one).

2. In (c), each side consists of elements which belong to either exactly one or to all three of the sets A, B, C.

3. $\dfrac{(3n)!}{6^n n!}$. (Solve the recurrence relation $a_n = \binom{3n-1}{2} a_{n-1}$.)

4. $35 (= \binom{7}{4})$. (First put the + signs in a row. Then insert the − signs.)

5. 485. (The number of man's female relatives at the party is from 0 to 3.)

6. $\sum_{i=5}^{9} \binom{9}{i}\binom{8}{12-i}$. (a) $\sum_{i=7}^{9} \binom{9}{i}\binom{8}{12-i}$ (b) $\binom{9}{5}\binom{8}{7}$.

7. ${}^{14}P_{12} = \frac{14!}{2!}$; $2 \times 12 \times {}^{11}P_9 = \dfrac{24 \times 11!}{2!}$; $\frac{1}{91}$. **8.** $n!$; $(n-1)!$; $\frac{1}{2}(n-1)!$. **9.** For an algebraic proof, write $r \times r!$ as $(r+1)! - r!$. For a combinatorial proof, classify the permutations of the set $\{1, 2, \ldots, n\}$ according to the smallest integer that is not in its original place. **11.** If $d_1 = 0$, then by Exercise (1.9), $k < n!$. $50 = 2 \times 4! + 1 \times 2!$, $100 = 4 \times 4! + 2 \times 2!$, $1000 = 1 \times 6! + 2 \times 5! + 1 \times 4! + 2 \times 3! + 2 \times 2!$. **12.** 7. ($3^n$ must be at least 900.) **13.** $10^5 - {}^{10}P_5$. (Complementary counting.) **14.** k^m if $n = 2m$; k^{m+1} if $n = 2m + 1$. **16.** (i) Denote the object types by integers from 1 to n. Arrange the objects in each selection in an ascending order of types. Then every selection corresponds to a monotonically increasing sequence of length k taking values in the set $\{1, 2, \ldots, n\}$. (ii) 2002 $(= \binom{10+5-1}{5})$. (The digits of the number selected as in the hint must occur in an ascending order. So every such number corresponds to a unique monotonically increasing sequence of length 5 taking values in the set $\{0, 1, \ldots, 9\}$.) **17.** Number the boxes from 1 to n. Each placement of

r objects amounts to a selection of r elements from the set $\{1, 2, \ldots, n\}$ with repetitions allowed freely. For the second part, first put one object in each box. The answer is $\binom{r-1}{r-n}$. **18.** $\binom{103}{2} - 3\binom{52}{2}$. (Complementary counting.)

19. (a) $(m-1)!\binom{m}{n}n!$. (b) $3n^2 + 3n + 1$. (Each bracelet is determined by the numbers, say x, y, z, of blue beads between the red beads. There are $\binom{6n+2}{2}$ i.e., $18n^2 + 9n + 1$ ordered triples (x, y, z) of non-negative integers with $x + y + z = 6n$. Classify them according to how many of the entries are distinct.) **20.** $\binom{m}{2}\binom{n}{2}$; $\binom{m+n-2}{m-1}$. (Each path covers $m + n - 2$ inter-road distances, of which $m - 1$ are East-West.) **21.** At a junction which is i roads to the East and $j - i$ roads to the North of the starting point, there will be $2^{k-j}\binom{j}{i}$ persons for $0 \le i \le j$.

22. $\binom{2}{2}\binom{4}{3}\binom{2}{1} + \binom{2}{2}\binom{4}{2}\binom{2}{2} + \binom{2}{1}\binom{4}{3}\binom{2}{2} + \binom{2}{1}\binom{4}{4}\binom{2}{1} = 8 + 6 + 8 + 4 = 26$.

23. 2^6. **24.** 6. **25.** (a) $\binom{n-m+k-1}{n-m}$. (Take n identical objects and k distinct boxes. Put i objects into the i-th box, for $i = 1, 2, \ldots, n$. Place the remaining $n - m$ objects using Exercise (1.17).) (b) 2^{n-1}. (Put $y_i = x_1 + x_2 + \ldots + x_i$. The subset $\{y_1, y_2, \ldots, y_{k-1}\}$ of $\{1, 2, \ldots, n - 1\}$ determines the composition uniquely.) (c) There are $\binom{n}{3}$ subsets of $\{1, 2, \ldots, n\}$ having three elements x, y, z with $x < y < z$. Exclude those in which $x + y \le z$.

26. 25 $(= \frac{150}{6})$. (Label the boxes. Put the balls. Then remove the labels.)

27. Let X be the set of the n distinct objects. For (iii), consider unordered pairs $(A, X - A)$, where A is a proper, non-empty subset of X. For (iv), exactly one box contains 2 elements.

28. Call the objects as $x_1, x_2, \ldots x_n$. Classify their placements according to whether the last object x_n has a company or not.

29. $S_{5,3} = 3S_{4,3} + S_{4,2} = 3\binom{4}{2} + (2^3 - 1) = 18 + 7 = 25$ by (iv) and (iii) of Exercise (1.27). **30.** $\binom{6}{3}\binom{7}{3}\binom{5}{3} = 7000$. (Apply first part of Exercise (1.17) to balls of each colour.) **31.** $\frac{1}{51}$. **32.** $\frac{1}{11}\binom{20}{10}$. (Arrange the players in a row in descending order of heights. Associate a left parenthesis with team A members and a right one with B.) **33.** Associate a '(' with i if $a_i - a_{i-1} = 1$ and a ')' if $a_i - a_{i-1} = -1$. **34.** Given any arrangement of n left and n right parentheses, for $x = 1, 2, \ldots, n$, let $f(x)$ be the number of left parentheses preceding the x-th right parenthesis.

36. The n-th line is cut into n segments by the earlier lines. Each of these cuts an old region into two subregions. In the space, the n-th plane cuts the earlier planes into lines which are in general position and is thus decomposed into a_{n-1} plane regions each of which cuts a space region into two. From $a_0 = 1$, we get $a_n = a_0 + \sum_{i=1}^{n}(a_i - a_{i-1}) = 1 + \sum_{i=1}^{n} i = 1 + \frac{n(n+1)}{2} = \frac{1}{2}(n^2 + n + 2)$. So $b_i - b_{i-1} = \frac{1}{2}(i^2 - i + 1)$. Adding, $b_n = \frac{1}{6}(n^3 + 5n + 6)$.

37. $\frac{1}{2}n(n-3)$ diagonals. They intersect in $\binom{n}{4}$ points, each of which occurs in two diagonals. A diagonal which contains k of these points is divided into $k+1$ segments. So the total number of segments is $2\binom{n}{4} + \frac{1}{2}n(n-3)$.

38. The removed squares are of the same colour. Hence in what is left there are 32 squares of one colour and 30 of the other. But each pair of adjacent squares has one square of each colour. For a solution without colouring, let V and H be the numbers of vertical and horizontal pairs in a supposed pairing. Then $V + H = 31$. In a vertical pair, call the upper square as a top and the lower one as a bottom. For $i = 1, 2, \ldots, 8$, let T_i and B_i be the numbers of tops and bottoms respectively in the i-th row (from the top). Then $B_1 = 0 = T_8$. Also $T_i = B_{i+1}$ for $i = 1$ to 7. Observe that $T_i + B_i$ is even for $i = 2$ to 7 and odd for $i = 1, 8$. Hence show that T_i is odd for every $i = 1$ to 7. So $V = \sum_{i=1}^{7} T_i$ is odd. Similarly show that H is odd. This means $31 = V + H$ would be even, a contradiction.

39. 25. (The incidence relation is that a given student reads a given paper.)

40. Call the persons $p_1, p_2, \ldots, p_5$. For every $i \neq j$, there is at least one lock, say $L_{ij} = L_{ji}$ which neither p_i nor p_j can open. For distinct pairs of persons, these locks must be distinct as otherwise there is a lock which none of three persons can open. So there are at least $\binom{5}{2} = 10$ locks. A solution with 10 locks is possible if we make 3 keys to each L_{ij} and give them to every person other than p_i and p_j. Every person has 6 keys.

41. Combine complementary counting, DeMorgan's laws and (13), (14).

42. $(m+1)!(m^2 - m)$. (The answer is $|A' \cap B' \cap C'|$. To find $|A|$, think of w_1 and w_2 as forming a single block. Then $|A| = 2 \times (m+2)!$. $|B|$ and $|C|$ also equal this. $|A \cap B| = |B \cap C| = |C \cap A| = 2 \times (m+1)!$. Finally, $A \cap B \cap C = \emptyset$.) **44.** 18 $(= 3^4 - 4 \times 3^3 + 6 \times 3^2 - 4 \times 3 + 3)$.

45. Note that $|A_i| = (n-1)!$ for all i because an element of A_i is uniquely determined by a permutation of the set $\{1, 2, \ldots, n\} - \{i\}$. Similarly, $|A_i \cap A_j| = (n-2)!$ whenever $i < j$, $|A_i \cap A_j \cap A_k| = (n-3)!$ whenever $i < j < k$, etc. Hence $s_1 = n(n-1)! = n!$, $s_2 = \binom{n}{2}(n-1)! = \frac{n!}{2}, s_3 = \binom{n}{3}(n-3)! = \frac{n!}{3!}$ etc.

46. Same as D_4 which equals $4!(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}) = 12 - 4 + 1 = 9$. For a direct solution, a permutation of four symbols, say, 1 to 4 in which no point is in its own place must either permute the four elements cyclically or else must split the set into two pairs and permute each pair. There are 6 permutations of the first kind and 3 of the second.

47. Let $\{a_i, b_i\}$, $i = 1, 2, 3, 4, 5$ be the first five pairs. Let S be the set of all ways to pair the cards again. For $i = 1$ to 5, let A_i be the set of those pairings in which $\{a_i, b_i\}$ is a pair. Then $|S| = 9 \times 7 \times 5 \times 3 \times 1 = 945$.

Each $|A_i|$ equals $7 \times 5 \times 3 \times 1 = 105$. For $i \neq j$, $|A_i \cap A_j| = 5 \times 3 \times 1 = 15$. Continuing like this the number of favorable cases is $945 - 5 \times 105 + 10 \times 15 - 10 \times 3 + 5 \times 1 - 1 = 544$. Hence the desired probability is $\frac{544}{945}$.

48. First note that if the fake coin is known to be among three coins and also whether it is lighter or heavier than a genuine coin, then it can be detected in just one weighing. Call this the finishing step. Now let W_i be the weight of the coin C_i. Let $A = W_1 + W_2 + W_3 + W_4$ and $B = W_5 + W_6 + W_7 + W_8$. Begin by comparing A and B. There are three cases.

(i) If $A > B$ then the true statement is in the set $\{H_1, H_2, H_3, H_4, L_5, L_6, L_7, L_8\}$. Now let $C = W_4 + W_9 + W_{10} + W_{11}$ and $D = W_1 + W_2 + W_3 + W_8$. Compare C and D. If $C < D$ then one of H_1, H_2, H_3 is true and apply the finishing step. If $C = D$, then one of L_5, L_6 and L_7 is true and again apply the finishing step. If $C > D$, then either H_4 or L_8 is true. Weighing C_4 (or C_8) against any other C_i will determine which one is true.

(ii) $A < B$. This is similar to (i).

(iii) $A = B$. Then the true statement is in the set $\{H_9, H_{10}, H_{11}, H_{12}, L_9, L_{10}, L_{11}, L_{12}\}$. Compare C and D as defined in case (i). Unequal pans lead to the finishing step. Equality of C and D implies that the fake coin is C_{12}. Weigh it against any other coin.

If there are 13 coins, then the true statement lies in the set $S = \{H_1, H_2, \ldots, H_{13}, L_1, L_2, \ldots, L_{13}\}$ with 26 elements. No matter what we do at the first weighing, one of the outcomes will give us a subset T of S with at least ten elements in which the true statement lies. Since $10 > 3^2$, two more weighings will not be always enough.

Chapter 2

1. 9. (Solve $120n + 5\frac{n(n-1)}{2} = 90(2n-4)$. Exclude $n = 16$ since some of the angles will exceed 180 degrees in that case.)

2. Write $\dfrac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}}$ as $\dfrac{\sqrt{a_i} - \sqrt{a_{i+1}}}{a_i - a_{i+1}}$ for $i = 1, 2, \ldots, n-1$.

3. (a) $4m^3 + 9m^2 + 6m + 1$, where $n = 2m + 1$. (b) $n + 1 - \frac{1}{2^n}$. (c) $n^2 + \frac{(n-1)n^2}{2} = \frac{n^2(n+1)}{2}$. (d) $(a-d)a(a+d)(a+2d) + d^4 = (a^2 + ad - d^2)^2$.

4. 3 and 6. **5.** $3r = p + 2q$ (e.g., $p = 1, q = 4, r = 3$). (Take logarithms.)

6. $a = 5, b = 8, c = 12$. ($50 = 2a + 2b + 2c = 3b + 2 + 2c = \frac{c^2}{6} + 2c + 2$.)

7. (B) and (D). (a, b, c are respectively, the A.M., the G.M. and the H.M. of the first and the $(2n-1)$-st terms.)

8. $\frac{1-\sqrt{2}}{2}$. (Solve $a+c=1$ with $ac=\pm\frac{1}{4}$. Discard $ac=\frac{1}{4}$.)

9. 8. ($\dfrac{10}{(\sqrt{2})^{n-1}}<1$.) **10.** (b) H.P. (c) 6. (d) 2. **11.** (C)

12. (i) $(x+y+z)(x^2+y^2+z^2-xy-yz-zx)$ (Standard result.)

(ii) $-(a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$. (The expression vanishes if any two of a,b,c are equal. Hence $(a-b)(b-c)(c-a)$ is a factor.)

13. The odd powers of $(x^2-1)^{\frac{1}{2}}$ cancel out. Degree $=5$.

15. (a) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. (b) $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. (c) $a^4+a^2b+4b^2=6ac$. (Use $x^2+z^2=(x+z)^2-2xz$ and $x^3+z^3=(x+z)(x^2+z^2)-xz(x+z)$.)

16. $X=\dfrac{1-\cos\theta}{2-2\cos\theta+3\sin^2\theta}$, $Y=\dfrac{-2\sin\theta}{2-2\cos\theta+3\sin^2\theta}$.

17. Square both the sides and take absolute values.

18. $0=iZ^3+Z^2-Z+i=Z^2(iZ+1)+i(iZ+1)=(iZ+1)(Z^2+i)$ gives $Z=i$ or $Z^2=-i$. In either case, $|Z|=1$.

19. $i-1$. (Any four consecutive powers of i add up to 0.)

20. $(r,-\theta)$ and $(\frac{1}{r},-\theta)$. $\frac{z_1}{z_2}$ is a complex number whose absolute value is $\frac{|z_1|}{|z_2|}$ and which lies on the ray obtained by rotating the ray Oz_1 clockwise through an angle equal to the argument of z_2.

21. Apply (46) with θ replaced by $\frac{\theta}{2}$ to write $\tan\frac{\theta}{2}$ as a ratio. Multiply both the numerator and the denominator by $e^{i\theta/2}+e^{-i\theta/2}$.

22. $(a^2+b^2)(|z_1|^2+|z_2|^2)$.

23. The distance between any two consecutive powers of $e^{i\theta}$ is $|e^{i\theta}-1|=|2\sin\frac{\theta}{2}|$. If m,n are integers then $e^{im\theta}=e^{in\theta}$ gives $\theta=\frac{2\pi k}{m-n}$ for some integer k. Hence $\frac{\theta}{\pi}$ is a rational. If $\theta=\frac{p\pi}{q}$, where p,q are integers with no common factors and $q>0$, then the various powers of $e^{i\theta}$ form the vertices of a regular $2q$-gon inscribed in the unit circle.

24. For the first part draw lines through O parallel to L and L' and note that they pass through the points represented by z_2-z_1 and w_2-w_1 respectively. The quadrilateral whose vertices are z_1,z_2,z_3 and z_4 (in order) is a parallelogram if and only if $z_1+z_3=z_2+z-4$; a rhombus if further $(z_1-z_3)(\bar{z}_2-\bar{z}_4)=0$; a rectangle if and only if, in addition to $z_1+z_3=z_2+z_4$ we also have $(z_1-z_2)(\bar{z}_2-\bar{z}_3)=0$ and finally a square if and only if $z_4-z_1=\pm i(z_2-z_1)$ and $z_4-z_3=\mp(z_2-z_3)$ (together).

25. (a) $\mathbf{A}=\begin{bmatrix}-1/2 & 17/2\\ 7/2 & -3/2\end{bmatrix}$, $\mathbf{B}=\begin{bmatrix}4 & -13\\ -13 & 7\end{bmatrix}$. (b) $f=-(a+d)$, $g=ad-bc$. (It is *not* a coincidence that $-f$ and g are respectively the trace and the determinant of $\mathbf{A}$.)

27. (i) 0. (Direct expansion.) (ii) 0. ($R_1 \longrightarrow R_1 + R_2$). (iii) 0. ($\log_x y = \frac{\log y}{\log x}$ etc.) (iv) $\dfrac{4d^4}{a(a+d)^2(a+2d)^3(a+3d)^2(a+4d)}$. (Take out the denominators of the entries. Then do $C_2 \longrightarrow C_2 - C_3, R_3 \longrightarrow R_3 - R_2, R_2 \longrightarrow R_2 - R_1$ and finally $R_3 \longrightarrow R_3 - R_2$ again.) (v) -2. (Subtract first row from the other two.) (vi) $-4(a-b)(b-c)(c-a)$. (Subtract the first row from the other two. Then do $R_3 \longrightarrow R_3 + R_2, R_2 \longrightarrow R_2 - \frac{1}{2}R_3$. Expand.)

28. (b). (Interchange of any two rows gives a bijection from B to C.)

29. (a) $t = 0$. (Put $\lambda = 0$ and expand.) (b) Do $R_2 \longrightarrow R_2 - (R_1 + R_3)$ and expand w.r.t. the second row.) (c) 0. (Since $C_1 + C_2 = C_3$, $f \equiv 0$.)

30. (i) (b). ($R_3 \longrightarrow R_3 - (pR_1 + R_2)$ gives a determinant which has $(xz - y^2)$ as a factor. (ii) (B) and (E). (By similar reasoning, the determinant equals $(b^2 - ac)(a\alpha^2 + 2b\alpha + c)$.

31. (a) With the reasoning used in the last problem of Comment No. 6 of Chapter 24, the determinant equals $(A - B)^2(A + 2B)$. The R.H.S. factors using Part (i) of Exercise (2.12). (b) The determinant factors as $\begin{vmatrix} 0 & b & c \\ a & 0 & c \\ a & b & 0 \end{vmatrix} \begin{vmatrix} 0 & a & a \\ b & 0 & b \\ c & c & 0 \end{vmatrix}$. Each factors equals $2abc$ by direct expansion. (c) By direct expansion and then multiplying the determinant by $a^2b^2c^2$ we get $2(abc(a^2+1)(b^2+1)(c^2+1) - a^2b^2c^2(2 + \sum a^2) - \sum a^2b^2)$ which vanishes if $abc = 1$ and hence has $(abc - 1)$ as a factor. (d) As the second and the third columns are the same for every a, the given sum of determinants equals the determinant $\begin{vmatrix} \sum_{i=1}^{n-1} i & n & 6 \\ \sum_{i=1}^{n-1} i^2 & 2n^2 & 4n-2 \\ \sum_{i=1}^{n-1} i^3 & 3n^3 & 3n^2 - 3n \end{vmatrix}$. Using the summation formulas, the first column is $\frac{n(n-1)}{12}$ times the last. So the determinant is identically 0, a constant.

32. In each case prove that a typical (i, j)-th term of the L.H.S. equals that of the R.H.S.

33. The calculations are straightforward. Note, however, that if α is a complex number then $(\alpha\mathbf{A})^*$ equals $\bar{\alpha}\mathbf{A}^*$ and not $\alpha\mathbf{A}^*$. For the last assertion, note that $\mathbf{A}^*$ can also be regarded as the transpose of the conjugate of $\mathbf{A}$, i.e., as $(\bar{\mathbf{A}})'$. So $|\mathbf{A}^*| = |(\bar{\mathbf{A}})'| = |\bar{\mathbf{A}}|$. The last term equals $\overline{|\mathbf{A}|}$ because a determinant is the sum of certain products of the entries and complex conjugation preserves both products and sums.

34. (i) $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$. (ii) Zero square matrices. (iii) $x + iy$ is real if and only if the matrix $\begin{bmatrix} x & -y \\ y & x \end{bmatrix}$ is symmetric, since this can happen

if and only if $y = 0$. Similarly the number $x + iy$ is purely imaginary if and only if the corresponding matrix is skew-symmetric. (iv) Instead of verifying this by comparing the entries of both the sides, use the properties of the transpose in Exercise (2.32). Thus $(\alpha\mathbf{A} + \beta\mathbf{B})' = \alpha\mathbf{A}' + \beta\mathbf{B}'$ which equals $\pm(\alpha\mathbf{A} + \beta\mathbf{B})$ depending upon whether $\mathbf{A}' = \pm\mathbf{A}, \mathbf{B}' = \pm\mathbf{B}$ (the same sign holding everywhere. (v) First note that for any matrix $\mathbf{A}$ we have $((\mathbf{A})')' = \mathbf{A}$ as can be verified directly. Verbally, the transpose of a transpose is the original matrix. Now, for any square matrix $\mathbf{A}$, $(\mathbf{A} + \mathbf{A}')' = \mathbf{A}' + (\mathbf{A}')' = \mathbf{A}' + \mathbf{A} = \mathbf{A} + \mathbf{A}'$ which shows that $\mathbf{A} + \mathbf{A}'$ is symmetric. Proof of skew-symmetry of $\mathbf{A} - \mathbf{A}'$ is similar. Now, for any square matrix $\mathbf{A}$, we have $\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}') + \frac{1}{2}(\mathbf{A} - \mathbf{A}')$ which is a sum of a symmetric and a skew-symmetric matrix. As for uniqueness, suppose $\mathbf{P} + \mathbf{R} = \mathbf{Q} + \mathbf{S}$, where $\mathbf{P}, \mathbf{Q}$ are symmetric and $\mathbf{R}, \mathbf{S}$ are skew-symmetric. Then $\mathbf{P} - \mathbf{Q} = \mathbf{S} - \mathbf{R}$ which is both symmetric and skew-symmetric and hence the zero matrix by (ii) above. (vi) The product of two symmetric matrices of the same order need not be symmetric as we see, for example, by taking $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Similarly, nothing can be said in general in other cases.

36. The asserted equalities about trace follow by straight calculations. For example, suppose $\mathbf{A} = (a_{ij})$, $\mathbf{B} = (b_{ij})$ are square matrices of order n. Let $\mathbf{C} = (c_{ij}) = \mathbf{AB}$ and $\mathbf{D} = (d_{ij}) = \mathbf{BA}$. Then $\text{tr}(\mathbf{C}) = \sum_{i=1}^{n} c_{ii} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}b_{ji}$ while $\text{tr}(\mathbf{D}) = \sum_{i=1}^{n} d_{ii} = \sum_{i=1}^{n}\sum_{j=1}^{n} b_{ij}a_{ji} = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ji}b_{ij}$. The additivity of trace follows by changing the order of the last double sum. To see it is not multiplicative, take $\mathbf{A} = \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Then each of $\mathbf{A}, \mathbf{B}$ and $\mathbf{AB}$ has trace 2.

37. Call the given determinant as $f(\lambda)$. This is a polynomial in λ with leading term $(-1)^n\lambda^n$. Then $f(0)$ is simply the determinant of $\mathbf{A}$. This proves the first part. Notice that in taking a transpose of any square matrix, the diagonal entries remain as they are. Since the transpose has the same determinant as the original matrix, the characteristic polynomials of $\mathbf{A}$ and $\mathbf{A}'$ are identical. Hence so are their characteristic roots. Finally, note that the characteristic polynomial $f(\lambda)$ of $\mathbf{A}$ is simply $|\mathbf{A} - \lambda\mathbf{I}|$, where $\mathbf{I}$ is the identity matrix of order n i.e., an $n \times n$ matrix with all diagonal entries 1 and other entries 0 each. Let $g(\lambda)$ be the characteristic polynomial of the Hermitian adjoint $\mathbf{A}^*$ of $\mathbf{A}$. Then using the properties of transposes and determinants, we have $\mathbf{A}^* - \lambda\mathbf{I} = (\overline{\mathbf{A}} - \lambda\mathbf{I})'$ and so, taking determinants of both the sides, $g(\lambda) = |(\overline{\mathbf{A}} - \lambda\mathbf{I})'| = |\overline{\mathbf{A}} - \lambda\mathbf{I}| = |\overline{\mathbf{A} - \bar{\lambda}\mathbf{I}}| = \overline{|\mathbf{A} - \bar{\lambda}\mathbf{I}|} = \overline{f(\bar{\lambda})}$. It follows that the roots of $g(\lambda)$ are the complex conjugates of the roots of $f(\lambda)$. In other words, the characteristic roots of $\mathbf{A}^*$ are the complex conjugates of those of $\mathbf{A}$.

Chapter 3

1. 0. (Neither side is defined at $x = 1$. So 1 is not a root.) **2.** We have $\beta+\gamma+\delta-\alpha = \sum\alpha - 2\alpha = \frac{b}{a} - 2\alpha$ etc. So, the given expression, say E, equals $a^4 f(\frac{b}{a})$, where $f(x) = (x-2\alpha)(x-2\beta)(x-2\gamma)(x-2\delta)$. But by Theorem 2, $2\alpha, 2\beta, 2\gamma, 2\delta$ are the roots of the polynomial $a(\frac{x}{2})^4 - b(\frac{x}{2})^3 + c(\frac{x}{2})^2 - d\frac{x}{2} + e$ whose leading coefficient is $\frac{a}{16}$. So we must have $f(x) = \frac{16}{a}(a(\frac{x}{2})^4 - b(\frac{x}{2})^3 + c(\frac{x}{2})^2 - d\frac{x}{2} + e)$. Compute $f(\frac{b}{a})$. **3.** Let $D = \prod_{i<j}(r_i - r_j)^2$ where $f(x) = x^n + ax + b = (x-r_1)(x-r_2)\ldots(x-r_n)$. Then $f'(r_i) = \lim_{x\to r_i}\dfrac{f(x)-f(r_i)}{x-r_i} = \prod_{j\neq i}(r_i - r_j)$ and so $D = \pm\prod_{i=1}^{n} f'(r_i) = \pm\prod_{i=1}^{n}(a + nr_i^{n-1})$ where the sign is that of $(-1)^{n(n-1)/2}$. Assume first that all r_i's are non-zero. Then for every i, $r_i^n + ar_i + b = 0$ gives $f'(r_i) = nr_i^{n-1} + a = a - n(a + \frac{b}{r_i}) = a(1-n) - \frac{nb}{r_i}$. Now, $\frac{1}{r_i}$, $i = 1, 2, \ldots, n$ are the roots of $bx^n + ax^{n-1} + 1 = 0$. Hence $\sum\frac{1}{r_i} = -\frac{a}{b}$ while $\prod_i \frac{1}{r_i} = (-1)^n\frac{1}{b}$. The other sums of the form $\sum_{i<j}\frac{1}{r_i}\frac{1}{r_j}$, $\sum_{i<j<k}\frac{1}{r_i}\frac{1}{r_j}\frac{1}{r_k}$ etc. all vanish. This makes it easy to compute $\prod_{i=1}^{n} f'(r_i) = \prod_{i=1}^{n}(a(1-n) - \frac{nb}{r_i})$. It comes equal to the given expression. In case one of the roots, say r_1, is 0, b must vanish and the remaining roots satisfy $x^{n-1} = -a$. So $f'(r_1) = a$ while $f'(r_i) = a(1-n)$ for $i \neq 1$. This makes the calculation even easier. **4.** $a_1 = a_2 = \ldots = a_n = -1$. Apply (14) and induction on r. **5.** (a) The reciprocals of the roots are the roots of $a_0x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n = 0$. (b) The reciprocals are the roots of $x^n + \frac{x^{n-1}}{1!} + \frac{x^{n-2}}{2!} + \ldots + \frac{x}{(n-1)!} + \frac{1}{n!} = 0$. Hence $t_1 = -1$. Find t_m for $2 \le m \le n$ inductively using (14). For t_{n+1}, apply (15). **6.** $A = -3, B = 77$. (Write $p = -1 - \sqrt{1-A}, q = -1 + \sqrt{1-A}, r = -9 - \sqrt{81-B}, s = -9 + \sqrt{81-B}$ and apply $q - p = r - q = s - r$. A rare problem where the answer comes straight from the quadratic formula.) **7.** (a) 3. ($f(x) = x^8(x-6) - 2x^6(x-6) + x^2(x-1)(x-6) + x - 3$.) (b) $\frac{7}{6}x^2 + \frac{3}{2}x - \frac{2}{3}$. (Let $p(x) = q(x)(x-1)(x+2)(x+1) + a_2x^2 + a_1x + a_0$. Then $p(1) = 2, p(-2) = 1$ and $p(-1) = -1$ gives a system of three linear equations in the unknowns a_0, a_1 and a_2.) **8.** (a) $x = 4$. ($\log_5(\sqrt{x+5} + \sqrt{x})$ must be 1. So $\sqrt{x+5} + \sqrt{x} = 5$.) (b) $x = \frac{5}{4}$. ($x + 1 = (1 + \sqrt{x-1})^2$.) (c) no solution. (d) $x = -\frac{1}{4}$. (Since $6x^2 + 23x + 21 = (2x+3)(3x+7)$, the L.H.S. equals $1 + \frac{1}{u}$ where $u = \log_{3x+7}(2x+3)$ while the R.H.S. equals $4 - 2u$. Solve for u. $u = 1$ gives $x = -4$ which is discarded as the base of logarithms becomes negative. $2u = 1$ leads to $x = -\frac{1}{4}$ or $x = -2$. Discard the latter.) (e) $x = a^{-4/3}$ or $x = a^{-1/2}$. (Convert the bases of all logarithms to a and put $u = \log_a x$.) **9.** (a) 4 or $\frac{1}{4}$. (Take the numbers as a and λa. Solve $\frac{10\lambda}{1+\lambda} = 4\sqrt{\lambda}$. Discard $\lambda = 0$.) (b) $x = 3$. ($2, 2^x - 5$ and $2^x - \frac{7}{2}$ are in G.P. Solve as a quadratic in 2^x and discard $2^x = 4$.) (c) $x = -\frac{1}{4}, y = -\frac{1}{4}$. ($x + y$ and $\frac{x}{y}$ are the roots of $\alpha^2 - \frac{1}{2}\alpha - \frac{1}{2} = 0$. Solving

and using the sign restrictions, $x+y=-\frac{1}{2}$ and $\frac{x}{y}=1$. Solve the system.)

10. (a) Take the roots as α and α^n. Then $\alpha^{n+1}=\frac{c}{a}$ gives $\alpha=(\frac{c}{a})^{1/(n+1)}$. Put into $\alpha^n+\alpha=-\frac{b}{a}$. (b) a and b. ($(x-a)(x-b)-c$ is a monic polynomial of degree 2 with α,β as roots. So $(x-\alpha)(x-\beta)=(x-a)(x-b)-c$.) (c) $\alpha^2\beta$ and $\alpha\beta^2$. (Let γ,δ be the roots of $a^3x^2+abcx+c^3$. Then $\gamma+\delta=-\frac{b}{a}\frac{c}{a}=(\alpha+\beta)\alpha\beta$ and $\gamma\delta=(\frac{c}{a})^3=\alpha^3\beta^3$. So γ,δ are also the roots of $x^2-(\alpha+\beta)\alpha\beta+\alpha^3\beta^3=0$.) (d) $(q-s)^2-pr(q+s)+r^2q+p^2s$. Vanishing of this expression ia a necessary and sufficient condition for a common root. ($(x-\gamma)(x-\delta)=x^2+rx+s$ gives $(\alpha-\gamma)(\alpha-\delta)=\alpha^2+r\alpha+s$ and $(\beta-\gamma)(\beta-\delta)=\beta^2+r\beta+s$. Multiply and put $\alpha+\beta=-p$ and $\alpha\beta=q$.) (e) -1. (Apply (d) above. A much simpler way is to equate $a\alpha+b=b\alpha+a$ where α is the common root. This gives $\alpha=1$. Either of the two equations gives $a+b=-1$. Note that the individual values of a and b cannot be found.) (f) 4. (If the roots are α,β, their H.M. is $\frac{2\alpha\beta}{\alpha+\beta}$.) **11.** (a) real and distinct. (Positive discriminant.) (b) Yes. (Instead of applying the discriminant criterion, apply the Intermediate Value Property in Exercise (3.15)(a) below, after noting that the given quadratic expression is positive at $x=a$ and also at $x=d$ and negative at $x=b$ (or at $x=c$).) (c) Yes. (Either ac or $-ac$ is positive. So at least one of $P(x)$ and $Q(x)$ has a positive discriminant.) (d) (B). (Even when $b^2-4ac>0$, its square root is numerically less than b and so $\frac{-b+\sqrt{b^2-4ac}}{2a}$ and $\frac{-b-\sqrt{b^2-4ac}}{2a}$ are both negative.) **12.** $k=2$. ($e^{2\ln k}$ is simply k^2. So $2k^2-1=7$ gives $k=\pm2$. Discard -2.) **13.** A.P. ($b^2=ac$ implies that the first quadratic has $-\frac{b}{a}$ as a double root. If this is also a root of the second quadratic, then $d(\frac{b}{a})^2-2\frac{eb}{a}+f=0$. Using $b^2=ac$ this shows that dc, eb, fa and hence $\frac{d}{ab},\frac{e}{ac},\frac{f}{bc}$ are in A.P. Use $ac=b^2$ again, and multiply every term by b.) bf 14. Write $p(x)$ as $a(x-\alpha)^2$, where $a\neq0$. Then $p(\alpha+h)=ah^2$ has the same sign for all $h\neq0$ regardless of whether it is positive or negative. So the graph of $y=p(x)$ lies entirely on one side of the x-axis. **15.** (a) First assume $a>0$. Since one of $f(x_1)$ and $f(x_2)$ is negative, the graph of $y=ax^2+bx+c$ must be of the first type shown in the figure in the text. Let α,β be the distinct real roots with $\alpha<\beta$. Then exactly one of x_1 and x_2 lies in the open interval (α,β) and the other outside it. In either case, either α or β lies in the open interval (x_1,x_2). If $a<0$ apply this reasoning to $-f$. Finally, for the more general result, apply the reasoning above to the quadratic $ax^2+bx+(c-m)$.
(b) Let $f(x)=a^2x^2+2bx+2c$. Because of the Intermediate Value Property, it is enough to show that $f(\alpha)<0$ and $f(\beta)>0$. But since $a^2\alpha^2=-b\alpha-c$, we have $f(\alpha)=a^2\alpha^2-2a^2\alpha^2=-a^2\alpha^2$ which is always negative. Similarly, $f(\beta)=3a^2\beta^2>0$. **16.** $7,2,-9$. (Putting $x=7$ makes the first and the last columns identical. Hence 7 is a root. Expand and factor the determinant.) **17.** The denominators and hence the numerators of the first three ratios add to 0. Hence $ax+by+cz$ equals $(a+b+c)(x+y+z)$. So the last ratio is simply $x+y+z$ (as-

suming $a+b+c \neq 0$). Equating it with each of the first ratios gives a system of three homogeneous linear equations in x, y, z whose determinant is $\begin{vmatrix} b-c & b-2c & -c \\ -a & c-a & c-2a \\ a-2b & -b & a-b \end{vmatrix}$. Putting $p = b-c, q = c-a$ and $r = a-b$, the determinant equals $2(pqr - abc)$ by direct expansion. **18.** (a) If $a_1, a_2, a_3, b_1, b_2, b_3$ are all 0 then there are infinitely many solutions or no solution depending upon whether (d_1, d_2) is $(0, 0)$ or not. If at least one of these six coefficients is non-zero, and all the three numbers $a_1b_2 - a_2b_1, b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ are 0, then there are infinitely many solutions or no solution depending upon whether all the three numbers $a_1d_2 - a_2d_1, b_1d_2 - b_2d_1$ and $c_1d_2 - c_2d_1$ are 0 or whether at least one of them is non-zero. Finally, if at least one of the three numbers $a_1b_2 - a_2b_1, b_1c_2 - b_2c_1$ and $c_1a_2 - c_2a_1$ is non-zero, then there are infinitely many solutions no matter what d_1, d_2 are. (b) None. (c) One (viz., $k = 1$.) **19.** (C). $(ax^3 + bx + c = (x^2 + px + 1)(ax + c)$ gives $ap + c = 0$ and $cp + a = b$.) **20.** (a) If say, $a_1b_2 - a_2b_1 \neq 0$, the system $a_1s + a_2t = a_3$; $b_1s + b_2t = b_3$ can be solved uniquely for the unknowns s and t. The row operation $R_3 \longrightarrow R_3 - (sR_1 + tR_2)$ gives the value of Δ as $(a_1b_2 - a_2b_1)(c_3 - sc_1 - tc_2)$. Vanishing of Δ forces $c_3 = sc_1 + tc_2$. (b) For infinitely many solutions, Δ must vanish first of all. By Corollary 9, one of its rows is a linear combination of the other two. So, if there is a solution then the corresponding d must be the *same* linear combination of the other two d's. So, a *necessary* condition for infinitely many solutions is that $\Delta, \Delta_1, \Delta_2, \Delta_3$ should all vanish, where for $i = 1, 2, 3$, Δ_i is the determinant obtained by replacing the i-th column of the coefficient matrix by the column vector $(d_1, d_2, d_3)^T$. But simple examples show that this condition is not sufficient (e.g. when everything except d_1 is 0). When at least one of the nine expressions of the form $a_1b_2 - a_2b_1$ is non-zero, the condition is sufficient too by (a). Suppose all the nine expressions of the form $a_1b_2 - a_2b_1$ are 0. In this case if all the nine coefficients $a_1, \ldots, c_3$ vanish then all the d's must vanish. If at least one of them, say c_2 is non-zero, then the condition for infinitely many solutions is that all the three determinants $\begin{vmatrix} u_1 & d_1 \\ u_2 & d_2 \end{vmatrix}$, where u can be a, b or c are zero and the same holds for all three determinants of the form $\begin{vmatrix} u_2 & d_2 \\ u_3 & d_3 \end{vmatrix}$, where u can be a, b, c. (The concept of the rank of a matrix allows us to express all these cases in a single succinct statement. But that is beyond our scope.)

21. (a) Let Δ be the determinant of the coefficients. Do $R_3 \longrightarrow R_3 - (R_1 + 3R_2)$ to get $\Delta = 7(\lambda + 5)$. So for $\lambda \neq -5$ the system has a unique solution. For $\lambda = -5$, choose z arbitrarily. The first two equations give $x = \frac{4-5z}{7}$ and $y = \frac{13z-9}{7}$. The set of solutions is $\{(\frac{4-5z}{7}, \frac{13z-9}{7}, z) : z \text{ arbitrary}\}$. (b) All $\lambda \neq 0$. ($\Delta = \lambda^3 + 3\lambda$.) (c) (i) Unique solution for $a \neq 16$ (any b) (ii) infinite number of solutions for $a = 16, b = 4$ (iii) no solution if

$a = 16, b \neq 4$. **22.** If $x = u, y = v$ is a solution, the the column operation $C_3 \longrightarrow C_3 - (uC_1 + vC_2)$ makes Δ vanish. **23.** The reasoning in the last exercise extends to show that if the system has a solution then $\Delta = 0$. Conversely assume $\Delta = 0$ and at least one Δ_i, say $\Delta_2 \neq 0$. Then by Cramer's rule the system

$$\begin{aligned} a_1\lambda_1 + a_3\lambda_3 + a_4\lambda_4 &= a_2 \\ b_1\lambda_1 + b_3\lambda_3 + b_4\lambda_4 &= b_2 \\ c_1\lambda_1 + c_3\lambda_3 + c_4\lambda_4 &= c_2 \end{aligned}$$

has a solution (in fact a unique solution) in the unknowns $\lambda_1, \lambda_3, \lambda_4$. Now in Δ, the row operation $R_2 \longrightarrow R_2 - (\lambda_1 R_1 + \lambda_3 R_3 + \lambda_4 R_4)$ and subsequent expansion w.r.t. the second row shows that Δ equals $(d_2 - \lambda_1 d_1 - \lambda_3 d_3 - \lambda_4 d_4)\Delta_2$. Since $\Delta = 0$ but $\Delta_2 \neq 0$, we get $d_2 = \lambda_1 d_1 + \lambda_3 d_3 + \lambda_4 d_4$. This means that the second equation of the system is a linear combination of the other three equations and hence any solution of the latter will also be a solution of the whole system. But since $\Delta_2 \neq 0$, we already know by Cramer's rule that the system of the first, third and fourth equation has a (unique) solution. So the original system of four equations is consistent.

25. I. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. II. $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$. The composite transformation is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$ $= \begin{bmatrix} 0 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$ i.e., $x_1 = 3z_2, x_2 = 3z_1 + 3z_2$. z_1, z_2 can be expressed in terms of x_1, x_2 by finding the inverse of the matrix $\begin{bmatrix} 0 & 3 \\ 3 & 3 \end{bmatrix}$. But a direct calculation is much easier and gives $z_1 = -\frac{1}{3}x_1 + \frac{1}{3}x_2, z_2 = \frac{1}{3}x_1$.

26. (a) The second assertion can also be proved without the first by a direct, inductive proof. Replace α_n by x and consider $D(\alpha_1, \alpha_2, \ldots, \alpha_{n-1}, x)$. This is a polynomial of degree $n-1$ in x since its leading coefficient is non-zero by the inductive hypothesis. The roots of this polynomial are $\alpha_1, \ldots, \alpha_{n-1}$. As α_n differs from all these, we have $D(\alpha_1, \ldots, \alpha_{n-1}, \alpha_n) \neq 0$. (b) Take the product of $D(\alpha_1, \alpha_2, \ldots, \alpha_n)$ and its transpose. (c) We assume that the α's are all non-zero and distinct as otherwise we modify the given system suitably by deleting or combining the respective terms. Now, for $j = 0, 1, 2, \ldots, n-1$, consider the block of $n+1$ consecutive equations in the given system starting from the j-th equation. This is a linear system in the unknowns β's. Eliminating $\beta_1, \beta_2, \ldots, \beta_n$ from these $n+1$ equations and taking out the (non-zero) factor $\alpha_1^j \alpha_2^j \ldots \alpha_n^j$, gives

$$D_j = \begin{vmatrix} 1 & 1 & \ldots & 1 & a_j \\ \alpha_1 & \alpha_2 & \ldots & \alpha_n & a_{j+1} \\ \vdots & \vdots & \vdots & & \vdots \\ \alpha_1^n & \alpha_2^n & \ldots & \alpha_n^n & a_{j+n} \end{vmatrix} = 0 \; ; \quad (j = 0, 1, 2, \ldots, n-1)$$

Now consider the Vandermonde determinant $D(\alpha_1, \alpha_2, \ldots, \alpha_n, x)$. When expanded w.r.t. its last column, this determinant is a polynomial in x of degree n with leading coefficient $D(\alpha_1, \alpha_2, \ldots, \alpha_n)$ and roots $\alpha_1, \alpha_2, \ldots, \alpha_n$. Hence we must have

$$D(\alpha_1, \alpha_2, \ldots, \alpha_n, x) = D_0(x^n - \sigma_1 x^{n-1} + \sigma_2 x^{n-2} + \ldots + (-1)^n \sigma_n) \quad (1)$$

where $D_0 = D(\alpha_1, \alpha_2, \ldots, \alpha_n)$. Note that the first n columns of all the determinants D_j's above are identical to those of the Vandermonde determinant $D(\alpha_1, \alpha_2, \ldots, \alpha_n, x)$. So, if we expand the D_j's w.r.t. their last columns and cancel the factor D_0, then instead of (1) we get

$$a_{j+n} - a_{j+n-1}\sigma_1 + a_{j+n-2}\sigma_2 + \ldots + (-1)^n a_j \sigma_n = 0 \quad (2)$$

for $j = 0, 1, \ldots, n-1$. This is a system of n linear equations in the unknowns $\sigma_1, \sigma_2, \ldots, \sigma_n$. Solving this linear system, we get the σ's in terms of the $a_0, a_1, \ldots, a_{2n-1}$ which are known. Substituting these into (1) (and dividing by D_0), we get a desired polynomial. If we can determine the α's from this, we can determine the β's too by solving the first n of the given system of equations.

27. The characteristic polynomial, say $f(\lambda)$, of $\mathbf{A}$ has $(-1)^n\lambda^n$ as its leading term. Let the next term be $b\lambda^{n-1}$. Then $\frac{b}{(-1)^{n-1}}$ is the sum of the characteristic roots of $\mathbf{A}$. If we expand the determinant given in Exercise (2.37) as a sum of $n!$ terms each of which is a product of certain n entries, we see that in all except one of these $n!$ terms, at most $n-2$ factors are of the form $(a_{ii} - \lambda)$ for some i. As a result, a term involving λ^{n-1} will come only from the product of the diagonal terms and its coefficient will be precisely $(-1)^{n-1}\,\mathrm{tr}(\mathbf{A})$. This proves that the sum of the characteristic roots of $\mathbf{A}$ is the trace of $\mathbf{A}$. The assertion about the determinant follows by noting that the constant term of $f(\lambda)$ is simply $f(0)$ which, in the present case, is nothing but the determinant of $\mathbf{A}$. **28.** Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where a, b, c, d are complex. Let the characteristic roots be λ_1 and λ_2. Then λ_1, λ_2 are the roots of the (complex) quadratic $\lambda^2 - (a+d) + (ad - bc) = 0$. Now, if $\mathbf{A}$ is Hermitian, then a, d are real and $b = \bar{c}$. So $ad - bc = ad - |b|^2$ is also real. Hence the quadratic has real coefficients. Moreover, its discriminant comes out to be $(a-d)^2 + |b|^2$ which can never be negative. So λ_1, λ_2 are real, proving (i). For (ii) the argument is similar and in fact simpler since now both a and d vanish and so the characteristic polynomial becomes $\lambda^2 + |b|^2 = 0$, whose roots are purely imaginary. **29.** Use (15). Note that $a_1 = 0 = s_1$. First show that $s_2 = -2p$ and $s_3 = 3q$. Hence R.H.S. equals $-pq$. But by (15), $s_5 = -5pq$.

Chapter 4

1. Yes. (The sum equals q modulo 2 on one hand and 1 modulo 2 on the other.) **2.** 3,050 $(= \sum_{i=1}^{50} 2i + \sum_{i=1}^{20} 5i - \sum_{i=1}^{10} 10i.)$ **3.** Put $x = 0$ to see that C is an integer. Hence $Ax^2 + Bx$ is also an integer for every integer x. Putting $x = 1$, $A + B$ is an integer. Also putting $x = -1$, $A - B$ is an integer. Hence $2A = (A+B)+(A-B)$ is an integer. For the converse, note first that since $2A, A+B$ are integers so are $A-B$ and hence $2B$. Now if x is even, say $x = 2m$, then $f(x) = 4Am^2 + 2Bm + C$ is an integer. When x is odd, say $x = 2m+1$, $f(x) = 4Am^2 + 4Am + 2Bm + (A+B)m + C$ is an integer. **4.** Assume $2^r < m < 2^{r+1}$. Then $[\frac{2m-1}{2^{r+1}}] = 1 > [\frac{m}{2^{r+1}}] + [\frac{m-1}{2^{r+1}}]$. **5.** The hypothesis means that $100A + 28,309 + 10B$ and $620 + C$ are all divisible by k. The row operation $R_2 \longrightarrow (R_2 + 100R_1 + 10R_3)$ makes all entries in the second row divisible by k. **6.** (a) The rules are based on the fact that 10 and hence all powers of 10 are congruent to 1 modulo 3. Powers of 11 are alternately congruent to ± 1 modulo 10. Hence a number is divisible by 11 if and only if the difference between the sums of alternate digits is divisible by 11. (b) Let the number be *abcd*. Its value (in the usual form) is $343a + 49b + 7c + d$. The hypothesis means that $343a + 49b + 7c + d = 686d + 98a + 14b + 2c$. Hence a equals $2d$ or $2d + 1$. This limits d to 0, 1, 2 or 3. Further 7 must divide $2c - d$ which is possible only if $2c = d$ or $2c = d + 7$ and correspondingly, 7 divides $2b - c$ or $2b + 1 - c$. So, for $d = 0, 1, 2, 3$ the corresponding values of c are $0, 4, 4, 5$ while those of b are $0, 5, 4, 2$ respectively. Eliminating various possibilities, the number comes out to be 2541 (in base 7). **7.** Denote the days of the week by congruence classes modulo 7 with [0] corresponding to the day on which 13th January falls. Then in an ordinary year, the 13ths of the subsequent months fall on $[3], [3], [6], [1], \ldots, [3], [5]$. Every class appears at least once and at most thrice in this list. The same holds for a leap year where the 13ths fall on $[0], [3], [4], [0]. \ldots, [4], [6]$.

8. (a) $5^{12} + 2^{10} = (5^6 + 2^5)^2 - 2 \times 5^6 \times 2^5 = (5^6 + 2^5)^2 - (1000)^2$. Factorise. (b) Or else the number is even. (c) Otherwise each p_i is of the form $6k \pm 1$ and hence $p_i^2 \equiv 1 \ (mod\ 3)$. But then the number is divisible by 3. (d) If $p_1, p_2, \ldots, p_k$ are primes, the prime factors of $p_1 p_2 \ldots p_k + 1$ must be different from any of the p_i's. For the second part, for each n, the hint gives a prime-free block of $n - 1$ consecutive integers. (e) Let $p_1, p_2, \ldots, p_n$ be primes of the form $4k - 1$ each. Then at least one of the prime factors of $4p_1 p_2 \ldots p_n - 1$ is of the form $4k - 1$. (f) Straight verification. (g) Suppose $f(1)$ is a prime p. Then $f(mp + 1) - f(1)$ and hence $f(mp+1)$ is divisible by p for every positive integer m. At least one of these differs from p. (h) For $i = 1, 3, \ldots, p - 2$, $i \equiv -(p - i) \ (mod\ p)$. So $\{1.3. \ldots (p - 2)\}^2 = (-1)^{(p-1)/2}\{1.3. \ldots (p - 2)\}\{2.4. \ldots (p - 1)\} \equiv (-1)^{(p-1)/2}(p - 1)! \equiv \ (mod\ p)$. Apply Wilson's theorem. For the second part, write $\binom{(p-2)}{r} r!(p-r-1)! = (p-2)!(p-r-1)$. By Wilson's theorem, the R.H.S. is congruent to $-(r + 1)$ modulo p. For the L.H.S., note that $r!$ is congruent to $(-1)^r(p-1)(p-2)\ldots(p-r)$ modulo p. Apply Wilson's

theorem again. For the last part, put $k = (p-1)/2$. Then $\binom{2k}{k} = \frac{A}{B}$ where $B = k!$ and $A = (p-1)(p-2)\dots(p-k) = (-1)^k k! + mp$ for some integer m. So $k!$ divides mp and hence m since $k!$ is relatively prime to p. (i) Take $c = (2n)!$. Then $(p-1)\dots(p-2n) \equiv c \ (mod\ p)$. Apply Wilson's theorem. (j) Call $4(p-3)! + p + 2$ as u. Then $u = 4((p-3)!+1) + (p-2)$ which is divisible by $p-2$ by Wilson's theorem. For showing divisibility by p, consider $(p-1)(p-2)u$ and apply Wilson's theorem again. **9.** (a) $\prod_{i=1}^{k} \binom{a_i+2}{2}$. (First consider the case where $n = p_i^{a_i}$.) For the second part observe that $p_i^{\alpha_i} > 2$ for every $i = 1, 2, \dots, k$ and take logarithms. (b) 4. (See Chapter 24, Comment No. 3(iv) for an explanation.) **10.** $2^{m-1}H_n = \sum_{k=1}^{n} \frac{2^{m-1}}{k}$. For every $1 \le k \le n$, except for $k = 2^m$, $\frac{2^{m-1}}{k}$ is a fraction with an odd denominator. The sum of such fractions is a rational, say $\frac{u}{v}$, where v is odd. Now we have $2^m v H_n = v + 2u$. If H_n is an integer this is a contradiction as the L.H.S. is even and the R.H.S. is odd. **11.** (a) The euclidean algorithm for finding d already shows that $d = am + bn$ for some integers a, b. Suppose there is some $e = um + vn$ with u, v integers such that $0 < e < d$. Then since d divides m, n, it divides e too, a contradiction. (b) Since d divides m, n, both $\frac{m}{d}, \frac{n}{d}$ are integers. If they have a non-trivial common factor k (say), then $|k|d$ also divides m, n, contradicting that d is their *greatest* common divisor. (c) This follows from the prime power factorisations of m and n. **12.** (a) We are given $m(x-y) = nk$ for some integer k. Take prime power factorisations of both the sides. A typical prime power p^r dividing n must divide $x-y$ since p cannot divide m. So n divides $x-y$. (b) Let S be the set of residue classes modulo n of integers that are relatively prime to n (e.g., for $n = 12$, $S = \{[1], [5], [7], [11]\}$. The class $[m]$ is also an element of S. Define a function $f : S \longrightarrow S$ by $f([x]) = [mx]$. By (a), the function f is one-to-one. Hence it is onto since the set S is finite. In particular there is some k such that $[mk] = [1]$. (c) Let S be as above. Then the classes $[m], [m^2], [m^3], \dots$ are all in S. But as S is finite, there are some r, s with $1 \le r < s$ such that $[m^r] = [m^s]$. Taking $k = s - r$, this means n divides $m^r(m^k - 1)$. Since m^r is also relatively prime to n, by (a) we get n divides $m^k - 1$. The orders of 6 and 2 modulo 35 are 2 and 12 respectively. (To find the latter, it is not necessary to take actual powers of 2. Powers modulo 35 would suffice. Thus $2^5 \equiv -3$ and hence $2^{10} \equiv 9$ which further gives $2^{12} \equiv 36 \equiv 1 \ (mod\ 35)$.) (d) If $rk = s - t$ then $1 \equiv (m^r)^k \equiv m^{rk} \equiv m^{s-t}$ gives $m^t \equiv m^t m^{s-t} \equiv m^s$. For the converse, first note that $m^s \equiv m^t$ implies that $m^{s-t}m^t = m^t \equiv 1.m^t$ and hence $m^{s-t} \equiv 1$ by (a) above. Now write $s - t$ as $rq + u$, where $0 \le u < r$. Then $1 \equiv m^{s-t} \equiv m^{rq}m^u \equiv m^u$. If $u \neq 0$ we get a contradiction as u is less than the order of m modulo n. So $u = 0$ and hence r divides $s - t$. (e) By Fermat's theorem, $m^n \equiv 1 \equiv m^0$ modulo n. Apply (d). (f) If $c^2 \equiv -1 \ (mod\ p)$, then the order of c modulo p is 4, contradticting (e). **13.** (a) Let $u_i = n_1 \dots n_{i-1} n_{i+1} \dots n_k$.

From the hint and part (a) of Exercise (4.11), there exist integers a, b such that $au_i + bn_i = 1$. Take $m_i = au_i$. (b) Take $m = r_1m_1 + \ldots + r_km_k$, where $m_1, m_2, \ldots, m_k$ are as in (a). For uniqueness, suppose m' is another integer such that $m' \equiv r_i \ (mod\ n_i)$ for every $i = 1, 2, \ldots, k$. Then each of $n_1, n_2, \ldots, n_k$ divides $m - m'$ and since they are relatively prime, $m - m'$ is divisible by the product $n_1n_2 \ldots n_k$. (c) A number m of the given type is congruent to 0 modulo 2, to 4 modulo 7 and to 3 modulo 9. Verify that -24 is one such number. Apply (b) with $126 = 2 \times 9 \times 7$. (d) 4 and 9. ($3^{110} \equiv (3^2)^{55} \equiv 1 \ (mod\ 4)$. Also $3^{10} \equiv 3 \times (3^3)^3 \equiv 3 \times 2^3 \equiv -1 \ (mod\ 25)$ and hence $3^{110} \equiv (-1)^{11} \equiv -1 \ (mod\ 25)$. Use (b) with $100 = 4 \times 25$. Or write 3^{110} as $(10 - 1)^{55}$ and take the last two terms of the binomial expansion.) (e) Show that $7^{100} \equiv 1 \ (mod\ 100)$ and also $3^{100} \equiv 1 \ (mod\ 100)$ and hence $x - [x] = y - [y] = \frac{1}{100}$. (f) No, for $k = 1, 2$. Yes, for $k = 3$. $T_{n,3} = \frac{1}{4}AB$ where $A = (n + 100)(n + 101) + n(n + 1)$ and $B = (n + 100)(n + 101) - n(n + 1)$. Since $n(n + 1)$ is always even, 4 divides A. Also $B \equiv 0 \ (mod\ 100)$.

14. Let the first common visit come after m more days. Then $m \equiv -1 \ (mod\ 3)$, $m \equiv -3 \ (mod\ 7)$ and $m \equiv -2 \ (mod\ 10)$. The third condition implies $m = 10k + 8$ for some k. Reducing modulo 3, the first condition implies that $k = 3r$ for some r. So $m = 30r + 8$. Reduce modulo 7 and use the second condition to get $2r \equiv -4$ and hence $r \equiv -2$ modulo 7. So r may be taken as 5, which gives $m = 158$. **15.** $\frac{1}{4}$. (Modulo 5, 7 is the same as 2. Now $2^m \equiv 1, 2, 4$ or 3 depending upon whether the residue of m modulo 4 is 0, 1, 2 or 3. The same holds for 2^n. Thus $2^m + 2^n$ is divisible by 4 in four cases : (i) $m \equiv 0, n \equiv 2 \ (mod\ 4)$ (ii) $m \equiv 1, n \equiv 3 \ (mod\ 4)$ etc. In each case there are 25×25 pairs (m, n).) **16.** Let $E_n = p^{n+1} + (p + 1)^{2n-1}$. Then $E_{n+1} - pE_n = (p^2 + p + 1)p^{2n-1}$. Apply induction on n.) **17.** This time consider $E_{n+1} - 25E_n$. **18.** By Fermat's theorem, each of p, q divides $p^{q-1} + q^{p-1} - 1$. **19.** Note that 437 is the product of the primes 19 and 23. By Fermat's theorem, 16^{18} and hence 16^{90} is congruent to 1 modulo 19. So $16^{99} \equiv 16^9 \equiv 2^{36} \equiv 1 \ (mod\ 19)$ again by Fermat's theorem. So 19 divides $16^{99} - 1$. By a similar argument, 23 also divides it. For the second part, divisibility of $(18)! + 1$ by 19 follows directly from Wilson's theorem. To prove divisibility by 23, note that modulo 23, $(22)! \equiv (18)!(23-4)(23-3)(23-2)(23-1) \equiv (18)!24 \equiv (18)!$. Now apply Wilson's theorem. **20.** By Fermat's theorem, 3 and 5 divide $m^4 - 1$. Also $m^4 - 1 = (m^2 + 1)(m + 1)(m - 1)$. Each factor is even and one of the last two is divisible by 4. Hence 16 too divides $m^4 - 1$. **21.** The hypothesis amounts to saying that $n^5 \equiv 5$ modulo 7 as well as modulo 13. By a direct check, modulo 7, $n = 3$ is the only value satisfying $n^5 \equiv 5$. So $n \equiv 3 \ (mod\ 7)$. Similarly, $n \equiv 5 \ (mod\ 13)$. Further $-13 \equiv 1 \ (mod\ 7)$ and $14 \equiv 1 \ (mod\ 13)$. So, $-13 \times 3 + 14 \times 5 = 31$ equals 3 modulo 7 and 5 modulo 13. Hence by the Chinese Remainder Theorem, $n \equiv 31 \ (mod\ 91)$. **22.** (a) $p^2 = (b + a)(b - a)$ and primality of p imply that $b - a = p^2, p$ or 1. The first two possibilities are ruled out since $(a + p^2)^2$ and $(a + p)^2$

both exceed $p^2 + a^2$. $b - a = 1$ gives $b + a = p^2$ and hence $a = \frac{p^2-1}{2}$ which immediately gives the second part. Also since $p > 3$, it is of the form $6k \pm 1$ which gives $a = 6k(3k \pm 1)$. Since $k(3k \pm 1)$ is even for all k, we get the first part. (b) Both a and b can't be even. But since every square is congruent to 0 or 1 modulo 4 they can't both be odd either. If b is even, then note that $\frac{c+a}{2}$ and $\frac{c-a}{2}$ are relatively prime integers as any common factor of them would be a common factor of a and c. The rest of the reasoning is similar to that in (a). $(\frac{b}{2})^2 = \frac{(c+a)}{2}\frac{(c-a)}{2}$ makes each of $\frac{c+a}{2}$ and $\frac{c-a}{2}$ a square. (c) A common prime factor of any two of a, b, c must also be a factor of the third. Let d be the g.c.d. of a, b, c. Apply (b) above to the Pythagorean triple $(\frac{a}{d}, \frac{b}{d}, \frac{c}{d})$.

23. The direct implication is trivial. For the converse suppose $c = ud$. By Exercise (4.11)(a), there exist integers m, n such that $d = ma + nb$. Then $x = um, y = un$ is a solution of $ax+by = c$. **24.** 80, 1 and 19 respectively. (Let x, y, z denote the numbers of fruit of type A, B, C respectively. Taking 5 paise as a unit cost, we have two equations: (i) $x + y + z = 100$ and (ii) $x + 20y + 100z = 2000$. So $19y + 99z = 1900$ which implies that 19 divides z. As x, y, z are positive integers, the only possible value of z is 19.) **25.** (i) $\dfrac{(mn+pq)!}{(m!)^n(p!)^q}$ is the number of ways to arrange $mn + pq$ balls in which there are m balls each of n colours and p balls of each of some other q colours. (ii) $\dfrac{(n^2)!}{(n!)^n}$ is the total number of arrangements of n^2 balls in which there are n balls each of n colours. Now the n colours themselves can be permuted in $n!$ ways. **26.** (b) Let $c_n = F_{n+1}F_{n-1} - F_n^2$. Writing F_{n+1} as $F_n + F_{n-1}$ and F_n^2 as $(F_{n-1} + F_{n-2})F_n$, we see that $c_n = -c_{n-1}$. Apply induction on n. (c) Let $q = kp$, where k is an integer ≥ 2. Put $n = kp - p$ and $m = p$ in (a). (d) A common factor of F_n and F_{n+1} is also a factor of F_k for every k. (e) From (b), $F_n^2 + F_{n+1}^2 = F_{n+2}F_n + F_{n+1}F_{n-1} = F_{2n+1}$ by (a). **27.** Otherwise $2^p = 7^q$ for some integers p and q. **28.** No. It is not symmetric, e.g., $\sqrt{2} * 0$ is true but $0 * \sqrt{2}$ is false. **29.** Modify the proof of the irrationality of $\sqrt{2}$. **30.** Let $x = a - p, y = b - q$ and $z = r - c$. Then $x + y\sqrt{2} = \sqrt{3}z$. Squaring, $2\sqrt{2}xy = 3z^2 - (x^2 + 2y^2)$. As $\sqrt{2}$ is irrational while x, y, z are rational, this implies $xy = 0$ and $x^2 + 2y^2 = 3z^2$. Depending upon whether $x = 0$ or $y = 0$ we get $2y^2 = 3z^2$ or $x^2 = 3z^2$. Clearing the denominators we may suppose that x, y, z are integers that are relatively prime. In either case, we get a contradiction by considering the prime power factorisations of the two sides. The answer to the second part is 'no'. Take $a = 2, b = 0, p = 0$ and $q = 1$, for example. **31.** Squaring and using irrationality of $\sqrt{2}$, we have $a^2 + 2b^2 = x$ and $2ab = y$. **32.** 41 $(=(\sqrt{2})^{10} + (3^{1/5})^{10})$. **33.** The square is $\frac{26-15\sqrt{3}}{88+5\sqrt{3}-10\sqrt{76+10\sqrt{3}}}$. Since $76 + 10\sqrt{3} = (5\sqrt{3} + 1)^2$, the square is simply $\frac{1}{3}$. **34.** Putting $u = x + y$ and $v = x - y$, the expression equals $\sqrt{2}(u + v + \sqrt{uv})(\sqrt{\frac{u+v}{2} - \sqrt{uv}})$ which equals $(u + v + \sqrt{uv})(\sqrt{u + v - 2\sqrt{uv}})$. Since the second factor is

$(\sqrt{u}-\sqrt{v})$, this further reduces to $(\sqrt{u})^3-(\sqrt{v})^3$. **35.** (b). (Multiply and divide by $3+\sqrt{5}-2\sqrt{2}$. Rationalise again.)

36. The given expression, say E, equals $b_1+b_2+\ldots+b_{p-1}$, where $b_m=\frac{(p-1)!}{m}$ for $m=1,2,\ldots,p-1$. So E is certainly an integer. Taking residue classes modulo p and using Wilson's theorem, we have $[m][b_m]=[-1]$ and hence $[b_m]=-[a_m]$ where a_m is as in the hint. So $[E]=-\sum_{m=1}^{p-1}[a_m]$, which, by the hint, equals $-\sum_{m=1}^{p-1}[m]=-[p(\frac{p-1}{2})]=[0]$. Hence p divides E. For the second part, note that $(2p)!-2(p!)^2=2p^2(p-1)!F$, where $F=(p+1)(p+2)\ldots(p+(p-1))-(p-1)!$. So it suffices to show that F is divisible by p^2. When we expand $(p+1)(p+2)\ldots(p+(p-1))$ it equals $(p-1)!+pE+$ terms divisible by p^2. So $F=pE+$ terms divisible by p^2. Since p divides E we get that p^2 divides F. **37.** (i) Among the integers from 1 to p^r, $1,p,p^2,\ldots,p^r$ divide p^r. Rest are relatively prime to it. (ii) For $i=1,2,\ldots,k$, let $A_i=\{x:1\le x\le n,\ p_i \text{ divides } x\}$. Then $\phi(n)$ is simply $|\bigcap_{i=1}^{k}A_i'|$. This equals $\sum_{i=0}^{k}(-1)^i s_i$ as in Exercise (1.43). Here s_i is the sum of the cardinalities of sets of the form $A_{j_1}\cap A_{j_2}\cap\ldots\cap A_{j_i}$ for some $1\le j_1<j_2<\ldots<j_i\le k$. It is easily seen that this intersection consists precisely of multiples of $p_{j_1}p_{j_2}\ldots p_{j_i}$. Consequently, $|A_{j_1}\cap A_{j_2}\cap\ldots\cap A_{j_i}|=\dfrac{n}{p_{j_1}p_{j_2}\ldots p_{j_i}}=p_1^{t_1}p_2^{t_2}\ldots p_k^{t_k}$, where $t_{j_1}=r_{j_1}-1, t_{j_2}=r_{j_2}-1,\ldots,t_{j_i}=r_{j_i}-1$ and $t_j=r_j$ for all $j\ne j_1,j_2,\ldots j_i$. But these are also precisely the terms in the expansion of the product $(p_1^{r_1}-p_1^{r_1-1})(p_2^{r_2}-p_2^{r_2-1})\ldots(p_k^{r_k}-p_k^{r_k-1})$. (iii) As m,n are relatively prime, the primes appearing in their factorisations are different. The result follows from (ii). (iv) If m,n are relatively prime, then every divisor of mn is of the form uv where u,v divide m,n respectively. For n as in (ii), we have $d(n)=(r_1+1)(r_2+1)\ldots(r_k+1)$. **38.** Let S be the set of residue classes modulo n of integers that are relatively prime to n. Then $|S|=\phi(n)$. Let $[m]\in S$ and suppose that the order of $[m]$, as defined in Exercise (4.12)(c), is k. It suffices to show that k divides $\phi(n)$. Note that the distinct powers of $[m]$ are $[1],[m],[m^2],\ldots,[m^{k-1}]$. Now define a binary relation $\sim$ on S as follows. For $[x],[y]\in S$, define $[x]\sim[y]$ to mean that there exists some i such that $[xm^i]=[y]$. Reflexivity and transitivity of $\sim$ are immediately checked. For symmetry use the fact that for any i, $[m^i][m^{k-i}]=[1]$. Thus $\sim$ is an equivalence relation on the set S with $\phi(n)$ elements. The proof is completed by showing that each equivalence class has exactly k elements. **39.** (a) Complementary counting. (b) Let R be the counter-clockwise rotation of the polygon through an angle $2\pi/p$. Given a particular colouring of vertices of the p-gon, let k be the least positive integer such that the same colouring is obtained when R is applied k times. Then the colouring gets identified with exactly k colourings (including itself). (This integer k may be called

the order of the colouring.) By an argument analogous in spirit to that in Exercise (4.11) for the order of a residue class, it follows that k must divide p. When p is prime this means $k = 1$ or p. As at least two colours are involved, k cannot be 1. But when p is not a prime, k could be less than p. For example, consider a colouring of the vertices of a square in which two diagonally opposite vertices are white and the other two are black. (c) Declare two colourings in the set S equivalent if and only if they are obtained from each other by some rotation. This is an equivalence relation with each equivalence class having exactly p elements. **40.** In (b), these are those elements $[(g_{i_1}, g_{i_2}, \ldots, g_{i_p})]$ of S in which the subscripts are in an A.P. (modulo p). As the common difference can be from 1 to $p-1$, there are $p-1$ such elements. For any other element, increasing each subscript by some $k = 1, 2, \ldots, p$ will give p equivalent elements, proving (c). As there are $p-1$ singleton equivalence classes, and the remaining equivalence classes have cardinalities p each, $(p-1)! = |S| = p - 1 + pm$ for some integer m. This proves (d).

Chapter 5

1. (i) Clearly we may assume $r \leq k \leq n$. The L.H.S. is the number of ways to select k objects from n ones by marking them with one tick (say) and then further selecting r out of these k objects by marking them with another tick. But we might as well choose these r doubly favoured objects directly from n ones and then select $k-r$ more (singly favoured) objects from the remaining $n-r$ objects. Algebraic proof is very straightforward. For (ii), an algebraic proof can be given by repeated use of the reduction formula after bringing $\binom{n}{k+1}$ to the left. For a combinatorial proof, the R.H.S. is the number of ways to form a committee of $k+1$ persons from a set with m men and $r+1$ women numbered 1 to $r+1$ so that there is at least one woman in the committee. If we classify these committees according to the highest numbered woman in them, we get the sum on the L.H.S. (iii) can be proved by writing $\binom{p}{j}$ as $\binom{p}{p-j}$ and then using (12) in the text. For a combinatorial proof, the L.H.S. is the number of committees that can be formed from p men and q women so that there are exactly n more women than men in the committee. For any such committee A, form another committee B consisting of all women in A and all men not in A. Then B always has $p+n$ members. Moreover A and B determine each other uniquely. Finally (iv) follows by applying (i) to each term and then (22) to sum the terms. Alternatively, look at the R.H.S. as the number of ways to select a team of m players from n players and then some subset of the remaining $n-m$ players as a reserve force.

2. $\binom{n-1}{k} + (-1)^r \binom{n-1}{k-r-1}$. (Consider the coefficient of x^k in $\sum_{j=0}^{r} (1+x)^n (-1)^j x^j$ which is a geometric series. The sum is 0 when $n = r = k$ which is

consistent with (23).) **3.** (i) The first sum equals the coefficient of x^m in $(1-x)^n(1+x)^n$, i.e., in $(1-x^2)^n$. The second sum comes out to be $(-1)^m\binom{n-1}{m}$. This can be proved by induction on m. Or, apply Exercise (5.2) with $k=r=m$. (ii) Note that the first term of the sum is the same, viz., 1 for every n. Rewriting $\binom{2n}{0}$ as $\binom{2n+1}{0}$ and $\binom{2n-k}{k}$ as $\binom{2n-k+1}{k}-\binom{2n-k}{k-1}$ for $k>0$, we get that for $n\geq 1$, $S_n=\sum_{k=0}^{n}(-1)^k\binom{2n-k}{k}=\sum_{k=0}^{n}(-1)^k\binom{2n+1-k}{k}+\sum_{k=0}^{n}(-1)^k\binom{2n-1-k}{k}$, where in the last sum a change of index has been applied and the term corresponding to $k=n$ is redundant. A similar manipulation changes the first sum on the right to $S_{n+1}+S_n$ and the second one to S_n+S_{n-1} for $n\geq 2$. This proves $S_n=S_{n+1}+2S_n+S_{n-1}$ and hence $S_{n+1}=-(S_n+S_{n-1})$ for $n\geq 2$. The last part now follows by induction on n after verifying the first two cases viz., $S_1=0$ and $S_2=-1$. **4.** (i) Using the hint, rewrite the sum as $\frac{1}{\binom{n}{m}}\sum_{k=0}^{m}\binom{n-k}{m-k}$. The latter sum equals $\sum_{k=0}^{m}\binom{n-k}{n-m}$ which is the coefficient of x^{n-m} in a certain geometric progression. (ii) The sum is only ostensibly infinite. Write $k\binom{n}{k}\binom{m}{k}$ as $m\binom{n}{n-k}\binom{m-1}{k-1}$ and apply Vandermonde convolution. (iii) (C). (Differentiaing $\sum_{k=0}^{n}(-1)^kC_kx^k=(1-x)^n$ and multiplying by x gives $\sum_{k=0}^{n}(-1)^kkC_kx^k=-nx(1-x)^{n-1}$ and hence $\sum_{k=0}^{n}(-1)^k(k+1)C_kx^k=(1-x)^{n-1}(1-x-nx)$. Now multiply by $\sum_{k=0}^{n}C_{n-k}x^{n-k}$ which equals $(1+x)^n$ to recognise the given sum as the coefficient of x^n in $(1-x^2)^n-$ the coefficient of x^{n-2} in $n(1-x^2)^{n-1}$.) (iv) The given sum equals $f(1)+2g(1)+h(1)$, where $f(x)=\sum_{k=0}^{n}(-1)^k\binom{n}{k}x^k$, $g(x)=\sum_{k=0}^{n}(-1)^kk\binom{n}{k}x^k$ and $h(x)=\sum_{k=0}^{n}(-1)^kk^2\binom{n}{k}x^k$. Clearly, $f(x)=(1-x)^n$. Differentiate and multiply by x to get $g(x)=-nx(1-x)^{n-1}$. Another differentiation and multiplication by x gives $h(x)=-nx(1-x)^{n-1}+n(n-1)x^2(1-x)^{n-2}$. Clearly $f(1)=g(1)=h(1)=0$. **5.** For the inductive step, use $\sum_{k=1}^{n+1}k^2\binom{n+1}{k}=\sum_{k=1}^{n+1}k^2\binom{n}{k}+\sum_{k=1}^{n+1}k^2\binom{n}{k-1}$. The last term in the first sum is 0 while the second sum is, by a change of index, the same as $\sum_{k=0}^{n}(k+1)^2\binom{n}{k}$. To complete the proof we need (22) and (33) besides the induction hypothesis. **6.** Integrate the given equation to get $C_0+C_1x+C_2x^2+\ldots+C_{2n}x^{2n}=(1+x)^{2n}$, where C_0 is some constant which is determined as 1 by setting $x=0$. As this holds for all x, we must have $C_r=\binom{2n}{r}$ for every $r=0,1,2,\ldots,2n$. Hence $C_r=C_{2n-r}$ for all r. So the given sum can be written as $-\sum_{r=0}^{2n}r\binom{2n}{r}(-1)^r\binom{2n}{2n-r}$ and

can be recognised as the coefficient of x^{2n} in $-2nx(1+x)^{2n-1} \times (1-x)^{2n}$ and hence as the coefficient of x^{2n-1} in $-2n(1-x^2)^{2n-1}(1-x)$ i.e., in $2nx(1-x^2)^{2n-1} - 2n(1-x^2)^{2n-1}$. The second term contributes nothing. The rest is routine. Note that we need not have been *given* the value of C_r. We have *found* it. **7.** $\frac{405}{256}$. (A typical term $(-1)^k\binom{10}{k}(\frac{x}{2})^k(\frac{3^{10-k}}{x^{20-2k}})$ will be a scalar multiple of x^4 if and only if $k-20+2k=4$, i.e., for $k=8$.) **8.** $\frac{na_n}{2}$. (Since ${}^nC_r = {}^nC_{n-r}$, $\sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n-r}{{}^nC_r}$. Add the two.) **9.** We have $(c_0 + c_1 + c_2 + \ldots + c_n)^2 = (2^n)^2 = 2^{2n}$. Expand the square and note that $\sum_{k=0}^{n} c_k^2 = \sum_{k=0}^{n} c_k c_{n-k}$ equals the coefficient of x^n in $(1+x)^{2n}$.

10. (a) $a=2, n=4$. (b) $r=3$. (Solve : $\frac{n-r+1}{r} = \frac{7}{3}$ and $\frac{n-r}{r+1} = \frac{3}{2}$.)
(c) $n = 2r+1$. (Assuming that the terms are counted from 0 onwards.)
(d) $n=7$. (Assuming that the terms are counted from 1 onwards.)
(e) $m=12$. (f) $n=3$. (The data means that $\sum_{k=1}^{n}\binom{2n+1}{k} = 63$ and hence that $\sum_{k=0}^{n}\binom{2n+1}{k} = 64$. Apply (28) to get $2^{2n}=64$.)

11. Let $n = 2m$. The given sum, say A, then equals $\frac{1}{i\sqrt{3}}f(i\sqrt{3})$, where $f(x) = \sum_{r=1}^{3m}\binom{6m}{2r-1}x^{2r-1}$. If we let $g(x) = \sum_{r=0}^{3m}\binom{6m}{2r}x^{2r}$, then $f(x)+g(x) = \sum_{j=0}^{6m}\binom{6m}{j}x^j = (1+x)^{6m}$ while $g(x) - f(x) = (1-x)^{6m}$. So $f(x) = \frac{(1+x)^{6m}-(1-x)^{6m}}{2}$. The problem amounts to showing that $(1+i\sqrt{3})^{6m} = (1-i\sqrt{3})^{6m}$ or equivalently, to showing that $(\frac{1+i\sqrt{3}}{2})^{6m} = (\frac{1-i\sqrt{3}}{2})^{6m}$. Note that $(\frac{1+i\sqrt{3}}{2})^2 = \omega$ and $(\frac{1-i\sqrt{3}}{2})^2 = \frac{1}{\omega}$, where ω is the complex cube root of unity. The result follows since $\omega^{3m} = 1$. **12.** Using $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$, the L. H. S. of (32) is $n\sum_{k=1}^{n}\binom{n-1}{k-1}x^{k-1}$. By a change of index of summation and the binomial theorem, this is simply $n(1+x)^{n-1}$. **13.** This is a special case of Exercise (5.1) (iv) with $m=1$. **14.** (a) Putting $m=3$ in (1) gives $4\sum_{k=1}^{n}(k^3-3k^2+2k) = (n+1)n(n-1)(n-2)$. Use the formulas for $\sum_{k=1}^{n}k$ and $\sum_{k=1}^{n}k^2$. (b) No. The identity and hence its differentiation is valid only for $x \neq 1$. (c) This is valid because since the equalities hold for all $x \neq 1$, their limits as $x \to 1$ will be equal provided they exist. (d) Let $f(x) = \left(\sum_{k=0}^{2m}x^k\right)^n$. Then $\sum_{k=0}^{2mn}a_kx^k$, $\sum_{k=0}^{2mn}ka_kx^k$ and $\sum_{k=0}^{2mn}k^2a_kx^k$, equal, respectively, $f(x), xf'(x)$ and $x(xf'(x))' = x^2f''(x) + xf'(x)$. The given

sum can be written as $\sum_{k=0}^{2mn} \frac{k^2-k}{2} a_k$ and hence equals $\frac{1}{2}f''(1)$. To find its value, differentiate $1 + x + x^2 + \ldots + x^{2m}$ term by term twice, put $x = 1$ and use the formulas for the sums $\sum_{k=0}^{2m} k$ and $\sum_{k=0}^{2m} k^2$. Alternately, one can write $f(x)$ as $\left(\frac{1-x^{2m+1}}{1-x}\right)^n$ and differentiate twice. But in order to get $f''(1)$ one has to take $\lim_{x\to 1} f''(x)$.

15. (a) The sum equals $-2^n f(\frac{1}{2})$, where $f(x) = \sum_{r=0}^{n} (-1)^r r^3 \binom{n}{r} x^r$. Start from $\sum_{r=0}^{n} (-1)^r \binom{n}{r} x^r = (1-x)^n$. Differentiate and multiply by x. Doing this thrice gives $f(x) = -n(n-1)(n-2)(1-x)^{n-3}x^3 + 3n(n-1)(1-x)^{n-2}x^2 - n(1-x)^{n-1}x$. (b) The given sum equals $-\frac{1}{2}(g(1) + h(1))$, where $g(x) = \sum_{r=0}^{n} (-1)^r \binom{n}{r} rx^r$ and $h(x) = \sum_{r=0}^{n} (-1)^r \binom{n}{r} r^2 x^r$. By the same argument as in (a), show that $g(1) = 0 = h(1)$. (c) Once again, starting with $f_0(x) = \sum_{r=0}^{n} (-1)^r \binom{n}{r} x^r = (1-x)^n$, define a sequence of functions inductively by $f_k(x) = xf'_{k-1}(x)$ for $k \geq 1$. Then S_k is simply $f_k(1)$ and the terms in $f_k(x)$ having $(1-x)$ as a factor contribute nothing to this sum. (d) Expand $(2n-r)^{n+1}$ by the binomial theorem to show that the given sum equals the sum $\sum_{k=0}^{n+1} (-1)^k \binom{n+1}{k} (2n)^{n+1-k} S_k$. Apply (c). (e) Once k of the unlike objects are chosen, the remaining $n-k$ objects from the first two types can be chosen in $n-k+1$ ways. So the problem amounts to finding the sum $\sum_{k=0}^{n} \binom{n}{k}(n+1-k)$. Use (22) and (33). (f) Integrate $\sum_{k=0}^{n} \binom{n}{k} x^k = (1+x)^n$ and set $x = 2$. (g) Here $s_k = \dfrac{q^{k+1}-1}{q-1}$ for $k \geq 1$. Setting $s_0 = 0$, the given sum, say A, equals $\sum_{k=1}^{n+1} \binom{n+1}{k} s_{k-1}$ which comes out to be $\frac{1}{q-1} \sum_{k=1}^{n+1} \binom{n+1}{k}(q^k - 1)$ which is the same as $\frac{1}{q-1} \sum_{k=0}^{n+1} \binom{n+1}{k}(q^k - 1)$. By the binomial theorem (once with $x = q$ and again with $x = 1$) this comes out to be be $\frac{1}{q-1}\left((q+1)^{n+1} - 2^{n+1}\right)$. By a direct computation, $2^n S_n$ also equals this. **16.** Call the given sum as a_n. The index k can be made to run from $-\infty$ to ∞ ostensibly. (Only the terms for $0 \leq k \leq \frac{n}{2}$ matter.) Then $a_n = \sum_k \binom{n-k}{k} = \sum_k \binom{n-k-1}{k} + \sum_k \binom{n-k-1}{k-1}$. With a change of index in the second summation, this becomes $\sum_k \binom{(n-1)-k}{k} + \sum_k \binom{(n-2)-k}{k} = a_{n-1} + a_{n-2}$. Apply induction. **17.** Let k be the number of two rupee coins. Then in all there are $n-k$ coins.

18. In (i) we could as well range k over all integers. For an algebraic proof, note first that $S_{n+1,m+1} = (m+1)S_{n,m+1} + S_{n,m}$ by Exercise (1.28). Apply induction on n to write $S_{n,m+1}$ as $\sum_k \binom{n-1}{k} S_{k,m}$ and $S_{n,m}$ as $\sum_k \binom{n-1}{k} S_{k,m-1}$. Then $S_{n+1,m+1}$ equals $\sum_k \binom{n-1}{k}(mS_{k,m} + S_{k,m-1}) + \sum_k \binom{n-1}{k} S_{k,m}$. Apply Exercise (1.28) again to rewrite the first sum as $\sum_k \binom{n-1}{k} S_{k+1,m}$ and, hence as $\sum_k \binom{n-1}{k-1} S_{k,m}$. The reduction formula $\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$ completes the proof. For (ii), the range of variation of k is crucial. Using Exercise (1.28) again, $S_{n+1,m+1} = (m+1)S_{n,m+1} + S_{n,m}$. By induction on n, this equals $(m+1)\sum_{k=m}^{n-1}(m+1)^{n-1-k}S_{k,m} + S_{n,m} = \sum_{k=m}^{n-1}(m+1)^{n-k}S_{k,m} + S_{n,m} = \sum_{k=m}^{n}(m+1)^{n-k}S_{k,m}$. In these proofs you don't have to know anything about the Stirling numbers other than the recurrence formula for them. The combinatorial proofs, on the other hand, crucially need that $S_{n+1,m+1}$ stands for the number of ways to put $n+1$ distinct objects, say $x_1, x_2, \ldots, x_n, x_{n+1}$ into $m+1$ boxes so that no box is empty. Let B be the box in which x_{n+1} is put. Let k be the number of elements in the remaining m boxes. Then k varies from m to n. Since for each such k, these k elements can be chosen in $\binom{n}{k}$ ways, we get (i). For (ii) suppose that the objects are put into the boxes one by one, in the order $x_1, x_2, \ldots, x_{n+1}$. Let x_{k+1} be the last object to fall into a box which is previously empty. Then the first k objects occupy the remaining m non-distinct boxes (with no box to remain empty) and the last $n-k$ objects can be put freely into any of the $m+1$ boxes (which are now distinguished by their contents).

19. The given sum, say S, equals $T + (-1)^n a_n + (-1)^{n+1}a_{n+1}$, where $T = \sum_{k=0}^{n-1}(-1)^k a_k$. From the symmetry relation $a_{2n-k} = a_k$ and the fact that $\sum_{k=0}^{2n}(-1)^k a_k = 0$ we get $T = (-1)^{n+1}\frac{a_n}{2}$ and hence $S = (-1)^n\frac{a_n}{2} + (-1)^{n+1}a_{n+1}$. Put $a_n = \binom{2n}{n}$ and $a_{n+1} = \binom{2n}{n+1}$ to get the result.

20. Follow the notations and the reasoning in the solution to the second problem in Comment No. 16. Call $(1+x+x^2)^n$ as $f(x)$. For (a), write $f(x)$ as $\sum_{k=0}^{n}\binom{n}{k}(1+x)^{n-k}(x)^{2k}$. So, a_n = the coefficient of x^n in $f(x)$ equals $\sum_{k=0}^{n}\binom{n}{k}\binom{n-k}{n-2k} = \sum_{k=0}^{n}\binom{n}{2k}\binom{2k}{k}$. (For $k > \frac{n}{2}$, the terms are 0.) In (b), because of the symmetry relation $a_r = a_{2n-r}$, the given sum equals $\sum_{k=0}^{2n-1}(-1)^k a_k a_{2n-k-1}$ which is precisely the coefficient of x^{2n-1} in the product $f(-x)f(x)$, i.e., in $(1+x^2+x^4)^n$ and hence is 0. For (c), too, because

of the symmetry relation, the given sum is half of $\sum_{k=0}^{2n}(-1)a_k^2-(-1)^n a_n^2$. The result follows by using (31). For (d), note that the given sums equal, respectively, $\frac{f(1)+f(-1)}{2}$ and $\frac{f(1)-f(-1)}{2}$. Finally, for (e), let S_1,S_2,S_3,S_4 denote the given sums. By (d), S_1+S_3 and S_2+S_4 are known. Also $f(i)=(1+i+i^2)^n=i^n=S_1-S_3+i(S_2-S_4)$. Equating the real and imaginary parts of the two sides gives two more equations in S_1,S_2,S_3,S_4. However, since i^n depends on the congruence class of n modulo 4, so do the values of these four sums. The truth of the assertion can be verified separately in each of the four cases. **21.** There are three types of solutions. Those in which both the sides vanish, have at least one vanishing factor on each side and this can happen in $(3^n-2^n)^2$ ways. Solutions in which each side equals 1 have an even number of negative factors on each side and this happens in E^2 ways, where $E=\sum_{m=0}^{\infty}\binom{n}{2m}=2^{n-1}$ by (24). Similarly count the number of solutions in which both sides equal -1 using (25). Add the three counts. **22.** Writing $a_r=a_0+rd$ for $r=0,1,2,\ldots,n$, the given sum equals $a_0^2\sum_{r=0}^{n}\binom{n}{r}+2a_0d\sum_{r=0}^{n}\binom{n}{r}r+d^2\sum_{r=0}^{n}\binom{n}{r}r^2$. The three sums equal, respectively, $f(1), f'(1)$ and $f''(1)+f'(1)$, where $f(x)=\sum_{r=0}^{n}\binom{n}{r}x^r=(1+x)^n$. **23.** The expression in the brackets equals $\sum_{k=1}^{m}\left(1-(\frac{1}{2})^k\right)^r$ which is not easy to evaluate. But the given sum equals $\sum_{r=0}^{n}(-1)^r\binom{n}{r}\sum_{k=1}^{m}\left(1-(\frac{1}{2})^k\right)^r$. Here the indices r and k vary independently of each other and so the order of summation can be changed. Then the sum is $\sum_{k=1}^{m}\left(\sum_{r=0}^{n}(-1)^r\binom{n}{r}(1-(\frac{1}{2})^k)^r\right)$ which is simply $\sum_{k=1}^{m}(1-1+(\frac{1}{2})^k)^n=\sum_{k=1}^{m}\left((\frac{1}{2})^n\right)^k=\frac{(\frac{1}{2})^n(1-(\frac{1}{2})^{nm})}{1-(\frac{1}{2})^n}$. **24.** For (i), write k as $n-(n-k)$ and proceed as in Comment No. 4. For (ii) and (iii) apply 'dumb' induction on p using the reduction formula. **25.** The given sum, say A, equals the sum of the coefficients of the powers of xyz in $(1+x)^n(1+y)^n(1+z)^n$. Taking $z=\frac{1}{xy}$, A is the constant term in the expansion of $(1+x)^n(1+y)^n(1+xy)^n/(xy)^n$.

Chapter 6

1. (i) Add the three rows to take out $(a+b+c)$ as a positive factor. The other factor $ab+bc+ca-a^2-b^2-c^2$ equals $-\frac{1}{2}((a-b)^2+(b-c)^2+(c-a)^2)$. (ii) (A). (Put $a=x,b=2y,c=3z$. Then $u=a^2+b^2+c^2-ab-bc-ca$. (iii) (C). (The discriminant is $4(a^2+b^2+c^2-ab-bc-ca)$.) (iv) The roots are $2q\pm\sqrt{2q^2+r}$. Since $r=\alpha^4+\beta^4$ and $\alpha\beta=q$, we have $r\geq 2q^2$ and

hence $\sqrt{2q^2+r} \geq |2q|$. So one root is positive and the other negative.

2. Yes. (Let $z = x + iy$. Then $\frac{1-z}{1+z} = a + ib$, where $a = \frac{1-x^2-y^2}{1+2x+x^2+y^2}$ and $b = \frac{-2y}{1+2x+x^2+y^2}$. Given $x \geq 1$ and $y \geq 0$, it follows that $a \leq 0$ and $b \leq 0$.)

3. (a) No. (The quadratic $x^2+4x+30-k(x^2-8x+18) = 0$ will have distinct real roots if $(2+4k)^2 > (1-k)(30-18k)$, i.e., if $k^2 - 32k + 13 < 0$. So values between $16 - \sqrt{243}$ and $16 + \sqrt{243}$ have two preimages under f.) (b) $[2, 14]$. (The equation $\alpha x^2+6x-8-k(\alpha+6x-8x^2) = 0$ has real roots if and only if $k^2(9+8\alpha) + k(\alpha^2+46) + (8\alpha+9) \geq 0$. If this is to hold for every k, then we must have $9 + 8\alpha \geq 0$ and $(\alpha^2+46)^2 - 4(9+8\alpha)^2 \leq 0$. The latter reduces to $2 \leq \alpha \leq 14$. Strictly speaking, the denominator in $f(x)$ has real roots for these values of α and so $f(x)$ is not defined on entire $\mathbb{R}$. But we ignore this point as otherwise, the answer would be the empty set.) The function f is not one-to-one for $\alpha = 3$ since in that case f vanishes at two distinct points.

4. $|a + b - 2c| \leq |a - b|$. ($(a + b + k)^2 \geq 4(ab + kc)$ must hold for every k.)

5. (a) Consider two cases, depending upon whether $x \geq y$ or $x < y$. The smaller of x and y equals $\frac{1}{2}(x + y - |x - y|)$. (b) Verify separately for $x \geq 0$ and $x < 0$. (c) The graph of $y = |f(x)|$ is obtained from that of $y = f(x)$ by replacing its portions below the x-axis by their reflections into the x-axis. That of $y = f^+(x)$ is obtained by replacing these portions by their projections onto the x-axis. (We also say this by saying that the graph of $y = f(x)$ is 'truncated' below by the x-axis.) To get the graph of $y = f^-(x)$ from that of $y = f(x)$, first replace portions above the x-axis by their projections and then reflect the resulting graph into the x-axis.

6. (a) Take $x = |m - c|$ and $y = \frac{1}{2}(b-a) = (m-a)$. (It is easier to see this if you draw a diagram.) (c) By (b) and the induction hypothesis, $A^{n-1} \geq a_2 a_3 \ldots a_{n-1}(a_1 + a_n - A)$. From this and the hint, $A^n \geq a_1 a_2 \ldots a_{n-1} a_n$. If equality holds then we must have $A^{n-1} = a_2 a_3 \ldots a_{n-1}(a_1 + a_n - A)$ and $A(a_1 + a_n - A) = a_1 a_n$. By induction hypothesis, $a_2 = a_3 = \ldots = a_{n-1} = a_1 + a_n - A$ from which one can prove that all a's are equal.

8. The hint gives $C(AC - |B|^2) \geq 0$. Divide by C. **9.** (C). (On one hand $(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$ while on the other $(a+b+c)^2 \geq 0$.)

10. (a) 2. (b) Among all rectangles of unit area, the square has the least perimeter. (c) If x, y, z are the dimensions of the box, the surface area is $2(xy + yz + zx)$ and the volume is xyz. Assume $xyz = V$. Since H. M. $\leq$ G.M., we have $\frac{3xyz}{xy+yz+zx} \leq (xyz)^{1/3}$, i.e., $2(xy + yz + zx) \geq 6V^{2/3}$, with equality holding only for $x = y = z$ i.e., for a cube. (d) Let z denote the height. Then the surface area, say S, equals $xy+2yz+2zx = xy+yw+wx$, where $w = 2z$. The volume V equals $\frac{1}{2}xyw$. So S is minimum when $x = y = w$, i.e., when $x = y = 2z$ i.e., when $x = y = 2(\frac{V}{4})^{1/3}$ and

$z = (\frac{V}{4})^{1/3}$. (e) The dual of (a) says that if x, y are positive and $x+y = 2$, then the maximum value of xy is 1. The dual of (c) is that among all boxes with a given surface area, the cube encloses the maximum volume.

11. $(-\infty, \frac{1}{3}]$ and $[-\frac{1}{27}, \infty)$ respectively. (See Comment No. 5 (vi) of Chapter 24 for details.) **12.** $x \in [-1, 1) \cup (2, 4]$. (Factorise both the quadratics.)

13. (a) By A.M.−G.M. inequality we have $a + b + c \geq 3(abc)^{1/3}$ and also $\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \geq \frac{3}{(abc)^{1/3}}$. Multiply the two. More generally, for any positive real numbers $a_1, a_2, \ldots, a_n$, we have $(a_1+a_2+\ldots+a_n)(\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}) \geq n^2$. (b) Follow the hint and apply the A.M.−G.M. inequality.

14. There is no minimum. (The expression equals $y+\frac{1}{y}$, where $y = \log_a x < 1$. As $0 < y < 1$, $y + \frac{1}{y}$ comes as close to 2 as we like but does not equal 2. If $x = a$ were allowed, then the minimum is 2.)

15. (a) Apply A.M. -G.M. inequality to each of the n pairs $\{k, n-k+1\}$, for $k = 1, 2, \ldots, n$ and multiply. (b) Apply A.M.-G.M. to the n numbers $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}$ to get $\left(\frac{n+1}{2n}\right)^n > \frac{n!}{n^n}$. Since $\frac{n!}{n^n} < 1$, we also have $\frac{n!}{n^n} > \left(\frac{n!}{n^n}\right)^{3/2}$. (c) From (a), we have $\left(\frac{n+1}{2}\right)^{3n} > (n!)^3$. Also $n^n > n!$. Multiply the two. (d) Call the product as a_n. Let $b_n = \prod_{k=1}^{n} \frac{2k}{2k+1}$. Then $a_n < b_n$ whence $a_n^2 < a_n b_n = \frac{1}{2n+1} < \frac{1}{2n}$. (Note that $a_n = \frac{(2n)!}{4^n n! n!}$. Using the Sterling approximation in Exercise (6.53), a_n asymptotically equals $\frac{1}{\sqrt{\pi n}}$ which is stronger than the inequality asked. But this is a highly sophisticated proof.) (e) The assertion is equivalent to proving that $2(1 + b + b^2 + \ldots + b^n) < (n + 1)(1 + b^n)$. Rewrite the L.H.S. as $\sum_{k=0}^{n} (b^k + b^{n-k})$ and observe that for $k = 1, 2, \ldots, n-1$, $b^k + b^{n-k} < 1 + b^n$ since $(1 - b^k)(1 - b^{n-k}) > 0$ as both the factors are of the same sign regardless of whether $0 < b < 1$ or $b > 1$. (f) The problem is equivalent to proving that $A_n < B_n$, where $A_n = \frac{a+a^3+a^5+\ldots+a^{2n-1}}{n}$ and $B_n = \frac{1+a^2+a^4+\ldots+a^{2n}}{n+1}$. From the inequalities $a < \frac{1+a^2}{2}$, $a^3 < \frac{a^2+a^4}{2}, \ldots, a^{2n-1} < \frac{a^{2n-2}+a^{2n}}{2}$, we get that $A_n < C_n$, where $C_n = \frac{1+2a^2+2a^4+\ldots+2a^{2n-2}+a^{2n}}{2n}$ and so it suffices to show that $C_n < B_n$. By direct calculation the latter is equivalent to proving that $\frac{1+a^2+a^4+\ldots+a^{2n}}{n+1} < \frac{1+a^{2n}}{2}$. But this is true from (e) (with $b = a^2$).

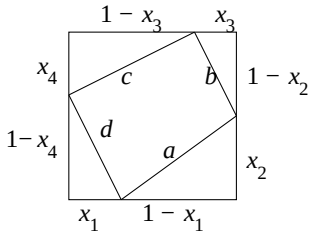
16. Take the three numbers as $\frac{b}{r}, b$ and br, where $r \neq 1$ as the numbers are distinct. We are given that $b(\frac{1}{r} + 1 + r) = \alpha S$ and $b^2(\frac{1}{r^2} + 1 + r^2) = \alpha^2 S^2$. Eliminate b and put $u = r + \frac{1}{r}$ to get $\alpha^2 = \frac{(u+1)}{u^2-1} = \frac{u+1}{u-1} = 1 + \frac{2}{u-1}$. Since $r \neq 1$, u varies either over $(2, \infty)$ or over $(-\infty, -2)$. Correspondingly, $\alpha^2 \in (1, 3)$ or $\alpha^2 \in (\frac{1}{3}, 1)$.

17. $(-\infty, 2)$. (The larger root, $-a + \sqrt{3-a}$, is less than 3. So $\sqrt{3-a} < 3 - a$. Hence $3 - a > 1$.) **18.** $(2, \infty)$. (Put $\alpha = \log_{0.3}(x - 1)$. Then

$\log_{0.09}(x-1) = \dfrac{\log_{0.03}(x-1)}{\log_{0.3}(0.09)} = \dfrac{\alpha}{2}$. The given condition reduces to $\alpha < 0$. Since the base 0.3 is less than 1, this gives $x - 1 > 1$.)

19. $x^2 + y^2 = x^2 + (1-x)^2 = 2(x - \frac{1}{2})^2 + \frac{1}{2}$. The first term varies from 0 to $\frac{1}{2}$. (A geometric argument is also possible by taking the line segment in the x-y plane from $(1, 0)$ to $(0, 1)$ and considering the points on it that are closest and farthest from the origin.)

20. Let $x_1, 1-x_1, x_2, 1-x_2$ etc. be the lengths of the segments of the sides of the square determined by the vertices of the quadrilateral as shown in the accompanying figure. Then $a^2 + b^2 + c^2 + d^2 = \sum_{i=1}^{4} x_i^2 + (1-x_i)^2$. Apply the last exercise to each term.



21. First assume $a > 0$. Since the leading coefficient is positive, the expression $ax^2 + bx + c < 0$ for all $x \in (\alpha, \beta)$ and hence in particular for $x = \pm 1$. This gives $a + b + c < 0$ and $a - b + c < 0$. $|b|$ equals either b or $-b$. In either case, the assertion holds. If $a < 0$, multiply the equation throughout by -1. This does not affect either the hypothesis or the conclusion.

22. (A) and (D). Group together the terms of $\alpha(n)$ as $(1) + (\frac{1}{2} + \frac{1}{3}) + (\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}) + (\frac{1}{8} + \ldots + \frac{1}{15}) + \ldots + (\frac{1}{2^{n-1}} + \ldots + \frac{1}{2^n - 1})$. Each group adds to at most 1. So $\alpha(n) \le n$. On the other hand replacing the first term by the smaller term $\frac{1}{2^n}$ and grouping as $(\frac{1}{2}) + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \ldots + \frac{1}{8}) + \ldots + (\frac{1}{2^{n-1}+1} + \ldots + \frac{1}{2^n})$ shows $\alpha(n) > \frac{n}{2}$ since each group adds to at least $\frac{1}{2}$.)

23. $a - \sqrt{2}a$ and $-a + \sqrt{6}a$. (For $x \ge a$, the equation is $x^2 - 2a(x-a) - 3a^2 = 0$. The roots are $a \pm \sqrt{2}a$, of which only $a - \sqrt{2}a$ is $\ge a$ since $a \le 0$. For $x < a$, the roots are $-a \pm \sqrt{6}a$ of which only $-a + \sqrt{6}a$ is to be taken.)

24. (a) $[-2, 1) - \{0\}$. (b) Domain $= (-\infty, \infty)$, range $= [0, 1)$. No. (e.g., $f(1) = f(-1)$.)

25. (a) $[-1, 2) \cup [3, \infty)$. ($(x+1)(x-3)$ must have the same sign as $x - 2$.) (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$. (Make two cases, depending upon whether $x \ge -2$ or $x < -2$.) (c) $(-\infty, -1) \cup (-\frac{1}{2}, 0) \cup (\frac{1}{2}, \infty)$. (The roots of the denominator are $-1, -\frac{1}{2}$ and 0.) (d) 4. (Solving as a quadratic in $|x-2|$, the roots are 1 and -2. $|x-1| = 1$ gives $x = 1$ or 3. $|x-2| = -2$ has no solution.) (e) $[1, \infty)$. ($|2^{y-1} - 1| = \pm(2^{y-1} - 1)$ depending upon whether $y \ge 1$ or $y < 1$. Since $|y|$ is also involved, make three cases: (i) $y \le 0$, (ii) $0 < y < 1$ and (iii) $1 \le y$.

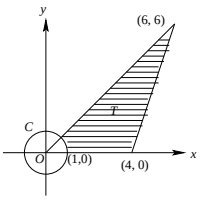
26. $(5, \infty)$. (The first two expressions are both positive only for $a > 0$. The third is always positive. The requirement $(a^2 + 2a) + (2a + 3) > a^2 + 3a + 8$ gives $a > 5$. The other two requirements hold for all $a > 0$.)

27. (a) Since $R = (5\sqrt{5}+11)^{2n+1} = \frac{4^{2n+1}}{(5\sqrt{5}-11)^{2n+1}}$, the problem reduces to showing that $f = (5\sqrt{5}-11)^{2n+1}$, or equivalently, to showing that $I = [R]$, the integral part of R, where $I = R - (5\sqrt{5}-11)^{2n+1}$. Certainly $I = (5\sqrt{5}+11)^{2n+1} - (5\sqrt{5}-11)^{2n+1}$ is an integer as the irrational terms cancel out. Moreover $R - I = (5\sqrt{5}-11)^{2n+1}$ which lies between 0 and 1 since $5\sqrt{5}$ lies between 11 and 12. Hence $I = [R]$. (b) By an analogous argument, when $n = 2m+1$, the integral part of $(\sqrt{5}+2)^n$ is $(\sqrt{5}+2)^n - (\sqrt{5}-2)^n$ which equals $\sum_{k=0}^{m} 2\binom{2m+1}{2k}5^k 2^{2m-2k+1}$. This is divisible by 4 as each term of the sum is so. When $n = 2m$ (say), the expression $I = (\sqrt{5}+2)^{2m} + (\sqrt{5}-2)^{2m} = \sum_{k=0}^{m} 2\binom{2m}{2k}5^k 2^{2m-2k}$ is an integer which is bigger than $(\sqrt{5}+2)^n$. But the difference (viz., $(\sqrt{5}-2)^n$) lies between 0 and 1. So the integral part of $(\sqrt{5}+2)^n$ is not I but $I-1$. The assertion follows from the fact that all except the last term in the sum are divisible by 4 while the last term, viz. $5^m 2$ is congruent to 2 modulo 4. (c) 9. (The expression falls short of an integer by $(\sqrt{2}-1)^{300}$ which is less than 10^{-100} since $(\sqrt{2}-1)^3 = \frac{1}{(\sqrt{2}+1)^3} = \frac{1}{7+5\sqrt{2}} < \frac{1}{10}$.)

28. (a) (i) Divide the interval $[0,3)$ into 12 equal subintervals of the form $[\frac{k-1}{12}, \frac{k}{12})$ for $k = 1, 2, \ldots, 12$. On each of these subintervals the function is constant. The values on the first four subintervals are $0, 1, 3$ and 4. Those over the next four subintervals are obtained by adding 7 to these values and this goes on. From this information a graph is easy to draw. (ii) The graph consists of the segment joining $(0,1)$ to $(1,0)$ and three more segments parallel to it. Both the graphs are only piecewise continuous.
(b) Let $f(x) = [x] + [2x] + [4x] + [8x] + [16x]$ and $g(x) = [x] + [2x] + [4x] + [8x] + [16x] + [32x]$. As in (i) above, the values of $f(x)$ are completely determined by its values over the interval $[0,1)$. Here f satisfies the periodicity-like condition $f(x+1) = f(x)+31$ for all x. Whether an integer n will be in the range of f depends only on the congruence class of n modulo 31. Since $12345 = 12400 - 55$, we easily see that $12345 \equiv 7\ (mod\ 31)$. Since $f(\frac{1}{4}) = 7$ we see that 7 and hence 12345 is in the range of f. As for $g(x)$, we have $g(x+1) = g(x) + 63$. Now, $12345 \equiv 60\ (mod\ 63)$. As $g(\frac{31}{32}) = 32$ while $g(1) = 63$, we see that 60 is not in the range of g. So there is no x for which $g(x) = 12345$. (c) Apply induction on $[\frac{n}{m}]$. Note that when n increases by m, each term on the R.H.S. increases by 1. For the ceiling function the sum on the right will equal $n + m - 1$.

29. The hint follows by writing a typical point on a line segment with end points at (u_1, v_1) and (u_2, v_2) in the form $(1-\lambda)u_1 + \lambda u_2, (1-\lambda)v_1 + \lambda v_2)$ for some $\lambda \in [0,1]$. As a result, all the boundary points of T satisfy the given inequality. As for an interior point, say P, let A be any vertex of T and let the line AP cut the opposite side at Q. Then P lies on the segment AQ whose end points satisfy the inequality.

30. The first inequality is satisfied by points outside the circle, say C, of radius 1 centred at $(1,0)$. The other three inequalities together determine a triangle with vertices $(0,0), (4,0)$ and $(6,6)$. The solution set is the intersection of this triangle with the exterior of the circle C.



31. $x = 3$ is the only value. **32.** $(-2,-1) \cup (-\frac{2}{3}, -\frac{1}{2})$. (Make two cases depending upon the sign of the product $(2x^2+5x+2)(x+1)$.)

33. This is best done multiplicatively rather than additively. Write $\frac{(n+1)^4}{n^4}$ as $1 + \frac{4}{n} + \frac{6}{n^2} + \frac{4}{n^3} + \frac{1}{n^4}$ and show that it is less than $\frac{10^{n+1}}{10^n}$ for $n \geq 2$.)

34. $m < -\frac{15}{2}$ and $m > 30$. ($x = \frac{25m}{2m+15}, y = \frac{2m-60}{2m+15}$. So, $m, 2m+15$ and $2m-60$ must have the same signs.) **35.** Let the figures be $x_1, x_2, \ldots, x_{12}$ clockwise. If $x_1+x_2, x_2+x_3, \ldots, x_{11}+x_{12}$ and $x_{12}+x_1$ are all ≤ 13, then their sum would be at most $13 \times 12 = 156$. But the sum $2\sum_{i=1}^{12} x_i$ is the same as the sum $2(1+2+\ldots+12) = 156$. So each $x_i + x_{i+1}$ must equal exactly 13. So both the neighbours of any figure would be the same, a contradiction. A simpler argument is to note that at least one of the two neighbours of 12 exceeds 1. The answer to the last part is 'no' as seen by putting the figures 1, 7, 6, 8, 5, 9, 4, 10, 3, 11, 2 and 12 clockwise (or anti-clockwise). **36.** Divide the triangle into 4 subtriangles by joining the midpoints of the sides. At least 2 of the given 5 points must lie in the same subtriangle. **37.** Every positive integer m can be written uniquely as $2^r u$, where $r \geq 0$ and u is odd. Let S be a set of $n+1$ integers from among 1 to $2n$. Define a function $f : S \longrightarrow \{1, 3, \ldots, 2n-1\}$ by $f(m) = u$. (e.g., $f(15) = 15, f(64) = 1, f(68) = 17$). By the pigeon hole principle, there exist $x, y \in S$ with $x < y$ such that $f(x) = f(y)$. But this means $\frac{y}{x}$ is an integer (in fact, a power of 2). For an inductive argument, suppose $S \subset \{1, 2, \ldots, 2n+1, 2n+2\}$ has $n+2$ (or more) elements. Obtain T from S by dropping $2n+1$ (in case it is in S) and replacing $2n+2$ (in case it is in S) by $n+1$. Apply induction hypothesis to T to get $x, y \in T$ with x dividing y. If y happens to be $n+1$ and $n+1 \notin S$, replace $n+1$ back by $2n+2$. Then $\{x, 2n+2\}$ is a desired pair in S. **38.** Call the classes A, B and C. At least one of them, say A, contains at least 6 integers, say $x_0 < x_1 < \ldots < x_5$. Let $y_i = x_i - x_0$ for $i = 1, 2, 3, 4, 5$. If any one of these five positive integers is in A we are done since we can take $x = x_0$ and $y = y_i$. Otherwise, three of them, say $y_{i_0} < y_{i_1} < y_{i_2}$ are in the same class, say B. Let $z_1 = y_{i_1} - y_{i_0}$ and $z_2 = y_{i_2} - y_{i_0}$. As before, if one of these is in B we are done. If one of them, say z_1, is in A then also we are done for we note that $z_1 = x_{i_1} - x_{i_0}$ and take $x = x_{i_0}$ and $y = z_1$. The only case left is that both z_1, z_2 are in C. In that case we let $w = z_2 - z_1$ and show that we are done no matter whether $w \in C, w \in B$ (by writing w as $y_{i_2} - y_{i_1}$) or $w \in A$ (by writing w as $x_{i_2} - x_{i_1}$).

39. Follow the hint. Suppose the sets S_1 and S_2 (say) have a common element. Let the binary sequences corresponding to the students 1 and 2 be $x = (x_1, x_2, \ldots, x_{10})$ and $y = (y_1, y_2, \ldots, y_{10})$ respectively. Let $z = (z_1, z_2, \ldots, z_{10})$ be a common element of S_1 and S_2. Let $A = \{i : x_i \neq z_i\}$. Verbally, A is the set of those indices where the entries of x and z differ from each other. By the very construction of the set S_1, $|A| \leq 2$. Similarly let $B = \{i : y_i \neq z_i\}$ and $C = \{i : x_i \neq y_i\}$. Then $|B| \leq 2$. Also, $C \subset A \cup B$ and so $|C| \leq 4$. This means that the answers given by students 1 and 2 differ for at most four questions. This implies the assertion. So the problem reduces to showing that at least two of these 20 sets $S_1, S_2, \ldots, S_{20}$ intersect. This is easy because otherwise they will all be mutually disjoint. But clearly, for every i, $|S_i| = \binom{10}{0} + \binom{10}{1} + \binom{10}{2} = 56$. A set with 1024 has no room for 20 mutually disjoint subsets of cardinality 56 each since $56 \times 20 > 1024$.

40. (a) If $x_n = y_n$, then we have both $x_n \leq y_n$ and $y_n \leq x_n$. So by (v) $L \leq M$ and also $M \leq L$ which together imply $L = M$. (vi) does not follow directly from (v) since it is not given that $\lim\limits_{n \longrightarrow \infty} z_n$ exists. Once its existence is proved we can apply (v) to conclude that it equals L. (b) Use $|\,|x_n| - |L|\,| < |x_n - L|$. The second part is false as shown by taking $x_n = (-1)^n$. This sequence can't converge to any real number. For suppose L were its limit. Take $\epsilon = \frac{1}{2}$. Since infinitely many terms of the sequence are 1 for every $\epsilon > 0$, the interval $(L - \frac{1}{2}, L + \frac{1}{2})$ will have to contain 1. Similarly it will have to contain -1. But then its length would be at least 2, a contradiction. However, $|x_n| = 1$ for all n and so $\{|x_n|\}$ converges to 1. (c) Apply induction on the number of terms in the summation. (d) The fallacy is that a_n is not the sum of a *fixed* number of sequences. For the extension of Theorem 3 to apply, the number of summands whose sum is a_n must be the same for all n. (e) $-\frac{1}{2}$. ($\lim\limits_{n\to\infty} \dfrac{n(n+1)}{2(1-n^2)} = \lim\limits_{n\to\infty} \dfrac{\frac{1}{n}+1}{2(\frac{1}{n^2}-1)} = \dfrac{0+1}{2(0-1)} = -\dfrac{1}{2}$.)

42. $\frac{5}{21}$ cc. of milk will come back along with $\frac{100}{21}$ cc. of water to the first glass. So $\alpha = \frac{100}{21}$. The second glass will retain $\frac{100}{21}$ cc. of milk. Hence $\beta = \frac{100}{21}$. So $\alpha = \beta$. A slicker way to see this is to note that since the volumes remain the same, as much milk goes into the second glass as water to the first.

43. a_n will approach 50 because each exchange reduces the difference in the percentages of water in the two glasses.

44. The recurrence relation is obtained by calculating how much water passes from the first glass to the second and vice versa in the n-th exchange. To solve it, write it as $a_n = a + ba_{n-1}$ with $a = \frac{100}{21}$ and $b = \frac{19}{21}$. Then a direct calculation gives $a_1 = a, a_2 = a(1+b), a_3 = a(1+b+b^2)$ which suggests that $a_n = a(1 + b + b^2 + \ldots + b^{n-1}) = a\dfrac{1-b^n}{1-b}$ which is easily verified by induction on n. Hence $a_n = 50\left(1 - \left(\frac{19}{21}\right)^n\right)$. Since $|\frac{19}{21}| < 1$, it follows that

$a_n \to 50$ as $n \to \infty$. The requirement $a_n > 20$ is equivalent to $(\frac{19}{21})^n < \frac{3}{5}$. The least n for which this happens is 6.

45. Let $L = \lim_{n\to\infty} \frac{F_n}{F_{n-1}}$. The recurrence relation gives $\frac{F_n}{F_{n-1}} = 1 + \frac{1}{F_{n-1}/F_{n-2}}$ and letting $n \to \infty, L = 1 + \frac{1}{L}$. The quadratic $L^2 - L - 1 = 0$ has $\frac{1+\sqrt{5}}{2}$ as the only positive root (the other root being $\frac{1-\sqrt{5}}{2}$). By Theorem 3, Part (v), $L \geq 0$ and so $L = \frac{1+\sqrt{5}}{2}$. For the second part, let $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Then $|\frac{\beta}{\alpha}| = |\frac{1-\sqrt{5}}{1+\sqrt{5}}| < 1$. So $\frac{\beta^n}{\alpha^n} \to 0$ as $n \to \infty$. Now,

$$\frac{F_n}{F_{n-1}} = \frac{\alpha^n - \beta^n}{\alpha^{n-1} - \beta^{n-1}} = \alpha \frac{1 - (\frac{\beta}{\alpha})^n}{1 - (\frac{\beta}{\alpha})^{n-1}} \to \alpha \frac{1-0}{1-0} = \alpha.$$

46. Since $b - a > 0$, there exists a positive integer n such that $\frac{1}{n} < b - a$. Consider the successive multiples $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{m}{n}, \ldots$ of $\frac{1}{n}$. At least one of these multiples lies in (a, b) as otherwise both a and b will be contained between two consecutive multiples, say $\frac{k}{n}$ and $\frac{k+1}{n}$. But this means $b - a < \frac{k+1}{n} - \frac{k}{n} = \frac{1}{n}$, a contradiction.

47. For each $n \in \mathbb{N}$, let x_n be any rational number in the interval $(\alpha - \frac{1}{n}, \alpha + \frac{1}{n})$. Then $x_n \to \alpha$ as $n \to \infty$.

48. The inductive verifications are routine. From $p_n^2 - 2q_n^2 = (-1)^n$ we get that $|\frac{p_n^2}{q_n^2} - 2| = \frac{1}{q_n^2}$ and hence that $|\frac{p_n}{q_n} - \sqrt{2}| = \frac{1}{(\frac{p_n}{q_n} + \sqrt{2})q_n^2} \leq \frac{1}{n}$ since $q_n^2 \geq n$ for all n as can be proved by induction on n. Hence $\frac{p_n}{q_n} \to \sqrt{2}$ as $n \to \infty$.

49. Since $0 < \sqrt{2} - 1 < 1$, by Theorem 4(i), $(\sqrt{2} - 1)^p \to 0$ as $p \to \infty$. Expanding $(\sqrt{2} - 1)^p$ gives the first part. For the second part, apply the first part with $\epsilon = b - a$ to get integers m, n such that $0 < m + n\sqrt{2} < b - a$. Then take successive multiples of $m + n\sqrt{2}$ and show that at least one of them lies in the interval (a, b).

50. As $n \to \infty$, $a_n - b_n \to \infty$, $a_n - c_n \to -\infty$, $a_n - d_n \to 1$ while $a_n - e_n = (-1)^{n+1}$ which has no limit as shown in the answer to Exercise (6.40) (b) above.

51. Let $r = 1 + h$, where $h > 0$. For $n \geq k+1$, $r^n = (1+h)^n > \binom{n}{k+1} h^{k+1}$ and so $\left|\frac{a_0 + a_1 n + a_2 n^2 + \ldots + a_k n^k}{r^n}\right| < \left|\frac{a_0 + a_1 n + \ldots + a_k n^k}{\binom{n}{k+1} h^{k+1}}\right|$ which tends to 0 by Exercise (6.41) (i) as the denominator is a polynomial of degree $k + 1$ in n.

52. Fix a positive integer $k > r$. Then for all $n > k$, $0 < \frac{r^n}{n!} < \frac{C}{n}$, where C is the constant $\frac{k^{k+1}}{k!}$. Apply the Sandwich Theorem.

53. Note that for all $n > 2$, $\frac{n^n}{n!} = (\frac{n}{1})(\frac{n}{2}) \ldots (\frac{n}{n}) > \frac{n}{1} \times 1 \times 1 \times \ldots \times 1 = n$.

Chapter 7

1. 4 $(= \frac{\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ}{\frac{1}{2} \sin 20^\circ \cos 20^\circ})$.

2. (a) Use (8) with $m = 4$. (b) Use (12). (c) Use $\sin^2(30^\circ + \theta) + \sin^2(30^\circ - \theta) = \frac{1}{2} + \sin^2\theta$. (d) Expand $\frac{1}{\cos 80^\circ} + \frac{1}{\cos 40^\circ} - \frac{1}{\cos 20^\circ}$. The denominator is $\frac{1}{8}$ by (b). For the numerator use $\cos 20^\circ + \cos 100^\circ = \cos 40^\circ$. (e) Begin with $\sin 2\alpha \sin 4\alpha \sin 6\alpha = \frac{\sqrt{7}}{8}$ which is obtained by putting $m = 7$ in (11) and noting that $\cos 6\alpha = \sin\alpha$ etc. For the second part, begin by putting $m = 3$ in (12). (f) Use (6) repeatedly. (g) Express $\cos^2\theta$ in terms of $\cos 2\theta$ and apply (21) after noting that $\cos(2\pi - 2\theta) = \cos 2\theta$.

3. (a) Consider $\cos 12^\circ - \cos 48^\circ$ and $\cos 24^\circ - \cos 84^\circ$. (b) The L.H.S. is $\cot 12^\circ \cot 24^\circ \cot 48^\circ \cot 84^\circ$. Multiply it by $\cot 36^\circ \cot 60^\circ \cot 72^\circ$ which has a known value and use (14) with $m = 7$. (c) Add to the L.H.S. $\cos\frac{6\pi}{15} + \cos\frac{10\pi}{15} + \cos\frac{12\pi}{15}$ whose value is known and use (21). (d) First find a quadratic with $\sin 18^\circ$ as a root and then put $\sin 18^\circ = \frac{2\tan 9^\circ}{1 - \tan^2 9^\circ}$.

4. From (9), one such polynomial is $(z-1)^n + (z-1)^{n-1} + (z-1) + 1$. When n is odd, its constant term, and hence the product of its roots is 1.

5. $n = 4$. (The given complex number is simply i.)

6. $(e^{\frac{2\pi k i}{n}})^r = 1$ if and only if n divides kr. If n is relatively prime to k then this means n divides r and hence that r is at least n. For the converse, suppose the g.c.d. of n and k is $d > 1$. Then $n/d < n$ and still $(e^{\frac{2\pi k i}{n}})^{(n/d)} = e^{2\pi(k/d)i} = 1$. The rest follows from the definition of $\phi(n)$.

7. Let m be the order of α. To prove that m divides n, write $n = mq + r$, where $0 \le r < m$. Then $\alpha^n = 1$ and also $\alpha^{mq} = (\alpha^m)^q = 1$ gives $\alpha^r = \alpha^{n-mq} = 1$. This would contradict the minimality of m unless $r = 0$. The other assertions are easy. For the last assertion, take the distinct powers of $e^{2\pi(n/d)i}$. (The argument given to show that m divides n is strikingly similar to that for Exercise (4.12)(d). It is indeed possible to give this argument only once in the abstract and then derive both Exercise (4.12)(d) and the present exercise as consequences. Such an abstraction is a part of what is called **group theory**.)

8. Since $z^{p+q} - z^p - z^q + 1 = (z^p - 1)(z^q - 1)$, α is either a p-th or a q-th root of unity. Apply (9) with $z = \alpha$. If both the possibilities hold then by the last exercise both p and q divide each other, contradicting that they are distinct primes.

9. (i) Apply (9) and the fact that in this case every n-th root of unity other than 1 is a primitive n-th root. (ii) For $n = 4$ and 6, the primitive n-th roots are $\pm i$ and $\frac{1 \pm \sqrt{3}i}{2}$ respectively. For $n = 8$ they are $\frac{1 \pm i}{\sqrt{2}}$ and $\frac{-1 \pm i}{\sqrt{2}}$. In general, if $n = 2^m$ then $\Phi_n(z) = z^{2^{m-1}} + 1$. (iii) By Exercise (7.7), every n-th root of unity is a primitive d-th root of unity for a unique divisor d of n.

Multiply together the factors of z^n-1 corresponding to all primitive d-th roots of unity to get $\Phi_d(z)$. (v) $\Phi_{15}(z) = \frac{z^{15}-1}{(z-1)(z^2+z+1)(z^4+z^3+z^2+z+1)} = \frac{z^{10}+z^5+1}{z^2+z+1}$. By long division this comes out to be $z^8-z^7+z^5-z^4+z^3-z+1$. (vi) Putting $w = z+\frac{1}{z}$, $\frac{\phi_{15}(z)}{z^4} = w^4-w^3-4w^2+4w+1$. Now put $w = \frac{2(1-x^2)}{1+x^2}$ (which comes from the fact that $2\cos\frac{2r\pi}{15} = \frac{2(1-\tan^2\frac{r\pi}{15})}{1+\tan^2\frac{r\pi}{15}}$).

10. (a) Start with $\frac{\Phi_7(z)}{z^3} = w^3+w^2-2w-1$ where $w = z+\frac{1}{z}$. $w = 2\cos\frac{2\pi}{7}$ is a root of this. Now express $2\cos\frac{2\pi}{7}$ in terms of $4\cos^2\frac{\pi}{7}$. (b) First show that $\Phi_{14}(z) = z^6-z^5+z^4-z^3+z^2-z+1 = z^3 f(w)$, where, as usual, $w = z+\frac{1}{z}$ and $f(w) = w^3-w^2-2w+1$ which has $2\cos\frac{\pi}{7}, 2\cos\frac{3\pi}{7}$ and $2\cos\frac{5\pi}{7}$ as roots. From this construct a polynomial $g(u)$ which has $\sec^2\frac{\pi}{14}, \sec^2\frac{3\pi}{14}$ and $\sec^2\frac{5\pi}{14}$ as roots. (Actually, since we are interested only in the sum of the roots, it is enough to compute only the top two terms of $g(u)$.) (c) Similar to (b) except that start with $\Phi_{11}(z)$. (d) It is enough to prove that $\sum_{k=1}^{7}\tan^4\frac{k\pi}{16} = 679$. Start with $\frac{z^{16}-1}{z^2-1} = (z^2+1)(z^4+1)(z^8+1)$ to get $f(w) = w(w^2-2)(w^4-2w^2+2)$ as a polynomial which has $2\cos\frac{\pi k}{8}$ as roots for $k = 1, 2, \ldots, 7$. From this construct a polynomial $g(u)$ having as its roots $\tan^2\frac{\pi k}{16}$ for $k = 1, 2, \ldots, 7$. Take the sum of the squares of the roots of $g(u)$ (for which only the top three terms of $g(u)$ are needed).

11. (i) $i\sqrt{3}$. (The expression equals $4+5\omega^{334}+3\omega^{365} = 4+5\omega+3\omega^2 = 1+2\omega$.) (ii) $\frac{1}{4}n^2(n+1)^2-n$. (For every k, $(k-1)(k-\omega)(k-\omega^2) = k^3-1$.)

12. (a) Take $\beta = \omega, \gamma = \omega^2$ or vice versa and expand. Use $\omega^2+\omega+1 = 0$. (b) $(A+B+C)(A+B\omega+C\omega^2)(A+B\omega^2+C\omega)$ is one possible factorisation. Because of symmetry, several other factorisations are possible. They are not essentially different. Note that the coefficients of A, B, C in each factor are the three powers of one of the cube roots of unity, viz., $1, \omega$ and ω^2. (c) The three factors of the L.H.S. as given in (b) equal, respectively, $(1+x)^n$, $(1+\omega x)^n$ and $(1+\omega^2 x)^n$ whose product is precisely the R.H.S.

13. Call the three sums as A, B, C respectively. Setting $x = 1$, $x = \omega$ and $x = \omega^2$ in $(1+x+x^2)^n = a_0x+a_1x+a_2x^2+\ldots+a_{2n}x^{2n}$ gives a system of three linear equations in A, B, C.

14. (a) ω or ω^2 depending upon whether we take $\beta = \omega^2\alpha$ and $\gamma = \omega\alpha$ or the other way. (The hypothesis $p < 0$ has no role except to ensure that the roots are non-zero.) (b) -1, $1-2\omega$ and $1-2\omega^2$.

15. (a) Call $x^{n+1}+(x+1)^{2n-1}$ as $f(x)$. Since $x^2+x+1 = (x-\omega)(x-\omega^2)$, in order to show that x^2+x+1 divides $f(x)$, it suffices to show that $f(\omega) = 0$ and $f(\omega^2) = 0$. Let $\xi = \frac{1+i\sqrt{3}}{2}$. Then ξ is a primitive sixth root of unity. Moreover, $\omega = \xi^2$ and $\omega+1 = \xi$. So $f(\omega) = \xi^{2n+2}+\xi^{2n-1} = \xi^{2n-1}(\xi^3+1) = 0$. Similarly, using $\omega^2+1 = \xi^5$ one shows that $f(\omega^2) = 0$. (b) The alternate solution would be valid provided it is shown not only

that x^2+x+1 divides $f(x)$ but also that the quotient polynomial $\frac{f(x)}{x^2+x+1}$ is an integer whenever x is an integer. This, in turn, would follow if it is shown that the quotient is a polynomial with integer coefficients. This can be seen by observing that since x^2+x+1 has all coefficients equal to 1, in performing the long division of $f(x)$ by x^2+x+1, at every stage the coefficients in the quotient will be integers.

16. All are true. (Begin by multiplying by $\frac{2\cos\frac{\theta}{2}}{2\cos\frac{\theta}{2}}$. Then $f_n(\theta)=\tan(2^n\theta)$.)

17. (a). (Use $\frac{\alpha}{2}+\frac{\beta}{2}+\frac{\gamma}{2}=\pi$. A twisted version of a standard result.)

18. (a) 0. (The expression equals $\cos A\cos B-\frac{1}{2}(\cos(A-B)+\cos(A+B))$.
(b) The expression equals the constant $-\frac{1}{2}(\cos 2-1)$. The slope is 0.

19. Yes. (Instead of computing $\tan 2A$, recognise the R.H.S. as $\tan\frac{b}{2}$.)

20. $\alpha+\beta=\pi+\gamma$ gives $\sin\gamma=-\sin(\alpha+\beta)$ and $\cos\gamma=-\cos(\alpha+\beta)$. The L.H.S. and the R.H.S. reduce to $\sin^2\alpha+\sin^2\beta-\sin^2(\alpha+\beta)$ and to $\cos^2(\alpha+\beta)-\frac{1}{2}(\cos 2\alpha+2\beta)$ respectively. Another method is to note that the conclusion is unaffected if we replace α,β,γ respectively by α',β',γ', where $\alpha'=2\pi r\pm\alpha$, $\beta'=2\pi s\pm\beta$ and $\gamma'=2\pi t\pm\gamma$, for any integers r,s,t. Choosing these integers and the signs suitably we can ensure that α',β' and γ' are the angles of a triangle. Using the circumradius of the triangle, the result to be proved translates into the cosine formula for a triangle.

21. (a) $n=6$. (The L.H.S. is a polynomial of degree 6 in $\sin x$. In the R.H.S., for even values of m, $\cos mx$ is a polynomial of degree m in $\sin x$. Match the degrees of the two sides.) (b) Infinite. (All triples of the form $(c,-c,-2c)$ will work.) (c) $b_0=0$ and $b_1=n$. (Put $\theta=0$ to get b_0. Differentiate both the sides and put $\theta=0$ again to get b_1. The result is intuitively clear from the observation that for small values of θ, $\sin n\theta$ is approximately $n\theta$.) (d) Follow the hint. Taking the real part, $\cos n\theta=\sum_{k=0}^{\infty}(-1)^k\binom{n}{2k}\cos^{n-2k}\theta\sin^{2k}\theta$. Writing $\sin^{2k}\theta$ as $(1-\cos^2\theta)^k$ we see that $\cos n\theta$ is a polynomial in $\cos\theta$, say $T_n(\cos\theta)$, as the sum is only ostensibly infinite. Each term is a polynomial of degree n in $\cos\theta$ with leading coefficient $\binom{n}{2k}$. So the coefficient of $\cos^n\theta$ in $T_n(\cos\theta)$ is the sum $\sum_{k=0}^{\infty}\binom{n}{2k}$ which equals 2^{n-1} by (24) in Chapter 5. In particular $T_n(x)$ has degree n. Similarly, by taking the imaginary part of the binomial expansion of $(\cos\theta+i\sin\theta)^{n+1}$, we get that $\sin(n+1)\theta=U_n(\cos\theta)\sin\theta$ for some polynomial U_n. To show that U_n has degree n, differentiate $\cos((n+1)\theta)=T_{n+1}(\cos\theta)$ to get $(n+1)\sin((n+1)\theta)=T'_{n+1}(\cos\theta)\sin\theta$ and use the fact that the degree of the derivative of a polynomial is one less than the degree of that polynomial. (In fact we also get that $U_n(x)=\frac{1}{n+1}T'_{n+1}(x)$.) So the two types of Chebyshev polynomials are interrelated.) Using the expressions for $\cos m\theta$ and $\sin m\theta$ for $m\le 3$, we get $T_0(x)=1, T_1(x)=$

$x, T_2(x) = 2x^2 - 1$ and $T_3(x) = 4x^3 - 3x$ while $U_0(x) = 1, U_1(x) = 2x, U_2(x) = 4x^2 - 1$ etc. (e) The recurrence relation for the $T_n(x)$'s follows from $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos n\theta \cos\theta$. A similar relation for the U's is $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$. (Thus both the sequences satisfy the same recurrence relation. But they differ because their initial terms differ.)

22. p. (Expand directly w.r.t. the first row. The coefficients of each of $1, a$ and a^2 are independent of p.) **23.** 1. (The expression equals $3(\cos^4\alpha + \sin^4\alpha) - 2(\cos^6\alpha + \sin^6\alpha) = 3(1 - 2\sin^2\alpha\cos^2\alpha) - 2(1 - 3\sin^2\alpha\cos^2\alpha)$.)

24. $\pm\sqrt{2}$. (From $\cos x = \frac{2\cos(x-y)\cos(x+y)}{\cos(x-y)+\cos(x-y)}$, we get $\cos^2 x \cos y(1 - \cos y) = -\sin^2 x \sin^2 y$ and further $-\frac{\cos y}{2\cos^2\frac{y}{2}} = \tan^2 x$. Add 1 to both the sides.)

25. This is a special case of (24).

26. Apply induction on n. Let S_n denote the given sum. Let $T_n = \sum \sin(\pm\alpha_1 \pm \ldots \pm \alpha_n)$. Clearly $T_n = 0$ since the terms with opposing sign combinations cancel each other. Then $S_{n+1} = A_n + B_n$, where $A_n = \sum \cos(\pm\alpha_1 \pm \alpha_2 \pm \ldots \pm \alpha_n + \alpha_{n+1}) = S_n \cos\alpha_{n+1} - T_n \sin\alpha_{n+1} = S_n \cos\alpha_{n+1}$ and similarly, $B_n = \sum \cos(\pm\alpha_1 \pm \alpha_2 \pm \ldots \pm \alpha_n - \alpha_{n+1}) = S_n \cos\alpha_{n+1}$. Adding the two gives the inductive step.

27. Put $A = \beta - \gamma, B = \gamma - \alpha, C = \alpha - \beta$ and $G = \alpha + \beta + \gamma$. Then the first expression equals $\sum(\sin(G - 2B) - \sin(G + 2C))$ and further $-2\sum \cos G \sin 2A$. Thus the problem reduces to showing that $-\sin 2A - \sin 2B - \sin 2C = 4\sin A \sin B \sin C$ which can be done using $A + B + C = 0$.

28. Let $u = \tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$. For a fixed θ, u is a constant and so $x = \tan\theta$ is a root of the cubic equation $x^3 - 3ux^2 - 3x - u = 0$. But since $\tan 3(\theta + \frac{\pi}{3}) = \tan 3\theta$, $x = \tan(\theta + \frac{\pi}{3})$ is also a root of this cubic. Similarly, $\tan(\theta + \frac{2\pi}{3})$ is a root. Apply (11) of Chapter 3 to get the sum of the squares of the roots as $9u^2 + 6$.

29. From $A + B + C = \pi$ first prove that $1 = \cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C$. (This is a standard result.) Substitute the given values of $\cos A, \cos B, \cos C$ and write 1 as $(\sin^2\alpha + \cos^2\alpha)(\sin^2\beta + \cos^2\beta)(\sin^2\gamma + \cos^2\gamma)$. This gives $(\sin\alpha\sin\beta\sin\gamma - \cos\alpha\cos\beta\cos\gamma)^2 = 0$.

Chapter 8

1. Note that as ABC is isosceles, so are EBD and FDC. Hence $ED = EB$ and $DF = FC$. **2.** D is the orthocentre of triangle ABC.

3. (A) and (B). ($\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = (p+q+r)(pq+qr+pr-p^2-q^2-r^2)$.)

4. Yes. (Apply the determinant criterion for concurrency.) **5.** This is a special case of the problem in Comment No. 10, in which the vertices of one of the two triangles happen to lie on the sides of the other. **6.** $(-4,-7)$. **7.** $(-a, a(t_1+t_2+t_3)+at_1t_2t_3)$. **8.** (a) No. e.g., take the numbers as $1, 0$ and -1. (In fact the three points would be collinear and hence will never lie on a circle unless they are the same.) (b) The points should be non-collinear (see e.g. the condition in Exercise (8.24)) and further $z_1+z_3=z_2+z_4$. (The diagonals of a parallelogram bisect each other.) (c) $0, \pi, -\pi$. **9.** $1-\frac{3}{2}i$ and $3-\frac{1}{2}i$. (Since $MA \perp MD$ and $MA=\frac{1}{2}MD$, A must be $2-i\pm\frac{i}{2}((1+i)-(2-i)$.)

10. Take the vertices at $3z_1, 3z_2$ and $3z_3$ traversed counterclockwise. Then the centres, say w_1, w_2, w_3, come out as $(1+\alpha)z_1+(2-\alpha)z_2$, $(1+\alpha)z_2+(2-\alpha)z_3$ and $(1+\alpha)z_3+(2-\alpha)z_1$ where $\alpha=e^{i\pi/3}$. The relation $w_3-w_1=e^{i\pi/3}(w_2-w_1)$ can be verified by a direct calculation.

11. The geometric data means that $z_2=e^{i\alpha}z_1$. Hence $z_1(1+e^{i\alpha})=-p$ and $z_1^2e^{i\alpha}=q$. Eliminating z_1, $p^2=qe^{-i\alpha}(1+e^{i\alpha})^2=q(e^{i\alpha}+2+e^{-i\alpha})=q(2+2\cos\alpha)=4q\cos^2\frac{\alpha}{2}$.

12. (a) (i) the x-axis (= perpendicular bisector of the line segment joining $5i$ and $-5i$) (ii) the x-axis again (note that $|1-iz|=|-i^2-iz|=|-i|\ |z+i|=|z+i|$.) (iii) ellipse with foci at $\pm i$ and major axis 3 (iv) hyperbola with foci at $\pm i$ and axis 3. (b) Let A, B be the points corresponding to z_1 and z_2. Let C and D be the points on the line AB which divide the segment AB internally and externally respectively, in the ratio $\lambda:1$. Then for any point P on the locus, PC and PD are bisectors of the angle APB and hence are at right angles. So the locus of P is a circle with CD as a diameter. (This circle is called the **Apollonius circle**. Let the complex numbers corresponding to C and D be, respectively, z_3 and z_4. Then $z_3=\frac{z_1+\lambda z_2}{1+\lambda}$, $z_4=\frac{z_1-\lambda z_2}{1-\lambda}$ and their midpoint, say M, is $\frac{z_1-\lambda^2 z_2}{1-\lambda^2}$. A direct algebraic proof that PM is of constant length is rather messy. But by writing $z-z_1=\lambda e^{i\alpha}(z-z_2)$, it is not hard to show that $\frac{z-z_3}{z-z_4}=\frac{1-\lambda}{1+\lambda}\frac{e^{i\alpha}+1}{e^{i\alpha}-1}$ which is purely imaginary for every (real) α, λ.)

13. $|z_0-z_1|^2=|z_2-z_0|\ |z_1-z_2|$ means that the length of the side z_0z_1 is the G.M. of those of the other two sides. By symmetry of the condition, this means the length of each side is the G.M. of the lengths of the other two. This forces all the sides to be equal.

14. The logic holds for a sum of squares of real numbers only. It fails for complex numbers as we see from $1^2+i^2=0$.

15. $a=b=2\pm\sqrt{3}$. (Write $a+i=e^{\pm i\pi/3}(1+ib)$ or use (8) to get a system of two equations in a and b.) **16.** -2 and $1-i\sqrt{3}$. (Multiply Z_1 by ω, ω^2.) **17.** All. (The data implies $z_1=e^{i\theta}, z_2=\pm iz_1$. Hence $w_1=z_1$ and $w_2=\pm z_2$.)

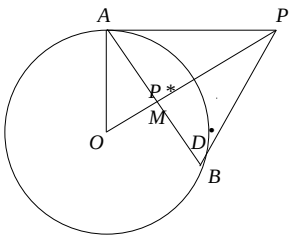
18. (A) and (D). (Either write the given ratio as $\frac{(z_1+z_2)(\overline{z_1}-\overline{z_2})}{|z_1-z_2|^2}$ or observe that z_1 lies on the circle with $z_2, -z_2$ as diametrically opposite points. The other parts of the hypothesis have no role.)

19. Let $A = e^{i\alpha}, B = e^{i\beta}, C = e^{i\gamma}$. Then the triangle ABC has its circumcentre at O and the data implies that its centroid is also at O. So ABC is equilateral. Hence we may take $\beta = \alpha + \frac{2\pi}{3}$ and $\gamma = \alpha - \frac{2\pi}{3}$. The result follows by a direct computation.

20. Take $A_j = ae^{i(\alpha+\frac{2\pi j}{m})}$ and $B_k = be^{i(\beta+\frac{2\pi k}{n})}$ for $j = 1, 2, \ldots, m$ and $k = 1, 2, \ldots, n$, where α, β are some fixed real numbers. (With a suitable choice of axes, one but not both of them may be assumed to be 0.) Then $|A_j - B_k|^2 = |A_j|^2 + |B_k|^2 - A_j\overline{B_k} - \overline{A_j}B_k = a^2 + b^2 - abe^{i(\alpha-\beta+\frac{2\pi j}{m}-\frac{2\pi k}{n})} - abe^{i(\beta-\alpha+\frac{2\pi k}{n}-\frac{2\pi j}{m})}$. Hence for a fixed $j = 1, 2, \ldots m$, the sum $\sum_{k=1}^{n} |A_j - B_k|^2$ equals $n(a^2+b^2) - abe^{i(\alpha-\beta+\frac{2\pi j}{m})} \sum_{k=1}^{n} e^{-\frac{2\pi ki}{n}} - \sum abe^{i(\beta-\alpha-\frac{2\pi j}{m})} \sum_{k=1}^{n} e^{\frac{2\pi ki}{n}}$. Each of the sums $\sum_{k=1}^{n} e^{-\frac{2\pi ki}{n}}$ and $\sum_{k=1}^{n} e^{\frac{2\pi ki}{n}}$ vanishes, being the sum of the roots of the polynomial $z^n - 1 = 0$. So, for each j, $\sum_{k=1}^{n} |A_j - B_k|^2$ is simply $n(a^2+b^2)$. Therefore the double sum $\sum_{j=1}^{m}\sum_{k=1}^{n} |A_j - B_k|^2$ equals $mn(a^2+b^2)$.

21. (a) Write $z^* = \frac{r^2}{\bar{z}} = \frac{r^2}{|z|^2}z$. Since $\frac{r^2}{|z|^2} > 0$, P and P^* have the same arguments. The second part follows from $\bar{z}z^* = r^2$ by taking the absolute values of both the sides. (b) $z^{**} = \dfrac{r^2}{\overline{z^*}} = \dfrac{r^2}{\overline{r^2/\bar{z}}} = \dfrac{r^2}{r^2/\bar{z}} = \dfrac{r^2}{r^2/z} = z$. Second part is clear from (a). (c) Let PA and PB be the tangents from P to C and M be the midpoint of the chord AB. Then the right-angled triangles OAP and OMA are similar and so $OM.OP = OA^2 = r^2$. But then by (a), $M = P^*$. The second part follows from this and (b).

(d) No. If this were the case, then for any point D on the shorter arc AB as above, the normal at D must divide the angle PDP^* externally, or equivalently, the tangent at D must divide it internally. But by taking D close to A (or B), we see that this is not so. (To avoid the consequent confusion, reflection w.r.t. a circle is often called **inversion** w.r.t. it.)



For (e), shift the origin to z_0. The first part of (f) follows by direct computation from (e). As $r \to \infty$, the expression $\frac{-ri}{r-1}$ tends to $-i$ which is the reflection of the point i in the x-axis. This is consistent with the

fact that the a straight line is the limiting case of a circle as its radius tends to ∞.

22. If $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ then $z_1 \cdot z_2 = x_1x_2 + y_1y_2 = \text{Re}\ (x_1 + iy_1)(x_2 - iy_2) = \text{Re}\ z_1\overline{z_2}$. The vector $z_1 \times z_2$ equals $(x_1y_2 - x_2y_1)\mathbf{k}$. But $x_1y_2 - x_2y_1$ is also the imaginary part of $(x_1 - iy_1)(x_2 + iy_2)$ i.e., of $\overline{z_1}z_2$.

23. Note that the vectors $z_2 - z_3$ and $z_1 - z_3$ represent the sides of the triangle meeting at the vertex z_3. Therefore, the area of the triangle equals, upto a $\pm$ sign, $\frac{1}{2}\text{Im}\ (\overline{z_2} - \overline{z_3})(z_1 - z_3) = \frac{1}{2}\text{Im}\ (\overline{z_3} - \overline{z_2})(z_3 - z_1) = \frac{1}{2}\text{Im}\ \frac{|z_3-z_2|^2(z_3-z_1)}{(z_3-z_2)} = \frac{1}{2}|z_3 - z_2|^2\text{Im}\left(\frac{z_3-z_1}{z_3-z_2}\right)$. The second part follows because the area of a triangle is 0 if and only if its vertices are collinear.

24. In the degenerate case where any two of the three numbers z_1, z_2, z_3 are equal, the assertion holds trivially. So assume that they are all distinct. Write $z_3 = (1 - \lambda)z_1 + \lambda z_2$ for a unique complex number λ. Let D be the value of the given determinant. Then the row operation $R_3 \longrightarrow R_3 - (1 - \lambda)R_1 - \lambda R_2$ followed by expansion w.r.t. the third row shows that $D = (\overline{z_3} - (1-\lambda)\overline{z_1} - \lambda\overline{z_2})(z_2 - z_1)$. Now z_1, z_2, z_3 are collinear if and only if λ is real. If λ is real then we also have $\overline{z_3} = (1 - \lambda)\overline{z_1} + \lambda\overline{z_2}$ and so $D = 0$. Conversely, if $D = 0$ then we must have $\overline{z_3} = (1 - \lambda)\overline{z_1} + \lambda\overline{z_2}$. But $\overline{z_3} = (1 - \bar{\lambda})\overline{z_1} + \bar{\lambda}\overline{z_2}$. This gives $(\lambda - \bar{\lambda})\overline{z_1} = (\lambda - \bar{\lambda})\overline{z_2}$. Since $\overline{z_1} \neq \overline{z_2}$ this means $\lambda = \bar{\lambda}$, i.e., λ is real and hence the three points are collinear. The last exercise gives a slicker argument because collinearity of z_1, z_2, z_3 is equivalent to saying that the ratio $\frac{z_3-z_1}{z_3-z_2}$ is real. It only remains to relate the area with the determinant D. The column operations $C_1 \longrightarrow C_1 + C_2$ followed by $C_2 \longrightarrow 2C_2 - C_1$ show that $D = -\frac{i}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ whose magnitude is the area of the triangle with vertices at z_1, z_2, z_3.

25. Performing similar row operations as in the last exercise, the given determinant equals $-\frac{i}{2}D$, where $D = \begin{vmatrix} x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \\ x_4^2 + y_4^2 & x_4 & y_4 & 1 \end{vmatrix}$. By the higher dimensional analogue of Theorem 8 of Chapter 3 the vanishing of D is equivalent to the existence of real numbers A, B, C, D not all 0, such that for every $j = 1, 2, 3, 4$, $A(x_j^2 + y_j^2) + Bx_j + Cy_j + D = 0$. This amounts to saying that the four points $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$ lie on the curve $A(x^2 + y^2) + Bx + Cy + D = 0$ which represents a circle if $A \neq 0$ and a straight line if $A = 0$.

26. Follow the hint. As ON is normal to the plane ABC, N is of the form $(\frac{t}{a}, \frac{t}{b}, \frac{t}{c})$ for some t. As N lies on this plane, we get $\frac{t}{a^2} + \frac{t}{b^2} + \frac{t}{c^2} = 1$. So $t = \frac{a^2b^2c^2}{b^2c^2+c^2a^2+a^2b^2}$ and $ON^2 = t^2(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}) = \frac{a^2b^2c^2}{b^2c^2+c^2a^2+a^2b^2}$. Finally, from $V = \frac{1}{6}abc = \frac{1}{3}\Delta.ON$, Δ^2 comes out as $\frac{1}{4}(b^2c^2 + c^2a^2 + a^2b^2)$.

Chapter 9

N. B. In locus problems (h, k) will denote the current coordinates except where they clash with the data, in which case they will be taken as (α, β).

1. (a) No. (See Comment No. 9 (ii) of Chapter 24.) (b) The ratio of the two areas is the same as the ratio of the perpendicular distances of the points P, A from the line BC. **2.** 10. (Rewrite the equation as $\frac{x^2}{25} + \frac{y^2}{16} = 1$ to see that it is an ellipse with foci at F_1 and F_2.) **3.** One. (The first circle touches the second one internally.) **4.** $(1, -2)$. (Write the condition $2b = a + c$ as $a - 2b + c = 0$ and further as $a \times 1 + b \times (-2) + c = 0$. Alternately, write the equation of the line as $2ax + 2by + 2c = 0$. This can be rewritten as $2ax + (a+c)y + 2c = 0$ and hence as $a(2x+y) + (y+2) = 0$ which is satisfied regardless of what a, c are when $2x+y = 0$ and $y+2 = 0$.)

5. A straight line passing through the origin. (The hypothesis implies that $\frac{y_2}{x_2} = \frac{y_1}{x_1} = \frac{y_3}{x_3}$.) **6.** (a) Yes. (In fact, they all lie on the line $ay = bx$.) (b) Yes. (Let $A = (-\frac{19}{2}, 0)$, $B = (0, -\frac{19}{3})$, $C = (\frac{17}{9}, 0)$ and $D = (0, \frac{17}{6})$ be the points in question. Note that the segments AC and BD intersect at O and so the concyclicity of A, B, C, D is equivalent to the condition that $AO.OC = BO.OD$ which is easily seen to hold. (c) Yes. (It passes through the centre $(3, -1)$ of the circle.) (d) No. (The given vertices from a right-angled triangle. Hence its circumcircle is centred at the midpoint of the hypotenuse i.e., at $(2, 0)$ and its radius equals half the hypotenuse i.e., 2. The point $(\frac{5}{2}, 1)$ lies inside this circle.)

7. $2\sqrt{3}$ sq. units. (Let $P = (1, \sqrt{3})$ and let the tangent at P meet the x-axis at A. Then APO is a right angled triangle with $\angle POA = 60°$. So $OA = 2OP = 4$.) **8.** $S = (2, 3)$. (cf. Exercise (8.8)(b).)

9. $7x - 4y = 3$, $7x - 4y = -25$ and $4x + 7y = 11$. (Consider lines through $(1, 1)$ and $(-3, 1)$ which are perpendicular to the given line and also the line through $(1, 1)$ parallel to the given line.)

10. $x^2 + (y - \frac{1}{\sqrt{3}})^2 = \frac{4}{3}$. (The third vertex is at $(0, \sqrt{3})$ and hence the circumcentre is at $(0, \frac{1}{\sqrt{3}})$. **11.** $b\sqrt{a^2 - b^2}$. (For maximum A, P must be farthest from the x-axis and hence at $(0, \pm b)$. Note that $F_1F_2 = 2ae$, where e = eccentricity $= \frac{\sqrt{a^2-b^2}}{a}$.)

12. (a) $a^2 > 2b^2$. (Similar reasoning to the first problem in Comment No. 6.) (b) $a \in (-\infty, -2) \cup (2, \infty)$. (Put $u = \frac{1+\sqrt{2}a}{2}$ and $v = \frac{1-\sqrt{2}a}{2}$. Take the midpoint of the chord as $(\lambda, -\lambda)$. Then the condition that $(2\lambda - u, -2\lambda - v)$ lies on the given circle for two distinct real values of λ comes out as $7(u + v)^2 + 4uv < 0$. In terms of a, this becomes $a^2 > 4$.)

13. (a) $x^2 + 2ax - b^2 + y^2 + 2py - q^2 = 0$; radius $= \sqrt{a^2 + b^2 + p^2 + q^2}$. (Let $A = (x_1, y_1)$ and $B = (x_2, y_2)$. Then the equation of the circle with AB as

a diameter is $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$ as can be obtained from the fact that for any point P on this circle, $\angle APB = 90°$. Note that it is not necessary to identify x_1, x_2, y_1, y_2 individually.) (b) All. (Eliminate y to see that the x_i's are the roots of the quartic $x^4 + c^4 = a^2x^2$. Similarly for the y_i's.)

14. $(-1, 0)$. (The tangents at $(1, \pm 2)$ are $y = \pm(x + 1)$.)

15. (a) the empty set since the L.H.S. is negative. (A teaser.) (b) A straight line parallel to the y-axis. (Even without writing out the distances, the quadratic terms will cancel out. So the locus is a straight line. To see it is vertical, note that whenever S lies on the curve so does its reflection in the x-axis.) (c) An ellipse with foci at the centres of the given circles. (C_2 is inside C_1. Let C be a circle of the given description. Let r, r_1, r_2 be the radii of C, C_1, C_2 respectively and M, M_1, M_2 their respective centres. Let d_1, d_2 be the distances of M from M_1, M_2 respectively. Then $d_1 = r_1 - r$ while $d_2 = r_2 + r$. So $d_1 + d_2 = r_1 + r_2$, a constant.)

16. 5. ($Q = (0, 3)$. The centre, say M, of the circle is $(-3, -3)$ while its radius is $\sqrt{20}$. So $PQ^2 = QM^2 - MP^2 = 45 - 20 = 25$.)

17. (a) $3 : 4$. (The lines being parallel, the ratio is independent of the intersecting line. So we may take it either as the x-axis or as the y-axis. If we rewrite the second line as $4x + 2y = -12$ then even without finding the intercepts on any of the axes explicitly, the ratio of their lengths must be $9 : 12$.) (b) $\frac{3}{4}$. (The lines are parallel. The radius is half the perpendicular distance between them.)

18. (a) Degenerate. (The points are collinear.) (b) Isosceles. (The vertices are $(1, 1), (2, -2)$ and $(-2, 2)$. Or apply symmetry of data in x and y.)

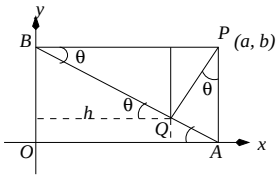
19. $\frac{192}{25}$ sq. units. (See the figure of the solution to Exercise (8.21). Here $r = 3$. Let $P = (4, 3)$. Then A can be taken as $(0, 3)$. $OM = \frac{9}{OP} = \frac{9}{5}$. Hence $PM = \frac{16}{5}$ and $MA = \sqrt{9 - OM^2} = \frac{12}{5}$. The desired area is twice that of the triangle AMP.)

20. Second quadrant. (The entire triangle is in the second quadrant and acute angled so that the orthocentre is inside the triangle.)

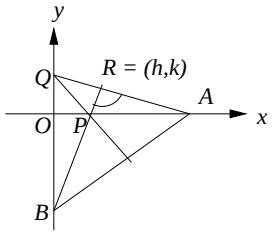
21. $c = -4$. The other two vertices are $(4, 4)$ and $(2, 0)$. (Call $(1, 3)$ and $(5, 1)$ as A and C respectively. Then the midpoint $M = (3, 2)$ of AC must lie on the given line $y = 2x + c$. This determines c as -4. To find the other vertices, one can consider a circle with AC as a diameter and take its points of intersection with the line $y = 2x-4$. But in the present problem, the line AC is perpendicular to the other diagonal and so the rectangle is a square. Hence one can proceed as in Exercise (8.9).) **22.** $x = 0$ and $(h^2 - r^2)x - 2rhy = 0$. (A line through O of the form $y = mx$ will touch the circle if the discriminant of the quadratic $x^2(1+m^2)-2rx-2hmx+h^2$

is 0. This gives only one value viz., $m = \frac{r^2-h^2}{2rh}$. But the line $x = 0$ is another tangent from O. **23.** $\frac{2b}{\sqrt{a^2-4b^2}}$. (The centres of the two circles are equidistant from the line $y = mx - b\sqrt{1+m^2}$ but lie on opposite sides of it. This gives $ma - b\sqrt{1+m^2} = b\sqrt{1+m^2}$.)

24. (i) $16x^2 + 10xy + y^2 = 2$. (Instead of working with the 1-parameter family $y = 4x + c$ of lines, it is just as easy to eliminate x_1, x_2, y_1, y_2 from the five equations : $h = \frac{2x_1+x_2}{3}$, $k = \frac{2y_1+y_2}{3}$, $x_1y_1 = 1$, $x_2y_2 = 1$ and $y_2 - y_1 = 4(x_2 - x_1)$. There are two points of trisection of the chord. The locus is a hyperbola, each branch containing one of these two points.) (ii) $x^{2/3} + y^{2/3} = c^{2/3}$. (Taking $A = (a, 0)$ and $B = (0, b)$ we get $\frac{h}{a} + \frac{k}{b} = 1$ and $\frac{k-a}{h-b} = \frac{a}{b}$. Also $a^2 + b^2 = c^2$. Eliminating a, b from these three equations is not easy, but a little trigonometry works wonders. Let $\theta = \angle OAB = \angle PBA$. From the figure it is easily seen that $a = c\cos\theta, b = c\sin\theta$ and further $h = BQ\cos\theta = a\cos^2\theta = c\cos^3\theta$ and similarly, $k = c\sin^3\theta$. Eliminating θ gives the locus.



(iii) $(2x-1)^2+4y^2 = 1$. (Similar to but easier than (i) since one end of the chord is always $(0, 0)$. Letting (x_1, y_1) be the other end, $x_1 = 2h, y_1 = 2k$ and $(x_1 - 1)^2 + y_1^2 = 1$.) (iv) $4x^2 + 4y^2 - 12x + 4y + \frac{31}{4} = 0$. (Here a pure geometry proof is the best. The locus is a concentric circle of half the radius as the given circle.) (v) $x^2 + y^2 = 2$. (Similar to (iv) and simpler.) (vi) $x^2 + y^2 = hx + ky$. (A method similar to (i) can be applied. But since we are dealing with the midpoint, say (α, β), of the chord, a simpler solution is possible. For, the equation of the chord is $x\alpha + y\beta = \alpha^2 + \beta^2$. Since it passes through (h, k), we get $h\alpha + k\beta = \alpha^2 + \beta^2$. The possibility of exploiting such simplifying features should always be kept in mind.) (vii) $\frac{y^2}{9} - \frac{x^2}{7} = 1$. (A direct computation is possible. Or recognise the curve as a vertical hyperbola with foci at $(0, \pm 4)$.) (viii) $x(x-7) + y(y+5) = 0$. (Let $P = (a, 0)$ and $Q = (0, b)$. Comparing various slopes gives $\frac{a}{5} = \frac{h}{k+5}, \frac{b}{7} = \frac{k}{7-h}$ and $\frac{b}{a} \times \frac{5}{7} = 1$. Eliminating a, b gives the answer. A pure geometry solution is also possible and can be conceived either as an afterthought having known the answer through coordinates or earlier by looking at the diagram. From the data, P is the orthocentre of the triangle ABQ and hence BP is perpendicular to AQ. So $\angle ARB = 90°$. Hence the locus of R is a circle with AB as a diameter.)



(ix) $y = (m-\frac{1}{m})x+a+mb+\frac{b}{m}$. (Taking $P = (c, a)$, get $Q = (b, a+m(b-c))$ and $R = (-b, a + \frac{1}{m}(b + c))$. Consider the centre of the rectangle to get $h + c = 0$ and $k + a = 2a + m(b - c) + \frac{1}{m}(b + c)$. Eliminate c.)
(x) $y^2 - 6y - 10x + 14 = 0$. (Equate h with $\sqrt{(h-3)^2 + (k-3)^2} - 2$.)

(xi) $2ax + 2by = a^2 + b^2 + k^2$. (Let the centre of the moving circle be (α, β) and let r be its radius. Then $(a-\alpha)^2 + (b-\beta)^2 = r^2$. The orthogonality condition gives $\alpha^2+\beta^2 = r^2+k^2$. Eliminate r.) (xii) $(x^2+y^2)^2(a^2y^2-b^2x^2) = (a^2+b^2)^2x^2y^2$. (The normal at a point $(a\sec\theta, b\tan\theta)$ is $a\sin\theta\, x+by = (a^2+b^2)\tan\theta$. This coincides with the line perpendicular to OR at R, whose equation is $hx + ky = h^2 + k^2$. Hence $\frac{a\sin\theta}{h} = \frac{b}{k} = \frac{(a^2+b^2)\tan\theta}{h^2+k^2}$, which gives $\sin\theta = \frac{bh}{ak}$ and $\cos\theta = \frac{h(a^2+b^2)}{a(h^2+k^2)}$.)

25. The locus is $(y-\frac{8}{9})^2 \frac{4}{9}(x-\frac{2}{9})$. This is a parabola with vertex at $(\frac{2}{9}, \frac{8}{9})$. (Work similar to (i) of the last exercise.) **26.** Very similar to (viii) of Exercise (9.24). **27.** The locus is $\frac{(x+3a)^2}{8a^2} - \frac{y^2}{8a^2} = 1$ which is a hyperbola. (The line $y-k = m(x-h)$ touches the parabola $y^2 = 4ax$ if and only if the discriminant of the quadratic $y^2 = 4a(\frac{y-k+mh}{m})$ vanishes. This reduces to $hm^2 - km + a = 0$. If m_1, m_2 are the roots of this quadratic then the angle between the two tangents from (h,k) is $45°$ if and only if $(\frac{m_1-m_2}{1+m_1m_2})^2 = 1$, i.e., $(m_1+m_2)^2 - 4m_1m_2 = 1 + 2m_1m_2 + (m_1m_2)^2$. Put $m_1 + m_2 = \frac{k}{h}, m_1m_2 = \frac{a}{h}$.) **28.** $4x^2 + 25y^2 = 4x^2y^2$. ($A = (2h, 0)$, $B = (0, 2k)$. Hence the equation of AB is $\frac{x}{2h} + \frac{y}{2k} = 1$. Condition for tangency of this line to the ellipse gives the locus. The first quadrant requirement restricts x, y to be positive. The family of circles has no role since every tangent to the ellipse in the first quadrant touches a unique member of the given family.)

29. In fact, the locus is $x - 3y + 5 = 0$. (Take $P = (x_1, y_1)$. Then $Q = (3x_1, 3y_1)$. Comparison of slopes gives $k - y_1 = 2(h - x_1)$ and $k - 3y_1 = -3(h - 3x_1)$. Also $x_1 + y_1 = 1$. Eliminating x_1, y_1 gives the locus. But since the problem only asks to show that the locus is a straight line, it suffices to note that here we have a system of three *linear* equations.)

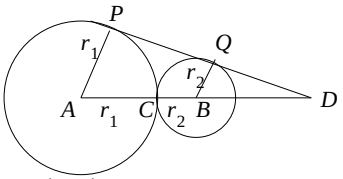
30. By Pythagorean theorem, $t_1^2 = \lambda - a^2$, $t_2^2 = \lambda - 2ax_1$ and $t_3^2 = \lambda - 2ay_1$, where $\lambda = x_1^2+y_1^2$. The given condition reduces to $2ax_1y_1 = \lambda(x_1+y_1-a)$ and hence to $2ax_1y_1 + a(x_1^2 + y_1^2) = (x_1^2 + y_1^2)(x_1 + y_1)$. Factoring, the locus is the union of the two given sets.

31. (a) $(0,0)$ or $(0, \frac{5}{2})$. (Take $A = (0, c)$. Let m be the slope of the diagonal AC. Then AC is either an internal or an external bisector of the angle at A between two lines with slopes 1 and 7. Hence $\frac{7-m}{1+7m} = \frac{m-1}{1+m}$. Solving, $m = 2$ or $-\frac{1}{2}$. Since $(1,2)$ satisfies $y = mx + c$, we get the corresponding values of c.) (b) $(2,-1), (2,-3), (0,-1)$ and $(0,-3)$. (Write the circle as $(x-1)^2 + (y+2)^2 = 2$.) (c) (i) No. (ii) Yes, since no three of them are concurrent. (The difference between every two equations can be put in the form $x + y = c$ for some constant c. But no two of these constants are the same.) (d) $(15, 8), (22, 31)$ and $(45, 24)$.

32. 32 sq. units. (The vertices C, D must be of the form $(5, a)$ and $(-3, a)$ for some a. The midpoint $(1, \frac{a+4}{2})$ of AC lies on the diameter. Find a.)

33. $(x-1)^2+(y+1)^2=49$. (Taking $\pi=\frac{22}{7}$, the area gives the radius as 7. The centre is the common point of the two given lines.) **34.** (a) Let the points asked be, respectively, (x_0,y_0) and (x_2,y_2). Then $x_0=\frac{b^2x_1-aby_1-ac}{a^2+b^2}, y_0=\frac{a^2y_1-abx_1-bc}{a^2+b^2}, x_2=\frac{(b^2-a^2)x_1-2aby_1-2ac}{a^2+b^2}$ and $y_2=\frac{(a^2-b^2)y_1-2abx_1-2bc}{a^2+b^2}$. (First find x_0,y_0 by solving together $ax_0+by_0+c=0$ and $(y_1-y_0)a=(x_1-x_0)b$. Then determine x_2,y_2 from $x_1+x_2=2x_0$ and $y_1+y_2=2y_0$.) (b) $14x+23y-40=0$. (B and C are the reflections of A in the given lines.) **35.** $\frac{15a^2}{4}$. (The quadrilateral is a trapezium with parallel vertical sides which are the chords of contacts of A w.r.t. the two curves. The tangent at a point $P=(at^2,2at)$ on the parabola is $y=\frac{1}{t}x+at$. The condition for its tangency to the circle $x^2+y^2=\frac{a^2}{2}$ gives $t=\pm1$ and the points of contact with the circle as $(-\frac{a}{2},\pm\frac{a}{2})$.) **36.** $2<r<8$. (Easier to find geometrically after noting that the radius of the second circle is 3 and the distance between their centres is 5.)

37. $(\frac{14}{5},\frac{8}{5})$; $y=0$ and $24x-7y-16=0$. (Here, too, pure geometry works fast. Note first that if two circles centred at A and B with radii r_1 and r_2 respectively touch (externally) at C and their common tangents other than the one at C meet at D, then C and D divide the segment AB internally and externally respectively in the ratio $r_1:r_2$. This is clear by considering the similar triangles APD and BQD in the figure.

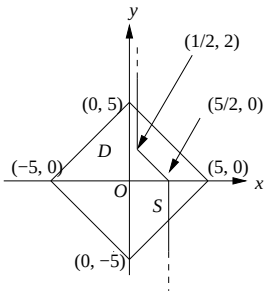


In the present problem, $A=(6,4), B=(2,1), r_1=4, r_2=1$. So, $C=(\frac{14}{5},\frac{8}{5})$ and $D(\frac{2}{3},0)$. For a line $y=m(x-\frac{2}{3})$ through D, the condition for tangency to either circle determines m as 0 or $\frac{24}{7}$. Alternately, the tangent $y=0$ can also be identified by inspection. Having done so, the other tangent can be determined as the reflection of the first tangent into the line joining the centres of the two circles.)

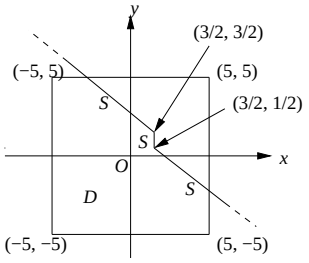
38. (a) Call the lines L_1 and L_2 respectively. Note that the vector $(-b_i,a_i)=-b_i\mathbf{i}+a_i\mathbf{j}$ is parallel to L_i for $i=1,2$. Let L_i' be the ray originating at P and parallel to the vector $(-b_i,a_i)$. Then $\frac{a_1a_2+b_1b_2}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}$ is the cosine of the angle with L_1', L_2' as arms and so this angle is acute or obtuse according as $a_1a_2+b_1b_2$ is positive or negative. Also, the equation of the bisector of this angle is $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}=-\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$, obtained by equating the algebraic distances of a point (x,y) from these two lines. For an alternate derivation based on trigonometry, first shift the origin to P so as to make $c_1=c_2=0$. Also, assume $a_1^2+b_1^2=1=a_2^2+b_2^2$ without loss of generality. Then $(-b_i,a_i)=(\cos\theta_i,\sin\theta_i)$ for $i=1,2$, where θ_i is the angle L_i' makes with the positive x-axis. The equation of the angle bisector is $(\sin\theta)x-(\cos\theta)y=0$ where $\theta=\frac{1}{2}(\theta_1+\theta_2)$. Multiplying throught by $2\cos(\frac{\theta_1-\theta_2}{2})$, rewrite this as $(\sin\theta_1+\sin\theta_2)x-(\cos\theta_1+\cos\theta_2)y=0$, and

hence as $a_1x+b_1y=-(a_2x+b_2y)$. (b) $y-\frac{14}{5}=-\frac{2+5\sqrt{5}}{11}(x-\frac{8}{5})$. (Let $m_1=\frac{4}{3}, m_2=\frac{1}{2}$ be the slopes of the given lines and m be the slope of either angle bisector. Then $\frac{m_1-m}{1+mm_1}=\frac{m-m_2}{1+mm_2}$ gives a quadratic $11m^2+4m-11=0$. As m_1, m_2 are both positive, the positive value of m corresponds to the bisector of the acute angle and is discarded. A much easier solution using (a) is $\sqrt{5}(x-2y+4)=4x-3y+2$. Both the equations represent the same line, as they ought to!) **39.** $2(al+bm)(ax+by+c)=(a^2+b^2)(lx+my+n)$. (Equation of L is of the form $(ax+by+c)+\lambda(lx+my+c)=0$ for some λ. The slopes of L_1, L_2 and L are $-\frac{a}{b}, -\frac{l}{m}$ and $-\frac{a+\lambda l}{b+\lambda m}$ respectively. The given condition implies $\dfrac{\frac{a}{b}-\frac{l}{m}}{1+\frac{al}{bm}}=\dfrac{\frac{a+\lambda l}{b+\lambda m}-\frac{a}{b}}{1+\frac{a(a+\lambda l)}{b(b+\lambda m)}}$. Simplifying and solving, $\lambda=-\dfrac{a^2+b^2}{2(al+bm)}$.) **40.** (a) $7x+y=36$. (Let m be the slope of the internal angle bisector at B. Note that ABC is a right angle. So to find m, it suffices to solve the linear equation $\frac{m-m_1}{1+mm_1}=1$, where $m=$ the slope of $AB=\frac{4}{3}$.) (b) $y=-\sqrt{3}\,x$. (Draw a diagram to see that the angle bisector makes an angle 120° with the positive x-axis.) **41.** $y+10=\frac{1}{3}(x-1)$. (The third side is parallel to the external angle bisector at the opposite vertex.) **42.** If (α, β) is a point on either of the two bisectors then $\frac{|\beta-m_1\alpha|}{\sqrt{1+m_1^2}}=\frac{|\beta-m_2\alpha|}{\sqrt{1+m_2^2}}$. Squaring, simplifying and cancelling m_1-m_2 (which is non-zero except in the degenerate case where the two lines coincide) gives $(m_1+m_2)(\beta^2-\alpha^2)+2\alpha\beta(1-m_1m_2)=0$. Replace α, β by x, y respectively and use $m_1+m_2=-\frac{2h}{b}$ and $m_1m_2=\frac{a}{b}$. **43.** Write the first pair as $Ax^2+2Hxy+By^2=0$, where $A=a^2-\lambda, B=b^2-\lambda$ and $H=h(a+b)$. The assertion is equivalent to proving that both the pairs have the same pair of angle bisectors. The equation for the pair of angle bisectors for the first pair is $H(x^2-y^2)+(B-A)xy=0$. By a direct substitution, this is seen to be the same as that for the second pair, viz., $h(x^2-y^2)+(b-a)xy=0$ regardless of λ. **44.** $(\frac{1}{2}, \pm\sqrt{2})$. (The centre lies on $x=\frac{1}{2}$ and its distance from O must be half the radius of the given circle.) **45.** The equation of the chord of contact of a point (x_0, y_0) is $xx_0+yy_0=1$. If (x_0, y_0) is on the given line then $2x_0+y_0=4$, i.e., $\frac{1}{2}x_0+\frac{1}{4}y_0=1$. So the chord of contact passes through $(\frac{1}{2}, \frac{1}{4})$ for every point (x_0, y_0) on the given line. (Exercise (9.4) is of a similar spirit.) bf 46. $x^2+y^2-10x-4y+4=0$. (Let $A=(h,k)$ be the centre of C_1. Since A is at a distance 3 from each of the two lines, we get $\frac{|5h+12k-10|}{13}=3=\frac{|5h-12k-40|}{13}$. Various sign combinations give $10h-50=0, 78$ or -78 and $24k+30=0, 78$ or -78. But $h>0$, $k>0$ imply $5h+12k-10=39$ and $5h-12k-40=-39$ which determine A as $(5,2)$ which is also the centre of C_2. The radius of C_2 is $\sqrt{3^2+(\frac{8}{2})^2}=5$.) **47.** $3(\sqrt{10}+3)$ units. (The two tangents are $y=x$ and $y=2x$. The centre (h,k) lies between these two at a distance 3 from each. Arguing as in the last exercise (but using the short cut $h<k<2h$ for sign

determination) the centre comes out as $(3(\sqrt{5}+\sqrt{2}), 3(\sqrt{5}+2\sqrt{2})$. As the radius is 3, $OA = \sqrt{h^2+k^2-9} = 3\sqrt{19+6\sqrt{10}} = 3(\sqrt{10}+3)$.) **48.** (i) $x^2+y^2-3x+1=0$. (Determine λ so that $x^2+y62-6x+8+\lambda(x^2+y^2-6) = 0$ passes through $(1,1)$.) (ii) $4x^2+4y^2+30x-13y-25=0$. **49.** (a) Let $E_3 = ax+by+c$. Then the centre of a typical circle of the family must lie on the perpendicular to L at (x_1, y_1) and hence must be of the form $(x_1+\lambda a, y_1+\lambda b)$ for some λ. Also its radius must equal $\sqrt{\lambda^2(a^2+b^2)}$. (b) $x^2+y^2-18x-16y+120=0$. (Apply (a) taking L to be the tangent at $(5,5)$. For a slicker solution, note that as the radii are equal, the point of contact must bisect the segment joining the two centres.) **50.** $x(x-1)+y(y-1)=0$. (A, B are $(0,0)$ and $(1,1)$. cf. Exercise (9.13)(a).) **51.** $y = -\frac{1}{5}x \pm \sqrt{2}$. **52.** (a) The set D consists of all points on or inside the square with vertices at $(\pm 5, 0)$ and $(0, \pm 5)$. The set S would also additionally include the vertically downward ray from $(\frac{5}{2}, 0)$. (b) The set S is the union of the regions $\{(x,y) : x \le 0, y \ge 2\}$ and $\{(x,y) : x \ge 2, y \le 0\}$ and the line segment joining $(0,2)$ to $(2,0)$.



Exercise (9.52)(a)



Exercise (9.53)

53. $D = \{(x,y) : |x| \le 5, |y| \le 5\}$. This is a square with vertices at $(\pm 5, \pm 5)$. The set S consists of points (x,y) for which $\max\{|x|, |y|\} = \max\{|x-3|, |y-2|\}$. This runs into four cases depending upon which of the members in each set is greater. The case $|y| \ge |x|$; $|y-2| \ge |x-3|$ is vacuous for in this case we must also have $|y| = |y-2|$ which is possible only for $y = 1$. But then we get $|x| \le 1$ and also $|x-3| \le 1$ contradicting the triangle inequality. Of the remaining three cases, the case $|y| \le |x|$; $|y-2| \le |x-3|$ leads to $|x| = |x-3|$ i.e., $x = \frac{3}{2}$. Further, $|y| \le \frac{3}{2}$ and $|y-2| \le \frac{3}{2}$ together give $\frac{1}{2} \le y \le \frac{3}{2}$. The other two cases are a little more complicated. If $|x| \le |y|$ and $|y-2| \le |x-3|$ then $|y| = |x-3|$ implies $x = 3 \pm y$. Further, $|x| \le |x-3|$ and $|y-2| \le |y|$ imply, respectively, $x \le \frac{3}{2}$ and $y \ge 1$. So we must have $x = 3-y$ and not $x = 3+y$. This gives a ray of slope -1 starting at $(\frac{3}{2}, \frac{3}{2})$. The fourth case, $|y| \le |x|$; $|x-3| \le |y-2|$ is handled similarly and gives the ray of slope -1 starting at $(\frac{3}{2}, \frac{1}{2})$. **54.** (c). (For example, take $(x_1, y_1) = (2,3)$ and $(x_2, y_2) = (2,-3)$. The property in question means that whenever a region contains two points it also contains the midpoint of the segment joining them. This is called the **midpoint**

convexity in analogy with the much stronger property, called **convexity**, which requires that whenever a region contains two points it contains *every* point on the segment joining them. The regions in (a), (b) and (d) are, in fact, convex as one can see by elementary sketches.) **55.** The boundary of a square with vertices on the two lines. (Taking the lines as the coordinate axes, the set is of the form $\{(x, y) : |x|+|y| = c\}$, where $c \geq 0$ is a constant. The square has its vertices at $(\pm c, 0)$ and $(0, \pm c)$.) **56.** (a) 2 sq. units. (b) 2 sq. units. (In both the cases, the area is that of a square with vertices at $(\pm 1, 0)$ and $(0, \pm 1)$. For (a) this follows from the last exercise. For (b) sketch the graphs of $y = |x| - 1$ and $y = -|x| + 1$.) **57.** $\pm\sqrt{2}$. (Let the chord be AB, where $A = (at_1^2, 2at_1)$ and $B = (at_2^2, 2at_2)$. Then the slope of the chord AB is $\frac{2}{t_1+t_2}$. If this coincides with the normal at A, then we get $t_1(t_1 + t_2) = -2$ since the slope of the normal at A is $-t_1$. Also $OA \perp OB$ gives $t_1t_2 = -4$. Solving, $t_1^2 = 2$.) **58.** The fixed point is $(a, 0)$ i.e., the focus. The locus is $y^2 = 2a(x-a)$. (Take $P = (at_1^2, 2at_1), Q = (at_2^2, 2at_2)$. As in the last exercise, PQ has slope $\frac{2}{t_1+t_2}$ and by a direct calculation, meets the x-axis at $(-at_1t_2, 0)$. But $t_1t_2 = -1$ since $OP \perp OQ$. For the locus write $2h = a(t_1^2 + t_2^2)$, $k = a(t_1 + t_2)$. Eliminate t_1 and t_2 using $t_1t_2 = -1$.) **59.** Let $A = (at_1^2, 2at_1)$ be a point on $y^2 = 4ax$. The focus F is $(a, 0)$ and so the slope of AF is $\frac{2t_1}{t_1^2-1}$. Since the normal at A has slope $-t_1$, a direct computation shows that it divides the angle between the ray AF and the direction of the positive x-axis. **60.** 2. (Take A, B, C as $(at_i^2, 2at_i),\ i = 1, 2, 3$ respectively. The tangent at $(at_1^2, 2at_i)$ is $t_iy = x + at_i^2$. Taking intersections two at a time we take $P = (at_2t_3, a(t_2 + t_3))$ $Q = (at_3t_1, a(t_3 + t_1))$ and $R = (at_1t_2, a(t_1 + t_2))$. The ratio of the areas of the triangle PQR and ABC equals $|\frac{2D_1}{D_2}|$, where $D_1 = \begin{vmatrix} t_1^2 & t_1 & 1 \\ t_2^2 & t_2 & 1 \\ t_3^2 & t_3 & 1 \end{vmatrix}$ and $D_2 = \begin{vmatrix} t_2t_3 & t_2+t_3 & 1 \\ t_3t_1 & t_3+t_1 & 1 \\ t_1t_2 & t_1+t_2 & 1 \end{vmatrix}$. The row operations $R_2 \longrightarrow R_2 - R_1$ and $R_3 \longrightarrow R_3 - R_1$ show that each of D_1 and D_2 equals $\pm(t_1 - t_2)(t_2 - t_3)(t_1 - t_3)$.)

61. Instead of applying brute force, let F be the focus $(a, 0)$. Then $FC \perp AB$ and this gives $\frac{2}{t_1+t_2} = -\frac{t_3^2-1}{2t_3}$ which can be rewritten as $t_1 + t_2 = \frac{4t_3}{1-t_3^2}$ and hence as $(t_1 + t_2 + t_3) = \frac{5t_3-t_3^3}{1-t_3^2}$. Call $t_1 + t_2 + t_3$ as u. Then t_3 is a root of the cubic $t^3 - ut^2 - 5t + u = 0$. By symmetry, t_1 and t_2 are also roots. The result follows by applying (11) of Chapter 3. For the second part, let the circumcircle of ABC be $(x - h)^2 + (y - k)^2 = r^2$ and let it meet the parabola at one more point, say $(at_4^2, 2at_4)$. Then t_1, t_2, t_3, t_4 are roots of the quartic equation $a^2t^4 + (4a^2 - 2ah)t^2 - 4akt + (h^2 + k^2 - r^2) = 0$. So $t_1 + t_2 + t_3 + t_4 = 0$. Since $t_1 + t_2 + t_3 = u$, this gives $t_4 = -u$. Similarly, from the coefficient of t and using $t_1t_2t_3 = -u$ and $t_1t_2 + t_2t_3 + t_3t_1 = -5$, we get $k = ua$. Further, $h^2 + k^2 - r^2 = a^2(t_1t_2t_3t_4) = a^2u^2$ which implies $h^2 = r^2$. But this means that the distance of the circumcentre from the y-axis equals the circumradius, which implies the second assertion.

Chapter 10

1. (a) $x = 0, \pm\frac{\pi}{2}$. (Putting $u = \cos x$, the equation is $u^7 + u^4 = 2u^2$. Since $|u| \leq 1$, $u^5 + u^2 = 2$ can hold only for $u = 1$.) (b) $n\pi,\ n\pi + (-1)^n \frac{\pi}{10}$ and $2n\pi \pm \frac{4\pi}{5}$. (Get a cubic in $u = \sin x$.) **2.** $n\pi + \frac{\pi}{4}$, n an integer. **3.** $\operatorname{cosec} 20°$. **4.** (a) Two. (Convert to a quadratic in $\sin x$. Discard $\sin x = -1$ which is necessary because when $\sin x = -1$, $x = \frac{3\pi}{2}$ and neither $\tan\frac{3\pi}{2}$ nor $\sec\frac{3\pi}{2}$ is defined.[1]) (b) None. ($5^x + 5^{-x} = 5^x + \frac{1}{5^x} \geq 2$ for all x.) (c) Two. (Letting $\sin\theta = \sqrt{x^2 + x + 1}$, the equation implies $\cot^2\theta = x(x + 1)$ and hence $\sin^2\theta = \operatorname{cosec}^2\theta$, giving $x^2 + x + 1 = 1$. This also follows more slickly from the inequalities $x(x + 1) \geq 0$ and $x^2 + x + 1 \leq 1$, implied, respectively, by the radical sign and the domain of the inverse sine function.) **5.** (a) $[0, \frac{\pi}{6}] \cup [\frac{5\pi}{6}, \pi] \cup \{\frac{\pi}{2}\}$. (b) (B). (Write $\sin^4 x + \cos^4 x$ as $(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = \frac{3}{4} + \frac{1}{4}\cos 4x$.) (c) (A). (Clear from the graphs of $y = \tan x$ and $y = x$ drawn together.) **6.** (D). (The discriminant criterion reads $\cos^2 p \geq 4(\cos p - 1)\sin p$. The L.H.S. is always non-negative while the factor $(\cos p - 1)$ on the R.H.S. can never be positive. Hence $\sin p \geq 0$ is a *sufficient* condition for the existence of solutions and this is satisfied by every $p \in (0, \pi)$. Determining the exact values of p for which the discriminant is non-negative is not easy. But if p is close to $-\frac{\pi}{2}$ (or to $\frac{3\pi}{2}$) then the discriminant is negative. So among the given choices, (D) is the only correct one.) **7.** The empty set. (Rewrite the second equation as $2\cos(\frac{x+y}{2})\cos(\frac{x-y}{2}) = \frac{3}{2}$.) **8.** (B). (The L.H.S. is at least 1 while the R.H.S. is at most 1 by the A.M.-G.M. inequality, with equality possible only for $x = y$. The question is poorly worded. 'The equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ has a solution in θ' would have been a better wording than '$\sec^2\theta = \frac{4xy}{(x+y)^2}$ is true'.) **9.** (a) No. ($1 \pm \sqrt{2}$ are the roots of the given equation treated as a quadratic in $\sin^2\theta$. But neither possibility can hold since $1 - \sqrt{2}$ is negative while $1 + \sqrt{2}$ is bigger than 1.) (b) $n\pi,\ n\pi + (-1)^n \sin^{-1}\frac{1}{4}$, n an integer. **10.** $e^{-i\pi/6}, e^{i\pi/2}$ and $e^{7i\pi/6}$, or, $\frac{\sqrt{3}}{2} - \frac{i}{2}, i$ and $-\frac{\sqrt{3}}{2} - \frac{i}{2}$. (Taking absolute values $|Z| = 1$ and hence $Z = e^{i\theta}$ for some θ. The equation becomes $e^{-i\theta} = ie^{2i\theta}$ or equivalently, $e^{i(3\theta + \frac{\pi}{2})} = 1$. Alternatively, multiply the equation by Z to get $iZ^3 = 1$ and hence $Z^3 = -i = e^{3i\pi/2}$ giving, $Z = i, i\omega$ and $i\omega^2$.)

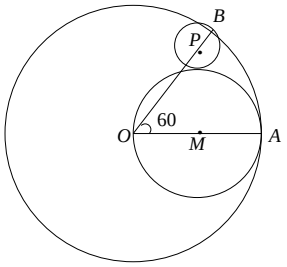
11. Putting $x = \frac{p}{q}$ and multiplying throughout by q^n we get $a_n p^n + a_{n-1}p^{n-1}q + a_{n-2}p^{n-2}q^2 + \ldots + a_1 pq^{n-1} + a_0 q^{n-1} = 0$. Every term except possibly the last one is divisible by p. Hence p divides $a_0 q^N$. Since p is relatively prime to q, p must divide a_0. The second assertion is proved similarly.

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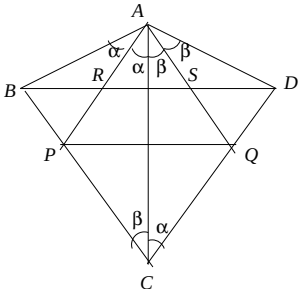
[1] Note, however, that both $\tan x$ and $\sec x$ are defined in a deleted neighbourhood of $\frac{3\pi}{2}$. (See Comment No. 7 of Chapter 13 for what this means.) Moreover, writing $\tan x + \sec x$ as $\frac{\cos x}{1 - \sin x}$ for x near but not equal to $\frac{3\pi}{2}$, one can show that as x tends to $\frac{3\pi}{2}$, both $\tan x + \sec x$ and $2\cos x$ have the same limit, viz. 0. So, the equality in the question holds in a limiting sense even at $\frac{3\pi}{2}$. If the question had asked to find the number of values of $\alpha \in [0, 2\pi]$ for which $\lim_{x\to\alpha} \tan x + \sec x = \lim_{x\to\alpha} 2\cos x$, then the answer would have been 'three' and not 'two'. But as the question stands, the correct answer is 'two'.

12. The only possible rational roots of the new polynomial, viz. y^3-y^2-2y+1 are ± 1. Both are ruled out by direct computation. Since y is rational if and only if x is, the original polynomial can have no rational roots either. **13.** Write the given equation as $2\cos\theta(1-\sin\theta)=\sqrt{3}\sin\theta$. Squaring, putting $u=\sin\theta$ and factoring, this gives $(2u-1)(2u^3-3u^2+4)=0$. The second factor is positive for all $u\in[0,1]$. The solution $u=\frac{1}{2}$ gives $\theta=\frac{\pi}{6}$ or $\theta=\frac{5\pi}{6}$. The latter does not satisfy the original equation. **14.** (a) $[\frac{3}{4},1]$. (Putting $u=\sin^2\theta$, the expression equals $u+(1-u)^2=(u-\frac{1}{2})^2+\frac{3}{4}$. u varies from 0 to 1. (b) $[0,\frac{3}{\sqrt{2}}]$. (x varies from $-\frac{\pi^2}{16}$ to $\frac{\pi^2}{16}$ and so $\sqrt{\frac{\pi^2}{16}-x^2}$ varies from 0 to $\frac{\pi}{4}$.) (c) $[-2,-1]\cup[1,2]$. ($-1\le\log_2\frac{x^2}{2}\le 1$ iff $\frac{1}{2}\le\frac{x^2}{2}\le 2$.) (d) Domain $=(-2,1)$, range $=[-1,1]$. ($\frac{\sqrt{4-x^2}}{1-x}>0$ iff $-2<x<1$. In this interval $\frac{\sqrt{4-x^2}}{1-x}$ can come as close to 0 as we like and hence its logarithm assumes all values less than some fixed negative number.) (e) $[\frac{1}{2}-\frac{4}{9}\pi^2,\sqrt{3}2-\frac{7}{36}\pi^2]$. (Both $\cos x$ and $-x(x+1)$ are strictly decreasing functions on the interval $[\frac{\pi}{6},\frac{\pi}{3}]$.) **15.** Write the given equation as $a\sin\theta+b-c\cos\theta=0$. Treat $\sin\alpha\ a+b-\cos\alpha\ c=0$ and $\sin\beta\ a+b-\cos\beta\ c=0$ as a system of linear equations in a,b,c. By Theorem 7 of Chapter 3, a,b,c are proportional, respectively, to $\cos\alpha-\cos\beta,\sin(\alpha-\beta)$ and $\sin\alpha-\sin\beta$ and hence to, $-\sin(\frac{\alpha+\beta}{2}),\cos(\frac{\alpha-\beta}{2})$ and $\cos(\frac{\alpha+\beta}{2})$. The result follows by direct substitution. **16.** (a) $\pm\frac{4}{5}$. (b) $\frac{17}{6}$. (If $\alpha=\cos^{-1}\frac{4}{5}$, then $\cos\alpha=\frac{4}{5}$ and this gives $\tan\alpha=\pm\frac{3}{4}$. But $0\le\alpha\le\pi$ implies that the positive sign must hold.) (c) $-\frac{\sqrt{24}}{5}$. (Let $\theta=\cos^{-1}\frac{1}{5}$. Then $\sin^{-1}\frac{1}{2}=\frac{\pi}{2}-\theta$ and so the expression equals $\cos(\frac{\pi}{2}+\theta)=-\sin\theta$.) (d) $\frac{56}{33}$. (From the data, $\tan(\alpha+\beta)=\frac{3}{4}$ and $\tan(\alpha-\beta)=\frac{5}{12}$. Write 2α as $(\alpha+\beta)+(\alpha-\beta)$.) (e) $-\frac{7}{17}$. **17.** A is bigger. (It is obvious that both A and B lie in the second quadrant and so it suffices to compare their tangents. $\tan A$ and $\tan B$ come out to be respectively $\frac{2\sqrt{2}-1}{2\sqrt{2}-4}$ and $\frac{92+30\sqrt{2}}{40\sqrt{2}-69}$. For a rigorous comparison, multiply both by $(2\sqrt{2}-4)(40\sqrt{2}-69)$ (which is positive). Then expand and compare the numerators. For a crude comparison, take $\sqrt{2}\approx 1.5$. Since with this approximation, the second number is way smaller than the first, the original numbers must have the same relationship.) **18.** $[0,3-2\sqrt{2}]$. (Put $x=\tan A$. Then $\tan B=\tan(\frac{\pi}{4}-A)$ gives $\tan A\tan B=f(x)$, where $f(x)=\frac{x-x^2}{1+x}$. The range of $f(x)$ as $x\in[0,1]$ can be found by first finding the extreme values of $f(x)$ or by identifying those k for which the quadratic $x^2-x+k(x+1)$ has a root in the interval $[0,1]$. A slicker way is to rewrite $1-\tan A\tan B=\tan A+\tan B$ as $uv=2$, where $u=\tan A+1$ and $v=\tan B+1$. Then $\tan A\tan B$ equals $uv-(u+v)+1=3-(u+v)$ whose maximum occurs when $u=v=\sqrt{2}$ by the A.M.-G.M. inequality. The minimum is clearly 0, occurring when one of A and B vanishes.) **19.** The inductive step reduces to proving that $\tan^{-1}\frac{n}{n+2}+\tan^{-1}\frac{1}{(n+1)^2+n+2}=\tan^{-1}\frac{n+1}{n+3}$, which can be done by showing that the tangents of both the sides are equal. (Note that all the ratios involved are between 0 and 1 and hence all the angles involved

are acute.) **20.** (i) $\tan^{-1}\frac{n}{n+1}$ (ii) $\frac{\pi}{2}+\tan^{-1}(\frac{n^2-n-2}{n^2+3n})$. (Experiment, guess and prove. Note that $\cot^{-1}x=\tan^{-1}\frac{1}{x}$.) **21.** Assume that $OA=OB=1$. Let M be the midpoint of OA. Let $OP=r$. Taking the x-axis along OA, $M=(\frac{1}{2},0)$ and $P=(\frac{r}{2},\frac{\sqrt{2}r}{2})$. Also $PB=1-r$. The conditions about P give $PM=(1-r)+\frac{1}{2}$. This gives an equation in r whose solution is $r=\frac{4}{5}$.



Exercise (10.21)



Exercise (10.22)

22. Let $\angle ACD=\alpha$ and $\angle ACB=\beta$. Then $\angle BAC=2\alpha$ and $\angle DAC=2\beta$. The problem is equivalent to showing that $\alpha=\beta$, for this would mean that AC is the perpendicular bisector of the segment BD. For a pure geometry solution,[2] let the angle bisectors AP and AQ of angles BAC and DAC meet BD at R and S respectively. Then $AQ\parallel BC$ and $AP\parallel DC$. Also, $BP:PC=AB:AC=AD:AC=DQ:QC$ whence PQ is parallel to BD. Hence $BS=PQ=RD$. This makes the triangles ABS and ADR congruent and gives $2\alpha+\beta=\alpha+2\beta$. A trigonometric solution is more straightforward. Applying the sine rule to the triangles ABC and ADC, we have $\frac{\sin\beta}{\sin(\pi-2\alpha-\beta)}=\frac{AB}{AC}=\frac{AD}{AC}=\frac{\sin\alpha}{\sin(\pi-2\beta-\alpha)}$, whence $\sin\alpha\sin(2\alpha+\beta)=\sin\beta\sin(2\beta+\alpha)$. This gives $\cos\frac{\alpha+\beta}{2}-\cos\frac{3\alpha+\beta}{2}=\cos\frac{\alpha+\beta}{2}-\cos\frac{3\beta+\alpha}{2}$. As α,β are acute angles, this can only mean $3\alpha+\beta=3\beta+\alpha$, i.e., $\alpha=\beta$. **23.** (D). ((A) and (B) are easily ruled out by the formulas for rotations. For a reflection, the line in question has to bisect $\angle QOP$. So the angle it makes with the x-axis is the A.M. of those made by OP and OQ.) **24.** (B). (The equation of L is $\frac{x}{a}+\frac{y}{b}=1$. Transforming this by (34) and (35), in the new coordinates it becomes $(\frac{\cos\alpha}{a}+\frac{\sin\alpha}{b})X+(\frac{\cos\alpha}{b}-\frac{\sin\alpha}{a})Y=1$. Hence $\frac{1}{p}=\frac{\cos\alpha}{a}+\frac{\sin\alpha}{b}$ and $\frac{1}{q}=\frac{\cos\alpha}{b}-\frac{\sin\alpha}{a}$. For a purely geometric argument, note that $\frac{1}{a^2}+\frac{1}{b^2}$ and $\frac{1}{p^2}+\frac{1}{q^2}$ both represent the reciprocal of the square of the distance of the origin from the line L, and this is independent of coordinates.) **25.** $(-\frac{1}{\sqrt{2}},\frac{7}{\sqrt{2}})$. (The intermediate positions are $(1,4)$ and $(3,4)$.) **26.** $\frac{x^2}{a^2}+\frac{(\lambda+1)^2y^2}{(\lambda a+b)^2}=1$, where λ is a constant. (If $P=(a\cos\theta,b\sin\theta)$, then $Q=(a\cos\theta,a\sin\theta)$. If (h,k) divides PQ in the

[2] I am indebted to Prof. D. G. Deshpande for this solution.

ratio $\lambda : 1$, then $h = a\cos\theta$ and $k = \frac{\lambda a+b}{\lambda+1}\sin\theta$. Eliminate θ.) **27.** The slope of the tangent at a point $P = (a\cos\theta, b\sin\theta)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $-\frac{b}{a}\cot\theta$. Hence the equation of the perpendicular from the focus $(ae, 0)$ to it is $y = \frac{a}{b}\tan\theta(x - ae)$. The equation of OP is $y = (\frac{b}{a}\tan\theta)x$. The corresponding directrix is $x = \frac{a}{e}$. The concurrence of these three lines can be proved using determinants and the relation $a^2 - b^2 = a^2e^2$. Or solve the first two equations to get $x = \frac{a}{e}$. **28.** $-\frac{a^2+b^2}{b}$. ($Q = (a\ \mathrm{cosec}\,\theta, b\cot\theta)$ and the slopes of the normals at P and Q are, respectively, $-\frac{a\sin\theta}{b}$ and $-\frac{a\cos\theta}{b}$.) **29.** $(4, 2)$ and $(-2, -6)$. (A typical point on the given circle is of the form $(1 + 5\cos\theta, -2 + 5\sin\theta)$. If this lies on the given line then $\tan\theta = \frac{4}{3}$.) **30.** Write both the ellipses in the standard form. A tangent to the first ellipse is of the form $x\cos\alpha + 2y\sin\alpha = 2$ for some α. This meets the second ellipse at a point $(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta)$ for two values of θ, say θ_1 and θ_2, one corresponding to P and the other to Q. The slope of the tangent to the second ellipse at a point $(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta)$ is $-\frac{\sqrt{3}}{\sqrt{6}}\cot\theta$. So the problem is equivalent to showing that $\cot\theta_1\cot\theta_2 = -2$. For this we need a quadratic in $\cot\theta$ which has $\cot\theta_1$ and $\cot\theta_2$ as roots. Such a quadratic is obtained by rewriting $\sqrt{6}\cos\alpha\cos\theta + 2\sqrt{3}\sin\alpha\sin\theta = 2$ as $\sqrt{6}\cos\alpha\cot\theta + 2\sqrt{3}\sin\alpha = 2\ \mathrm{cosec}\,\theta$ and squaring both the sides. **31.** Let $P = (a\cos\theta, b\sin\theta)$ be a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci at $F_1 = (ae, 0)$ and $F_2 = (-ae, 0)$. Then the slopes of PF_1 and PF_2 are, respectively, $\frac{b\sin\theta}{a(\cos\theta - e)}$ and $\frac{b\sin\theta}{a(\cos\theta + e)}$ while the slope of the normal at P is $\frac{a\tan\theta}{b}$. A direct computation (using $a^2(1 - e^2) = b^2$) shows that the normal at P bisects $\angle F_1PF_2$. **32.** With the same notation as in the solution to the last exercise, the equation of the tangent at P is $bx\cos\theta + ay\sin\theta = ab$. Hence $d = \frac{ab}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$. A direct computation gives the result. **33.** $2x + 3y + 22 = 0$. (As the problem involves the distances AB, AC and AD, it is preferable to parametrise the line in terms of the (algebraic) distance, say r of a point P from $(-5, -4)$. So, let $P = (-5 + r\cos\theta, -4 + r\sin\theta)$, where $\tan\theta$ is the slope of the line and let B, C, D correspond to $r = r_1, r = r_2$ and $r = r_3$ respectively. Then $-5 + r_1\cos\theta + 3(-4 + r_1\sin\theta) + 2 = 0$ which gives $\frac{15}{r_1} = \cos\theta + 3\sin\theta$. Similarly, $\frac{10}{r_2} = 2\cos\theta + \sin\theta$ and $\frac{6}{r_3} = \cos\theta - \sin\theta$. The given equation then reduces to a quadratic in $\tan\theta$ which has $-\frac{2}{3}$ as a double root.) **34.** $(\pm\frac{9}{5}, \mp\frac{12}{5})$. (Let $M = (h, k)$ be the centre of C_2. The maximum length of the common chord is 8, the diameter of C_1. Its midpoint O is at a perpendicular distance $\sqrt{25 - 16} = 3$ from M. Also OM has slope $-\frac{4}{3}$.) **35.** By a direct substitution, the point $(x_1 + r\cos\theta, y_1 + r\sin\theta)$ lies on the conic iff r is a root of the quadratic $Ur^2 + 2Vr + W = 0$, where $U = a^2\cos^2\theta + 2h\cos\theta\sin\theta + b^2\sin^2\theta, V = (ax_1 + hy_1 + g)\cos\theta + (by_1 + hx_1 + f)\sin\theta$ and $W = S_1$. Since P is the midpoint of AB, the sum of the roots vanishes, giving $V = 0$ and hence $\tan\theta = -\frac{ax_1+hy_1+g}{by_1+hx_1+f}$. So the equation of the chord with

midpoint P is $(by_1 + hx_1 + f)(y - y_1) + (ax_1 + hy_1 + g)(x - x_1) = 0$, which reduces to the given one. The computations involved in the other approach are complicated. **36.** As the angle in a semicircle is a right angle, the intercepts of the given chords by the circle of diameter $2a$ are $2a\cos\frac{\pi}{2n}, 2a\cos\frac{2\pi}{2n}, \ldots, 2a\frac{(n-1)\pi}{2n}$. To find their product, use (11) of Chapter 7 (with n there replaced by $2n$).

37. Let $k = \tan\theta$. Then one of the vertices of the triangle is the origin while the other two are $(\frac{p\cos\theta}{\cos(\alpha-\theta)}, \frac{p\sin\theta}{\cos(\alpha-\theta)})$ and $(\frac{p\cos\theta}{\cos(\alpha+\theta)}, -\frac{p\sin\theta}{\cos(\alpha+\theta)})$. These may be found algebraically or purely trigonometrically by recognising that the perpendicular from O to the third line has length p and makes an angle α with the x-axis. The perpendicular bisectors of the sides passing through O are easy to find and so this approach is not prohibitive. But a better way is to note that since one of the vertices is at the origin, the equation of the circumcircle is of the form $x^2 + y^2 + gx + fy = 0$ for some constants f, g to be determined. Homogenising this using $\frac{x\cos\alpha + y\sin\alpha}{p} = 1$ gives $p(x^2 + y^2) + (gx + fy)(x\cos\alpha + y\sin\alpha) = 0$ as the equation of the pair of straight lines $y = \pm kx$. Comparing this with $y^2 - k^2x^2 = 0$ gives a system of two equations in f and g, viz., $g\sin\alpha + f\cos\alpha = 0$ and $g\cos\alpha + k^2 f\sin\alpha = p + k^2$ which can be solved to give the result.

38. Let the vertices of the triangle be $(a\cos\theta_i, b\sin\theta_i), i = 1, 2, 3$. If (h, k) is the circumcentre and r the circumradius, then $a^2\cos^2\theta_i + b^2\sin^2\theta_i - 2ah\cos\theta_1 - 2bk\sin\theta_i = r^2 - h^2 - k^2$ for $i = 1, 2, 3$. Adding these three equations and using Exercise (8.19), $r^2 - h^2 - k^2$ comes out as $\frac{1}{2}(a^2 + b^2)$ and so with a little simplification we get $4ah\cos\theta_i + 4bk\sin\theta_i = (a^2 - b^2)\cos 2\theta_i$ for $i = 1, 2, 3$. But as in the solution to Exercise (8.19) we may suppose $\theta_2 = \theta_1 + \frac{2\pi}{3}$. The desired locus is obtained by eliminating θ_1 from the two equations : $4ah\cos\theta_1 + 4bk\sin\theta_1 = (a^2 - b^2)\cos 2\theta_1$ and $4ah\cos(\theta_1 + \frac{2\pi}{3}) + 4bk\sin(\theta_1 + \frac{2\pi}{3}) = (a^2 - b^2)\cos(2\theta_1 + \frac{4\pi}{3})$.

39. 11 (and not 12 as may appear at first sight). (Let R and r be the radii of the orbits of the earth (E) and the moon (M) respectively, where $0 < r < R$. Measure time t in years starting from some lunar eclipse. Taking the sun as the origin and the earth at $(R, 0)$ when $t = 0$, the position of the earth at time t is $(R\cos 2\pi t, R\sin 2\pi t)$ while that of the moon is $(R\cos 2\pi t + r\cos 24\pi t, R\sin 2\pi t + r\sin 24\pi t)$. At a solar eclipse, M must divide OE internally in the ratio $(R - r) : r$ which reduces to $\cos 24\pi t = -\cos 2\pi t$ and $\sin 24\pi t = -\sin 2\pi t$. Rewriting these as $\cos 24\pi t = \cos(\pi + 2\pi t)$ and $\sin 24\pi t = \sin(\pi + 2\pi t)$, we get $22\pi t = k\pi$ for some odd integer k. For $0 < t < 1$, the possible values of k are $1, 3, 5, \ldots, 21$.)

40. (a) In a standard form the equation of a hyperbola can be written as $AX^2 - Y^2 = k$ where $k \neq 0$ and $A > 0$. Its asymptotes are the lines $\sqrt{A}X + Y = 0$ and $\sqrt{A}X - Y = 0$. The general case also follows from this because in the reduction from it, X and Y are some linear expressions in the original variables x and y. (b) $(3x + 2y + 1)(20x + 11y + 47) = -54$.

(Take the equation of the hyperbola as $(3x+2y+1)(ax+by+c)=k$ where $ax+by+c=0$ is the other asymptote. Since $(-1,0)$ and $(-2,1)$ lie on the curve, we get $2a-2c=k$ and $6a-3b-3c=k$. Differentiating the equation w.r.t. x, and putting $x=-2, y=1$ and $y'=2$ gives $-17a+b+7c=k$. Solve this homogeneous system of three equations in four unknowns to express a, b, c as multiples of k.) (c) The tangent at a point $(a\sec\theta, b\tan\theta)$ on C passes through P if and only if $\frac{x_1\sec\theta}{a}-\frac{y_1\tan\theta}{b}=1$. This leads to a quadratic in $\sec\theta$, viz. $E\sec^2\theta-\frac{2x_1}{a}\sec\theta+(1+\frac{y_1^2}{b^2})=0$. The discriminant criterion for the roots to be real and distinct reduces to $E-1<0$, i.e. to $E<1$. Further, the roots are of the same sign if $0<E<1$ and of opposite signs if $E<0$. The assertion follows because $\sec\theta$ is positive for all points on one branch of C and negative for all points on the other branch. If $E=0$, then the point P lies on one (or both) of the asymptotes of the hyperbola C. In this case the quadratic degenerates to an equation of degree 1 (or 0). Accordingly, there is only one or no tangent from P, unless we treat an asymptote as a 'tangent at infinity'.

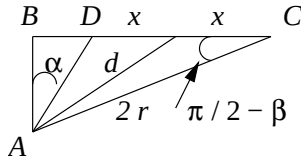
Chapter 11

1. As the angles depend only on the relative proportions of the sides, we assume $b+c=11, c+a=12$ and $a+b=13$ which gives $a=7, b=6$ and $a=5$. Use cosine formula to get $\cos A=\frac{1}{5}, \cos B=\frac{19}{35}$ and $\cos C=\frac{5}{7}$.)

2. (a) $5\pm\sqrt{6}$. (Letting the third side be x, we have $\frac{\sin(\frac{\pi}{3}-\theta)}{x}=\frac{\sin\frac{\pi}{3}}{9}=\frac{\sin(\frac{\pi}{3}+\theta)}{10}$ for some θ. Expanding and solving, $\cos\theta=\frac{x+10}{18}$ and $\sin\theta=\frac{\sqrt{3}(10-x)}{18}$. This leads to a quadratic in x, viz., $x^2-10x+19=0$.) (b) $A=45^\circ, B=60^\circ$ and $C=75^\circ$. (The data is superfluous. The single piece of information $\sin(C-A)=\frac{1}{2}$ gives $C-A=30^\circ$, which along with $A+C=120^\circ$ determines all angles.)

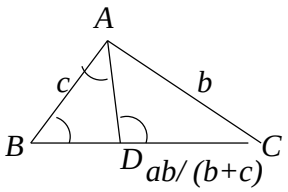
3. 75°. ($B=60^\circ$ and $\frac{\sin B}{\sin C}=\frac{\sqrt{3}}{\sqrt{2}}$ give $\sin C=\frac{1}{\sqrt{2}}$. So $C=45^\circ$ or 135°. But the latter would give a negative A.) **4.** $\frac{16}{7}$. (Take $a=4, b=5$ and $c=6$. Then with usual notations, $s=\frac{15}{2}$ and $\Delta=\frac{15\sqrt{7}}{4}$. So $R=\frac{abc}{4\Delta}=\frac{8}{\sqrt{7}}$ and $r=\frac{\Delta}{s}=\frac{\sqrt{7}}{2}$.)

5. The hypothesis is a twisted way of saying that $\angle ABC$ is a right angle and the conclusion is a twisted way of asking us to find the radius, which equals $\frac{1}{2}AC$. Call AC as $2r$ and CD as $2x$. Then $2x=BC-BD=2r\sin\beta-AB\tan\alpha=2r\sin\beta-2r\cos\beta\tan\alpha$, giving $x=r(\sin\beta-\cos\beta\tan\alpha)=\frac{\sin(\beta-\alpha)}{\cos\alpha}$. Further, since $AD\cos\alpha=AB=2r\cos\beta$, we have $AD=\frac{2r\cos\beta}{\cos\alpha}$. By Apollonius theorem, $AC^2+AD^2=2(d^2+x^2)$, i.e., $2r^2(1+\frac{\cos^2\beta}{\cos^2\alpha})=d^2+x^2$. Substituting for x and simplifying gives the result.



6. A.P. (See Comment No. 4(ii) of Chapter 24 for the reasoning.)

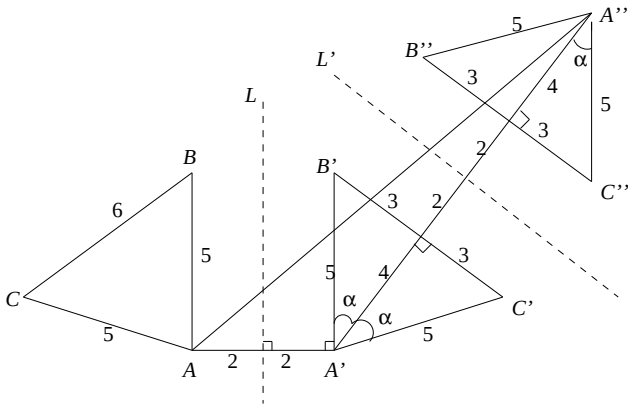
7. Let the internal angle bisector at A meet BC at D. Then $\angle ADC = B + \frac{A}{2}$ and so the condition $\angle A = 2\angle B$ is equivalent to $\angle ADC = \angle A$, which, in turn, amounts to saying that the triangles ABC and DAC are similar. This happens iff $BC : AC = CA : CD$. The result follows from the fact that $CD = \frac{ab}{b+c}$ (a property of angle bisectors that they divide the opposite sides in the ratio of the adjacent sides).



8. 113°. ($AD = c\sin B$ gives $\sin B = \frac{ab}{b^2-c^2} = \frac{\sin A \sin B}{\sin^2 B - \sin^2 C} = \frac{\sin A \sin B}{\sin(B+C)\sin(B-C)}$ $= \frac{\sin B}{\sin(B-C)}$. Hence $\sin(B-C) = 1$. So $B = C + 90°$.) **9.** $\frac{a^2}{6}$. (The radius of the inscribed circle is $\frac{\sqrt{3}}{6}a$. Hence the side of the square is $\frac{\sqrt{2}\sqrt{3}}{6}a$.)

10. $\angle A = 75°, \angle B = 45°$ and $c = \sqrt{6}$ cm. (First find c by the cosine formula and then find B using the sine rule.)

11. $\frac{8\sqrt{17}}{\sqrt{5}}$. (Let L denote the line in which the first reflection is taken and L' be the second line in which the reflection of $A'B'C'$ is taken. Then $AA' = 4$ cm and $A'A'' = 12$ cm. So, to find AA'' it suffices to know $\angle AA'A''$. Clearly this angle equals $90° + \alpha$, where $\alpha = \sin^{-1}\frac{3}{5}$. This gives $\cos\angle AA'A'' = -\sin\alpha = -\frac{3}{5}$ and hence $AA'' = \sqrt{16 + 144 + 96 \times \frac{3}{5}}$ cm.s.)



12. $\frac{\sqrt{3}+1}{2}$ cm^2. (Let h be the altitude on the given side. Then $h\cot 45° + h\cot 30° = \sqrt{3} + 1$.)

13. The formula for P follows by letting $BC = 2x$ and noting that the power of the midpoint of BC w.r.t. the circle is x^2 and also $h(2r - h)$. The area

A equals $hx = h\sqrt{h(2r-h)}$. Hence $\dfrac{A}{P^3} = \dfrac{h\sqrt{h(2r-h)}}{8\left(\sqrt{2hr-h^2}+\sqrt{2hr}\right)^3} = \dfrac{\sqrt{2r-h}}{8\left(\sqrt{2r-h}+\sqrt{2r}\right)^3} \to \frac{1}{64}$ as $h \to 0$.

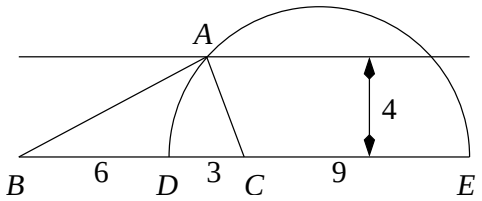
14. 90°. (The cosine formula reduces the condition to $a^2 = b^2 + c^2$.)

15. (i) BC. (Similar triangles.) (ii) $(AP)^2$. (Similar triangles.) (iii) $\angle B$.

16. Only (A) and (D) are feasible. (The relation $\sin B = \frac{b \sin A}{a}$ forces $\sin B$ to exceed 1 in (B) and in (C). In (E) $a = b$ gives $\angle A = \angle B$ and hence both are obtuse. (A) is feasible in many cases, e.g., if $A = 60°$ and $\angle B = 90°$. For (D), one possible choice is $\angle A = 60°, \angle B = 80°$.)

17. (D). (Since $\frac{P}{2} + \frac{Q}{2} = \frac{\pi}{2}$, $\tan\frac{P}{2}\tan\frac{Q}{2} = 1$.)

18. Draw a segment $BC = 9$ cm and let D, E divide it internally and externally in the ratio 2 : 1. Draw a semicircle with DE as a diameter. Draw a line parallel to BC at a distance 4 cm from it and let it cut the semicircle at A. Then ABC is a desired triangle.

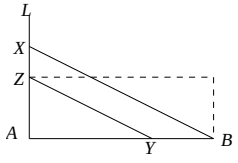


19. No. The construction does not specify how to draw a line through B whose intercept between the circle and the line OA equals the radius. There is, in fact, no construction for such a line.

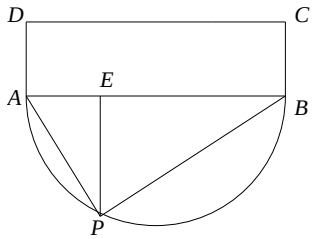
20. The proposed construction is inadmissible. Although the point of contact of the circle with the line L will indeed be the foot of the perpendicular from P to L, drawing such a circle only by trial and error is not permissible. For a valid construction draw any (sufficiently large) circle centred at P and let it cut L at A and B. Take Q to be the midpoint of the segment AB (for which ruler and compass constructions are available.)

21. Draw a line parallel to L and at a distance half the given length. From either of its points of intersection with the circle, draw a line perpendicular to L. (Use standard school geometry constructions to draw such lines.)

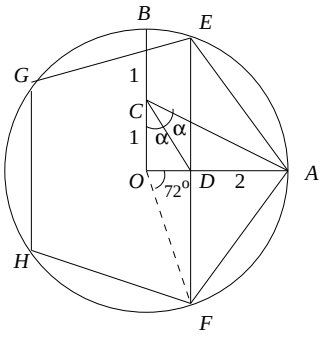
22. Let AB be the segment given as a base and let $PQRS$ be the given rectangle. Draw a line L through A perpendicular to AB. On L mark a point X such that $AX = PQ$. On AB take a point Y such that $AY = PS$. Through Y draw a line parallel to BX and let it meet L at Z. Then the rectangle with sides AB and AZ is a desired rectangle. (The triangles ABX and AYZ are similar. So $AZ.AB = AX.AY = PQ.PS$.)



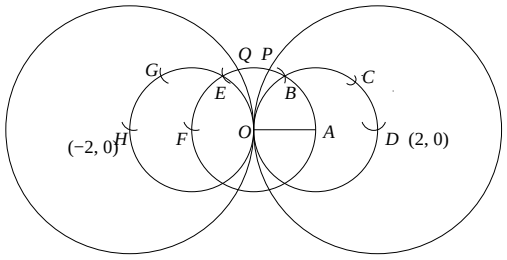
23. Let the given rectangle be $ABCD$ with $AB \geq AD$. On AB take E so that $AE = AD$. Draw a semi-circle with AB as a diameter and let the perpendicular to AB at E meet this semi-circle at P. Then the square with AP as a side is a desired square. (By similarity of the triangles APB and AEP, $AP^2 = AB.AE$.) The construction is a geometric way to find the G.M. of two (positive) numbers (whence probably the name).



24. By the last exercise, a segment of length $\sqrt{5}$ can be constructed as the side of a square having the same area as a rectangle with sides 5 and 1. One can then extend it to get a segment of length $\sqrt{5}+1$. Using either Exercise (11.21) or (11.22) draw a right-angled triangle in which the hypotenuse is $\sqrt{5}+1$ and one arm is 1. This gives a construction for an angle of measure $72°$. A quicker construction can be based on the division property of an angle bisector of a triangle. Take two mutually perpendicular radii OA and OB of a circle of radius 2 and let C be the midpoint of OB. Let the internal angle bisector of $\angle OCA$ meet OA at D. Then $OD : DA = 1 : \sqrt{5}$ and so $OD = \frac{2}{1+\sqrt{5}}$. Through D draw a line parallel to OB and let it meet the circle at E and F. Then A, E, F are vertices of a regular pentagon inscribed in the circle. (The remaining two vertices can be obtained by drawing arcs EG and FH each equal to the arc AE.)



25. Without loss of generality, let the given segment, say OA, be of unit length. Draw a circle of radius 1 with A as the centre and starting from O, mark off point B, C, D with $OB = BC = CD = 1$. Similarly draw the unit circle, C_1, with centre at O and get E and F on it with $AB = BE = EF$.



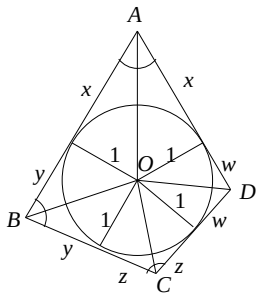
Finally draw a unit circle with F as the centre and get G, H on it with $OE = EG = GH = 1$. It is easily seen that taking O as the origin and A as $(1, 0)$, the points D, H are $(2, 0)$ and $(-2, 0)$ respectively. Now let the circles of radii 2 each with D, H as centres meet C_1 at P, Q respectively

(all points of intersections are to be taken on the positive side of the x-axis). A little calculation gives $P = \left(\frac{1}{4}, \frac{\sqrt{15}}{4}\right)$ and $Q = \left(-\frac{1}{4}, \frac{\sqrt{15}}{4}\right)$. So PQ is a segment of length $\dfrac{1}{2}$. The circle with O as the centre and radius equal to PQ will cut the segment OA at its midpoint.[1]

26. Only $\sin 15^\circ \cos 15^\circ$ is rational. (It is not necessary to actually compute the expressions. It is enough to relate them to $\sin 30^\circ$ and $\cos 30^\circ$ which are known to be rational and irrational respectively. The last number equals $\sin^2 15^\circ$ which is related to $\cos 30^\circ$ in a rational manner, i.e., by a linear equation whose coefficients are rational. So irrationality of $\cos 30^\circ$ forces $\sin^2 15^\circ$ to be irrational. But then so is $\cos^2 15^\circ$. And this forces their square roots, i.e., $\sin 15^\circ$ and $\cos 15^\circ$ to be irrational too.)

27. 3. ($A_0A_1 = 1, A_0A_2 = A_0A_4 = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$.)

28. Let O be the centre of the inscribed circle (whose radius may be taken to be 1). Let A, B, C, D be the vertices of the quadrilateral and x, y, z, w be the lengths of the tangents from A, B, C, D respectively to this circle.



From the figure, $x = \cot\frac{A}{2}$ etc. and $a = x + y = \cot\frac{A}{2} + \cot\frac{B}{2}$ etc. Hence $s = x + y + z + w = \cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} + \cot\frac{D}{2}$. Note that $\frac{A}{2} + \frac{C}{2} = \frac{\pi}{2} = \frac{B}{2} + \frac{D}{2}$ as $ABCD$ is cyclic. So $ab = (x + y)(y + z) = \cot^2\frac{B}{2} + \cot\frac{B}{2}(\cot\frac{A}{2} + \cot\frac{C}{2}) + 1 = s\cot\frac{B}{2} - \cot\frac{B}{2}\cot\frac{D}{2} + 1 = sy$. This proves $\frac{ab}{s} = y$. The other three equalities are proved similarly.

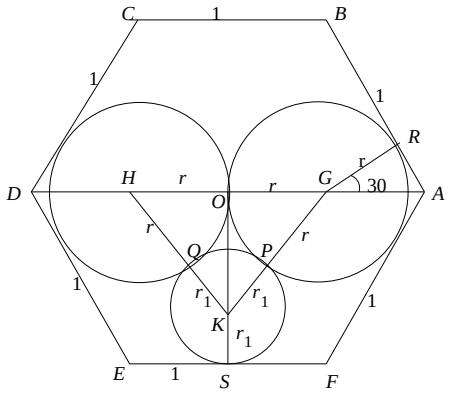
29. Here coordinates are more convenient. The distances of the diagonal from the centre of the circle are $52(= \sqrt{65^2 - 39^2})$ and 60 cm.s respectively. Take the circle as $x^2 + y^2 = 65^2$ and the first diagonal parallel to the x-axis. Then the slope of the other diagonal is $\pm\frac{3}{4}$. Choosing the $+$ sign, its equation is $y = \frac{3}{4}x \pm 75$. Choose the $-$ sign. By a direct calculation, it meets the circle at $(16, -63)$ and $(56, -33)$. The other diagonal is $y = -52$ and meets the circle at $(\pm 39, -52)$. A direct calculation gives the sides.

30. Here, too, coordinates can be used. But a direct solution involves the same work. By symmetry, the first two circles have the same radius , say r and their centres, say G, H, lie on the line AD. (See the figure on the next page.) Moreover, they must touch each other at the centre, say O, of the hexagon. Let K and r_1 be the centre and the radius of the third

[1] I am indebted to Prof. A. R. Shastri for this solution.

circle and assume that it touches the first two circles at P and Q. Draw perpendiculars GR and KS from G to AB and from K to EF respectively. Take each side of the hexagon to equal 1. Then $OA, \ldots, OE, OF$ also equal 1 each.

In the right-angled triangle GRA, $\angle RGA = 30^\circ$ and so $r = GR = \frac{\sqrt{3}}{2}GA$. Hence $1 = OA = OG + GA = r + \frac{2}{\sqrt{3}}r$ which gives $r = \frac{\sqrt{3}}{\sqrt{3}+2}$. Also in the equilateral triangle OEF, $OS = \frac{\sqrt{3}}{2}$ and so $OK = OS - KS = \frac{\sqrt{3}}{2} - r_1$. Hence the right-angled triangle KOG gives $(r+r_1)^2 = (\frac{\sqrt{3}}{2} - r_1)^2 + r^2$. This gives $r_1 = \frac{3}{8r+4\sqrt{3}}$. Substituting the value of r obtained earlier gives the value of r_1 as $\frac{3(\sqrt{3}+2)}{4(4\sqrt{3}+3)}$. Rationalise the denominator to get the result.



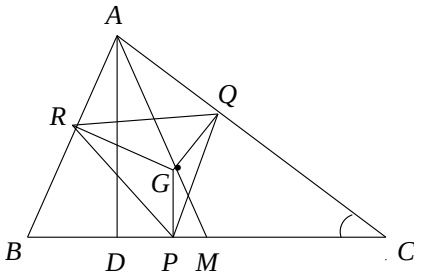
31. Take the vertices of the n-gon at $(a\cos\frac{2\pi k}{n}, a\sin\frac{2\pi k}{n})$ for $k = 1, 2, \ldots, n$ and P at $(\frac{a}{2}, 0)$. Then the sum asked is $a^4 \sum_{k=1}^{n} \left[(\frac{1}{2} - \cos\frac{2\pi k}{n})^2 + \sin^2\frac{2\pi k}{n}\right]^2$ which further becomes $a^4 \sum_{k=1}^{n} \left(\frac{5}{4} - \cos\frac{2\pi k}{n}\right)^2$. On squaring the terms this becomes $a^4 \sum_{k=1}^{n} \left(\frac{25}{16} - \frac{5}{2}\cos\frac{2\pi k}{n} + \frac{1}{2} + \frac{1}{2}\cos\frac{4\pi k}{n}\right)$. The result follows since $\sum_{k=1}^{n} \cos\frac{2\pi k}{n}$ and $\sum_{k=1}^{n} \cos\frac{4\pi k}{n}$ both vanish by (21) or (24) of Chapter 7.

32. For $k = 1, 2, \ldots, n-1$, the sine rule applied to the triangle CP_kA gives $\frac{1}{CP_k} = \frac{1}{b\sin A}\sin(\pi - \frac{\pi k}{2n} - A) = \frac{1}{b\sin A}\sin(A + \frac{\pi k}{2n})$. Hence the given sum equals $\frac{1}{b\sin A}\sum_{k=1}^{n-1} \sin(A + \frac{\pi k}{2n})$. By (23) in Chapter 7, this comes out as $\dfrac{\sin(\frac{\pi}{4} + A)\sin(\frac{\pi}{4} - \frac{\pi}{4n})}{b\sin A \sin\frac{\pi}{4n}}$ which equals $\dfrac{1}{2b}(\cot A + 1)(\cot\dfrac{\pi}{4n} - 1)$. Put $\cot A = \frac{b}{a}$ to complete the proof. The case $n = 2$ gives a formula for the length of the angle bisector of the right angle of a right-angled triangle.

33. The identity $\cot A\cot B + \cot B\cot C + \cot C\cot A = 1$ for a triangle ABC suggests that we put $l = \cot A, m = \cot B, n = \cot C$. Then the equality to be proved reduces to proving $\sin 2A + \sin 2B + \sin 2C = 4\sin A\sin B\sin C$. This is a standard result. (p. 134 of *Plane Trigonometry - Part I* by S. L. Loney.) **34.** Geometrically the data means that the line $x\cos\alpha + y\cos\beta = c$ touches the curve $x^2 - 2xy\cos\gamma + y^2 = c^2$ (which, incidentally, is an ellipse as we see by taking $A = B = 1$ and

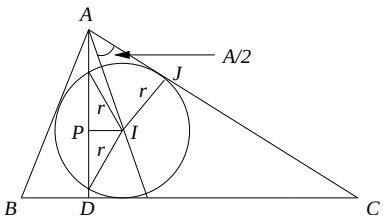
$H = -\cos\gamma$ in the general equation of a conic). The condition for tangency gives $\cos^2\alpha + 2\cos\alpha\cos\beta\cos\gamma + \cos^2\beta = \sin^2\gamma$ which is equivalent to $(\cos\gamma + \cos\alpha\cos\beta)^2 = \sin^2\alpha\sin^2\beta$. Taking square roots, this means $\cos\gamma = -\cos(\alpha \pm \beta) = \cos(\pi + \alpha \pm \beta)$. Hence $\gamma = 2n\pi \pm (\pi + \alpha \pm \beta)$ which implies the result. Alternatively, we may assume c is positive as otherwise we replace the first equation by its negative. We may further assume that x, y are also positive as otherwise we can replace the corresponding coefficient ($\cos\alpha$ etc.) by its negative by changing the angle suitably (i.e., replacing α by $\alpha + \pi$ etc.) Finally we may suppose $0 \le \gamma \le \pi$ as otherwise we replace γ by $\cos^{-1}(\cos\gamma)$. We now have a triangle with sides x, y and c such that the angle between the sides x and y is γ. Relating α, β to the other two angles of the triangle gives the result. (The two solutions are conceptually different because x, y represent different things in them.)

35. Let the altitudes of the triangle be AD, BE, CF. Then GP is parallel to and one-third of AD. So $GP = \frac{1}{3}c\sin B = \frac{bc}{6\rho}$, where ρ is the circumradius of ABC. Similarly $GQ = \frac{ca}{6\rho}$ and $GR = \frac{ab}{6\rho}$. Also, $\angle PGQ = 180^\circ - \angle C$. So the area of ΔPGQ is $\frac{1}{2}GP.GQ\sin C = \frac{abc^2\sin C}{72\rho^2} = \frac{abc^3}{144\rho^3}$. Similarly finding the areas of the triangles QGR and RGP and adding, the area of the triangle PQR equals $\frac{abc}{144\rho^3}(a^2 + b^2 + c^2)$. Putting $a = 2\rho\sin A$ etc. completes the solution.

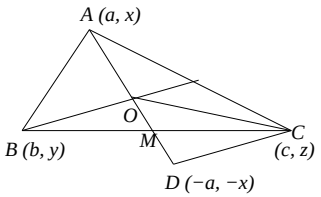


36. The form of the L.H.S. of the equality to be proved suggests that the key lies in relating x, y, z to r. To see this relationship, draw the perpendicular IP from the incentre I to the altitude AD of the triangle ABC. (See the figure next page.) Also draw a perpendicular IJ to the side CA. Then from the triangle AIJ, we have $AI = \frac{r}{\sin\frac{A}{2}}$. From the figure given below, it is easily seen that $\angle PAI = \frac{|B-C|}{2}$. So, $PI = AI\sin\frac{|B-C|}{2} = \frac{r\sin\frac{|B-C|}{2}}{\sin\frac{A}{2}}$. Hence $x^2 = 4(r^2 - PI^2) = \frac{4r^2(\sin^2\frac{A}{2} - \sin^2\frac{B-C}{2})}{\sin^2\frac{A}{2}} = \frac{4r^2(1-\cos A - 1 + \cos(B-C))}{1-\cos A} = \frac{4r^2(\cos(B+C) + \cos(B-C)}{1-\cos A} = \frac{8r^2\cos B\cos C}{1-\cos A}$. Hence $\frac{1}{x^2} = \frac{1-\cos A}{8r^2\cos B\cos C}$. Deriving similar expressions for $\frac{1}{y^2}$ and $\frac{1}{z^2}$ and adding, $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{\cos A + \cos B + \cos C - \cos^2 A - \cos^2 B - \cos^2 C}{8r^2\cos A\cos B\cos C}$. Also, since $R = \frac{r}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$, we have $\frac{1}{8rR\cos A\cos B\cos C} = \frac{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}{8r^2\cos A\cos B\cos C}$. Thus the problem is now reduced to showing that $\cos A + \cos B + \cos C - \cos^2 A - \cos^2 B - \cos^2 C = 2\cos A\cos B\cos C + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$, or equivalently, to showing that $\cos A + \cos B + \cos C + \sin^2 A + \sin^2 B + \sin^2 C = 3 + 2\cos A\cos B\cos C + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$. This is done by putting together two standard identities, viz., (i) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2\cos A\cos B\cos C$ and (ii)

$\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ (see pages 135 and 137 of *Plane Trigonometry - Part I* by S. L. Loney).



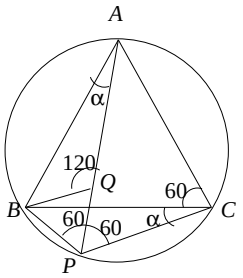
Exercise (11.36)



Exercise (11.37)

37. Consider the triangle with vertices $A = (a, x), B = (b, y), C = (c, z)$. Then the origin is the centroid of the this triangle. Let M be the midpoint of BC and let $D = (-a, -x)$. Then the triangles OMB and DMC are congruent and so in the triangle OCD, $OD = \sqrt{a^2 + x^2}, OC = \sqrt{c^2 + z^2}$ and $CD = OB = \sqrt{b^2 + y^2}$. Since the coordinates of the vertices of OCD are all rational, so is its area. Moreover the area equals $\pm\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ -a & -x & 1 \\ c & z & 1 \end{vmatrix}$ i.e., $\pm\frac{1}{2}(cx - az)$. On the other hand, $-bcx^2 - cay^2 - abz^2 = -bcx^2 - ca(x + z)^2 - caz^2 = x^2(-bc - ca) + z^2(-ab - bc) - 2acxz = c^2x^2 + a^2z^2 - 2acxz = (cx - az)^2$. This completes the proof.

38. For a trigonometric proof, let $\angle BAP = \alpha$ and R be the circumradius. Then the problem reduces to proving $\sin(\alpha + 60^\circ) = \sin\alpha + \sin(60^\circ - \alpha)$. For a pure geometry solution, take Q on AP so that $PB = PQ$. Then since $\angle BPQ = \angle BCA = 60^\circ$, the triangle BPQ is equilateral. This makes $\angle BQA = 120^\circ$ and $\Delta BQA \equiv \Delta BPC$. Ptolemy's theorem also gives the result directly. But a solution using complex numbers appears awkward.



39. (i) Put $a = \sin^2 A$ etc. (ii) Put $\frac{a}{a+d} = \sin^2\alpha$ and $\frac{b}{b+c} = \sin^2\beta$. (iii) Put $x = \tan A, y = \tan B$ to reduce the inequality to $\sin 2(A + B) \le 1$.

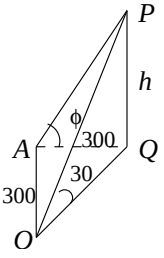
Chapter 12

1. $60 \cot 15^\circ = 120 + 60\sqrt{3}$ m. **2.** $h = \dfrac{d}{\sqrt{\cot^2\alpha + \cot^2\beta - 2\cot\alpha\cot\beta\cos\gamma}}$. (Eliminate a, b from $a = h\cot\alpha, b = h\cot\beta$ and $a^2 + b^2 - 2ab\cos\gamma = d^2$.)

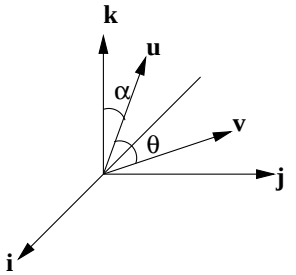
3. Draw a diagram and notice that $\beta = \tan^{-1}\frac{1}{n} - \tan^{-1}\frac{1}{2n}$.

4. Let l denote the length of the ladder. From the projections of the ladder on the ground we get $l\cos\beta - l\cos\alpha = a$ while from its projections on the wall we get $l\sin\alpha - l\sin\beta = b$. Rewriting these equations as $2l\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = a$ and $2l\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = b$ and dividing the first by the second gives the result.

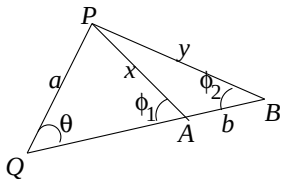
5. $\phi = \tan^{-1}\frac{\sqrt{2}}{\sqrt{3}}$ radians $= \frac{180\sqrt{2}}{\pi\sqrt{3}}$ degrees. Height of tower $= 100\sqrt{6}$ meters. (Let P and Q be the top and the bottom of the tower and h be its height. Then OAQ is a right angled isosceles triangle with $OA = 300$. So $OQ = 300\sqrt{2}$ and $h = 300\sqrt{2}\tan 30^\circ = 300\frac{\sqrt{2}}{\sqrt{3}} = 100\sqrt{6}$. Also $\tan\phi = \frac{h}{300} = \frac{\sqrt{2}}{\sqrt{3}}$.)



6. The crucial point is to find the angle, say θ, between the tower and the road. This is best done using vectors. Let the unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ point, respectively, due East, due North and vertically upwards. The tower is given to lie in the plane spanned by $\mathbf{j}$ and $\mathbf{k}$ and inclined to the latter at an angle α as in Figure (a) below. So a unit vector, say $\mathbf{u}$ in the direction of the tower is $\cos\alpha\mathbf{k} + \sin\alpha\mathbf{j}$. The unit vector, say $\mathbf{v}$, in the direction of the Northwest is $-\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$. So $\cos\theta = \mathbf{u}\cdot\mathbf{v} = \frac{\sin\alpha}{\sqrt{2}}$.



(a)



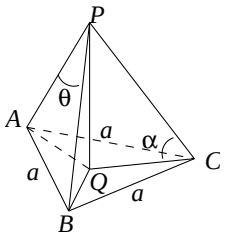
(b)

Having found θ, the rest of the work is completely in the plane containing the tower and the road as shown in Figure (b) above. Let P and Q be top and the bottom of the tower and A, B be the points on the road at which the tower subtends angles ϕ_1, ϕ_2 respectively. Let x, y denote the sides PA, PB respectively. Then from triangle PAQ, we have $\frac{x}{\sin\theta} = \frac{a}{\sin\phi_1}$, i.e., $x = \frac{a\sin\theta}{\sin\phi_1}$. Similarly from triangle PBQ, we get $y = \frac{a\sin\theta}{\sin\phi_2}$. Finally, from triangle PAB, $\frac{b}{\sin(\phi_1-\phi_2)} = \frac{x}{\sin\phi_2} = \frac{y}{\sin\phi_1}$ and hence $\frac{b^2}{\sin^2(\phi_1-\phi_2)} = \frac{xy}{\sin\phi_1\sin\phi_2}$. Substituting the values of x and y obtained

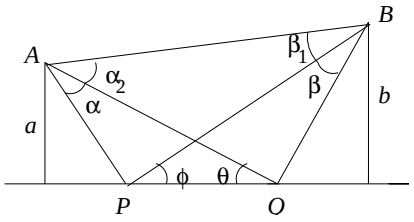
earlier, we get $\dfrac{b^2}{\sin^2(\phi_1-\phi_2)} = \dfrac{a^2\sin^2\theta}{\sin^2\phi_1\sin^2\phi_2} = \dfrac{a^2(1-\cos^2\theta)}{\sin^2\phi_1\sin^2\phi_2}$. With a little manipulation, this gives $\cos\theta = \dfrac{[a^2\sin^2(\phi_1-\phi_2) - b^2\sin^2\phi_1\sin^2\phi_2]^{1/2}}{a\sin(\phi_1-\phi_2)}$.

The solution is completed using $\cos\theta = \frac{\sin\alpha}{\sqrt{2}}$.

7. Let the legs meet the ground at A, B, C. Let P be the point of intersection of the lines along the legs and Q be the projection of P on the ground. Then Q is at the centre of the equilateral triangle ABC and so $QA = QB = QC = \frac{a}{\sqrt{3}}$, a being the side of the the triangle ABC. Also $PA = PB = PC = QC\sec\alpha = \frac{a\sec\alpha}{\sqrt{3}}$. The desired angle, say θ, is the angle between PA and PB and is given by $\cos\theta = \frac{PA^2+PB^2-AB^2}{2PA.PB}$. Substituting the values of PA and PB gives the result.

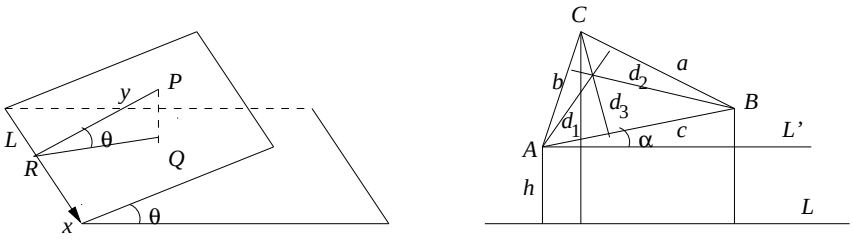


8. Note that $\alpha = \pm(\alpha_1 - \alpha_2)$ and $\beta = \pm(\beta_1 - \beta_2)$, the signs depending on the locations of P and Q. But this does not affect the reasoning. From triangle APB, we have $\frac{\sin\beta_1}{\sin\alpha_1} = \frac{AP}{BP}$. Similarly, from triangle AQB, $\frac{\sin\beta_2}{\sin\alpha_2} = \frac{AQ}{BQ}$. To get hold of the ratio $\frac{\sin\alpha}{\sin\beta}$, introduce $\theta = \angle AQP$ and $\phi = \angle BPQ$. Applying the sine rule to the triangles APQ and BPQ, and dividing, we have $\frac{\sin\alpha}{\sin\beta} = \frac{BQ\sin\theta}{AP\sin\phi}$. Multiplying the three ratios together and using $a = AQ\sin\theta, b = BP\sin\phi$ completes the solution.



9. Because of the identity $\cos A+\cos B+\cos C = 1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ which holds for every triangle, the problem amounts to showing that $\cos^2\theta = 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$. If the plane of the triangle ABC is horizontal then the hypothesis implies that it is equilateral and the conclusion holds with $\theta = 0$. Otherwise, let L be the line in which the plane of the triangle meets the given horizontal plane. The trick is to get rid of the horizontal plane and thereby convert the hypothesis to one involving only the plane of the triangle. For this, note that if P is any point in the plane of the triangle and PQ, PR are perpendiculars from P to the horizontal plane and to the line L respectively (see Figure (a) on the next page), then $\angle PRQ = \theta$ and so $PQ = PR\sin\theta$. So the vertical distance of P above the horizontal plane is $\sin\theta$ times its perpendicular distance from the line L.

In this new interpretation, the hypothesis now becomes that the altitudes of the triangle ABC are proportional to the perpendicular distances of A, B, C respectively from a fixed line L in the plane of the triangle, the constant of proportionality being $\sin\theta$. (See figure (b) next page.)



(a) (b)

Draw a line L' through A parallel to L and let α be the angle between L' and AB. If h is the distance between L and L', then the distances of A, B, C from L are, respectively, $h, h + c\sin\alpha$ and $h + b\sin(A+\alpha)$. So we have $d_1 = h\sin\theta, d_2 = (h + c\sin\alpha)\sin\theta$ and $d_3 = (h + b\sin(A+\alpha))\sin\theta$. Subtracting and using formulas like $d_1 = b\sin C = c\sin B$, we get

$$d_2 - d_1 = (a-b)\sin C = c\sin\alpha\sin\theta \quad (1)$$

$$\text{and} \quad d_3 - d_1 = (a-c)\sin B = b\sin(A+\alpha)\sin\theta \quad (2)$$

which can be replaced by

$$\sin A - \sin B = \sin\alpha\sin\theta \quad (3)$$

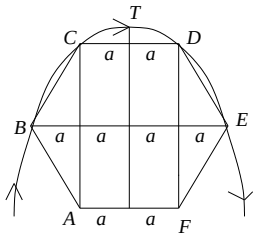
$$\text{and} \quad \sin A - \sin C = \sin(A+\alpha)\sin\theta \quad (4)$$

respectively. Eliminating α from these two equations gives the result. To carry out the elimination, multiply (3) by $\cos A$ and subtract it from (4) to get $\cos\alpha\sin A\sin\theta = (\sin A - \sin C) - \cos A(\sin A - \sin B)$. Square the expressions for $\sin\alpha\cos A\sin\theta$ and $\cos\alpha\sin A\sin\theta$ and add. Then use the fact that $\sin^2 B + \sin^2 C - 2\sin B\sin C\cos A = \sin^2 A$ (which is a version of the cosine rule) and, after some more simplification, the standard identity $\sin B + \sin C - \sin A = 4\cos\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$.

10. Apply (11) with $d' = d$ and $u = \lambda\sqrt{g(h + \sqrt{h^2+d^2})}$ which simplifies to $u^2 = \lambda^2 gd(\sec\beta + \tan\beta)$. This leads to a quadratic inequality in $\tan\alpha$, viz., $\tan^2\alpha - 2\lambda^2\tan\alpha(\sec\beta + \tan\beta) \le -1 - 2\tan\beta(\sec\beta + \tan\beta)$. Complete the square and simplify.

11. Let t_1, t_2 be the times at which the particles pass through B, Let h denote the horizontal displacement from A to B. Then we have $t_1 = \dfrac{h}{V\cos\alpha}$ and $t_2 = \dfrac{h}{V\cos\alpha'}$ and so the time interval $t_2 - t_1$ equals $\dfrac{h(\cos\alpha' - \cos\alpha)}{V\cos\alpha\cos\alpha'}$. Equating the vertical displacements we get, $\dfrac{h\sin\alpha}{\cos\alpha} - \dfrac{gh^2}{V^2\cos^2\alpha} = \dfrac{h\sin\alpha'}{\cos\alpha'} - \dfrac{gh^2}{V^2\cos^2\alpha'}$. So $h\sin(\alpha - \alpha') = \dfrac{gh^2}{2V^2}\dfrac{(\cos^2\alpha' - \cos^2\alpha)}{\cos\alpha\cos\alpha'}$. This factors as $\frac{gh}{V}(t_2 - t_1)(\cos\alpha + \cos\alpha')$. A routine trigonometric simplification gives the answer.

12. Let $2a$ be the side of the hexagon $ABCDEF$. We may assume that the particle is projected from the ground at a speed in a direction making angle α with the horizontal. Then u will also be its speed when it hits the ground while its minimum speed occurs when it is at the top, say T, of the path, at which time its speed is $u\cos\alpha$ (there being no vertical component of the velocity). So the problem amounts to showing that $\frac{u}{u\cos\alpha} = \frac{\sqrt{31}}{\sqrt{3}}$, i.e., $\sec^2\alpha = \frac{31}{3}$.



Let t_1, t_2, t_3, t_4, t_5 be the times (from the start) when the particle reaches B, C, T, D, E respectively. From the horizontal displacements we get that t_1, t_2, t_3, t_4, t_5 are in an A.P. with common difference $\frac{a}{u\cos\alpha}$. Also, considering the vertical displacements, we have $(u\sin a)t_1 - \frac{1}{2}gt_1^2 = a\sqrt{3}$ and $(u\sin a)t_2 - \frac{1}{2}gt_2^2 = 2a\sqrt{3}$. The result could be obtained by eliminating t_1 and t_2 from these two equations along with $(t_2 - t_1) = \frac{a}{u\cos\alpha}$. But this is a little cumbersome. Instead, note that since the particle is at the same vertical levels at t_1 and t_5, they are roots of the quadratic $\frac{1}{2}gt^2 - (u\sin\alpha)t + a\sqrt{3} = 0$. Noting that the square of the difference of the roots of a quadratic $Ax^2 + Bx + C = 0$ is $\frac{B^2-4AC}{A^2}$ and using $t_5 - t_1 = \frac{4a}{u\cos\alpha}$,

$$\frac{4(u^2\sin^2\alpha - 2ag\sqrt{3})}{g^2} = \frac{16a^2}{u^2\cos^2\alpha} \tag{5}$$

Similarly, t_2 and t_4 are the roots of the quadratic $\frac{1}{2}gt^2 - (u\sin\alpha)t + 2a\sqrt{3} = 0$ and this gives

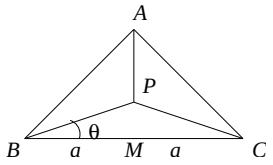
$$\frac{4(u^2\sin^2\alpha - 4ag\sqrt{3})}{g^2} = \frac{4a^2}{u^2\cos^2\alpha} \tag{6}$$

Dividing (5) by (6) we get $ag = \frac{\sqrt{3}u^2\sin^2\alpha}{14}$. Putting this into (5) (or in (6)), we get $4u^2\cos^2\alpha(u^2\sin^2\alpha - \frac{3}{7}u^2\sin^2\alpha) = \frac{48}{196}u^4\sin^4\alpha$, which, upon simplification gives $\tan^2\alpha = \frac{28}{3}$ whence $\sec^2\alpha = \frac{31}{3}$ as was to be shown.

13. $\frac{u^2}{g(1-\sin\beta)}$ (as measured on the inclined plane). (Let the plane of the motion intersect the inclined plane in a line L and let θ be the angle which the forward direction of L makes with the horizontal plane. Then $-\beta \le \theta \le \beta$. Let the initial velocity make an angle α with L. Then the range as measured along L is $\dfrac{u^2(\sin(2\alpha+\theta) - \sin\theta)}{g\cos^2\theta}$. The maximum value of this for a given θ is $\frac{u^2}{g(1+\sin\theta)}$. For $\theta \in [-\beta, \beta]$ this is maximum when $\theta = -\beta$. When this happens, the line L is at right angles to the line of intersection of the given inclined plane with the horizontal plane and the motion is above that portion of L which is descending along the inclined plane.)

Chapter 13

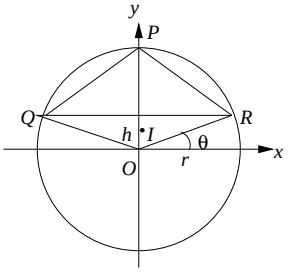
1. Let $A = (0,1), M = (0,0), B = (-a,0)$ and $C = (a,0)$. Suppose Let $\theta = \angle PBM$. In all there are three tours, two of which, viz., $P-A-B-C-P$ and $P-A-C-B-P$ have lengths $(1-\tan\theta+\sqrt{2}+2+\sec\theta)a$ each while the third one, viz., $P-C-A-B-P$ has length $(2\sqrt{2}+2\sec\theta)a$. The three tours are of equal length if $(3-\sqrt{2}-\tan\theta)^2 = \sec^2\theta$, which gives $\tan\theta = \frac{9-4\sqrt{2}}{7}$. For $0 \le \tan\theta < \frac{9-4\sqrt{4}}{7}$, the third tour is the shortest.



2. $8 : (\pi+4)$. (If $2a$ and b denote the horizontal and the vertical sides of the rectangle then the problem amounts to maximising $2ab + \frac{\pi}{4}a^2$ subject to the constraint that $2b + (\pi+2)a = c$, a constant. This reduces to maximising the function $f(a) = ac - 2a^2 - \frac{3\pi}{4}a^2$ for $0 \le a \le \frac{c}{\pi+2}$.)

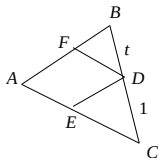
3. (a) $2hk$. (If a, b are the intercepts of the line on the axes, then the problem amounts to minimising $\frac{1}{2}ab$ subject to the constraint $\frac{h}{a} + \frac{y}{b} = 1$ with $a > 0, b > 0$ or equivalently, to minimising the function $f(b) = \frac{b^2 h}{2(b-k)}$ for $k < b < \infty$.) (b) 18. (More generally, if instead of $(8,2)$ we had a point (h,k) in the first quadrant, then the answer would be $h+k+2\sqrt{hk}$. Very similar to (a) except that now we want to minimise $a+b$.)

4. (i) $\frac{3\sqrt{3}}{4}r^2$. (Take the circle as $x^2+y^2 = r^2$, P as $(0,r)$ and the equation of QR as $y = h$, where $-r \le h \le r$. Then the area of ΔPQR equals $(r-h)\sqrt{r^2-h^2} = r^2\cos\theta(1-\sin\theta)$, where $h = r\sin\theta$. The critical points of this function in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ are $\theta = \frac{\pi}{2}$ and $\theta = -\frac{\pi}{6}$. (ii) $\frac{r}{2}\sqrt{10\sqrt{5}-22}$. (Here we may suppose that h and hence θ are non-negative. Let $I = (0,k)$ be the incentre of the triangle OQR. Equating the distances of I from the three sides, we get $k = \frac{r\sin\theta}{1+\cos\theta}$. So the inradius of ΔOQR is $h-k = \frac{r\sin\theta\cos\theta}{1+\cos\theta} = r\cot\theta - r\cos\theta\cot\theta = f(\theta)$ (say). The critical points of f over $[0, \frac{\pi}{2}]$ correspond to $\cos\theta = 1$ and $\cos\theta = \frac{-1+\sqrt{5}}{2}$. The latter gives the maximum inradius of $\triangle OQR$.)



5. $(0,0)$. (Let $f(x) = \dfrac{d}{dx}\left(\dfrac{x}{1+x^2}\right) = \dfrac{1-x^2}{(1+x^2)^2}$. Then $f'(x) = \dfrac{2x^3-6x}{(1+x^2)^3}$. Hence f is decreasing on $(-\infty, -\sqrt{3})$, increasing on $(-\sqrt{3}, 0)$, decreasing on $(0, \sqrt{3})$ and again increasing on $(\sqrt{3}, \infty)$. So f has a local minimum at 0. Since $\lim\limits_{x\to\infty} f(x) = 0 < f(0)$, f has, in fact, a global minimum at 0. Putting $x = \tan\theta$ simplifies the calculations but only marginally.) **6.** The expression attains its minimum $\frac{(a+b)^2}{a}$ at $x = \frac{a^2}{a+b}$.

7. Let D, F divide the segments BC, BA respectively in the ratio $t:1$ each. Let Δ, Δ_1 and Δ_2 denote the areas of the triangles ABC, BFD and CED respectively. Then $\frac{\Delta_1}{\Delta} = \frac{t^2}{(t+1)^2}$ and $\frac{\Delta_2}{\Delta} = \frac{1}{(t+1)^2}$. Hence the area of the parallelogram $AFDE = \Delta - \Delta_1 - \Delta_2 = \frac{(t+1)^2 - t^2 - 1}{(t+1)^2}\Delta$. This is maximum when $t = 1$. Calculation of Δ completes the proof. (As nothing is given about the signs of p, q, r, the expression for the area ought to have been replaced by its absolute value. The vertices A, B, C lie on the parabola $y^2 = x$. But this has little bearing on the problem.)

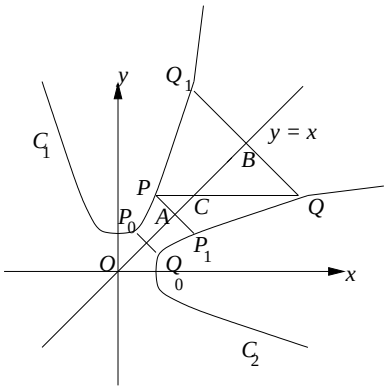


8. Let the critical points be $c_1, c_2, \ldots, c_n$ with $a \le c_1 < c_2 < \ldots < c_{n-1} < c_n \le b$. Apply Theorem 3 to as many of the subintervals $[a, c_1], [c_1, c_2], \ldots, [c_{n-1}, c_n], [c_n, b]$ as needed. For example, to prove (i), suppose $x < y$ and $c_i, c_{i+1}, \ldots, c_j$ are the critical points between x and y. Then $f(x) \le f(c_i) \le f(c_{i+1}) \le \ldots f(c_j) \le f(y)$ which implies $f(x) \le f(y)$. **9.** (A) (The data implies that $h(x)$ is decreasing whence $h(x) \le h(0) = 0$ for all $x \in [0, \infty)$. But f and hence h takes only non-negative values. This forces $h(x) = 0$ for all $x \in [0, \infty)$. **10.** All values in $(-2, -1) \cup [-1, \infty)$. (Regardless of the value of b, the function is decreasing on $[0, 1)$ and increasing on $[1, 3]$. For the minimum to occur at 1, the expression in b must be non-negative.) **11.** (B) ((D) is also correct but is already subsumed by (B).) (Let $f(x) = ax^2 + bx + c$. Then $a > 0$ and $b^2 < 4ac$. $g(x)$ comes out as $ax^2 + (b+2a)x + (b+c+2a)$ and $(b+2a)^2 < 4a(b+c+2a)$.) **12.** (D). (Write $f(x)$ as $1 - \frac{2}{x^2+1}$.) **13.** (C). (Assuming $n > 0$, every term except the first has a minimum at 0.) **14.** Increasing on $[-\frac{1}{2}, 0)$ and on $[\frac{1}{2}, \infty)$; decreasing on $(-\infty, -\frac{1}{2}]$ and on $(0, \frac{1}{2}]$. **15.** f is increasing and g is decreasing. (Use $x < \tan x$ for $x \in (0, \frac{\pi}{2})$ to show $f'(x) > 0$ for $x \in (0, 1]$. For g, use $\sin x \cos x < \sin x < x$ for $x \in (0, \frac{\pi}{2})$.)

16. (a) $\{x : x \ge 0$. (b) $\cos(\ln \theta)$. (This expression is positive for the given range of θ. The other expression is negative.) (c) Using second derivative, show that the function $f(x) = x \ln x$ is concave upwards on $(0, \infty)$ and apply (20) with $x_1 = 1 - x, x_2 = 1 + x, \lambda_1 = \frac{1}{2}, \lambda_2 = \frac{1}{2}$. (d) With $x = \tan\theta$ and rewriting $\sec\theta - 1$ as $\frac{\tan^2\theta}{\sec\theta + 1}$, the inequality to be proved reduces to $\ln(\tan\theta + \sec\theta) \ge \frac{\tan\theta}{\sec\theta+1}$ for $\theta \in [0, \frac{\pi}{2})$. Take derivatives. (e) $x + 1$ and $1 - x$. (A linear function is of the form $Ax + B$ for some constants A and B. The extrema occur only at the end points.) **17.** (a) $f(x) = x^{1/x} = e^{\ln x/x}$. So $f'(x) = x^{1/x}\frac{d}{dx}(\frac{\ln x}{x}) = x^{1/x}\frac{1 - \ln x}{x^2}$. Hence f is maximum at $x = e$. Thus $e^{1/e} > \pi^{1/\pi}$ and so $e^\pi > \pi^e$. (b) $n = 3$. (From the hint and (a) the comparison is reduced to $2^{1/2}$ and $3^{1/3}$.) **18.** (i) $a = 2, b = -\frac{1}{2}$. (Solve the equations $-a - 2b + 1 = 0$ and $\frac{a}{2} + 4b + 1 = 0$. The question ought to have contained $\ln|x|$ in place of $\ln x$.) (ii) No extrema if $0 < b \le 1$; if $b > 1$ then there is a local maximum at $\frac{b - \sqrt{b^2-1}}{4}$ and a local minimum at $\frac{b + \sqrt{b^2-1}}{4}$. **19.**

$\lambda \in [-\frac{3}{2}, 0) \cup (0, \frac{3}{2}]$. (Put $u = \sin x$. Then the problem amounts to finding the extrema of the function $u^3 + \lambda u^2$ for $-1 \le u \le 1$.) **20.** $|c|$ if $c \le \frac{1}{2}$ and $\sqrt{c - \frac{1}{4}}$ if $c \ge \frac{1}{2}$. (Minimise $x^2 + (x^2 - c)^2$, $x \in \mathbb{R}$.) **21.** $(\pm\frac{a^2}{\sqrt{a^2+b^2}}, \pm\frac{b^2}{\sqrt{a^2+b^2}})$. (First assume P is in the first quadrant. Take it as $(a\cos\theta, b\sin\theta)$, where $0 \le \theta \le \frac{\pi}{2}$. From the equation of the tangent at P, ON comes out to be $\frac{ab}{\sqrt{b^2\cos^2\theta + a^2\sin^2\theta}}$. The area, say Δ, of the triangle OPN is $\frac{1}{2}ON.\sqrt{OP^2 - ON^2}$. Since $OP = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}$, after a little simplification (based on the identity $\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$) we get $\Delta = \frac{ab(a^2-b^2)}{2} f(\theta)$, where $f(\theta) = \frac{\sin\theta\cos\theta}{b^2\cos^2\theta + a^2\sin^2\theta}$ $= \frac{\sin 2\theta}{a^2+b^2+(b^2-a^2)\cos 2\theta}$. $f'(\theta) = 0$ gives $\cos 2\theta = \frac{a^2-b^2}{a^2+b^2}$ and hence $\cos\theta = \frac{a}{\sqrt{a^2+b^2}}$ and $\sin\theta = \frac{b}{\sqrt{a^2+b^2}}$. This gives P as $(\frac{a^2}{\sqrt{a^2+b^2}}, \frac{b^2}{\sqrt{a^2+b^2}})$. Because of symmetry of the ellipse about both its axes, there are three other positions of P, where Δ is maximum.) **22.** $\frac{\sqrt{3}}{2}\frac{av}{\sqrt{u^2+uv+v^2}}$. (If d is the distance at time t, then by the cosine formula, $d^2 = (u^2 + uv + v^2)t^2 - a(2u + v)t + a^2$ which can be minimised by completing squares or by taking derivatives.)

23. Let PP_1, QQ_1 and PQ meet the line $y = x$ at A, B, C respectively. As the trapezium PP_1QQ_1 is symmetric about the line $y = x$, we have $PC \ge PA = \frac{1}{2}PP_1$ and $CQ \ge QB = \frac{1}{2}QQ_1$. Adding the two implies the desired inequality. Because of this inequality, the points P_0 and Q_0 must be the reflections of each other in the line $y = x$ for otherwise, replacing one of these two points by the reflection of the other would give us a pair of points, say P^* and Q^* such that $P^*Q^* < P_0Q_0$.



The problem now reduces to minimising the distance between points of the form $(t, t^2 + 1)$ and $(t^2 + 1, t)$ for $t \in \mathbb{R}$. By a direct calculation the minimum occurs when $t^2 - t + 1$ is minimum, i.e., when $t = \frac{1}{2}$. This gives $P_0 = (\frac{1}{2}, \frac{5}{4})$ and $Q_0 = (\frac{5}{4}, \frac{1}{2})$.

To do the problem by the method given at the end of Comment No. 16, i.e., by observing that the line P_0Q_0 must be normal to C_1 at P_0 and to C_2 at Q_0, let $P_0 = (s, s^2 + 1)$ and $Q_0 = (t^2 + 1, t)$ for some s, t. The normalcy condition just given gives $-\frac{1}{2s} = \frac{t-s^2-1}{t^2+1-s} = -2t$. This gives $t = \frac{1}{4s}$ and further a quintic in s, viz., $32s^5 + 48s^3 - 16s^2 - 8s + 1 = 0$. By trial, $s = \frac{1}{2}$ is a root. This indeed corresponds to $P_0 = (\frac{1}{2}, \frac{5}{4})$ and $Q_0 = (\frac{5}{4}, \frac{1}{2})$. To complete the solution, the remaining roots of the quintic in s will have to be found and this is not easy. The solution given above

is simpler because it crucially uses the fact that the curves C_1 and C_2 are reflections of each other in the line $y = x$. **24.** (a) Let $b = \lambda a + (1-\lambda)c$ where $0 < \lambda < 1$. Then $a - b + c = (1-\lambda)a + \lambda c$. Apply (20) to both and add. (b) Linear functions. (i.e., functions of the form $Ax + B$, where A, B are constants). **25.** Without loss of generality assume that $f(x)$ is concave upwards on $[c-\delta, c]$ and concave downwards on $[c, c+\delta]$ for some $\delta > 0$. Then $f'(x)$ increases on $[c-\delta, c]$ and decreases on $[c, c+\delta]$ and hence has a local maximum at $x = c$. So the derivative of $f'(x)$ at c, in case it exists, must vanish. The points of inflection of the sine function are $n\pi$, where n is an integer. Those of the cosine function are of the form $n\pi + (-1)^n \frac{\pi}{2}$, where n is an integer. (These are also precisely the zeros of the respective functions. But that is only a coincidence.) 0 is not a point of inflection of $f(x) = x^4$ since f is concave upwards on $(-\infty, 0]$ and also on $[0, \infty)$ and so there is no change of behaviour. A sufficient condition for c to be a point of inflection is that $f''(c) = 0$ but $f'''(c) \neq 0$. If $f'''(c) > 0$, then $f''(x) < 0$ for x slightly left of c and $f''(x) > 0$ for x slightly bigger than c. So the function changes from concave downward to upward at c. Similarly, if $f'''(c) < 0$ then it changes from concave upward to downward.

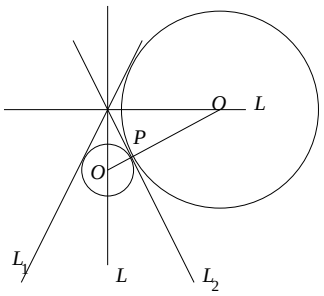
Chapter 14

1. If all the three factors in $(a+b-c)(b+c-a)(c+a-b)$ are positive then (1) applies. Otherwise, exactly one of them is ≤ 0, in which case the product is also ≤ 0 and the inequality holds trivially since $abc > 0$. **2.** When $CA = CB$. (The base AB is fixed. Maximise the height.) **3.** $-\frac{\sqrt{3}}{2}$. (The said mean, say M, equals $-\frac{1}{3}(\sin\alpha + \sin\beta + \sin\gamma)$. If the triangle is acute-angled, then $\alpha + \beta + \gamma = 2\pi$. For a fixed γ, the expression $-\sin\alpha - \sin\beta$ equals $-2\sin(\frac{\alpha+\beta}{2})\cos(\frac{\alpha-\beta}{2})$ and hence is minimum when $\alpha = \beta$. So among acute-angled triangles, M is minimum when $\alpha = \beta = \gamma = \frac{2\pi}{3}$ and the minimum value is $-\frac{\sqrt{3}}{2}$. For a right-angled or an obtuse-angled triangle, one of α, β and γ equals the sum of the other two; say $\gamma = \alpha + \beta$. Again for a fixed γ, M is minimum when $\alpha = \beta = \frac{\gamma}{2}$ and its value is $-\frac{2\sin\frac{\gamma}{2} + \sin\gamma}{3}$. Here $0 \leq \gamma \leq \pi$. M is minimum in the degenerate case where $\gamma = \pi$ and its value is $-\frac{2}{3}$. But this is bigger than $-\frac{\sqrt{3}}{2}$ for acute-angled triangles. Hence the minimum of M for all triangles is $-\frac{\sqrt{3}}{2}$. **4.** The first inequality is equivalent to showing that $ab + bc + ca \leq a^2 + b^2 + c^2$ and follows from $(a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$ with equality holding iff a, b, c are all equal. (Here we do not need that a, b, c are sides of a triangle.) For the second inequality, by the cosine rule $a^2 + b^2 - c^2 \leq 2ab$. Write two similar inequalities and add the three. **5.** $\frac{1}{3}$. (Since $\frac{\tan A + \tan B}{1 - \tan A \tan B} = \sqrt{3}$, a constant, the problem is equivalent to minimising $\tan A + \tan B$. As the function $\tan x$ is concave upward on $[0, \frac{\pi}{3}]$, we have $\tan A + \tan B \geq 2\tan(\frac{A+B}{2}) = \frac{2}{\sqrt{3}}$ with equality only if $A = B = \frac{\pi}{6}$.) **6.** The only change needed in the proof is that the inequality (19) of the last chapter will be strict since the

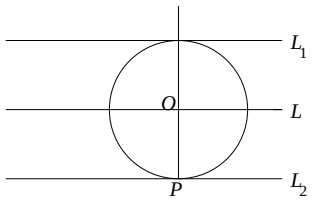
point $P_3 = (x_3, f(x_3))$ on the graph of $y = f(x)$ lies strictly below the point on the chord joining $(x_1, f(x_1))$ and $(x_2, f(x_2))$. In case $h'' < 0$ on (a, b), the inequality has to be reversed. **8.** The result is equivalent to proving that $(m+n)\ln(ma+nb) > (m+n)\ln(m+n) + m\ln a + n\ln b$, i.e., to proving that $(\frac{m}{m+n})\ln a + (\frac{n}{m+n})\ln b < \ln(\frac{ma+nb}{m+n})$. This follows by taking $h(x) = \ln x$, $\lambda_1 = \frac{m}{m+n}$, $\lambda_2 = \frac{n}{m+n}$ and applying the last part of Exercise (14.6). **9.** The inequality reduces to proving that $(n^2+1)^{2n} > (1-\frac{1}{n})^{n-1}(1+\frac{1}{n})^{n+1}$ which follows directly from the last exercise by taking $a = 1-\frac{1}{n}$, $b = 1+\frac{1}{n}$, $m = n-1$ and $n = n+1$. **10.** Without loss of generality assume $a > b$. Apply Exercise (13.16)(c) with $x = \frac{a-b}{a+b}$ to get $u^{2/(a+b)} > 1$, where $u = (\frac{2a}{a+b})^a(\frac{2b}{a+b})^b$. As the exponent $2/(a+b)$ is positive, this means that $u > 1$ which is equivalent to the inequality to be proved. This result is unlikely to come from Exercise (14.6) because the downward concavity of the $\ln x$ function will yield an upper bound (viz., $(a+b)^{a+b}$) on $a^a b^b$. What we need here is a lower bound on $a^a b^b$ and that is provided by the upward concavity of the function $x\ln x$. **11.** If $\cot A + \cot B + \cot C = \sqrt{3}$, then squaring and using the identity $\cot A\cot B + \cot B\cot C + \cot C\cot A = 1$ gives $(\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 = 0$ which implies $\cot A = \cot B = \cot C$. The same identity also works for the second characterisation. The third one follows from Exercise (14.1). Reduce the last one to the inequliaty $\sin^2 A + \sin^2 B + \sin^2 C \geq 2\sqrt{3}\sin A\sin B\sin C$. The A.M.-G.M. inequality reduces this further to maximising $\sin A\sin B\sin C$. For many other characterisations, too, it is convenient to view them as optimisation problems. For example, if it is given that $\cos\frac{A}{2} + \cos\frac{B}{2} + \cos\frac{C}{2} = \frac{3\sqrt{3}}{2}$, then to conclude that ABC is equilateral, show that $\cos\frac{A}{2} + \cos\frac{B}{2} + \cos\frac{C}{2} \leq \frac{3\sqrt{3}}{2}$, with equality holding only when $A = B = C$. Wherever possible, replace products by sums which are easier to optimise (e.g., replace $\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ by $\frac{1}{4}(\sin A + \sin B + \sin C)$.) It is easy to create characterisations in terms of values of various trigonometric functions of the three angles. But the more appealing ones are those in terms of certain geometric arrtributes of the triangle, such as the inradius, the circumradius and the area. The more intricate characterisations of equilateral triangles involve some variable parameter such as a point inside the traingle. Exercise (14.18) is a challenging example of this type.) **12.** Put $a = \tan\frac{A}{2}, b = \tan\frac{B}{2}$ and $c = \tan\frac{C}{2}$ where A, B, C are angles lying in $(0, \pi)$. Then $ab + bc + ca = 1$ implies that $\frac{A+B+C}{2} = \pi/2$. This reduces the problem to that solved at the beginning of Comment No. 2. **13.** The cosine formula and the identities $h = b\sin C = c\sin B$ reduce inequality to $\cos A + 1 \geq 2\sin B\sin C$. The R.H.S. equals $\cos(B-C) + \cos A$. Therefore the ineqality is true, with equality holding if and only if $B = C$.

14. (i) For notational simplicity, call P_1 as P. First do the degenerate case when P lies on one of the two given lines, say, on L_2. Take L to be either of the two angle bisectors of the angle between L_1 and L_2 in case they

intersect (see Figure (a)). If they are parallel, take L to be the line which is parallel to them and lying exactly midway between them (see Figure (b)). (The latter can be thought of as the limiting case of the former as the point of intersection of L_1 and L_2 moves to infinity. But we treat it separately.) Easy constructions are available to draw L in either case. Draw the perpendicular to L_2 at P (by yet another standard construction) and let it meet L at O. Then the circle with O as the centre and OP as a radius is a desired circle.

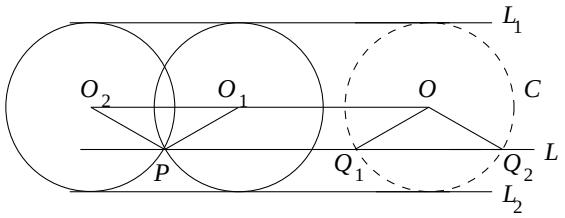


(a)



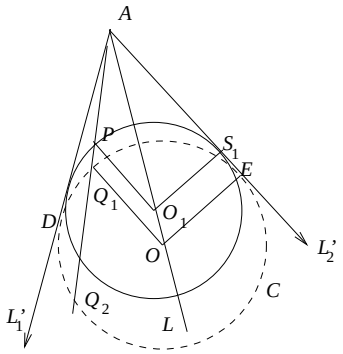
(b)

Now suppose P lies on neither of the two lines L_1 and L_2. First suppose that L_1 and L_2 are parallel to each other. Then a desired circle exists if and only if P lies in the region bounded by L_1 and L_2. Assume this is the case. First draw *any* circle, say C, touching L_1 and L_2. The idea is to shift this circle suitably. Suppose O is the centre of this circle. Let the line L through P parallel to L_1 (and hence also to L_2) meet C at Q_1 and Q_2. Complete the parallelograms OQ_1PO_1 and OQ_2PO_2. Then either of the two circles, one with centre O_1 and the other with centre O_2 and radius O_1P (or O_2P) will touch L_1, L_2 and pass through P.



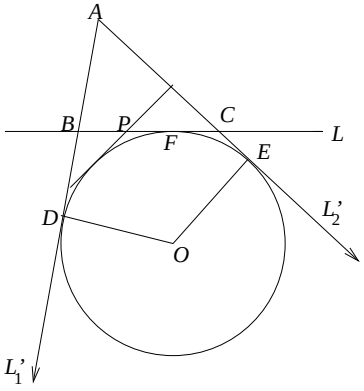
Finally, suppose L_1 and L_2 intersect at A (say). Let L_1', L_2' be their rays originating at A which include the point P. Here also the idea is to first draw *any* circle C which touches the rays L_1', L_2' and then shift it suitably. But in shifting, we also have to change the radius suitably. Let L be the angle bisector of the angle with arms L_1', L_2'. Take any point O on L and draw a circle C with centre O touching L_1', L_2' at D and E respectively. Let AP meet this circle C at Q_1 and Q_2. Through P draw a line parallel to Q_1O and let it meet AO at O_1. Then the circle centred at O_1 and radius O_1P will touch L_1' and L_2'. To see this, let S_1 be the foot of the perpendicular from O_1 to L_2. Then the triangles AO_1S_1

and AOE are similar and so $O_1S_1 : OE = AO_1 : AO$. But the triangles APO_1 and AQ_1O are also similar. Hence $AO_1 : AO = O_1P : OQ_1$. Hence $O_1S_1 : OE = O_1P : OQ_1$. But $OE = OQ_1$. So, $O_1S_1 = O_1P$ which proves that the circle centred at O_1 and passing through P touches L_2. By a similar argument it touches L_1. (Starting from Q_2 instead of Q_1 we shall get another desired circle. It is not shown in the figure.)



Part (ii) can be reduced to (i) as follows. We assume that P_1, P_2 lie on the same side of L_1 as otherwise no such circle exists. Let L be the perpendicular bisector of the segment P_1P_2. Then P_1 and P_2 are the reflections of each other in L. Let L_1' be the reflection of L_1 in L. Then the desired circle touches L_1, L_1' and passes through P_1. One can similarly reduce (i) to (ii) by taking P_2 to be the reflection of P_1 in the angle bisector of L_1 and L_2. A direct solution to (ii) can be given using the concept of power of a point w.r.t. a circle. Let the line P_1P_2 meet L_1 at P and let the desired circle touch L_1 at T. Then $PT^2 = PP_1.PP_2$. T can be marked off on L_1 using the construction in Exercise (11.23) and from T, P_1 and P_2, the desired circle can easily be drawn. (Note that there are two possible positions for T and hence two possible circles in general.)

15. This is one of those gems where you may spend hours to hit the key idea, but once you do, the solution comes in seconds. And in the present problem, the key concept[2] is the least expected, viz., that of the ex-circle of a triangle! Let the lines L_1 and L_2 meet at A and let L_1', L_2' be their rays originating at A which include the given point P as shown in the figure. Suppose a typical line L through P meets L_1' and L_2' at B and C respectively. Let the ex-circle of the triangle ABC opposite to the vertex A touch L_1', L_2' and L at D, E and F respectively. Since $BF = BD$, $CF = CE$ and $AD = AE$, AD and AE are each half the perimeter of the triangle ABC.

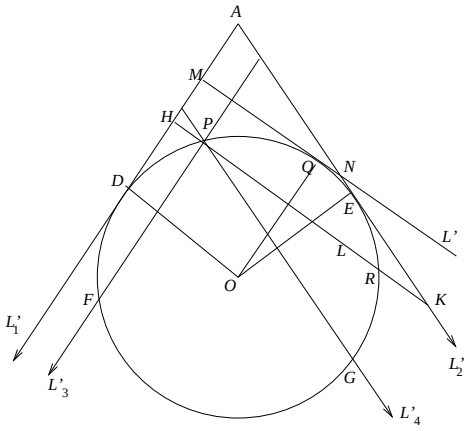


So, to get the desired line L for part (i) of the exercise, first mark off D and E on L_1' and L_2' respectively so that $AD = AE = p/2$. Draw perpendiculars to L_1, L_2 at D, E respectively and let them meet at O.

[2] I am indebted to Tanya Khovanova for this solution.

Draw the circle with centre O and radius OD. Finally, draw tangents from P to this circle. Either of these two tangents is a desired line. (It is assumed here that P lies inside the region bounded by the half rays L_1', L_2' and the shorter arc DE. Otherwise the problem has no solution as p would be too small.)

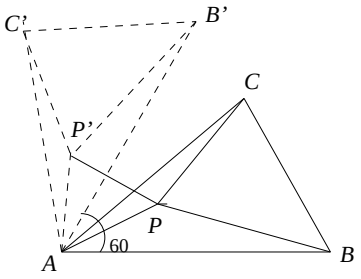
If P is on the circle described above then an interesting thing happens. The tangent at P along with L_1', L_2' forms a triangle of perimeter p. In fact, this is true of every point on the shorter arc DE of the circle. Moreover, in this case it can be shown that p is the least perimeter of a triangle that can be formed by a line through P and L_1', L_2'. For, consider any other line, say L, through P which meets the rays L_1' and L_2' at H and K respectively (say). We claim that the perimeter of the triangle AHK is greater than p.



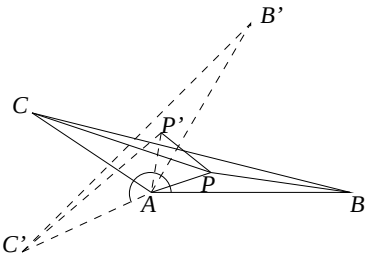
For this, first draw rays L_3', L_4' from P that are parallel to L_1', L_2' respectively. Let L, L_3', L_4' intersect the circle at R, F, G respectively. The very fact that the points H, K lie on the *rays* L_1', L_2' (and not just on the *lines* L_1 and L_2) means that the point R must lie either on the arc PF or on the arc PG. Without loss of generality suppose the latter is the case. Note that since the chord PG is parallel to L_2', the point of contact of L_2', viz. E, is the midpoint of the arc PG. As we are assuming that R lies on this arc, the midpoint, say Q, of the arc PR must lie on the arc PE. Let L' be the tangent at Q and let it cut L_1', L_2' at M, N respectively. Then MN is parallel to L and hence the triangles AMN and AHK are similar. As the latter contains the former, it has a greater perimeter than the former. But as noted earlier, the perimeter of the triangle AMN is also p. Thus we have shown that ΔAHK has perimeter greater than p.

This important observation reduces part (ii) of the exercise to constructing a circle which touches L_1', L_2' and passes through the given point P. Part (i) of the last exercise gives a construction for such a circle. (There are, in fact, two such circles. But we take the one for which the point P lies on the shorter arc joining the points of contact of the circle with L_1', L_2'. This is also the circle shown in the figure accompanying the construction. For the other circle, P lies on the longer arc joining the points of contact with the rays L_1', L_2'.) Once this circle is drawn, the tangent to it at the point P is the desired line, i.e. the line which along with L_1' and L_2' forms a triangle of the least possible perimeter.

16. Here the function to be minimised, viz. $PA + PB + PC$ is a function of two variables, viz., the coordinates of the point P. Optimisation of such functions is beyond our scope in general. However, a geometric argument can be given as follows. Rotate the triangle ABC around the point A through 60° to get a triangle $A'B'C'$. Obviously $A' = A$. Let P' be the point obtained from P under this rotation. Since $\angle PAP' = 60^\circ$ and $AP = AP'$, the triangle PAP' is equilateral. Hence $AP = PP'$. Also $PC = P'C'$. Therefore $PA + PB + PC = BP + PP' + P'C'$. This is the length of a path consisting of three line segments joining the fixed points B and C'. So its least value will occur when all the three segments lie along the same straight line. This will happen when the angles $\angle APB$ and $\angle AP'C'$ both equal 120°. Since $\angle AP'C' = \angle APC$, this means that P is the point in the triangle ABC at which the three sides subtend equal angles, viz. 120° each.



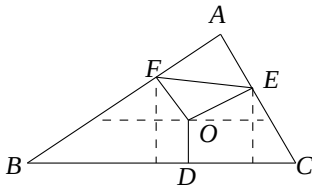
However, if one of the angles, say $\angle A$, of the triangle ABC exceeds 120 degrees, then there is no such point inside the triangle. The argument above breaks down because although $PA+PB+PC$ still equals the length of the path $B-P-P'-C'$, this path now lies in the (outer) angle BAC' which exceeds 180°. Hence it can never be a straight line path. Its length will decrease as P (and hence P' too) move closer to A. So the minimum value of $PA + PB + PC$ occurs in the degenerate case when P is A.



17. $\sqrt{10}$. (Observe that $f(x) = |PA - PB|$ where $A = (3, 2), B = (0, 1)$ and $P = (x, x^2)$ is a point on the parabola $y = x^2$. The maximum equals AB and occurs when P lies on the line AB but outside the segment AB.)

18. The first part of the hint follows by noting that OA is a diameter of the circumcircle of ΔAEF. Now, $EF \geq$ the projection of EF on BC. But the latter is the sum of the projections of EO and OF. These projections are, respectively, $OE \sin C$ and $OF \sin B$. Hence

$$OA \geq OE \frac{\sin C}{\sin A} + OF \frac{\sin B}{\sin A}.$$

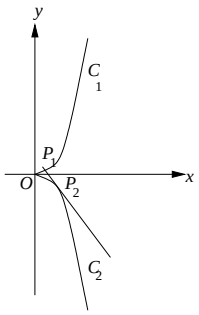


Similar inequalities hold for OB and OC. The result follows by adding the three inequalities and noting that $\dfrac{\sin B}{\sin C} + \dfrac{\sin C}{\sin B} \geq 2$ etc.

Chapter 15

1. $x+y-3=0$. (The normal at $(x_0, \frac{x_0^2}{4})$ has equation $y-\frac{x_0^2}{4}+\frac{2}{x_0}(x-x_0)=0$.) **2.** $(-\frac{16}{5}, \frac{53}{10})$. (Let (h,k) be the centre. Then $(h-2)^2+(k-4)^2 = h^2+(k-1)^2$. Also the radius through $(2,4)$ is perpendicular to the tangent to $y=x^2$ at $(2,4)$ and hence has slope $-\frac{1}{4}$. So $\frac{k-4}{h-2}=-\frac{1}{4}$.) **3.** None. (A normal to $xy=1$ must have positive slope. Hence all that can be said is that a, b are of opposite signs. So, (A) and (D) are definitely false and exactly one of (B) and (C) is true.) **4.** $y \pm \frac{1}{\sqrt{2}}x = 1$. (Let the normal at a point (x_0, x_0^2) cut the curve again at (x_1, x_1^2) and let L be the length of the chord joining these two points. Then $x_1 = -x_0 - \frac{1}{2x_0}$ and hence $L^2 = (x_0-x_1)^2(1+(x_0+x_1)^2)$ comes out to be $\dfrac{(4x_0^2+1)^3}{16x_0^4}$. This is minimum when $48x_0^4(4x_0^2+1)^2 \times 8x_0 = 64x_0^3(4x_0^2+1)^3$ which reduces to $2x_0^2=1$, since $x_0=0$ is discarded as in that case the normal never cuts the curve again. So there are two chords of minimum lengths corresponding to $x_0 = \pm\frac{1}{\sqrt{2}}$. Note incidentally, that the corresponding points on the parabola are the ends of its latus rectum.)

5. (a) Let $P_i = (x_i, x_i^3)$ for $i = 1, 2, 3, \ldots$. From the equation of the tangent at P_1 and the fact that it passes through P_2 we get $x_2^2 + x_1x_2 + x_1^2 = 3x_1^2$ which implies that $x_2 = -2x_1$ since $x_2 \neq x_1$. By the same reasoning, $x_3 = -2x_2, x_4 = -2x_3$, etc. This proves the first part. From this and the determinant formula for the area of a triangle, the given ratio equals 16. (It may appear strange that the tangent at P_1 cuts the curve again at only one more point, because as the curve is the graph of a cubic equation, normally a straight line should meet it either in one or in three points inasmuch as a cubic polynomial has either one or three real roots. In the present case, however, the line happens to be a tangent and hence its point of contact with the curve corresponds to a double root. If you get confused like this, you are potentially a good thinker. If you can resolve the confusion, you are already a good thinker!) (b) Implicit differentiation gives the equation of the tangent at (x_0, y_0) as $x_0^2 + y_0^2 m = a(mx_0 + y_0)$, where $m = \frac{x_0^2 - ay_0}{ax_0 - y_0^2}$. After some simplification, β comes out as $\frac{ax_0y_0}{y_0^2-ax_0}$. By symmetry, $\alpha = \frac{ax_0y_0}{x_0^2-ay_0}$. The result follows by a direct computation. (c)



The curve is the union of two curves, say C_1 and C_2, which are respectively, the graphs of the functions $y = x^{\frac{3}{2}}$ and $y = -x^{\frac{3}{2}}$, where $x \geq 0$. Since C_1 is concave upwards throughout, a tangent to it at a point on it cannot meet it again. The same holds for C_2 which is concave downwards throughout. So if P_1 and P_2 are points on the given curve such that the tangent at P_1 is also the normal at P_2 then either P_1 is on C_1 and

P_2 on C_2 or vice versa. As C_1 and C_2 are the reflections of each other in the x-axis, by symmetry, we suppose that the first possibility holds. So let $P_1 = (x_1, x_1^{3/2})$ and $P_2 = (x_2, -x_2^{3/2})$. Then $y - x_1^{3/2} = \frac{3}{2}x_1^{1/2}(x - x_1)$ and $y + x_2^{3/2} = \frac{2}{3}x_2^{-1/2}(x - x_2)$ represent the same line. This gives two equations in x_1 and x_2, viz., (i) $\frac{3}{2}x_1^{1/2} = \frac{2}{3}x_2^{-1/2}$ and (ii) $x_1^{3/2} = 2x_2^{3/2} + \frac{4}{3}x_2^{1/2}$. (i) simplifies to $x_1x_2 = \frac{16}{81}$. Putting $x_1 = \frac{16}{81x_2}$ reduces (ii) to a cubic in x_2, viz., $2x_2^3 + \frac{4}{3}x_2^2 = \frac{64}{729}$. By inspection, $x_2 = \frac{8}{9}$ is a root and there are no other positive roots as the function $2x_2^3 + \frac{4}{3}x_2^2$ increases strictly with x_2 for $x_2 \geq 0$. $x_2 = \frac{8}{9}$ determines x_1 as $\frac{2}{9}$.

6. (a) $y = x+4$ and $y = x-2$. (The tangent must have slope 1. Differentiating implicitly and setting $y' = 1$ gives $x+8y = 17$. Solving simultaneously with the equation of the ellipse gives $(\frac{11}{3}, \frac{5}{3})$ and $(-\frac{5}{3}, \frac{7}{3})$ as points of contact. The work is simplified if the origin is shifted to $(1, 2)$. The problem can also be done without calculus by finding c such that the line $y = x + c$ touches the ellipse. This gives a quadratic in c, viz., $c^2 - 2c - 8 = 0$.) (b) $y = -\frac{1}{2}(x - \frac{\pi}{2})$ and $y = -\frac{1}{2}(x + \frac{3\pi}{2})$. (Differentiating implicitly and setting $y' = -\frac{1}{2}$ gives $\sin(x + y) = 1$ and hence $\cos(x + y) = 0$, whence $y = 0$. So the points of contact are $(\frac{\pi}{2}, 0)$ and $(-\frac{3\pi}{2}, 0)$.) (c) There are no points where the tangent is horizontal. It is vertical at $(1, 1)$. (Implicit differentiation w.r.t. x gives $y' = \frac{y}{y^2 - x}$. $y' = 0$ gives $y = 0$ which is impossible for any point on the curve. For a vertical tangent, set the denominator $y^2 - x$ equal to 0. If that sounds awkward, differentiate implicitly w.r.t. y and set $\frac{dx}{dy}$ equal to 0.) (d) $y = x + 2$. (Let the tangent to the first curve at $(\frac{y_1^2}{8}, y_1)$ be also the tangent to the second curve at $(-\frac{1}{y_2}, y_2)$. Compare the two equations of this common tangent to get a system of two equations in y_1 and y_2.) **7.** (a) $\frac{15-5x}{3\sqrt[3]{(1-x)^5}} - 2\sin(4x + 2)$. (b) 1. (Implicit differentiation gives $e^{xy} + xe^{xy}(y + xy') = y' + \sin(2x)$.) (c) $x + y = 1$. (When $x = 0$, y equals 1. Since $(1+x)^y = e^{y\ln(1+x)}$, implicit differentiation w.r.t. x gives $y' = (1 + x)^y(\frac{y}{(1+x)} + y'\ln(1 + x)) + \frac{2\sin x \cos x}{\sqrt{1-\sin^4 x}}$. Put $x = 0, y = 1$ to get $y' = 1$. So the slope of the normal at $(0, 1)$ is -1.) **8.** (A), (B) and (D). (y equals x for $x \geq 0$ and $\frac{1}{3}x$ for $x \leq 0$.) **9.** $g(x) = -1 + \sqrt{x}$. (Let $y = f(x)$. Then $y = (x+1)^2$. Interchange x and y and then express y in terms of x.) **10.** For example, take $f_1(x) = 1, f_2(x) = g_1(x) = x$ and $g_2(x) = x^2$. Then $D(x) = 0$, but $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix} = -1 \neq D'(x)$. The second assertion follows by expanding the determinants and the product rule. For a 3×3 determinant $D(x)$ with entries $f_i(x), g_i(x), h_i(x)$ in the i-th column for $i = 1, 2, 3$, $D'(x)$ is the sum of three determinants obtained from $D(x)$ by differentiating one row at a time, i.e., in a compact form

$$\frac{d}{dx}\begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix} = \begin{vmatrix} f_1' & f_2' & f_3' \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1' & g_2' & g_3' \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1' & h_2' & h_3' \end{vmatrix}$$

When the entries in a particular row are all constants, the corresponding summand vanishes.

11. (a) $-\frac{8}{\ln 2} + \frac{32}{\pi^2+16}$. (The function equals $(\frac{\ln \sin x}{\ln \cos x})^2 + 2\tan^{-1} x$.) (b) $\frac{1}{e}$. ($f(x)$ equals $\frac{\ln(\ln x)}{\ln x}$.) (c) Evidently the question deals with $p'(1)$. Putting $x = 1$ in the given inequality, we get $|p(1)| \le 0$ and hence $p(1) = 0$. Hence for all $x \ge 0, x \ne 1$, $|\frac{p(x)-p(1)}{x-1}| = |\frac{p(x)}{x-1}| \le |\frac{e^{x-1}-1}{x-1}|$. As $x \to 1$, the first term approaches $|p'(1)|$ while the last term approaches the absolute value of the derivative of the function e^{x-1} at $x = 1$. As this derivative is 1, the result follows. (d) 0. (By Exercise (15.10), $F'(a)$ is a sum of three determinants. By the hypothesis, each of these vanishes as two of its rows are identical. All we need is differentiability of the nine functions. That they are polynomials is not needed.) (e) 0. (Three applications of the last part of Exercise (15.10) give $f'''(x) = \begin{vmatrix} 6 & -\cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$. At $x = 0$, the first two rows are equal. The parameter p has no role in the problem.)

12. (a) Let $y_n = \sec^n \theta - \cos^n \theta$. Since $x^2 + 4 = (\sec\theta + \cos\theta)^2$ and $y_n^2 + 4 = (\sec^n \theta + \cos^n \theta)^2$, the assertion is equivalent to proving that $y_n' = \frac{n(\sec^n \theta + \cos^n \theta)}{\sec\theta + \cos\theta}$, where $'$ indicates differentiation w.r.t. x. The cases $n = 1, 2$ are easily verified by direct computation. For higher n, first prove the recurrence relation $y_n = (\sec\theta + \cos\theta)y_{n-1} - y_{n-2} = \sqrt{x^2+4}\, y_{n-1} - y_{n-2}$ and apply induction on n. Differentiation w.r.t. x gives $y_n' = \frac{x}{\sqrt{x^2+4}} y_{n-1} + \sqrt{x^2+4}\, y_{n-1}' - y_{n-2}'$. Substituting in terms of θ gives the result. (b) Apply induction on n. The case $n = 1$ is verified by finding y' and y'' and substituting. The inductive step follows immediately by differentiation. (This is one of those rare problems where the inductive step takes a lot less work than the start of the induction.) (c) First take logs and multiply by x to get $y = x\ln x - x\ln(a + bx)$. Differentiation and simplification gives $xy' - y = \frac{ax}{a+bx}$. Another differentiation gives $xy'' = \frac{a^2}{(a+bx)^2}$. Multiplying by x^2 gives the result. Implicit differentiation would be very cumbersome here. (d) The result follows by directly substituting $y' = e^x(\tan^{-1} x + \frac{1}{1+x^2})$ and $y'' = e^x(\tan^{-1} x + \frac{2}{1+x^2} - \frac{2x}{(1+x^2)^2})$ and the fact that e^x never vanishes. (e) Apply induction on n. For notational brevity, denote $\dfrac{d^k u}{dx^k}$ by $u^{(k)}$ etc. Assume $(uv)^{(n)} = \sum_{k=0}^{k=n} \binom{n}{k} u^{(k)} v^{(n-k)}$ holds. Differentiating, $(uv)^{((n+1))} = \sum_{k=0}^{k=n} \binom{n}{k} u^{(k+1)} v^{(n-k)} + \sum_{k=0}^{k=n} \binom{n}{k} u^{(k)} v^{(n+1-k)}$. In the first summation, write $v^{(n-k)}$ as $v^{((n+1)-(k+1))}$ and call $k + 1$ as k. Then the first sum equals $\sum_{k=1}^{k=n+1} \binom{n}{k-1} u^{(k)} v^{(n+1-k)}$. Adding terms of this sum to the corresponding terms of the second sum and using the identity $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ gives $(uv)^{((n+1))} = \sum_{k=0}^{n+1} \binom{n+1}{k} u^{(k)} v^{((n+1-k))}$ as

desired. (f) Apply induction on n. For the inductive step, differentiate $\frac{(-1)^n n!}{x^{n+1}}(\ln x - H_n)$ to get $\frac{(-1)^n n!}{x^{n+2}} + \frac{(-1)^{n+1}(n+1)!}{x^{n+2}}(\ln x - H_n)$. Rewrite the first term as $-\frac{(-1)^{n+1}(n+1)!}{x^{n+2}(n+1)}$ and use $H_{n+1} = H_n + \frac{1}{n+1}$. (g) Using $y_2' = y_1's + y_1s'$ and $y_3' = y_1't + y_1t'$, split the second row of the determinant on the L.H.S. into two parts as $(y_1', y_1's, y_1't) + (0, y_1s', y_2t')$. This splits the determinant into two determinants of which the first one vanishes since the second row is a multiple of the first. Thus the L.H.S. equals the determinant $\begin{vmatrix} y_1 & y_1s & y_1t \\ 0 & y_1s' & y_1t' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$. Now using $y_2'' = y_1''s + 2y_1's' + y_1s''$ and $y_3'' = y_1''t + 2y_1't' + y_1t''$, split the third row into three parts as $(y_1'', y_1''s, y_1''t) + (0, 2y_1's', 2y_1't') + (0, y_1s'', y_1t'')$. Then the determinant splits into three determinants. The first two of these vanish while the third one has value $y_1^3(s't'' - s''t')$.

13. (a) For the inductive step, write $F_n = f_1f_2 \dots f_n$. Then $F_n = F_{n-1}f_n$ for $n > 1$. So $F_n' = F_{n-1}'f_n + F_{n-1}f_n'$ and hence $\frac{F_n'}{F_n} = \frac{F_{n-1}'}{F_{n-1}} + \frac{f_n'}{f_n}$. Apply the induction hypothesis to $\frac{F_{n-1}'}{F_{n-1}}$. (b) 24. (By (a), $21 = 4 - 7 + k$.) (c) Go on adding the terms from the right to left to get $y = \frac{x}{x-a}\frac{x}{x-b}\frac{x}{x-c}$. Apply (a). (d) Let $f(x) = \sin(x^2+1)$. Then $f(x+h) - f(x) = 2\cos(\frac{(x+h)^2+x^2+2}{2})\sin(\frac{2xh+h^2}{2})$. As $h \to 0$, the first factor tends to $\cos(x^2+1)$ by continuity of the cosine function. Rewriting the second factor as $\sin(xh)\cos(\frac{h^2}{2}) + \cos(xh)\sin(\frac{h^2}{2})$, we get $\dfrac{\sin(\frac{2xh+h^2}{2})}{h} = x\frac{\sin(xh)}{xh}\cos(\frac{h^2}{2}) + \cos(xh)\frac{\sin(\frac{h^2}{2})}{\frac{h^2}{2}}\frac{h}{2}$ which tends to $x \times 1 \times 1 + 1 \times 1 \times 0 = x$ as $h \to 0$. Hence $f'(x) = 2x\cos(x^2+1)$.

14. (a) $\frac{1}{2\sqrt{6}}$. (The limit equals $G'(1)$.) (b) 5. (Add and subtract $f(a)g(a)$ from the numerator to see that the limit equals $f(a)g'(a) - f'(a)g(a)$.) (c) $a - b$. (In disguise, the question asks to find $\lim\limits_{x\to 0}\frac{\ln(1+ax)-\ln(1+bx)}{x}$ which is precisely the derivative of $\ln(1+ax) - \ln(1+bx)$ at 0.) (d) $2\ln 2$. (Divide and multiply the ratio by x. Then the given limit is simply $\frac{f'(0)}{g'(0)}$, where $f(x) = 2^x$ and $g(x) = \sqrt{1+x}$.) (e) 4. (Put $y = \sqrt{3}$. Then the given limit is the derivative (w.r.t. y) of the function $\sqrt{f(y^2)}$ at $y = 3$. By the chain rule, this is the value of $\frac{1}{2\sqrt{f(y^2)}}f'(y^2)2y$ at $y = 3$.) (f) $\frac{2}{3\sqrt{3}}$. (Rationalising both the numerator and the denominator, the ratio equals $\dfrac{a-x}{\sqrt{a+2x}+\sqrt{3x}}\dfrac{\sqrt{3a+x}+2\sqrt{x}}{3a-3x}$. No derivative is needed.)

15. (a) $\frac{n(n+1)}{2}$. (Differentiate the numerator and the denominator twice.) (b) -1. (One application of the l'Hôpital's rule reduces the limit to $\lim\limits_{h\to 0}\frac{-1}{(1+2h)(1+h)}$.) (c) $\frac{1}{2}$. (l'Hôpital's rule is not advisable here since

it will have to be applied four times! Instead, simplify the given ratio trigonometrically as $\dfrac{x}{\sin x}\dfrac{\tan 2x - 2\tan x}{4\sin^3 x}$. The first factor tends to 1. Write the second factor as $\dfrac{\sin 2x\cos x - 2\sin x\cos 2x}{4\sin^3 x\cos x\cos 2x}$ which further reduces to simply $\frac{1}{2\cos x\cos 2x}$. (d) e^2. (Put $y = x^2$ and take logs. The limit $\lim\limits_{y\to 0^+}\frac{\ln(1+5y)-\ln(1+3y)}{y}$ equals 2 by (c) of the last exercise.) (e) e^2. (Take logs. $\lim\limits_{x\to 0}\frac{\ln f(1+x)-\ln 3}{x}$ equals $\frac{f'(1)}{f(1)}$ by l'Hôpital's rule.) (f) b. (Put $y = \frac{b}{a^x}$. Then the given limit equals $\lim\limits_{y\to 0^+}\frac{b\sin y}{y}$.) (g) $\frac{2}{3}$. (Write the function as $\frac{\tan x+x}{\tan x}\times\frac{\tan x-x}{x^2\tan x}$. The first factor tends to 2. Apply l'Hôpital's rule to the second and divide both the numerator and the denominator by $\tan^2 x$.) (h) -1. (As $x\to-\infty$, $x\sin\frac{1}{x}$ tends to 1. Hence the first term of the numerator behaves like x^3 and in comparison to it the second term is insignificant. The denominator equals $1 - x^3$ for $x < 0$ and the significant term is $-x^3$. For a formal proof, divide both the numerator and the denominator by x^3.) (i) The limit does not exist. (For $0 < x < 1$, $f(x) = 0$ while $f(x) = \sin 1$ for $-1 \le x < 0$. So the right-hand limit is 0 while the left-hand limit is $\sin 1$.) (j) 1. (For $0 < |x| < \pi$, $f(x) = \sin x$ and hence $g(f(x)) = \sin^2 x + 1$.)

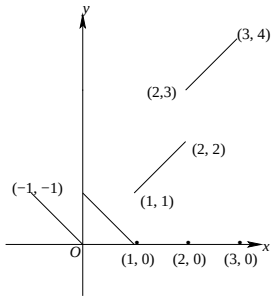
16. (a) 7. (There is an implied hint that $x - 2$ is a factor of the numerator. Taking it out and factoring further, $f(x) = x + 5$ for $x \neq 2$.) (b) $a = -1, b = 1$. (Rewrite the expression as $\dfrac{ax\cos x + b\sin x}{x^2\sin x}$. By l'Hôpital's rule the given limit equals $\lim\limits_{x\to 0}\dfrac{a\cos x - ax\sin x + b\cos x}{2x\sin x + x^2\cos x}$. For x near 0, the denominator behaves like $3x^2$. In the numerator the middle term behaves like $-ax^2$. But $(a + b)\cos x$ behaves like $a + b$. So, for the limit to exist, $a + b$ must vanish. In that case the limit is $-\frac{a}{3}$, giving $a = -1$.) (c) $f(x) = (-\frac{1}{9} + \frac{2}{3}\ln\frac{2}{3})x$ for $x \le 0$ and $f(x)$ as given for $x > 0$. (Let $g(x) = ax + b$. We need two equations to find a and b. Continuity of $f(x)$ at 0 gives $b = \lim\limits_{x\to 0^+}\left[\frac{(1+x)}{(2+x)}\right]^{1/x}$. The logarithm of the expression is $\frac{\ln(1+x)-\ln(2+x)}{x}$. Here the numerator tends to $-\ln 2$ which is a negative number and so the the ratio tends to $-\infty$ as $x\to 0^+$. Hence the expression $\left[\frac{(1+x)}{(2+x)}\right]^{1/x}$ tends to 0, giving $b = 0$. (l'Hôpital's rule is not applicable here because the ratio is not of an indeterminate form as x approaches 0.) To find a, note that $f(-1) = -a$. $f'(1)$ can be found by first taking log. In a neighbourhood of 1, $\ln f(x) = \frac{\ln(1+x)-\ln(2+x)}{x}$ and so $\dfrac{f'(x)}{f(x)} = \dfrac{x(\frac{1}{1+x} - \frac{1}{2+x}) - \ln(1+x) + \ln(2+x)}{x^2}$. Since $f(1) = \frac{2}{3}$, we get $f'(1) = \frac{2}{3}(\frac{1}{2} - \frac{1}{3} - \ln\frac{2}{3}) = \frac{1}{9} + \frac{2}{3}\ln\frac{3}{2}$. Hence $a = -\frac{1}{9} + \frac{2}{3}\ln\frac{2}{3}$. (d) $a = \frac{\pi}{6}, b = -\frac{\pi}{12}$. (Continuity at $\frac{\pi}{4}$ gives $\frac{\pi}{4} + a = \frac{\pi}{2} + b$ while that at $\frac{\pi}{2}$ gives $b = -a - b$.) (e) $a = 8$. (By l'Hôpital's rule, the

left-handed limit at 0 is 8. By rationalisation, the right-hand limit is also 8.) (f) $a = \frac{2}{3}, b = e^{2/3}$. (Taking logs and applying l'Hôpital's rule, the left and right-handed limits are e^a and $e^{2/3}$ respectively.) (g) $f(x) = \begin{cases} x+a+1 & \text{if } x < -a \\ (x+a-1)^2+b & \text{if } -a \le x < 0 \\ x^2+b & \text{if } 0 \le x < 1 \\ (x-2)^2+b & \text{if } x \ge 1 \end{cases}$. For continuity, $a = 1$ and $b = 0$. For these values, $g \circ f$ is differentiable at 0. (Since g changes its definition at 0, to find $g(f(x))$ first determine where $f(x) < 0$ and where $f(x) \ge 0$. The former occurs for $x < -a$. But since $-a \le 0$, the values $-a \le x < 0$ have to be handled differently from those where $0 \le x$. Another break up of definition is necessary at 1 since $|x-1|$ is involved. Continuity of $g \circ f$ at $-a$ gives $1 = 1 + b$ i.e., $b = 0$ while its continuity at 0 gives $(a-1)^2 + b = b$ giving $a = 1$. For these values, $(g \circ f)(x) = x^2$ for all $x \in (-1, 1)$, whence it is differentiable at 0.

17. $\delta = \frac{0.01}{169}$ will work. $(|x^3 - 7^3| = |x-7|\ |x^2 + 7x + 49|$. For $0 \le x \le 8, |x^2 + 7x + 49| \le 169$. Although it is not asked, the best δ that will work is the smaller of the two numbers $(343.01)^{1/3} - 7$ and $7 - (342.99)^{1/3}$. The former is the smaller. See Exercise (16.17) (d).) **18.** No. A simple counter-example is to take $f(x)$ to be identically 0 and $g(x)$ to be any function for which $\lim_{x \to a} g(x)$ does not exist. As a less trivial example, let $a = 0$ and $f(x) = g(x) = 1$ for $x > 0$ and -1 for $x < 0$. **19.** First suppose c is rational. Take $\epsilon = \frac{1}{2}$. Then $f(c) - \epsilon = \frac{1}{2}$ and $f(c) + \epsilon = \frac{3}{2}$. If f were continuous at c, then there would be some $\delta > 0$ (corresponding to this ϵ) such that $f(x)$ would lie in the interval $(\frac{1}{2}, \frac{3}{2})$ for all $x \in (c-\delta, c+\delta)$. But no matter how small δ is, as long as it is positive, this interval will contain at least one (in fact infinitely many) irrational number, say x_0. (The property needed here is the denseness of the set of irrational numbers, see Exercise (6.46).) But then $f(x_0) = 0 \notin (\frac{1}{2}, \frac{3}{2})$, a contradiction. So f is discontinuous at c. A similar argument works if c is irrational except that we now use the denseness of rationals. **20.** (i) Otherwise $g(x) = (f(x)+g(x)) - f(x)$ would be continuous. (ii) If $g(c) \neq 0$, then by continuity, $g(x) \neq 0$ for all x in a neighbourhood of c. If $f(x)g(x)$ were continuous at c, so would be its ratio with $g(x)$, a contradiction. For the examples, take $f(x) = [x], g(x) = x$ and $c = 0$. Here the product $f(x)g(x)$ is continuous at c. But if we let $f(x) = \frac{1}{x}$ for $x \neq 0$ with $f(0) = 0$ and $g(x) = x$, the product $f(x)g(x)$ is discontinuous at 0. (iii) Let $f(x)$ and $g(x)$ both equal the function in (15.19). **21.** (a) (i) undefined for $x < 0$ and hence discontinuous at 0; however, right differentiable at 0. (ii) differentiable. (The function is a constant in a neighbourhood of 0.) (iii) continuous but not differentiable at 0. (The right-handed derivative is 0 while the left-handed derivative is 1 since $f(x) = xe^{-2/x}$ for $x > 0$ and $f(x) = x$ for $x < 0$. (b) $\frac{3}{2}$. (Apply l'Hôpital's rule twice to find $\lim_{x \to 0} \dfrac{e^{2x} \cos x - 2 \sin x - 1}{x^2}$.) (c) 0 and 1 respectively. (Note that

$e^{\frac{1}{x}} \to \infty$ as $x \to 0^+$ but it tends to 0 as $x \to 0^-$.) (d) $g(x) = f(f(x) = \begin{cases} x+2 & 0 \le x \le 1 \\ 2-x & 1 < x \le 2 \\ 4-x & 2 < x \le 3 \end{cases}$. Discontinuous at 1 and 2. (Proceed as in Exercise (15.16)(g) above by first determining where $0 \le f(x) \le 2$ and where $2 < f(x) \le 3$.) (e) 1. ($g(x) \equiv x - 4$ in a for all x near 20, in fact for all $x \ge 4$.) (f) $-\frac{2}{9}$. ($f(x) = \frac{1}{2x-5}$ for *all* $x \ne \frac{5}{2}$.) **22.** (a) Differentiable everywhere. (Note that $f'_-(0) = 1 = f'_+(0)$.) (b) Continuous everywhere, differentiable at all $x \ne n\pi$ for n an integer. (c) Discontinuous at all non-zero values of x for which $x \sin \pi x$ is an integer. Differentiable at all other points (including 0). (The function is an even function and equals 0 for $-1 < x < 1$. It is not easy to identify the points where $x \sin \pi x$ is a non-zero integer. But studying the increasing/decreasing behaviour of the function $x \sin \pi x$, it is not hard to show that for every integer $n \ge 0$, the interval $(n, n+1)$ contains $2n$ such points.) (d) Discontinuous at all integers and for $-1 \le x < 0$ (where the function is undefined); differentiable everywhere else. (The function is constant on all intervals of the form $(n, n+1)$, where n is an integer other than -1.) (e) Continuous everywhere, differentiable everywhere except at 0 and 1. ($g(x)$ comes out to be $-x$ for $-2 \le x \le 0$, 0 for $0 \le x \le 1$ and $2x - 2$ for $1 \le x \le 2$.) (f) Continuous everywhere, differentiable everywhere except at 3. (For behaviour at 1, note that $f(x) = 3 - x$ for $1 < x \le 3$.) (g) Differentiable everywhere except at $x = 2$, discontinuous at $x = 2$. (h) Discontinuous at 0 and all points of the form $\pm\sqrt{n}$, where n is a positive integer. Differentiable everywhere else. (i) Discontinuous at ± 1, differentiable everywhere else. (j) Continuous everywhere, differentiable everywhere except at $x = 2$. (Because of the factor $(x-1)^2$, the first term is differentible at 1. The second term equals $\cos x$. (k) Continuous at all points of $[0, 2]$. Differentiable at all points except 1. (As f is strictly increasing on $[0, 1]$, $g(x) = x^3 - x^2 + x + 1$ for all $0 \le x \le 1$.)

23. (a) The function is differentiable everywhere in $[-1, 3]$ except at $0, 1, 2$ and 3. (At the end point -1, the function is differentiable, because only right-sided differentiability is needed. But at 3, it is not even left continuous and hence not differentiable either.) (b) All points except 0. (The function equals x^2 for $x \ge 0$ and $-x^2$ for $x < 0$. So $f'_+(0) = 2$ while $f'_-(0) = -2$.) (c) $[-\frac{2}{a}, \frac{a}{3}]$. (A direct calculation gives $f''(x) = ae^{ax}(ax + 2)$ for $x \le 0$ and $f''(x) = 2a - 6x$ for $x \ge 0$.)



24. f, f' are continuous at all points. f'' is continuous at all points except at $x = 1$ as the left and right-handed derivatives are 1 and 4 respectively.

25. (a) (C) and (D) are true. ($\tan[f(x)]$ would be discontinuous when $\frac{x}{2} - 1$

is an odd multiple of $\frac{\pi}{2}$, i.e., when $x-2$ is an odd multiple of π. This cannot happen for $0 \le x \le \pi$. $\frac{1}{f(x)}$ is undefined and hence discontinuous at $x = 2$. $f^{-1}(x) = 2x+2$ is continuous. It is assumed here that $\tan[f(x)]$ simply means $\tan(f(x))$. If it is taken to mean the tangent of the integral part of $f(x)$, then it will not be continuous at points where $f(x)$ is an integer, i.e., when x is an even integer.) (b) (i) $f^{-1}(x) = \frac{x+5}{3}$. (ii) $f^{-1}(x) = \frac{1}{2}(1+4\log_2 x)$ for $x \ge 1$. (Note, however, that f is not one-to-one on $\mathbb{R}$. In fact, $f(x) = f(1-x)$ for all $x \in \mathbb{R}$.) (iii) $f^{-1} = f$. **26.** All are true. (For (A), note that $6x + 12 > 0$ even for $-1 \le x \le 0$. For (B), both the right and left limits at 2 equal 35. For (C), the left-handed derivative at 2 is 24 while the right-handed derivative is -1. For (D), the function is rising till 2 and then falling.) **27.** The presumption that $\lim_{n\to\infty} a_n$ exists is wrong. a_n does tend to 1 as n tends to ∞ through even values of n and to -1 as n tends to ∞ through odd values. Call these numbers as L_1 and L_2 respectively. Now, when n is even, $n+1$ is odd and vice versa. So, the correct argument would be $\lim_{n\to\infty} a_n + a_{n+1} = L_1 + L_2 = 0$ and similarly $L_1 L_2 = -1$. Both these are true statements. But neither L_1 nor L_2 equals L. In fact, L does not exist.

Chapter 16

1. Assuming that the z-axis points vertically upwards, let the solid lie between the planes $z = a$ and $z = b$, where $a < b$. For $c \in [a, b]$, let $f(c)$ denote the volume of the portion of the solid lying between the planes $z = a$ and $z = c$. It is intuitively clear that f is a continuous function on $[a, b]$. (For a rigorous proof, we need that the solid is bounded. Specifically, $f(c+h) - f(c)$ is the volume of the portion of the solid between the two parallel planes $z = c$ and $z = c + h$. This is at most h times the maximum area of cross-section of the solid by a horizontal plane. The latter is finite as the solid is bounded. So $f(c + h) - f(c)$ tends to 0 as $h \to 0$. A similar reasoning shows that f is differentiable at c and moreover, that $f'(c)$ is the area of cross section of the solid by the plane $z = c$.) Also $f(a) = 0$ and $f(b) = V$, the volume of the solid. By IVP, $f(c) = \frac{V}{2}$ for some $c_1 \in (a, b)$. The plane $z = c_1$ is the desired plane.

2. Let the equation of L be $a_0x + b_0y = c_0$. For any $c \in \mathbb{R}$, let L_c be the line $a_0x + b_0y = c$ and let $f(c)$ be the length of the chord of the ellipse cut off by L_c. (Take $f(c) = 0$ if L_c does not intersect the ellipse.) As the ellipse is bounded, there exist some $p > 0$ such that $f(c) = 0$ for $c < -p$ and for $c > p$. Note further that $f(0) \ge 2b$. Use continuity of f and apply IVP to $[-p, 0]$ and to $[0, p]$ to get a desired line (in fact, two of them).

3. Follow the hint. Note that since $f(a) \ge a$ we have $g(a) \le 0$. Similarly $g(b) \ge 0$. Neither $\mathbb{R}$ nor $(0, 1)$ has the fixed point property as is shown, respectively, by the functions $f(x) = x + 1$ and $f(x) = \frac{x}{2}$.

4. Define $g(x) = \begin{cases} f(x+\frac{1}{2}) - f(x) & \text{if } 0 \le x \le \frac{1}{2} \\ f(x-\frac{1}{2}) - f(x) & \text{if } \frac{1}{2} < x \le 1 \end{cases}$. Since $f(1) = f(0)$, g is continuous at $\frac{1}{2}$ and moreover $g(0)$ and $g(1)$ are of opposite signs. Apply IVP to g.

5. Parametrise C as $(a\cos\theta, b\sin\theta), 0 \le \theta \le 2\pi$. Let the given triangle be ABD with $AB \le BD \le DA$. For a point $X = (a\cos\theta, b\sin\theta)$ on C, other than P, there are two points, say $Y_1 = (x_1, y_1)$ and $Y_2 = (x_2, y_2)$ in the plane of C such that the triangles PXY_1 and PXY_2 are similar to ABD. These two points lie on opposite sides of the line PX. Moreover, they vary continuously with X and hence with θ. So if we let $f(\theta) = (\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1)(\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} - 1)$, then f is a continuous function of θ. If X is close to P, then the line PX makes a very small angle with the tangent at P. (Draw a figure to see this.) Therefore one of Y_1 and Y_2 is inside C and the other is outside C and so $f(\theta) < 0$. But if X is the point on C which is farthest to P then neither of Y_1, Y_2 is inside C because $PY_1 = PY_2 \ge PX$. (Here we need that $AB \le AD$.) Therefore $f(\theta) \ge 0$. By Theorem (1.3), $f(\alpha) = 0$ for some $\theta = \alpha$. This means that out of two the corresponding points Y_1 and Y_2, at least one, say Y_1, is on C. If $Q = (a\cos\alpha, b\sin\alpha)$, and $R = Y_1$, then PQR is a desired triangle.

6. From its derivative, f is strictly increasing and hence one-to-one. Also $f(x) \to \pm\infty$ as $x \to \pm\infty$. Hence by IVP, f is onto. (Given any $y_0 \in \mathbb{R}$, one can find a, b such that $f(a) < y_0 < f(b)$ and apply IVP to the interval $[a, b]$. In the present problem, a, b can be given explicitly in terms of y_0. But that is not vital.)

7. f' is either 1 or -1. There is no point at which it is 0. Here f fails to be differentiable at one point, viz., 0, and so the theorem cannot be applied. (Incidentally, this shows that in mathematics, the conclusion can be thwarted even if the hypothesis fails even at just one point, in sharp contrast with real life where a layman would simply tend to ignore such a 'bad' point since it is in vast minority!)

8. Here $\frac{f(1)-f(0)}{1-0} = 1$. But the derivative, wherever it exists is 0. The function f is not continuous at one point of $[0, 1]$.

9. By Rolle's theorem, for every positive integer n, the interval $(\frac{1}{(n+1)\pi}, \frac{1}{n\pi})$ contains at least one zero of f'.

10. Divide the interval $[a, b]$ into n equal subintervals by points $x_i = a + \frac{i(b-a)}{n}$ for $i = 0, 1, 2, \ldots, n$. For $i = 1, 2, \ldots, n$, apply the MVT to get $c_i \in (x_{i-1}, x_i)$ such that $f'(c_i)\frac{b-a}{n} = f(x_1) - f(x_{i-1})$. Add these n equations.

11. Follow the hint. Note that the point c_x depends on x but is not unique for a given x. So c_x may not vary continuously with x. Still, $f'(c_x) \to L$ as $x \to a^+$, because given any $\epsilon > 0$, first find a $\delta > 0$ (which exists since

it is given that $\lim_{x\to a^+} f'(x) = L$) such that $|f'(x) - L| < \epsilon$ whenever $a < x < a + \delta$. But then, whenever $a < x < a + \delta$, we also have $a < c_x < a + \delta$ and so $f'(c_x) - L| < \epsilon$.

12. Suppose f is differentiable in a deleted neighbourhood of a point a and $\lim_{x\to a} f'(x)$ exists and equals L (say). Then by the last exercise (and its 'left-handed' analogue), $f'_+(a)$ and $f'_-(a)$ also equal L each. So $f'(a)$ exists and equals L. In other words, f' is continuous at a.

13. Let $f(x) = x^4 + 3x + 1$. Then $f(-2) = 11 > 0$ and $f(-1) = -1 < 0$. So by the IVP, f has at least one zero in $(-2, -1)$. If it had more than one zeros, then Rolle's theorem would give that f' vanishes at least once in $(-2, -1)$. But this is impossible since $f'(x) = 4x^3 + 3$ vanishes only at $x = -\sqrt[3]{\frac{3}{4}}$ which does not lie in $(-2, -1)$.

14. (B). (For $x > 0$, the MVT gives $\tan^{-1} x = \frac{x}{1+c^2}$ for some c. But $\frac{1}{1+c^2} \leq 1$.) **15.** $h(10) = 11$. ($h' \equiv 0$ and hence h is a constant.) **16.** By Rolle's theorem, there is some $c \in (0, 1)$ such that $f'(c) = 0$. Now for any $x \in [0, 1]$, the MVT applied to f' gives that $|f'(x) - f'(c)| = |f''(c_1)|\ |x - c|$ for some c_1 (depending on x). The result follows since $|f''(c_1)| \leq 1$ and $|x - c| < 1$. **17.** (a) Simply replace f by g defined by $g(x) = f(x) - f(a)$. Note that $f' = g'$. (b) A direct computation gives $\psi(a) = \psi(b)$. Apply Part (a) to ψ. (c) No. In fact to think so is one of the most common mistakes. Application of Lagrange's MVT will give points $c_1, c_2 \in (a, b)$ for which $\dfrac{f'(c_1)}{g'(c_2)} = \dfrac{f(b) - f(a)}{g(b) - g(a)}$. But there is no guarantee that $c_1 = c_2$. (As an actual example, try $f(x) = x^2, g(x) = x^3$ on $[0, 1]$.) (d) Apply MVT to $f(x) = x^{1/3}$ on the intervals $[343, 343.01]$ and $[342.99, 343]$ which are of equal lengths and use the hint. **18.** Let n be the number of distinct roots of $p(x)$ in (a, b) having odd multiplicities. Note that if α is a root of $p(x)$ with multiplicity m, then $f(x)$ reverses or retains its sign as x passes through α according as m is odd or even. So $p(a)p(b)$ will be positive if n is even and negative if n is odd. The result follows by observing that n has the same parity as the total number of roots of $p(x)$ in (a, b) (counted according to their multiplicities). **19.** Let $x_1 < x_2 < \ldots < x_n$ be the distinct roots of $p(x)$ in (a, b) with multiplicities $m_1, m_2, \ldots, m_n$ respectively. Then $k = m_1 + m_2 + \ldots + m_n$. Apply the hint to those x_i's for which $m_i > 1$. Then it follows that $p'(x)$ has $k - n$ roots among these x_i's. In addition, between every two successive x_j's, there is a root of $p'(x)$ by Rolle's theorem. So in all, $p'(x)$ has at least $(k - n) + (n - 1)$ roots in (a, b).

20. (a) Denote the roots of $p(x)$ by $r_1, r_2, \ldots, r_n$. If all r_i's are real then by repeated applications of the last exercise, $p^{(n-2)}(x)$ has two real roots. But $p^{(n-2)}(x)$ is the real quadratic $Ax^2 + Bx + c$, where $A = n(n - 1) \ldots 3 = \frac{n!}{2}$, $B = (n-1)!a_1$ and $C = (n-2)!a_2$. The discriminant criterion

gives a contradiction. For an algebraic proof, express a_1, a_2 in terms of $r_1, r_2, \ldots, r_n$. Then the given condition reduces to $(n-1)\sum_{i=1}^{n} r_i^2 < 2\sum_{i<j} r_i r_j$ and hence to $\sum_{i<j}(r_i - r_j)^2 < 0$ which is a contradiction if all r_i's are real. (b) $p'_{2n-1}(x) = p_{2n-2}(x) > 0$ shows that $p_{2n-1}(x)$ is strictly increasing. Let $\alpha = \alpha_n$ be its unique real root. Since $p'_{2n}(x) = p_{2n-1}(x)$, p_{2n} has its minimum at α. But $p_{2n}(\alpha) = p_{2n-1}(\alpha) + \frac{\alpha^{2n}}{(2n)!}$. So $p_{2n}(x) \geq \frac{\alpha^{2n}}{(2n)!}$ for all real x. **21.** Call the determinant as $g(x)$. $f(x)$ is of the form $k(x-\alpha)^2$ for some constant k. Divisibility of $g(x)$ by $f(x)$ is equivalent to saying that α is a root of $g(x)$ having multiplicity at least two, which, in turn, amounts to saying that α is a root of $g(x)$ and also of $g'(x)$. These follow respectively, by putting $x = \alpha$ in the given determinant and then in its derivative which is obtained by simply differentiating the first row, by Exercise (15.10). (The hypothesis about the degrees of $A(x), B(x), C(x)$ is not needed.)

22. $(1.4)^2 = 1.96 < 2$. Taking $f(x)$ as $\sqrt{x}$ and applying the MVT to the interval $[1.96, 2]$, we get $2 - 1.4 = (0.04) \times \frac{1}{2\sqrt{c}}$ for some $c \in (1.96, 2)$. Since $\frac{1}{\sqrt{c}} < \frac{1}{\sqrt{1.96}}$, we get that the error is less than $(0.04) \times \frac{1}{2.8} \approx 0.01429$.

23. If $f'(a)$ exists, take $L(x) = f(a) + f'(a)(x-a) = f'(a)x + (f(a) - af'(a))$. Then $\frac{|f(x)-L(x)|}{|x-a|} = \left|\frac{f(x)-L(x)}{x-a}\right| = \left|\frac{f(x)-f(a)}{x-a} - f'(a)\right| \to 0$ from the very definition of $f'(a)$. For the converse, suppose $\frac{|f(x)-Ax-B|}{|x-a|}$ and hence $\frac{f(x)-Ax-B}{x-a}$ tends to 0 as $x \to a$. Then, in particular, the numerator tends to 0 and so by continuity of f at a, we have $f(a) = Aa + B$. Rewriting $\frac{f(x)-Ax-B}{x-a}$ as $\frac{f(x)-Ax+Aa-f(a)}{x-a} = \frac{f(x)-f(a)}{x-a} - A$, we get that f is differentiable at a with derivative A. If f is not given to be continuous at a, the direct implication remains intact. In the converse, we still get that f has a removable discontinuity at a and when it is removed (by redefining $f(a)$ as $Aa + B$), the function becomes differentiable at a. **24.** Let $Q(x) = Ax^2 + Bx + C$. Solve the system : (i) $Aa^2 + Ba + C = \alpha$ (ii) $2Aa + B = \beta$ and (iii) $2A = \gamma$ for A, B, C.

25. (a) Apply Rolle's theorem to $g, g', g'', \ldots$ in succession to get $c_1 \in (a, b)$ such that $g'(c_1) = 0$, then $c_2 \in (a, c_1)$ such that $g''(c_2) = 0$, then $c_3 \in (a, c_2)$ such that $g'''(c_3) = 0$ and so on. The desired c is c_n. (b) Without loss of generality, let $c \geq 0$. Then g' has some zero in $(c, 1)$. If 0 is the only zero of g' in $(-1, c)$, then there is a local extremum at 0 and so $g'''(0) = 0$ as we are given that $g''(0) = 0$. Otherwise g' has at least three distinct zeros in $(-1, 1)$. Apply Rolle's theorem to g' and then to g''. **26.** Note that if r and k are positive integers then the r-th derivative of $(x-a)^k$ vanishes at a except when $r = k$, in which case its value at a (and, in fact, everywhere) is $k!$. With this, it is easy to verify that the function g in the hint satisfies the conditions in Exercise (16.25).

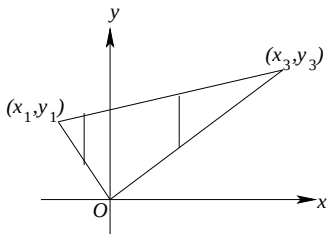
27. Taking $f(x) = 3\sin x + \tan x$, we have $f(x) \approx 4x - \frac{1}{6}x^3$ with an error term of order at most x^4. So for small positive values of x, $f(x) < 4x$. Put differently, $\frac{3}{4}\sin x + \frac{1}{4}\tan x < x$. The former is the point which divides the interval $[\sin x, \tan x]$ in the ratio $1:3$. However, by (23), $\frac{2}{3}\sin x + \frac{1}{3}\tan x > 3x$ for small $x > 0$. In fact, the given inequality holds for all $x \in (0, \frac{\pi}{2})$ and not just for all small positive values of x, because by Taylor's theorem for any $x \in (0, \frac{\pi}{2})$, we have $f(x) = 4x - \frac{1}{6}x^3 + \frac{x^4}{24}f^{(4)}(c)$ for some $c \in (0, x)$. A direct computation gives that $f^{(4)} > 0$ on $(0, \frac{\pi}{2})$. The last part follows by applying Taylor's theorem to the function $f(x) = \tan^{-1} x$, to get that for every $x > 0$, there exists some $c \in (0, x)$ such that $\tan^{-1} x = x - \frac{c}{1+c^2}$. This shows that $\tan^{-1} x < x$ for all $x > 0$ and hence that $\tan^{-1} x > x$ for all $x < 0$. Together, $|\tan^{-1} x| \leq |x|$ for all x.

28. Apply Taylor's theorem to write $f(x) - f(c) = \frac{f^{(n)}(c_1)}{n!}(x-c)^n$ for some c_1 between x and c. Since $f^{(n)}$ is continuous at c, if x is sufficiently close to c, then $f^{(n)}(c_1)$ has the same sign as $f^{(n)}(c)$. For odd n, $(x-c)^n$ is positive for $x > c$ and negative for $x < c$. So the expression $f(x) - f(c)$ has opposite signs on the two sides of the point c. Hence there is neither a local maximum nor a local minimum at c, proving (i). But if n is even, then $(x-c)^n > 0$ for all $x \neq c$ and so $f(x) - f(c)$ has the same sign (viz., that of $f^{(n)}(c)$) on both the sides of c. This proves (ii). To apply this theorem to the function $f(x) = x^5 - 5x^4 + 5x^3$ at 0, a direct computation gives that $f'(0) = f''(0) = 0$ while $f'''(0) = 30$. So the first non-vanishing derivative is of order 3. As 3 is odd, there is neither a local maximum nor a local minimum at 0. However, near 0 the function is approximately $5x^3$ and hence is increasing at 0.

29. Instead of (1) we shall have $0 \leq \left|\frac{f(x+h)-f(x)}{h}\right| \leq |h|$ which will still imply that $f'(x) = 0$ for all x. However, the inequality $|f(x) - f(y)| \leq |x - y|$ does not imply that f is constant as is seen by the function $f(x) = \frac{1}{2}x$.

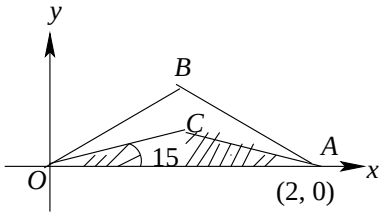
Chapter 17

1. First taking $x_2 = y_2 = 0$, write down the equations of the three sides of the triangle. Then its area equals $\int_{x_1}^{0} |y_1 + \frac{y_3 - y_1}{x_3 - x_1}(x - x_1) - \frac{y_1 x}{x_1}|dx + \int_{0}^{x_3} |y_1 + \frac{y_3 - y_1}{x_3 - x_1}(x - x_1) - \frac{y_3 x}{x_3}|dx$. The expressions inside the absolute value signs maintain their signs over the intervals of integration. Keeping in mind that $x_1 \leq 0$ and $x_3 \geq 0$, the two integrals come out to be, respectively, $\frac{-x_1|x_1y_3 - x_3y_1|}{2(x_3 - x_1)}$ and $\frac{x_3|x_1y_3 - x_3y_1|}{2(x_3 - x_1)}$. Hence the area is $\frac{1}{2}|x_1y_3 - x_3y_1|$.

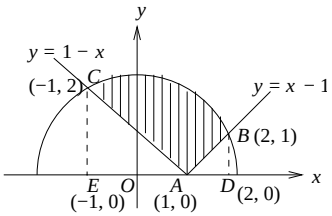


For a general triangle, we apply the formula above after transferring the origin to (x_2, y_2). Then the area is $\frac{1}{2}|(x_1-x_2)(y_3-y_2)-(x_3-x_2)(y_1-y_2)|$ which equals the desired formula as seen by subtracting the second row of the determinant from the other two.

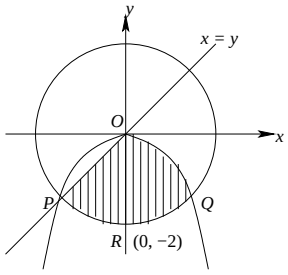
2. The triangle OAB is isosceles with $\angle BOA = \angle BAO = \tan^{-1}\frac{1}{\sqrt{3}} = 30°$. The region R is the portion of the triangle lying below the angle bisectors of angles BOA and BAO. It is a triangle with vertices at O, A and $C = (1, \tan 15°)$. Its area is $\frac{1}{2}\times 2\times \tan 15° = 2-\sqrt{3}$.



3. (i) There are two such regions. One of them has area $\sin^{-1}\alpha - \frac{\alpha^3}{3}$, where $\alpha = \frac{-1+\sqrt{5}}{2}$ and the other has area $\frac{\pi}{4} - \sin^{-1}\alpha + \frac{\alpha^3}{3}$. (ii) $\frac{125}{24}a^2$. (Integrate $-\frac{3}{4}x + a - \frac{x^2}{4a}$ from $x = -4a$ to $x = a$.) (iii) $\frac{11}{8}$. $(= \int_{-1}^{2} \frac{x+2}{4} - \frac{x^2}{4}\,dx.)$ (iv) $\frac{3}{\pi} + \frac{1}{\pi^2}$ $(= \int_{-1}^{1} x\sin\pi x\,dx + \int_{1}^{3/2} -x\sin\pi x\,dx)$. (v) $\ln 2$ $(= \int_0^{\pi/4} \tan x\,dx + \int_{\pi/4}^{\pi/2} \cot x\,dx)$. (vi) $y = \sqrt{5-x^2}$ is the upper arc of the circle $x^2+y^2=5$. It cuts the line $y = x-1$ at $B = (2,1)$ and the line $y = 1-x$ at $C = (-1,2)$. Hence the desired area is the area under the arc CB of the circle minus the areas of the triangles ABD and ACE. This comes out as $\int_{-1}^{2} \sqrt{5-x^2}\quad dx \;-\; \frac{1}{2} \;-\; 2$, i.e., $\frac{5}{2}(\sin^{-1}(\frac{2}{\sqrt{5}}) + \sin^{-1}(\frac{1}{\sqrt{5}}) + \frac{4}{5}) - \frac{5}{2} = \frac{5\pi}{4} - \frac{1}{2}$. (The area under the arc CB can also be found by adding the area of the right-angled sector OBC and those of the right-angled triangles OCE and OBD.)

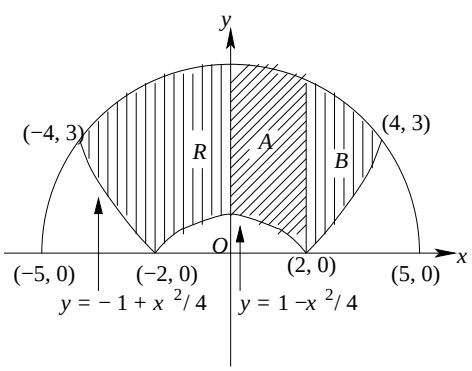


(vii) Identifying the points of intersection of the given curves, taken two at a time, the region bounded by the three curves is the shaded region in the figure. Its area splits into two (unequal) parts, PRO and ORQ. They are respectively, $\int_{-\sqrt{2}}^{0} x + \sqrt{4-x^2}dx$ and $\int_{0}^{\sqrt{2}} -\frac{1}{\sqrt{2}}x^2 + \sqrt{4-x^2}dx$ which come out to be $\frac{\pi}{2}$ and $\frac{\pi}{2}+\frac{1}{3}$. So total area is $\pi + \frac{1}{3}$.

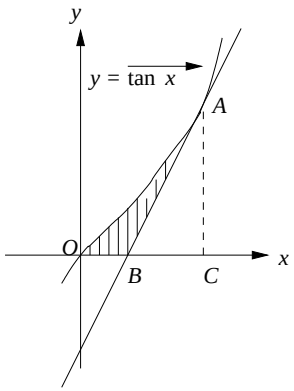


(viii) The second curve is a part of the parabola $y = 1 - \frac{x^2}{4}$ for $-2 \le x \le 2$ and a part of the parabola $y = \frac{x^2}{4} - 1$ for $|x| \ge 2$. So the desired region, say R, shown in the figure below, is bounded above by the circle $x^2+y^2=25$,

below by the portion of the parabola $y = 1 - \frac{x^2}{4}$ and on the two sides by portions of the parabola $y = \frac{x^2}{4} - 1$. The latter intersect the circle at $(\pm 4, 3)$. By symmetry about the y-axis, the area of R is $2(A + B)$, where $A = \int_0^2 \sqrt{25 - x^2} - 1 + \frac{x^2}{4}\, dx$ and $B = \int_2^4 \sqrt{25 - x^2} + 1 - \frac{x^2}{4}\, dx$. This comes out to be $50 \sin^{-1} \frac{4}{5} - 8$.

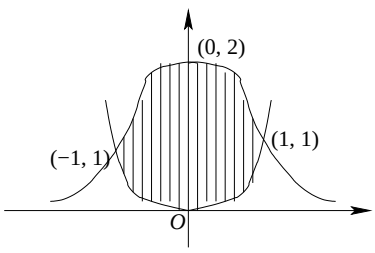


(viii)

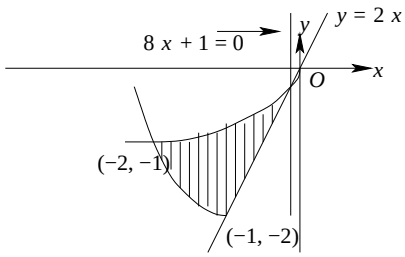


(ix)

(ix) The tangent to $y = \tan x$ at $(\frac{\pi}{4}, 1)$ (shown as A in the figure above) cuts the x-axis at $(\frac{\pi}{4} - \frac{1}{2}, 0)$ (shown as B). The desired area is obtained by subtracting the area of the triangle ABC from the area under the graph of $y = \tan x$ between $x = 0$ and $x = \frac{\pi}{4}$. It comes out as $\ln 2 - \frac{1}{4}$. (x) $\frac{10}{3}$. (First sketch the graph of the cubic $y = x(x-1)^2$. It meets $y = 2$ at $(2, 2)$. So the given region is bounded above by $y = 2$, below by $y = x(x-1)^2$, and on the two sides by $x = 0$ and $x = 2$.) (xi) $\frac{e^2-5}{4e}$. (The two curves intersect at $(1, 0)$ and $(\frac{1}{e}, -1)$. For $x \in (\frac{1}{e}, 1)$, $\ln x < 0$ and so $ex \ln x < \frac{\ln x}{ex}$. The area equals $\int_{1/e}^1 (\frac{\ln x}{ex} - ex \ln x)\, dx = \int_{-1}^0 (\frac{u}{e} - eue^{2u})\, du$, where $u = \ln x$. Integrate the second term by parts.) (xii) The two curves meet at $(\pm 1, 1)$ as shown in the figure below. For $-1 \le x \le 1$, we have $x^2 \le \frac{2}{1+x^2}$. Hence the area between them equals $\int_{-1}^1 \frac{2}{1+x^2} - x^2$. By a direct computation this is $\pi - \frac{2}{3}$.



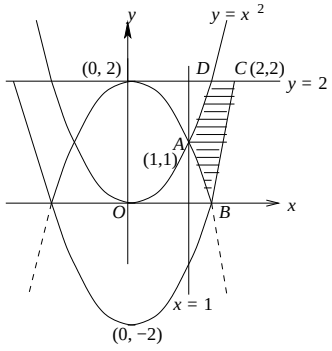
(xii)



(xiii)

(xiii) Continuity of f at $x = 1$ implies $1 + a + b = 2$ while that at $x = -1$ implies $1 - a + b = -2$. Solving, $a = 2$ and $b = -1$. Hence, in the third quadrant the graph of $y = f(x)$ consists of the portion of the parabola $y = x^2 + 2x - 1$ for $-1 - \sqrt{2} \le x \le -1$ and of the straight line $y = 2x$ for $-1 \le x \le 0$. The former meets the parabola $x = -2y^2$ in the third quadrant at the point $(-2, -1)$ while the latter meets it at $(-\frac{1}{8}, -\frac{1}{4})$ which is also a point on the line $8x + 1 = 0$. So the region is as shown in the figure above. Its area is $A + B$, where $A = \int_{-2}^{-1} -\sqrt{\frac{-x}{2}} - x^2 - 2x + 1 \, dx$ and $B = \int_{-1}^{-1/8} -\sqrt{\frac{-x}{2}} - 2x \, dx$. Together this equals $\int_{-2}^{-1/8} -\sqrt{\frac{-x}{2}} - 2x \, dx + \int_{-2}^{-1} -x^2 + 1 \, dx$ and comes out as $\frac{383}{192}$.

(xiv) $\frac{17}{27}$. ($f(x)$ equals $(1 - x)^2$ for $0 \le x \le \frac{1}{3}$; equals $2x(1 - x)$ for $\frac{1}{3} \le x \le \frac{2}{3}$ and equals x^2 for $\frac{2}{3} \le x \le 1$. So the area is the sum of the three integrals $\int_0^{1/3} (1 - x)^2 \, dx$, $\int_{1/3}^{2/3} 2x(1 - x) \, dx$ and $\int_{2/3}^1 x^2 \, dx$. The first and the last of these are equal.) (xv) The curve $y = |2 - x^2|$ consists of the portion of the parabola $y = 2 - x^2$ for $-\sqrt{2} \le x \le \sqrt{2}$ and that of the parabola $y = x^2 - 2$ for $|x| \ge \sqrt{2}$. The former meets the curve $y = x^2$ at $(\pm 1, 1)$. On the right of $x = 1$, the parabolas $y = x^2$ and $y = x^2 - 2$ meet the line $y = 2$ at the points $D = (\sqrt{2}, 2)$ and $C = (2, 2)$ respectively. Hence the desired region is the region $ABCD$ in the figure. It is more convenient to slice it horizontally. Its area equals the sum $\int_0^1 (\sqrt{2 + y} - \sqrt{2 - y}\ ) dy + \int_1^2 (\sqrt{2 + y} - \sqrt{y}\ ) dy$ which can be rewritten as $\int_0^2 \sqrt{2 + y} \, dy - \int_0^1 \sqrt{2 - y} \, dy - \int_1^2 \sqrt{y} \, dy$. It comes out to be $\frac{20}{3} - 4\sqrt{2}$.



4. Area = 4; $a = 2\sqrt{2}$. (Solve $(x - \frac{8}{x})\big|_2^a = 2$, i.e., $a - \frac{8}{a} = 0$.)

5. $m = 4$ and $m = -2$. (The line $y = mx$ cuts the curve $y = x - x^2$ at $(0, 0)$ and $(1 - m, m(1 - m))$. The area between the two is the absolute value of $\int_0^{1-m} x - x^2 - mx \, dx$. Equating this with $\frac{9}{2}$ gives $|(1 - m)^3| = 27$.)

6. (B) and (C). (The side of the triangle is $\sqrt{x^2 + (g(x))^2}$ and hence its area is $\frac{\sqrt{3}}{4}(x^2 + (g(x))^2)$. This gives $x^2 + (g(x))^2 = 1$. (A) is ruled out as a function has to be single valued.)

7. The first part is a special case of (24) in Chapter 7 and can also be proved

directly by induction. Using the first part, the integrand in the second part equals $\sum_{i=1}^{k} 2\cos(2i-1)x\cos x = \sum_{i=1}^{k}(\cos 2ix + \cos(2i-2)x)$. For any even $j > 0$, $\int_0^{\pi/2} \cos jx\ dx = \frac{1}{j}\sin\frac{j\pi}{2} = 0$. Therefore, the only non-zero term in the integral is $\int_0^{\pi/2}\cos 0x\ dx$ which equals $\frac{\pi}{2}$.

8. (a) Note that $\sin by$ maintains its sign throughout the interval $[\frac{j\pi}{b}, \frac{(j+1)\pi}{b}]$ for every integer j. Hence $S_j = |\int_{j\pi/b}^{(j+1)\pi/b} e^{-ay}\sin by\ dy|$. Using (53), this comes out as $\frac{b(1+e^{-a\pi/b})}{a^2+b^2}(e^{-a\pi/b})^j$ which proves that the S_j's form a G.P. with the initial term $\frac{b(1+e^{-a\pi/b})}{a^2+b^2}$ and common ratio $e^{-a\pi/b}$. For $a = -1$ and $b = \pi$ these quantities are, respectively, $\frac{\pi(1+e)}{1+\pi^2}$ and e. So $\sum_{j=0}^{n} S_j = \frac{\pi(1+e)}{1+\pi^2}\frac{e^{n+1}-1}{e-1}$. (b) $A_n = \frac{n}{2}r^2\sin\frac{2\pi}{n}$. Since $\frac{2\pi}{n} \to 0$ as $n \to \infty$, we have $A_n = \pi r^2\frac{\sin(2\pi/n)}{2\pi/n} \to \pi r^2$ as $n \to \infty$. Since the polygon covers more and more area of the circle as n increases, its limit is the area of the circle. For a sector with angle θ at the centre (as at the end of Comment No. 5), if we divide it into n equal sectors and approximately treat each one as a triangle, the area is approximately $\frac{n}{2}r^2\sin(\frac{\theta}{n})$. By a similar argument, this tends to $\frac{1}{2}r^2\theta$ as n tends to infinity.

9. (a) $\frac{2}{\pi}$ $(= \int_0^1 \sin\pi x\ dx.)$ (b) $\frac{\pi}{2}$ $(= \int_0^1 \frac{dx}{\sqrt{2x-x^2}} = \int_0^1 \frac{du}{\sqrt{1-u^2}}$, where $u = 1-x)$. (c) 1. (Rewrite the expression as $\frac{1}{n}\sum_{r=1}^{n}\ln(1+\frac{r}{n})$. Then the limit equals $\int_0^1 \ln(1+x)\ dx$. To evaluate it, integrate by parts.) (d) $\sqrt{5}-1$. (Rewrite the sum as $\dfrac{1}{2n}\displaystyle\sum_{r=1}^{2n}\frac{4(\frac{r}{2n})}{\sqrt{1+4(\frac{r}{2n})^2}}$. Then its limit as n tends to ∞ equals $\int_0^1 \frac{4x\ dx}{\sqrt{1+4x^2}} = \int_1^5 \frac{du}{2\sqrt{u}}$, where $u = 1+4x^2$.)

10. $10^{-6}S_{10000} = \frac{1}{10000}\sum_{r=1}^{10000}\sqrt{\frac{r}{10000}}$ which can be recognised as a Riemann sum of the function $f(x) = \sqrt{x}$ over the interval $[0,1]$ partitioned into 10000 equal parts. As 10000 is a fairly large number, $10^{-6}S_{10000} \approx \int_0^1 \sqrt{x}\ dx = \frac{2}{3}$, so that $S_{10000} \approx 6666.6666....$ To obtain an upper bound on the error, note that since $\sqrt{x}$ is an increasing function on $[0,1]$, for every $r = 1, 2, \ldots, 10000$, $\sqrt{\frac{r-1}{10000}} \le \int_{(r-1)/10000}^{r/10000}\sqrt{x}\ dx \le \sqrt{\frac{r}{10000}}$. Adding these 10000 inequalities, we have $\frac{S_{9999}}{100} \le \int_0^1\sqrt{x}\ dx \le \frac{S_{10000}}{100}$, or in other words, $10^{-6}(S_{10000}-100) \le \frac{2}{3} \le 10^{-6}S_{10000}$ which implies $\frac{2}{3}10^6 \le S_{10000} \le \frac{2}{3}10^6 + 100$. As the two ends of the inequality differ by 100, the error in estimating S_{10000} as $\frac{2}{3}10^6$ is at most 100. (With

a FORTRAN programme, S_{10000} came out to be $666,716.459$ which is almost the A.M. of the two bounds.) For the second part, note that $2\sqrt{n}-2 = \int_1^n \frac{dx}{\sqrt{x}}$. Partition the interval $[1,n]$ into $n-1$ equal parts. Since the integrand is strictly decreasing, a similar reasoning as above gives $T_n - 1 < 2\sqrt{n}-2 < T_{n-1} < T_n$ which implies the result. Finally, $198 < [T_{10000}] < 199$ whence $[T_{10000}] = 198$.

11. Let $S_n = 1^{10} + 2^{10} + \ldots + n^{10}$. Recognise $\frac{1}{n^{11}}S_n$ as a Riemann sum of $f(x) = x^{10}$ over the interval $[0,1]$. So, for $k = 11$, the expression $\frac{S_n}{n^k}$ tends to a definite limit, viz., $\int_0^1 x^{10}dx = \frac{1}{11}$. (This can also be obtained if we notice that S_n is a polynomial of degree 11 in n with leading coefficient $\frac{1}{11}$.) For $k \neq 11$, write $\frac{S_n}{n^k}$ as $\frac{S_n}{n^{11}} \times n^{11-k}$ which tends to 0 if $k > 11$ and to ∞ if $k < 11$. **12.** (i) $\int_0^2 \rho(x)dx$ (ii) $\int_0^2 \sigma(x)\rho(x)dx$ (iii) $\dfrac{\int_0^2 \sigma(x)\rho(x)dx}{\int_0^2 \rho(x)dx}$.

13. Let $a = z_0 < z_1 < z_2 < \ldots < z_{n-1} < z_n$ be a partition of $[a,b]$. This induces a partition of the solid by a family of parallel planes $z = z_i, i = 0,1,\ldots,n$ in which the portion between two adjacent planes $z = z_{i-1}$ and $z = z_i$ has volume approximately $A(z_i)(z_i - z_{i-1})$ for $i = 1,2,\ldots,n$. So the approximate volume is a Riemann sum of $A(z)$ on $[a,b]$ and the exact volume is the limit of this as the mesh of the partition tends to 0.

14. (i) $\frac{2\pi}{3}r^3$ $(= \int_0^r \pi(r^2 - z^2)\,dz)$ (ii) $\frac{1}{3}r^2h$ $(= \int_0^h \pi(\frac{r}{h}(h-z))^2\,dz)$.

15. Split the integral into $n+1$ integrals as $\sum_{j=1}^{n} \int_{(j-1)\pi}^{j\pi} |\sin x|\,dx + \int_{n\pi}^{n\pi+v} |\sin x|\,dx$. Each of the first n integrals equals 2. The last one equals $1 - \cos v$.

16. (a) Let $f(x)$ denote the given function. Then $f'(x) = 2(x-1)(x-2)^3 + 3(x-1)^2(x-2)^2 = (x-1)(x-2)^2(5x-7)$. Hence local maximum at $x = 1$ and local minimum at $x = \frac{7}{5}$. (Neither at 2 since $f'(x)$ does not change sign at 2.) (b) Local minimum at 3. (argument similar to (a). Note that at $x = 0$, there is no change of sign of f' since $x(e^x - 1)$ is positive on both the sides of 0.)

17. (a) 80. (As far as the integral is concerned, x is a constant and so the integral equals $\frac{(f(x))^2-16}{x-1} = \frac{f(x)-4}{x-1}(f(x)+4)$ which tends to $f'(1)(f(1)+4)$ as $x \to 1$.) (b) $f(x)\sin 2x + \int_0^x f(u)\cos(x+u)\,du$. $(\int_0^x f(u)\sin(x+u)\,du = \sin x \int_0^x f(u)\cos u\,du + \cos x \int_0^x f(u)\sin u\,du$. Differentiate each term using the product rule and the fundamental theorem of calculus. It is tempting to get the answer by putting $u = x$ in the integrand. This would give the derivative incorrectly as $f(x)\sin 2x$. The extra term, viz., $\int_0^x f(u)\cos(x+u)\,du$ comes from the partial derivative of the integrand w.r.t. x. The proof is beyond our scope.)

18. (i) $\dfrac{\sin(x_0^2)}{2x_0}$. (Write the expression as $\frac{1}{x+x_0}\dfrac{\int_{x_0}^{x}\sin(t^2)\,dt}{x-x_0}$. The first factor tends to $\frac{1}{2x_0}$ as $x \to x_0$. To find the limit of the second factor apply the fundamental theorem. In case you need elaboration, let $G(t)$ be an antiderivative of $\sin(t^2)$. $G(t)$ definitely exists although it cannot be expressed in a closed form. Then the second factor is simply $\frac{G(x)-G(x_0)}{x-x_0}$.) (ii) $\frac{1}{3}$. (Again, let $F(t)$ be an antiderivative of $\frac{t^2}{t^6+1}$. This time $F(t)$ can be expressed in a closed form. But that is not needed. Then the given ratio equals $\frac{F(x^2)-F(0)}{x^6}$ which is of an indeterminate form. By l'Hôpital's rule, its limit as $x \to 0$ is the same as $\lim\limits_{x\to 0}\dfrac{2xF'(x^2)}{6x^5} = \lim\limits_{x\to 0}\dfrac{2x}{6x^5}\dfrac{x^4}{x^{12}+1}$.)

19. By the fundamental theorem, $\frac{dx}{dy} = \frac{1}{\sqrt{1+4y^2}}$. Hence $\frac{dy}{dx} = \sqrt{1+4y^2}$ which gives $\frac{d^2y}{dx^2} = \frac{4y\frac{dy}{dx}}{\sqrt{1+4y^2}} = 4y$.

20. $A = -\frac{3}{2}, B = \frac{35}{36}$, C can be any real number. (Write the integrand as $\frac{4e^{2x}+6}{9e^{2x}-4}$ and equate it with the derivative of the R.H.S. Comparing the numerators, $9A + 18B = 4$ and $-4A = 6$. C cannot be determined from the data. The way the the problem is asked it can be done simply by diffrentiation. It is, of course, a good idea to verify the answer by *finding* an antiderivative. For this, first put $u = e^x$ to convert the integral to $\int \frac{4u^2+6}{9u^2-4}\frac{du}{u}$ and thence to $\int \frac{2v+3}{v(9v-4)}\,dv$, where $v = u^2$. Use partial fractions.)

21. From the form of the conclusion, it is clear that we should make $(b-a)$ as a single variable, say u. The data then gives $a = \frac{4-u}{2}, b = \frac{4+u}{2}$ and $u > 0$. The given sum of integrals becomes $\int_0^{(4-u)/2} g(x)dx + \int_0^{(4+u)/2} g(x)dx = F(u)$ (say). Using (18), $F'(u) = -\frac{1}{2}g(\frac{4-u}{2}) + \frac{1}{2}g(\frac{4+u}{4})$. This is positive since g is monotonically increasing and $\frac{4-u}{2} < \frac{4+u}{2}$ (as $u > 0$).

22. (a) $\frac{1}{2}$. (Differentiate using (19) to get $f(x) = 1 - xf(x)$ which gives $f(x) = \frac{1}{x+1}$. This gives the value of $f(1)$. But the question is vacuous as the function $f(x) = \frac{1}{1+x}$ does not satisfy the hypothesis.) (b) ± 1. (Solve $x^2 = \sqrt{2-x^2}$.)

23. $a = \frac{1}{4}$, $b = -\frac{5}{4}$ and $f(x) = \frac{1}{4}x^2 - \frac{5}{4}x + 2$. (Simplifying the determinant (using $C_2 \longrightarrow C_2 - C_1$, followed by $R_2 \longrightarrow R_2 + R_1$, $R_3 \longrightarrow R_3 + R_1$ and finally, $R_3 \longrightarrow R_3 - 2R_2$) gives $f'(x) = 2ax + b$ and hence $f(x) = ax^2 + bx + c$ for some constant c. $f(0) = 2$ gives $c = 2$. Condition (b) implies $f'(\frac{5}{2}) = 0$. $f(1) = 1$ gives one more equation in a and b.)

24. $\int \sec^3\theta\ d\theta = \int \sec\theta\sec^2\theta\ d\theta = \sec\theta\tan\theta - \int \tan\theta \times \tan\theta\sec\theta\ d\theta = \tan\theta\sec\theta - \int(\sec^2\theta - 1)\sec\theta\ d\theta = \tan\theta\sec\theta + \int \sec\theta\ d\theta - \int \sec^3\theta\ d\theta = \tan\theta\sec\theta + \ln(\tan\theta + \sec\theta) - \int \sec^3\theta\ d\theta$. Hence $\int \sec^3\theta\ d\theta = \frac{1}{2}(\tan\theta\sec\theta + \ln(\tan\theta + \sec\theta))$. To find $\int \sqrt{a^2+x^2}\ dx$ from this put $x = a\tan\theta$. The

integral then comes out as $\frac{1}{2}x\sqrt{a^2+x^2}+\frac{a^2}{2}\ln(x+\sqrt{a^2+x^2})$ (minus a constant which may be dropped).

25. (i) $\int \sin^2 x dx = \int \frac{d}{dx}(-\cos x)\sin x dx = -\cos x \sin x + \int \cos x \cos x dx = -\cos x \sin x + - \int \sin^2 x dx$. So $\int \sin^2 x dx = \frac{x}{2} - \frac{\sin x \cos x}{2}$. Writing $\sin^2 x$ as $\frac{1}{2}(1-\cos 2x)$ gives the same anti-derivative. Integration by parts applied twice to $\int \sin(x^2)dx$ gives $\int(1-\frac{4x^4}{3})\sin(x^2)dx = x\sin(x^2) - \frac{2}{3}x^3\cos(x^2)$. But this is no good for finding $\int \sin(x^2)dx$ since $1-\frac{4x^2}{3}$is not a constant which can be pulled out of the integral sign. (ii) The three antiderivatives are $\frac{1}{2}\sin^2 x$, $-\frac{1}{2}\cos^2 x$ and $\frac{1}{4}(\sin^2 x - \cos^2 x)$. They differ by constants and so there is nothing wrong.

26. (a) follows by routine computations. For (b), note that $\frac{d}{dx}e^{ibx} = \frac{d}{dx}(\cos bx + i\sin bx) = -b\sin bx + bi\cos bx = ibe^{ibx}$. For the second part, write $c = a+ib$, where a, b are real. Then $e^{cx} = e^{ax}e^{ibx}$. Apply the product rule. Finally, to prove (c), from (b) we have $\int e^{(a+ib)x}\,dx = \frac{1}{a+ib}e^{(a+ib)x} = \frac{a-ib}{a^2+b^2}e^{ax}(\cos bx + i\sin bx)$. Equating the real parts of the two sides gives (54) while equating the imaginary parts gives (53).

27. (i) $-4\sqrt{1-\sin(\frac{x}{2})}$. (Multiply and divide by $\sqrt{1-\sin(\frac{x}{2})}$ and then put $u = \sin(\frac{x}{2})$. More simply, rewrite the integrand as $|\sin\frac{x}{2} + \cos\frac{x}{2}|$.) (ii) $-\frac{2}{5}(1-x)^{5/2} + \frac{4}{3}(1-x)^{3/2} - 2(1-x)^{1/2}$. (Put $u = 1-x$.) (iii) $-2\sqrt{1-x} - \sin^{-1}x + \sqrt{x}\sqrt{1-x}$. (Rewrite the integrand as $\frac{\sqrt{1-x}}{1+\sqrt{x}}$ and then put $x = \sin^2\theta$. The substitution $x = \cos^2\theta$ is also tempting since it reduces the original integrand to a succinct $\tan(\frac{\theta}{2})$. But then the integral becomes $-2\int \tan\frac{\theta}{2}\sin\theta\cos\theta\,d\theta$. This can be further converted to the integral of a rational function of $u = \tan\frac{\theta}{2}$. But the partial fractions are complicated as the denominator is $(1+u^2)^3$. If you think of such an approach first, it is advisable to abandon it temporarily in time and look for a better one.) (iv) $\frac{1}{2}\ln\frac{\sqrt{\cos 2x}-\cos x}{\sqrt{\cos 2x}+\cos x} - \frac{1}{\sqrt{2}}\ln\frac{\sqrt{\cos 2x}-\sqrt{2}\cos x}{\sqrt{\cos 2x}+\sqrt{2}\cos x}$. (The integrand equals $\frac{(1-\tan^2 x)^{1/2}}{\tan x}$. Put $u = \tan x$ and then $1-u^2 = v^2$.) (v) $\sqrt{2}\tan^{-1}(\sqrt{2\tan x}+1) + \sqrt{2}\tan^{-1}(\sqrt{2\tan x}-1)$. (Rewrite the integrand as $\frac{1+\tan x}{\sqrt{\tan x}}$ and put $u = \sqrt{\tan x}$.) (vi) $\ln(xe^x) - \ln(1+x)e^x + \frac{1}{1+xe^x}$. ($u = xe^x$ converts the integral to $\int \frac{du}{u(1+u)^2}$. Use partial fractions.) (vii) $\frac{4}{\pi}(x\sin^{-1}x + \sqrt{1-x^2}) - x$. (The integrand equals $\frac{2\sin^{-1}x - \pi/2}{\pi/2}$. Integrate by parts.) (viii)$\ln x - \ln(2-x+2\sqrt{1-x}) - 2\sin^{-1}\sqrt{x}$. (Similar to (iii).) (ix) $\sum_{k=1}^{8}\frac{u^k}{k} + \ln(u+1) + 6v(\frac{e^{3v}}{3} - \frac{3e^{2v}}{2} + 3e^v - v) - 6(\frac{e^{3v}}{9} - \frac{3e^{2v}}{4} + 3e^v - \frac{v^2}{2})$, where $u = \sqrt[12]{x}$ and $v = \ln(1+u^2) = \ln(1+\sqrt[6]{x})$. (The substitution $u = \sqrt[12]{x}$ gets rid of all the radicals and converts the integral to $12(I_1+I_2)$, where $I_1 = \int \frac{u^8}{u+1}du$ and $I_2 = \int \frac{\ln(1+u^2)}{1+u^2}u^7 du$. To find I_1, write $\frac{u^8}{u+1}$ as $\frac{u^8-1}{u+1} + \frac{1}{u+1}$. The substitution $v = \ln(1+u^2)$ reduces

I_2 to $\int \frac{v}{2}(e^v-1)^3 dv$. Expand the cube and integrate by parts.) (x) $\frac{1+\sin 2\theta}{4}\ln(1+\sin 2\theta) + \frac{1-\sin 2\theta}{4}\ln(1-\sin 2\theta)$. (Rewrite $\ln(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta})$ as $\frac{1}{2}(\ln(1+\sin 2\theta) - \ln(1-\sin 2\theta))$ and put $t = \sin 2\theta$. Alternately, split the integral as $I_1 + I_2$, where $I_1 = \int(\cos^2\theta - \sin^2\theta)\ln(\cos\theta+\sin\theta)$ and $I_2 = \int(\cos^2\theta - \sin^2\theta)\ln(\cos\theta-\sin\theta)$. Put $u = \cos\theta+\sin\theta$ to find I_1 and $u = \cos\theta - \sin\theta$ to find I_2. Yet another approach is to rewrite the integrand as $\cos 2\theta \ln(\sec 2\theta + \tan 2\theta)$ and integrate by parts.) (xi) $x\tan^{-1}(\frac{2x+2}{3}) - \frac{3}{2}\ln((2x+2)^2+9) + \tan^{-1}(\frac{2x+2}{3})$. (Recognise the integrand as $\tan^{-1}(\frac{2x+2}{3})$ and then integrate by parts.) (xii) $xe^x \ln x - e^x$. (Recognise $(x+1)e^x$ as the derivative of xe^x and integrate by parts.)

28. Let $F(x)$ be an antiderivative of $f(x)$. Then $\frac{d}{dt}(F(h(t))) = F'(h(t))h'(t) = f(h(t))h'(t)$. Applying the fundamental theorem of calculus (first form).

29. (i) $1 - \frac{\sqrt{3}}{2}$. (Put $x = \sin u$.) (ii) 4. (The integrand is an even function. $\int_0^2 |x^2-1|\ dx = \int_0^1 1-x^2\ dx + \int_1^2 x^2-1\ dx$.) (iii) $\ln 2 - \frac{1}{10} + \frac{1}{2}\ln(\frac{3}{2})$. (Rewriting the numerator as $2x^3(x^2-1)+(x^2+1)^2$, the integrand becomes $\frac{2x^3}{(x^2+1)^2} + \frac{1}{x^2-1}$. To integrate the first term, put $x^2+1 = u$. Resolve the second term into partial fractions.) (iv) 2. (Put $u = \ln x$.) **30.** $k = 16$ is a possible value. (Putting $u = x^2$ the integral becomes $\int_1^{16} \frac{e^{\sin u}}{u}\ du$ which equals $F(16) - F(1)$.) In fact, this is the only (positive) value since F is strictly increasing on $(0,\infty)$, its derivative being positive. **31.** $T_{10} = 0.693771,\ S_{10} = 0.69315$.

Chapter 18

1. For the first part, integrate by parts twice. For the second part calculate I_1 directly as 1 and apply the first part with $n = 3$.

2. Denoting the integral by I_m in each case, the reduction formulas are:
(a) $I_m = x^m \sin x + mx^{m-1}\cos x - m(m-1)I_{m-2}$ (b) $I_m = x^m e^x - m_{m-1}$
(c) $I_m = x(\ln x)^m - mI_{m-1}$ (d) $I_m = \dfrac{x^{a+1}(\ln x)^m}{a+1} - \dfrac{m}{a+1}I_{m-1}$.

3. $n = 3$. (The given integral, say I_n, equals $e\int_{-1}^{0} e^u u^n du$, where $u = x-1$. Prove the reduction formula : $I_n = (-1)^{n+1} - nI_{n-1}$ for $n \geq 1$ and go on computing starting with $I_0 = e-1$.)

4. Since $0 \leq \sin x \leq 1$ for all $x \in [0, \frac{\pi}{2}], \sin^{2m+1}(x) \leq \sin^{2m}(x) \leq \sin^{2m-1}(x)$ for all $x \in [0, \frac{\pi}{2}]$. Hence $I_{2m+1} \leq I_{2m} \leq I_{2m-1}$. Divide throughout by I_{2m+1} and use (3) with $n = 2m+1$.

5. Let $a_n = \dfrac{2.2}{1.3}\cdot\dfrac{4.4}{3.5}\cdot\dfrac{6.6}{5.7}\cdots\dfrac{(2n)(2n)}{(2n-1)(2n+1)}$. Then $a_n = \dfrac{(2^n n!)^2}{(2n)!}\dfrac{(2^n n!)^2}{(2n+1)!} =$

$\frac{\pi}{2}\frac{I_{2n+1}}{I_{2n}}$. By the last exercise and the Sandwich Theorem, $\lim_{n\to\infty}\frac{I_{2n}}{I_{2n+1}}=1$. So $\frac{I_{2n+1}}{I_{2n}}\to 1$ as $n\to\infty$. Hence $a_n\to\frac{\pi}{2}$ as $n\to\infty$.

6. (a) fg and f/g are even when f,g are both even or both odd. They are odd when one of f and g is even and the other odd. If g is even, then the composite $f\circ g$ is even regardless of f. If g is odd then $f\circ g$ is even or odd according as f. For (d) note first that a function f is even if and only if $f=f\circ g$, where g is the function $g(x)=-x$. So by the chain rule, $f'(x)=(f\circ g)'(x)=f'(g(x))g'(x)=f'(-x)(-1)=-f'(-x)$ which proves that f' is an odd function. Geometrically, the tangents to the graph $y=f(x)$ at the points $(x,f(x))$ and $(-x,f(-x))$ have opposite slopes which is obvious since the graph is symmetric about the y-axis. For an odd function f, the argument is similar except that this time $f=-(f\circ g)$. Geometrically, the tangents at the points $(x,f(x))$ and $(-x,f(-x))$ are parallel to each other. (e) 0. (The integrand is odd since the first factor is even and the second is odd.)

7. (a) If T_1,T_2 are periods of f then so is T_1+T_2 since $f(x+(T_1+T_2))=f((x+T_1)+T_2)=f(x+T_1)=f(x)$. Also, T_1-T_2 is a period since $f(x+(T_1-T_2))=f((x+T_1-T_2)+T_2)=f(x+T_1)=f(x)$. For (b), follow the hint. By (a), mT^* and hence $T-mT^*$ is a period of f. If $T>mT^*$, then $T-mT^*$ will be a positive period which is less than T^* since $T-mT^*<(m+1)T^*-mT^*$, a contradiction. **8.** (a) $3I$. (First put $u=2x$. Then use the fact that the integral of f over any interval of length $6T$ is $6I$.) (b) (A). (The integrand is a periodic function with period π. Hence $g(\pi)=\int_0^\pi\cos^4 t\,dt=\int_x^{x+\pi}\cos^4 t\,dt$. Add $g(x)=\int_0^x\cos^4 t\,dt$ to both the sides.) **9.** $f(x)+f(\frac{1}{x})=\frac{(\ln x)^2}{2}$. ($f(\frac{1}{x})=\int_1^{1/x}\frac{\ln t}{1+t}dt$. Putting $u=\frac{1}{t}$, this becomes $\int_1^x\frac{\ln u}{u(1+u)}du$. But since u is a dummy variable of integration, we can as well replace it by t. So $f(\frac{1}{x})=\int_1^x\frac{\ln t}{t(1+t)}dt$. So $f(x)+f(\frac{1}{x})=\int_1^x\frac{\ln t}{1+t}(1+\frac{1}{t})dt=\int_1^x\frac{\ln t}{t}dt=\int_0^{\ln v}v\,dv$, where $v=\ln t$.) **10.** $\int_0^\pi f(\sin x)dx=\int_0^\pi(\pi-x)f(\sin(\pi-x))dx=\pi\int_0^\pi f(\sin x)dx-\int_0^\pi f(\sin x)dx$.

11. It is no use applying (37) to $\int_0^1\tan^{-1}\frac{1}{1-x+x^2}dx$. Instead, let $I=\int_0^1\tan^{-1}x\,dx$. Then by (37) I also equals $\int_0^1\tan^{-1}(1-x)\,dx$. Adding, $2I=\int_0^1\tan^{-1}x+\tan^{-1}(1-x)dx=\int_0^1\tan^{-1}\frac{1}{1-x+x^2}dx$. Integration by parts gives $I=\frac{\pi}{4}-\frac{1}{2}\ln 2$. Since $\tan^{-1}(1-x+x^2)+\tan^{-1}\left(\frac{1}{1-x+x^2}\right)=\frac{\pi}{2}$ for all $x\in[0,1]$, we have $\int_0^1\tan^{-1}(1-x+x^2)=\frac{\pi}{2}-\int_0^1\tan^{-1}\left(\frac{1}{1-x+x^2}\right)dx=\frac{\pi}{2}-2I=\ln 2$.

12. Yes, as can be seen by applying (37) and adding as in the last exercise.

13. (c). (Apply (38) and add to get $2I_1 = I_2$.) **14.** Apply (37) and add.

15. In all explanations, I denotes the given definite integral.

(i) 0. (Apply (37) and note that $\cos[(2n+1)(\pi - x)] = -\cos(2n+1)x$.) (ii) $\frac{\pi^2}{16}$. (By (37), $2I = \pi J$, where $J = \int_0^{\pi/2} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$. To find J, write the integrand as $\frac{\sin 2x}{1+\cos^2 2x}$. Now put $\cos 2x = u$.) (iii) $\pi\sqrt{2}$. (Applying (38), $2I = \pi J$, where $J = \int_{\pi/4}^{3\pi/4} \frac{d\phi}{1+\sin\phi}$. Write the integrand of J as $\frac{1-\sin\phi}{\cos^2\phi} = \sec^2\phi - \tan\phi\sec\phi$. (iv) $\frac{\pi}{4}$. (By (37), $2I = \int_0^{\pi/2} 1 dx$.) (v) $\frac{\pi}{2}$. (vi) 2. (By (38), $2I = \int_{\pi/4}^{3\pi/4} (\frac{1}{1+\cos x} + \frac{1}{1-\cos x}) dx = \int_{\pi/4}^{3\pi/4} 2\text{cosec}^2 x dx$.) (vii) $\frac{\pi}{8}\ln 2$. (Once again, by (37), $2I = \int_0^{\pi/4} \ln 2 \; dx$.) (viii) 1. (Split $\int_{-1}^{1} [x] \; dx$ as $\int_{-1}^{0} -1 \; dx + \int_0^1 0 \; dx$.) (ix) $-\frac{1}{2}$. (Handle the integral of the first term as in (viii). The second term is an odd function.) (x) $-\frac{\pi}{2}$. (Break the interval of integration at $\frac{5\pi}{6}, \pi$ and $\frac{7\pi}{6}$.) (xi) $\frac{\pi\alpha}{\sin\alpha}$. (By (37), $2I = \pi J$, where $J = \int_0^{\pi} \frac{dx}{1+\cos\alpha\sin x}$. The substitution $t = \tan\frac{x}{2}$ changes J to the improper integral $\int_0^{\infty} \frac{2dt}{(t+\cos\alpha)^2 + \sin^2\alpha}$. An antiderivative of the integrand is $\frac{2}{\sin\alpha}\tan^{-1}\left(\frac{t+\cos\alpha}{\sin\alpha}\right)$. Its value at $t = \infty$ is to be taken as $\lim_{t\to\infty} \frac{2}{\sin\alpha}\tan^{-1}\left(\frac{t+\cos\alpha}{\sin\alpha}\right)$ which equals $\frac{2}{\sin\alpha}\frac{\pi}{2}$ since $\sin\alpha > 0$. The value of the antiderivative at $t = 0$ is $\frac{2}{\sin\alpha}\tan^{-1}\cot\alpha = \frac{2}{\sin\alpha}(\frac{\pi}{2} - \alpha)$.) (xii) $-\frac{\pi}{4} - \frac{8}{15}$. (The row operation $R_1 \longrightarrow R_1 - \sec x R_3$ gives $f(x) = (\sec^2 x + \cot x\text{cosec}\, x - \cos x)(\cos^4 x - \cos^2 x)$ which simplifies to $\cos^2 x - 1 - \cos^5 x$. Integrate term by term using the substitution $u = \sin x$ to integrate $\cos^5 x$. There is apparently no elegant way to get the answer.) (xiii) $\frac{1}{2}\ln 2 + \frac{1}{2} - \frac{\pi}{4}$. (Integrating by parts, $I = \frac{1}{2}\ln 2 - \frac{1}{2}J$, where $J = \int_0^1 \frac{x(\sqrt{1-x} - \sqrt{1+x})}{(\sqrt{1+x} + \sqrt{1-x})\sqrt{1-x^2}} dx$. If we multiply and divide the integrand of J by $(\sqrt{1+x} - \sqrt{1-x})$, it simplifies to $1 - \frac{1}{\sqrt{1-x^2}}$. Alternately, put $x = \cos 2\theta$ to get rid of both the radicals $\sqrt{1+x}$ and $\sqrt{1-x}$.) (xiv) $-\frac{\pi}{\sqrt{3}} + \frac{\pi^2}{12} + \frac{\pi}{4}\ln\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$. (Put $x = \tan\theta$ to see that $\cos^{-1}(\frac{2x}{1+x^2}) = \cos^{-1}(\sin 2\theta) = \frac{\pi}{2} - 2\tan^{-1} x$. Since $\tan^{-1} x$ is an odd function, $I = \frac{\pi}{2}\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4} dx$ which can be evaluated by resolving the integrand into partial fractions as $-1 + \frac{1}{2}\frac{1}{1+x^2} + \frac{1}{4}\frac{1}{1+x} + \frac{1}{4}\frac{1}{1-x}$.) (xv)$\pi^2$. (By (37), $2I = 2\pi J$, where $J = \int_0^{2\pi} \frac{\sin^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$. Apply (32).) J also equals $\int_0^{2\pi} \frac{\cos^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$. Hence $2J = 2\pi$.)

16. Let $I = \int_0^{\pi/2} \ln(\sin x) dx$. Then by (37), $2I$ equals $-\frac{\pi}{2}\ln 2 + J$, where $J = \int_0^{\pi/2} \ln(\sin 2x) dx = \frac{1}{2}\int_0^{\pi} \ln(\sin x) dx$. From the hint, J equals I. The last integral in the question is 0 as the integrand equals $\ln 2 - \ln(\sin 2x)$. **17.** $\frac{1}{e}$. (Let $a_n = \frac{(n!)^{1/n}}{n}$. Then $\ln a_n = \frac{1}{n}(\sum_{k=1}^{n} (\ln k - \ln n)) = \frac{1}{n}\sum_{k=1}^{n} \ln(\frac{k}{n}) \to \int_0^1 \ln x \; dx$ as $n \to \infty$.) **18.** Integrating by parts, the given integral, say I_n, equals $\frac{1}{2} - J_n$, where $J_n = \int_0^1 \frac{2x^{n+1}}{(1+x^2)^2} dx \leq \int_0^1 2x^{n+1} dx = \frac{2}{n+2}$.

Hence $J_n \to 0$ as $n \to \infty$. It is not easy to evaluate I_n. Nor is it needed! **19.** Use the method of Comment No. 4 to express the sum as $\int_0^1 (1-x)^n \sum_{k=0}^{n^2} (-1)^k (n+k+1)\binom{n^2}{k} x^k dx$. Split this as $I + J$ where $I = (n+1)\int_0^1 (1-x)^{n^2+n} dx$ and $J = \int_0^1 (1-x)^n x \sum_{k=1}^{n^2} (-1)^k k\binom{n^2}{k} x^{k-1} dx$. Then J equals $\int_0^1 (1-x)^n x \frac{d}{dx}(1-x)^{n^2} dx = -n^2 \int_0^1 x(1-x)^n (1-x)^{n^2-1} dx$.

Chapter 19

1. $\frac{dr}{dt} = \alpha$, where t is the time, r the radius (at time t) and α is some constant. (We are given $\frac{dV}{dt} = 4\pi r^2 \alpha$ for some constant α. Since $V = \frac{4\pi}{3} r^3$, we have $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.) **2.** (C). (By trying each choice by direct substitution.) **3.** Solving the given differential equation, $x(t) = x(0)e^{-\lambda t}$. So $x(t) = \frac{1}{2}$ when $t = \dfrac{\ln 2}{\lambda}$. This is independent of $x(0)$. **4.** Let t = time since death, $x = f(t)$ = amount of C_{14} at time t and $y = g(t)$ = amount of $C_{12} = k$ (a constant). Then $\frac{dx}{dt} = e^{-\lambda x}$, where $\lambda = \frac{\ln 2}{T}$ by the last exercise. Also $x(0) = \alpha k$. So $x = f(t) = \alpha k e^{-(t \ln 2)/T} \cdot f(t) = \beta k$ gives $\frac{\beta}{\alpha} = e^{-(t \ln 2)/T}$, i.e., $t = \frac{\ln \alpha - \ln \beta}{\ln 2} T$. **5.** The function $f(x) = \ln(1+x)$ is defined for all $x \in (-1, 1)$. Since $f'(x) = \frac{1}{1+x}$ and $f''(x) = -\frac{1}{(1+x)^2}$, by Taylor's approximation of the first order at $x = 0$, we have $\ln(1+x) = x - \frac{x^2}{2(1+c)^2}$ for some $c \in (-|x|, |x|)$. No matter what c is, the last term is negligible if $|x|$ is small. **6.** If one rupee is invested, the total at the end of the n-th year is $(1 + \frac{\alpha}{n})^n$ rupees. This will equal 2 when $n = \frac{\ln 2}{\ln(1+\frac{\alpha}{100})} \approx \frac{100 \ln 2}{\alpha}$ by the last exercise. The validity of the rule is based on the fact that $\ln 2 \approx 0.6931471 \approx 0.70$. **7.** The conclusion is a clumsy way of saying that the graphs of $y = u(x)$ and $y = v(x)$ do not intersect for $x > x_1$. Let $P(x)$ be an antiderivative of $p(x)$ so chosen that $P(x_1) = 0$. Then by applying (40) (with a change of notation) to the two given linear differential equations, we have :

$$u(x) = e^{-P(x)} \left[u(x_1) + \int_{x_1}^{x} e^{P(t)} f(t) dt \right] \tag{1}$$

$$\text{and} \quad v(x) = e^{-P(x)} \left[v(x_1) + \int_{x_1}^{x} e^{P(t)} g(t) dt \right] \tag{2}$$

for all $x \geq x_1$. Subtracting (2) from (1),

$$u(x) - v(x) = e^{-P(x)}(u(x_1) - v(x_1)) + e^{-P(x)} \int_{x_1}^{x} e^{P(t)} (f(t) - g(t)) dt \tag{3}$$

for all $x \geq x_1$. Since the exponential function is always positive, the hypothesis $u(x_1) > v(x_1)$ makes the first term of the R.H.S. positive while,

the hypothesis $f(t) > g(t)$ for all $t > x_1$ makes the integrand and hence the second term positive for all $x > x_1$. So the graph of $y = u(x)$ lies above that of $y = v(x)$ for all $x \geq x_1$.

8. Let $P = (x_0, y_0)$ be a point on the curve. Then the tangent at P meets the x-axis at $(\frac{mx_0 - y_0}{m}, 0)$ and the y-axis at $(0, y_0 - mx_0)$, where m is the value of $\frac{dy}{dx}$ at $x = x_0$. Therefore the condition in the problem translates as $(mx_0 - y_0)^2 = -4m$. Writing this as a quadratic in m, the curve satisfies the d.e. $x^2(\frac{dy}{dx})^2 + (4 - 2xy)\frac{dy}{dx} + y^2 = 0$. Solving as a quadratic, $\frac{dy}{dx} = \frac{xy - 2 \pm 2\sqrt{1 - xy}}{x^2}$, which has $xy = 1$ as a solution by inspection. To find the general solution, put $u = 1 - xy$. Then, $x\frac{du}{dx} = 2u \pm 2\sqrt{u}$. Again, $u = 0$ is a solution (corresponding to $xy = 1$). Otherwise, separating the variables, $\frac{du}{2(u \pm \sqrt{u})} = \frac{dx}{x}$. Putting $u = v^2$, the integral of the L.H.S. comes out as $\ln(\sqrt{u} \pm 1)$. Hence the general solution to (6) is $\ln(\sqrt{u} \pm 1) = \ln x + k$, where k is a constant, or equivalently, $u = (cx \pm 1)^2$ where c is a positive constant. Therefore the solutions of (5) are (i) $xy = 1$ and (ii) $y = \pm 2c - c^2x$. (i) represents a hyperbola passing through $(1, 1)$. (ii) is a straight which will pass through $(1, 1)$ only if $c = 1$ and the $+$ sign is chosen. Thus the only curves satisfying the condition in the problem are the hyperbola $xy = 1$ and the straight line $x + y = 2$. **9.** We have $\frac{dy}{dx} = \alpha y$ for some constant α. From the given equation of the normal at P, α comes out as a. Solving $\frac{dy}{dx} = ay$ with the initial condition $y = 1$ when $x = 1$ gives $y = e^{-a}e^{ax}$. The desired area is $\int_0^1 -\frac{x}{a} + 1 + \frac{1}{a} - \frac{1}{e^a}e^{ax}\,dx$ which comes out as $1 - \frac{1}{2a} + \frac{1}{ae^a}$.

10. The equation of the normal at a point (x_0, y_0) on the curve is $x - x_0 + m(y - y_0) = 0$, where m is the slope of the tangent at (x_0, y_0). So the data implies $|my_0 + x_0| = |y_0|\sqrt{1 + m^2}$. On squaring, and noting that $m = -\frac{dx}{dy}$, this leads to the differential equation $y^2(\frac{dx}{dy})^2 = 2xy\frac{dx}{dy} + x^2(\frac{dx}{dy})^2$. So, at every point of the curve, one of the two possibilities holds, viz. either (i) $\frac{dx}{dy} = 0$ or (ii) $y^2\frac{dx}{dy} = 2xy + x^2\frac{dx}{dy}$. The second possibility can be tackled by the substitution $x = uy$ and leads to the general solution $x^2 + y^2 = cx$ and then to the particular solution $x^2 + y^2 = 2x$. But the first possibility also gives a valid solution, viz. the vertical line $x = 1$. The complete solution, therefore, is the union of the circle $x^2 + y^2 = 2x$ and the straight line $x = 1$. If one insists upon a single equation in x and y whose solution set is this union of two curves, it is $(x - 1)(x^2 + y^2 - 2x) = 0$. (Actually, the reasoning needed here is much more subtle. All we can get immediately from the equation $y^2(\frac{dx}{dy})^2 = 2xy\frac{dx}{dy} + x^2(\frac{dx}{dy})^2$ is that at every point of the curve, *either* (i) *or* (ii) holds. It *does not* follow automatically that either (i) holds throughout for the curve or (ii) does. To conclude this would be as absurd as saying that since every human being is either a man or a woman, therefore, either every human being is a man or else every human being is a woman! To solve either (i) or (ii), we need to know that it

not only holds at a point of the curve, but in some neighbourhood of that point. To deduce this from the equation $y^2(\frac{dx}{dy})^2 = 2xy\frac{dx}{dy} + x^2(\frac{dx}{dy})^2$ rigorously requires theoretical calculus of functions of two variables and completeness of the real line. It is well beyond the JEE level.) **11.** (a) With the notations in the solution to the last exercise, if $P = (x_0, y_0)$, then $Q = (x_0 + my_0, 0)$. The condition $PQ = k$ means $m^2y_0^2 + y_0^2 = k^2$ which translates into the given d.e. It can be solved by separating the variables to give $(x \pm c)^2 + y^2 = k^2$ as the general solution. This is a family of circles with centres on the x-axis and radii k. The member that passes through $(0, k)$ is $x^2 + y^2 = k^2$. (b) In fact there are two singular solutions, viz., $y = k$ and $y = -k$. Both are evident geometrically.

12. As there is only one arbitrary constant, the order is 1, without any calculations. But to find the degree we need to find the d.e. Differentiating $y^2 = 2cx + 2c\sqrt{c}$ and eliminating c we get $y^2 = 2xy\frac{dy}{dx} + 2y\frac{dy}{dx}\sqrt{y\frac{dy}{dx}}$ which can be rewritten as $(y^2 - 2xy\frac{dy}{dx})^2 = 4y^3(\frac{dy}{dx})^3$ from which the degree is 3.

13. Multiplying both the sides of (15) by $2\dot{x}$, we get $\frac{d}{dt}(\dot{x})^2 = -\alpha\frac{d}{dt}(x^2)$ which gives $(\dot{x})^2 = -\alpha x^2 + a$ for some constant a. Since α is positive, so must be a. Writing α as b^2 and a as c^2, we get $\dot{x} = \pm\sqrt{c^2 - b^2x^2} = \pm c\sqrt{1 - \lambda^2x^2}$, where $\lambda = \frac{b}{c}$. Separating the variables and integrating, the general solution is $\lambda x = \sin(\lambda ct + p\lambda) = \sin(bt + \frac{pb}{c})$, where p is some other constant. This can be rewritten as $x = A\cos bt + B\sin bt$, where A, B are arbitrary constants. **14.** (i) $y^2 - x^2 = c$, which is also a family of rectangular hyperbolas. (The d.e. of the original family is $x\frac{dy}{dx} + y = 0$. Hence that of its family of trajectories is $x\frac{dx}{dy} = y$.) (ii) $x^2 + y^2 = cy$ which is the family of circles passing through the origin and centres on the y-axis. (The equation of a typical member of the given family can be written as $\frac{x^2+y^2}{x} = k$, where k is a constant. Hence its d.e. is $\frac{d}{dx}(\frac{x^2+y^2}{x}) = 0$ which gives $2xy\frac{dy}{dx} = y^2 - x^2$. The d.e. of the family of orthogonal trajectories is $\frac{dy}{dx} = \frac{2xy}{x^2-y^2}$ which can be solved by putting $y = vx$. The orthogonality of members of the two families can be verified geometrically by noting that at one of the points of their intersection, viz., at the origin, the tangents lie along the two axes.) (iii) $x^2 + \frac{y^2}{2} = c$ which is a family of concentric ellipses. (The d.e. of the given family is $\frac{dy}{dx} = \frac{y}{2x}$. Hence that of the family of trajectories is $\frac{dy}{dx} = -\frac{2x}{y}$.) (iv) $e^{-x}\cos y = k$, k a constant.

15. A typical member of this family has equation of the form $(x - 2\cos\theta)^2 + (y - 2\sin\theta)^2 = 1$. Differentiating, $x - 2\cos\theta = -(y - 2\sin\theta)\frac{dy}{dx}$. Putting this into the first equation, $y - 2\sin\theta = \pm\frac{1}{\sqrt{1+(\frac{dy}{dx})^2}}$ and hence $x - 2\cos\theta = \mp\frac{\frac{dy}{dx}}{\sqrt{1+(\frac{dy}{dx})^2}}$. Eliminating θ gives $\left(y \mp \frac{1}{\sqrt{1+(\frac{dy}{dx})^2}}\right)^2 + \left(x \pm \frac{\frac{dy}{dx}}{\sqrt{1+(\frac{dy}{dx})^2}}\right)^2 = 4$ as the d.e. of the family of circles of radii 1 with centres lying on the circle $x^2 + y^2 = 4$. This simplifies to $(x^2 + y^2 - 3)^2(1 + (\frac{dy}{dx})^2) = 4\left(y - x\frac{dy}{dx}\right)^2$

Geometrically, it is evident that this family has two envelopes, viz., the circles $x^2+y^2=9$ and $x^2+y^2=1$. Both these can be verified by direct substitution.

16. (i) $x+xy=cy$. (Rewrite as $x\frac{dy}{dx}=y+y^2$ and separate variables or rewrite as $\frac{xy'-y}{y^2}=1$ and recognise the L.H.S. as $-\frac{d}{dx}(\frac{x}{y})$.) (ii) $y=\frac{x^4}{2}+cx^2$. (Divide by x and solve as a linear differential equation.) (iii) $x^2+xe^y=c$. (Exact equation.) (iv) $e^{x/y}+\ln y=c$. ($\frac{1}{y^2}$ is an integrating factor. To see this note that the term ydy causes no problem and can be safely put aside temporarily. The remaining part is $e^{x/y}(ydx-xdy)$. Recognise $ydx-xdy$ as the numerator of $d(\frac{x}{y})$.) (v) $y-\tan^{-1}(x+y)=c$. (Putting $u=x+y$ converts the equation to $u^2(\frac{du}{dx}-1)=1$, i.e., $u^2\frac{du}{dx}=1+u^2$ which can be solved by separating the variables.) (vi) $x^4+2x^2y^2=c$. (Write as $\frac{dy}{dx}=-\frac{x^2+y^2}{xy}$ and put $y=vx$ as the numerator and the denominator of the R.H.S. are homogeneous functions of the same degree.) (vii) $\frac{1}{y^2}=1-x^2+ce^{-x^2}$. (The given substitution reduces the equation to a linear d.e., viz., $\frac{dv}{dx}+2xv=-2x^3$.) (viii) $\frac{1}{x^2}=1-y^2+ce^{-y^2}$. (Interchange x and y in (vii).) (ix) $y=e^x+x-\ln(c-e^{e^x})$. (The hint reduces the d.e. to $u'+(e^x+1)u=e^x$ which is a linear d.e. with integrating factor e^{e^x+x}. To find $\int e^xe^{e^x}e^xdx$, put $t=e^x$ and integrate by parts.)

17. The degenerate cases when either all a_1,b_1,c_1 or all a_2,b_2,c_2 vanish have $y=0$ and $x=0$ as solutions respectively. Otherwise follow the hint. Let $\Delta=a_1b_2-a_2b_1$. If $\Delta\neq0$ then by Crammer's rule h and k can be chosen so as to satisfy the system $a_1h+b_1k+c_1=0$ and $a_2h+b_2k+c_2=0$. In this case the substitutions $u=x+h$ and $v=y+k$ transform the given d.e. to $\frac{dv}{du}=\frac{a_1u+b_1v}{a_2u+b_2v}$ which can be solved by putting $v=zu$ as in Comment No. 15. The only case left is when $a_1b_2=a_2b_1$. Here, if a_2 and b_2 both vanish then the original d.e. can be written as $c_2\frac{dy}{dx}-b_1y=a_1x+c_1$ which is a linear d.e. If at least one of a_2 and b_2 is non-zero, then $a_1b_2=a_2b_1$ implies that there exists some λ such that $a_1=\lambda a_2$ and $b_1=\lambda b_2$. Putting $u=a_2x+b_2y$, the original d.e. changes to $\frac{du}{dx}=a_2+\frac{b_2(\lambda u+c_1)}{u+c_2}$ which is in the separate variables form.

18. (i) $y=-c_1\ln(c_1-x)+c_2$. (Put $p=y'$. Then the equation becomes $xp'+p=p^2$ or $\frac{dp}{dx}=\frac{p^2-p}{x}$ whose general solution is $p=\frac{c_1}{c_1-x}$ for some constant c_1. Put back $p=\frac{dy}{dx}$ and solve. (ii) $y=c_2e^{c_1x}$. (Putting $y'=p$, the equation becomes $y\frac{dp}{dx}=p^2$. To get rid of x write $\frac{dp}{dx}$ as $\frac{dp}{dy}\frac{dy}{dx}=p\frac{dp}{dy}$. The equation now becomes $py\frac{dp}{dy}=p^2$. $p=0$ leads to the solution $y=c$, a constant. Otherwise, the equation becomes $ydp=pdy$ whose solution is $p=c_1y$. Put back $p=\frac{dy}{dx}$ and solve again.) (iii) $e^y(y-1)=c_1x+c_2$. (Once again write $p=y'$ and $y''=p\frac{dp}{dy}$ to reduce the d.e. to $yp\frac{dp}{dy}+(1+y)p^2=0$. Again, $p=0$ is a possibility which leads to $y=c$. Otherwise, $\frac{dp}{p}=-\frac{y+1}{y}dy$ gives $pye^y=c_1$, i.e., $ye^ydy=c_1dx$.)

Chapter 20

1. Note that if $f(y) = 0$ for some y then for every x, writing x as $(x-y)+y$ we get $f(x) = f(x-y)f(y) = 0$, i.e., f vanishes identically. Hence if $f(x) \neq 0$ for at least one value of x, then it is non-zero for every x. For positivity, note that $f(x) = f(\frac{x}{2} + \frac{x}{2}) = (f(\frac{x}{2}))^2$ which is a square of a non-zero real number. For the second part, let $g(x) = \ln f(x)$. Then g is differentiable at 0 and $g'(0) = \frac{f'(0)}{f(0)} = f'(0)$ since $f(0) = f(0+0) = (f(0))^2$ and $f(0) \neq 0$ implies $f(0) = 1$. Also, g satisfies the functional equation $g(x+y) = g(x) + g(y)$ for all x, y. Hence by Comment No. 3, $g(x) = cx$ for all x, where $c = g'(0) = f'(0)$. In the Main Problem, $c = 2$.

2. Suppose f is continuous at x_0. Then $f(x_0 + h) \to f(x_0)$ as $h \to 0$. This means $f(x_0) + f(h) \to f(x_0)$ and hence $f(h) \to 0$ as $h \to 0$. So, f is continuous at 0 and hence at all points. For a direct proof of continuity at $x \in \mathbb{R}$, write $f(x+h) = f(x-x_0) + f(x_0+h)$. Taking limits as $h \to 0$, we get $\lim_{h\to 0} f(x+h) = f(x-x_0) + f(x_0) = f(x)$.

3. Using the given conditions, for every x and $h \neq 0$ we have

$$\frac{f(x+h) - f(x)}{h} = \frac{f(x)f(h) - f(x)}{h} = f(x)\frac{f(h)-1}{h} = f(x)g(h)$$

which tends to $f(x)$ as $h \to 0$. Hence $f'(x)$ exists and equals $f(x)$.

4. Define $g : \mathbb{R} \longrightarrow \mathbb{R}$ by $g(x) = \ln(f(e^x))$. Then g is continuous (since f is so) and satisfies $g(x+y) = g(x) + g(y)$. So by Comment No. 4, $g(x) = cx$ for some constant c. This means $f(e^x) = e^{cx} = (e^x)^c$ for all x. Hence $f(x) = x^c$ for all positive x. **5.** Functions of the form $f(x) = c \ln x$ for some constant c. (Apply the last exercise to the function $g(x) = e^{f(x)}$.)

6. (a) Let $x_0 \in \mathbb{R}$. Taking $x = 2x_0$ and $y = 2h$ in the given functional equation, we have $f(x_0+h) = \frac{1}{2}(f(2x_0)+f(2h))$ which tends to $\frac{1}{2}(f(2x_0)+f(0))$ as $h \to 0$, since f is given to be continuous at 0. But again by the functional equation, $\frac{1}{2}(f(2x_0) + f(0)) = f(\frac{2x_0+0}{2}) = f(x_0)$. So f is continuous at x_0. (b) $f(2) = -1$. (For a fixed x, applying the given functional equation repeatedly with $y = 0$ (and using $f(0) = 1$), we get $f(\frac{x}{2}) = \frac{1}{2}f(x) + \frac{1}{2}$, $f(\frac{x}{4}) = \frac{1}{2}f(\frac{x}{2}) + \frac{1}{2} = \frac{1}{4}f(x) + \frac{1}{4} + \frac{1}{2}$, $f(\frac{x}{8}) = \frac{1}{8}f(x) + \frac{1}{8} + \frac{1}{4} + \frac{1}{2}$. Continuing, (or more rigorously, by induction on n) we have

$$f(\frac{x}{2^n}) = \frac{f(x)}{2^n} + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^n} = \frac{f(x)}{2^n} + \frac{2^n - 1}{2^n}$$

and hence $\dfrac{f(\frac{x}{2^n}) - 1}{\frac{x}{2^n}} = \dfrac{f(x)-1}{x}$ for every positive integer n. As $n \to \infty$, the L.H.S. tends to $f'(0)$ which is given as -1. So $f(x) = 1 - x$ for all x. The given functional equation resembles linearity. If $f(x)$ is a linear function, i.e., a function of the form $f(x) = Ax + B$, then it satisfies a

stronger functional equation, viz., $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ whenever x, y, α, β are real numbers and $\alpha + \beta = 1$. This can be verified by direct calculation. The geometric significance of this is that if a point z divides the segment from x to y in a given ratio, then $f(z)$ also divides the segment from $f(x)$ to $f(y)$ in the same ratio. The functional equation given in this exercise satisfies only a special case of this, viz., where $\alpha = \beta = \frac{1}{2}$. Geometrically, f takes mid-points to mid-points. This is sometimes called **mid-point linearity**. Rather complicated examples can be given to show that mid-point linearity does not imply linearity. The present exercise shows that if, in addition, we have continuity and differentiability even at one point, then mid-point linearity implies linearity.) **7.** (A) (with period 1). **8.** (a) Follow the hint to get a strictly descending sequence $\{T_n\}$ of positive periods of f. As this sequence is bounded below (by 0, for example), by completeness of $\mathbb{R}$ it has a limit, say T^*. Now, given any $\epsilon > 0$, there exists some $m \in \mathbb{N}$ such that $T_n - T^* < \epsilon$ for every $n \geq m$. Since $T^* < T_{n+1} < T_n$, this gives $T_n - T_{n+1} < \epsilon$. But by Exercise (18.7)(a), $T_n - T_{n+1}$ is a period of f. (b) Follow the hint. Let $c \in \mathbb{R}$. For every $n \in \mathbb{N}$, by (a), there exists a positive period T_n such that $T_n < \frac{1}{n}$. Since the interval $(c, c + \frac{1}{n})$ has length exceeding T_n, it must contain some integral multiple, say kT_n of T_n (as otherwise both c and $c + \frac{1}{n}$ will lie between two consecutive multiples of T_n). Then kT_n is also a period of f and hence $f(kT_n) = f(0 + kT_n) = f(0)$. Take x_n to be kT_n. By the Sandwich theorem, $x_n \to c$ as $n \to \infty$. So by Theorem 1 of Chapter 16, $f(c) = \lim_{n\to\infty} f(x_n) = f(0)$. As this holds for every $c \in \mathbb{R}$, f is a constant function, a contradiction. (c) Take $f(x) = x - [x]$ and $g(x) = |\sin x|$. Let $h(x) = f(x) + g(x)$. Then f, g have 1 and π as their periods. But h is not periodic. For, if $T \neq 0$ were a period of h, then we must, in particular, have $h(T) = h(0 + T) = h(0) = 0$. But since f, g only take non-negative values, this would mean $f(T) = 0$ and $g(T) = 0$. The former implies that T is an integer while the latter implies that T is an integral multiple of π. As π is irrational, this is impossible unless $T = 0$. For the second part, assume $T_1/T_2 = \frac{m}{n}$, where m, n are positive integers. Then $nT_1 = mT_2$ is a common period of $f(x)$ and $g(x)$ and hence of $h(x)$ too. **9.** Since $\frac{1}{\sqrt{2}}(\sin x + \cos x) = \sin(\frac{\pi}{4} + x)$, the graph is obtained by shifting the graph of $y = \sin x$ (shown in Figure (a) of Comment No. 10) to the left by $\frac{\pi}{4}$ units. **10.** If there exists a function h such that $g \circ f = h$, then trivially, $f(x_1) = f(x_2)$ would imply $g(f(x_1)) = g(f(x - 2))$, i.e., $h(x_1) = h(x_2)$. Conversely, suppose that this condition holds. We need to define g only on the range of the function f. So suppose y_1 is a point in the range of f. Then, there exists some x_1 such that $f(x_1) = y_1$. Define $g(y_1) = h(x_1)$. The trouble with this definition is that the same y_1 may also equal $f(x_2)$ for some $x_2 \neq x_1$. (If f is a one-to-one function, then this cannot happen. But we are not given that f is one-to-one.) In that case, $g(y_1)$ will have another definition, viz., as $f(x_2)$. So unless these two definitions coincide, the function g will not be a **well-defined function**. The given condition

is exactly what we need to ensure that g is well-defined.

11. The functions in (A) satisfy the given equations. But so would those in (C) if we replace $g(x) = \sin\sqrt{x}$ by $g(x) = |\sin\sqrt{x}|$. So the composites $f \circ g$ and $g \circ f$ given in the question do not determine f and g uniquely and (D) is the correct statement. **12.** None. ((A) is false for $x = 3$; (B) is false for $x = y = 0$ and (C) fails for $x = -1$.) **13.** (A) and (D). ((A) follows by direct calculation and (D) from the very definition of a rational function. To see that (C) is false is a little easier if $f(x)$ is written as $1 + \frac{3}{x-1}$. (B) is false as $f(1)$ is undefined.) **14.** The best proof is by considering the complex-valued function $h(x) = g(x) + if(x)$. Then (i) and (ii) together imply that h satisfies the functional equation

$$h(x + y) = h(x)h(y) \tag{1}$$

for all real x, y while (iii) means $h'(0) = i$. Duplicating the work in the Main Problem for a complex-valued function, we get

$$h'(x) = ih(x) \tag{2}$$

for all real x. Even though i is a complex constant, (2) can be solved to get $h(x) = ke^{ix}$, where k is a possibly complex constant. $h'(0) = i$ determines k as 1. The result follows by taking the real and imaginary parts of e^{ix}. In the work done upto reaching (2), the use of complex numbers is essentially a matter of convenience and can be avoided by parallel (albeit lengthier) work without them. But to solve (2) without complex numbers is not so easy. Note first that the single equation (2) is equivalent to the system

$$g'(x) = -f(x) \tag{3}$$

$$\text{and} \quad f'(x) = g(x) \tag{4}$$

of two differential equations. These two equations are interlinked. To solve such a system, one method is to 'delink' them first. For this, differentiate (4) and (3) to get $f''(x) = -f(x)$ and $g''(x) = -g(x)$. In other words, both $y = f(x)$ and $y = g(x)$ are solutions of the second order d.e.

$$\frac{d^2y}{dx^2} = -y \tag{5}$$

which, except for a change of notation, is the same equation as (15) in Chapter 19 with $\alpha = 1$. From the solution to Exercise (19.13), we get that $f(x) = A\cos x + B\sin x$ and $g(x) = C\cos x + D\sin x$ for some real constants A, B, C, D. Determining these constants from the initial conditions completes the solution. **15.** Neither equality is true. In fact, it may happen that $R_1 \cap R_2$ is the empty relation in which case its domain as well as range is $\emptyset$, but still R_1 and R_2 have the same, non-empty domain, e.g., take $S = \mathbb{R}$, and R_1, R_2 to be two unit discs centered at, say $(0, 0)$ and $(0, 10)$. A similar counterexample works for the range. All that can be said is that the domain (range) of $R_1 \cap R_2$ is a *subset* of the intersection of the domains (ranges) of R_1 and R_2.

16. (a) and (b) are immediate from the definitions. For (c), note that the (i,j)-th entry is $\sum_{k=1}^{n} a_{ik}a_{kj}$. Now the term $a_{ik}a_{kj}$ equals 0 or 1, the latter possibility holding if and only if a_{ik} and a_{kj} both equal 1. In particular, to say that the (i,j)-th entry is non-zero means that there is at least one k such that $a_{ik} = 1$ and $a_{kj} = 1$. This translates as $x_i R x_k$ and $x_k R x_j$. The transitivity condition is equivalent to saying that whenever this happens, we must also have $x_i R x_j$, i.e., $a_{ij} = 1$. This proves (d).

17. (i) $2^{n(n+1)/2}$. (There are $\frac{n(n+1)}{2}$ positions on and above the diagonal. Every subset of these determines a symmetric relation.) (ii) 2^{n^2-n}. (All diagonal entries have to be included. Every subset of the non-diagonal positions, along with these diagonal entries, determines a reflexive relation.) (iii) $2^{n(n-1)/2}$. (iv) $2^{n(n+1)/2} + 2^{n^2-n} - 2^{n(n-1)/2}$. (Use the principle of inclusion and exclusion.)

18. That R is an equivalence relation is too trivial. The equivalence classes are in one-to-one correspondence with elements of the range set of f, i.e., with the set $f(S)$. For the last part, let S be the set of all straight lines in the plane, T be the set $\mathbb{R} \cup \{\infty\}$, where ∞ is a symbol not in $\mathbb{R}$ and define $f : S \longrightarrow T$ by $f(L) =$ slope of L if L is not parallel to the y-axis and $f(L) = \infty$ otherwise. Then $L_1 \parallel L_2$ if and only if $f(L_1) = f(L_2)$.

19. (a) Reflexivity follows by taking $\alpha = 1$. For symmetry, note that if α is a non-zero real number then so is $\frac{1}{\alpha}$. For transitivity, use that the product of two non-zero real numbers is a non-zero real number. (b) The complex number 0 sits alone in its equivalence class. The non-zero complex numbers get divided according to the lines through O, they lie, that is all complex numbers lying on a given line through the origin O (except the origin itself) form one equivalence class. (c) The proof that R is an equivalence relation will be essentially similar. But the equivalence classes will be the (half) rays from O. Again $\{0\}$ will be a singleton equivalence class.

20. To prove that R is an equivalence relation, use the facts that 0 is a purely imaginary number and further that the negative and the sum of two purely imaginary numbers is also a purely imaginary number. The equivalence classes are lines parallel to the y-axis.

21. In both the cases reflexivity and symmetry are immediate. For transitivity, suppose $(a,b)R(c,d)$ and $(c,d)R(e,f)$ in the first part. Then $a+d = b+c$ and $c+f = d+e$. So $a+d+f = b+c+f = b+d+e$ which implies $a+f = b+e$, i.e., $(a,b)R(e,f)$. An entirely analogous argument works for the other relation. Note that in the first case, if $(a,b)R(c,d)$, then $a-b = c-d$. In other words, the integer $a-b$ depends only on the equivalence class of the element (a,b) and not on a particular representative of it. So the equivalence classes correspond to integers in the first case. Similarly, in

the second case, they correspond to rational numbers. (In essence, this problem outlines the construction of integers from natural numbers and that of rational numbers from the integers. We already saw in Comment No. 19 of Chapter 2 how the complex numbers can be constructed from the real numbers. So the only missing link now is the construction of the real numbers from the rational numbers. This is far more involved and beyond our scope.)

22. If two (non-zero) integers x and y divide each other, then $y = ax$ and $x = by$ for some integers a, b. We then have $x = abx$ whence $ab = 1$. If a, b are positive integers then this forces $a = b = 1$, i.e., $x = y$. So the relation $|$ is antisymmetric on the set of positive integers. However, on the set of all integers, $ab = 1$ can also mean $a = b = -1$. In that case $x \neq y$. So antisymmetry fails. The law of dichotomy fails for $|$. For example, neither $3|5$ nor $5|3$.

Chapter 21

1. (a) Two. (Specifically, they are $\pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.) (b) $\frac{\sqrt{21}}{2}$ $(= \frac{1}{2}|(2\mathbf{i}-\mathbf{k})\times(\mathbf{j}+2\mathbf{k})|)$. (c) $\frac{1}{4\sqrt{6}}(8\mathbf{i}+4\mathbf{j}+4\mathbf{k})$ $(= \frac{\overrightarrow{PQ}\times\overrightarrow{PR}}{|\overrightarrow{PQ}\times\overrightarrow{PR}|}$.) (d) $k = 6$. ($q = |\vec{a}\times\vec{b}|$. Area of $OABC$ is the sum of the areas of the triangles OAC and ABC and so $p = \frac{1}{2}(|\vec{a}\times\vec{b}|+|(9\vec{a}+2\vec{b})\times(10\vec{a}+\vec{b})|) = \frac{1}{2}(|\vec{a}\times\vec{b}|+|11(\vec{b}\times\vec{a})|) = 6|\vec{a}\times\vec{b}|$.) (e) (i) Since $\overrightarrow{PA_i} = \overrightarrow{PO} + \overrightarrow{OA_k}$ for every $k = 1, 2, \ldots, n$, the problem reduces to proving that $\sum_{k=1}^{n} \overrightarrow{OA_k} = \mathbf{0}$. For this, suppose without loss of generality that $A_1, A_2, \ldots, A_n$ represent the complex n-th roots of unity. Then their sum is the complex number 0 which corresponds to the vector $\mathbf{0}$. (ii) Let r be the circumradius of the polygon and $\mathbf{u}$ be a unit vector perpendicular to the plane of the polygon. Then for every $k = 1, 2, \ldots, n-1$, $\overrightarrow{OA_k} \times \overrightarrow{OA_{k+1}}$ is the vector $\pm r^2 \sin\frac{2\pi}{n}\mathbf{u}$ with the sign depending only on the choice of the direction of the vector $\mathbf{u}$ but not on the index k. The result follows by adding these $(n-1)$ equalities.

2. (a) (C). (Express all the box products as determinants.) (b) 0. (All the three vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$ lie in a plane perpendicular to $\mathbf{X}$ and hence are coplanar.) (c) $\sqrt{13}$. (d) 0. (e) 0. (The first ratio equals 1 while the second one equals -1.) (f) By (13), we have $(\vec{B}\times\vec{C})\times(\vec{C}\times\vec{A}) = ((\vec{B}\times\vec{C})\cdot\vec{A})\vec{C} - ((\vec{B}\times\vec{C})\cdot\vec{C})\vec{A} = (\vec{A}\cdot(\vec{B}\times\vec{C}))\vec{C}$ since $\vec{B}\times\vec{C}$ is perpendicular to C. Take the dot product of both the sides with $\vec{A}\times\vec{B}$. (g) $\sqrt{59}$. ($[\vec{U}\ \vec{V}\ \vec{W}] = \vec{U}\cdot(\vec{V}\times\vec{W}) = 1\times|\vec{V}\times\vec{W}|\cos\theta$, where θ is the angle between $\vec{U}$ and $\vec{V}\times\vec{W}$. The maximum value of this is $|\vec{V}\times\vec{W}|$.) (h) 0. (Since $(\vec{B}+\vec{C})\times(\vec{B}+\vec{C}) = \vec{0}$, we have $(\vec{B}+\vec{C})\times(\vec{A}+\vec{B}+\vec{C}) = (\vec{B}+\vec{C})\times\vec{A}$ which is perpendicular to A.)

3. (a) This follows because the Vandermonde determinant $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ is non-zero as a, b, c are distinct. (Or compute the determinant directly as $(b-a)(c-a)(c-b)$.) (b) $abc = -1$. (Split the given determinant as $\Delta_1 + \Delta_2$, where $\Delta_1 = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix}$. Then $\Delta_2 = abc\Delta$ and permuting the columns, $\Delta_2 = \Delta$, where Δ is the Vandermonde determinant above. The result follows since $\Delta \neq 0$.)

4. One possible choice is $\vec{v}_1 = 2\mathbf{i}, \vec{v}_2 = -\mathbf{i} + \mathbf{j}$ and $\vec{v}_3 = 3\mathbf{i} - 2\mathbf{j} \pm 4\mathbf{k}$. (The problem merely asks for one such set of vectors. One of the simplest beginnings is to take $\vec{v}_1 = 2\mathbf{i}$. Taking $\vec{v}_2$ to lie in the **i**-**j** plane, the requirement $\vec{v}_2 \cdot \vec{v}_2 = 2$ means $\vec{v}_2$ is of the form $\sqrt{2}(\cos\theta\mathbf{i} + \sin\theta\mathbf{j})$ for some θ. The requirement $\vec{v}_1 \cdot \vec{v}_2 = -2$ determines θ as $\frac{3\pi}{4}$ or $\frac{5\pi}{4}$. Taking the first possibility, $\vec{v}_2 = -\mathbf{i} + \mathbf{j}$. Now let $v_3 = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$. The remaining three conditions give a system of three equations in a, b, c.)

5. (a) $\frac{\pi}{3}$. ($(\vec{a} + 2\vec{b}) \cdot (5\vec{a} - 4\vec{b}) = 0$ along with $|\vec{a}| = 1 = |\vec{b}|$ gives $\vec{a} \cdot \vec{b} = \frac{1}{2}$.) (b) Yes. (As $\vec{A}$ is perpendicular to both $\vec{B}$ and to $\vec{C}$, it equals $\alpha(\vec{B} \times \vec{C})$ for some α. $|\vec{B} \times \vec{C}| = \sin\frac{\pi}{6} = \frac{1}{2}$ and $|\vec{A}| = 1$ imply that $\alpha = \pm 2$.) (c) $\frac{\pi}{6}$. (Use (13) to get $(\mathbf{a}\cdot\mathbf{c})\mathbf{a} - (\mathbf{a}\cdot\mathbf{a})\mathbf{c} + \mathbf{b} = \mathbf{0}$, i.e., $(\mathbf{a}\cdot\mathbf{c})\mathbf{a} = \mathbf{c} - \mathbf{b}$. Take dot products of both the sides with $\mathbf{a}, \mathbf{b}, \mathbf{c}$ to get a system of equations in $\mathbf{a}\cdot\mathbf{b}, \mathbf{a}\cdot\mathbf{c}$ and $\mathbf{b}\cdot\mathbf{c}$, solving which we get $\mathbf{a}\cdot\mathbf{c} = \pm\sqrt{3}$.) (d) $\frac{\pi}{4}$. (On one hand, $\vec{a}$ is a linear combination of $\hat{i}$ and $\hat{i} + \hat{j}$ and hence a linear combination of $\hat{i}$ and $\hat{j}$. But on the other, it is also a linear combination of $\hat{i} - \hat{j}$ and $\hat{i} + \hat{k}$, say $\vec{a} = \alpha(\hat{i} - \hat{j}) + \beta(\hat{i} + \hat{k})$. But then $\beta = 0$ as otherwise $\vec{a}$ will not be a linear combination of $\hat{i}$ and $\hat{j}$. Hence $\vec{a}$ is a multiple of $\hat{i} - \hat{j}$. Since we are interested only in the angle between $\vec{a}$ and $\hat{i} - 2\hat{j} + 2\hat{k}$, we may as well take $\vec{a} = \hat{i} - \hat{j}$.) (e) $\frac{1}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$. (Take the square of the box product $\vec{c} \cdot (\vec{a} \times \vec{b})$ which equals $\pm|\vec{c}|\ |\vec{a}|\ |\vec{b}| \sin\frac{\pi}{6}$ since $\vec{c}$ is parallel to $\vec{a} \times \vec{b}$.) (f) $\frac{3}{2}$. (By direct calculation, $|\vec{a} \times \vec{b}| = |2\hat{i} - 2\hat{j} + \hat{k}| = 3$. Also $|\vec{c} - \vec{a}|^2 = 8$ and $\vec{a} \cdot \vec{c} = |\vec{c}|$ gives $|\vec{c}|^2 = |\vec{a}|^2 - 8 = 1$.)

6. $p = 1$ and $p = -\frac{1}{3}$. (Let $\{\hat{i}, \hat{j}\}$ and $\{\hat{u}, \hat{v}\}$ be orthonormal bases for the old and the new system respectively. Then $\hat{u} = \cos\alpha\ \hat{i} + \sin\alpha\ \hat{j}$ and $\hat{v} = -\sin\alpha\ \hat{i} + \cos\alpha\ \hat{j}$, where α is the angle of rotation. Equating the two expressions for $\vec{a}$ we get $2p\hat{i} + \hat{j} = (p+1)\hat{u} + \hat{v}$ which gives a system of two equations : (i) $(p+1)\cos\alpha - \sin\alpha = 2p$ and (ii) $(p+1)\sin\alpha + \cos\alpha = 1$. Eliminating α, we get $(p+1)^2 + 1 = 4p^2 + 1$.)

7. (a) We have $|\vec{C}|^2 = \vec{C} \cdot (\vec{A} \times \vec{B})$ and $|\vec{B}|^2 = \vec{B} \cdot (\vec{A} \times \vec{C})$. But $\vec{B} \cdot (\vec{A} \times \vec{C}) = -\vec{C} \cdot (\vec{A} \times \vec{B})$. Hence $|\vec{B}|^2 = -|\vec{C}|^2$ which can happen only when $|\vec{B}| = |\vec{C}| = 0$. (b) The result is trivially true when $\mathbf{b} \times \mathbf{c} = \mathbf{0}$ since in that case both the sides represent the zero vector. So assume $\mathbf{b} \times \mathbf{c} \neq \mathbf{0}$.

Note that if the result holds with $\mathbf{a} = \mathbf{u}, \mathbf{a} = \mathbf{v}$ and $\mathbf{a} = \mathbf{w}$ then it holds with $\mathbf{a} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$ for any scalars α, β, γ. So it suffices to verify it for elements of any basis. We take the basis consisting of the vectors $\mathbf{b}, \mathbf{c}$ and $\mathbf{b} \times \mathbf{c}$. The verification is trivial for $\mathbf{a} = \mathbf{b} \times \mathbf{c}$ as both the sides reduce to $\mathbf{0}$ in that case. To verify it when $\mathbf{a} = \mathbf{b}$, note that no matter what $\mathbf{a}$ is, the vector $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is perpendicular to $\mathbf{b} \times \mathbf{c}$ and hence lies in the plane spanned by $\mathbf{b}$ and $\mathbf{c}$. In particular, there exist scalars α, β such that

$$\mathbf{b} \times (\mathbf{b} \times \mathbf{c}) = \alpha\mathbf{b} + \beta\mathbf{c} \tag{1}$$

To find α and β, we need two equations in them. As the vector on the L.H.S. is perpendicular to $\mathbf{b}$, taking dot product with $\mathbf{b}$ we get

$$\alpha(\mathbf{b} \cdot \mathbf{b}) + \beta(\mathbf{b} \cdot \mathbf{c}) = 0 \tag{2}$$

On taking dot product with $\mathbf{c}$, the L.H.S. of (1) becomes $\mathbf{c} \cdot (\mathbf{b} \times (\mathbf{b} \times \mathbf{c}))$ which is the same as $(\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{c} \times \mathbf{b}) = -|\mathbf{b} \times \mathbf{c}|^2$. So,

$$\alpha(\mathbf{b} \cdot \mathbf{c}) + \beta(\mathbf{c} \cdot \mathbf{c}) = -|\mathbf{b} \times \mathbf{c}|^2 \tag{3}$$

It is very easy to show that $-|\mathbf{b} \times \mathbf{c}|^2 = (\mathbf{b} \cdot \mathbf{c})^2 - (\mathbf{b} \cdot \mathbf{b})(\mathbf{c} \cdot \mathbf{c})$. It is then easy to solve (2) and (3) together to get $\alpha = \mathbf{b} \cdot \mathbf{c}$ and $\beta = -\mathbf{b} \cdot \mathbf{b}$. Putting these into (1), we see that Equation (13) in the text holds when $\mathbf{a} = \mathbf{b}$. Interchanging the roles of $\mathbf{b}$ and $\mathbf{c}$, it also holds for $\mathbf{a} = \mathbf{c}$. As noted before this proves that (13) holds for all $\mathbf{a}$. (c) If there exists a vector $\mathbf{X}$ such that $\mathbf{A} \times \mathbf{X} = \mathbf{B}$, then taking the dot product of both the sides with $\mathbf{A}$, it follows that $\mathbf{A} \cdot \mathbf{B} = 0$, i.e., $\mathbf{A} \perp \mathbf{B}$. Conversely, suppose $\mathbf{A} \cdot \mathbf{B} = 0$. Then by (13) in the text, $\mathbf{A} \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{A} \cdot \mathbf{B})\mathbf{A} - (\mathbf{A} \cdot \mathbf{A})\mathbf{B} = -|\mathbf{A}|^2\mathbf{B}$. So $\mathbf{X} = \mathbf{U}$, where $\mathbf{U} = -\frac{1}{|\mathbf{A}|^2}(\mathbf{A} \times \mathbf{B})$ is a solution of $\mathbf{A} \times \mathbf{X} = \mathbf{B}$. Further for every scalar λ, $\mathbf{A} \times \lambda\mathbf{A} = \mathbf{0}$ and so $\mathbf{X} = \mathbf{U} + \lambda\mathbf{A}$ is also a solution. For the converse, if $\mathbf{Y}$ is a solution, then $\mathbf{A} \times (\mathbf{Y} - \mathbf{U}) = \mathbf{0}$ and hence $\mathbf{Y} - \mathbf{U} = \lambda\mathbf{A}$ for some λ. To do the problem algebraically, resolve $\mathbf{A}, \mathbf{B}, \mathbf{X}$ into components as $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{B} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ and $\mathbf{X} = x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}$. Then the single vector equation $\mathbf{A} \times \mathbf{X} = \mathbf{B}$ is equivalent to a system of three linear equations in x_1, x_2, x_3, viz.,

$$a_2x_3 - a_3x_2 = b_1, \quad a_3x_1 - a_1x_3 = b_2 \quad \text{and} \quad a_1x_2 - a_2x_1 = b_3 \tag{4}$$

Note that here Δ, the determinant of the coefficients of x_1, x_2, x_3, viz., the determinant $\begin{vmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{vmatrix}$ vanishes and so the system has either no solutions or infinitely many solutions. Multiplying the three equations in (4) by a_1, a_2, a_3 respectively and adding we see that a necessary condition for a solution to exist is $a_1b_1 + a_2b_2 + a_3b_3 = 0$. Conversely, if this condition is satisfied, then by direct substitution we see that $x_1 = u_1, x_2 = u_2, x_3 = u_3$ is a solution, where $u_1 = -\frac{a_2b_3 - a_3b_2}{L}, u_2 = -\frac{a_3b_1 - a_1b_3}{L}$ and $u_3 = -\frac{a_1b_2 - a_2b_1}{L}$ and $L = a_1^2 + a_2^2 + a_3^2$ which is non-zero by assumption.

(This corresponds to $\mathbf{X} = \mathbf{U}$ above.) Further if $x_1 = y_1, x_2 = y_2, x_3 = y_3$ is any other solution then $x_1 = y_1 - u_1, x_2 = y_2 - u_2, x_3 = y_3 - u_3$ is a solution of the corresponding homogeneous system, viz.,

$$a_2x_3 - a_3x_2 = 0, \quad a_3x_1 - a_1x_3 = 0 \quad \text{and} \quad a_1x_2 - a_2x_1 = 0 \tag{5}$$

As we are assuming that $a_1^2 + a_2^2 + a_3^2 \neq 0$, at least one of the a_i's, say $a_3 \neq 0$. Then applying Theorem 7 of Chapter 3 to the first two equations above, we get that $y_1 - u_1, y_2 - u_2, y_3 - u_3$ are proportional to a_1a_3, a_2a_3, a_3^2, and hence proportional to a_1, a_2, a_3 respectively. Hence there is some λ such that $y_i = u_i + \lambda a_i$ for $i = 1, 2, 3$. Conversely every y_1, y_2, y_3 of this form give a solution of (4). This completes the algebraic solution to the problem. As you can see the solution using vectors is far more concise (even though the essential ideas are not very different). (d) $\vec{B} = (\frac{5}{3}, \frac{2}{3}, \frac{2}{3})$. (By (c), the general solution of $\vec{A} \times \vec{B} = \vec{C}$ is $\vec{B} = -\frac{\vec{A}\times\vec{C}}{|\vec{A}|^2} + \lambda\vec{A} = (\lambda + \frac{2}{3}, \lambda - \frac{1}{3}, \lambda - \frac{1}{3})$. The condition $\vec{A} \cdot \vec{B} = 3$ determines λ as 1.) (e) From the hint, $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]^2 = ||\mathbf{a}||^2||\mathbf{b} \times \mathbf{c}||^2 - ||\mathbf{a} \times (\mathbf{b} \times \mathbf{c})||^2 = ||\mathbf{a}||^2(||\mathbf{b}||^2||\mathbf{c}||^2 - (\mathbf{b}\cdot\mathbf{c})^2) - ||(\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}||^2$. A further expansion of the second term as $(\mathbf{a}\cdot\mathbf{c})^2||\mathbf{b}||^2 + ||(\mathbf{a}\cdot\mathbf{b})^2||\mathbf{c}||^2 - 2(\mathbf{a}\cdot\mathbf{c})(\mathbf{a}\cdot\mathbf{b})(\mathbf{b}\cdot\mathbf{c})$ gives the result. (f) $\mathbf{u} - \mathbf{v}$ is both perpendicular and parallel to $\mathbf{a}$. Hence it must vanish. For examples, note that even though $\mathbf{j} \neq \mathbf{k}$, $\mathbf{i}\cdot\mathbf{j} = \mathbf{i}\cdot\mathbf{k}$ and $(\mathbf{j} - \mathbf{k}) \times \mathbf{j} = (\mathbf{j} - \mathbf{k}) \times \mathbf{k}$. (g) $\mathbf{u} - \mathbf{v}$ is perpendicular to three non-coplanar vectors. So it must vanish.

8. 3. (Note that $\vec{a}\cdot\vec{p} = \vec{b}\cdot\vec{q} = \vec{c}\cdot\vec{r} = 1$ while $\vec{b}\cdot\vec{p} = \vec{c}\cdot\vec{q} = \vec{a}\cdot\vec{r} = 0$.) **9.** $\frac{1}{2}(\mathbf{p} + \mathbf{q} + \mathbf{r})$. (Let $|\mathbf{p}| = |\mathbf{q}| = |\mathbf{r}| = \lambda$. Then $(\frac{\mathbf{p}}{\lambda}, \frac{\mathbf{q}}{\lambda}, \frac{\mathbf{r}}{\lambda})$ is an orthonormal basis. So for every vector $\mathbf{x}$, we have $(\frac{\mathbf{p}}{\lambda}\cdot\mathbf{x})\frac{\mathbf{p}}{\lambda} + (\frac{\mathbf{q}}{\lambda}\cdot\mathbf{x})\frac{\mathbf{q}}{\lambda} + (\frac{\mathbf{r}}{\lambda}\cdot\mathbf{x})\frac{\mathbf{r}}{\lambda} = \mathbf{x}$, i.e.,

$$(\mathbf{p}\cdot\mathbf{x})\ \mathbf{p} + (\mathbf{q}\cdot\mathbf{x})\ \mathbf{q} + (\mathbf{r}\cdot\mathbf{x})\ \mathbf{r} = \lambda^2\mathbf{x} \tag{6}$$

Now the first term of the L.H.S. of the given equation equals the vector $(\mathbf{p}\cdot\mathbf{p})(\mathbf{x} - \mathbf{q}) - (\mathbf{p}\cdot\mathbf{x} - \mathbf{p}\cdot\mathbf{q})\mathbf{p} = \lambda^2\mathbf{x} - \lambda^2\mathbf{q} - (\mathbf{p}\cdot\mathbf{x})\mathbf{p}$. Similarly, simplify the remaining two terms and add. Then because of (10), we get

$$3\lambda^2\mathbf{x} - \lambda^2(\mathbf{p} + \mathbf{q} + \mathbf{r}) - \lambda^2\mathbf{x} = \mathbf{0} \tag{7}$$

Cancelling λ^2 gives the answer. **10.** The expression does not simplify to anything as it is not given that $\vec{b}$ and $\vec{c}$ are perpendicular to each other. If they are, then $\vec{b}, \vec{c}, \vec{b} \times \vec{c}$ is an orthonormal basis and the given vector simply equals $\vec{a}$. **11.** (A) and (C). (If θ is the angle between $\vec{a}$ and $\vec{b}$, then $|\vec{v}| = \sin\theta$, $\vec{a}\cdot\vec{b} = \cos\theta$ and so $|\vec{u}|^2 = |\vec{a}|^2 + (\vec{a}\cdot\vec{b})^2|\vec{b}|^2 - 2(\vec{a}\cdot\vec{b})^2 = 1 + \cos^2\theta - 2\cos^2\theta = \sin^2\theta$, while $\vec{u}\cdot\vec{a} = 1 - \cos^2\theta = \sin^2\theta$ and $\vec{u}\cdot\vec{b} = \cos\theta - \cos\theta = 0$.)

12. (a) $a = -40$. (For collinearity, $(60\mathbf{i} + 3\mathbf{j}) - (40\mathbf{i} - 8\mathbf{j}) = \lambda((40\mathbf{i} - 8\mathbf{j}) - (a\mathbf{i} - 52\mathbf{j}))$ for some λ. This gives the system : $20 = (40 - a)\lambda$ and $11 = 44\lambda$.)

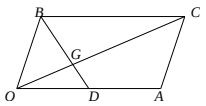
(b) $\alpha = \pm 1, \beta = 1$. (Setting $\begin{vmatrix} 1 & 1 & 1 \\ 4 & 1 & 4 \\ 1 & \alpha & \beta \end{vmatrix} = 0$ gives $\beta = 1$. $|\vec{c}| = \sqrt{3}$ gives $1 + \alpha^2 + \beta^2 = 3$.) (c) c is the G.M. of a and b. (Setting $\begin{vmatrix} a & a & a \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$ gives $c^2 = ab$.) (d) $\lambda = -\frac{37}{4}$. (Set $\overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD}) = 0$.)

13. (a) $\vec{c} = \pm\frac{1}{\sqrt{2}}(\hat{j} - \hat{k})$. (Let $\vec{c} = \alpha\vec{a} + \beta\vec{b} = (2\alpha + \beta)\hat{i} + (\alpha + 2\beta)\hat{j} + (\alpha - \beta)\hat{k}$. $\vec{a} \cdot \vec{c} = 0$ gives $\beta = -2\alpha$ while $|\vec{c}| = 1$ gives $6(\alpha^2 + \beta^2 + \alpha\beta) = 1$.) (b) (A) and (C). (The problem can be done by verifying case by case. But to get the general form of a vector, say $\vec{u}$, which satisfies the conditions in the problem, suppose $\vec{u} = \alpha\vec{b} + \beta\vec{c} = (\alpha + \beta)\hat{i} + (2\alpha + \beta)\hat{j} - (\alpha + 2\beta)\hat{k}$. The condition about the projection of $\vec{u}$ on $\vec{a}$ means $|\vec{u} \cdot \vec{a}| = \sqrt{\frac{2}{3}}|\vec{a}| = 2$. This simplifies to $|\alpha + \beta| = 2$, i.e., to $\alpha + \beta = \pm 2$. In the first case $\vec{u}$ is of the form $2\hat{i} + 2\hat{j} - 4\hat{k} + \alpha(\hat{j} + \hat{k})$ for a scalar α. Taking $\alpha = 1$ gives (A). The case $\alpha + \beta = -2$ gives $\vec{u} = -2\hat{i} - 2\hat{j} + 4\hat{k} + \alpha(\hat{j} + \hat{k})$ for some α. Again, putting $\alpha = 1$ gives (C).)

14. $3\hat{i} - \hat{j} - \hat{k}$ or $-\hat{i} + 3\hat{j} + 3\hat{k}$. (The position vector of the mid-point of BC, say M, is $2\hat{i}$. Since E lies on the line AM, $\overrightarrow{AE} = \lambda\overrightarrow{AM} = \lambda(\hat{i} - \hat{j} - \hat{k})$ for some λ. The area of the face ABC is $\frac{1}{2}|(\hat{j} + \hat{k}) \times 2\hat{i}| = |\hat{j} - \hat{k}| = \sqrt{2}$. As the volume is $\frac{2\sqrt{2}}{3}$, the length of the altitude DE equals $\frac{3}{\sqrt{2}}\frac{2\sqrt{2}}{3} = 2$. As the triangle AED is right-angled at E and $AD = 4$, we have $AE = 2\sqrt{3}$. But $\overrightarrow{AE} = \lambda(\hat{i} - \hat{j} - \hat{k})$. This gives $3\lambda^2 = 12$. So $\lambda = \pm 2$. Hence the possible position vectors of E are $(\hat{i} + \hat{j} + \hat{k}) \pm 2(\hat{i} - \hat{j} - \hat{k})$.) **15.** $\frac{4}{3}$. (Let PQ be the shortest distance with P lying on the line AB and Q on the line CD. Then the vector $\overrightarrow{PQ}$ is perpendicular to both $\overrightarrow{AB}$ and to $\overrightarrow{CD}$ and hence parallel to their cross product, which comes out to be $-20\mathbf{i} + 10\mathbf{j} - 20\mathbf{k}$. Now, $\overrightarrow{AC} = \overrightarrow{AP} + \overrightarrow{PQ} + \overrightarrow{QC}$ and so $\overrightarrow{AC} \cdot \overrightarrow{PQ} = |\overrightarrow{PQ}|^2$. Hence PQ is the length of the projection of $\overrightarrow{AC}$ on the vector $-20\mathbf{i} + 10\mathbf{j} - 20\mathbf{k}$ or equivalently, on the vector $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ which is a little easier to handle. Note that the problem does not require us to identify the points P and Q, but only the distance PQ. Instead of A and C we could have taken any points X and Y on the two lines respectively. Then the projection of $\overrightarrow{XY}$ on $\overrightarrow{PQ}$ would have been the same. If we are also to identify the points P and Q, one way to do it would be to take P as $(1, -2, -1) + s(3, 2, -2)$, i.e., as $(3s + 1, 2s - 2, -2s - 1)$ and Q as $(t + 1, -6t + 2, -4t - 1)$ for some s, t. Since $\overrightarrow{PQ}$ is parallel to $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, we get a system of equations $\frac{3s-t}{2} = \frac{2s+6t-4}{-1} = \frac{4t-2s}{2}$ solving which we get $P = (\frac{21}{9}, -\frac{10}{9}, -\frac{17}{9})$ and $Q = (\frac{13}{9}, -\frac{6}{9}, -\frac{25}{9})$. One can then directly compute the shortest distance as the distance between P and Q.) **16.** The angle between the two

lines is 60° and the equation of the plane containing them is $x+y-z=0$. (Parametric equations of the second line are $x=2t+9, y=-t-4, z=t+5$. Putting these into the equations of the first line we get $(3,-1,2)$ (corresponding to $t=-3$) as the point of intersection of the two lines. So they do lie in the same plane. Let $\mathbf{u},\mathbf{v}$ be vectors parallel to the first and the second line respectively. $\mathbf{v}$ is already given as $2\mathbf{i}-\mathbf{j}+\mathbf{k}$. $\mathbf{u}$ can be taken as $(6\mathbf{i}+4\mathbf{j}-5\mathbf{k})\times(\mathbf{i}-5\mathbf{j}+2\mathbf{k})$ i.e., as $-17\mathbf{i}-17\mathbf{j}-34\mathbf{k}$ or as $\mathbf{i}+\mathbf{j}+2\mathbf{k}$ which is easier to handle. Then $\mathbf{u}\times\mathbf{v}$ is a vector perpendicular to the plane containing the two lines.)

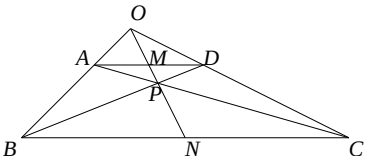
17. Assume BD and CO meet at G. Let $BG:GD=\lambda$ and $CG:GO=\mu$. Letting the origin lie at O, denote the position vectors of A,B,C,D,G by $\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d},\mathbf{g}$ respectively. Then since $\mathbf{d}=\frac{1}{2}\mathbf{a}$ and $\mathbf{c}=\mathbf{a}+\mathbf{b}$, we have $\mathbf{g}=\frac{\lambda\mathbf{d}+\mathbf{b}}{\lambda+1}=\frac{\lambda\mathbf{a}+2\mathbf{b}}{2(\lambda+1)}$ on one hand and $\mathbf{g}=\frac{\mathbf{c}}{\mu+1}=\frac{\mathbf{a}+\mathbf{b}}{\mu+1}$ on the other. Since $\mathbf{a},\mathbf{b}$ are linearly independent, we get the system : $\frac{\lambda}{2(\lambda+1)}=\frac{1}{\mu+1}$ and $\frac{2}{2(\lambda+1)}=\frac{1}{\mu+1}$. Solving, we get $\lambda=2$ and $\mu=2$. (A pure geometry proof is much simpler since the triangles OGD and CGB are similar and $BC=2AD$.)



18. Denote the position vectors of the vertices A,B,C of a triangle by $\mathbf{a},\mathbf{b}$ and $\mathbf{c}$ respectively. Let D,E,F be the mid-points of the sides BC,CA,AB respectively. Then the position vector, say $\mathbf{d}$, of D is $\frac{1}{2}(\mathbf{b}+\mathbf{c})$. Since $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ can be written as $\frac{1}{3}\mathbf{a}+\frac{2}{3}\mathbf{d}$ we see that the point, say G, whose position vector is $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ lies on the median AD and divides it in the ratio $2:1$. By symmetry, the same holds for the other two medians too. (The diagrams for this and the next exercise are too standard. Draw them yourself!) **19.** Let AD,BE,CF be the angle bisectors of the triangle ABC. Assume that BE meets AD at I. Let $\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d},\mathbf{e},\mathbf{f}$ and $\mathbf{i}$ be the position vectors of A,B,C,D,E,F,I respectively. The sides a,b,c of the triangle are then $|\mathbf{b}-\mathbf{c}|,|\mathbf{c}-\mathbf{a}|$ and $|\mathbf{a}-\mathbf{b}|$ respectively. Since D divides BC in the ratio $c:b$, we have $\mathbf{d}=\frac{b\mathbf{b}+c\mathbf{c}}{b+c}$. Further, since $BD=\frac{ca}{b+c}$, I divides AD in the ratio $c:\frac{ca}{c+b}$ which is the same as $b+c:a$. So $\mathbf{i}=\frac{a\mathbf{a}+b\mathbf{b}+c\mathbf{c}}{a+b+c}$. As this expression is symmetric in a,b,c, I also lies on the bisector CF. Replacing a by $|\mathbf{b}-\mathbf{c}|$ etc gives an expression for $\mathbf{i}$ in terms of $\mathbf{a},\mathbf{b},\mathbf{c}$. **20.** Take the origin at O and assume that $OP:PD=\lambda:1$ and $AP:PE=\mu:1$. Let $\mathbf{a},\mathbf{b},\mathbf{d},\mathbf{e}$ be the position vectors of A,B,D,E respectively. Then $\mathbf{d}=\frac{\mathbf{a}+2\mathbf{b}}{3}$ and $\mathbf{e}=\frac{\mathbf{b}}{2}$. Calculations similar to those in the answer to Exercise (21.17), give the system of equations : $\frac{\lambda}{3(\lambda+1)}=\frac{1}{\mu+1}$ and $\frac{2\lambda}{3(\lambda+1)}=\frac{\mu}{2(\mu+1)}$. Solving, $\mu=4$ and $\lambda=\frac{3}{2}$. Hence $OP:PD=\frac{3}{2}:1=3:2$. For a slicker solution, eliminate $\mathbf{b}$ to get $\frac{3}{5}\mathbf{d}=\frac{4}{5}\mathbf{e}+\frac{1}{5}\mathbf{a}$, which implies that each side equals $\mathbf{p}$.

21. Let $ABCD$ be a trapezium whose non-parallel sides AB and CD meet at O. Let M and N be the mid-points of the parallel sides AD and BC respectively. Let the diagonals meet at P. Let the position vec-

tors of A, B, C, D, P, M, N w.r.t. O be $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{p}, \mathbf{m}$ and $\mathbf{n}$ respectively. Then $\mathbf{b} = \lambda\mathbf{a}$ and $\mathbf{c} = \lambda\mathbf{d}$ for some $\lambda \neq 1$. Further $\mathbf{m} = \frac{1}{2}(\mathbf{a} + \mathbf{d})$ and $\mathbf{n} = \frac{\lambda}{2}(\mathbf{a} + \mathbf{d})$. Suppose $AP : PC = \alpha : 1$ and $DP : PB = \beta : 1$. Then equating the two expressions for $\mathbf{p}$ we get a system of two equations in α and β, whose solution is $\alpha = \beta = \frac{1}{1+\lambda}$. This makes $\mathbf{p} = \frac{\lambda}{1+\lambda}(\mathbf{a} + \mathbf{d})$. Thus, $\mathbf{p}, \mathbf{m}$ and $\mathbf{n}$, are all multiples of the vector $\mathbf{a} + \mathbf{d}$. So the points O, M, P, N are all collinear.



22. Following the hint, take $\mathbf{b} = \mathbf{i}$, $\mathbf{c} = \cos\alpha\,\mathbf{i} + \sin\alpha\,\mathbf{j}$ and $\mathbf{d} = \cos\beta\,\mathbf{i} + \sin\beta\,\mathbf{j}$ for some α, β. Also we may assume $\mathbf{w} = \mathbf{k}$. Then by a direct computation, $(\mathbf{w}\ \mathbf{c}\ \mathbf{d}) = \sin(\beta - \alpha)$, $(\mathbf{w}\ \mathbf{d}\ \mathbf{b}) = -\sin\beta$ and $(\mathbf{w}\ \mathbf{b}\ \mathbf{c}) = \sin\alpha$. The result follows by direct substitution.

23. Let the equations of the two lines be $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. (The same line can be represented by many equations obtained from each other by multiplying all the coefficients by a non-zero constant. To avoid this, we may further require that $a_1^2 + b_1^2 + c_1^2 = 1$ and $a_2^2 + b_2^2 + c_2^2 = 1$. Now each line has only two possible equations.) If these two lines are parallel, then $a_1b_2 - a_2b_1 = 0$. We can easily 'perturb' a_1, b_1, c_1 slightly to, say a_1', b_1', c_1' so that $a_1'^2 + b_1'^2 + c_1'^2 = 1$. (For example, by taking $a_1' = a_1 + \epsilon_1, b_1' = b_1 + \epsilon_1$ and $c_1' = \pm\sqrt{1 - a_1'^2 - b_1'^2}$, where the sign is chosen so that c_1' and c_1 do not have opposite signs and ϵ_1 is a real number such that $|\epsilon_1|$ is small. Similarly, we can perturb the second line to a line of the form $a_2'x + b_2'y + c_2' = 0$ using another numerically small number ϵ_2. Since ϵ_1 and ϵ_2 are independent of each other, in general $a_1'b_2' - a_2'b_1'$ will not be zero even though $a_1b_2 - a_2b_1$ is 0. This shows that being parallel is not a stable property. On the other hand, not being parallel is a stable property, because if $a_1b_2 - a_2b_1 \neq 0$, and a_1', a_2', b_1', b_2' are sufficiently close to a_1, a_2, b_1, b_2 respectively, then $a_1'b_2' - a_2'b_1'$ will also be non-zero. A similar argument shows that for three lines in a plane, concurrency is an unstable property, but non-concurrency is a stable property.

24. (a) It suffices to show that $||\mathbf{u} + \mathbf{v}||^2 \leq (||\mathbf{u}|| + ||\mathbf{v}||)^2$. As the L.H.S. is $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = ||\mathbf{u}||^2 + ||\mathbf{v}||^2 + 2\mathbf{u} \cdot \mathbf{v}$, the inequality reduces to the Cauchy−Schwarz inequality. (b) Call $\mathbf{u} - \mathbf{v}$ as $\mathbf{w}$. Then $\mathbf{u} = \mathbf{v} + \mathbf{w}$ and so by (a), $||\mathbf{u}|| \leq ||\mathbf{v}|| + ||\mathbf{w}||$. So $||\mathbf{u}|| - ||\mathbf{v}|| \leq ||\mathbf{w}||$. But since $\mathbf{v} = \mathbf{u} + (-\mathbf{w})$ and $|| - \mathbf{w}|| = ||\mathbf{w}||$, by (a) we also have $||\mathbf{v}|| - ||\mathbf{u}|| \leq ||\mathbf{w}||$. These two inequalities together imply the result since $|\ ||\mathbf{u}|| - ||\mathbf{v}||\ |$ has to equal either $||\mathbf{u}|| - ||\mathbf{v}||$ or $||\mathbf{v}|| - ||\mathbf{u}||$.

25. The proof of distributivity of addition over multiplication requires the distributivity of the dot and the cross products, i.e., the facts that for any three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ and similarly, $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$. (In addition, we of course also need other things like the

distributivity of the scalar multiplication. But these properties are used too frequently to be mentioned explicitly every time.) The only non-trivial verification is that of the associativity of quaternionic multiplication. For this, let $u = (u_0, \mathbf{u}), v = (v_0, \mathbf{v}), w = (w_0, \mathbf{w})$ be three quaternions. Then

$$\begin{aligned} u(vw) &= (u_0, \mathbf{u})(v_0w_0 - \mathbf{v}\cdot\mathbf{w}, v_0\mathbf{w} + w_0\mathbf{v} + \mathbf{v}\times\mathbf{w}) = (a_0, \mathbf{a}) \\ \text{where } a_0 &= u_0v_0w_0 - u_0(\mathbf{v}\cdot\mathbf{w}) - \mathbf{u}\cdot(v_0\mathbf{w} + w_0\mathbf{v} + \mathbf{v}\times\mathbf{w}) \\ &= u_0v_0w_0 - u_0(\mathbf{v}\cdot\mathbf{w}) - v_0(\mathbf{w}\cdot\mathbf{u}) - w_0(\mathbf{u}\cdot\mathbf{v}) - (\mathbf{u}\ \mathbf{v}\ \mathbf{w}) \\ \text{and } \mathbf{a} &= u_0v_0\mathbf{w} + v_0w_0\mathbf{u} + w_0u_0\mathbf{v} - (\mathbf{v}\cdot\mathbf{w})\mathbf{u} \\ &\quad + u_0\mathbf{v}\times\mathbf{w} + v_0\mathbf{u}\times\mathbf{w} + w_0\mathbf{u}\times\mathbf{v} + \mathbf{u}\times(\mathbf{v}\times\mathbf{w}) \end{aligned}$$

By a similar calculation, $(uv)w = (b_0, \mathbf{b})$, where

$$\begin{aligned} b_0 &= u_0v_0w_0 - w_0(\mathbf{u}\cdot\mathbf{w}) - v_0(\mathbf{u}\cdot\mathbf{w}) - v_0(\mathbf{u}\cdot\mathbf{w}) - (\mathbf{w}\ \mathbf{u}\ \mathbf{v}) \\ \text{and } \mathbf{b} &= u_0v_0\mathbf{w} + v_0w_0\mathbf{u} + w_0u_0\mathbf{v} - (\mathbf{u}\cdot\mathbf{v})\mathbf{w} \\ &\quad + u_0\mathbf{v}\times\mathbf{w} + v_0\mathbf{u}\times\mathbf{w} + w_0\mathbf{u}\times\mathbf{v} + (\mathbf{u}\times\mathbf{v})\times\mathbf{w} \end{aligned}$$

Equality of a_0 and b_0 follows from that of $(\mathbf{u}\ \mathbf{v}\ \mathbf{w})$ and $(\mathbf{w}\ \mathbf{u}\ \mathbf{v})$. To see that $\mathbf{a} = \mathbf{b}$, expand $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})$ as $(\mathbf{u}\cdot\mathbf{w})\mathbf{v} - (\mathbf{u}\cdot\mathbf{v})\mathbf{w}$ and $(\mathbf{u}\times\mathbf{v})\times\mathbf{w} = -\mathbf{w}\times(\mathbf{u}\times\mathbf{v})$ as $(\mathbf{w}\cdot\mathbf{u})\mathbf{v} - (\mathbf{w}\cdot\mathbf{v})\mathbf{u}$. This proves that the quaternionic multiplication is associative. To show that it is not commutative, note for example, that $(0, \mathbf{i})(0, \mathbf{j}) = (0, \mathbf{k})$ but $(0, \mathbf{j})(0, \mathbf{i}) = (0, -\mathbf{k})$.

26. (a) to (c) are routine verifications. For (d), a succinct proof is possible. Note that $N(uv) = \sqrt{(uv)\overline{uv}} = \sqrt{(uv)(\bar{v}\bar{u})} = \sqrt{u(v\bar{v})\bar{u}}$ by associativity of multiplication. Now, even though the multiplication is not commutative, the quaternion $v\bar{v}$ is a real number and hence commutes with every quaternion. So, $u(v\bar{v})\bar{u} = (v\bar{v})(u\bar{u})$. But $v\bar{v}$, $u\bar{u}$ are non-negative real numbers and hence they commute with each other. So ultimately, $N(uv) = \sqrt{(u\bar{u})}\sqrt{(v\bar{v})} = N(u)N(v)$. To prove (e), suppose that $u = (u_0, \mathbf{u})$ is a non-zero quaternion. Then $u\bar{u}$ is a positive real number. Take v to be the quaternion $\frac{1}{u\bar{u}}\bar{u}$, i.e., the quaternion $(\frac{u_0}{u\bar{u}}, \frac{-\mathbf{u}}{u\bar{u}})$. Then a direct computation gives that $uv = vu = 1$. (Again, this is very analogous to saying that the complex number $\dfrac{x}{x^2+y^2} - i\dfrac{y}{x^2+y^2}$ is the reciprocal of $x + iy$, or as ordered pairs, $\dfrac{1}{(x,y)} = (\dfrac{x}{x^2+y^2}, \dfrac{-y}{x^2+y^2})$.)

27. Consider the quaternions $u = (1, \mathbf{u})$ and $v = (1, \mathbf{v})$. Then uv is the quaternion $(1 - \mathbf{u}\cdot\mathbf{v}, \mathbf{u} + \mathbf{v} + \mathbf{u}\times\mathbf{v})$. By (d) of the last exercise, $(N(uv))^2 = (N(u))^2(N(v))^2$ which implies the result from the definition of the norm. For a direct proof, note that $|\mathbf{u} + \mathbf{v} + \mathbf{u}\times\mathbf{v}|^2 = (\mathbf{u} + \mathbf{v} + \mathbf{u}\times\mathbf{v})\cdot(\mathbf{u} + \mathbf{v} + \mathbf{u}\times\mathbf{v}) = |\mathbf{u}|^2 + |\mathbf{v}|^2 + |\mathbf{u}\times\mathbf{v}|^2 + 2\mathbf{u}\cdot\mathbf{v}$ since $\mathbf{u}\cdot(\mathbf{u}\times\mathbf{v})$ and $\mathbf{v}\cdot(\mathbf{u}\times\mathbf{v})$ are both 0. The result follows by expanding both the sides and noting that $|\mathbf{u}|^2\ |\mathbf{v}|^2 = (\mathbf{u}\cdot\mathbf{v})^2 + |\mathbf{u}\times\mathbf{v}|^2$.

28. (a) Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ be a unit vector parallel to L. As $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ is an orthonormal basis, we have $u_1 = \mathbf{u} \cdot \mathbf{i} = \cos\alpha$. Similarly for β, γ. Given two of these angles, say α and β, γ is given by $\cos\gamma = \pm\sqrt{1 - \cos^2\alpha - \cos^2\beta}$ which determines γ upto a supplementary angle. (b) $\cos^{-1} \dfrac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{l_2^2 + m_2^2 + n_2^2}}$. (c) (i) $\dfrac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$ (Let $P_0 = (x_0, y_0, z_0)$ be the foot of the perpendicular from P_1 to the plane. The direction ratios of P_0P_1 are (a, b, c). Hence $x_1 - x_0 = \frac{ar}{\sqrt{a^2+b^2+c^2}}$ etc. for some r. Also $a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0) = 0$. The deisred distance is $|r|$.) (ii) $\sqrt{A - B^2}$ where $A = (x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2$ and $B = l(x_1 - x_0) + m(y_1 - y_0) + n(z_1 - z_0)$. (Let the desired distance be d. $\mathbf{u} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$ is a unit vector along L. Resolving $\overrightarrow{P_0P_1}$ along and perpendicular to L, $d^2 = |\overrightarrow{P_0P_1} - (\overrightarrow{P_0P_1} \cdot \mathbf{u})\mathbf{u}|^2 = |\overrightarrow{P_0P_1}|^2 - (\overrightarrow{P_0P_1} \cdot \mathbf{u})^2$.) (d) If the position vectors of the vertices are $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{d}$, then each segment passes through the point $\frac{1}{4}(\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d})$ and is divided in the ratio $3 : 1$ at this point. (e) From the hint, the four altitudes are concurrent if and only if there exist scalars $\alpha, \beta, \gamma, \delta$ such that

$$(*)\quad \mathbf{a} + \alpha(\mathbf{b} \times \mathbf{c}) = \mathbf{b} + \beta(\mathbf{c} \times \mathbf{a}) = \mathbf{c} + \gamma(\mathbf{a} \times \mathbf{b}) = \delta(\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b})$$

If this holds then taking the dot products of the first three expression with $\mathbf{a}, \mathbf{b}, \mathbf{c}$ gives $\mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{b}$ which is easily seen to be equivalent to the given condition. Conversely, if this condition holds, let $U = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{b}$ and $V = [\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$. Take $\delta = \dfrac{U}{V}, \alpha = \dfrac{U - ||\mathbf{a}||^2}{V}, \beta = \dfrac{U - ||\mathbf{b}||^2}{V}$ and $\gamma = \dfrac{U - ||\mathbf{c}||^2}{V}$. With these values, each of the first three expressions in (*) equals the last one as can be seen by applying Exercise (21.7)(g).

29. Comparing the coefficients of $\vec{i}, \vec{j}, \vec{k}$, the given equation is equivalent to the system of three homogeneous linear equations :

$$(1 - \lambda)x + 3y - 4z = 0,\quad x - (3 + \lambda)y + 5z = 0 \quad \text{and} \quad 3x + y - \lambda z = 0.$$

By Theorem 8 in Chapter 3, this system has a non-trivial solution if and only if $\Delta = 0$, where Δ is the determinant $\begin{vmatrix} 1-\lambda & 3 & -4 \\ 1 & -3-\lambda & 5 \\ 3 & 1 & -\lambda \end{vmatrix}$. By direct expansion, $\Delta = -\lambda^3 - 2\lambda^2 - \lambda = -\lambda(\lambda+1)^2$. Hence $\Delta = 0$ for $\lambda = 0$ and $\lambda = -1$. So these are the values for which the given equation has a non-trivial solution. (Although not asked in the problem, it is instructive to identify these solutions. For $\lambda = 0$, the solutions are all multiples of $x = 1, y = -3, z = -1$. It is tempting to think that since $\lambda = -1$ is a double root, for this value of λ, there will be two linearly independent solutions. But that is not always the case. For example, in the present problem, for $\lambda = -1$, the only solutions of the system are all multiples of the solution $x = 1, y = -2, z = -1$.)

Chapter 22

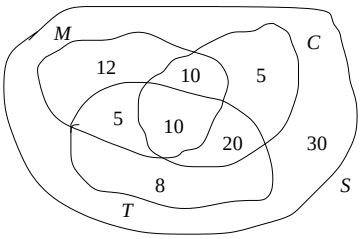
1. $\frac{1}{1260}$ $(= \frac{2!4!3!}{9!})$. **2.** (i) yes (ii) no. (x and y can occur together.)

3. $1-(\frac{1}{4})^{10}$ and $(\frac{3}{16})^{10}$. (For the first part, take the complementary probability. For the second, note that for each question the probability that both the students answer it identically and wrongly is $\frac{3}{16}$.)

4. (a) 0.39. (Use Equation (5) and take complementary probability.) (b) 1.2 $(= 2-(P(A)+P(B)) = 2-(P(A\cup B)+P(A\cap B)) = 2-(0.6+0.2))$. (c) (A), (B), (C) are true. ((C) is just Equation (5). (A) and (B) follow from (C) and the fact that $0 \le P(A\cup B) \le 1$.) (d) (A) and (D). (From the data, $P(E\cap F) = \frac{1}{12}$ and $P(E\cup F) = 1-P(E'\cap F') = 1-\frac{1}{2} = \frac{1}{2}$. Hence by (5), $P(E)+P(F) = \frac{1}{12}+\frac{1}{2} = \frac{7}{12}$.) (e) (B), (C) and (D). ((B) and (C) are obvious by common sense. For a mathematical proof, $P(E\cap F^c) = P(E)-P(E\cap F) = P(E)(1-P(F)) = P(E)P(F^c)$. For (C), note that by complementary probability and Equation (5), $P(E^c\cap F^c) = 1-(P(E)+P(F)+P(E\cap F)) = (1-P(E))(1-P(F)) = P(E^c)P(F^c)$. For (D) write the L.H.S. as $\frac{1}{P(F)}(P(E\cap F)+P(E^c\cap F))$ and use the linear independence of E and F and hence of E^c and F. But once again, the result is obvious by common sense too because given the occurrence of F, either E occurs or E^c occurs but not both.) (f) 0.25. (From the data, $P(A) = 1-0.3 = 0.7$ and $P(B^c) = 0.6$. So $P(A\cup B^c) = P(A)+P(B^c)-P(A\cap B^c) = 0.8$. Also $B\cap(A\cup B^c)$ is the same event as $A\cap B$. So $P(B\cap(A\cup B^c)) = P(A\cap B) = P(A)-P(A\cap B^c) = 0.7-0.5 = 0.2$.) (g) (A) and (D). (Essentially same as (D) of Part (e) above. (B) and (C) imply the equalities $P(EF) = P(F)P(\bar{F})$ and $P(\bar{E}F)P(\bar{F})+P(E\bar{F})P(F) = P(F)P(\bar{F})$. As nothing is given about the relationship between E and F, these equalities need not hold in general.) (h) (C). (Apply DeMorgan's laws or common sense.) (i) No (unless it is given that both A and B can never fail simultaneously.)

5. The necessary and sufficient condition is that the events A and B are identical, in the sense that when either one of them occurs so does the other. Strictly speaking this is a relationship between the events and not between their probabilities. In fact, no relationship between the numbers $P(A)$ and $P(B)$ can characterise the statement $P(A\cup B) = P(A\cap B)$.

6. (a) 30 students. (Let M, C, T also denote the respective subsets of the set, say S, of all 100 students interviewed. Subtract $|M\cap C\cap T|$ from $|M\cap C|$ to get $|M\cap C\cap T'| = 10$. Similarly fill the other regions of the Venn diagram with the cardinalities of the respective subsets. The total 70 equals $|M\cup C\cup T|$. So $|S-(M\cup C\cup T)| = 100-70 = 30$. (Equivalently, use the principle of inclusion and exclusion.)



(b) From the principle of inclusion and exclusion (for probabilities),

$$P(A \cup B \cup C) = P(A)+P(B)+P(C)-P(AB)-P(BC)-P(AC)+P(ABC).$$

Filling in the given values, we have $P(BC) = 1.23 - P(A \cup B \cup C)$. The result follows from the fact that $0.75 \leq P(A \cup B \cup C) \leq 1$. (Actually, the data is inconsistent since $P(ABC)$ can never exceed a$P(AB)$ as is given.)
7. 0.784 ($= 1 - (0.6)^3$. Take complementary probability.)

8. (a) $\frac{1}{14}$ ($= \frac{60}{840}$). (Imitate the solution to the problem at the beginning of Chapter 1, Comment No. 5. First arrange the remaining letters in 12 possible ways. Then put the four S's in 5 possible places.) (b) $\frac{7}{15}$ ($= \frac{56}{120}$, by similar reasoning). (c) $\frac{1}{924}$ ($= \frac{26\times 7!\times 6!}{14!}$) and $\frac{3}{1001}$ ($= \frac{36\times 6!\times 6!}{13!}$). (In each case, count the number of favourable arrangements by classifying them according to the location of the red ball.)

9. (a) $\frac{1}{36}$ ($= \frac{6}{6\times 6\times 6}$). (b) $\frac{16}{81}$ ($= (\frac{2}{3})^4$. Each score has to lie between 2 to 5. The probability of this is $\frac{2}{3}$.) (c) $(\frac{3}{5})^7 - (\frac{8}{15})^7$. (First find the probability that no number exceeds 9. Then subtract the probability that no number exceeds 8. It is assumed here that the problem means that a number greater than 9 never occurs and that the number 9 must occur at least once.) (d) $\frac{11}{40}$. (Let S be the set of all 3-selections from $\{1, 2, \ldots, 10\}$. Then $|S| = \binom{10}{3} = 120$. Let A be the set of those 3-selections in which the minimum chosen element is 3 and B be the set of those in which the maximum chosen element is 7. Then $|A| = \binom{7}{2} = 21$, $|B| = \binom{6}{2} = 15$ and $|A \cap B| = 3$. Hence $|A \cup B| = 21 + 15 - 3 = 33$.)

10. $\frac{5}{21}$ ($= \frac{0.1}{0.1+0.32}$, by taking the conditional probability).

11. $\frac{5}{9}$. (Because of the second statement, we might as well suppose that we have only 10 tickets numbered 1 to 10 to begin with. For a formal argument, take conditional probability, letting B be the event that neither of the chosen tickets has a number exceeding 10 and A be the event that one of the tickets bears the number 5 and the other ticket bears a higher number. Then $P(B) = \dfrac{\binom{10}{2}}{\binom{100}{2}}$ and $P(AB) = \frac{25}{\binom{100}{2}}$. So $P(A|B) = \frac{25}{\binom{10}{2}}$.)

12. $\frac{1}{3} \leq p \leq \frac{1}{2}$. (The requirement that each of the three expressions is non-negative forces p to lie in $[-\frac{1}{3}, \frac{1}{2}]$. Also the sum of the three expressions is at most 1, being the probability of some event. This gives $4(1 + 3p) + 3(1 - p) + 6(1 - 2p) \leq 12$, i.e., $p \geq \frac{1}{3}$.)

13. (C). (Let A, B, C denote the events of passing tests I, II, III respectively. Then $P(AB+AC)$ is given to be $\frac{1}{2}$. But (assuming A, B, C to be mutually independent), $P(AB+AC) = P(AB)+P(AC)-P(ABC) = pq+\frac{1}{2}p-\frac{1}{2}pq$.)

14. (B) and (C). (Subtracting the third parobability from the second we get $pmc = \frac{1}{10}$. Also, the third part of the data gives $pm(1 - c) + mc(1 - p) +$

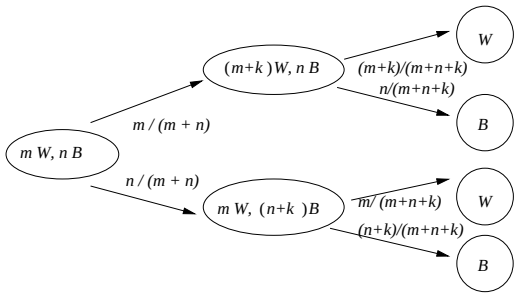
$pc(1-m) = \frac{1}{10}$, i.e., $pm+mc+pc-3pmc = \frac{4}{10}$. Hence $pm+mc+pc = \frac{7}{10}$. The first part of the data gives $p+m+c-(pm+mc+pc)+pmc = \frac{3}{4}$. As the values of pmc and of $pm+mc+pc$ are already known, we get the value of $p+m+c$. Note that it is not possible from the data to determine the individual values of p, m and c. Nor are they needed. All we can say is that they are roots of the cubic $x^3 - \frac{27}{20}x^2 + \frac{7}{10}x - \frac{1}{10} = 0$. Of course, this is not needed either in the present problem. But it is always a good idea to determine just how much you can get from the data.)

15. $\frac{\binom{N+5}{5} + 2\binom{N}{4} + 11\binom{N}{3} + 25\binom{N}{2} + 20\binom{N}{1}}{\binom{N+7}{5}}$. (The data may appear incomplete and hence baffling at first sight since it is not given how many of the N coins are of 10 paise and how many are of 5 paise value. But the requirements that only five coins are to be chosen and that their total value is to be less than 1 rupee and fifty paise severely restrict the choice. In all there are $\binom{N+7}{5}$ ways to select 5 coins from the given $N+7$ coins. Let a and b be the numbers of the fifty and twenty five paise coins in the selection. If $a = 0$ then the condition about the total value will hold no matter how the remaining 5 coins are chosen and this choice can be done in $\binom{N+5}{5}$ ways. If $a = 1$ then b can be at most 3 and the remaining $5-a-b$ small coins can be chosen arbitrarily. So the total number of selections of this type is $2\binom{N}{4} + 10\binom{N}{3} + 20\binom{N}{2} + 20\binom{N}{1}$ ways, where the coefficients of the terms indicate the number of ways to choose from the two larger types of coins. Finally, suppose $a = 2$. Then b has to be either 0 or 1 and so the number of selections of this type is $\binom{N}{3} + 5\binom{N}{2}$.)

16. (a) $\frac{3}{38}$. (There are $\binom{20}{3}$ selections in all. If the three numbers are in an A.P. then either the first two lie in the set $\{1, 2, \ldots, 10\}$ and uniquely determine the third or else the last two lie in the set $\{11, 12, \ldots, 20\}$ and uniquely determine the first. The number of selections of each type is $\binom{10}{2}$. The key idea in this problem is to classify the progressions into two classes according to the location of the middle term.) (b) 0.62. (The ordered pair (p, q) can be chosen in 100 ways. The condition about real roots implies $p^2 \geq 4q$, i.e., $q \leq \frac{p^2}{4}$. But since q is an integer from 1 to 10, we further have $1 \leq q \leq \min\{[\frac{p^2}{4}], 10\}$, where $[x]$ denotes the integral part of x. By direct calculation, for $p = 1, 2, 3, 4, 5$ and 6, the possible numbers of values of q are $0, 1, 2, 4, 6$ and 9 respectively, while for $p = 7, 8, 9, 10$, q can be any integer from 1 to 10. So the total number of favourable cases is 62.)

17. $\frac{13}{32}$. (Denote the selections by ordered triples with entries b and w, where b denotes a black ball and w denotes a while ball. There are three cases: $(w, w, b), (w, b, w)$ and (b, w, w). The probability of the first case is $\frac{3}{4} \times \frac{1}{2} \times \frac{3}{4} = \frac{9}{32}$. Similarly compute the probabilities of the other two cases and add. See (i) in Comment No. 4 of Chapter 24 for a diagram which explains how the solution is arrived at.)

18. $\frac{m}{m+n}$. (From the tree diagram shown, the desired probability is the sum $\frac{m}{m+n}\frac{m+k}{m+n+k} + \frac{n}{m+n}\frac{m}{m+n+k} = \frac{m(m+k)+mn}{(m+n)(m+n+k)}$. It is surprising that the answer is independent of k and is the same as the probability that the first ball drawn is white. In other words, the additional balls put do not affect the probability!)



19. $\frac{9m}{m+8N}$. (The sample space, say S, consists of 8 ordered triplets of the form (X, Y, Z), where X has two possible values, say F and B depending upon whether the coin drawn is fair or biased and Y, Z denote the outcomes of the first two tosses with H for a head and T for a tail as usual. The triplets (F, H, T) and (B, H, T) occur with probabilities $\frac{m}{N} \times \frac{1}{2} \times \frac{1}{2} = \frac{m}{4N}$ and $\frac{N-m}{N} \times \frac{2}{3} \times \frac{1}{3} = \frac{2(N-m)}{9N}$. The desired probability is the conditional probability that $X = F$ (i.e., that the coin is fair) given that one of these two triplets occur. This conditional probability, therefore, equals the ratio of $\frac{m}{4N}$ to the sum $\frac{m}{4N} + \frac{2(N-m)}{9N}$.) **20.** Imitate the solution to the problem solved in Comment No. 14. For $n \geq 3$, a sequence of H's and T's of length n in which no three consecutive entries are H must begin with a t or with HT or with HHT. Hence the recurrence relation for p_n is $p_n = \frac{1}{4}p_{n-1} + \frac{3}{16}p_{n-2} + \frac{9}{64}p_{n-3}$ for $n \geq 4$ with the initial conditions $p_1 = p_2 = 1$ and $p_3 = 1 - (\frac{3}{4})^3 = \frac{37}{64}$. With this recurrence relation, one can verify by induction that for every $n \geq 1$, $p_n = (\frac{3}{4})^{n+1}(2 + \alpha^{n+1} + \beta^{n+1})$. The cases $n = 1, 2, 3$ are verified using $\alpha + \beta = -\frac{2}{3}$ and $\alpha\beta = \frac{1}{3}$. For the inductive step, it is a little more convenient to write $(\frac{3}{4})^{n+1}(2 + \alpha^{n+1} + \beta^{n+1})$ as $2(\frac{3}{4})^{n+1} + \gamma^{n+1} + \delta^{n+1}$, where $\gamma = \frac{3\alpha}{4}$ and $\delta = \frac{3\beta}{4}$. Then γ and δ are roots of the quadratic $16x^2 + 8x + 3 = 0$ and hence also of the cubic $(4x - 3)(16x^2 + 8x + 3) = 0$. Expanding this cubic gives $\gamma^3 = \frac{1}{4}\gamma^2 + \frac{3}{16}\gamma + \frac{9}{64}$. So, for every $n \geq 3$, $\gamma^{n+1} = \frac{1}{4}\gamma^n + \frac{3}{16}\gamma^{n-1} + \frac{9}{64}\gamma^{n-2}$ and a similar relationship holds for δ. It is now easy to verify by induction that $p_n = 2(\frac{3}{4})^{n+1} + \gamma^{n+1} + \delta^{n+1}$. **21.** $\frac{32}{55}$. (Let R stand for a red ball and B for a black ball. After any transfer, the contents of the urn A determine those of urn B. So it suffices to draw a tree diagram showing only the contents of the urn A. This diagram is very similar to that in the answer to Exercise (22.18) except that the contents of the first 'ellipse' are $(6R, 4B)$, those of the ellipses in the next column are $(5R, 4B)$ and $(6R, 3B)$ respectively and those of the 'circles' in the last column are $(6R, 4B), (5R, 5B), (6R, 4B)$ and $(7R, 3B)$ respectively. Calculating the probabilities representing the various arrows, we get the desired probability as the sum $\frac{3}{5} \times \frac{5}{11} \times \frac{6}{10} + \frac{3}{5} \times \frac{6}{11} \times \frac{5}{10} + \frac{2}{5} \times \frac{7}{11} \times \frac{6}{10} + \frac{2}{5} \times \frac{4}{11} \times \frac{7}{10} = \frac{90+90+84+56}{550}$.) **22.** $\frac{1}{3}$ (and not

$\frac{1}{6}$ as might appear at first sight. The catch is that the faulty machines get identified even when the first two machines tested are good.) **23.** The twentieth person. (p_n, the probability that the n-th person gets a free ticket, equals $\frac{(n-1)\times 364\times 363\times \ldots \times (367-n)}{(365)^n}$. A direct comparison gives that $p_n < p_{n+1}$ if and only if $n^2 - n < 365$. So p_n increases till $n = 20$ and decreases thereafter.)

Chapter 23

1. (B). (The three series are geometric series with common ratios $\cos^2\phi, \sin^2\phi$ and $\sin^2\phi\cos^2\phi$ respectively, all of whixh are numerically less than 1 since $0 < \phi < \frac{\pi}{2}$. So, by Equation (5) in the text, $x = \frac{1}{1-\cos^2\phi} = \frac{1}{\sin^2\phi}$, $y = \frac{1}{1-\sin^2\phi} = \frac{1}{\cos^2\phi}$ and $z = \frac{1}{1-\sin^2\phi\cos^2\phi}$. So $z = \dfrac{1}{1-\frac{1}{xy}} = \frac{xy}{xy-1}$.)

2. $\frac{2355}{999}$. (Denote the given number by α. The decimal expansion of α is $2.357357357357\ldots$. That is, α is the sum of the infinite series $2+\frac{3}{10}+\frac{5}{10^2}+\frac{7}{10^3}+\frac{3}{10^2}+\frac{5}{10^5}+\frac{7}{10^6}+\frac{3}{10^7}+\ldots$, or equivalently, of the infinite series $2+\frac{357}{1000}+\frac{357}{(1000)^2}+\frac{357}{(1000)^3}+\ldots$ which, from the second term onwards, is a geometric series with common ratio $\frac{357}{1000}$ and the first term also equal to $\frac{357}{1000}$. By (5), $\alpha = 2+\dfrac{\frac{357}{1000}}{1-\frac{357}{1000}} = 2+\frac{357}{999}$. An alternate solution can be given by writing α as $2+\beta$, where $\beta = 0.\overline{357} = 0.357357357357\ldots$. Multiplying a decimal number by 1000 shifts the decimal points by three places to the right. So $1000\beta = 357.357357357\ldots$ which is nothing but $357+\beta$. Hence $1000\beta = 357+\beta$ from which we get β and then α. This alternate solution is often given in high schools and is quite popular. Its fallacy is that the rule about shifting of the decimal point on multiplication by powers of 10, which is valid for terminating decimals has been extended without justification to an infinite decimal. A valid justification comes only after recognising an infinite decimal as an infinite series.) **3.** The angle between the two hands is 120° at 4.00 p.m. and decreases at the rate of $(6-\frac{1}{2})^\circ$ per minute. So the two hands will coincide at $\frac{120}{11/2}$, i.e., $\frac{240}{11}$ or 21.818 ... minutes past 4.00 p.m. To get the answer using infinite series, it takes 20 minuets for the minute hand to come to where the hour hand was initially (i.e., at 4.00 p.m.). By that time the hour hand moves some distance to cover which the minute hand needs $20\times\frac{1}{12}$ minutes, since the minute hand is 12 times as fast as the hour hand. This process continues and gives an infinite geometric series $20+20.\frac{1}{12}+20.\frac{1}{144}+\ldots$, the sum of which is $\frac{20}{1-\frac{1}{12}}$, i.e., $\frac{240}{11}$. **4.** When the fly is moving towards B, the distance between the fly and B decreases at the rate of 24 km/hr, and the fly covers $\frac{20}{24}$th of their initial distance, and during this time A moves by $\frac{5}{24}$th of that distance. A similar calculation holds when the fly starts moving towards A. So in the first round trip, the fly covers a distance of $(\frac{20}{24}+\frac{15}{24}.\frac{20}{25})$ i.e.,

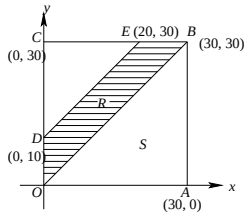
$\frac{4}{3}$ km and at the end of it A and B are at a distance $\frac{15}{24} - (\frac{15}{24}.\frac{5}{25} + \frac{15}{24}.\frac{4}{25})$, i.e., $\frac{2}{5}$ km. Applying the same reasoning to the subsequent round trips of the fly, the total distance travelled by it is an infinite geometric series with the first term $\frac{4}{3}$ and common ratio $\frac{2}{5}$. So its sum is $\frac{20}{9}$. **5.** The mathematicians will meet after $\frac{1}{9}$ hr. The fly, which never rests, travels $\frac{20}{9}$ km in this time. **6.** A typical term of the product of the n factors $(1+x+x^2+x^3+\ldots)\times(1+x+x^2+x^3+\ldots)\times\ldots\times(1+x+x^2+x^3+\ldots)$ is of the form $x^{k_1}x^{k_2}\ldots x^{k_n}$, where $k_1, k_2, \ldots, k_n$ are non-negative integers. This will equal x^r if and only if $k_1+k_2+\ldots+k_n = r$. It follows that for a fixed non-negative integer r, the coefficient of x^r in the product above is the number of ways of splitting r as a sequence of length n of non-negative integers adding up to r. But this is also precisely the number of putting r identical balls into n distinct boxes. (Specifically, the placement in which the i-th box contains k_i balls for $i = 1, 2, \ldots, n$ corresponds to the term $x^{k_1+k_2+\ldots+k_n}$ of the product above.) By the solution to Exercise (1.17), the number of such placements is also the number of ways to select r objects from n types of objects with repetitions allowed freely. **7.** We have $a_0+a_1x+a_2x^2+\ldots+a_rx^r+\ldots = (1-x)^{-n}$. But, by the binomial theorem, we have $(1-x)^{-n} = b_0 + b_1x + b_2x^2 + \ldots + b_rx^r + \ldots$, where for $r \geq 0$, $b_r = (-1)^r\frac{(-n)(-n-1)\ldots(-n-r+1)}{r!} = \frac{n(n+1)(n+2)\ldots(n+r-1)}{r!} = \binom{n+r-1}{r}$. **8.** In any selection of the given type, there will be k objects from the $2n$ unlike objects for a unique k, $0 \leq k \leq n$. The number of ways to choose these k objects is $\binom{2n}{k}$ which is precisely the coefficient of x^k in $(1+x)^{2n}$. Now the remaining $n-k$ objects from the n like objects can be chosen in only one way, because as the objects are identical it does not matter which ones you choose, all that matters is how many you choose. Hence for every k from 0 to n, the number of ways to choose $n-k$ objects from the n like objects is the coefficient of x^{n-k} in $1+x+x^2+\ldots+x^n$. Multiplying together, the coefficient of x^n in the product $(1+x)^{2n}(1+x+x^2+x^3+\ldots+x^n)$ is the total number of selections of the given type. Now, even if we replace the factor $1+x+x^2+\ldots+x^n$ by the infinite series $1+x^2+\ldots+x^n+x^{n+1}+\ldots$, the coefficient of x^n in the product is unaffected because the extra terms contribute nothing to it. **9.** Rewriting $\frac{(1+x)^n}{1-x}$ as $(1+x)^n(1+x+x^2+x^n+x^{n+1}+\ldots$ it follows, by essentially the same reasoning as in the last exercise, that the coefficient of x^n in it is the sum of the coefficients of the various powers of x in the expansion of $(1+x)^n$. But this sum is precisely 2^n by Equation (22) in Chapter 5 (and can also be seen by putting $x = 1$ in the expansion of $(1+x)^n$). For the second part, the coefficient of x^n in $\frac{(1+x)^{2n+1}}{1-x}$ is the sum $\sum_{k=0}^{n}\binom{2n+1}{k}$ which equals 2^{2n}, i.e., 4^n by (28) in Chapter 5. **10.** (i) The trick is to write 2^r as $2^n \times (\frac{1}{2})^{n-r}$ and thereby recognise $2^r\binom{2n-r}{n}$ as $2^n\times$ the coefficient of x^n in $(x+\frac{1}{2})^{2n-r}$. The given sum then equals 2^nS, where S is the coefficient of x^n in the sum $\sum_{r=0}^{n}(x+\frac{1}{2})^{2n-r}$. This is

a geometric series which simplifies to $\dfrac{2(1+2x)^n - \frac{1}{2^n}(1+2x)^{2n+1}}{2^n(1-2x)}$. Put $y = 2x$. Then by the last exercise, $2^n S$ equals $2.2^n 2^n - \frac{1}{2^n}.2^n.4^n = 4^n$. (ii) This is similar but more straightforward. The given sum, say S, is $\sum_{r=0}^{n-1} \binom{(n-1)+r}{n-1}(\frac{1}{2})^r$ which is the coefficient of x^{n-1} in $\sum_{r=0}^{n-1}(x+\frac{1}{2})^{(n-1)+r}$. The answer comes by summing this geometric series, putting $y = 2x$ and applying the last exercise. (iii) results from (ii) if we replace $n-1$ by n. **11.** (a) $\frac{n}{2}$. (Classifying the subsets of S according to their cardinaliries, the sum $\sum_{A \subset S} |A|$ equals $\sum_{k=0}^{n} k\binom{n}{k}$ which equals $n2^{n-1}$ by (33) in Chapter 5. An easier way to find this sum is to pair off each subset with its complement. Together they have n elements and there are $\frac{2^n}{2}$ such pairs.) (b) Let k be the least element of an r-subset A of $S = \{1, 2, \ldots, n\}$. Then $k \leq n - r + 1$ and the remaining $r-1$ elements come from the set $\{k+1, k+2, \ldots, n\}$. It follows that for every $k = 1, 2, \ldots, (n-r+1)$, there are exactly $\binom{n-k}{r-1}$ r-subsets of S which have k as their least element. Hence the expected value of the function f is $\frac{1}{\binom{n}{r}} \sum_{k=1}^{n-r+1} k\binom{n-k}{r-1}$. Using (2) of Chapter 5 (with a slight change of notations) the sum comes out as $\binom{n+1}{r+1} = \frac{n+1}{r+1}\binom{n}{r}$. For a much slicker solution, for an $(r+1)$-subset B of the set $\{0, 1, 2, \ldots, n\}$, let $g(B)$ be the set obtained from B by removing its least element. Then $g(B)$ is an r-subset of $S = \{1, 2, \ldots, n\}$. The function g so defined associates to each $(r+1)$-subset of $\{0, 1, \ldots, n\}$ an r-subset of S. If A is any r-subset of S with least element k, then A has precisely k pre-images under g, viz., the sets obtained by adding to A any one of the elements $0, 1, \ldots, k-1$. It follows that the sum $\sum_{A \subset S} f(A)$ as A varies over all r-subsets of S is the number of $(r+1)$-subsets of the set $\{0, 1, 2, \ldots, n\}$. **12.** (i) 13, (ii) $\frac{53}{5}$. ((i) is essentially part (ii) of the first problem of Comment No. 8 with $p = \frac{4}{52}$. For (ii), simplify by replacing the aces with white balls and the other cards by black balls. More generally, suppose balls are drawn, one by one, without replacement, from an urn containing m white and n black balls. Then the average number of draws till the first white ball comes is $\frac{m+n+1}{m+1}$. To see this, keep on drawing the balls all the way. This gives $\binom{m+n}{m}$ equally likely binary sequences. For $k = 0, 1, 2, \ldots, n$, there are $\binom{m+n-k-1}{m-1}$ sequences in which the first white ball appears after $k+1$ draws. Hence the answer is $\frac{1}{\binom{m+n}{m}} \sum_{k=0}^{n} (k+1)\binom{m+n-k-1}{m-1}$. Use (2) in Chapter 5 to evaluate the sum. For a fixed m, the answer can also be verified by induction on n. Calling the average as $E_{m,n}$ and considering the two possible outcomes of the first draw, we have the relation $E_{m,n} = \frac{m}{m+n} + (1 + E_{m,n-1})\frac{n}{m+n}$.)

13. There are $\frac{1}{2}(n^2+1)(n^2+2)$ tickets in all. Of these only m^2 tickets carry a reward of m rupees for $m = 1, 2, \ldots, n$. So the probability of getting

a reward of m rupees is $\frac{2m^2}{(n^2+1)(n^2+2)}$. Hence the average amount of reward is $\frac{2(1^3+2^3+\ldots+n^3)}{(n^2+1)(n^2+2)}$. Use the well-known formula $\sum_{m=1}^{n} m^3 = \frac{n^2(n+1)^2}{4}$.

14. $\frac{8}{3}$ $(= \frac{1}{3} \times 2 + \frac{2}{3} \times 3)$. Denote by $E_{n,m}$ the average number of testings needed when it is known that exactly m out of the n machines are faulty (where $0 \le m \le n$). Then $E_{n,0} = 0$ since no testing is needed. A recurrence relation for $E_{n,m}$ can be written down by looking at the possible outcomes of the first testing. Thus, for $n \ge 2$ and $m \ge 1$ we have $E_{n,m} = \frac{n-m}{n}(1 + E_{n-1,m}) + \frac{m}{n}(1 + E_{n-1,m-1})$ with the initial condition that $E_{n,0} = 0$ for all $n \ge 1$. **15.** (a) Rupees $-\frac{2q^2+q}{u}$, $\frac{q-q^3}{u}$ and $\frac{2q^2+q^3}{u}$ respectively where $q = 1 - p$ and $u = (1 + q + q^2)^2$. (If the game ends on the n-th toss, then A's net gain is $2k$ if $n = 3k + 1$, $-k$ if $n = 3k - 1$ and $-k$ if $n = 3k$. So, the expected net gain of A equals $\sum_{k=0}^{\infty} 2kq^{3k}p - \sum_{k=1}^{\infty} kq^{3k-2}p - \sum_{k=1}^{\infty} kq^{3k-1}p$ which comes to $-\frac{2q^2+q}{(1+q+q^2)^2}$. Calculations for B and C are similar. The game is poised in B's favour if $1 > 2q + 2q^2$, i.e. if $p > \frac{3-\sqrt{3}}{2}$ and in C's favour if $p < \frac{3-\sqrt{3}}{2}$. If $p = \frac{3-\sqrt{3}}{2}$, it is poised equally between B and C. Note that the game is never poised in A's favour. (b) No, because the average gain is $\frac{3\times1+2\times3\times5+1\times3\times25-1\times125}{216} = -\frac{17}{216}$ which is negative. But a layman may fall for it by (incorrectly) thinking that the probability of getting his rupee back is at least $\frac{3}{6}$, and hence that he stands more than an even chance of getting richer. The freedom to choose the lucky number is only an additional lure since the value of x has no bearing on the outcome as long as the die is fair. **16.** $\frac{5}{18}$. (Assume B and C call x and y minutes respectively after 8:00 a.m. Then x, y vary uniformly and independently over the interval $[0, 30]$ and so (x, y) varies uniformly over the square S shown in the figure. The requirement that the phone is busy when C calls is that $x \le y \le x + 10$. So the favourable region R is the trapezium $ODEB$. Divide the area of R by that of S.)



17. (a) The favourable set of the event $x = y$ is a line segment, specifically, a diagonal of the square $S = [a, b] \times [a, b]$. Its area is 0. The other two cases follow from the fact that the area of a triangle is unaffected by whether you include a side of it or not. (b) Two lines are parallel if and only if the angles they make with a fixed direction are equal. Apply (a) to the interval $[0, 2\pi]$ (or, more accurately, to the interval $[0, 2\pi)$). **18.** For (i) follow the hint and note that the only zeros of the sine function are of the form $n\pi$, where n is an integer. Note further that even though 0 is a zero of $\sin y$, it is not a zero of $\frac{\sin y}{y}$ since the latter is to be treated as 1 by continuity. For (ii) apply Exercise (3.5)(a) with $a_0 = 1$ and $a_1 = -\frac{1}{6}$.

Chapter 24

1. The statement is false, e.g., take D_1 and D_2 to be the sets of positive and negative real numbers respectively, let $f_1(x)\sqrt{x}$ and $f_2(x) = \sqrt{-x}$. (The statement would be true if $D_1 \cup D_2$ is replaced by $D_1 \cap D_2$.)

2. The statement is false. The simplest counter-example is to take $f(x) = x^2$ and $r = 0$. (The correct assertion is that r is a root of $f'(x)$ repeated $m - 1$ times.)

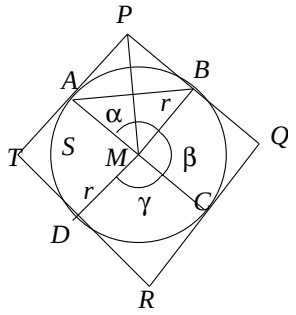
3. None of the three statements is correct since f is neither one-to-one nor onto. (The question was asked when functions were newly introduced in the JEE syllabus and there is little more to the question than knowing the definitions of the terms involved.)

4. By induction on n, it is easy to prove that for every $k \geq 0$, $\dfrac{d^k}{dx^k}(e^{\alpha x}) = \alpha^k x$. By a direction substitution, the L.H.S. is $(a_n\alpha^n + a_{n-1}\alpha^{n-1} + \ldots + a_1\alpha + a_0)e^{\alpha x}$ and the result follows from the fact that the factor $e^{\alpha x}$ never vanishes.

5. (C) (a standard result in geometry).

6. This results as a special case of the problem done in Comment No. 10 of Chapter 8, when the vertices of one of the two triangles happen to lie on the sides of the other.

7. Let M and r be respectively, the centre and the radius of the circle S. Let $\alpha, \beta, \gamma, \delta$ be the angles subtended at M by the sides AB, BC, CD and DA respectively. Also let the circumscribing quadrilateral be $PQRT$ as shown in the figure. Then $\angle P = \pi - \alpha, \angle Q = \pi - \beta, \angle R = \pi - \gamma$ and $\angle T = \pi - \delta$. Since $PQRT$ is given to be cyclic, we have $\angle P + \angle R = \angle Q + \angle T = \pi$ and hence $\alpha + \gamma = \beta + \delta = \pi$. Now the quadrilateral $MAPB$ is symmetric about the diagonal MP. Hence $AB = 2r\sin\frac{\alpha}{2}$. Similarly, $CD = 2r\sin\frac{\gamma}{2}$. But $\frac{\alpha}{2} + \frac{\gamma}{2} = \frac{\pi}{2}$. So we have $AB^2 + CD^2 = 4r^2(\sin^2\frac{\alpha}{2} + \cos^2\frac{\alpha}{2}) = 4r^2$. By a similar reasoning, $BC^2 + DA^2 = 4r^2$ which proves the result.



8. Let G be the set of all girls at the party. For a boy b, let G_b be the subset of girls who dance with b. The problem is equivalent to showing that there exist b, b' such that neither $G_b \subset G_{b'}$ nor $G_{b'} \subset G_b$. If this does not hold, then there will be a boy b^* such that $G_{b^*} \subset G_b$ for every boy b. Since $G_{b^*} \neq \emptyset$, there is some girl g in G_{b^*}. But then g is also in G_b for every boy b, contradicting that no girl dances with every boy. (We are tacitly

assuming here that the number of boys is finite. The argument will also work if, instead, the number of girls is finite, for then we could let b^* be a boy who dances with the least number of girls. If neither set is finite, then the result need not hold always. For example, suppose, that the boys are numbered $b_1, b_2, \ldots, b_n, \ldots$ and similarly for the girls. Assume that b_i dances with g_j whenever $j > i$. This exercise is yet another proof how things can go wrong when we extrapolate from the finite to the infinite. Exercise (24.15) below is yet another illustration.)

9. Denote the stones by $S_1, S_2, \ldots, S_{11}$ and their weights by $x_1, x_2, \ldots x_{11}$ respectively. The condition in the problem means that for every $i = 1, 2, \ldots, 11$, there exist distinct integers $j_1, j_2, \ldots, j_5, k_1, \ldots, k_5$ all different from i such that $x_{j_1} + x_{j_2} + \ldots + x_{j_5} - x_{k_1} - \ldots - x_{k_5} = 0$. This gives a system of 11 linear equations in 11 variables $x_1, x_2, \ldots, x_{11}$. When written in the matrix form, in each row the diagonal entry is 0 and out of the remaining 10 entries, 5 equal 1 and the remaining equal -1. It is trivial to check that for every real number λ, $x_1 = x_2 \ldots = x_{11} = \lambda$ is a solution of this system. The problem asks to show that these are the only possible solutions.

10. For $n \geq 1$, $A^{n+1} = A^n A = A^n \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ $= \begin{pmatrix} F_{n+1} + F_n & F_{n+1} \\ F_n + F_{n-1} & F_n \end{pmatrix} = \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix}$. Apply induction on n.

11. $\cap$ is a partial but not a total order. (Neither $1 \cap i$ nor $i \cap 1$ holds true.)

12. The set is the region bounded by an ellipse with foci at a and b and major axis of length r. (Here a, b are complex numbers representing certain points in the complex plane. Do not confuse them with the real numbers a, b you see in the equation of an ellipse in the standard form.) To prove its convexity, suppose $|z_1 - a| + |z_1 - b| \leq r$, $|z_2 - a| + |z_2 - b| \leq r$ and $z_3 = (1 - \lambda)z_1 + \lambda z_2$, where $0 \leq \lambda \leq 1$ is a real number. Splitting a as $(1 - \lambda)a + \lambda a$ and applying the triangle inequality, we get $|z_3 - a| \leq (1 - \lambda)|z_1 - a| + \lambda|z_2 - a|$. Write a similar inequality for $|z_3 - b|$ and add to get $|z_3 - a| + |z_3 - b| \leq r$.

13. The cubic $f(x) = x^3 - 3x^2 + 5x$ is strictly increasing and so the given conditions indeed define α and β uniquely. But to identify them individually is not easy. Nor is it needed. Letting $g(x) = x^3 - 3x^3 + 5x - 3$, the given conditions give $g(\alpha) = -g(\beta)$. The cubic $g(x)$ is also strictly increasing and hence one-to-one. If $g(x)$ were an odd function, we could conclude that $\alpha = -\beta$, i.e., $\alpha + \beta = 0$. But $g(x)$ is not an odd function because of the x^2 and the constant terms. They can be gotten rid of with a substitution. Put $u = x + 1$. Then $g(x) = (u+1)^3 - 3(u+1)^2 + 5(u+1) - 3 = u^3 + 2u = h(u)$ (say). Then $h(u)$ is an odd function of u. The given conditions imply $h(\alpha - 1) = -h(\beta - 1)$. Hence $\alpha - 1 = -(\beta - 1)$ which gives $\alpha + \beta = 2$. Of course, the method will not work if we replace 17 and -11 by some

arbitrary numbers. It works only for the particular given numbers. But then, that is the name of the game.

14. For notational simplicity, denote $\alpha, \beta, \gamma, \delta$ by $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ respectively. Put $t_i = \tan\frac{\alpha_i}{2}$ for $i = 1, 2, 3, 4$. Then by a direct substitution, t_1, t_2, t_3, t_4 are the roots of the quartic equation

$$(1 - a + c)t^4 + 2bt^3 + (2c - 6)t^2 + 2bt + (a + c - 1) = 0 \tag{1}$$

By a well-known formula (given on p. 132 of *Plane Trigonometry - Part I* by S. L. Loney),

$$\tan\left(\frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{2}\right) = \frac{s_1 - s_3}{1 - s_2 + s_4} \tag{2}$$

where $s_1 = \sum t_i = t_1 + t_2 + t_3 + t_4$, $s_2 = \sum t_i t_j$, $s_3 = \sum t_i t_j t_k$ and $s_4 = t_1 t_2 t_3 t_4$. All these values can be found from the coefficients of the various powers of t in (1). In particular, we have $s_1 = s_3 = \frac{2b}{a-c-1}$. So by (2), $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 2k\pi$ for some integer k.

15. For each fixed n, $\sum\limits_{m=1}^{\infty} a_{m,n} = -1 + \sum\limits_{m=n+1}^{\infty} \frac{1}{2^{m-n}} = -1 + \sum\limits_{k=1}^{\infty} \frac{1}{2^k} = -1 + 1 = 0$. However, for a fixed m, only the first m terms of the series $\sum\limits_{n=1}^{\infty} a_{m,n}$ are non-zero and so its sum is $\frac{1}{2^{m-1}} + \frac{1}{2^{m-2}} + \ldots + \frac{1}{4} + \frac{1}{2} - 1 = -\frac{1}{2^{m-1}}$ and hence $\sum\limits_{m=1}^{\infty} \sum\limits_{n=1}^{\infty} a_{m,n} = -\sum\limits_{m=1}^{\infty} \frac{1}{2^{m-1}} = -2$, again using the formula for the sum of an infinite geometric series. (Both the verifications become very vivid if you picture points of $\mathbb{N} \times \mathbb{N}$ by an (infinite) array and take the sums of the entries in a particular row or a column.)

16. Let t be the time measured in hours from 6:00 a.m.. Then the length of the shadow at time t is proportional to $\cot\frac{\pi t}{12}$ for $0 \le t \le 6$ and to $\cot(\pi - \frac{\pi t}{12})$ for $6 \le t \le 12$. Suppose the work is over at $t = T$. Equating the work done in the forenoon and in the afternoon gives

$$\int_2^6 \cot\left(\frac{\pi t}{12}\right) dt = -\int_8^T \cot\left(\frac{\pi t}{12}\right) dt \tag{3}$$

A simple integration gives $\ln(\sin\frac{\pi}{6}) = \ln(\sin\frac{\pi T}{12}) - \ln(\sin\frac{2\pi}{3})$, whence $\sin\frac{\pi T}{12} = \frac{\sqrt{3}}{4}$. Since $6 \le T \le 12$, we get $T = \frac{12}{\pi}(\pi - \sin^{-1}\frac{\sqrt{3}}{4})$. (With a calculator, this is almost $4{:}16'30''$ p.m.)

17. $(0, 1]$. (Putting $b = \tan\theta$, the expression equals $(\sec\theta x + \sin\theta)^2 + \cos^2\theta$ whose minimum value is $\cos^2\theta$. θ varies over $(-\frac{\pi}{2}, \frac{\pi}{2})$.

18. (B). (Without loss of generality take A as $(r, 0)$ and B as $(0, r)$. If $P = (r\cos\theta, r\sin\theta)$, then the centroid of the triangle PAB is (x, y), where $x = \frac{r + r\cos\theta}{3}$ and $y = \frac{r + r\sin\theta}{3}$.)

19. $\frac{a}{b} = \frac{n-5}{6}$. (Solve : $\binom{n}{5}a^{n-5}b^5 = \binom{n}{6}a^{n-6}b^6$. It is assumed here that a^n is taken as the 0-th term. If it is taken as the first term, the answer would be $\frac{n-4}{5}$.) **20.** $n = 7$. (Solve $\binom{n+1}{3} - \binom{n}{3} = 21$.)

21. 14. (Complementary counting. Of the 16 possible functions the two constant ones are not onto.)

22. (C). (Write the expressions as $\frac{K}{d}, \frac{K}{c}, \frac{K}{b}$ and $\frac{K}{a}$, where $K = abcd$.)

23. $n = 11$. (Solve : $2 \times 2n + \frac{2n(2n-1)}{2} \times 3 = 57n + \frac{n(n-1)}{2} \times 2$.)

24. (D). (Without loss of generality, take $z_2 = iz_1$. Then $z_1^n = z_2^n = 1$ imply $i^n = 1$.) **25.** (C). (Write $\frac{1-i\sqrt{3}}{2}$ as $\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}$.)

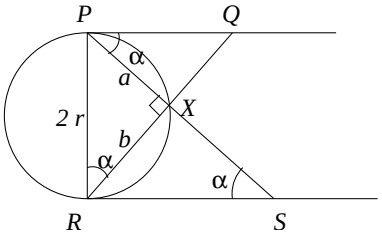
26. $x = 1$. (Solve : $\dfrac{x}{1+\frac{x}{2}} = \dfrac{x^2}{1+\frac{x^2}{2}}$.)

27. $(\frac{1}{\sqrt{2}})^n$. (The problem is equivalent to minimising the product $(\sec^2\alpha_1)(\sec^2\alpha_2)\ldots(\sec^2\alpha_n)$ subject to the constraint $(\tan\alpha_1)\ldots(\tan\alpha_n) = 1$. Because of many problems of a similar type, intuitively the minimum will occur when all α_i's are equal. For a rigorous proof, put $y_i = \tan\alpha_i$. Then y_i can take any positive value. The problem then amounts to minimising $(1+y_1^2)(1+y_2^2)\ldots(1+y_n^2)$ subject to $y_1y_2\ldots y_n = 1$. If, say $y_1 \neq y_2$, then set $z_1 = z_2 = \sqrt{y_1y_2}$. Then $z_1z_2 = y_1y_2$. But $(1+z_1^2)(1+z_2^2) < (1+y_1^2)(1+y_2^2)$. So the minimum must occur when all y_i's are equal and hence each equals 1. A proof using the method of Lagrange multipliers is also possible.)

28. (C). (Note that $\tan\alpha\tan\beta = 1$. So $\tan\alpha(1 - \tan\beta\tan\gamma) = \tan\beta + \tan\gamma$ gives $\tan\alpha - \tan\gamma = \tan\beta + \tan\gamma$.) **29.** $\frac{200}{\sqrt{3}}$ $(= 100\cot 30^\circ - 100\cot 60^\circ)$ meters. **30.** 2. ($x = \frac{5}{3+4m}$ is an integer only for $m = -1, -2$.)

31. $x = \frac{3}{2}$. (Write the equation as $Y^2 = -4X$, where $Y = y+2$ and $X = x-\frac{1}{2}$.)

32. (A). (Let a, b be the lengths of the sides XP and XR of the right-angled triangle PXR and let $\alpha = \angle XRP$. Then $a = PQ\cos\alpha = PQ\frac{b}{2r}$ gives $PQ = \frac{2ra}{b}$. Interchanging the roles of a and b, $RS = \frac{2rb}{a}$. So $PQ.RS = 4r^2$.)



33. $\frac{1}{|m-n|}$. (The point of intersection, say A, of $y = mx$ and $y = nx + 1$ is $(\frac{1}{m-n}, \frac{m}{m-n})$ while the point, say B, of intersection of $y = nx$ and $y = mx + 1$ is $(\frac{1}{n-m}, \frac{n}{n-m})$. $|\overrightarrow{OA} \times \overrightarrow{OB}|$ gives the desired area.)

34. $y = \frac{1}{\sqrt{3}}x + \sqrt{3}$. (Let $y = mx + c$ be a common tangent. The conditions for tangency to the two curves give a system (i) $|3m + c| = 3\sqrt{1 + m^2}$ and (ii) $c = \frac{1}{m}$. Solving, $m = \pm\frac{1}{\sqrt{3}}$. The sign is determined by the requirement that the point of contact be above the x-axis.)

35. $b = -3$. (The two intercepts are $\frac{b+1}{b+2}$ and $-b - 1$. Hence the area is $-\frac{(b+1)^2}{b+2}$. In the original problem the correct answer was to be picked from $\{-1, 3, -3, 1\}$. This could be done from the positivity of the two intercepts, without calculating the area. An alertness of this sort can save you time.)

36. $f^{-1}(x) = \frac{x+\sqrt{x^2-4}}{2}$. (Solve $y = x + \frac{1}{x}$ for x to get $x = \frac{y \pm \sqrt{y^2-4}}{2}$. The negative sign gives $x < 1$ for $y > 2$ and hence is ruled out.)

37. (B). ($g(x)$ is always positive.) **38.** π (l'Hôpitals's rule.)

39. $(-1)^k(k-1)\pi$. ($f(k) = 0$ while for a small $\delta > 0$, $f(k - \delta)$ equals $(k-1)\sin(k\pi - \pi\delta) = -(k-1)(-1)^k \sin \pi\delta$. Divide by $-\delta$ and let $\delta \to 0^+$.)

40. 4. (For $x \geq 0$, $F(x) = x(1 + \sqrt{x}) = x + x^{3/2}$. Also, $f(x) = F'(x)$.)

41. At $-1, 0$ and 1. (From $x^3 - x = (x-1)x(x+1)$, $f(x)$ equals x for $x \in (-\infty, -1] \cup [0, 1]$ and $f(x) = x^3$ for $x \in [-1, 0] \cup [1, \infty)$. The left- and right-handed derivatives of $f(x)$ do not match at $-1, 0$ and 1.)

42. (A). ($f'(x) = e^{x-x^2}(1 + x - 2x^2)$. The first factor is always positive, the second is positive between the roots of the quadratic expression in it.)

43. $\frac{\pi}{2}$. ($I = I_1 + I_2$, where I_1 and I_2 are the integrals (with the same integrand) over $[-\pi, 0]$ and $[0, \pi]$ respectively. Put $y = -x$ in I_1 to see that $I_1 + I_2$ equals $\int_0^\pi \cos^2 x dx$.)

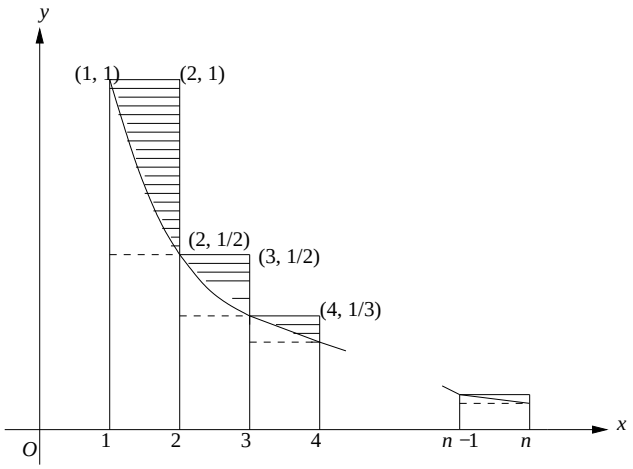
44. (D). ($\cos|x| = \cos x$ is differentiable everywhere. So non-differentiability of $|x|$ at 0 rules out (A) and (B). In (C), the non-differentiability of $\sin|x|$ at 0 adds to that of $|x|$ at 0, while in (D), they cancel each other.)

45. 1. (Rewriting the L.H.S. as $\dfrac{\log_2(x-1)}{\log_2 4} = \dfrac{\log_2(x-1)}{2}$, the equation reduces to $x - 1 = (x-3)^2$, which has 2 and 5 as roots, of which the first is discarded since the logarithms of negative numbers are not defined.)

46. $\alpha = -1$. (Equate $\dfrac{\alpha\frac{\alpha x}{x+1}}{\frac{\alpha x}{x+1} + 1}$ with x to get $\alpha^2 x = (\alpha+1)x^2 + x$. If this is to hold for all x, $\alpha + 1$ must vanish and α^2 must equal 1.)

47. (C). ($[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix} = 1$ by direct expansion.)

48. Consider the graph of the function $y = \frac{1}{x}$ for $x \geq 1$ as shown in the figure below. It is clear that H_{n-1} represents the sum of the areas of the rectangles with unit bases and heights $1, \frac{1}{2}, \ldots, \frac{1}{n-1}$. Obviously, this is greater than the area below the curve $y = \frac{1}{x}$, the x-axis and the lines $x = 1, x = n$, which equals $\ln n$. So $H_{n-1} > \ln n$. On the other hand, if we consider the lower rectangles, we get, $\frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} < \ln n$, i.e., $H_n - \ln n < 1$. The two inequalities together prove (i). Moreover, $H_{n-1} - \ln n$ equals the shaded area in the figure and increases with n. Further, $H_{n-1} - \ln n < 1$ (since $H_{n-1} < H_n$). So the sequence $\{H_{n-1} - \ln n\}_{n=1}^{\infty}$ is monotonically increasing and bounded above by 1. This proves (ii).



49. Depending upon whether $m \geq n$ or $n \geq m$, the number $\frac{(2m)!}{(m+n)!}$ or the number $\frac{(2n)!}{(m+n)!}$ is an integer. (It is tempting to involve a binomial coefficient here. But that is of no help. This is a good illustration of a problem which can be tackled directly with no specialised knowledge.)

50. (i) $\pm\dfrac{\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}$. (To obtain this vector note that $\vec{d}\cdot\vec{a} = \vec{d}\cdot\vec{b}$ holds if and only if $\vec{d} \perp (\vec{a} - \vec{b})$. Similarly, $\vec{d} \perp (\vec{b} - \vec{c})$. The two hold together if and only if $\vec{d}$ is parallel to $(\vec{a} - \vec{b}) \times (\vec{b} - \vec{c})$.) (ii) By (i), we may suppose that all the three vectors $\vec{a}, \vec{b}, \vec{c}$ are inclined at an angle α to some unit vector, which may be taken to be $\mathbf{k}$. Then they have the stated form. By a direct calculation, $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ comes out as $4\sin^2\alpha(\sin^2\frac{\theta_1-\theta_2}{2} + \sin^2\frac{\theta_2-\theta_3}{2} + \sin^2\frac{\theta_3-\theta_1}{2})$. For this to be maximum, first of all α must be $\frac{\pi}{2}$. Then the vectors $\vec{a}, \vec{b}, \vec{c}$ all lie in the xy-plane and the problem reduces to that of maximising $a^2 + b^2 + c^2$ for a triangle ABC with circumradius 1. This, in turn, amounts to maximising $\sin^2 A + \sin^2 B + \sin^2 C$. The answer is an equilateral triangle by any of the methods discussed in Chapter 14. (iii) Simply expand $|\vec{a} - \vec{b}|^2 =$

$|\vec{a}|^2+|\vec{b}|^2-2(\vec{a}\cdot\vec{b})=2-\vec{a}\cdot\vec{b}$. Similarly, expand $|\vec{b}-\vec{c}|^2$ and $|\vec{c}-\vec{a}|^2$. Also, $|\vec{a}+\vec{b}+\vec{c}|^2=3+2(\vec{a}\cdot b+\vec{b}\cdot\vec{b}+\vec{c}\cdot\vec{a})$.

51. Let x, y, z be the numbers at the first, the second and the third draw respectively. Then $|S|=p^3$, where $S=\{(x,y,z): 1\le x\le p, 1\le y\le p, 1\le z\le p\}$. Let $A=\{(x,y,z)\in S: x+y+z=2p$. Because of the restrictions, if $(x,y,z)\in A$, then $x+y=p+k$ for some $k=0,1,\ldots,p-1$. For any such k, let $A_k=\{(x,y,z)\in A: x+y=p+k\}$. Then A is the disjoint union of $A_0, A_1,\ldots,A_{p-1}$. Now $|A_0|=p-1$ since x can take any value from 1 to $p-1$. However, for $k=1,2,\ldots,p-1$, x can vary only from k to p and so $|A_k|=p-k+1$. Hence $|A|=\sum\limits_{k=0}^{p-1}|A_k|=p-1+\sum\limits_{k=1}^{p-1}(p-k+1)$ $=p-1+\sum\limits_{j=2}^{p}j=p-1+\frac{p(p+1)}{2}-1=\frac{p^2+3p-4}{2}=\frac{(p-1)(p+4)}{2}$. The desired probability is $\frac{|A|}{|S|}$.

52. 0.0875 $(=4\times(0.5)^3\times(0.05)+(0.5)^4)$. (For a total of at least 7 points, either 3 matches must give 2 points each and the fourth one 1 point or else all 4 matches must give 2 points each. The division of matches according to countries is an irrelevant piece of data.)

53. (a) $\frac{1}{2}$, (b) $\frac{23}{30}$. (Let A be the event that S_1 wins and B be the event that S_2 wins. As all players are of equal strength, $P(A)=P(B)=\frac{1}{2}$. The answer to (b) is $P(A)+P(B)-P(A\cap B)$. Note that for $A\cap B$ to occur, S_1 must, first of all, be paired with some player other than S_2. The probability of this is $\frac{14}{15}$. This follows easily by taking complementary probability without having to count all possible pairings. So, $P(A\cap B)=\frac{14}{15}P(A)P(B)=\frac{7}{30}$. Hence the desired probability is $\frac{1}{2}+\frac{1}{2}-\frac{7}{30}$.

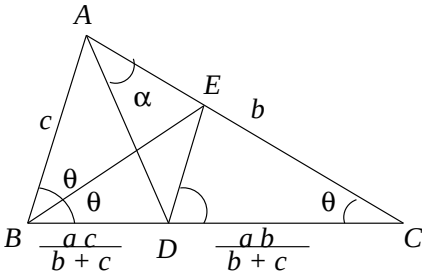
54. $\dfrac{97}{25^4}$. (E occurs if and only if the selected number is $29, 36, 63$ or 92. So $P(E)=\frac{1}{25}$. Call this p. Then the desired probability is the binomial probability $4p^3q+p^4$, where $q=1-p=\frac{24}{25}$.)

55. $\frac{4^5+4\times4^4\times3+3\times4^3\times3^2}{7^5}=\frac{832}{2401}$. (Letting A and B also denote the wins of the respective players, out of the 32 possible binary sequences of length 5 with entries A and B, the favorable ones can be listed explicitly. There is one sequence with all five A's, four sequences with 4 A's and 1 B and three sequence with 3 A's and 2 B's.)

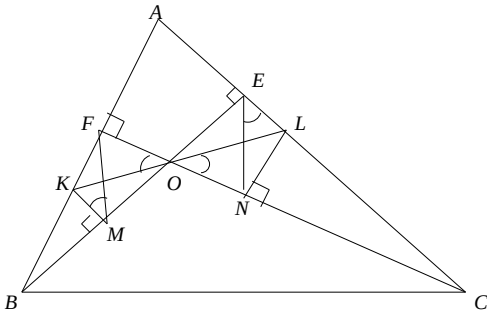
56. $\frac{23}{36}$. (Complementary counting.)

57. An ordered pair (P,Q) of subsets is of the 'favourable type' i.e., $P\cap Q=\emptyset$ if and only if E_i occurs for every $i=1,2,\ldots,n$. So the desired probability is $P(E_1)P(E_2)\ldots P(E_n)$. Now, for each fixed i, $E_i=(A_iB_i)'$, where A_i is the event that $x_i\in P$ and B_i is the event that $x_i\in Q$. Note that $P(A_i)=P(B_i)=\frac{1}{2}$ and they are independent. So $P(E_i)=\frac{3}{4}$ for every i.

58. For a trigonometric solution, since D divides BC in the ratio $c : b$, we have $BD = \frac{ac}{b+c}$ and $DC = \frac{ab}{b+c}$. So, $AB = DC$ gives $c^2 + bc = ab$. Also, $\angle B = 2\angle C$ gives $\frac{b}{c} = \frac{\sin 2C}{\sin C} = 2\cos C = \frac{a^2+b^2-c^2}{2ab}$. The two equations together imply $bc = b^2 - c^2$, i.e., $\sin B \sin C = \sin^2 B - \sin^2 C$. Using $B = 2C$ again, this gives a quadratic for $\cos C$, viz., $4\cos^2 C - 2\cos C - 1 = 0$, from which $\cos C = \frac{1+\sqrt{5}}{4} = \cos 36°$. But a pure geometry solution is much easier. Let the angle bisector of $\angle B$ meet AC at E. Then $\angle EBC = \angle ECB$ whence $EB = EC$. Since $AB = DC$ and $\angle ABE = \angle DCE$, the triangles ABE and DCE are congruent. So, $\angle BAE = \angle CDE$, which makes the quadrilateral $ABDE$ cyclic. But then $\frac{1}{2}\angle A = \angle DAE = \angle DBE = \frac{1}{2}\angle B = \angle C$ whence $\angle A = \angle B = 2\angle C$ which gives $5\angle C = 180°$.



59. There are too many lines and points of intersection to make a solution based on coordinates practicable. Instead, pure geometry works fine. Note that the quadrilaterals $OFKM$ and $OELN$ are cyclic. So $\angle KMF = \angle KOF = \angle NOL = \angle NEL$. So the angles $\angle FMO$ and $\angle MEN$ are equal, being complementary to the angles $\angle KMF$ and $\angle NEL$ respectively. Hence the lines FM and EN are parallel as the inverse angles made by the transversal FN with them are equal.



60. Here the cyclic symmetry of the data makes coordinates the right tool. Take the line as the x-axis and A, B, C as $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ respectively. Then D is $(x_1, 0)$ and the equation of the perpendicular from D to BC is $(x - x_1)(x_2 - x_3) + y(y_2 - y_3) = 0$. Write the equations of the other two perpendiculars by cyclic symmetry and prove the concurrency by showing that the determinant of the coefficients vanishes as the row sum does.

61. Note that P, Q, R are the centres of the equilateral triangles described on the segments AB, BC, CA. This suggests that complex numbers would come in handy. In fact, the problem is a limiting case of Exercise (8.10).

62. Let $P = 2 \times 4 \times 6 \times \ldots \times 2000$ and $Q = 1 \times 3 \times 5 \times \ldots \times 1999$. Write Q as $(2001-2) \times (2001-4) \times \ldots \times (2001-2000)$.

63. 2499. (Let $\left[\frac{x}{99}\right] = \left[\frac{x}{101}\right] = k$. Then $x = 99k+u = 101k+v$, where u, v are integers with $0 \le u \le 98$ and $0 \le v \le 100$. This gives $u-v = 2k$. Also $k \ge 0$. For each $k \ge 0$, let $S_k = \{(u,v) : 0 \le u \le 98; 0 \le v \le 100$ and $u - v = 2k\}$. Then $|S_0| = 99, |S_1| = 97, |S_2| = 95, \ldots, |S_{49}| = 1$. The desired number is obtained by adding these $|S_i|$'s and subtracting 1 so as to exclude the case $x = 0$. Alternately, simply list the integers x which satisfy the given condition as $1, 2, \ldots, 98, 101, 102, \ldots, 197, 202, \ldots, 296, 302, \ldots, 4949$. The second solution is simpler and also identifies the integers satisfying the given condition.)

64. Observe that $3x < x + y + z < 3z$ and $3x^2 < x^2 + y^2 + z^2 < 3z^2$.

65. Factoring, $|x^2(y-z) + y^2(z-x) + z^2(x-y)| = |(x-y)(y-z)(z-x)|$ $= |x-y||y-z||z-x| \le |z||x||y|$ by the triangle inequality.

66. (i) The assertion is equivalent to $(abc)^{a+b+c} \le a^a b^b c^c$, i.e. to $a^a b^b c^c \ge 1$ (since $abc = 1$), which in turn, reduces to $a \ln a + b \ln b + c \ln c \ge 0$. By symmetry, we may assume $a \le b \le c$. Since $abc = 1$, this means $a \le 1$ and hence $\ln a \le 0$ and similarly $c \ge 1$ and hence $\ln c \ge 0$. The assertion follows by noting that $a \ln a + b \ln b + c \ln c = a \ln a - b(\ln a + \ln c) + c \ln c =$ $(a-b) \ln a + (c-b) \ln c$. (ii) Let $x = a - 1 + \dfrac{1}{b}, y = b - 1 + \dfrac{1}{c}$ and $z = c - 1 + \dfrac{1}{a}$. We have to show $xyz \le 1$. Since a, b, c are all positive and $abc = 1$, at least one of them is ≥ 1 and so at least one of x, y, z is positive. If one of x, y, z, say x is negative, then the other two must be positive. For, $x < 0$ gives $a < 1$ and $b > 1$ which respectively imply that $z > 0$ and $y > 0$. So in this case, xyz is negative and we are through. The only case left is when x, y, z are all positive. In this case, because of $abc = 1$, we can rewrite y as $b(1 + a - \dfrac{1}{b})$. This gives $xy = b(a - 1 + \dfrac{1}{b})(1 + a - \dfrac{1}{b}) \le b(\frac{2a}{2})^2 = ba^2$ by the A.M.$-$G.M. inequality. Similarly, $yz \le cb^2$ and $zx \le ac^2$. Multiplying the three inequalities gives the result.

67. Let $A = (a_{ij})$. Fix any k from 1 to n. Then for every $i = 1, 2, \ldots, n$, there is a unique integer $j(i)$ such that $a_{ij(i)} = k$. By symmetry of the matrix, $j(j(i)) = i$ for every i. If k does not occur on the diagonal, then $j(i) \ne i$ for every i. But then all the indices from 1 to n will get paired off into pairs of the form $\{i, j(i)\}$ and hence n would be even. So every k occurs at least once on the diagonal. (More precisely, every k occurs on the diagonal an odd number of times.) As there are only n places on the diagonal, by the pigeon hole principle, no k can occur more than once.

68. Consider $x^4 - 2ax^2 + x + a^2 - a = 0$ as a quadratic in a. Then its roots are $a = x^2 + x$ and $a = x^2 - x + 1$, which gives a factorisation as $(a - x^2 - x)(a - x^2 + x - 1) = 0$. Hence either $x^2 + x = a$ or $x^2 - x + 1 = a$.

The possible values of x are, therefore, $\dfrac{-1 \pm \sqrt{1+4a}}{2}$ and $\dfrac{-1 \pm \sqrt{4a-3}}{2}$. These are all real if and only if $a \geq \frac{3}{4}$.

69. f is identically 0, 1 or -1. (First taking $x = 0 = y$, show that $f(0) = (f(0))^3$ and hence that $f(0)$ equals 0, 1 or -1. The first case easily leads to $f \equiv 0$. In the second case, taking $y = -x$, we get

$$1 = f(0) = f(x - x) = f(x)f(-x)f(-x^2)$$

Hence $f(x) \neq 0$ for every $x \in \mathbb{R}$. Also from the given functional equation, for every x we have

$$f(x) = f(1 + (x - 1) = f(1)[f(x - 1)]^2 = f(1)[f(x)f(-1)f(-x)]^2$$

Cancelling $f(x)$, this gives $1 = kf(x)(f(-x))^2$, where $k = f(1)f(-1))^2$. Replacing x by $-x$ we get $1 = kf(-x)(f(x))^2$. Hence $f(x) = f(-x)$ for all $x \in \mathbb{R}$. It then follows that $1 = k(f(x))^3$, whence $f(x)$ is a constant. The analysis in the third case, viz., $f(0) = -1$ is similar.)

70. We assume the a_i's are all distinct (as otherwise, we merely combine the corresponding terms together). Also, without loss of generality, assume $a_1 < a_2 < \ldots < a_n$. Let $f(x) = \dfrac{A_1}{x - a_1} + \dfrac{A_2}{x - a_2} + \ldots + \dfrac{A_n}{x - a_n}$ and $g(x) = f(x)(x - a_1)(x - a_2)\ldots(x - a_n)$. Then $g(x)$ is a polynomial of degree $n - 1$ and has the same roots as $f(x)$. So $f(x)$ has at most $n - 1$ distinct real roots. Now, $f(x)$ also equals $\dfrac{F'(x)}{F(x)}$ where $F(x) = (x - a_1)^{A_1}(x - a_2)^{A_2}\ldots(x - a_n)^{A_n}$ as seen easily by logarithmic differentiation. Since F vanishes at all a_i's, by Rolle's theorem F' has at least one root in two consecutive a's and hence at least $n - 1$ real roots. Put together, $f(x)$ has exactly $n - 1$ real roots.

71. The determinant form of the question provides little clue. Instead, writing $A + 2B$ as $A + B + C + B - C = \pi + B - C$ and hence $\cos(A + 2B)$ as $-\cos(B - C)$ etc. and expanding the determinant, the problem reduces to proving

$$\begin{aligned} 2\cos A\cos B\cos C \quad &+ \quad \sum \cos A\cos B\cos(A - B) \\ &= \quad \cos(B - C)\cos(C - A)\cos(A - B) \qquad (4) \end{aligned}$$

The trick now is to transfer any one term of the summation, say the term $\cos C\cos A\cos(C - A)$ to the R.H.S. Writing $\cos A\cos B\cos C + \cos A\cos B\cos(A-B)$ as $\cos A\cos B(-\cos(A+B)+\cos(A-B))$ and hence as $\frac{1}{2}\sin 2A\sin 2B$ and similarly $\cos A\cos B\cos C + \cos B\cos C\cos(B - C)$ as $\frac{1}{2}\sin 2B\sin 2C$ the problem is reduced to showing that

$$\frac{1}{2}\sin 2B(\sin 2A+\sin 2C) = \cos(C-A)(\cos(A-B)\cos(B-C)-\cos C\cos A)$$

Factorising $\sin 2A + \sin 2C$ as $2\sin(A+C)\cos(A-C)$, the problem further reduces to showing that

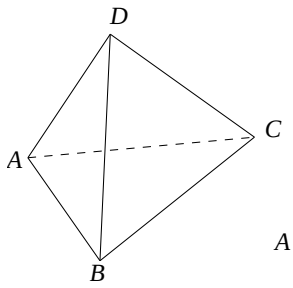
$$\sin 2B \sin(A+C) = \cos(A-B)\cos(B-C) - \cos C \cos A$$

which is proved by writing the R.H.S. as $\frac{1}{2}[\cos(A-C)+\cos(A+C-2B)] - \frac{1}{2}[\cos(A+C) - \cos(C-A)]$ and using $A+C = \pi - B$ again.

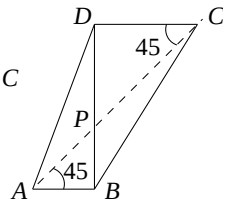
72. Take the sides as $n-1, n, n+1$. Express Δ and s in terms of n. Solve $\Delta = 4s$ to get $n = 14$. Finally, $R = \frac{abc}{4\Delta} = \frac{65}{8}$.

73. For $1 \le i \le 50$, let $S_i = \{i, 100-i\}$. Then $|A \cap S_i| \le 1$ for $1 \le i \le 49$. Coupled with $|A| = 50$, this means $|A \cap S_i| \ge 1$ for $1 \le i \le 50$. Put $i = 36$ to get the result.

74. Here the insufficiency of the data is itself a clue. Let the quadrilateral be $ABCD$. Without loss of generality, we are given that $AB + CD + BD = 16$ and that the area of $ABCD$ is 32 sq. units and are asked to find AC. In general two pieces of data do not determine a quadrilateral uniquely (for which 5 pieces of information are needed). So, this must be a case where some optimum value implies several equalities. The equality $AB + CD + BD = 16$ implies an upper bound on the area of the quadrilateral $ABCD$. Decomposing it into two triangles, viz. ΔABD and ΔCBD as shown in (a), we get



(a)



(b)

$$\begin{aligned} 32 &= \text{area of} ABCD \\ &= \frac{1}{2}AB.BD.\sin\angle ABD + \frac{1}{2}CD.BD.\sin\angle BDC \\ &\le \frac{1}{2}BD.(AB+CD) \\ &\le \frac{1}{2}\left[\frac{BD+(AB+CD)}{2}\right]^2 \\ &= \frac{1}{2}\left(\frac{16}{2}\right)^2 = 32 \end{aligned}$$

which forces equality throughout. In particular, $\angle ABD$ and $\angle BDC$ are right angles and also $BD = AB + CD = 8$ as shown in (b). The latter requirement makes $\angle CAB = \angle BDC = 45°$. This does not determine the quadrilateral $ABCD$ uniquely, because the point of intersection, say P,

of the diagonals can vary. But no matter where P is, we have $AC = AP + PC = \sqrt{2}(AB + CD) = 8\sqrt{2}$.

75. 969. (For $1 \leq n \leq 1000$, the n-th door will change its state as many times as the number of positive divisors of n, including 1 and n. Note that whenever d is a divisor, so is n/d and that they are distinct unless $n = d^2$. So, the number of divisors of n is odd if and only if n is a perfect square. There are 31 perfect squares from 1 to 1000.)

76. (a) Let $m = p_1^{s_1} p_2^{s_2} \dots p_k^{s_k}$ be the prime factor decomposition of m where $p_1, p_2, \dots, p_k$ are distinct primes and $s_1, s_2, \dots, s_k$ are positive integers. Let v be the product of those p_i's for which s_i is odd. Then v is squarefree and $m = u^2 v$ where $u = p_1^{[s_1/2]} p_2^{[s_2/2]} \dots p_k^{[s_k/2]}$. For the second part, assume that in the prime factorisation of n, the prime p_i appears with exponent t_i, for $i = 1, 2, \dots, k$. If m divides n^2 then $2t_i \geq s_i$ for $i = 1, 2, \dots, k$. Hence $t_i \geq s_i/2$ which implies that $t_i \geq [s_i/2]$ if s_i is even and $t_i \geq [s_i/2]+1$ if s_i is odd. But these are precisely the exponents of p_i in uv.
(b) 77. (By Exercise (11.7), the problem amounts to minimising the sum $a+b+c$ subject to the constraints : (i) a, b, c are all positive integers, (ii) $a^2 - b^2 = bc$, (iii) $c^2 > a^2 + b^2$ and (iv) $a + b > c$. In the solution, a, b, c can have no common factor, or else cancelling it we get a similar but smaller triangle. To prove the hint, write b as $u^2 v$ where v is squarefree. By (ii), b divides a^2 and hence uv divides a. By (ii) again, $u^2 v^2$ divides bc and so v^2 divides c. Hence v is a common factor of a, b, c. So $v = 1$ and we take b as u^2. Since u^2 divides a^2, u divides a. Let $a = ku$ for a positive integer k. The condition $\angle A = 2\angle B$ gives $\sin B < \sin A < 2 \sin B$ and hence $b < a < 2b$, which translates into $u < k < 2u$. (That $a < 2b$ can also be derived from (ii) and (iv) together.) Also, from (ii), $c = k^2 - u^2$. (iii) now reduces to $k^2 > 3u^2$. Putting it all together, we get that the interval $(3u^2, 4u^2)$, contains a perfect square, viz., k^2. By trial, the least u for which this happens is $u = 4$ and then $k = 7$. This gives $a = 28, b = 16$ and $c = 33$. The problem is a good combination of trigonometry and number theory, two seemingly unrelated topics.)

77. First observe that $f(2k) = 2f(k)$. Repeated applications give that if $n = 2^r m$ where m is odd then $f(n) = 2^r f(m)$. To find $f(m)$ when m is odd and ≥ 3, suppose $q = [\log_2 m]$. Then $2^q < m < 2^{q+1}$ and a little experimentation suggests that $f(m) = 2(m - 2^q)$. This can be written more compactly as $f(m) = 2m - 2^{\lceil \log_2 m \rceil}$ for all odd $m > 1$, where $\lceil x \rceil$ is the ceiling of x, defined in Comment No. 2 of Chapter 4. A proof of this by induction on m is a little messy as it runs into two cases depending upon the residue class of m modulo 4. But if you work out special cases such as $m = 23, 25, 27, 29, 31$ you will know what to do. In each case, while computing $f(m)$ by striking off various integers, let S_m be the set of integers that remain after the second round of cancellations. Then S_m forms an initial segment of the progression $2, 6, 10, 14, \dots$. When $m = 4k+1$, S_{m+2} has one more element than S_m, viz. $4k+2$. In the third round,

the cancellation begins with the second term 6 for S_m and with the first term 2 for S_{m+2}. As a result, the integers that get cancelled from S_{m+2} from the the fourth round onwards are obtained by adding 4 to those that get cancelled from S_m. Hence, the integer that is left out at the end for $m+2$ is 4 higher than that left for m. In other words, $f(m+2) = 4+f(m)$. Note that since $m = 4k+1$, there is no power of 2 between m and $m+2$. So, $\lceil \log_2(m+2) \rceil$ is the same as $\lceil \log_2 m \rceil$. Since $f(m) = 2m - 2^{\lceil \log_2 m \rceil}$, we have $f(m+2) = 4 + f(m) = 2(m+2) - 2^{\lceil \log_2 (m+2) \rceil}$. So the proof of the inductive step is complete in this case.

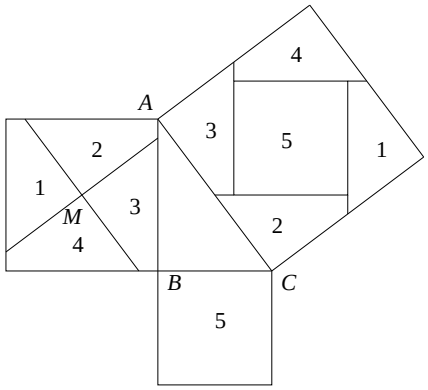
Next, suppose $m = 4k-1$. Then $S_{m+2} = S_m$, but the cancellation in the third round begins with 2 for S_m and with 6 for S_{m+2}. As the cancellation goes on cyclically, the integer that is left in S_{m+2} will be the one which is cyclically next to the integer that is left in S_m. Usually, this will mean $f(m+2) = 4 + f(m)$ as we would like to have for the inductive step. The only exception is when $f(m)$ is the very last member of the progression S_m, viz. $m-1 = 4k-2$, for, in that case, $4+f(m) = 4k+2$ is not in the set S_{m+2} (which is the same as S_m). In this case, the member of the progression S_m that comes cyclically next to $m-1$ is the very first term of the progression, viz. 2. So, in this case $f(m+2)$ slumps to 2. By induction hypothesis, applied to m, this occurs if and only if $2m - 2^{\lceil \log_2 m \rceil} = m-1$, i.e. $m = 2^{\lceil \log_2 m \rceil} - 1$. In this case there is a power of 2 between m and $m+2$. (As an illustration, work out what happens when $m = 31$.) So now $\lceil \log_2(m+2) \rceil = \lceil \log_2 m \rceil + 1$ and we do have $f(m+2) = 2 = 2(m+2) - 2^{\lceil \log_2 (m+2) \rceil}$. This completes the proof of the inductive step in all cases.

To compute $f(1000)$, write 1000 as $2^3 \times 125$. Then $f(1000) = 8f(125)$. Since $2^6 < 125 < 2^7$, we have $\lceil \log_2 125 \rceil = 7$ and so $f(125) = 250 - 2^7 = 250 - 128 = 122$. Hence $f(1000) = 8 \times 122 = 976$.

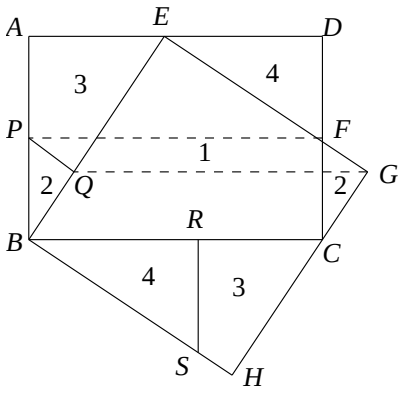
78. Suppose first that m, n are relatively prime, with $n \leq m$ (say). Taking the ends of the diagonal as $(0,0)$ and (m,n), it cuts the vertical line $y = j$ at $(j, \frac{jn}{m})$ for $j = 1, 2, \ldots, m$. Since $n \leq m$, the interval $(\frac{(j-1)n}{m}, \frac{jn}{m})$ contains either none or one integer and depending upon that, the diagonal passes through one or two of the squares in the j-th column. As the interval $(0, n)$ contains $n-1$ integers, the diagonal passes through $m + (n-1)$ squares in all. For the general case, let $m = dm'$ and $n = dn'$. Then m' and n' are relatively prime. Divide the board into d^2 chessboards of size $m' \times n'$ each by lines parallel to the sides. Only d of these smaller boards contain points of the diagonal. Apply the first case to each one of them.

79. (i) For transitivity, suppose A is equidecomposable to B and B is equidecomposable to C. Then there is some decomposition, say, $\{D_1, D_2, \ldots, D_n\}$ of A into polygonal figures and some mutually disjoint figures $D_1', D_2', \ldots, D_n'$, congruent to $D_1, D_2, \ldots, D_n$ respectively, whose union is B. (Here by mutually disjoint we only mean that the interiors are disjoint.) Then

$A = \bigcup_{i=1}^{n} D_i$ and $B = \bigcup_{i=1}^{n} D'_i$. The catch is that the same decomposition may not work to build C. However, we are given some other decompositions, say $\{E_1, E_2, \ldots, E_m\}$ and $\{E'_1, E'_2, \ldots, E'_m\}$ of B and C respectively such that E'_j is congruent to E_j for $j = 1, 2, \ldots, m$. Superimpose the two decompositions of B. That is, for $1 \leq i \leq n$ and $1 \leq j \leq m$, let $F_{ij} = D'_i \cap E_j$. Then each F_{ij} is a polygonal figure, which may be empty or degenerate sometimes. For each $i = 1, 2, \ldots, m$, $\bigcup_{j=1}^{m} F_{ij}$ is a decomposition of D'_i. Let $\bigcup_{j=1}^{n} G_{ij}$ be the corresponding decomposition of D_i (which is congruent to D'_i). Similarly, for $j = 1, 2, \ldots, m$, let $\bigcup_{i=1}^{n} H_{ij}$ be the decomposition of E'_j corresponding to the decomposition $\bigcup_{i=1}^{n} F_{ij}$ of the polygon E_j. Then for all i, j, G_{ij} is congruent to H_{ij}, each being congruent to F_{ij}. Further, $A = \bigcup_{i,j} G_{ij}$ and $C = \bigcup_{i,j} H_{ij}$, which shows that A and C are equidecomposable. (ii) Form a right angled triangle ABC whose sides AB and BC equal those of the given two squares. Through the centre, say M, of the outwardly described square on the side AB, draw lines parallel and perpendicular to the hypotenuse AC. These lines divide the square into four mutually congruent quadrilaterals. Through the midpoints of the sides of the square outwardly described on AC, draw lines alternately parallel to the sides AB and BC as shown in the figure. It is easy to show that these lines lie along the boundary of a square whose side equals BC. This gives a decomposition of the square on AC into five parts, four of which are congruent to the four parts of the square on AB. (Note that in order to show the congruency of two quadrilaterals it is enough to show that all their angles match and that one pair of corresponding sides is equal.)

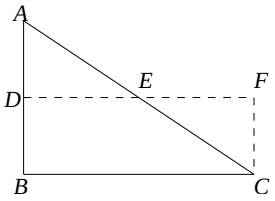


(iii) Let $ABCD$ be a rectangle with AB as a shorter side. If $AD \geq 2AB$, then cut off sufficient number of squares with sides equal to AB so as to be left with a rectangle $A'B'CD$ in which $A'B' < A'D < 2A'B$. If the problem can be solved for the rectangle $A'B'CD$ then by applying (ii) repeatedly, it can be solved for the original rectangle $ABCD$. So we suppose $AB < AD < 2AB$.



Following the hint, the hypothesis $AD < 2AB$ ensures that the point C lies on the side GH of the square $EGHB$. Let EG and CD meet at F. Through F and G draw lines parallel to AD ro meet AB and BE at P and Q respectively. Take R on BC so that $RC = AE$ (and hence $BR = ED$). Finally, through R draw a line parallel to AB to meet the side BH at S. Then the quadrilateral $BEFC$ is common to the rectangle $ABCD$ and to the square $BEGH$. The triangles PQB and EDF are congruent to the triangles FGC and BRS respectively. Finally, the quadrilaterals $QPAE$ and $HSRC$ are congruent to each other. It is interesting to note that both in (ii) and (iii) we have only shifted the various pieces of the figures. We have neither rotated nor flipped them any time.

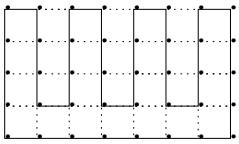
(iv) By decomposing the given triangle into two right-angled ones (by the altitude on its longest side) and using (iii) and (ii), it suffices to show that every right-angled triangle is equidecomposable to a rectangle. This can be done easily as shown in the figure by drawing a line through the centre of the hypotenuse parallel to one of the other two sides. Note that this time we do need to rotate the triangle ADE through $180°$ around E to get the triangle CFE.



80. Follow the hint. For (i), note that a function $f : X \longrightarrow Y$ is onto if and only if it is in none of the sets A_k's, i.e. if and only if it is in the intersection $\bigcap_{k=1}^{m} A'_k$. Apply Exercise (1.43). For (ii), every surjective function $f : X \longrightarrow Y$ induces a composition $(n_1, n_2, \ldots, n_m)$ of n where n_k is the cardinality of the set $f^{-1}(\{y_k\})$. Conversely, given one such composition of n, the n_1 elements that go to y_1 can be chosen in $\binom{n}{n_1}$ ways, having done this, those that go to y_2 can be chosen in $\binom{n-n_1}{n_2}$ ways and so on. Hence the number of surjective functions that induce the same composition $(n_1, n_2, \ldots, n_m)$ is $\frac{n!}{n_1!(n-n_1)!} \times \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \times \frac{(n-n_1-n_2)!}{n_3!(n-n_1-n_2-n_3)!} \times \ldots \times \frac{(n-n_1-n_2-\ldots-n_{m-1})!}{n_m!}$

(where the last factor actually equals 1). This is simply $\frac{n!}{n_1!n_2!\ldots n_m!}$.

81. A necessary and sufficient condition is that mn should be even, or equivalently, that at least one of m and n is even. The sufficiency is proved by actually drawing a Hamiltonian cycle in the manner illustrated in the figure (where $m = 8$ and $n = 5$). For the converse, assume that mn is odd. Without loss of generality assume that the distance between any two consecutive parallel roads is the same, say 1 unit. Then the houses can be assigned co-ordinates (x, y) where x, y are integers with $1 \leq x \leq m$ and $1 \leq y \leq n$. Each move changes the parity of $x + y$ from even to odd or vice versa. In any Hamiltonian cycle, there are exactly mn moves. As mn is odd, when you return to the starting point, the parity of $x+y$ will be reversed, a contradiction.



82. Assume $k = 3$ and let $d = a \oplus b \oplus c$ with $a = \sum_{i=0}^{m} a_i 2^i, b = \sum_{i=0}^{m} b_i 2^i, c = \sum_{i=0}^{m} c_i 2^i$ and $d = \sum_{i=0}^{m} d_i 2^i$ where at least one of a_m, b_m, c_m is non-zero. As $d \neq 0$, at least one d_i is non-zero. Let r be the highest index such that $d_r \neq 0$. Then $r \leq m$ and $d_{r+1} = d_{r+2} = \ldots = d_m = 0$. As $a_r + b_r + c_r$ is odd, at least one of them, say a_r is 1. Now define a' as $a' = \sum_{i=0}^{m} a'_i 2^i$ where we let $a'_i = a_i$ for all $i > r$, $a'_r = 0$ and for $i < r$, a'_i is 0 or 1 according as $b_i + c_i$ is even or odd. Then $a' \oplus b \oplus c = 0$. Also $a' < a$. Take $x = a - a'$.

83. (a) Suppose a player reaches 89 exactly. If his opponent selects x subsequently, he can then select $11-x$ to get exactly 100. (Note that $1 \leq x \leq 10$ ensures that $1 \leq (11 - x) \leq 10$.) By the same reasoning, a player who first reaches 78 can ensure his victory. Working backwards, the player who starts the game can ensure his victory by choosing 1 at the start. (b) Assume A plays first. Let r be the remainder when n is divided by $k+1$. If $r > 0$, then A has a winning strategy by removing r sticks at the start. But if $r = 0$, then B has a winning strategy. (c) Again assume that A starts the game. Suppose first that there are only two piles. If they are of unequal sizes then A has a winning strategy. He begins by equalising the piles. No matter what B does then, the two piles will be unequal. So A again makes them equal. If, however, the two piles are of the same size to begin with, then B has the winning strategy. The winning strategy here consists of making the digital sum of the sizes of the two piles vanish. If we have k (> 2) piles of sizes $a_1, a_2, \ldots, a_k$ to begin with and $a_1 \oplus a_2 \oplus \ldots \oplus a_k \neq 0$, then by Exercise (24.82), A can make the digital sum vanish by removing a suitable number of sticks from one of the piles. He continues to do this every time his turn comes up. If, however, $a_1 \oplus a_2 \oplus \ldots \oplus a_k = 0$, then by a similar reasoning, B has the winning strategy. (d) Call a position (x, y) on the chess board a winning position if a

player who plays first from it has a winning strategy, i.e., can ensure his victory. For example, for every $k \in \mathbb{N}$, $(k,0),(0,k)$ and (k,k) are winning positions. Call a position a losing position if every single move from it leads to a winning position. For example, $(2,1)$ and $(1,2)$ are losing positions. Note that every position which leads to some losing position in one move is a winning position. Both types of positions are symmetrically located about the diagonal $y = x$. The losing positions below the diagonal can then be determined recursively as $(2,1),(5,3),(7,4),(10,6),(13,8),\ldots$. In general, the k-th losing position is (a_k, b_k) where b_k is the least positive integer different from $a_1, b_1, \ldots, a_{k-1}, b_{k-1}$ and $a_k = b_k + k$. A closed form expression for b_k comes out, quite surprisingly, as $b_k = [k\alpha]$ where $\alpha = \frac{1+\sqrt{5}}{2}$ is the golden ratio of Exercise (6.45). For a proof and an alternate formulation in terms of the Fibonacci numbers, see p. 108 of *Challenging Mathematical Problems with Elementary Solutions, Volume II* by A. M. Yaglom and I. M. Yaglom.

84. (a) Trivial if $r_1 = r_2$. If $r_1 > r_2$ (say), drop a perpendicular M_2Q from M_2 to M_1P_1 where M_1, M_2 are the centres of the circles. Then $P_1P_2 = QM_2 = \sqrt{M_1M_2^2 - QM_1^2} = \sqrt{(r_1+r_2)^2 - (r_1-r_2)^2}$. (b) Let the tangent touch C at P. Then $P_1P_2 = P_1P + PP_2$. Apply (a) to each segment to get $\sqrt{r_1r_2} = \sqrt{r_1r} + \sqrt{rr_2}$. (c) First rewrite the relation to be proved as $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{2}{r}(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}) + 2(\frac{1}{r_1r_2} + \frac{1}{r_2r_3} + \frac{1}{r_3r_1})$ and hence as

$$(\sigma_2^2 - 4\sigma_1\sigma_3)r^2 - 2\sigma_2\sigma_3 r + \sigma_3^2 = 0 \tag{1}$$

The cosine formula gives $\cos\alpha = \dfrac{(r+r_2)^2 + (r+r_3)^2 - (r_2+r_3)^2}{2(r+r_2)(r+r_3)} = 1 - a$ where $a = \dfrac{2r_2r_3}{(r+r_2)(r+r_3)}$. Define b, c similarly and write $\cos\beta = 1-b$ and $\cos\gamma = 1-c$. In terms of a, b, c the trigonometric identity becomes

$$a^2 + b^2 + c^2 = 2(ab + bc + ca) - 2abc \tag{2}$$

Let $P = (r+r_1)(r+r_2)(r+r_3)$. Then direct computations give

$$\begin{aligned} a^2+b^2+c^2 &= \frac{4r_2^2r_3^2(r+r_1)^2 + 4r_3^2r_1^2(r+r_2)^2 + 4r_1^2r_2^2(r+r_3)^2}{P^2} \\ &= \frac{4r^2(r_2^2r_3^2 + r_3^2r_1^2 + r_1^2r_2^2) + 8r\sigma_2\sigma_3 + 12\sigma_3^2}{P^2} \\ &= \frac{4r^2(\sigma_2^2 - 2\sigma_1\sigma_3) + 8r\sigma_2\sigma_3 + 12\sigma_3^2}{P^2} \qquad (3) \\ ab+bc+ca &= \frac{4\sigma_3(r_1(r+r_2)(r+r_3) + r_2(r+r_3)(r+r_1) + r_3(r+r_1)(r+r_2))}{P^2} \\ &= \frac{4\sigma_3(\sigma_1r^2 + 2\sigma_2r + 3\sigma_3)}{P^2} \qquad (4) \end{aligned}$$

and finally,

$$abc = \frac{8\sigma_3^2}{P^2} \tag{5}$$

Substituting these into (2) gives (1) as was to be proved.

85. $n/2$. (Take advantage of cyclicity of the expression. By the A.M.-G.M. inequality, the maximum values of the terms are $\frac{1}{2}(\sin^2 x_1 + \cos^2 x_2)$, $\frac{1}{2}(\sin^2 x_2 + \cos^2 x_3)$, $\ldots$, $\frac{1}{2}(\sin^2 x_n + \cos^2 x_1)$. Add. Maximum occurs when every x_i equals $\pi/4$.

86. $\frac{k}{m+n+k}$. (The natural sample space here, say X, is very clumsy to describe. Let the symbols a, b, c correspond to the colours white, black and red respectively. Then X consists of ordered r-tuples of the form $(x_1, x_2, \ldots, x_r)$ where each x_i is either a, b or c and a, b, c occur at most m, n, k times respectively (which, in particular, means $r \leq m + n + k$). The integer r is not the same for all elements of X. It is an integer such that the r terms $x_1, x_2, \ldots, x_r$ contain exactly two of a, b, c the maximum number of times (viz., m, n, k respectively) and x_r is one of these two symbols. The last requirement ensures that we stop drawing the balls as soon as two of the three colours are exhausted. The favourable set consists of all such r-tuples in which x_r is either black or white. Elements of X are not equally likely and that makes this approach even more unwieldy. The trick lies in switching to a more manageable sample space, say Y, consisting of all sequences of length $m+n+k$ in which a, b, c appear exactly m, n, k times respectively. Physically, this means that we continue to draw the balls even after the balls left are of only one colour. Then Y has $\dfrac{(m+n+k)!}{m!n!k!}$ elements and these are all equally likely. The favourable set now consists of all elements of Y whose last term is c. So the problem is equivalent to finding the probablility that the last ball is red, when all the balls are drawn one by one without replacement. From the reasoning given at the end of Comment No. 5, Chapter 22, this probability is simply $\frac{k}{m+n+k}$. A more straightforward way is to express the number of favourable cases as $\dfrac{(m+n)!}{m!n!} \displaystyle\sum_{t=1}^{k} \binom{m+n+k-t-1}{k-t}$ where the parameter t denotes the number of red balls left at the end. This sum can be evaluated using the binomial identity (1) in Chapter 5, with a slight change of notation. In fact, the shorter proof above is akin to the combinatorial proof of this identity.)

87. (i) $\dfrac{\pi}{4}$ (ii) $\dfrac{3\pi}{4}$ (iii) $\tan\theta$. (For (iii), use $\dfrac{\tan\theta}{\cos 2\theta} = \tan 2\theta - \tan\theta$ to write the series as a telescopic series. Note that in order to go from the partial sums to the infinite sums, the first two series require the continuity of the arctan function at 1 along with Theorem 1 of Chapter 16, while the last one needs the continuity of the tan function at 0.)

88. From (4) in Chapter 11, with $A = \pi/2$ (say), $r = \frac{2R}{\sqrt{2}}(\cos\frac{B-C}{2} - \frac{1}{\sqrt{2}}) \leq \sqrt{2}R(1 - \frac{1}{\sqrt{2}})$. Simplify. For a more direct solution, take the sides as $1, \sin\theta, \cos\theta$, where θ is an acute angle. The desired inequality reduces to $1 + \sin\theta + \cos\theta \geq (1 + \sqrt{2})\sin 2\theta$, which follows by combining $1 \geq \sin 2\theta$ with $\sin\theta + \cos\theta \geq 2\sqrt{\sin\theta\cos\theta}$. As expected, equality holds when $\theta = \pi/4$, i.e. when the triangle is right-angled isosceles.

89. The conditions imply that d and $b - c$ are odd integers. If all the three zeros of $f(x)$, say r_1, r_2, r_3, are integers, then $d = -r_1r_2r_3$ implies that r_1, r_2, r_3 are all odd. But then $b - c = -(r_1 + r_2 + r_3) + r_1r_2 + r_2r_3 + r_3r_1$ would be even, a contradiction. For the second part, if r is an integer such that $r^3 + br^2 + cr + d = 0$, then r cannot be even since d is odd. But since b and c have different parities, r cannot be odd either.

89. Putting $t = \tan\theta$ reduces the equation to the quartic $t^4 + 4at^3 + 18t^2 + 4at + 1 = 0$. The positivity of all the roots implies $a < 0$. Because of the symmetry of the coefficients around the centre, the substitution $u = t + \frac{1}{t}$ reduces this further to $f(u) = 0$ where $f(u) = u^2 + 4au + 16$. The condition that the original equation has four distinct roots translates into $f(u)$ having two real zeros both greater than 2. As the leading coefficient of $f(u)$ is positive, this gives $f(2) > 0$, whence $a > -\frac{5}{2}$. Also, the discriminant criterion, coupled with $a < 0$ gives $a < -2$. Alternately, recast the given equation as $\sec^4\theta + 4a\tan\theta\sec\theta + 16\tan^2\theta = 0$ and hence as $g(x) = 0$ where $x = \sin 2\theta$ and $g(x) = 4x^2 + 2ax + 1 = 0$. Then $g(x)$ has two ditinct zeros in $(0, 1)$, whence $g(1) > 0$.

91. (i) $-\frac{28}{3}$ (ii) -11. (For (i) simply complete the squares. For (ii) first get rid of the xy term by rotating the axes through an angle α given by $\tan 2\alpha = \frac{2}{1-3} = -1$, as shown in Comment No. 18 in Chapter 10. Specifically, put $x = \dfrac{\sqrt{2-\sqrt{2}}}{2}X - \dfrac{\sqrt{2+\sqrt{2}}}{2}Y$ and $y = \dfrac{\sqrt{2+\sqrt{2}}}{2}X + \dfrac{\sqrt{2-\sqrt{2}}}{2}Y$. An alternate method is to find, for each fixed y, the minimum, say $m(y)$ of $f(x, y)$, treated as a function of x only, y being a constant. Then the minimum value of $m(y)$, treated as a function of y, will give the answer. For (i), this is trivial because $m(y) = 3y^2 - 2y - 9$ for all $y \in \mathbb{R}$. For (ii), $m(y) = 3y^2 - 2y - (y - 3)^2 = 2y^2 + 4y - 9$ whose minimum is -11. Here the second method is simpler and also has the advantage that it works for all kinds of functions $f(x, y)$ of two variables. Furhter, it also works when the point (x, y) is allowed to vary only over some subset, say S, of the plane. A similar strategy works for maximising a function too. Still, the method is seldom used because in general the calculation of $m(y)$ is complicated and even if we can do it, minimising it as a function of y is still more formidble. Therefore the methods based on partial derivatives of $f(x, y)$, as indicated in Comment No. 15 of Chapter 13 are used.)

92. (a) If P is an interior point of S, any line through it will have the desired property. Suppose P is a boundary point of S. If P lies on an edge,

say E of S, but is not a vertex of S, then take L to be the line which contains this edge. Now suppose P is a vertex. Let L be a straight line through P. If L passes through the angle of the polygon at the vertex P, then it contains points of S only on one side of P, because this angle is less than $180°$, as the polygon is convex. Otherwise, L meets S only at P. (b) Let L be a line through P as given by (a). Let Q_1 and Q_2 be the points of intersection of the line L with S. Then, restricted to this line, f increases on one side and decreases on the other as you move away from P. (f may remain constant on L, if L happens to be a level curve of f, i.e. a line of the form $ax + by = c$ for some c.) So we may suppose $f(Q_1) \le f(P) \le f(Q_2)$. Note that Q_1 and Q_2 lie on some edges of S. Moving further along these edges if necessary, we reach the desired vertices. (c) The vertex set of S is finite and so there exist vertices at which f has its maximum and minimum among the vertices. By (b), f also attains its maximum and minimum on S at these vertices.

93. (a) $x = \frac{1}{2}(b+c-a)$ which is positive if and only if $b+c > a$. Similarly for y and z. If a, b, c are the sides of a triangle, then x, y, z are the lengths of the tangents to its incircle from its vertices. (b) With Ravi substitution, $s = x + y + z$ and $\Delta = \sqrt{xyz(x+y+z)}$. The inequality to be proved reduces to showing that $(xy+yz+zx)^2 \ge 3xyz(x+y+z)$, which follows since $(xy-yz)^2 + (yz-zx)^2 + (zx-xy)^2 \ge 0$ with equality holding if and only if $x = y = z$.

94. (a) The first equality follows by equating the area Δ with rs and also with $\frac{abc}{4R}$. For the second, the Ravi substitution gives $ab + bc + ca = s^2 + xy + yz + zx$. Also, $r^2 = \frac{\Delta^2}{s^2} = \frac{xyz}{x+y+z}$ and $4Rr = \frac{abc}{s} = \frac{(x+y)(y+z)(z+x)}{x+y+z}$. The result follows by equating $xyz + (x+y)(y+z)(z+x)$ with $(x+y+z)(xy+yz+zx)$. For the second part, note that a, b, c are the roots of the cubic $t^3 - 2st + (s^2 + 4Rr + r^2)t - 4Rrs = 0$. So, R, r and s determine a, b, c. (Note that each of the triplets R, r, Δ and R, s, Δ also determines the triangle uniquely, but r, s, Δ does not because these attributes are not independent since $\Delta = rs$.) (b) Let the triangle be ABC with I as its incentre and without loss of generality assume that the line cuts the sides AB, AC in points B', C' respectively. Let the angle bisector of $\angle A$ meet BC and $B'C'$ at D and D' respectively. Let the sides of the triangle $AB'C'$ be a', b', c'. Then $AD' = \dfrac{2b'c'\cos\frac{A}{2}}{b'+c'}$. The data implies that $b'c' = \frac{1}{2}bc$ and $b' + c' = s$, using which it is easy to show that $AD' = AI$.

95. (a) A calculus-based proof can be given by considering the function $f(\theta_1) = \cot\theta_1 + \cot(\theta - \theta_1)$. For a purely trigonometric proof, write $\cot\theta_1 + \cot\theta_2$ as $\frac{\sin\theta}{\sin\theta_1 \sin\theta_2}$ and hence as $\frac{2\sin\theta}{\cos(\theta_1-\theta_2)-\cos\theta}$. Maximise the denominator. The calculus based proof works for (b) too. (c) Let $\theta_1 = \angle BAP$ and $\theta_2 = \angle PAC$. Then $\frac{b}{y} + \frac{c}{z} = \cot B + \cot C + \cot\theta_1 + \cot\theta_2$. Apply (a) with $\theta = A$. (d) The minimum occurs when P is the incentre of the triangle

ABC. (For a fixed x, P moves on a line parallel to BC and by (c), P has to lie on the angle bisector of the angle A. The same holds for the other two angles.)

96. (a) Fix the first triangle. Then the problem is of triangular optimisation for the second triangle. For a fixed C', apply part (b) of the last exercise with $\theta = \pi - C'$, to show that $a^2 \cot A' + b^2 \cot B'$ is minimum when $\frac{\sin A'}{\sin B'} = \frac{\sin A}{\sin B}$. So, for the minimum to occur, $\sin A', \sin B', \sin C'$ must be proportional to $\sin A, \sin B, \sin C$ respectively, giving $A' = A, B' = B$ and $C' = C$. The equality in this case follows from the identity $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$. (b) Multiply the L.H.S. of the inequality in (a) by $\frac{a'b'c'}{R'}$ and the R.H.S. by $4\Delta'$.

97. This is a booby trap. Solving the given differential equation is not easy. Nor is it needed! Instead, if the assertion fails, then without loss of generality, the maximum of f on $[a, b]$ is positive. Let the maximum occur at some $c \in (a, b)$. Then $f'(c) = 0$ and so $f''(c) = f(c) > 0$. But this means that f has a local minimum at c, contradicting that it has a maximum at c. (In essence, we have duplicated the proof of Rolle's theorem and then applied Theorem 8 of Chapter 16. Note that mere application of Rolle's theorem will not suffice. So, sometimes it pays to know the proof and not just the statement of a theorem.)

98. (a) Or else, A, B, C would all lie on the prependicular bisector of the segment PQ and hence would be collinear. (b) A circle centred at $A(= O)$ must be mapped to itself by T. Also $BC = \sqrt{2}$. Hence $B = (\cos\alpha, \sin\alpha)$ and $C = (\cos\beta, \sin\beta)$ for some α, β with $\beta = \alpha \pm \frac{\pi}{2}$. In the first case let S be the (counterclockwise) rotation around O through the angle α, while in the second case, let S be the reflection in the line $y = \tan(\frac{\alpha}{2})x$. Then S preserves orientation in the first case and reverses it in the second. But in either case, S is an isometry which takes $(0, 0), (1, 0)$ and $(0, 1)$ to A, B, C respectively. For any point P in the plane, the distances of $S(P)$ from A, B, C are the same as those of $T(P)$ from A, B, C respectively. So $T(P) = S(P)$. (c) Use polar coordinates. The geometric significance of α follows from the definition of S in the solution to (b). (See also Exercise (10.23).) (d) Let $A = T(O)$ and let S be the parallel translation by the vector $\overrightarrow{AO}$. Then the composite $S \circ T$ is an isometry which fixes the origin. Apply (b) to it and then get back T as $S^{-1} \circ (S \circ T)$. (e) Let t be the complex number representing the point $T(O)$. Apply (c) to $f(z) - t$ after identifying the column vector $\begin{pmatrix} x \\ y \end{pmatrix}$ with the complex number $x + iy$ and multiplication by the matrix $\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$ with multiplication by the complex number $a = e^{i\alpha} = \cos\alpha + i \sin\alpha$. Bijectivity of T follows by explictly constructing the inverse function, which is $f^{-1}(z) = \overline{a}(z - t)$ in the first case and $f^{-1}(z) = a(\overline{z} - \overline{t})$ in the second. Note that from the very

definition of an isometry, it is immediate that it is a one-to-one function. But its surjectivity is not so obvious. Indeed, a translation of a half plane into itself is not onto, even though it is an isometry. For the last part, suppose $a = e^{i\alpha} \neq 1$. The (unique) fixed point of f is $z_0 = \frac{t}{1-a}$. And, $f(z) = a(z - z_0) + z_0$, which shows that T is a counterclockwise rotation through the angle α around the point z_0. (f) From the problem solved in Comment No. 16 of Chapter 8, the reflection of z in L is $\dfrac{c - b\overline{z}}{\overline{b}} = bc - b^2\overline{z}$ since $|b| = 1$. Also the vector ib is parallel to L. So the translating vector is of the form ibd for some real d. In terms of matrices, this glide reflection is given by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c\cos\beta - d\sin\beta \\ d\cos\beta + c\sin\beta \end{pmatrix} - \begin{pmatrix} \cos 2\beta & -\sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

(g) All that remains is to show that given complex numbers a, t with $|a| = 1$, we can find real numbers c, d and a complex number b with $|b| = 1$ such that $a\overline{z} + t = bc - b^2\overline{z} + ibd$ for all z. For this, let b be either one of the two (complex) square roots of $-a$. Having chosen b, let c and d be, respectively, the real and imaginary parts of the complex number t/b. (h) This follows from (g) in view of the fact that the composite of two orientation preserving or two orientation reversing isometries is always orientation preserving. For a direct proof based on (e), suppose T_1, T_2 are rotations with corresponding functions $f_1(z) = a_1z + t_1$ and $f_2(z) = a_2z + t_2$. Then $T_2 \circ T_1$ is the isometry given by the function $f_3(z) = a_2(a_1z + t_1) + t_2 = a_1a_2z + (a_2t_1 + t_2)$. Note incidentally that $T_2 \circ T_1 \neq T_1 \circ T_2$ in general. However, if we let $a_k = e^{i\alpha_k}$ for $k = 1, 2$, then each composite is a rotation through the angle $\alpha_1 + \alpha_2$ which degenerates into a translation if $\alpha_1 + \alpha_2$ is a multiple of 2π (or equivalently, T_1, T_2 are rotations through the same angle but one clockwise and the other anticlockwise). A similar calculation shows that if $f_k(z) = a_k\overline{z} + t_k$ for $k = 1, 2$, then the composite $T_2 \circ T_1$ is given by the function $f_3(z) = a_2\overline{a_1}z + a_2\overline{t_1} + t_2$. This is a rotation which degenerates into a translation when $a_1 = a_2$, i.e. when both the glide reflections are w.r.t. lines that are parallel to each other.

99. (a) 298. ($100x + 10y + z = 2(x^2 + y^2 + z^2)$ implies z is even. Hence $y^2 + z^2 \leq 9^2 + 8^2 = 145$ which forces $x \leq 2$. Then apply trial and error.) (b) Let r be the leading digit of 2^n as well as of 5^n. Then $r2^a \leq 2^n < (r + 1)2^a$ and $r2^b \leq 5^n < (r + 1)2^b$ for some non-negative integers a and b. Multiplying these inequalities, show that the interval $[r^2, (r + 1)^2)$ must contain some power of 10. This is possible only for $r = 3$.

100. 2,3 and 5. (For a desired prime p, $(p-1)! + 1 = p^m$ for some $m \geq 1$. This gives $(p-1)! = ((p-1) + 1)^m - 1$. Expand by the binomial theorem and note that for $p > 5$, $(p-1)!$ is divisible by $(p-1)^2$ to get that if $p > 5$ then $p - 1$ is divisible by $(p-1)^2$, a contradiction.

Appendix 1

1. If the diagonal entries are $a_1, a_2, \ldots, a_n$, a direct computation shows that the characteristic polynomial is simply $(a_1 - \lambda)(a_2 - \lambda)\ldots(a_n - \lambda)$.

2. As the linear transformation is defined by multiplication of a matrix and a column vector, the result follows from Exercise (3.24).

3. Write P as $\begin{bmatrix} \cos 30^\circ & \sin 30^\circ \\ -\sin 30^\circ & \cos 30^\circ \end{bmatrix}$. Then, for every column vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, we have

$$P \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \cos 30^\circ - x_2 \sin 30^\circ \\ x_1 \sin 30^\circ + x_2 \cos 30^\circ \end{bmatrix}$$

which shows that the transformation induced by P is a rotation through 30°. So, for every positive integer n, P^n represents a rotation through $\frac{n\pi}{6}$. Therefore, P^{12} is the identity matrix. Since $2005 \equiv 1 \ (mod\ 12)$, $P^{2005} = P$.

4. An odd integer. (The equality $(A+B)(A-B) = (A-B)(A+B)$ reduces to $BA - AB = AB - BA$ which means $AB = BA$. Taking transposes, $(AB)^t = B^tA^t = -BA = -AB = (-1)^k AB$.)

5. There are no values of α for which $A^2 = B$. ($A^2 = B$ gives $\begin{bmatrix} \alpha^2 & 0 \\ \alpha + 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$. The equations $\alpha^2 = 1$ and $\alpha + 1 = 5$ have no solution.)

6. All three statements are true. (Write the system as $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ k \\ 1 \end{bmatrix}$ where $A = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 1 & -3 & 4 \end{bmatrix}$. Then $\det(A) = 0$ and so A is singular. In such a case, for a solution to exist, it is necessary that all the matrices obtained by replacing one column of A by the column vector on the right must also be singular. In the present case, if we replace the second column, the resulting matrix is $\begin{bmatrix} 1 & -1 & 3 \\ -1 & k & -2 \\ 1 & 1 & 4 \end{bmatrix}$ whose determinant is $k - 3$ and hence non-zero for $k \neq 3$. The determinant in the question is obtained by interchanging the second and the third column of this matrix. So, it, too, is non-zero for $k \neq 3$. Hence the system has no solution for $k \neq 3$.

Appendix 2

1. $\frac{9}{2}$ cu. units. (Take the plane as $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. Then the desired volume is $\frac{1}{6}abc$. As the plane passes through $(1, 1, 1)$, we have $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$. Also

its being parallel to the vectors $\mathbf{i}-\mathbf{k}$ and $-\mathbf{i}+\mathbf{j}$ gives two more equations in a, b, c.)

2. $\frac{17}{5\sqrt{3}}$ units. (Take the projection of $3\hat{i}+4\hat{k}$ on $(\hat{i}+2\hat{j}+3\hat{k})\times(3\hat{i}+\hat{j}+2\hat{k})$.)

3. $x+y-2z=3$, $Q=(6,5,-2)$. (Let $Q=(a,b,c)$. Then the segment joining $(2,1,6)$ to (a,b,c) is perpendicular to the plane and its midpoint lies on the plane.)

4. $\{v_1\hat{i}+\hat{j}-v_1\hat{k} : v_1 \in I\!R\}$. (Let $\mathbf{v}=v_1\hat{i}+v_2\hat{j}+v_3\hat{k}$ be a vector with given properties. Perpendicularity of $\mathbf{v}$ to $\mathbf{a}\times\mathbf{b}$ gives $v_1=v_3$ while the second requirement gives $\mathbf{v}\cdot\mathbf{c}=1$.)

5. 45° (or, 135°). ($\vec{A}$ is parallel to $\mathbf{n}_1\times\mathbf{n}_2$ where $\mathbf{n}_1=(2\hat{j}+3\hat{k})\times(4\hat{j}-3\hat{k})$ and $\mathbf{n}_2=(\hat{j}-\hat{k})\times(3\hat{i}+3\hat{j})$. This gives that $\vec{A}$ is parallel to $-\hat{i}+\hat{k}$.)

6. (i) $\frac{\sqrt{5}}{3}$. (The given line, say L, is parallel to $\hat{i}+2\hat{j}+2\hat{k}$. So, a typical point on it is $P=(t,1+2t,2t)$. Determine t so that OP is perpendicular to L. Or, mimimise the distance OP which equals $\sqrt{9t^2+4t+1}$.) (ii) $2\sqrt{2}$. (The given plane comes out to be $x+y+1=0$.)

7. $\cos^{-1}(\frac{1}{3})$. (More generally, if a, b, c are the lengths of the sides of a rectangulalr parallelopiped, then the cosines of the angles between the four diagonals are all possible values of the expression $\dfrac{\pm a^2\pm b^2\pm c^2}{a^2+b^2+c^2}$ where out of the 8 possible sign combinations, those where all three signs are the same are to be excluded. For a vector based proof, take the origin at the centre of the parallelopiped and the vertices as $\frac{1}{2}(\pm a\hat{i}\pm b\hat{j}\pm c\hat{k})$.)

8. Call the points (x_1,y_1,z_1) and (a,b,c) as P and A respectively. Let the perpendicular from P to the line fall at Q. A unit vector $\mathbf{u}$ along the line L is $l\hat{i}+m\hat{j}+n\hat{k}$. Then AQ is the projection of AP on L and so $d^2=\|\overrightarrow{AP}\|^2-\|\overrightarrow{AQ}\|^2=\|\overrightarrow{AP}\|^2-(\overrightarrow{AP}\cdot\mathbf{u})^2=\|\overrightarrow{AP}\times\mathbf{u}\|^2$.